

GRADUATE STUDIES
IN MATHEMATICS | 8 |

A First Course in Sobolev Spaces

Second Edition

Giovanni Leoni



American Mathematical Society

A First Course
in Sobolev Spaces
Second Edition

GRADUATE STUDIES
IN MATHEMATICS **| 8 |**

A First Course in Sobolev Spaces

Second Edition

Giovanni Leoni



American Mathematical Society
Providence, Rhode Island

EDITORIAL COMMITTEE

Dan Abramovich
Daniel S. Freed (Chair)
Gigliola Staffilani
Jeff A. Viaclovsky

2010 *Mathematics Subject Classification*. Primary 46E35; Secondary 26A27, 26A30, 26A42, 26A45, 26A46, 26A48, 26B30, 30H25.

For additional information and updates on this book, visit
www.ams.org/bookpages/gsm-181

Library of Congress Cataloging-in-Publication Data

Names: Leoni, Giovanni, 1967–

Title: A first course in Sobolev spaces / Giovanni Leoni.

Description: Second edition. | Providence, Rhode Island : American Mathematical Society, [2017]

| Series: Graduate studies in mathematics; volume 181 | Includes bibliographical references and index.

Identifiers: LCCN 2017009991 | ISBN 9781470429218 (alk. paper)

Subjects: LCSH: Sobolev spaces.

Classification: LCC QA323 .L46 2017 | DDC 515/.782–dc23

LC record available at <https://lcn.loc.gov/2017009991>

Copying and reprinting. Individual readers of this publication, and nonprofit libraries acting for them, are permitted to make fair use of the material, such as to copy select pages for use in teaching or research. Permission is granted to quote brief passages from this publication in reviews, provided the customary acknowledgment of the source is given.

Republication, systematic copying, or multiple reproduction of any material in this publication is permitted only under license from the American Mathematical Society. Permissions to reuse portions of AMS publication content are handled by Copyright Clearance Center's RightsLink® service. For more information, please visit: <http://www.ams.org/rightslink>.

Send requests for translation rights and licensed reprints to reprint-permission@ams.org.

Excluded from these provisions is material for which the author holds copyright. In such cases, requests for permission to reuse or reprint material should be addressed directly to the author(s). Copyright ownership is indicated on the copyright page, or on the lower right-hand corner of the first page of each article within proceedings volumes.

© 2017 by the American Mathematical Society. All rights reserved.

The American Mathematical Society retains all rights
except those granted to the United States Government.
Printed in the United States of America.

⊗ The paper used in this book is acid-free and falls within the guidelines
established to ensure permanence and durability.
Visit the AMS home page at <http://www.ams.org/>

10 9 8 7 6 5 4 3 2 1 22 21 20 19 18 17

In memory of my Ph.D. advisor, James Serrin

Contents

Preface	xiii
Preface to the Second Edition	xiii
Preface to the First Edition	xv
Acknowledgments	xxi
Second Edition	xxi
First Edition	xxii
Part 1. Functions of One Variable	
Chapter 1. Monotone Functions	3
§1.1. Continuity	3
§1.2. Differentiability	9
Chapter 2. Functions of Bounded Pointwise Variation	29
§2.1. Pointwise Variation	29
§2.2. Continuity	34
§2.3. Differentiability	40
§2.4. Monotone Versus BPV	44
§2.5. The Space $BPV(I; Y)$	47
§2.6. Composition in $BPV(I; Y)$	55
§2.7. Banach Indicatrix	59
Chapter 3. Absolutely Continuous Functions	67
§3.1. $AC(I; Y)$ Versus $BPV(I; Y)$	67
§3.2. The Fundamental Theorem of Calculus	71

§3.3.	Lusin (N) Property	84
§3.4.	Superposition in $AC(I; Y)$	91
§3.5.	Chain Rule	95
§3.6.	Change of Variables	100
§3.7.	Singular Functions	103
Chapter 4.	Decreasing Rearrangement	111
§4.1.	Definition and First Properties	111
§4.2.	Function Spaces and Decreasing Rearrangement	126
Chapter 5.	Curves	133
§5.1.	Rectifiable Curves	133
§5.2.	Arclength	143
§5.3.	Length Distance	146
§5.4.	Curves and Hausdorff Measure	149
§5.5.	Jordan's Curve Theorem	152
Chapter 6.	Lebesgue–Stieltjes Measures	157
§6.1.	Measures Versus Increasing Functions	157
§6.2.	Vector-valued Measures Versus $BPV(I; Y)$	168
§6.3.	Decomposition of Measures	177
Chapter 7.	Functions of Bounded Variation and Sobolev Functions	183
§7.1.	$BV(\Omega)$ Versus $BPV(\Omega)$	183
§7.2.	Sobolev Functions Versus Absolutely Continuous Functions	188
§7.3.	Interpolation Inequalities	196
Chapter 8.	The Infinite-Dimensional Case	205
§8.1.	The Bochner Integral	205
§8.2.	L^p Spaces on Banach Spaces	212
§8.3.	Functions of Bounded Pointwise Variation	220
§8.4.	Absolute Continuous Functions	224
§8.5.	Sobolev Functions	229
Part 2. Functions of Several Variables		
Chapter 9.	Change of Variables and the Divergence Theorem	239
§9.1.	Directional Derivatives and Differentiability	239
§9.2.	Lipschitz Continuous Functions	242
§9.3.	The Area Formula: The C^1 Case	249

§9.4. The Area Formula: The Differentiable Case	262
§9.5. The Divergence Theorem	273
Chapter 10. Distributions	281
§10.1. The Spaces $\mathcal{D}_K(\Omega)$, $\mathcal{D}(\Omega)$, and $\mathcal{D}'(\Omega)$	281
§10.2. Order of a Distribution	288
§10.3. Derivatives of Distributions and Distributions as Derivatives	290
§10.4. Rapidly Decreasing Functions and Tempered Distributions	298
§10.5. Convolutions	302
§10.6. Convolution of Distributions	305
§10.7. Fourier Transforms	309
§10.8. Littlewood–Paley Decomposition	316
Chapter 11. Sobolev Spaces	319
§11.1. Definition and Main Properties	319
§11.2. Density of Smooth Functions	325
§11.3. Absolute Continuity on Lines	336
§11.4. Duals and Weak Convergence	344
§11.5. A Characterization of $W^{1,p}(\Omega)$	349
Chapter 12. Sobolev Spaces: Embeddings	355
§12.1. Embeddings: $mp < N$	356
§12.2. Embeddings: $mp = N$	372
§12.3. Embeddings: $mp > N$	378
§12.4. Superposition	387
§12.5. Interpolation Inequalities in $\mathbb{R}^N$	399
Chapter 13. Sobolev Spaces: Further Properties	411
§13.1. Extension Domains	411
§13.2. Poincaré Inequalities	430
§13.3. Interpolation Inequalities in Domains	449
Chapter 14. Functions of Bounded Variation	459
§14.1. Definition and Main Properties	459
§14.2. Approximation by Smooth Functions	462
§14.3. Bounded Pointwise Variation on Lines	468
§14.4. Coarea Formula for BV Functions	478
§14.5. Embeddings and Isoperimetric Inequalities	482

§14.6. Density of Smooth Sets	489
§14.7. A Characterization of $BV(\Omega)$	493
Chapter 15. Sobolev Spaces: Symmetrization	497
§15.1. Symmetrization in L^p Spaces	497
§15.2. Lorentz Spaces	502
§15.3. Symmetrization of $W^{1,p}$ and BV Functions	504
§15.4. Sobolev Embeddings Revisited	510
Chapter 16. Interpolation of Banach Spaces	517
§16.1. Interpolation: K -Method	517
§16.2. Interpolation: J -Method	526
§16.3. Duality	530
§16.4. Lorentz Spaces as Interpolation Spaces	535
Chapter 17. Besov Spaces	539
§17.1. Besov Spaces $B_q^{s,p}$	539
§17.2. Some Equivalent Seminorms	545
§17.3. Besov Spaces as Interpolation Spaces	551
§17.4. Sobolev Embeddings	561
§17.5. The Limit of $B_q^{s,p}$ as $s \rightarrow 0^+$ and $s \rightarrow m^-$	565
§17.6. Besov Spaces and Derivatives	571
§17.7. Yet Another Equivalent Norm	578
§17.8. And More Embeddings	585
Chapter 18. Sobolev Spaces: Traces	591
§18.1. The Trace Operator	592
§18.2. Traces of Functions in $W^{1,1}(\Omega)$	598
§18.3. Traces of Functions in $BV(\Omega)$	605
§18.4. Traces of Functions in $W^{1,p}(\Omega)$, $p > 1$	606
§18.5. Traces of Functions in $W^{m,1}(\Omega)$	621
§18.6. Traces of Functions in $W^{m,p}(\Omega)$, $p > 1$	626
§18.7. Besov Spaces and Weighted Sobolev Spaces	626
Appendix A. Functional Analysis	635
§A.1. Topological Spaces	635
§A.2. Metric Spaces	638
§A.3. Topological Vector Spaces	639

§A.4. Normed Spaces	643
§A.5. Weak Topologies	645
§A.6. Hilbert Spaces	648
Appendix B. Measures	651
§B.1. Outer Measures and Measures	651
§B.2. Measurable and Integrable Functions	655
§B.3. Integrals Depending on a Parameter	662
§B.4. Product Spaces	663
§B.5. Radon–Nikodym’s and Lebesgue’s Decomposition Theorems	665
§B.6. Signed Measures	666
§B.7. L^p Spaces	668
§B.8. Modes of Convergence	673
§B.9. Radon Measures	676
§B.10. Covering Theorems in $\mathbb{R}^N$	678
Appendix C. The Lebesgue and Hausdorff Measures	681
§C.1. The Lebesgue Measure	681
§C.2. The Brunn–Minkowski Inequality	683
§C.3. Mollifiers	687
§C.4. Maximal Functions	694
§C.5. BMO Spaces	695
§C.6. Hardy’s Inequality	698
§C.7. Hausdorff Measures	699
Appendix D. Notes	703
Appendix E. Notation and List of Symbols	711
Bibliography	717
Index	729

Preface

The Author List, I: giving credit where credit is due. The first author: Senior grad student in the project. Made the figures.

—Jorge Cham, www.phdcomics.com

Preface to the Second Edition

There are a lot of changes in the second edition. In the first part of the book, starting from Chapter 2, instead of considering real-valued functions, I treat functions $u : I \rightarrow Y$, where $I \subseteq \mathbb{R}$ is an interval and Y is a metric space. This change is motivated by the addition of a new chapter, Chapter 8, where I introduce the Bochner integral and study functions mapping time into Banach spaces. This type of functions plays a crucial role in evolution equations.

Another important addition in the first part of the book is Section 7 in Chapter 7, which begins the study of interpolation inequalities for Sobolev functions of one variable. One wants to estimate some appropriate norm of an intermediate derivative in terms of the norms of the function and the highest-order derivative.

Except for Chapter 8, **the first part of the book is meant as a textbook for an advanced undergraduate course or beginning graduate course on real analysis or functions of one variable.** One should simply take Y to be the real line $\mathbb{R}$ so that $\mathcal{H}^1$ reduces to the Lebesgue measure $\mathcal{L}^1$. All the results needed from measure theory are listed in the appendices at the end of the book.

The second part of the book starts with Chapter 9, which went through drastic changes. In the revised version I give an overview of classical results for functions of several variables, which are somehow scattered in the literature. These include Rademacher's and Stepanoff's differentiability theorems,

Whitney's extension theorem, and Brouwer's fixed point theorem. The focus of the chapter is now the divergence theorem for Lipschitz domains. While this fundamental result is quoted and used in every book on partial differential equations, it's hard to find a thorough proof in the literature. To introduce the surface integral on the boundary I start by proving the area formula, first in the C^1 case, and then, using Whitney's extension theorem in the Lipschitz case.

In the chapter on distributions, Chapter 10, I added rapidly decreasing functions, tempered distributions, and Fourier transforms. This was long overdue.

The book is structured in such a way that an instructor of a course on Sobolev spaces could actually skip Chapters 9 and 10, which serve mainly as reference chapters and jump to Chapter 11.

Chapters 11, 12, and 13, the first part of Chapter 18 could be used as a textbook on a course on Sobolev spaces. They are mostly self-contained.

One of the main changes in these chapters is that I caved in and decided to include higher order derivatives. The reason why I did not do it in the first edition was because standard operations like the product rule and the chain formula become incredibly messy for higher order derivatives and there are so many multi-indices to take into account that even elementary proofs become unnecessarily complicated. My compromise is to present proofs first in $W^{1,p}(\Omega)$ (first-order derivatives) or in $W^{2,p}(\Omega)$ (second-order derivatives) and only after, when the idea of the proof is clear, to do the general case $W^{m,p}(\Omega)$. I did not always follow this rule, since sometimes there was no significant change in difficulty in treating $W^{m,p}(\Omega)$ rather than $W^{1,p}(\Omega)$.

The advantage in having higher order derivatives is that I can now prove the classical interpolation inequalities of Gagliardo and Nirenberg. These are done in Section 12.5 in Chapter 12 for $\mathbb{R}^N$ and in Section 13.3 in Chapter 13 for uniformly Lipschitz domains. The main novelty with respect to other textbooks is that in the case of uniformly Lipschitz domains I do not assume that functions are in $W^{m,p}(\Omega)$ but only that u is in $L^q(\Omega)$ and the weak derivatives of order m are in $L^p(\Omega)$.

Another new section is Section 12.4 in Chapter 12, where I study superposition in Sobolev spaces.

The last major departure from the first edition is the chapter on Besov spaces, Chapter 17. This chapter was completely rewritten in collaboration with Ian Tice. The main motivation behind the changes was the proof that the trace space of functions in $W^{2,1}(\mathbb{R}^N)$ is given by the Besov space $B^{1,1}(\mathbb{R}^{N-1})$ (see Chapter 18). I am only aware of one complete and simple

proof, which is in a recent paper of Mironescu and Russ [173]. It makes use of two equivalent norms for $B^{1,1}(\mathbb{R}^{N-1})$, one using second order difference quotients and the other the Littlewood-Paley norm. To introduce the second norm, I went through several different versions of Chapter 17. Eventually, to study Besov spaces I used heavily the K method of real interpolation introduced by Peetre. The interpolation theory needed was added in a new chapter, Chapter 16.

Webpage for mistakes, comments, and exercises: The AMS is hosting a webpage for this book at

<http://www.ams.org/bookpages/gsm-181/>

where updates, corrections, and other material may be found.

Preface to the First Edition

There are two ways to introduce Sobolev spaces: The first is through the elegant (and abstract) theory of distributions developed by Laurent Schwartz in the late 1940s; the second is to look at them as the natural development and unfolding of monotone, absolutely continuous, and BV functions¹ of one variable.

To my knowledge, this is one of the first books to follow the second approach. I was more or less forced into it: This book is based on a series of lecture notes that I wrote for the graduate course “Sobolev Spaces”, which I taught in the fall of 2006 and then again in the fall of 2008 at Carnegie Mellon University. In 2006, during the first lecture, I found out that half of the students were beginning graduate students with no background in functional analysis (which was offered only in the spring) and very little in measure theory (which, luckily, was offered in the fall). At that point I had two choices: continue with a classical course on Sobolev spaces and thus lose half the class after the second lecture or toss my notes and rethink the entire operation, which is what I ended up doing.

I decided to begin with monotone functions and with the Lebesgue differentiation theorem. To my surprise, none of the students taking the class had actually seen its proof.

I then continued with functions of bounded pointwise variation and absolutely continuous functions. While these are included in most books on real analysis/measure theory, here the perspective and focus are rather different, in view of their applications to Sobolev functions. Just to give an example, most books study these functions when the domain is either the closed interval $[a, b]$ or $\mathbb{R}$. I needed, of course, open intervals (possibly unbounded).

¹BV functions are functions of bounded variation.

This changed things quite a bit. A lot of the simple characterizations that hold in $[a, b]$ fall apart when working with arbitrary unbounded intervals.

After the first three chapters, in the course I actually jumped to Chapter 7, which relates absolutely continuous functions with Sobolev functions of one variable, and then started with Sobolev functions of several variables. In the book I included three more chapters: Chapter 5 studies curves and arclength. I think it is useful for students to see the relation between rectifiable curves and functions with bounded pointwise variation.

Some classical results on curves that most students in analysis have heard of, but whose proof they have not seen, are included, among them Peano's filling curve and the Jordan curve theorem.

Section 5.4 is more advanced. It relates rectifiable curves with the $\mathcal{H}^1$ Hausdorff measure. Besides Hausdorff measures, it also makes use of the Vitali–Besicovitch covering theorem. All these results are listed in Appendices B and C.

Chapter 6 introduces Lebesgue–Stieltjes measures. The reading of this chapter requires some notions and results from abstract measure theory. Again it departs slightly from modern books on measure theory, which introduce Lebesgue–Stieltjes measures only for right continuous (or left) functions. I needed them for an arbitrary function, increasing or with bounded pointwise variation. Here, I used the monograph of Saks [201]. I am not completely satisfied with this chapter: I have the impression that some of the proofs could have been simplified more using the results in the previous chapters. Readers' comments will be welcome

Chapter 4 introduces the notion of decreasing rearrangement. I used some of these results in the second part of the book (for Sobolev and Besov functions). But I also thought that this chapter would be appropriate for the first part. The basic question is how to modify a function that is not monotone into one that is, keeping most of the good properties of the original function. While the first part of the chapter is standard, the results in the last two sections are not covered in detail in classical books on the subject.

As a final comment, the first part of the book could be used for an advanced undergraduate course or beginning graduate course on real analysis or functions of one variable.

The second part of the book starts with one chapter on absolutely continuous transformations from domains of $\mathbb{R}^N$ into $\mathbb{R}^N$. I did not cover this chapter in class, but I do think it is important in the book in view of its ties with the previous chapters and their applications to the change of variables formula for multiple integrals and of the renewed interest in the subject in recent years. I only scratched the surface here.

Chapter 10 introduces briefly the theory of distributions. The book is structured in such a way that an instructor could actually skip it in case the students do not have the necessary background in functional analysis (as was true in my case). However, if the students do have the proper background, then I would definitely recommend including the chapter in a course. It is really important.

Chapter 11 starts (at long last) with Sobolev functions of several variables. Here, I would like to warn the reader about two quite common misconceptions. Believe it or not, if you ask a student what a Sobolev function is, often the answer is “A Sobolev function is a function in L^p whose derivative is in L^p .” This makes the Cantor function a Sobolev function :(

I hope that the first part of the book will help students to avoid this danger.

The other common misconception is, in a sense, quite the opposite, namely to think of weak derivatives in a very abstract way not related to the classical derivatives. One of the main points of this book is that weak derivatives of a Sobolev function (but not of a function in BV!) are simply (classical) derivatives of a good representative. Again, I hope that the first part of the volume will help here.

Chapters 11, 12, and 13 cover most of the classical theorems (density, absolute continuity on lines, embeddings, chain rule, change of variables, extensions, duals). This part of the book is more classical, although it contains a few results published in recent years.

Chapter 14 deals with functions of bounded variation of several variables. I covered here only those parts that did not require too much background in measure theory and geometric measure theory. This means that the fundamental results of De Giorgi, Federer, and many others are not included here. Again, I only scratched the surface of functions of bounded variation. My hope is that this volume will help students to build a solid background, which will allow them to read more advanced texts on the subject.

Chapter 17 is dedicated to the theory of Besov spaces. There are essentially three ways to look at these spaces. One of the most successful is to see them as an example/by-product of interpolation theory (see [7], [232], and [233]). Interpolation is very elegant, and its abstract framework can be used to treat quite general situations well beyond Sobolev and Besov spaces.

There are two reasons for why I decided not to use it: First, it would depart from the spirit of the book, which leans more towards measure theory and real analysis and less towards functional analysis. The second reason is that in recent years in calculus of variations there has been an increased

interest in nonlocal functionals. I thought it could be useful to present some techniques and tricks for fractional integrals.

The second approach is to use tempered distributions and Fourier theory to introduce Besov spaces. This approach has been particularly successful for its applications to harmonic analysis. Again it is not consistent with the remainder of the book.

This left me with the approach of the Russian school, which relies mostly on the inequalities of Hardy, Hölder, and Young, together with some integral identities. The main references for this chapter are the books of Besov, Il'in, and Nikol'skiĭ [26], [27].

I spent an entire summer working on this chapter, but I am still not happy with it. In particular, I kept thinking that there should be easier and more elegant proofs of some of the results, but I could not find one.

In Chapter 18 I discuss traces of Sobolev and BV functions. Although in this book I only treat first-order Sobolev spaces, the reason I decided to use Besov spaces over fractional Sobolev spaces (note that in the range of exponents treated in this book these spaces coincide, since their norms are equivalent) is that the traces of functions in $W^{k,1}(\Omega)$ live in the Besov space $B^{k-1,1}(\partial\Omega)$, and thus a unified theory of traces for Sobolev spaces can only be done in the framework of Besov spaces.

Finally, Chapter 15 is devoted to the theory of symmetrization in Sobolev and BV spaces. This part of the theory of Sobolev spaces, which is often missing in classical textbooks, has important applications in sharp embedding constants, in the embedding $N = p$, as well as in partial differential equations.

In Appendices A, B, and C I included essentially all the results from functional analysis and measure theory that I used in the text. I only proved those results that cannot be found in classical textbooks.

What is missing in this book: For didactic purposes, when I started to write this book, I decided to focus on first-order Sobolev spaces. In my original plan I actually meant to write a few chapters on higher-order Sobolev and Besov spaces to be put at the end of the book. Eventually I gave up: It would have taken too much time to do a good job, and the book was already too long.

As a consequence, interpolation inequalities between intermediate derivatives are missing. They are treated extensively in [7].

Another important theorem that I considered adding and then abandoned for lack of time was Jones's extension theorem [122].

Chapter 14, the chapter on BV functions of several variables, is quite minimal. As I wrote there, I only touched the tip of the iceberg. Good

reference books of all the fundamental results that are not included here are [10], [72], and [251].

References: The rule of thumb here is simple: I only quoted papers and books that I actually read at some point (well, there are a few papers in German, and although I do have a copy of them, I only “read” them in a weak sense, since I do not know the language). I believe that misquoting a paper is somewhat worse than not quoting it. Hence, if an important and relevant paper is not listed in the references, very likely it is because I either forgot to add it or was not aware of it. While most authors write books because they are experts in a particular field, I write them because I want to learn a particular topic. I claim no expertise on Sobolev spaces.

Webpage for mistakes, comments, and exercises: In a book of this length and with an author a bit absent-minded, typos and errors are almost inevitable. I will be very grateful to those readers who write to giovanni@andrew.cmu.edu indicating those errors that they have found. The AMS is hosting a webpage for this book at

<http://www.ams.org/bookpages/gsm-105/>

where updates, corrections, and other material may be found.

The book contains more than 200 exercises, but they are not equally distributed. There are several on the parts of the book that I actually taught, while other chapters do not have as many. If you have any interesting exercises, I will be happy to post them on the web page.

Giovanni Leoni

Acknowledgments

The Author List, II. The second author: Grad student in the lab that has nothing to do with this project, but was included because he/she hung around the group meetings (usually for the food). The third author: First year student who actually did the experiments, performed the analysis and wrote the whole paper. Thinks being third author is “fair”.

— Jorge Cham, www.phdcomics.com

Second Edition

I am profoundly indebted to Ian Tice for months spent discussing several parts of the book, especially Section 12.5 and Chapters 16 and 17. In particular, Chapter 17 was really a collaborative effort. Any mistake is purely due to me. Thanks, Ian, I owe you a big one.

I would like to thank all the readers who sent corrections and suggestions to improve the first edition over the years. I would also like to thank the Friday afternoon reading club (Riccardo Cristoferi, Giovanni Gravina, Matteo Rinaldi) for reading parts of the book.

I am really grateful to to Sergei Gelfand, AMS publisher, and to the all the AMS staff I interacted with, especially to Christine Thivierge, for her constant help and technical support during the preparation of this book, and to Luann Cole and Mike Saitas for editing the manuscript.

I would like to acknowledge the Center for Nonlinear Analysis (NSF PIRE Grant No. OISE-0967140) for its support during the preparation of this book. This research was partially supported by the National Science Foundation under Grants No. DMS-1412095 and DMS-1714098.

Also for this edition, many thanks must go to all the people who work at the interlibrary loan of Carnegie Mellon University for always finding in a timely fashion all the articles I needed.

The picture on the back cover of the book was taken by Adella Guo, a student from Carnegie Mellon School of Design, whom I would like to thank.

Finally, I would like to thank Jorge Cham for giving me permission to continue to use quotes from www.phdcomics.com for the second edition. They are of course the best part of the book :)

First Edition

I am profoundly indebted to Pietro Siorpaes for his careful and critical reading of the manuscript and for catching $2^{\aleph_0}$ mistakes in previous drafts. All remaining errors are, of course, mine.

Several iterations of the manuscript benefited from the input, suggestions, and encouragement of many colleagues and students, in particular, Filippo Cagnetti, Irene Fonseca, Nicola Fusco, Bill Hrusa, Bernd Kawohl, Francesco Maggi, Jan Malý, Massimiliano Morini, Roy Nicolaides, Ernest Schimmerling, and all the students who took the Ph.D. courses “Sobolev spaces” (fall 2006 and fall 2008) and “Measure and Integration” (fall 2007 and fall 2008) taught at Carnegie Mellon University. A special thanks to Eva Eggeling who translated an entire paper from German for me (and only after I realized I did not need it; sorry, Eva!).

The picture on the back cover of the book was taken by Monica Montagnani with the assistance of Alessandrini Alessandra (always trust your high school friends for a good laugh... at your expense).

I am really grateful to Edward Dunne and Cristin Zannella for their constant help and technical support during the preparation of this book. I would also like to thank Arlene O’Sean for editing the manuscript, Lori Nero for drawing the pictures, and all the other staff at the AMS I interacted with.

I would like to thank three anonymous referees for useful suggestions that led me to change and add several parts of the manuscript. Many thanks must go to all the people who work at the interlibrary loan of Carnegie Mellon University for always finding in a timely fashion all the articles I needed.

I would like to acknowledge the Center for Nonlinear Analysis (NSF Grant Nos. DMS-9803791 and DMS-0405343) for its support during the preparation of this book. This research was partially supported by the National Science Foundation under Grant No. DMS-0708039.

Finally, I would like to thank Jorge Cham for giving me permission to use some of the quotes from www.phdcomics.com. They are really funny.

Part 1

Functions of One Variable

Monotone Functions

Undergradese, I: “Is it going to be an open book exam?” Translation: “I don’t have to actually memorize anything, do I?”

— Jorge Cham, www.phdcomics.com

In this chapter we study continuity and differentiability properties of monotone functions. The central result of this chapter is the Lebesgue differentiation theorem.

1.1. Continuity

In this section we study regularity properties of monotone functions.

Definition 1.1. Let $E \subseteq \mathbb{R}$. A function $u : E \rightarrow \mathbb{R}$ is called

- (i) *increasing* if $u(x) \leq u(y)$ for all $x, y \in E$ with $x < y$,
- (ii) *strictly increasing* if $u(x) < u(y)$ for all $x, y \in E$ with $x < y$,
- (iii) *decreasing* if $u(x) \geq u(y)$ for all $x, y \in E$ with $x < y$,
- (iv) *strictly decreasing* if $u(x) > u(y)$ for all $x, y \in E$ with $x < y$,
- (v) *monotone* if any of the above holds.

A monotone function is not continuous in general, so a natural question is how discontinuous it can be. The answer is given by the following theorem. In what follows an *interval* $I \subseteq \mathbb{R}$ is a set of $\mathbb{R}$ such that if $x, y \in I$ and $x < y$, then $[x, y] \subseteq I$.

Theorem 1.2. *Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be a monotone function. Then u has at most countably many discontinuity points. Conversely, given a countable set $E \subset \mathbb{R}$, there exists a monotone function $u : \mathbb{R} \rightarrow \mathbb{R}$ whose set of discontinuity points is exactly E .*

We begin with a preliminary result. Given a set X and a function $v : X \rightarrow [0, \infty]$, the *infinite sum of v over X* is defined as

$$\sum_{x \in X} v(x) := \sup \left\{ \sum_{x \in E} v(x) : E \subseteq X, E \text{ finite} \right\}.$$

The following lemma shows that every infinite sum which is finite can be written as a series.

Lemma 1.3. *Given a set X and a function $v : X \rightarrow [0, \infty]$, if*

$$\sum_{x \in X} v(x) < \infty,$$

then the set $\{x \in X : v(x) > 0\}$ is countable. Moreover, v does not take the value ∞ .

Proof. Define

$$L := \sum_{x \in X} v(x) < \infty.$$

For $n \in \mathbb{N}$ set $X_n := \{x \in X : v(x) > 1/n\}$ and let E be a finite subset of X_n . Then

$$\frac{1}{n} \text{number of elements of } E \leq \sum_{x \in E} v(x) \leq L,$$

which shows that E cannot have more than $\lfloor nL \rfloor$ elements, where $\lfloor \cdot \rfloor$ is the integer part. In turn, X_n has a finite number of elements, and so

$$\{x \in X : v(x) > 0\} = \bigcup_n X_n$$

is countable. □

We turn to the proof of Theorem 1.2.

Proof. Step 1: Assume that $I = [a, b]$ and, without loss of generality, that u is increasing. For every $x \in (a, b)$ there exist

$$\lim_{y \rightarrow x^+} u(y) =: u_+(x), \quad \lim_{y \rightarrow x^-} u(y) =: u_-(x).$$

Let $S(x) := u_+(x) - u_-(x) \geq 0$ be the jump of u at x . Then u is continuous at x if and only if $S(x) = 0$. Moreover, since u is increasing, for every $a \leq x_1 < \cdots < x_n \leq b$,

$$\begin{aligned} u(a) &\leq u_-(x_1) \leq u_+(x_1) \leq u_-(x_2) \leq u_+(x_2) \\ &\leq \cdots \leq u_-(x_n) \leq u_+(x_n) \leq u(b), \end{aligned}$$

and so,

$$\sum_{x \in [a, b]} S(x) \leq u(b) - u(a).$$

It now follows by the previous lemma that u has at most countably many discontinuity points.

Step 2: If I is an arbitrary interval, construct an increasing sequence of intervals $[a_n, b_n]$ such that $I = \bigcup_n [a_n, b_n]$. Since the union of countable sets is countable and on each interval $[a_n, b_n]$, the set of discontinuity points of u is at most countable; by the previous step it follows that the set of discontinuity points of u in I is at most countable.

Step 3: Conversely, let $E \subset I$ be a countable set. If E is finite, then an increasing function with discontinuity set E may be constructed by hand. Consider now the more interesting case in which E is denumerable, so that $E = \{x_n : n \in \mathbb{N}\}$. For each $n \in \mathbb{N}$ define the increasing function $u_n : \mathbb{R} \rightarrow \mathbb{R}$ as

$$u_n(x) := \begin{cases} -1/n^2 & \text{if } x < x_n, \\ 1/n^2 & \text{if } x \geq x_n. \end{cases}$$

Note that u_n is discontinuous only at the point x_n . Set

$$u(x) := \sum_{n=1}^{\infty} u_n(x), \quad x \in \mathbb{R}.$$

Since $|u_n(x)| \leq 1/n^2$ for all $x \in \mathbb{R}$, the series of functions is uniformly convergent, and so it is continuous at every point at which all the functions u_n are continuous. In particular, u is continuous in $\mathbb{R} \setminus E$.

We now prove that u is discontinuous at every point of E . Indeed, for every $m \in \mathbb{N}$ write

$$u = u_m + \sum_{n \neq m} u_n.$$

Then $\sum_{n \neq m} u_n$ is continuous at x_m while u_m is not. Hence, u is discontinuous at each $x \in E$. To conclude, observe that u is increasing, since it is the pointwise limit of a sequence of increasing functions. $\square$

By taking $E := \mathbb{Q}$, we obtain the following result.

Corollary 1.4. *There exists an increasing function $u : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous at all irrational points and discontinuous at all rational points.*

Definition 1.5. Given an interval $I \subseteq \mathbb{R}$, an increasing function $u : I \rightarrow \mathbb{R}$ is called an *increasing saltus* or *jump function* if for every $[a, b] \subseteq I$ it can be written in the form

$$u(x) = c + \sum_{n \in E} u_n(x), \quad x \in [a, b],$$

where $E \subseteq \mathbb{N}$, $c \in \mathbb{R}$, and

$$u_n(x) = \begin{cases} 0 & \text{if } x < a_n, \\ b_n & \text{if } x = a_n, \\ c_n & \text{if } x > a_n, \end{cases}$$

with $a_n \in [a, b]$ and $0 \leq b_n \leq c_n$, with at least one of the two inequalities strict, for all $n \in E$.

Exercise 1.6 (The jump function). Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be increasing. For each $x \in I$ define

$$u_J(x) := \sum_{y \in I, y < x} (u_+(y) - u_-(y)) + u(x) - u_-(x),$$

where $u_-(\inf I) := u(\inf I)$ if $\inf I \in I$ and $u_+(\sup I) := u(\sup I)$ if $\sup I \in I$. Prove that u_J is a jump function, that u_J and $u - u_J$ are increasing, and that $u - u_J$ is continuous. The function u_J is called the *jump function* of u .

Exercise 1.7. Let $I \subseteq \mathbb{R}$ be an interval and let $x_0 \in I$. Prove that every increasing function $u : I \rightarrow \mathbb{R}$ can be written uniquely as the sum of an increasing jump function $v : I \rightarrow \mathbb{R}$ and of an increasing continuous function $w : I \rightarrow \mathbb{R}$ with $w(x_0) = 0$.

We conclude this section by studying the continuity of the inverse of an increasing function.

Theorem 1.8 (Inverse of an increasing function). *Let $I \subset \mathbb{R}$ be an interval bounded from below, let $u : I \rightarrow \mathbb{R}$ be an increasing function, let $J \subseteq \mathbb{R}$ be the smallest interval that contains $u(I)$, and let $v : J \rightarrow \mathbb{R}$ be the function defined by (see Figure 1)*

$$(1.1) \quad v(y) := \inf\{x \in I : u(x) \geq y\}, \quad y \in J.$$

Then

- (i) v is increasing and left continuous,
- (ii) v jumps at some point $y_0 \in J \setminus \{\sup_I u\}$ if and only if $u(x) \equiv y_0$ for all x in some interval $(x_1, x_2) \subseteq I$, with $x_1 < x_2$,
- (iii) $v(u(x)) \leq x$ for every $x \in I$, with the strict inequality holding if and only if u is constant on some interval $[x_1, x] \subseteq I$, with $x_1 < x$,
- (iv) $v(y) \equiv x_0$ for all y in some interval $(y_1, y_2) \subseteq J$, with $y_1 < y_2$, and for some $x_0 \in I^\circ$ if and only if u jumps at x_0 and $(y_1, y_2) \subseteq (u_-(x_0), u_+(x_0))$.

In particular, if the function u is strictly increasing, then v is a left inverse of u and is continuous.

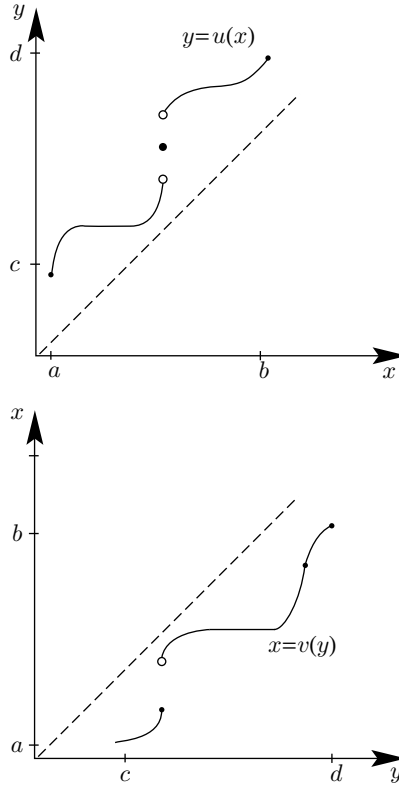


Figure 1. The graphs of u and its generalized inverse v .

Proof. We begin by observing that if u is constant, then $u(I)$ is a singleton and there is not much to prove. Thus, in what follows we assume that u is not constant.

(i) If $y_1, y_2 \in J$, with $y_1 \leq y_2$, then

$$\{x \in I : u(x) \geq y_2\} \subseteq \{x \in I : u(x) \geq y_1\},$$

and so

$$v(y_1) = \inf\{x \in I : u(x) \geq y_1\} \leq \inf\{x \in I : u(x) \geq y_2\} = v(y_2),$$

which shows that v is increasing.

To prove that v is left continuous, fix $y_0 \in J$, with $y_0 > \inf J$. If $v(y) = v(y_0)$ for all $y \in J$ with $y < y_0$, then there is nothing to prove. Thus assume that there is $y_1 \in J$ with $y_1 < y_0$ and $v(y_1) < v(y_0)$. Let $0 < \varepsilon < v(y_0) - v(y_1)$ and fix $x_0 \in [v(y_0) - \varepsilon, v(y_0))$. Then $u(x_0) < y_0$ by the definition of $v(y_0)$. Hence, if $y \in (u(x_0), y_0)$, we have that $u(x) \leq u(x_0) < y$ for all $x \in I$ with $x < x_0$, since u is increasing, and so

$$v(y) = \inf\{x \in I : u(x) \geq y\} \geq x_0 \geq v(y_0) - \varepsilon,$$

which shows that v is left continuous at y_0 .

(ii) Assume that $v(y_0) < v_+(y_0)$ for some $y_0 \in J$, with $y_0 < \sup J = \sup_I u$. By the definition of $v(y_0)$ and the fact that u is increasing, we have that $u(x) \geq y_0$ for all $x > v(y_0)$. On the other hand, if $v(y_0) < x < v_+(y_0)$, then, since v is increasing, $v(y) > x$ for all $y > y_0$, and so $u(x) < y$ for all $y > y_0$, which implies that $u(x) \leq y_0$. Thus, $u(x) \equiv y_0$ for all $x \in (v(y_0), v_+(y_0))$.

Conversely, assume that $u(x) \equiv y_0$ for all x in some interval $(x_1, x_2) \subseteq I$, with $x_1 < x_2$ and $y_0 < \sup J = \sup_I u$. Then $v(y_0) \leq x_1$, by the definition of $v(y_0)$. On the other hand, if $y \in (y_0, \sup J)$, then for every $x \in (x_1, x_2)$, we have that $u(x) = y_0 < y$, and so $v(y) \geq x$ by the definition of $v(y)$. Letting $x \rightarrow x_2^+$, we get that $v(y) \geq x_2$ for all $y \in (y_0, \sup J)$. In particular, $v_+(y_0) \geq x_2$. Thus, we have shown that $v(y_0) \leq x_1 < x_2 \leq v_+(y_0)$.

(iii) Taking $y = u(x)$ in the definition of $v(y)$, yields $v(u(x)) \leq x$.

If $v(u(x_0)) < x_0$ for some $x_0 \in I$, then there exists $x \in I$, with $x < x_0$, such that $u(x) \geq u(x_0)$, but since u is increasing, it follows that $u \equiv u(x_0)$ in $[x, x_0]$. Conversely, if $u \equiv \text{const.}$ in some interval $[x, x_0] \subseteq I$, then $v(u(x_0)) \leq x < x_0$.

(iv) Assume that $v \equiv x_0 \in I^\circ$ in some interval $(y_1, y_2) \subseteq J$, with $y_1 < y_2$. If $x \in I$ with $x > x_0$, then $u(x) \geq y$ for all $y \in (y_1, y_2)$ by the definition of $v(y)$. Letting $y \rightarrow y_2^-$, we get that $u(x) \geq y_2$, and so $u_+(x_0) \geq y_2$. On the other hand, if $x \in I$ with $x < x_0$, then $u(x) < y$ for all $y \in (y_1, y_2)$ by the definition of $v(y)$. Letting $y \rightarrow y_1^+$, we get that $u(x) \leq y_1$, and so $u_-(x_0) \leq y_1$.

Conversely, let $y \in (u_-(x_0), u_+(x_0))$. If $x < x_0$, then $u(x) < y$, and so $v(y) \geq x_0$. On the other hand, since u and v are increasing and by part (iii), for every $x \in I$ with $x > x_0$,

$$x_0 \leq v(y) \leq v(u_+(x_0)) \leq v(u(x)) \leq x.$$

Letting $x \rightarrow x_0^+$, we get that $v(y) = x_0$ for all $y \in (u_-(x_0), u_+(x_0))$.

Finally, assume that u is strictly increasing. Then by parts (i) and (ii) the function v is continuous. By part (iii), $v(u(x)) = x$ for every $x \in I$, which proves that v is a left inverse of u . This concludes the proof. $\square$

Remark 1.9. The fact that I is bounded from below is used to guarantee that the function v does not take the value $-\infty$. Thus, (a modified version of) the previous theorem continues to hold if I is unbounded from below, provided the function v is allowed to take the value $-\infty$. We leave the details as an exercise.

Exercise 1.10. Let $I = [a, b]$, let $u : [a, b] \rightarrow \mathbb{R}$ be increasing and left continuous, and let $v : [u(a), u(b)] \rightarrow \mathbb{R}$ be defined as in the previous theorem.

Prove that

$$u(x) = \inf\{y \in [u(a), u(b)] : v(y) \geq x\}, \quad x \in [a, b].$$

Remark 1.11. The previous theorem continues to hold if u is decreasing, with the only changes being that I is assumed to be bounded from above, the function $v : J \rightarrow \mathbb{R}$ is now defined by

$$v(y) := \inf\{x \in I : u(x) \leq y\},$$

that in part (i) one should replace increasing and left continuous with decreasing and right continuous, that in part (ii) one should take $J \setminus \{\inf_I u\}$ in place of $J \setminus \{\sup_I u\}$, and that in part (iv) the interval $(u_-(x_0), u_+(x_0))$ should be replaced by $(u_+(x_0), u_-(x_0))$.

1.2. Differentiability

We now study the differentiability of monotone functions.

Definition 1.12. Let $E \subseteq \mathbb{R}$ and let $x_0 \in E$ be an accumulation point of E . Given a function $u : E \rightarrow \mathbb{R}$, if there exists in $[-\infty, \infty]$ the limit

$$\lim_{x \rightarrow x_0} \frac{u(x) - u(x_0)}{x - x_0},$$

then the limit is called the *derivative* of u at x_0 and is denoted $u'(x_0)$ or $\frac{du}{dx}(x_0)$. The function u is *differentiable* at x_0 if $u'(x_0)$ exists in $\mathbb{R}$.

The central theorem of this section is Lebesgue's theorem, which shows that a monotone function is differentiable everywhere except possibly on a set of Lebesgue measure zero. Note that this result relies strongly on monotonicity. Indeed, most (continuous) functions are nowhere differentiable. A classical example is due to Weierstrass and is given by

$$(1.2) \quad u(x) := \sum_{n=0}^{\infty} a^n \cos(b^n \pi x), \quad x \in \mathbb{R},$$

where $0 < a < 1$ and $b \in \mathbb{N}$ is an odd integer such that $ab > 1 + \frac{3}{2}\pi$ (see [108]). In Theorem 1.15 below we present a simpler example.

Definition 1.13. Given two metric spaces (X, d_X) and (Y, d_Y) and $E \subseteq X$, a function $u : E \rightarrow Y$ is

(i) *Lipschitz continuous* if

$$\text{Lip}(u; E) = \sup_{x, y \in E, x \neq y} \frac{d_Y(u(x), u(y))}{d_X(x, y)} < \infty,$$

(ii) Hölder continuous of exponent $\alpha > 0$ if

$$|u|_{C^{0,\alpha}} := \sup_{x,y \in E, x \neq y} \frac{d_Y(u(x), u(y))}{(d_X(x, y))^\alpha} < \infty.$$

The number $\text{Lip}(u; E)$ is called the *Lipschitz constant* of u . We usually write $\text{Lip } u$ in place of $\text{Lip}(u; E)$, whenever the underlying set is clear.

Exercise 1.14. Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be Hölder continuous of exponent $\alpha > 1$. Prove that u is constant.

Theorem 1.15. Let $v(x) = |x|$ for $x \in [-1, 1]$ and extend v to $\mathbb{R}$ as a periodic function of period 2. Then the function

$$(1.3) \quad u(x) = \sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n v(4^n x), \quad x \in \mathbb{R},$$

is real-valued, continuous, and nowhere differentiable.

Proof. Let $v_n(x) = \left(\frac{3}{4}\right)^n v(4^n x)$, $x \in \mathbb{R}$. Since v is periodic, $\sup_{y \in \mathbb{R}} |v(y)| = \sup_{y \in [-1, 1]} |v(y)| = 1$, and so $|v_n(x)| \leq \left(\frac{3}{4}\right)^n$. It follows that the series of functions converges uniformly to a continuous bounded function u .

Next we prove that u is nowhere differentiable. Fix $x \in \mathbb{R}$. We are going to construct a sequence $h_m \rightarrow 0$ such that $\frac{u(x+h_m) - u(x)}{h_m} \rightarrow \infty$ as $m \rightarrow \infty$. We take $h_m = \pm \frac{1}{2} \frac{1}{4^m}$, where the sign is chosen in such a way that in the open interval of endpoints $4^m x$ and $4^m(x + h_m)$ there is no integer. Let's prove that we can always do this. We have $4^m(x + \frac{1}{2} \frac{1}{4^m}) - 4^m(x - \frac{1}{2} \frac{1}{4^m}) = 1$. If both $4^m(x + \frac{1}{2} \frac{1}{4^m})$ and $4^m(x - \frac{1}{2} \frac{1}{4^m})$ are integers, then in the interval $(4^m(x - \frac{1}{2} \frac{1}{4^m}), 4^m(x + \frac{1}{2} \frac{1}{4^m}))$ there is no integer and so we can take the sign of h_m as we like. If $4^m(x + \frac{1}{2} \frac{1}{4^m})$ and $4^m(x - \frac{1}{2} \frac{1}{4^m})$ are not both integers, then in the interval $(4^m(x - \frac{1}{2} \frac{1}{4^m}), 4^m(x + \frac{1}{2} \frac{1}{4^m}))$ there is exactly one integer. If this integer is $4^m x$, then we take the sign of h_m as we like. If the integer is in $(4^m(x - \frac{1}{2} \frac{1}{4^m}), 4^m x)$, then we take $h_m = \frac{1}{2} \frac{1}{4^m}$, while if the integer is in $(4^m x, 4^m(x + \frac{1}{2} \frac{1}{4^m}))$, then we take $h_m = -\frac{1}{2} \frac{1}{4^m}$.

We now study

$$(1.4) \quad \begin{aligned} \frac{v_n(x + h_m) - v_n(x)}{h_m} &= \frac{\left(\frac{3}{4}\right)^n v(4^n(x + h_m)) - \left(\frac{3}{4}\right)^n v(4^n x)}{h_m} \\ &= \left(\frac{3}{4}\right)^n \frac{v(4^n x \pm \frac{1}{2} 4^{n-m}) - v(4^n x)}{\pm \frac{1}{2} \frac{1}{4^m}}. \end{aligned}$$

If $n > m$, then $\frac{1}{2} 4^{n-m}$ is an even integer and so by the periodicity of v the difference quotient is zero. If $n = m$, then since by construction in the open interval of endpoints $4^m x$ and $4^m(x + h_m)$ there is no integer, we have that

the points $(x + h_m, v(4^m(x + h_m)))$ and $(x, v(4^m x))$ lie in the same line of the graph of v with slope either 1 or -1 . Hence by (1.4),

$$(1.5) \quad \frac{|v_m(x + h_m) - v_m(x)|}{|h_m|} = \left(\frac{3}{4}\right)^m \frac{4^m |h_m|}{|h_m|} = 3^m.$$

Finally, if $n < m$, then using the fact that v is Lipschitz continuous with Lipschitz constant 1, we get

$$(1.6) \quad \frac{|v_n(x + h_m) - v_n(x)|}{|h_m|} \leq \left(\frac{3}{4}\right)^n \frac{4^n |h_m|}{|h_m|} = 3^n.$$

Hence,

$$\frac{u(x + h_m) - u(x)}{h_m} = \sum_{n=1}^m \frac{v_n(x + h_m) - v_n(x)}{h_m}$$

and using the inequality $|a + b| \geq |b| - |a|$, (1.5), and (1.6), we get

$$\begin{aligned} \left| \frac{u(x + h_m) - u(x)}{h_m} \right| &= \left| \frac{v_m(x + h_m) - v_m(x)}{h_m} + \sum_{n=1}^{m-1} \frac{v_n(x + h_m) - v_n(x)}{h_m} \right| \\ &\geq \left| \frac{v_m(x + h_m) - v_m(x)}{h_m} \right| - \sum_{n=1}^{m-1} \left| \frac{v_n(x + h_m) - v_n(x)}{h_m} \right| \\ &\geq 3^m - \sum_{n=1}^{m-1} 3^n = 3^m - \frac{1}{2}3^m + \frac{3}{2} = \frac{1}{2}3^m + \frac{3}{2} \rightarrow \infty \end{aligned}$$

as $m \rightarrow \infty$. This concludes the proof. $\square$

Exercise 1.16. Study the Hölder continuity of the function u in (1.3).

Remark 1.17. It is actually possible to construct a continuous function $u : (-1, 1) \rightarrow \mathbb{R}$ such that

$$\liminf_{h \rightarrow 0^+} \left| \frac{u(x + h) - u(x)}{h} \right| < \limsup_{h \rightarrow 0^+} \left| \frac{u(x + h) - u(x)}{h} \right| = \infty$$

for all $x \in (-1, 1)$ (see [177]).

In contrast to Theorem 1.15, a monotone function is differentiable everywhere except possibly on a set of Lebesgue measure zero. Note that this result implies, in particular, that the Weierstrass function and the function u in (1.3) are monotone on no interval.

Theorem 1.18 (Lebesgue). *Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be a monotone function. Then there exists a set $E \subset \mathbb{R}$, with $\mathcal{L}^1(E) = 0$, such that u is everywhere differentiable on $I \setminus E$.*

We recall that the *Lebesgue outer measure* of a set $E \subseteq \mathbb{R}$ is defined as

$$\mathcal{L}_o^1(E) := \inf \left\{ \sum_{n=1}^{\infty} (b_n - a_n) : E \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n) \right\}.$$

The proof will use the following properties of the outer measure $\mathcal{L}_o^1$ (see Proposition C.4):

(L_1) If $\{E_n\}_{n \in \mathbb{N}}$ is a sequence of subsets of $\mathbb{R}$, then

$$\mathcal{L}_o^1 \left(\bigcup_{n=1}^{\infty} E_n \right) \leq \sum_{n=1}^{\infty} \mathcal{L}_o^1(E_n).$$

(L_2) If $\{E_n\}_{n \in \mathbb{N}}$ is a sequence of subsets of $\mathbb{R}$, with $E_n \subseteq E_{n+1}$ for every n , then

$$\lim_{n \rightarrow \infty} \mathcal{L}_o^1(E_n) = \mathcal{L}_o^1 \left(\bigcup_{n=1}^{\infty} E_n \right).$$

(L_3) If $E \subseteq \mathbb{R}$, then

$$\mathcal{L}_o^1(E) = \inf \{ \mathcal{L}_o^1(U) : U \text{ open}, E \subseteq U \}.$$

(L_4) If $I_1, \dots, I_m$ are disjoint intervals, then

$$\mathcal{L}_o^1(I_1 \cup \dots \cup I_m) = \sum_{n=1}^m \text{length } I_n.$$

We begin with some auxiliary results.

Lemma 1.19. *Let $E \subset \mathbb{R}$ be a bounded set and let $\mathcal{F}$ be a family of open intervals with the property that each $x \in E$ is the left endpoint of an interval $(x, x + h_x)$ in $\mathcal{F}$. Then for every $\varepsilon > 0$ there exist disjoint intervals $I_1, \dots, I_n \in \mathcal{F}$ and a set*

$$F \subseteq E \cap \bigcup_{k=1}^n I_k$$

such that $\mathcal{L}_o^1(F) \geq \mathcal{L}_o^1(E) - \varepsilon$.

Proof. Define $E_n := \{x \in E : h_x > 1/n\}$. Then $E_n \subseteq E_{n+1}$ and $\bigcup_{n=1}^{\infty} E_n = E$. By property (L_2), $\mathcal{L}_o^1(E_n) \rightarrow \mathcal{L}_o^1(E)$ as $n \rightarrow \infty$. Hence we can find $m \in \mathbb{N}$ so large that

$$\mathcal{L}_o^1(E_m) > \mathcal{L}_o^1(E) - \frac{\varepsilon}{2}.$$

Let $a_1 := \inf E_m$, $b_1 := \sup E_m$, and let $\ell := b_1 - a_1 > 0$. Given

$$\eta = \frac{\varepsilon}{2(m\ell + 1)},$$

by the definition of infimum we can find $x_1 \in E_m$ with $a_1 < x_1 < a_1 + \eta$. By the definition of E_m there exists an interval $I_1 = (x_1, x_1 + h_1)$ in $\mathcal{F}$, with $h_1 > 1/m$. If $x_1 + h_1 \geq b_1$, then we can take $F = E_m \cap (x_1, x_1 + h_1)$.

If $x_1 + h_1 < b_1$, let

$$a_2 := \inf\{x \in E_m : x \geq x_1 + h_1\}.$$

Then by the definition of infimum we can find $x_2 \in E_m$ with $a_2 < x_2 < a_2 + \eta$. By the definition of E_m there exists an interval $I_2 = (x_2, x_2 + h_2)$ in $\mathcal{F}$, with $h_2 > 1/m$. If $x_2 + h_2 \geq b_1$, we stop, while if $x_2 + h_2 < b_1$ we define

$$a_3 := \inf\{x \in E_m : x \geq x_2 + h_2\}.$$

We continue in this way constructing intervals I_k until $x_k + h_k < b_1$. Since each interval I_k has length larger than $\frac{1}{m}$, we have that we will find at most n intervals with $n < m\ell + 1$. Let

$$S := \bigcup_{k=1}^n I_k, \quad T := \bigcup_{k=1}^n (x_k - \eta, x_k].$$

Then $x_1 - \eta \leq a_1 = \inf E_m$ and so $E_m \subseteq S \cup T$. Moreover, the intervals I_k are disjoint by construction. Now by property (L_1) ,

$$\begin{aligned} \mathcal{L}_o^1(E) - \frac{\varepsilon}{2} &< \mathcal{L}_o^1(E_m) \leq \mathcal{L}_o^1(E_m \cap S) + \mathcal{L}_o^1(E_m \cap T) \\ &\leq \mathcal{L}_o^1(E_m \cap S) + \mathcal{L}_o^1(T) \leq \mathcal{L}_o^1(E_m \cap S) + \sum_{k=1}^n \mathcal{L}_o^1((x_k - \eta, x_k]) \\ &= \mathcal{L}_o^1(E_m \cap S) + n\eta \leq \mathcal{L}_o^1(E_m \cap S) + \frac{\varepsilon}{2}, \end{aligned}$$

and so $E_m \cap S$ has the desired property. $\square$

Lemma 1.20. *Let $E \subset \mathbb{R}$ be a bounded set and let $\mathcal{F}$ be a family of open intervals with the property that each $x \in E$ is the left endpoint of an interval $(x, x + h_x)$ in $\mathcal{F}$ with h_x arbitrarily small (that is, for every $\eta > 0$ there is at least one interval with $h_x < \eta$). Then for every $\varepsilon > 0$ there exist disjoint intervals $I_1, \dots, I_n \in \mathcal{F}$ and a set*

$$F \subseteq E \cap \bigcup_{k=1}^n I_k$$

such that

$$\mathcal{L}_o^1(F) \geq \mathcal{L}_o^1(E) - \varepsilon, \quad \sum_{k=1}^n \text{length } I_k \leq \mathcal{L}_o^1(E) + \varepsilon.$$

Proof. By property (L_3) there exists an open set $U \supseteq E$ such that

$$\mathcal{L}_o^1(U) \leq \mathcal{L}_o^1(E) + \varepsilon.$$

Let $\mathcal{F}'$ be the subfamily of intervals $(x, x + h_x)$ in $\mathcal{F}$ contained in U . Note that for each $x \in E \subseteq U$ there must exist at least one such interval, since U contains a ball centered at x and there are intervals of arbitrarily small length.

Apply Lemma 1.19 to the family $\mathcal{F}'$ to find disjoint intervals $I_1, \dots, I_n \in \mathcal{F}'$ and a set

$$F \subseteq E \cap \bigcup_{k=1}^n I_k$$

such that

$$\mathcal{L}_o^1(F) \geq \mathcal{L}_o^1(E) - \varepsilon.$$

Since the intervals are disjoint and contained in U , it follows from properties (L_1) and (L_4) that

$$\sum_{k=1}^n \text{length } I_k \leq \mathcal{L}_o^1(U) \leq \mathcal{L}_o^1(E) + \varepsilon.$$

This concludes the proof. $\square$

Given a set $E \subseteq \mathbb{R}$ and function $u : E \rightarrow \mathbb{R}$, for every $x_0 \in E$ such that x_0 is an accumulation point for the sets $E \cap (-\infty, x_0)$ and $E \cap (x_0, \infty)$, the four *Dini's derivatives* of u are given by

$$\begin{aligned} \overline{D}_- u(x_0) &:= \limsup_{x \rightarrow x_0^-} \frac{u(x) - u(x_0)}{x - x_0}, & \overline{D}_+ u(x_0) &:= \limsup_{x \rightarrow x_0^+} \frac{u(x) - u(x_0)}{x - x_0}, \\ \underline{D}_- u(x_0) &:= \liminf_{x \rightarrow x_0^-} \frac{u(x) - u(x_0)}{x - x_0}, & \underline{D}_+ u(x_0) &:= \liminf_{x \rightarrow x_0^+} \frac{u(x) - u(x_0)}{x - x_0}. \end{aligned}$$

We now turn to the proof of Lebesgue's differentiation theorem.

Proof. Step 1: Assume that u is increasing and that I is bounded and let

$$E := \{x \in I^\circ : \underline{D}_+ u(x) < \overline{D}_+ u(x)\}.$$

We claim that E has Lebesgue measure zero. To see this we write E as a countable union of sets

$$E = \bigcup_{r,s \in \mathbb{Q}, 0 < r < s} E_{r,s}, \quad E_{r,s} := \{x \in E : \underline{D}_+ u(x) < r < s < \overline{D}_+ u(x)\}.$$

It is enough to prove that each set $E_{r,s}$ has Lebesgue measure zero. Since $\underline{D}_+ u(x) < r$ for each $x \in E_{r,s}$, we can find an open interval $(x, x+h)$, where $h > 0$ is arbitrarily small and such that

$$(1.7) \quad \frac{u(x+h) - u(x)}{h} < r.$$

Let $\mathcal{F}$ be the family of all such intervals. By Lemma 1.20 for every $\varepsilon > 0$ there exist disjoint intervals $I_1, \dots, I_n \in \mathcal{F}$ and a set

$$F_{r,s} \subseteq E_{r,s} \cap \bigcup_{k=1}^n I_k$$

such that

$$(1.8) \quad \mathcal{L}_o^1(F_{r,s}) \geq \mathcal{L}_o^1(E_{r,s}) - \varepsilon, \quad \sum_{k=1}^n \text{length } I_k \leq \mathcal{L}_o^1(E_{r,s}) + \varepsilon.$$

Write $I_k = (x_k, x_k + h_k)$. Then by (1.7) and (1.8),

$$(1.9) \quad \sum_{k=1}^n u(x_k + h_k) - u(x_k) < r \sum_{k=1}^n h_k \leq r \mathcal{L}_o^1(E_{r,s}) + r\varepsilon.$$

Let $V := \bigcup_{k=1}^n I_k$. Note that V is open. Since $F_{r,s} \subseteq E_{r,s} \cap V$, we have that $\overline{D}_+ u(x) > s$ for each $x \in F_{r,s}$, and so we can find an open interval $(x, x+t) \subseteq V$, where $t > 0$ is arbitrarily small and such that

$$(1.10) \quad \frac{u(x+t) - u(x)}{t} > s.$$

Let $\mathcal{G}$ be the family of all such intervals. By Lemma 1.20 for every $\varepsilon > 0$ there exist disjoint intervals $J_1, \dots, J_m \in \mathcal{G}$ and a set

$$(1.11) \quad G_{r,s} \subseteq F_{r,s} \cap \bigcup_{i=1}^m J_i$$

such that

$$(1.12) \quad \mathcal{L}_o^1(G_{r,s}) \geq \mathcal{L}_o^1(F_{r,s}) - \varepsilon.$$

Write $J_i = (y_i, y_i + t_i)$. Then by (1.8), (1.10), (1.11), and (1.12),

$$(1.13) \quad \begin{aligned} \sum_{i=1}^m u(y_i + t_i) - u(y_i) &> s \sum_{i=1}^m t_i \geq s \mathcal{L}_o^1(G_{r,s}) \\ &\geq s \mathcal{L}_o^1(F_{r,s}) - s\varepsilon \geq s \mathcal{L}_o^1(E_{r,s}) - 2s\varepsilon. \end{aligned}$$

But since each J_i is contained in $V = \bigcup_{k=1}^n I_k$, and since the intervals I_k are disjoint, it follows that each interval J_i is contained in some interval I_k . Since u is increasing it follows that

$$\sum_{i=1}^m u(y_i + t_i) - u(y_i) \leq \sum_{k=1}^n u(x_k + h_k) - u(x_k).$$

Combining this inequality with (1.9) and (1.13) gives

$$\begin{aligned} s \mathcal{L}_o^1(E_{r,s}) - 2s\varepsilon &< \sum_{i=1}^m u(y_i + t_i) - u(y_i) \\ &\leq \sum_{k=1}^n u(x_k + h_k) - u(x_k) < r \mathcal{L}_o^1(E_{r,s}) + r\varepsilon, \end{aligned}$$

that is,

$$(s - r) \mathcal{L}_o^1(E_{r,s}) < 2s\varepsilon + r\varepsilon.$$

Since $s - r > 0$, letting $\varepsilon \rightarrow 0^+$ we conclude that $\mathcal{L}_o^1(E_{r,s}) = 0$.

Hence, we have shown that $\mathcal{L}_o^1(E) = 0$. It follows that for all $x \in I^\circ \setminus E$ there exists the right derivative $u'_+(x)$ (possibly infinite).

With a similar proof we can show that the left derivative exists (possibly infinite) for all $x \in I^\circ$ except for a set of Lebesgue measure zero.

Step 2: Let

$$E_\infty := \{x \in I^\circ : u'_+(x) = \infty\}.$$

We leave as an exercise to prove that $\mathcal{L}_o^1(E_\infty) = 0$.

Step 3: If I is an arbitrary interval, construct an increasing sequence of intervals $[a_n, b_n]$ such that $I = \bigcup_n [a_n, b_n]$. Since the union of sets of Lebesgue outer measure zero still has Lebesgue outer measure zero and since on each interval $[a_n, b_n]$ the set of points in which u is not differentiable has Lebesgue outer measure zero by the previous step, it follows that the set of points of I in which u is not differentiable has Lebesgue outer measure zero. $\square$

The proof is concluded by the following exercise.

Exercise 1.21. Let $I \subseteq \mathbb{R}$ be an interval and let $u, v : I \rightarrow \mathbb{R}$.

- (i) Prove that the set of $x \in I^\circ$ such that there is $\varepsilon > 0$ such that $v(y) > v(x)$ for all $y \in I \cap (x - \varepsilon, x + \varepsilon)$, $y \neq x$, is countable.
- (ii) Prove that the sets $\{x \in I^\circ : \overline{D}_+ u(x) < \underline{D}_- u(x)\}$ and $\{x \in I^\circ : \overline{D}_- u(x) < \underline{D}_+ u(x)\}$ are countable. Hint: Apply part (i) to the function $v(x) = u(x) - rx$ to prove that the set $\{x \in I^\circ : \overline{D}_- u(x) < r < \underline{D}_+ u(x)\}$ is countable.
- (iii) Prove that the set $\{x \in I^\circ : u'_+(x) \neq u'_-(x)\}$ is countable.

Exercise 1.22. Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be continuous in I and differentiable for all $x \in I$ except at most a countable set E .

- (i) Prove that if $u'(x) > 0$ for all $x \in I \setminus E$, then u is strictly increasing. Hint: If $u(b) < u(a)$, let $u(b) < t < u(a)$ with $t \notin u(E)$ and consider the set $\{x \in [a, b] : u(x) = t\}$.
- (ii) Prove that if $u'(x) \geq 0$ for all $x \in I \setminus E$, then u is increasing.

The next theorem shows that Lebesgue's differentiation theorem is sharp, to be precise, that there exist monotone functions that are not differentiable on sets of Lebesgue measure zero.

Theorem 1.23. *For every set $E \subset \mathbb{R}$ of Lebesgue (outer) measure zero there exists a continuous increasing function that is not differentiable at every point of E .*

Proof. Since $\mathcal{L}_o^1(E) = 0$, for every $l \in \mathbb{N}$ we may cover E with a countable family of open intervals $\{(a_{l,m}, b_{l,m})\}_m$ such that

$$(1.14) \quad \sum_{m=1}^{\infty} (b_{l,m} - a_{l,m}) \leq \frac{1}{2^l}.$$

Let $\mathcal{I} := \{(a_{l,m}, b_{l,m})\}_{l,m} = \{I_n\}_n$, with $I_n = (a_n, b_n)$. Note that by (1.14),

$$(1.15) \quad \sum_{n=1}^{\infty} (b_n - a_n) \leq 1.$$

Moreover, for every $x \in E$ we may find infinitely many I_n such that $x \in I_n$. For each $n \in \mathbb{N}$ define the continuous increasing function $u_n : \mathbb{R} \rightarrow [0, \infty)$ as follows:

$$u_n(x) := \begin{cases} 0 & \text{if } x < a_n, \\ x - a_n & \text{if } a_n \leq x \leq b_n, \\ b_n - a_n & \text{if } x > b_n. \end{cases}$$

Note that $0 \leq u_n(x) \leq b_n - a_n$ for all $x \in \mathbb{R}$. Set

$$(1.16) \quad u(x) := \sum_{n=1}^{\infty} u_n(x), \quad x \in \mathbb{R}.$$

Then for all $x \in \mathbb{R}$ and $l \in \mathbb{N}$,

$$0 \leq u(x) - \sum_{n=1}^l u_n(x) \leq \sum_{n=l+1}^{\infty} (b_n - a_n),$$

and so the series of functions is uniformly convergent by (1.15). In particular, this implies that u is continuous. Since each u_n is nonnegative and increasing, it follows that u has the same properties.

It remains to show that u is not differentiable in E . Thus, fix $x_0 \in E$ and $l \in \mathbb{N}$. By the way we constructed the sequence $\mathcal{I}$, we may find l positive integers $n_1 < n_2 < \dots < n_l$ such that $x_0 \in I_{n_i}$ for all $i = 1, \dots, l$. Let $x \in I_{n_1} \cap \dots \cap I_{n_l}$, $x \neq x_0$. Since each u_n is increasing, the difference quotient is nonnegative, and so

$$\frac{u(x) - u(x_0)}{x - x_0} \geq \sum_{i=1}^l \frac{u_{n_i}(x) - u_{n_i}(x_0)}{x - x_0} = \sum_{i=1}^l \frac{x - a_{n_i} - (x_0 - a_{n_i})}{x - x_0} = l.$$

It follows that

$$\limsup_{x \rightarrow x_0} \frac{u(x) - u(x_0)}{x - x_0} \geq l$$

for every $l \in \mathbb{N}$. Hence,

$$\limsup_{x \rightarrow x_0} \frac{u(x) - u(x_0)}{x - x_0} = \infty,$$

which implies that u is not differentiable at x_0 . □

Remark 1.24. Note that the set of points at which the increasing function u (defined in (1.16)) is not differentiable contains the set E , but in general it may be larger. To the author's knowledge an exact characterization of the class of sets that are sets of nondifferentiability for increasing functions is still an open problem (see Exercise 1.29 below for some properties of this set). We refer to [37] and [237] for more information on this subject.

As a simple consequence of Lebesgue's theorem we obtain that the derivative of a monotone function is always locally Lebesgue integrable.

Corollary 1.25. *Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be a monotone function. Then u' is a Lebesgue measurable function and for every $[a, b] \subseteq I$,*

$$(1.17) \quad \int_a^b |u'(x)| dx \leq |u(b) - u(a)|.$$

Moreover, if u is bounded, then u' is Lebesgue integrable and

$$(1.18) \quad \int_I |u'(x)| dx \leq \sup_I u - \inf_I u.$$

Proof. Assume that u is increasing, fix $[a, b] \subseteq I$, and let

$$(1.19) \quad v(x) := \begin{cases} u(x) & \text{if } x \leq b, \\ u(b) & \text{if } x \geq b. \end{cases}$$

Let $E := \{x \in [a, b] : v \text{ is not differentiable at } x\}$. By Lebesgue's theorem, $\mathcal{L}^1(E) = 0$. Moreover,

$$(1.20) \quad v'(x) = u'(x) \quad \text{for all } x \in (a, b) \setminus E.$$

Since v is increasing, we have that $v'(x) \geq 0$ for all $x \in [a, b] \setminus E$ and

$$v'(x) = \lim_{n \rightarrow \infty} \frac{v(x + 1/n) - v(x)}{1/n}.$$

By Theorem 1.2 the nonnegative functions,

$$w_n(x) := \frac{v(x + 1/n) - v(x)}{1/n}, \quad x \in [a, b],$$

are continuous except for a countable set, and so they are Borel measurable functions. Hence, $v' : [a, b] \setminus E \rightarrow [0, \infty)$ is a Lebesgue measurable function.

Since v is increasing, by (1.19) for every $h > 0$ we have

$$(1.21) \quad \begin{aligned} \frac{1}{h} \int_a^b [v(x + h) - v(x)] dx &= \frac{1}{h} \left\{ \int_b^{b+h} v(x) dx - \int_a^{a+h} v(x) dx \right\} \\ &\leq \frac{1}{h} \{(u(b) - u(a))h\} = u(b) - u(a). \end{aligned}$$

Taking $h := 1/n$, it follows from Fatou's lemma, (1.20), and (1.21) that

$$\begin{aligned} 0 &\leq \int_a^b u'(x) dx = \int_a^b v'(x) dx \\ &\leq \liminf_{n \rightarrow \infty} n \int_a^b [v(x + 1/n) - v(x)] dx \leq u(b) - u(a). \end{aligned}$$

This concludes the proof of the first part.

Assume next that u is bounded and construct an increasing sequence of intervals $[a_n, b_n]$ such that $I = \bigcup_n [a_n, b_n]$. By the previous part,

$$0 \leq \int_{a_n}^{b_n} u'(x) dx \leq u(b_n) - u(a_n) \leq \sup_I u - \inf_I u,$$

and so, letting $n \rightarrow \infty$ and using the Lebesgue monotone convergence theorem, we obtain (1.18). $\square$

Remark 1.26. In Exercise 1.29 below we will show that u' is actually a Borel function.

Exercise 1.27. Prove that if equality holds in (1.17) (respectively, in (1.18)), then u is continuous in $[a, b]$, (respectively, in I).

Using the previous corollary, we obtain the following result. We will see its usefulness for functions of bounded pointwise variation in Corollary 2.51.

Corollary 1.28. *Let $I \subseteq \mathbb{R}$ be an interval, let $u : I \rightarrow \mathbb{R}$ be a monotone function, and let $h > 0$. Then for every $[a, b] \subseteq I$, with $b - a \geq h$,*

$$(1.22) \quad \frac{1}{h} \int_a^{b-h} |u(x+h) - u(x)| dx \leq |u(b) - u(a)|.$$

Moreover, if u is bounded, then

$$(1.23) \quad \frac{1}{h} \int_{I_h} |u(x+h) - u(x)| dx \leq \sup_I u - \inf_I u,$$

where $I_h := \{x \in I : x + h \in I\}$.

Proof. Assume that u is increasing, fix $[a, b] \subseteq I$, with $b - a \geq h$, and let v be defined as in (1.19). Then by (1.21),

$$\begin{aligned} \frac{1}{h} \int_a^{b-h} [u(x+h) - u(x)] dx &\leq \frac{1}{h} \int_a^b [v(x+h) - v(x)] dx \\ &\leq u(b) - u(a). \end{aligned}$$

Assume next that u is bounded and construct an increasing sequence of intervals $[a_n, b_n]$ such that $I = \bigcup_n [a_n, b_n]$. Then

$$0 \leq \frac{1}{h} \int_{a_n}^{b_n-h} [u(x+h) - u(x)] dx \leq u(b_n) - u(a_n) \leq \sup_I u - \inf_I u,$$

and so, letting $n \rightarrow \infty$ and using the Lebesgue monotone convergence theorem, we obtain (1.23). $\square$

In Corollary 1.25 we have shown that the derivative of a monotone function is Lebesgue measurable. The next exercise shows that the derivative of an arbitrary function is actually Borel measurable. We recall that a set $E \subseteq \mathbb{R}$ is called an F_σ set if it is a countable union of closed sets, a G_δ set if it is a countable intersection of open sets.

Exercise 1.29. Let $I \subseteq \mathbb{R}$ be an open interval, let $u : I \rightarrow \mathbb{R}$ be an arbitrary function, and let

$$E := \{x \in I : u \text{ is continuous at } x\},$$

$$F := \{x \in I : u \text{ is differentiable at } x\}.$$

Prove that

(i) for each $n \in \mathbb{N}$ the set

$$U_n := \{x \in I : \text{there is } \delta > 0 \text{ such that } (x - \delta, x + \delta) \subseteq I$$

$$\text{and if } y, z \in (x - \delta, x + \delta), \text{ then } |u(y) - u(z)| < 1/n\}$$

is open,

(ii) the set E is a G_δ set,

(iii) for each $\varepsilon > 0$, $\delta > 0$ the set

$$U_{\varepsilon, \delta} := \{x \in E : \text{there is } y \in I \text{ such that } 0 < |x - y| < \delta$$

$$\text{and } \frac{u(y) - u(x)}{y - x} > \varepsilon\}$$

is a relatively open subset of E , that is, the intersection of E with an open set,

$$(iv) \{x \in E : \overline{D}u(x) > 0\} = \bigcup_{\varepsilon > 0, \varepsilon \in \mathbb{Q}} \bigcap_{\delta > 0, \delta \in \mathbb{Q}} U_{\varepsilon, \delta}, \text{ where}$$

$$\overline{D}u(x) := \limsup_{y \rightarrow x} \frac{u(y) - u(x)}{y - x},$$

(v) the sets $\{x \in E : \overline{D}u(x) > \alpha\}$, $\{x \in E : \underline{D}u(x) < \alpha\}$ are Borel sets for every $\alpha \in \mathbb{R}$, where

$$\underline{D}u(x) := \liminf_{y \rightarrow x} \frac{u(y) - u(x)}{y - x},$$

(vi) the set F is a Borel set,

(vii) the function $v : I \rightarrow \mathbb{R}$, defined by

$$v(x) := \begin{cases} u'(x) & \text{if } x \in F, \\ 0 & \text{otherwise,} \end{cases}$$

is Borel measurable.

Exercise 1.30 (Expansion in base b). Let $b \in \mathbb{N}$ be such that $b \geq 2$ and let $x \in [0, 1]$.

- (i) Prove that there exists a sequence $\{a_n\}_n$ of nonnegative integers such that $0 \leq a_n < b$ for all $n \in \mathbb{N}$ and

$$x = \sum_{n=1}^{\infty} \frac{a_n}{b^n}.$$

Prove that the sequence $\{a_n\}_n$ is uniquely determined by x , unless x is of the form $x = \frac{k}{b^m}$ for some $k, m \in \mathbb{N}$, in which case there are exactly two such sequences.

- (ii) Conversely, given a sequence $\{a_n\}_n$ of nonnegative integers such that $0 \leq a_n < b$ for all $n \in \mathbb{N}$, prove that the series $\sum_{n=1}^{\infty} \frac{a_n}{b^n}$ converges to a number $x \in [0, 1]$.

The next example shows that the inequality in (1.17) may be strict, and so for continuous monotone functions the fundamental theorem of calculus for Lebesgue integration fails.

Example 1.31 (The Cantor function). Divide $[0, 1]$ into three equal subintervals, and remove the middle interval $I_{1,1} := (\frac{1}{3}, \frac{2}{3})$. Divide each of the two remaining closed intervals $[0, \frac{1}{3}]$ and $[\frac{2}{3}, 1]$ into three equal subintervals, and remove the middle intervals $I_{1,2} := (\frac{1}{9}, \frac{2}{9})$ and $I_{2,2} := (\frac{7}{9}, \frac{8}{9})$. Continuing in this fashion, at each step n remove 2^{n-1} middle intervals $I_{1,n}, \dots, I_{2^{n-1},n}$, each of length $\frac{1}{3^n}$. The *Cantor set* $\mathbb{D}$ is defined as

$$\mathbb{D} := [0, 1] \setminus \bigcup_{n=1}^{\infty} \bigcup_{k=1}^{2^{n-1}} I_{k,n}.$$

The set $\mathbb{D}$ is closed (since its complement is given by a family of open intervals), and

$$\mathcal{L}^1(\mathbb{D}) = 1 - \sum_{n=1}^{\infty} \sum_{k=1}^{2^{n-1}} \text{diam } I_{k,n} = 1 - \sum_{n=1}^{\infty} \sum_{k=1}^{2^{n-1}} \frac{1}{3^n} = 1 - \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} = 0.$$

Thus, $\mathbb{D}$ has Lebesgue measure zero. To prove that $\mathbb{D}$ is uncountable, observe that $x \in \mathbb{D}$ if and only if

$$(1.24) \quad x = \sum_{n=1}^{\infty} \frac{c_n}{3^n},$$

where each $c_n \in \{0, 2\}$ (see the previous exercise). For every $x \in \mathbb{D}$ of the form (1.24) define

$$u(x) := \sum_{n=1}^{\infty} \frac{d_n}{2^n},$$

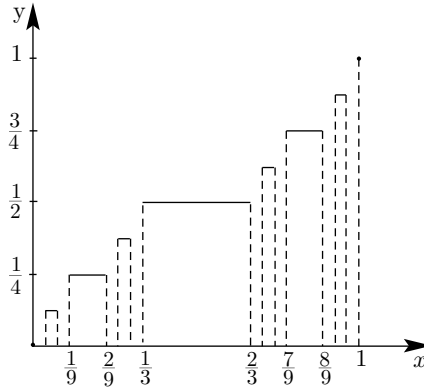


Figure 2. The Cantor function.

where

$$d_n := \begin{cases} 1 & \text{if } c_n = 2, \\ 0 & \text{if } c_n = 0. \end{cases}$$

The function $u : \mathbb{D} \rightarrow [0, 1]$ is well-defined (why?), increasing, continuous, and has the same values at the endpoints of each removed interval $I_{k,n}$ and thus u extends to a continuous function on $[0, 1]$. This function is called the *Cantor function* (see Figure 2). Since $u(\mathbb{D}) = [0, 1]$, it follows that $\mathbb{D}$ is uncountable. Note also that $u'(x) = 0$ for all $x \in I_{k,n}$, $k, n \in \mathbb{N}$, so that $u' = 0$ except on a set of Lebesgue measure zero. Since $u(0) = 0$ and $u(1) = 1$, it follows that the inequality in (1.17) is strict, and so the fundamental theorem of calculus for Lebesgue integration fails.

Exercise 1.32. Consider the complete metric space

$$X := \{u : [0, 1] \rightarrow \mathbb{R} : u \text{ is continuous, } u(0) = 0, \text{ and } u(1) = 1\},$$

where we take the metric induced by supremum norm

$$\|u\|_\infty := \max_{x \in [0, 1]} |u(x)|.$$

Consider the operator $T : X \rightarrow X$, defined by

$$T(u)(x) := \begin{cases} \frac{1}{2}u(3x) & \text{if } 0 \leq x \leq \frac{1}{3}, \\ \frac{1}{2} & \text{if } \frac{1}{3} < x < \frac{2}{3}, \\ \frac{1}{2} + \frac{1}{2}u(3x - 2) & \text{if } \frac{2}{3} \leq x \leq 1. \end{cases}$$

(i) Prove that T is a contraction, to be precise, that

$$\|T(u_1) - T(u_2)\|_\infty \leq \frac{1}{2}\|u_1 - u_2\|_\infty$$

for all $u_1, u_2 \in X$.

(ii) Prove that the Cantor function is the unique fixed point of T .

- (iii) Define recursively the sequence of functions $u_0(x) := x$, $u_{n+1}(x) := T(u_n)(x)$, $x \in [0, 1]$. Prove that for all $n \in \mathbb{N}$,

$$|u_n(x_1) - u_n(x_2)| \leq |x_1 - x_2|^\alpha$$

for all $x_1, x_2 \in [0, 1]$, where $\alpha := \frac{\log 2}{\log 3}$. Deduce that the Cantor function u is Hölder continuous with exponent α .

- (iv) Prove that the exponent α and the constant one are the best possible for the Cantor function u , in the sense that the inequality

$$|u(x_1) - u(x_2)| \leq c|x_1 - x_2|^\beta$$

does not hold for all $x_1, x_2 \in [0, 1]$ if either $\beta > \alpha$ or $\beta = \alpha$ but $c < 1$.

Using the Cantor set, we can construct Lebesgue measurable sets that are not Borel sets.

Exercise 1.33 (A non-Borel, Lebesgue measurable set). Consider the function $v : [0, 1] \rightarrow \mathbb{R}$ defined by

$$v(x) := x + u(x), \quad x \in [0, 1].$$

- (i) Prove that $\mathcal{L}^1(v(\mathbb{D})) = 1$.
- (ii) Let $E \subseteq v(\mathbb{D})$ be a set that is not Lebesgue measurable (see Exercise C.2 in Appendix C) and consider the set $F := v^{-1}(E) \subseteq \mathbb{D}$. Prove that F is Lebesgue measurable but not a Borel set.

We have seen that the Cantor function is increasing with $u' = 0$, $\mathcal{L}^1$ -a.e. in $[0, 1]$. It is actually possible to construct *strictly increasing* continuous functions whose derivative is zero except on a set of Lebesgue measure zero. To prove this result, we will need the following result.

Theorem 1.34 (Fubini). *Let $I \subseteq \mathbb{R}$ be an interval and let $\{u_n\}_n$ be a sequence of increasing functions, $u_n : I \rightarrow \mathbb{R}$. Assume that the series of functions $\sum_{n=1}^\infty u_n$ converges pointwise in I . Then $\sum_{n=1}^\infty u_n$ converges uniformly on compact sets of I , the sum of the series*

$$u(x) := \sum_{n=1}^{\infty} u_n(x), \quad x \in I,$$

is differentiable $\mathcal{L}^1$ -a.e. in I , and

$$u'(x) = \sum_{n=1}^{\infty} u'_n(x) \quad \text{for } \mathcal{L}^1\text{-a.e. } x \in I.$$

Proof. Since the result is local, without loss of generality, we may assume that $I = [a, b]$.

Step 1: Assume that $u_n(a) = 0$ for all $n \in \mathbb{N}$ so that $u_n \geq 0$. For every $l \in \mathbb{N}$ set

$$s_l := \sum_{n=1}^l u_n.$$

Then for $x \in [a, b]$,

$$0 \leq u(x) - s_l(x) = \sum_{n=l+1}^{\infty} u_n(x) \leq \sum_{n=l+1}^{\infty} u_n(b),$$

and so

$$0 \leq \sup_{[a,b]} |u(x) - s_l(x)| \leq \sum_{n=l+1}^{\infty} u_n(b) \rightarrow 0$$

as $l \rightarrow \infty$. Hence, the series is uniformly convergent.

Since u and s_l are increasing, by Lebesgue's differentiation theorem (Theorem 1.18) they are differentiable $\mathcal{L}^1$ -a.e. in $[a, b]$, and since the countable union of sets of Lebesgue zero measure still has Lebesgue measure zero, we may find a Lebesgue measurable set $E \subset [a, b]$, with $\mathcal{L}^1(E) = 0$, such that u and all the functions s_l are everywhere differentiable in $[a, b] \setminus E$.

Since the functions $s_{l+1} - s_l$ and $u - s_l$ are increasing for all $l \in \mathbb{N}$, we have that $(s_{l+1} - s_l)' \geq 0$ and $(u - s_l)' \geq 0$ in $[a, b] \setminus E$. Hence,

$$0 \leq s'_1(x) \leq s'_2(x) \leq \cdots \leq u'(x)$$

for all $x \in [a, b] \setminus E$, and so there exists

$$\lim_{n \rightarrow \infty} s'_n(x) \leq u'(x) < \infty.$$

In turn,

$$(1.25) \quad \lim_{n \rightarrow \infty} u'_n(x) = \lim_{n \rightarrow \infty} [s'_n(x) - s'_{n-1}(x)] = 0$$

for all $x \in [a, b] \setminus E$.

It remains to show that $\lim_{n \rightarrow \infty} s'_n(x) = u'(x)$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$. Since $\lim_{n \rightarrow \infty} s'_n(x)$ exists for all $x \in [a, b] \setminus E$, it is enough to show this for a subsequence $\{s'_{n_k}\}_k$ of $\{s'_n\}_n$. So let $n_k \nearrow \infty$ be such that

$$0 \leq u(b) - s_{n_k}(b) \leq \frac{1}{2^k}$$

and define $v_k(x) := u(x) - s_{n_k}(x) \leq \frac{1}{2^k}$. Then v_k is increasing and

$$0 \leq \sum_{k=1}^{\infty} v_k(x) \leq \sum_{k=1}^{\infty} \frac{1}{2^k}.$$

Hence, we can apply the first part of the proof to the sequence $\{v_k\}_k$ to conclude that

$$\lim_{k \rightarrow \infty} v'_k(x) = 0$$

for $\mathcal{L}^1$ -a.e. $x \in [a, b]$ by (1.25). This completes the proof.

Step 2: Write $u_n = u_n - u_n(a) + u_n(a) =: w_n + u_n(a)$ and apply Step 1 to w_n . $\square$

Exercise 1.35. Let $C \subseteq \mathbb{R}$ be a closed set and let

$$u(x) := \text{dist}(x, C) = \inf\{|x - y| : y \in C, \quad x \in \mathbb{R}\}.$$

Prove that

(i) u is 1-Lipschitz continuous, that is,

$$|u(x) - u(y)| \leq |x - y|$$

for all $x, y \in \mathbb{R}$,

(ii) there exists an increasing function $v : \mathbb{R} \rightarrow \mathbb{R}$ such that v' is continuous and

$$C = \{x \in \mathbb{R} : v'(x) = 0\},$$

(iii) if C contains no intervals, then every such v is strictly increasing.

Exercise 1.36. Let $I \subseteq \mathbb{R}$ be an interval, let $u : I \rightarrow \mathbb{R}$ be an increasing function, and let u_J be the jump function of u . Prove that $u'_J(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in I$.

Theorem 1.37. *There exists a strictly increasing continuous function $u : \mathbb{R} \rightarrow \mathbb{R}$ whose derivative is zero except on a set of Lebesgue measure zero.*

Proof. Let $v : \mathbb{R} \rightarrow \mathbb{R}$ be the Cantor function extended to be 1 for $x \geq 1$ and 0 for $x \leq 0$. Consider a countable dense set $\{a_n\}_n$ in $\mathbb{R}$ (e.g., the rational numbers) and define

$$u(x) := \sum_{n=1}^{\infty} \frac{1}{2^n} v(2^n(x - a_n)), \quad x \in \mathbb{R}.$$

Since $0 \leq v \leq 1$, the series of functions converges uniformly to a continuous bounded function.

To prove that u is strictly increasing, let $x < y$. By density there exists $x < a_m < y$. Then $y - a_m > 0 > x - a_m$ and so

$$v(2^m(y - a_m)) > 0 = v(2^m(x - a_m)),$$

since the Cantor function is positive for $x > 0$, while $v(2^n(y - a_n)) \geq v(2^n(x - a_n))$, since v is increasing.

It now follows by Fubini's theorem that $v'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in \mathbb{R}$. $\square$

Exercise 1.38. Let $u : [a, b] \rightarrow \mathbb{R}$ be differentiable at $x_0 \in (a, b)$ and let $x_n, y_n \in (a, b) \setminus \{x_0\}$ be such that $x_n \neq y_n$ and $x_n, y_n \rightarrow x_0$.

- (i) Prove that if $x_n < x_0 < y_n$ for all $n \in \mathbb{N}$, then

$$\lim_{n \rightarrow \infty} \frac{u(y_n) - u(x_n)}{y_n - x_n} = u'(x_0).$$

- (ii) Prove that if the condition $x_n < x_0 < y_n$ is violated, then the limit $\lim_{n \rightarrow \infty} \frac{u(y_n) - u(x_n)}{y_n - x_n}$ may not exist or may be different from $u'(x_0)$.

Exercise 1.39. For each $x \in (0, 1]$ find positive integers $a_0 < \cdots < a_n < \cdots$ such that

$$x = \sum_{n=0}^{\infty} \frac{1}{2^{a_n}}.$$

(see Exercise 1.30). Fix $r > 0$, $r \neq 1$. For $x = 0$ define $u(0) := 0$, while for each $x \in (0, 1]$ define

$$u(x) := \sum_{n=0}^{\infty} \frac{2^n}{3^{a_n}}.$$

- (i) Prove that u is strictly increasing.
(ii) Given $x \in (0, 1]$, for every $m \in \mathbb{N}_0$ let k_m be the number of subscripts $n = 0, \dots, m$ for which $a_n \leq m$. Define

$$x_m := \sum_{n=0}^{k_m} \frac{1}{2^{a_n}}, \quad z_m := x_m + \frac{1}{2^m}.$$

Prove that $x_m < x < z_m$ and that $u(z_m) - u(x_m) \rightarrow 0$ as $m \rightarrow \infty$.

- (iii) Prove that u is continuous.
(iv) Prove that if u is differentiable at some $x \in (0, 1]$, then necessarily $u'(x) = 0$. Hint: Compute

$$\frac{u(z_m) - u(x_m)}{z_m - x_m}.$$

Exercise 1.40. Let $\{r_n\}_n$ be an enumeration of the rationals in $[a, b]$ and define

$$u(x) := \sum_{n=1}^{\infty} \frac{1}{2^n} (x - r_n)^{1/3}, \quad x \in [a, b].$$

- (i) Prove that u is continuous and strictly increasing.
(ii) Prove that if the series

$$\frac{1}{3} \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{1}{(x - r_n)^{2/3}}$$

converges, then there exists

$$u'(x) = \frac{1}{3} \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{1}{(x - r_n)^{2/3}}.$$

-
- (iii) Prove that at all other points there exists $u'(x) = \infty$.
- (iv) Let $u([a, b]) = [\alpha, \beta]$. Prove that the inverse of $u^{-1} : [\alpha, \beta] \rightarrow \mathbb{R}$ is differentiable at every point and the set $\{x \in [\alpha, \beta] : (u^{-1})'(x) = 0\}$ is dense in $[\alpha, \beta]$.

Functions of Bounded Pointwise Variation

*Undergradese, II: “Is grading going to be curved?” Translation:
“Can I do a mediocre job and still get an A?”*

— Jorge Cham, www.phdcomics.com

Let $I \subseteq \mathbb{R}$ be an interval. The set of monotone functions $u : I \rightarrow \mathbb{R}$ is not a vector space, since in general the difference of monotone functions is not monotone. In this chapter we characterize the smallest vector space of functions $u : I \rightarrow \mathbb{R}$ that contains all monotone functions.

In many applications such as abstract evolution equations, gradient flows (see, e.g., [33], [71]), optimal transport (see, e.g., [11]), length spaces, (see, e.g., [38]), it is important to consider functions $u : I \rightarrow Y$, where Y is a metric space, possibly infinite-dimensional. In this book we will focus mainly on the cases in which Y is $\mathbb{R}^M$ or an infinite-dimensional normed space.

For readers without a solid background in functional analysis, simply take $Y = \mathbb{R}^M$ in the entire chapter.

2.1. Pointwise Variation

In what follows, given an interval $I \subseteq \mathbb{R}$, a *partition* of I is a finite set $P := \{x_0, \dots, x_n\} \subset I$, where

$$x_0 < \dots < x_n.$$

Definition 2.1. Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u : I \rightarrow Y$ be a function. The *pointwise variation* of u on the interval I is

$$(2.1) \quad \text{Var } u := \sup \left\{ \sum_{i=1}^n d(u(x_i), u(x_{i-1})) \right\},$$

where the supremum is taken over all partitions $P := \{x_0, \dots, x_n\}$ of I , $n \in \mathbb{N}$. A function $u : I \rightarrow Y$ has *finite* or *bounded pointwise variation* if $\text{Var } u < \infty$.

The space of all functions $u : I \rightarrow Y$ of bounded pointwise variation is denoted by $BPV(I; Y)$.

When $Y = \mathbb{R}$, for simplicity we write $BPV(I)$ for $BPV(I; \mathbb{R})$ ¹.

Remark 2.2. Note that if I reduces to a singleton, then it has no partitions. In this case we set

$$\text{Var } u := 0.$$

In the remaining of the chapter it is tacitly understood that

$$\inf I < \sup I,$$

with some obvious exceptions.

Remark 2.3. Note that if one of the endpoints, say, $b := \sup I$, belongs to I , then in the definition of $\text{Var } u$ it suffices to restrict our attention to partitions $P := \{x_0, \dots, x_n\}$ such that $x_0 < \dots < x_n = b$. Indeed, if $P := \{x_0, \dots, x_n\}$ is a partition with $x_n < b$, then $Q := \{x_0, \dots, x_n, b\}$ is also a partition of I and so

$$\sum_{i=1}^n d(u(x_i), u(x_{i-1})) \leq \sum_{i=1}^n d(u(x_i), u(x_{i-1})) + d(u(b), u(x_n)) \leq \text{Var } u.$$

In the sequel we will often use this property without further mention.

To highlight the dependence on the interval I , we will sometimes write $\text{Var}_I u$.

Given an interval $I \subseteq \mathbb{R}$ and a metric space (Y, d) , a function $u : I \rightarrow Y$ has *locally finite* or *locally bounded pointwise variation* if $\text{Var}_{[a,b]} u < \infty$ for all intervals $[a, b] \subseteq I$. The space of all functions $u : I \rightarrow Y$ of locally bounded pointwise variation is denoted by $BPV_{\text{loc}}(I; Y)$. When $Y = \mathbb{R}$, as before we write $BPV_{\text{loc}}(I)$ for $BPV_{\text{loc}}(I; \mathbb{R})$.

It almost goes without saying that if $I = [a, b]$, then

$$BPV_{\text{loc}}([a, b]; Y) = BPV([a, b]; Y).$$

If $\Omega \subseteq \mathbb{R}$ is an open set, then we can write Ω as a countable union of pairwise disjoint open intervals

$$\Omega = \bigcup_n I_n.$$

¹This is not the standard notation for this space, which is usually denoted $BV(I)$ in books on real analysis and measure theory. Although we do not like changing standard notation, unfortunately, in the literature the notation $BV(I)$ is also used for a quite different (although strictly related) function space. This book studies both spaces, so we really had to change the notation for one of them.

Given a function $u : \Omega \rightarrow Y$, where (Y, d) is a metric space, we define the *pointwise variation* of u on Ω as

$$\text{Var } u := \sum_n \text{Var}_{I_n} u,$$

and we say that u has *bounded pointwise variation in Ω* if $\text{Var } u < \infty$. The space of functions $u : \Omega \rightarrow Y$ of bounded pointwise variation is denoted by $BPV(\Omega; Y)$.

Exercise 2.4. Given $u : [a, b] \rightarrow \mathbb{R}$, the extended real numbers

$$\text{PVar } u := \sup \left\{ \sum_{i=1}^n (u(x_i) - u(x_{i-1}))^+ \right\}$$

and

$$\text{NVar } u := \sup \left\{ \sum_{i=1}^n (u(x_i) - u(x_{i-1}))^- \right\},$$

where the suprema are taken over all partitions $P := \{x_0, \dots, x_n\}$ of $[a, b]$, $n \in \mathbb{N}$, are called the *positive* and *negative pointwise variation* of u in $[a, b]$, respectively. Prove that if $u \in BPV([a, b])$, then

$$\text{Var } u = \text{PVar } u + \text{NVar } u$$

and that

$$u(b) - u(a) = \text{PVar } u - \text{NVar } u.$$

Remark 2.5. Note that a function $u : I \rightarrow \mathbb{R}^M$ belongs to $BPV(I; \mathbb{R}^M)$ (respectively, $BPV_{\text{loc}}(I; \mathbb{R}^M)$) if and only if each component $u_i : I \rightarrow \mathbb{R}$, $i = 1, \dots, M$, belongs to $BPV(I)$ (respectively, $BPV_{\text{loc}}(I)$). Indeed, for all $i = 1, \dots, M$ and for every interval $J \subseteq I$ we have

$$(2.2) \quad \text{Var}_J u_i \leq \text{Var}_J u \leq \text{Var}_J u_1 + \dots + \text{Var}_J u_M.$$

We now discuss the relation between functions of bounded pointwise variation with other important classes of functions. We begin by showing that Lipschitz continuous functions have locally bounded pointwise variation.

Proposition 2.6. *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u : I \rightarrow Y$ be a Lipschitz continuous function. Then u belongs to $BPV_{\text{loc}}(I; Y)$. Moreover, if I is bounded, then u belongs to $BPV(I; Y)$.*

Proof. Since u is Lipschitz continuous, there exists $L \geq 0$ such that

$$d(u(x), u(z)) \leq L|x - z|$$

for all $x, z \in I$. Given $[a, b] \subseteq I$ and a partition $x_0 < \cdots < x_n$ of $[a, b]$, we have that

$$\sum_{i=1}^n d(u(x_i), u(x_{i-1})) \leq L \sum_{i=1}^n |x_i - x_{i-1}| = L(x_n - x_0) \leq L(b - a).$$

Taking the supremum over all partitions of $[a, b]$ gives

$$\text{Var}_{[a,b]} u \leq L(b - a) < \infty.$$

Similarly, if I is bounded, then $\text{Var}_I u \leq L \text{length } I < \infty$. $\square$

The next exercise shows that continuous functions of bounded pointwise variation need not be Lipschitz continuous.

Exercise 2.7. Let $u : [0, 1] \rightarrow \mathbb{R}$ be defined by

$$u(x) := \begin{cases} x^a \sin \frac{1}{x^b} & \text{if } 0 < x \leq 1, \\ 0 & \text{if } x = 0, \end{cases}$$

where $a, b \in \mathbb{R}$. Study for which a, b the function u has bounded pointwise variation.

In contrast to the previous proposition, there is no relation between a Hölder continuous function and a function of bounded pointwise variation. This fact is illustrated by the following two exercises.

Exercise 2.8. Let

$$u(x) := \begin{cases} \frac{1}{\log \frac{2}{x}} & \text{if } 0 < x \leq 1, \\ 0 & \text{if } x = 0. \end{cases}$$

Prove that u has finite pointwise variation in $[0, 1]$ but it is not Hölder continuous of any exponent $0 < \alpha \leq 1$.

Exercise 2.9. Let $1 < \alpha < 2$ and

$$L := \sum_{n=1}^{\infty} \frac{1}{n^\alpha} < \infty.$$

Subdivide the interval $[0, L]$ into countably many intervals with $x_0 = 0$ and $x_n - x_{n-1} = \frac{1}{n^\alpha}$. In each interval $[x_{n-1}, x_n]$ define

$$u(x) := \begin{cases} \sqrt{x - x_{n-1}} & \text{if } x_{n-1} \leq x \leq \frac{x_n + x_{n-1}}{2}, \\ \sqrt{x_n - x} & \text{if } \frac{x_n + x_{n-1}}{2} \leq x \leq x_n. \end{cases}$$

Prove that the function u is Hölder continuous of exponent $1/2$ but $\text{Var}_{[0,L]} u = \infty$.

Exercise 2.10. Let $\{r_n\}_n$ be an enumeration of $[0, 1] \cap \mathbb{Q}$ and define

$$u(x) := \begin{cases} \frac{1}{2^n} & \text{if } x = r_n \text{ for some } n, \\ 0 & \text{otherwise in } [0, 1]. \end{cases}$$

Compute $\text{Var}_{[0,1]} u$.

The next proposition shows that monotone functions have locally finite pointwise variation.

Proposition 2.11. *Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be a monotone function. Then for every interval $J \subseteq I$,*

$$\text{Var}_J u = \sup_J u - \inf_J u.$$

In particular, u belongs to $BPV_{\text{loc}}(I)$. Moreover, u belongs to $BPV(I)$ if and only if it is bounded.

Proof. Assume that u is increasing and let $J \subseteq I$ be an interval. Then for every partition $P := \{x_0, \dots, x_n\}$ of J ,

$$\begin{aligned} \sum_{i=1}^n |u(x_i) - u(x_{i-1})| &= \sum_{i=1}^n (u(x_i) - u(x_{i-1})) \\ &= u(x_n) - u(x_0) \leq \sup_J u - \inf_J u. \end{aligned}$$

Taking the supremum over all partitions of J yields

$$\text{Var}_J u \leq \sup_J u - \inf_J u.$$

To prove the opposite inequality, it suffices to consider the case in which J is nondegenerate. Consider the partition $P := \{a, b\} \subset J$, where $\inf J \leq a < b \leq \sup J$. Then

$$u(b) - u(a) = |u(b) - u(a)| \leq \text{Var}_J u.$$

If $\sup J \in J$, then taking $b = \sup J$ gives $u(b) = \sup_J u$, while if $\sup J \notin J$, letting $b \nearrow \sup J$ gives $u(b) \rightarrow \sup_J u$. Reasoning in a similar way for the left endpoint, we obtain that

$$\sup_J u - \inf_J u \leq \text{Var}_J u.$$

This concludes the proof. □

The previous proposition will be complemented by Theorem 2.36 below.

Finally we discuss the relation between bounded functions and functions of bounded pointwise variation.

Proposition 2.12. *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u : I \rightarrow Y$. For every $x_0 \in I$ and every interval $J \subseteq I$ containing x_0 ,*

$$d(u(x), u(x_0)) \leq \text{Var}_J u$$

for all $x \in J$. In particular, if u belongs to $BPV_{\text{loc}}(I; Y)$ (respectively, $BPV(I; Y)$), then u is locally bounded (respectively, bounded).

Proof. Fix an interval $J \subseteq I$ containing x_0 . For every $x \in J$ with $x \neq x_0$, consider the partition $P := \{x_0, x\}$. Then

$$d(u(x), u(x_0)) \leq \text{Var}_J u,$$

which concludes the proof. $\square$

Remark 2.13. If Y is a normed space, then it follows from the previous proposition that

$$\sup_{x \in J} \|u(x)\| \leq \|u(x_0)\| + \text{Var}_J u.$$

Example 2.14. The function $u(x) := \sin x$, $x \in \mathbb{R}$, is bounded and belongs to $BPV_{\text{loc}}(\mathbb{R})$, but it does not belong to $BPV(\mathbb{R})$ (why?).

2.2. Continuity

In this section we study continuity properties of functions of bounded pointwise variation.

Proposition 2.15. *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space and let $u : I \rightarrow Y$ be a function. If $c \in I$, then*

$$\text{Var}_{I \cap (-\infty, c]} u + \text{Var}_{I \cap [c, \infty)} u = \text{Var } u.$$

Proof. Let $I_1 := I \cap (-\infty, c]$ and $I_2 := I \cap [c, \infty)$. Let $P := \{x_0, \dots, x_n\}$ and $Q := \{y_0, \dots, y_m\}$ be partitions of I_1 and I_2 , respectively. Then $P \cup Q$ is a partition of I , and so

$$\sum_{i=1}^n d(u(x_i), u(x_{i-1})) + \sum_{i=1}^m d(u(y_i), u(y_{i-1})) \leq \text{Var } u.$$

Taking first the supremum over all partitions P of I_1 and then over all partitions Q of I_2 yields

$$\text{Var}_{I_1} u + \text{Var}_{I_2} u \leq \text{Var } u.$$

Conversely, let $P := \{x_0, \dots, x_n\}$ be a partition of I . Let $m \in \{1, \dots, n\}$ be such that $x_{m-1} \leq c \leq x_m$. Then $P_1 := \{x_0, \dots, x_{m-1}, c\}$ and $P_2 :=$

$\{c, x_m, \dots, x_n\}$ are partitions of I_1 and I_2 , respectively, and so

$$\begin{aligned} \sum_{i=1}^n d(u(x_i), u(x_{i-1})) &\leq \sum_{i=1}^{m-1} d(u(x_i), u(x_{i-1})) + d(u(c), u(x_{m-1})) \\ &\quad + d(u(x_m), u(c)) + \sum_{i=m+1}^n d(u(x_i), u(x_{i-1})) \\ &\leq \text{Var}_{I_1} u + \text{Var}_{I_2} u. \end{aligned}$$

Taking the supremum over all partitions P of I gives $\text{Var } u \leq \text{Var}_{I_1} u + \text{Var}_{I_2} u$. $\square$

Exercise 2.16. Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space and let $u : I \rightarrow Y$ be a function. Prove that if I does not contain its right endpoint, then

$$\lim_{x \rightarrow (\sup I)^-} \text{Var}_{I \cap (-\infty, x]} u = \text{Var } u,$$

while if I does not contain its left endpoint, then

$$\lim_{x \rightarrow (\inf I)^+} \text{Var}_{I \cap [x, \infty)} u = \text{Var } u.$$

Proposition 2.15 implies that the function $x \in I \mapsto \text{Var}_{I \cap (-\infty, x]} u$ is increasing. However, for functions in $BPV_{\text{loc}}(I)$ it may be infinite. Thus, we modify it as follows.

Theorem 2.17 (Indefinite pointwise variation). *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, let $t_0 \in I$, and let $u \in BPV_{\text{loc}}(I; Y)$. For every $x \in I$ define*

$$(2.3) \quad \mathbf{V}(x) := \begin{cases} \text{Var}_{[t_0, x]} u & \text{if } x \geq t_0, \\ -\text{Var}_{[x, t_0]} u & \text{if } x < t_0. \end{cases}$$

Then for all $x, y \in I$, with $x < y$,

$$(2.4) \quad d(u(y), u(x)) \leq \mathbf{V}(y) - \mathbf{V}(x) = \text{Var}_{[x, y]} u.$$

In particular, $\mathbf{V}$ is increasing and u is continuous at all but countably many points of I . Moreover, if Y is a complete metric space, then there exist

$$u_-(x) = \lim_{y \rightarrow x^-} u(y), \quad u_+(t) = \lim_{y \rightarrow t^+} u(y)$$

for all $x \in I$.² Finally, if $Y = \mathbb{R}$, the functions $\mathbf{V} \pm u$ are increasing.

Proof. Fix $x, t \in I$, with $x < t$. By Proposition 2.15,

$$(2.5) \quad \text{Var}_{[x, t]} u = \begin{cases} \text{Var}_{[t_0, t]} u - \text{Var}_{[t_0, x]} u = \mathbf{V}(t) - \mathbf{V}(x) & \text{if } t_0 \leq x < t, \\ \text{Var}_{[x, t_0]} u - \text{Var}_{[t, t_0]} u = -\mathbf{V}(x) + \mathbf{V}(t) & \text{if } x < t \leq t_0, \\ \text{Var}_{[x, t_0]} u + \text{Var}_{[t_0, t]} u = -\mathbf{V}(x) + \mathbf{V}(t) & \text{if } x \leq t_0 \leq t. \end{cases}$$

²With the obvious changes if x is an endpoint of I .

Since $d(u(t), u(x)) \leq \text{Var}_{[x,t]} u$, the inequality (2.4) follows.

Note that (2.4) implies that $\mathbf{V}(x) \leq \mathbf{V}(t)$ for all $x, t \in I$, with $x < t$. Hence, the function $\mathbf{V}$ is increasing. It follows by Theorem 1.2 that $\mathbf{V}$ is continuous at all but countably many points of I . Let $x_0 \in I$ be such that $\mathbf{V}$ is continuous at x_0 . Then for every $\varepsilon > 0$ there exists $\delta = \delta(x_0, \varepsilon) > 0$ such that

$$|\mathbf{V}(x) - \mathbf{V}(x_0)| \leq \varepsilon$$

for all $x \in I$ with $|x_0 - x| \leq \delta$. By (2.4), u is continuous at x_0 . Thus, u is continuous at all but countably many $x \in I$.

Next assume that Y is complete. Since $\mathbf{V}$ is increasing, it has a left and a right limit at every interior point of I . Hence, given $x_0 \in I^\circ$ for every $\varepsilon > 0$ there exists $\delta = \delta(x_0, \varepsilon) > 0$ such that

$$\mathbf{V}(x) - \mathbf{V}_+(x_0) \leq \varepsilon$$

for all $x \in I$ with $x_0 < x < x_0 + \delta$. Taking $x_0 < x < t < x_0 + \delta$, we have that

$$\mathbf{V}(t) - \mathbf{V}(x) \leq \mathbf{V}(t) - \mathbf{V}_+(x_0) \leq \varepsilon$$

and so by (2.4), $d(u(t), u(x)) \leq \varepsilon$ for all $x_0 < x < t < x_0 + \delta$. Since Y is complete, this implies that there exists $u_+(x_0)$. Similarly, we can show that there exists $u_-(x_0)$.

Finally, if $Y = \mathbb{R}$,

$$(2.6) \quad \pm(u(t) - u(x)) \leq |u(t) - u(x)| \leq \mathbf{V}(t) - \mathbf{V}(x).$$

Hence, the functions $\mathbf{V} \pm u$ are increasing. $\square$

The function $\mathbf{V}$ is called the *indefinite pointwise variation* of the function u .

Remark 2.18. To highlight the dependence on t_0 or on u , when necessary, we will write $\mathbf{V}_{t_0}$ or $\mathbf{V}_{t_0, u}$.

Remark 2.19. Note that when $\inf I \in I$, the choice $t_0 = \inf I$ gives the simpler function

$$(2.7) \quad \mathbf{V}_\infty(x) := \text{Var}_{I \cap (-\infty, x]} u, \quad x \in I.$$

However, if $\inf I$ does not belong to I or is $-\infty$, then the function $\mathbf{V}_\infty$ may not be finite, unless $u \in \text{BPV}(I)$.

Using the function $\mathbf{V}$ we can establish the analog of Corollary 1.28.

Theorem 2.20. Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u \in \text{BPV}_{\text{loc}}(I; Y)$. Then for $h > 0$,

$$(2.8) \quad \frac{1}{h} \int_{I_h} d(u(x+h), u(x)) \, dx \leq \text{Var } u,$$

where $I_h := \{x \in I : x+h \in I\}$.

Proof. If $\text{Var } u = \infty$, then there is nothing to prove. Thus, assume that $\text{Var } u < \infty$. Fix $[a, b] \subseteq I$ with $0 < h \leq b - a$. By (2.4) and Corollary 1.28 applied to $\mathbf{V}$ we have

$$\begin{aligned} \frac{1}{h} \int_a^{b-h} d(u(x+h), u(x)) dx &\leq \frac{1}{h} \int_a^{b-h} (\mathbf{V}(x+h) - \mathbf{V}(x)) dx \\ &\leq \mathbf{V}(b) - \mathbf{V}(a) = \text{Var}_{[a,b]} u. \end{aligned}$$

Construct an increasing sequence of intervals $[a_n, b_n]$ such that $I = \bigcup_n [a_n, b_n]$. If $\text{diam } I_h > 0$, then for all n sufficiently large we have that $0 < h < b_n - a_n$, and so by the previous inequality we have that

$$\frac{1}{h} \int_{a_n}^{b_n-h} d(u(x+h), u(x)) dx \leq \text{Var}_{[a_n, b_n]} u \leq \text{Var } u.$$

Letting $n \rightarrow \infty$ and using the Lebesgue monotone convergence theorem gives

$$\frac{1}{h} \int_{I_h} d(u(x+h), u(x)) dx \leq \text{Var } u.$$

The previous inequality continues to hold if $\text{diam } I_h = 0$, since in this case the integral on the left-hand side is zero. This completes the proof. $\square$

In Corollary 2.51 we will prove a converse of Theorem 2.20.

Inequality (2.4) shows that if the indefinite pointwise variation $\mathbf{V}$ (see (2.3)) is right continuous at some $x \in I$ (respectively, left continuous), then so is u . Next we show that the opposite is also true.

Proposition 2.21. *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u \in \text{BPV}_{\text{loc}}(I; Y)$. If u is right continuous at some $x \in I$ (respectively, left continuous), then so is $\mathbf{V}$.*

Proof. Assume that u is right continuous at $x_0 \in I$. Fix $r > 0$ such that $x_0 + r \in I$. Then given $\varepsilon > 0$ there exists $0 < \delta \leq r$ such that

$$(2.9) \quad d(u(x), u(x_0)) \leq \varepsilon$$

for all $x \in I$ with $x_0 \leq x \leq x_0 + \delta$. Let $x_0 < x_1 < \cdots < x_n = x_0 + r$ be a partition of $[x_0, x_0 + r]$ such that

$$\text{Var}_{[x_0, x_0+r]} u \leq \sum_{i=1}^n d(u(x_i), u(x_{i-1})) + \varepsilon.$$

By adding the point $x_0 + \delta$ to the partition, without loss of generality, we may assume that $x_0 < x_1 \leq x_0 + \delta$. Then by (2.9),

$$\begin{aligned} \text{Var}_{[x_0, x_0+r]} u &\leq d(u(x_1), u(x_0)) + \sum_{i=2}^n d(u(x_i), u(x_{i-1})) + \varepsilon \\ &\leq \text{Var}_{[x_1, x_0+r]} u + 2\varepsilon. \end{aligned}$$

By (2.4), it follows that

$$\mathbf{V}(x_1) - \mathbf{V}(x_0) = \text{Var}_{[x_0, x_1]} u \leq 2\varepsilon.$$

Note that x_1 depends on δ and so on ε . Let $\delta_1 := x_1 - x_0 > 0$. Since $\mathbf{V}$ is increasing, for all $x_0 < x < x_0 + \delta_1$,

$$\mathbf{V}(x_0) \leq \mathbf{V}(x) \leq \mathbf{V}(x_0 + \delta_1) = \mathbf{V}(x_1) \leq \mathbf{V}(x_0) + 2\varepsilon,$$

which shows that $\mathbf{V}$ is right continuous at x_0 . $\square$

Exercise 2.22. Let $I \subseteq \mathbb{R}$ be an interval and let $u \in BPV(I)$. Prove that if I contains its right endpoint $b := \sup I$ and u is left continuous at b , then

$$\lim_{x \rightarrow b^-} \text{Var}_{I \cap (-\infty, x]} u = \text{Var } u.$$

Exercise 2.23. Let $I \subseteq \mathbb{R}$ be an interval and let $u \in BPV(I)$. Let $\mathbf{V}$ be the function defined in (2.3). Prove that $\sup_I \mathbf{V} - \inf_I \mathbf{V} = \text{Var } u$.

The following result will be useful in the parametrization of curves in Chapter 5.

Corollary 2.24. *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u \in BPV_{\text{loc}}(I; Y)$ be a continuous function. Then we can write*

$$u = w \circ \mathbf{V},$$

where $w : J \rightarrow Y$ is Lipschitz continuous with Lipschitz constant at most 1 and $\text{Var}_I u = \text{Var}_J w$, where $J := \mathbf{V}(I)$.

Proof. By Theorem 2.17 and Proposition 2.21 the function $\mathbf{V} : I \rightarrow \mathbf{V}(I)$ is continuous, increasing, and onto. Hence, $J = \mathbf{V}(I)$ is an interval. Note that the function $\mathbf{V}$ is constant on an interval if and only if u is constant there. Hence, for every $s \in J$ there exists either a single point x_s such that $\mathbf{V}(x_s) = s$ or there exists a maximal interval of endpoints α_s and β_s such that $\mathbf{V}(x) = s$ for all $x \in [\alpha_s, \beta_s] \cap I$. In the latter case, we let x_s be any element in $[\alpha_s, \beta_s] \cap I$. For $s \in J$ define

$$w(s) := u(x_s).$$

Note that by construction

$$(2.10) \quad w(\mathbf{V}(x)) = u(x)$$

for all $x \in I$. We claim that w is Lipschitz continuous with Lipschitz constant less than or equal to 1. To see this, note that for all $s, t \in J$, with $s < t$, we have

$$d(w(t), w(s)) = d(u(x_t), u(x_s)) \leq \mathbf{V}(x_t) - \mathbf{V}(x_s) = t - s,$$

by (2.4) and the facts that $\mathbf{V}(x_t) = t$ and $\mathbf{V}(x_s) = s$. Thus, w is Lipschitz continuous with Lipschitz constant less than or equal to 1.

In particular, if $s_0 < \cdots < s_n$ is a partition of J , then

$$\sum_{i=1}^n d(w(s_i), w(s_{i-1})) \leq \sum_{i=1}^n (s_i - s_{i-1}) \leq \text{length } J = \text{Var}_I u.$$

On the other hand, since $\mathbf{V}$ is increasing, for every partition of I , $x_0 < \cdots < x_n$, we have that $\mathbf{V}(x_0) \leq \cdots \leq \mathbf{V}(x_n)$ and so if at least one of these inequalities is strict, then by (2.10),

$$\sum_{i=1}^n d(u(x_i), u(x_{i-1})) = \sum_{i=1}^n d(w(\mathbf{V}(x_i)), w(\mathbf{V}(x_{i-1}))) \leq \text{Var}_J w.$$

On the other hand, if $\mathbf{V}(x_0) = \mathbf{V}(x_n)$, then u is constant in $[x_0, x_n]$ and so the left-hand side of the previous inequality is zero. Thus the previous inequality continues to hold. In turn, $\text{Var}_I u \leq \text{Var}_J w$, which completes the proof. $\square$

Exercise 2.25. Let $p \geq 1$. Given a function $u : [a, b] \rightarrow \mathbb{R}$, we define the p -variation of u as

$$\text{Var}_p u := \sup \left\{ \left(\sum_{i=1}^n |u(x_i) - u(x_{i-1})|^p \right)^{1/p} \right\},$$

where the supremum is taken over all partitions $P = \{x_0, \dots, x_n\}$ of I , $n \in \mathbb{N}$.

- (i) Prove that if $\text{Var}_p u < \infty$, then u is bounded and for every $x \in (a, b)$ there exist in $\mathbb{R}$ the left and right limits $u_+(x)$ and $u_-(x)$.
- (ii) Prove that if $\text{Var}_p u < \infty$, then u has at most a countable number of discontinuity points $\{t_n\}_n$ and that

$$\text{Var}_p u \geq \left\{ \sum_n |u_+(t_n) - u_-(t_n)|^p \right\}^{1/p}.$$

- (iii) Prove that if $\text{Var}_p u < \infty$ and $q > p$, then

$$\text{Var}_q u \leq (2\|u\|_\infty)^{\frac{q-p}{q}} (\text{Var}_p u)^{\frac{p}{q}}.$$

- (iv) Prove that for all $u, v : [a, b] \rightarrow \mathbb{R}$,

$$\text{Var}_p(u + v) \leq \text{Var}_p u + \text{Var}_p v.$$

- (v) Prove that if $u : [a, b] \rightarrow \mathbb{R}$ is Hölder continuous of exponent $\frac{1}{p}$, then $\text{Var}_p u < \infty$.

Exercise 2.26. Let $p \geq 1$. Given a function $u : [a, b] \rightarrow \mathbb{R}$ for every $\varepsilon > 0$ let

$$\text{Var}_p^{(\varepsilon)} u := \sup \left\{ \left(\sum_{i=1}^n |u(x_i) - u(x_{i-1})|^p \right)^{1/p} \right\},$$

where the supremum is taken over all partitions $P = \{x_0, \dots, x_n\}$ of I such that $0 < x_i - x_{i-1} \leq \varepsilon$, $i = 1, \dots, n$, $n \in \mathbb{N}$. Define

$$\text{Var}_p^{(0)} u := \lim_{\varepsilon \rightarrow 0^+} \text{Var}_p^{(\varepsilon)} u.$$

- (i) Prove that $\text{Var}_p^{(0)} u < \infty$ if and only if $\text{Var}_p u < \infty$.
- (ii) Prove that there exists a function u for which $\text{Var}_p^{(0)} u < \text{Var}_p u < \infty$.
- (iii) Prove that if u is continuous and $\text{Var}_p^{(0)} u < \infty$, then $\text{Var}_q^{(0)} u = 0$ for $q > p$. Hint: Use uniform continuity.
- (iv) Find a continuous function $u : [0, 1] \rightarrow \mathbb{R}$ such that $\text{Var} u = \infty$ and $\text{Var}_2 u < \infty$.

Exercise 2.27. Let $u \in BPV([0, 2\pi])$ and let $a_k := \int_0^{2\pi} u(x) e^{-ikx} dx$, $k \in \mathbb{Z}$, be its Fourier coefficients. Prove that there exists a constant $c > 0$ such that $|a_k| \leq c/|k|$ for all $k \in \mathbb{Z} \setminus \{0\}$. Hint: Consider a special partition of $[0, 2\pi]$ and construct a suitable step function.

2.3. Differentiability

In this section we discuss the differentiability of functions of bounded pointwise variation in the case in which Y is a finite dimensional vector space. We begin by studying the relation between u' and the derivative of the indefinite pointwise variation $\mathbf{V}$ of u (see (2.3)).

Theorem 2.28. Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, let $u \in BPV_{\text{loc}}(I; Y)$, and let $E := \{x \in I : u \text{ is differentiable at } x\}$. Then

$$(2.11) \quad \|u'(x)\| = \mathbf{V}'(x) \quad \text{for } \mathcal{L}^1\text{-a.e. } x \in E.$$

Proof. Since $\mathbf{V}$ is increasing by Theorem 2.17, it is differentiable at $\mathcal{L}^1$ -a.e. $x \in I$, in view of the Lebesgue differentiation theorem (Theorem 1.18).

Let $x \in E$ be such that $\mathbf{V}$ has a derivative (possibly infinite) at x . Consider the case in which $x \in I^\circ$. By (2.4),

$$\|u(x+h) - u(x)\| \leq \mathbf{V}(x+h) - \mathbf{V}(x)$$

for all $h > 0$ such that $x + h \in I$. By dividing both sides by $h > 0$ and letting $h \rightarrow 0^+$ we obtain

$$(2.12) \quad \|u'(x)\| \leq \mathbf{V}'(x).$$

To prove the converse inequality, assume first that $I = [a, b]$, so that $\text{Var } u < \infty$, and let

$$E_0 := \{x \in I : u \text{ and } \mathbf{V} \text{ are differentiable at } x \text{ and } \mathbf{V}'(x) > \|u'(x)\|\}.$$

For every $k \in \mathbb{N}$, let E_k be the set of all points $x \in E_0$ such that

$$(2.13) \quad \frac{\mathbf{V}(x_2) - \mathbf{V}(x_1)}{x_2 - x_1} \geq \frac{\|u(x_2) - u(x_1)\|}{x_2 - x_1} + \frac{1}{k}$$

for all intervals $[x_1, x_2]$ such that $x \in [x_1, x_2]$ and $0 < x_2 - x_1 \leq 1/k$. Since

$$E_0 = \bigcup_{k=1}^{\infty} E_k,$$

to prove (2.11), it suffices to show that $\mathcal{L}^1(E_k) = 0$. Let $\varepsilon > 0$ and consider a partition $P := \{x_0, \dots, x_n\}$ of I such that

$$(2.14) \quad \sum_{i=1}^n \|u(x_i) - u(x_{i-1})\| > \text{Var } u - \varepsilon.$$

By adding more points to the partition, without loss of generality, we may assume that $0 < x_i - x_{i-1} \leq \frac{1}{k}$ for all $i = 1, \dots, n$. Fix $i \in \{1, \dots, n\}$. We distinguish two cases. If the interval $[x_{i-1}, x_i]$ contains points of E_k , then multiplying (2.13) by $x_i - x_{i-1}$ yields

$$\mathbf{V}(x_i) - \mathbf{V}(x_{i-1}) \geq \|u(x_i) - u(x_{i-1})\| + \frac{x_i - x_{i-1}}{k},$$

while if $[x_{i-1}, x_i]$ does not intersect E_k , then

$$\mathbf{V}(x_i) - \mathbf{V}(x_{i-1}) \geq \|u(x_i) - u(x_{i-1})\|$$

by (2.4). Summing these two inequalities and using (2.14) gives

$$\begin{aligned} \text{Var } u &= \sum_{i=1}^n (\mathbf{V}(x_i) - \mathbf{V}(x_{i-1})) \geq \sum_{i=1}^n \|u(x_i) - u(x_{i-1})\| + \frac{1}{k} \mathcal{L}_o^1(E_k) \\ &\geq \text{Var } u - \varepsilon + \frac{1}{k} \mathcal{L}_o^1(E_k), \end{aligned}$$

which implies that $\mathcal{L}_o^1(E_k) \leq k\varepsilon$. Given the arbitrariness of $\varepsilon > 0$, we conclude that $\mathcal{L}^1(E_k) = 0$. This proves (2.11) in the case $I = [a, b]$.

In the general case it suffices to write I as an increasing countable union of closed intervals and apply the previous case in each closed interval. Since the countable union of sets of Lebesgue measure zero has Lebesgue measure zero, we conclude the proof. $\square$

In the case $Y = \mathbb{R}$ an alternative proof is given in the following exercise.

Exercise 2.29. Let $u \in BPV([a, b])$ and for every $a \leq x \leq b$ define

$$\mathbf{V}(x) := \text{Var}_{[a, x]} u.$$

Let P be a partition of $[a, b]$, with $a = x_0 < x_1 < \cdots < x_n = b$, and define $v : [a, b] \rightarrow \mathbb{R}$ inductively as follows. If $u(x_0) \leq u(x_1)$, for every $x \in [x_0, x_1]$ set $v(x) := u(x) - u(x_0)$, while if $u(x_0) > u(x_1)$, for every $x \in [x_0, x_1]$ set $v(x) := u(x_0) - u(x)$. Let $1 \leq k < n$ and suppose that v has been defined on $[x_0, x_k]$. If $u(x_k) \leq u(x_{k+1})$, for every $x \in (x_k, x_{k+1}]$ set $v(x) := u(x) + v(x_k) - u(x_k)$, while if $u(x_k) > u(x_{k+1})$, for every $x \in (x_k, x_{k+1}]$ set $v(x) := -u(x) + v(x_k) + u(x_k)$. Prove that

- (i) on each interval $[x_{k-1}, x_k]$, $1 \leq k \leq n$, either $v - u$ or $v + u$ is constant,
- (ii) $\mathbf{V}(x) = \text{Var}_{[a, x]} v$ for every $a \leq x \leq b$ and

$$v(b) = \sum_{k=1}^n |u(x_k) - u(x_{k-1})|,$$

- (iii) for every $n \in \mathbb{N}$ there is a function $v_n : [a, b] \rightarrow \mathbb{R}$ such that $\mathbf{V} - v_n$ is increasing, $\mathbf{V}(a) = v_n(a)$,

$$\mathbf{V}(b) - v_n(b) \leq \frac{1}{2^n},$$

and $|v'_n(x)| = |u'(x)|$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$,

- (iv) $\mathbf{V}'(x) = |u'(x)|$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$.

The next exercise shows that in general the inequality (2.12) can be strict.

Exercise 2.30. Let $u : [0, 1] \rightarrow \mathbb{R}$ be defined by

$$u(x) := \begin{cases} x^a \cos \frac{1}{x} & \text{if } 0 < x \leq 1, \\ 0 & \text{if } x = 0, \end{cases}$$

where $a > 0$.

- (i) Prove that if $a = 2$, then $u'(0) = 0 \neq \mathbf{V}'(0) < \infty$.
- (ii) Prove that if $1 < a < 2$, then $u'(0) = 0$, while $\mathbf{V}'(0) = \infty$.

Corollary 2.31. Let $I \subseteq \mathbb{R}$ be an interval and let $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$. Then u is differentiable $\mathcal{L}^1$ -a.e. in I , and for every interval $J \subseteq I$,

$$(2.15) \quad \int_J \|u'(x)\| dx \leq \text{Var}_J u.$$

In particular, if $u \in BPV(I; \mathbb{R}^M)$, then u' is Lebesgue integrable.

Proof. Step 1: By (2.2), if $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$, then $u_i \in BPV_{\text{loc}}(I)$ for all $i = 1, \dots, M$. Let $\mathbf{V}_{u_i}$ be the indefinite pointwise variation of u_i (see (2.3)). By Theorem 2.17 the functions $\mathbf{V}_{u_i} \pm u_i$ are increasing, and so

$$u_i = \frac{1}{2}(\mathbf{V}_{u_i} + u_i) - \frac{1}{2}(\mathbf{V}_{u_i} - u_i)$$

is the difference of two increasing functions. It follows from Lebesgue's differentiation theorem (Theorem 1.18) that u_i is differentiable $\mathcal{L}^1$ -a.e. in I for all $i = 1, \dots, M$. In turn, u is differentiable $\mathcal{L}^1$ -a.e. in I .

To prove (2.15), let $[a, b] \subseteq I$ and let $\mathbf{V}_u$ be the indefinite pointwise variation of u (see (2.3)). Since $\mathbf{V}_u$ is increasing, by Corollary 1.25 applied to $\mathbf{V}_u$, Theorem 2.17, and (2.11), we obtain

$$(2.16) \quad \int_a^b \|u'\| dx = \int_a^b \mathbf{V}'_u dx \leq \mathbf{V}_u(b) - \mathbf{V}_u(a) = \text{Var}_{[a,b]} u.$$

Now, if $J \subseteq I$ is any nonempty interval, construct an increasing sequence of intervals $[a_n, b_n]$ such that $J = \bigcup_n [a_n, b_n]$. Then from (2.15),

$$\int_{a_n}^{b_n} \|u'\| dx \leq \text{Var}_{[a_n, b_n]} u \leq \text{Var}_J u.$$

Letting $n \rightarrow \infty$ and using the Lebesgue monotone convergence theorem concludes the proof. $\square$

Remark 2.32. Note that if $u \in BPV(I; \mathbb{R}^M)$, one could have used the function $\mathbf{V}_\infty$ defined in (2.7) instead of $\mathbf{V}_u$.

Exercise 2.33. Let $u_n(x) = \frac{1}{n^2}e^{-n^2x^2}$, $x \in \mathbb{R}$.

- (i) Calculate $\text{Var } u_n$.
- (ii) Let $u(x) := \sum_{n=1}^{\infty} u_n(x)$, $x \in \mathbb{R}$. Prove that $u \in BPV(\mathbb{R})$.
- (iii) Prove that $\sum_{n=1}^{\infty} u'_n(x)$ does not converge uniformly in $[-1, 1]$.
- (iv) Find a formula for u' .
- (v) What is the relevance of this exercise?

Exercise 2.34. Let $u : [a, b] \rightarrow \mathbb{R}^M$.

- (i) Let $g : [\alpha, \beta] \rightarrow \mathbb{R}^M$ be a continuous function. Prove that if $x_0 \in [\alpha, \beta]$, then

$$\left\| \int_{\alpha}^{\beta} g(x) dx \right\| \geq \int_{\alpha}^{\beta} \|g(x)\| dx - 2 \int_{\alpha}^{\beta} \|g(x) - g(x_0)\| dx.$$

- (ii) Using part (i), prove that if u is of class $C^1([a, b]; \mathbb{R}^M)$, then

$$\text{Var } u = \int_a^b \|u'(x)\| dt.$$

- (iii) Prove that if $u : [a, b] \rightarrow \mathbb{R}^M$ is (everywhere) differentiable and u' is Riemann integrable, then

$$\text{Var } u = \int_a^b \|u'(x)\| \, dx.$$

Compare this with Theorem 2.40 below.

Exercise 2.35. Let $I = [a, b]$.

- (i) Prove that if $v : I \rightarrow \mathbb{R}$ has bounded pointwise variation and the intermediate value property (that is, if it takes two values, it takes also all the values in between), then v is continuous.
- (ii) Prove that if $u : I \rightarrow \mathbb{R}$ is differentiable and u' has bounded pointwise variation, then u' is continuous.

2.4. Monotone Versus BPV

Another consequence of Theorem 2.17 is the result mentioned at the beginning of the chapter, namely, the characterization of the smallest vector space of functions $u : I \rightarrow \mathbb{R}$ that contains all monotone functions.

Theorem 2.36. *Let $I \subseteq \mathbb{R}$ be an interval. The smallest vector space of functions $u : I \rightarrow \mathbb{R}$ that contains all monotone functions (respectively, bounded monotone functions) is given by the space $BPV_{\text{loc}}(I)$ (respectively, $BPV(I)$). Moreover, every function in $BPV_{\text{loc}}(I)$ (respectively, $BPV(I)$) may be written as a difference of two increasing functions (respectively, two bounded increasing functions).*

Proof. Let $u, v : I \rightarrow \mathbb{R}$ and let $t \in \mathbb{R}$. By (2.1) for every interval $J \subseteq I$ we have

$$(2.17) \quad \text{Var}_J(tu) = |t| \text{Var}_J u, \quad \text{Var}_J(u + v) \leq \text{Var}_J u + \text{Var}_J v.$$

Hence, the functions in $BPV_{\text{loc}}(I)$ (respectively, $BPV(I)$) form a vector space.

By Propositions 2.11 and 2.12, the space $BPV_{\text{loc}}(I)$ (respectively, $BPV(I)$) contains all monotone functions (respectively, bounded monotone functions). To prove that it is the smallest, it suffices to show that every function u in $BPV_{\text{loc}}(I)$ (respectively, in $BPV(I)$) may be written as the difference of two increasing functions (respectively, bounded increasing). In view of Theorem 2.17, it suffices to write

$$u = \mathbf{V} - (\mathbf{V} - u) \quad \text{or} \quad u = \frac{1}{2}(\mathbf{V} + u) - \frac{1}{2}(\mathbf{V} - u).$$

This concludes the proof. □

Exercise 2.37. Let $I \subseteq \mathbb{R}$ be an interval and let $u \in BPV_{\text{loc}}(I)$ be right continuous (respectively, left continuous). Prove that for every $x \in I^\circ$ the functions $\frac{1}{2}(\mathbf{V} + u)$ and $\frac{1}{2}(\mathbf{V} - u)$ cannot both be discontinuous at x .

Exercise 2.38. Let $u, v \in BPV([a, b])$. Prove the following.

- (i) $u \pm v \in BPV([a, b])$.
- (ii) $uv \in BPV([a, b])$.
- (iii) If $v(x) \geq c > 0$ for all $x \in [a, b]$ and for some $c > 0$, then $u/v \in BPV([a, b])$.
- (iv) $\max\{u, v\}, \min\{u, v\} \in BPV([a, b])$.
- (v) What happens if we replace $[a, b]$ with an arbitrary interval $I \subseteq \mathbb{R}$ (possibly unbounded)?

Theorem 2.36 shows that every function in $BPV_{\text{loc}}(I)$ may be written as the difference of two increasing functions. The next result shows that the difference of two increasing functions can lose every kind of monotonicity and be rather pathological.

Exercise 2.39. Let (X, d) be a metric space and let E be the intersection of countably many G_δ sets dense in X . Prove that E is dense in X .

Theorem 2.40 (Weil). *There exists a Lipschitz continuous function $u : [0, 1] \rightarrow \mathbb{R}$, which is everywhere differentiable, monotone on no interval of $[0, 1]$, and such that u' is bounded but not Riemann integrable over any interval of $[0, 1]$.*

Proof. Step 1: Let X be the vector space of all bounded functions $v : [0, 1] \rightarrow \mathbb{R}$ with the property that there exists a differentiable function $u : [0, 1] \rightarrow \mathbb{R}$ such that $u' = v$. Then X is a Banach space when endowed with the sup norm. To see this, it suffices to show that it is a closed subspace of the space of bounded functions. Let $\{v_n\}_n$ be a sequence in X such that $\|v_n - v\|_\infty \rightarrow 0$, for some bounded function $v : [0, 1] \rightarrow \mathbb{R}$. Then $\|u'_n - v\|_\infty \rightarrow 0$. By replacing u_n with $u_n - u_n(0)$, we may assume that $u_n(0) = 0$ for all n . It follows (why?) that there exists a differentiable function u such that $u' = v$. This shows that $v \in X$.

Let Z be the subspace of all $v \in X$ such that the set $v^{-1}(\{0\})$ is dense in $[0, 1]$. Note that the set $v^{-1}(\{0\})$ is a G_δ set. Indeed, let u be such that $u' = v$. Then

$$\begin{aligned} v^{-1}(\{0\}) &= \left\{x \in [0, 1] : \lim_{n \rightarrow \infty} n(u(x + \frac{1}{n}) - u(x)) = 0\right\} \\ &= \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} \left\{x \in [0, 1] : n|u(x + \frac{1}{n}) - u(x)| < \frac{1}{k}\right\} =: \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} U_{n,k}. \end{aligned}$$

Since u is continuous, the sets $U_{n,k}$ are relatively open, which shows that $v^{-1}(\{0\})$ is a countable intersection of relatively open sets.

We claim that Z is a Banach space. Indeed, let $\{v_n\}_n$ be a sequence in Z such that $\|v_n - v\|_\infty \rightarrow 0$ for some $v \in X$. Then $v_n^{-1}(\{0\})$ is a G_δ set dense in $[0, 1]$ for every n . It follows by the previous exercise that $E_0 := \bigcap_{n=1}^\infty v_n^{-1}(\{0\})$ is dense in $[0, 1]$. But since $E_0 \subseteq v^{-1}(\{0\})$, we have that $v^{-1}(\{0\})$ is dense in $[0, 1]$, which shows that v belongs to Z .

Step 2: Let W be the set of all $v \in Z$ such that there exists a proper interval $I \subseteq [0, 1]$ such that either $v \geq 0$ in I or $v \leq 0$ in I . Let $\{I_n\}_n$ be the family of all closed proper intervals with rational endpoints and set

$$W_n^+ := \{v \in W : v \geq 0 \text{ in } I_n\}, \quad W_n^- := \{v \in W : v \leq 0 \text{ in } I_n\}.$$

Then the sets $W_n^\pm$ are closed and

$$W := \bigcup_{n=1}^\infty (W_n^+ \cup W_n^-).$$

Fix n . We claim that $W_n^\pm$ is nowhere dense. We only prove it for W_n^+ . Let $v \in W_n^+$ and let $\varepsilon > 0$. Since $v^{-1}(\{0\})$ is dense in $[0, 1]$ and $\inf I_n < \sup I_n$, there exists $x_0 \in I_n^\circ$ such that $v(x_0) = 0$. By Exercise 1.40 the space Z contains a nonzero function $w \in Z$ with $w(x_0) \neq 0$. By multiplying w by $\varepsilon/(2\|w\|_\infty)$, without loss of generality, we may assume that $\|w\|_\infty < \varepsilon$. Moreover, by possibly changing sign to w , we can also assume that $w(x_0) < 0$. Consider the function $v + w$. Since $\|w\|_\infty < \varepsilon$, we have that $v + w \in B(v, \varepsilon)$. Moreover, $(v + w)(x_0) < 0$. Thus, $v + w \notin W_n^+$. This shows that W_n^+ has empty interior.

In turn, by the Baire category theorem (see Corollary A.13) we have that W is a proper subset of Z . Let $v \in Z \setminus W$. Given any proper interval $I \subseteq [0, 1]$, the function v must change sign in I . Thus, if u is such that $u' = v$, necessarily u cannot be monotone in I .

By the mean value theorem, u is Lipschitz continuous, and so by Proposition 2.6, u is of bounded pointwise variation on bounded sets.

The proof of the fact that u' is not Riemann integrable on any closed interval $[a, b]$ will make use of absolutely continuous functions, and so it will be postponed until after Theorem 3.12. $\square$

Exercise 2.41. Let $I_n := (a_n, b_n)$, $n \in \mathbb{N}$, be a sequence of pairwise disjoint intervals in $[0, 1]$ and choose $a_n < \alpha_n < \beta_n < b_n$ such that $\alpha_n - a_n = b_n - \beta_n$.

- (i) For each $n \in \mathbb{N}$ construct a differentiable function $u_n : I_n \rightarrow \mathbb{R}$ such that

- (a) $u_n = 0$ in $I_n \setminus (\alpha_n, \beta_n)$,
- (b) $0 \leq u_n \leq (\alpha_n - a_n)^2$,

$$(c) \max_{I_n} |u'_n| = 1.$$

(ii) Define

$$u(x) := \begin{cases} u_n(x) & \text{if } x \in I_n \text{ for some } n \in \mathbb{N}, \\ 0 & \text{elsewhere.} \end{cases}$$

Prove that if $x_0 \in [0, 1] \setminus \bigcup_n I_n$, then for all $x \in [0, 1]$, $x \neq x_0$,

$$\frac{|u(x) - u(x_0)|}{|x - x_0|} \leq |x - x_0|.$$

(iii) Prove that u is differentiable and that $|u'| \leq 1$.

(iv) Prove that the intervals I_n can be chosen so that $[0, 1] \setminus \bigcup_n I_n$ is closed, has positive Lebesgue measure, and is nowhere dense (that is, its closure has empty interior).

(v) Prove that with such a choice of $\{I_n\}_n$ the function u' is discontinuous at every accumulation point of $[0, 1] \setminus \bigcup_n I_n$ and u' is not Riemann integrable.

2.5. The Space $BPV(I; Y)$

In this section we prove that $BPV(I; Y)$ is a Banach space, provided Y is a Banach space.

Let $I \subseteq \mathbb{R}$ be an interval and let $(Y, \|\cdot\|)$ be a normed space. By the properties of the supremum of a set and of the norm $\|\cdot\|$ we have that

$$\text{Var}(u + v) \leq \text{Var } u + \text{Var } v$$

and

$$\text{Var}(su) = |s| \text{Var } u$$

for all functions $u : I \rightarrow Y$ and $v : I \rightarrow Y$, and for all $s \in \mathbb{R}$.

However, the pointwise variation Var is not a norm since a constant function has zero variation. To overcome this problem, it suffices to consider $\|u(x_0)\| + \text{Var } u$ for some $x_0 \in I$. Indeed, the following result holds.

Theorem 2.42. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a Banach space, and let $x_0 \in I$. Then the space $BPV(I; Y)$ endowed with the norm*

$$(2.18) \quad \|u\|_{BPV} := \|u(x_0)\| + \text{Var } u$$

is a Banach space.

Proof. Let $\{u_n\}_n$ be a Cauchy sequence in $BPV(I; Y)$. Given $\varepsilon > 0$, there exists $n_\varepsilon \in \mathbb{N}$ such that

$$\|u_n(x_0) - u_m(x_0)\| + \text{Var}(u_n - u_m) \leq \varepsilon$$

for all $n, m \geq n_\varepsilon$. Hence, by Remark 2.13, $\|u_n - u_m\|_\infty \leq \varepsilon$ for all $n, m \geq n_\varepsilon$. This implies that $\{u_n\}_n$ is a Cauchy sequence in the space

$$B(I; Y) = \{v : I \rightarrow Y, v \text{ bounded}\}$$

with the norm $\|\cdot\|_\infty$. Using the completeness of Y , we have that $B(I; Y)$ is a Banach space (exercise). Hence, the sequence $\{u_n\}_n$ converges uniformly to some bounded function $u : I \rightarrow Y$.

Given a partition $x_0 < x_1 < \cdots < x_\ell$ of I we have that

$$\sum_{i=1}^{\ell} \|(u_n - u_m)(x_i) - (u_n - u_m)(x_{i-1})\| \leq \text{Var}(u_n - u_m) \leq \varepsilon$$

for all $n, m \geq n_\varepsilon$. Letting $m \rightarrow \infty$ and using the fact that $u_m(x_i) \rightarrow u(x_i)$ for all $i = 0, \dots, \ell$ (since uniform convergence implies pointwise convergence) gives

$$\sum_{i=1}^{\ell} \|(u_n - u)(x_i) - (u_n - u)(x_{i-1})\| \leq \varepsilon$$

for all $n \geq n_\varepsilon$. Taking the supremum over all partitions of I gives

$$\text{Var}(u_n - u) \leq \varepsilon$$

for all $n \geq n_\varepsilon$. Since $\|u_n(x_0) - u_m(x_0)\| \rightarrow \|u_n(x_0) - u(x_0)\|$ as $m \rightarrow \infty$, we have shown that

$$\|u_n - u\|_{BPV} \leq 2\varepsilon$$

for all $n \geq n_\varepsilon$. Moreover, $\text{Var } u \leq \text{Var}(u_n - u) + \text{Var } u_n < \infty$. Thus, $u \in BPV(I; Y)$ and $u_n \rightarrow u$ in $BPV(I; Y)$. This completes the proof. $\square$

The next exercise shows that the space $BPV(I)$ endowed with the norm defined in (2.18) is not separable.

Exercise 2.43. Consider the space $BPV([0, 1])$ endowed with the norm (2.18). For every $u \in BPV([0, 1])$ and $r > 0$ let

$$B_{BPV}(u, r) := \{v \in BPV([0, 1]) : \|u - v\| < r\}.$$

For every $t \in [0, 1]$ define $u_t := \chi_{\{t\}}$.

- (i) Prove that $B_{BPV}(u_t, 1) \cap B_{BPV}(u_s, 1) = \emptyset$ for every $t, s \in [0, 1]$, with $t \neq s$.
- (ii) Prove that for every $t \in [0, 1]$, $B_{BPV}(u_t, 1) \subseteq B_{BPV}(0, 3)$.
- (iii) Prove that the Banach space $(BPV([0, 1]), \|\cdot\|)$ is not separable.

We conclude this section with an important compactness result.

Theorem 2.44 (Helly's selection theorem). *Let $I \subseteq \mathbb{R}$ be an interval and let $\mathcal{F}$ be an infinite family of functions in $BPV(I)$. Assume that there exist $x_0 \in I$ and a constant $c > 0$ such that*

$$(2.19) \quad |u(x_0)| + \text{Var } u \leq c$$

for all $u \in \mathcal{F}$. Then there exist a sequence $\{u_n\}_n$ in $\mathcal{F}$ and a function $v \in BPV(I)$ such that

$$u_n(x) \rightarrow v(x)$$

for all $x \in I$.

We divide the proof into a few lemmas, which are of independent interest.

Lemma 2.45. *Let $I \subseteq \mathbb{R}$ be an interval and let $\mathcal{F}$ be an infinite family of functions $u : I \rightarrow \mathbb{R}$. Assume that there exists a constant $c > 0$ such that $|u(x)| \leq c$ for all $x \in I$ and all $u \in \mathcal{F}$. Then for every countable set $E \subset I$, there exists a sequence $\{u_n\}_n$ in $\mathcal{F}$ such that the limit $\lim_{n \rightarrow \infty} u_n(x)$ exists in $\mathbb{R}$ for all $x \in E$.*

Proof. The proof makes use of the *Cantor diagonal argument*. Without loss of generality, we can assume that E is denumerable, so that $E = \{x_k : k \in \mathbb{N}\}$. Since the set $\{u(x_1) : u \in \mathcal{F}\}$ is bounded, by the Bolzano–Weierstrass theorem we can find a sequence $\{u_n^{(1)}\}_n$ in $\mathcal{F}$ for which there exists the limit

$$\lim_{n \rightarrow \infty} u_n^{(1)}(x_1) = \ell_1 \in \mathbb{R}.$$

Since the set $\{u_n^{(1)}(x_2) : n \in \mathbb{N}\}$ is bounded, again by the Bolzano–Weierstrass theorem we can find a subsequence $\{u_n^{(2)}\}_n$ of $\{u_n^{(1)}\}_n$ for which there exists the limit

$$\lim_{n \rightarrow \infty} u_n^{(2)}(x_2) = \ell_2 \in \mathbb{R}.$$

By induction for every $k \in \mathbb{N}$, $k > 1$, we can find a subsequence $\{u_n^{(k)}\}_n$ of $\{u_n^{(k-1)}\}_n$ for which there exists the limit

$$\lim_{n \rightarrow \infty} u_n^{(k)}(x_k) = \ell_k \in \mathbb{R}.$$

We now consider the diagonal elements of the infinite matrix, that is, the sequence $\{u_n^{(n)}\}_n$. For every fixed $x_k \in E$ we have that the sequence $\{u_n^{(n)}(x_k)\}_{n=k}^\infty$ is a subsequence of $\{u_n^{(k)}(x_k)\}_n$, and thus it converges to ℓ_k as $n \rightarrow \infty$. This completes the proof. $\square$

Lemma 2.46. *Let $I \subseteq \mathbb{R}$ be an interval and let $\mathcal{F}$ be an infinite family of increasing functions $u : I \rightarrow \mathbb{R}$. Assume that there exists a constant $c > 0$ such that*

$$|u(x)| \leq c$$

for all $x \in I$ and all $u \in \mathcal{F}$. Then there exist a sequence $\{u_n\}_n$ in $\mathcal{F}$ and an increasing function $v : I \rightarrow \mathbb{R}$ such that $\lim_{n \rightarrow \infty} u_n(x) = v(x)$ for all $x \in I$.

Proof. Apply the previous lemma with E given by the union of all the rationals in I and the endpoints of I that belong to I (if any) to find a sequence $\{u_n\}_n$ in $\mathcal{F}$ such that the limit $\lim_{n \rightarrow \infty} u_n(x)$ exists in $\mathbb{R}$ for all $x \in E$. Define $w : E \rightarrow \mathbb{R}$ by

$$(2.20) \quad w(x) := \lim_{n \rightarrow \infty} u_n(x), \quad x \in E.$$

Since each function u_n is increasing, if $x, y \in E$ and $x < y$, then $u_n(x) \leq u_n(y)$ for all $n \in \mathbb{N}$, and so letting $n \rightarrow \infty$, we conclude that $w(x) \leq w(y)$. Thus, w is increasing.

To extend w to $I \setminus E$, for $x \in I \setminus E$ we define

$$w(x) := \sup_{y \in E \cap (-\infty, x)} w(y).$$

Note that w is increasing by construction. We claim that

$$(2.21) \quad \lim_{n \rightarrow \infty} u_n(x) = w(x)$$

for all those $x \in I$ such that w is continuous at x . To see this, fix $\varepsilon > 0$ and any such x . In view of (2.20) it is enough to consider the case in which $x \in I \setminus E$. Since x is not an endpoint of I , using the continuity of w at x , together with the density of the rational numbers, we may find $x_1, x_2 \in E$ such that $x_1 < x < x_2$ and

$$(2.22) \quad w(x_2) - w(x_1) < \frac{\varepsilon}{2}.$$

By (2.20) there exists an integer $n_0 \in \mathbb{N}$ such that

$$(2.23) \quad |u_n(x_1) - w(x_1)| < \frac{\varepsilon}{2}, \quad |u_n(x_2) - w(x_2)| < \frac{\varepsilon}{2}$$

for all $n \geq n_0$. Since w and u_n are increasing,

$$\begin{aligned} w(x) - \varepsilon &\leq w(x_2) + w(x_1) - w(x_1) - \varepsilon \\ &\leq w(x_1) - \frac{\varepsilon}{2} \leq u_n(x_1) \leq u_n(x) \leq u_n(x_2) \leq w(x_2) + \frac{\varepsilon}{2} \\ &= w(x_2) + w(x_1) - w(x_1) + \frac{\varepsilon}{2} \\ &\leq w(x) + w(x_2) - w(x_1) + \frac{\varepsilon}{2} \leq w(x) + \varepsilon \end{aligned}$$

for all $n \geq n_0$, where we have used (2.22) and (2.23). This implies that

$$|u_n(x) - w(x)| \leq \varepsilon$$

for all $n \geq n_0$. Thus the claim is proved.

Now let E_1 be the set of points of discontinuity of w , which is at most countable. Applying the previous lemma once more, with E_1 and $\{u_n\}_n$ in

place of E and $\mathcal{F}$, we may find a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ that converges at every point of E_1 . The function

$$v(x) := \begin{cases} w(x) & \text{if } x \in I \setminus E_1, \\ \lim_{k \rightarrow \infty} u_{n_k}(x) & \text{if } x \in E_1 \end{cases}$$

has all the desired properties. Indeed, in view of (2.20) and (2.21), we have that

$$w(x) = \lim_{n \rightarrow \infty} u_n(x) = \lim_{k \rightarrow \infty} u_{n_k}(x)$$

for all $x \in I \setminus E_1$. □

We now turn to the proof of the Helly theorem.

Proof of Helly's selection theorem. For every $u \in \mathcal{F}$ write

$$u = \mathbf{V}_u - (\mathbf{V}_u - u),$$

where $\mathbf{V}_u$ is the function defined in (2.3). By Theorems 2.17 and 2.36, $\mathbf{V}_u$ and $\mathbf{V}_u - u$ are increasing functions, and by (2.19),

$$|\mathbf{V}_u(x)| \leq c, \quad |\mathbf{V}_u(x) - u(x)| \leq |\mathbf{V}_u(x)| + |u(x) - u(x_0)| + |u(x_0)| \leq 2c$$

for all $x \in I$, and thus we may apply the previous lemma to the family $\{\mathbf{V}_u : u \in \mathcal{F}\}$ to find a sequence $\{u_n\}_n$ in $\mathcal{F}$ and an increasing function $v_1 : I \rightarrow \mathbb{R}$ such that

$$\lim_{n \rightarrow \infty} \mathbf{V}_{u_n}(x) = v_1(x)$$

for all $x \in I$. Applying the previous lemma to the family $\{\mathbf{V}_{u_n} - u_n\}$, we may find a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and an increasing function $v_2 : I \rightarrow \mathbb{R}$ such that

$$\lim_{k \rightarrow \infty} (\mathbf{V}_{u_{n_k}} - u_{n_k})(x) = v_2(x)$$

for all $x \in I$. By Theorem 2.36, the function $v := v_1 - v_2$ belongs to $BPV(I)$ and

$$\begin{aligned} \lim_{k \rightarrow \infty} u_{n_k}(x) &= \lim_{k \rightarrow \infty} [\mathbf{V}_{u_{n_k}}(x) - (\mathbf{V}_{u_{n_k}} - u_{n_k})(x)] \\ &= v_1(x) - v_2(x) = v(x) \end{aligned}$$

for all $x \in I$. □

The fact that the function v belongs to $BPV(I)$ can also be derived by the lower semicontinuity of the variations with respect to pointwise convergence. To be precise, we have the following result.

Proposition 2.47. *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space and let $\{u_n\}_n$ be a sequence of functions $u_n : I \rightarrow Y$ converging pointwise to a function $u : I \rightarrow Y$. Then*

$$(2.24) \quad \text{Var } u \leq \liminf_{n \rightarrow \infty} \text{Var } u_n.$$

In particular, if the right-hand side of the previous inequality is finite, then u belongs to $BPV(I; Y)$.

Proof. Let $P = \{x_0, \dots, x_m\}$ be a partition of I . Since $u_n(x_i) \rightarrow u(x_i)$ as $n \rightarrow \infty$ for all $i = 0, \dots, m$, for every $\varepsilon > 0$ we may find $\bar{n} \in \mathbb{N}$ such that

$$d(u_n(x_i), u(x_i)) \leq \frac{\varepsilon}{2m}$$

for all $n \geq \bar{n}$ and $i = 0, \dots, m$. Hence, for $n \geq \bar{n}$,

$$\begin{aligned} \sum_{i=1}^m d(u(x_i), u(x_{i-1})) &\leq \sum_{i=1}^m (d(u_n(x_i), u_n(x_{i-1})) + \frac{\varepsilon}{m}) \\ &= \sum_{i=1}^m d(u_n(x_i), u_n(x_{i-1})) + \varepsilon \leq \text{Var } u_n + \varepsilon. \end{aligned}$$

Taking the limit inferior on both sides, we obtain

$$\sum_{i=1}^m d(u(x_i), u(x_{i-1})) \leq \liminf_{n \rightarrow \infty} \text{Var } u_n + \varepsilon.$$

Letting $\varepsilon \rightarrow 0^+$ and taking the supremum over all partitions yields the desired result. $\square$

Remark 2.48. If u is continuous, then it is enough to assume that $u_n(x) \rightarrow u(x)$ for all x in a dense set E of I . To see this, let $P = \{x_0, \dots, x_m\}$ be as in the previous proof. By the continuity of u and the density of E there exists a partition $P' = \{y_0, \dots, y_m\} \subseteq E$ such that

$$d(u(y_i), u(x_i)) \leq \frac{\varepsilon}{m}$$

for all $i = 0, \dots, m$. We may now continue as in the previous proof (working with P' in place of P) to conclude that

$$\sum_{i=1}^m d(u(x_i), u(x_{i-1})) \leq \liminf_{n \rightarrow \infty} \text{Var } u_n + 2\varepsilon.$$

The next exercise shows that the equality sign does not necessarily hold in (2.24).

Exercise 2.49. For $x \in [0, 1]$, let $u(x) := (x, 0)$ and $u_n(x) := (x, \frac{1}{n} \sin 2\pi n^\alpha x)$, $n \in \mathbb{N}$, $\alpha > 0$.

(i) Estimate $\text{Var } u_n$ for all $n \in \mathbb{N}$ and $\alpha > 0$.

(ii) Prove that if $\alpha = \frac{1}{2}$, then

$$\text{Var } u = \lim_{i \rightarrow \infty} \text{Var } \gamma_{2i}.$$

(iii) Prove that for $\alpha = 1$ or $\alpha = 2$ we have strict inequality in (2.24).

Exercise 2.50. Consider the space $BPV([0, 1])$ endowed with the metric

$$d(u, v) := \int_0^1 |u(x) - v(x)| dx + |\text{Var } u - \text{Var } v|.$$

- (i) Prove that if $\{u_n\}_n$ is in $BPV([0, 1])$ and $\sup_n d(u_n, 0) < \infty$, then there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and a function $u \in BPV([0, 1])$ such that $d(u_{n_k}, u) \rightarrow 0$. Hint: Use Helly's theorem.
- (ii) Prove that $(BPV([0, 1]), d)$ is a complete metric space.
- (iii) Prove that $(BPV([0, 1]), d)$ is separable. Hint: Any function can be approximated by an inscribed polygonal function.
- (iv) Is there any relation between convergence in the norm (2.18) and in the metric d ? Is there any difference in the answer if the limit function u is continuous?

An important consequence of the Helly selection theorem is the following result.

Corollary 2.51. *Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be an integrable function such that*

$$c := \limsup_{h \rightarrow 0^+} \frac{1}{h} \int_{I_h} |u(x+h) - u(x)| dx < \infty,$$

where

$$I_h := \{x \in I : x+h \in I\}.$$

Then there exists a function $v \in BPV(I)$ such that $v(x) = u(x)$ for $\mathcal{L}^1$ -a.e. $x \in I$ and $\text{Var } v = c$.

Proof. For every $n \in \mathbb{N}$ we partition the interval I into subintervals of equal length $\delta_n > 0$, with $\delta_n \rightarrow 0$ as $n \rightarrow \infty$. Let $\{x_{j,n}\}_j$ be the family of endpoints of the intervals. For every $x \in (x_{j-1,n}, x_{j,n})$ define the piecewise constant function

$$u_n(x) := \frac{1}{\delta_n} \int_{x_{j-1,n}}^{x_{j,n}} u(t) dt.$$

Extend u_n at the endpoints in such a way that it is right continuous and if $b \in I$, extend u_n at b in such a way that it is left continuous at b . Since u_n

is piecewise constant,

$$\begin{aligned}
 \text{Var } u_n &= \sum_j \frac{1}{\delta_n} \left| \int_{x_{j,n}}^{x_{j+1,n}} u(t) dt - \int_{x_{j-1,n}}^{x_{j,n}} u(t) dt \right| \\
 &= \sum_j \frac{1}{\delta_n} \left| \int_{x_{j-1,n}}^{x_{j,n}} u(x + \delta_n) dx - \int_{x_{j-1,n}}^{x_{j,n}} u(x) dx \right| \\
 &\leq \sum_j \frac{1}{\delta_n} \int_{x_{j-1,n}}^{x_{j,n}} |u(x + \delta_n) - u(x)| dx \\
 &\leq \frac{1}{\delta_n} \int_{I_{\delta_n}} |u(x + \delta_n) - u(x)| dx,
 \end{aligned}$$

and so, by hypothesis,

$$(2.25) \quad \limsup_{n \rightarrow \infty} \text{Var } u_n \leq c.$$

Moreover, since $u_n \rightarrow u$ in $L^1_{\text{loc}}(I)$ (why?), we may find $x_0 \in I$ and a subsequence, not relabeled, such that $\sup_n |u_n(x_0)| < \infty$,

$$\sup_n \text{Var } u_n \leq c + \varepsilon,$$

and $u_n(x) \rightarrow u(x)$ for $\mathcal{L}^1$ -a.e. $x \in I$. Thus, we are in a position to apply Helly's selection theorem to find a further subsequence, not relabeled, and a function $v \in BPV(I)$ such that

$$u_n(x) \rightarrow v(x)$$

for all $x \in I$. Hence $v(x) = u(x)$ for $\mathcal{L}^1$ -a.e. $x \in I$, and so

$$\frac{1}{h} \int_{I_h} |u(x+h) - u(x)| dx = \frac{1}{h} \int_{I_h} |v(x+h) - v(x)| dx \leq \text{Var } v$$

by Theorem 2.20. Letting $h \rightarrow 0^+$, we conclude that $c \leq \text{Var } v$. On the other hand, by Proposition 2.47 and (2.25),

$$\text{Var } v \leq \liminf_{n \rightarrow \infty} \text{Var } u_n \leq c,$$

and so $\text{Var } v = c$. □

Results of this type are often useful in the regularity theory of solutions of differential equations (see, e.g., [33] and [71]).

Exercise 2.52. Prove that the sequence $\{u_n\}_n$ constructed in the previous corollary converges to u in $L^1_{\text{loc}}(I)$.

2.6. Composition in $BPV(I; Y)$

In this section we study the composition of functions with bounded pointwise variation. Exercise 2.38 shows that $BPV([a, b])$ is closed under the operations of addition and multiplication. The next exercise shows that it is not closed under composition (see, however, Exercise 2.54).

Exercise 2.53. Consider the functions $f : [0, 1] \rightarrow \mathbb{R}$ and $u : [0, 1] \rightarrow [0, 1]$, defined by

$$f(z) := \sqrt{z}, \quad u(x) := \begin{cases} x^2 \sin^2 \frac{1}{x} & \text{if } 0 < x \leq 1, \\ 0 & \text{if } x = 0. \end{cases}$$

Prove that f and u have bounded pointwise variation, but $f \circ u$ does not.

Exercise 2.54. Let $I, J \subseteq \mathbb{R}$ be two intervals, let $f \in BPV_{\text{loc}}(J)$, and let $u : I \rightarrow J$ be monotone. Prove that $f \circ u$ belongs to $BPV_{\text{loc}}(I)$.

Next we discuss necessary and sufficient conditions for a given function f to be such that $f \circ u$ is a function of bounded pointwise variation for *every* function u of bounded pointwise variation.

Theorem 2.55 (Josephy). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : \mathbb{R}^M \rightarrow \mathbb{R}$. Then $f \circ u$ belongs to $BPV_{\text{loc}}(I)$ (respectively, $BPV(I)$) for all functions $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$ (respectively, $BPV(I; \mathbb{R}^M)$) if and only if f is locally Lipschitz continuous.*

Proof. We only prove the result in the case of $BPV_{\text{loc}}(I; \mathbb{R}^M)$, the case of $BPV(I; \mathbb{R}^M)$ being completely similar.

Step 1: Assume that $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$ and that f is locally Lipschitz continuous. Fix an interval $[a, b] \subseteq I$. Since u is bounded in $[a, b]$ (see Proposition 2.12), we have that $m := \sup_{x \in [a, b]} \|u(x)\| < \infty$, and so we may find $L > 0$ such that

$$|f(z_1) - f(z_2)| \leq L \|z_1 - z_2\|$$

for all $z_1, z_2 \in \overline{B(0, m)}$. Let $P := \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. Then

$$\begin{aligned} \sum_{i=1}^n |(f \circ u)(x_i) - (f \circ u)(x_{i-1})| &= \sum_{i=1}^n |f(u(x_i)) - f(u(x_{i-1}))| \\ &\leq L \sum_{i=1}^n \|u(x_i) - u(x_{i-1})\| \leq L \text{Var}_{[a, b]} u, \end{aligned}$$

which implies that $\text{Var}_{[a, b]}(f \circ u) \leq L \text{Var}_{[a, b]} u < \infty$.

Step 2: Conversely, assume that $f \circ u \in BPV_{\text{loc}}(I)$ for every $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$. Fix $[a, b] \subseteq I$. We begin by showing that f is locally bounded. Fix $r > 0$. We claim that f is bounded in $\overline{B(0, r)}$. Indeed, assume by

contradiction that there exists $\{z_n\}_n$ in $\overline{B(0, r)}$ such that $|f(z_n)| \rightarrow \infty$ as $n \rightarrow \infty$. By compactness we can assume that $z_n \rightarrow z_0$, and by taking a further subsequence, also that

$$\|z_n - z_0\| \leq 1/n^2$$

for every n . For every $n \in \mathbb{N}$, set $I_n := [a + \frac{b-a}{n+1}, a + \frac{b-a}{n})$. Consider the function

$$(2.26) \quad u(x) := \begin{cases} z_n & \text{if } x \in I_n^\circ, n \in \mathbb{N}, \\ z_0 & \text{otherwise in } I. \end{cases}$$

Then $\text{Var}_{\overline{I_n}} u = 2\|z_n - z_0\|$ and hence, by Exercise 2.22 (which holds for the left endpoint with the obvious changes) and Proposition 2.15 we have that

$$\text{Var}_{[a, b]} u = \lim_{n \rightarrow \infty} \text{Var}_{[a + \frac{b-a}{n+1}, b]} u = \lim_{n \rightarrow \infty} \sum_{k=1}^n \text{Var}_{\overline{I_k}} u \leq \sum_{k=1}^{\infty} \frac{2}{k^2} < \infty.$$

Since u is constant outside $[a, b]$, it follows that u belongs to $BPV_{\text{loc}}(I; \mathbb{R}^M)$, and so by hypothesis the function $f \circ u$ belongs to $BPV([a, b])$. However,

$$\text{Var}_{[a, b]}(f \circ u) \geq \text{Var}_{\overline{I_n}}(f \circ u) = 2|f(z_n) - f(z_0)| \rightarrow \infty$$

as $n \rightarrow \infty$, which is a contradiction.

Step 3: Next we claim that f is Lipschitz continuous in $\overline{B(0, r)}$. Indeed, assume by contradiction that this is not the case. Then we may find two sequences $\{y_n\}_n, \{z_n\}_n$ in $\overline{B(0, r)}$ such that $y_n \neq z_n$ and

$$(2.27) \quad \frac{|f(y_n) - f(z_n)|}{\|y_n - z_n\|} > 2(n^2 + n)$$

for all $n \in \mathbb{N}$. Since $\{y_n\}_n$ is bounded, we may extract a subsequence (not relabeled) such that $y_n \rightarrow y_\infty$. Take a further subsequence (not relabeled) such that (2.27) continues to hold and

$$(2.28) \quad \|y_n - y_\infty\| < \frac{1}{(n+1)^2}.$$

Since f is bounded in $\overline{B(0, r)}$ by some constant $c_r > 0$ by Step 2, by (2.27) for all $n \in \mathbb{N}$, we have

$$(2.29) \quad 2c_r \geq |f(y_n) - f(z_n)| > 2(n^2 + n)\|y_n - z_n\|.$$

Hence,

$$(2.30) \quad 0 < \delta_n := \frac{\|y_n - z_n\|(b-a)}{2c_r} < \frac{b-a}{2(n^2 + n)} \rightarrow 0.$$

We now define the piecewise constant function $u : I \rightarrow \mathbb{R}^M$ in the following way:

$$(2.31) \quad u(x) := \begin{cases} y_\infty & \text{if } x \leq a, \\ z_n & \text{if } x \in I_n, x - \left(a + \frac{b-a}{n+1}\right) \text{ is not a multiple of } \delta_n, n \in \mathbb{N}, \\ y_n & \text{if } x \in I_n, x - \left(a + \frac{b-a}{n+1}\right) \text{ is a multiple of } \delta_n, n \in \mathbb{N}, \\ y_1 & \text{if } x \geq b. \end{cases}$$

Here I_n is as in Step 2. For every $n \in \mathbb{N}$, let

$$(2.32) \quad \ell_n := \frac{\text{diam } I_n}{\delta_n} = \frac{b-a}{\delta_n(n^2+n)} = \frac{2c_r}{\|y_n - z_n\|(n^2+n)} > 2$$

by (2.29) and set $m_n := \max\{j \in \mathbb{N}_0 : j < \ell_n\}$. Note that m_n is the number of times that the function u takes the value y_n in the interval I_n . Since $\ell_n > 2$, we have

$$(2.33) \quad \frac{\ell_n}{2} \leq m_n < \ell_n.$$

Consider the partition P_n of $\bar{I}_n = \left[a + \frac{b-a}{n+1}, a + \frac{b-a}{n}\right]$ given by

$$\begin{aligned} P_n &:= \left\{a + \frac{b-a}{n+1} + \frac{1}{2}j\delta_n : j = 0, \dots, 2m_n\right\} \\ &=: \{x_{0,n}, \dots, x_{2m_n,n}\}. \end{aligned}$$

Then by (2.29)–(2.33),

$$\begin{aligned} \text{Var}_{\bar{I}_n}(f \circ u) &\geq \sum_{i=1}^{2m_n} |f(u(x_{i,n})) - f(u(x_{i-1,n}))| \\ &\geq 2m_n |f(y_n) - f(z_n)| \geq 2\ell_n(n^2+n)\|y_n - z_n\| = 4c_r, \end{aligned}$$

and so for every $n \in \mathbb{N}$ by Proposition 2.15 we obtain that

$$\text{Var}_{[a,b]}(f \circ u) \geq \sum_{k=1}^n \text{Var}_{\bar{I}_k}(f \circ u) \geq 4c_r n \rightarrow \infty$$

as $n \rightarrow \infty$.

On the other hand, since u is a step function on $\bar{I}_n$, by (2.28)–(2.33) we have that (why?)

$$\begin{aligned} \text{Var}_{\bar{I}_n} u &\leq 2m_n \|y_n - z_n\| + \|y_{n+1} - y_n\| + \|y_{n+1} - z_n\| \\ &< 2\ell_n \|y_n - z_n\| + 2\|y_{n+1} - y_\infty\| + 2\|y_\infty - y_n\| + \|y_n - z_n\| \\ &< \frac{4c_r}{n^2+n} + \frac{4}{(n+1)^2} + \frac{c_r}{n^2+n} \leq \frac{5c_r+4}{n^2}. \end{aligned}$$

Since $y_n \rightarrow y_\infty$ and $z_n \rightarrow y_\infty$, it follows by (2.31) that u is right continuous at a . Hence, by Exercise 2.22 (which holds for the left endpoint with the

obvious changes) and Proposition 2.15 we have that

$$\mathrm{Var}_{[a,b]} u = \lim_{n \rightarrow \infty} \mathrm{Var}_{\left[a + \frac{b-a}{n+1}, b\right]} u = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mathrm{Var}_{\bar{I}_k} u \leq \sum_{k=1}^{\infty} \frac{5c_r + 4}{k^2} < \infty,$$

which shows that $u \in BPV_{\mathrm{loc}}(I)$. Hence, we have obtained a contradiction. $\square$

Remark 2.56. The previous proof continues to hold if $f : \mathbb{R}^M \rightarrow Z$, where (Z, d_Z) is a metric space. On the other hand, reasoning as in Step 1 we have that if (Y, d_Y) and (Z, d_Z) are metric spaces and $f : Y \rightarrow Z$ is locally Lipschitz continuous, then $f \circ u$ belongs to $BPV_{\mathrm{loc}}(I; Z)$ (respectively, $BPV(I; Z)$) for all functions $u \in BPV_{\mathrm{loc}}(I; Y)$ (respectively, $BPV(I; Y)$).

The next exercise gives necessary and sufficient conditions for a function h to have the property that $u \circ h$ belongs to $BPV(I)$ for *every* function $u \in BPV(I)$.

Exercise 2.57. For each $\ell \in \mathbb{N}$ let

$$\mathcal{J}_\ell := \{E \subseteq [0, 1] : E \text{ can be written as a union of } \ell \text{ intervals}\},$$

where we allow singletons as degenerate closed intervals. A function $g : [0, 1] \rightarrow \mathbb{R}$ is said to be of ℓ -bounded pointwise variation if $g^{-1}([a, b]) \in \mathcal{J}_\ell$ for all $[a, b] \subset \mathbb{R}$.

- (i) Prove that if $g : [0, 1] \rightarrow \mathbb{R}$ is of ℓ -bounded pointwise variation and if $|g(x)| \leq c$ for all $x \in [0, 1]$, then $\mathrm{Var} g \leq 4c(\ell + 1)$.
- (ii) Prove that if $u : [0, 1] \rightarrow \mathbb{R}$ is increasing and if $g : [0, 1] \rightarrow [0, 1]$ is of ℓ -bounded pointwise variation for some $\ell \in \mathbb{N}$, then $u \circ g$ is bounded and of ℓ -bounded pointwise variation.
- (iii) Prove that if $u : [0, 1] \rightarrow \mathbb{R}$ is of bounded pointwise variation and if $g : [0, 1] \rightarrow [0, 1]$ is of ℓ -bounded pointwise variation for some $\ell \in \mathbb{N}$, then $u \circ g$ is of bounded pointwise variation.
- (iv) For each $n \in \mathbb{N}$ let $g_n : [0, 1] \rightarrow \{0, \frac{1}{3^n}\}$ and let $g := \sum_{n=1}^{\infty} g_n$. Prove that $\mathrm{Var} g \geq \frac{1}{6} \sum_{n=1}^{\infty} \mathrm{Var} g_n$.
- (v) Prove that if $g : [0, 1] \rightarrow [0, 1]$ is such that $u \circ g$ is of bounded pointwise variation for all $u : [0, 1] \rightarrow \mathbb{R}$ of bounded pointwise variation, then $g : [0, 1] \rightarrow \mathbb{R}$ is of ℓ -bounded pointwise variation for some $\ell \in \mathbb{N}$. Hint: If not, then for each $n \in \mathbb{N}$ find an interval $I_n \subseteq [0, 1]$ such that $g^{-1}(I_n) \notin \mathcal{J}_{3^n}$ and take $u := \sum_{n=1}^{\infty} u_n$ for an appropriate choice of u_n .

2.7. Banach Indicatrix

We conclude this chapter with a characterization of continuous functions of bounded pointwise variation due to Banach [18]. It expresses the fact that the graph of a function in $BPV(I)$ cannot intersect a horizontal line too many times. This result paved the way to the area formula (see [10], [72], [76], and [251]).

Definition 2.58. Given two nonempty sets X, Y , a function $u : X \rightarrow Y$, and a subset $E \subseteq X$, for every $y \in Y$ let $N_u(y; E)$ be the number of elements (if any) of the set $\{x \in E : u(x) = y\}$. The function $N_u(\cdot; E) : Y \rightarrow \mathbb{N}_0 \cup \{\infty\}$ is called the *counting function* or *Banach indicatrix* of u in the set E .

Note that $N_u(y; E) = \mathcal{H}^0(E \cap u^{-1}(\{y\}))$. When $E = X$, we sometimes omit the dependence on E ; namely, we write $N_u := N_u(\cdot; X)$.

We begin with a preliminary result due to Federer [75].

Theorem 2.59 (Federer). *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, let $u : I \rightarrow Y$ be such that $u(J)$ is a Borel set for every interval $J \subseteq I$, and let $\mu : \mathcal{B}(Y) \rightarrow [0, \infty]$ be a measure. For every $n \in \mathbb{N}$ let $\mathcal{F}_n$ be a countable family of pairwise disjoint intervals with the properties that $\mathcal{F}_n$ is a partition of I and*

$$\sup_{J \in \mathcal{F}_n} \text{diam } J \rightarrow 0$$

as $n \rightarrow \infty$. Then $N_u(\cdot; I)$ is a Borel function and

$$\int_Y N_u(y; I) d\mu(y) = \lim_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \mu(u(J)).$$

Proof. We divide the proof into two parts.

Step 1: We claim that for every $y \in Y$,

$$(2.34) \quad N_u(y; I) = \lim_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \chi_{u(J)}(y).$$

To see this, if $N_u(y; I) > 0$, let $m \in \mathbb{N}$ be such that $m \leq N_u(y; I)$. Then there exist m distinct points $x_1, \dots, x_m \in I$ such that $u(x_i) = y$. Assume that $x_1 < \dots < x_m$ and let $\delta_m := \min\{x_i - x_{i-1} : i = 2, \dots, m\} > 0$. Since $\sup_{J \in \mathcal{F}_n} \text{diam } J \rightarrow 0$ as $n \rightarrow \infty$, there exists $n_\delta \in \mathbb{N}$ such that

$$\sup_{J \in \mathcal{F}_n} \text{diam } J < \delta_m$$

for all $n \geq n_\delta$. Using the fact that each $\mathcal{F}_n$ is a partition of I , it follows that for $n \geq n_\delta$ there must exist m distinct intervals $J_{1,n}, \dots, J_{m,n} \in \mathcal{F}_n$ such

that $x_i \in J_{i,n}$ for all $i = 1, \dots, m$ and $n \geq n_\delta$. Thus, for all $n \geq n_\delta$,

$$m = \sum_{i=1}^m \chi_{u(J_{i,n})}(y) \leq \sum_{J \in \mathcal{F}_n} \chi_{u(J)}(y),$$

and consequently,

$$m \leq \liminf_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \chi_{u(J)}(y).$$

If $N_u(y; I) < \infty$, take $m := N_u(y; I)$, while if $N_u(y; I)$ is infinite, let $m \rightarrow N_u(y; I)$ in the previous inequality to conclude that

$$N_u(y; I) \leq \liminf_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \chi_{u(J)}(y).$$

This inequality continues to hold if $N_u(y; I) = 0$. On the other hand, by the definition of $N_u(y; I)$, for every integer $n \in \mathbb{N}$, we have that

$$(2.35) \quad \sum_{J \in \mathcal{F}_n} \chi_{u(J)}(y) \leq N_u(y; I),$$

where we have used again the fact that $\mathcal{F}_n$ is a partition of I . Letting $n \rightarrow \infty$ in the previous inequality gives

$$\limsup_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \chi_{u(J)}(y) \leq N_u(y; I).$$

Hence (2.34) holds. Note that since by hypothesis $\chi_{u(J)}$ is a Borel function for every $J \in \mathcal{F}_n$, (2.34) implies that $N_u(\cdot; I)$ is a Borel function.

Step 2: We claim that

$$\liminf_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \mu(u(J)) \geq \int_Y N_u(y; I) d\mu(y).$$

By Corollary B.42 in Appendix B, for every $n \in \mathbb{N}$,

$$(2.36) \quad \sum_{J \in \mathcal{F}_n} \mu(u(J)) = \sum_{J \in \mathcal{F}_n} \int_Y \chi_{u(J)} d\mu(y) = \int_Y \sum_{J \in \mathcal{F}_n} \chi_{u(J)} d\mu(y).$$

It now follows from (2.35) that

$$\limsup_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \mu(u(J)) \leq \int_Y N_u(y; I) d\mu(y).$$

On the other hand, from (2.36), Fatou's lemma, and Step 1, in this order, we get

$$\begin{aligned} \liminf_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \mu(u(J)) &= \liminf_{n \rightarrow \infty} \int_Y \sum_{J \in \mathcal{F}_n} \chi_{u(J)} d\mu(y) \\ &\geq \int_Y \liminf_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \chi_{u(J)} d\mu(y) = \int_Y N_u(y; I) d\mu(y). \end{aligned}$$

This completes the proof. $\square$

The following theorem was first proved by Banach [18] in the case $Y = \mathbb{R}$ and extended by Federer [75] to metric spaces.

Theorem 2.60 (Banach). *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u : I \rightarrow Y$ be a continuous function. Then $N_u(\cdot; I)$ is a Borel function and*

$$(2.37) \quad \int_Y N_u(y; I) d\mathcal{H}^1(y) = \text{Var } u.$$

In particular, u belongs to BPV(I) if and only if $N_u(\cdot; I)$ is Lebesgue integrable with respect to $\mathcal{H}^1$.

Compare the following lemma with Exercise 2.26.

Lemma 2.61. *Let (Y, d) be a metric space and let $u : [a, b] \rightarrow Y$ be a continuous function. For every $\delta > 0$ let $P_\delta := \{x_{0,\delta}, \dots, x_{n_\delta,\delta}\}$ be a partition of $[a, b]$, with*

$$x_{0,\delta} := a < x_{1,\delta} < \dots < x_{n_\delta,\delta} =: b,$$

such that $x_{i,\delta} - x_{i-1,\delta} \leq \delta$ for all $i = 1, \dots, n_\delta$. Then

$$\text{Var } u = \lim_{\delta \rightarrow 0^+} \sum_{i=1}^{n_\delta} d(u(x_{i,\delta}), u(x_{i-1,\delta})).$$

Proof. Without loss of generality, we may assume that $\text{Var } u > 0$, since otherwise u is constant and there is nothing to prove. Fix any $0 < t < \text{Var } u$ and find a partition $P = \{x_0, \dots, x_m\}$ of $[a, b]$, with $x_0 = a$ and $x_m = b$ (see Remark 2.3), such that

$$(2.38) \quad S := \sum_{j=1}^m d(u(x_j), u(x_{j-1})) > t.$$

Since u is uniformly continuous, there exists $\eta > 0$ such that

$$(2.39) \quad d(u(x), u(x')) \leq \frac{S-t}{2m}$$

for all $x, x' \in [a, b]$ with $|x - x'| \leq \eta$. Let

$$0 < \delta_0 := \frac{1}{2} \min \left\{ \eta, \min_{j=1, \dots, m} (x_j - x_{j-1}) \right\}.$$

If $0 < \delta \leq \delta_0$, then for each $j = 1, \dots, m-1$ there exists a unique index $i_j \in \{1, \dots, n_\delta\}$ such that $x_j \in (x_{i_j-1,\delta}, x_{i_j,\delta}]$. Moreover, if $j \neq k$, then $i_j \neq i_k$, and $i_1 < \dots < i_{m-1}$. Set $i_0 := 0$ and $i_m := n_\delta$.

Since

$$|x_{i_j,\delta} - x_j| \leq x_{i_j,\delta} - x_{i_j-1,\delta} \leq \delta \leq \eta,$$

by (2.39) we have

$$\begin{aligned} d(u(x_j), u(x_{j-1})) &\leq d(u(x_j), u(x_{i_j, \delta})) \\ &\quad + d(u(x_{i_j, \delta}), u(x_{i_{j-1}, \delta})) + d(u(x_{i_{j-1}, \delta}), u(x_{j-1})) \\ &\leq \frac{S-t}{m} + d(u(x_{i_j, \delta}), u(x_{i_{j-1}, \delta})). \end{aligned}$$

Summing over all $j = 1, \dots, m$ and using (2.38), we get

$$\begin{aligned} S &= \sum_{j=1}^m d(u(x_j), u(x_{j-1})) \leq S - t + \sum_{j=1}^m d(u(x_{i_j, \delta}), u(x_{i_{j-1}, \delta})) \\ &\leq S - t + \sum_{i=1}^{n_\delta} d(u(x_{i, \delta}), u(x_{i-1, \delta})), \end{aligned}$$

and so

$$t \leq \sum_{i=1}^{n_\delta} d(u(x_{i, \delta}), u(x_{i-1, \delta})) \leq \text{Var } u.$$

In turn,

$$\begin{aligned} t &\leq \liminf_{\delta \rightarrow 0^+} \sum_{i=1}^{n_\delta} d(u(x_{i, \delta}), u(x_{i-1, \delta})) \\ &\leq \limsup_{\delta \rightarrow 0^+} \sum_{i=1}^{n_\delta} d(u(x_{i, \delta}), u(x_{i-1, \delta})) \leq \text{Var } u. \end{aligned}$$

Letting $t \nearrow \text{Var } u$ gives the desired result. $\square$

Note that in the previous lemma we are not assuming that $\text{Var } u < \infty$.

Exercise 2.62. Prove that the previous lemma continues to hold if $u : [a, b] \rightarrow Y$ is only assumed to be left continuous (respectively, right continuous). Hint: Use left continuity (respectively, right continuity) at each point x_j to find δ_0 .

Lemma 2.63. Let (Y, d) be a metric space and let $u : [a, b] \rightarrow Y$ be a continuous function. Then

$$d(u(a), u(b)) \leq \mathcal{H}^1(u([a, b])) \leq \text{Var } u.$$

Proof. By the triangle inequality the function $f(y) := d(y, u(a))$, $y \in Y$, is Lipschitz continuous with Lipschitz constant 1. Hence, by Proposition C.44,

$$\mathcal{H}^1(f(u([a, b]))) \leq \mathcal{H}^1(u([a, b])).$$

The function $f \circ u : [a, b] \rightarrow \mathbb{R}$ is continuous and so $f(u([a, b]))$ contains the interval of endpoints $(f \circ u)(a)$ and $(f \circ u)(b)$. Since $\mathcal{H}^1 = \mathcal{L}^1$ in $\mathbb{R}$, we have

$$\begin{aligned} d(u(a), u(b)) &= (f \circ u)(b) - (f \circ u)(a) \\ &\leq \mathcal{L}^1(f(u([a, b]))) = \mathcal{H}^1(f(u([a, b]))) \leq \mathcal{H}^1(u([a, b])). \end{aligned}$$

This proves the first inequality.

To prove the second, it suffices to assume that $\text{Var } u < \infty$, since otherwise there is nothing to prove. By Corollary 2.24 we can write

$$u = w \circ V,$$

where $w : V([a, b]) \rightarrow Y$ is Lipschitz continuous with Lipschitz constant at most 1. Hence, by Proposition C.44,

$$\begin{aligned} \mathcal{H}^1(u([a, b])) &= \mathcal{H}^1(w(V([a, b]))) \leq \mathcal{H}^1(V([a, b])) \\ &= \mathcal{L}^1([0, \text{Var } u]) = \text{Var } u, \end{aligned}$$

where we used the fact that $\mathcal{H}^1 = \mathcal{L}^1$ in $\mathbb{R}$. □

Proof of Banach's theorem. Step 1: Assume that $I = [a, b]$. For every $n \in \mathbb{N}$ let $\mathcal{F}_n$ be the partition of $[a, b]$ given by

$$\begin{aligned} I_{1,n} &:= [a, a + \frac{b-a}{2^n}], \\ I_{k,n} &:= [a + (k-1)\frac{b-a}{2^n}, a + k\frac{b-a}{2^n}] \quad \text{for } k = 2, \dots, 2^n. \end{aligned}$$

By Lemma 2.63 and Proposition 2.15, we get

$$\sum_{J \in \mathcal{F}_n} d(u(\max J), u(\min J)) \leq \sum_{J \in \mathcal{F}_n} \mathcal{H}^1(u(J)) \leq \sum_{J \in \mathcal{F}_n} \text{Var}_J u = \text{Var}_{[a,b]} u.$$

Letting $n \rightarrow \infty$ and using Lemma 2.61, we obtain

$$\begin{aligned} \text{Var}_{[a,b]} u &= \lim_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} d(u(\max J), u(\min J)) \\ &\leq \lim_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \mathcal{H}^1(u(J)) \leq \text{Var}_{[a,b]} u, \end{aligned}$$

and so, by Theorem 2.59 we have that

$$\int_Y N_u(y; [a, b]) d\mathcal{H}^1(y) = \lim_{n \rightarrow \infty} \sum_{J \in \mathcal{F}_n} \mathcal{H}^1(u(J)) = \text{Var}_{[a,b]} u.$$

Step 2: If I is an arbitrary interval, construct an increasing sequence of intervals $[a_k, b_k]$ such that $I = \bigcup_k [a_k, b_k]$. By the definition of $N_u(y; \cdot)$ for every $y \in \mathbb{R}$ we have that

$$N_u(y; [a_k, b_k]) \leq N_u(y; [a_{k+1}, b_{k+1}])$$

and $N_u(y; [a_k, b_k]) \rightarrow N_u(y; I)$. Hence, by Proposition 2.15, Exercise 2.22, the previous step, and the Lebesgue monotone convergence theorem, we get

$$\begin{aligned} \text{Var}_I u &= \lim_{k \rightarrow \infty} \text{Var}_{[a_k, b_k]} u = \lim_{k \rightarrow \infty} \int_Y N_u(y; [a_k, b_k]) d\mathcal{H}^1(y) \\ &= \int_Y N_u(y; I) d\mathcal{H}^1(y). \end{aligned}$$

This concludes the proof. $\square$

Corollary 2.64. *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u \in BPV(I; Y)$ be continuous. Then the set of values $y \in Y$ for which $u^{-1}(\{y\})$ is infinite has $\mathcal{H}^1$ measure zero.*

Proof. Since by the previous theorem the function N_u is integrable, it follows that it must be finite for $\mathcal{H}^1$ -a.e. $y \in Y$. $\square$

The previous corollary expresses the fact that a function in $BPV(I; Y)$ cannot oscillate too much.

Exercise 2.65. Consider the Cantor set

$$\mathbb{D} := [0, 1] \setminus \bigcup_{n=1}^{\infty} \bigcup_{k=1}^{2^{n-1}} I_{k,n}.$$

For every interval $I_{k,n} = (a_{k,n}, b_{k,n})$, $n \in \mathbb{N}$, $k = 1, \dots, 2^{n-1}$, define

$$g_{k,n}(x) := \frac{1}{2^n} \sin \frac{2\pi(x - a_{k,n})}{b_{k,n} - a_{k,n}}, \quad x \in [a_{k,n}, b_{k,n}].$$

(i) Prove that the function $g : [0, 1] \rightarrow \mathbb{R}$, defined by

$$g(x) := \begin{cases} g_{k,n}(x) & \text{if } x \in I_{k,n}, n \in \mathbb{N}, k = 1, \dots, 2^{n-1}, \\ 0 & \text{if } x \in \mathbb{D}, \end{cases}$$

is continuous.

(ii) Let $f := g + u$, where u is the Cantor function. Prove that for every $n \in \mathbb{N}$,

$$[0, 1] = f \left(\bigcup_{k \leq 2^{n-1}} I_{k,n} \right).$$

(iii) Prove that the counting function $N_f : \mathbb{R} \rightarrow \mathbb{N} \cup \{0, \infty\}$ is identically ∞ in $[0, 1]$.

Exercise 2.66. Let $u : [a, b] \rightarrow \mathbb{R}$ be continuous. Using Banach's theorem, give an alternative proof of the fact that if $c \in (a, b)$, then

$$\text{Var}_{[a,c]} u + \text{Var}_{[c,b]} u = \text{Var} u.$$

Exercise 2.67. Let (Y, d) be a metric space and let $u : [a, b] \rightarrow Y$. Prove that

$$\text{Var } u = \int_Y N_u(y; I) d\mathcal{H}^1(y) + \sum_{a \leq x < b} v_+(x) + \sum_{a < y \leq b} v_-(y),$$

where

$$v_+(x) := \limsup_{t \rightarrow x^+} d(u(t), u(x)), \quad v_-(x) := \limsup_{t \rightarrow x^-} d(u(t), u(x)).$$

Absolutely Continuous Functions

*Undergradese, III: “Is this going to be on the text?” Translation:
“Tell us what’s going to be on the test”*

— Jorge Cham, www.phdcomics.com

The Cantor function shows that for monotone functions the fundamental theorem of calculus fails for Lebesgue integration. Indeed,

$$\int_0^1 u'(x) dx = 0 < u(1) - u(0) = 1.$$

To recover it in the context of Lebesgue integration, we need to restrict ourselves to a subclass of functions of bounded pointwise variation. This leads us to the notion of absolute continuity.

For readers without a solid background in functional analysis, simply take $Y = \mathbb{R}^M$ in the entire chapter.

3.1. $AC(I; Y)$ Versus $BPV(I; Y)$

Definition 3.1. Let $I \subseteq \mathbb{R}$ be an interval and let (Y, d) be a metric space. A function $u : I \rightarrow Y$ is said to be *absolutely continuous* on I if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$(3.1) \quad \sum_{i=1}^n d(u(b_i), u(a_i)) \leq \varepsilon$$

for every finite number of nonoverlapping intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq I$ and

$$\sum_{i=1}^n (b_i - a_i) \leq \delta.$$

The space of all absolutely continuous functions $u : I \rightarrow Y$ is denoted by $AC(I; Y)$.

When $Y = \mathbb{R}$ we write $AC(I)$ for $AC(I; \mathbb{R})$.

Remark 3.2. Note that since n is arbitrary, we can also take $n = \infty$, namely, replace finite sums by series.

Let $I \subseteq \mathbb{R}$ be an interval and let (Y, d) be a metric space. A function $u : I \rightarrow Y$ is *locally absolutely continuous* if it is absolutely continuous in $[a, b]$ for every interval $[a, b] \subseteq I$. The space of all locally absolutely continuous functions $u : I \rightarrow Y$ is denoted by $AC_{\text{loc}}(I; Y)$. As before, when $Y = \mathbb{R}$ we write $AC_{\text{loc}}(I)$ for $AC_{\text{loc}}(I; \mathbb{R})$. Note that

$$AC_{\text{loc}}([a, b]; Y) = AC([a, b]; Y).$$

If $\Omega \subseteq \mathbb{R}$ is an open set and (Y, d) is a metric space, then we define the notion of absolute continuity for a function $u : \Omega \rightarrow Y$ as in Definition 3.1, with the only change that we now require the intervals $[a_i, b_i]$ to be contained in Ω in place of I . The space of all absolutely continuous functions $u : \Omega \rightarrow Y$ is denoted by $AC(\Omega; Y)$.

Exercise 3.3. Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$.

- (i) Prove that u belongs to $AC(I)$ if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\left| \sum_{i=1}^n (u(b_i) - u(a_i)) \right| \leq \varepsilon$$

for every finite number of nonoverlapping intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq I$ and $\sum_{i=1}^n (b_i - a_i) \leq \delta$.

- (ii) Assume that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\left| \sum_{i=1}^n (u(b_i) - u(a_i)) \right| \leq \varepsilon$$

for every finite number of intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq I$ and $\sum_{i=1}^n (b_i - a_i) \leq \delta$. Prove that u is locally Lipschitz continuous.

Part (ii) of the previous exercise shows that in Definition 3.1 we cannot remove the condition that the intervals (a_i, b_i) are pairwise disjoint.

Remark 3.4. By taking $n = 1$ in Definition 3.1, it follows that an absolutely continuous function $u : I \rightarrow Y$ is uniformly continuous.

The next exercise shows that the converse is false.

Exercise 3.5. Let $u : (0, 1] \rightarrow \mathbb{R}$ be defined by

$$u(x) := x^a \sin \frac{1}{x^b},$$

where $a, b \in \mathbb{R}$. Determine for which a, b the function u is absolutely continuous. Prove that there exist a, b for which u is uniformly continuous but not absolutely continuous.

Exercise 3.6. Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be differentiable with bounded derivative. Prove that u belongs to $AC(I)$.

Exercise 3.7. Let $u, v \in AC([a, b])$. Prove the following.

- (i) $u \pm v \in AC([a, b])$.
- (ii) $uv \in AC([a, b])$.
- (iii) If $v(x) > 0$ for all $x \in [a, b]$, then $u/v \in AC([a, b])$.
- (iv) $\max\{u, v\}, \min\{u, v\} \in AC([a, b])$.
- (v) What happens if the interval $[a, b]$ is replaced by an arbitrary interval $I \subseteq \mathbb{R}$ (possibly unbounded)?

We now turn to the relation between absolutely continuous functions and functions of bounded pointwise variation. In Corollary 2.31 we have proved that if $u : I \rightarrow \mathbb{R}$ has bounded pointwise variation, then u is bounded and u' is Lebesgue integrable. However, the function $u(x) := x$, $x \in \mathbb{R}$, is absolutely continuous, but it is unbounded and $u'(x) = 1$, which is not Lebesgue integrable. Also the function $u(x) := \sin x$, $x \in \mathbb{R}$, is absolutely continuous, bounded, but u' is not Lebesgue integrable. These simple examples show that an absolutely continuous function may not have bounded pointwise variation. Proposition 3.9 below will show that this can happen only on unbounded intervals.

Exercise 3.8. Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a complete metric space, and let $u : I \rightarrow Y$ be uniformly continuous.

- (i) Prove that u may be extended uniquely to $\bar{I}$ in such a way that the extended function is still uniformly continuous.
- (ii) Prove that if u belongs to $AC(I; Y)$, then its extension belongs to $AC(\bar{I}; Y)$.
- (iii) Let Y be a normed space. Prove that there exist $A, B > 0$ such that for all $x \in I$,

$$\|u(x)\| \leq A + B|x|.$$

The previous exercise shows that although an absolutely continuous function may be unbounded, it cannot grow faster than linear when $|x| \rightarrow \infty$.

Hence, the functions $u(x) = |x|^p$, $p > 1$, and $v(x) = |x| \log(1 + |x|)$ cannot be absolutely continuous in an unbounded interval I .

Next we show that $AC_{\text{loc}}(I; Y) \subseteq BPV_{\text{loc}}(I; Y)$.

Proposition 3.9. *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u \in AC_{\text{loc}}(I; Y)$. Then u belongs to $BPV_{\text{loc}}(I; Y)$. In particular, if $Y = \mathbb{R}^M$, then u is differentiable $\mathcal{L}^1$ -a.e. in I and u' is locally Lebesgue integrable.*

Proof. Let $[a, b] \subseteq I$. Take $\varepsilon = 1$, and let $\delta > 0$ be as in Definition 3.1. Let n be the integer part of $2(b - a)/\delta$ and partition $[a, b]$ into n intervals $[x_{i-1}, x_i]$ of equal length $(b - a)/n$. Since $(b - a)/n \leq \delta$, in view of (3.1), on each interval $[x_{i-1}, x_i]$ we have that $\text{Var}_{[x_{i-1}, x_i]} u \leq 1$, and so by Proposition 2.15,

$$\text{Var}_{[a, b]} u = \sum_{i=1}^n \text{Var}_{[x_{i-1}, x_i]} u \leq n \leq 2(b - a)/\delta < \infty,$$

where we have used the fact that $(b - a)/n \geq \delta/2$. The last part of the statement follows by Corollary 2.31. $\square$

Corollary 3.10. *Let $I \subseteq \mathbb{R}$ be a bounded interval and let (Y, d) be a metric space. Then $AC(I; Y) \subseteq BPV(I; Y)$. In particular, if $u \in AC([a, b]; \mathbb{R}^M)$, then u is differentiable $\mathcal{L}^1$ -a.e. in $[a, b]$ and u' is Lebesgue integrable.*

The converse of Corollary 3.10 is false, since absolutely continuous functions are always continuous while monotone functions may not be. Even more, there exist continuous monotone functions that are not absolutely continuous. The Cantor function and the function constructed in Exercise 1.39 are such examples. What is missing for a continuous function in $BPV(I; Y)$ to belong to $AC(I; Y)$ is the fundamental theorem of calculus.

In view of Proposition 3.9, if $u \in AC_{\text{loc}}(I; Y)$, then u belongs to $BPV_{\text{loc}}(I; Y)$ and thus its indefinite pointwise variation $\mathbf{V}$ (see (2.3)) is continuous in view of Proposition 2.21. We now prove that $\mathbf{V}$ is actually in $AC_{\text{loc}}(I)$.

Theorem 3.11. *Let $I \subseteq \mathbb{R}$ be an interval, let (Y, d) be a metric space, and let $u : I \rightarrow Y$. Then $u \in AC_{\text{loc}}(I; Y)$ (respectively, $AC(I; Y)$) if and only if $\mathbf{V} \in AC_{\text{loc}}(I)$ (respectively, $AC(I)$).*

Proof. Assume that $u \in AC_{\text{loc}}(I; Y)$ and let $J \subseteq I$ be an interval such that $u \in AC(J; Y)$ (in particular any closed interval in I will have this property). Given $\varepsilon > 0$, let $\delta > 0$ be as in Definition 3.1 with J in place of I . Consider a finite number of nonoverlapping intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq J$ and $\sum_{i=1}^n (b_i - a_i) \leq \delta$. Given a partition $P_i = \{x_{0,i}, \dots, x_{m_i,i}\}$ of $[a_i, b_i]$,

since

$$\sum_{i=1}^n \sum_{k=1}^{m_i} |x_{k,i} - x_{k-1,i}| = \sum_{i=1}^n (b_i - a_i) < \delta,$$

it follows from the fact that $u \in AC([a, b]; Y)$ that

$$\sum_{i=1}^n \sum_{k=1}^{m_i} d(u(x_{k,i}), u(x_{k-1,i})) \leq \varepsilon.$$

Taking the supremum over every partition P_i of $[a_i, b_i]$ for each i , we have that

$$\sum_{i=1}^n \text{Var}_{[a_i, b_i]} u \leq \varepsilon,$$

that is, by (2.4),

$$\sum_{i=1}^n (V(b_i) - V(a_i)) \leq \varepsilon,$$

which shows that V belongs to $AC(J)$. By the arbitrariness of J it follows that V belongs to $AC_{\text{loc}}(I)$.

Conversely, assume that V belongs to $AC_{\text{loc}}(I)$ and let $J \subseteq I$ be an interval such that V belongs to $AC(J)$. Given $\varepsilon > 0$, let $\delta > 0$ be as in Definition 3.1, but now for V . Consider a finite number of nonoverlapping intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq J$ and $\sum_{i=1}^n (b_i - a_i) \leq \delta$. Then by (2.4),

$$\sum_{i=1}^n d(u(b_i), u(a_i)) \leq \sum_{i=1}^n (V(b_i) - V(a_i)) \leq \varepsilon,$$

which concludes the proof. $\square$

3.2. The Fundamental Theorem of Calculus

In this section we will show that absolutely continuous functions $u : I \rightarrow \mathbb{R}^M$ can be characterized as the family of all functions for which the fundamental theorem of calculus holds (for the Lebesgue integral).

We have seen that the Cantor function $u : [0, 1] \rightarrow \mathbb{R}$ has zero derivative $\mathcal{L}^1$ -a.e. in I , but $u(1) = 1$, $u(0) = 0$. The following theorem shows that this cannot happen for absolutely continuous functions.

Theorem 3.12. *Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a normed space, and let $u \in AC_{\text{loc}}(I; Y)$ be such that there exists $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in I$. Then u is constant.*

Proof. Let $a, b \in I$ with $a < b$. We claim that $u(a) = u(b)$. Given $\varepsilon > 0$, let $\delta > 0$ be the number given in the definition of absolute continuity in $[a, b]$. Let $E := \{x \in (a, b) : u'(x) = 0\}$. Then $\mathcal{L}_o^1(E) = b - a$.

For every $x \in E$, we have that

$$\lim_{h \rightarrow 0} \frac{u(x+h) - u(x)}{h} = 0,$$

and so there exists $h_x > 0$ such that $[x - h_x, x + h_x] \subset (a, b)$ and

$$(3.2) \quad \|u(x+h) - u(x)\| \leq \varepsilon|h|$$

for all $x+h \in I$ with $|h| \leq h_x$. Consider the family $\mathcal{F}$ of intervals $(x, x+h)$, where $x \in E$ and $0 < h \leq h_x$. By Lemma 1.20 there exist disjoint intervals $(x_1, x_1 + h_1), \dots, (x_n, x_n + h_n) \in \mathcal{F}$ and a set

$$F \subseteq E \cap \bigcup_{i=1}^n (x_i, x_i + h_i)$$

such that

$$\mathcal{L}_o^1(F) \geq b - a - \delta, \quad \sum_{i=1}^n h_i \leq b - a + \delta.$$

Without loss of generality, assume that $x_1 < x_2 < \dots < x_n$. Since

$$\sum_{i=1}^n h_i \geq \mathcal{L}_o^1(F) \geq b - a - \delta,$$

the sum of the lengths of the intervals $[a, x_1], [x_1 + h_1, x_2], \dots, [x_n + h_n, b]$ is less than or equal to δ . Since u is absolutely continuous, we have that

$$\|u(a) - u(x_1)\| + \sum_{i=1}^{n-1} \|u(x_{i+1}) - u(x_i + h_i)\| + \|u(b) - u(x_n + h_n)\| \leq \varepsilon.$$

On the other hand, by (3.2), $\|u(x_i + h_i) - u(x_i)\| \leq \varepsilon h_i$, and so

$$\begin{aligned} \|u(a) - u(b)\| &\leq \|u(a) - u(x_1)\| + \sum_{i=1}^{n-1} \|u(x_{i+1}) - u(x_i + h_i)\| \\ &\quad + \sum_{i=1}^n \|u(x_i) - u(x_i + h_i)\| + \|u(b) - u(x_n + h_n)\| \\ &\leq \varepsilon + \varepsilon \sum_{i=1}^n h_i \leq \varepsilon + \varepsilon(b - a). \end{aligned}$$

Letting $\varepsilon \rightarrow 0^+$ gives $u(a) = u(b)$. Hence, u is constant. $\square$

Using the previous theorem, we are in a position to complete the proof of Theorem 2.40. We begin with some well-known results on Riemann integration.

Exercise 3.13 (Riemann integration, I). Let $u : [a, b] \rightarrow \mathbb{R}$ be a bounded function and for $x \in [a, b]$ define

$$\omega(x) := \lim_{\delta \rightarrow 0^+} \sup\{|u(x_1) - u(x_2)| : x_1, x_2 \in [a, b], \\ |x_1 - x| \leq \delta, |x_2 - x| \leq \delta\}.$$

- (i) Prove that u is continuous at $x \in [a, b]$ if and only if $\omega(x) = 0$.
- (ii) Prove that if the set

$$E := \{x \in [a, b] : u \text{ is discontinuous at } x\}$$

has positive Lebesgue outer measure, then there exists a constant $\alpha > 0$ such that the set

$$E_1 := \{x \in E : \omega(x) \geq \alpha\}$$

has positive Lebesgue outer measure.

- (iii) Let $t := \mathcal{L}_o^1(E_1) > 0$ and prove that for every partition P the intervals containing points of E_1 in their interior have total length greater than or equal to t .
- (iv) Deduce that if u is Riemann integrable, then E has Lebesgue measure zero.

Exercise 3.14 (Riemann integration, II). Let $u : [a, b] \rightarrow \mathbb{R}$ be a bounded function.

- (i) Let

$$g(x) := \int_a^{\overline{x}} u(t) dt - \int_{\underline{a}}^x u(t) dt, \quad a < x \leq b, \quad g(a) := 0,$$

where $\int_a^{\overline{x}}$ and $\int_{\underline{a}}^x$ are the upper and lower Riemann integrals. Prove that g is Lipschitz continuous.

- (ii) Prove that if u is continuous at x , then g is differentiable at x and $g'(x) = 0$.
- (iii) Deduce that if u is continuous $\mathcal{L}^1$ -a.e. in $[a, b]$, then u is Riemann integrable.

Proof of Theorem 2.40, continued. We claim that u' cannot be Riemann integrable on any closed interval $[a, b]$. Indeed, by the previous exercises, this would imply that u' is continuous except for a set of Lebesgue outer measure zero. But, since u is nowhere monotone, if u' is continuous at x , then necessarily $u'(x) = 0$. Hence, $u' = 0$ $\mathcal{L}^1$ -a.e. on $[a, b]$. By the previous theorem, we would get that u is a constant in $[a, b]$, which is a contradiction. $\square$

Remark 3.15. Note that the function constructed in Theorem 2.40 is Lipschitz continuous. Hence, it provides an example of a differentiable function with bounded derivative for which the fundamental theorem of calculus holds for Lebesgue integration (see Theorem 3.20 below) but not for Riemann integration.

We now show that the primitive of an integrable function is absolutely continuous.

Theorem 3.16. *Let $I \subseteq \mathbb{R}$ be an interval and let $v : I \rightarrow \mathbb{R}^M$ be a Lebesgue integrable function. Fix $x_0 \in I$ and let*

$$u(x) := \int_{x_0}^x v(t) dt, \quad x \in I.$$

Then the function u is absolutely continuous in I and $u'(x) = v(x)$ for $\mathcal{L}^1$ -a.e. $x \in I$.

We begin with an auxiliary result.

Lemma 3.17. *Let $v : [a, b] \rightarrow \mathbb{R}$ be a Lebesgue integrable function such that*

$$\int_a^x v(t) dt = 0$$

for all $x \in [a, b]$. Then $v(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$.

Proof. Observe that for every $a \leq \alpha < \beta \leq b$ we have

$$\int_{\alpha}^{\beta} v(t) dt = \int_a^{\beta} v(t) dt - \int_a^{\alpha} v(t) dt = 0.$$

Let $U \subset [a, b]$ be open. Then U can be written as a countable union of disjoint open intervals (α_n, β_n) . Since

$$\chi_{\bigcup_{n=1}^{\ell} (\alpha_n, \beta_n)}(t) |v(t)| \leq |v(t)|$$

for every $t \in [a, b]$ and $\lim_{\ell \rightarrow \infty} \chi_{\bigcup_{n=1}^{\ell} (\alpha_n, \beta_n)}(t) v(t) = \chi_U(t) v(t)$ we can apply the Lebesgue dominated convergence theorem to conclude that

$$\begin{aligned} \int_U v(t) dt &= \int_a^b \chi_U(t) v(t) dt = \lim_{\ell \rightarrow \infty} \int_a^b \chi_{\bigcup_{n=1}^{\ell} (\alpha_n, \beta_n)}(t) v(t) dt \\ &= \lim_{\ell \rightarrow \infty} \int_a^b \sum_{n=1}^{\ell} \chi_{(\alpha_n, \beta_n)}(t) v(t) dt = \lim_{\ell \rightarrow \infty} \sum_{n=1}^{\ell} \int_{\alpha_n}^{\beta_n} v(t) dt = 0. \end{aligned}$$

In turn, for every compact set $K \subseteq [a, b]$,

$$\int_K v(t) dt = \int_a^b v(t) dt - \int_{[a, b] \setminus K} v(t) dt = 0.$$

Now let $E^+ := \{x \in [a, b] : v(x) > 0\}$. Since

$$\mathcal{L}^1(E^+) = \sup\{\mathcal{L}^1(K) : K \subseteq E^+, K \text{ compact}\},$$

if $\mathcal{L}^1(E^+) > 0$, then there exists a compact set $K \subseteq E^+$ with $\mathcal{L}^1(K) > 0$. Define $K_n := \{x \in K : v(x) \geq 1/n\}$. Then $K = \bigcup_n K_n$. Since $v > 0$ in K ,

$$0 = \int_K v(t) dt \geq \int_{K_n} v(t) dt \geq \frac{1}{n} \mathcal{L}^1(K_n),$$

and so $\mathcal{L}^1(K_n) = 0$ for all n , which implies that $\mathcal{L}^1(K) = 0$. Thus, we have a contradiction, and so $\mathcal{L}^1(E^+) = 0$. Similarly, we can prove that the set $E^- := \{x \in [a, b] : v(x) < 0\}$ has Lebesgue measure zero. Hence, we have shown that $v(x) = 0$ for all $x \in [a, b] \setminus (E^+ \cup E^-)$. $\square$

Remark 3.18. The proof is simpler if v is Riemann integrable. In this case v is continuous for $\mathcal{L}^1$ -a.e. $x \in [a, b]$ (see Exercise 3.13). Take a point $x_0 \in (a, b)$ such that v is continuous at x_0 . If $v(x_0) > 0$, then taking $\varepsilon = v(x_0)/2$ we can find $\delta > 0$ such that $v(x) \geq v(x_0) - \varepsilon = \varepsilon$ for all $x \in (x_0 - \delta, x_0 + \delta) \subseteq (a, b)$. Then

$$0 = \int_{x_0 - \delta}^{x_0 + \delta} v(t) dt \geq 2\delta\varepsilon > 0,$$

which is a contradiction. Similarly, if $v(x_0) < 0$, then we get a contradiction. Hence, $v = 0$ at every continuity point of v .

We now turn to the proof of Theorem 3.16.

Proof. It is enough to give the proof for $M = 1$.

Step 1: Assume that v is bounded, with $|v(x)| \leq c$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$. Then u is absolutely continuous, since

$$\begin{aligned} \sum_{i=1}^n |u(b_i) - u(a_i)| &= \sum_{i=1}^n \left| \int_{a_i}^{b_i} v(t) dt \right| \leq \sum_{i=1}^n \int_{a_i}^{b_i} |v(t)| dt \\ &\leq c \sum_{i=1}^n (b_i - a_i) \leq \varepsilon \end{aligned}$$

for every finite number of nonoverlapping intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq I$ and $\sum_{i=1}^n (b_i - a_i) \leq \delta := \varepsilon/(c + 1)$. Extend v to be zero outside $[a, b]$ and define

$$w_n(x) := \frac{u(x + 1/n) - u(x)}{1/n} = n \int_x^{x+1/n} v(t) dt.$$

Then $|w_n(x)| \leq nc(x + 1/n - x) = c$ for all $x \in [a, b]$. Since u is absolutely continuous, it is differentiable for all $x \in [a, b]$ except a set of measure zero

and so $w_n(x) \rightarrow u'(x)$ as $n \rightarrow \infty$ for all $x \in [a, b]$ except a set of measure zero. Hence, by the Lebesgue dominated convergence theorem, for all $x_0 \in [a, b]$,

$$\int_a^{x_0} u'(x) \, dx = \lim_{n \rightarrow \infty} \int_a^{x_0} w_n(x) \, dx.$$

On the other hand,

$$\begin{aligned} \int_a^{x_0} w_n(x) \, dx &= n \int_a^{x_0} (u(x + 1/n) - u(x)) \, dx \\ &= n \left[\int_{x_0}^{x_0+1/n} u(x) \, dx - \int_a^{a+1/n} u(x) \, dx \right] \\ &\rightarrow u(x_0) - u(a) = u(x_0) - 0, \end{aligned}$$

where we used the fact that u is continuous. Hence,

$$\int_a^{x_0} u'(x) \, dx = u(x_0) = \int_a^{x_0} v(x) \, dx,$$

that is,

$$\int_a^{x_0} (u'(x) - v(x)) \, dx = 0$$

for all $x_0 \in [a, b]$. By the previous lemma, it follows that $u'(x) = v(x)$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$.

Step 2: Assume that $v \geq 0$ and define

$$v_n(x) := \begin{cases} v(x) & \text{if } v(x) \leq n, \\ 0 & \text{if } v(x) > n. \end{cases}$$

Then

$$u(x) = \int_a^x v(t) \, dt = \int_a^x v_n(t) \, dt + \int_a^x (v(t) - v_n(t)) \, dt =: u_n(x) + s_n(x).$$

By Step 1 we have that $u'_n(x) = v_n(x)$ for all $x \in [a, b]$ except a set of Lebesgue measure zero. On the other hand, since $v \geq v_n$ we have that s_n is increasing and so $s'_n(x) \geq 0$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$. Hence, since $u = u_n + s_n$, by differentiating, we obtain that

$$u'(x) = u'_n(x) + s'_n(x) = v_n(x) + s'_n(x) \geq v_n(x) + 0$$

for $\mathcal{L}^1$ -a.e. $x \in [a, b]$, say, for all $x \in [a, b] \setminus E_n$, where $\mathcal{L}^1(E_n) = 0$. Since countable union of sets of Lebesgue measure zero have Lebesgue measure zero, we have that $E := \bigcup_n E_n$ has Lebesgue measure zero. If $x \in [a, b] \setminus E$, then $u'(x) \geq v_n(x)$ for all n and so, letting $n \rightarrow \infty$ we obtain that $u'(x) \geq v(x)$. In turn,

$$\int_a^b u'(x) \, dx \geq \int_a^b v(x) \, dx = u(b) - u(a).$$

On the other hand, by Theorem 1.25,

$$\int_a^b u'(x) \, dx \leq u(b) - u(a),$$

which shows that

$$\int_a^b u'(x) \, dx = \int_a^b v(x) \, dx.$$

Hence,

$$\int_a^b (u'(x) - v(x)) \, dx = 0,$$

but since $u' \geq v$, it follows that $u'(x) = v(x)$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$.

Step 3: In the general case it is enough to write $v = v^+ - v^-$, where s^+ and s^- are the positive and negative parts of $s \in \mathbb{R}$, and observe that since

$$u(x) = \int_{x_0}^x v^+(t) \, dt - \int_{x_0}^x v^-(t) \, dt =: u_1(x) + u_2(x),$$

the result follows by applying Step 2 to u_1 and u_2 . $\square$

By applying the previous theorem to each $[a, b] \subseteq I$ we obtain

Corollary 3.19. *Let $I \subseteq \mathbb{R}$ be an interval and let $v : I \rightarrow \mathbb{R}^M$ be a locally Lebesgue integrable function. Fix $x_0 \in I$ and let*

$$u(x) := \int_{x_0}^x v(t) \, dt, \quad x \in I.$$

Then the function u is locally absolutely continuous in I and $u'(x) = v(x)$ for $\mathcal{L}^1$ -a.e. $x \in I$.

We are now ready to prove the main theorem of this section.

Theorem 3.20 (Fundamental theorem of calculus). *Let $I \subseteq \mathbb{R}$ be an interval. A function $u : I \rightarrow \mathbb{R}^M$ belongs to $AC_{\text{loc}}(I; \mathbb{R}^M)$ if and only if*

- (i) u is continuous in I ,
- (ii) u is differentiable $\mathcal{L}^1$ -a.e. in I , and u' belongs to $L^1_{\text{loc}}(I; \mathbb{R}^M)$,
- (iii) the fundamental theorem of calculus is valid; that is, for all $x, x_0 \in I$,

$$u(x) = u(x_0) + \int_{x_0}^x u'(t) \, dt.$$

Proof. Assume that $u \in AC_{\text{loc}}(I; \mathbb{R}^M)$. In view of Proposition 3.9, it remains to prove (iii). By Corollary 3.19 the function

$$(3.3) \quad w(x) := \int_{x_0}^x u'(t) \, dt, \quad x \in I,$$

belongs to $AC_{\text{loc}}(I; \mathbb{R}^M)$ with $w'(x) = u'(x)$ for $\mathcal{L}^1$ -a.e. $x \in I$. Since $u - w \in AC_{\text{loc}}(I)$ and

$$(u - w)'(x) = u'(x) - w'(x) = 0$$

for $\mathcal{L}^1$ -a.e. $x \in I$, by Theorem 3.12, we have that $u - w$ is constant in I . Thus, there exists $c \in \mathbb{R}$ such that

$$u(x) = c + \int_{x_0}^x u'(t) dt$$

for all $x \in I$. Taking $x = x_0$ gives $c = u(x_0)$.

Conversely, assume that (i)–(iii) are satisfied. Then by Theorem 3.16, the function w defined in (3.3) belongs to $AC_{\text{loc}}(I; \mathbb{R}^M)$, and hence, by (iii), so does u . $\square$

Exercise 3.21. Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be a convex function.

- (i) Prove that the left and right derivatives $u'_-(x)$ and $u'_+(x)$ exist in $\mathbb{R}$ for all $x \in I^\circ$.
- (ii) Prove that the functions u'_- and u'_+ are increasing.
- (iii) Prove that

$$(3.4) \quad u(x) - u(x_0) = \int_{x_0}^x u'_-(t) dt = \int_{x_0}^x u'_+(t) dt$$

for all $x, x_0 \in I^\circ$ with $x_0 < x$.

Exercise 3.22. Prove that there exists an increasing absolutely continuous function $u : [0, 1] \rightarrow \mathbb{R}$ which is not constant and with the property that for every $x \in [0, 1]$ with $u'(x) > 0$, u is not strictly increasing in a neighborhood of x .

As a corollary of Theorem 3.20 we recover the formula for integration by parts.

Corollary 3.23 (Integration by parts). *Let $I \subseteq \mathbb{R}$ be an interval and let $u, v \in AC_{\text{loc}}(I)$. Then for all $x, x_0 \in I$,*

$$\int_{x_0}^x uv' dt = u(x)v(x) - u(x_0)v(x_0) - \int_{x_0}^x u'v dt.$$

Proof. Since $u, v \in AC_{\text{loc}}(I)$, by Exercise 3.7, $uv \in AC_{\text{loc}}(I)$, and thus, by part (iii) of Theorem 3.20,

$$u(x)v(x) - u(x_0)v(x_0) = \int_{x_0}^x (uv)' dt.$$

Since the functions u, v , and uv are differentiable $\mathcal{L}^1$ -a.e. in I , the standard calculus rule for a product now gives the desired result. $\square$

Another consequence of Theorem 3.20 is the following result, which says that for monotone functions, to prove absolute continuity in a fixed interval $[a, b]$, it is sufficient to test the fundamental theorem of calculus only at the endpoints a and b .

Corollary 3.24 (Tonelli). *Let $I \subseteq \mathbb{R}$ be an interval, let $u : I \rightarrow \mathbb{R}$ be a monotone function, and let $[a, b] \subseteq I$. Then u belongs to $AC([a, b])$ if and only if*

$$\int_a^b |u'(t)| dt = |u(b) - u(a)|.$$

Moreover, if u is bounded, then u belongs to $AC(I)$ if and only if

$$\int_I |u'(t)| dt = \sup_I u - \inf_I u.$$

Proof. Without loss of generality, we may assume that u is increasing.

Step 1: If $u \in AC([a, b])$, then by Theorem 3.20 we have

$$(3.5) \quad \int_a^b u'(t) dt = u(b) - u(a).$$

Conversely, assume that (3.5) holds and let

$$w(x) := \int_a^x u'(t) dt, \quad x \in [a, b].$$

If $a \leq x_0 < x \leq b$, then by Corollary 1.25,

$$(3.6) \quad w(x) - w(x_0) = \int_{x_0}^x u'(t) dt \leq u(x) - u(x_0),$$

and so the function $u - w$ is increasing in $[a, b]$. But by (3.5),

$$u(b) - u(a) = \int_a^b u'(t) dt = w(b) - w(a),$$

and so $(u - w)(b) = (u - w)(a)$. This implies that the increasing function $u - w$ must be constant in $[a, b]$. Since $w \in AC([a, b])$ by Theorem 3.16, it follows that $u \in AC([a, b])$.

Step 2: If $u \in AC(I)$, then by the previous step for every $\inf I < x_0 < x < \sup I$, we have

$$\int_{x_0}^x u'(t) dt = u(x) - u(x_0).$$

Letting $x_0 \rightarrow (\inf I)^+$ and $x \rightarrow (\sup I)^-$ and using the Lebesgue monotone convergence theorem and the fact that u is continuous and increasing gives

$$(3.7) \quad \int_I u'(t) dt = \sup_I u - \inf_I u.$$

Conversely, assume that (3.7) holds. Then u' is Lebesgue integrable. Moreover, by Exercise 1.27, u is continuous in I . Fix $c \in \mathbb{R}$ and define

$$w(x) := c + \int_{\inf I}^x u'(t) dt, \quad x \in I.$$

As in the previous step we have that $u - w$ is increasing (and bounded) and that (3.6) holds for all $\inf I < x_0 < x < \sup I$. Letting $x_0 \rightarrow (\inf I)^+$ and $x \rightarrow (\sup I)^-$ in (3.6) and using the Lebesgue monotone convergence theorem, the monotonicity and the continuity of u and w , we obtain that

$$\begin{aligned} \lim_{x \rightarrow (\sup I)^-} u(x) - \lim_{x \rightarrow (\inf I)^+} u(x) &= \int_{\inf I}^{\sup I} u'(t) dt \\ &= \lim_{x \rightarrow (\sup I)^-} w(x) - \lim_{x \rightarrow (\inf I)^+} w(x), \end{aligned}$$

and so, since $u - w$ is increasing and continuous,

$$\begin{aligned} \sup_{x \in I} (u(x) - w(x)) &= \lim_{x \rightarrow (\sup I)^-} (u(x) - w(x)) \\ &= \lim_{x \rightarrow (\inf I)^+} (u(x) - w(x)) \\ &= \inf_{x \in I} (u(x) - w(x)), \end{aligned}$$

which implies as before that $u - w$ must be constant. By choosing c appropriately, we have that $u = w$, and so u belongs to $AC(I)$, since w does (see Theorem 3.16). $\square$

Since functions with bounded pointwise variation are differences of bounded monotone functions, the previous corollary implies the following result.

Theorem 3.25 (Tonelli). *Let $I \subseteq \mathbb{R}$ be an interval, let $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$, and let $[a, b] \subseteq I$. Then u belongs to $AC([a, b]; \mathbb{R}^M)$ if and only if*

$$(3.8) \quad \int_a^b \|u'(t)\| dt = \text{Var}_{[a,b]} u.$$

In addition, if u belongs to $BPV(I; \mathbb{R}^M)$, then u belongs to $AC(I; \mathbb{R}^M)$ if and only if

$$(3.9) \quad \int_I \|u'(t)\| dt = \text{Var } u.$$

Proof. Step 1: Let $\mathbf{V}$ be the indefinite pointwise variation of u (see (2.3)). By Corollary 1.25 and Theorem 2.28,

$$(3.10) \quad \int_a^b \|u'(t)\| dt = \int_a^b \mathbf{V}'(t) dt \leq \mathbf{V}(b) - \mathbf{V}(a) = \text{Var}_{[a,b]} u.$$

Hence, if

$$\int_a^b \|u'(t)\| dt = \text{Var}_{[a,b]} u,$$

then all the previous inequalities are equalities, and so

$$\int_a^b \mathbf{V}'(t) dt = \mathbf{V}(b) - \mathbf{V}(a).$$

Since $\mathbf{V}$ is increasing, it follows by Corollary 3.24 that the function $\mathbf{V}$ belongs to $AC([a, b])$. In turn, by Theorem 3.11, $u \in AC([a, b])$.

Conversely, if $u \in AC([a, b]; \mathbb{R}^M)$, then by Theorem 3.20 for every partition $\{x_0, \dots, x_n\}$ of $[a, b]$,

$$\sum_{i=1}^n \|u(x_i) - u(x_{i-1})\| = \sum_{i=1}^n \left\| \int_{x_{i-1}}^{x_i} u' dt \right\| \leq \sum_{i=1}^n \int_{x_{i-1}}^{x_i} \|u'\| dt = \int_a^b \|u'\| dt.$$

Taking the supremum over all partitions, we get

$$\text{Var}_{[a,b]} u \leq \int_a^b \|u'\| dt,$$

which, together with (3.10) yields the desired equality.

Step 2: Assume that $u \in BPV(I; \mathbb{R}^M)$. Then by Corollary 1.25 and Theorem 2.28,

$$\int_I \|u'(t)\| dt = \int_I \mathbf{V}'(t) dt \leq \sup_I \mathbf{V} - \inf_I \mathbf{V} = \text{Var } u,$$

and so if (3.9) holds, then

$$\int_I \mathbf{V}'(t) dt = \sup_I \mathbf{V} - \inf_I \mathbf{V}.$$

In turn, by the previous corollary, $\mathbf{V} \in AC(I)$. As in the previous step we conclude that $u \in AC(I; \mathbb{R}^M)$.

Conversely, if $u \in AC(I; \mathbb{R}^M)$, then by the previous step, (3.8) holds for every $[a, b] \subseteq I$. If $\inf I \in I$, define $a_n := \inf I$, and otherwise construct a sequence $a_n \searrow \inf I$. Similarly, if $\sup I \in I$, define $b_n := \sup I$, and otherwise construct a sequence $b_n \nearrow \sup I$. It suffices to apply the previous step in $[a_n, b_n]$ and then to let $n \rightarrow \infty$ using Proposition 2.15 and the Lebesgue monotone convergence theorem. $\square$

Theorem 3.20 characterizes functions in $AC_{\text{loc}}(I; \mathbb{R}^M)$ in terms of the fundamental theorem of calculus. We now prove a similar result for functions in $AC(I; \mathbb{R}^M)$.

We recall the following definition.

Definition 3.26. If $E \subseteq \mathbb{R}$ is a Lebesgue measurable set and $v : E \rightarrow \mathbb{R}^M$ is a Lebesgue measurable function, then v is *equi-integrable* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\int_F \|v(x)\| dx \leq \varepsilon$$

for every Lebesgue measurable set $F \subseteq E$, with $\mathcal{L}^1(F) \leq \delta$.

Exercise 3.27. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set, let $1 \leq p \leq \infty$, and let $v \in L^p(E; \mathbb{R}^M)$. Prove that v is equi-integrable. Prove that if we only assume that $v \in L^1_{\text{loc}}(E; \mathbb{R}^M)$, then the result may no longer be true.

Exercise 3.28. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set with finite measure and let $v : E \rightarrow \mathbb{R}^M$ be equi-integrable. Prove that $v \in L^1(E; \mathbb{R}^M)$.

Theorem 3.29 (Fundamental theorem of calculus, II). *Let $I \subseteq \mathbb{R}$ be an interval. A function $u : I \rightarrow \mathbb{R}^M$ belongs to $AC(I; \mathbb{R}^M)$ if and only if*

- (i) u is continuous in I ,
- (ii) u is differentiable $\mathcal{L}^1$ -a.e. in I , and u' belongs to $L^1_{\text{loc}}(I; \mathbb{R}^M)$ and is equi-integrable,
- (iii) the fundamental theorem of calculus is valid; that is, for all $x, x_0 \in I$,

$$u(x) = u(x_0) + \int_{x_0}^x u'(t) dt.$$

Proof. Assume that $u \in AC(I; \mathbb{R}^M)$. In view of Theorem 3.20 it remains to show that u' is equi-integrable. By Exercise 3.8 we may assume, without loss of generality, that I is closed. Fix $\varepsilon > 0$ and let $\delta > 0$ be as in Definition 3.1. Consider a Lebesgue measurable set $E \subseteq I$, with $\mathcal{L}^1(E) \leq \frac{\delta}{2}$. By the outer regularity of the Lebesgue measure we may find an open set $U \supseteq E$ such that $\mathcal{L}^1(U) < \delta$. Decompose U into a countable family $\{J_i\}_i$ of pairwise disjoint intervals. By replacing each J_i with $J_i \cap I$, we may assume that $J_i \subseteq I$. Let $\overline{J}_i = [a_i, b_i]$. Using the fact that $\mathcal{L}^1(U) < \delta$, we have that $\sum_i |b_i - a_i| < \delta$. Reasoning as in the first part of the proof of Theorem 3.11 we have that

$$\sum_i \text{Var}_{[a_i, b_i]} u \leq \varepsilon.$$

Hence, by Tonelli's theorem applied to each interval $[a_i, b_i]$,

$$\int_E \|u'\| dt \leq \int_U \|u'\| dt = \sum_i \int_{[a_i, b_i]} \|u'\| dt = \sum_i \text{Var}_{[a_i, b_i]} u \leq \varepsilon.$$

This proves that u' is equi-integrable.

Conversely, assume that u satisfies (i)–(iii). Since u' is equi-integrable, given $\varepsilon > 0$ let $\delta > 0$ be the number given in Definition 3.26. Given a finite

number of nonoverlapping intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq I$ and $\sum_{i=1}^n (b_i - a_i) \leq \delta$, by Theorem 3.20 in $[a_i, b_i]$ we have

$$\begin{aligned} \sum_{i=1}^n \|u(b_i) - u(a_i)\| &= \sum_{i=1}^n \left\| \int_{a_i}^{b_i} u' dt \right\| \\ &\leq \sum_{i=1}^n \int_{a_i}^{b_i} \|u'\| dt = \int_{\bigcup_{i=1}^n (a_i, b_i)} \|u'\| dt \leq \varepsilon. \end{aligned}$$

This shows that u is absolutely continuous. $\square$

Remark 3.30. If I is bounded, then in view of Exercises 3.27 and 3.28 in part (iii) we can replace u' equi-integrable with u' integrable.

The next corollary will be useful in the study of Sobolev spaces in Chapter 7. The proof follows from Exercise 3.27 and Theorem 3.29.

Corollary 3.31. *Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}^M$ be such that*

- (i) u is continuous on I ,
- (ii) u is differentiable $\mathcal{L}^1$ -a.e. in I , and $u' \in L^p(I; \mathbb{R}^M)$ for some $1 \leq p \leq \infty$,
- (iii) the fundamental theorem of calculus is valid; that is, for all $x, x_0 \in I$,

$$u(x) = u(x_0) + \int_{x_0}^x u'(t) dt.$$

Then u belongs to $AC(I; \mathbb{R}^M)$.

Exercise 3.32. Let $p > 1$ and let $AC_p([a, b])$ be the class of all functions $u : [a, b] \rightarrow \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\left(\sum_{i=1}^n |u(b_i) - u(a_i)|^p \right)^{1/p} \leq \varepsilon$$

for every finite number of nonoverlapping intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq I$ and

$$\left(\sum_{i=1}^n (b_i - a_i)^p \right)^{1/p} \leq \delta.$$

- (i) Prove that if $u \in AC_p([a, b])$, then $\text{Var}_p u < \infty$ (see Exercise 2.25).
- (ii) Prove that the function

$$u(x) = \sum_{n=0}^{\infty} \frac{1}{2^{n/p}} \cos 2^n \pi x, \quad x \in [0, 1],$$

is such that $\text{Var}_p u < \infty$, but it does not belong to $AC_p([0, 1])$.

In the discussion before Exercise 3.8 we have shown that an absolutely continuous function defined in an unbounded interval may not have bounded pointwise variation. As a corollary of the previous theorem we can now characterize the absolutely continuous functions that have bounded pointwise variation.

Corollary 3.33. *Let $I \subseteq \mathbb{R}$ be an interval and let $u \in AC_{\text{loc}}(I; \mathbb{R}^M)$. Then u belongs to $BPV(I; \mathbb{R}^M)$ if and only if u' belongs to $L^1(I; \mathbb{R}^M)$. In this case u belongs to $AC(I; \mathbb{R}^M)$.*

Proof. In view of Corollary 2.31, it remains to show that if u' is Lebesgue integrable, then u belongs to $BPV(I; \mathbb{R}^M)$. By Proposition 3.9 we have that $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$. Hence by Theorem 3.25, for every $[a, b] \subseteq I$,

$$\int_a^b \|u'\| dt = \text{Var}_{[a,b]} u.$$

Taking a_n and b_n as in Step 2 of the proof of Theorem 3.25 and using Proposition 2.15 (note that in that proposition the function u is completely arbitrary), Exercise 2.22, and the Lebesgue monotone convergence theorem, we have that

$$\int_I \|u'\| dt = \text{Var}_I u.$$

Since u' is Lebesgue integrable, it follows that $\text{Var}_I u < \infty$.

The last statement follows from Corollary 3.31. □

3.3. Lusin (N) Property

In this section we show that absolutely continuous functions map sets of measure zero into sets of measure zero and that this property characterizes them among functions of bounded variation. We recall that $\mathcal{H}^1$ is the one-dimensional Hausdorff measure (see Appendix C).

Definition 3.34. Let $E \subseteq \mathbb{R}$, let (Y, d) be a metric space, and let $u : E \rightarrow Y$. We say that u satisfies the *Lusin (N) property* if

$$\mathcal{H}^1(u(F)) = 0$$

for every set $F \subseteq E$ with $\mathcal{L}^1(F) = 0$.

In particular, when $Y = \mathbb{R}$, then $\mathcal{H}^1 = \mathcal{L}^1$ and so u satisfies the Lusin (N) property if

$$\mathcal{L}^1(u(F)) = 0$$

for every set $F \subseteq E$ with $\mathcal{L}^1(F) = 0$.

Example 3.35. The Cantor function $u : [0, 1] \rightarrow \mathbb{R}$ does not satisfy the Lusin (N) property since $\mathcal{L}^1(\mathbb{D}) = 0$ but $\mathcal{L}^1(u(\mathbb{D})) = 1$.

The next result shows that everywhere differentiable functions with bounded derivative satisfy the Lusin (N) property.

Theorem 3.36. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, and let $u : I \rightarrow Y$. Assume that there exist a set $E \subseteq I$ (not necessarily measurable) and $c \geq 0$ such that u is differentiable for all $x \in E$, with*

$$(3.11) \quad \|u'(x)\| \leq c \quad \text{for all } x \in E.$$

Then

$$\mathcal{H}_o^1(u(E)) \leq c\mathcal{L}_o^1(E).$$

Proof. Without loss of generality, we may assume that $E \subseteq I^\circ$. Fix $\varepsilon > 0$. By the properties of the Lebesgue outer measure we may find an open set U such that $U \supseteq E$ and

$$\mathcal{L}^1(U) \leq \mathcal{L}_o^1(E) + \varepsilon.$$

By replacing U with $U \cap I^\circ$, if necessary, we may suppose that $U \subseteq I^\circ$. Take $E_0 := \emptyset$ and inductively for each $n \in \mathbb{N}$ let E_n be the set of points $x \in U \setminus E_{n-1}$ such that $(x - 1/n, x + 1/n) \subseteq U$ and

$$(3.12) \quad \|u(x+h) - u(x)\| \leq (c + \varepsilon)|h|$$

for every $0 < |h| < 1/n$. The sets E_n are Borel sets (exercise). We claim that

$$(3.13) \quad E \subseteq \bigcup_{n=1}^{\infty} E_n.$$

To see this, fix $x \in E$. Since $\|u'(x)\| \leq c$, there exists $\delta > 0$ such that

$$\|u(x+h) - u(x)\| \leq (c + \varepsilon)|h|$$

for all $x+h \in I$, with $|h| < \delta$. This implies that $x \in E_n$ for every integer $n > 1/\delta$ so large that $(x - 1/n, x + 1/n) \subseteq U$ and so the claim is proved.

Decompose U as a countable family $\{J_k\}_k$ of pairwise disjoint intervals with $0 < \text{length } J_k < 1/n$. Define $E_{k,n} := E_n \cap J_k$. Note that the sets $E_{k,n}$ are disjoint Borel sets contained in U . Hence,

$$(3.14) \quad \sum_{k,n} \mathcal{L}^1(E_{k,n}) = \mathcal{L}^1\left(\bigcup_{k,n} E_{k,n}\right) \leq \mathcal{L}^1(U).$$

It follows from (3.12) and the fact that $\text{length } J_k < 1/n$ that if $E_{k,n}$ is nonempty, the function $u : E_{k,n} \rightarrow Y$ is Lipschitz continuous with Lipschitz constant at most $c + \varepsilon$. Hence, by Proposition C.44,

$$\mathcal{H}_o^1(u(E_{k,n})) \leq (c + \varepsilon)\mathcal{L}_o^1(E_{k,n}).$$

Then by (3.13) and (3.14),

$$\begin{aligned}\mathcal{H}_o^1(u(E)) &\leq \mathcal{H}_o^1\left(\bigcup_{k,n} u(E_{k,n})\right) \leq \sum_{k,n} \mathcal{H}_o^1(u(E_{k,n})) \\ &\leq (c + \varepsilon) \sum_{k,n} \mathcal{L}^1(E_{k,n}) \leq (c + \varepsilon) \mathcal{L}^1(U) \\ &\leq (c + \varepsilon) (\mathcal{L}_o^1(E) + \varepsilon).\end{aligned}$$

It now suffices to let $\varepsilon \rightarrow 0^+$. $\square$

As a consequence of the previous theorem, we have the following result.

Corollary 3.37. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, and let $u : I \rightarrow Y$. Assume that there exists a set $E \subseteq I$ such that u is differentiable for all $x \in E$. If the set E has $\mathcal{H}^1$ measure zero or if $u' = 0$ in E , then $\mathcal{H}^1(u(E)) = 0$.*

Proof. Assume that E has Lebesgue measure zero. For every $n \in \mathbb{N}$ write $E_n := \{x \in E : \|u'(x)\| \leq n\}$. Since u is differentiable in E , it follows that $E = \bigcup_{n=1}^{\infty} E_n$, while by the previous theorem,

$$\mathcal{H}_o^1(u(E_n)) \leq n \mathcal{L}_o^1(E_n) = 0.$$

By the countable subadditivity of $\mathcal{H}_o^1$ we obtain that $\mathcal{H}_o^1(u(E)) = 0$.

If $u' = 0$ in E , then we may take $c := 0$ in the previous theorem. $\square$

Remark 3.38. The Cantor function shows that the previous corollary does not hold if we replace everywhere differentiability in E with $\mathcal{L}^1$ -a.e. differentiability in E .

Another important consequence of Theorem 3.36 is the following.

Theorem 3.39. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, let $u : I \rightarrow Y$, and let $E \subseteq I$ be a Lebesgue measurable set on which u is differentiable. Then $u(E)$ is $\mathcal{H}^1$ measurable and*

$$(3.15) \quad \mathcal{H}^1(u(E)) \leq \int_E \|u'(x)\| dx.$$

Proof. Step 1: By the properties of the Lebesgue measure we may write E as the union of a set of Lebesgue measure zero and countably many compact sets, precisely, $E = E_0 \cup \bigcup_n K_n$, where K_n is compact and $\mathcal{L}^1(E_0) = 0$. By the previous corollary $\mathcal{H}^1(u(E_0)) = 0$, while $u(K_n)$ is compact since $u : K_n \rightarrow \mathbb{R}$ is continuous (since differentiable). Hence, $u(E)$ is $\mathcal{H}^1$ measurable.

Step 2: Assume that $\mathcal{L}^1(E) < \infty$. Fix $n \in \mathbb{N}$. For every $k \in \mathbb{N}$ write

$$E_{k,n} := \left\{x \in E : \frac{k-1}{2^n} \leq \|u'(x)\| < \frac{k}{2^n}\right\}.$$

Then $E = \bigcup_{k=1}^{\infty} E_{k,n}$, and so by Theorem 3.36,

$$\begin{aligned} \mathcal{H}^1(u(E)) &= \mathcal{H}^1\left(\bigcup_k u(E_{k,n})\right) \leq \sum_k \mathcal{H}^1(u(E_{k,n})) \\ &\leq \sum_k \frac{k}{2^n} \mathcal{L}^1(E_{k,n}) = \sum_k \frac{k-1}{2^n} \mathcal{L}^1(E_{k,n}) + \frac{1}{2^n} \sum_k \mathcal{L}^1(E_{k,n}) \\ &\leq \sum_k \int_{E_{k,n}} \|u'(x)\| dx + \frac{1}{2^n} \mathcal{L}^1(E) \leq \int_E \|u'(x)\| dx + \frac{1}{2^n} \mathcal{L}^1(E), \end{aligned}$$

and it suffices to let $n \rightarrow \infty$.

Step 3: If $\mathcal{L}^1(E) = \infty$, for every $k \in \mathbb{N}$ write

$$E_k := E \cap ([-k-1, -k] \cup (k, k+1]).$$

Then by the previous step applied to E_k we have

$$\begin{aligned} \mathcal{H}^1(u(E)) &= \mathcal{H}^1\left(u\left(\bigcup_k E_k\right)\right) \leq \sum_k \mathcal{H}^1(u(E_k)) \\ &\leq \sum_k \int_{E_k} \|u'(x)\| dx = \int_E \|u'(x)\| dx, \end{aligned}$$

where we used the fact that the sets E_k are disjoint. $\square$

Remark 3.40. Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, and let $u : I \rightarrow Y$. If u is differentiable on the interval $[a, b] \subseteq I$, then, in particular, it is continuous on $[a, b]$, and so by Lemma 2.63 and Theorem 3.39,

$$\|u(b) - u(a)\| \leq \mathcal{H}^1(u([a, b])) \leq \int_a^b \|u'(x)\| dx.$$

We are now ready to present the main theorem of this section.

Theorem 3.41 (Lusin (N) property). *Let $I \subseteq \mathbb{R}$ be an interval. A function $u : I \rightarrow \mathbb{R}^M$ belongs to $AC_{\text{loc}}(I; \mathbb{R}^M)$ if and only if*

- (i) u is continuous on I ,
- (ii) u is differentiable $\mathcal{L}^1$ -a.e. in I , and $u' \in L^1_{\text{loc}}(I; \mathbb{R}^M)$,
- (iii) u satisfies the Lusin (N) property.

Proof. Step 1: Assume that u satisfies (i)–(iii) and fix $[a, b] \subseteq I$. We claim that u belongs to $AC([a, b]; \mathbb{R}^M)$. Let $\{(a_k, b_k)\}_{k=1}^n$ be a finite number of nonoverlapping intervals of $[a, b]$ and let

$$E_k := \{x \in [a_k, b_k] : u'(x) \text{ exists}\}.$$

By (ii), $\mathcal{L}^1([a_k, b_k] \setminus E_k) = 0$, and so by (iii), $\mathcal{H}^1(u([a_k, b_k] \setminus E_k)) = 0$. Since u is continuous, by Theorem 3.39 and by Lemma 2.63,

$$\begin{aligned} \sum_{k=1}^n \|u(b_k) - u(a_k)\| &\leq \sum_{k=1}^n \mathcal{H}^1(u([a_k, b_k])) = \sum_{k=1}^n \mathcal{H}^1(u(E_k)) \\ &\leq \sum_{k=1}^n \int_{E_k} \|u'(x)\| dx = \sum_{k=1}^n \int_{a_k}^{b_k} \|u'(x)\| dx. \end{aligned}$$

Since, by (ii), u' is Lebesgue integrable in $[a, b]$, in view of Exercise 3.27, u' is equi-integrable in $[a, b]$. Hence, given $\varepsilon > 0$, we may find $\delta > 0$ such that if $\sum_{k=1}^n (b_k - a_k) \leq \delta$, then $\sum_{k=1}^n \int_{a_k}^{b_k} \|u'(x)\| dx \leq \varepsilon$. The absolute continuity of u in $[a, b]$ follows from the previous inequality.

Step 2: Conversely, assume that $u \in AC_{\text{loc}}(I; \mathbb{R}^M)$. In view of Theorem 3.20, it remains to show that u satisfies property (iii). Fix $[a, b] \subseteq I$ and a Lebesgue measurable set $E \subset [a, b]$ with $\mathcal{L}^1(E) = 0$. Without loss of generality, we may assume that $E \subseteq (a, b)$. Fix $\varepsilon > 0$ and let $\delta > 0$ be as in Definition 3.1. By the properties of the outer measure we may find an open set U such that $U \supseteq E$ and $\mathcal{L}^1(U) \leq \delta$. By replacing U with $U \cap (a, b)$, we can assume that $U \subseteq (a, b)$. Decompose U into a countable family $\{J_k\}_k$ of pairwise disjoint intervals. Since u is continuous, for every k we may find $a_k, b_k \in \overline{J_k}$ such that

$$\text{diam } u(\overline{J_k}) = \|u(b_k) - u(a_k)\|.$$

Using the fact that $\mathcal{L}^1(U) \leq \delta$, we have that $\sum_k |b_k - a_k| \leq \delta$, and so, by the absolute continuity of u and Remark 3.2,

$$\sum_k \|u(b_k) - u(a_k)\| \leq \varepsilon.$$

Hence, by (C.28),

$$\begin{aligned} \mathcal{H}_\varepsilon^1(u(E)) &\leq \mathcal{H}_\varepsilon^1(u(U)) \\ &\leq \sum_k \text{diam } u(\overline{J_k}) = \sum_k \|u(b_k) - u(a_k)\| \leq \varepsilon. \end{aligned}$$

Letting $\varepsilon \rightarrow 0^+$ and using (C.29) we conclude that $\mathcal{H}_o^1(u(E)) = 0$. $\square$

Remark 3.42. Note that since the sets E_k defined in the first part of the proof are Borel sets (why?), the previous theorem continues to hold if in place of the (N) property we only require that u maps Borel sets of Lebesgue measure zero into sets of $\mathcal{H}^1$ measure zero.

Corollary 3.43. *Let $I \subseteq \mathbb{R}$ be an interval. A function $u : I \rightarrow \mathbb{R}^M$ belongs to $AC(I; \mathbb{R}^M)$ if and only if*

- (i) u is continuous on I ,

- (ii) u is differentiable $\mathcal{L}^1$ -a.e. in I , and u' belongs to $L^1_{\text{loc}}(I; \mathbb{R}^M)$ and is equi-integrable,
- (iii) u satisfies the *Lusin (N) property*.

Proof. It is enough to repeat Step 1 of the proof of Theorem 3.41 word for word, with the only differences being that $[a, b]$ should be replaced by I and that Exercise 3.27 is no longer needed, since, by (ii), u' is assumed to be equi-integrable. $\square$

Remark 3.44. Note that if I is a bounded interval, then $u : I \rightarrow \mathbb{R}^M$ belongs to $AC(I; \mathbb{R}^M)$ if and only if (i)–(iii) of the previous corollary hold. Indeed, if $u \in AC(I; \mathbb{R}^M)$, then by Corollary 3.10, u' is integrable, and so equi-integrable by Exercise 3.27. Thus, property (ii) of the previous corollary holds. Properties (i) and (iii) follow from Theorem 3.41.

Exercise 3.45. Let $u : [a, b] \rightarrow \mathbb{R}$ be continuous and strictly increasing.

- (i) Prove that u is absolutely continuous if and only if it maps the set

$$E := \{x \in [a, b] : u'(x) = \infty\}$$

into a set of Lebesgue measure zero.

- (ii) Is part (i) still true if u is only assumed to be continuous and increasing instead of strictly increasing?

Exercise 3.46. Let $u : [a, b] \rightarrow \mathbb{R}$ be continuous and strictly increasing. Prove that its inverse $u^{-1} : [u(a), u(b)] \rightarrow \mathbb{R}$ is absolutely continuous if and only if the set

$$E := \{x \in [a, b] : u'(x) = 0\}$$

has Lebesgue measure zero.

In view of Remark 3.40 and Step 1 of the proof of Theorem 3.41, we have the following.

Corollary 3.47. *Let $I \subseteq \mathbb{R}$ be an interval. If $u : I \rightarrow \mathbb{R}^M$ is everywhere differentiable in I and $u' \in L^1_{\text{loc}}(I; \mathbb{R}^M)$, then u belongs to $AC_{\text{loc}}(I; \mathbb{R}^M)$.*

Exercise 3.48. Prove that if $u : [a, b] \rightarrow \mathbb{R}^M$ is continuous, differentiable on $[a, b]$ except for at most a countable number of points, and if u' is Lebesgue integrable, then u belongs to $AC([a, b]; \mathbb{R}^M)$.

As a consequence of Theorem 3.41 we can characterize those functions with locally bounded pointwise variation that are locally absolutely continuous functions. Precisely, we have the following result.

Corollary 3.49. *Let $I \subseteq \mathbb{R}$ be an interval. A function $u : I \rightarrow \mathbb{R}^M$ belongs to $AC_{\text{loc}}(I; \mathbb{R}^M)$ if and only if*

- (i) u is continuous on I ,
- (ii) $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$,
- (iii) u satisfies the Lusin (N) property.

Proof. In view of Corollary 2.31 we are in a position to apply Theorem 3.41. $\square$

The next exercise gives an example of a function that satisfies the Lusin (N) property, but it is not of bounded pointwise variation in any interval of $(0, 1)$.

Exercise 3.50. Let $\{(a_n, b_n)\}_n$ be a base for the topology of $(0, 1)$ and let $\{r_n\}_n$ be a sequence of positive numbers such that $\sum_{n=1}^{\infty} r_n < \infty$. We construct inductively a sequence of functions $u_n : [0, 1] \rightarrow \mathbb{R}$ and a sequence of intervals (c_n, d_n) as follows. Let $u_0(x) := x$. Assume that $u_{n-1} : [0, 1] \rightarrow \mathbb{R}$ has been defined and that u_{n-1} is a continuous piecewise affine function. Let $(c_n, d_n) \subseteq (a_n, b_n)$ be an interval such that u_{n-1} restricted to (c_n, d_n) is linear and $\mathcal{L}^1(u_{n-1}((c_n, d_n))) < r_n$. We define the continuous function u_n to be u_{n-1} outside (c_n, d_n) , while in (c_n, d_n) we define it as a continuous piecewise affine function such that $\mathcal{L}^1(u_n((c_n, d_n))) < r_n$ and $\text{Var}_{(c_n, d_n)} u_n > n$.

- (i) Prove that $\{u_n\}_n$ converges uniformly to a continuous function $u : [0, 1] \rightarrow \mathbb{R}$.
- (ii) Prove that for every interval $[a, b] \subset (0, 1)$ with $a < b$, $\text{Var}_{[a, b]} u = \infty$.
- (iii) Prove that u satisfies the Lusin (N) property. Hint: Let

$$E_0 := [0, 1] \setminus \bigcup_{i=1}^{\infty} (c_i, d_i), \quad E_n := (c_n, d_n) \setminus \bigcup_{i=n+1}^{\infty} (c_i, d_i)$$

and

$$E_{\infty} := \bigcap_{n=1}^{\infty} \bigcup_{i=n}^{\infty} (c_i, d_i).$$

For every set $E \subset [0, 1]$, with $\mathcal{L}^1(E) = 0$, write

$$E = (E \cap E_{\infty}) \cup \bigcup_{n=0}^{\infty} (E \cap E_n).$$

Exercise 3.51 (The Cantor set and the Lusin (N) property). Let $\mathbb{D}$ be the Cantor set.

- (i) Prove that every number $x \in [0, 2]$ can be written as

$$x = 2 \sum_{n=1}^{\infty} \frac{c_n(x)}{3^n},$$

where $c_n(x) \in \{0, 1, 2\}$, and deduce that every element in $[0, 2]$ can be written as the sum of two elements of $\mathbb{D}$.

- (ii) Prove that if $v : \mathbb{D} \rightarrow \mathbb{R}$ is continuous and $\mathcal{L}^1(v(\mathbb{D})) = 0$, then v can be extended to a continuous function on $[0, 1]$ that has the Lusin (N) property. Hint: Make v differentiable outside $\mathbb{D}$.
- (iii) For each $x \in \mathbb{D}$ write

$$x = 2 \sum_{n=1}^{\infty} \frac{c_n(x)}{3^n},$$

where $c_n(x) \in \{0, 1\}$ and prove that there exist two continuous functions $u_1 : [0, 1] \rightarrow \mathbb{R}$ and $u_2 : [0, 1] \rightarrow \mathbb{R}$ with the Lusin (N) property and such that for all $x \in \mathbb{D}$,

$$u_1(x) = \sum_{n=1}^{\infty} \frac{c_{2n}(x)}{3^n}, \quad u_2(x) = \sum_{n=1}^{\infty} \frac{c_{2n+1}(x)}{3^n}.$$

- (iv) Prove that $u_1 + u_2$ does not have the Lusin (N) property.
- (v) Why is this example important for absolute continuity?

Exercise 3.52. Given an absolutely continuous function $u : [a, b] \rightarrow \mathbb{R}$, define

$$v(x) := \max_{a \leq y \leq x} u(y), \quad x \in [a, b].$$

Prove that v is absolutely continuous and that $v'(x) = u'(x)\chi_{\{v=u\}}(x)$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$.

3.4. Superposition in $AC(I; Y)$

In this section we discuss the analog of Theorem 2.55. The following exercise (see Exercise 2.53) shows that the composition of absolutely continuous functions is not absolutely continuous (see however Exercise 3.67).

Exercise 3.53. Consider the functions $f : [0, 1] \rightarrow \mathbb{R}$ and $u : [0, 1] \rightarrow [0, 1]$, defined by

$$f(z) := \sqrt{z}, \quad u(x) := \begin{cases} x^2 \sin^2 \frac{1}{x} & \text{if } 0 < x \leq 1, \\ 0 & \text{if } x = 0. \end{cases}$$

Prove that f and u are absolutely continuous but their composition $f \circ u$ is not.

Remark 3.54. In Theorem 3.83 we will see that given f and u absolutely continuous, $f \circ u$ is absolutely continuous if and only if $(f' \circ u)u'$ is integrable, where $f'(u(x))u'(x)$ is interpreted to be zero whenever $u'(x) = 0$.

The next result gives necessary and sufficient conditions on $f : \mathbb{R} \rightarrow \mathbb{R}$ for $f \circ u$ to be absolutely continuous for *all* absolutely continuous functions $u : [a, b] \rightarrow \mathbb{R}$.

Theorem 3.55 (Superposition). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : \mathbb{R}^M \rightarrow \mathbb{R}$. Then $f \circ u \in AC_{\text{loc}}(I)$ for all functions $u \in AC_{\text{loc}}(I; \mathbb{R}^M)$ if and only if f is locally Lipschitz continuous. In particular, if $M = 1$, f is locally Lipschitz continuous and $u \in AC_{\text{loc}}(I)$, then the chain rule (3.24) holds.*

Proof. Step 1: Assume that f is locally Lipschitz continuous and let $u \in AC_{\text{loc}}(I)$. Fix an interval $[a, b]$. Since u is continuous,

$$m := \max_{x \in [a, b]} \|u(x)\| < \infty,$$

and so there exists $L > 0$ such that

$$(3.16) \quad \|f(z_1) - f(z_2)\| \leq L \|z_1 - z_2\|$$

for all $z_1, z_2 \in \overline{B(0, m)}$. We claim that $f \circ u$ is absolutely continuous in $[a, b]$. Indeed, since $u \in AC([a, b]; \mathbb{R}^M)$, for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\sum_k \|u(b_k) - u(a_k)\| \leq \varepsilon/L$ for every finite number of nonoverlapping intervals $(a_k, b_k) \subseteq [a, b]$, with $\sum_k (b_k - a_k) \leq \delta$. Hence, by (3.16),

$$\sum_k |(f \circ u)(b_k) - (f \circ u)(a_k)| \leq L \sum_k \|u(b_k) - u(a_k)\| \leq \varepsilon,$$

which proves the claim.

The validity of the chain rule follows from Corollary 3.68.

Step 2: Assume that $f \circ u \in AC_{\text{loc}}(I)$ for all functions $u \in AC_{\text{loc}}(I; \mathbb{R}^M)$. We claim that f is locally Lipschitz continuous. The proof follows closely that of Theorem 2.55, with the only difference that instead of discontinuous functions u (see (2.26), (2.31)) we will use piecewise affine functions. Fix $[a, b] \subseteq I$. We begin by showing that f is locally bounded. Fix $r > 0$. We claim that f is bounded in $\overline{B(0, r)}$. Indeed, assume by contradiction that there exists $\{z_n\}_n$ in $\overline{B(0, r)}$ such that $|f(z_n)| \rightarrow \infty$ as $n \rightarrow \infty$. By compactness we can assume that $z_n \rightarrow z_\infty$, and by taking a further subsequence, also that

$$\|z_n - z_\infty\| \leq 1/n^2$$

for every n . For every $n \in \mathbb{N}$, set $I_n := [a + \frac{b-a}{n+1}, a + \frac{b-a}{n}]$ and let x_n be the middle point of I_n and $\ell_n := (\text{diam } I_n)/2 = (b-a)/(2n(n+1))$. Consider the function

$$(3.17) \quad u(x) := \begin{cases} z_n + z_\infty - z_n(x_n - x)/\ell_n & \text{if } x \in I_n^\circ, x \leq x_n, n \in \mathbb{N}, \\ z_n + z_\infty - z_n(x - x_n)/\ell_n & \text{if } x \in I_n^\circ, x \geq x_n, n \in \mathbb{N}, \\ z_\infty & \text{otherwise in } I. \end{cases}$$

We claim that $u \in AC(I; \mathbb{R}^M)$. Since $z_n \rightarrow z_\infty$, we have that u is continuous at $x = a$. Hence, the function u is continuous and differentiable except for a countable number of points, and so, in view of Exercise 3.48, to prove that it is locally absolutely continuous in I , it remains to show that u' is integrable. By (3.17) we have

$$\int_I \|u'\| dx = \int_a^b \|u'\| dx \leq \sum_{n=1}^{\infty} \|z_n - z_\infty\| \leq \sum_{n=1}^{\infty} 1/n^2 < \infty.$$

It follows that u belongs to $AC(I; \mathbb{R}^M)$, and so by hypothesis the function $f \circ u$ belongs to $AC_{\text{loc}}(I)$. On the other hand,

$$\text{Var}_{[a,b]}(f \circ u) \geq \text{Var}_{\bar{I}_n}(f \circ u) \geq |f(z_n) - f(z_0)| \rightarrow \infty$$

as $n \rightarrow \infty$, which is a contradiction.

Step 3: Next we claim that f is Lipschitz continuous in $\overline{B(0, r)}$. Indeed, assume by contradiction that this is not the case. Then we may find two sequences $\{z_n\}_n, \{y_n\}_n$ in $\overline{B(0, r)}$ such that $z_n \neq y_n$ and

$$(3.18) \quad \frac{|f(z_n) - f(y_n)|}{\|z_n - y_n\|} > 2(n^2 + n)$$

for all $n \in \mathbb{N}$. Since $\{z_n\}_n$ is bounded, we may extract a subsequence (not relabeled) such that $z_n \rightarrow z_\infty$. Take a further subsequence (not relabeled) such that (3.18) continues to hold and

$$(3.19) \quad \|z_n - z_\infty\| < 1/(n+1)^2.$$

By Step 2 f is bounded in $\overline{B(0, r)}$ by some constant $c_r > 0$. Hence, by (3.18) for all $n \in \mathbb{N}$ we have

$$(3.20) \quad 2c_r \geq |f(z_n) - f(y_n)| > 2(n^2 + n)\|z_n - y_n\|,$$

which gives

$$0 < \delta_n := \frac{\|z_n - y_n\|(b-a)}{2c_r} < \frac{b-a}{2(n^2 + n)} \rightarrow 0.$$

For every $n \in \mathbb{N}$, we divide the interval $I_n := [a + \frac{b-a}{n+1}, a + \frac{b-a}{n}]$ into subintervals of length δ_n . Thus, let

$$(3.21) \quad \ell_n := \frac{\text{diam } I_n}{\delta_n} = \frac{b-a}{\delta_n(n^2 + n)} = \frac{2c_r}{\|z_n - y_n\|(n^2 + n)} > 2$$

and set $m_n := \max\{j \in \mathbb{N}_0 : j < \ell_n\}$. Since $\ell_n > 2$, we have

$$(3.22) \quad \ell_n/2 \leq m_n < \ell_n.$$

Consider the partition P_n of I_n given by

$$\begin{aligned} P_n &:= \left\{ a + \frac{b-a}{n+1} + \frac{1}{2}j\delta_n : j = 0, \dots, 2m_n \right\} \cup \left\{ a + \frac{b-a}{n} \right\} \\ &=: \{x_{0,n}, \dots, x_{2m_n+1,n}\}. \end{aligned}$$

Let $I_{i,n} := [x_{i,n}, x_{i+1,n}]$, $i = 0, \dots, 2m_n$. We now define the piecewise affine function $u : I \rightarrow \mathbb{R}$ in the following way. Set $z_0 := z_1$. Define $u(x) := z_\infty$ if $x \leq a$, $u(x) := z_0$ if $x \geq b$, while in each interval I_n , $n \in \mathbb{N}$, set

$$(3.23) \quad u(x) := \begin{cases} 2(z_n - y_n)(x - x_{2i-1,n})/\delta_n + y_n & \text{if } x \in I_{2i-1,n}, \ 1 \leq i \leq m_n - 1, \\ 2(y_n - z_n)(x - x_{2i,n})/\delta_n + z_n & \text{if } x \in I_{2i,n}, \ 0 \leq i \leq m_n - 1, \\ (z_{n-1} - y_n)(x - x_{2m_n-1,n})/d_n + y_n & \text{if } x \in I_{2m_n-1,n}, \end{cases}$$

where $d_n := x_{2m_n+1,n} - x_{2m_n-1,n}$. Note that $\delta_n/2 \leq d_n \leq 2\delta_n$. We claim that $u \in AC(I; \mathbb{R}^M)$. Since $z_n \rightarrow z_\infty$ and $y_n \rightarrow z_\infty$, we have that u is continuous at $x = a$. Hence, the function u is continuous and differentiable except for a countable number of points, and so, in view of Exercise 3.48, to prove that it is locally absolutely continuous in I , it remains to show that u' is integrable. By (3.19), (3.20), and (3.22) we have

$$\begin{aligned} \int_I \|u'\| dx &= \int_a^b \|u'\| dx \leq \sum_{n=1}^{\infty} (2m_n \|z_n - y_n\| + \|y_n - z_{n-1}\|) \\ &\leq \sum_{n=1}^{\infty} (2\ell_n \|z_n - y_n\| + \|y_n - z_n\| + \|z_n - z_\infty\| + \|z_\infty - z_{n-1}\|) \\ &\leq \sum_{n=1}^{\infty} \frac{4c_r + 2}{n^2} < \infty. \end{aligned}$$

On the other hand, by (3.20)–(3.22),

$$\begin{aligned} \sum_{i=1}^{2m_n+1} |f(u(x_{i,n})) - f(u(x_{i-1,n}))| &\geq 2m_n |f(z_n) - f(y_n)| \\ &> 2\ell_n(n^2 + n) \|z_n - y_n\| = 4c_r, \end{aligned}$$

and so for every $n \in \mathbb{N}$ by Proposition 2.15 we obtain that

$$\text{Var}_{[a,b]}(f \circ u) \geq \sum_{k=1}^n \text{Var}_{I_k}(f \circ u) \geq 4c_r n \rightarrow \infty$$

as $n \rightarrow \infty$. Hence, we have obtained a contradiction. $\square$

Remark 3.56. Note that in the necessity part of the theorem we have actually proved a much stronger result, namely that if $f : \mathbb{R}^M \rightarrow \mathbb{R}$ is such that $f \circ u \in BPV_{\text{loc}}(I)$ for all functions $u \in AC(I; \mathbb{R}^M) \subseteq AC_{\text{loc}}(I; \mathbb{R}^M)$, then f is locally Lipschitz continuous.

Remark 3.57. The previous proof continues to hold if $f : \mathbb{R}^M \rightarrow Z$, where (Z, d_Z) is a metric space. On the other hand, reasoning as in Step 1 we have that if (Y, d_Y) and (Z, d_Z) are metric spaces and $f : Y \rightarrow Z$ is locally Lipschitz continuous, then $f \circ u$ belongs to $BPV_{\text{loc}}(I; Z)$ (respectively, $BPV(I; Z)$) for all functions $u \in BPV_{\text{loc}}(I; Y)$ (respectively, $BPV(I; Y)$).

Exercise 3.58. Let $I \subseteq \mathbb{R}$ be an interval, let $f : \mathbb{R}^M \rightarrow \mathbb{R}$, and let $1 \leq p < \infty$. Modify the proof of Theorem 3.55 to show that $f \circ u \in AC_{\text{loc}}(I)$ for all functions $u \in AC(I; \mathbb{R}^M)$ with $u' \in L^p(I; \mathbb{R}^M)$ if and only if f is locally Lipschitz continuous.

3.5. Chain Rule

Next we discuss the validity of the chain rule for absolutely continuous functions. The next result establishes the validity of the chain rule under very weak hypotheses.

Theorem 3.59 (Chain rule). *Let $I, J \subseteq \mathbb{R}$ be two intervals, let $(Y, \|\cdot\|)$ be a normed space, and let $f : J \rightarrow Y$ and $u : I \rightarrow J$ be such that f, u , and $f \circ u$ are differentiable $\mathcal{L}^1$ -a.e. in their respective domains. If f satisfies the Luzin (N) property, then for $\mathcal{L}^1$ -a.e. $x \in I$,*

$$(3.24) \quad (f \circ u)'(x) = f'(u(x))u'(x),$$

where $f'(u(x))u'(x)$ is interpreted to be zero whenever $u'(x) = 0$ (even if f is not differentiable at $u(x)$).

In the proof we will show that $(f \circ u)'(x) = 0$ and $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in I$ such that f is not differentiable at $u(x)$.

To prove Theorem 3.59, we need an auxiliary result, which is a converse of Corollary 3.37.

Lemma 3.60. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, and let $u : I \rightarrow Y$. Assume that u is differentiable on a set $E \subseteq I$ (not necessarily measurable), with $\mathcal{H}^1(u(E)) = 0$. Then $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in E$.*

Proof. Let $E_0 := \{x \in E : \|u'(x)\| > 0\}$. We claim that $\mathcal{L}^1(E_0) = 0$. For every integer $k \in \mathbb{N}$ let

$$E_k := \{x \in E_0 : \|u(x) - u(t)\| \geq |x - t|/k \text{ for all } t \in (x - 1/k, x + 1/k) \cap I\}.$$

Noting that $E_0 = \bigcup_k E_k$, we fix k and we let $F := J \cap E_k$, where J is an interval of length less than $1/k$. To prove that $\mathcal{L}^1(E_0) = 0$, it suffices to show that $\mathcal{L}^1(F) = 0$. Since $\mathcal{H}^1(u(E)) = 0$ and $F \subseteq E$, for every $\varepsilon > 0$ we may find a sequence $\{G_n\}_n$ of subsets of Y such that

$$u(F) \subseteq \bigcup_n G_n, \quad \sum_n \text{diam } G_n \leq \varepsilon.$$

Let $F_n := u^{-1}(G_n) \cap F$. Since $\{F_n\}_n$ covers F , we have

$$\begin{aligned} \mathcal{L}_o^1(F) &\leq \sum_n \mathcal{L}_o^1(F_n) \leq \sum_n \sup_{x,t \in F_n} |x - t| \\ &\leq \sum_n k \sup_{x,t \in F_n} \|u(x) - u(t)\| \leq k \sum_n \text{diam } G_n \leq k\varepsilon, \end{aligned}$$

where we have used definition of F_n . It now suffices to let $\varepsilon \rightarrow 0^+$. $\square$

Remark 3.61. In the case $Y = \mathbb{R}$ the previous lemma holds even if we assume in place of differentiability that u has a derivative at every $x \in E$.

Corollary 3.62. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, and let $u : I \rightarrow Y$, $v : I \rightarrow Y$. Assume that there exists a set $E \subseteq I$ such that u and v are differentiable for all $x \in E$ and $u(x) = v(x)$ for all $x \in E$. Then $u'(x) = v'(x)$ for $\mathcal{L}^1$ -a.e. $x \in E$.*

Proof. Let $w := u - v$. Then $w(E) = \{0\}$, and so $\mathcal{H}^1(w(E)) = 0$. By the previous lemma, $u'(x) - v'(x) = w'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in E$. $\square$

We turn to the proof of Theorem 3.59.

Proof of Theorem 3.59. Let

$$\begin{aligned} G &:= \{z \in J : f \text{ is not differentiable at } z\}, \\ F &:= \{x \in I : u \text{ or } f \circ u \text{ is not differentiable at } x\}. \end{aligned}$$

By hypothesis $\mathcal{L}^1(G) = \mathcal{L}^1(F) = 0$. Let

$$E := \{x \in I^\circ \setminus F : u(x) \in G\}.$$

Since $u(E) \subseteq G$, we have $\mathcal{L}^1(u(E)) = 0$, and since f satisfies the Lusin (N) property, we obtain that

$$\mathcal{H}^1((f \circ u)(E)) = 0.$$

By Lemma 3.60 applied to u and to $f \circ u$, we conclude that

$$u'(x) = (f \circ u)'(x) = 0$$

for $\mathcal{L}^1$ -a.e. $x \in E$.

On the other hand, if $x \in I^\circ \setminus F$ and $u(x) \notin G$, then we may apply the standard chain rule to conclude that $f \circ u$ is differentiable at x with $(f \circ u)'(x) = f'(u(x))u'(x)$. $\square$

The next example shows the importance of the Lusin (N) property.

Example 3.63. Let $u : [0, 1] \rightarrow \mathbb{R}$ be a strictly increasing function such that $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in [0, 1]$ (see Theorem 1.37) and let $f := u^{-1}$. Note that f is strictly increasing, and so by Lebesgue's differentiation theorem (see Theorem 1.18) it is differentiable for $\mathcal{L}^1$ -a.e. $x \in [0, 1]$, despite the

fact that $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in [0, 1]$. Moreover, $(f \circ u)(x) = x$ for all $x \in [0, 1]$, and so $(f \circ u)'(x) = 1$ for all $x \in [0, 1]$, while $f'(u(x))u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in [0, 1]$, since $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in [0, 1]$.

Corollary 3.64. *Let $I, J \subseteq \mathbb{R}$ be two intervals, let $(Y, \|\cdot\|)$ be a normed space, and let $f : J \rightarrow Y$ and $u : I \rightarrow J$ be such that f and u are differentiable $\mathcal{L}^1$ -a.e. in their respective domains. Suppose that u' is zero at most on a set of Lebesgue measure zero. Then $f \circ u$ is differentiable $\mathcal{L}^1$ -a.e. in I and the chain rule (3.24) holds.*

Proof. Let G and F be as in the proof of Theorem 3.59 and let

$$E := \{x \in I^\circ : u(x) \in G\}.$$

Since $u(E) \subseteq G$, we have $\mathcal{L}^1(u(E)) = 0$, and so by Lemma 3.60 we conclude that $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in E$. But then, according to our assumption that u' is zero at most on a set of Lebesgue measure zero, E must have Lebesgue measure zero. If $x \in I^\circ \setminus E$, then f is differentiable at $u(x)$, and so the chain rule (3.24) holds $\mathcal{L}^1$ -a.e. in $I^\circ \setminus E$. $\square$

Corollary 3.65. *Let $I, J \subseteq \mathbb{R}$ be two intervals, let $f : J \rightarrow \mathbb{R}^M$ and $u : I \rightarrow J$ be such that u and $f \circ u$ are differentiable $\mathcal{L}^1$ -a.e. in their respective domains. If $f \in AC_{\text{loc}}(J; \mathbb{R}^M)$, then the chain rule (3.24) holds.*

Proof. In view of Theorem 3.41, we have that f is differentiable $\mathcal{L}^1$ -a.e. in I and satisfies the Lusin (N) property and so all the hypotheses of Theorem 3.59 are satisfied. $\square$

A less trivial consequence of Theorem 3.59 is the following result.

Corollary 3.66. *Let $I, J \subseteq \mathbb{R}$ be two intervals, let $f \in AC_{\text{loc}}(J; \mathbb{R}^M)$, and let $u : I \rightarrow J$ be monotone. Then $f \circ u$ is differentiable $\mathcal{L}^1$ -a.e. in I and the chain rule (3.24) holds.*

Proof. Note that the composite $f \circ u$ belongs to $BPV_{\text{loc}}(I; \mathbb{R}^M)$ by Exercise 2.54, and so by Corollary 2.31, $f \circ u$ is differentiable $\mathcal{L}^1$ -a.e. in I . We are now in a position to apply the previous corollary. $\square$

Exercise 3.67. Let $I, J \subseteq \mathbb{R}$ be two intervals, let $f \in AC_{\text{loc}}(J)$, and let $u : I \rightarrow J$ be monotone and $AC'_{\text{loc}}(I)$. Prove that $f \circ u$ belongs to $AC'_{\text{loc}}(I)$.

Corollary 3.68. *Let $I, J \subseteq \mathbb{R}$ be two intervals, let $f : J \rightarrow \mathbb{R}$ be locally Lipschitz continuous, and let $u \in BPV_{\text{loc}}(I)$. Then $f \circ u$ is differentiable $\mathcal{L}^1$ -a.e. in I and the chain rule (3.24) holds.*

Proof. By Theorem 2.55, we have that $f \circ u \in BPV_{\text{loc}}(I)$. Hence we can apply Corollary 2.31 to conclude that $f \circ u$ is differentiable $\mathcal{L}^1$ -a.e. in I .

Since f is locally Lipschitz continuous, it is locally absolutely continuous, and so the result follows from Corollary 3.65. $\square$

To conclude this section, we observe that if $f : \mathbb{R}^M \rightarrow \mathbb{R}$ is a Lipschitz continuous function with $M \geq 2$ and if $u : I \rightarrow \mathbb{R}^M$ is absolutely continuous, then $f \circ u \in AC(I)$ (see Step 1 of the proof of Theorem 3.55), but the analog of (3.24), which is

$$(3.25) \quad (f \circ u)'(x) = \sum_{i=1}^M \frac{\partial f}{\partial z_i}(u(x)) u'_i(x),$$

where $\frac{\partial f}{\partial z_i}(u(x)) u'_i(x)$ is interpreted to be zero whenever $u'_i(x) = 0$, may fail. This is illustrated by the next example.

Example 3.69. Let $M = 2$, and consider the functions

$$f(z) = f(z_1, z_2) := \max\{z_1, z_2\}, \quad z \in \mathbb{R}^2,$$

and $u(x) := (x, x)$ for $x \in \mathbb{R}$. Then $v(x) := (f \circ u)(x) = x$ so that $v'(x) = 1$, while the right-hand side of (3.25) is *nowhere defined*, since $u'(x) = (1, 1)$.

The problem lies in the fact that for a Lipschitz continuous function $f : \mathbb{R}^M \rightarrow \mathbb{R}$, the set

$$\Sigma^f := \{z \in \mathbb{R}^M : f \text{ is not differentiable at } z\}$$

has $\mathcal{L}^M$ -measure zero by Rademacher's theorem (see Theorem 9.14), but for $M \geq 2$ it is easy to construct a set of $\mathcal{L}^M$ -measure zero that contains the range of an absolutely continuous curve (see Example 3.69). Thus, to recover (3.25), stronger restrictions on the set Σ^f are needed.

Definition 3.70. Let (Y, d) be a metric space. A set $E \subseteq Y$ is called $\mathcal{H}^1$ -rectifiable if there exists a sequence of Lipschitz continuous functions $u_n : \mathbb{R} \rightarrow Y$, $n \in \mathbb{N}$, such that

$$\mathcal{H}^1\left(E \setminus \bigcup_n u_n(\mathbb{R})\right) = 0.$$

A set $E \subseteq Y$ is *purely $\mathcal{H}^1$ -unrectifiable* if

$$\mathcal{H}^1(E \cap u(\mathbb{R})) = 0$$

for every Lipschitz continuous function $u : \mathbb{R} \rightarrow Y$.

Exercise 3.71. Let $E \subseteq \mathbb{R}^M$ be purely $\mathcal{H}^1$ -unrectifiable. Prove that $\mathcal{H}^1(E \cap u(I)) = 0$ for every interval $I \subseteq \mathbb{R}$ and for all $u \in AC(I; \mathbb{R}^M)$.

A useful characterization of purely $\mathcal{H}^1$ -unrectifiable sets is given by the following result.

Theorem 3.72. Let $E \subseteq \mathbb{R}^M$. Then the following properties are equivalent:

(i) E is purely $\mathcal{H}^1$ -unrectifiable.

(ii) For every interval $I \subseteq \mathbb{R}$ and for all $u \in AC(I; \mathbb{R}^M)$,

$$(3.26) \quad u'(x) = 0 \quad \text{for } \mathcal{L}^1\text{-a.e. } x \in u^{-1}(E).$$

Proof. Step 1: Assume that E is purely $\mathcal{H}^1$ -unrectifiable, let $I \subseteq \mathbb{R}$ be an interval, and consider a function $u \in AC(I; \mathbb{R}^M)$. By the previous exercise, $\mathcal{H}^1(E \cap u(I)) = 0$. Let $F := u^{-1}(E \cap u(I))$. By Lemma 3.60, $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in F$, which gives (ii).

Step 2: Assume that (ii) holds. We claim that E is purely $\mathcal{H}^1$ -unrectifiable. Indeed, assume by contradiction that there exists a Lipschitz continuous function $w : \mathbb{R} \rightarrow \mathbb{R}^M$ such that

$$(3.27) \quad \mathcal{H}^1(E \cap w(\mathbb{R})) > 0.$$

Then $\mathcal{H}^1(E \cap w([-n, n])) \rightarrow \mathcal{H}^1(E \cap w(\mathbb{R})) > 0$ as $n \rightarrow \infty$. Fix $n \in \mathbb{N}$ so large that

$$\mathcal{H}^1(E \cap w([-n, n])) > 0.$$

Since w is Lipschitz continuous, $\mathcal{H}^1(w([-n, n])) < \infty$. By property (ii) there exists a set $G \subset [-n, n]$ with $\mathcal{L}^1(G) = 0$ such that $w'(x) = 0$ for all $x \in [-n, n] \setminus G$. Since w is Lipschitz continuous, w satisfies the Lusin (N) property. Hence, $\mathcal{H}^1(w(G)) = 0$. On the other hand, by Corollary 3.37, $\mathcal{H}^1(w([-n, n] \setminus G)) = 0$. Thus, $\mathcal{H}^1(w([-n, n])) = 0$, which is a contradiction. $\square$

The following theorem is due to Marcus and Mizel [159].

Theorem 3.73 (Chain rule, II). *Let $f : \mathbb{R}^M \rightarrow \mathbb{R}$ be a locally Lipschitz continuous function and let $I \subseteq \mathbb{R}$ be an interval. Then $f \circ u \in AC_{\text{loc}}(I)$ for every $u \in AC_{\text{loc}}(I; \mathbb{R}^M)$. Moreover, if the set Σ^f is purely $\mathcal{H}^1$ -unrectifiable, then*

$$(3.28) \quad (f \circ u)'(x) = \sum_{i=1}^M \frac{\partial f}{\partial z_i}(u(x)) u'_i(x)$$

for $\mathcal{L}^1$ -a.e. $x \in I$, where $\frac{\partial f}{\partial z_i}(u(x)) u'_i(x)$ is interpreted to be zero whenever $u'_i(x) = 0$.

Proof. The fact that $f \circ u \in AC_{\text{loc}}(I)$ follows by Theorem 3.55. By Theorem 3.20, we have that u and $f \circ u$ are differentiable for $\mathcal{L}^1$ -a.e. $x \in I$. Thus, it suffices to prove (3.28) for $\mathcal{L}^1$ -a.e. $x \in I$ such that $f \circ u$ and u are differentiable at x . If at any of these points $x \in I$, $u(x) \in \mathbb{R}^M \setminus \Sigma^f$, then f is differentiable at $u(x)$, and so (3.28) follows by the classical chain rule.

Thus, it remains to consider those $x \in I$ such that $x \in u^{-1}(\Sigma^f)$ and $f \circ u$ and u are differentiable at x . By Theorem 3.72, we have that

$$u'(x) = 0 \quad \text{for } \mathcal{L}^1\text{-a.e. } x \in u^{-1}(\Sigma^f).$$

Fix $x \in I^\circ$ such that $x \in u^{-1}(\Sigma^f)$, $u'(x) = 0$, and $f \circ u$ and u are differentiable at x . To conclude the proof, we need to show that $(f \circ u)'(x) = 0$. Since u is bounded in $[x - \delta, x + \delta] \subseteq I$, let L_1 be the Lipschitz constant of f in the ball containing $u([x - \delta, x + \delta])$. Then, for all $0 < |h| < \delta$,

$$\frac{|(f \circ u)(x + h) - (f \circ u)(x)|}{|h|} \leq L_1 \frac{|u(x + h) - u(x)|}{|h|} \rightarrow L_1 |u'(x)| = 0$$

as $h \rightarrow 0$. This concludes the proof. $\square$

Remark 3.74. Necessary and sufficient conditions for the validity of the chain rule (3.28) have been proved in [140] and [175]. If $f : \mathbb{R}^M \rightarrow \mathbb{R}$ is only assumed to be a locally Lipschitz continuous function, then in place of (3.28) it is still possible to prove a weaker form of the chain rule. This has been established by Ambrosio and Dal Maso in [9].

3.6. Change of Variables

As a corollary of Theorem 3.59 we have the following change of variables formula.

Theorem 3.75 (Change of variables). *Let $g : [p, q] \rightarrow \mathbb{R}$ be an integrable function and let $u : [a, b] \rightarrow [p, q]$ be differentiable $\mathcal{L}^1$ -a.e. in $[a, b]$. Then $(g \circ u)u'$ is integrable and the change of variables*

$$(3.29) \quad \int_{u(\alpha)}^{u(\beta)} g(t) dt = \int_{\alpha}^{\beta} g(u(x))u'(x) dx$$

holds for all $\alpha, \beta \in [a, b]$ if and only if the function $f \circ u$ belongs to $AC([a, b])$, where

$$f(z) := \int_p^z g(t) dt, \quad z \in [p, q].$$

Proof. If $f \circ u \in AC([a, b])$, then since f is absolutely continuous (see Theorem 3.16), we can apply Corollary 3.65 to obtain the chain rule formula

$$(3.30) \quad (f \circ u)'(x) = g(u(x))u'(x)$$

for $\mathcal{L}^1$ -a.e. $x \in [a, b]$. Since $f \circ u \in AC([a, b])$, it follows from Corollary 3.10 and (3.30) that $(g \circ u)u'$ is integrable, and by the fundamental theorem of

calculus (see Theorem 3.20), for all $\alpha, \beta \in [a, b]$,

$$\begin{aligned} \int_{u(\alpha)}^{u(\beta)} g(t) dt &= (f \circ u)(\beta) - (f \circ u)(\alpha) \\ &= \int_{\alpha}^{\beta} (f \circ u)'(x) dx = \int_{\alpha}^{\beta} g(u(x))u'(x) dx. \end{aligned}$$

Conversely, if $(g \circ u)u'$ is integrable and the identity

$$(f \circ u)(\beta) - (f \circ u)(\alpha) = \int_{\alpha}^{\beta} g(u(x))u'(x) dx$$

holds for all $\alpha, \beta \in I$, then, since the right-hand side is absolutely continuous by Theorem 3.16, it follows that $f \circ u$ is absolutely continuous. $\square$

Remark 3.76. The previous proof shows, in particular, that the function $x \in I \mapsto g(u(x))u'(x)$ is measurable (see also Exercise 1.29), since it is the derivative of the function $f \circ u$, but this does not imply that the function $x \in I \mapsto g(u(x))$ is measurable (see the next exercise).

Exercise 3.77. Let $I, J \subseteq \mathbb{R}$ be intervals. Prove that there exist a continuous increasing function $u : I \rightarrow J$ and a Lebesgue measurable function $g : J \rightarrow \mathbb{R}$ such that $g \circ u : I \rightarrow \mathbb{R}$ is not Lebesgue measurable. Hint: See Exercise 1.33.

Corollary 3.78. Assume that $g : [p, q] \rightarrow \mathbb{R}$ is an integrable function and that $u : [a, b] \rightarrow [p, q]$ is monotone and absolutely continuous. Then $(g \circ u)u'$ is integrable and the change of variables formula (3.29) holds.

Exercise 3.79. Prove that under the hypotheses of the previous corollary the function $f \circ u$ is absolutely continuous and then prove the corollary.

Corollary 3.80. Assume that $g : [p, q] \rightarrow \mathbb{R}$ is a measurable, bounded function and that $u : [a, b] \rightarrow [p, q]$ is absolutely continuous. Then $(g \circ u)u'$ is integrable and the change of variables formula (3.29) holds.

Exercise 3.81. Prove that under the hypotheses of the previous corollary the function $f \circ u$ is absolutely continuous and then prove the corollary.

Corollary 3.82. Assume that $g : [p, q] \rightarrow \mathbb{R}$ is an integrable function, that $u : [a, b] \rightarrow [p, q]$ is absolutely continuous, and that $(g \circ u)u'$ is integrable. Then the change of variables formula (3.29) holds.

Proof. Let

$$g_n(z) := \begin{cases} n & \text{if } g(z) > n, \\ g(z) & \text{if } -n \leq g(z) \leq n, \\ -n & \text{if } g(z) < -n. \end{cases}$$

Applying the Lebesgue dominated convergence theorem and the previous corollary, we obtain

$$\begin{aligned} \int_{u(\alpha)}^{u(\beta)} g(z) dz &= \lim_{n \rightarrow \infty} \int_{u(\alpha)}^{u(\beta)} g_n(z) dz \\ &= \lim_{n \rightarrow \infty} \int_{\alpha}^{\beta} g_n(u(x)) u'(x) dx = \int_{\alpha}^{\beta} g(u(x)) u'(x) dx. \end{aligned}$$

□

Theorem 3.83. *Assume that $f : [p, q] \rightarrow \mathbb{R}$ and $u : [a, b] \rightarrow [p, q]$ are absolutely continuous. Then $f \circ u$ belongs to $AC([a, b])$ if and only if $(f' \circ u)u'$ is integrable, where $f'(u(x))u'(x)$ is interpreted to be zero whenever $u'(x) = 0$.*

Proof. If $f \circ u$ belongs to $AC([a, b])$, then by Proposition 3.9, $f \circ u$ is differentiable $\mathcal{L}^1$ -a.e. in $[a, b]$ and so by Theorem 3.59, (3.24) holds. In turn, $(f' \circ u)u'$ is integrable again by Proposition 3.9.

Conversely, assume that $(f' \circ u)u'$ is integrable. By replacing f with $f - f(p)$, without loss of generality, we may assume that $f(p) = 0$. Since f is absolutely continuous, by Theorem 3.20,

$$f(z) = \int_p^z f'(t) dt$$

for all $z \in [p, q]$. Extend f' to be zero at points where f is not differentiable and let g be the resulting function. Then we are in a position to apply the previous corollary to obtain that the change of variables formula (3.29) holds. In turn, by Theorem 3.75, $f \circ u$ belongs to $AC([a, b])$. □

To extend the previous results to arbitrary intervals, we consider two functions $g : J \rightarrow \mathbb{R}$ and $u : I \rightarrow J$ and we assume that there exist in $\mathbb{R}$ the limits

$$(3.31) \quad \lim_{x \rightarrow (\inf I)^+} u(x) = \ell, \quad \lim_{x \rightarrow (\sup I)^-} u(x) = L.$$

In this case, the analog of (3.29) becomes

$$(3.32) \quad \int_{\ell}^L g(z) dz = \int_{\inf I}^{\sup I} g(u(x)) u'(x) dx.$$

Indeed, we have the following result.

Theorem 3.84. *Let $I, J \subseteq \mathbb{R}$ be two intervals, let $g : J \rightarrow \mathbb{R}$ be an integrable function, and let $u : I \rightarrow J$ be differentiable $\mathcal{L}^1$ -a.e. in I and such that there exist in $\mathbb{R}$ the limits (3.31). Then $(g \circ u)u'$ is integrable on I and the changes*

of variables (3.29) and (3.32) hold for all $[\alpha, \beta] \subseteq I$ if and only if the function $f \circ u$ belongs to $AC(I) \cap BPV(I)$, where

$$f(z) := \int_{\inf J}^z g(t) dt, \quad z \in J.$$

Proof. Assume that $f \circ u \in AC(I) \cap BPV(I)$. Then, we can proceed as in the proof of Theorem 3.75 to show that (3.29) holds for all $[\alpha, \beta] \subseteq I$. Since $f \circ u \in BPV(I)$, by Corollary 2.31 and (3.30) its derivative $(g \circ u)u'$ is integrable on I . Hence, by taking limits as α and β approach the endpoints of I and using (3.31) and the Lebesgue dominated convergence theorem, we conclude that (3.32) holds.

Conversely, if $(g \circ u)u'$ is integrable and (3.29) and (3.32) hold for all $[\alpha, \beta] \subseteq I$, then

$$(f \circ u)(\beta) - (f \circ u)(\alpha) = \int_{\alpha}^{\beta} g(u(x))u'(x) dx$$

for all $\alpha, \beta \in I$. As in Theorem 3.75 we deduce that $f \circ u \in AC_{\text{loc}}(I)$. In turn, by Corollary 3.33, it follows that $f \circ u \in AC(I) \cap BPV(I)$. $\square$

Corollary 3.85. *Let $I, J \subseteq \mathbb{R}$ be two intervals, let $f \in AC_{\text{loc}}(J; \mathbb{R})$, and let $u : I \rightarrow J$ be $AC_{\text{loc}}(I)$. Then $f \circ u$ belongs to $AC_{\text{loc}}(I; \mathbb{R})$ if and only if $(f' \circ u)u'$ is locally integrable, where $f'(u(x))u'(x)$ is interpreted to be zero whenever $u'(x) = 0$.*

Exercise 3.86. Prove the previous corollary.

3.7. Singular Functions

In this section we prove that every function of bounded pointwise variation may be decomposed into the sum of an absolutely continuous function and a singular function.

Definition 3.87. Let $I \subseteq \mathbb{R}$ be an interval and let $(Y, \|\cdot\|)$ be a normed space. A nonconstant function $u : I \rightarrow Y$ is said to be *singular* if it is differentiable at $\mathcal{L}^1$ -a.e. $x \in I$ with $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in I$.

The jump function u_J of a function $u \in BPV_{\text{loc}}(I)$ is an example of a singular function. Another example is the Cantor function or the function given in Theorem 1.37.

The following theorem provides a characterization of singular functions.

Theorem 3.88 (Singular functions). *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, and let $u : I \rightarrow Y$ be a nonconstant function such that $u'(x)$ exists for $\mathcal{L}^1$ -a.e. $x \in I$. Then u is a singular function if and only if there exists a Lebesgue measurable set $E \subseteq I$ such that $\mathcal{L}^1(I \setminus E) = 0$ and $\mathcal{H}^1(u(E)) = 0$.*

Proof. Assume that u is singular and let $E := \{x \in I : u'(x) = 0\}$. Then $\mathcal{L}^1(I \setminus E) = 0$. By Corollary 3.37 we have that $\mathcal{H}^1(u(E)) = 0$. Conversely, assume that there exists a Lebesgue measurable set $E \subseteq I$ such that $\mathcal{L}^1(I \setminus E) = 0$ and $\mathcal{H}^1(u(E)) = 0$. Then by Lemma 3.60, $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in E$. Since $\mathcal{L}^1(I \setminus E) = 0$, we have that $u'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in I$. $\square$

As an application of Theorem 3.16 we obtain the standard decomposition of a monotone function into an absolutely continuous monotone function and a singular monotone function.

Theorem 3.89. *Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}$ be an increasing function. Then u may be decomposed as the sum of three increasing functions*

$$(3.33) \quad u = u_{AC} + u_C + u_J,$$

where $u_{AC} \in AC_{loc}(I)$, u_C is continuous and singular, and u_J is the jump function of u .

Proof. Define $v := u - u_J$. By Exercises 1.6 and 1.36 we have that v is increasing, continuous, and $v'(x) = u'(x)$ for $\mathcal{L}^1$ -a.e. $x \in I$. Fix $x_0 \in I$ and for every $x \in I$ define

$$(3.34) \quad u_{AC}(x) := \int_{x_0}^x v'(t) dt = \int_{x_0}^x u'(t) dt, \quad u_C(x) := v(x) - u_{AC}(x).$$

Then the decomposition (3.33) holds. Moreover, by Theorem 3.16, we have $u'_C(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in I$. It remains to show that u_C is increasing. Let $s, x \in I$, with $s < x$. By Corollary 1.25,

$$u_{AC}(x) - u_{AC}(s) = \int_s^x v'(t) dt \leq v(x) - v(s),$$

and so $u_C(x) \geq u_C(s)$ by (3.34)₂. $\square$

The function u_C is called the *Cantor part* of u .

Since every function with bounded pointwise variation may be written as a difference of two increasing functions, an analogous result holds for functions of bounded pointwise variation.

Corollary 3.90. *Let $I \subseteq \mathbb{R}$ be an interval and let $u \in BPV_{loc}(I; \mathbb{R}^M)$. Then u may be decomposed as the sum of three functions in $BPV_{loc}(I; \mathbb{R}^M)$, i.e.,*

$$(3.35) \quad u = u_{AC} + u_C + u_J,$$

where $u_{AC} \in AC_{loc}(I; \mathbb{R}^M)$, u_C is continuous and singular, and

$$(3.36) \quad u_J(x) := \begin{cases} \sum_{t \in I, t_0 \leq t < x} (u_+(t) - u_-(t)) + u(x) - u_-(x) & \text{if } x \geq t_0, \\ -\sum_{t \in I, x < t \leq t_0} (u_+(t) - u_-(t)) + u(x) - u_+(x), & \text{if } x \leq t_0, \end{cases}$$

where $t_0 \in I$. Moreover, for every interval $[a, b] \subseteq I$,

$$(3.37) \quad \text{Var}_{[a,b]} u = \text{Var}_{[a,b]} u_{\text{AC}} + \text{Var}_{[a,b]} u_{\text{C}} + \text{Var}_{[a,b]} u_{\text{J}},$$

where

$$(3.38) \quad \text{Var}_{[a,b]} u_{\text{AC}} = \int_a^b \|u'(x)\| dx,$$

$$(3.39) \quad \begin{aligned} \text{Var}_{[a,b]} u_{\text{J}} = & \sum_{x \in (a,b)} (\|u_+(x) - u(x)\| + \|u(x) - u_-(x)\|) \\ & + \|u(a) - u_+(a)\| + \|u(b) - u_-(b)\|. \end{aligned}$$

If, in addition, $u \in \text{BPV}(I)$, then

$$\begin{aligned} \text{Var } u &= \text{Var } u_{\text{AC}} + \text{Var } u_{\text{C}} + \text{Var } u_{\text{J}} \\ &= \int_I \|u'(x)\| dx + \text{Var } u_{\text{C}} + \sum_{x \in I} (\|u_+(x) - u(x)\| + \|u(x) - u_-(x)\|), \end{aligned}$$

where $u_-(\inf I) := u(\inf I)$ if $\inf I \in I$ and $u_+(\sup I) := u(\sup I)$ if $\sup I \in I$.

Proof. The decomposition (3.35) follows by modifying the proof of the previous theorem. We leave the details as an exercise.

By (2.17) for every interval $J \subseteq I$ we have

$$\text{Var}_J u \leq \text{Var}_J u_{\text{AC}} + \text{Var}_J u_{\text{C}} + \text{Var}_J u_{\text{J}}.$$

Thus, to prove (3.37), it remains to show

$$(3.40) \quad \text{Var}_{[a,b]} u \geq \text{Var}_{[a,b]} u_{\text{AC}} + \text{Var}_{[a,b]} u_{\text{C}} + \text{Var}_{[a,b]} u_{\text{J}}.$$

We divide the proof of (3.40) into five steps.

Step 1: Assume first that u is continuous, so that $u_{\text{J}} = 0$. We claim that for every interval $[\alpha, \beta] \subseteq I$,

$$(3.41) \quad \text{Var}_{[\alpha,\beta]} u \geq \text{Var}_{[\alpha,\beta]} u_{\text{AC}} + \|u_{\text{C}}(\beta) - u_{\text{C}}(\alpha)\|.$$

To see this, let

$$E := \{x \in (\alpha, \beta) : u'_{\text{C}}(x) = 0\}.$$

Fix $\varepsilon > 0$ and let $\delta > 0$ be as in Definition 3.1 for the absolutely continuous function u_{AC} . Using the definition of differentiability, for every $x \in E$ we may find $h_x > 0$ such that $x + h_x < \beta$ and

$$(3.42) \quad \|u_{\text{C}}(x + h) - u_{\text{C}}(x)\| \leq \varepsilon h / (\beta - \alpha)$$

for all $0 < h \leq h_x$. Let $\mathcal{F}$ be the family of all intervals $(x, x + h)$, $x \in E$, $0 < h \leq h_x$. Since, $\mathcal{L}^1(E) = \beta - \alpha$, by Lemma 1.20 there exist disjoint

intervals $(a_i, b_i) \in \mathcal{F}$, $i = 1, \dots, n$ such that

$$(3.43) \quad \mathcal{L}^1\left(\bigcup_{i=1}^n (a_i, b_i)\right) \geq \beta - \alpha - \delta.$$

By relabeling the points, if necessary, we may assume that $b_{i-1} \leq a_i$ for all $i = 2, \dots, n$. Set $b_0 := \alpha$, $a_{n+1} := \beta$ and consider the partition

$$P = \{b_0, a_1, b_1, \dots, a_n, b_n, a_{n+1}\}.$$

By Proposition 2.15,

$$(3.44) \quad \text{Var}_{[\alpha, \beta]} u = \sum_{i=1}^n \text{Var}_{[a_i, b_i]} u + \sum_{i=1}^{n+1} \text{Var}_{[b_{i-1}, a_i]} u.$$

By Corollary 2.31 and the fact that $u'_{\text{AC}} = u'$ $\mathcal{L}^1$ -a.e. in I ,

$$(3.45) \quad \begin{aligned} \sum_{i=1}^n \text{Var}_{[a_i, b_i]} u &\geq \sum_{i=1}^n \int_{a_i}^{b_i} \|u'\| dx \\ &= \sum_{i=1}^n \int_{a_i}^{b_i} \|u'_{\text{AC}}\| dx = \sum_{i=1}^n \text{Var}_{[a_i, b_i]} u_{\text{AC}}. \end{aligned}$$

Using (2.17), we obtain that

$$\begin{aligned} \text{Var}_{[b_{i-1}, a_i]} u &\geq \text{Var}_{[b_{i-1}, a_i]} u_{\text{C}} - \text{Var}_{[b_{i-1}, a_i]} u_{\text{AC}} \\ &\geq \|u_{\text{C}}(a_i) - u_{\text{C}}(b_{i-1})\| - \text{Var}_{[b_{i-1}, a_i]} u_{\text{AC}}, \end{aligned}$$

which, together with (3.44) and (3.45), yields

$$(3.46) \quad \begin{aligned} \text{Var}_{[\alpha, \beta]} u &\geq \sum_{i=1}^n \text{Var}_{[a_i, b_i]} u_{\text{AC}} + \sum_{i=1}^{n+1} \|u_{\text{C}}(a_i) - u_{\text{C}}(b_{i-1})\| \\ &\quad - \sum_{i=1}^{n+1} \text{Var}_{[b_{i-1}, a_i]} u_{\text{AC}}. \end{aligned}$$

By (3.43) we have

$$\sum_{i=1}^{n+1} (a_i - b_{i-1}) = \beta - \alpha - \sum_{i=1}^n (b_i - a_i) \leq \delta.$$

Hence (see the proof of Theorem 3.29),

$$(3.47) \quad \sum_{i=1}^{n+1} \text{Var}_{[b_{i-1}, a_i]} u_{\text{AC}} \leq \varepsilon.$$

On the other hand, by (3.42),

$$\begin{aligned}
 \sum_{i=1}^{n+1} \|u_C(a_i) - u_C(b_{i-1})\| &\geq \|u_C(\beta) - u_C(\alpha)\| - \sum_{i=1}^n \|u_C(b_i) - u_C(a_i)\| \\
 (3.48) \qquad \qquad \qquad &\geq \|u_C(\beta) - u_C(\alpha)\| - \frac{\varepsilon}{\beta - \alpha} \sum_{i=1}^n (b_i - a_i) \\
 &\geq \|u_C(\beta) - u_C(\alpha)\| - \varepsilon.
 \end{aligned}$$

Combining (3.46), (3.47), and (3.48) and using Proposition 2.15 for u_{AC} , we obtain

$$\text{Var}_{[\alpha, \beta]} u \geq \text{Var}_{[\alpha, \beta]} u_{AC} + \|u_C(\beta) - u_C(\alpha)\| - 3\varepsilon.$$

By letting $\varepsilon \rightarrow 0^+$, we obtain (3.41).

Step 2: Assume that u is continuous. Fix an interval $[a, b] \subseteq I$ and consider a partition P of $[a, b]$, with $a = x_0 < \cdots < x_m = b$. Applying (3.41) in each interval $[x_{i-1}, x_i]$ and using Proposition 2.15 yields

$$\begin{aligned}
 \text{Var}_{[a, b]} u &= \sum_{i=1}^m \text{Var}_{[x_{i-1}, x_i]} u \\
 &\geq \sum_{i=1}^m \text{Var}_{[x_{i-1}, x_i]} u_{AC} + \sum_{i=1}^m \|u_C(x_i) - u_C(x_{i-1})\| \\
 &= \text{Var}_{[a, b]} u_{AC} + \sum_{i=1}^m \|u_C(x_i) - u_C(x_{i-1})\|.
 \end{aligned}$$

Taking the supremum over all partitions of $[a, b]$ gives

$$\text{Var}_{[a, b]} u \geq \text{Var}_{[a, b]} u_{AC} + \text{Var}_{[a, b]} u_C.$$

Thus, we have proved that (3.40) holds if u is continuous.

Step 3: Assume that u has a finite number of discontinuity points in (a, b) , say, $a < x_1 < \cdots < x_\ell < b$. Let $x_0 := a$, $x_{\ell+1} := b$ and fix $\delta > 0$ so small that

$$0 < \delta < \min\{x_i - x_{i-1} : i = 1, \dots, \ell + 1\}.$$

Using Proposition 2.15, we have

$$\begin{aligned}
\text{Var}_{[a,b]} u &= \sum_{i=1}^{\ell+1} \text{Var}_{[x_{i-1}+\delta, x_i-\delta]} u + \sum_{i=0}^{\ell+1} \text{Var}_{[x_i-\delta, x_i+\delta] \cap [a,b]} u \\
&\geq \sum_{i=1}^{\ell+1} \text{Var}_{[x_{i-1}+\delta, x_i-\delta]} u \\
&\quad + \sum_{i=1}^{\ell} (\|u(x_i + \delta) - u(x_i)\| + \|u(x_i) - u(x_i - \delta)\|) \\
&\quad + \|u(a + \delta) - u(a)\| + \|u(b) - u(b - \delta)\| \\
&\geq \sum_{i=1}^{\ell+1} (\text{Var}_{[x_{i-1}+\delta, x_i-\delta]} u_{\text{AC}} + \text{Var}_{[x_{i-1}+\delta, x_i-\delta]} u_{\text{C}}) \\
&\quad + \sum_{i=1}^{\ell} (\|u(x_i + \delta) - u(x_i)\| + \|u(x_i) - u(x_i - \delta)\|) \\
&\quad + \|u(a + \delta) - u(a)\| + \|u(b) - u(b - \delta)\|,
\end{aligned}$$

where we have used the previous step together with the fact that in each interval $[x_{i-1}+\delta, x_i-\delta]$ the jump function u_{J} is constant, and so the variation of u in those intervals reduces to the one of $u_{\text{C}} + u_{\text{AC}}$. Letting $\delta \rightarrow 0^+$ in the previous inequality and using Proposition 2.15 and Exercise 2.22 yields

$$\begin{aligned}
\text{Var}_{[a,b]} u &\geq \sum_{i=1}^{\ell+1} (\text{Var}_{[x_{i-1}, x_i]} u_{\text{AC}} + \text{Var}_{[x_{i-1}, x_i]} u_{\text{C}}) \\
&\quad + \sum_{i=1}^{\ell} (\|u_+(x_i) - u(x_i)\| + \|u(x_i) - u_-(x_i)\|) \\
&\quad + \|u_+(a) - u(a)\| + \|u(b) - u_-(b)\| \\
&= \text{Var}_{[a,b]} u_{\text{AC}} + \text{Var}_{[a,b]} u_{\text{C}} + \text{Var}_{[a,b]} u_{\text{J}}.
\end{aligned}$$

Step 4: Finally, if u has an infinite number of discontinuity points in (a, b) , say $\{x_i\}_i$, consider the saltus function $u_{\text{J},k}$ corresponding to the points x_i with $i < k$. For each $k \in \mathbb{N}$ and $x \in [a, b]$ define

$$u_k(x) := u_{\text{AC}}(x) + u_{\text{C}}(x) + u_{\text{J},k}.$$

Since the discontinuity points of u_k in (a, b) are $\{x_1, \dots, x_k\}$, by the previous step

$$\begin{aligned} \text{Var}_{[a,b]} u_k &= \text{Var}_{[a,b]} u_{AC} + \text{Var}_{[a,b]} u_C \\ &\quad + \sum_{i=1}^k (\|u_+(x_i) - u(x_i)\| + \|u(x_i) - u_-(x_i)\|) \\ &\quad + \|u_+(a) - u(a)\| + \|u(b) - u_-(b)\|. \end{aligned}$$

Since

$$\begin{aligned} \text{Var}_{[a,b]} u_k &= \text{Var}_{[a,b]}(u_k - u + u) \leq \text{Var}_{[a,b]}(u_k - u) + \text{Var}_{[a,b]} u \\ &\leq \sum_{i=k}^{\infty} (\|u_+(x_i) - u(x_i)\| + \|u(x_i) - u_-(x_i)\|) + \text{Var}_{[a,b]} u, \end{aligned}$$

by the previous equality we have

$$\begin{aligned} \text{Var}_{[a,b]} u_{AC} + \text{Var}_{[a,b]} u_C &+ \sum_{i=1}^k (\|u_+(x_i) - u(x_i)\| + \|u(x_i) - u_-(x_i)\|) \\ &+ \|u_+(a) - u(a)\| + \|u(b) - u_-(b)\| \\ &\leq \sum_{i=k}^{\infty} (\|u_+(x_i) - u(x_i)\| + \|u(x_i) - u_-(x_i)\|) + \text{Var}_{[a,b]} u. \end{aligned}$$

Letting $k \rightarrow \infty$, we obtain (3.40).

Step 5: If, in addition, $u \in BPV(I; \mathbb{R}^M)$, taking a_n and b_n as in Step 2 of the proof of Theorem 3.25, we apply (3.37) in $[a_n, b_n]$ and use Proposition 2.15 and the Lebesgue monotone convergence theorem. $\square$

Reasoning as in Steps 3–5 of Corollary 3.90 we have the following result.

Corollary 3.91. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space and let $u : I \rightarrow Y$ be a jump function. Then for every $[a, b] \subseteq I$,*

$$\begin{aligned} \text{Var}_{[a,b]} u &= \sum_{x \in (a,b)} (\|u_+(x) - u(x)\| + \|u(x) - u_-(x)\|) \\ &\quad + \|u(a) - u_+(a)\| + \|u(b) - u_-(b)\|. \end{aligned}$$

Remark 3.92. We refer to the paper [41] of Cater for necessary and sufficient conditions on two continuous functions $u : [a, b] \rightarrow \mathbb{R}$ and $v : [a, b] \rightarrow \mathbb{R}$ with finite pointwise variation for the equality $\text{Var}_{[a,b]}(u + v) = \text{Var}_{[a,b]} u + \text{Var}_{[a,b]} v$ to hold.

Decreasing Rearrangement

Undergradese, IV: “Can I get an extension?” Translation: “Can you re-arrange your life around mine?”

— Jorge Cham, www.phdcomics.com

In this chapter we introduce the notion of decreasing rearrangement u^* of a Lebesgue measurable function $u : E \rightarrow [0, \infty]$, where $E \subseteq \mathbb{R}$ is a Lebesgue measurable set. Roughly speaking, u^* is a function defined on the interval $[0, \mathcal{L}^1(E))$, which is obtained from u by rearranging the superlevel sets $\{x \in E : u(x) > s\}$ in such a way that u^* becomes a decreasing function. For more information on this topics we refer to the monographs of Kawohl [130] and Kesavan [131]. For simplicity, in this chapter we deal only with nonnegative functions, although the theory can also be carried out for real-valued functions.

Except for the last section, most of the results of this chapter continue to hold if we consider a measure space $(X, \mathfrak{M}, \mu)$ and replace the Lebesgue measurable set $E \subseteq \mathbb{R}$ and the Lebesgue measure $\mathcal{L}^1$ by a set $E \in \mathfrak{M}$ of X and by μ , respectively. We leave the details as an exercise.

4.1. Definition and First Properties

Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty]$ be a Lebesgue measurable function. The *distribution function* of u is the function $\varrho_u : [0, \infty) \rightarrow [0, \mathcal{L}^1(E)]$, defined by

$$(4.1) \quad \varrho_u(s) := \mathcal{L}^1(\{x \in E : u(x) > s\}), \quad s \geq 0.$$

The function u is said to *vanish at infinity* if it is Lebesgue measurable and $\varrho_u(s) < \infty$ for every $s > 0$. Note that for a function u vanishing at infinity

the value $\varrho_u(0)$ could be infinite. For example, if E has infinite measure and u is everywhere positive, then $\varrho_u(0) = \infty$. Moreover,

$$(4.2) \quad \varrho_u(s) = 0 \quad \text{for all } s \geq \operatorname{ess\,sup}_E u,$$

$$(4.3) \quad \varrho_u(s) = \mathcal{L}^1(E) \quad \text{for all } s < \operatorname{ess\,inf}_E u,$$

where we recall that the *essential supremum* and *essential infimum* of u are defined as

$$(4.4) \quad \operatorname{ess\,sup}_E u := \inf\{s : u(x) \leq s \text{ for } \mathcal{L}^1\text{-a.e. } x \in E\},$$

$$(4.5) \quad \operatorname{ess\,inf}_E u := \sup\{s : s \leq u(x) \text{ for } \mathcal{L}^1\text{-a.e. } x \in E\}.$$

Note that if E has infinite measure and $u : E \rightarrow [0, \infty]$ vanishes at infinity, then $\operatorname{ess\,inf}_E u = 0$.

Some important properties of ϱ_u are summarized in the next proposition.

Proposition 4.1. *Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u, v, u_n : E \rightarrow [0, \infty]$, $n \in \mathbb{N}$, be Lebesgue measurable functions. Then the following properties hold:*

- (i) *The function $\varrho_u : [0, \infty) \rightarrow [0, \mathcal{L}^1(E)]$ is decreasing and right continuous.*
- (ii) *If u vanishes at infinity, then*

$$\lim_{s \rightarrow \infty} \varrho_u(s) = \mathcal{L}^1(\{x \in E : u(x) = \infty\})$$

and ϱ_u is continuous at $s > 0$ if and only if

$$\mathcal{L}^1(\{x \in E : u(x) = s\}) = 0.$$

- (iii) *If $u(x) \leq v(x)$ for $\mathcal{L}^1$ -a.e. $x \in E$, then $\varrho_u \leq \varrho_v$. In particular, if $u(x) = v(x)$ for $\mathcal{L}^1$ -a.e. $x \in E$, then $\varrho_u = \varrho_v$.*
- (iv) *If $u_n(x) \nearrow u(x)$ for $\mathcal{L}^1$ -a.e. $x \in E$, then $\varrho_{u_n} \nearrow \varrho_u$.*

Proof. In what follows, for $s \geq 0$ we set

$$(4.6) \quad E_s := \{x \in E : u(x) > s\}.$$

- (i) If $0 \leq s \leq r$, then $E_r \subseteq E_s$, and so $\varrho_u(r) \leq \varrho_u(s)$, which shows that ϱ_u is decreasing.

Since ϱ_u is decreasing, to prove that ϱ_u is right continuous at $s \geq 0$ it is enough to consider a decreasing sequence $s_n \rightarrow s^+$. Then $E_{s_n} \subseteq E_{s_{n+1}}$ and $\bigcup_{n=1}^{\infty} E_{s_n} = E_s$, and so by Proposition B.9(i) we have that

$$(\varrho_u)_+(s) = \lim_{n \rightarrow \infty} \varrho_u(s_n) = \lim_{n \rightarrow \infty} \mathcal{L}^1(E_{s_n}) = \mathcal{L}^1\left(\bigcup_{n=1}^{\infty} E_{s_n}\right) = \varrho_u(s).$$

(ii) Assume that u vanishes at infinity. Since ϱ_u is decreasing, there exists

$$\lim_{s \rightarrow \infty} \varrho_u(s) = \inf_{s > 0} \varrho_u(s),$$

and thus, to evaluate it, it suffices to consider the sequence $s = n \rightarrow \infty$. Then $E_n \supseteq E_{n+1}$ and

$$\bigcap_{n=1}^{\infty} E_n = \{x \in E : u(x) = \infty\} =: E_{\infty}.$$

Since $\mathcal{L}^1(E_n) < \infty$, it follows by Proposition B.9(ii) that

$$\lim_{n \rightarrow \infty} \varrho_u(s_n) = \lim_{n \rightarrow \infty} \mathcal{L}^1(E_n) = \mathcal{L}^1(E_{\infty}).$$

Similarly, if $s > 0$, consider an increasing sequence $\{s_n\}_n$ such that $s_n \rightarrow s^-$. Then $E_{s_n} \supseteq E_{s_{n+1}}$ and

$$\bigcap_{n=1}^{\infty} E_{s_n} = \{x \in E : u(x) \geq s\}.$$

Since $\mathcal{L}^1(E_{s_n}) < \infty$, we conclude as before that

$$(4.7) \quad \begin{aligned} (\varrho_u)_-(s) &= \lim_{n \rightarrow \infty} \varrho_u(s_n) = \mathcal{L}^1(\{x \in E : u(x) \geq s\}) \\ &= \varrho_u(s) + \mathcal{L}^1(\{x \in E : u(x) = s\}). \end{aligned}$$

Hence, ϱ_u is continuous at s if and only if $\mathcal{L}^1(\{x \in E : u(x) = s\}) = 0$.

(iii) For every $s \geq 0$,

$$\{x \in E : u(x) > s\} \subseteq \{x \in E : v(x) > s\} \cup F,$$

where $\mathcal{L}^1(F) = 0$, and so $\varrho_u(s) \leq \varrho_v(s)$.

(iv) In view of part (iii), by modifying each u_n on a set of measure zero, we can assume that $u_n(x) \nearrow u(x)$ for all $x \in E$. For every $s \geq 0$ and $n \in \mathbb{N}$ set $E_{s,n} := \{x \in E : u_n(x) > s\}$. Since $u_n \leq u_{n+1}$, we have that $E_{s,n} \subseteq E_{s,n+1}$ for all $n \in \mathbb{N}$, and since $u_n(x) \rightarrow u(x)$ for all $x \in E$, it follows that $\bigcup_{n=1}^{\infty} E_{s,n} = E_s$. Hence, again by Proposition B.9(i),

$$(4.8) \quad \lim_{n \rightarrow \infty} \varrho_{u_n}(s) = \lim_{n \rightarrow \infty} \mathcal{L}^1(E_{s,n}) = \mathcal{L}^1(E_s) = \varrho_u(s).$$

□

Exercise 4.2. Give an example of a function u not vanishing at infinity for which ϱ_u is not left continuous.

Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty]$ be a Lebesgue measurable function. Since by the previous proposition the distribution function $\varrho_u : [0, \infty) \rightarrow [0, \infty]$ is decreasing, it admits a left

inverse $u^* : [0, \infty) \rightarrow [0, \infty]$ (see Theorem 1.8 and Remark 1.11), called the *decreasing rearrangement* of u . Precisely, for $t \geq 0$ we set

$$(4.9) \quad u^*(t) := \inf\{s \in [0, \infty) : \varrho_u(s) \leq t\}.$$

Note that if u vanishes at infinity, then by the previous proposition $\varrho_u(s) \rightarrow t_\infty$ as $s \rightarrow \infty$, where $t_\infty := \mathcal{L}^1(\{x \in E : u(x) = \infty\})$, and so for $t > t_\infty$ the set

$$\{s \in [0, \infty) : \varrho_u(s) \leq t\}$$

is nonempty. Thus, $u^*(t) < \infty$ for all $t > t_\infty$.

Observe that if $\mathcal{L}^1(E) < \infty$, then it follows from the definition of u^* that

$$(4.10) \quad u^*(t) = 0 \quad \text{for all } t \geq \mathcal{L}^1(E).$$

In what follows, for every Lebesgue measurable set $F \subseteq \mathbb{R}$ we define

$$(4.11) \quad F^* := [0, \mathcal{L}^1(F)).$$

In particular, if $\mathcal{L}^1(F) = \infty$, then $F^* = [0, \infty)$.

In view of Proposition 4.1 we have the following result.

Proposition 4.3. *Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty]$ be a Lebesgue measurable function. Then the following properties hold:*

- (i) *The function $u^* : [0, \infty) \rightarrow [0, \infty]$ is decreasing and right continuous,*

$$u^*(0) = \operatorname{ess\,sup}_E u,$$

and

$$u^*(t) \geq \operatorname{ess\,inf}_E u \quad \text{for all } t < \mathcal{L}^1(E).$$

- (ii) *Assume that u vanishes at infinity and let $t_\infty := \mathcal{L}^1(\{x \in E : u(x) = \infty\})$. Then u^* jumps at some point $t_0 \in (t_\infty, \mathcal{L}^1(E))$ if and only if $\varrho_u(s) \equiv t_0$ for all s in some interval $(s_1, s_2) \subset [0, \infty)$, with $s_1 < s_2$. Moreover,*

$$u^*(\varrho_u(s)) \leq s \quad \text{for every } s > 0,$$

with strict inequality holding if and only if ϱ_u is constant in some interval $[s', s] \subset [0, \infty)$, with $s' < s$, while $u^(t) \equiv s_0$ for all t in some interval $(t_1, t_2) \subseteq J$, with $t_1 < t_2$, and for some $s_0 > 0$ if and only if*

$$\mathcal{L}^1(\{x \in E : u(x) = s_0\}) > 0$$

and $(t_1, t_2) \subseteq (\varrho_u(s_0), (\varrho_u)_-(s_0))$.

- (iii) *For all $s, t \geq 0$, $u^*(t) > s$ if and only if $\varrho_u(s) > t$.*

(iv) The functions u and u^* are equi-measurable, that is, for all $s \geq 0$,

$$\mathcal{L}^1(\{t \in E^* : u^*(t) > s\}) = \mathcal{L}^1(\{x \in E : u(x) > s\}).$$

In particular, if u vanishes at infinity, then so does u^* .

(v) $\mathcal{L}^1(\{t \in E^* : u^*(t) = 0\}) \leq \mathcal{L}^1(\{x \in E : u(x) = 0\})$ with equality holding if and only if either

$$\mathcal{L}^1(\{x \in E : u(x) > 0\}) < \infty$$

or

$$\mathcal{L}^1(\{x \in E : u(x) > 0\}) = \infty \text{ and } \mathcal{L}^1(\{x \in E : u(x) = 0\}) = 0.$$

If $\mathcal{L}^1(\{x \in E : u(x) > 0\}) = \infty$, then $\{t \in E^* : u^*(t) = 0\}$ is empty.

Proof. (i) If $0 \leq t_1 \leq t_2$, then

$$\{s \in [0, \infty) : \varrho_u(s) \leq t_1\} \subseteq \{s \in [0, \infty) : \varrho_u(s) \leq t_2\},$$

and so

$$\begin{aligned} u^*(t_1) &= \inf\{s \in [0, \infty) : \varrho_u(s) \leq t_1\} \\ &\geq \inf\{s \in [0, \infty) : \varrho_u(s) \leq t_2\} = u^*(t_2). \end{aligned}$$

Thus, u^* is decreasing.

To prove that u^* is right continuous, fix $t_0 \geq 0$. If $u^*(t_0) = 0$, then, since u^* is decreasing, we have that $u^*(t) = 0$ for all $t \geq t_0$ and there is nothing to prove. Thus, assume that $u^*(t_0) > 0$ and fix $0 < s_0 < u^*(t_0)$. It follows from (4.9) that $\varrho_u(s_0) > t_0$. Hence, we can find $\delta > 0$ such that $\varrho_u(s_0) > t_0 + \delta$. Using (4.9) once more, we have that $u^*(t) \geq s_0$ for all $t \in [t_0, t_0 + \delta]$, and so

$$s_0 \leq u_+^*(t_0) \leq u^*(t_0).$$

Letting $s_0 \rightarrow u^*(t_0)$ proves that u^* is right continuous. Note that if ϱ_u were finite, then we could have applied Remark 1.11.

Next we claim that

$$(4.12) \quad u^*(0) = u_+^*(0) = \operatorname{ess\,sup}_E u.$$

We begin by showing that

$$(4.13) \quad 0 \leq u^*(t) \leq \operatorname{ess\,sup}_E u \quad \text{for all } t \geq 0.$$

If $\operatorname{ess\,sup}_E u = \infty$, then there is nothing to prove. Thus, assume that $\operatorname{ess\,sup}_E u < \infty$. Then $\varrho_u(\operatorname{ess\,sup}_E u) = 0$ by (4.2). Hence, (4.13) holds by (4.9).

It follows from (4.13) that if $\operatorname{ess\,sup}_E u = 0$, then $u^* = 0$, and so (4.12) follows. Thus, assume that $\operatorname{ess\,sup}_E u > 0$ and fix any $0 < s_0 < \operatorname{ess\,sup}_E u$. By the definition of essential supremum (see (4.4)) there exists a Lebesgue

measurable subset $F \subseteq E$, with $t_0 := \mathcal{L}^1(F) > 0$, such that $u(x) > s_0$ for all $x \in F$. Hence $\varrho_u(s_0) \geq t_0$. In turn, by (4.9), for all $0 \leq t < t_0$ we have that $u^*(t) \geq s_0$, which shows that

$$s_0 \leq u_+^*(0) \leq u^*(0) \leq \operatorname{ess\,sup}_E u.$$

Letting $s_0 \rightarrow \operatorname{ess\,sup}_E u$, we conclude that (4.12) holds.

On the other hand, if $\operatorname{ess\,inf}_E u > 0$, then $\varrho_u(s) = \mathcal{L}^1(E)$ for all $s < \operatorname{ess\,inf}_E u$ by (4.3), and so if $t < \mathcal{L}^1(E)$, then $u^*(t) \geq \operatorname{ess\,inf}_E u$ by (4.9).

(ii) The first two statements follow from Remark 1.11 and the last from Remark 1.11 and Proposition 4.1.

(iii) Assume that $u^*(t) > s$ for some $s, t \geq 0$. Then $\varrho_u(s) > t$ by (4.9).

Conversely, assume that $\varrho_u(s) > t$. Since ϱ_u is right continuous, we have that $\varrho_u(r) > t$ for all $r \in [s, s + \delta]$ for some $\delta > 0$. Hence, $u^*(t) > s$ by (4.9).

(iv) By part (iii) for $s \geq 0$, we have that $u^*(t) > s$ for $t \geq 0$ if and only if $\varrho_u(s) > t$, and so

$$\{t \geq 0 : u^*(t) > s\} = \{t \geq 0 : \varrho_u(s) > t\} = [0, \varrho_u(s)).$$

Hence,

$$\mathcal{L}^1(\{t \geq 0 : u^*(t) > s\}) = \varrho_u(s) = \mathcal{L}^1(\{x \in E : u(x) > s\})$$

by the definition of ϱ_u . The result now follows from (4.10) and (4.11).

(v) If $\mathcal{L}^1(\{x \in E : u(x) > 0\}) < \infty$, then by part (iv) and (4.11),

$$\begin{aligned} \mathcal{L}^1(\{x \in E : u(x) = 0\}) &= \mathcal{L}^1(E) - \mathcal{L}^1(\{x \in E : u(x) > 0\}) \\ &= \mathcal{L}^1(E^*) - \mathcal{L}^1(\{s \in E^* : u^*(s) > 0\}) \\ &= \mathcal{L}^1(\{s \in E^* : u^*(s) = 0\}). \end{aligned}$$

On the other hand, if $\mathcal{L}^1(\{x \in E : u(x) > 0\}) = \infty$, then by part (iv),

$$\mathcal{L}^1(\{t \in E^* : u^*(t) > 0\}) = \mathcal{L}^1(\{x \in E : u(x) > 0\}) = \infty.$$

Since u^* is decreasing, it follows that $u^*(t) > 0$ for all $t \geq 0$. Hence, the set $\{t \in E^* : u^*(t) = 0\}$ is empty. $\square$

Remark 4.4. Note that parts (iv) and (v) of the previous proposition imply, in particular, that (4.10) can be strengthened as follows. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty]$ be a Lebesgue measurable function. Recall that

$$\varrho_u(0) = \mathcal{L}^1(\{x \in E : u(x) > 0\}).$$

If $\varrho_u(0) = \infty$, then $u^*(t) > 0$ for all $t \geq 0$, while if $\varrho_u(0) < \infty$, then $u^*(t) > 0$ for all $t \in [0, \varrho_u(0))$ and $u^*(t) = 0$ for all $t \geq \varrho_u(0)$.

Exercise 4.5. What happens to part (ii) of the previous proposition if we remove the assumption that u vanishes at infinity?

Corollary 4.6. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty)$ be a function vanishing at infinity. Then

$$(4.14) \quad \mathcal{L}^1(\{x \in E : u(x) \in B\}) = \mathcal{L}^1(\{t \in E^* : u^*(t) \in B\})$$

for all Borel sets $B \subseteq (0, \infty)$. Moreover, if either $\mathcal{L}^1(\{x \in E : u(x) > 0\}) < \infty$ or

$$\mathcal{L}^1(\{x \in E : u(x) > 0\}) = \infty \text{ and } \mathcal{L}^1(\{x \in E : u(x) = 0\}) = 0,$$

then (4.14) holds for all Borel sets $B \subseteq [0, \infty)$.

Proof. Define the measures

$$\begin{aligned} \mu(B) &:= \mathcal{L}^1(\{x \in E : u(x) \in B\}), \\ \nu(B) &:= \mathcal{L}^1(\{t \in E^* : u^*(t) \in B\}), \end{aligned}$$

where B belongs to the Borel σ -algebra $\mathcal{B}((0, \infty))$. Since

$$\begin{aligned} \{x \in E : u(x) \in (0, \infty)\} &= \bigcup_{n=1}^{\infty} \{x \in E : u(x) > \tfrac{1}{n}\}, \\ \{t \in E^* : u^*(t) \in (0, \infty)\} &= \bigcup_{n=1}^{\infty} \{t \in E^* : u^*(t) > \tfrac{1}{n}\}, \end{aligned}$$

and both u and u^* vanish at infinity (see Proposition 4.3(iv)), it follows that μ and ν are σ -finite. By Proposition 4.3(iv), μ and ν coincide on all intervals (a, ∞) , where $a > 0$. Since the family of these intervals generates the Borel σ -algebra $\mathcal{B}((0, \infty))$, it follows by Corollary B.16 and Remark B.17 that μ and ν coincide on $\mathcal{B}((0, \infty))$.

The last part of the statement follows from Proposition 4.3(v). $\square$

Corollary 4.7. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u \in L^\infty(E)$ be nonnegative. Then u^* belongs to $L^\infty(E^*)$ and

$$\|u^*\|_{L^\infty(E^*)} = \|u\|_{L^\infty(E)}.$$

Proof. By Proposition 4.3 we have that

$$\|u^*\|_{L^\infty(E^*)} = u^*(0) = \|u\|_{L^\infty(E)} < \infty.$$

$\square$

Exercise 4.8. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set, let $u : E \rightarrow [0, \infty)$ be a function vanishing at infinity, and let $\Psi : [0, \infty) \rightarrow [0, \infty)$ be an increasing function. Prove that $(\Psi \circ u)^*(t) = (\Psi \circ u^*)(t)$ for all but countably many $t \in E^*$.

Exercise 4.9. Let $a > 0$ and $b \in \mathbb{R}$ and consider the function $u : \mathbb{R} \rightarrow \mathbb{R}$, defined by

$$u(x) := \begin{cases} \frac{1}{(b-x+a)^2} & \text{if } x \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

Find u^* .

Corollary 4.10. Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow [0, \infty)$ be a continuous function vanishing at infinity. Then u^* is continuous in I^* .

Proof. By part (i) of Proposition 4.3, u^* jumps at some point $t_0 \in (0, \mathcal{L}^1(I))$ if and only if $\varrho_u(s) \equiv t_0$ for all s in some interval $(s_1, s_2) \subset [0, \infty)$, with $s_1 < s_2$. In view of (4.2) we have that

$$\text{ess inf}_I u \leq s_1 < s_2 \leq \text{ess sup}_I u.$$

By replacing the interval (s_1, s_2) with a smaller one, we can assume that $\varrho_u(s_1) = \varrho_u(s_2) = t_0$. Hence,

$$\mathcal{L}^1(\{y \in I : s_1 < u(y) \leq s_2\}) = 0.$$

On the other hand, since u is continuous, we have that $u(I)$ is an interval containing $(\text{ess inf}_I u, \text{ess sup}_I u)$. In particular, $u(I)$ contains (s_1, s_2) , which is a contradiction. $\square$

The next proposition shows that decreasing rearrangement preserves inequalities between functions and behaves well with respect to monotone convergence.

Proposition 4.11. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u, v, u_n : E \rightarrow [0, \infty]$, $n \in \mathbb{N}$, be Lebesgue measurable functions.

- (i) If $u(x) \leq v(x)$ for $\mathcal{L}^1$ -a.e. $x \in E$, then $u^* \leq v^*$. In particular, if $u(x) = v(x)$ for $\mathcal{L}^1$ -a.e. $x \in E$, then $u^* = v^*$.
- (ii) If $u_n(x) \nearrow u(x)$ for $\mathcal{L}^1$ -a.e. $x \in E$, then $u_n^* \nearrow u^*$.

Proof. (i) By Proposition 4.1(iii), for every $t \geq 0$,

$$\{s \in [0, \infty) : \varrho_v(s) \leq t\} \subseteq \{s \in [0, \infty) : \varrho_u(s) \leq t\}$$

and hence $u^*(t) \leq v^*(t)$.

(ii) By part (i) we have that $u_n^* \leq u_{n+1}^* \leq u^*$ for all $n \in \mathbb{N}$. Hence,

$$(4.15) \quad \lim_{n \rightarrow \infty} u_n^*(t) \leq u^*(t)$$

for all $t \geq 0$. Suppose that $u^*(t) > 0$ for some $t \geq 0$ and fix $s \in [0, u^*(t))$. By Proposition 4.3(iii), $\varrho_u(s) > t$. Hence, by Proposition 4.1(iv) there exists $k \in \mathbb{N}$ such that $\varrho_{u_k}(s) > t$. Again by Proposition 4.3(iii) we have that $u_k^*(t) > s$ and since $u_n^* \leq u_{n+1}^* \leq u^*$ for all $n \in \mathbb{N}$, it follows that

$$\lim_{n \rightarrow \infty} u_n^*(t) > s.$$

Letting $s \nearrow u^*(t)$, we conclude that $\lim_{n \rightarrow \infty} u_n^*(t) \geq u^*(t)$, which, together with (4.15), implies that

$$\lim_{n \rightarrow \infty} u_n^*(t) = u^*(t)$$

for all $t \geq 0$. □

Remark 4.12. It follows from part (i) of Proposition 4.11 that modifying a nonnegative Lebesgue measurable function on a set of measure zero does not change its decreasing rearrangement. In what follows, we will use this fact without further mention. For the same reason, we can replace the set E with a measurable subset $F \subseteq E$ such that $\mathcal{L}^1(E \setminus F) = 0$.

The next exercise shows that the decreasing rearrangement of a simple function is still a simple function.

Exercise 4.13. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty)$ be a simple function vanishing at infinity (see Figure 1), that is,

$$u = \sum_{i=0}^k c_i \chi_{E_i},$$

where $c_0 > \dots > c_k > 0$, $E_i \subseteq E$ are pairwise disjoint Lebesgue measurable sets with $0 < \mathcal{L}^1(E_i) < \infty$, $i = 1, \dots, k$. For every $i = 1, \dots, k$ set

$$F_i := \bigcup_{j=0}^i E_j.$$

(i) Let $r_i := \mathcal{L}^1(F_i)$, $i = 1, \dots, k$. Prove that

$$\varrho_u = \sum_{i=0}^k \left[\sum_{j=0}^i \mathcal{L}^1(E_j) \right] \chi_{[c_{i+1}, c_i)},$$

where $c_{k+1} := 0$, and that

$$u^* = \sum_{i=0}^k c_i \chi_{[r_{i-1}, r_i)},$$

where $r_{-1} := 0$.

(ii) Let $\{p_0, \dots, p_k\}, \{q_0, \dots, q_k\} \subset \mathbb{R}$ and prove that

$$\sum_{i=0}^k p_i q_i = \sum_{i=0}^k P_i Q_i,$$

where $P_i := p_i - p_{i+1}$ for $i = 0, \dots, k-1$, $P_k := p_k$, and $Q_i = \sum_{j=0}^i q_j$ for $i = 0, \dots, k$.

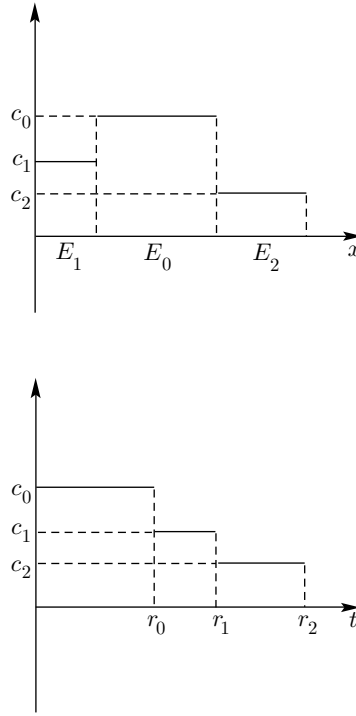


Figure 1. The graphs of u and u^* .

(iii) Prove that

$$(4.16) \quad u = \sum_{i=0}^k \alpha_i \chi_{F_i},$$

where $\alpha_i := c_i - c_{i+1}$, $i = 1, \dots, k$, and that

$$u^* = \sum_{i=0}^k \alpha_i \chi_{F_i^*}.$$

Using the previous exercise, it is possible to give a simple proof of a classical result of Hardy and Littlewood.

Theorem 4.14 (Hardy–Littlewood’s inequality). *Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u, v : E \rightarrow [0, \infty)$ be Lebesgue measurable functions. Then*

$$\int_E u(x)v(x) dx \leq \int_{E^*} u^*(t)v^*(t) dt.$$

Proof. Step 1: Assume first that $u, v : E \rightarrow [0, \infty)$ are simple functions vanishing at infinity and, using the notation (4.16), write

$$u = \sum_{i=0}^k \alpha_i \chi_{F_i}, \quad v = \sum_{j=0}^l \beta_j \chi_{G_j},$$

so that by part (iii) of the previous exercise,

$$u^* = \sum_{i=0}^k \alpha_i \chi_{F_i^*}, \quad v^* = \sum_{j=0}^l \beta_j \chi_{G_j^*}.$$

Then

$$\begin{aligned} \int_E uv \, dx &= \sum_{i=0}^k \sum_{j=0}^l \alpha_i \beta_j \int_E \chi_{F_i} \chi_{G_j} \, dx \\ &= \sum_{i=0}^k \sum_{j=0}^l \alpha_i \beta_j \mathcal{L}^1(F_i \cap G_j) \leq \sum_{i=0}^k \sum_{j=0}^l \alpha_i \beta_j \min\{\mathcal{L}^1(F_i), \mathcal{L}^1(G_j)\}, \end{aligned}$$

while

$$\begin{aligned} \int_{E^*} u^* v^* \, dt &= \sum_{i=0}^k \sum_{j=0}^l \alpha_i \beta_j \int_{E^*} \chi_{F_i^*} \chi_{G_j^*} \, dt \\ &= \sum_{i=0}^k \sum_{j=0}^l \alpha_i \beta_j \mathcal{L}^1(F_i^* \cap G_j^*) \\ &= \sum_{i=0}^k \sum_{j=0}^l \alpha_i \beta_j \min\{\mathcal{L}^1(F_i^*), \mathcal{L}^1(G_j^*)\} \\ &= \sum_{i=0}^k \sum_{j=0}^l \alpha_i \beta_j \min\{\mathcal{L}^1(F_i), \mathcal{L}^1(G_j)\}, \end{aligned}$$

where we have used (4.11) and the fact that

$$\mathcal{L}^1(F_i^* \cap G_j^*) = \min\{\mathcal{L}^1(F_i^*), \mathcal{L}^1(G_j^*)\},$$

since both sets are half-open intervals of the form $[0, a)$ for some $a > 0$. **Step 2:** Now let $u, v : E \rightarrow [0, \infty)$ be two Lebesgue measurable functions and construct two increasing sequences $\{u_n\}_n$ and $\{v_n\}_n$ of nonnegative simple functions vanishing at infinity, below u and v , and converging pointwise to u and v , respectively (see Corollary B.37). By Proposition 4.11 we have that $u_n^* v_n^* \leq u_{n+1}^* v_{n+1}^*$ for all $n \in \mathbb{N}$ and that $\{u_n^* v_n^*\}_n$ converges pointwise to $u^* v^*$. By Step 1, for each $n \in \mathbb{N}$ we have

$$\int_E u_n v_n \, dx \leq \int_{E^*} u_n^* v_n^* \, dt$$

and the result follows by the Lebesgue monotone convergence theorem. $\square$

Exercise 4.15. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set and let $u, v : E \rightarrow [0, \infty)$ be Lebesgue measurable functions. Prove that for every $r \geq 0$,

$$\int_E u(x) \chi_{\{v \leq r\}}(x) dx \geq \int_{E^*} u^*(t) \chi_{\{v^* \leq r\}}(t) dt.$$

Hint: Assume first that u is integrable.

We now turn to one of the most important properties of decreasing rearrangement.

Theorem 4.16. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set, let $u : E \rightarrow [0, \infty)$ be a function vanishing at infinity, and let $f : [0, \infty) \rightarrow [0, \infty)$ be a Borel function. Then

$$(4.17) \quad \int_{E^*} f(u^*(t)) dt \leq \int_E f(u(x)) dx,$$

with equality holding if $f(0) = 0$ or $\mathcal{L}^1(\{x \in E : u(x) > 0\}) < \infty$ or

$$(4.18) \quad \mathcal{L}^1(\{x \in E : u(x) > 0\}) = \infty \text{ and } \mathcal{L}^1(\{x \in E : u(x) = 0\}) = 0.$$

In particular, for any $p > 0$,

$$\int_{E^*} (u^*(t))^p dt = \int_E (u(x))^p dx.$$

Proof. Step 1: We claim that

$$\int_{\{u>0\}} f(u(x)) dx = \int_{\{u^*>0\}} f(u^*(t)) dt.$$

Let $\tilde{f}$ be the restriction of f to $(0, \infty)$ and construct an increasing sequence $\{f_n\}_n$ of nonnegative Borel simple functions defined on $(0, \infty)$ and converging pointwise to $\tilde{f}$. Since each f_n may be written as

$$f_n = \sum_{i=0}^{k_n} c_{i,n} \chi_{B_{i,n}},$$

where $c_{i,n} \neq c_{j,n}$ for $i \neq j$ and the Borel sets $B_{i,n} \subseteq (0, \infty)$ are pairwise disjoint, by applying Corollary 4.6, we conclude that

$$\begin{aligned} \int_{\{u>0\}} f_n(u(x)) dx &= \sum_{i=0}^{k_n} c_{i,n} \mathcal{L}^1(\{x \in E : u(x) \in B_{i,n}\}) \\ &= \sum_{i=0}^{k_n} c_{i,n} \mathcal{L}^1(\{t \in E^* : u^*(t) \in B_{i,n}\}) \\ &= \int_{\{u^*>0\}} f_n(u^*(t)) dt. \end{aligned}$$

The claim now follows by the Lebesgue monotone convergence theorem.

Step 2: Write

$$(4.19) \quad \begin{aligned} \int_E f(u(x)) dx \\ = \int_{\{u>0\}} f(u(x)) dx + f(0)\mathcal{L}^1(\{x \in E : u(x) = 0\}), \end{aligned}$$

where $f(0)\mathcal{L}^1(\{x \in E : u(x) = 0\})$ is understood to be zero if $f(0) = 0$, independently of the value of $\mathcal{L}^1(\{x \in E : u(x) = 0\})$. Similarly,

$$(4.20) \quad \begin{aligned} \int_{E^*} f(u^*(t)) dt \\ = \int_{\{u^*>0\}} f(u^*(t)) dt + f(0)\mathcal{L}^1(\{t \in E^* : u^*(t) = 0\}). \end{aligned}$$

Hence, it follows from the previous step, (4.19), (4.20), and Proposition 4.3(v) that (4.17) holds.

Moreover, if $f(0) = 0$ or $f(0) > 0$ and

$$\mathcal{L}^1(\{t \in E^* : u^*(t) = 0\}) = \mathcal{L}^1(\{x \in E : u(x) = 0\}),$$

then

$$(4.21) \quad \int_E f(u(x)) dx = \int_{E^*} f(u^*(t)) dt.$$

Thus, the result now follows by Proposition 4.3(v) once more. $\square$

Remark 4.17. It follows from the previous proof that equality holds in (4.17) if and only if one of the following conditions holds:

- (i) $\int_{\{u>0\}} f(u(x)) dx = \infty$.
- (ii) $\int_{\{u>0\}} f(u(x)) dx < \infty$ and $f(0) = 0$.
- (iii) $\int_{\{u>0\}} f(u(x)) dx < \infty$, $f(0) > 0$, and $\mathcal{L}^1(\{x \in E : u(x) > 0\}) < \infty$.
- (iv) $\int_{\{u>0\}} f(u(x)) dx < \infty$, $f(0) > 0$, and (4.18) holds.

Exercise 4.18. Let $u(x) = e^{-x}\chi_{[0,\infty)}(x)$ and let $f = \chi_{\{0\}}$. Find u^* and prove directly that

$$0 = \int_0^\infty f(u^*(t)) dt < \int_{\mathbb{R}} f(u(x)) dx = \infty.$$

The previous theorem shows that the operator $u \mapsto u^*$ preserves the norm in L^p for $1 \leq p < \infty$. Next we show that it is a continuous operator from $L^p(E)$ into $L^p(E^*)$. More generally, we have the following result.

Theorem 4.19. *Let $\Psi : \mathbb{R} \rightarrow [0, \infty)$ be a convex function such that $\Psi(0) = 0$, let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set, and let $u, v : E \rightarrow [0, \infty)$ be two functions vanishing at infinity. Then*

$$(4.22) \quad \int_{E^*} \Psi(u^*(t) - v^*(t)) dt \leq \int_E \Psi(u(x) - v(x)) dx.$$

In particular,

$$\int_{E^*} |u^*(t) - v^*(t)|^p dt \leq \int_E |u(x) - v(x)|^p dx$$

for all $1 \leq p < \infty$ and the operator $u \mapsto u^*$ is continuous from $L^p(E)$ into $L^p(E^*)$.

Proof. Define

$$\Psi_+(s) := \begin{cases} \Psi(s) & \text{if } s \geq 0, \\ 0 & \text{otherwise,} \end{cases} \quad \Psi_-(s) := \begin{cases} \Psi(s) & \text{if } s < 0, \\ 0 & \text{otherwise.} \end{cases}$$

Then Ψ_+ and Ψ_- are still convex (why?) and $\Psi = \Psi_+ + \Psi_-$. Thus, to prove (4.22), it suffices to prove it for Ψ_+ and Ψ_- separately. We only prove (4.22) for Ψ_+ , since the proof of (4.22) for Ψ_- is very similar. By Exercise 3.21, the right derivative $(\Psi_+)'_+$ of Ψ_+ exists in $\mathbb{R}$ and is increasing and for every $s \in \mathbb{R}$,

$$\Psi_+(s) = \int_0^s (\Psi_+)'_+(\tau) d\tau,$$

where we have used the fact that $\Psi_+(0) = 0$. Hence, for $x \in E$,

$$\begin{aligned} \Psi_+(u(x) - v(x)) &= \int_0^{u(x)-v(x)} (\Psi_+)'_+(\tau) d\tau \\ &= \int_{v(x)}^{u(x)} (\Psi_+)'_+(u(x) - r) dr \\ &= \int_0^\infty (\Psi_+)'_+(u(x) - r) \chi_{\{v \leq r\}}(x) dr, \end{aligned}$$

where in the second equality we have used the change of variable $\tau = u(x) - r$ and in the third equality we have used the facts that $(\Psi_+)'_+(s) = 0$ for $s < 0$ and that $v \geq 0$. Integrating both sides over E and using Tonelli's theorem, we get

$$(4.23) \quad \int_E \Psi_+(u(x) - v(x)) dx = \int_0^\infty \int_E (\Psi_+)'_+(u(x) - r) \chi_{\{v \leq r\}}(x) dx dr.$$

Similarly,

$$(4.24) \quad \int_{E^*} \Psi_+(u^*(t) - v^*(t)) dt = \int_0^\infty \int_{E^*} (\Psi_+)'_+(u^*(t) - r) \chi_{\{v^* \leq r\}}(t) dt dr.$$

Since for every fixed t the function $g(s) := (\Psi_+)'_+(s - t)$ is increasing, by Exercise 4.8 we have that $(g \circ u)^*(t) = (g \circ u^*)(t)$ for all but countably many $t \in E^*$, and so, also by Exercise 4.15,

$$(4.25) \quad \int_E (\Psi_+)'_+(u(x) - t) \chi_{\{v \leq r\}}(x) dx \geq \int_{E^*} (\Psi_+)'_+(u^*(t) - r) \chi_{\{v^* \leq r\}}(t) dt.$$

The result now follows by combining (4.23), (4.24), and (4.25). $\square$

In the next exercise we present an alternative proof in the case in which Ψ is even.

Exercise 4.20. Let $\Psi : [0, \infty) \rightarrow [0, \infty)$ be a convex function such that $\Psi(0) = 0$, let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set, and let $u, v : E \rightarrow [0, \infty)$ be two functions vanishing at infinity.

- (i) Prove that if $r_1 \geq r_2 \geq 0$ and $s_1 \geq s_2 \geq 0$, then

$$\Psi(|r_1 - s_1|) + \Psi(|r_2 - s_2|) \leq \Psi(|r_1 - s_2|) + \Psi(|r_2 - s_1|).$$

- (ii) Prove that if $r_1 \geq r_2 \geq \cdots \geq r_n \geq 0$, $s_1 \geq s_2 \geq \cdots \geq s_n \geq 0$, $n \in \mathbb{N}$, and if

$$f : \{1, \dots, n\} \rightarrow \{1, \dots, n\}, \quad g : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$$

are two bijections, then

$$\sum_{i=1}^n \Psi(|r_i - s_i|) \leq \sum_{i=1}^n \Psi(|r_{f(i)} - s_{g(i)}|).$$

- (iii) Prove that if $u, v : E \rightarrow [0, \infty)$ are simple functions of the form

$$(4.26) \quad u = \sum_{i=0}^n \alpha_i \chi_{E_i}, \quad v = \sum_{i=0}^n \beta_i \chi_{E_i},$$

where $\mathcal{L}^1(E_i) = k$ for all $i = 1, \dots, n$, then

$$(4.27) \quad \int_{E^*} \Psi(|u^*(t) - v^*(t)|) dt \leq \int_E \Psi(|u(x) - v(x)|) dx.$$

- (iv) Prove that if $u, v : E \rightarrow [0, \infty)$ are simple functions of the form (4.26), where $\mathcal{L}^1(E_i) = r_i \in \mathbb{Q} \cap [0, \infty)$ for all $i = 1, \dots, n$, then (4.27) holds.
- (v) Prove that (4.27) holds for arbitrary $u, v : E \rightarrow [0, \infty)$ vanishing at infinity.

4.2. Function Spaces and Decreasing Rearrangement

In this section we prove that if u belongs to some of the function spaces studied in the previous chapters, then so does u^* and its corresponding norm decreases.

We begin by showing that the pointwise variation of u^* is less than the pointwise variation of u .

Theorem 4.21 (Bounded variation). *Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow [0, \infty)$ be a function vanishing at infinity. Then for all $t_1, t_2 \in I^*$, with $t_1 \leq t_2$,*

$$(4.28) \quad u^*(t_1) - u^*(t_2) \leq \text{Var}_I u.$$

In particular,

$$(4.29) \quad \text{Var}_{I^*} u^* = \sup_{I^*} u^* - \inf_{I^*} u^* \leq \text{Var}_I u.$$

Proof. If either $\text{Var}_I u = \infty$ or u is a constant function, then (4.28) holds. Thus assume that $\text{Var}_I u < \infty$ and that u is not a constant function. It follows by Proposition 2.11 that u is bounded with

$$\sup_I u - \inf_I u \leq \text{Var}_I u.$$

On the other hand, by Proposition 4.3, if $t_1, t_2 \in I^*$, with $t_1 \leq t_2$, then $\inf_I u \leq u^*(t_2) \leq u^*(t_1) \leq u^*(0) = \sup_I u$, and so (4.28) holds. The inequality (4.29) follows by the arbitrariness of t_1 and t_2 . $\square$

Next we show that if u vanishes at infinity and is absolutely continuous, then so is u^* .

Theorem 4.22 (Absolute continuity). *Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow [0, \infty)$ be a function vanishing at infinity. If u belongs to $AC(I)$, then u^* belongs to $AC(I^*)$ and for every $1 \leq p \leq \infty$,*

$$(4.30) \quad \|(u^*)'\|_{L^p(I^*)} \leq \|u'\|_{L^p(I)}.$$

Proof. Step 1: Since u is uniformly continuous (see Remark 3.4) and vanishes at infinity, if $\sup I = \infty$, then there exists $\lim_{x \rightarrow \infty} u(x) = 0$ (why?). Similarly, if $\inf I = -\infty$, then $\lim_{x \rightarrow -\infty} u(x) = 0$. On the other hand, if one of the endpoints of I is finite, then by uniform continuity u can be extended to $\bar{I}$ as a uniform continuous function. Thus, in all cases, u is bounded. In turn, by Proposition 4.3, $u^*(0) = \sup_I u < \infty$ and u^* is right continuous at 0.

By Theorem 3.29, we have that u' is equi-integrable in I and thus for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$(4.31) \quad \int_E |u'(x)| dx \leq \varepsilon$$

for every Lebesgue measurable set $E \subseteq I$ with $\mathcal{L}^1(E) \leq \delta$.

Let $a, b \in I^*$ with $0 < a < b$ and for $x \in I$ define

$$v(x) := \begin{cases} u^*(b) & \text{if } u(x) < u^*(b), \\ u(x) & \text{if } u^*(b) \leq u(x) \leq u^*(a), \\ u^*(a) & \text{if } u(x) \geq u^*(a). \end{cases}$$

Since $u(I)$ is an interval and $\inf_I u \leq u^*(b) \leq u^*(a) \leq u^*(0) = \sup_I u$ by Proposition 4.3, by Proposition 2.11, Theorems 3.25 and 3.59, we have

$$(4.32) \quad u^*(a) - u^*(b) \leq \text{Var}_I v = \int_I |v'(x)| dx = \int_{\{u^*(b) < u < u^*(a)\}} |u'(x)| dx.$$

On the other hand, by (4.1) and Proposition 4.1(i),

$$(4.33) \quad \mathcal{L}^1(\{x \in I : u^*(b) < u(x) < u^*(a)\}) = \varrho_u(u^*(b)) - (\varrho_u)_-(u^*(a)) \leq b - a,$$

where in the last inequality we have used the fact that $\varrho_u(u^*(b)) \leq b$ and $(\varrho_u)_-(u^*(a)) \geq a$ by Proposition 4.3(iii). Since u^* is right continuous at 0, by taking $a_n \searrow 0$ and using by Proposition B.9(i) it follows that (4.33) continues to hold for $a = 0$.

Hence, for every finite number of nonoverlapping intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq I^*$ and $\sum_{i=1}^n (b_i - a_i) \leq \delta$, by (4.33) we have that

$$\sum_{i=1}^n \mathcal{L}^1(\{x \in I : u^*(b_i) < u(x) < u^*(a_i)\}) \leq \sum_{i=1}^n (b_i - a_i) \leq \delta$$

and so by (4.31),

$$\sum_{i=1}^n (u^*(a_i) - u^*(b_i)) \leq \int_{\bigcup_{i=1}^n \{u^*(b_i) < u < u^*(a_i)\}} |u'(x)| dx \leq \varepsilon,$$

which shows that u^* is absolutely continuous in I^* .

Step 2: By (4.32) and (4.33),

$$\begin{aligned} u^*(a) - u^*(b) &\leq \int_{\{u^*(b) < u < u^*(a)\}} |u'(x)| dx \\ &\leq \|u'\|_{L^\infty(I)} \mathcal{L}^1(\{x \in I : u^*(b) < u(x) < u^*(a)\}) \\ &\leq \|u'\|_{L^\infty(I)} (b - a). \end{aligned}$$

Dividing both sides by $b - a$ and letting $b \rightarrow a^+$, we obtain (4.30) for $p = \infty$.

Step 3: Let $1 \leq p < \infty$ and for $t \in I^*$ define

$$g(t) := \int_{\{u > u^*(t)\}} |u'(x)|^p dx.$$

We claim that

$$(4.34) \quad g'(t) \geq |(u^*)'(t)|^p \quad \text{for } \mathcal{L}^1\text{-a.e. } t \in I^*.$$

Since u^* is decreasing by Proposition 4.3 and g is increasing, it follows from Lebesgue's differentiation theorem that u^* and g are differentiable at $\mathcal{L}^1$ -a.e. $t \in I^*$. Fix a point $t_0 \in (I^*)^\circ$ at which both u^* and g are differentiable. If $(u^*)'(t_0) = 0$, then (4.34) holds for t_0 because $g'(t_0) \geq 0$. Thus assume $(u^*)'(t_0) < 0$. Assume also that t_0 is a Lebesgue point of $(u^*)'$. We claim that there exists a sequence $h_n \rightarrow 0^+$ such that

$$(4.35) \quad \mathcal{L}^1(\{x \in I : u^*(t_0 + h_n) < u(x) \leq u^*(t_0)\}) = h_n$$

for all n . By (4.1),

$$(4.36) \quad \mathcal{L}^1(\{x \in I : u^*(t_0 + h) < u(x) \leq u^*(t_0)\}) = \varrho_u(u^*(t_0 + h)) - \varrho_u(u^*(t_0)).$$

Given $s_0 > 0$ consider a strictly increasing sequence $s_n \rightarrow s_0$. By Propositions 4.3(v), B.9 and the fact that u vanishes at infinity we have that

$$\begin{aligned} & \mathcal{L}^1(\{x \in I : u(x) = s_0\}) \\ &= \bigcap_{n=1} \mathcal{L}^1(\{x \in I : u(x) > s_n\}) - \mathcal{L}^1(\{x \in I : u(x) > s_0\}) \\ &= \bigcap_{n=1} \mathcal{L}^1(\{t \in I^* : u^*(t) > s_n\}) - \mathcal{L}^1(\{t \in I^* : u^*(t) > s_0\}) \\ &= \mathcal{L}^1(\{t \in I^* : u^*(t) = s_0\}). \end{aligned}$$

Note that we could also have used Corollary 4.6. In turn, by (4.7),

$$\begin{aligned} (\varrho_u)_-(s_0) &= \varrho_u(s_0) + \mathcal{L}^1(\{x \in I : u(x) = s_0\}) \\ &= \varrho_u(s_0) + \mathcal{L}^1(\{t \in I^* : u^*(t) = s_0\}) \end{aligned}$$

for $s_0 > 0$ and so taking $s_0 = u^*(t_0 + h)$ we get

$$(\varrho_u)_-(u^*(t_0 + h)) = \varrho_u(u^*(t_0 + h)) + \mathcal{L}^1(\{t \in I^* : u^*(t) = u^*(t_0 + h)\}).$$

Since t_0 is a Lebesgue point of $(u^*)'$, we have

$$\lim_{r \rightarrow 0^+} \frac{1}{2r} \int_{t_0-r}^{t_0+r} |(u^*)'(t) - (u^*)'(t_0)| = 0,$$

and since $(u^*)'(t_0) < 0$, we can find a sequence $h_n \rightarrow 0^+$ such that $(u^*)'(t_0 + h_n) < 0$ for every n . In turn, $\{t \in I^* : u^*(t) = u^*(t_0 + h_n)\}$ is a singleton. Thus,

$$(4.37) \quad (\varrho_u)_-(u^*(t_0 + h_n)) = \varrho_u(u^*(t_0 + h_n)).$$

By Proposition 4.3(iii), we have $\varrho_u(u^*(t_0 + h_n)) \leq t_0 + h_n$ and $(\varrho_u(u^*(t_0 + h_n)))_- \geq t_0 + h_n$. It follows from (4.37) that $\varrho_u(u^*(t_0 + h_n)) = t_0 + h_n$ and similarly, $\varrho_u(u^*(t_0)) = t_0$. Thus (4.35) holds by (4.36).

In turn, by Jensen's inequality (see Theorem B.50) and (4.32),

$$\begin{aligned} \frac{g(t_0 + h_n) - g(t_0)}{h_n} &\geq \left| \frac{1}{h_n} \int_{\{u^*(t_0 + h_n) < u \leq u^*(t_0)\}} |u'(x)| dx \right|^p \\ &\geq \left| \frac{u^*(t_0 + h_n) - u^*(t_0)}{h_n} \right|^p. \end{aligned}$$

Letting $n \rightarrow \infty$ and using the fact that both u^* and g are differentiable at t_0 gives (4.34) at t_0 .

This shows that the claim (4.34) holds. By integrating (4.34) over $[a, b] \subseteq I^*$ and using Corollary 1.25 we get

$$\int_I |u'(x)|^p dx \geq g(b) - g(a) \geq \int_a^b g'(t) dt \geq \int_a^b |(u^*)'(t)|^p dt.$$

It now suffices to let $a \rightarrow (\inf I^*)^+$ and $b \rightarrow (\sup I^*)^-$. $\square$

Remark 4.23. When $I = \mathbb{R}$ it is possible to show that decreasing rearrangement preserves also Hölder's continuity and moduli of continuity (see [64]).

The next exercise shows that decreasing rearrangement does not preserve higher-order regularity.

Exercise 4.24. Consider the function $u : [0, 2] \rightarrow [0, \infty)$ defined by

$$u(x) := 3 + (x - \tfrac{1}{2})^2 - (x - \tfrac{1}{2})^4.$$

Prove that $(u^*)'$ is discontinuous.

We refer to the papers of Cianchi [47] and Dahlberg [56] for more information on the regularity of $(u^*)'$.

Theorem 4.25 (Singular functions). *Let $I \subseteq \mathbb{R}$ be a bounded interval and let $u : I \rightarrow [0, \infty)$ be a function vanishing at infinity and differentiable $\mathcal{L}^1$ -a.e. in I . Then u is a singular function if and only if $u^* : (I^*)^\circ \rightarrow [0, \infty)$ is a singular function.*

Proof. If u is singular, then by Theorem 3.88, there exists a Lebesgue measurable set $E \subseteq I$ such that $\mathcal{L}^1(I \setminus E) = 0$ and $\mathcal{L}^1(u(E)) = 0$. Let $B \subset [0, \infty)$ be a Borel set such that $\mathcal{L}^1(B) = 0$ and $B \supseteq u(E)$. Then by Corollary 4.6,

$$\mathcal{L}^1(\{x \in I : u(x) \in B\}) = \mathcal{L}^1(\{t \in I^* : u^*(t) \in B\}).$$

Hence, $\mathcal{L}^1(I^* \setminus (u^*)^{-1}(B)) = \mathcal{L}^1(I \setminus u^{-1}(B)) = \mathcal{L}^1(I \setminus E) = 0$. Since $\mathcal{L}^1(B) = 0$, it follows from Theorem 3.88 that u^* is singular.

The other implication is similar. $\square$

Exercise 4.26. What happens if I is unbounded?

Definition 4.27. Let $E \subseteq \mathbb{R}^N$ and $F \subseteq [-\infty, \infty]$ be two Lebesgue measurable sets. A Lebesgue measurable function $\varphi : E \rightarrow F$ is called *measure-preserving* if

$$\mathcal{L}^N(\varphi^{-1}(G)) = \mathcal{L}^1(G)$$

for every Lebesgue measurable set $G \subseteq F$.

Exercise 4.28. Let $E \subset \mathbb{R}$ be a Lebesgue measurable set with $\mathcal{L}^1(E) < \infty$ and let $a \geq 0$. Prove that the function $\varphi : E \rightarrow [a, a + \mathcal{L}^1(E)]$, defined by

$$\varphi(x) := a + \mathcal{L}^1(E \cap (-\infty, x]), \quad x \in E,$$

is measure-preserving.

Exercise 4.29. Given a sequence of measure-preserving functions $\varphi_n : E_n \rightarrow F_n$, where $\{E_n\}_n, \{F_n\}_n$ are two sequences of pairwise disjoint Lebesgue measurable subsets of $\mathbb{R}$, prove that the function

$$\varphi : \bigcup_{n=1}^{\infty} E_n \rightarrow \bigcup_{n=1}^{\infty} F_n,$$

defined by

$$\varphi(x) := \varphi_n(x) \quad \text{if } x \in E_n,$$

is measure-preserving. Prove that the result continues to hold if the sets F_n are only assumed to be pairwise disjoint up to sets of Lebesgue measure zero.

Exercise 4.30. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set with finite measure and let $u : E \rightarrow [0, \infty)$ be a Lebesgue measurable function. Let $\varphi : E \rightarrow [0, \mathcal{L}^1(E)]$ be defined by

$$(4.38) \quad \varphi(x) := \varrho_u(u(x)) + \mathcal{L}^1(\{y \in E : y \leq x, u(y) = u(x)\}).$$

(i) Prove that φ is Lebesgue measurable.

(ii) Let $0 < \alpha \leq \mathcal{L}^1(E)$. Prove that there exist two extended real numbers $x_0 \in [-\infty, \infty]$ and $s_0 \in [0, \infty)$ such that

$$\varrho_u(s_0) + \mathcal{L}^1(\{y \in E : y \leq x_0, u(y) = s_0\}) = \alpha$$

and

$$\varrho_u(s) + \mathcal{L}^1(\{y \in E : y \leq x, u(y) = s\}) > \alpha$$

if either $x \in E$, with $x > x_0$, and $s = s_0$, or $x \in E$ is arbitrary and $0 \leq s < s_0$.

(iii) Prove that

$$\{x \in E : \varphi(x) \leq \alpha\} = \{x \in E : u(x) > s_0\} \cup \{x \in E : x \leq x_0, u(x) = s_0\}.$$

(iv) Consider the measure $\mu : \mathcal{B}([0, \mathcal{L}^1(E)]) \rightarrow [0, \infty]$, defined by

$$\mu(B) := \mathcal{L}^1(\varphi^{-1}(B)), \quad B \in \mathcal{B}([0, \mathcal{L}^1(E)]).$$

Prove that for every $0 \leq \alpha \leq \mathcal{L}^1(E)$ we have that

$$\mu([0, \alpha]) = \mathcal{L}^1(\varphi^{-1}([0, \alpha])) = \mathcal{L}^1(\{x \in E : \varphi(x) \leq \alpha\}) = \alpha$$

and deduce that φ is measure-preserving.

(v) Prove that for $\mathcal{L}^1$ -a.e. $x \in E$,

$$u^*(\varphi(x)) = u(x).$$

(vi) Remove the assumption that E has finite measure.

Exercise 4.31. Let $E \subseteq \mathbb{R}$ be a Lebesgue measurable set, let $u : E \rightarrow [0, \infty)$ be a function vanishing at infinity, and let

$$(4.39) \quad E_{>} := \{x \in E : u(x) > 0\}, \quad F_{>} := \{t \in E^* : u^*(t) > 0\}.$$

Prove that the function $\varphi : E_{>} \rightarrow F_{>}$ given in (4.38) is measure-preserving and that

$$u^*(\varphi(x)) = u(x)$$

for $\mathcal{L}^1$ -a.e. $x \in E_{>}$.

Exercise 4.32 (Lusin (N) property). Let $I \subseteq \mathbb{R}$ be an interval, let $u : I \rightarrow [0, \infty)$ be a function vanishing at infinity. Prove that if u maps sets of Lebesgue measure zero into sets of Lebesgue measure zero, then so does u^* .

Exercise 4.33. Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow [0, \infty)$ be a function vanishing at infinity and differentiable $\mathcal{L}^1$ -a.e. in I . Let $I_{>}$ and $F_{>}$ be defined as in (4.39) and let φ be the function in (4.38) with $E = I$.

(i) Prove that if u is differentiable at $x_0 \in I_{>}$ and $u'(x_0) \neq 0$, then there exists $\delta > 0$ such that $\varphi(x) \neq \varphi(x_0)$ for all $x \in I_{>} \cap (x_0 - \delta, x_0 + \delta)$.

(ii) Let $I_{\text{diff}} \subseteq I_{>}$ be the set of all $x \in I_{>}$ such that u is differentiable at x and u^* is differentiable at $\varphi(x)$. Prove that if $\varphi|_{I_{\text{diff}}}$ is differentiable at $x_0 \in I_{\text{diff}}$, that is, if there exists in $\mathbb{R}$ the limit

$$\ell := \lim_{x \in I_{\text{diff}}, x \rightarrow x_0} \frac{\varphi(x) - \varphi(x_0)}{x - x_0},$$

then $|\ell| \geq 1$.

(iii) Prove that if $x_0 \in I_{\text{diff}}$ is such that $(u^*)'(\varphi(x_0)) \neq 0$, then $u'(x_0) \neq 0$ and $\varphi|_{I_{\text{diff}}}$ is continuous at x_0 .

(iv) Prove that if $x_0 \in I_{\text{diff}}$ is such that $(u^*)'(\varphi(x_0)) \neq 0$, then $\varphi|_{I_{\text{diff}}}$ is differentiable at $x_0 \in I_{\text{diff}}$ with

$$(u^*)'(\varphi(x_0))(\varphi|_{I_{\text{diff}}})'(x_0) = u'(x_0).$$

(v) Prove that for any $p > 0$,

$$\int_{I^*} |(u^*)'(t)|^p dt \leq \int_I |u'(x)|^p dx.$$

Curves

Undergradese, V: “Hmmm, what do you mean by that?” Translation: “What’s the answer so we can all go home?”

— Jorge Cham, www.phdcomics.com

In this chapter we will study rectifiable curves and their relation with functions of bounded pointwise variation.

5.1. Rectifiable Curves

In the literature there are many different definitions of curves. Let $I \subseteq \mathbb{R}$ be an interval and let $u : I \rightarrow \mathbb{R}^M$ be a (possibly discontinuous) function. As the *parameter* t traverses I , $u(t)$ traverses a curve in $\mathbb{R}^M$. Rather than calling u a curve, it is better to regard any vector function v obtained from u by a suitable change of parameter as representing the same curve as u . Thus, one should define a curve as an equivalence class of equivalent parametric representations.

Definition 5.1. Given two intervals $I, J \subseteq \mathbb{R}$, a metric space (Y, d) , and two functions $u : I \rightarrow Y$ and $v : J \rightarrow Y$, we say that they are (*Lebesgue*) *equivalent* if there exists a continuous, bijective function $\phi : I \rightarrow J$ such that

$$(5.1) \quad u(t) = v(\phi(t))$$

for all $t \in I$. We write $u \sim v$ and we call u and v *parametric representations* and the function ϕ a *parameter change*.

Note that in view of Theorem 1.8, $\phi^{-1} : J \rightarrow I$ is also continuous.

Exercise 5.2. Prove that $\sim$ is an equivalence relation.

Definition 5.3. A *curve* γ is an equivalence class of parametric representations. The curve γ is said to be *continuous* if one (and so all) of its parametric representations is continuous.

When X is a normed space, by restricting parametric representations, parameter changes and their inverses in Definitions 5.1 and 5.3 to be differentiable $\mathcal{L}^1$ -a.e., or Lipschitz continuous, or of class C^n , $n \in \mathbb{N}_0$, etc., we may define curves γ that are differentiable $\mathcal{L}^1$ -a.e., or Lipschitz continuous, or of class C^n , $n \in \mathbb{N}_0$, respectively.

Given a metric space (Y, d) and a curve γ with parametric representation $u : I \rightarrow Y$, where $I \subseteq \mathbb{R}$ is an interval, the *multiplicity* of a point $y \in Y$ is the (possibly infinite) number of points $t \in I$ such that $u(t) = y$. Note that the multiplicity of a curve can also be expressed in terms of the Banach indicatrix (see Definition 2.58). Since every parameter change $\phi : I \rightarrow J$ is bijective, the multiplicity of a point does not depend on the particular parametric representation. The *range* of γ is the set of points of Y with positive multiplicity, that is, $u(I)$. A point in the range of γ with multiplicity one is called a *simple point*. If every point of the range is simple, then γ is called a *simple arc*.

If $I = [a, b]$ and $u(a) = u(b)$, then a continuous curve γ is called a *closed curve*. A closed curve is called *simple* if every point of the range is simple, with the exception of $u(a)$, which has multiplicity two.

Example 5.4. The curves γ_n with parametric representations $u_n : [0, \frac{\pi}{2}] \rightarrow \mathbb{R}^2$, given by

$$u_n(t) := (\sin^2 nt, 0), \quad t \in [0, \frac{\pi}{2}],$$

$n \in \mathbb{N}$, all have the same range, but are different curves. Indeed, γ_n covers n times the segment of endpoints $(0, 0)$ and $(1, 0)$.

Exercise 5.5. Given two intervals $I, J \subseteq \mathbb{R}$ and two functions $u : I \rightarrow \mathbb{R}^M$ and $v : J \rightarrow \mathbb{R}^M$, we say that they are *Fréchet equivalent* if for every $\varepsilon > 0$ there exists a continuous, bijective function $\phi : I \rightarrow J$ such that

$$|u(t) - v(\phi(t))| < \varepsilon$$

for all $t \in I$. We write $u \stackrel{F}{\sim} v$. Prove that $\stackrel{F}{\sim}$ is an equivalence relation.

Exercise 5.6. A *Fréchet curve* γ is an equivalence class (with respect to $\stackrel{F}{\sim}$) of parametric representations. Prove that if a Fréchet curve γ has a continuous representative, then all of its representatives are continuous.

The next exercise shows that two functions $u : I \rightarrow \mathbb{R}^M$ and $v : J \rightarrow \mathbb{R}^M$ may be Fréchet equivalent but not (Lebesgue) equivalent.

Exercise 5.7. For $t, \tau \in [0, 1]$ let $u(t) := (t, 0)$ and

$$v(\tau) := \begin{cases} (\frac{3}{2}\tau, 0) & \text{if } 0 \leq \tau \leq \frac{1}{3}, \\ (\frac{1}{2}, 0) & \text{if } \frac{1}{3} \leq \tau \leq \frac{2}{3}, \\ (1 + \frac{3}{2}(\tau - 1), 0) & \text{if } \frac{2}{3} \leq \tau \leq 1. \end{cases}$$

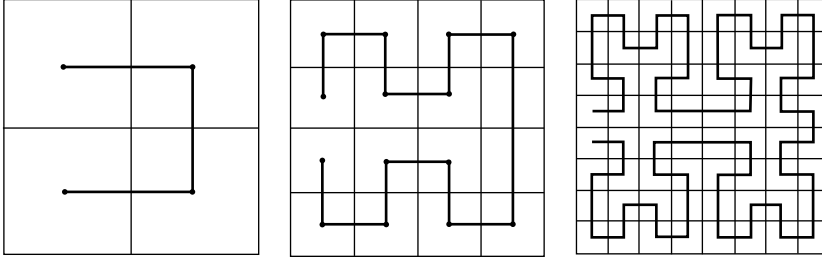


Figure 1. Peano's curve.

Prove that u and v are Fréchet equivalent but not (Lebesgue) equivalent.

The next result shows that the class of continuous curves is somehow too large for our intuitive idea of curve. Indeed, we construct a continuous curve that fills the unit square in $\mathbb{R}^2$. The first example of this type was given by Peano in 1890 [188]. We present here another example given by Hilbert [119]. The proof is taken from [174].

Theorem 5.8 (Hilbert). *There exists a continuous function $u : [0, 1] \rightarrow \mathbb{R}^2$ such that $u([0, 1]) = [0, 1]^2$.*

Proof. For every $n \in \mathbb{N}$ divide the interval $[0, 1]$ into 4^n closed intervals $I_{k,n}$, $k = 1, \dots, 4^n$, of length $\frac{1}{4^n}$ and the square $[0, 1]^2$ into 4^n closed squares $Q_{k,n}$, $k = 1, \dots, 4^n$, of side length $\frac{1}{2^n}$. Construct a bijective correspondence between the 4^n intervals $I_{k,n}$ and the 4^n squares $Q_{k,n}$ in such a way that (see Figure 1)

- (i) to two adjacent intervals there correspond adjacent squares,
- (ii) to the four intervals of length $\frac{1}{4^{n+1}}$ contained in some interval $I_{k,n}$ there correspond the four squares of side length $\frac{1}{2^{n+1}}$ contained in the square corresponding to $I_{k,n}$.

By relabeling the squares, if necessary, we will assume that the square $Q_{k,n}$ corresponds to the interval $I_{k,n}$. Let

$$\mathcal{F} := \{I_{k,n} : n \in \mathbb{N}, k = 1, \dots, 4^n\},$$

$$\mathcal{G} := \{Q_{k,n} : n \in \mathbb{N}, k = 1, \dots, 4^n\}.$$

By the axiom of continuity of the reals, if $\{J_i\}_i$ is any infinite sequence of intervals in $\mathcal{F}$ such that $J_{i+1} \subseteq J_i$ for all $i \in \mathbb{N}$ and $\{R_i\}_i$ is the corresponding sequence of squares in $\mathcal{G}$, then there exist a unique $t \in [0, 1]$ and $(x, y) \in [0, 1]^2$ such that

$$\{t\} = \bigcap_i J_i, \quad \{(x, y)\} = \bigcap_i R_i.$$

We set the point t and the point (x, y) in correspondence.

We claim that this correspondence defines a continuous function $u : [0, 1] \rightarrow \mathbb{R}^2$ with all the desired properties. Indeed, a point $t \in [0, 1]$ that is not an endpoint of any interval determines uniquely a sequence $\{J_i\}_i$ to which it belongs and hence a point (x, y) belonging to the corresponding sequence $\{R_i\}_i$. The same is true for $t = 0$ and $t = 1$. A point t that is common to two different intervals $I_{k_1, m}$ and $I_{k_2, m}$ for some $m \in \mathbb{N}$ is also common to two different intervals $I_{k, n}$, $k = 1, \dots, 4^n$, for all $n \geq m$. Hence, it belongs to two different sequences $\{J_i\}_i$, $\{J'_i\}_i$. Since the squares R_i and R'_i , corresponding to J_i and J'_i , respectively, are adjacent by property (i), it follows that

$$\bigcap_i R_i = \bigcap_i R'_i.$$

Thus, to every $t \in [0, 1]$ there corresponds a unique $(x, y) \in [0, 1]^2$ that we denote $u(t)$.

Since every $(x, y) \in [0, 1]^2$ belongs to one, two, three, or four sequences $\{R_i\}_i$, there exists one, two, three, or four $t \in [0, 1]$ such that $u(t) = (x, y)$. Hence, $u([0, 1]) = [0, 1]^2$.

To prove that u is continuous, write $u(t) = (x(t), y(t))$, $t \in [0, 1]$. By conditions (i) and (ii) we have that

$$|x(t_1) - x(t_2)| \leq 2^{1-n}, \quad |y(t_1) - y(t_2)| \leq 2^{1-n}$$

for all $t_1, t_2 \in [0, 1]$, with $|t_1 - t_2| \leq 1/4^n$. This proves the uniform continuity of u . $\square$

In view of the previous result, to recover the intuitive idea of a curve, we restrict the class of continuous curves to those with finite or σ -finite length.

Definition 5.9. Given a metric space (Y, d) and a curve γ in Y , let $u : I \rightarrow Y$ be a parametric representation of γ , where $I \subseteq \mathbb{R}$ is an interval. We define the *length* of γ as

$$L(\gamma) := \text{Var } u = \sup \left\{ \sum_{i=1}^n d(u(t_i), u(t_{i-1})) \right\},$$

where the supremum is taken over all partitions $P = \{t_0, \dots, t_n\}$ of I , $n \in \mathbb{N}$. We say that the curve γ is *rectifiable* if $L(\gamma) < \infty$.

Remark 5.10. Many authors define the notion of length only for continuous curves. This is motivated by the fact that if γ is a curve with parametric representation $u : I \rightarrow Y$, where $I \subseteq \mathbb{R}$ is an interval and u is discontinuous at $t \in I$, then the length of γ also measures the length of the segment joining the points $u(t)$ and $u_-(t)$, as well as the length of the segment joining the points $u(t)$ and $u_+(t)$ (see Corollary 5.14 below), although these segments may not belong to the curve. However, see Theorem 5.28.

Given a metric space (Y, d) and a curve γ in Y , with parametric representation $u : I \rightarrow Y$, where $I \subseteq \mathbb{R}$ is an interval, we say that γ is *locally rectifiable*, or that it has *σ -finite length*, if $\text{Var}_{[a,b]} u < \infty$ for every interval $[a, b] \subseteq I$.

Exercise 5.11. Given a metric space (Y, d) , prove that if two parametric representations $u : I \rightarrow Y$ and $v : J \rightarrow Y$ are equivalent, then $\text{Var}_I u = \text{Var}_J v$. Hence, the definitions of rectifiability and of length of a curve do not depend on the particular parametric representation.

In view of Theorem 3.25 we have

Theorem 5.12. *Given a rectifiable curve γ in $\mathbb{R}^M$ with parametric representation $u : I \rightarrow \mathbb{R}^M$, where $I \subseteq \mathbb{R}$ is an interval, then*

$$(5.2) \quad \int_I \|u'(t)\| dt \leq L(\gamma).$$

The equality holds in (5.2) if and only if $u \in AC(I; \mathbb{R}^M)$.

Example 5.13. In view of the previous theorem, if $f \in AC([a, b]; \mathbb{R}^N)$, then its graph is given by the curve γ with parametric representation

$$\begin{aligned} u : [a, b] &\rightarrow \mathbb{R}^{N+1}, \\ t &\mapsto (t, f(t)), \end{aligned}$$

and

$$L(\text{Gr } f) = L(\gamma) = \int_a^b \sqrt{1 + \|f'(t)\|^2} dt.$$

When there is strict inequality in (5.2), given a rectifiable curve γ in $\mathbb{R}^M$ with parametric representation $u : I \rightarrow \mathbb{R}^M$, as in Corollary 3.90, we may decompose $u : I \rightarrow \mathbb{R}^M$ into an absolutely continuous part, a Cantor part, and a jump part. Thus, we obtain

$$u = u_{\text{AC}} + u_{\text{C}} + u_{\text{J}},$$

where for a fixed $z_0 \in I$ and for all $t \in I$,

$$\begin{aligned} u_{\text{AC}}(t) &:= \int_{z_0}^t u'(s) ds, \\ u_{\text{J}}(t) &:= \begin{cases} \sum_{z \in I, z_0 \leq z < t} (u_+(z) - u_-(z)) + u(t) - u_-(t) & \text{if } t \geq z_0, \\ -\sum_{z \in I, t < z \leq z_0} (u_+(z) - u_-(z)) + u(t) - u_+(t), & \text{if } t \leq z_0, \end{cases} \end{aligned}$$

and $u_{\text{C}} := u - u_{\text{AC}} - u_{\text{J}}$ is continuous, with $u'_{\text{C}}(t) = 0$ for $\mathcal{L}^1$ -a.e. $t \in I$. As usual, we set $u_-(\inf I) := u(\inf I)$ if $\inf I \in I$, and $u_+(\sup I) := u(\sup I)$ if $\sup I \in I$.

Let γ_{AC} , γ_C , and γ_J be the rectifiable curves with parametric representations u_{AC} , u_C , and u_J , respectively. Then we have the following result, whose proof follows from Corollary 3.90.

Corollary 5.14. *Given a rectifiable curve γ with parametric representation $u : I \rightarrow \mathbb{R}^M$, where $I \subseteq \mathbb{R}$ is an interval, then*

$$\begin{aligned} L(\gamma) &= L(\gamma_{AC}) + L(\gamma_C) + L(\gamma_J) \\ &= \int_I \|u'(t)\| dt + \text{Var } u_C + \sum_{t \in I} (\|u_+(t) - u(t)\| + \|u(t) - u_-(t)\|), \end{aligned}$$

where $u_-(\inf I) := u(\inf I)$ if $\inf I \in I$ and $u_+(\sup I) := u(\sup I)$ if $\sup I \in I$.

Remark 5.15. If $f \in BPV([a, b]; \mathbb{R}^N)$, then its graph is given by the curve γ with parametric representation

$$\begin{aligned} u : [a, b] &\rightarrow \mathbb{R}^{N+1}, \\ t &\mapsto (t, f(t)). \end{aligned}$$

Thus, if we decompose $f = f_{AC} + f_C + f_J$ as in Corollary 3.90, we have that $u_{AC}(t) = (t, f_{AC}(t))$ and $u_C(t) = (0, f_C(t))$ for all $t \in [a, b]$. Thus,

$$\begin{aligned} L(\text{Gr } f) &= L(\gamma) = \int_a^b \sqrt{1 + \|f'(t)\|^2} dt + \text{Var } f_C \\ &\quad + \sum_{t \in [a, b]} (\|f_+(t) - f(t)\| + \|f(t) - f_-(t)\|). \end{aligned}$$

In particular, if f is continuous and increasing and if $f'(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in [a, b]$, then $f = f_C$, and so by Proposition 2.11,

$$L(\text{Gr } f) = \int_a^b 1 dt + \text{Var } f_C = b - a + f(b) - f(a).$$

Hence, if we consider the Cantor function $f : [0, 1] \rightarrow \mathbb{R}$ defined in Example 1.31, then $L(\text{Gr } f) = 2$.

Given a normed space $(Y, \|\cdot\|)$ and a curve γ , with parametric representation $u : I \rightarrow Y$, where $I \subseteq \mathbb{R}$ is an interval, if u is differentiable at $t_0 \in I$ and $u'(t_0) \neq 0$, the straight line parametrized by

$$w(t) := u(t_0) + u'(t_0)(t - t_0), \quad t \in \mathbb{R},$$

is called a *tangent line* to γ at the point $u(t_0)$, and the vector $u'(t_0)$ is called a *tangent vector* to γ at the point $u(t_0)$. While the tangent line is independent of the parametrization, the tangent vector changes with the parametrization. Indeed, by (5.1), if v is differentiable at $\phi(t_0)$ and ϕ at t_0 , then by the chain rule,

$$u'(t_0) = v'(\phi(t_0))\phi'(t_0),$$

and thus both length and orientation can change with the parametrization.

Note that if γ is a simple arc, then the tangent line at a point y of the range Γ of γ , if it exists, is unique, while if the curve is self-intersecting at some point $y \in \Gamma$, then there could be more than one tangent line at y .

We conclude this section by studying lower semicontinuity properties of the length of a curve. In view of Proposition 2.47 we have

Proposition 5.16. *Let (Y, d) be a metric space and let $\gamma, \gamma_n, n \in \mathbb{N}$, be curves with parametric representations $u, u_n : I \rightarrow Y, n \in \mathbb{N}$, respectively, where $I \subseteq \mathbb{R}$ is an interval. If*

$$(5.3) \quad u(t) = \lim_{n \rightarrow \infty} u_n(t)$$

for all $t \in I$, then

$$(5.4) \quad L(\gamma) \leq \liminf_{n \rightarrow \infty} L(\gamma_n).$$

Remark 5.17. As in Remark 2.48, if u is continuous, then it is enough to assume that (5.3) holds for all t in a dense set E of I .

Exercise 5.18. Given two intervals $I, J \subseteq \mathbb{R}$ and two functions $u : I \rightarrow \mathbb{R}^M$ and $v : J \rightarrow \mathbb{R}^M$, prove that if u and v are Fréchet equivalent, then $\text{Var}_I u = \text{Var}_J v$.

In the special case of graphs of absolutely continuous functions (see Example 5.13), equality in (5.4) implies convergence in L^1 of $\{f'_n\}_n$. Indeed, we have the following result.

Theorem 5.19. *Let $f, f_n : [a, b] \rightarrow \mathbb{R}^N, n \in \mathbb{N}$, be absolutely continuous functions such that*

$$(5.5) \quad f(t) = \lim_{n \rightarrow \infty} f_n(t)$$

for all $t \in [a, b]$ and

$$(5.6) \quad \lim_{n \rightarrow \infty} \int_a^b \sqrt{1 + \|f'_n\|^2} dt = \int_a^b \sqrt{1 + \|f'\|^2} dt.$$

Then

$$\lim_{n \rightarrow \infty} \int_a^b \left| \sqrt{1 + \|f'_n\|^2} - \sqrt{1 + \|f'\|^2} \right| dt = 0$$

and

$$\lim_{n \rightarrow \infty} \int_a^b \|f'_n - f'\| dt = 0.$$

Proof. For every $t \in [a, b]$ and $n \in \mathbb{N}$, define

$$g_n(t) := \sqrt{1 + \|f'(t)\|^2} + \sqrt{1 + \|f'_n\|^2} - 2\sqrt{1 + \frac{1}{4}\|f'(t) + f'_n(t)\|^2}.$$

Since the function $y \mapsto \sqrt{1+y^2}$ is strictly convex, we have that $g_n(t) > 0$ unless $f'(t) = f'_n(t)$. Consider the sequence

$$v_n(t) := \frac{1}{2}(f(t) + f_n(t)), \quad t \in [a, b].$$

By hypothesis, $v_n(t) \rightarrow f(t)$ for every $t \in [a, b]$, and so, by the previous proposition applied to the sequence $\{v_n\}_n$,

$$\liminf_{n \rightarrow \infty} \int_a^b \sqrt{1 + \|\frac{1}{2}(f' + f'_n)\|^2} dt \geq \int_a^b \sqrt{1 + \|f'\|^2} dt,$$

which, together with (5.6), implies that

$$\begin{aligned} \limsup_{n \rightarrow \infty} \int_a^b g_n dt &= \lim_{n \rightarrow \infty} \int_a^b \left(\sqrt{1 + \|f'\|^2} + \sqrt{1 + \|f'_n\|^2} \right) dt \\ &\quad - 2 \liminf_{n \rightarrow \infty} \int_a^b \sqrt{1 + \frac{1}{4}\|f' + f'_n\|^2} dt \leq 0. \end{aligned}$$

Since $g_n \geq 0$, we have that $g_n \rightarrow 0$ in $L^1([a, b])$. Extract a subsequence $\{g_{n_i}\}_i$ of $\{g_n\}_n$ such that $g_{n_i}(t) \rightarrow 0$ for $\mathcal{L}^1$ -a.e. $t \in [a, b]$. Let E be the set of $t \in [a, b]$ for which $g_{n_i}(t) \rightarrow 0$.

We claim that $f'_{n_i}(t) \rightarrow f'(t)$ for all $t \in E$. Indeed, assume by contradiction that this is not true. Then there exist $t_0 \in E$ and a subsequence (not relabeled) such that either $f'_{n_i}(t_0) \rightarrow \ell \in \mathbb{R}$, with $\ell \neq f'(t_0)$, or $\|f'_{n_i}(t_0)\| \rightarrow \infty$. In the first case, we have that

$$g_{n_i}(t_0) \rightarrow \sqrt{1 + \|f'(t_0)\|^2} + \sqrt{1 + \ell^2} - 2\sqrt{1 + \frac{1}{4}\|f'(t_0) + \ell\|^2} > 0,$$

which contradicts the fact that $g_{n_i}(t_0) \rightarrow 0$. Thus, assume that $\|f'_{n_i}(t_0)\| \rightarrow \infty$. Fix $y, z \in \mathbb{R}^N$, and consider the function

$$g(s) := \sqrt{1 + \|y + sz\|^2}, \quad s \in (0, \infty).$$

Since g is convex and differentiable, the function $s \in (0, \infty) \mapsto g(s) - 2g(\frac{s}{2})$ is increasing. To see this, it suffices to differentiate and use the fact that g' is an increasing function. Using this fact with $y = f'(t_0)$ and $z = f'_{n_i}(t_0) - f'(t_0)$, we have that for all $0 \leq s < 1$,

$$\begin{aligned} g_{n_i}(t_0) &= \sqrt{1 + \|f'(t_0)\|^2} + g(1) - 2g(\frac{1}{2}) \\ &\geq \sqrt{1 + \|f'(t_0)\|^2} + g(s) - 2g(\frac{s}{2}) \\ &= \sqrt{1 + \|f'(t_0)\|^2} + \sqrt{1 + \|f'(t_0) + s(f'_{n_i}(t_0) - f'(t_0))\|^2} \\ &\quad - 2\sqrt{1 + \|f'(t_0) + \frac{s}{2}(f'_{n_i}(t_0) - f'(t_0))\|^2}. \end{aligned}$$

Taking $s := \|f'_{n_i}(t_0) - f'(t_0)\|^{-1} \in (0, 1)$ for all i sufficiently large, we have that

$$g_{n_i}(t_0) \geq \sqrt{1 + \|f'(t_0)\|^2} + \sqrt{1 + \|f'(t_0) + \xi_{n_i}\|^2} - 2\sqrt{1 + \|f'(t_0) + \frac{1}{2}\xi_{n_i}\|^2},$$

where $\xi_{n_i} := \frac{f'_{n_i}(t_0) - f'(t_0)}{\|f'_{n_i}(t_0) - f'(t_0)\|}$. Since $\|\xi_{n_i}\| = 1$, we may find a subsequence (not relabeled) such that $\xi_{n_i} \rightarrow \xi$, with $\|\xi\| = 1$. Letting $i \rightarrow \infty$ in the previous inequality yields

$$0 = \lim_{i \rightarrow \infty} g_{n_i}(t_0) \geq \sqrt{1 + \|f'(t_0)\|^2} + \sqrt{1 + \|f'(t_0) + \xi\|^2} - 2\sqrt{1 + \|f'(t_0) + \frac{1}{2}\xi\|^2} > 0,$$

where the last inequality follows from the fact that $\xi \neq 0$. Thus, we have reached a contradiction even in this case, and so the claim is proved.

Next we show that

$$(5.7) \quad \lim_{i \rightarrow \infty} \int_a^b \left| \sqrt{1 + \|f'_{n_i}\|^2} - \sqrt{1 + \|f'\|^2} \right| dt = 0.$$

Using the fact that $||x| - |x - y|| \leq |y|$ for all $x, y \in \mathbb{R}$, we have

$$\left| \sqrt{1 + \|f'_{n_i}\|^2} - \left| \sqrt{1 + \|f'_{n_i}\|^2} - \sqrt{1 + \|f'\|^2} \right| \right| \leq \sqrt{1 + \|f'\|^2},$$

and thus, by the Lebesgue dominated convergence theorem and the fact that $f'_{n_i}(t) \rightarrow f'(t)$ for $\mathcal{L}^1$ -a.e. $t \in [a, b]$,

$$\begin{aligned} \lim_{i \rightarrow \infty} \int_a^b \left(\sqrt{1 + \|f'_{n_i}\|^2} - \left| \sqrt{1 + \|f'_{n_i}\|^2} - \sqrt{1 + \|f'\|^2} \right| \right) dt \\ = \int_a^b \sqrt{1 + \|f'\|^2} dt, \end{aligned}$$

which, together with (5.6), implies (5.7).

Finally, we prove that

$$(5.8) \quad \lim_{i \rightarrow \infty} \int_a^b \|f'_{n_i}\| dt = \int_a^b \|f'\| dt.$$

By Fatou's lemma,

$$\liminf_{i \rightarrow \infty} \int_a^b \|f'_{n_i}\| dt \geq \int_a^b \|f'\| dt.$$

Thus, if (5.8) fails, then there is a subsequence (not relabeled) and $\varepsilon > 0$ such that

$$(5.9) \quad \int_a^b \|f'\| dt + 3\varepsilon(b - a) \leq \int_a^b \|f'_{n_i}\| dt$$

for all $i \in \mathbb{N}$. Since $\int_a^b \sqrt{1 + \|f'\|^2} dt < \infty$, we may find $\delta = \delta(\varepsilon) > 0$ such that

$$(5.10) \quad \int_F \sqrt{1 + \|f'\|^2} dt < \varepsilon(b - a)$$

for every Lebesgue measurable set $F \subset I$, with $\mathcal{L}^1(F) < \delta$. By Egoroff's theorem there is a Lebesgue measurable set $E_\varepsilon \subset [a, b]$, with $\mathcal{L}^1(E_\varepsilon) < \delta$, such that $\{f'_{n_i}\}_i$ converges uniformly in $[a, b] \setminus E_\varepsilon$. Hence, we may find an integer $i_\varepsilon \in \mathbb{N}$ such that $\|f'_{n_i}(t)\| \leq \|f'(t)\| + \varepsilon$ for all $t \in [a, b] \setminus E_\varepsilon$ and for all $i \geq i_\varepsilon$. In turn, by (5.9) for all $i \geq i_\varepsilon$,

$$\begin{aligned} \int_a^b \|f'\| dt + 3\varepsilon(b-a) &\leq \int_a^b \|f'_{n_i}\| dt = \int_{[a,b] \setminus E_\varepsilon} \|f'_{n_i}\| dt + \int_{E_\varepsilon} \|f'_{n_i}\| dt \\ &\leq \int_a^b \|f'\| dt + \varepsilon(b-a) + \int_{E_\varepsilon} \sqrt{1 + \|f'_{n_i}\|^2} dt. \end{aligned}$$

It follows that

$$\begin{aligned} 2\varepsilon(b-a) &\leq \int_{E_\varepsilon} \sqrt{1 + \|f'_{n_i}\|^2} dt \\ &\leq \int_{E_\varepsilon} \sqrt{1 + \|f'\|^2} dt + \int_a^b \left| \sqrt{1 + \|f'_{n_i}\|^2} - \sqrt{1 + \|f'\|^2} \right| dt \\ &\leq \varepsilon(b-a) + \int_a^b \left| \sqrt{1 + \|f'_{n_i}\|^2} - \sqrt{1 + \|f'\|^2} \right| dt \end{aligned}$$

for all $i \geq i_\varepsilon$, where we have used the triangle inequality and (5.10). Letting $i \rightarrow \infty$ and using (5.7), we obtain a contradiction. Hence, (5.8) holds, and in turn, reasoning as in the proof of (5.7), we have that

$$(5.11) \quad \lim_{i \rightarrow \infty} \int_a^b \|f'_{n_i} - f'\| dt = 0.$$

To conclude, we observe that, by the uniqueness of the limit, (5.7) and (5.11) actually hold for the whole sequence (why?). $\square$

Exercise 5.20. Let $g : \mathbb{R}^N \rightarrow [0, \infty)$ be strictly convex.

- (i) Prove that if $g(s) \rightarrow \infty$ as $\|s\| \rightarrow \infty$, then there exist $c > 0$ and $S > 0$ such that $g(s) \geq c\|s\|$ for all $\|s\| \geq S$.
- (ii) Prove that if $f, f_n : [a, b] \rightarrow \mathbb{R}^N$, $n \in \mathbb{N}$, are absolutely continuous functions such that

$$f(t) = \lim_{n \rightarrow \infty} f_n(t)$$

for all $t \in [a, b]$ and

$$\lim_{n \rightarrow \infty} \int_a^b g(f'_n(t)) dt = \int_a^b g(f'(t)) dt < \infty,$$

then

$$\lim_{n \rightarrow \infty} \int_a^b |g(f'_n(t)) - g(f'(t))| dt = 0.$$

(You may use the fact that by (5.5),

$$\liminf_{n \rightarrow \infty} \int_a^b g(f'_n(t)) dt \geq \int_a^b g(f'(t)) dt;$$

see [8]).

- (iii) Prove that if in part (ii) we also assume that $g(s) \rightarrow \infty$ as $\|s\| \rightarrow \infty$, then

$$\lim_{n \rightarrow \infty} \int_a^b \|f'_n - f'\| dt = 0.$$

5.2. Arclength

Next we introduce the notion of arclength of a locally rectifiable curve.

Definition 5.21. Let (Y, d) be a metric space. Given a locally rectifiable curve γ in Y , we say that γ is parametrized by *arclength* if it admits a parametric representation $w : J \rightarrow Y$ such that

$$(5.12) \quad \text{Var}_{[s_1, s_2]} w = s_2 - s_1$$

for all $s_1, s_2 \in J$, with $s_1 < s_2$.

Remark 5.22. Note that the function w is Lipschitz continuous with Lipschitz constant less than or equal to 1. Indeed, by (2.4),

$$(5.13) \quad d(w(s_1), w(s_2)) \leq \text{Var}_{[s_1, s_2]} w = s_2 - s_1$$

for all $s_1, s_2 \in J$, with $s_1 < s_2$.

Proposition 5.23. Let γ be a curve in $\mathbb{R}^M$. Assume that γ can be parametrized by arclength and let $w : J \rightarrow \mathbb{R}^M$ be a parametric representation satisfying (5.12). Then $\|w'(s)\| = 1$ for $\mathcal{L}^1$ -a.e. $s \in J$.

Proof. Since w is Lipschitz continuous, it is differentiable for $\mathcal{L}^1$ -a.e. $s \in J$ by Corollary 2.31. It follows from (5.13) that $\|w'(s)\| \leq 1$ for $\mathcal{L}^1$ -a.e. $s \in J$. On the other hand, by Theorem 5.12 and (5.12) for every interval $J_1 \subseteq J$,

$$\mathcal{L}^1(J_1) = \text{Var}_{J_1} w = \int_{J_1} \|w'(s)\| ds.$$

Hence,

$$0 = \mathcal{L}^1(J_1) - \int_{J_1} \|w'(s)\| ds = \int_{J_1} (1 - \|w'(s)\|) ds$$

and since $\|w'(s)\| \leq 1$ for $\mathcal{L}^1$ -a.e. $s \in J_1$, necessarily $\|w'(s)\| = 1$ for $\mathcal{L}^1$ -a.e. $s \in J_1$ and the proof is complete. $\square$

The main theorem of this section is the following.

Theorem 5.24 (Arclength). *Let (Y, d) be a metric space and let γ be a continuous locally rectifiable curve. Assume that γ admits a parametric representation $u : I \rightarrow Y$ such that u is not constant on any proper subinterval of I . Then γ can be parametrized by arclength. In particular, if γ is a continuous simple locally rectifiable arc in Y , then γ can be parametrized by arclength.*

Proof. By Theorem 2.17 and Proposition 2.21 the indefinite pointwise variation $\mathbf{V} : I \rightarrow \mathbf{V}(I)$ of u , given by

$$(5.14) \quad \mathbf{V}(t) := \begin{cases} \text{Var}_{[t_0, t]} u & \text{if } t \geq t_0, \\ -\text{Var}_{[t, t_0]} u & \text{if } t < t_0, \end{cases}$$

is continuous, increasing, and onto. Moreover, $J := \mathbf{V}(I)$ is an interval. We claim that $\mathbf{V}$ is strictly increasing. Indeed, if there exist $t_1 < t_2$ such that $\mathbf{V}(t_1) = \mathbf{V}(t_2)$, then either $t_1, t_2 \geq t_0$ or $t_1, t_2 \leq t_0$ and so by Proposition 2.15, $\text{Var}_{[t_1, t_2]} u = 0$, which implies that u is constant in $[t_1, t_2]$, which contradicts the hypothesis on u . Hence, $\mathbf{V} : I \rightarrow \mathbf{V}(I)$ is invertible, and so $\mathbf{V}^{-1} : J \rightarrow I$ is continuous. Thus, the function $w : J \rightarrow X$, defined by

$$(5.15) \quad w(s) := u(\mathbf{V}^{-1}(s)), \quad s \in J,$$

is equivalent to u .

Moreover, by Theorem 2.17, for every $s_1, s_2 \in J$, with $s_1 < s_2$, we have that

$$\text{Var}_{[s_1, s_2]} w = \text{Var}_{[\mathbf{V}^{-1}(s_1), \mathbf{V}^{-1}(s_2)]} u = \mathbf{V}(\mathbf{V}^{-1}(s_2)) - \mathbf{V}(\mathbf{V}^{-1}(s_1)) = s_2 - s_1.$$

This concludes the proof. $\square$

Remark 5.25. If γ is rectifiable, then in place of the function $\mathbf{V}$ one can use the function

$$\mathbf{V}_\infty(t) := \text{Var}_{I \cap (-\infty, t]} u$$

so that J becomes the interval of endpoints 0 and $L(\gamma)$.

The next exercise shows that unlike (Lebesgue) curves, Fréchet curves can always be parametrized by arclength.

Exercise 5.26. Prove that if γ is a continuous rectifiable Fréchet curve in $\mathbb{R}^M$ with $L(\gamma) > 0$, then γ can be parametrized by arclength

Exercise 5.27. Given two intervals $I, J \subseteq \mathbb{R}$ and two continuous functions $u : I \rightarrow \mathbb{R}^M$ and $v : J \rightarrow \mathbb{R}^M$, we say that $u \approx v$ if there exists an increasing function (possibly nonstrictly increasing) $\phi : I \rightarrow J$ such that $\phi(I) = J$ and for every discontinuity point of $t \in I$ of ϕ , the interval $[\phi_-(t), \phi_+(t)]$ is contained in an interval $[\tau_1, \tau_1] \subseteq J$ on which v is constant.

(i) Prove that $\approx$ is an equivalence relation.

- (ii) Given a continuous function $u : I \rightarrow \mathbb{R}^M$ of bounded pointwise variation, prove that the inverse function $\mathbf{V}^{-1} : J \rightarrow I$ (properly defined) satisfies the conditions for an admissible parameter change and that the function

$$w(s) := u(\mathbf{V}^{-1}(s)), \quad s \in J,$$

is Lipschitz continuous.

By modifying the proof of Theorem 5.24, we can show that if a simple arc γ is not continuous in a normed space, then the range of the curve γ is contained in the range of a (different) curve parametrized by arclength and with the same length.

Theorem 5.28. *Let $(Y, \|\cdot\|)$ be a normed space and let γ be a rectifiable simple arc γ in Y with $L(\gamma) > 0$. Then there exists a curve parametrized by arclength whose range contains the range of Γ and whose length is $L(\gamma)$.*

Proof. Let $u : I \rightarrow \mathbb{R}^M$ be a parametric representation of γ , where $I \subseteq \mathbb{R}$ is an interval. Since γ may be discontinuous, the indefinite pointwise variation $\mathbf{V} : I \rightarrow \mathbf{V}(I)$ given in (5.14) is still strictly increasing and invertible, but it may be discontinuous. To be precise, in view of Proposition 2.21, u and $\mathbf{V}$ have the same set of discontinuity points. In particular $\mathbf{V}(I)$ may not be an interval. Let w be the function defined in (5.15). Then (5.13) continues to hold. Hence, the function w is continuous in $\mathbf{V}(I)$, and so it may be uniquely extended to $\overline{\mathbf{V}(I)}$ by continuity. Let J be the smallest interval containing $\mathbf{V}(I)$ and let $\{(a_n, b_n)\}_n$ be the family of connected components of $J^\circ \setminus \overline{\mathbf{V}(I)}$.

Let $t_0 \in I$ be a discontinuity point of u . Then by Corollary 3.90,

$$(5.16) \quad \mathbf{V}(t_0) - \mathbf{V}_-(t_0) = \|u(t_0) - u_-(t_0)\|,$$

$$(5.17) \quad \mathbf{V}_+(t_0) - \mathbf{V}(t_0) = \|u(t_0) - u_+(t_0)\|.$$

If $\mathbf{V}(t_0) - \mathbf{V}_-(t_0) > 0$, then there exists an interval (a_n, b_n) such that

$$(5.18) \quad a_n = \mathbf{V}_-(t_0), \quad b_n = \mathbf{V}(t_0).$$

Hence, $w(a_n) = u_-(t_0)$ and $w(b_n) = u(t_0)$. In this case, we extend the function w to be affine in the interval (a_n, b_n) . To be precise, we set

$$w(s) := \frac{u(t_0) - u_-(t_0)}{b_n - a_n}(s - a_n) + u_-(t_0), \quad s \in (a_n, b_n).$$

Then w is continuous in $[a_n, b_n]$ and $\text{Var}_{[s_1, s_2]} w = s_2 - s_1$ for all $s_1, s_2 \in (a_n, b_n)$, with $s_1 < s_2$, by (5.16) and (5.18).

Similarly, if $V^+(t_0) - V(t_0) > 0$, then there exists an interval (a_m, b_m) such that $a_m = V(t_0)$ and $b_m = V^+(t_0)$, and we define

$$w(s) := \frac{u_+(t_0) - u(t_0)}{b_m - a_m}(s - a_m) + u(t_0), \quad s \in (a_m, b_m).$$

Thus, we have extended w to J in such a way that w satisfies (5.12). The curve $\tilde{\gamma}$ with parametric representation $w : J \rightarrow \mathbb{R}^M$ has length $L(\gamma)$, and its range contains the range of γ . $\square$

5.3. Length Distance

Let (Y, d) be a metric space. Given a set $E \subseteq Y$ and two points $y_1, y_2 \in E$, consider the family $\mathcal{F}_{y_1, y_2}$ of all continuous curves with range in E and endpoints y_1 and y_2 . We are interested in finding a curve of minimal length, that is, a solution of the problem

$$(5.19) \quad \inf\{L(\gamma) : \gamma \in \mathcal{F}_{y_1, y_2}\}.$$

Any solution (if it exists) of the previous minimization problem is called a *shortest path* joining y_1 and y_2 .

Remark 5.29. Let (Y, d) be a metric space and let $E \subseteq Y$. For every $y_1, y_2 \in E$ let

$$d_L(x_1, x_2) := \inf\{L(\gamma) : \gamma \in \mathcal{F}_{y_1, y_2}\}.$$

Note that $0 \leq d_L(y_1, y_2) \leq \infty$, $d_L(y_1, y_2) = d_L(y_2, y_1)$. Moreover, since $d(y_1, y_2) \leq d_L(y_1, y_2)$, we have that $d_L(y_1, y_2) = 0$ if and only if $y_1 = y_2$. Finally, if $y_1, y_2, y_3 \in E$, and

$$d_L(y_1, y_3) + d_L(y_3, y_2) < \infty,$$

let $\gamma_1 \in \mathcal{F}_{y_1, y_3}$ and $\gamma_2 \in \mathcal{F}_{y_3, y_2}$. Then we can join γ_1 to γ_2 to get a curve γ with range in E and joining y_1 to y_2 . Hence,

$$d_L(y_1, y_2) \leq L(\gamma) \leq L(\gamma_1) + L(\gamma_2).$$

Taking first the infimum over all curves γ_1 and then over all curves γ_2 gives

$$d_L(y_1, y_2) \leq d_L(y_1, y_3) + d_L(y_3, y_2).$$

Thus, $d_L : Y \times Y \rightarrow [0, \infty]$ has all the properties of a distance but it can be infinite. It is called the *intrinsic* or *length distance with respect to E*.

A metric space (Y, d) is called a *length space* if $d_L = d$, where d_L is the length distance with respect to Y . The Euclidean space $\mathbb{R}^M$ is a length space, while the unit sphere $\partial B(0, 1)$ in $\mathbb{R}^M$ is not.

Exercise 5.30. Find the length distance of the metric space $(\mathbb{R}^2, d)$, where

$$d((x_1, y_1), (x_2, y_2)) := |x_1 - x_2| + \sqrt{|y_1 - y_2|}.$$

We recall the following definitions.

Definition 5.31. Given a metric space (Y, d) , a set $E \subseteq Y$ is said to be

- (i) *disconnected* if there exist two open sets U and V in Y such that $U \cap E$ and $V \cap E$ are nonempty, $U \cap V = \emptyset$, and $E \subseteq U \cup V$,
- (ii) *connected* if E is not disconnected,
- (iii) *pathwise connected* if for all $z_1, z_2 \in E$ there exists a continuous curve joining z_1 and z_2 and with range contained in E .

Given a metric space (Y, d) and a set $E \subseteq Y$, a *connected component* of E is a maximal connected subset of E , that is, a connected subset of E that is properly contained in no other connected subset of E .

Exercise 5.32. Let (Y, d) be a metric space and let $E \subseteq Y$. Prove that if E is pathwise connected, then E is connected.

Exercise 5.33. Let $(Y, \|\cdot\|)$ be a normed space and let $E \subseteq Y$ be an open connected set. Consider the sets

$$U := \{y \in E : \text{there exists a continuous curve} \\ \text{joining } y \text{ and } y_0 \text{ and with range contained in } E\}$$

and $B := E \setminus A$. Prove that U and V are open and conclude that E is pathwise connected.

Exercise 5.34. Let (Y, d) be a metric space and let $E_1, E_2 \subseteq Y$ be two connected sets. Prove that if $E_1 \cap \overline{E_2}$ is nonempty, then $E_1 \cup E_2$ is connected.

Exercise 5.35. Prove that the set $E = E_1 \cup E_2$ of $\mathbb{R}^2$, where

$$E_1 := \{(0, x_2) : -1 \leq x_2 \leq 1\}, \quad E_2 := \{(x_1, \sin 1/x_1) : x_1 > 0\},$$

is connected but not pathwise connected.

Exercise 5.36. Prove that in a length space every ball $B(y, r)$ contains a ball $B(y, \varepsilon)$ which is path connected.

To prove the existence of shortest paths we will rely on the following compactness theorem.

Theorem 5.37 (Ascoli–Arzelà). *Let $I \subset \mathbb{R}$ be a closed bounded interval and let $(Y, \|\cdot\|_Y)$ be a Banach space. Then $\mathcal{U} \subseteq C(I; Y)$ is relatively compact in $C(I; Y)$ if and only*

- (i) *for every $x \in I^\circ$, the set $\{u(x) : u \in \mathcal{U}\}$ is relatively compact in Y ,*
- (ii) *$\mathcal{U}$ is uniformly equicontinuous, that is, for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|u(x_1) - u(x_2)\|_Y \leq \varepsilon$ for all $x_1, x_2 \in I$ with $|x_1 - x_2| \leq \delta$ and for all $u \in \mathcal{U}$.*

Proof. We only prove that (i) and (ii) implies that $\mathcal{U} \subseteq C(I; Y)$ is relatively compact in $C(I; Y)$ and leave the other implication as an exercise.

Let $I = [a, b]$. We claim that the set $\{u(a) : u \in \mathcal{U}\}$ is relatively compact in Y . Indeed, fix $\varepsilon > 0$ and let $0 < \delta < b - a$ be as in (ii). Since the set $\{u(a + \delta) : u \in \mathcal{U}\}$ is relatively compact in Y , there exist $y_1, \dots, y_m \in Y$ such that for every $u \in \mathcal{U}$ there is $i \in \{1, \dots, m\}$ with $\|u(a + \delta) - y_i\|_Y \leq \varepsilon$. In turn, by (ii) and the triangle inequality, $\|u(a) - y_i\|_Y \leq 2\varepsilon$, which proves the claim. Similarly, $\{u(b) : u \in \mathcal{U}\}$ is relatively compact in Y .

Next, given $\varepsilon > 0$ let $\delta > 0$ be the number given in (ii). Consider a partition $\{x_0, \dots, x_n\}$ of $[a, b]$ into subintervals of length less than or equal to δ . Since for each i the set $\{u(x_i) : u \in \mathcal{U}\}$ is relatively compact, there exist $y_{1,i}, \dots, y_{m_i,i} \in Y$ such that for every $u \in \mathcal{U}$ there is $j \in \{1, \dots, m_i\}$ with $\|u(x_i) - y_{j,i}\|_Y \leq \varepsilon$.

For every $u \in \mathcal{U}$ let u_n be a continuous function such that $u_n(x_i) = u(x_i)$ for all $i = 1, \dots, n$ and u_n is affine in $[x_{i-1}, x_i]$. If $x \in [x_{i-1}, x_i]$, then by (ii)

$$\begin{aligned} \|u(x) - u_n(x)\|_Y &\leq \|u(x) - u(x_i)\|_Y + \|u(x_i) - u_n(x)\|_Y \\ &\leq \varepsilon + \|u(x_i) - u(x_{i-1})\|_Y \leq 2\varepsilon. \end{aligned}$$

Hence, $\|u - u_n\|_{C(I; Y)} \leq 2\varepsilon$.

Consider the finite family $\mathcal{V} \subset C(I; Y)$ of all piecewise affine functions v such that for every $i = 1, \dots, n$, $v(x_i) \in \{y_{1,i}, \dots, y_{m_i,i}\}$, and v is affine in $[x_{i-1}, x_i]$.

For every $u \in \mathcal{U}$ there exists $v \in \mathcal{V}$ such that $\|u(x_i) - v(x_i)\|_Y \leq \varepsilon$ for all $i = 1, \dots, n$. In turn, by (ii), for every $i = 1, \dots, n$, if $x \in [x_{i-1}, x_i]$,

$$\|u(x) - v(x)\|_Y \leq \|u(x) - u_n(x)\|_Y + \|u_n(x) - v(x)\|_Y \leq 4\varepsilon.$$

This completes the proof. $\square$

Theorem 5.38 (Existence of shortest paths). *Let $C \subseteq \mathbb{R}^M$ be a closed set, let $y_1, y_2 \in C$, and let $\mathcal{F}_{y_1, y_2}$ be the family of all continuous curves with range in C and endpoints y_1 and y_2 . If the family $\mathcal{F}_{y_1, y_2}$ is nonempty, then problem (5.19) admits a solution.*

Proof. Let

$$L := \inf\{L(\gamma) : \gamma \in \mathcal{F}_{y_1, y_2}\}.$$

If $L = \infty$, then any $\gamma \in \mathcal{F}_{y_1, y_2}$ will do, while if $L = 0$, then $y_1 = y_2$ and the problem becomes trivial. Thus, assume that $0 < L < \infty$ and let $\{\gamma_n\}_n$ be a sequence of curves in $\mathcal{F}_{y_1, y_2}$ such that

$$\lim_{n \rightarrow \infty} L(\gamma_n) = L.$$

Let $u_n : [a_n, b_n] \rightarrow \mathbb{R}^M$ be a parametric representation of γ_n with $u_n(a_n) = y_1$ and $u_n(b_n) = y_2$. Note that in general we are not in a position to apply

Theorem 5.24 to γ_n , since u_n may be constant on a proper interval. However, by Corollary 2.24 and the fact that $L(\gamma_n) < \infty$, we can write

$$u_n = w_n \circ \mathbf{V}_n,$$

where $\mathbf{V}_n(t) := \text{Var}_{[a_n, t]} u$, $w_n : [0, L(\gamma_n)] \rightarrow \mathbb{R}^M$ is Lipschitz continuous with Lipschitz constant at most 1, and $L(\gamma_n) = \text{Var}_{I_n} u_n$. Define

$$v_n(s) := w_n(sL(\gamma_n)), \quad s \in [0, 1].$$

Then v_n is Lipschitz continuous with Lipschitz constant at most $L(\gamma_n)$, $v_n(0) = y_1$, $v_n(1) = y_2$ and $\text{Var}_{[0, 1]} v_n = L(\gamma_n) \leq L + 1$ for all n sufficiently large. By the Ascoli–Arzelà theorem there exists a function $v : [0, 1] \rightarrow \mathbb{R}^M$ and a subsequence (not relabeled) such that $v_n \rightarrow v$ uniformly in $[0, 1]$ as $n \rightarrow \infty$. Using the facts that

$$\|v_n(s_1) - v_n(s_2)\| \leq L(\gamma_n)|s_1 - s_2|$$

for all $s_1, s_2 \in [0, 1]$ and that $L(\gamma_n) \rightarrow L$, letting $n \rightarrow \infty$, we obtain that v is Lipschitz continuous with Lipschitz constant at most L . Moreover, $v(0) = y_1$, $v(1) = y_2$, and $v([0, 1]) \subseteq C$, since C is closed. Thus, the curve γ parametrized by w belongs to $\mathcal{F}_{y_1, y_2}$, and so $L(\gamma) \geq L$. On the other hand, by Proposition 5.16,

$$L(\gamma) \leq \liminf_{n \rightarrow \infty} L(\gamma_n) = \lim_{n \rightarrow \infty} L(\gamma_n) = L,$$

and so $L(\gamma) = L$ and the proof is concluded. $\square$

Exercise 5.39. Prove that the previous theorem continues to hold if $\mathbb{R}^M$ is replaced by a proper metric space (Y, d) , that is, a metric space in which $\{y \in Y : d(y, y_0) \leq r\}$ is compact for every $y_0 \in Y$ and $r > 0$.

5.4. Curves and Hausdorff Measure

In this section we study the relation between curves and the Hausdorff measure $\mathcal{H}^1$.

Theorem 5.40. *Let γ be a locally rectifiable continuous simple arc in $\mathbb{R}^M$ with range Γ and let $f : \Gamma \rightarrow [0, \infty]$ be an $\mathcal{H}^1$ measurable function. Then*

$$(5.20) \quad \int_{\Gamma} f(y) d\mathcal{H}^1(y) = \int_I f(u(t)) \|u'(t)\| dt,$$

where $u : I \rightarrow \mathbb{R}^M$ is any Lipschitz continuous parametric representation of γ .

Note that γ admits a Lipschitz continuous parametric representation in view of Theorem 5.24.

Proof. Step 1: Assume first that γ is rectifiable and that $f = \chi_W$, where $W \subseteq \Gamma$ is a relatively open set in Γ . Then $u^{-1}(W)$ is relatively open in I and thus we can find countably many intervals I_k such that

$$(5.21) \quad u^{-1}(W) = \bigcup_k I_k.$$

We claim that

$$(5.22) \quad \mathcal{H}^1(u(I_k)) = \int_{I_k} \|u'(t)\| dt.$$

To see this, fix a partition $P := \{t_0, \dots, t_n\}$ of I_k . Since γ is a simple arc, u is injective. Moreover, $\mathcal{H}^1(\{y\}) = 0$ for all $y \in Y$. Hence, by Lemma 2.63, we have that

$$\begin{aligned} \mathcal{H}^1(u(I_k)) &\geq \mathcal{H}^1(u([t_0, t_n])) = \mathcal{H}^1\left(\bigcup_{i=1}^n u([t_{i-1}, t_i])\right) \\ &= \sum_{i=1}^n \mathcal{H}^1(u([t_{i-1}, t_i])) \geq \sum_{i=1}^n \|u(t_i) - u(t_{i-1})\|. \end{aligned}$$

Taking the supremum over all partitions P of I_k gives $\mathcal{H}^1(u(I_k)) \geq \text{Var}_{I_k} u$. On the other hand, reasoning exactly as in the second part of the proof of Lemma 2.63, we have that $\mathcal{H}^1(u(I_k)) \leq \text{Var}_{I_k} u$. Thus, $\mathcal{H}^1(u(I_k)) = \text{Var}_{I_k} u$. The claim (5.22) now follows from Theorem 5.12.

Summing both sides of (5.22) and using (5.21) and the fact that u is injective gives

$$\begin{aligned} \int_{\Gamma} f(y) d\mathcal{H}^1(y) &= \mathcal{H}^1(W) = \mathcal{H}^1\left(\bigcup_k u(I_k)\right) \\ &= \sum_k \mathcal{H}^1(u(I_k)) = \sum_k \int_{I_k} \|u'(t)\| dt \\ &= \int_{u^{-1}(W)} \|u'(t)\| dt = \int_I f(u(t)) \|u'(t)\| dt. \end{aligned}$$

Step 2: Assume next that γ is rectifiable and that $f = \chi_E$, where $E \subseteq \Gamma$ is an $\mathcal{H}^1$ measurable set. Then for every $k \in \mathbb{N}$ there exists a relatively open set $W_k \subseteq \Gamma$ containing E such that

$$\mathcal{H}^1(W_k \setminus E) \leq 1/k.$$

Setting $B := \bigcap_k W_k$, we have that $\mathcal{H}^1(B \setminus E) = 0$. Since $V_n := \bigcap_{k=1}^n W_k$ is relatively open in Γ , by Step 1,

$$\int_{\Gamma} \chi_{V_n}(y) d\mathcal{H}^1(y) = \int_I \chi_{V_n}(u(t)) \|u'(t)\| dt.$$

Since γ is rectifiable, by Step 1, $\mathcal{H}^1(\Gamma) = \text{Var}_I u < \infty$. Letting $n \rightarrow \infty$ it follows by the Lebesgue dominated convergence theorem applied to both sides that

$$(5.23) \quad \int_{\Gamma} \chi_B(y) d\mathcal{H}^1(y) = \int_I \chi_B(u(t)) \|u'(t)\| dt.$$

Let $F := \{t \in I : u \text{ is differentiable at } t\}$. Then $\mathcal{L}^1(I \setminus F) = 0$ and so by Theorem 3.41, $\mathcal{H}^1(u(I \setminus F)) = 0$.

On the other hand, since $\mathcal{H}^1(B \setminus E) = 0$, by Lemma 3.60, $u'(t) = 0$ for $\mathcal{L}^1$ -a.e. $t \in u^{-1}(B \setminus E) \cap F$. Hence,

$$\int_I \chi_{B \setminus E}(u(t)) \|u'(t)\| dt = \int_{I \cap F} \chi_{B \setminus E}(u(t)) \|u'(t)\| dt = 0.$$

Since $\mathcal{H}^1(B \setminus E) = 0$ it follows from the previous equality and (5.23) that

$$(5.24) \quad \int_{\Gamma} \chi_E(y) d\mathcal{H}^1(y) = \int_I \chi_E(u(t)) \|u'(t)\| dt.$$

Step 3: The case in which Γ is rectifiable and f is a simple function follows by (5.23) and the additivity of the integrals. If f is an $\mathcal{H}^1$ measurable function, find an increasing sequence of simple functions converging pointwise to f and apply the Lebesgue monotone convergence theorem to both sides.

Finally, if γ is locally rectifiable, consider an increasing sequence of closed intervals $[a_n, b_n] \subseteq I$ whose union gives I , by applying (5.20) to the rectifiable curve γ_n parametrized by $u : [a_n, b_n] \rightarrow \mathbb{R}^M$ and then applying the Lebesgue monotone convergence theorem to both sides. $\square$

Remark 5.41. Since for a simple arc γ we have that $N_u(y; I) = 1$ for all y in the range of γ , the equality $\mathcal{H}^1(u(I_k)) = \text{Var}_{I_k} u$ is just a special case of Banach's theorem, Theorem 2.60.

Remark 5.42. It follows from the previous result with $f = 1$ that the Hausdorff dimension of a rectifiable continuous simple arc is one. This is in sharp contrast to the Peano curve whose Hausdorff dimension is two.

Remark 5.43. If γ is parametrized by arclength $v : J \rightarrow \mathbb{R}^M$, then by Proposition 5.23, $\|v'(s)\| = 1$ for $\mathcal{L}^1$ -a.e. $s \in J$ and so (5.20) takes the simpler form

$$\int_{\Gamma} f(y) d\mathcal{H}^1(y) = \int_J f(v(s)) ds.$$

The right-hand side is called *curve* or *line integral* and is often denoted $\int_{\gamma} f ds$.

Remark 5.44. As usual, if $f : \Gamma \rightarrow \mathbb{R}$ is an $\mathcal{H}^1$ integrable function, then by writing $f = f^+ - f^-$, where f^+ and f^- are the positive and negative parts of f , and applying (5.20) to f^+ and f^- , we conclude that (5.20) holds for f .

5.5. Jordan's Curve Theorem

We conclude this chapter with the Jordan curve theorem. We recall that a pathwise connected set is connected, but the converse is not true in general (see Exercise 5.35). However, if a set is open, then it is connected if and only if it is pathwise connected. In what follows, we will use this fact without further notice. In this section we use the notation $z = (x, y)$ for a generic point of $\mathbb{R}^2$. Given two points $z_1, z_2 \in \mathbb{R}^2$, we denote by $\overrightarrow{z_1 z_2}$ the segment joining them, that is,

$$\overrightarrow{z_1 z_2} := \{tz_1 + (1-t)z_2 : t \in [0, 1]\}.$$

We recall that a closed curve is continuous.

Theorem 5.45 (Jordan's curve theorem). *Given a (continuous) closed simple curve γ in $\mathbb{R}^2$ with range Γ , the set $\mathbb{R}^2 \setminus \Gamma$ consists of two connected components.*

We call the unbounded connected component the *exterior component* of γ and the other one the *interior component* of γ .

We begin with some preliminary results.

Lemma 5.46. *Let $a < b$, $c < d$ and let $u : [-1, 1] \rightarrow [a, b] \times [c, d]$ and $v : [-1, 1] \rightarrow [a, b] \times [c, d]$ be two continuous functions such that $u_1(-1) = a$, $u_1(1) = b$, and $v_2(-1) = c$, $v_2(1) = d$. Then there exists $t_0, s_0 \in [-1, 1]$ such that $u(s_0) = v(t_0)$.*

Proof. Assume by contradiction that $u(s) \neq v(t)$ for all $s, t \in [-1, 1]$. Let $Q := [-1, 1] \times [-1, 1]$ and consider the continuous function $h : Q \rightarrow Q$ defined by

$$h(s, t) := \left(\frac{v_1(t) - u_1(s)}{\|u(s) - v(t)\|_\infty}, \frac{u_2(s) - v_2(t)}{\|u(s) - v(t)\|_\infty} \right),$$

where $\|z\|_\infty := \max\{|x|, |y|\}$. Note that $h(s, t) \in \partial Q$. By the Brouwer fixed point theorem (see Theorem 9.36 below) there exists $(s_0, t_0) \in Q$ such that $h(s_0, t_0) = (s_0, t_0)$. But since $h(Q) \subseteq \partial Q$, necessarily, $(s_0, t_0) \in \partial Q$. Hence, $s_0 = \pm 1$ or $t_0 = \pm 1$. If $s_0 = 1$, then

$$1 = \frac{v_1(t_0) - u_1(1)}{\|u(1) - v(t_0)\|_\infty} = \frac{v_1(t_0) - b}{\|u(1) - v(t_0)\|_\infty} \leq 0,$$

while if $s_0 = -1$, then

$$-1 = \frac{v_1(t_0) - u_1(-1)}{\|u(-1) - v(t_0)\|_\infty} = \frac{v_1(t_0) - a}{\|u(-1) - v(t_0)\|_\infty} \geq 0,$$

which gives a contradiction.

On the other hand, if $t_0 = 1$, then

$$1 = \frac{u_2(s_0) - v_2(1)}{\|u(s_0) - v(1)\|_\infty} = \frac{u_2(s_0) - d}{\|u(s_0) - v(1)\|_\infty} \leq 0,$$

while if $t_0 = -1$, then

$$-1 = \frac{u_2(s_0) - v_2(-1)}{\|u(s_0) - v(-1)\|_\infty} = \frac{u_2(s_0) - c}{\|u(s_0) - v(1)\|_\infty} \geq 0,$$

which gives again a contradiction. This completes the proof. $\square$

Remark 5.47. Since Γ is bounded, the set $\mathbb{R}^2 \setminus \Gamma$ has only one unbounded component. Indeed, let $\Gamma \subseteq B(0, r)$. The set $\mathbb{R}^2 \setminus \overline{B(0, r)}$ is open and pathwise connected and so it is connected. Moreover, $\mathbb{R}^2 \setminus \overline{B(0, r)} \subseteq \mathbb{R}^2 \setminus \Gamma$, and so there is a connected component that contains $\mathbb{R}^2 \setminus \overline{B(0, r)}$.

Lemma 5.48. *Given a (continuous) closed simple curve γ in $\mathbb{R}^2$ with range Γ , if U is a connected component of $\mathbb{R}^2 \setminus \Gamma$, then $\partial U = \Gamma$.*

Proof. If V is any other connected component of $\mathbb{R}^2 \setminus \Gamma$, then by Exercise 5.34, $\overline{U} \cap V = \emptyset$, since otherwise $U \cup V$ would be connected, and this would contradict the maximality of U . Thus $\partial U \subseteq \Gamma$. Assume by contradiction that $\partial U \subset \Gamma$. Then there exists a simple arc whose range $C \subset \Gamma$ is closed and contains ∂U .

Moreover, since $\partial U \subset \Gamma$, there must be at least another connected component. Since there is only one unbounded connected component, there is a bounded connected component. Take p to be a point in this bounded connected component. If U is bounded, take p in U . Let $r > 0$ be so large that $B(p, r) \supset \Gamma$. Then $\partial B(p, r)$ is contained in the unbounded connected component of $\mathbb{R}^2 \setminus \Gamma$.

Define $u(z) := z$ for all $z \in C$. Since C is homeomorphic to $[0, 1]$, by Tietze's extension theorem (see Theorem A.10 in the appendix) we can extend u to a continuous function $u : \overline{B(p, r)} \rightarrow C$. If U is bounded, define

$$v(z) := \begin{cases} u(z) & \text{if } z \in \overline{U}, \\ z & \text{if } z \in \mathbb{R}^2 \setminus U, \end{cases}$$

while if U is unbounded, define

$$v(z) := \begin{cases} z & \text{if } z \in \overline{U}, \\ u(z) & \text{if } z \in \mathbb{R}^2 \setminus U. \end{cases}$$

Then $v : \overline{B(p, r)} \rightarrow \overline{B(p, r)} \setminus \{p\}$. Indeed, if U is bounded, then p belongs to U and v is mapped into $C \subset \Gamma$, while if U is unbounded, then p belongs to $\mathbb{R}^2 \setminus U$ and v is mapped into $C \subset \Gamma$. Moreover, since $\partial U \subseteq C$ and $v(z) = z$ for all $z \in C$, we have that v is continuous. Note that $v(z) = z$ for all $z \in \partial B(p, r)$. Consider the projection $\pi : \overline{B(p, r)} \setminus \{p\} \rightarrow \partial B(p, r)$ given

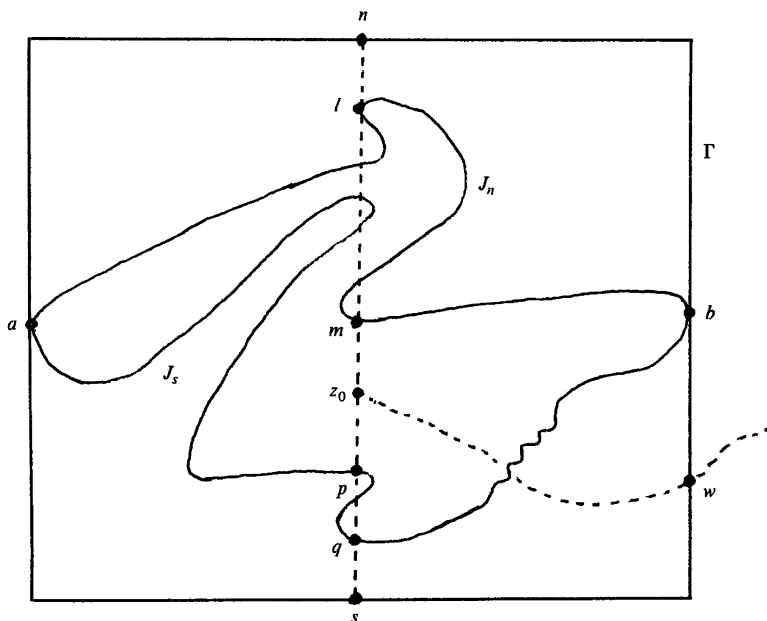


Figure 2. Figure from R. Maehara, The Jordan curve theorem via the Brouwer fixed point theorem, Amer. Math. Monthly 91 (1984), no. 10, 641–643. Copyright 1984 Mathematical Association of America. All Rights Reserved. Reprinted with permission.

by $\pi(z) := p + r \frac{z-p}{\|z-p\|}$ and let $h : \partial B(p, r) \rightarrow \partial B(p, r)$ be a rotation by 180 degrees. Then the continuous function $h \circ \pi \circ v : \overline{B(p, r)} \rightarrow \partial B(p, r) \subset \overline{B(p, r)}$ has no fixed point. This contradicts the Brouwer fixed point theorem (see Theorem 9.36 below). $\square$

We now turn to the proof of Jordan's curve theorem. Since γ is a simple curve, in what follows, with an abuse of notation, we will often confuse γ with its range Γ .

Proof of Jordan's curve theorem. Since Γ is compact, there exist $a, b \in \Gamma$ such that $\|a - b\| = \text{diam } \Gamma$. By changing coordinates, without loss of generality we may assume that $a = (-1, 0)$ and $b = (1, 0)$. Then the rectangle $R = [-1, 1] \times [-2, 2]$ contains Γ and has only a and b on its boundary (see Figure 2). Let $n = (0, 2)$ and $s = (0, -2)$. By Lemma 5.46 the segment $\overrightarrow{ns}$ intersects Γ . Let l be the point in $\Gamma \cap \overrightarrow{ns}$ with maximal y -component.

The points a and b divide Γ in two arcs, let Γ_n be the one containing l and let Γ_s be the other one. Let m be the point in $\Gamma_n \cap \overrightarrow{ns}$ with minimal y -component.

Then the segment $\overrightarrow{ms}$ intersects Γ_s , since otherwise, denoting by $\widehat{lm}$ the subarc contained in Γ_n with endpoints l and m , the curve given by $\overrightarrow{nl} + \widehat{lm} + \overrightarrow{ms}$ would not intersect the curve Γ_s , contradicting Lemma 5.46. Let p and q be the points in $\Gamma_s \cap \overrightarrow{ms}$ be the points with maximal and minimal y -component, respectively. Finally, let z_0 be the middle point of the segment $\overrightarrow{mp}$. Note that $m \neq p$ since γ is simple. Hence, z_0 does not belong to Γ . Let U be the connected component of $\mathbb{R}^2 \setminus \Gamma$ which contains z_0 . We claim that U is bounded.

Assume by contradiction that U is unbounded. Since U is open and connected, it is path-connected and so we can find a polygonal path in U joining z_0 to a point outside R . Let w be the point at which this polygonal path first intersects ∂R and denote with γ_1 the portion of this polygonal arc joining z_0 and w . If w is on the lower half of R , let $\widehat{ws}$ be the subarc contained in ∂R with endpoints w and s and not intersecting a, b . Then the curve given by $\overrightarrow{nl} + \widehat{lm} + \overrightarrow{mz_0} + \gamma_1 + \widehat{ws}$ would not intersect the curve Γ_s , contradicting Lemma 5.46.

On the other hand, if w is on the upper half of R , let $\widehat{wn}$ be the subarc contained in ∂R with endpoints w and n and not intersecting a, b . Then the curve $\overrightarrow{sz_0} + \gamma_1 + \widehat{wn}$ would not intersect the curve Γ_n , contradicting again Lemma 5.46.

This shows that U is bounded. It remains to show that U is the only bounded connected component of $\mathbb{R}^2 \setminus \Gamma$. Assume by contradiction that there is another one, say, V . Then $V \subset R$, since $\mathbb{R}^2 \setminus R$ is pathwise connected and thus contained in the unbounded connected component of $\mathbb{R}^2 \setminus \Gamma$. Since the segments $\overrightarrow{nl} \setminus \{l\}$ and $\overrightarrow{qs} \setminus \{q\}$ are contained in the unbounded connected component of $\mathbb{R}^2 \setminus \Gamma$, they do not intersect V . Similarly, since $z_0 \in U$, the segment $\overrightarrow{mp} \setminus \{m, p\}$ is contained in U , and thus it does not intersect V . It follows that the curve γ_2 given by $\overrightarrow{nl} + \widehat{lm} + \overrightarrow{mp} + \widehat{pq} + \overrightarrow{qs}$, where $\widehat{pq}$ the subarc contained in Γ_s with endpoints p and q , does not intersect V . Since a and b are not in γ_2 , there are balls $B(a, r)$ and $B(b, r)$ which do not intersect γ_2 . By Lemma 5.48, $\partial V = \Gamma$ and so a and b belong to ∂V . Hence, there exist $a_1 \in V \cap B(a, r)$ and $b_1 \in V \cap B(b, r)$. Let $\widehat{a_1 b_1}$ be a polygonal path in V joining a_1 and b_1 . Then the curve $\overrightarrow{aa_1} + \widehat{a_1 b_1} + \overrightarrow{b_1 b}$ does not intersect γ_2 , contradicting again Lemma 5.46. This concludes the proof. $\square$

Lebesgue–Stieltjes Measures

*Undergradese, VI: “Are you going to have office hours today?”
Translation: “Can I do my homework in your office?”*

— Jorge Cham, www.phdcomics.com

In the literature there have been several attempts to extend the notion of pointwise variation to functions $u : R \rightarrow \mathbb{R}^M$, where now R is a rectangle of $\mathbb{R}^N$, but they have never been quite satisfactory. As we will see in the next chapters, a more successful approach is to give a different characterization of functions of bounded pointwise variation, namely as the class of functions whose “derivative” is a finite measure. We will make this concept more precise in Chapter 7. In this chapter we describe the correspondence between functions of bounded pointwise variation and Radon measures. We will start with increasing functions. We will show that there is a one-to-one correspondence between increasing, right continuous functions and (positive) Radon measures. In this chapter

$I \subseteq \mathbb{R}$ is an open interval

and $\mathcal{L}^1$ stands for the Lebesgue measure restricted to Lebesgue measurable subsets of I . Given an interval $J \subseteq \mathbb{R}$ we denote by $\mathcal{S}(J)$ the semiring of all intervals $(a, b]$, where $a, b \in J$, with $a \leq b$, and by $\mathcal{R}(J)$ the ring generated by $\mathcal{S}(J)$. The smallest σ -algebra containing $\mathcal{R}(J)$ is the Borel σ -algebra $\mathcal{B}(J)$. Finally, we recall that a *Radon measure* $\mu : \mathcal{B}(J) \rightarrow [0, \infty]$ is a measure finite on compact subsets of J .

6.1. Measures Versus Increasing Functions

The next result shows that every (positive) Radon measure gives rise to a right (or left) continuous increasing function.

Theorem 6.1. *Let $I \subseteq \mathbb{R}$ be an open interval and let $\mu : \mathcal{B}(I) \rightarrow [0, \infty]$ be a Radon measure. Then for a fixed $t_0 \in I$ the function $u_\mu : I \rightarrow \mathbb{R}$, defined by*

$$(6.1) \quad u_\mu(x) := \begin{cases} \mu((t_0, x]) & \text{if } x \geq t_0, \\ -\mu((x, t_0]) & \text{if } x < t_0, \end{cases}$$

is increasing and right continuous and

$$(6.2) \quad u_\mu(b) - u_\mu(a) = \mu((a, b])$$

for all $a, b \in I$, with $a \leq b$.

Proof. By the properties of measures, if $E, F \in \mathcal{B}(I)$, with $E \subseteq F$, we have that $\mu(E) \leq \mu(F)$, and so, if $t_0 \leq x_1 \leq x_2$, then

$$u_\mu(x_1) = \mu((t_0, x_1]) \leq \mu((t_0, x_2]) = u_\mu(x_2),$$

while if $x_1 \leq x_2 \leq t_0$, then $\mu((x_1, t_0]) \geq \mu((x_2, t_0])$, and so

$$u_\mu(x_1) = -\mu((x_1, t_0]) \leq -\mu((x_2, t_0]) = u_\mu(x_2).$$

Finally, if $x_1 \leq t_0 \leq x_2$, then $u_\mu(x_1) \leq 0 \leq u_\mu(x_2)$. Hence, u_μ is increasing.

To prove that u_μ is right continuous, fix $x \in I$ and let $x_n \rightarrow x^+$. Since u_μ is increasing, by extracting a subsequence it suffices to assume that $\{x_n\}_n$ is decreasing. There are two cases. If $x \geq t_0$, then, since $\mu((t_0, x]) < \infty$, by Proposition B.9(ii) we have that

$$\lim_{n \rightarrow \infty} \mu((t_0, x_n]) = \mu\left(\bigcap_{n=1}^{\infty} (t_0, x_n]\right) = \mu((t_0, x]),$$

while if $x < t_0$, then by Proposition B.9(i),

$$\lim_{n \rightarrow \infty} \mu((x_n, t_0]) = \mu\left(\bigcup_{n=1}^{\infty} (x_n, t_0]\right) = \mu((x, t_0]).$$

Thus, u_μ is right continuous.

Property (6.2) follows from the definition of u_μ . □

Remark 6.2. (i) Note that if we work with intervals of the type $[a, b)$, then the function u_μ defined correspondingly is left continuous.

(ii) If $\mu : \mathcal{B}(I) \rightarrow [0, \infty)$ is a finite measure, then we can work with the simpler function

$$u(x) := \mu(I \cap (-\infty, x]), \quad x \in I.$$

Next we show the relation between the properties of the measure μ and of the increasing function u_μ . Set

$$I_{\text{dis}} := \{x \in I : u_\mu \text{ is discontinuous at } x\},$$

$$I_{\text{der0}} := \{x \in I : u'_\mu(x) = 0\}.$$

Theorem 6.3. *Let $I \subseteq \mathbb{R}$ be an open interval and let $\mu : \mathcal{B}(I) \rightarrow [0, \infty]$ be a Radon measure and let u_μ be the function given in (6.1). Then*

- (i) *The measure μ is purely atomic if and only if u_μ is a jump function. Moreover, in this case, for every Borel set $E \subseteq I$,*

$$(6.3) \quad \mu(E) = \sum_{x \in I_{\text{dis}}} \mu(E \cap \{x\}).$$

- (ii) *The measure μ is absolutely continuous with respect to $\mathcal{L}^1$ if and only if u_μ is locally absolutely continuous. Moreover, in this case, for every Borel set $E \subseteq I$,*

$$(6.4) \quad \mu(E) = \int_E u'_\mu dx.$$

- (iii) *The measure μ is nonatomic and $\mu \perp \mathcal{L}^1$ if and only if u_μ is a singular function. Moreover, in this case,*

$$(6.5) \quad \mu(E) = \mu(E \setminus I_{\text{der}0})$$

for every Borel set $E \subseteq I$.

Lemma 6.4. *Let $I \subseteq \mathbb{R}$ be an open interval and let $\mu : \mathcal{B}(I) \rightarrow [0, \infty]$ be a Radon measure. Then $E \in \mathcal{B}(I)$ is an atom if and only if, up to a set of measure zero, $E = \{x\}$ with $x \in I$ a discontinuity point of u_μ .*

Proof. Let $E \in \mathcal{B}(I)$ be an atom. In what follows we will neglect sets of measure zero. Then $\ell := \mu(E) > 0$ and for every Borel set $F \subseteq E$, either $\mu(F) = 0$ or $\mu(F) = \ell$. Partition I into intervals $I_{i,1}$ of the form $(a, b]$ with endpoints in I and of length less than or equal to 1. Since $\mu(\overline{I_{i,1}}) < \infty$ for every i and $I_{i,1} \cap E$ has either measure zero or ℓ , it follows that $\ell < \infty$. In turn, since the sets $I_{i,1} \cap E$, $i \in \mathbb{N}$, are disjoint, there exists only one $i_1 \in \mathbb{N}$ such that $\mu(I_{i_1,1} \cap E) = \ell$, while $\mu(I_{i,1} \cap E) = 0$ for all $i \neq i_1$. This implies that $E \subseteq I_{i_1,1} =: I_1$.

Inductively, assume that $E \subseteq I_n \subseteq \cdots \subseteq I_1$ have been selected with $\text{diam } I_n \leq \frac{1}{n}$ and $\mu(I_n) = \ell$. Partition I_n into intervals $I_{i,n+1}$ of the form $(a, b]$ and of length less than or equal to $\frac{1}{n+1}$ and let i_{n+1} be the only integer such that $\mu(I_{i_{n+1},n+1} \cap E) = \ell$, while $\mu(I_{i,n+1} \cap E) = 0$ for $i \neq i_{n+1}$. Then $E \subseteq I_{i_{n+1},n+1} =: I_{n+1}$. Thus we have constructed a decreasing sequence of intervals I_n containing E , with $\mu(I_n) = \ell$ and $\text{diam } I_n \leq \frac{1}{n}$. But then $E \subseteq \bigcap_{n=1}^{\infty} I_n$. Since $\text{diam } I_n \rightarrow 0$, it follows that E must be a singleton, say, $E = \{x\}$. Moreover, by (6.2) we have that

$$u_\mu(x) - u_\mu(x-h) = \mu((x-h, x])$$

for all $h > 0$ such that $x-h \in I$. Letting $h \rightarrow 0^+$ gives

$$(6.6) \quad u_\mu(x) - (u_\mu)_-(x) = \mu(\{x\}).$$

Thus, $\{x\}$ is an atom of μ if and only if x is a discontinuity point of u_μ . $\square$

Lemma 6.5. *Let $I \subseteq \mathbb{R}$ be an open interval and let $\mu : \mathcal{B}(I) \rightarrow [0, \infty]$ be a Radon measure. Then for every Borel set $E \subseteq I$,*

$$(6.7) \quad \mathcal{L}_o^1(u_\mu(E)) \leq \mu(E).$$

Moreover, if u_μ is continuous in I , then equality holds in (6.7).

Proof. Given $\varepsilon > 0$ we can find an open set $U \subseteq I$ containing E such that $\mu(U \setminus E) \leq \varepsilon$. Decompose U into countably many disjoint intervals of the form $(a_i, b_i]$. Then the set $\bigcup_i [u_\mu(a_i), u_\mu(b_i)]$ covers $u_\mu(E)$, and so

$$\begin{aligned} \mathcal{L}_o^1(u_\mu(E)) &\leq \mathcal{L}^1\left(\bigcup_i [u_\mu(a_i), u_\mu(b_i)]\right) \leq \sum_i \mathcal{L}^1([u_\mu(a_i), u_\mu(b_i)]) \\ &= \sum_i (u_\mu(b_i) - u_\mu(a_i)) = \sum_i \mu((a_i, b_i]) \\ &= \mu(U) = \mu(E) + \mu(U \setminus E) \leq \mu(E) + \varepsilon, \end{aligned}$$

where we have used (6.2). Letting $\varepsilon \rightarrow 0$ concludes the first part of the proof.

Assume next that u_μ is continuous in I , let $E \subseteq I$ be a Borel set, and let $V \subseteq \mathbb{R}$ be an open set such that $u_\mu(E) \subseteq V$. We claim that $\mu(E) \leq \mathcal{L}^1(V)$. Since I is open and u_μ is continuous, the set $U := u_\mu^{-1}(V)$ is open and thus we can decompose it into countably many disjoint intervals of the form $(a_i, b_i]$. It follows from (6.2) and the fact that $E \subseteq U$ that

$$\begin{aligned} \mu(E) &\leq \mu(U) = \sum_i \mu((a_i, b_i]) = \sum_i (u_\mu(b_i) - u_\mu(a_i)) \\ &= \sum_i \mathcal{L}^1(u_\mu((a_i, b_i))) \leq \mathcal{L}^1(V), \end{aligned}$$

where we used the facts that $u_\mu((a_i, b_i))$ is an interval of endpoints $u_\mu(a_i)$ and $u_\mu(b_i)$ because u_μ is continuous and increasing, and that the sets $(u_\mu(a_i), u_\mu(b_i))$ are disjoint and contained in V . This proves the claim. By taking the infimum over all open sets V containing $u_\mu(E)$ we get $\mu(E) \leq \mathcal{L}^1(u_\mu(E))$. $\square$

Remark 6.6. Note that in general the inequality (6.7) is strict. Indeed, let x be a discontinuity point of u_μ . Take $E = \{x\}$. Then

$$\mathcal{L}_o^1(\{u_\mu(x)\}) = 0 < u_\mu(x) - (u_\mu)_-(x) = \mu(\{x\}),$$

where we have used (6.6).

We now turn to the proof of Theorem 6.3.

Proof of Theorem 6.3. (i) Assume that μ is purely atomic. Then every set of positive measure contains an atom. Since I_{dis} is countable, we can write $I_{\text{dis}} = \{a_n : n \in G\}$ for some $G \subseteq \mathbb{N}$. Hence, for every Borel set $E \subseteq I$,

$$\mu(E) = \mu(E \setminus I_{\text{dis}}) + \mu(E \cap I_{\text{dis}}) = 0 + \sum_n \mu(E \cap \{a_n\}),$$

where we have used Lemma 6.4. Hence, if $x \in I$ is such that $x \geq t_0$, by (6.1),

$$(6.8) \quad u_\mu(x) = \mu((t_0, x]) = \sum_n \mu((t_0, x] \cap \{a_n\}) = \sum_{t_0 < a_n \leq x} \mu(\{a_n\}),$$

while if $x < t_0$,

$$u_\mu(x) = - \sum_{x < a_n \leq t_0} \mu(\{a_n\}),$$

which shows that u_μ is a jump function (see Definition 1.5).

Conversely, assume that u_μ is a jump function. Since u_μ is right continuous, if $[a, b] \subseteq I$ we can write

$$u_\mu(x) = c + \sum_{a \leq a_n \leq x} b_n, \quad x \in [a, b],$$

where $c \in \mathbb{R}$ and $b_n > 0$. It follows from (6.2) that for $a \leq t < x \leq b$,

$$(6.9) \quad \mu((t, x]) = u_\mu(x) - u_\mu(t) = \sum_{t < a_n \leq x} b_n.$$

Fix $\varepsilon > 0$ and find n_ε so large that

$$(6.10) \quad \sum_{n > n_\varepsilon} b_n \leq \varepsilon.$$

By relabeling a_n we can assume that $a_n \leq a_{n+1}$ for all $n < n_\varepsilon$. Then by (6.9) we have

$$\mu((a, b] \setminus I_{\text{dis}}) \leq \mu((a, a_1)) + \sum_{n=1}^{n_\varepsilon-1} \mu((a_n, a_{n+1})) + \mu((a_{n_\varepsilon}, b]) \leq \sum_{n > n_\varepsilon} b_n \leq \varepsilon.$$

Letting $\varepsilon \rightarrow 0^+$, we have that $\mu((a, b] \setminus I_{\text{dis}}) = 0$. By the arbitrariness of $[a, b]$, it follows that $\mu(I \setminus I_{\text{dis}}) = 0$, which shows that every set of positive measure contains an atom. Thus μ is purely atomic.

(ii) Assume that $\mu \ll \mathcal{L}^1$. Since singletons have Lebesgue measure zero, it follows that $\mu(\{x\}) = 0$ for all $x \in I$. Hence, by Lemma 6.4, μ has no atoms and u_μ is continuous in I . In view of Theorem 3.41 to prove that u_μ is locally absolutely continuous, it suffices to show that it satisfies the Lusin (N) property. Let $E \subset I$ be such that $\mathcal{L}^1(E) = 0$. Find a Borel set F containing E such that $\mathcal{L}^1(F) = 0$. Since $\mu \ll \mathcal{L}^1$, it follows that

$\mu(F) = 0$. For every $\varepsilon > 0$ we can find an open set $U \subseteq I$ such that $F \subseteq U$ and $\mu(U) \leq \varepsilon$. By decomposing U into countably many disjoint intervals of the form $(a_i, b_i]$, it follows from (6.2) and the fact that u_μ is increasing and continuous that

$$\begin{aligned} \mathcal{L}_o^1(u_\mu(E)) &\leq \mathcal{L}_o^1(u_\mu(U)) \leq \sum_i \mathcal{L}_o^1(u_\mu((a_i, b_i])) \\ &= \sum_i (u_\mu(b_i) - u_\mu(a_i)) = \sum_i \mu((a_i, b_i]) = \mu(U) \leq \varepsilon. \end{aligned}$$

Letting $\varepsilon \rightarrow 0^+$ shows that $\mathcal{L}_o^1(u_\mu(E)) = 0$.

Conversely, assume that u_μ is locally absolutely continuous. Then by (6.2) and Theorem 3.20 we have

$$\mu((a, b]) = u_\mu(b) - u_\mu(a) = \int_a^b u'_\mu dx.$$

Given an open set $U \subseteq I$, by decomposing U into countably many disjoint intervals of the form $(a_i, b_i]$, it follows that

$$(6.11) \quad \mu(U) = \int_U u'_\mu dx.$$

Since u'_μ is locally integrable, for every Borel set $E \subset I$ with $\mathcal{L}^1(E) = 0$, we have that $\int_E u'_\mu dx = 0$. In turn, given $\varepsilon > 0$ we can find an open set $U \subseteq I$ containing E such that $\int_U u'_\mu dx \leq \varepsilon$. It follows that $\mu(E) \leq \mu(U) \leq \varepsilon$ and by letting $\varepsilon \rightarrow 0^+$ we conclude that $\mu \ll \mathcal{L}^1$.

Now, let $E \subseteq I$ be a bounded Borel set, with $\overline{E} \subset I$. Then we can find a sequence of decreasing open sets U_n and a set $F \subset I$ with $\mathcal{L}^1(F) = 0$ such that

$$E \cup F = \bigcap_{n=1}^{\infty} U_n.$$

Since $\overline{E} \subset I$ is compact and I open, we can assume that the sets U_n are compactly contained in I so that $\mu(U_n) < \infty$. Using the fact that $\mu \ll \mathcal{L}^1$, we have that $\mu(F) = 0$. Taking U to be U_n in (6.11) and using Proposition B.9(ii) and the Lebesgue dominated convergence theorem, we get

$$\begin{aligned} (6.12) \quad \mu(E) &= \mu(E \cup F) = \lim_{n \rightarrow \infty} \mu(U_n) = \lim_{n \rightarrow \infty} \int_{U_n} u'_\mu dx \\ &= \int_{E \cup F} u'_\mu dx = \int_E u'_\mu dx, \end{aligned}$$

where we used the fact that $\mu(F) = \mathcal{L}^1(F) = 0$. The case of an arbitrary Borel set E is left as an exercise.

(iii) Assume that μ is nonatomic and $\mu \perp \mathcal{L}^1$. As in part (ii) it follows that u_μ is continuous. Since $\mu \perp \mathcal{L}^1$, there exists a Borel set $E_\mu \subseteq I$ such that for every Borel set $E \subseteq I$ we have

$$\mu(E) = \mu(E \cap E_\mu), \quad \mathcal{L}^1(E) = \mathcal{L}^1(E \setminus E_\mu).$$

In particular, $\mu(I \setminus E_\mu) = 0$ and $\mathcal{L}^1(E_\mu) = 0$. Hence, $\mathcal{L}^1(u_\mu(I \setminus E_\mu)) = 0$ by Lemma 6.5. Since u_μ is increasing, by Lebesgue's differentiation theorem (see Theorem 1.18), the set $F := \{x \in I : u_\mu \text{ is not differentiable at } x\}$ satisfies $\mathcal{L}^1(F) = 0$. By Lemma 3.60 and the fact that $\mathcal{L}_o^1(u_\mu(I \setminus (E_\mu \cup F))) = 0$ we have that $u'_\mu(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in I \setminus (E_\mu \cup F)$. But $\mathcal{L}^1(E_\mu \cup F) = 0$, and so u_μ is a singular function.

Conversely, assume that u_μ is a singular function. Then u_μ is continuous, and so by Lemma 6.4, μ is nonatomic. Since u_μ is singular, we have that $\mathcal{L}^1(I \setminus I_{\text{der}0}) = 0$. By Corollary 3.37 and Lemma 6.5,

$$\mu(I_{\text{der}0}) = \mathcal{L}^1(u(I_{\text{der}0})) = 0$$

and thus for every Borel set $E \subseteq I$,

$$\mu(E) = \mu(E \setminus I_{\text{der}0}),$$

which shows that $\mu \perp \mathcal{L}^1$. □

Next we prove the converse of Theorem 6.1; that is, we show that to every increasing function we may associate a unique Radon measure.

Theorem 6.7. *Let $I \subseteq \mathbb{R}$ be an open interval and let $u : I \rightarrow \mathbb{R}$ be an increasing function. Then there exists a unique Radon measure $\mu_u : \mathcal{B}(I) \rightarrow [0, \infty]$ such that*

$$(6.13) \quad \mu_u((a, b]) = u_+(b) - u_+(a)$$

for all $a, b \in I$, with $a \leq b$. Moreover,

$$(6.14) \quad \mu_u(I) = \sup_I u - \inf_I u.$$

In particular, μ_u is finite if and only if u is bounded.

The measure μ_u given by the previous theorem is called the *Lebesgue–Stieltjes measure* generated by u .

To construct μ_u , we begin by defining μ_u on the semiring $\mathcal{S}(I)$ of all intervals $(a, b]$, where $a, b \in I$, with $a \leq b$. In view of Theorem 6.1, we set

$$(6.15) \quad \rho_u((a, b]) := u_+(b) - u_+(a).$$

Lemma 6.8. *Let $I \subseteq \mathbb{R}$ be an open interval and let $u : I \rightarrow \mathbb{R}$ be an increasing function. Then the set function $\rho_u : \mathcal{S}(I) \rightarrow [0, \infty)$ is countably subadditive, that is,*

$$\rho_u((a, b]) \leq \sum_{n=1}^{\infty} \rho_u((a_n, b_n])$$

for every $(a, b], (a_n, b_n] \in \mathcal{S}(I)$, $n \in \mathbb{N}$, such that $(a, b] \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n]$.

Proof. Let $(a, b], (a_n, b_n] \in \mathcal{S}(I)$ be such that $(a, b] \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n]$. If $a = b$ there is nothing to prove, thus assume $a < b$. Fix $\varepsilon > 0$. Since u_+ is right-continuous, there exist $0 < \delta < b - a$ and $0 < \delta_n < \text{dist}(b_n, \mathbb{R} \setminus I)$ such that $u_+(x) - u_+(a) \leq \varepsilon$ for all $a \leq x \leq a + \delta$ and $u_+(x) - u_+(b_n) \leq \frac{\varepsilon}{2^n}$ for all $b_n \leq x \leq b_n + \delta_n$. Let $a < \alpha \leq a + \delta$ and $b_n < \beta_n \leq b_n + \delta_n$. Then

$$[\alpha, b] \subset (a, b] \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n] \subseteq \bigcup_{n=1}^{\infty} (a_n, \beta_n)$$

and so by compactness we can find ℓ such that

$$[\alpha, b] \subset \bigcup_{n=1}^{\ell} (a_n, \beta_n).$$

Then

$$\begin{aligned} \rho_u((a, b]) &= u_+(b) - u_+(a) \leq u_+(b) - u_+(\alpha) + \varepsilon \\ &\leq \sum_{n=1}^{\ell} (u_+(\beta_n) - u_+(a_n)) + \varepsilon \\ &\leq \sum_{n=1}^{\ell} (u_+(b_n) - u_+(a_n)) + 2\varepsilon \leq \sum_{n=1}^{\infty} \rho_u((a_n, b_n]) + 2\varepsilon. \end{aligned}$$

It suffices to let $\varepsilon \rightarrow 0^+$. □

The outer measure $\mu_u^* : \mathcal{P}(I) \rightarrow [0, \infty]$, given by

$$(6.16) \quad \mu_u^*(E) := \inf \left\{ \sum_{n=1}^{\infty} \rho_u((a_n, b_n]) : (a_n, b_n] \in \mathcal{S}(I), E \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n] \right\},$$

for $E \subseteq I$, is called the *Lebesgue–Stieltjes outer measure* generated by u .

Exercise 6.9. Let $I \subseteq \mathbb{R}$ be an open interval and let $u : I \rightarrow \mathbb{R}$ be an increasing function. Prove that μ_u^* is a metric outer measure.

We are now ready to prove Theorem 6.7.

Proof of Theorem 6.7. It follows from Carathéodory's theorem (see Theorem B.13) that the restriction of μ_u^* to the σ -algebra of all μ_u^* -measurable sets is a complete measure. Moreover, by the previous exercise and Proposition B.19 every Borel set in I is μ_u^* -measurable.

In view of Lemma 6.8 and Remark B.3, (6.13) holds.

Note that if $[a, b] \subset I$, then, since I is open, we can find $a_1 \in I$ with $a_1 < a$. Then $[a, b] \subset (a_1, b]$, and so (6.13) and the monotonicity of μ_u with respect to inclusion imply that μ_u is finite on $[a, b]$. Hence, μ_u is a Radon measure.

To prove (6.14), it is enough to let $a \searrow \inf I$ and $b \nearrow \sup I$ in (6.13) and to use Proposition B.9(i).

Uniqueness follows from Corollary B.16. This concludes the proof. $\square$

Remark 6.10. (i) If $I \subseteq \mathbb{R}$ is an open interval and $u, v : I \rightarrow \mathbb{R}$ are two increasing functions, then by the uniqueness established in the previous theorem,

$$\mu_{u+v} = \mu_u + \mu_v.$$

(ii) If the interval I is not open, then it contains one of its endpoints, say $a := \inf I$. In this case, if $u : I \rightarrow \mathbb{R}$ is increasing, then we can extend u to $(-\infty, a)$ by setting $u(x) := u(a)$ for all $x < a$. Thus, we can construct the Lebesgue–Stieltjes measure of the extension of u defined on the Borel σ -algebra of $(-\infty, \sup I)$ and then consider its restriction to $\mathcal{B}(I)$.

(iii) If $\mu : \mathcal{B}(I) \rightarrow [0, \infty]$ is a Radon measure and $u_\mu : I \rightarrow \mathbb{R}$ is the function defined by (6.1), then by the uniqueness proved in the previous theorem, it follows that μ is the Lebesgue–Stieltjes measure generated by u_μ , that is, $\mu_{u_\mu} = \mu$.

Remark 6.11. Note that under the hypotheses of Theorem 6.7, if $b \in I$ taking $a \in I$ with $a < b$ in (6.13) and letting $a \nearrow b^-$ along a sequence yields

$$(6.17) \quad \mu_u(\{b\}) = u_+(b) - u_-(b).$$

Hence, again by (6.13), if $a, b \in I$, with $a \leq b$, then

$$(6.18) \quad \mu_u([a, b]) = \mu_u((a, b]) + \mu_u(\{a\}) = u_+(b) - u_-(a),$$

$$(6.19) \quad \mu_u((a, b)) = \mu_u((a, b]) - \mu_u(\{b\}) = u_-(b) - u_+(a).$$

Exercise 6.12. Construct the Lebesgue–Stieltjes outer measures corresponding to

$$u_1(x) = \begin{cases} [x] & \text{for } x \geq 0, \\ 0 & \text{for } x < 0, \end{cases} \quad u_2(x) = \begin{cases} 1 & \text{for } x \geq 0, \\ -1 & \text{for } x < 0, \end{cases}$$

where $[x]$ is the integer part of x .

Exercise 6.13. Let $I \subseteq \mathbb{R}$ be an open interval and let $u : I \rightarrow \mathbb{R}$ be an increasing function. Prove that if in place of (6.15), we define

$$\varrho_u((a, b]) := u(b) - u(a), \quad (a, b] \in \mathcal{S}(I),$$

and in (6.16) we use ϱ_u instead of ρ_u , then

$$\mu_u((a, b]) = u(b) - u^+(a) - \sum_{x \in (a, b)} (u^+(x) - u(x))$$

for every $(a, b] \in \mathcal{S}(I)$.

The following change of variables formula transforms a Lebesgue–Stieltjes integral into an integral with respect to the Lebesgue measure.

Theorem 6.14. *Let $I \subset \mathbb{R}$ be an open interval bounded from below, let $u : I \rightarrow \mathbb{R}$ be a right continuous increasing function, and let $f : I \rightarrow [0, \infty)$ be a Borel function. Then for every $a, b \in I$, with $a < b$,*

$$(6.20) \quad \int_{(a, b]} f(x) d\mu_u(x) = \int_{u(a)}^{u(b)} f(v(t)) dt,$$

where μ_u is the Lebesgue–Stieltjes measure generated by u and $v : J \rightarrow \mathbb{R}$ is the inverse of u defined as in Theorem 1.8.

Proof. Step 1: We claim that if $\alpha, \beta \in I$, with $\alpha < \beta$, then $\alpha < v(t) \leq \beta$ if and only if $u(\alpha) < t \leq u(\beta)$. Indeed, if $\alpha < v(t)$, then by the definition of v (see (1.1)) we have that $u(\alpha) < t$. On the other hand, if $v(t) \leq \beta$, let $\varepsilon > 0$ be so small that $\beta + \varepsilon \in I$. Again by the definition of v and the fact that u is increasing we have that $u(\beta + \varepsilon) \geq t$. Letting $\varepsilon \rightarrow 0^+$ and using the fact that u is right continuous gives $u(\beta) \geq t$. Thus, if $\alpha < v(t) \leq \beta$, then $u(\alpha) < t \leq u(\beta)$.

Conversely, if $u(\alpha) < t$, using the fact that u is right continuous, we may find $x > \alpha$ such that $u(x) < t$. Hence, $v(t) \geq x > \alpha$ by the definition of v . Finally, if $u(\beta) \geq t$, then $v(t) \leq \beta$, again by the definition of v . This shows that if $u(\alpha) < t \leq u(\beta)$, then $\alpha < v(t) \leq \beta$. Thus, the claim holds.

Step 2: Assume that $f = \chi_{(\alpha, \beta]}$, where $\alpha, \beta \in I$, with $\alpha < \beta$. There are now two cases. If $(\alpha, \beta] \cap (a, b] = \emptyset$, then the integral on the left-hand side of (6.20) is zero. On the other hand, by Step 1, if $t \in (u(a), u(b)]$, then $v(t) \in (a, b]$, and so $\chi_{(\alpha, \beta]}(v(t)) = 0$, since $(\alpha, \beta] \cap (a, b] = \emptyset$. Thus, the integral on the right-hand side of (6.20) is zero also.

If $(\alpha, \beta] \cap (a, b] \neq \emptyset$, then $(\alpha, \beta] \cap (a, b] = (s, t]$, where $s := \max\{\alpha, a\}$ and $t := \min\{\beta, b\}$. Thus,

$$\int_{(a, b]} \chi_{(\alpha, \beta]}(x) d\mu_u(x) = \mu_u((s, t]) = u(t) - u(s),$$

where we have used (6.13) and the fact that u is right continuous. On the other hand, again by Step 1, $v(t) \in (\alpha, \beta]$ if and only if $t \in (u(\alpha), u(\beta)]$. Thus,

$$\begin{aligned} \int_{u(a)}^{u(b)} \chi_{(\alpha, \beta]}(v(t)) dt &= \int_J \chi_{(u(a), u(b)]}(t) \chi_{(u(\alpha), u(\beta)]}(t) dt \\ &= \min\{u(\beta), u(b)\} - \max\{u(\alpha), u(a)\} \\ &= u(b) - u(a), \end{aligned}$$

where in the last equality we have used the fact that u is increasing. This proves (6.20) in the case that $f = \chi_{(\alpha, \beta]}$.

Step 3: For every Borel set $B \subseteq I$, define

$$\mu_1(B) := \int_{(a, b]} \chi_B(x) d\mu_u(x), \quad \mu_2(B) := \int_{u(a)}^{u(b)} \chi_B(v(t)) dt.$$

Note that if $a_n \searrow \inf I$ and $b_n \nearrow \sup I$, then for all n sufficiently large, $(a, b] \subseteq (a_n, b_n]$, and so by the previous step

$$\mu_1((a_n, b_n]) = \mu_2((a_n, b_n]) = u(b) - u(a) < \infty.$$

Then $\mu_1 : \mathcal{B}(I) \rightarrow [0, \infty]$ and $\mu_2 : \mathcal{B}(I) \rightarrow [0, \infty]$ are two Radon measures that coincide on intervals of the form $(\alpha, \beta]$. It follows from Corollary B.16 that $\mu_1 = \mu_2$. This shows that (6.20) holds in the case that $f = \chi_B$, where $B \subseteq I$ is a Borel set.

A standard argument (using simple functions and the Lebesgue monotone convergence theorem) shows that (6.20) holds for a Borel function. $\square$

Remark 6.15. A similar result holds if u is right continuous and decreasing.

The following exercises discuss integration by parts for Lebesgue–Stieltjes measures.

Exercise 6.16. Let $\mu : \mathcal{B}((a, b)) \rightarrow [0, \infty)$ and $\nu : \mathcal{B}((a, b)) \rightarrow [0, \infty)$ be two finite Borel measures. Prove that

- (i) $\int_{(a, b)} \mu([t, b)) d\nu(t) = \int_{(a, b)} \nu((a, x]) d\mu(x).$
- (ii) $\int_{(a, b)} \mu((a, t)) d\nu(t) + \int_{(a, b)} \nu((a, x]) d\mu(x) = \mu((a, b))\nu((a, b)).$
- (iii) Let $f(x) := \frac{1}{2}(\mu((a, x]) + \mu((a, x)))$ and $g(x) := \frac{1}{2}(\nu((a, x]) + \nu((a, x)))$, $x \in (a, b)$. Prove that

$$\int_{(a, b)} f d\nu + \int_{(a, b)} g d\mu = \mu((a, b))\nu((a, b)).$$

Exercise 6.17. Let $I \subseteq \mathbb{R}$ be an open interval and let $u, v : I \rightarrow \mathbb{R}$ be increasing functions. Prove that for every $[a, b] \subset I$,

$$\frac{1}{2} \int_{(a, b)} (u_+ + u_-) d\mu_v + \frac{1}{2} \int_{(a, b)} (v_+ + v_-) d\mu_u = u_-(b)v_-(b) - v_+(a)u_+(a).$$

Exercise 6.18. Let $I \subseteq \mathbb{R}$ be an open interval and let $u : I \rightarrow \mathbb{R}$ be an increasing function. Prove that for every $\varphi \in C_c^1(I)$,

$$\int_I u \varphi' dx = - \int_I \varphi d\mu_u.$$

6.2. Vector-valued Measures Versus $BPV(I; Y)$

Next we extend Theorem 6.1 to vector-valued measures. As we have seen in Theorem 6.7 an increasing function $u : I \rightarrow \mathbb{R}$ uniquely determines a Radon measure $\mu_u : \mathcal{B}(I) \rightarrow [0, \infty]$. The measures μ_u is finite if and only if u is bounded. Hence, if now $u : I \rightarrow \mathbb{R}^M$ belongs to $BPV_{\text{loc}}(I; \mathbb{R}^M)$ we cannot expect u to determine a vector-valued measure $\lambda_u : \mathcal{B}(I) \rightarrow \mathbb{R}^M$ unless u belongs to $BPV(I; \mathbb{R}^M)$. To overcome this difficulty we restrict our attention to vector-valued measures defined on the σ -field

$$(6.21) \quad \mathcal{B}_c(I) := \{E \in \mathcal{B}(I) : E \text{ is compactly contained in } I\},$$

that is, $\mathcal{B}_c(I)$ consists of all Borel sets whose closure is a compact subset of I . In particular, I does not belong to $\mathcal{B}_c(I)$ since I is open by assumption.

Given an open interval $I \subseteq \mathbb{R}$, a normed space $(Y, \|\cdot\|)$, and a vector-valued measure $\lambda : \mathcal{B}_c(I) \rightarrow Y$, we recall that the total variation of λ is the measure $\|\lambda\| : \mathcal{B}_c(I) \rightarrow [0, \infty]$ given by

$$(6.22) \quad \|\lambda\|(E) := \left\{ \sum_{i=1}^n \|\lambda(E_i)\| \right\},$$

where the supremum is taken over all finite families of disjoint sets $E_i \in \mathcal{B}_c(I)$, $i = 1, \dots, n$, such that $E = \bigcup_{i=1}^n E_i$ and over all $n \in \mathbb{N}$. In particular,

$$(6.23) \quad \|\lambda\|(\{x\}) = \|\lambda(\{x\})\|$$

for every $x \in I$.

Theorem 6.19. Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a normed space and let $\lambda : \mathcal{B}_c(I) \rightarrow Y$ be a vector-valued measure. Then for every fixed $t_0 \in I$ the function $u_\lambda : I \rightarrow Y$, defined by

$$(6.24) \quad u_\lambda(x) := \begin{cases} \lambda((t_0, x]) & \text{if } x \geq t_0, \\ -\lambda((x, t_0]) & \text{if } x < t_0, \end{cases}$$

is right continuous,

$$(6.25) \quad u_\lambda(b) - u_\lambda(a) = \lambda((a, b])$$

for all $a, b \in I$, with $a \leq b$, and for every interval $[a, b] \subseteq I$,

$$(6.26) \quad \text{Var}_{[a,b]} u_\lambda = \|\lambda\|([a, b]), \quad \text{Var}_{(a,b]} u_\lambda = \|\lambda\|((a, b]).$$

In particular, $\|\lambda\| : \mathcal{B}_c(I) \rightarrow [0, \infty]$ is a Radon measure if and only if $u_\lambda \in BPV_{\text{loc}}(I; Y)$ and it is a finite measure if and only if $u_\lambda \in BPV(I; Y)$.

Proof. Step 1: The fact that u_λ is right continuous and (6.25) follow as in the proof of Theorem 6.1. We only prove (6.26)₁.

To prove (6.26)₁, consider a partition $x_0 < x_1 < \dots < x_n$ of $[a, b]$. By (6.25) we have

$$\sum_{i=1}^n \|u_\lambda(x_i) - u_\lambda(x_{i-1})\| = \sum_{i=1}^n \|\lambda((x_{i-1}, x_i])\| \leq \|\lambda\|([a, b]).$$

Taking the supremum over all partitions gives $\text{Var}_{[a,b]} u_\lambda \leq \|\lambda\|([a, b])$.

Step 2: To prove the opposite inequality it suffices to assume that $\text{Var}_{[a,b]} u_\lambda < \infty$. Then the function $\mathbf{V}(x) := \text{Var}_{[a,x]} u_\lambda$ is finite in $[a, b]$. Since u_λ is right continuous, by Proposition 2.21, so is $\mathbf{V}$. Extend $\mathbf{V}$ to be $\mathbf{V}(a)$ for $x < a$.

We claim that

$$(6.27) \quad \|\lambda(E)\| \leq \mu_{\mathbf{V}}(E)$$

for every $E \in \mathcal{R}([a, b])$, where $\mu_{\mathbf{V}}$ is the measure given in Theorem 6.7 (see also Remark 6.10(ii)). To see this, let $E \in \mathcal{R}([a, b])$. Write $E = \bigcup_{i=1}^n (a_i, b_i]$, where $(a_i, b_i] \in \mathcal{S}([a, b])$, $i = 1, \dots, n$ are pairwise disjoint intervals. By the additivity of λ , the right continuity of $\mathbf{V}$, (2.4) and (6.25), we get

$$\begin{aligned} \|\lambda(E)\| &\leq \sum_{i=1}^n \|\lambda((a_i, b_i])\| = \sum_{i=1}^n \|u_\lambda(b_i) - u_\lambda(a_i)\| \leq \sum_{i=1}^n \mathbf{V}(b_i) - \mathbf{V}(a_i) \\ &= \sum_{i=1}^n \mathbf{V}_+(b_i) - \mathbf{V}_+(a_i) = \sum_{i=1}^n \mu_{\mathbf{V}}((a_i, b_i]) = \mu_{\mathbf{V}}(E). \end{aligned}$$

Thus (6.27) holds.

Next we show that (6.27) holds for all $E \in \mathcal{B}([a, b])$. For $E, F \in \mathcal{B}([a, b])$ define

$$d(E, F) := \mu_{\mathbf{V}}(E \Delta F),$$

where $E \Delta F := (E \setminus F) \cup (F \setminus E)$ is the *symmetric difference* between the sets E and F . It can be checked that d is a semimetric in $\mathcal{B}([a, b])$, that is, $d(E, E) = 0$, $d(E, F) = d(F, E)$ and $d(E, F) \leq d(E, G) + d(G, F)$ for all $E, F, G \in \mathcal{B}([a, b])$.

For every $E \in \mathcal{B}([a, b])$ and $\varepsilon > 0$ there exists $F \in \mathcal{R}([a, b])$ such that $\mu_{\mathbf{V}}(E \Delta F) \leq \varepsilon$. Hence, $\mathcal{R}([a, b])$ is dense in $\mathcal{B}([a, b])$. Moreover, for $E, F \in \mathcal{R}([a, b])$, by the finite additivity of λ ,

$$\lambda(E) - \lambda(F) = \lambda(E \setminus (E \cap F)) - \lambda(F \setminus (E \cap F)),$$

and so by (6.27),

$$\|\lambda(E) - \lambda(F)\| \leq \mu_{\mathbf{V}}(E \setminus (E \cap F)) + \mu_{\mathbf{V}}(F \setminus (E \cap F)) = d(E, F).$$

It follows that $\lambda : \mathcal{R}([a, b]) \rightarrow Y$ is Lipschitz continuous with respect to the semimetric d . Thus, by the density of $\mathcal{R}([a, b])$ in $\mathcal{B}([a, b])$ we have that $\|\lambda(E)\| \leq \mu_{\mathbf{V}}(E)$ for all $E \in \mathcal{B}([a, b])$.

Now, let $E \in \mathcal{B}([a, b])$ and let $E_i \in \mathcal{B}([a, b])$, $i = 1, \dots, n$, be disjoint sets such that $E = \bigcup_{i=1}^n E_i$. By what we just proved and by the additivity of $\mu_{\mathbf{V}}$,

$$\sum_{i=1}^n \|\lambda(E_i)\| \leq \sum_{i=1}^n \mu_{\mathbf{V}}(E_i) = \mu_{\mathbf{V}}(E).$$

Taking the supremum over all such partitions of E (see (6.22)) gives

$$\|\lambda\|(E) \leq \mu_{\mathbf{V}}(E).$$

In particular, by (6.18),

$$\|\lambda\|([a, b]) \leq \mu_{\mathbf{V}}([a, b]) = \text{Var}_{[a, b]} u_{\lambda},$$

which, together with Step 1, proves (6.26)₁.

The last part of the statement follows from (6.26). $\square$

It follows from the previous theorem that if $\|\lambda\| : \mathcal{B}_c(I) \rightarrow [0, \infty)$, then $u_{\lambda} \in BPV_{\text{loc}}(I; Y)$. Next we show that the function $u_{\|\lambda\|}$ defined in (6.1) with μ replaced by $\|\lambda\|$ coincides with the indefinite variation $\mathbf{V}_{u_{\lambda}}$ of u_{λ} (see (2.3)).

Corollary 6.20. *Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a normed space, and let $\lambda : \mathcal{B}_c(I) \rightarrow Y$ be a vector-valued measure such that $\|\lambda\| : \mathcal{B}_c(I) \rightarrow [0, \infty)$. Then $u_{\|\lambda\|} = \mathbf{V}_{u_{\lambda}}$.*

Proof. In view of the previous theorem u_{λ} is right continuous, but then by Proposition 2.21, so is $\mathbf{V}_{u_{\lambda}}$. Hence, $(\mathbf{V}_{u_{\lambda}})_+(t_0) = \mathbf{V}_{u_{\lambda}}(t_0) = 0$ by Remark 2.2 and (2.3). Let $x \in I$ with $t_0 \leq x$. By Exercise 2.16, (2.4), (6.2), and (6.26)₂ for $\|\lambda\|$ and the fact that $u_{\|\lambda\|}(t_0) = 0$, in this order, we have

$$\mathbf{V}_{u_{\lambda}}(x) = \text{Var}_{(t_0, x]} u_{\lambda} = \|\lambda\|((t_0, x]) = u_{\|\lambda\|}(x).$$

The same holds for $x \leq t_0$. $\square$

In the same spirit of Theorem 6.3 we have the following results.

Theorem 6.21. *Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a normed space and let $\lambda : \mathcal{B}_c(I) \rightarrow Y$ be a vector-valued measure such that $\|\lambda\| : \mathcal{B}_c(I) \rightarrow [0, \infty)$. Then the measure λ is purely atomic if and only if u_{λ} is a jump function. Moreover, in this case, for every Borel set $E \in \mathcal{B}_c(I)$,*

$$(6.28) \quad \lambda(E) = \sum_n \lambda(E \cap \{a_n\}), \quad \|\lambda\|(E) = \sum_n \|\lambda(E \cap \{a_n\})\|,$$

where $\{a_n\}_n$ is the set of discontinuity points of u_{λ} .

Lemma 6.22. *Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a normed space and let $\lambda : \mathcal{B}_c(I) \rightarrow Y$ be a vector-valued measure such that $\|\lambda\| : \mathcal{B}_c(I) \rightarrow [0, \infty)$. Then λ and $\|\lambda\|$ have the same atoms.*

Proof. Let $E \in \mathcal{B}_c(I)$ be an atom for λ . Then $\lambda(E) = y \neq 0$ and for every subset $F \in \mathcal{B}_c(I)$ of E either $\lambda(F) = 0$ or $\lambda(F) = \lambda(E)$. Let $F \in \mathcal{B}_c(I)$ be a subset of E and let $F_i \in \mathcal{B}_c(I)$, $i = 1, \dots, n$, be a family of disjoint sets such that

$$F = \bigcup_{i=1}^n F_i.$$

If $\lambda(F) = 0$, then since each F_i can only have measure either 0 or y , it follows that all the sets F_i have measure zero. Hence, $\|\lambda\|(F) = 0$. If $\lambda(F) = y$, then only one F_i can have measure y and all others must have measure zero. It follows that

$$\sum_{i=1}^n \|\lambda(F_i)\| = \|\lambda(E)\|$$

and taking the supremum over all such families gives $\|\lambda\|(F) = \|\lambda(E)\| > 0$. Thus, E is an atom for $\|\lambda\|$. It follows from Lemma 6.4 applied to $\|\lambda\|$ restricted to $\mathcal{B}([a, b])$ for some $E \subseteq [a, b] \subseteq I$, that E must be a singleton.

Conversely, if $E \in \mathcal{B}_c(I)$ is an atom for $\|\lambda\|$, then again by Lemma 6.4, E must be a singleton, $E = \{x\}$. Since the only subsets of E are E and the empty set, necessarily $\lambda(E) \neq 0$, and so E is an atom for λ . $\square$

We now turn to the proof of Theorem 6.21.

Proof of Theorem 6.21. Assume that λ is purely atomic. In view of the previous lemma, λ and $\|\lambda\|$ have the same atoms. By Lemma 6.4 the atoms of $\|\lambda\|$ are given by the singletons corresponding to the discontinuity points of $u_{\|\lambda\|}$, but since $u_{\|\lambda\|} = \mathbf{V}_{u_\lambda}$ by Corollary 6.20, it follows from Proposition 2.21 that the discontinuity points of u_λ are the same as the ones of $u_{\|\lambda\|}$. Reasoning as in the first part of the proof of Theorem 6.3 with λ in place of μ (see (6.8)) we see that u_λ is a jump function and (6.28)₁ holds.

Let $E \in \mathcal{B}_c(I)$ be such that $\|\lambda\|(E) > 0$. Then by (6.22) there exist disjoint sets $E_i \in \mathcal{B}_c(I)$, $i = 1, \dots, n$, such that $E = \bigcup_{i=1}^n E_i$ and $\sum_{i=1}^n \|\lambda(E_i)\| > 0$. Hence, $\lambda(E_i) \neq 0$ for at least one i . Since λ is purely atomic, the set E_i contains an atom. In turn, so does E . Thus $\|\lambda\|$ is purely atomic, and so by Theorem 6.3, $u_{\|\lambda\|} = \mathbf{V}_{u_\lambda}$ is a jump function and (6.28)₂ holds (see also (6.23)).

Conversely, assume that u_λ is a jump function. Then by (2.4) and Corollary 3.91, $\mathbf{V}_{u_\lambda}$ is a jump function. In turn, by Theorem 6.3 and again the fact that $u_{\|\lambda\|} = \mathbf{V}_{u_\lambda}$ we have that $\|\lambda\|$ is purely atomic. Since $\|\lambda(E)\| \leq \|\lambda\|(E)$ for every $E \in \mathcal{B}_c(I)$, it follows that if $\lambda(E) \neq 0$, then $\|\lambda\|(E) > 0$ and so

E contains an atom for $\|\lambda\|$, but again by Lemma 6.4, the atom is also an atom for λ . Thus, λ is also purely atomic. This concludes the proof. $\square$

Theorem 6.23. *Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a normed space and let $\lambda : \mathcal{B}_c(I) \rightarrow Y$ be a vector-valued measure such that $\|\lambda\| : \mathcal{B}_c(I) \rightarrow [0, \infty)$. Then $\lambda \ll \mathcal{L}^1$ if and only if u_λ is locally absolutely continuous. Moreover, in this case, if $Y = \mathbb{R}^M$, then for every $E \in \mathcal{B}_c(I)$,*

$$(6.29) \quad \lambda(E) = \int_E u'_\lambda dx, \quad \|\lambda\|(E) = \int_E \|u'_\lambda\| dx.$$

Proof. Assume that $u_\lambda \in AC_{\text{loc}}(I; Y)$. By Theorem 3.11 the indefinite variation $\mathbf{V}_{u_\lambda}$ of u_λ belongs to $AC_{\text{loc}}(I)$. Since $u_{\|\lambda\|} = \mathbf{V}_{u_\lambda}$ by Corollary 6.20, it follows from Theorem 6.3 that $\|\lambda\| \ll \mathcal{L}^1$. Since $\|\lambda(E)\| \leq \|\lambda\|(E)$ for every $E \in \mathcal{B}_c(I)$, it follows that $\lambda \ll \mathcal{L}^1$.

Conversely, assume that $\lambda \ll \mathcal{L}^1$. Let $E \in \mathcal{B}_c(I)$ be such that $\mathcal{L}^1(E) = 0$. Then for every $F \subseteq E$ with $F \in \mathcal{B}_c(I)$ we have that $\mathcal{L}^1(F) = 0$ and so $\lambda(F) = 0$, since $\lambda \ll \mathcal{L}^1$. This implies that $\|\lambda\|(E) = 0$. Hence, $\|\lambda\| \ll \mathcal{L}^1$ and since $u_{\|\lambda\|} = \mathbf{V}_{u_\lambda}$ by the previous corollary, it follows from Theorem 6.3 that $\mathbf{V}_{u_\lambda} \in AC_{\text{loc}}(I)$. Again by Theorem 3.11 we obtain that $u_\lambda \in AC_{\text{loc}}(I; Y)$.

Next, assume that $Y = \mathbb{R}^M$ and that $\lambda \ll \mathcal{L}^1$, or, equivalently, $u_\lambda \in AC_{\text{loc}}(I; \mathbb{R}^M)$. Then $\mathbf{V}_{u_\lambda} \in AC_{\text{loc}}(I)$ and so by Theorem 6.3 (see also Remark 6.10), for every $E \in \mathcal{B}_c(I)$,

$$\|\lambda\|(E) = \int_E u'_{\|\lambda\|} dx = \int_E \mathbf{V}'_{u_\lambda} dx.$$

Since $\mathbf{V}'_{u_\lambda}(x) = \|u'_\lambda(x)\|$ for $\mathcal{L}^1$ -a.e. $x \in I$ by Theorem 2.28, (6.29)₂ holds. Reasoning as in the proof of (6.11) we get that (6.29)₁ holds for every open set $U \in \mathcal{B}_c(I)$. Since $\lambda \ll \mathcal{L}^1$ we can proceed as in (6.12) to conclude that (6.29)₁ holds for every $E \in \mathcal{B}_c(I)$. This completes the proof. $\square$

Theorem 6.24. *Let $I \subseteq \mathbb{R}$ be an open interval and let $\lambda : \mathcal{B}_c(I) \rightarrow \mathbb{R}^M$ be a vector-valued measure such that $\|\lambda\| : \mathcal{B}_c(I) \rightarrow [0, \infty)$. Then λ is nonatomic and $\lambda \perp \mathcal{L}^1$ if and only if u_λ is a singular function.*

Proof. Assume that λ is nonatomic and $\lambda \perp \mathcal{L}^1$. By Lemma 6.22, $\|\lambda\|$ is nonatomic. Since $u_{\|\lambda\|} = \mathbf{V}_{u_\lambda}$ by Corollary 6.20, it follows from Theorem 6.3 that $\mathbf{V}_{u_\lambda}$ is continuous. In turn, by Proposition 2.21, so is u_λ .

Since $\lambda \perp \mathcal{L}^1$, there exists a Borel set $E_\lambda \subseteq I$ such that for every $E \in \mathcal{B}_c(I)$ we have

$$\lambda(E) = \lambda(E \cap E_\lambda), \quad \mathcal{L}^1(E) = \mathcal{L}^1(E \setminus E_\lambda).$$

In turn, given $E \in \mathcal{B}_c(I)$, let $E_i \in \mathcal{B}([a, b])$, $i = 1, \dots, n$, be disjoint sets such that $E = \bigcup_{i=1}^n E_i$. Then

$$\sum_{i=1}^n \|\lambda(E_i)\| = \sum_{i=1}^n \|\lambda(E_i \cap E_\lambda)\|$$

and so taking the supremum over all such families of sets gives

$$\|\lambda\|(E) = \|\lambda\|(E \cap E_\lambda),$$

which shows that $\|\lambda\| \perp \mathcal{L}^1$. It follows from Theorem 6.3 that $\mathbf{V}_{u_\lambda}$ is a singular function, and so $\mathbf{V}'_{u_\lambda}(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in I$. Since $\mathbf{V}'_{u_\lambda}(x) = \|u'_\lambda(x)\|$ for $\mathcal{L}^1$ -a.e. $x \in I$ by Theorem 2.28, we have that u_λ is also a singular function.

Conversely, assume that u_λ is a singular function. Then by Proposition 2.21 and Theorem 2.28, $\mathbf{V}_{u_\lambda}$ is a singular function, and so by Theorem 6.3 we have that $\|\lambda\|$ is nonatomic and $\|\lambda\| \perp \mathcal{L}^1$. By Lemma 6.22, λ is nonatomic.

Since $\|\lambda\| \perp \mathcal{L}^1$ there exists a Borel set $E_\lambda \subseteq I$ such that for every $E \in \mathcal{B}_c(I)$ we have

$$\|\lambda\|(E) = \|\lambda\|(E \cap E_\lambda), \quad \mathcal{L}^1(E) = \mathcal{L}^1(E \setminus E_\lambda).$$

It follows that $\|\lambda\|(E \setminus E_\lambda) = 0$ for every $E \in \mathcal{B}_c(I)$, and since $\|\lambda(F)\| \leq \|\lambda\|(F)$ for every $F \in \mathcal{B}_c(I)$, we have that $\lambda(E) = \lambda(E \cap E_\lambda)$ for every $E \in \mathcal{B}_c(I)$. Thus, $\lambda \perp \mathcal{L}^1$. $\square$

Next we discuss the converse of Theorem 6.19, namely the analog of Theorem 6.7.

Theorem 6.25. *Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a Banach space, and let $u \in BPV_{\text{loc}}(I; Y)$. Then there exists a unique vector-valued measure $\lambda_u : \mathcal{B}_c(I) \rightarrow Y$ such that*

$$(6.30) \quad \lambda_u((a, b]) = u_+(b) - u_+(a)$$

for all $a, b \in I$, with $a \leq b$. Moreover, $\|\lambda_u\| \leq \mu_{\mathbf{V}}$ and equality holds if u is right continuous. Finally, if $u \in BPV(I; Y)$, then λ_u can be uniquely extended to a vector-valued measure $\lambda_u : \mathcal{B}(I) \rightarrow Y$ and

$$\|\lambda_u\|(I) \leq \text{Var } u.$$

In the previous statement $\mathbf{V}$ is the indefinite variation of u (see (2.3)) and $\mu_{\mathbf{V}}$ is the Lebesgue–Stieltjes measure generated by $\mathbf{V}$ (see Theorem 6.7). The vector-valued measure λ_u given by the previous theorem is called the *Lebesgue–Stieltjes vector-valued measure* generated by u .

Lemma 6.26. *Let $I \subseteq \mathbb{R}$ be an open interval, let $\mathcal{S} \subseteq \mathcal{P}(I)$ be a semiring, let $(Y, \|\cdot\|)$ be a normed space, and let $\rho : \mathcal{S} \rightarrow Y$ be a finitely additive*

measure. Then ρ can be uniquely extended to a finitely additive measure $\lambda : \mathcal{R} \rightarrow Y$, where $\mathcal{R}$ is the ring generated by $\mathcal{S}$.

Proof. Given $E \in \mathcal{R}$, let $E_1, \dots, E_n \in \mathcal{S}$ be pairwise disjoint sets such that

$$(6.31) \quad E = \bigcup_{i=1}^n E_i.$$

We define

$$(6.32) \quad \lambda(E) := \sum_{i=1}^n \rho(E_i).$$

To prove that this is a good definition, let $F_1, \dots, F_m \in \mathcal{S}$ be pairwise disjoint sets such that $E = \bigcup_{k=1}^m F_k$. Then for each i , $E_i = \bigcup_{k=1}^m (E_i \cap F_k)$ and the sets $E_i \cap F_1, \dots, E_i \cap F_m$ are pairwise disjoint. Since ρ is finitely additive, we have that

$$\sum_{i=1}^n \rho(E_i) = \sum_{i=1}^n \sum_{k=1}^m \rho(E_i \cap F_k) = \sum_{k=1}^m \sum_{i=1}^n \rho(E_i \cap F_k) = \sum_{k=1}^m \rho(F_k),$$

which shows that the extension of ρ to $\mathcal{R}$ does not depend on the particular partition of E . In particular, if $E \in \mathcal{S}$, then $\lambda(E) = \rho(E)$.

It follows from the finite additivity of ρ and the definition of λ , that λ is finitely additive.

Now, let $\lambda_1 : \mathcal{R} \rightarrow Y$ be another finitely additive extension of ρ . Given $E \in \mathcal{R}$, let E_i be as in (6.31). Then by the additivity of λ and λ_1 ,

$$\lambda_1(E) = \sum_{i=1}^n \lambda_1(E_i) = \sum_{i=1}^n \rho(E_i) = \sum_{i=1}^n \lambda(E_i) = \lambda(E),$$

which proves the uniqueness of the extension. $\square$

We turn to the proof of Theorem 6.25.

Proof of Theorem 6.25. Step 1: We apply the previous lemma with $\mathcal{S}(I)$ the semiring of all intervals $(a, b]$, where $a, b \in I$, with $a \leq b$, and with $\rho_u((a, b]) := u_+(b) - u_+(a)$, to extend ρ_u to a finitely additive measure $\rho_u : \mathcal{R}(I) \rightarrow Y$, where $\mathcal{R}(I)$ is the ring generated by $\mathcal{S}(I)$.

We claim that $\|\rho_u(E)\| \leq \mu_{\mathbf{V}}(E)$ for all $E \in \mathcal{R}(I)$. To see this, let $(a, b] \in \mathcal{S}(I)$ with $a < b$, and let $a < x < b < t < \sup I$. It follows from (2.4) that

$$\|u(t) - u(x)\| \leq \mathbf{V}(t) - \mathbf{V}(x).$$

Letting $x \searrow a$ and $t \searrow b$, by Corollary 2.31 and the fact that $\mathbf{V}$ is increasing, we get

$$\|u_+(b) - u_+(a)\| \leq \mathbf{V}_+(b) - \mathbf{V}_+(a),$$

or, equivalently, by (6.15),

$$(6.33) \quad \|\rho_u((a, b])\| \leq \mu_{\mathbf{V}}((a, b]).$$

By (6.32) and the additivity of $\mu_{\mathbf{V}}$ it follows that $\|\rho_u(E)\| \leq \mu_{\mathbf{V}}(E)$ for every $E \in \mathcal{R}(I)$, which proves the claim.

For $E, F \in \mathcal{B}_c(I)$ define

$$d(E, F) := \mu_{\mathbf{V}}(E \Delta F).$$

Reasoning as in Step 2 of the proof of Theorem 6.19 with λ replaced by ρ_u we have that d is a semimetric in $\mathcal{B}_c(I)$, that $\mathcal{R}(I)$ is dense in $\mathcal{B}_c(I)$, and that $\rho_u : \mathcal{R}(I) \rightarrow Y$ is Lipschitz continuous with respect to the semimetric d . Thus, by the density of $\mathcal{R}(I)$ in $\mathcal{B}_c(I)$ we can uniquely extend ρ_u to a function $\lambda_u : \mathcal{B}_c(I) \rightarrow Y$. Moreover, $\|\lambda_u(E)\| \leq \mu_{\mathbf{V}}(E)$ for all $E \in \mathcal{B}_c(I)$.

In $\mathcal{B}_c(I) \times \mathcal{B}_c(I)$ consider the semimetric $d_2((E_1, F_1), (E_2, F_2)) := d(E_1, E_2) + d(F_1, F_2)$. Since the function $f : \mathcal{B}_c(I) \times \mathcal{B}_c(I) \rightarrow \mathcal{B}_c(I)$ given by $f(E, F) := E \cup F$ is uniformly continuous, again by the density of $\mathcal{R}(I)$ in $\mathcal{B}_c(I)$ and the finite additivity of ρ_u , we obtain that λ_u is finitely additive.

To prove that λ_u is countably additive, let $E_n \in \mathcal{B}_c(I)$ be pairwise disjoint sets with $\bigcup_n E_n = E \in \mathcal{B}_c(I)$. Define $F_n := E \setminus \bigcup_{i=1}^n E_i$. Then $F_{n+1} \subseteq F_n$ for every n and $\bigcap_n F_n = \emptyset$. By the linearity of λ_u ,

$$\lambda_u(E) - \sum_{i=1}^n \lambda_u(E_i) = \lambda_u(F_n).$$

Since $\|\lambda_u(F_n)\| \leq \mu_{\mathbf{V}}(F_n) \rightarrow 0$ as $n \rightarrow \infty$ by Proposition B.9(ii) applied to $\mu_{\mathbf{V}}$, it follows that λ_u is countably additive.

Step 2: We prove that $\|\lambda_u\| \leq \mu_{\mathbf{V}}$. Given $E \in \mathcal{B}_c(I)$ let $(a_i, b_i] \in \mathcal{S}(I)$, $i = 1, \dots, n$, be pairwise disjoint with $\bigcup_{i=1}^n (a_i, b_i] \subseteq E$, by the additivity and monotonicity of $\mu_{\mathbf{V}}$ we have

$$\sum_{i=1}^n \|\lambda_u((a_i, b_i])\| \leq \sum_{i=1}^n \mu_{\mathbf{V}}((a_i, b_i]) \leq \mu_{\mathbf{V}}(E).$$

Taking the supremum over all sets gives $\|\lambda_u\|(E) \leq \mu_{\mathbf{V}}(E)$. In particular, if $u \in BPV(I; Y)$, then

$$\|\lambda_u\|(I) \leq \mu_{\mathbf{V}}(I) = \sup_I \mathbf{V} - \inf_I \mathbf{V} = \text{Var } u,$$

where we have used Theorem 6.1 and (2.4).

Step 3: Assume that u is right continuous. We claim that $\|\lambda_u\| = \mu_{\mathbf{V}}$. Fix $a, b \in I$, with $a \leq b$, and consider a partition $a = x_0 < x_1 < \dots < x_n = b$ of

I. By (6.30) and the fact that u is right continuous we have

$$\sum_{i=1}^n \|u(x_i) - u(x_{i-1})\| = \sum_{i=1}^n \|\lambda_u((x_{i-1}, x_i])\| \leq \|\lambda_u\|((a, b]),$$

and so

$$\text{Var}_{[a,b]} u \leq \|\lambda_u\|((a, b]).$$

In view of Proposition 2.21 the function $\mathbf{V}$ is right continuous. Hence by (2.4) and (6.13) we obtain that

$$(6.34) \quad \mu_{\mathbf{V}}((a, b]) = \mathbf{V}(b) - \mathbf{V}(a) = \text{Var}_{[a,b]} u \leq \|\lambda_u\|((a, b]).$$

Reasoning as in the previous step, we conclude that $\mu_{\mathbf{V}} \leq \|\lambda_u\|$, which, together with Step 2, gives $\|\lambda_u\| = \mu_{\mathbf{V}}$.

Step 4: Finally, if $u \in BPV(I; Y)$, then $\mu_{\mathbf{V}}(I) < \infty$ by (6.14) and so it suffices to repeat Steps 1–3 with $\mathcal{B}(I)$ in place of $\mathcal{B}_c(I)$. $\square$

Remark 6.27. If $I \subseteq \mathbb{R}$ is an open interval, $(Y, \|\cdot\|)$ is a Banach space, and $u, v \in BPV_{\text{loc}}(I; Y)$, then by the uniqueness established in the previous theorem,

$$\lambda_{u+v} = \lambda_u + \lambda_v.$$

Remark 6.28. If $I \subseteq \mathbb{R}$ is an open interval, $(Y, \|\cdot\|)$ is a normed space and $\lambda : \mathcal{B}_c(I) \rightarrow Y$ is a vector-valued measure such that $\|\lambda\| : \mathcal{B}_c(I) \rightarrow [0, \infty)$, then by the uniqueness established in the previous theorem, λ is the Lebesgue–Stieltjes measure generated by u_λ , that is, $\lambda_{u_\lambda} = \lambda$.

Exercise 6.29. Let $a \in \mathbb{R}$ and let $u := \chi_{\{a\}}$. Find λ_u and $\mu_{\mathbf{V}}$ and show that in general the inequality $\|\lambda_u\| \leq \mu_{\mathbf{V}}$ may be strict.

Exercise 6.30. Let $I \subseteq \mathbb{R}$ be an open interval and let $u \in BPV_{\text{loc}}(I)$ be a right continuous function. Prove that

$$(\lambda_u)^+ = \mu_{\frac{1}{2}(\mathbf{V}+u)}, \quad (\lambda_u)^- = \mu_{\frac{1}{2}(\mathbf{V}-u)},$$

where $(\lambda_u)^+$ and $(\lambda_u)^-$ are the (positive) Radon measures given in Proposition B.70.

Exercise 6.31. Let $I \subseteq \mathbb{R}$ be an open interval and let $u, v \in BPV_{\text{loc}}(I)$. Prove that for every $[a, b] \subset I$,

$$\frac{1}{2} \int_{(a,b)} (u_+ + u_-) d\lambda_v + \frac{1}{2} \int_{(a,b)} (v_+ + v_-) d\lambda_u = u_-(b)v_-(b) - v_+(a)u_+(a).$$

Exercise 6.32. Let $I \subseteq \mathbb{R}$ be an open interval and let $u \in BPV_{\text{loc}}(I)$. Prove that for every $\varphi \in C_c^1(I)$,

$$(6.35) \quad \int_I u \varphi' dx = - \int_I \varphi d\lambda_u.$$

6.3. Decomposition of Measures

In Corollary 3.90 we have shown that every function in $BPV_{\text{loc}}(I; \mathbb{R}^M)$ can be decomposed as the sum of three functions in $BPV_{\text{loc}}(I; \mathbb{R}^M)$, a locally absolutely continuous function, a continuous singular function, and a saltus function. In this section we prove that the corresponding Lebesgue–Stieltjes measures are mutually singular.

Let $I \subseteq \mathbb{R}$ be an open interval and let $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$. By Corollary 3.90, u may be decomposed as the sum of three functions in $BPV_{\text{loc}}(I; \mathbb{R}^M)$,

$$(6.36) \quad u = u_{\text{AC}} + u_{\text{C}} + u_{\text{J}},$$

where

$$u_{\text{AC}}(x) := \int_{t_0}^x u' dt, \quad x \in I,$$

belongs to $AC'_{\text{loc}}(I; \mathbb{R}^M)$, u_{C} is a singular function, and u_{J} is the jump function of u , that is,

$$(6.37) \quad u_{\text{J}}(x) := \begin{cases} \sum_{t \in I, t_0 \leq t < x} (u_+(t) - u_-(t)) + u(x) - u_-(x) & \text{if } x \geq t_0, \\ -\sum_{t \in I, x < t \leq t_0} (u_+(t) - u_-(t)) + u(x) - u_+(x), & \text{if } x \leq t_0, \end{cases}$$

where $t_0 \in I$.

Moreover, again by Corollary 3.90,

$$(6.38) \quad \mathbf{V}_u = \mathbf{V}_{u_{\text{AC}}} + \mathbf{V}_{u_{\text{C}}} + \mathbf{V}_{u_{\text{J}}},$$

where for a function $v : I \rightarrow \mathbb{R}^M$ we denote by $\mathbf{V}_v$ the indefinite variation of v (see (2.3)), with

$$(6.39) \quad \mathbf{V}_{u_{\text{AC}}}(x) = \int_{t_0}^x \|u'\| dt, \quad x \in I,$$

and

$$\begin{aligned} \mathbf{V}_{u_{\text{J}}}(x) = & \sum_{t \in (t_0, x)} (\|u_+(t) - u(t)\| + \|u(t) - u_-(t)\|) \\ & + \|u_+(x) - u(x)\| + \|u(x) - u_-(x)\| \end{aligned}$$

if $x \geq t_0$ and

$$\begin{aligned} \mathbf{V}_{u_{\text{J}}}(x) = & - \sum_{t \in (x, t_0)} (\|u_+(t) - u(t)\| + \|u(t) - u_-(t)\|) \\ & - \|u_+(x) - u(x)\| - \|u(x) - u_-(x)\| \end{aligned}$$

if $x < t_0$.

Let $\lambda_u, \lambda_{u_{\text{AC}}}, \lambda_{u_{\text{C}}}, \lambda_{u_{\text{J}}} : \mathcal{B}_c(I) \rightarrow \mathbb{R}^M$ be the Lebesgue–Stieltjes measures generated by $u, u_{\text{AC}}, u_{\text{C}}, u_{\text{J}}$, respectively. Using (6.36) and Remark 6.27,

we have that

$$(6.40) \quad \lambda_u = \lambda_{u_{AC}} + \lambda_{u_C} + \lambda_{u_J}.$$

Similarly, by (6.38) and by Remark 6.10,

$$(6.41) \quad \mu_{\mathbf{V}_u} = \mu_{\mathbf{V}_{u_{AC}}} + \mu_{\mathbf{V}_{u_C}} + \mu_{\mathbf{V}_{u_J}}.$$

We want to prove that the measures $\lambda_{u_{AC}}$, λ_{u_C} , λ_{u_J} (respectively, $\mu_{\mathbf{V}_{u_{AC}}}$, $\mu_{\mathbf{V}_{u_C}}$, $\mu_{\mathbf{V}_{u_J}}$) are mutually singular. Since u_{AC} and u_C are continuous by Corollary 3.90, it follows from Theorem 6.25 that

$$(6.42) \quad \|\lambda_{u_{AC}}\| = \mu_{\mathbf{V}_{u_{AC}}}, \quad \|\lambda_{u_C}\| = \mu_{\mathbf{V}_{u_C}}.$$

Set

$$(6.43) \quad I_{\text{dis}} := \{x \in I : u \text{ is discontinuous at } x\}.$$

We begin by proving that the measures λ_{u_J} and $\mu_{\mathbf{V}_{u_J}}$ are concentrated on the set I_{dis} .

Theorem 6.33. *Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a normed space, let $u \in BPV_{\text{loc}}(I; Y)$, and let u_J be as in (6.37). Then for every $E \in \mathcal{B}_c(I)$,*

$$(6.44) \quad \lambda_{u_J}(E) = \sum_{x \in E} (u_+(x) - u_-(x)), \quad \|\lambda_{u_J}\|(E) = \sum_{x \in E} \|u_+(x) - u_-(x)\|,$$

$$(6.45) \quad \mu_{\mathbf{V}_{u_J}}(E) = \sum_{x \in E} (\|u_+(x) - u(x)\| + \|u(x) - u_-(x)\|).$$

Moreover,

$$(6.46) \quad \|\lambda_u\|(E) = \mu_{\mathbf{V}_u}(E)$$

for every $E \in \mathcal{B}_c(I)$, with $E \subseteq I \setminus I_{\text{dis}}$.

Proof. Step 1: If $x \in I$, then by (6.37),

$$\begin{aligned} (u_J)_+(x) &= \sum_{t \in I, t \leq x} (u_+(t) - u_-(t)), \\ (u_J)_-(x) &= \sum_{t \in I, t < x} (u_+(t) - u_-(t)), \end{aligned}$$

where we have used the fact that

$$\lim_{t \rightarrow x^\pm} (u(t) - u_-(t)) = 0.$$

Hence, also by (6.17),

$$(6.47) \quad \lambda_{u_J}(\{x\}) = (u_J)_+(x) - (u_J)_-(x) = u_+(x) - u_-(x).$$

Since by (6.30) and (6.24), $u_{\lambda_{u_J}} = (u_J)_+$ and $(u_J)_+$ is a jump function, we can apply Theorem 6.21 to obtain (6.28), which, together with (6.47) gives (6.44).

On the other hand, taking $x_1, x_2 \in I \setminus I_{\text{dis}}$ such that $x_1 < x < x_2$, by (2.3) and Corollary 3.91, we obtain

$$\begin{aligned} \mathbf{V}_{u_J}(x_2) - \mathbf{V}_{u_J}(x_1) &= \text{Var}_{[x_1, x_2]} u_J \\ &= \sum_{t \in (x_1, x_2)} (\|u_+(t) - u(t)\| + \|u(t) - u_-(t)\|). \end{aligned}$$

Letting $x_1 \nearrow x$ and $x_2 \searrow x$, by (6.17) we get

$$\begin{aligned} (6.48) \quad \mu_{\mathbf{V}_{u_J}}(\{x\}) &= (\mathbf{V}_{u_J})_+(x) - (\mathbf{V}_{u_J})_-(x) \\ &= \|u_+(x) - u(x)\| + \|u(x) - u_-(x)\|. \end{aligned}$$

Since by (6.1) and (6.13), $u_{\mu_{\mathbf{V}_{u_J}}} = (\mathbf{V}_{u_J})_+$ and $(\mathbf{V}_{u_J})_+$ is a jump function by Proposition 2.21, we can apply Theorem 6.3 to obtain (6.3), which, together with (6.48) gives (6.45).

Step 2: Fix $E \in \mathcal{B}_c(I)$, with $E \subseteq I \setminus I_{\text{dis}}$. By the previous step and the fact that $\|\lambda_{u_J}(E)\| \leq \mu_{\mathbf{V}_{u_J}}(E)$ (see Theorem 6.25) for every Borel subset $F \subseteq E$, we have that $\lambda_{u_J}(F) = 0$ and $\mu_{\mathbf{V}_{u_J}}(E) = 0$, and so by (6.40) and (6.41),

$$\lambda_u(F) = (\lambda_{u_{AC}} + \lambda_{u_C})(F), \quad \mu_{\mathbf{V}_u}(E) = (\mu_{\mathbf{V}_{u_{AC}}} + \mu_{\mathbf{V}_{u_C}})(E),$$

which implies that

$$\|\lambda_u\|(E) = \|\lambda_{u_{AC}} + \lambda_{u_C}\|(E).$$

Since the function $u_{AC} + u_C$ is continuous, applying Theorem 6.25 and (6.38) to $u_{AC} + u_C$, we get

$$\|\lambda_u\|(E) = \|\lambda_{u_{AC} + u_C}\|(E) = \mu_{\mathbf{V}_{u_{AC} + u_C}}(E) = \mu_{\mathbf{V}_{u_{AC}} + \mathbf{V}_{u_C}}(E) = \mu_{\mathbf{V}_u}(E).$$

This concludes the proof. $\square$

Next, we show that the measures $\lambda_{u_{AC}}, \lambda_{u_C}, \lambda_{u_J}$ (respectively, $\mu_{\mathbf{V}_{u_{AC}}}, \mu_{\mathbf{V}_{u_C}}, \mu_{\mathbf{V}_{u_J}}$) are mutually singular. Set

$$(6.49) \quad I_{\text{der}0} := \{x \in I : \mathbf{V}'_{u_C}(x) = 0\}.$$

Theorem 6.34. *Let $I \subseteq \mathbb{R}$ be an open interval, let $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$. Then the measures $\lambda_{u_{AC}}, \lambda_{u_C}, \lambda_{u_J}$ (respectively, $\mu_{\mathbf{V}_{u_{AC}}}, \mu_{\mathbf{V}_{u_C}}, \mu_{\mathbf{V}_{u_J}}$) are*

mutually singular. Moreover, for every $E \in \mathcal{B}_c(I)$,

$$(6.50) \quad \lambda_{u_{AC}}(E) = \int_E u' dx, \quad \mu_{\mathbf{V}_{u_{AC}}}(E) = \|\lambda_{u_{AC}}\|(E) = \int_E \|u'\| dx,$$

$$(6.51) \quad \lambda_{u_C}(E) = \lambda_u(E \setminus (I_{\text{der}0} \cup I_{\text{dis}})),$$

$$(6.52) \quad \mu_{\mathbf{V}_{u_C}}(E) = \|\lambda_{u_C}\|(E) = \|\lambda_u\|(E \setminus (I_{\text{der}0} \cup I_{\text{dis}})).$$

Proof. In view of Theorems 2.17 and 6.33 the measures λ_{u_J} , $\|\lambda_{u_J}\|$ and $\mu_{\mathbf{V}_{u_J}}$ are purely atomic and concentrated on the countable set I_{dis} (see (6.43)). On the other hand, since u_{AC} is absolutely continuous and u_C is a singular function, it follows from Theorems 6.23 and 6.24 that the measures $\lambda_{u_{AC}}$ and $\|\lambda_{u_{AC}}\| = \mu_{\mathbf{V}_{u_{AC}}}$ are absolutely continuous with respect to $\mathcal{L}^1$, while the measures λ_{u_C} and $\|\lambda_{u_C}\| = \mu_{\mathbf{V}_{u_C}}$ are nonatomic and mutually singular with respect to $\mathcal{L}^1$.

Thus, the measures $\lambda_{u_{AC}}$, λ_{u_C} , λ_{u_J} are mutually singular, and so are the measures $\|\lambda_{u_J}\|$, $\|\lambda_{u_{AC}}\|$, $\|\lambda_{u_C}\|$ and the measures $\mu_{\mathbf{V}_{u_{AC}}}$, $\mu_{\mathbf{V}_{u_C}}$, $\mu_{\mathbf{V}_{u_J}}$.

Formulas (6.50), (6.51), and (6.52) follow from Theorems 6.23 and 6.24 and Remark 6.28. $\square$

Set

$$I_{\text{diff}} := \{x \in I : u \text{ is differentiable at } x\}.$$

The following theorem, which is known as the De la Vallée Poussin decomposition theorem, is the central result of this section.

Theorem 6.35 (De la Vallée Poussin). *Let $I \subseteq \mathbb{R}$ be an open interval, let $u \in BPV_{\text{loc}}(I; \mathbb{R}^M)$. Then for every $E \in \mathcal{B}_c(I)$,*

$$\begin{aligned} \lambda_u(E) &= \int_E u' dx + \lambda_u(E \setminus (I_{\text{diff}} \cup I_{\text{dis}})) + \sum_{x \in E \cap I_{\text{dis}}} (u_+(x) - u_-(x)), \\ \|\lambda_u\|(E) &= \int_E \|u'\| dx + \|\lambda_u\|(E \setminus (I_{\text{diff}} \cup I_{\text{dis}})) + \sum_{x \in E \cap I_{\text{dis}}} \|u_+(x) - u_-(x)\|, \end{aligned}$$

while

$$\begin{aligned} \mu_{\mathbf{V}_u}(E) &= \int_E \|u'\| dx + \|\lambda_u\|(E \setminus (I_{\text{diff}} \cup I_{\text{dis}})) \\ &\quad + \sum_{x \in E \cap I_{\text{dis}}} (\|u_+(x) - u(x)\| + \|u(x) - u_-(x)\|). \end{aligned}$$

Proof. Let $E \in \mathcal{B}_c(I)$. We begin by showing that

$$\mu_{\mathbf{V}_u}((E \cap I_{\text{der}0}) \setminus (I_{\text{diff}} \cup I_{\text{dis}})) = 0.$$

By the Lebesgue differentiation theorem and the fact that I_{dis} is countable, $\mathcal{L}^1(I \setminus (I_{\text{diff}} \cup I_{\text{dis}})) = 0$. In particular, $\mathcal{L}^1(I_{\text{der}0} \setminus (I_{\text{diff}} \cup I_{\text{dis}})) = 0$. Since

$\mu_{\mathbf{V}_{u_{\text{AC}}}} \ll \mathcal{L}^1$, it follows that $\mu_{\mathbf{V}_{u_{\text{AC}}}}((E \cap I_{\text{der}0}) \setminus (I_{\text{diff}} \cup I_{\text{dis}})) = 0$. On the other hand, by (6.52), $\mu_{\mathbf{V}_{u_{\text{C}}}}((E \cap I_{\text{der}0}) \setminus (I_{\text{diff}} \cup I_{\text{dis}})) = 0$, while by (6.45), $\mu_{\mathbf{V}_{u_{\text{J}}}}((E \cap I_{\text{der}0}) \setminus (I_{\text{diff}} \cup I_{\text{dis}})) = 0$. The claim now follows from (6.38).

In particular, since $\|\lambda_u\| \leq \mu_{\mathbf{V}_u}$ by Theorem 6.25, it follows that for every $F \in \mathcal{B}_c(I)$, with $F \subseteq E$,

$$(6.53) \quad \lambda_u((F \cap I_{\text{der}0}) \setminus (I_{\text{diff}} \cup I_{\text{dis}})) = 0.$$

Using (6.53) we get

$$\begin{aligned} \lambda_u(E) &= \lambda_u((E \cap I_{\text{diff}} \cap I_{\text{der}0}) \setminus I_{\text{dis}}) + \lambda_u(E \setminus (I_{\text{diff}} \cup I_{\text{der}0} \cup I_{\text{dis}})) \\ &\quad + \lambda_u(E \cap I_{\text{dis}}) \\ &= \lambda_u((E \cap I_{\text{diff}} \cap I_{\text{der}0}) \setminus I_{\text{dis}}) + \lambda_u(E \setminus (I_{\text{diff}} \cup I_{\text{dis}})) + \lambda_u(E \cap I_{\text{dis}}). \end{aligned}$$

By Theorem 6.33 and Theorem 6.34,

$$\lambda_{u_{\text{J}}}(E \cap I_{\text{dis}}) = \sum_{x \in E \cap I_{\text{dis}}} (u_+(x) - u_-(x))$$

and

$$\begin{aligned} \lambda_u((E \cap I_{\text{diff}} \cap I_{\text{der}0}) \setminus I_{\text{dis}}) &= \lambda_{u_{\text{AC}}}((E \cap I_{\text{diff}} \cap I_{\text{der}0}) \setminus I_{\text{dis}}) + 0 + 0 \\ &= \int_{(E \cap I_{\text{diff}} \cap I_{\text{der}0}) \setminus I_{\text{dis}}} u' dx = \int_E u' dx, \end{aligned}$$

where we used (6.40) and the facts that I_{dis} is countable, $\mathcal{L}^1(I \setminus I_{\text{diff}}) = 0$ by the Lebesgue differentiation theorem, and $\mathcal{L}^1(I \setminus I_{\text{der}0}) = 0$, since $\mathbf{V}_{u_{\text{C}}}$ is a singular function by Theorem 2.28.

The proof for $\|\lambda_u\|$ and $\mu_{\mathbf{V}_u}$ is similar and is left as an exercise. $\square$

Remark 6.36. An alternative proof of Theorem 6.35 is to use the Lebesgue decomposition theorem to decompose λ_u into the sum of three mutually singular measures: a measure absolutely continuous with respect to the Lebesgue measure, a nonatomic measure singular with respect to the Lebesgue measure, and a purely atomic measure. Then one can use the Radon–Nikodym theorem and the Vitali–Besicovitch differentiation theorem to show that the absolutely continuous measure with respect to the Lebesgue measure takes the form

$$\int_E u' dx.$$

This approach is taken up in [201]. Here, we arrived at the same conclusion using instead the decomposition of u proved in Corollary 3.90.

Functions of Bounded Variation and Sobolev Functions

Prospective Grad Students, I: “Will my teachers take personal interest in my learning, or will I be spending all-nighters working on problem sets made by frustrated faculty who would rather be doing research instead?”

— Jorge Cham, www.phdcomics.com

In Chapter 6 we have seen that to every function $u : I \rightarrow \mathbb{R}^M$ of bounded pointwise variation we may associate a vector-valued measure λ_u for which the integration by parts formula (6.35) holds. From the distributional point of view (see Chapter 10), this implies that the distributional or weak derivative of u is the vector-valued measure λ_u . In the following section we will see that this leads us to an “alternative” characterization of the space of functions of bounded pointwise variation, which has the advantage of carrying over to functions defined in $\mathbb{R}^N$. In this chapter we will consider functions from an open set $\Omega \subseteq \mathbb{R}$ with values in $\mathbb{R}^M$.

7.1. $BV(\Omega)$ Versus $BPV(\Omega)$

Definition 7.1. Given an open set $\Omega \subseteq \mathbb{R}$, the *space of functions of bounded variation* $BV(\Omega; \mathbb{R}^M)$ is defined as the space of all functions $u \in L^1(\Omega; \mathbb{R}^M)$ for which there exists a vector-valued measure $\lambda : \mathcal{B}(\Omega) \rightarrow \mathbb{R}^M$ such that

$$\int_{\Omega} u \varphi' dx = - \int_{\Omega} \varphi d\lambda$$

for all $\varphi \in C_c^1(\Omega)$. The measure λ is called the *weak* or *distributional derivative* of u and is denoted Du .

We recall that $C_c^1(\Omega)$ is the space of all C^1 functions $\varphi : \Omega \rightarrow \mathbb{R}$ with compact support contained in Ω .

Exercise 7.2. Let $\Omega \subseteq \mathbb{R}$ be an open set and let $u \in BV(\Omega; \mathbb{R}^M)$. Prove that the weak distributional derivative Du of u is unique.

The next result establishes the relationship between functions of bounded variation and functions of bounded pointwise variation. In what follows, $\|\lambda\|$ is the total variation measure of λ (see (6.22)).

Theorem 7.3. Let $\Omega \subseteq \mathbb{R}$ be an open set. If $u : \Omega \rightarrow \mathbb{R}^M$ is integrable and belongs to $BPV(\Omega; \mathbb{R}^M)$, then $u \in BV(\Omega; \mathbb{R}^M)$ and

$$\|Du\|(\Omega) \leq \text{Var } u.$$

Conversely, if $u \in BV(\Omega; \mathbb{R}^M)$, then u admits a right continuous representative $\bar{u}$ in $BPV(\Omega; \mathbb{R}^M)$ such that

$$\text{Var } \bar{u} = \|Du\|(\Omega).$$

We begin with an auxiliary result.

Lemma 7.4. Given an open interval $I \subseteq \mathbb{R}$, let $u \in L_{\text{loc}}^1(I; \mathbb{R}^M)$ be such that

$$\int_I u \varphi' dx = 0$$

for all $\varphi \in C_c^1(I)$. Then there exists a constant $c \in \mathbb{R}^M$ such that $u(x) = c$ for $\mathcal{L}^1$ -a.e. $x \in I$.

Proof. Step 1: The proof is simpler in the case in which $u \in L^1(I)$ and $I = (a, b)$, and thus we begin with this case. A density argument shows that

$$(7.1) \quad \int_a^b u \varphi' dx = 0$$

for all $\varphi \in C^1([a, b])$ with $\varphi(a) = \varphi(b) = 0$. We claim that

$$(7.2) \quad \int_a^b uw dx = 0$$

for all $w \in C([a, b])$ with $\int_a^b w dt = 0$. To see this, fix any such w and define the function

$$\varphi(x) := \int_a^x w(t) dt, \quad x \in I.$$

Since $\varphi(a) = 0$ and $\varphi(b) = \int_a^b w(t) dt = 0$, we have that φ is admissible in (7.1), and so the claim holds.

Now, let $w \in C([a, b])$. Taking $w - \frac{1}{b-a} \int_a^b w dt$ in (7.2), we get

$$\int_a^b u \left(w - \frac{1}{b-a} \int_a^b w dt \right) dx = 0,$$

which can be written as

$$\int_a^b w \left(u - \frac{1}{b-a} \int_a^b u \, dt \right) dx = 0.$$

Since this is true for all $w \in C([a, b])$, again by density we obtain that

$$\int_a^b \chi_E \left(u - \frac{1}{b-a} \int_a^b u \, dt \right) dx = 0$$

for all Lebesgue measurable sets $E \subseteq (a, b)$. This implies that (why?)

$$u(x) - \frac{1}{b-a} \int_a^b u \, dt = 0$$

for $\mathcal{L}^1$ -a.e. $x \in I$.

Step 2: In the general case, let (a_n, b_n) be an increasing sequence of open intervals such that $[a_n, b_n] \subset (a_{n+1}, b_{n+1})$ and

$$I = \bigcup_{n=1}^{\infty} (a_n, b_n).$$

Then for every n ,

$$\int_{(a_n, b_n)} u \varphi' \, dx = 0$$

for all $\varphi \in C_c^1((a_n, b_n))$. Hence, by Step 1, there exists $c_n \in \mathbb{R}$ such that $u(x) = c_n$ for $\mathcal{L}^1$ a.e. $x \in (a_n, b_n)$. But since, $[a_n, b_n] \subset (a_{n+1}, b_{n+1})$, it follows that $c_n = c_{n+1}$ for all n . $\square$

Proof of Theorem 7.3. By working on each connected component of Ω , we may assume that $\Omega = I$, with $I \subseteq \mathbb{R}$ an open interval.

Assume that $u : I \rightarrow \mathbb{R}^M$ is integrable and belongs to $BPV(I; \mathbb{R}^M)$. Let λ_u be vector-valued measure given in Theorem 6.25. Then $\|\lambda_u\|$ is finite with

$$\|\lambda_u\|(I) \leq \text{Var } u.$$

Moreover, by Exercise 6.32 we have that

$$\int_I u \varphi' \, dx = - \int_I \varphi \, d\lambda_u$$

for all $\varphi \in C_c^1(I)$. Hence, $u \in BV(I; \mathbb{R}^M)$.

Conversely, if $u \in BV(I; \mathbb{R}^M)$, let $\lambda : \mathcal{B}(I) \rightarrow \mathbb{R}^M$ be the vector-valued Radon measure given in Definition 7.1. Then

$$\int_I u \varphi' \, dx = - \int_I \varphi \, d\lambda$$

for all $\varphi \in C_c^1(I)$. Define u_λ as in (6.24). Then by Theorem 6.19, the function u_λ is right continuous, has bounded pointwise variation, and $\text{Var } u_\lambda = \|\lambda\|(I)$. Moreover, by Exercise 6.32,

$$\int_I u_\lambda \varphi' dx = - \int_I \varphi d\lambda$$

for all $\varphi \in C_c^1(I)$. It follows that

$$\int_I (u - u_\lambda) \varphi' dx = 0$$

for all $\varphi \in C_c^1(I)$. By the previous lemma, this implies that $u - u_\lambda$ is constant for $\mathcal{L}^1$ -a.e. $x \in I$. It follows that $u_\lambda + \text{const}$ is a representative of u . $\square$

Remark 7.5. Since a function of bounded pointwise variation is bounded (see Corollary 2.31), if I is bounded, then u is integrable, and so in the first part of the statement of Theorem 7.3 the hypothesis that u is integrable is redundant. However, when I is unbounded, the function $u \equiv 1$ has bounded pointwise variation, but it is not integrable and thus does not belong to $BV(I)$.

An alternative proof of the first part of the previous theorem is given by the next exercise.

Exercise 7.6. Let $u \in BVP((a, b))$.

- (i) Prove that u is Riemann integrable and that for every $\varphi \in C_c^1((a, b))$ the Riemann integral $\int_a^b u \varphi' dx$ is well-defined.
- (ii) Prove that for every $\varphi \in C_c^1((a, b))$ and for every $\varepsilon > 0$ there exists a partition of $[a, b]$, $x_0 := a < x_1 < \cdots < x_n := b$, such that

$$\int_a^b u \varphi' dx \leq \sum_{i=1}^n u(x_i) \varphi'(x_i) (x_i - x_{i-1}) + \varepsilon.$$

- (iii) Prove that the partition in part (ii) may be chosen so that

$$\sum_{i=1}^n u(x_i) \varphi'(x_i) (x_i - x_{i-1}) \leq \sum_{i=1}^n u(x_i) [\varphi(x_i) - \varphi(x_{i-1})] + \varepsilon.$$

- (iv) Prove that for every $\varphi \in C_c^1((a, b))$,

$$\int_a^b u \varphi' dx \leq \sup_{[a, b]} |\varphi| \text{Var}_{(a, b)} u.$$

- (v) Prove that the functional

$$L : C_c^1((a, b)) \rightarrow \mathbb{R}$$

$$\varphi \mapsto L(\varphi) := \int_a^b u \varphi' dx$$

is linear and that it can be extended uniquely to a linear continuous functional $L : C_0((a, b)) \rightarrow \mathbb{R}$, with $\|L\|_{(C_0((a, b)))'} \leq \text{Var}_{(a, b)} u$.

Exercise 7.7. Let $\Omega \subseteq \mathbb{R}$ be an open set. Prove that the space $BV(\Omega; \mathbb{R}^M)$ is a Banach space with the norm

$$\|u\|_{BV(\Omega; \mathbb{R}^M)} := \|u\|_{L^1(\Omega; \mathbb{R}^M)} + \|Du\|(\Omega).$$

Next we introduce a notion of pointwise variation that does not change if we change the representative of a function $u \in L^1(I; \mathbb{R}^M)$.

Definition 7.8. Let $\Omega \subseteq \mathbb{R}$ be an open set and let $u : \Omega \rightarrow \mathbb{R}^M$ be a locally integrable function. The *essential pointwise variation* of u is defined as

$$\text{essVar } u := \inf \{ \text{Var } v : v : \Omega \rightarrow \mathbb{R}, v(x) = u(x) \text{ for } \mathcal{L}^1\text{-a.e. } x \in \Omega \}.$$

It follows from the previous definition that if $\Omega \subseteq \mathbb{R}$ is an open set and $u : \Omega \rightarrow \mathbb{R}^M$ is a locally integrable function, then for every function $v : \Omega \rightarrow \mathbb{R}^M$ that coincides with u except on a set of Lebesgue measure zero,

$$\text{essVar } u = \text{essVar } v.$$

Theorem 7.9. Let $\Omega \subseteq \mathbb{R}$ be an open set and let $u \in L^1(\Omega; \mathbb{R}^M)$. Then $\text{essVar } u < \infty$ if and only if u belongs to $BV(\Omega; \mathbb{R}^M)$. Moreover,

$$\|Du\|(\Omega) = \text{essVar } u.$$

Proof. Assume that $\text{essVar } u < \infty$. Then for every $\varepsilon > 0$ there exists a representative v of u such that

$$\text{Var } v \leq \text{essVar } u + \varepsilon < \infty.$$

By Theorem 7.3, it follows that v (and hence u) belongs to $BV(\Omega; \mathbb{R}^M)$ and

$$\|Du\|(\Omega) \leq \text{Var } v \leq \text{essVar } u + \varepsilon.$$

Given the arbitrariness of $\varepsilon > 0$, we conclude that $\|Du\|(\Omega) \leq \text{essVar } u$.

Conversely, assume that $u \in BV(\Omega; \mathbb{R}^M)$. Then by Theorem 7.3, u admits a right continuous representative $\bar{u}$, which has bounded pointwise variation and

$$\text{Var } \bar{u} = \|Du\|(\Omega).$$

In particular,

$$\text{essVar } u \leq \text{Var } \bar{u} = \|Du\|(\Omega).$$

This concludes the proof. $\square$

Exercise 7.10. Let $I \subseteq \mathbb{R}$ be an open interval and let $u \in BV(I; \mathbb{R}^M)$. Prove that

$$\text{essVar } u = \sup \left\{ \sum_{i=1}^n \|u(x_i) - u(x_{i-1})\| \right\},$$

where the supremum is taken over all partitions $P := \{x_0, \dots, x_n\}$ of I such that each x_i is a Lebesgue point of u (see Corollary B.119), $i = 1, \dots, n$, $n \in \mathbb{N}$.

In the remainder of this section we extend some of the previous results to functions that have only locally bounded variation. The *space of functions of locally bounded variation* $BV_{\text{loc}}(\Omega; \mathbb{R}^M)$ is defined as the space of all functions $u \in L^1_{\text{loc}}(\Omega; \mathbb{R}^M)$ such that $u \in BV(U; \mathbb{R}^M)$ for all open sets $U \Subset \Omega$. We recall that $\mathcal{B}_c(I)$ is the σ -field of all Borel subsets compactly contained in I (see (6.21)).

Exercise 7.11. Let $\Omega \subseteq \mathbb{R}$ be an open set and let $u \in BV_{\text{loc}}(\Omega; \mathbb{R}^M)$. Prove that there is a vector-valued measure $\lambda : \mathcal{B}_c(\Omega) \rightarrow \mathbb{R}^M$ such that for all $\varphi \in C_c^1(\Omega)$,

$$\int_{\Omega} u \varphi' dx = - \int_{\Omega} \varphi d\lambda.$$

Exercise 7.12. Let $\Omega \subseteq \mathbb{R}$ be an open set and let $u : \Omega \rightarrow \mathbb{R}^M$.

- (i) Prove that if $u : \Omega \rightarrow \mathbb{R}^M$ has locally bounded pointwise variation, then u has locally bounded variation.
- (ii) Prove that if $u \in L^1_{\text{loc}}(\Omega; \mathbb{R}^M)$ has locally bounded variation, then u admits a right continuous representative that has locally bounded pointwise variation.

7.2. Sobolev Functions Versus Absolutely Continuous Functions

In this section we introduce the notion of Sobolev functions and study their relation to absolutely continuous functions.

Definition 7.13. Given an open set $\Omega \subseteq \mathbb{R}$, $n \in \mathbb{N}$, and $1 \leq p \leq \infty$, we say that a function $u \in L^1_{\text{loc}}(\Omega; \mathbb{R}^M)$ admits a *weak* or *distributional derivative of order n in $L^p(\Omega; \mathbb{R}^M)$* if there exists a function $v \in L^p(\Omega; \mathbb{R}^M)$ such that

$$\int_{\Omega} u \varphi^{(n)} dx = (-1)^n \int_{\Omega} v \varphi dx$$

for all $\varphi \in C_c^\infty(\Omega)$. The function v is denoted $u^{(n)}$.

A similar definition can be given when $L^p(\Omega; \mathbb{R}^M)$ is replaced by $L^p_{\text{loc}}(\Omega; \mathbb{R}^M)$.

Definition 7.14. Given an open set $\Omega \subseteq \mathbb{R}$, $m \in \mathbb{N}$, and $1 \leq p \leq \infty$, the *Sobolev space* $W^{m,p}(\Omega; \mathbb{R}^M)$ is the space of all functions $u \in L^p(\Omega; \mathbb{R}^M)$

which admit weak derivatives of order n in $L^p(\Omega; \mathbb{R}^M)$ for every $n = 1, \dots, m$. The space $W^{m,p}(\Omega; \mathbb{R}^M)$ is endowed with the norm

$$\|u\|_{W^{m,p}(\Omega; \mathbb{R}^M)} := \|u\|_{L^p(\Omega; \mathbb{R}^M)} + \sum_{n=1}^m \|u^{(n)}\|_{L^p(\Omega; \mathbb{R}^M)}.$$

The space $W_{\text{loc}}^{m,p}(\Omega; \mathbb{R}^M)$ is defined as the space of all functions $u \in L_{\text{loc}}^p(\Omega; \mathbb{R}^M)$ which admit weak derivatives of order n in $L_{\text{loc}}^p(\Omega; \mathbb{R}^M)$ for every $n = 1, \dots, m$.

When $M=1$ for simplicity we write $W^{m,p}(\Omega)$ and $W_{\text{loc}}^{m,p}(\Omega)$ for $W^{m,p}(\Omega; \mathbb{R})$ and $W_{\text{loc}}^{m,p}(\Omega; \mathbb{R})$, respectively.

Exercise 7.15. Let $\Omega \subseteq \mathbb{R}$ be an open set, let $n, M \in \mathbb{N}$, and let $u \in L_{\text{loc}}^1(\Omega; \mathbb{R}^M)$ be such that its weak derivative $u^{(n)}$ of order n exists in $L_{\text{loc}}^1(\Omega; \mathbb{R}^M)$. Prove that $u^{(n)}$ is unique.

Note that $W^{1,1}(\Omega; \mathbb{R}^M) \subset BV(\Omega; \mathbb{R}^M)$. Indeed, if $u \in W^{1,1}(\Omega; \mathbb{R}^M)$, then in the definition of $BV(\Omega; \mathbb{R}^M)$, we can take the vector-valued measure

$$\lambda(E) := \int_E u' dx, \quad E \subseteq \mathcal{B}(\Omega).$$

The following theorem, together with Corollary 3.90, shows that the inclusion $W^{1,1}(\Omega; \mathbb{R}^M) \subset BV(\Omega; \mathbb{R}^M)$ is strict.

Theorem 7.16. *Let $\Omega \subseteq \mathbb{R}$ be an open set and let $1 \leq p \leq \infty$. Then a function $u : \Omega \rightarrow \mathbb{R}^M$ belongs to $W^{1,p}(\Omega; \mathbb{R}^M)$ if and only if it admits an absolutely continuous representative $\bar{u} : \Omega \rightarrow \mathbb{R}^M$ such that $\bar{u}$ and its classical derivative $\bar{u}'$ belong to $L^p(\Omega; \mathbb{R}^M)$. Moreover, if $p > 1$, then $\bar{u}$ is Hölder continuous of exponent $1/p'$.*

Proof. By working on each connected component of Ω , we may assume that $\Omega = I$, with $I \subseteq \mathbb{R}$ an open interval. If $\bar{u} : I \rightarrow \mathbb{R}^M$ is absolutely continuous with $\bar{u} \in L^p(I; \mathbb{R}^M)$ and $\bar{u}' \in L^p(I; \mathbb{R}^M)$, then by Corollary 3.23,

$$\int_I u \varphi' dx = - \int_I u' \varphi dx$$

for all $\varphi \in C_c^1(I)$. It follows that $\bar{u}'$ is the weak derivative of $\bar{u}$ and that $\bar{u} \in W^{1,p}(I; \mathbb{R}^M)$.

Conversely, assume that $u \in W^{1,p}(I; \mathbb{R}^M)$ and let v be its weak derivative. Fix a Lebesgue point $x_0 \in I$ of u (see Corollary B.119) and define

$$(7.3) \quad \bar{u}(x) := u(x_0) + \int_{x_0}^x v(t) dt, \quad x \in I.$$

Since $v \in L^p(\Omega; \mathbb{R}^M)$, we have that v is locally integrable, and thus, by applying Corollary 3.19, we have that $\bar{u}$ is locally absolutely continuous,

with $\bar{u}'(x) = v(x)$ for $\mathcal{L}^1$ -a.e. $x \in I$, and, in turn, by Corollary 3.31 we have that $\bar{u}$ is actually absolutely continuous in I (when $p = \infty$ observe that since v is bounded, by (7.3) the function $\bar{u}$ is Lipschitz continuous). Using Corollary 3.23, we get

$$\int_I \bar{u} \varphi' dx = - \int_I \bar{u}' \varphi dx = - \int_I v \varphi dx$$

for all $\varphi \in C_c^1(I)$. Hence, we have that

$$\int_I (u - \bar{u}) \varphi' dx = 0$$

for all $\varphi \in C_c^1(I)$. By Lemma 7.4 it follows that $u - \bar{u}$ is constant $\mathcal{L}^1$ -a.e. in I , and since $x_0 \in I$ is a Lebesgue point of u and $u(x_0) = \bar{u}(x_0)$, we have that $u = \bar{u}$ $\mathcal{L}^1$ -a.e. in I . This shows that u has an absolutely continuous representative.

To prove that $\bar{u}$ is Hölder continuous of exponent $1/p'$, let $x, y \in I$, with $x < y$. By Theorem 3.20,

$$\bar{u}(x) - \bar{u}(y) = \int_y^x u'(t) dt,$$

and so, by Hölder's inequality,

$$\begin{aligned} (7.4) \quad \|\bar{u}(x) - \bar{u}(y)\| &\leq \int_y^x \|u'(t)\| dt \leq (x - y)^{1/p'} \left(\int_y^x \|u'(t)\|^p dt \right)^{1/p} \\ &\leq (x - y)^{1/p'} \left(\int_\Omega \|u'(t)\|^p dt \right)^{1/p}, \end{aligned}$$

which shows that $\bar{u}$ is Hölder continuous of exponent $1/p'$. $\square$

Exercise 7.17. Let $\Omega \subseteq \mathbb{R}$ be an open set, let $u : \Omega \rightarrow \mathbb{R}^M$, let $m \in \mathbb{N}$, with $m \geq 2$, and let $1 \leq p < \infty$. Prove that $u \in W^{m,p}(\Omega; \mathbb{R}^M)$ if and only if it admits a representative $\bar{u} : \Omega \rightarrow \mathbb{R}^M$ of class $C^{m-1}(\Omega; \mathbb{R}^M)$ such that its classical derivative of order $m - 1$ is absolutely continuous and $\bar{u}$ and all its classical derivatives $\bar{u}^{(n)}$, $n = 1, \dots, m$, belong to $L^p(\Omega; \mathbb{R}^M)$.

Exercise 7.18. Given an open set $\Omega \subseteq \mathbb{R}$, $m, M \in \mathbb{N}$, and $1 \leq p \leq \infty$, prove that $W^{m,p}(\Omega; \mathbb{R}^M)$ is a Banach space.

Exercise 7.19. Given an open set $\Omega \subseteq \mathbb{R}$, $m, M \in \mathbb{N}$, prove that

$$H^m(\Omega; \mathbb{R}^M) := W^{m,2}(\Omega; \mathbb{R}^M)$$

is a Hilbert space.

As usual, when $M = 1$ we write $H^m(\Omega)$ for $H^m(\Omega; \mathbb{R}^M)$.

Corollary 7.20. *Let $\Omega \subseteq \mathbb{R}$ be an open bounded set and let $u : \Omega \rightarrow \mathbb{R}^M$. Then $u \in W^{1,1}(\Omega; \mathbb{R}^M)$ if and only if it admits an absolutely continuous representative $\bar{u} : \Omega \rightarrow \mathbb{R}^M$.*

Proof. If $\bar{u} : \Omega \rightarrow \mathbb{R}^M$ is absolutely continuous, then by Corollary 3.10, $\bar{u}$ is bounded, and so integrable, and $\bar{u}'$ is integrable. We are now in a position to apply the previous theorem. $\square$

Corollary 7.21. *Let $\Omega \subseteq \mathbb{R}$ be an open bounded set, let $u : \Omega \rightarrow \mathbb{R}^M$, and let $1 < p < \infty$. Then $u \in W^{1,p}(\Omega; \mathbb{R}^M)$ if and only if it admits an absolutely continuous representative $\bar{u} : \Omega \rightarrow \mathbb{R}^M$ such that its classical derivative $\bar{u}'$ belongs to $L^p(\Omega; \mathbb{R}^M)$.*

Proof. Since for sets of finite measure $L^p(\Omega; \mathbb{R}^M) \subset L^1(\Omega; \mathbb{R}^M)$, we have that $W^{1,p}(\Omega; \mathbb{R}^M) \subset W^{1,1}(\Omega; \mathbb{R}^M)$. Thus, the result follows from Corollary 7.20. $\square$

The next exercise gives an example of an absolutely continuous function in $L^1(\mathbb{R}) \setminus W^{1,1}(\mathbb{R})$.

Exercise 7.22. Let

$$g(x) := \begin{cases} \frac{1}{n} & \text{if } x \in (n-1 + \frac{2k}{2n}, n-1 + \frac{2k+1}{2n}], \quad n \in \mathbb{N}, 0 \leq k < n, \\ -\frac{1}{n} & \text{if } x \in (n-1 + \frac{2k+1}{2n}, n-1 + \frac{2k+2}{2n}], \quad n \in \mathbb{N}, 0 \leq k < n, \end{cases}$$

and let

$$u(x) := \begin{cases} \int_0^x g(t) dt & \text{if } x \geq 0, \\ u(-x) & \text{if } x < 0. \end{cases}$$

- (i) Prove that u is absolutely continuous.
- (ii) Prove that $u \in L^1(\mathbb{R})$.
- (iii) Prove that $u' \in \bigcap_{\varepsilon > 0} L^{1+\varepsilon}(\mathbb{R}) \setminus L^1(\mathbb{R})$.

Theorem 7.23 (Superposition). *Let $\Omega \subseteq \mathbb{R}$ be an open set, let $1 \leq p < \infty$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$. Then $f \circ u \in W^{1,p}(\Omega)$ for all functions $u \in W^{1,p}(\Omega)$ if and only if f is locally Lipschitz continuous and, when $\mathcal{L}^1(\Omega) = \infty$, $f(0) = 0$.*

Proof. The proof relies on Theorems 3.55, Exercise 3.58, and Theorem 7.16 and is left as an exercise. $\square$

Exercise 7.24. Given $m, M \in \mathbb{N}$, and $1 \leq p \leq \infty$, let $u \in W^{m,p}((a, b); \mathbb{R}^M)$. Prove that there exist $a_1, \dots, a_m \in \mathbb{R}$ and $b_1, \dots, b_m \in (0, 1]$ such that the function

$$v(x) := \begin{cases} \sum_{n=1}^m a_n u(a + b_n(a - x)) & \text{if } 2a - b < x < a, \\ u(x) & \text{if } a < x < b, \\ \sum_{n=1}^m a_n u(a + b_n(x - b)) & \text{if } b < x < 2b - a, \end{cases}$$

is well-defined and belongs to $W^{m,p}((2a-b, 2b-a); \mathbb{R}^M)$. Prove also that

$$\|v\|_{W^{m,p}((2a-b, 2b-a))} \leq c\|u\|_{W^{m,p}((a,b))}$$

for some constant $c = c(m, M, p) > 0$.

Exercise 7.25. Let $I \subset \mathbb{R}$ be an open interval with infinite length, let $m, M \in \mathbb{N}$, let $1 \leq p \leq \infty$, and let $u \in W^{m,p}(I; \mathbb{R}^M)$. Prove that u can be extended to a function $v \in W^{m,p}(\mathbb{R}; \mathbb{R}^M)$ with

$$\|v\|_{W^{m,p}(\mathbb{R})} \leq c\|u\|_{W^{m,p}(I)}$$

for some constant $c = c(m, M, p) > 0$.

Theorem 7.26. Let $I \subseteq \mathbb{R}$ be an open interval, let $m, M \in \mathbb{N}$, and let $1 \leq p < \infty$. Then functions in $C^\infty(I; \mathbb{R}^M) \cap W^{m,p}(I; \mathbb{R}^M)$ are dense in $W^{m,p}(I; \mathbb{R}^M)$.

Proof. Let $u \in W^{m,p}(I; \mathbb{R}^M)$. By Exercises 7.24 and 7.25 we can extend u to a function $v \in W^{m,p}(J; \mathbb{R}^M)$ with

$$\|v\|_{W^{m,p}(J)} \leq c\|u\|_{W^{m,p}(I)},$$

where J is an interval containing $\bar{I}$ and such that $\text{dist}(I, \partial J) > 0$. For $0 < \varepsilon < \text{dist}(I, \partial J)$ let $v_\varepsilon := v * \varphi_\varepsilon$, where with $\varphi \in C_c^\infty(\mathbb{R})$ satisfies $\text{supp } \varphi \subseteq \overline{B(0, 1)}$, $\int_{\mathbb{R}} \varphi(x) dx = 1$ and $\varphi_\varepsilon(x) := \frac{1}{\varepsilon} \varphi(\frac{x}{\varepsilon})$. By Theorem C.20, $v_\varepsilon \in C^\infty(\mathbb{R}; \mathbb{R}^M)$, with

$$v_\varepsilon^{(n)}(x) = \int_J \frac{d^n \varphi_\varepsilon}{dx^n}(x-t)v(t) dt = (-1)^n \int_J \frac{d^n \varphi_\varepsilon}{dt^n}(x-t)v(t) dt$$

for $x \in \mathbb{R}$ and $n \in \mathbb{N}$. In particular, if $x \in I$, then since $\varepsilon < \text{dist}(I, \partial J)$ we have that

$$\text{supp } \varphi_\varepsilon(x - \cdot) \subseteq \overline{B(x, \varepsilon)} \subseteq J,$$

and so we can integrate by parts to get

$$v_\varepsilon^{(n)}(x) = \int_J \varphi_\varepsilon(x-t)v^{(n)}(t) dt = (v^{(n)} * \varphi_\varepsilon)(x).$$

It now follows from Theorem C.16 applied to v and to $v^{(n)}$ that

$$\|v_\varepsilon - u\|_{W^{m,p}(I)} = \|v_\varepsilon - v\|_{W^{m,p}(I)} \rightarrow 0$$

as $\varepsilon \rightarrow 0^+$. □

Exercise 7.27. Prove that the previous theorem continues to hold when the interval I is replaced by an open set $\Omega \subseteq \mathbb{R}$.

Finally, we consider the case $p = \infty$.

Theorem 7.28. Let $I \subseteq \mathbb{R}$ be an open interval and let $u : I \rightarrow \mathbb{R}^M$. Then $u \in W^{1,\infty}(I; \mathbb{R}^M)$ if and only if it admits a bounded, Lipschitz continuous representative $\bar{u} : I \rightarrow \mathbb{R}^M$.

Proof. If $\bar{u} : I \rightarrow \mathbb{R}^M$ is Lipschitz continuous, then it is absolutely continuous, and so, as in the proof of Theorem 7.16, we conclude that $\bar{u} \in W^{1,\infty}(I; \mathbb{R}^M)$.

Conversely, assume that $u \in W^{1,\infty}(I; \mathbb{R}^M)$ and define the function $\bar{u}$ as in (7.3). Since $v \in L^\infty(I; \mathbb{R}^M)$, we have that v is Lipschitz continuous and locally integrable, and thus we may proceed as in the proof of Theorem 7.16 to conclude that $\bar{u} = u$ $\mathcal{L}^1$ -a.e. in I . $\square$

We conclude this section by showing that in bounded domains, up to constants, the L^p norm of a Sobolev function can be estimated by the L^p norm of its weak derivative.

Theorem 7.29 (Poincaré's inequality). *Let $I = (a, b)$ and let $1 \leq p < \infty$. Then*

$$(7.5) \quad \int_a^b \|u(x) - u_I\|^p dx \leq (b-a)^p \int_a^b \|u'(x)\|^p dx$$

for all $u \in W^{1,p}((a, b); \mathbb{R}^M)$, where

$$u_I := \frac{1}{b-a} \int_a^b u(x) dx.$$

Proof. Fix $u \in W^{1,p}((a, b); \mathbb{R}^M)$. By Corollaries 7.20 and 7.21 there exists an absolutely continuous representative of u , $\bar{u} : (a, b) \rightarrow \mathbb{R}^M$, such that $\bar{u}'$ belongs to $L^p((a, b); \mathbb{R}^M)$. By Exercise 3.8 we may extend $\bar{u}$ to $[a, b]$ in such a way that the extension is still absolutely continuous. Since $\bar{u}$ is continuous, by the mean value theorem for every $i = 1, \dots, M$ there exists $x_i \in [a, b]$ such that

$$(u_I)_i = \frac{1}{b-a} \int_a^b \bar{u}_i(x) dx = \bar{u}_i(x_i).$$

By Theorem 3.20, for all $x \in (a, b)$,

$$\bar{u}_i(x) - (u_I)_i = \int_{x_i}^x \bar{u}'_i(t) dt,$$

and so as in (7.4),

$$\|\bar{u}(x) - u_I\| \leq (b-a)^{1/p'} \left(\int_a^b \|\bar{u}'(t)\|^p dt \right)^{1/p}.$$

Raising both sides to exponent p and integrating in x gives

$$\int_a^b \|u(x) - u_I\|^p dx \leq (b-a)^p \int_a^b \|\bar{u}'(t)\|^p dt,$$

where we have used the fact that $p/p' = p - 1$. $\square$

Exercise 7.30. State and prove the analog of Poincaré's inequality in $W^{m,p}((a, b))$.

Exercise 7.31. Let $I = (a, b)$, let $J = [c, d] \subset (a, b)$ and let $1 \leq p < \infty$. Prove that there exists a constant $\kappa > 0$ such that

$$\int_a^b |u(x) - u_J|^p dx \leq \kappa(b-a)^p \int_a^b |u'(x)|^p dx$$

for all $u \in W_{\text{loc}}^{1,p}((a, b))$.

Exercise 7.32. Let $u \in W^{1,1}((a, b))$.

(i) Prove that for all $x \in (a, b)$,

$$\bar{u}(x) - u_I = \frac{1}{b-a} \left[\int_a^x (s-a)u'(s) ds - \int_x^b (b-s)u'(s) ds \right].$$

(ii) Prove that

$$\max_{a \leq s \leq b} (s-a)(b-s) = \frac{1}{4}(b-a)^2.$$

(iii) Prove that

$$\int_a^b |u(x) - u_I| dx \leq \frac{1}{2}(b-a) \int_a^b |u'(x)| dx.$$

(iv) Prove that the constant $\frac{1}{2}(b-a)$ is sharp.

We conclude this section with a weighted Poincaré inequality, which will be used to prove Poincaré's inequality in convex domains (see Theorem 13.36).

Proposition 7.33 (Weighted Poincaré's inequality). *Let $g : [0, d] \rightarrow [0, \infty)$, $g \neq 0$, be increasing in $[0, c]$ and decreasing in $[c, d]$ for some $0 \leq c \leq d$ and let $1 \leq p < \infty$. Then there exists a constant $c_p > 0$ depending only on p such that*

$$(7.6) \quad \int_0^d \|u(x) - \bar{u}_{(0,d)}\|^p g(x) dx \leq c_p d^p \int_0^d \|u'(x)\|^p g(x) dx$$

for all $u \in W^{1,p}((0, d); \mathbb{R}^M)$, where

$$\bar{u}_{(0,d)} := \frac{1}{\int_0^d g(x) dx} \int_0^d u(x)g(x) dx.$$

Proof. If $g = 0$ on some interval $[0, b]$ (respectively, $[b, d]$), then all the integrals involved reduce to integrals over $[b, d]$ (respectively, $[0, b]$). Thus, we can assume that g is strictly positive in $(0, d)$. Also, by a scaling argument, it is enough to prove the result for $d = 1$. Finally, by dividing the inequality (7.6) by $\int_0^1 g(x) dx$, we may assume that

$$(7.7) \quad \int_0^1 g(x) dx = 1.$$

By Corollaries 7.20 and 7.21, without loss of generality, we may assume that u is absolutely continuous. By Theorem 3.20 and (7.7),

$$\begin{aligned}
 u(x) - \bar{u}_{(0,1)} &= u(x) - \int_0^1 u(t)g(t) dt = \int_0^1 [u(x) - u(t)]g(t) dt \\
 &= \int_0^1 \int_t^x u'(s)g(t) ds dt \\
 &= \int_0^x \int_t^x u'(s)g(t) ds dt - \int_x^1 \int_x^t u'(s)g(t) ds dt \\
 &= \int_0^x u'(s) \int_0^s g(t) dt ds - \int_x^1 u'(s) \int_s^1 g(t) dt ds,
 \end{aligned}$$

where we have used Fubini's theorem. Hence,

$$\|u(x) - \bar{u}_{(0,1)}\| \leq \int_0^x \|u'(s)\| \int_0^s g(t) dt ds + \int_x^1 \|u'(s)\| \int_s^1 g(t) dt ds.$$

Raising both sides to the power p , multiplying by $g(x)$, and integrating over $[0, 1]$ gives

$$\begin{aligned}
 \int_0^1 \|u(x) - \bar{u}_{(0,1)}\|^p g(x) dx &\leq \int_0^1 \left(\int_0^x \|u'(s)\| \int_0^s g(t) dt ds \right. \\
 &\quad \left. + \int_x^1 \|u'(s)\| \int_s^1 g(t) dt ds \right)^p g(x) dx \\
 &\leq 2^{p-1} \int_0^1 \left(\int_0^x \|u'(s)\| \int_0^s g(t) dt ds \right)^p g(x) dx \\
 &\quad + 2^{p-1} \int_0^1 \left(\int_x^1 \|u'(s)\| \int_s^1 g(t) dt ds \right)^p g(x) dx \\
 &=: A + B,
 \end{aligned}$$

where in the second inequality we have used the inequality

$$(7.8) \quad \left(\sum_{i=1}^n a_i \right)^p \leq n^{p-1} \sum_{i=1}^n a_i^p,$$

where $a_i \geq 0$, $i = 1, \dots, n$, which follows from the convexity of the function $f(t) := |t|^p$, $t \in \mathbb{R}$.

We estimate A . If $0 \leq s \leq c$, then by hypothesis $g(t) \leq g(s)$ for all $0 \leq t \leq s$, so for $0 \leq x \leq c$ we have that

$$\begin{aligned} \left(\int_0^x \|u'(s)\| \int_0^s g(t) dt ds \right)^p &\leq \left(\int_0^x \|u'(s)\| (g(s))^{1/p} \int_0^s (g(t))^{1/p'} dt ds \right)^p \\ &\leq \left(\int_0^x \|u'(s)\| (g(s))^{1/p} ds \right)^p \\ &\leq \int_0^1 \|u'(s)\|^p g(s) ds, \end{aligned}$$

where we have used Hölder's inequality twice and (7.7). On the other hand, if $c \leq s \leq 1$, then by hypothesis $g(s) \geq g(x)$ for all $s \leq x \leq 1$, and so for $c \leq x \leq 1$ we have that

$$\begin{aligned} &\left(\int_0^x \|u'(s)\| \int_0^s g(t) dt ds \right)^p g(x) \\ &= \left(\int_0^c \|u'(s)\| \int_0^s g(t) dt ds + \int_c^x \|u'(s)\| \int_0^s g(t) dt ds \right)^p g(x) \\ &\leq 2^{p-1} \left(\int_0^c \|u'(s)\| \int_0^s g(t) dt ds \right)^p g(x) + 2^{p-1} \left(\int_c^x \|u'(s)\| ds \right)^p g(x) \\ &\leq 2^{p-1} \left(\int_0^1 \|u'(s)\|^p g(s) ds \right) g(x) + 2^{p-1} \left(\int_c^x \|u'(s)\| (g(s))^{1/p} ds \right)^p \\ &\leq 2^{p-1} \left(\int_0^1 \|u'(s)\|^p g(s) ds \right) g(x) + 2^{p-1} \left(\int_0^1 \|u'(s)\|^p g(s) ds \right), \end{aligned}$$

again by Hölder's inequality and (7.7). Thus, for all $0 \leq x \leq 1$,

$$\begin{aligned} &\left(\int_0^x \|u'(s)\| \int_0^s g(t) ds ds \right)^p g(x) \\ &\leq 2^{p-1} \left(\int_0^1 \|u'(s)\|^p g(s) ds \right) (g(x) + 1), \end{aligned}$$

and, in turn, using (7.7) once more,

$$A \leq c(p) \int_0^1 \|u'(s)\|^p g(s) ds \int_0^1 (g(x) + 1) dx = 2c(p) \int_0^1 \|u'(s)\|^p g(s) ds.$$

A similar estimate holds for B , thus giving the desired result for $p > 1$. The proof in the case $p = 1$ is simpler, since there is no need to use Hölder's inequality. $\square$

7.3. Interpolation Inequalities

In this section we prove that if a function u belongs to some L^q space and its highest order weak derivative $u^{(m)}$ belongs some L^p space, with p and q

possibly different, then we can obtain additional information on the function itself and on all the intermediate derivatives of u .

For simplicity we consider only scalar-valued functions, that is, $M = 1$, although all the results hold for vector-valued functions. Also, when there is no possibility of confusion, we write $\|u\|_{L^p}$ for $\|u\|_{L^p(I)}$.

We begin with the case $m = 1$.

Theorem 7.34. *Let $I \subseteq \mathbb{R}$ be an open interval, let $1 \leq p, q, r \leq \infty$ be such that $r \geq q$, and let $u \in W_{\text{loc}}^{1,1}(I)$. Then*

$$(7.9) \quad \|u\|_{L^r(I)} \leq \ell^{1/r-1/q} \|u\|_{L^q(I)} + \ell^{1-1/p+1/r} \|u'\|_{L^p(I)}$$

for every $0 < \ell < \mathcal{L}^1(I)$.

Proof. It is enough to prove (7.9) in the case $\mathcal{L}^1(I) < \infty$ and then use the Lebesgue monotone convergence theorem to obtain the case $\mathcal{L}^1(I) = \infty$.

If $\|u'\|_{L^p} = \infty$, then the right-hand side of (7.9) is infinite and so there is nothing to prove. Thus, without loss of generality we may assume that $\|u'\|_{L^p} < \infty$. By Exercise 7.31 we can assume that $u \in W^{1,p}(I)$. In turn, by Corollaries 7.20 and 7.21 we have that (a representative of) u is absolutely continuous in I , and by Exercise 3.8 we may extend u to $\bar{I}$ in such a way that the extension is still absolutely continuous.

Step 1: Assume that $r = \infty$. For every $x \in I$, let $I_x := [x - \ell/2, x + \ell/2] \cap I$. By the fundamental theorem of calculus (see Theorem 3.20),

$$u(x) = u(x_1) + \int_{x_1}^x u'(t) dt,$$

where $x_1 \in \bar{I}$ is taken such that $|u(x_1)| = \min_{I_x} |u|$. Then by Hölder's inequality,

$$\begin{aligned} |u(x)| &\leq \min_{I_x} |u| + \int_{I_x} |u'(t)| dt \leq \ell^{-1/q} \|u\|_{L^q(I_x)} + \ell^{1/p'} \|u'\|_{L^p(I)} \\ &\leq \ell^{-1/q} \|u\|_{L^q(I)} + \ell^{1-1/p} \|u'\|_{L^p(I)}. \end{aligned}$$

This proves (7.9) when $r = \infty$.

Step 2: Assume next that $r < \infty$ and $r > q$ (if $r = q$ there is nothing to prove). Then by Young's inequality (B.17) with exponent r/q we can write

$$\begin{aligned} \|u\|_{L^r} &= \left(\int_I |u|^{r-q} |u|^q dx \right)^{1/r} \leq \|u\|_{L^\infty}^{1-q/r} \|u\|_{L^q}^{q/r} \\ &\leq \ell^{1/r} \|u\|_{L^\infty} + \ell^{1/r-1/q} \|u\|_{L^q}. \end{aligned}$$

By Step 1,

$$\|u\|_{L^\infty} \leq \ell^{-1/q} \|u\|_{L^q} + \ell^{1-1/p} \|u'\|_{L^p}.$$

Combining the last two inequalities gives (7.9). $\square$

Exercise 7.35. Prove that if $p = 1$ and $u \in BV_{\text{loc}}(I)$, then inequality (7.9) continues to hold with $\|u'\|_{L^1(I)}$ replaced by the total variation $\|Du\|(I)$.

Corollary 7.36. Let $I = (a, b)$, let $1 \leq p, q, r \leq \infty$ be such that $1 + 1/r \geq 1/p$ and $r \geq q$ and let $u \in W_{\text{loc}}^{1,1}(I)$ with $u' \in L^p(I)$. Let $x_0 \in [a, b]$ be such that $|u(x_0)| = \min_{[a,b]} |u|$. Then

$$\|u - u(x_0)\|_{L^r(I)} \leq 8\|u\|_{L^q(I)}^\alpha \|u'\|_{L^p(I)}^{1-\alpha},$$

where $\alpha := 0$ if $r = q$ and $1 - 1/p + 1/r = 0$ and otherwise

$$\alpha := \frac{1 - 1/p + 1/r}{1 - 1/p + 1/q}.$$

Note that as in the proof of Theorem 7.34 we can assume that u is absolutely continuous in $[a, b]$ and so by the Weierstrass theorem x_0 exists.

Proof. Define $v(x) := u(x) - u(x_0)$, $x \in (a, b)$. If $\|u'\|_{L^p} = 0$, then u is constant and so $v = 0$ and there is nothing to prove. Thus, assume that $\|u'\|_{L^p} > 0$ and that $\|u\|_{L^q} > 0$.

Note that $0 \leq \alpha \leq 1$. Moreover, by the definition of x_0 , $\|v\|_{L^q} \leq 2\|u\|_{L^q}$. Let $I_1 := (-\infty, b)$ and $I_2 := (a, \infty)$. Since $v \in C^1([a, b])$ and $v(x_0) = 0$, the functions

$$v_1(x) := \begin{cases} v(x) & \text{for } x_0 < x < b, \\ 0 & \text{for } x \leq x_0, \end{cases} \quad v_2(x) := \begin{cases} v(x) & \text{for } a < x < x_0, \\ 0 & \text{for } x \geq x_0, \end{cases}$$

belong to $W^{1,1}(I_1)$ and $W^{1,1}(I_2)$, respectively, with $\|v_i\|_{L^q(I_i)} \leq \|v\|_{L^q(I)} \leq 2\|u\|_{L^q(I)}$ and $\|v'_i\|_{L^p(I_i)} \leq \|u'\|_{L^p(I)}$, $i = 1, 2$. By Theorem 7.34 applied to v_i , $i = 1, 2$, we get

$$\|v\|_{L^r(I)} \leq \|v_1\|_{L^r(I_1)} + \|v_2\|_{L^r(I_2)} \leq 4\ell^{1/r-1/q}\|u\|_{L^q(I)} + 2\ell^{1-1/p+1/r}\|u'\|_{L^p(I)}$$

for every $0 < \ell < \infty$. It suffices to take

$$\ell := \left(\|u\|_{L^q(I)} \|u'\|_{L^p(I)}^{-1} \right)^{1/(1-1/p+1/q)}$$

in the previous inequality. □

Next we consider the case $m = 2$ and $k = 1$.

Theorem 7.37. Let $I \subseteq \mathbb{R}$ be an open interval, let $1 \leq p, q, r \leq \infty$ be such that

$$(7.10) \quad \frac{1}{2q} + \frac{1}{2p} \geq \frac{1}{r},$$

and let $u \in W_{\text{loc}}^{2,1}(I)$. Then there exists $c = c(p, q, r) > 0$ such that

$$(7.11) \quad \|u'\|_{L^r(I)} \leq c\ell^{1/r-1-1/q}\|u\|_{L^q(I)} + c\ell^{1-1/p+1/r}\|u''\|_{L^p(I)}$$

for every $0 < \ell < \mathcal{L}^1(I)$.

We begin with an auxiliary result which is of interest in itself.

Lemma 7.38. *Let $I = (a, b)$, let $1 \leq p, q, r \leq \infty$ satisfy (7.10) and let $u \in W_{\text{loc}}^{2,1}(I)$ be such $u'' \in L^p(I)$ and u' has at least one zero in $[a, b]$. Then there exists a constant $c = c(p, q, r) > 0$ such that*

$$(7.12) \quad \|u'\|_{L^r(I)} \leq c \|u\|_{L^q(I)}^\theta \|u''\|_{L^p(I)}^{1-\theta},$$

where

$$(7.13) \quad \theta = \frac{1 - 1/p + 1/r}{2 - 1/p + 1/q}.$$

Note that as in the proof of Theorem 7.34, but with u replaced by u' , we can assume that u' is absolutely continuous in $[a, b]$.

Proof. If $\|u''\|_{L^p} = 0$, then u' is constant in I and since it has a zero, then u' must be zero. Thus, assume that $\|u''\|_{L^p} > 0$. Moreover, reasoning as in the beginning of the proof of Theorem 7.34 we can assume that u' is absolutely continuous in $[a, b]$. In turn, $u \in C^1([a, b])$.

Step 1: Assume first that either u' is nonnegative in I or nonpositive in I . Then by the fundamental theorem of calculus, for every $x, x_1 \in (a, b)$,

$$\begin{aligned} |u(x) - u(x_1)| &\leq \int_a^b |u'(t)| dt = |u(b) - u(a)| \\ &\leq |u(b) - u(x_1)| + |u(a) - u(x_1)| \leq 2\|u - u(x_1)\|_{L^\infty}, \end{aligned}$$

which shows that

$$(7.14) \quad \|u - u(x_1)\|_{L^\infty} \leq \|u'\|_{L^1} \leq 2\|u - u(x_1)\|_{L^\infty}.$$

If $r > 1$ we can apply Corollary 7.36 to both u and u' to obtain

$$(7.15) \quad \|u - u(x_0)\|_{L^\infty} \leq 8\|u\|_{L^q}^{\alpha_1} \|u'\|_{L^r}^{1-\alpha_1},$$

$$(7.16) \quad \|u'\|_{L^r} \leq 8\|u'\|_{L^1}^{\alpha_2} \|u''\|_{L^p}^{1-\alpha_2},$$

where $x_0 \in [a, b]$ is such that $|u(x_0)| = \min_{[a,b]} |u|$ and

$$\alpha_1 = \frac{1 - 1/r}{1 - 1/r + 1/q}, \quad \alpha_2 = \frac{1 - 1/p + 1/r}{2 - 1/p}.$$

Combining (7.14)–(7.16) we get

$$\begin{aligned} \|u'\|_{L^r} &\leq 8\|u'\|_{L^1}^{\alpha_2} \|u''\|_{L^p}^{1-\alpha_2} \leq c\|u - u(x_0)\|_{L^\infty}^{\alpha_2} \|u''\|_{L^p}^{1-\alpha_2} \\ &\leq c\|u\|_{L^q}^{\alpha_1\alpha_2} \|u'\|_{L^r}^{(1-\alpha_1)\alpha_2} \|u''\|_{L^p}^{1-\alpha_2}. \end{aligned}$$

Since $u \in C^1([a, b])$, it follows that

$$\|u'\|_{L^r} \leq c\|u\|_{L^q}^{\alpha_1\alpha_2/(1-(1-\alpha_1)\alpha_2)} \|u''\|_{L^p}^{(1-\alpha_2)/(1-(1-\alpha_1)\alpha_2)},$$

which gives (7.12).

On the other hand, if $r = 1$, then by (7.10) necessarily $q = p = 1$. In this case by (7.14) and (7.15) with $r = \infty$,

$$\|u'\|_{L^1} \leq 2\|u - u(x_1)\|_{L^\infty} \leq 8\|u\|_{L^1}^{1/2}\|u'\|_{L^\infty}^{1/2}.$$

Since u' has at least one zero in $[a, b]$, by the fundamental theorem of calculus (see Theorem 3.20), $\|u'\|_{L^\infty} \leq \|u''\|_{L^1}$, which together with the previous inequality gives (7.12) even in this case.

Step 2: In the general case, we write the open set $\{x \in I : u'(x) \neq 0\}$ as a union of countably many open sets I_n in which u' is either positive or negative. By applying Step 1 in each interval I_n we obtain

$$\begin{aligned} \int_I |u'|^r dx &= \int_{\{u' \neq 0\}} |u'|^r dx = \sum_n \int_{I_n} |u'|^r dx \\ &\leq c \sum_n \|u\|_{L^q(I_n)}^{r\theta} \|u''\|_{L^p(I_n)}^{r(1-\theta)} \\ &\leq c \left(\sum_n \|u\|_{L^q(I_n)}^q \right)^{r\theta/q} \left(\sum_n \|u''\|_{L^p(I_n)}^p \right)^{r(1-\theta)/p} \\ &\leq c \|u\|_{L^q}^{r\theta} \|u''\|_{L^p}^{r(1-\theta)}, \end{aligned}$$

where we used the inequality

$$(7.17) \quad \sum_n a_n^s b_n^t \leq \left(\sum_n a_n \right)^s \left(\sum_n b_n \right)^t,$$

which holds for $s+t \geq 1$ and $a_n, b_n > 0$ (exercise). Note that $\frac{r\theta}{q} + \frac{r(1-\theta)}{p} \geq 1$ by (7.10) and (7.13). $\square$

Exercise 7.39. Let $(Y, \|\cdot\|_Y)$ be a Banach space and let $P \subset Y$ be a finite dimensional subspace.

- (i) Prove that for every $y \in Y$ there exists $p \in P$ such that

$$\|y - p\|_Y = \text{dist}(y, P).$$

- (ii) Let $Y = L^q([a, b])$, let P be the set of polynomials of degree one and let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous increasing function. Prove that the polynomial p given in part (i) with y given by f touches f at least twice.

We are now ready to prove Theorem 7.37.

Proof of Theorem 7.37. Reasoning as in the beginning of the proof of Theorem 7.34 we can assume that $I = (a, b)$, that $\|u''\|_{L^p} < \infty$, and that u' is absolutely continuous in $[a, b]$. In turn, $u \in C^1([a, b])$.

By Exercise 7.39 with $Y = L^q(I)$ and P the subspace of polynomials of degree one, we can find a polynomial p_1 of degree one such that

$$\|u - p_1\|_{L^q} = \min_{p \in P} \|u - p\|_{L^q}.$$

Let $v := u - p_1$. Then

$$(7.18) \quad \|v\|_{L^q} \leq \|u - 0\|_{L^q}, \quad \|v''\|_{L^p} = \|u''\|_{L^p},$$

$$(7.19) \quad \|p_1\|_{L^q} \leq \|u - p_1\|_{L^q} + \|u\|_{L^q} \leq 2\|u\|_{L^q}.$$

Assume for simplicity that $a = 0$ (the general case can be obtained by translation). Fix $s \in (0, \frac{1}{3}b)$ and $t \in (\frac{2}{3}b, b)$. Then for every $x \in (0, b)$,

$$p_1'(x) = \frac{p_1(t) - p_1(s)}{t - s}.$$

By averaging first in s over $(0, \frac{1}{3}b)$ and then in t over $(\frac{2}{3}b, b)$ and by Hölder's inequality we get

$$|p_1'(x)| \leq 9b^{-2} \int_{2b/3}^b |p_1(t)| dt + 9b^{-2} \int_0^{b/3} |p_1(s)| ds \leq cb^{-1-1/q} \|p_1\|_{L^q}.$$

Finally, taking the L^r norm in the variable x on both sides gives

$$(7.20) \quad \|p_1'\|_{L^r} \leq cb^{1/r-1-1/q} \|p_1\|_{L^q} \leq c\ell^{1/r-1-1/q} \|p_1\|_{L^q},$$

where in the second inequality we used the fact that $1 + 1/q - 1/r \geq 0$ by (7.10).

Since the function v has at least two zeros by Exercise 7.39, its derivative vanishes at least once. Hence, we are in a position to apply Lemma 7.38 to obtain

$$\begin{aligned} \|u'\|_{L^r} &\leq \|v'\|_{L^r} + \|p_1'\|_{L^r} \leq c\|v\|_{L^q}^\theta \|v''\|_{L^p}^{1-\theta} + \|p_1'\|_{L^r} \\ &\leq c\|u\|_{L^q}^\theta \|u''\|_{L^p}^{1-\theta} + c\ell^{1/r-1-1/q} \|u\|_{L^q} \\ &\leq c\ell^{1/r-1-1/q} \|u\|_{L^q} + c\ell^{1-1/p+1/r} \|u''\|_{L^p}, \end{aligned}$$

where we have used (7.18), (7.19), and (7.20) and Young's inequality. This completes the proof. $\square$

We now consider the general case $m \geq 2$.

Theorem 7.40. *Let $I \subseteq \mathbb{R}$ be an open interval, let $1 \leq p, q, r \leq \infty$, $m \in \mathbb{N}$, $k \in \mathbb{N}_0$, with $0 \leq k < m$, be such that*

$$(7.21) \quad \left(1 - \frac{k}{m}\right) \frac{1}{q} + \frac{k}{m} \frac{1}{p} \geq \frac{1}{r},$$

and let $u \in W_{\text{loc}}^{m,1}(I)$. Then there exists a constant $c = c(k, m, p, q, r) > 0$ such that

$$(7.22) \quad \|u^{(k)}\|_{L^r(I)} \leq c\ell^{1/r-k-1/q} \|u\|_{L^q(I)} + c\ell^{m-k-1/p+1/r} \|u^{(m)}\|_{L^p(I)}$$

for every $0 < \ell < \mathcal{L}^1(I)$. In particular, for $p = q = r$,

$$(7.23) \quad \|u^{(k)}\|_{L^p(I)} \leq c\ell^{-k}\|u\|_{L^p(I)} + c\ell^{m-k}\|u^{(m)}\|_{L^p(I)}.$$

Proof. The proof is by induction on m . We consider the cases $k = 0$ and $1 \leq k < m$ separately. For $m = 2$ and $k = 1$, (7.22) holds in view of Theorem 7.37. Let $n \in \mathbb{N}$ and assume that (7.22) holds for all $m \leq n$ and all $1 \leq k < m$ and for all $1 \leq p, q, r \leq \infty$ satisfying (7.21). We claim that (7.22) holds with $m = n + 1$. Let $1 \leq p, q, r \leq \infty$ and $1 \leq k < n + 1$, be such that

$$(7.24) \quad \left(1 - \frac{k}{n+1}\right) \frac{1}{q} + \frac{k}{n+1} \frac{1}{p} \geq \frac{1}{r},$$

and let $u \in W_{\text{loc}}^{n+1,1}(I)$. We claim that there exists $c = c(k, n+1, p, q, r) > 0$ such that

$$(7.25) \quad \|u^{(k)}\|_{L^r} \leq c\ell^{1/r-k-1/q}\|u\|_{L^q} + c\ell^{n+1-k-1/p+1/r}\|u^{(n+1)}\|_{L^p}$$

for every $0 < \ell < \mathcal{L}^1(I)$. It is enough to prove this inequality in the case in which I is bounded. The general case follows from the Lebesgue monotone convergence theorem. Moreover, we can assume that $\|u\|_{L^q} < \infty$ and that $\|u^{(n+1)}\|_{L^p} < \infty$, since otherwise there is nothing to prove. Reasoning as in the beginning of the proof of Theorem 7.34, with u replaced by $u^{(n-1)}$ we can assume that $u^{(n-1)} \in C^1(\bar{I})$, which implies that $u \in C^n(\bar{I})$.

Step 1: Assume that $k \geq 2$. In view of (7.24) we may find $r_1 \geq 1$ such that

$$\left(1 - \frac{1}{k}\right) \frac{1}{q} + \frac{1}{k} \frac{1}{r} = \frac{1}{r_1}, \quad \left(1 - \frac{k-1}{n}\right) \frac{1}{r_1} + \frac{k-1}{n} \frac{1}{p} \geq \frac{1}{r}.$$

Since $k \leq n$, by the induction hypothesis, for $0 < \delta < 1$ and $0 < \ell < \mathcal{L}^1(I)$,

$$\|u'\|_{L^{r_1}} \leq c(\delta\ell)^{1/r_1-1-1/q}\|u\|_{L^q} + (\delta\ell)^{k-1-1/r+1/r_1}\|u^{(k)}\|_{L^r},$$

while again by the induction hypothesis applied to the function u' in place of u ,

$$\|u^{(k)}\|_{L^r} \leq c\ell^{1/r-k+1-1/r_1}\|u'\|_{L^{r_1}} + \ell^{n+1-k-1/p+1/r}\|u^{(n+1)}\|_{L^p}.$$

Combining the last two inequalities gives

$$\begin{aligned} \|u^{(k)}\|_{L^r} &\leq c\delta^{1/r_1-1-1/q}\ell^{1/r-k-1/q}\|u\|_{L^q} \\ &\quad + c\delta^{k-1-1/r+1/r_1}\|u^{(k)}\|_{L^r} + \ell^{n+1-k-1/p+1/r}\|u^{(n+1)}\|_{L^p}. \end{aligned}$$

Since $u \in C^n(\bar{I})$ and I is bounded, it suffices to take δ so small that $c\delta^{k-1-1/r+1/r_1} \leq \frac{1}{2}$.

In the case $k = 1$ we can assume that $n \geq 3$ and proceed as in Step 1 estimating u' in terms of u and $u^{(n-1)}$ and $u^{(n-1)}$ in terms of u' and $u^{(n)}$. The details are left as an exercise.

Step 2: The case $k = 0$ is similar and is left as an exercise. Observe that for $m = 1$ and $k = 0$, (7.22) holds in view of Theorem 7.34. $\square$

As a corollary of Theorem 7.40 we have the following.

Theorem 7.41. *Let $I \subseteq \mathbb{R}$ be an open interval, let $1 \leq p, q, r \leq \infty$, $m \in \mathbb{N}$, $k \in \mathbb{N}_0$, with $0 \leq k < m$, satisfy (7.21) and let $u \in W_{\text{loc}}^{m,1}(I)$ with $u \in L^q(I)$ and $u^{(m)} \in L^p(I)$. Then there exists a constant $c = c(k, m, p, q, r) > 0$ such that*

$$(7.26) \quad \|u^{(k)}\|_{L^r(I)} \leq c \|u\|_{L^q(I)}^\theta \|u^{(m)}\|_{L^p(I)}^{1-\theta},$$

if I has infinite length and

$$(7.27) \quad \|u^{(k)}\|_{L^r(I)} \leq c \ell^{1/r-k-1/q} \|u\|_{L^q(I)} + c \|u\|_{L^q(I)}^\theta \|u^{(m)}\|_{L^p(I)}^{1-\theta}$$

for every $0 < \ell < \mathcal{L}^1(I)$ if I has finite length, where

$$(7.28) \quad \theta := \frac{m - k - 1/p + 1/r}{m - 1/p + 1/q}$$

and $\theta = 1$ and $r = \infty$ if $m = 1$, $q = \infty$, and $p = 1$.

Proof. If $\|u^{(m)}\|_{L^p} = 0$, then u is a polynomial of degree $m - 1$. If $\mathcal{L}^1(I) = \infty$, then $u = 0$, while if $\mathcal{L}^1(I) < \infty$, (7.27) follows from direct calculations. Thus, in what follows, we assume that $\|u^{(m)}\|_{L^p} > 0$ and $\|u\|_{L^q} > 0$. Consider the function

$$(7.29) \quad g(t) = At^{-a} + Bt^b,$$

where $a := k + 1/q - 1/r$, $A := \|u\|_{L^q}$, $b := m - k - 1/p + 1/r$, $B := \|u^{(m)}\|_{L^p}$ and $0 < t \leq t_0 := \ell$. By Theorem 7.40 we have that

$$(7.30) \quad \|u^{(k)}\|_{L^r} \leq \inf_{0 < t \leq t_0} g(t).$$

By differentiating g we find that

$$(7.31) \quad \inf_{0 < t \leq t_0} g(t) = cA^{b/(a+b)}B^{a/(a+b)},$$

if

$$(7.32) \quad t_1 := [aA/(bB)]^{1/(a+b)} \leq t_0,$$

while

$$(7.33) \quad \inf_{0 < t \leq t_0} g(t) = At_0^{-a} + Bt_0^b$$

if $t_1 > t_0$. Hence, if $t_0 \geq t_1$ (in particular, by taking ℓ large enough this always holds if $\mathcal{L}^1(I) = \infty$), then the inequalities (7.26) and (7.27) follow from (7.30) and (7.31) and the definition of a , A , b , B , and t_0 .

On the other hand, if $t_0 < t_1$, then by the definition of t_1 (see (7.32)), $B \leq cAt_0^{-a-b}$, and so from (7.33),

$$\inf_{0 < t \leq t_0} g(t) \leq cAt_0^{-a}.$$

Hence, (7.27) follows from the previous inequality, (7.30), and the definition of a , A , and t_0 . $\square$

Exercise 7.42. Prove that the inequality (7.21) is necessary for (7.26) to hold.

Exercise 7.43. Let $1 \leq p, q, r \leq \infty$, $m \in \mathbb{N}$, $k \in \mathbb{N}_0$, with $0 \leq k < m$, satisfy (7.21). Given $T > 0$, let $W_{\#}^{m,1}((0, T))$ be the space of all functions in $W_{\text{loc}}^{m,1}(\mathbb{R})$ that are T -periodic. Prove that the inequality (7.26) holds for every $u \in W_{\#}^{m,1}((0, T))$.

The Infinite-Dimensional Case

Prospective Grad Students, II: “When you look at me, do you see a young, creative mind, or a series of potential journal papers?”
— Jorge Cham, www.phdcomics.com

8.1. The Bochner Integral

We begin by introducing the Bochner integral. We refer to [62], [63], [67], [69], [83], and [206] for more information on the subject and for the proofs omitted here.

Definition 8.1. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $(Y, \|\cdot\|)$ be a Banach space. A *simple function* s is a function $s : X \rightarrow Y$ of the form

$$s = \sum_{i=1}^{\ell} c_i \chi_{E_i},$$

where $\ell \in \mathbb{N}$, $c_1, \dots, c_\ell \in Y$ are distinct and the sets $E_i \subseteq X$ belong to $\mathfrak{M}$ and are mutually disjoint.

- (i) A function $u : X \rightarrow Y$ is said to be *strongly measurable* if there exists a sequence $\{s_n\}_n$ of simple functions $s_n : X \rightarrow Y$ such that

$$\lim_{n \rightarrow \infty} \|s_n(x) - u(x)\| = 0 \quad \text{for } \mu\text{-a.e. } x \in X.$$

- (ii) A function $u : X \rightarrow Y$ is said to be *weakly measurable* if for every $T \in Y'$ the function

$$x \in X \mapsto \langle T, u(x) \rangle_{Y', Y} = T(u(x)) \text{ is measurable,}$$

where $\langle \cdot, \cdot \rangle_{Y', Y} : Y' \times Y \rightarrow \mathbb{R}$ is the duality pairing defined by

$$\langle T, y \rangle_{Y', Y} := T(y)$$

for $T \in Y'$ and $y \in Y$.

- (iii) A function $u : X \rightarrow Y'$ is said to be *weakly star measurable* if for every $y \in Y$ the function

$$x \in X \mapsto \langle u(x), y \rangle_{Y', Y} = u(x)(y) \text{ is measurable.}$$

Exercise 8.2. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $(Y, \|\cdot\|)$ be a Banach space. Prove that if a function $u : X \rightarrow Y$ is weakly measurable, then the function $x \in X \mapsto \|u(x)\|$ is measurable.

The relation between weak and strong measurability is given by the following theorem.

Theorem 8.3 (Pettis). *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $(Y, \|\cdot\|)$ be a Banach space. A function $u : X \rightarrow Y$ is strongly measurable if and only if it is weakly measurable and there exists $E \in \mathfrak{M}$, with $\mu(E) = 0$, such that the set $u(X \setminus E)$ is a separable set of Y (in the norm sense).*

Proof. Step 1: Assume that $u : X \rightarrow Y$ is strongly measurable. Then there exist a sequence $\{s_n\}_n$ of simple functions $s_n : X \rightarrow Y$ and a set $E \in \mathfrak{M}$, with $\mu(E) = 0$, such that

$$\lim_{n \rightarrow \infty} \|s_n(x) - u(x)\| = 0 \quad \text{for all } x \in X \setminus E.$$

Hence, for every $T \in Y'$ we have that

$$\lim_{n \rightarrow \infty} T(s_n(x)) = T(u(x)) \quad \text{for all } x \in X \setminus E.$$

Since the function $T \circ s_n : X \rightarrow \mathbb{R}$ is a simple function, it is measurable, and in turn, the real-valued function $x \in X \mapsto T(u(x))$ is measurable as a pointwise limit of measurable functions. Hence, u is weakly measurable.

Since the range of each s_n is finite, the set $\bigcup_{n=1}^{\infty} s_n(X)$ is countable. In turn, the set $C := \overline{\bigcup_n s_n(X)}$ is separable. If $x \in X \setminus E$, then $s_n(x) \rightarrow u(x)$, and so $u(x) \in C$. It follows that $u(X \setminus E) \subseteq C$. Using the fact that a subset of a separable set (in a metric space) is separable, we have that $u(X \setminus E)$ is separable.

Step 2: Assume that u is weakly measurable and that there exists $E \in \mathfrak{M}$, with $\mu(E) = 0$, such that the set $u(X \setminus E)$ is a separable set of Y . Let $\{y_n\}_n$ be a dense sequence in $u(X \setminus E)$ and for each n let $x_n \in X \setminus E$ be such that $u(x_n) = y_n$. By the Hahn–Banach theorem (see Theorem A.32), there exists $T_n \in Y'$ such that $T_n(y_n) = \|y_n\|$ and $\|T_n\|_{Y'} = 1$. We claim that

$$\|u(x)\| = \sup_n T_n(u(x))$$

for every $x \in X \setminus E$. Indeed,

$$T_n(u(x)) \leq \|T_n\|_{Y'} \|u(x)\| = 1 \|u(x)\|$$

and so

$$\sup_n T_n(u(x)) \leq \|u(x)\|.$$

On the other hand, find a subsequence $\{x_{n_k}\}_k$ of $\{x_n\}_n$ such that $y_{n_k} = u(x_{n_k}) \rightarrow u(x)$. Then

$$\begin{aligned} \|u(x)\| &= \lim_{k \rightarrow \infty} \|u(x_{n_k})\| = \lim_{k \rightarrow \infty} T_{n_k}(u(x_{n_k})) \\ &= \lim_{k \rightarrow \infty} [T_{n_k}(u(x)) + T_{n_k}(u(x_{n_k}) - u(x))] \\ &\leq \sup_n T_n(u(x)) + \limsup_{k \rightarrow \infty} \|T_{n_k}\|_{Y'} \|u(x_{n_k}) - u(x)\| \\ &= \sup_n T_n(u(x)) + 0. \end{aligned}$$

This proves the claim. Since u is weakly measurable, each function $x \in X \mapsto T_n(u(x))$ is measurable, and so the function $x \in X \mapsto \|u(x)\|$ is measurable. Similarly, we can show that the functions $g_n(x) := \|u(x) - y_n\|$ are measurable.

For every $m \in \mathbb{N}$, consider the function $t_m : Y \rightarrow \{0, y_1, \dots, y_m\}$ defined by

$$t_m(y) := \begin{cases} y_{n_{y,m}} & \text{if } y \in u(X \setminus E), \\ 0 & \text{otherwise,} \end{cases}$$

where for $y \in u(X \setminus E)$, $n_{y,m} \in \{1, \dots, m\}$ is the smallest number such that

$$\|y - y_{n_{y,m}}\| = \min_{1 \leq n \leq m} \|y - y_n\|.$$

Note that as $m \rightarrow \infty$ we have that $y_{n_{y,m}} \rightarrow y$ by the density of the sequence $\{y_n\}_n$ in $u(X \setminus E)$. Hence, $t_m(y) \rightarrow y$ for every $y \in u(X \setminus E)$. Define

$$s_m(x) := \begin{cases} t_m(u(x)) & \text{if } x \in X \setminus E, \\ 0 & \text{otherwise.} \end{cases}$$

The function s_m takes only a finite number of values. Moreover, $s_m(x) \rightarrow u(x)$ for every $x \in X \setminus E$. It remains to show that s_m is measurable. Observe that for $1 \leq n \leq m$,

$$\begin{aligned} \{x \in X : s_m(x) = y_n\} \\ = \{x \in X \setminus E : \|u(x) - y_n\| = \min_{1 \leq j \leq m} \|u(x) - y_j\|\} \end{aligned}$$

$$\bigcap \{x \in X \setminus E : \|u(x) - y_l\| > \|u(x) - y_n\| \text{ for all } l = 1, \dots, n-1\}.$$

This set belongs to $\mathfrak{M}$ since the function g_l is measurable for every l . This shows that s_m is measurable and concludes the proof. $\square$

Exercise 8.4. Consider the Hilbert space

$$\ell^2([0, 1]) := \left\{ f : [0, 1] \rightarrow \mathbb{R} : \sum_{x \in [0, 1]} f^2(x) < \infty \right\}$$

with the inner product

$$(f, g)_{\ell^2([0, 1])} := \sum_{x \in [0, 1]} f(x)g(x).$$

Define $u : [0, 1] \rightarrow \ell^2([0, 1])$ by $u(x) := \chi_{\{x\}}$. Prove that u is weakly measurable but not strongly measurable.

Exercise 8.5. Consider the Banach $\ell^\infty := L^\infty(\mathbb{N})$ with the counting measure. Define the function $u : [0, 1] \rightarrow \ell^\infty$ by

$$u(x) := \{(\operatorname{sgn}(\sin(2^n \pi x)) + 1)/2\}_{n \in \mathbb{N}}, \quad x \in [0, 1].$$

Prove that u is weakly star measurable but not strongly measurable.

Definition 8.6. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let Y be a Banach space. A simple function $s : X \rightarrow Y$ is *Bochner integrable* if it has the form

$$s = \sum_{i=1}^{\ell} c_i \chi_{E_i},$$

where $c_1, \dots, c_\ell \in Y$, $\ell \in \mathbb{N}$, are distinct, the sets $E_i \in \mathfrak{M}$ are mutually disjoint, and $c_i = 0$ whenever $\mu(E_i) = \infty$. For every $E \in \mathfrak{M}$, the *Bochner integral* of s over E is defined by

$$\int_E s \, d\mu := \sum_{i=1}^{\ell} c_i \mu(E_i \cap E),$$

where $c_i \mu(E_i \cap E)$ is set to be zero whenever $c_i = 0$ and $\mu(E_i \cap E) = \infty$.

Definition 8.7. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $(Y, \|\cdot\|)$ be a Banach space. A strongly measurable function $u : X \rightarrow Y$ is *Bochner integrable* if there exists a sequence $\{s_n\}_n$ of simple Bochner integrable functions such that

$$\lim_{n \rightarrow \infty} \|s_n(x) - u(x)\| = 0 \quad \text{for } \mu\text{-a.e. } x \in X$$

and

$$\lim_{n \rightarrow \infty} \int_X \|s_n - u\| \, d\mu = 0.$$

For every $E \in \mathfrak{M}$ the *Bochner integral* of u over E is defined by

$$(8.1) \quad \int_E u \, d\mu := \lim_{n \rightarrow \infty} \int_E s_n \, d\mu.$$

It may be verified that this limit exists and that it is independent of the particular sequence $\{s_n\}_n$.

Remark 8.8. Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $(Y, \|\cdot\|_Y)$ be a Banach space, and let $u : X \rightarrow Y$ be a Bochner integrable function. If $Y \hookrightarrow Z$, where $(Z, \|\cdot\|_Z)$ is a Banach space and $i : Y \rightarrow Z$ is the immersion (see Definition A.36), then the function $i \circ u : X \rightarrow Z$ is Bochner integrable and

$$\int_E i \circ u \, d\mu = i \left(\int_E u \, d\mu \right)$$

for every $E \in \mathfrak{M}$. Thus, without loss of generality, we can identify $\int_E i \circ u \, d\mu$ with $\int_E u \, d\mu$.

Theorem 8.9. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $(Y, \|\cdot\|)$ be a Banach space. A strongly measurable function $u : X \rightarrow Y$ is Bochner integrable if and only if $\|u\|$ is Lebesgue integrable over X . Moreover, if $u : X \rightarrow Y$ is Bochner integrable, then*

$$(8.2) \quad \left\| \int_E u \, d\mu \right\| \leq \int_E \|u\| \, d\mu$$

for every $E \in \mathfrak{M}$.

Proof. Step 1: Assume that the real-valued function $x \in X \mapsto \|u(x)\|$ is Lebesgue integrable. Since u is strongly measurable, there exists a sequence $\{s_n\}_n$ of Bochner integrable simple functions such that

$$(8.3) \quad \lim_{n \rightarrow \infty} \|s_n(x) - u(x)\| = 0 \quad \text{for } \mu\text{-a.e. } x \in X.$$

Define

$$t_n(x) := \begin{cases} s_n(x) & \text{if } \|s_n(x)\| \leq \frac{3}{2}\|u(x)\|, \\ 0 & \text{if } \|s_n(x)\| > \frac{3}{2}\|u(x)\|. \end{cases}$$

Then t_n is still a simple Bochner integrable function, with $\|t_n(x)\| \leq \frac{3}{2}\|u(x)\|$ for all $x \in X$ and all n . Moreover, by (8.3), if $\|u(x)\| > 0$, taking $\varepsilon = \frac{1}{2}\|u(x)\|$, we have that $\|s_n(x) - u(x)\| \leq \varepsilon = \frac{1}{2}\|u(x)\|$ for all n sufficiently large, which implies that $\|s_n(x)\| \leq \frac{3}{2}\|u(x)\|$ for all n sufficiently large. Hence, $t_n(x) = s_n(x)$ for all n large and so $t_n(x) \rightarrow u(x)$ as $n \rightarrow \infty$. On the other hand, if $\|u(x)\| = 0$, then $t_n(x) = 0 = u(x)$ for all n . This shows that $\lim_{n \rightarrow \infty} \|t_n(x) - u(x)\| = 0$ for μ -a.e. $x \in X$. In turn,

$$\lim_{n \rightarrow \infty} \int_X \|t_n - u\| \, d\mu = 0$$

by the Lebesgue dominated convergence theorem. It follows that u is Bochner integrable.

Step 2. Conversely, assume that $u : X \rightarrow Y$ is Bochner integrable and consider a sequence $\{s_n\}_n$ of Bochner integrable simple functions such that

$$\lim_{n \rightarrow \infty} \|s_n(x) - u(x)\| = 0 \quad \text{for } \mu\text{-a.e. } x \in X$$

and

$$\lim_{n \rightarrow \infty} \int_X \|s_n - u\| d\mu = 0.$$

Then for $x \in X$ and for $\ell, n \in \mathbb{N}$,

$$\begin{aligned} 0 &\leq \| \|s_n(x)\| - \|s_\ell(x)\| \| \leq \|s_n(x) - s_\ell(x)\| \\ &\leq \|s_n(x) - u(x)\| + \|s_\ell(x) - u(x)\|, \end{aligned}$$

and so

$$\int_X \| \|s_n\| - \|s_\ell\| \| d\mu \leq \int_X \|s_n - u\| d\mu + \int_X \|s_\ell - u\| d\mu \rightarrow 0$$

as $\ell, n \rightarrow \infty$. Thus, $\{\|s_n\|\}_n$ is a Cauchy sequence in $L^1(X, \mathfrak{M}, \mu)$, and so it converges in $L^1(X, \mathfrak{M}, \mu)$ (and up to a subsequence also pointwise μ -a.e. in X) to a function $v : X \rightarrow \mathbb{R}$. On the other hand, for μ -a.e. $x \in X$,

$$0 \leq \| \|s_n(x)\| - \|u(x)\| \| \leq \|s_n(x) - u(x)\| \rightarrow 0$$

as $n \rightarrow \infty$. Hence, $v(x) = \|u(x)\|$ for μ -a.e. $x \in X$, which shows that $\|u\|$ is integrable.

Since (8.2) holds for simple functions, by the properties of the norm, the general case follows by (8.1), the continuity of the norm, and Step 2. $\square$

The proofs of the next three theorems follow the corresponding ones for Lebesgue integration and are left as an exercise.

Theorem 8.10 (Lebesgue dominated convergence theorem). *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $(Y, \|\cdot\|)$ be a Banach space. Let $u_n : X \rightarrow Y$, $n \in \mathbb{N}$, and $u : X \rightarrow Y$ be strongly measurable and assume that $\|u_n(x)\| \leq v(x)$ for all $n \in \mathbb{N}$ and for μ -a.e. $x \in X$, for some Lebesgue integrable function $v : X \rightarrow [0, \infty)$, and that $u_n(x) \rightarrow u(x)$ as $n \rightarrow \infty$ for μ -a.e. $x \in X$. Then u is Bochner integrable and*

$$\lim_{n \rightarrow \infty} \int_X \|u_n - u\| d\mu = 0.$$

In particular,

$$\lim_{n \rightarrow \infty} \int_X u_n d\mu = \int_X u d\mu.$$

Theorem 8.11 (Fatou's lemma). *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $(Y, \|\cdot\|)$ be a Banach space. Let $u_n : X \rightarrow Y$, $n \in \mathbb{N}$, and $u : X \rightarrow Y$ be strongly measurable and assume that $u_n(x) \rightarrow u(x)$ as $n \rightarrow \infty$ for μ -a.e. $x \in X$. If*

$$\sup_n \int_X \|u_n\| d\mu < \infty,$$

then u is Bochner integrable and

$$\int_X \|u\| d\mu \leq \liminf_{n \rightarrow \infty} \int_X \|u_n\| d\mu.$$

Theorem 8.12 (Egoroff). *Let $(X, \mathfrak{M}, \mu)$ be a measure space with μ finite, let $(Y, \|\cdot\|)$ be a Banach space, and let $u, u_n : X \rightarrow Y$, $n \in \mathbb{N}$, be strongly measurable functions such that*

$$\lim_{n \rightarrow \infty} \|u_n(x) - u(x)\| = 0$$

for μ -a.e. $x \in X$. Then for every $\varepsilon > 0$ there exists a measurable set $E \in \mathfrak{M}$, with $\mu(X \setminus E) \leq \varepsilon$, such that

$$\lim_{n \rightarrow \infty} \sup_{x \in E} \|u_n(x) - u(x)\| = 0.$$

Next we prove that the Bochner integral commutes with elements of Y' .

Theorem 8.13. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $(Y, \|\cdot\|)$ be a Banach space. Let $u : X \rightarrow Y$ be Bochner integrable. Then for every $T \in Y'$,*

$$T\left(\int_X u \, d\mu\right) = \int_X T(u) \, d\mu.$$

Proof. We proceed as in the first part of the proof of Theorem 8.9 to construct a sequence $\{t_n\}_n$ of simple functions $t_n : X \rightarrow Y$ such that $\|t_n(x)\| \leq \frac{3}{2}\|u(x)\|$ for all $x \in X$ and all n , $\lim_{n \rightarrow \infty} \|t_n(x) - u(x)\| = 0$ for μ -a.e. $x \in X$, and

$$\lim_{n \rightarrow \infty} \int_X \|t_n - u\| \, d\mu = 0.$$

Write

$$t_n = \sum_{i=1}^{m_n} c_{i,n} \chi_{E_{i,n}}.$$

Now, let $T \in Y'$. Then by the continuity and linearity of T , and the fact that $\lim_{n \rightarrow \infty} \int_X t_n \, d\mu = \int_X u \, d\mu$,

$$\begin{aligned} T\left(\int_X u(x) \, d\mu\right) &= \lim_{n \rightarrow \infty} T\left(\int_X t_n(x) \, d\mu\right) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^{m_n} T(c_{i,n}) \mu(E_{i,n}) = \lim_{n \rightarrow \infty} \int_X T(t_n(x)) \, d\mu. \end{aligned}$$

Since

$$|T(t_n(x))| \leq \|T\|_{Y'} \|t_n(x)\| \leq \frac{3}{2} \|T\|_{Y'} \|u(x)\|,$$

we can apply the Lebesgue dominated convergence theorem to conclude that

$$\lim_{n \rightarrow \infty} \int_X T(t_n(x)) \, d\mu = \int_X \lim_{n \rightarrow \infty} T(t_n(x)) \, d\mu = \int_X T(u(x)) \, d\mu.$$

This concludes the proof. $\square$

8.2. L^p Spaces on Banach Spaces

Definition 8.14. Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $(Y, \|\cdot\|)$ be a Banach space, and let $1 \leq p < \infty$. Then

$$L^p(X; Y) := \{u : X \rightarrow Y : u \text{ strongly measurable,} \\ \|u\|_{L^p(X; Y)} < \infty\},$$

where

$$\|u\|_{L^p(X; Y)} := \left(\int_X \|u\|^p d\mu \right)^{1/p}.$$

If $p = \infty$, then

$$L^\infty(X; Y) := \{u : X \rightarrow Y : u \text{ strongly measurable,} \\ \|u\|_{L^\infty(X; Y)} < \infty\},$$

where

$$\|u\|_{L^\infty(X; Y)} = \operatorname{ess\,sup}_{x \in X} \|u(x)\| \\ := \inf \{t \geq 0 : \|u(x)\| \leq t \text{ for } \mu\text{-a.e. } x \in X\}.$$

As in the case $Y = \mathbb{R}$ we identify functions with their equivalence classes.

Theorem 8.15. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let Y be a Banach space.*

- (i) $L^p(X; Y)$ is a Banach space for $1 \leq p \leq \infty$;
- (ii) the family of all Bochner integrable simple functions is dense in $L^p(X; Y)$ for $1 \leq p < \infty$;
- (iii) if X is a separable metric space, μ is a σ -finite Radon measure, and if Y is separable, then $L^p(X; Y)$ is separable for $1 \leq p < \infty$.

A major problem in the theory of L^p spaces on Banach spaces is the identification of the dual of $L^p(X; Y)$. Here we will consider only the two important special cases in which Y is either separable or reflexive.

Definition 8.16. Let $(X, \mathfrak{M}, \mu)$ be a measure space, let Y be a Banach space, and let $1 \leq p \leq \infty$. Then the space $L_w^p(X; Y')$ is the space of all (equivalence classes of) weakly star measurable functions $u : X \rightarrow Y'$ such that $\|u\|_{Y'} \in L^p(X; \mathbb{R})$. The space $L_w^p(X; Y')$ is endowed with the norm

$$\|u\|_{L_w^p(X; Y')} := \left(\int_X \|u\|_{Y'}^p d\mu \right)^{1/p} \quad \text{for } 1 \leq p < \infty,$$

and

$$\|u\|_{L_w^\infty(X; Y')} := \operatorname{ess\,sup}_{x \in X} \|u(x)\|_{Y'} \quad \text{for } p = \infty.$$

Theorem 8.17 (Riesz representation theorem in L^p). *Let $(X, \mathfrak{M}, \mu)$ be a measure space with μ σ -finite, let Y be a Banach space, let $1 \leq p < \infty$, and let q be its Hölder conjugate exponent.*

- (i) *Assume that Y is separable. If $T \in (L^p(X; Y))'$, then there exists a unique $v \in L_w^q(X; Y')$ such that*

$$(8.4) \quad T(u) = \int_X \langle v, u \rangle_{Y', Y} d\mu$$

for every $u \in L^p(X; Y)$. Moreover, the norm of T coincides with $\|v\|_{L_w^q(X; Y')}$. Conversely, every functional of the form (8.4), where $v \in L_w^q(X; Y')$, is a bounded linear functional on $L^p(X; Y)$.

- (ii) *Assume that Y is reflexive. Then for $T \in (L^p(X; Y))'$ there exists a unique $v \in L^q(X; Y')$ such that*

$$(8.5) \quad T(u) = \int_X \langle v, u \rangle_{Y', Y} d\mu$$

for every $u \in L^p(X; Y)$. Moreover, the norm of T coincides with $\|v\|_{L^q(X; Y')}$. Conversely, every functional of the form (8.5), where $v \in L^q(X; Y')$, is a bounded linear functional on $L^p(X; Y)$.

Corollary 8.18. *Let $(X, \mathfrak{M}, \mu)$ be a measure space with μ σ -finite, let Y be a reflexive Banach space, and let $1 < p < \infty$. Then $L^p(X; Y)$ is reflexive.*

In what follows we restrict our attention to the case in which the measure space $(X, \mathfrak{M}, \mu)$ is given by an interval $I \subseteq \mathbb{R}$ with the Lebesgue measure.

Theorem 8.19. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a Banach space, and let $v : I \rightarrow Y$ be locally Bochner integrable. Then for $\mathcal{L}^1$ -a.e. $x \in I$,*

$$\lim_{r \rightarrow 0^+} \frac{1}{r} \int_{x-r}^{x+r} \|v(t) - v(x)\| dt = 0.$$

Proof. By the Pettis theorem (see Theorem 8.3) there exists a set $E_0 \subset I$ with $\mathcal{L}^1(E_0) = 0$ such that $v(I \setminus E_0)$ is separable. Let $\{y_n\}_n$ be dense in $v(I \setminus E_0)$. Since the function $w_n(x) := \|v(x) - y_n\|$, $x \in I$, is locally Lebesgue integrable, by Theorem 3.16 there exists $E_n \subset I$ with $\mathcal{L}^1(E_n) = 0$ such that

$$\lim_{r \rightarrow 0^+} \frac{1}{2r} \int_{x-r}^{x+r} \|v(t) - y_n\| dt = \|v(x) - y_n\|$$

for all $x \in I \setminus E_n$. Let $E := \bigcup_{n=0}^{\infty} E_n$. Then $\mathcal{L}^1(E) = 0$ and for $x \in I \setminus E$ and $n \in \mathbb{N}$,

$$\begin{aligned} \limsup_{r \rightarrow 0^+} \frac{1}{2r} \int_{x-r}^{x+r} \|v(t) - v(x)\| dt &\leq \lim_{r \rightarrow 0^+} \frac{1}{2r} \int_{x-r}^{x+r} \|v(t) - y_n\| dt \\ &\quad + \|v(x) - y_n\| = 2\|v(x) - y_n\|. \end{aligned}$$

Since $\{y_n\}_n$ is dense in $v(I \setminus E_0)$ and $x \notin E_0$, we can find a subsequence $\{y_{n_k}\}_k$ of $\{y_n\}_n$ such that $y_{n_k} \rightarrow v(x)$. By taking $n = n_k$ in the previous inequality and letting $k \rightarrow \infty$ we conclude the proof. $\square$

As a consequence of this theorem, we can extend the mollification theory (see Section C.3 in Appendix C) to Banach-valued functions. Given a nonnegative bounded function $\varphi \in L^1(\mathbb{R})$ with

$$(8.6) \quad \text{supp } \varphi \subseteq \overline{B(0, 1)}, \quad \int_{\mathbb{R}} \varphi(x) \, dx = 1,$$

for every $\varepsilon > 0$ we define $\varphi_\varepsilon(x) := \frac{1}{\varepsilon} \varphi(x/\varepsilon)$, $x \in \mathbb{R}$. The functions φ_ε are called *mollifiers*. Note that $\text{supp } \varphi_\varepsilon \subseteq \overline{B(0, \varepsilon)}$. Hence, given an open interval $I \subseteq \mathbb{R}$, a Banach space Y , and a function $u \in L^1_{\text{loc}}(I; Y)$, we may define

$$u_\varepsilon(x) := (u * \varphi_\varepsilon)(x) = \int_I \varphi_\varepsilon(x - t) u(t) \, dt$$

for $x \in I_\varepsilon$, where the open set I_ε is given by

$$I_\varepsilon := \{x \in I : \text{dist}(x, \partial I) > \varepsilon\}.$$

The function $u_\varepsilon : I_\varepsilon \rightarrow Y$ is called a *mollification* of u . Note that if u is Bochner integrable, then $u_\varepsilon(x)$ is well-defined for every $x \in \mathbb{R}$.

The following theorems can be proved exactly as in Theorems C.16 and C.20.

Theorem 8.20. *Let $I \subseteq \mathbb{R}$ be an open interval, let $\varphi \in L^1(I)$ be a non-negative bounded function satisfying (8.6), let Y be a Banach space, and let $u \in L^1_{\text{loc}}(I; Y)$.*

- (i) *For every Lebesgue point $x \in I$, $u_\varepsilon(x) \rightarrow u(x)$ as $\varepsilon \rightarrow 0^+$.*
- (ii) *If $u \in L^p(I; Y)$, $1 \leq p \leq \infty$, then*

$$\|u_\varepsilon\|_{L^p(\mathbb{R}; Y)} \leq \|u\|_{L^p(I; Y)}$$

for every $\varepsilon > 0$ and

$$\|u_\varepsilon\|_{L^p(\mathbb{R}; Y)} \rightarrow \|u\|_{L^p(I; Y)} \text{ as } \varepsilon \rightarrow 0^+.$$

- (iii) *If $u \in L^p(I; Y)$, $1 \leq p < \infty$, then*

$$\lim_{\varepsilon \rightarrow 0^+} \|u_\varepsilon - u\|_{L^p(I; Y)} = 0.$$

Theorem 8.21. *Let $I \subseteq \mathbb{R}$ be an open interval, let $\varphi \in C_c^\infty(\mathbb{R})$ be a non-negative bounded function satisfying (8.6), let Y be a Banach space, and let $u \in L^1_{\text{loc}}(I; Y)$. Then $u_\varepsilon \in C^\infty(I_\varepsilon; Y)$ for all $0 < \varepsilon < 1$, and for every $n \in \mathbb{N}$,*

$$(8.7) \quad u_\varepsilon^{(n)}(x) = (u * \varphi_\varepsilon^{(n)})(x) = \int_I \varphi_\varepsilon^{(n)}(x - t) u(t) \, dt$$

for all $x \in I_\varepsilon$. Moreover, if $u \in L^p(I; Y)$, $1 \leq p \leq \infty$, then $u_\varepsilon \in C^\infty(\mathbb{R}; Y)$.

Exercise 8.22. Let $I \subseteq \mathbb{R}$ be an open interval, let Y be a Banach space, and let $v : I \rightarrow Y$ be a locally Bochner integrable function such that

$$\int_I \varphi(x)v(x) \, dx = 0$$

for every function $\varphi \in C_c^\infty(I)$. Prove that $v(x) = 0$ for $\mathcal{L}^1$ -a.e. $x \in I$.

Exercise 8.23. Let $I \subseteq \mathbb{R}$ be an open bounded interval, let Y be a Banach space, and let $1 \leq p < \infty$. Prove that functions in $C^\infty(\bar{I}; Y)$ are dense in $L^p(I; Y)$.

Next we characterize compact sets in $L^p(I; Y)$.

Theorem 8.24 (Simon). *Let $I \subset \mathbb{R}$ be a bounded interval, let $1 \leq p < \infty$, and let $(Y, \|\cdot\|)$ be a Banach space. Then $\mathcal{U} \subseteq L^p(I; Y)$ is relatively compact in $L^p(I; Y)$ if and only*

(i) *for every $x_1, x_2 \in I^\circ$, with $x_1 < x_2$, the set*

$$\left\{ \int_{x_1}^{x_2} u(x) \, dx : u \in \mathcal{U} \right\}$$

is relatively compact in Y ,

(ii) $\|\tau_h(u) - u\|_{L^p(I^h; Y)} \rightarrow 0$ as $h \rightarrow 0^+$ uniformly for $u \in \mathcal{U}$, where $\tau_h(u)(x) := u(x+h)$ and $I^h := \{x \in I : x+h \in I\}$.

Proof. Step 1: Assume that $\mathcal{U} \subseteq L^p(I; Y)$ is relatively compact in $L^p(I; Y)$. Since for every $x_1, x_2 \in I^\circ$, with $x_1 < x_2$, the function

$$u \in L^p(I; Y) \mapsto \int_{x_1}^{x_2} u(x) \, dx$$

is continuous from $L^p(I; Y)$ into Y and since continuous functions map compact sets into compact sets, it follows that (i) holds.

On the other hand, by the compactness of $\overline{\mathcal{U}}$, given $\varepsilon > 0$, we may find $u_1, \dots, u_n \in L^p(I; Y)$ such that for every $u \in \mathcal{U}$ there exists $i \in \{1, \dots, n\}$ with $\|u - u_i\|_{L^p(I; Y)} \leq \varepsilon$. By the density of $C(\bar{I}; Y)$ into $L^p(I; Y)$ (see Exercise 8.23), without loss of generality, we may assume that the functions u_i are in $C(\bar{I}; Y)$. Since each function u_i is uniformly continuous, there exists $\delta_i > 0$ such that $\|u_i(x) - u_i(s)\| \leq \varepsilon$ for all $s, x \in \bar{I}$ with $|s - x| \leq \delta_i$. In turn, for $0 < h \leq \delta_i$,

$$\int_{I^h} \|\tau_h(u_i)(x) - u_i(x)\|^p \, dx = \int_{I^h} \|u_i(x+h) - u_i(x)\|^p \, dx \leq \varepsilon^p \mathcal{L}^1(I^h),$$

and so $\|\tau_h(u_i) - u_i\|_{L^p(I^h; Y)} \leq \varepsilon(\mathcal{L}^1(I))^{1/p}$. Let $\delta := \min_i \delta_i > 0$. For $0 < h \leq \delta$ and $u \in \mathcal{U}$, let $i \in \{1, \dots, n\}$ be such that $\|u - u_i\|_{L^p(I; Y)} \leq \varepsilon$. Writing

$$\tau_h(u) - u = \tau_h(u - u_i) + (\tau_h(u_i) - u_i) + (u_i - u),$$

it follows that $\|\tau_h(u) - u\|_{L^p(I^h; Y)} \leq 2\varepsilon + \varepsilon(\mathcal{L}^1(I))^{1/p}$, which proves (ii).

Step 2: Assume that $\mathcal{U} \subseteq L^p(I; Y)$ satisfies properties (i) and (ii). Let $0 < \eta < \mathcal{L}^1(I)$ and for every $u \in \mathcal{U}$ define the right mean function

$$M_\eta(u)(x) := \frac{1}{\eta} \int_x^{x+\eta} u(s) ds, \quad x \in I^\eta.$$

For every $x_1, x_2 \in I^\eta$, with $x_1 < x_2$, by a change of variables and Hölder's inequality we have

$$\begin{aligned} \|M_\eta(u)(x_2) - M_\eta(u)(x_1)\| &\leq \frac{1}{\eta} \int_{x_1}^{x_1+\eta} \|\tau_{x_2-x_1}(u) - u\| ds \\ &\leq \frac{1}{\eta^{1/p}} \|\tau_{x_2-x_1}(u) - u\|_{L^p(I^{x_2-x_1}; Y)}. \end{aligned}$$

It follows from property (ii) that the family of functions $\{M_\eta(u) : u \in \mathcal{U}\}$ belongs to $C(\overline{I^\eta}; Y)$ and is uniformly equicontinuous.

Moreover, by property (i) for every $x \in (I^\eta)^\circ$ we have that the set $\{M_\eta(u)(x) : u \in \mathcal{U}\}$ is relatively compact in Y . Thus, we are in a position to apply the Ascoli–Arzelà theorem (see Theorem 5.37) to obtain that the family $\{M_\eta(u) : u \in \mathcal{U}\}$ is relatively compact in $C(\overline{I^\eta}; Y)$.

Since by property (ii) the function $h \mapsto \tau_h(u) - u$ is continuous in $L^p(I^\eta; Y)$, we can write

$$\begin{aligned} \int_{I^\eta} \|M_\eta(u) - u\|^p dx &= \int_{I^\eta} \left\| \frac{1}{\eta} \int_0^\eta (\tau_h(u)(x) - u(x)) dh \right\|^p dx \\ &\leq \sup_{0 < h \leq \eta} \int_{I^\eta} \|\tau_h(u)(x) - u(x)\|^p dx. \end{aligned}$$

Using again property (ii) for every $\varepsilon > 0$ we can find $0 < \eta_\varepsilon < \frac{1}{2}\mathcal{L}^1(I)$ (independent of u) such that for every $0 < \eta \leq \eta_\varepsilon$, the right-hand side of the previous inequality is bounded from above by ε^p for every $u \in \mathcal{U}$. Hence,

$$(8.8) \quad \int_{I^\eta} \|M_\eta(u) - u\|^p dx \leq \varepsilon^p$$

for every $0 < \eta \leq \eta_\varepsilon$ and for every $u \in \mathcal{U}$. Fix $0 < \eta \leq \eta_\varepsilon$. Since $\{M_\eta(u) : u \in \mathcal{U}\}$ is relatively compact in $C(\overline{I^\eta}; Y)$ there exist $u_1, \dots, u_n \in C(\overline{I^\eta}; Y)$ such that for every $u \in \mathcal{U}$ there is $i \in \{1, \dots, n\}$ with

$$(8.9) \quad \|M_\eta(u) - u_i\|_{C(\overline{I^\eta}; Y)} \leq \varepsilon.$$

In turn, by (8.8) and (8.9),

$$\begin{aligned} \|u - u_i\|_{L^p(I^\eta; Y)} &\leq \|u - M_\eta(u)\|_{L^p(I^\eta; Y)} + \|M_\eta(u) - u_i\|_{L^p(I^\eta; Y)} \\ &\leq \|M_\eta(u) - u_i\|_{C(\overline{I^\eta}; Y)} (\mathcal{L}^1(I))^{1/p} + \varepsilon \leq (1 + (\mathcal{L}^1(I))^{1/p}) \varepsilon. \end{aligned}$$

In particular, since $\eta < \frac{1}{2}\mathcal{L}^1(I)$ we have that $I^\eta \supset J_1$, where J_1 is the interval of endpoints $\inf I$ and the middle point of I , and so the set $\mathcal{U}_1 := \{u : J_1 \rightarrow Y : u \in \mathcal{U}\}$ is relatively compact in $L^p(J_1; Y)$.

With a similar proof, working with h negative, we can show that the set $\mathcal{U}_2 := \{u : J_2 \rightarrow Y : u \in \mathcal{U}\}$ is relatively compact in $L^p(J_2; Y)$, where J_2 is the interval of endpoints the middle point of I and $\sup I$. It follows (why?) that $\mathcal{U}$ is relatively compact in $L^p(I; Y)$. $\square$

Exercise 8.25. Let $I \subset \mathbb{R}$ be a bounded interval and let Y be a Banach space. Prove that $\mathcal{U} \subseteq L^\infty(I; Y)$ is relatively compact in $C(\overline{I}; Y)$ if and only if properties (i) and (ii) (with $p = \infty$) are satisfied.

The following corollary will be useful in Section 8.5 below.

Corollary 8.26. Let $I \subset \mathbb{R}$ be a bounded interval, let $(Y_0, \|\cdot\|_{Y_0})$, $(Y_1, \|\cdot\|_{Y_1})$, and $(Y_2, \|\cdot\|_{Y_2})$ be Banach spaces with $Y_0 \hookrightarrow Y_1 \hookrightarrow Y_2$. Assume that the embedding $Y_0 \hookrightarrow Y_1$ is compact. Let $1 \leq p < \infty$ and let $\mathcal{U} \subseteq L^p(I; Y_0)$ be a bounded set such that $\|\tau_h(u) - u\|_{L^p(I^h; Y_2)} \rightarrow 0$ as $h \rightarrow 0^+$ uniformly for $u \in \mathcal{U}$. Then $\mathcal{U}$ is relatively compact in $L^p(I; Y_1)$.

Proof. Step 1: We claim that for every $\varepsilon > 0$ there exists $c_\varepsilon > 0$ such that

$$(8.10) \quad \|y\|_{Y_1} \leq \varepsilon \|y\|_{Y_0} + c_\varepsilon \|y\|_{Y_2}$$

for every $y \in Y_0$. By hypothesis there exist $c_1, c_2 > 0$ such that

$$(8.11) \quad \|y\|_{Y_1} \leq c_1 \|y\|_{Y_0}$$

for all $y \in Y_0$ and

$$(8.12) \quad \|y\|_{Y_2} \leq c_2 \|y\|_{Y_1}$$

for all $y \in Y_1$. Assume that (8.10) fails. Then for every n there exists $y_n \in Y_0$ such that

$$\|y_n\|_{Y_1} > \varepsilon \|y_n\|_{Y_0} + n \|y_n\|_{Y_2}.$$

Note that by (8.11) the previous inequality implies that $\|y_n\|_{Y_0} > 0$. Let $w_n := y_n / \|y_n\|_{Y_0}$. Then

$$(8.13) \quad \|w_n\|_{Y_1} > \varepsilon \|w_n\|_{Y_0} + n \|w_n\|_{Y_2} = \varepsilon + n \|w_n\|_{Y_2}.$$

Since the sequence $\{w_n\}_n$ is bounded in Y_0 , by (8.11), it is also bounded in Y_1 . In turn, by (8.13), $w_n \rightarrow 0$ in Y_2 . On the other hand, since the embedding $Y_0 \hookrightarrow Y_1$ is compact, there exist a subsequence $\{w_{n_k}\}_k$ and

$w_0 \in Y_1$ such that $w_{n_k} \rightarrow w_0$ in Y_1 . Necessarily, $w_0 = 0$ by (8.12). Hence, taking $n = n_k$ in (8.13) and letting $k \rightarrow \infty$ we get

$$0 = \lim_{k \rightarrow \infty} \|w_{n_k}\|_{Y_1} \geq \varepsilon > 0,$$

which is a contradiction.

Step 2: Since $\mathcal{U} \subseteq L^p(I; Y_0)$ is bounded, by Hölder's inequality for every $x_1, x_2 \in I^\circ$, with $x_1 < x_2$, the set

$$Y_{x_1, x_2} := \left\{ \int_{x_1}^{x_2} u(x) dx : u \in \mathcal{U} \right\}$$

is bounded in Y_0 . On the other hand, the embedding $Y_0 \hookrightarrow Y_1$ is compact, and so the set Y_{x_1, x_2} is relatively compact in Y_1 . By (8.12), Y_{x_1, x_2} is also relatively compact in Y_2 . It follows by Theorem 8.24 that $\mathcal{U}$ is relatively compact in $L^p(I; Y_2)$.

It remains to show that $\mathcal{U}$ is relatively compact in $L^p(I; Y_1)$. Let $\{u_n\}_n$ be a sequence in $\mathcal{U}$. Since $\mathcal{U}$ is relatively compact in $L^p(I; Y_2)$, by sequential compactness, there exists a subsequence, not relabeled, such that $u_n \rightarrow u_0$ in $L^p(I; Y_2)$ for some $u_0 \in L^p(I; Y_2)$. In particular, $\{u_n\}_n$ is a Cauchy sequence in $L^p(I; Y_2)$. Let $\varepsilon > 0$. By Step 1 and the Minkowski inequality for every $n, m \in \mathbb{N}$,

$$\begin{aligned} \|u_n - u_m\|_{L^p(I; Y_1)} &\leq \varepsilon \|u_n - u_m\|_{L^p(I; Y_0)} + c_\varepsilon \|u_n - u_m\|_{L^p(I; Y_2)} \\ &\leq 2\varepsilon K + c_\varepsilon \|u_n - u_m\|_{L^p(I; Y_2)}, \end{aligned}$$

where $K := \sup_{u \in \mathcal{U}} \|u\|_{L^p(I; Y_0)} < \infty$. Letting $n, m \rightarrow \infty$, by the fact that $\{u_n\}_n$ is a Cauchy sequence in $L^p(I; Y_2)$ we get

$$\limsup_{n, m \rightarrow \infty} \|u_n - u_m\|_{L^p(I; Y_1)} \leq 2\varepsilon K.$$

By letting $\varepsilon \rightarrow 0^+$ we obtain that a subsequence of $\{u_n\}_n$ is a Cauchy sequence in $L^p(I; Y_1)$. This shows that the closure of $\mathcal{U}$ is sequentially compact in $L^p(I; Y_1)$ and concludes the proof. $\square$

Exercise 8.27. Let $I \subset \mathbb{R}$ be a bounded interval, let $(Y_0, \|\cdot\|_{Y_0})$, $(Y_1, \|\cdot\|_{Y_1})$, and $(Y_2, \|\cdot\|_{Y_2})$ be Banach spaces with $Y_0 \hookrightarrow Y_1 \hookrightarrow Y_2$. Assume that the embedding $Y_0 \hookrightarrow Y_1$ is compact. Let $\mathcal{U} \subseteq L^\infty(I; Y_0)$ be a bounded set such that $\|\tau_h(u) - u\|_{L^\infty(I^h; Y_2)} \rightarrow 0$ as $h \rightarrow 0^+$ uniformly for $u \in \mathcal{U}$. Prove that $\mathcal{U}$ is relatively compact in $C(\bar{I}; Y_1)$.

We conclude this section with a useful result.

Theorem 8.28. Let $I \subseteq \mathbb{R}$ be an open interval, let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set, and let $1 \leq p < \infty$. Then $L^p(I; L^p(E))$ can be identified with $L^p(E \times I)$.

Proof. Given $u \in L^p(E \times I)$, by Fubini's theorem, the function $t \in I \mapsto \int_E |u(x, t)|^p dx$ is Lebesgue measurable with

$$\int_I \left(\int_E |u(x, t)|^p dx \right) dt = \|u\|_{L^p(E \times I)}^p < \infty.$$

It follows that $\int_E |u(x, t)|^p dx < \infty$ for $\mathcal{L}^1$ -a.e. $t \in I$. Hence, setting $v(t) := u(\cdot, t)$, by Theorem 8.9, we have that $v : I \rightarrow L^p(E)$ with

$$\|v\|_{L^p(I; L^p(E))}^p = \int_I \|v(t)\|_{L^p(E)}^p dt = \int_I \int_E |u(x, t)|^p dx dt = \|u\|_{L^p(E \times I)}^p < \infty,$$

which shows that $v \in L^p(I; L^p(E))$ with $\|v\|_{L^p(I; L^p(E))} = \|u\|_{L^p(E \times I)}$. Consider the linear operator

$$T : L^p(E \times I) \rightarrow L^p(I; L^p(E))$$

$$u \mapsto v.$$

Then T preserves the norms and so it is injective. By identifying u with $T(u)$, this shows that $L^p(E \times I) \subseteq L^p(I; L^p(E))$.

Conversely, given $v \in L^p(I; L^p(E))$, there exists a sequence $\{s_n\}$ of Bochner integrable simple functions $s_n : I \rightarrow L^p(E)$ such that

$$\lim_{n \rightarrow \infty} \|s_n(t) - v(t)\|_{L^p(E)} = 0 \quad \text{for } \mathcal{L}^1\text{-a.e. } t \in I$$

and

$$\lim_{n \rightarrow \infty} \int_I \|s_n(t) - v(t)\|_{L^p(E)}^p dt = 0.$$

Write

$$s_n = \sum_{i=1}^{m_n} u_{i,n} \chi_{E_{i,n}},$$

where $u_{i,n} \in L^p(E)$ and $E_{i,n} \subseteq I$ are Lebesgue measurable pairwise disjoint sets. Define

$$g_n(x, t) := \sum_{i=1}^{m_n} u_{i,n}(x) \chi_{E_{i,n}}(t), \quad x \in E, t \in I.$$

Then $g_n : E \times I \rightarrow \mathbb{R}$ is Lebesgue measurable, since it is given by sums and products of Lebesgue measurable functions. Moreover,

$$\begin{aligned} \|s_n\|_{L^p(I; L^p(E))}^p &= \int_I \|s_n(t)\|_{L^p(E)}^p dt \\ &= \int_I \left(\sum_{i=1}^{m_n} \chi_{E_{i,n}}(t) \int_E |u_{i,n}(x)|^p dx \right) dt \\ &= \|g_n\|_{L^p(E \times I)}^p. \end{aligned}$$

A similar computation shows that $\|s_n - s_m\|_{L^p(I; L^p(E))} = \|g_n - g_m\|_{L^p(E \times I)}$. Since $s_n \rightarrow v$ in $L^p(I; L^p(E))$, we have that $\{s_n\}_n$ is a Cauchy sequence in

$L^p(I; L^p(E))$. In turn, $\{g_n\}_n$ is a Cauchy sequence in $L^p(E \times I)$. Hence, there exists $u \in L^p(E \times I)$ such that $g_n \rightarrow u$ in $L^p(E \times I)$. However, from the first part of the theorem $L^p(E \times I) \subseteq L^p(I; L^p(E))$. Moreover, $T(g_n) = s_n$. Hence, $T(u) = v$, which shows that $L^p(I; L^p(E)) \subseteq L^p(E \times I)$. $\square$

8.3. Functions of Bounded Pointwise Variation

In this section we discuss the differentiability of functions of bounded pointwise variation in the case in which Y is an infinite dimensional normed space.

We begin by establishing the analog of Corollary 1.28.

Theorem 8.29. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, and let $u \in BV_{\text{loc}}(I; Y)$. Then $u(I)$ is separable, u is strongly measurable, and for $h > 0$,*

$$(8.14) \quad \frac{1}{h} \int_{I^h} \|u(x+h) - u(x)\| dx \leq \text{Var } u,$$

where $I^h := \{x \in I : x+h \in I\}$.

Proof. Let's prove that u is weakly measurable. Let $T \in Y'$ and consider the function $v(x) := T(u(x))$, $x \in I$. Let $[a, b] \subseteq I$ and let $x_0 < \dots < x_n$ be a partition of $[a, b]$. Then by the linearity and continuity of T ,

$$\begin{aligned} \sum_{i=1}^n |v(x_i) - v(x_{i-1})| &= \sum_{i=1}^n |T(u(x_i) - u(x_{i-1}))| \\ &\leq \|T\|_{Y'} \sum_{i=1}^n \|u(x_i) - u(x_{i-1})\| \leq \|T\|_{Y'} \text{Var}_{[a,b]} u. \end{aligned}$$

Taking the supremum over all partitions gives $\text{Var}_{[a,b]} v \leq \|T\|_{Y'} \text{Var}_{[a,b]} u < \infty$. Thus, $v \in BV_{\text{loc}}(I)$, and so it is measurable. Given the arbitrariness of $T \in Y'$, it follows that u is weakly measurable.

Next we show that $u(I)$ is separable. By Theorem 2.17 the set of jump points of u is countable. Let $\{x_n\}_n$ be the sequence obtained by taking the union of the rational numbers in I and the set of jump points of u . Consider the countable set $F := \{u(x_n) : n \in \mathbb{N}\}$. Given $y \in u(I)$, if $y \in F$, then there is nothing to prove. If $y \in u(I) \setminus F$, then there exists $x \in I$ such that $u(x) = y$ and u is continuous at x . By the density of the rationals, there exists a subsequence $\{x_{n_k}\}_k$ of $\{x_n\}_n$ such that $x_{n_k} \rightarrow x$ as $k \rightarrow \infty$. By continuity $u(x_{n_k}) \rightarrow y$. Hence, $u(I)$ is separable. It now follows by the Pettis Theorem (see Theorem 8.3) that u is strongly measurable.

Next we prove (8.14). If $\text{Var } u = \infty$, then there is nothing to prove. Thus, assume that $\text{Var } u < \infty$. If $\text{diam } I^h > 0$, let $[a, b] \subseteq I$ be such that

$0 < h \leq b - a$. By (2.4) and Corollary 1.28 applied to $\mathbf{V}$ we have

$$\begin{aligned} \frac{1}{h} \int_a^{b-h} \|u(x+h) - u(x)\| dx &\leq \frac{1}{h} \int_a^{b-h} (\mathbf{V}(x+h) - \mathbf{V}(x)) dx \\ &\leq \mathbf{V}(b) - \mathbf{V}(a) = \text{Var}_{[a,b]} u. \end{aligned}$$

Construct an increasing sequence of intervals $[a_n, b_n]$ such that $a_n \searrow \inf I$, $b_n \nearrow \sup I$. If $\text{diam } I^h > 0$, then for all n sufficiently large we have that $0 < h < b_n - a_n$, and so by the previous inequality we have that

$$\frac{1}{h} \int_{a_n}^{b_n-h} \|u(x+h) - u(x)\| dx \leq \text{Var}_{[a_n, b_n]} u \leq \text{Var } u.$$

Letting $n \rightarrow \infty$ and using the Lebesgue monotone convergence theorem gives

$$\frac{1}{h} \int_{I^h} \|u(x+h) - u(x)\| dx \leq \text{Var } u.$$

The previous inequality continues to hold if $\text{diam } I^h = 0$, since in this case the integral on the left-hand side is zero. This completes the proof. $\square$

The following two exercises show that when Y is an infinite-dimensional normed space, functions of bounded pointwise variation may be nowhere differentiable.

Exercise 8.30. Let $Y = L^1([a, b])$ and let $u : [a, b] \rightarrow Y$ be defined by

$$(u(x))(y) := \begin{cases} 1 & \text{if } a \leq y < x, \\ 0 & \text{if } x \leq y \leq b. \end{cases}$$

Prove that u has bounded pointwise variation but it is nowhere differentiable.

Exercise 8.31. Let $Y = c_0$ and let $u : [0, 1] \rightarrow Y$ be defined by

$$u(x) := \{a_n(x)\}_n, \quad a_n(x) := \int_0^x \sin(2n\pi s) ds.$$

Prove that u has bounded pointwise variation but it is nowhere differentiable.

We recall that c_0 is the space of all sequences $\{a_n\}_n$ of real numbers converging to zero, endowed with the norm $\sup_n |a_n|$.

Exercise 8.32. Let $Y = L^\infty((0, 1))$ and let $u : [0, 1] \rightarrow Y$ be defined by

$$(u(x))(y) := y \sin(x/y), \quad x \in [0, 1], y \in (0, 1).$$

Prove that u has bounded pointwise variation but it is nowhere differentiable.

The following theorem shows that if Y is reflexive, then a function of bounded pointwise variation is differentiable $\mathcal{L}^1$ -a.e.

Theorem 8.33. *Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a reflexive Banach space, and let $u \in BV_{\text{loc}}(I; Y)$. Then u is differentiable at $\mathcal{L}^1$ -a.e. $x \in I$, the function u' is locally Bochner integrable and for every interval $J \subseteq I$,*

$$\int_J \|u'(x)\| dx \leq \text{Var}_J u.$$

In particular, if $u \in BPV(I; Y)$, then u' is Bochner integrable.

The proof relies on the fact that if Y is a reflexive Banach space, Lipschitz continuous functions $u : I \rightarrow Y$ are differentiable $\mathcal{L}^1$ -a.e. We will prove this result in Theorem 8.38 below.

Proof. Step 1: Assume $I = [a, b]$. By Theorem 8.29 we have that $u(I)$ is separable. Hence, by replacing Y with the closure of the subspace generated by the set $u(I)$, without loss of generality, we may assume that Y is reflexive and separable. (Exercise. Hint: Use Theorem A.64.)

By Theorem 2.36, the indefinite pointwise variation $\mathbf{V}$ of u is increasing and

$$\|u(x+h) - u(x)\| \leq |\mathbf{V}(x+h) - \mathbf{V}(x)|$$

for all $x, x+h \in I$. Dividing by $|h|$ and letting $h \rightarrow 0$, it follows by Lebesgue's differentiation theorem (see Theorem 1.18) applied to $\mathbf{V}$, that for $\mathcal{L}^1$ -a.e. $x \in I$,

$$(8.15) \quad \limsup_{h \rightarrow 0} \frac{\|u(x+h) - u(x)\|}{|h|} \leq \mathbf{V}'(x) < \infty.$$

Hence, the set

$$(8.16) \quad E_0 := \left\{ x \in I : \limsup_{h \rightarrow 0} \frac{\|u(x+h) - u(x)\|}{|h|} = \infty \right\}$$

satisfies $\mathcal{L}^1(E_0) = 0$. For every $n \in \mathbb{N}$ let C_n be the set of points $x \in I$ such that

$$(8.17) \quad \|u(x+h) - u(x)\| \leq n|h|$$

for all $x \in I$ and for all $|h| < 1/n$ with $x+h \in I$. Observe that $I \setminus E_0 = \bigcup_n C_n$. Hence, it suffices to prove that u is differentiable for $\mathcal{L}^1$ -a.e. $x \in C_n$.

Fix n . We claim that C_n is closed. To see this, let $x_k \in C_n$ be such that $x_k \rightarrow x \in I$. Let $|h| < 1/n$ be such that $x+h \in I$ (if no such h exists, then x belongs to C_n automatically). Fix $0 < \varepsilon < 1/n - |h|$ and find k so large that $|x_k - x| < \varepsilon$. Then

$$|x+h - x_k| \leq |h| + |x - x_k| < |h| + \varepsilon < 1/n,$$

and so, since $x_k \in C_n$,

$$\begin{aligned} \|u(x+h) - u(x)\| &\leq \|u(x+h) - u(x_k)\| + \|u(x_k) - u(x)\| \\ &\leq n|x+h-x_k| + n|x_k-x| < n(|h| + 2\varepsilon). \end{aligned}$$

Letting $\varepsilon \rightarrow 0^+$ in the previous inequality shows that $x \in C_n$. Thus, C_n is closed.

Hence, the set $I \setminus C_n$ is relatively open in I . Thus we can write $I \setminus C_n$ as the disjoint union of countably many intervals I_k . These intervals will all be open with the possible exception of the two intervals that contain an endpoint of I . Extend $u : I \setminus C_n \rightarrow Y$ in each interval I_k of endpoints $a_k < b_k$ by setting

$$u_n(x) := u(a_k) + \frac{u(b_k) - u(a_k)}{b_k - a_k}(x - a_k).$$

Note that if $|b_k - a_k| < 1/n$, then, since $a_k \in C_n$, it follows from (8.17) that $\|u(b_k) - u(a_k)\| \leq n(b_k - a_k)$. On the other hand, if $|b_k - a_k| \geq 1/n$, then $\|u(b_k) - u(a_k)\| \leq 2n\|u\|_\infty(b_k - a_k)$ (we recall that functions in $BPV(I; Y)$ are bounded; see Proposition 2.12). Thus, $u_n : I_k \rightarrow Y$ is Lipschitz continuous with Lipschitz constant at most $n(1 + 2\|u\|_\infty)$.

It follows also from (8.17) that $u_n : I \rightarrow Y$ is Lipschitz continuous (exercise).

By Theorem 8.38 the function u_n is differentiable at $\mathcal{L}^1$ -a.e. $x \in I$. Moreover, since the function $v_n(x) := \text{dist}(x, C_n)$, $x \in I$, is Lipschitz continuous, by Proposition 2.6 and Corollary 2.31, v_n is differentiable at $\mathcal{L}^1$ -a.e. $x \in I$.

Let $x \in C_n$ be such that both u_n and v_n are differentiable at x and x is not an endpoint of any I_k . We claim that u is differentiable at x . Since x is not an isolated point of C_n , necessarily, $v'_n(x) = 0$ (why?). Fix $0 < \varepsilon < 1/2$ and find $0 < \delta < 1/n$ such that

$$\begin{aligned} (8.18) \quad &\|u_n(x+h) - u_n(x) - u'_n(x)h\| \leq \varepsilon|h|, \\ &\text{dist}(x+h, C_n) = |v_n(x+h) - v_n(x) - v'_n(x)h| < \varepsilon|h| \end{aligned}$$

for all $x+h \in I$ with $0 < |h| \leq \delta$. Fix $0 < |h| \leq \delta/2$. Since $\text{dist}(x+h, C_n) < \varepsilon|h|$, there exists $t \in C_n$ such that $|x+h-t| < \varepsilon|h| < 1/n$. Note that $|x-t| \leq |x+h-t| + |h| \leq \delta$. Using the fact that $u = u_n$ on C_n , we have that

$$\begin{aligned} \|u(x+h) - u(x) - u'_n(x)h\| &\leq \|u(x+h) - u(t)\| \\ &\quad + \|u_n(t) - u_n(x) - u'_n(x)(t-x)\| + \|u'_n(x)\||x+h-t| \\ &\leq n|x+h-t| + \varepsilon|h| + \|u'_n(x)\||x+h-t| \leq (n+1+\|u'_n(x)\|)\varepsilon|h|, \end{aligned}$$

where we used (8.17) and (8.18). This shows that u is differentiable at x .

Step 2: For an arbitrary interval I , it suffices to take an increasing sequence of closed intervals whose union is I and to apply Step 1 in each closed interval. $\square$

Remark 8.34. The previous theorem applies, in particular, for $Y = L^p(E)$, $1 < p < \infty$, where $E \subseteq \mathbb{R}^N$ is a Lebesgue measurable set.

The Banach spaces Y for which Theorem 8.33 holds, namely, for which every function $u : I \rightarrow Y$ of bounded pointwise variation is differentiable $\mathcal{L}^1$ -a.e. in I , are said to have the Radon–Nikodym property.

Definition 8.35. A Banach space Y has the *Radon–Nikodym property* (RNP) if for every interval $I \subseteq \mathbb{R}$ and every function $u : I \rightarrow Y$ of bounded pointwise variation, u is differentiable $\mathcal{L}^1$ -a.e. in I .

Exercises 8.30 and 8.32 show that $Y = L^1(I)$ and $Y = L^\infty(I)$ do not have the RNP. For spaces without the RNP property one can use a weaker notion of derivative, namely, the metric derivative.

Definition 8.36. Given $E \subseteq \mathbb{R}$, an accumulation point $x_0 \in E$ of E and a metric space (Y, d) , the *metric derivative* $\frac{d_m u}{dx}(x_0)$ of a function $u : E \rightarrow Y$ at x_0 is defined as the limit

$$(8.19) \quad \lim_{h \rightarrow 0} \frac{d(u(x+h), u(x))}{|h|}$$

provided this limit exists in $[0, \infty]$. We say that u is *metric differentiable* at x_0 if there exists $\frac{d_m u}{dx}(x_0) < \infty$.

Exercise 8.37. Let $I \subseteq \mathbb{R}$ be an interval and let Y be a Banach space. Using the fact that every Lipschitz continuous function $v : I \rightarrow Y$ admits a finite metric derivative $\mathcal{L}^1$ -a.e. in I ,¹ modify the proof of Theorem 8.33 to prove that if $u \in BV_{\text{loc}}(I; Y)$, then for $\mathcal{L}^1$ -a.e. $x \in I$, there exists in $\mathbb{R}$ the metric derivative $\frac{d_m u}{dx}(x)$, that the function $\frac{d_m u}{dx}$ is locally integrable and that for every interval $J \subseteq I$,

$$\int_J \frac{d_m u}{dx}(x) dx \leq \text{Var}_J u.$$

8.4. Absolute Continuous Functions

In this section we extend some of the results of Section 3.2 in Chapter 3 to functions $u : I \rightarrow Y$, where Y is a normed space.

In view of Exercise 8.30 we cannot expect Theorem 3.20 to hold for arbitrary Banach spaces. However, we will show that it holds if Y is reflexive.

¹We will prove this in Exercise 8.40.

Theorem 8.38. *Let $I \subseteq \mathbb{R}$ be an interval and let $(Y, \|\cdot\|)$ be a reflexive Banach space. A function $u : I \rightarrow Y$ belongs to $AC_{\text{loc}}(I; Y)$ if and only if*

- (i) u is continuous in I ,
- (ii) u is differentiable $\mathcal{L}^1$ -a.e. in I , and u' is locally Bochner integrable,
- (iii) the fundamental theorem of calculus is valid; that is, for all $x, x_0 \in I$,

$$u(x) = u(x_0) + \int_{x_0}^x u'(t) dt.$$

Proof. Step 1: Assume that $u \in AC_{\text{loc}}(I; Y)$. For $T \in Y'$ consider the function $w := T \circ u$. We claim that w belongs to $AC_{\text{loc}}(I)$. To see this, let $[a, b] \subseteq I$. Since $u \in AC([a, b]; Y)$, for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\sum_{i=1}^n \|u(b_i) - u(a_i)\| \leq \varepsilon$$

for every finite number of nonoverlapping intervals (a_i, b_i) , $i = 1, \dots, n$, with $[a_i, b_i] \subseteq [a, b]$ and $\sum_{i=1}^n (b_i - a_i) \leq \delta$. In turn, by the linearity of T ,

$$\begin{aligned} \sum_{i=1}^n |w(b_i) - w(a_i)| &= \sum_{i=1}^n |T(u(b_i) - u(a_i))| \\ &\leq \|T\|_{Y'} \sum_{i=1}^n \|u(b_i) - u(a_i)\| \leq \varepsilon \|T\|_{Y'}. \end{aligned}$$

This proves the claim.

Step 2: Assume that $u \in AC_{\text{loc}}(I; Y)$. As in the proof of Theorem 8.29 we may assume that Y is reflexive and separable. In turn, its dual Y' is reflexive and separable. Let $\{T_k\}_k$ be dense in Y' . Consider the function $w_k := T_k \circ u$. By Step 1, w_k belongs to $AC_{\text{loc}}(I)$. Hence, by Proposition 3.9, it is differentiable for all $x \in I \setminus E_k$, where $\mathcal{L}^1(E_k) = 0$. By the linearity of T_k , this implies that there exists

$$(8.20) \quad \lim_{h \rightarrow 0} T_k \left(\frac{u(x+h) - u(x)}{h} \right) = w'_k(x) \in \mathbb{R}$$

for all $x \in I \setminus E_k$. Set $E := \bigcup_{k=0}^{\infty} E_k$, where

$$E_0 := \left\{ x \in I : \limsup_{h \rightarrow 0} \frac{\|u(x+h) - u(x)\|}{|h|} = \infty \right\}.$$

As in the proof of Theorem 8.29 we have that $\mathcal{L}^1(E_0) = 0$. In turn, $\mathcal{L}^1(E) = 0$. Let $x \in I \setminus E$. Since $x \notin E_0$, there exists $L_0 > 0$ such that

$$(8.21) \quad \frac{\|u(x+h) - u(x)\|}{|h|} \leq L_0$$

for all h sufficiently small. Using the fact that Y is reflexive, by Corollary A.65 we can find a sequence $h_j \rightarrow 0$ and $v(x) \in Y$ such that

$$\frac{u(x + h_j) - u(x)}{h_j} \rightharpoonup v(x).$$

It follows, also from (8.20), that $w'_k(x) = T_k(v(x))$. We claim that

$$(8.22) \quad \frac{u(x + h) - u(x)}{h} \rightharpoonup v(x)$$

as $h \rightarrow 0^+$. Fix $T \in Y'$ and $\eta > 0$. By the density of $\{T_k\}_k$ in Y' , there exists T_m such that $\|T - T_m\|_{Y'} \leq \eta$. Since $v'_m(x) = T_m(v(x))$, there exists $h_0 > 0$ such that

$$\left| T_m\left(\frac{u(x + h) - u(x)}{h}\right) - T_m(v(x)) \right| \leq \eta$$

for all $0 < |h| \leq h_0$. In turn, also by (8.21),

$$\begin{aligned} \left| T\left(\frac{u(x + h) - u(x)}{h}\right) - T(v(x)) \right| &\leq \|T - T_m\|_{Y'}(L_0 + \|v(x)\|) \\ &+ \left| T_m\left(\frac{u(x + h) - u(x)}{h}\right) - T_m(v(x)) \right| \leq (L_0 + \|v(x)\| + 1)\eta \end{aligned}$$

for all $0 < |h| \leq h_0$. This proves the claim.

The function

$$x \in I \setminus E \mapsto v(x)$$

is weakly measurable, since each function $x \in I \mapsto \frac{u(x+h)-u(x)}{h}$ is strongly measurable. Since Y is separable, it follows by Pettis' theorem (see Theorem 8.3) that it is strongly measurable. Moreover, by the lower semicontinuity of the norm with respect to weak convergence (see Proposition A.67) and (8.15),

$$\|v(x)\| \leq \liminf_{h \rightarrow 0} \frac{\|u(x + h) - u(x)\|}{|h|} \leq \mathbf{V}'(x)$$

for $\mathcal{L}^1$ -a.e. $x \in I$. It follows from Fatou's lemma that

$$\begin{aligned} \int_J \|v(x)\| dx &\leq \int_J \liminf_{h \rightarrow 0^+} \frac{\|u(x + h) - u(x)\|}{h} dx \\ &\leq \int_J \mathbf{V}'(x) dx \leq \text{Var}_J u, \end{aligned}$$

where in the last inequality we have used Corollary 1.28. Hence, v is locally Bochner integrable.

Since for $T \in Y'$ the function $w := T \circ u$ belongs to $AC_{\text{loc}}(I)$, by Step 1, by Proposition 3.9, w is differentiable for $\mathcal{L}^1$ -a.e. $x \in I$. It follows from (8.22) and the linearity of T that for $\mathcal{L}^1$ -a.e. $x \in I$,

$$\frac{w(x + h) - w(x)}{h} = T\left(\frac{u(x + h) - u(x)}{h}\right) \rightarrow T(v(x)),$$

which shows that $w'(x) = T(v(x))$ for $\mathcal{L}^1$ -a.e. $x \in I$.

Hence, by Theorem 3.20 applied to the function w , we have that

$$\begin{aligned} T(u(x)) - T(u(x_0)) &= w(x) - w(x_0) = \int_{x_0}^x w'(t) dt = \int_{x_0}^x T(v(t)) dt \\ &= T\left(\int_{x_0}^x v(t) dt\right), \end{aligned}$$

where we have used Theorem 8.13. Hence,

$$T\left(u(x) - u(x_0) - \int_{x_0}^x v(t) dt\right) = 0.$$

Taking the supremum over all T with $\|T\|_{Y'} \leq 1$ and using the fact that for $y \in Y$, $\|y\| = \sup_{\|T\|_{Y'} \leq 1} T(y)$, we get

$$u(x) - u(x_0) - \int_{x_0}^x v(t) dt = 0.$$

It now follows from Theorem 8.19 that u is differentiable for $\mathcal{L}^1$ -a.e. $x \in I$ with $u'(x) = v(x)$.

Step 3. Conversely, assume that u satisfies (i)–(iii). Then $\|u'\|$ is Lebesgue integrable on every $[a, b] \subseteq I$ and so equi-integrable in $[a, b]$. Reasoning as in the second part of the proof of Theorem 3.29 we can show that u is absolutely continuous in $[a, b]$.

This concludes the proof. $\square$

Remark 8.39. In view of the previous theorem, Theorems 3.25, 3.29, and Corollaries 3.31 and 3.33 continue to hold with $\mathbb{R}^M$ replaced by a reflexive Banach space.

Exercise 8.40. Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a Banach space, and let $u : I \rightarrow Y$ be a Lipschitz continuous function.

(i) Let $\{y_n\}_n \subseteq u(I)$ be dense in $u(I)$ and define

$$v_n(x) := \|u(x) - y_n\|, \quad x \in I.$$

Prove that for $x, x+h \in I$,

$$(8.23) \quad |v_n(x+h) - v_n(x)| \leq \|u(x+h) - u(x)\|$$

and deduce that

$$\sup_n |v'_n(x)| \leq \liminf_{h \rightarrow 0} \frac{\|u(x+h) - u(x)\|}{|h|}$$

for $\mathcal{L}^1$ -a.e. $x \in I$.

(ii) Prove that for $x, x+h \in I$,

$$\|u(x+h) - u(t)\| = \sup_n |v_n(x+h) - v_n(x)| \leq \left| \int_x^{x+h} \sup_n |v'_n(t)| dt \right|$$

and deduce that

$$\limsup_{h \rightarrow 0} \frac{\|u(x+h) - u(t)\|}{|h|} \leq \sup_n |v'_n(x)|$$

for $\mathcal{L}^1$ -a.e. $x \in I$. Deduce that u is metrically differentiable at $\mathcal{L}^1$ -a.e. $x \in I$.

Exercise 8.41. Let $I \subseteq \mathbb{R}$ be an interval, let $(Y, \|\cdot\|)$ be a normed space, let $u : I \rightarrow Y$. Prove that Theorems 3.36 and 3.39 continue to hold if we assume metric differentiability in place of differentiability (see (8.19)) and replace $\|u'(x)\|$ with $\frac{d_m u}{dx}(x)$ in (3.11) and (3.15).

Theorem 3.41 continues to hold for functions with values in a reflexive Banach space. To be precise, we have

Theorem 8.42 (Lusin (N) property, II). *Let $I \subseteq \mathbb{R}$ be an interval, let Y be a reflexive Banach space. A function $u : I \rightarrow Y$ belongs to $AC_{\text{loc}}(I; Y)$ if and only if*

- (i) u is continuous on I ,
- (ii) u is differentiable $\mathcal{L}^1$ -a.e. in I , and $u' \in L^1_{\text{loc}}(I; Y)$,
- (iii) u satisfies the Lusin (N) property.

Proof. The proof is exactly the same of Theorem 3.41, with the only change that we use Theorem 8.38 in place of Theorem 3.20. $\square$

Remark 8.43. Let $I \subseteq \mathbb{R}$ be an interval and let Y be a normed space. If a function $u : I \rightarrow Y$ belongs to $AC_{\text{loc}}(I; Y)$ and is differentiable $\mathcal{L}^1$ -a.e. in I , then we can still show that u satisfies the Lusin (N) property.

In turn, Corollaries 3.43 and 3.49 continue to hold for functions $u : I \rightarrow Y$, with Y a reflexive Banach space.

Corollary 8.44. *Let $I \subseteq \mathbb{R}$ be an interval and let Y be a reflexive Banach space. A function $u : I \rightarrow Y$ belongs to $AC(I; Y)$ if and only if*

- (i) u is continuous on I ,
- (ii) u is differentiable $\mathcal{L}^1$ -a.e. in I , and u' belongs to $L^1_{\text{loc}}(I; Y)$ and is equi-integrable,
- (iii) u satisfies the Lusin (N) property.

Corollary 8.45. *Let $I \subseteq \mathbb{R}$ be an interval and let Y be a reflexive Banach space. A function $u : I \rightarrow Y$ belongs to $AC_{\text{loc}}(I; Y)$ if and only if*

- (i) u is continuous on I ,
- (ii) $u \in BPV_{\text{loc}}(I; Y)$,
- (iii) u satisfies the Lusin (N) property.

Exercise 8.46. Let $I \subseteq \mathbb{R}$ be an interval, let Y be a Banach space, and let $u : I \rightarrow Y$. Using Exercises 8.37 and 8.41 prove that Corollary 8.45 continues to hold. State and prove the analog of Theorems 8.42 and Corollaries 8.44.

Remark 8.47. In view of Theorems 8.33 and 8.42, Corollaries 3.65, 3.66, 3.68 continue to hold if $f : I \rightarrow Y$, where Y is a reflexive Banach space.

Remark 8.48. In view of Theorems 8.33 and 8.42, Corollary 3.90 continues to hold for functions $u \in BPV_{\text{loc}}(I; Y)$, where Y is a reflexive Banach space.

8.5. Sobolev Functions

In this section we extend the notion of Sobolev spaces to Banach-valued functions.

Definition 8.49. Given an open set $\Omega \subseteq \mathbb{R}$, a Banach space Y , $n \in \mathbb{N}$, and $1 \leq p \leq \infty$, we say that a function $u \in L^1_{\text{loc}}(\Omega; Y)$ admits a *weak or distributional derivative of order n in $L^p(\Omega; Y)$* if there exists a function $v \in L^p(\Omega; Y)$ such that

$$\int_{\Omega} u \varphi^{(n)} dx = (-1)^n \int_{\Omega} v \varphi dx$$

for all $\varphi \in C_c^\infty(\Omega)$. The function v is denoted $u^{(n)}$.

A similar definition can be given when $L^p(\Omega; Y)$ is replaced by $L^p_{\text{loc}}(\Omega; Y)$.

Remark 8.50. Let $\Omega \subseteq \mathbb{R}$ be an open set, let Y be a Banach space, and let $u : \Omega \rightarrow Y$ be a locally Bochner integrable function. If $Y \hookrightarrow Z$, where Z is a Banach space, with immersion $i : Y \rightarrow Z$ (see Definition A.36), and if the function $i \circ u : \Omega \rightarrow Z$ admits a weak derivative of order n in $L^p_{\text{loc}}(\Omega; Z)$, then in view of Remark 8.8, we have that

$$i\left(\int_{\Omega} \varphi^{(n)} u dx\right) = \int_{\Omega} \varphi^{(n)} (i \circ u) dx = (-1)^n \int_{\Omega} \varphi (i \circ u)^{(n)} dx.$$

Since we agreed (see Remark 8.8) to identify $\int_{\Omega} \varphi^{(n)} u dx$ with $i\left(\int_{\Omega} \varphi^{(n)} u dx\right)$, with an abuse of notation, we can write

$$\int_{\Omega} \varphi^{(n)} u dx = - \int_{\Omega} \varphi u^{(n)} dx.$$

Exercise 8.51. Let $\Omega \subseteq \mathbb{R}$ be an open set, let Y be a Banach space, let $n \in \mathbb{N}$, and let $u \in L^1_{\text{loc}}(\Omega; Y)$ be such that its weak derivative $u^{(n)}$ of order n exists in $L^1_{\text{loc}}(\Omega; Y)$. Prove that $u^{(n)}$ is unique.

We now define Sobolev spaces for Banach-valued functions.

Definition 8.52. Given an open set $\Omega \subseteq \mathbb{R}$, a Banach space Y , $m \in \mathbb{N}$, and $1 \leq p \leq \infty$, the *Sobolev space* $W^{m,p}(\Omega; Y)$ is the space of all functions $u \in L^p(\Omega; Y)$ which admit weak derivatives of order n in $L^p(\Omega; Y)$ for every $n = 1, \dots, m$. The space $W^{m,p}(\Omega; Y)$ is endowed with the norm

$$\|u\|_{W^{m,p}(\Omega; Y)} := \|u\|_{L^p(\Omega; Y)} + \sum_{n=1}^m \|u^{(n)}\|_{L^p(\Omega; Y)}.$$

The space $W_{\text{loc}}^{m,p}(\Omega; Y)$ is defined as the space of all functions $u \in L_{\text{loc}}^p(\Omega; Y)$ which admit weak derivatives of order n in $L_{\text{loc}}^p(\Omega; Y)$ for every $n = 1, \dots, m$.

Exercise 8.53. Given an open set $\Omega \subseteq \mathbb{R}$, a Banach space Y , $m \in \mathbb{N}$, and $1 \leq p \leq \infty$, prove that $W^{m,p}(\Omega; Y)$ is a Banach space.

Exercise 8.54. Given an open set $\Omega \subseteq \mathbb{R}$, a Hilbert space Y , $m \in \mathbb{N}$, prove that $H^m(\Omega; Y) := W^{m,2}(\Omega; Y)$ is a Hilbert space.

We have seen in Theorem 7.16 that $W^{1,1}((a, b); \mathbb{R}^M)$ coincides with $AC((a, b); \mathbb{R}^M)$. In the infinite-dimensional case this is no longer true, unless Y is reflexive or, more generally, satisfies the RNP property (see Definition 8.35). Indeed, in the next theorem we will show that if a function has a weak derivative, then it is strongly differentiable and locally absolutely continuous. However, when the Banach space Y does not satisfy the RNP, then there exist absolutely continuous functions that are nowhere differentiable (see Exercises 8.30 and 8.32). Thus the inclusion

$$W^{1,1}((a, b); Y) \subseteq AC((a, b); Y)$$

is strict unless Y satisfies the RNP.

Theorem 8.55. Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a Banach space, and let $u \in L_{\text{loc}}^1(I; Y)$. Then u has a weak derivative $u' \in L_{\text{loc}}^1(I; Y)$ if and only if there exist a representative $\bar{u}$ of u and a function $v \in L_{\text{loc}}^1(I; Y)$ such that

$$(8.24) \quad \bar{u}(x) = \bar{u}(x_0) + \int_{x_0}^x v(s) \, ds$$

for all $x, x_0 \in I$. Moreover, $\bar{u}$ is (strongly) differentiable at $\mathcal{L}^1$ -a.e. $x \in I$, its strong derivative coincides with v , and $\bar{u}$ is locally absolutely continuous.

Lemma 8.56. Let $I \subseteq \mathbb{R}$ be an open interval, let $(Y, \|\cdot\|)$ be a Banach space, and let $u : I \rightarrow Y$ be a locally integrable function whose weak derivative is zero. Then u is equivalent to a constant function.

The proof is identical to the one of Lemma 7.4 and we omit it.

Proof of Theorem 8.55. Assume that $\bar{u}$ is given by the formula (8.24), then

$$\frac{\bar{u}(x+h) - \bar{u}(x)}{h} = \frac{1}{h} \int_x^{x+h} v(s) \, ds \rightarrow v(x)$$

as $h \rightarrow 0$ for $\mathcal{L}^1$ -a.e. $x \in I$ by Theorem 8.19. Hence, $\bar{u}$ is strongly differentiable for $\mathcal{L}^1$ -a.e. $x \in I$ with (strong) derivative given by v . Let's prove that v is also the weak derivative of $\bar{u}$. Given $\varphi \in C_c^1(I)$, let $(a, b) \Subset I$ be such that the support of φ is contained in (a, b) . Let

$$w(x) := \begin{cases} v(x) & \text{if } x \in (a, b), \\ 0 & \text{otherwise.} \end{cases}$$

Then $w \in L^1(\mathbb{R}; Y)$. Hence, by Theorem 8.20 (with the mollifier $\chi_{(-1,0)}$),

$$(8.25) \quad \frac{1}{h} \int_x^{x+h} w(s) \, ds \rightarrow w \text{ in } L^1(\mathbb{R}; Y) \text{ as } h \rightarrow 0^+.$$

On the other hand, for $x \in \text{supp } \varphi$ and $h > 0$ small,

$$(8.26) \quad \frac{\bar{u}(x+h) - \bar{u}(x)}{h} = \frac{1}{h} \int_x^{x+h} v(s) \, ds = \frac{1}{h} \int_x^{x+h} w(s) \, ds.$$

Let $0 < h < \text{dist}(\text{supp } \varphi, I \setminus \{a, b\})$. Then

$$(8.27) \quad \int_a^b \frac{\varphi(x) - \varphi(x-h)}{h} \bar{u}(x) \, dx = \frac{1}{h} \int_a^b \varphi(x) (\bar{u}(x) - \bar{u}(x+h)) \, dx.$$

Since,

$$\begin{aligned} & \left\| \int_a^b \varphi(x) \left(\frac{\bar{u}(x) - \bar{u}(x+h)}{h} - v(x) \right) \, dx \right\| \\ &= \left\| \int_a^b \varphi(x) \left(\frac{1}{h} \int_x^{x+h} w(s) \, ds - w(x) \right) \, dx \right\| \\ &\leq \|\varphi\|_\infty \int_a^b \left\| \frac{1}{h} \int_x^{x+h} w(s) \, ds - w(x) \right\| \, dx \rightarrow 0 \end{aligned}$$

by (8.25), it follows from the Lebesgue dominated convergence theorem, (8.26), and (8.27),

$$\begin{aligned} \int_I \varphi'(x) \bar{u}(x) \, dx &= \int_a^b \varphi'(x) \bar{u}(x) \, dx \\ &= \lim_{h \rightarrow 0^+} \int_a^b \frac{\varphi(x-h) - \varphi(x)}{-h} \bar{u}(x) \, dx \\ &= - \lim_{h \rightarrow 0^+} \int_a^b \varphi(x) \frac{\bar{u}(x+h) - \bar{u}(x)}{h} \, dx = - \int_a^b \varphi(x) v(x) \, dx. \end{aligned}$$

Together with Exercise 8.51 this shows that v is the weak derivative of $\bar{u}$.

Conversely, assume that v is the weak derivative of u and define $w(x) = u(x_0) + \int_{x_0}^x v(s) ds$. Then by the first part of the proof, v is the weak derivative of w . Hence, for every $\varphi \in C_c^1(I)$,

$$\int_I \varphi'(x)u(x) dx = - \int_a^b \varphi(x)v(x) dx = - \int_a^b \varphi'(x)w(x) dx,$$

and so

$$\int_I \varphi'(x)(u(x) - w(x)) dx = 0$$

for all $\varphi \in C_c^1(I)$. It follows by Lemma 8.56 that $u - w$ is constant. Since $u(x_0) - w(x_0) = 0$, it follows that $u = w$. $\square$

As a consequence of the following theorem we can now characterize elements in the space $W^{1,p}(\Omega; Y)$.

Theorem 8.57. *Let $\Omega \subseteq \mathbb{R}$ be an open set, let Y be a Banach space, and let $1 \leq p < \infty$. A function $u : I \rightarrow Y$ belongs to $W^{1,p}(\Omega; Y)$ if and only if it admits an absolutely continuous representative $\bar{u} : \Omega \rightarrow Y$ which is differentiable $\mathcal{L}^1$ -a.e. in Ω and such that $\bar{u}$ and its classical derivative $\bar{u}'$ belong to $L^p(\Omega; Y)$.*

Proof. Exercise. $\square$

Exercise 8.58. State and prove the analogous result for $W^{m,p}(\Omega; Y)$.

Next we discuss evolution triples. Let $(Y, \|\cdot\|_Y)$ be a Banach space embedded in a Hilbert space $(H, (\cdot, \cdot)_H)$. Assume that Y is dense in H . We claim that

$$(8.28) \quad H' \hookrightarrow Y'.$$

To see this, given $h \in H$, consider the linear functional $T_h : Y \rightarrow \mathbb{R}$ defined by $T_h(y) := (h, y)_H$, $y \in Y$. Then

$$|T_h(y)| = |(h, y)_H| \leq \|h\|_H \|y\|_H \leq c \|h\|_H \|y\|_Y.$$

Hence, $T_h \in Y'$, with $\|T_h\|_{Y'} \leq c \|h\|_H$. Note that

$$(8.29) \quad (h, y)_H = \langle T_h, y \rangle_{Y', Y}$$

for all $y \in Y$. The mapping $h \in H \mapsto T_h$ is linear and continuous. It is also injective, since if $T_h = 0$, then $(h, y)_H = 0$ for all $y \in Y$ and so, by the density of Y in H , we have that $h = 0$. The claim (8.28) follows by identifying h with T_h . In particular, we can rewrite (8.29) as

$$(8.30) \quad (h, y)_H = \langle h, y \rangle_{Y', Y}$$

for all $h \in H$ and $y \in Y$.

By identifying H with its dual, we have that

$$(8.31) \quad Y \hookrightarrow H \cong H' \hookrightarrow Y'.$$

This is called an *evolution triple*.

Remark 8.59. If $h \in Y \hookrightarrow H$, then we identify h with T_h and so in Y' the norm of h is

$$\|T_h\|_{Y'} = \sup_{y \in Y \setminus \{0\}, \|y\|_Y \leq 1} |(h, y)_H| \leq c\|h\|_H \leq c'\|h\|_Y.$$

In particular, if $Y = H^1(\Omega)$ and $H = L^2(\Omega)$, then given $u \in H^1(\Omega)$, we have that u is identified with $T_u \in (H^1(\Omega))'$, where

$$\|T_u\|_{(H^1(\Omega))'} = \sup_{v \in H^1(\Omega) \setminus \{0\}, \|v\|_{H^1(\Omega)} \leq 1} \left| \int_{\Omega} uv \, dx \right| \leq \|u\|_{L^2(\Omega)} < \|u\|_{H^1(\Omega)}$$

provided u is not constant on each connected component of Ω .

The following theorem strengthens Theorem 8.57.

Theorem 8.60. *Let $I \subset \mathbb{R}$ be an open bounded interval, let $(Y, \|\cdot\|_Y)$ be a Banach space embedded in a Hilbert space $(H, (\cdot, \cdot)_H)$, with Y dense in H , and let $1 \leq p < \infty$. If $u \in L^p(I; Y)$ has a weak derivative $u' \in L^{p'}(I; Y')$, then u has a continuous representative $\bar{u} \in C(\bar{I}; H)$ with*

$$\|\bar{u}\|_{C(\bar{I}; H)} \leq c(\|u\|_{L^p(I; Y)} + \|u'\|_{L^{p'}(I; Y')})$$

for some constant $c > 0$ independent of u . Moreover, the function $x \in \bar{I} \mapsto \|u(x)\|_H^2$ is absolutely continuous, with

$$\frac{d}{dx} \|u(x)\|_H^2 = 2u'(x)(u(x)) = 2\langle u'(x), u(x) \rangle_{Y', Y}$$

for $\mathcal{L}^1$ -a.e. $x \in I$.

We begin with a preliminary lemma.

Lemma 8.61. *Let $I \subseteq \mathbb{R}$ be an open interval, let $(H, (\cdot, \cdot)_H)$ be a Hilbert space, and let $u, v \in C^1(I; H)$. Then the function*

$$g(x) := (u(x), v(x))_H$$

is of class $C^1(I)$ and

$$g'(x) = (u'(x), v(x))_H + (u(x), v'(x))_H.$$

Proof. Let $x \in I$ and $h \neq 0$ small. Then using the bilinearity of the inner product,

$$\begin{aligned} \frac{g(x+h) - g(h)}{h} &= \frac{(u(x+h), v(x+h))_H - (u(x), v(x))_H}{h} \\ &= \frac{(u(x+h) - u(x) + u(x), v(x+h))_H - (u(x), v(x))_H}{h} \\ &= \left(\frac{u(x+h) - u(x)}{h}, v(x+h) \right)_H + \left(u(x), \frac{v(x+h) - v(x)}{h} \right)_H. \end{aligned}$$

Letting $h \rightarrow 0$ and using the continuity of the inner product gives the desired result. $\square$

We now turn to the proof of Theorem 8.60.

Proof. Let $I = (a, b)$. As in Exercise 7.24 we can extend u to $J := (2a - b, 2b - a)$. Let $u_\varepsilon := u * \varphi_\varepsilon$, where φ_ε is a standard mollifier, with $\varphi \in C_c^\infty(\mathbb{R})$ and $0 < \varepsilon < b - a$. Reasoning as in the proof of Theorem 7.26 and using Theorem 8.21 we have that $u_\varepsilon \in C_c^\infty(\mathbb{R}; Y)$, $u_\varepsilon \rightarrow u$ in $L^p(I; Y)$ and $u'_\varepsilon \rightarrow u'$ in $L^p(I; Y')$. Since $u_\varepsilon \in C_c^\infty(\mathbb{R}; Y)$ and $Y \hookrightarrow H$, we have that $u_\varepsilon \in C_c^\infty(\mathbb{R}; H)$ and so by the previous lemma, the fundamental theorem of calculus, and property (8.30), for $x, x_0 \in I$,

$$\begin{aligned} \|u_\varepsilon(x)\|_H^2 &= \|u_\varepsilon(x_0)\|_H^2 + \int_{x_0}^x \frac{d}{ds} \|u_\varepsilon(s)\|_H^2 ds \\ (8.32) \quad &= \|u_\varepsilon(x_0)\|_H^2 + 2 \int_{x_0}^x (u'_\varepsilon(s), u_\varepsilon(s))_H ds \\ &= \|u_\varepsilon(x_0)\|_H^2 + 2 \int_{x_0}^x \langle u'_\varepsilon(s), u_\varepsilon(s) \rangle_{Y', Y} ds. \end{aligned}$$

By Hölder's inequality

$$\begin{aligned} \|u_\varepsilon(x)\|_H^2 &\leq \|u_\varepsilon(x_0)\|_H^2 + 2 \int_I \|u'_\varepsilon(s)\|_{Y'} \|u_\varepsilon(s)\|_Y ds \\ &\leq \|u_\varepsilon(x_0)\|_H^2 + 2 \|u'_\varepsilon\|_{L^{p'}(I; Y')} \|u_\varepsilon\|_{L^p(I; Y)}. \end{aligned}$$

In turn,

$$\begin{aligned} \|u_\varepsilon(x) - u_\delta(x)\|_H^2 &\leq \|u_\varepsilon(x_0) - u_\delta(x_0)\|_H^2 \\ &\quad + 2 \|u'_\varepsilon - u'_\delta\|_{L^{p'}(I; Y')} \|u_\varepsilon - u_\delta\|_{L^p(I; Y)}. \end{aligned}$$

Since by Theorems 8.20 and 8.21, $u_\varepsilon(x) \rightarrow u(x)$ for every Lebesgue point of u , $u_\varepsilon \rightarrow u$ in $L^p(I; Y)$ and $u'_\varepsilon \rightarrow u'$ in $L^p(I; Y')$, letting x_0 be a Lebesgue point of u , it follows from the previous inequality that

$$\sup_{x \in I} \|u_\varepsilon(x) - u_\delta(x)\|_H^2 \rightarrow 0$$

as $\varepsilon, \delta \rightarrow 0^+$. Hence, $\{u_\varepsilon\}_\varepsilon$ is a Cauchy sequence in $C(\bar{I}; H)$ and so it converges uniformly to a function $w \in C(\bar{I}; H)$. On the other hand, since $u_\varepsilon(x) \rightarrow u(x)$ for every Lebesgue point of u , we have that $u(x) = w(x)$ for $\mathcal{L}^1$ -a.e. $x \in I$.

Letting $\varepsilon \rightarrow 0^+$ in (8.32) gives

$$\|w(x)\|_H^2 = \|u(x_0)\|_H^2 + 2 \int_{x_0}^x \langle u'(s), u(s) \rangle_{Y', Y} ds$$

for all $x, x_0 \in I$. In turn,

$$\|u(x)\|_H^2 = \|u(x_0)\|_H^2 + 2 \int_{x_0}^x \langle u'(s), u(s) \rangle_{Y', Y} ds$$

for $\mathcal{L}^1$ -a.e. $x \in I$. This implies that the function $x \in [a, b] \mapsto \|u(x)\|_H^2$ is absolutely continuous and concludes the proof. $\square$

We conclude this section with an important compactness theorem.

Theorem 8.62 (Aubin–Lions–Simon). *Let $I \subset \mathbb{R}$ be an open bounded interval, let $(Y_0, \|\cdot\|_{Y_0})$, $(Y_1, \|\cdot\|_{Y_1})$, and let $(Y_2, \|\cdot\|_{Y_2})$ be Banach spaces with $Y_0 \hookrightarrow Y_1 \hookrightarrow Y_2$. Assume that the embedding $Y_0 \hookrightarrow Y_1$ is compact. Let $1 \leq p < \infty$, $1 \leq q \leq \infty$, and let $\mathcal{V}$ be the Banach space of all functions $u \in L^p(I; Y_0)$ whose distributional derivative u' belongs to $L^q(I; Y_2)$ endowed with the norm*

$$(8.33) \quad \|u\|_{\mathcal{V}} := \|u\|_{L^p(I; Y_0)} + \|u'\|_{L^q(I; Y_2)}.$$

Then the embedding $\mathcal{V} \hookrightarrow L^p(I; Y_1)$ is compact.

Proof. Let $\{u_n\}_n$ be a bounded sequence in $\mathcal{V}$. Using (8.33) and the fact that $q \geq 1$ and I is bounded, by Hölder's inequality we obtain that

$$(8.34) \quad \{u_n\}_n \text{ is bounded in } L^p(I; Y_0),$$

$$(8.35) \quad \{u'_n\}_n \text{ is bounded in } L^1(I; Y_2).$$

In view of Corollary 8.26, it suffices to show that $\|\tau_h(u_n) - u_n\|_{L^p(I^h; Y_2)} \rightarrow 0$ as $h \rightarrow 0^+$ uniformly for $n \in \mathbb{N}$. Since $u_n \in W^{1,1}(I; Y_2)$, by Theorem 8.55, for every $0 < h < \mathcal{L}^1(I)$ and $x \in I^h$,

$$u_n(x+h) - u_n(x) = \int_x^{x+h} u'_n(s) ds.$$

Hence,

$$\|\tau_h(u_n)(x) - u(x)\|_{Y_2} \leq \int_x^{x+h} \|u'_n(s)\|_{Y_2} ds.$$

If $q \geq p$, raising both sides to power p and using Hölder's inequality gives

$$\|\tau_h(u_n)(x) - u(x)\|_{Y_2}^p \leq h^{p-1} \int_x^{x+h} \|u'_n(s)\|_{Y_2}^p ds.$$

Integrating in x over I^h and using Fubini's theorem yields

$$\begin{aligned} \int_{I^h} \|\tau_h(u_n)(x) - u(x)\|_{Y_2}^p dx &\leq h^{p-1} \int_{I^h} \int_x^{x+h} \|u'_n(s)\|_{Y_2}^p ds dx \\ &\leq h^p \int_{\inf I+h}^{\sup I} \|u'_n(s)\|_{Y_2}^p ds. \end{aligned}$$

It follows that

$$\int_{I^h} \|\tau_h(u_n)(x) - u(x)\|^p dx \leq Mh^p,$$

where $M := \sup_n \|u'_n\|_{L^p(I; Y_2)}$. This shows that $\|\tau_h(u_n) - u_n\|_{L^p(I^h; Y_2)} \rightarrow 0$ as $h \rightarrow 0^+$ uniformly for $n \in \mathbb{N}$ and completes the proof in the case $q \geq p$. The case $q < p$ is left as an exercise. (Hint: Use Young's inequality, Theorem 10.48.) $\square$

Exercise 8.63. Assume that all the hypotheses of the previous theorem are satisfied with $p = \infty$. Prove that the embedding $\mathcal{V} \hookrightarrow C(\bar{I}; Y_1)$ is compact.

Part 2

Functions of Several Variables

Change of Variables and the Divergence Theorem

Prospective Grad Students, III: “Will your qualifying exams procedure utterly destroy my dignity and sense of self-respect?”

—Jorge Cham, www.phdcomics.com

In this chapter we extend some of the concepts and results of Chapter 3 to functions of several variables, including the change of variables formulas, the fundamental theorem of calculus, and integration by parts. We will prove the area formula and the divergence theorem.

9.1. Directional Derivatives and Differentiability

Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be normed spaces, let $E \subseteq X$, let $u : E \rightarrow Y$, and let $x_0 \in E$. Given a direction $v \in X \setminus \{0\}$, let L_v be the line through x_0 in the direction v , that is, $L_v := \{x \in X : x = x_0 + tv, t \in \mathbb{R}\}$, and assume that x_0 is an accumulation point of the set $E \cap L_v$. The *directional derivative* of u at x_0 in the direction v is defined as

$$\frac{\partial u}{\partial v}(x_0) := \lim_{t \rightarrow 0} \frac{u(x_0 + tv) - u(x_0)}{t},$$

provided the limit exists in Y . If $Y = \mathbb{R}$, then we allow the limit to exist in $[-\infty, \infty]$.

When $X = \mathbb{R}^N$ and $v = e_i$, where e_i is a vector of the canonical basis, the directional derivative $\frac{\partial u}{\partial e_i}(x_0)$, if it exists, is called the *partial derivative* of u with respect to x_i and is denoted $\frac{\partial u}{\partial x_i}(x_0)$ or $\partial_i u(x_0)$. We recall that the

standard, or canonical, orthonormal basis of $\mathbb{R}^N$ is given by the unit vectors $e_1 := (1, 0, \dots, 0)$, $e_2 := (0, 1, 0, \dots, 0)$, $\dots$, $e_N := (0, \dots, 0, 1)$.

Definition 9.1. Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be normed spaces, let $E \subseteq X$, let $u : E \rightarrow Y$, and let $x_0 \in E$ be an accumulation point of E . The function u is *differentiable at x_0* if there exists a continuous linear function $T : X \rightarrow Y$ (depending on u and x_0) such that

$$(9.1) \quad \lim_{x \rightarrow x_0} \frac{u(x) - u(x_0) - T(x - x_0)}{\|x - x_0\|_X} = 0,$$

provided the limit exists. The function T , if it exists, is called the *differential* of u at x_0 and is denoted $du(x_0)$ or du_{x_0} .

Exercise 9.2. Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be normed spaces, let $E \subseteq X$, let $u : E \rightarrow Y$, and let $x_0 \in E$ be an accumulation point of E . Prove that if u is differentiable at x_0 , then u is continuous at x_0 and for every direction v such that x_0 is an accumulation point of the set $E \cap L_v$ (if any), there exists $\frac{\partial u}{\partial v}(x_0) = du_{x_0}(v)$.

In particular, if $Y = \mathbb{R}$ and $u : E \rightarrow \mathbb{R}$ is differentiable at an interior point of E , then u is continuous at x_0 , all partial and directional derivatives exist at x_0 and for every direction v ,

$$(9.2) \quad \frac{\partial u}{\partial v}(x_0) = \sum_{i=1}^N \frac{\partial u}{\partial x_i}(x_0) v_i.$$

The following exercise shows that these conditions are not enough to guarantee differentiability.

Exercise 9.3. Let $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$u(x, y) := \begin{cases} x & \text{if } y = x^2, x \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Prove that u is continuous at $(0, 0)$, all partial and directional derivatives exist at $(0, 0)$ and that (9.2) holds but that u is not differentiable at $(0, 0)$.

Exercise 9.4. Let $E \subseteq \mathbb{R}^N$, let $u : E \rightarrow \mathbb{R}$, let $x_0 \in E$, and let $i \in \{1, \dots, N\}$. Assume that there exists $r > 0$ such that $B(x_0, r) \subseteq E$ and for all $j \neq i$ and for all $x \in B(x_0, r)$ the partial derivative $\partial_j u$ exists in $\mathbb{R}$ at x and is continuous at x_0 . Assume also that $\partial_i u(x_0)$ exists in $\mathbb{R}$. Prove that u is differentiable at x_0 . Hint: Use the mean value theorem on segments parallel to the axes.

To define higher-order derivatives, for simplicity, we restrict ourselves to the case $X = \mathbb{R}^N$ and $Y = \mathbb{R}$, which is all we will need in this book. Let $E \subseteq \mathbb{R}^N$, let $u : E \rightarrow \mathbb{R}$ and let $x_0 \in E$. Let $i \in \{1, \dots, N\}$ and assume that there exists in $\mathbb{R}$ the partial derivative $\partial_i u(x)$ for all $x \in E$. If $j \in \{1, \dots, N\}$

and x_0 is an accumulation point of $E \cap L_j$, where L_j is the line through x_0 in the direction e_j , then we can consider the partial derivative of the function $\partial_i u$ with respect to x_j , that is,

$$\frac{\partial}{\partial x_j} \left(\frac{\partial u}{\partial x_i} \right) (x_0) =: \frac{\partial^2 u}{\partial x_j \partial x_i} (x_0).$$

Inductively, one can define higher-order partial derivatives $\frac{\partial^m u}{\partial x_{i_m} \cdots \partial x_{i_1}}$.

Note that in general the order in which we take derivatives is important.

Exercise 9.5. Let $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$u(x, y) := \begin{cases} \frac{x^3 y - x y^3}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Prove that $\frac{\partial^2 u}{\partial x \partial y}(0, 0) \neq \frac{\partial^2 u}{\partial y \partial x}(0, 0)$.

The Schwartz theorem shows that if the second-order mixed partial derivatives are continuous, then they coincide.

Theorem 9.6 (Schwartz). *Let $E \subseteq \mathbb{R}^N$, let $u : E \rightarrow \mathbb{R}$, let $x_0 \in E$, and let $i, j \in \{1, \dots, N\}$. Assume that there exists $r > 0$ such that $B(x_0, r) \subseteq E$ and for all $x \in B(x_0, r)$, the partial derivatives $\frac{\partial u}{\partial x_i}(x)$, $\frac{\partial u}{\partial x_j}(x)$, and $\frac{\partial^2 u}{\partial x_j \partial x_i}(x)$ exist in $\mathbb{R}$. Assume also that $\frac{\partial^2 u}{\partial x_j \partial x_i}$ is continuous at x_0 . Then there exists $\frac{\partial^2 u}{\partial x_i \partial x_j}(x_0)$ and*

$$\frac{\partial^2 u}{\partial x_i \partial x_j}(x_0) = \frac{\partial^2 u}{\partial x_j \partial x_i}(x_0).$$

Exercise 9.7. Let $f : ((-r, r) \setminus \{0\}) \times ((-r, r) \setminus \{0\}) \rightarrow \mathbb{R}$. Assume that the double limit $\lim_{(s,t) \rightarrow (0,0)} f(s, t)$ exists in $\mathbb{R}$ and that the limit $\lim_{t \rightarrow 0} f(s, t)$ exists in $\mathbb{R}$ for all $s \in (-r, r) \setminus \{0\}$. Prove that the iterated limit $\lim_{s \rightarrow 0} \lim_{t \rightarrow 0} f(s, t)$ exists and

$$\lim_{s \rightarrow 0} \lim_{t \rightarrow 0} f(s, t) = \lim_{(s,t) \rightarrow (0,0)} f(s, t).$$

Exercise 9.8. Prove the Schwartz theorem. Hint: For $N = 2$ take the function f in the previous exercise to be an appropriate second-order difference quotient and apply repeatedly the mean value theorem on segments parallel to the axes.

In view of Schwartz's theorem, for smooth functions the order in which we take partial derivatives is not important, and thus we can use multi-indices.

Given $N \in \mathbb{N}$, for a *multi-index* $\alpha = (\alpha_1, \dots, \alpha_N) \in \mathbb{N}_0^N$ we set

$$\frac{\partial^{|\alpha|}}{\partial x^\alpha} := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_N^{\alpha_N}}, \quad |\alpha| := \alpha_1 + \cdots + \alpha_N,$$

where $x = (x_1, \dots, x_N) \in \mathbb{R}^N$. For simplicity we will often write ∂^α for $\frac{\partial^{|\alpha|}}{\partial x^\alpha}$ and ∂_i^n for $\frac{\partial^n}{\partial x_i^n}$ when the variables of differentiation are clear. Given two multi-indices $\alpha = (\alpha_1, \dots, \alpha_N)$ and $\beta = (\beta_1, \dots, \beta_N)$, we write $\alpha \leq \beta$ if $\alpha_i \leq \beta_i$ for all $i = 1, \dots, N$. Also $\alpha! := \alpha_1! \dots \alpha_N!$ and

$$(9.3) \quad \binom{\alpha}{\beta} := \frac{\alpha!}{\beta!(\alpha - \beta)!}.$$

Given $x \in \mathbb{R}^N$ and $\alpha \in \mathbb{N}_0^N$ we write $x^\alpha := x_1^{\alpha_1} \dots x_N^{\alpha_N}$.

Given an open set $\Omega \subseteq \mathbb{R}^N$, for every nonnegative integer $m \in \mathbb{N}_0$ we denote by $C^m(\Omega)$ the vector space of all functions that are continuous together with their partial derivatives up to order m . We set $C^\infty(\Omega) := \bigcap_{m=0}^\infty C^m(\Omega)$ and we define $C_c^m(\Omega)$ and $C_c^\infty(\Omega)$ as the subspaces of $C^m(\Omega)$ and $C^\infty(\Omega)$, respectively, consisting of all functions that have compact support in Ω .

In the following chapters we will often use Taylor's formula.

Theorem 9.9 (Taylor's formula with integral remainder). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $u \in C^m(\Omega)$, $m \in \mathbb{N}$, and let $x_0 \in \Omega$. Then for every $x \in B(x_0, r) \subseteq \Omega$,*

$$\begin{aligned} u(x) = & u(x_0) + \sum_{1 \leq |\alpha| \leq m} \frac{1}{\alpha!} \partial^\alpha u(x_0) (x - x_0)^\alpha \\ & + \sum_{|\alpha|=m} \frac{m}{\alpha!} (x - x_0)^\alpha \int_0^1 (1-t)^{m-1} (\partial^\alpha u(tx + (1-t)x_0) - \partial^\alpha u(x_0)) dt. \end{aligned}$$

Exercise 9.10. Prove the previous theorem. Hint: Use the one-dimensional function $g(t) := u(tx_0 + (1-t)x)$, $0 \leq t \leq 1$.

Exercise 9.11. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $u \in C^m(\Omega)$, $m \in \mathbb{N}$, and let $x_0 \in \Omega$. Prove that for every direction $v \in \mathbb{R}^N \setminus \{0\}$,

$$(9.4) \quad \frac{\partial^n u}{\partial v^n}(x_0) = \sum_{|\alpha|=n} \frac{1}{\alpha!} \partial^\alpha u(x_0) v^\alpha,$$

where $\frac{\partial^n u}{\partial v^n}$ stands for the repeated directional derivative of u in the direction v .

For more information on differentiability of functions of several variables, we refer the reader to the books of Bartle [20] and Fleming [80].

9.2. Lipschitz Continuous Functions

In this section we prove two important theorems for Lipschitz continuous functions. We recall that if $E \subseteq \mathbb{R}^N$ and $u : E \rightarrow \mathbb{R}$, then u is *Lipschitz*

continuous if

$$\text{Lip}(u; E) = \sup_{x, y \in E, x \neq y} \frac{|u(x) - u(y)|}{\|x - y\|} < \infty.$$

We usually write $\text{Lip } u$ in place of $\text{Lip}(u; E)$, whenever the underlying set is clear.

The next exercise shows that a Lipschitz continuous function can always be extended to $\mathbb{R}^N$ with the same Lipschitz constant.

Exercise 9.12. Let $E \subseteq \mathbb{R}^N$ and let $u : E \rightarrow \mathbb{R}$ be a Lipschitz continuous function. Define

$$v(x) := \inf\{u(y) + \text{Lip}(u; E)\|x - y\| : y \in E\}, \quad x \in \mathbb{R}^N.$$

- (i) Prove that v is Lipschitz continuous with Lipschitz constant $\text{Lip}(u; E)$.
- (ii) Prove that $v(x) = u(x)$ for all $x \in E$.
- (iii) Prove that there exists a function $w : \mathbb{R}^N \rightarrow \mathbb{R}$ with Lipschitz constant $\text{Lip}(u; E)$ such that $w(x) = u(x)$ for all $x \in E$ and

$$\inf_{\mathbb{R}^N} w = \inf_E u, \quad \sup_{\mathbb{R}^N} w = \sup_E u.$$

Exercise 9.13. Let $u : \mathbb{R}^N \rightarrow \mathbb{R}$ be a Lipschitz continuous function. Let $D \subseteq S^{N-1}$ be dense in S^{N-1} and let $x_0 \in \mathbb{R}^N$ be such that there exist all partial derivatives $\frac{\partial u}{\partial x_i}(x_0)$, $i = 1, \dots, N$, all directional derivatives $\frac{\partial u}{\partial v}(x_0)$ for $v \in D$, and

$$\frac{\partial u}{\partial v}(x_0) = \sum_{i=1}^N \frac{\partial u}{\partial x_i}(x_0) v_i$$

for all $v = (v_1, \dots, v_N) \in D$. Prove that u is differentiable at x_0 .

We recall that S^{N-1} is the *unit sphere* $\partial B(0, 1)$. Compare the previous exercise with Exercise 9.3.

Theorem 9.14 (Rademacher). *Let $u : \mathbb{R}^N \rightarrow \mathbb{R}$ be a Lipschitz continuous function. Then u is differentiable at $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$.*

Proof. If $N = 1$, then u is absolutely continuous and the result follows from Lebesgue's differentiation theorem (Theorem 1.18) and Proposition 3.9. Thus, assume that $N \geq 2$. For every $v \in S^{N-1}$ let

$$E_v := \left\{ x \in \mathbb{R}^N : \text{there exists } \frac{\partial u}{\partial v}(x) \in \mathbb{R} \right\}$$

and let H_v be the hyperplane orthogonal to v . For every $x_0 \in H_v$, the function $g(t) := u(x_0 + tv)$, $t \in \mathbb{R}$, is Lipschitz continuous and thus, by the result for $N = 1$, g is differentiable for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$. Hence, $\frac{\partial u}{\partial v}(x_0 + tv)$

exists in $\mathbb{R}$ for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$. This shows that $x_0 + tv \in E_v$ for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$. Since the set E_v is a Borel set (why?), it follows by Tonelli's theorem that $\mathcal{L}^N(\mathbb{R}^N \setminus E_v) = 0$. Next consider a countable set $D \subset S^{N-1}$ dense in S^{N-1} . By what we just proved, for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ there exist $\frac{\partial u}{\partial v}(x)$ for all $v \in D$ and $\frac{\partial u}{\partial x_i}(x_0)$ for all $i = 1, \dots, N$.

We claim that

$$(9.5) \quad \frac{\partial u}{\partial v}(x_0) = \sum_{i=1}^N \frac{\partial u}{\partial x_i}(x_0) v_i$$

for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ and for all $v \in D$. To see this, fix $v \in D$ and let $\varphi \in C_c^1(\mathbb{R}^N)$. Since u is Lipschitz continuous, by the Lebesgue dominated convergence theorem,

$$\begin{aligned} \lim_{t \rightarrow 0^+} \int_{\mathbb{R}^N} \frac{u(x + tv) - u(x)}{t} \varphi(x) dx &= \int_{\mathbb{R}^N} \frac{\partial u}{\partial v}(x) \varphi(x) dx, \\ \lim_{t \rightarrow 0^+} \int_{\mathbb{R}^N} u(x) \frac{\varphi(x - tv) - \varphi(x)}{t} dx &= - \int_{\mathbb{R}^N} \frac{\partial \varphi}{\partial v}(x) u(x) dx. \end{aligned}$$

By using the change of variables $y = x + tv$ on the left-hand side of the first equality, we see that the two left-hand sides coincide. Hence,

$$\int_{\mathbb{R}^N} \frac{\partial u}{\partial v}(x) \varphi(x) dx = - \int_{\mathbb{R}^N} \frac{\partial \varphi}{\partial v}(x) u(x) dx.$$

Similarly,

$$\int_{\mathbb{R}^N} \sum_{i=1}^N \frac{\partial u}{\partial x_i}(x) v_i \varphi(x) dx = - \int_{\mathbb{R}^N} \sum_{i=1}^N \frac{\partial \varphi}{\partial x_i}(x) v_i u(x) dx.$$

Since the two right-hand sides coincide, we obtain that

$$\int_{\mathbb{R}^N} \frac{\partial u}{\partial v}(x) \varphi(x) dx = \int_{\mathbb{R}^N} \sum_{i=1}^N \frac{\partial u}{\partial x_i}(x) v_i \varphi(x) dx$$

for all $\varphi \in C_c^1(\mathbb{R}^N)$. Given the arbitrariness of $\varphi \in C_c^1(\mathbb{R}^N)$, it follows that (9.5) holds for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ (why?).

We are now in a position to apply Exercise 9.13 to obtain the desired result. $\square$

Exercise 9.15. Let $E \subseteq \mathbb{R}^N$ and let $u : E \rightarrow \mathbb{R}$ be a Lipschitz continuous function. Prove that for every set $F \subseteq \mathbb{R}$ with $\mathcal{L}^1(F) = 0$, $\nabla u(x) = 0$ for $\mathcal{L}^N$ -a.e. $x \in u^{-1}(F)$. (Hint: Use Lemma 3.60 and Tonelli's theorem).

As a corollary of Rademacher's theorem we can show the following result.

Theorem 9.16 (Stepanoff). *Every function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ is differentiable at $\mathcal{L}^N$ -a.e. x in the (possibly empty) set*

$$(9.6) \quad E_u := \left\{ x \in \mathbb{R}^N : \limsup_{y \rightarrow x} \frac{|u(y) - u(x)|}{\|y - x\|} < \infty \right\}.$$

Proof. Assume that E_u is nonempty. Let $\mathcal{F}$ be the family of all balls $B(x, r)$, with $x \in \mathbb{Q}^N$ and $r \in \mathbb{Q} \cap (0, \infty)$, such that $u : B(x, r) \rightarrow \mathbb{R}$ is bounded. Write $\mathcal{F} = \{B_n : n \in \mathbb{N}\}$. It follows from the definition of E_u that $E_u \subseteq \bigcup_{n=1}^{\infty} B_n$. For every $x \in B_n$ define

$$\begin{aligned} v_n(x) &:= \sup\{v(x) : v : B_n \rightarrow \mathbb{R} \text{ is Lipschitz continuous,} \\ &\quad \text{Lip } v \leq n, \text{ and } v \leq u|_{B_n}\}, \\ w_n(x) &:= \inf\{w(x) : w : B_n \rightarrow \mathbb{R} \text{ is Lipschitz continuous,} \\ &\quad \text{Lip } w \leq n, \text{ and } w \geq u|_{B_n}\}, \end{aligned}$$

where we recall that $u|_{B_n}$ denotes the restriction of u to B_n . Then (exercise) $\text{Lip } v_n \leq n$, $\text{Lip } w_n \leq n$, and $v_n \leq u|_{B_n} \leq w_n$ in B_n .

By Exercise 9.12 and Rademacher's theorem, v_n and w_n are differentiable at $\mathcal{L}^N$ -a.e. $x \in B_n$. Define

$$F_n := \{x \in B_n : v_n \text{ and } w_n \text{ are differentiable at } x\}.$$

Then $\mathcal{L}^N(B_n \setminus F_n) = 0$. By Exercise 9.15 applied to $v_n - w_n$, we have that $\nabla v_n(x) = \nabla w_n(x)$ for $\mathcal{L}^N$ -a.e. $x \in B_n$ such that $v_n(x) = w_n(x)$. Set

$$G_n := \{x \in F_n : v_n(x) = w_n(x) \text{ and } \nabla v_n(x) \neq \nabla w_n(x)\}.$$

Then $\mathcal{L}^N(G_n) = 0$. Hence, the set

$$(9.7) \quad E := \bigcup_{n=1}^{\infty} (B_n \setminus F_n) \cup G_n$$

has Lebesgue measure zero.

We claim that u is differentiable at all $x \in E_u \setminus E$. To see this, fix $x \in E_u \setminus E$. By definition of E_u , there exist $r > 0$ and $L > 0$ such that

$$(9.8) \quad |u(y) - u(x)| \leq L\|y - x\|$$

for all $y \in B(x, r)$. By the density of the rational numbers, we can find $n \geq L$ such that $B_n \subseteq B(x, r)$ and $x \in B_n$. Since $x \notin E$, it follows that $x \in F_n$. By definition of v_n and w_n and by (9.8) we have that

$$u(x) - n\|y - x\| \leq v_n(y) \leq u(y) \leq w_n(y) \leq u(x) + n\|y - x\|$$

for all $y \in B_n$. In particular, for $y = x$ we obtain $v_n(x) = u(x) = w_n(x)$. Moreover, since $x \notin G_n$, we have that $\nabla v_n(x) = \nabla w_n(x) =: A$. Hence,

using the previous inequality, together with the facts that v_n and w_n are differentiable at x and $v_n(x) = u(x) = w_n(x)$, it follows that

$$\begin{aligned} 0 &= \lim_{y \rightarrow x} \frac{v_n(y) - u(x) - A(y - x)}{\|y - x\|} \leq \liminf_{y \rightarrow x} \frac{u(y) - u(x) - A(y - x)}{\|y - x\|} \\ &\leq \limsup_{y \rightarrow x} \frac{u(y) - u(x) - A(y - x)}{\|y - x\|} \leq \lim_{y \rightarrow x} \frac{w_n(y) - u(x) - A(y - x)}{\|y - x\|} = 0, \end{aligned}$$

which shows that u is differentiable at x . This concludes the proof. $\square$

Although it was not needed in the previous theorem, it turns out that the set E_u given in (9.6) is Lebesgue measurable and so are the partial derivatives of u when restricted to E_u .

Theorem 9.17. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u : \Omega \rightarrow \mathbb{R}$ be a function. Then the set*

$$E := \{x \in \Omega : u \text{ is differentiable at } x\}$$

is Lebesgue measurable and so are the partial derivatives

$$\partial_i u : E \rightarrow \mathbb{R}, i = 1, \dots, N.$$

Proof. Let E_u be the set given in (9.6) with $\mathbb{R}^N$ replaced by Ω and for $k \in \mathbb{N}$ write

$$E_k := \{x \in \Omega : |u(x + h) - u(x)| \leq k\|h\| \text{ for all } h \text{ with } \|h\| < 1/k, \\ \text{dist}(x, \partial\Omega) > 1/k\}.$$

Then $E_u = \bigcup_k E_k$. Note that the sets E_k are relatively closed in the open set $\{x \in \Omega : \text{dist}(x, \partial\Omega) > 1/k\}$, and so E_u is a Borel set. In turn, by Stepanoff's theorem the set E is Lebesgue measurable. Fix k and subdivide E_k into sets $E_{k,\ell}$ of diameter $1/k$. The function $u : E_{k,\ell} \rightarrow \mathbb{R}$ is Lipschitz continuous and thus by Exercise 9.12 it can be extended to $\mathbb{R}^N$ to a Lipschitz continuous function $u_{k,\ell}$ with the same Lipschitz constant. For $i = 1, \dots, N$ and for $n \geq k$ consider the function $g_{n,k,\ell} : \mathbb{R}^N \rightarrow \mathbb{R}$ defined by

$$g_{n,k,\ell}(x) := \frac{u_{k,\ell}(x + e_i/n) - u_{k,\ell}(x)}{1/n}.$$

Since $u_{k,\ell}$ is Lipschitz continuous, it follows that $g_{n,k,\ell}$ is Lebesgue measurable, and so are the functions

$$\partial_i^+ u_{k,\ell} := \limsup_{n \rightarrow \infty} g_{n,k,\ell}, \quad \partial_i^- u_{k,\ell} := \liminf_{n \rightarrow \infty} g_{n,k,\ell}.$$

In turn the set $F_{k,\ell} := \{x \in \mathbb{R}^N : \partial_i^+ u_{k,\ell} = \partial_i^- u_{k,\ell}\}$ is Lebesgue measurable and so is the function $\partial_i u_{k,\ell} : F_{k,\ell} \rightarrow \mathbb{R}$. It remains to show that $\partial_i u_{k,\ell} = \partial_i u$

$\mathcal{L}^N$ -a.e. in $E_{k,\ell} \cap E$. Let x be a point of density one of $E_{k,\ell} \cap E$ (see Corollary B.121). Then for $y_n \in E_{k,\ell}$,

$$\begin{aligned}
 & |u_{k,\ell}(x + \tfrac{1}{n}e_i) - u_{k,\ell}(x) - \tfrac{1}{n}\partial_i u(x)| \\
 &= |u_{k,\ell}(x + \tfrac{1}{n}e_i) - u(x) - \tfrac{1}{n}\partial_i u(x)| \\
 (9.9) \quad &\leq |u_{k,\ell}(x + \tfrac{1}{n}e_i) - u_{k,\ell}(y_n)| + |u(y_n) - u(x) - \nabla u(x) \cdot (y_n - x)| \\
 &\quad + |\nabla u(x) \cdot (x + \tfrac{1}{n}e_i - y_n)| \\
 &\leq (k + \|\nabla u(x)\|)\|x + \tfrac{1}{n}e_i - y_n\| \\
 &\quad + |u(y_n) - u(x) - \nabla u(x) \cdot (y_n - x)|,
 \end{aligned}$$

where we have used the facts that $u(y_n) = u_{k,\ell}(y_n)$ and $u(x) = u_{k,\ell}(x)$ and that $u_{k,\ell}$ is Lipschitz continuous with constant less than or equal to k . Fix $0 < \varepsilon < 1$. Since u is differentiable at x , there is $\delta_1 > 0$ such that

$$(9.10) \quad |u(y) - u(x) - \nabla u(x) \cdot (y - x)| \leq \varepsilon \|y - x\|$$

for all y with $\|y - x\| < \delta_1$. On the other hand, since x is a point of density one of $E_{k,\ell} \cap E$, there exists $\delta_2 < 1/(2k)$ such that

$$\mathcal{L}^N(E_{k,\ell} \cap E \cap B(x, r)) > (1 - (\tfrac{\varepsilon}{2})^N) \mathcal{L}^N(B(x, r))$$

for all $r < \delta_2$. This implies that

$$(9.11) \quad \mathcal{L}^N(B(x, r) \setminus (E_{k,\ell} \cap E)) < (\tfrac{\varepsilon}{2})^N \mathcal{L}^N(B(x, r)) = \alpha_N(\tfrac{\varepsilon r}{2})^N.$$

Let n be so large that $1/n < \min\{\delta_1, \delta_2\}/2$ and find $y_n \in E_{k,\ell} \cap E$ such that $\|x + e_i/n - y_n\| < \varepsilon/n$. Note that if no such y_n exists, then $B(x + e_i/n, \varepsilon/n) \subseteq \mathbb{R}^N \setminus (E_{k,\ell} \cap E)$. But since $B(x + e_i/n, \varepsilon/n) \subset B(x, 2/n)$ and $1/n < \delta_2/2$, by (9.11) we would have that

$$B(x + \tfrac{1}{n}e_i, \tfrac{\varepsilon}{n}) \subseteq B(x, \tfrac{2}{n}) \setminus (E_{k,\ell} \cap E),$$

so that

$$\alpha_N(\tfrac{\varepsilon}{n})^N \leq \mathcal{L}^N(B(x, \tfrac{2}{n}) \setminus (E_{k,\ell} \cap E)) < \alpha_N(\tfrac{\varepsilon}{n})^N,$$

which is a contradiction. Hence, y_n exists and so from (9.9) and (9.10) we get

$$\begin{aligned}
 & |u_{k,\ell}(x + \tfrac{1}{n}e_i) - u_{k,\ell}(x) - \tfrac{1}{n}\partial_i u(x)| \\
 &\leq (k + \|\nabla u(x)\|)\|x + \tfrac{1}{n}e_i - y_n\| + \varepsilon\|y_n - x\| \\
 &\leq (k + \|\nabla u(x)\|)\tfrac{\varepsilon}{n} + \varepsilon\|y_n - x - \tfrac{1}{n}e_i\| + \tfrac{\varepsilon}{n} \\
 &\leq (k + \|\nabla u(x)\| + 2)\tfrac{\varepsilon}{n},
 \end{aligned}$$

and so

$$\left| \frac{u_{k,\ell}(x + e_i/n) - u_{k,\ell}(x)}{1/n} - \partial_i u(x) \right| \leq (k + \|\nabla u(x)\| + 2)\varepsilon.$$

Letting $n \rightarrow \infty$ and then $\varepsilon \rightarrow 0$ shows that $\partial_i u_{k,\ell}(x) = \partial_i u(x)$. Since this is true for every k and ℓ , we have that $\partial_i u : E \rightarrow \mathbb{R}$ is Lebesgue measurable. $\square$

In contrast to the previous theorem, the following exercise shows that the set where one of the partial derivatives $\partial_i u$ exists may not be Lebesgue measurable.

Exercise 9.18. Let $G \subset \mathbb{R}$ be a non-Lebesgue measurable set and let

$$u(x, y) := \chi_{\mathbb{Q}}(x)\chi_G(y), \quad (x, y) \in \mathbb{R}^2.$$

- (i) Prove that u is Lebesgue measurable.
- (ii) Prove that the set $\{(x, y) \in \mathbb{R}^2 : \partial_1 u \text{ exists at } (x, y)\}$ is not Lebesgue measurable.

In what follows, given $x = (x_1, \dots, x_N) \in \mathbb{R}^N$, we denote by x' the $(N-1)$ -dimensional vector obtained by removing the N th component from x and with an abuse of notation we write

$$(9.12) \quad x = (x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R}.$$

The next exercise provides a sufficient condition to recover measurability.

Exercise 9.19. Let $R = I_1 \times \dots \times I_N \subseteq \mathbb{R}^N$, where I_i are open intervals, let $u : R \rightarrow \mathbb{R}$ be a Lebesgue measurable function such that for $\mathcal{L}^{N-1}$ -a.e. $x' \in I_1 \times \dots \times I_{N-1}$,

$$(9.13) \quad \min \left\{ \limsup_{t \rightarrow x_N^+} u(x', t), \limsup_{t \rightarrow x_N^-} u(x', t) \right\} \leq u(x', x_N) \\ \leq \max \left\{ \liminf_{t \rightarrow x_N^+} u(x', t), \liminf_{t \rightarrow x_N^-} u(x', t) \right\}$$

for every $x_N \in I_N$, and let $E_N := \{x \in R : \text{there exists } \partial_N u(x) \text{ in } \mathbb{R}\}$.

- (i) For $x \in R$ define

$$\partial_N^{\text{rat}} u(x) := \lim_{r \rightarrow 0^+, r \in \mathbb{Q}} \frac{u(x + r e_N) - u(x)}{r}$$

whenever the limit exists and let $E_N^{\text{rat}} := \{x \in R : \text{there exists } \partial_N^{\text{rat}} u(x) \text{ in } \mathbb{R}\}$. Prove that E_N^{rat} is Lebesgue measurable and $\partial_N^{\text{rat}} u : E_N^{\text{rat}} \rightarrow \mathbb{R}$ is a Lebesgue measurable function.

- (ii) Let $x = (x', x_N) \in E_N^{\text{rat}}$ be such that (9.13) holds for x' and let $h_n \rightarrow 0$ with $h_n \neq 0$ for every $n \in \mathbb{N}$. Prove that there exist two sequences $\{r_n\}_n$ and $\{s_n\}_n$ of rational numbers such that $\frac{r_n}{h_n} \rightarrow 1$, $\frac{s_n}{h_n} \rightarrow 1$ and

$$u(x + r_n e_N) > u(x + h_n e_N) - |h_n|/n, \\ u(x + s_n e_N) < u(x + h_n e_N) + |h_n|/n$$

for every n . Deduce that

$$\partial_N^{\text{rat}} u(x) = \lim_{n \rightarrow \infty} \frac{u(x + h_n e_N) - u(x)}{h_n}$$

and that there exists $\partial_N u(x) = \partial_N^{\text{rat}} u(x)$.

- (iii) Prove that the set E_N is Lebesgue measurable and $\partial_N u : E_N \rightarrow \mathbb{R}$ is a Lebesgue measurable function.

Observe that (9.13) holds if for $\mathcal{L}^{N-1}$ -a.e. $x' \in I_1 \times \cdots \times I_{N-1}$, the function $u(x', \cdot)$ is right continuous or left continuous at x_N or is monotone in a neighborhood of x_N .

Exercise 9.20. Let $R = I_1 \times \cdots \times I_N \subseteq \mathbb{R}^N$, where I_i are open intervals, let $u : R \rightarrow \mathbb{R}$ be a Lebesgue measurable function such that for $\mathcal{L}^{N-1}$ -a.e. $x' \in I_1 \times \cdots \times I_{N-1}$ the function $u(x', \cdot)$ has bounded pointwise variation in I_N . Prove that there exists a Lebesgue measurable function $v : R \rightarrow \mathbb{R}$ such that $v(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in R$, the set $E_N := \{x \in R : \text{there exists } \partial_N v(x) \text{ in } \mathbb{R}\}$ is Lebesgue measurable and $\partial_N v : E_N \rightarrow \mathbb{R}$ is a Lebesgue measurable function.

9.3. The Area Formula: The C^1 Case

In this section we prove a change of variables formula for surface integrals. We begin by studying linear transformations. Given $N \in \mathbb{N}$ we denote by $\|\cdot\|_N$ the Euclidean norm in $\mathbb{R}^N$.

Given $k, N \in \mathbb{N}$ and a linear function $L : \mathbb{R}^k \rightarrow \mathbb{R}^N$, the *adjoint* of L is the linear function $L^t : \mathbb{R}^N \rightarrow \mathbb{R}^k$ such that

$$(9.14) \quad y \cdot L^t(x) = L(y) \cdot x$$

for all $y \in \mathbb{R}^k$ and $x \in \mathbb{R}^N$. The matrix representing L^t is simply the transpose of the matrix representing L .

Given $k, N \in \mathbb{N}$ and a linear function $L : \mathbb{R}^k \rightarrow \mathbb{R}^N$, we say that L is *orthogonal* if $L(y_1) \cdot L(y_2) = y_1 \cdot y_2$ for all $y_1, y_2 \in \mathbb{R}^k$.

We recall that $\mathcal{H}_o^k$ stands for the k -dimensional Hausdorff outer measure and $\mathcal{H}^k$ is the k -dimensional Hausdorff measure obtained by restricting $\mathcal{H}_o^k$ to the σ -algebra of all $\mathcal{H}_o^k$ -measurable sets (see the Carathéodory Theorem B.13). We refer to Section C.7 for more details on the Hausdorff measure.

Exercise 9.21. Given $k, N \in \mathbb{N}$ with $1 \leq k \leq N$ and a linear function $L : \mathbb{R}^k \rightarrow \mathbb{R}^N$, prove that if $E \subseteq \mathbb{R}^k$ is Lebesgue measurable, then $L(E)$ is $\mathcal{H}_o^k$ -measurable.

Remark 9.22. An orthogonal function $L : \mathbb{R}^k \rightarrow \mathbb{R}^N$ preserves inner products and distances, since

$$\begin{aligned} \|L(y_1) - L(y_2)\|_N &= \sqrt{L(y_1 - y_2) \cdot L(y_1 - y_2)} \\ &= \sqrt{(y_1 - y_2) \cdot (y_1 - y_2)} = \|y_1 - y_2\|_k. \end{aligned}$$

Thus, L is Lipschitz continuous with Lipschitz constant one and it is injective with $L^{-1} : L(\mathbb{R}^k) \rightarrow \mathbb{R}^k$ Lipschitz continuous with Lipschitz constant one. If $E \subseteq \mathbb{R}^k$ is Lebesgue measurable, then by the previous exercise $L(E)$ is $\mathcal{H}_o^k$ -measurable. Hence, by Propositions C.44 and C.50,

$$\mathcal{H}^k(L(E)) \leq \mathcal{H}^k(E) = \mathcal{L}^k(E).$$

On the other hand, since $L^{-1} : L(\mathbb{R}^k) \rightarrow \mathbb{R}^k$ is Lipschitz continuous with Lipschitz constant one,

$$\mathcal{L}^k(E) = \mathcal{H}^k(E) = \mathcal{H}^k(L^{-1}(L(E))) \leq \mathcal{H}^k(L(E))$$

and thus

$$(9.15) \quad \mathcal{H}^k(L(E)) = \mathcal{L}^k(E).$$

Observe also that $L^t \circ L = I_k$.

Definition 9.23. Given $N \in \mathbb{N}$ and a linear function $L : \mathbb{R}^N \rightarrow \mathbb{R}^N$, we say that L is

- (i) a *rotation* if it is orthogonal and the corresponding matrix has positive determinant,
- (ii) *symmetric* if $L = L^t$,
- (iii) *diagonal* if the corresponding matrix is diagonal,
- (iv) *positive definite* if $L(x) \cdot x > 0$ for all $x \in \mathbb{R}^N \setminus \{0\}$.

Theorem 9.24 (Decomposition). *Let $1 \leq k \leq N$ and let $L : \mathbb{R}^k \rightarrow \mathbb{R}^N$ be a linear function. Assume that the corresponding matrix has rank k . Then there exist an orthogonal linear function $P : \mathbb{R}^k \rightarrow \mathbb{R}^k$, a positive definite, diagonal linear function $D : \mathbb{R}^k \rightarrow \mathbb{R}^k$, and an orthogonal linear function $Q : \mathbb{R}^k \rightarrow \mathbb{R}^N$ such that*

$$L = Q \circ D \circ P.$$

Proof. We claim that the function $L^t \circ L : \mathbb{R}^k \rightarrow \mathbb{R}^k$ is symmetric and positive definite. Indeed, by (9.14),

$$\begin{aligned} (L^t \circ L)(y_1) \cdot y_2 &= (L^t(L(y_1))) \cdot y_2 \\ &= L(y_1) \cdot L(y_2) = y_1 \cdot (L^t(L(y_2))) = y_1 \cdot (L^t \circ L)(y_2) \end{aligned}$$

and

$$(L^t \circ L)(y) \cdot y = (L^t(L(y))) \cdot y = L(y) \cdot L(y) = \|L(y)\|_N^2 > 0$$

for all $y \in \mathbb{R}^k \setminus \{0\}$, since the matrix corresponding to L has rank k . It follows that the eigenvalues μ_i of $L^t \circ L$ are all positive and that there exists an orthonormal basis $\{b_1, \dots, b_k\}$ of eigenvectors. Let $\lambda_i := \sqrt{\mu_i}$. Then $(L^t \circ L)(b_i) = \lambda_i^2 b_i$ for all $i = 1, \dots, k$. Let $\{e_1, \dots, e_k\}$ be the canonical basis in $\mathbb{R}^k$.

Given any two vector spaces of dimension k each with a given basis, there is a linear function between these two vector spaces that maps one basis into the other. Let $P : \mathbb{R}^k \rightarrow \mathbb{R}^k$ be the linear function that maps $\{b_1, \dots, b_k\}$ into $\{e_1, \dots, e_k\}$, let $D : \mathbb{R}^k \rightarrow \mathbb{R}^k$ be the linear function that maps $\{e_1, \dots, e_k\}$ into $\{\lambda_1 e_1, \dots, \lambda_k e_k\}$, and let $Q : \mathbb{R}^k \rightarrow \mathbb{R}^N$ be the linear function that maps $\{\lambda_1 e_1, \dots, \lambda_k e_k\}$ into $\{L(b_1), \dots, L(b_k)\}$. Note that since J_L has rank k , the vector space $L(\mathbb{R}^k) \subseteq \mathbb{R}^N$ has dimension k and $\{L(b_1), \dots, L(b_k)\}$ is a basis in $L(\mathbb{R}^k)$.

Since $P(b_i) = e_i$, the function P is orthogonal. To verify that Q is orthogonal, note that

$$Q(e_i) \cdot Q(e_j) = \frac{1}{\lambda_i \lambda_j} L(b_i) \cdot L(b_j) = \frac{1}{\lambda_i \lambda_j} b_i \cdot (L^t(L(b_j))) = \frac{\lambda_j^2}{\lambda_i \lambda_j} b_i \cdot b_j = \delta_{i,j}.$$

It remains to show that $L = Q \circ D \circ P$. We have

$$(Q \circ D \circ P)(b_i) = (Q \circ D)(e_i) = Q(\lambda_i e_i) = L(b_i),$$

and thus the result follows by linearity. $\square$

Given $E \subseteq \mathbb{R}^k$ and $\Psi : E \rightarrow \mathbb{R}^N$, assume that Ψ is differentiable at some point $y \in E$. We recall that the *Jacobian matrix* of Ψ at y is the $N \times k$ matrix given by

$$(9.16) \quad J_\Psi(y) = \nabla \Psi(y) := \begin{pmatrix} \nabla \Psi_1(y) \\ \vdots \\ \nabla \Psi_N(y) \end{pmatrix}$$

and the *Jacobian* of Ψ at y is the number

$$(9.17) \quad |||J_\Psi(y)||| := \sqrt{\det(J_\Psi^t(y) J_\Psi(y))},$$

where $J_\Psi^t(y)$ is the transpose of $J_\Psi(y)$. Note that when $k = N$,

$$(9.18) \quad |||J_\Psi(y)||| = |\det J_\Psi(y)|.$$

Theorem 9.25 (Cauchy–Binet formula). *Let $1 \leq k \leq N$, let A be a $N \times k$ matrix, and let B be a $k \times N$ matrix. Then*

$$(9.19) \quad \det BA = \sum_{\alpha \in \Lambda_{N,k}} \det(a_{\alpha_i, j})_{i,j=1}^k \det(b_{i, \alpha_j})_{i,j=1}^k,$$

where

$$\Lambda_{N,k} := \{\alpha \in \mathbb{N}^k : 1 \leq \alpha_1 < \alpha_2 < \dots < \alpha_k \leq N\}.$$

In particular,

$$\det A^t A = \sum_{\alpha \in \Lambda_{N,k}} (\det(a_{\alpha_i,j})_{i,j=1}^k)^2.$$

Proof. We only give a sketch of the proof. Using the fact that for square matrices the determinant of the product of two matrices is given by the product of the determinants of the two matrices, we have

$$\begin{aligned} \det(I + AB) &= \det \begin{pmatrix} I & A \\ 0 & I \end{pmatrix} \begin{pmatrix} I & -A \\ B & I \end{pmatrix} = \det \begin{pmatrix} I & A \\ 0 & I \end{pmatrix} \det \begin{pmatrix} I & -A \\ B & I \end{pmatrix} \\ &= \det \begin{pmatrix} I & -A \\ B & I \end{pmatrix} \begin{pmatrix} I & A \\ 0 & I \end{pmatrix} = \det(I + BA), \end{aligned}$$

where I and 0 are identity matrices and zero matrices of whatever dimension is needed to make sense of the previous expressions. The identity

$$\det(I_N + AB) = \det(I_k + BA)$$

is called the *Sylvester determinant identity*. If we now let $t \in \mathbb{R}$ and rescale everything, we obtain

$$\det(tI_N + AB) = t^{N-k} \det(tI_k + BA).$$

Since both the left-hand and right-hand sides are polynomials of degree N in t , by equating the coefficients of the t^{N-k} terms we get (9.19). $\square$

We are now ready to prove the area formula for injective linear functions.

Theorem 9.26 (Area formula for linear functions). *Let $1 \leq k \leq N$ and let $L : \mathbb{R}^k \rightarrow \mathbb{R}^N$ be a linear function. Assume that J_L has rank k . Then for every Lebesgue measurable $E \subseteq \mathbb{R}^k$, $L(E)$ is $\mathcal{H}_o^k$ -measurable and*

$$\mathcal{H}^k(L(E)) = \int_E |||J_L||| dy = |||J_L||| \mathcal{L}^k(E).$$

Proof. Step 1: Assume first that $k = N$ and that L is a positive definite diagonal linear function $D : \mathbb{R}^k \rightarrow \mathbb{R}^k$. Then

$$D(y) = (\lambda_1 y_1, \dots, \lambda_k y_k).$$

Consider a rectangle $R = I_1 \times \dots \times I_k$. Then by Fubini's theorem and (9.18),

$$\mathcal{L}^k(D(R)) = \lambda_1 \cdots \lambda_k \mathcal{L}^k(R) = \det J_D \mathcal{L}^k(R) = |||J_D||| \mathcal{L}^k(R).$$

If $V \subseteq \mathbb{R}^k$ is an open set, then we can write V as a countable union of disjoint rectangles R_n , and since D is injective, the sets $D(R_n)$ are also disjoint and so $\mathcal{L}^k(D(V)) = |||J_D||| \mathcal{L}^k(V)$. By approximating Lebesgue measurable sets with open sets we obtain that $\mathcal{L}^k(D(E)) = |||J_D||| \mathcal{L}^k(E)$ for every Lebesgue measurable set $E \subseteq \mathbb{R}^k$.

Step 2: Given now $L : \mathbb{R}^k \rightarrow \mathbb{R}^N$ such that J_L has rank k , by Theorem 9.24 there exist an orthogonal linear function $P : \mathbb{R}^k \rightarrow \mathbb{R}^k$, a positive definite

diagonal linear function $D : \mathbb{R}^k \rightarrow \mathbb{R}^k$, and an orthogonal linear function $Q : \mathbb{R}^k \rightarrow \mathbb{R}^N$ such that $L = Q \circ D \circ P$. For every Lebesgue measurable set $E \subseteq \mathbb{R}^k$, by Exercise 9.21, Proposition C.50, (9.15), and Step 1,

$$(9.20) \quad \begin{aligned} \mathcal{H}^k(L(E)) &= \mathcal{H}^k(Q(D(P(E)))) = \mathcal{H}^k(D(P(E))) \\ &= \mathcal{L}^k(D(P(E))) = |||J_D||| \mathcal{L}^k(P(E)) = |||J_D||| \mathcal{L}^k(E). \end{aligned}$$

Now since $Q^t \circ Q = I_k$,

$$\begin{aligned} L^t \circ L &= (Q \circ D \circ P)^t \circ (Q \circ D \circ P) = P^t \circ D^t \circ Q^t \circ Q \circ D \circ P \\ &= P^t \circ D^t \circ D \circ P, \end{aligned}$$

and so

$$\det(L^t \circ L) = \det P^t \det(D^t \circ D) \det P.$$

Since $\det P^t = \det P = \pm 1$ we have $\det(L^t \circ L) = \det(D^t \circ D)$. Hence,

$$|||J_L||| = \sqrt{\det(L^t \circ L)} = \sqrt{\det(D^t \circ D)} = |||J_D|||,$$

which concludes the proof of the formula. $\square$

Next we extend the area formula to C^1 functions.

Theorem 9.27 (Area formula, C^1 case). *Let $1 \leq k \leq N$, let $V \subseteq \mathbb{R}^k$ be an open set and let $\Psi : V \rightarrow \mathbb{R}^N$ be a function of class C^1 . Let $E \subseteq V$ be a Lebesgue measurable set and assume that Ψ is injective in E . Then $\Psi(E)$ is $\mathcal{H}_o^k$ -measurable and*

$$\mathcal{H}^k(\Psi(E)) = \int_E |||J_\Psi(y)||| dy.$$

We divide the proof in a few lemmas.

Lemma 9.28. *Let $1 \leq k \leq N$, let $V \subseteq \mathbb{R}^k$ be an open set and let $\Psi : V \rightarrow \mathbb{R}^N$ be a function of class C^1 and let $y_0 \in V$. Assume that $J_\Psi(y_0)$ has rank k . Then for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$(1 - \varepsilon) \|L(y_1 - y_2)\|_N \leq \|\Psi(y_1) - \Psi(y_2)\|_N \leq (1 + \varepsilon) \|L(y_1 - y_2)\|_N$$

for all $y_1, y_2 \in B_k(y_0, \delta) \subseteq V$, where $L(y) := J_\Psi(y_0)y^t$, $y \in \mathbb{R}^k$.

Proof. Since $J_\Psi(y_0)$ has rank k there exists $c \in (0, 1)$ such that

$$(9.21) \quad \|L(y)\|_N \geq c \|y\|_k \quad \text{for all } y \in \mathbb{R}^k.$$

Since Ψ is of class C^1 there exists $\delta > 0$ such that

$$(9.22) \quad \|J_\Psi(y) - J_\Psi(y_0)\|_{N \times k} \leq c\varepsilon$$

for every $y \in B_k(y_0, \delta)$. Let $y_1, y_2 \in B_k(y_0, \delta)$. By the fundamental theorem of calculus applied to the function

$$h(t) := \Psi(y_1 t + (1 - t)y_2) - L(y_1 t + (1 - t)y_2)$$

we have

$$\Psi(y_1) - \Psi(y_2) - L(y_1 - y_2) = \int_0^1 (J_\Psi(y_1 t + (1-t)y_2) - J_\Psi(y_0))(y_1 - y_2)^t dt.$$

Hence, by (9.21) and (9.22),

$$\begin{aligned} \|\Psi(y_1) - \Psi(y_2)\|_N &\leq \|L(y_1 - y_2)\|_N + \|\Psi(y_1) - \Psi(y_2) - L(y_1 - y_2)\|_N \\ &\leq \|L(y_1 - y_2)\|_N + c\varepsilon\|y_1 - y_2\|_k \leq (1 + \varepsilon)\|L(y_1 - y_2)\|_N, \end{aligned}$$

while

$$\begin{aligned} \|\Psi(y_1) - \Psi(y_2)\|_N &\geq \|L(y_1 - y_2)\|_N - \|\Psi(y_1) - \Psi(y_2) - L(y_1 - y_2)\|_N \\ &\geq \|L(y_1 - y_2)\|_N - c\varepsilon\|y_1 - y_2\|_k \geq (1 - \varepsilon)\|L(y_1 - y_2)\|_N, \end{aligned}$$

which completes the proof. $\square$

Lemma 9.29. *Let $1 \leq k \leq N$, let $V \subseteq \mathbb{R}^k$ be an open set, let $\Psi : V \rightarrow \mathbb{R}^N$ be a function of class C^1 , and let $y_0 \in V$. Assume that $J_\Psi(y_0)$ has rank k . Then for every $0 < \varepsilon < 1$ there exists $\delta > 0$ such that for every Lebesgue measurable set $E \subseteq B_k(y_0, \delta) \subseteq V$, $\Psi(E)$ is $\mathcal{H}_o^k$ -measurable and*

$$(1 - \varepsilon)^{k+1} \int_E |||J_\Psi(y)||| dy \leq \mathcal{H}^k(\Psi(E)) \leq (1 + \varepsilon)^{k+1} \int_E |||J_\Psi(y)||| dy.$$

Proof. Let L be as in the previous lemma. Since $J_\Psi(y_0)$ has rank k , the linear function $L : \mathbb{R}^k \rightarrow \mathbb{R}^N$ is injective, and so there exists $L^{-1} : L(\mathbb{R}^k) \rightarrow \mathbb{R}^k$. Given $0 < \varepsilon < 1$, let $\delta > 0$ be so small that the conclusions of the previous lemma hold and also so that

$$(9.23) \quad (1 + \varepsilon)^{-1} |||J_\Psi(y)||| \leq |||J_\Psi(y_0)||| \leq (1 + \varepsilon) |||J_\Psi(y)|||$$

for all $y \in B_k(y_0, \delta)$, where we used the fact that Ψ is of class C^1 . Since by the previous lemma,

$$(9.24) \quad (1 - \varepsilon)\|L(y_1 - y_2)\|_N \leq \|\Psi(y_1) - \Psi(y_2)\|_N \leq (1 + \varepsilon)\|L(y_1 - y_2)\|_N$$

for all $y_1, y_2 \in B_k(y_0, \delta) \subseteq V$, taking $y_1 = L^{-1}(x_1)$ and $y_1 = L^{-1}(x_2)$ we get

$$\|(\Psi \circ L^{-1})(x_1) - (\Psi \circ L^{-1})(x_2)\|_N \leq (1 + \varepsilon)\|x_1 - x_2\|_N$$

for all $x_1, x_2 \in L(B_k(y_0, \delta))$. Thus, $\Psi \circ L^{-1}$ is Lipschitz continuous with Lipschitz constant less than or equal to $1 + \varepsilon$. It follows by Proposition C.44, the area formula for L , and (9.23),

$$\begin{aligned} (9.25) \quad \mathcal{H}_o^k(\Psi(E)) &= \mathcal{H}_o^k((\Psi \circ L^{-1})(L(E))) \leq (1 + \varepsilon)^k \mathcal{H}_o^k(L(E)) \\ &= (1 + \varepsilon)^k \int_E |||J_L||| dy \leq (1 + \varepsilon)^{k+1} \int_E |||J_\Psi(y)||| dy. \end{aligned}$$

Similarly, by (9.24) and the fact that L is injective it follows that Ψ is injective, and so taking $y_1 = \Psi^{-1}(x_1)$ and $y_1 = \Psi^{-1}(x_2)$ we get

$$\|(L \circ \Psi^{-1})(x_1) - (L \circ \Psi^{-1})(x_2)\|_N \leq (1 - \varepsilon)^{-1}\|x_1 - x_2\|_N$$

for all $x_1, x_2 \in \Psi(B_k(y_0, \delta))$. Thus, $L \circ \Psi^{-1}$ is Lipschitz continuous with Lipschitz constant less than or equal to $(1 - \varepsilon)^{-1}$. It follows by Proposition C.44, the area formula for L , and (9.23),

$$\begin{aligned} (1 + \varepsilon)^{-1} \int_E |||J_\Psi(y)||| dy &\leq \int_E |||J_L||| dy = \mathcal{H}_o^k(L(E)) \\ &= \mathcal{H}_o^k((L \circ \Psi^{-1})(\Psi(E))) \leq (1 - \varepsilon)^{-k} \mathcal{H}_o^k(\Psi(E)), \end{aligned}$$

which gives the other inequality since $(1 - \varepsilon) \leq (1 + \varepsilon)^{-1}$.

We leave as an exercise to prove that $\Psi(E)$ is $\mathcal{H}_o^k$ -measurable. $\square$

Next we prove the area formula under the additional assumption that J_Ψ has rank k everywhere in V .

Lemma 9.30. *Let $1 \leq k \leq N$, let $V \subseteq \mathbb{R}^k$ be an open set and let $\Psi : V \rightarrow \mathbb{R}^N$ be a function of class C^1 such that $J_\Psi(y)$ has rank k for every $y \in V$. Let $E \subseteq V$ be a Lebesgue measurable set and assume that Ψ is injective in E . Then*

$$\mathcal{H}^k(\Psi(E)) = \int_E |||J_\Psi(y)||| dy.$$

Proof. Fix $\varepsilon > 0$ and cover V with countably many balls $B_i \subseteq V$ such that for every $E \subseteq B_i$ the previous lemma applies. Given a Lebesgue measurable set $E \subseteq V$, define inductively, $E_1 := E \cap B_1$, $E_i := (E \cap B_i) \setminus \bigcup_{j=1}^{i-1} B_j$. Then the sets E_i are disjoint and their union is E . By the previous lemma applied to each E_i we get

$$(1 - \varepsilon)^{k+1} \int_{E_i} |||J_\Psi(y)||| dy \leq \mathcal{H}^k(\Psi(E_i)) \leq (1 + \varepsilon)^{k+1} \int_{E_i} |||J_\Psi(y)||| dy.$$

Summing over i and using the fact that Ψ is injective in E gives

$$(1 - \varepsilon)^{k+1} \int_E |||J_\Psi(y)||| dy \leq \mathcal{H}^k(\Psi(E)) \leq (1 + \varepsilon)^{k+1} \int_E |||J_\Psi(y)||| dy.$$

We now let $\varepsilon \rightarrow 0^+$. $\square$

To remove the condition that J_Ψ has rank k in V we use a variation of Sard's theorem (see Theorem 14.48).

Lemma 9.31. *Let $1 \leq k \leq N$, let $V \subseteq \mathbb{R}^k$ be an open set, and let $\Psi : V \rightarrow \mathbb{R}^N$ be a function of class C^1 . Let $\Sigma := \{y \in V : \Psi \text{ is differentiable at } y \text{ and } J_\Psi(y) \text{ has rank less than } k\}$. Then $\mathcal{H}^k(\Psi(\Sigma)) = 0$.*

Proof. Step 1: Assume that V is bounded and that J_Ψ is bounded in V , say, $\|J_\Psi(y)\| \leq M$ for all $y \in V$. For $\varepsilon > 0$ consider the function $\Psi_\varepsilon : V \rightarrow \mathbb{R}^N \times \mathbb{R}^k$ given by $\Psi_\varepsilon(y) := (\Psi(y), \varepsilon y)$. Then

$$J_{\Psi_\varepsilon}(y) = \begin{pmatrix} J_\Psi(y) \\ \varepsilon I_k \end{pmatrix},$$

and so $J_{\Psi_\varepsilon}(y)$ has rank k for every $y \in V$. Then by the Cauchy–Binet formula (see Theorem 9.25),

$$\begin{aligned} |||J_{\Psi_\varepsilon}(y)|||^2 &= \sum_{\alpha \in \Lambda_{N,k}} \left(\det \frac{\partial(\Psi_\varepsilon)_\alpha}{\partial y}(y) \right)^2 \\ &\leq |||J_\Psi(y)|||^2 + c(1 + M^2)\varepsilon^2 \end{aligned}$$

for some constant $c > 0$. In particular, if $y \in \Sigma$, then

$$(9.26) \quad |||J_{\Psi_\varepsilon}(y)|||^2 \leq c\varepsilon^2,$$

where as usual the constant c changes from line to line. Since $\Psi = \Pi \circ \Psi_\varepsilon$, where $\Pi : \mathbb{R}^N \times \mathbb{R}^d \rightarrow \mathbb{R}^N$ is the projection operator given by $\Pi(x, y) := x$ and since Π is Lipschitz continuous with Lipschitz constant one, it follows from Proposition C.44, the area formula, and (9.26),

$$\begin{aligned} \mathcal{H}^k(\Psi(\Sigma)) &= \mathcal{H}^k(\Pi(\Psi_\varepsilon(\Sigma))) \leq 1\mathcal{H}^k(\Psi_\varepsilon(\Sigma)) \\ &= \int_\Sigma |||J_{\Psi_\varepsilon}(y)||| dy \leq c\varepsilon \mathcal{L}^k(V). \end{aligned}$$

Letting $\varepsilon \rightarrow 0$ gives $\mathcal{H}^k(\Psi(\Sigma)) = 0$.

Step 2: The general case follows by writing V as an increasing sequence of open bounded sets V_n with $\overline{V_n} \subseteq V_{n+1} \subseteq V$ for all n and by applying Step 1 in each set V_n . $\square$

We now turn to the proof of Theorem 9.27.

Proof of Theorem 9.27. Let Σ be as in Lemma 9.31. Then $\mathcal{H}^k(\Psi(\Sigma)) = 0$, and so by Lemma 9.30, applied in the open set $V \setminus \Sigma$,

$$\mathcal{H}^k(\Psi(E)) = \mathcal{H}^k(\Psi(E \setminus \Sigma)) = \int_{E \setminus \Sigma} |||J_\Psi(y)||| dy = \int_E |||J_\Psi(y)||| dy,$$

where the last equality follows from the fact that $|||J_\Psi(y)||| = 0$ for $y \in \Sigma$. $\square$

Exercise 9.32. Let $1 \leq k \leq N$, let $V \subseteq \mathbb{R}^k$ be an open set and let $\Psi : V \rightarrow \mathbb{R}^N$ be a function of class C^1 . Let $E \subseteq V$ be a Lebesgue measurable set. Prove that

$$\int_{\mathbb{R}^N} \mathcal{H}^0(E \cap \Psi^{-1}(\{x\})) d\mathcal{H}^k(x) = \int_E |||J_\Psi(y)||| dy.$$

As a consequence of the area formula we have the following change of variables formula for surface integrals.

Theorem 9.33. Let $1 \leq k \leq N$, let $V \subseteq \mathbb{R}^k$ be an open set and let $\Psi : V \rightarrow \mathbb{R}^N$ be a function of class C^1 . Let $H \subseteq \Psi(V)$ be a Borel set and let

$u : H \rightarrow [-\infty, \infty]$ be a Borel function, which is either $\mathcal{H}^k$ integrable or has a sign. Assume that Ψ is injective in $\Psi^{-1}(H)$. Then

$$(9.27) \quad \int_H u(x) d\mathcal{H}^k(x) = \int_{\Psi^{-1}(H)} u(\Psi(y)) |||J_\Psi(y)||| dy.$$

Proof. Assume first that $u = \chi_G$, where $G \subseteq H$ is a Borel set. Then $\Psi^{-1}(G)$ is a Borel set and so by Theorem 9.27,

$$\begin{aligned} \int_H u(x) d\mathcal{H}^k(x) &= \mathcal{H}^k(G) = \mathcal{H}^k(\Psi(\Psi^{-1}(G))) \\ &= \int_{\Psi^{-1}(G)} |||J_\Psi(y)||| dy = \int_{\Psi^{-1}(H)} \chi_G(\Psi(y)) |||J_\Psi(y)||| dy. \end{aligned}$$

Next take u to be a simple function, $u = \sum_{i=1}^n c_i \chi_{G_i}$, where the Borel sets G_i are disjoint. Then by what we just proved and the linearity of integrals

$$\begin{aligned} \int_H u(x) d\mathcal{H}^k(x) &= \sum_{i=1}^n c_i \mathcal{H}^k(G_i) = \sum_{i=1}^n c_i \int_{\Psi^{-1}(G_i)} |||J_\Psi(y)||| dy \\ &= \int_{\Psi^{-1}(H)} u(\Psi(y)) |||J_\Psi(y)||| dy. \end{aligned}$$

For a nonnegative Borel function u , construct an increasing sequence of Borel simple functions converging pointwise to u and apply the Lebesgue monotone convergence theorem on both sides.

Finally, if the Borel function $u : H \rightarrow [-\infty, \infty]$ is $\mathcal{H}^k$ integrable, then as usual we can write $u = u^+ - u^-$, apply (9.27) to u^+ and u^- , and use the linearity of integrals to deduce (9.27) for u . $\square$

Exercise 9.34. Prove that the previous theorem continues to hold if we assume that Ψ is injective in $\Psi^{-1}(H) \setminus E$, where $\mathcal{L}^k(E) = 0$.

Since $\mathcal{H}^N = \mathcal{L}^N$ (see Proposition C.50) in the case $k = N$ we obtain the classical change of variables for Lebesgue integration.

Corollary 9.35 (Change of variables, the C^1 case). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $\Psi : \Omega \rightarrow \mathbb{R}^N$ be a function of class C^1 . Let $H \subseteq \Psi(\Omega)$ be a Borel set and let $u : H \rightarrow [-\infty, \infty]$ be a Borel function, which is either Lebesgue integrable or has a sign. Assume that Ψ is injective in $\Psi^{-1}(H)$. Then*

$$\int_H u(x) dx = \int_{\Psi^{-1}(H)} u(\Psi(y)) |\det J_\Psi(y)| dy.$$

Using the change of variables formula we can prove Brouwer's fixed point theorem [36].

Theorem 9.36 (Brouwer's fixed point theorem). *Let $K \subset \mathbb{R}^N$ be a non-empty compact convex set and let $\Psi : K \rightarrow K$ be a continuous transformation. Then there exists $x \in K$ such that $\Psi(x) = x$.*

We begin with a preliminary lemma.

Lemma 9.37. *There is no function $\Psi : \overline{B(0, 1)} \rightarrow S^{N-1}$ such that $\Psi(x) = x$ for all $x \in S^{N-1}$ and which is continuous together with all its partial derivatives*

Proof. Assume by contradiction that Ψ exists and for $t \in [0, 1]$ define

$$\Psi_t(x) := t\Psi(x) + (1 - t)x.$$

Then for every $x \in \overline{B(0, 1)}$,

$$\|\Psi_t(x)\| \leq t\|\Psi(x)\| + (1 - t)\|x\| \leq 1,$$

thus $\Psi_t : \overline{B(0, 1)} \rightarrow \overline{B(0, 1)}$. Moreover, $\Psi_t(x) = x$ for every $x \in S^{N-1}$.

Define $\Phi(x) := \Psi(x) - x$. Let $M > 0$ be such that $\|J_\Phi(x)\|_{N \times N} \leq M$ for all $x \in \overline{B(0, 1)}$. Then by the mean value theorem applied to each component, we obtain that Φ is Lipschitz continuous with Lipschitz constant at most $\sqrt{N}M$. We claim that Ψ_t is injective for every $0 < t < \min\{1, 1/(\sqrt{N}M)\} =: t_0$. Indeed, assume by contradiction that there exist $x_1, x_2 \in \overline{B(0, 1)}$ such that $\Psi_t(x_1) = \Psi_t(x_2)$ and $x_1 \neq x_2$. Since $\Psi_t(x) = x + t\Phi(x)$ it follows that

$$\|x_2 - x_1\| = \|t(\Phi(x_2) - \Phi(x_1))\| \leq t\sqrt{N}M\|x_2 - x_1\| < \|x_2 - x_1\|,$$

which is a contradiction. Hence, the claim holds.

Since $J_{\Psi_t} = I_N + tJ_\Phi$ and J_Φ is bounded in $\overline{B(0, 1)}$, by taking t_0 smaller, if necessary, we can assume that $\det J_{\Psi_t}(x) > 0$ for all $x \in \overline{B(0, 1)}$. It follows by the inverse function theorem that the set $U_t := \Psi_t(B(0, 1))$ is open for all $0 < t < t_0$.

We claim that $U_t = B(0, 1)$ for every $0 < t < t_0$. Indeed, assume that this is not the case. Since $U_t \subseteq \overline{B(0, 1)}$ by what we proved above, then $\partial U_t \subseteq \overline{B(0, 1)}$, and so if $U_t \neq B(0, 1)$, then there must exist $y_0 \in \partial U_t$ such that $y_0 \in B(0, 1)$. Let $y_n \in U_t$ be such that $y_n \rightarrow y_0$ and find $x_n \in B(0, 1)$ such that $y_n = \Psi_t(x_n)$. By compactness, up to a subsequence, we may assume that $x_n \rightarrow x_0 \in \overline{B(0, 1)}$ with $\Psi_t(x_0) = y_0$, by the continuity of Ψ_t . Note that x_0 cannot belong to $B(0, 1)$ as otherwise $y_0 = \Psi_t(x_0)$ would belong to $\Psi_t(B(0, 1)) = U_t$, so, necessarily $x_0 \in S^{N-1}$. But then $\Psi_t(x_0) = x_0$ and so $x_0 = y_0$, which is again a contradiction since $y_0 \in B(0, 1)$. This proves that $U_t = B(0, 1)$ for every $0 < t < t_0$.

For $t \in [0, 1]$ define the function

$$g(t) := \int_{B(0, 1)} \det J_{\Psi_t}(x) \, dx = \int_{B(0, 1)} \det(I_N + tJ_\Phi(x)) \, dx.$$

Since for every $0 < t < t_0$ the function $\Psi_t : B(0, 1) \rightarrow B(0, 1)$ is a bijection and $\det J_{\Psi_t} > 0$, it follows by Corollary 9.35 that $g(t) = \mathcal{L}^N(B(0, 1))$ for all $0 < t < t_0$. Since g is a polynomial of degree N , we have that $g(t) = \mathcal{L}^N(B(0, 1))$ for all $0 \leq t \leq 1$. In particular,

$$(9.28) \quad 0 < \mathcal{L}^N(B(0, 1)) = g(1) = \int_{B(0, 1)} \det J_{\Psi}(x) \, dx.$$

On the other hand, since by hypothesis $\|\Psi(x)\|^2 = \Psi(x) \cdot \Psi(x) = 1$ for every $x \in \overline{B(0, 1)}$, if $x \in B(0, 1)$ and $v \in \mathbb{R}^N$, then by replacing x with $x + sv$ differentiating with respect to s we get

$$(J_{\Psi}(x)v) \cdot \Psi(x) = 0,$$

which shows that the range of $J_{\Psi}(x)$ is orthogonal to the vector $\Psi(x)$. In turn, $J_{\Psi}(x)$ has rank less than or equal to $N - 1$ and so $\det J_{\Psi}(x) = 0$, which contradicts (9.28). This concludes the proof. $\square$

We now turn to the proof of Brouwer's fixed point theorem.

Proof of Theorem 9.36. Step 1: Assume first that $K = \overline{B(0, 1)}$ and that $\Psi : \overline{B(0, 1)} \rightarrow \overline{B(0, 1)}$ is continuous. Extend Ψ to a continuous function outside $\overline{B(0, 1)}$. Using standard mollifiers, we may construct a sequence $\{\Psi_n\}_n$ of functions in $C^1(\mathbb{R}^N; \mathbb{R}^N)$ such that $\Psi_n \rightarrow \Psi$ uniformly on compact sets. By selecting a subsequence, not relabeled, we may assume that $\|\Psi_n - \Psi\|_{C(\overline{B(0, 1)})} \leq 1/n$. Then for $x \in \overline{B(0, 1)}$,

$$\|\Psi_n(x)\| \leq \|\Psi(x)\| + \|\Psi_n(x) - \Psi(x)\| \leq 1 + 1/n,$$

and so the function $\Phi_n := \Psi_n/(1 + 1/n)$ maps $\overline{B(0, 1)}$ into $\overline{B(0, 1)}$. We claim that Φ_n has a fixed point. Indeed, if not, then $\Phi_n(x) \neq x$ for all $x \in \overline{B(0, 1)}$. Define $\Phi : \overline{B(0, 1)} \rightarrow S^{N-1}$ as follows. For each $x \in \overline{B(0, 1)}$ let $\Theta_n(x)$ be the intersection with the sphere S^{N-1} of the ray from $\Phi_n(x)$ to x , precisely¹,

$$\Theta_n(x) := x + F_n(x)(x - \Phi_n(x)),$$

¹To obtain F_n , we consider the line

$$s\Phi_n(x) + (1 - s)x, \quad s \in \mathbb{R},$$

through the distinct points $\Phi_n(x)$ and x and then find $s \in \mathbb{R}$ such that

$$\begin{aligned} 1 &= \|s\Phi_n(x) + (1 - s)x\|^2 \\ &= s^2\|x - \Phi_n(x)\|^2 + 2s(x \cdot \Phi_n(x) - \|x\|^2) + \|x\|^2. \end{aligned}$$

It suffices to solve for s .

where

$$F_n(x) := \frac{-x \cdot \Phi_n(x) + \|x\|^2}{\|x - \Phi_n(x)\|^2} + \frac{\sqrt{(x \cdot \Phi_n(x) - \|x\|^2)^2 + \|x - \Phi_n(x)\|^2(1 - \|x\|^2)}}{\|x - \Phi_n(x)\|^2}.$$

Then $\Theta_n : \overline{B(0,1)} \rightarrow S^{N-1}$ is continuous and is the identity on the unit sphere, which contradicts the previous lemma. Hence, the claim is true and thus there exists $x_n \in \overline{B(0,1)}$ such that $\Phi_n(x_n) = x_n$. By compactness we may assume that $x_n \rightarrow x_0 \in \overline{B(0,1)}$ and so since $\Psi_n(x_n) = (1 + 1/n)x_n$, letting $n \rightarrow \infty$ and using uniform convergence we obtain that $\Psi(x_0) = x_0$.

Step 2: Let $K = \overline{B(0,R)}$ for some $R > 0$ and let $\Psi : \overline{B(0,R)} \rightarrow \overline{B(0,R)}$ be a continuous transformation. To obtain a fixed point, it suffices to apply the previous step to the rescaled function $\Psi_R(x) := R^{-1}\Psi(Rx)$, $x \in \overline{B(0,1)}$.

Step 3: Let $K \subset \mathbb{R}^N$ be a nonempty compact convex set and let $\Psi : K \rightarrow K$ be a continuous transformation. Find $R > 0$ such that $K \subseteq \overline{B(0,R)}$ and for each $x \in \overline{B(0,R)}$ consider the continuous transformation $\Phi(x) := \Psi(\Pi(x))$, $x \in \overline{B(0,R)}$, where $\Pi : \mathbb{R}^N \rightarrow K$ is the projection onto the convex set K . Note that $\Phi(K) \subseteq K \subseteq \overline{B(0,R)}$, and so by the continuity of Π we have that $\Phi : \overline{B(0,R)} \rightarrow \overline{B(0,R)}$ is continuous. By the previous step there exists $x \in \overline{B(0,R)}$ such that $x = \Phi(x) = \Psi(\Pi(x))$. On the other hand, since $\Phi(K) \subseteq K$, we have that $x \in K$, and so $\Pi(x) = x$. Thus, the previous identity reduces to $x = \Psi(x)$ and the proof is completed. $\square$

Exercise 9.38. Let $\Psi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be of class C^2 and let $M_1, \dots, M_N$ be the cofactors of the first row of the Jacobian matrix J_Ψ .²

(i) Prove that $\sum_{i=1}^N \partial_i M_i$ can be written symbolically as

$$\det \begin{pmatrix} \nabla \\ \nabla \Psi_2 \\ \vdots \\ \nabla \Psi_N \end{pmatrix}.$$

(ii) Let $N = 2$. Using part (i), prove that $\partial_1 M_1 + \partial_2 M_2 = 0$.

²If $A = (a_{ij})_{i,j=1,\dots,N}$ is an $N \times N$ matrix, the *cofactor* of the entry a_{ij} is defined by

$$M_{ij} := (-1)^{i+j} \det C_{ij}, \quad i, j = 1, \dots, N,$$

where C_{ij} is the $(N-1) \times (N-1)$ matrix obtained from A by removing the i th row and the j th column.

(iii) Let $N > 2$ and prove that

$$\det \begin{pmatrix} \nabla \\ \nabla \Psi_2 \\ \vdots \\ \nabla \Psi_N \end{pmatrix} = \sum_{k=2}^N \det \begin{pmatrix} \nabla \\ \nabla \Psi_2 \\ \vdots \\ \nabla \Psi_N \end{pmatrix}_k,$$

where the subscript k means that the differential operator ∇ in the first row acts only on the k th row.

(iv) Prove that

$$\det \begin{pmatrix} \nabla \\ \nabla \Psi_2 \\ \vdots \\ \nabla \Psi_N \end{pmatrix}_k = 0$$

for all $k = 2, \dots, N$ and conclude that $\sum_{i=1}^N \partial_i M_i \equiv 0$.

Exercise 9.39. Let $\Psi \in C^2(\mathbb{R}^N; \mathbb{R}^N)$ be such that $\Psi(x) = x$ for $\|x\| \geq 1$ and let $u \in C_c^1(\mathbb{R}^N)$. Using the notation (E.2) in the appendix,

$$v(y) := \int_{-\infty}^{y_1} u(y'_1, t) dt, \quad y \in \mathbb{R}^N.$$

(i) Prove that

$$(u \circ \Psi) \det J_\Psi = \det \begin{pmatrix} \nabla(v \circ \Psi) \\ \nabla \Psi_2 \\ \vdots \\ \nabla \Psi_N \end{pmatrix}.$$

(ii) Prove that

$$(u \circ \Psi) \det J_\Psi = M_1 \partial_1(v \circ \Psi) + \dots + M_N \partial_N(v \circ \Psi),$$

where $M_1, \dots, M_N$ are the cofactors of the first row of the Jacobian matrix J_Ψ .

(iii) Let $r > 0$ be such that $u = 0$ outside the cube $Q(0, r)$. Prove that

$$\int_{Q(0, r)} M_i(x) \partial_i(v \circ \Psi)(x) dx = - \int_{Q(0, r)} (v \circ \Psi)(x) \partial_i M_i(x) dx$$

for all $i = 2, \dots, N$ and

$$\int_{Q(0, r)} M_1 \partial_1(v \circ \Psi) dx = - \int_{Q(0, r)} (v \circ \Psi) \partial_1 M_1 dx + \int_{Q(0, r)} u dx.$$

(iv) Prove that

$$(9.29) \quad \int_{\mathbb{R}^N} u(y) dy = \int_{\mathbb{R}^N} u(\Psi(x)) \det J_\Psi(x) dx.$$

Exercise 9.40. Using mollifiers (see Appendix C) prove that (9.29) continues to hold if $\Psi \in C^1(\mathbb{R}^N; \mathbb{R}^N)$ is the identity outside some ball and $u \in C_c(\mathbb{R}^N)$.

Exercise 9.41. Let $\Psi \in C^1(\mathbb{R}^N; \mathbb{R}^N)$ be the identity outside the unit ball. Prove that Ψ is onto.

Exercise 9.42. Use the previous exercises to give an alternative proof of the Brouwer fixed point theorem.

9.4. The Area Formula: The Differentiable Case

In this section we will prove that the area formula and the change of variables continue to hold for differentiable transformations. The proof relies on an important extension theorem due to Whitney [244].

Theorem 9.43 (Whitney extension). *Let $C \subseteq \mathbb{R}^N$ be a closed set and let $u : C \rightarrow \mathbb{R}$ and $w : C \rightarrow \mathbb{R}^N$ be two continuous functions. Assume that for every $x_0 \in C$ and every $\varepsilon > 0$ there exists $\delta = \delta(x_0, \varepsilon) > 0$ such that*

$$(9.30) \quad |u(x) - u(y) - w(y) \cdot (x - y)| \leq \varepsilon \|x - y\|$$

for all $x, y \in C \cap B(x_0, \delta)$. Then there exists a function $v \in C^1(\mathbb{R}^N)$ such that $v = u$ and $\nabla v = w$ in C .

Whitney's extension theorem is based on the following decomposition result, which turned out to be quite useful in other extension theorems (see Exercise 13.7; see also the paper of Jones [122] and the monograph of Stein [220]).

Theorem 9.44 (Whitney decomposition). *Given a nonempty closed set $C \subset \mathbb{R}^N$, there exists a countable family $\mathcal{F} = \{Q_n : n \in \mathbb{N}\}$ of closed cubes such that*

- (i) $\mathbb{R}^N \setminus C = \bigcup_n Q_n$,
- (ii) $Q_n^\circ \cap Q_m^\circ = \emptyset$ for all $n \neq m$,
- (iii) $\text{diam } Q_n \leq \text{dist}(Q_n, C) \leq 4 \text{diam } Q_n$,
- (iv) if Q_n and Q_m touch, then $\frac{1}{4} \text{diam } Q_n \leq \text{diam } Q_m \leq 4 \text{diam } Q_n$,
- (v) for every cube Q_n in $\mathcal{F}$ there are at most $(12)^N$ cubes in $\mathcal{F}$ that touch Q_n ,
- (vi) for every fixed $0 < \varepsilon < \frac{1}{4}$ and every n let Q_n^* be the closed cube with the same center of Q_n and diameter $(1 + \varepsilon) \text{diam } Q_n$, then for every $x \in \mathbb{R}^N \setminus C$ there exist at most $(12)^N$ cubes Q_n^* that contain x .

Proof. Let $\mathcal{G}_0 := \{\overline{Q(x, 1)} : x \in \mathbb{Z}^N\}$. The family $\mathcal{G}_0$ leads to a family $\{\mathcal{G}_k\}_{k \in \mathbb{Z}}$ of collections of cubes with the property that each cube in the family $\mathcal{G}_k$ gives rise to 2^N cubes in the family $\mathcal{G}_{k+1}$ by bisecting the sides. The cubes in the family $\mathcal{G}_k$ have side length 2^{-k} and, in turn, diameter $\sqrt{N}2^{-k}$.

For $k \in \mathbb{Z}$ define

$$U_k := \{x \in \mathbb{R}^N \setminus C : \sqrt{N}2^{-k+1} < \text{dist}(x, C) \leq \sqrt{N}2^{-k+2}\}.$$

Then

$$(9.31) \quad \mathbb{R}^N \setminus C = \bigcup_{k \in \mathbb{Z}} U_k.$$

Consider the family

$$\mathcal{F}_0 := \bigcup_{k \in \mathbb{Z}} \{Q \in \mathcal{G}_k : Q \cap U_k \neq \emptyset\}.$$

If $x \in \mathbb{R}^N \setminus C$, then by (9.31), $x \in U_k$ for some $k \in \mathbb{Z}$. Since the family $\mathcal{G}_k$ is a partition of $\mathbb{R}^N$, there exists $Q \in \mathcal{G}_k$ such that $x \in Q$. Thus, $Q \in \mathcal{F}_0$. This shows that

$$(9.32) \quad \mathbb{R}^N \setminus C \subseteq \bigcup_{Q \in \mathcal{F}_0} Q.$$

Next we claim that for every $Q \in \mathcal{F}_0$,

$$(9.33) \quad \text{diam } Q < \text{dist}(Q, C) \leq 4 \text{diam } Q.$$

Indeed, if $Q \in \mathcal{F}_0$, then $Q \in \mathcal{G}_k$ for some $k \in \mathbb{Z}$, so that $\text{diam } Q = \sqrt{N}2^{-k}$. Moreover, since $Q \cap U_k \neq \emptyset$, there exists $x \in Q \cap U_k$. Thus, also by the definition of U_k ,

$$\text{dist}(Q, C) \leq \text{dist}(x, C) \leq 4\sqrt{N}2^{-k}.$$

On the other hand,

$$\text{dist}(Q, C) \geq \text{dist}(x, C) - \text{diam } Q > \sqrt{N}2^{-k+1} - \sqrt{N}2^{-k} = \sqrt{N}2^{-k} = \text{diam } Q,$$

which proves (9.33).

Note that the first inequality in (9.33) shows, in particular, that every $Q \in \mathcal{F}_0$ is contained in $\mathbb{R}^N \setminus C$, so that also, by (9.32),

$$\mathbb{R}^N \setminus C = \bigcup_{Q \in \mathcal{F}_0} Q.$$

Thus, properties (i) and (iii) are satisfied. To obtain (ii), we construct an appropriate subfamily of $\mathcal{F}_0$. We begin by observing that if $Q_1 \in \mathcal{G}_{k_1}$ and $Q_2 \in \mathcal{G}_{k_2}$ intersect, with $k_1 < k_2$, then, necessarily, $Q_1 \supset Q_2$.

Start from any cube $Q \in \mathcal{F}_0$ and consider the maximal cube Q' (with respect to inclusion) in $\mathcal{F}_0$ that contains Q . Such a cube exists, since, by

(9.33), the diameter of any cube in $\mathcal{F}_0$ containing Q cannot exceed $4 \operatorname{diam} Q$. Note that by the previous observation there is only one such maximal cube Q' . Let $\mathcal{F}$ be the subfamily of maximal cubes in $\mathcal{F}_0$. Then (i)–(iii) hold.

To prove (iv), assume that $Q_1, Q_2 \in \mathcal{F}$ touch. Then

$$\operatorname{diam} Q_2 \leq \operatorname{dist}(Q_2, C) \leq \operatorname{dist}(Q_1, C) + \operatorname{diam} Q_1 \leq 5 \operatorname{diam} Q_1,$$

where we have used (9.33). On the other hand, since $\operatorname{diam} Q_2 = 2^k \operatorname{diam} Q_1$ for some integer $k \in \mathbb{Z}$, then, necessarily, $\operatorname{diam} Q_2 \leq 4 \operatorname{diam} Q_1$. By reversing the roles of Q_1 and Q_2 , we obtain (iv).

Next we show that (v) holds. Fix a cube $Q \in \mathcal{F}$ and let $k \in \mathbb{Z}$ be such that $Q \in \mathcal{G}_k$. In the family $\mathcal{G}_k$ there are only $3^N - 1$ cubes that touch Q . Each cube in $\mathcal{G}_k$ can contain at most 4^N cubes of $\mathcal{F}$ with diameter greater than or equal to $\frac{1}{4} \operatorname{diam} Q$. Hence (v) follows from (iv).

Finally, we prove (vi). For every $x \in \mathbb{R}^N \setminus C$ let $Q \in \mathcal{F}$ be such that $x \in Q$. We claim that if $Q_n \in \mathcal{F}$, then Q_n^* intersects Q only if Q_n touches Q . Indeed, consider the union of all the cubes in $\mathcal{F}$ that touch Q . By (iv), the diameter of each of these cubes is greater than or equal to $\frac{1}{4} \operatorname{diam} Q$. Since $0 < \varepsilon < \frac{1}{4}$, the union of these cubes contains Q_n^* . By the maximality of the family $\mathcal{F}$, it follows that Q_n must be one of these cubes and the claim is proved. Property (vi) now follows from (v). $\square$

The family $\mathcal{F}$ is called a *Whitney decomposition* of $\mathbb{R}^N \setminus C$.

Lemma 9.45. *Let $C \subset \mathbb{R}^N$ be a closed set. Then there exists a locally finite family of functions $\psi_n \in C_c^\infty(\mathbb{R}^N \setminus C)$ such that $\sum_{n=1}^\infty \psi = 1$ in $\mathbb{R}^N \setminus C$, for every n ,*

$$(9.34) \quad \operatorname{diam} \operatorname{supp} \psi_n \leq 2 \operatorname{dist}(\operatorname{supp} \psi_n, C),$$

$$(9.35) \quad \|\nabla \psi_n(x)\| \leq \frac{c}{\operatorname{dist}(x, C)} \quad \text{for all } x \in \mathbb{R}^N \setminus C$$

and for some constant $c = c(N) > 0$, and for every $x \in \mathbb{R}^N \setminus C$ there are at most $(12)^N$ functions ψ_n such that x belongs to $\operatorname{supp} \psi_n$.

Proof. Let $\{Q_n\}_n$ be the family given in the previous theorem. Construct a nonnegative function $\varphi \in C^\infty(\mathbb{R}^N)$ such that $\varphi = 0$ outside $(-3/4, 3/4)^N$ and $\varphi = 1$ in $(-1/2, 1/2)^N$. If Q_n is centered at y_n and has side-length ℓ_n , define

$$(9.36) \quad \varphi_n(x) := \varphi((x - y_n)/\ell_n).$$

Then $\varphi_n = 1$ in Q_n . Hence, if we define $\Phi := \sum_{n=1}^\infty \varphi_n$ we have that $\Phi \geq 1$ in $\mathbb{R}^N \setminus C$ and so the function $\psi_n := \varphi_n/\Phi$ is well-defined and $\sum_{n=1}^\infty \psi_n = 1$ in $\mathbb{R}^N \setminus C$.

Moreover, by the properties of φ , (9.36), and the fact that $\text{dist}(Q_n, C) \geq \sqrt{N}\ell_n$,

$$\text{dist}(\text{supp } \psi_n, C) \geq \frac{3}{4}\sqrt{N}\ell_n \geq \frac{1}{2} \text{diam supp } \psi_n,$$

which proves (9.34).

By property (vi), for every $x \in \mathbb{R}^N \setminus C$ there are at most $(12)^N$ functions φ_n whose support intersect a small neighborhood of x . Hence, $\Phi \in C^\infty(\mathbb{R}^N \setminus C)$ and for $x \in \mathbb{R}^N \setminus C$,

$$\nabla \psi_n(x) = \frac{\nabla \varphi_n(x)}{\Phi(x)} - \frac{\varphi_n(x) \nabla \Phi(x)}{\Phi^2(x)}.$$

By (9.36),

$$\nabla \varphi_k(x) = \nabla \varphi((x - x_k)/\ell_k)/\ell_k,$$

and so

$$\|\nabla \Phi(x)\| \leq \|\nabla \varphi\|_\infty \sum_{x \in \text{supp } \varphi_k} \frac{1}{\ell_k} \leq \frac{(12)^N 4\sqrt{N} \|\nabla \varphi\|_\infty}{\text{dist}(x, C)},$$

where we used the inequality $\text{dist}(x, C) \leq \text{dist}(Q_k, C) \leq 4\sqrt{N}\ell_k$, which follows from property (iii) and the fact that $x \in \text{supp } \psi_k$. Since for $x \in \mathbb{R}^N \setminus C$, $1 \leq \Phi(x) \leq (12)^N$, (9.35) holds and the proof is complete. $\square$

We now turn to the proof of Whitney's theorem.

Proof. For every $x_0 \in C$ and $\delta > 0$ let

$$(9.37) \quad \rho_{x_0}(\delta) := \sup \left\{ \frac{|u(x) - u(y) - w(y) \cdot (x - y)|}{\|x - y\|} : x, y \in C \cap B(x_0, \delta) \right\}.$$

Then by (9.30),

$$(9.38) \quad \lim_{\delta \rightarrow 0^+} \rho_{x_0}(\delta) = 0 \quad \text{for every } x_0 \in C.$$

Step 1: For each n let $x_n \in C$ be a point with closest distance from $\text{supp } \psi_n$, and for $x \in \mathbb{R}^N \setminus C$ define

$$(9.39) \quad v(x) := \sum_{n=0}^{\infty} (u(x_n) + w(x_n) \cdot (x - x_n)) \psi_n(x),$$

while for $x \in C$ set $v(x) := u(x)$.

We claim that if $x_0 \in C$, then

$$(9.40) \quad \lim_{x \rightarrow x_0} \frac{v(x) - u(x_0) - w(x_0) \cdot (x - x_0)}{\|x - x_0\|} = 0.$$

In view of (9.38) it suffices to verify (9.40) for all $x \in \mathbb{R}^N \setminus C$ with $x \rightarrow x_0$. By (9.39) and the fact that $\sum_{n=0}^{\infty} \psi = 1$ in $\mathbb{R}^N \setminus C$, for $x \in \mathbb{R}^N \setminus C$, with $\|x - x_0\| \leq 1$, we have

$$\begin{aligned} v(x) - u(x_0) - w(x_0) \cdot (x - x_0) \\ = \sum_{n=0}^{\infty} (u(x_n) - u(x_0) - w(x_0) \cdot (x_n - x_0)) \psi_n(x) \\ + \sum_{n=0}^{\infty} (w(x_n) - w(x_0)) \cdot (x - x_n) \psi_n(x) =: I + II. \end{aligned}$$

For every n with $x \in \text{supp } \psi_n$, let $y_n \in \text{supp } \psi_n$ be such that

$$\text{dist}(\text{supp } \psi_n, C) = \|x_n - y_n\|.$$

Then by (9.34),

$$\begin{aligned} (9.41) \quad \text{dist}(x, C) &\leq \|x - x_n\| \leq \|x - y_n\| + \|y_n - x_n\| \leq \text{diam supp } \psi_n \\ &\quad + \text{dist}(\text{supp } \psi_n, C) \leq 3 \text{dist}(\text{supp } \psi_n, C) \leq 3 \text{dist}(x, C). \end{aligned}$$

Since $\text{dist}(x, C) \leq \|x - x_0\|$, it follows that

$$(9.42) \quad \|x_n - x_0\| \leq \|x - x_n\| + \|x - x_0\| \leq 4\|x - x_0\|.$$

In turn,

$$\begin{aligned} |I| &\leq \sum_{n=0}^{\infty} \frac{|u(x_n) - u(x_0) - w(x_0) \cdot (x_0 - x_n)|}{\|x_n - x_0\|} \|x_n - x_0\| \psi_n(x) \\ &\leq \sum_{n=0}^{\infty} \rho_{x_0}(\|x_n - x_0\|) \|x_n - x_0\| \psi_n(x) \\ &\leq \rho_{x_0}(4\|x - x_0\|) 4\|x - x_0\| \sum_{n=0}^{\infty} \psi_n(x) \\ &= \rho_{x_0}(4\|x - x_0\|) 4\|x - x_0\|. \end{aligned}$$

On the other hand, let ρ_w be a modulus of continuity for w . Then by (9.40),

$$|II| \leq \sum_{n=0}^{\infty} \rho_w(\|x_n - x_0\|) \|x - x_n\| \psi_n(x) \leq \rho_w(4\|x - x_0\|) 3\|x - x_0\|.$$

Combining the last two inequalities and letting $x \rightarrow x_0$ gives (9.40).

Hence, we have shown that v is differentiable in C with $\nabla v = w$. On the other hand, since each Q_n^* intersects at most $(12)^N$ cubes Q_n^* , for every $x \in \mathbb{R}^N \setminus C$ there exists a neighborhood V of x such that in V only $(12)^N$ terms in the series (9.39) are nonzero. Hence, the function v belongs to

$C^\infty(\mathbb{R}^N \setminus C)$ with

$$(9.43) \quad \nabla v(x) = \sum_{n=0}^{\infty} (u(x_n) + w(x_n) \cdot (x - x_n)) \nabla \psi_n(x) + \sum_{n=0}^{\infty} w(x_n) \psi_n(x)$$

for $x \in \mathbb{R}^N \setminus C$.

Step 2: In view of Step 1, to prove that $v \in C^1(\mathbb{R}^N)$, it remains to show that for $x_0 \in C$,

$$(9.44) \quad \lim_{x \rightarrow x_0} \nabla v(x) = w(x_0).$$

By the continuity of w in C and Step 1 again, it suffices to verify (9.44) for all $x \in \mathbb{R}^N \setminus C$ with $x \rightarrow x_0$.

For every $y \in C$ we have that

$$\begin{aligned} w(y) &= \nabla_x (u(y) + w(y) \cdot (x - y)) = \nabla_x \left(\sum_{n=0}^{\infty} (u(y) + w(y) \cdot (x - y)) \psi_n(x) \right) \\ &= \sum_{n=0}^{\infty} (u(y) + w(y) \cdot (x - y)) \nabla \psi_n(x) + \sum_{n=0}^{\infty} w(y) \psi_n(x). \end{aligned}$$

Given $x \in \mathbb{R}^N \setminus C$, with $\|x - x_0\| \leq 1$, let $y \in C$ be such that $\text{dist}(x, C) = \|x - y\|$. Then by (9.43),

$$\begin{aligned} \nabla v(x) - w(y) &= \sum_{n=0}^{\infty} (u(x_n) - u(y) - w(y) \cdot (x_n - y)) \nabla \psi_n(x) \\ &\quad + \sum_{n=0}^{\infty} (w(x_n) - w(y)) \cdot (x - x_n) \nabla \psi_n(x) \\ &\quad + \sum_{n=0}^{\infty} (w(x_n) - w(y)) \psi_n(x) = I + II + III. \end{aligned}$$

If n is such that $x \in \text{supp } \psi_n$, then by (9.41),

$$(9.45) \quad \|x_n - y\| \leq \|x - x_n\| + \|x - y\| \leq 4 \text{dist}(x, C) \leq 4\|x - x_0\|.$$

Hence, by (9.35), (9.37), and (9.45),

$$\begin{aligned} \|I\| &\leq \sum_{n=0}^{\infty} \frac{|u(x_n) - u(y) - w(y) \cdot (x_n - y)|}{\|x_n - y\|} \|x_n - y\| \|\nabla \psi_n(x)\| \\ &\leq (12)^N 4c\rho_{x_0}(4\|x - x_0\|). \end{aligned}$$

On the other hand, by (9.35) and (9.45),

$$\|II\| \leq \sum_{n=0}^{\infty} \rho_w(\|x_n - y\|) \|x - x_n\| \|\nabla \psi_n(x)\| \leq (12)^N 3c\rho_w(4\|x - x_0\|),$$

while

$$\|III\| \leq \sum_{n=0}^{\infty} \rho_w(\|x_n - y\|) \psi_n(x) \leq (12)^N \rho_w(4\|x - x_0\|).$$

By combining the last three inequalities we have

$$\begin{aligned} \|\nabla v(x) - w(x_0)\| &\leq \|\nabla v(x) - w(y)\| + \|w(y) - w(x_0)\| \\ &\leq c\rho_{x_0}(4\|x - x_0\|) + c\rho_w(4\|x - x_0\|) \rightarrow 0 \end{aligned}$$

as $x \rightarrow x_0$. This concludes the proof. $\square$

Whitney's theorem continues to hold for C^m and C^∞ functions.

Theorem 9.46 (Whitney extension, II). *Let $C \subseteq \mathbb{R}^N$ be a closed set, let $m \in \mathbb{N}_0$ and let $u : C \rightarrow \mathbb{R}$ and $u_\alpha : C \rightarrow \mathbb{R}^N$, $\alpha \in \mathbb{N}_0^N$, $0 < |\alpha| \leq m$, be continuous functions. Assume that for every multi-index α , with $0 \leq |\alpha| \leq m$, for every $x_0 \in C$, and for every $\varepsilon > 0$, there exists $\delta = \delta(x_0, \varepsilon, \alpha) > 0$ such that*

$$(9.46) \quad \left| u_\alpha(x) - \sum_{|\beta| \leq m - |\alpha|} \frac{u_\beta(y)}{\beta!} (x - y)^\beta \right| \leq \varepsilon \|x - y\|^{m - |\alpha|}$$

for all $x, y \in C \cap B(x_0, \delta)$, and where $u_0 := u$. Then there exists a function $v \in C^m(\mathbb{R}^N)$ such that $v = u$ in C , $\partial^\alpha v = u_\alpha$ in C for every multi-index α , with $0 < |\alpha| \leq m$, and v is analytic in $\mathbb{R}^N \setminus C$.

Proof. The proof is similar to the previous one. In (9.39) one should replace the polynomial of degree one $u(x_n) + v(x_n) \cdot (x - x_n)$ with

$$u(x_n) + \sum_{|\alpha| \leq m} \frac{u_\alpha(x_n)}{\alpha!} (x - x_n)^\alpha.$$

The details are left as an exercise. $\square$

Remark 9.47. The long-standing problem of determining whether a function $u : C \rightarrow \mathbb{R}$ defined on a compact set $C \subseteq \mathbb{R}^N$ can be extended to a function of class $C^m(\mathbb{R}^N)$ has recently been solved by C. Fefferman (see [77]).

In view of Whitney's theorem, if $\Omega \subseteq \mathbb{R}^N$ is an open set and $m \in \mathbb{N}_0 \cup \{\infty\}$, we define the space $C^m(\overline{\Omega})$ as the space of all functions $u \in C^m(\Omega)$ that can be extended to $C^m(\mathbb{R}^N)$.

Using Whitney's extension theorem we can prove the following approximation result.

Theorem 9.48. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u : \Omega \rightarrow \mathbb{R}$ be such that the set*

$$E := \{x \in \Omega : u \text{ is differentiable at } x\}$$

is nonempty. Then there exists a sequence of disjoint closed sets $C_n \subseteq E$ and a sequence of C^1 functions $g_n : \mathbb{R}^N \rightarrow \mathbb{R}$ such that $g_n = u$ and $\nabla g_n = \nabla u$ in C_n and $\mathcal{L}^N(E \setminus C) = 0$, where $C := \bigcup_{n=1}^{\infty} C_n$.

Proof. For $k \in \mathbb{N}$ and $x \in E$ set

$$v_k(x) := \sup \left\{ \frac{|u(y) - u(x) - \nabla u(x) \cdot (y - x)|}{\|x - y\|} : y \in \Omega, 0 < \|x - y\| \leq \frac{1}{k} \right\}.$$

Then $v_k(x) \rightarrow 0$ as $k \rightarrow \infty$. Hence, by Lusin's and Egoroff's theorems there exists a sequence of disjoint closed sets C_n such that $\frac{\partial u}{\partial x_i}$ is continuous in C_n for every $i = 1, \dots, N$, $v_k \rightarrow 0$ uniformly in each C_n , and $\mathcal{L}^N(E \setminus C) = 0$, where $C := \bigcup_{n=1}^{\infty} C_n$. We can now apply Whitney's extension theorem to find a C^1 function g_n such that $g_n = u$ and $\nabla g_n = \nabla u$ in C_n . $\square$

We can now extend Theorem 9.33 to differentiable functions.

Theorem 9.49 (Area formula, the differentiable case). *Let $1 \leq k \leq N$, let $V \subseteq \mathbb{R}^k$ be an open set and let $\Psi : V \rightarrow \mathbb{R}^N$ be a differentiable function. Let $H \subseteq \Psi(V)$ be a Borel set and let $u : H \rightarrow [-\infty, \infty]$ be a Borel function, which is either $\mathcal{H}^k$ integrable or has a sign. Assume that Ψ is injective in $\Psi^{-1}(H)$. Then*

$$\int_H u(x) d\mathcal{H}^k(x) = \int_{\Psi^{-1}(H)} u(\Psi(y)) |||J_\Psi(y)||| dy.$$

Proof. In view of Theorem 9.48 applied to each component of Ψ we can find a sequence of disjoint closed sets $C_n \subseteq V$ and a sequence of C^1 functions $\Psi_n : \mathbb{R}^k \rightarrow \mathbb{R}^N$ such that $\Psi_n = \Psi$ and $J_{\Psi_n} = J_\Psi$ in C_n and

$$\mathcal{L}^k(V \setminus C) = 0,$$

where $C := \bigcup_{n=1}^{\infty} C_n$. We claim that $\mathcal{H}^k(\Psi(V \setminus C)) = 0$. To see this, write

$$V \setminus C = \bigcup_{l=1}^{\infty} E_l,$$

where

$$E_l := \left\{ y \in V \setminus C : \frac{\|\Psi(y) - \Psi(z)\|_N}{\|y - z\|_k} \leq l \text{ for all } z \in V \text{ with } 0 < \|z - y\|_k \leq \frac{1}{l} \right\}.$$

Then $\Psi : E_l \rightarrow \mathbb{R}^N$ is locally Lipschitz continuous. Indeed, if z, y belong to a compact subset $K \subseteq E_l$ and $\|z - y\|_k \leq 1/l$, then $\|\Psi(y) - \Psi(z)\|_N \leq l\|y - z\|_k$ by the definition of E_l . On the other hand, if $\|z - y\|_k > 1/l$, then

$$\|\Psi(y) - \Psi(z)\|_N \leq 2 \max_K \|\Psi\|_N \leq 2l \max_K \|\Psi\|_N \|y - z\|_k.$$

Hence, by Proposition C.44,

$$\mathcal{H}^k(\Psi(K)) \leq L_K \mathcal{H}^k(K) = L_K \mathcal{L}^k(K) = 0,$$

where $L_K := l(1 + 2 \max_K \|\Psi\|_N)$. By letting $K \nearrow E_l$ we obtain that $\mathcal{H}^k(\Psi(E_l)) = 0$ and since this is true for every l , it follows that $\mathcal{H}^k(\Psi(V \setminus C)) = 0$.

Hence, if $u \geq 0$,

$$\begin{aligned} \int_H u(x) d\mathcal{H}^k(x) &= \int_{H \cap \Psi(C)} u(x) d\mathcal{H}^k(x) = \sum_n \int_{H \cap \Psi(C_n)} u(x) d\mathcal{H}^k(x) \\ &= \sum_n \int_{H \cap \Psi_n(C_n)} u(x) d\mathcal{H}^k(x) = \sum_n \int_{\Psi_n^{-1}(H) \cap C_n} u(\Psi_n(y)) |||J_{\Psi_n}(y)||| dy \\ &= \sum_n \int_{\Psi^{-1}(H) \cap C_n} u(\Psi(y)) |||J_{\Psi}(y)||| dy = \int_{\Psi^{-1}(H) \cap C} u(\Psi(y)) |||J_{\Psi}(y)||| dy \\ &= \int_{\Psi^{-1}(H)} u(\Psi(y)) |||J_{\Psi}(y)||| dy, \end{aligned}$$

where we have used the fact that the sets $H \cap \Psi(C_n)$ are disjoint since Ψ is injective in $\Psi^{-1}(H)$ and Theorem 9.33 for Ψ_n .

If $u : H \rightarrow [-\infty, \infty]$ is $\mathcal{H}^k$ integrable, write $u = u^+ - u^-$, apply the previous identity to u^+ and u^- , and use the linearity of integrals to conclude the proof. $\square$

Exercise 9.50. Prove that the previous theorem continues to hold if in place of everywhere differentiability of Ψ we assume that $\Psi : V \rightarrow \mathbb{R}^N$ is continuous and there exists a set $F \subseteq V$, with $\mathcal{L}^k(V \setminus F) = 0$ and $\mathcal{H}^k(\Psi(V \setminus F)) = 0$, such that Ψ is differentiable in F .

Corollary 9.51 (Change of variables, II). *Let $V \subseteq \mathbb{R}^N$ be an open set and let $\Psi : V \rightarrow \mathbb{R}^N$ be a differentiable function. Let $H \subseteq \Psi(V)$ be a Borel set and let $u : H \rightarrow [-\infty, \infty]$ be a Borel function, which is either Lebesgue integrable or has a sign. Assume that Ψ is injective in $\Psi^{-1}(H)$. Then*

$$\int_H u(x) dx = \int_{\Psi^{-1}(H)} u(\Psi(y)) |\det J_{\Psi}(y)| dy.$$

Using Exercise 9.50 we obtain the standard change of variables formula for multiple integrals.

Theorem 9.52 (Change of variables for multiple integrals). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $\Psi : \Omega \rightarrow \mathbb{R}^N$ be continuous. Assume that there exist two Lebesgue measurable sets $F, G \subseteq \Omega$ on which Ψ is differentiable and one-to-one, respectively. If*

- (i) $\mathcal{L}^N(\Omega \setminus F) = 0$,
- (ii) $\mathcal{L}^N(\Psi(\Omega \setminus F)) = 0$,
- (iii) $\mathcal{L}^N(\Psi(\Omega \setminus G)) = 0$,

then the change of variables formula

$$\int_{\Psi(E)} u(y) dy = \int_E u(\Psi(x)) |\det J_\Psi(x)| dx$$

holds for every Borel set $E \subseteq \Omega$ and for every Borel function $u : \Psi(E) \rightarrow [-\infty, \infty]$, which is either Lebesgue integrable or has a sign.

Remark 9.53. (i) The hypothesis that Ω is open may be replaced by the hypothesis that Ω is any set whose boundary has Lebesgue measure zero, provided we assume $\mathcal{L}^N(\Psi(\Omega \setminus \Omega^\circ)) = 0$. Then $\mathcal{L}^N(\Omega \setminus \Omega^\circ) = 0$ and it is enough to apply the theorem to Ω° .

(ii) Note that under the hypotheses of the previous theorem we obtain, as a by-product, that $|\det J_\Psi|$ is a locally integrable function such that

$$\int_E |\det J_\Psi(x)| dx < \infty$$

for every Borel set $E \subseteq \Omega$ for which $\mathcal{L}^N(\Psi(E)) < \infty$.

Exercise 9.54. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $\Psi : \Omega \rightarrow \mathbb{R}^N$ be a Lipschitz continuous function. Prove that Ψ has the (N) property, that is, that if $E \subseteq \Omega$ has Lebesgue measure zero, then so does $\Psi(E)$. Deduce that if $\Psi : \Omega \rightarrow \mathbb{R}^N$ is a Lipschitz continuous function, which is one-to-one except on a set of measure zero, then Theorem 9.52 holds.

We will see in Chapter 12 (see Exercise 12.54), under suitable regularity hypotheses on the open set Ω , an important class of transformations satisfying the hypotheses of Theorem 9.52 is given by transformations $\Psi : \Omega \rightarrow \mathbb{R}^N$ belonging to the Sobolev space $W^{1,p}(\Omega; \mathbb{R}^N)$, $p > N$, and that are one-to-one except at most on a set of Lebesgue measure zero. Another extremely useful class of transformations is given by (locally) Lipschitz continuous functions $\Psi : \Omega \rightarrow \mathbb{R}^N$, which are again one-to-one except at most on a set of Lebesgue measure zero.

Perhaps one of the most important applications of Theorem 9.52 and Remark 9.53 is given by *spherical coordinates* in $\mathbb{R}^N$. We proceed by induction. For $N = 2$ we consider the standard polar coordinate system. Given a point $x = (x_1, \dots, x_N) \in \mathbb{R}^N$, let $r := \|x\|$, let θ_1 be the angle from the positive x_1 -axis to x , precisely,

$$\theta_1 := \cos^{-1}(x_1/r), \quad 0 < \theta_1 < \pi,$$

and let $(\rho, \theta_2, \dots, \theta_{N-1})$ be the spherical coordinates for $x'_1 = (x_2, \dots, x_N) \in \mathbb{R}^{N-1}$, where $\rho := \|x'_1\|_{N-1} = r \sin \theta_1$. The coordinates of x are

$$\begin{aligned} x_1 &= r \cos \theta_1, \\ x_2 &= r \sin \theta_1 \cos \theta_2, \\ &\vdots \\ x_{N-1} &= r \sin \theta_1 \sin \theta_2 \cdots \sin \theta_{N-2} \cos \theta_{N-1}, \\ x_N &= r \sin \theta_1 \sin \theta_2 \cdots \sin \theta_{N-2} \sin \theta_{N-1}. \end{aligned}$$

This defines a coordinate system $(r, \theta_1, \dots, \theta_{N-1})$, which is invertible except on a set of $\mathcal{L}^N$ -measure zero. The Jacobian is

$$\det J_\Psi(r, \theta_1, \dots, \theta_{N-1}) = r^{N-1} \sin^{N-2} \theta_1 \sin^{N-3} \theta_2 \cdots \sin \theta_{N-2}.$$

Example 9.55. Let $u(x) = g(\|x\|)$, where $g : [r_1, r_2] \rightarrow \mathbb{R}$ is continuous and $0 \leq r_1 < r_2$. Then

$$\int_{r_1 < \|x\| < r_2} u(x) dx = \beta_N \int_{r_1}^{r_2} g(r) r^{N-1} dr.$$

To find β_N , take $g = 1$. Then

$$\alpha_N(r_2^N - r_1^N) = \mathcal{L}^N(B(0, r_2)) - \mathcal{L}^N(B(0, r_1)) = \beta_N \frac{r_2^N - r_1^N}{N},$$

where $\alpha_N = \mathcal{L}^N(B(0, 1))$, and so $\beta_N = N\alpha_N$. It turns out that β_N is the $N - 1$ surface measure of $S^{N-1} = \partial B(0, 1)$.

Example 9.56. Given an open set $V \subseteq \mathbb{R}^k$ and a Lipschitz function $f : V \rightarrow \mathbb{R}$, consider the graph of f ,

$$\text{Gr } f := \{(y, t) \in V \times \mathbb{R} : t = f(y)\}$$

and let $\Psi : V \rightarrow \mathbb{R}^{k+1}$ be the function $\Psi(y) := (y, f(y))$. Then,

$$J_\Psi(y) = \begin{pmatrix} I_k \\ \nabla f(y) \end{pmatrix}$$

and

$$|||J_\Psi(y)||| = \sqrt{1 + \|\nabla f(y)\|_k^2}.$$

Given an $\mathcal{H}^k$ integrable Borel function $u : \text{Gr } f \rightarrow \mathbb{R}$, the surface integral of u over $\text{Gr } f$ is

$$(9.47) \quad \int_{\text{Gr } f} u d\mathcal{H}^k = \int_V u(y, f(y)) \sqrt{1 + \|\nabla f(y)\|_k^2} dy.$$

Taking $u = 1$, we get that the k -dimensional surface measure of $\text{Gr } f$ is

$$\mathcal{H}^k(\text{Gr } f) = \int_U \sqrt{1 + \|\nabla f(y)\|_k^2} dy.$$

In particular, if f is constant, say, $f(y) = c$ for all $y \in V$, then

$$(9.48) \quad \int_{\text{Gr } f} u \, d\mathcal{H}^k = \int_V u(y, c) \, dy.$$

In the applications, usually we are given an open set $\Omega \subset \mathbb{R}^N$ and the boundary $\partial\Omega$ is locally given by the graph of a function f . In this case $k = N - 1$.

9.5. The Divergence Theorem

We recall that a rigid motion $T : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is an affine function given by $T(x) = c + R(x)$, $x \in \mathbb{R}^N$, where R is a rotation and $c \in \mathbb{R}^N$.

Definition 9.57. Given an open set $\Omega \subseteq \mathbb{R}^N$ we say that its boundary $\partial\Omega$ is *Lipschitz continuous* if for all $x_0 \in \partial\Omega$ there exist a rigid motion $T : \mathbb{R}^N \rightarrow \mathbb{R}^N$ with $T(x_0) = 0$, a Lipschitz continuous function $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$, and $r > 0$ such that, setting $y := T(x)$, we have

$$(9.49) \quad T(\Omega \cap B(x_0, r)) = \{y \in B(0, r) : y_N > f(y')\}.$$

We say that $\partial\Omega$ is of class C^m , $m \in \mathbb{N}_0$, if the functions f are of class C^m .

Observe that R , f , and r depend on x_0 . The coordinates x are called *background coordinates* while the coordinates y are called *local coordinates*.

Remark 9.58. Without loss of generality, in the previous definition one can replace the ball $B(x_0, r)$ with any small (open) neighborhood of x_0 . We will use this fact without further notice.

We remark that if $\partial\Omega$ is of class C^m for $m \in \mathbb{N}$, then $\partial\Omega$ is an $(N - 1)$ -dimensional surface of class C^m .

Remark 9.59. The previous definition implies in particular that $\partial\Omega = \partial(\overline{\Omega})$, so that the open set Ω cannot lie on both sides of its boundary. Sets like

$$\mathbb{R}^N \setminus \{x \in \mathbb{R}^N : x_N = 0\}$$

or

$$B(0, 1) \setminus \{x = (x_1, x_2) \in \mathbb{R}^2 : 0 \leq x_1 < 1, x_2 = 0\}$$

are excluded.

Exercise 9.60. Let $\Omega \subset \mathbb{R}^2$ be a bounded open set with the property that for each point $x_0 \in \partial\Omega$ there exist local coordinates $y = (y_1, y_2) \in \mathbb{R} \times \mathbb{R}$, with $y = 0$ at $x = x_0$, a Lipschitz continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$, and $r > 0$ such that $\partial\Omega \cap B(x_0, r)$ in local coordinates becomes

$$\{y \in B(0, r) : y_2 = f(y_1)\}.$$

Prove that $\partial\Omega = \partial(\overline{\Omega})$. Hint: Prove that if $\partial\Omega \neq \partial(\overline{\Omega})$, then there exists a component Γ of $\partial\Omega$ that is a closed curve and such that $\mathbb{R}^2 \setminus \Gamma$ is connected. Prove that this violates the Jordan curve theorem (see Theorem 5.45).

The following example shows that domains with continuous boundary may have exponential cusps.

Exercise 9.61. Prove that the domain

$$\{x = (x_1, x_2) \in \mathbb{R}^2 : 0 < x_1 < 1, -\exp(-1/x_1) < x_2 < \exp(-1/x_1)\}$$

is of class C^0 and that the function f in Definition 9.57 cannot be taken of class $C^{0,\alpha}$ for any $\alpha > 0$.

Definition 9.62. Given an open set $\Omega \subseteq \mathbb{R}^N$ with Lipschitz continuous boundary and $x_0 \in \partial\Omega$, a vector $\tau \in \mathbb{R}^N$ is called a *tangent vector to $\partial\Omega$ at the point x_0* if there exists a function $\psi : (-\delta, \delta) \rightarrow \mathbb{R}^N$ differentiable at 0 such that $\psi((-\delta, \delta)) \subseteq \partial\Omega$, $\psi(0) = x_0$ and $\psi'(0) = \tau$. The set of all tangent vectors to $\partial\Omega$ at x_0 is called the *tangent space* to $\partial\Omega$ at x_0 and is denoted $T_{\partial\Omega}(x_0)$.

Given an open set with Lipschitz continuous boundary, in a neighborhood of $x_0 \in \partial\Omega$ we have that $\partial\Omega \cap B(x_0, r)$ in local coordinates becomes $\{y \in B(0, r) : y_N = f(y')\}$. In view of Rademacher's theorem (see Theorem 9.14), the function f is differentiable for $\mathcal{L}^{N-1}$ -a.e. $y' \in \mathbb{R}^{N-1}$. At a point $x \in \partial\Omega$ at which f is differentiable, there exists the tangent space $T_{\partial\Omega}(x)$. Note that $T_{\partial\Omega}(x)$ is given by the $(N-1)$ -dimensional vector space given by $\ker \nabla g$, where $g(y) = y_N - f(y')$ (exercise).

Definition 9.63. Given an open set $\Omega \subseteq \mathbb{R}^N$ with Lipschitz continuous boundary and a point $x_0 \in \partial\Omega$ for which there exists a tangent space $T_{\partial\Omega}(x_0)$ of dimension $N-1$, a vector $\nu \in \mathbb{R}^N \setminus \{0\}$ is called a *normal vector to $\partial\Omega$ at x_0* if it is orthogonal to all vectors in $T_{\partial\Omega}(x_0)$, that is, $\nu \cdot \tau = 0$ for all $\tau \in T_{\partial\Omega}(x_0)$. A unit normal vector ν to $\partial\Omega$ at x_0 is called a *unit outward normal to Ω at x_0* if there exists $\delta > 0$ such that $x_0 - t\nu \in \Omega$ and $x_0 + t\nu \in \mathbb{R}^N \setminus \overline{\Omega}$ for all $0 < t < \delta$.

We are ready to prove the divergence theorem.

Theorem 9.64 (Divergence theorem). *Let $\Omega \subset \mathbb{R}^N$ be an open, bounded set with Lipschitz continuous boundary and let $u : \overline{\Omega} \rightarrow \mathbb{R}^N$ be a Lipschitz continuous function. Then*

$$(9.50) \quad \int_{\Omega} \operatorname{div} u(x) \, dx = \int_{\partial\Omega} u(x) \cdot \nu(x) \, d\mathcal{H}^{N-1}(x),$$

where $\nu(x)$ is the outward unit normal to $\partial\Omega$ at x and $\operatorname{div} u := \sum_{i=1}^N \partial_i u_i$.

Note that in view of Rademacher's theorem (see Theorem 9.14) we have that u is differentiable at $\mathcal{L}^N$ -a.e. $x \in \Omega$. Thus both integrals in (9.50) are well-defined.

Proof. Step 1: We first prove that the divergence theorem holds when Ω is given by a rectangle,

$$R := (a_1, b_1) \times \cdots \times (a_N, b_N).$$

Write $R' := \Pi_{i=1}^{N-1}(a_i, b_i) \subset \mathbb{R}^{N-1}$. Then by Fubini's theorem,

$$\begin{aligned} \int_R \partial_N u_N(x) dx &= \int_{R'} \left(\int_{a_N}^{b_N} \partial_N u_N(x', x_N) dx_N \right) dx' \\ &= \int_{R'} (u_N(x', b_N) - u_N(x', a_N)) dx' \\ &= \int_{R'} u(x', b_N) \cdot e_N dx' + \int_{R'} u(x', a_N) \cdot (-e_N) dx' \\ &= \int_{R' \times \{b_N\}} u(x) \cdot \nu d\mathcal{H}^{N-1}(x) + \int_{R' \times \{a_N\}} u(x) \cdot \nu d\mathcal{H}^{N-1}(x), \end{aligned}$$

where in the third equality we have used the fundamental theorem of calculus (see Theorem 3.20) applied to the Lipschitz function of one variable $x_N \in [a_N, b_N] \mapsto u_N(x', x_N)$ with x' fixed, and in the last equality we have used formula (9.48). Summing the resulting identities gives the desired result.

Step 2: Next we prove that the divergence theorem holds when Ω is a set of the form

$$\Omega := \{(x', x_N) \in R' \times \mathbb{R} : f(x') < x_N < b_N\},$$

where R' is as before and $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ is a Lipschitz continuous function and $u = 0$ in an open neighborhood of $\partial\Omega \setminus \text{graph } f$. The change of variables $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ given by

$$(9.51) \quad y = \Phi(x) = (x', x_N - f(x')),$$

maps the set Ω into the set

$$W := \{(y', y_N) \in R' \times \mathbb{R} : 0 < y_N < b_N - f(y')\}.$$

Let $R := R' \times (0, b_N)$ and consider the function

$$w(y) := \begin{cases} u(y', y_N + f(y')) & \text{if } y \in W, \\ 0 & \text{if } y \in R \setminus W. \end{cases}$$

Since $u = 0$ near $x_N = b_N$, it follows that w is Lipschitz continuous in $\overline{R}$. Hence, by Step 1,

$$\begin{aligned} \int_W \operatorname{div} w(y) dy &= \int_R \operatorname{div} w(y) dy = \int_{\partial R} w(y) \cdot \nu(y) d\mathcal{H}^{N-1}(y) \\ &= - \int_{R'} w_N(0, y') dy'. \end{aligned}$$

By the chain rule, for $\mathcal{L}^N$ -a.e. $y \in W$,

$$\begin{aligned} \operatorname{div} w(y) &= \frac{\partial u_N}{\partial x_N}(y', y_N + f(y')) \\ &\quad + \sum_{i=1}^{N-1} \left(\frac{\partial u_i}{\partial x_N}(y', y_N + f(y')) \frac{\partial f}{\partial y_i}(y') + \frac{\partial u_i}{\partial x_i}(y', y_N + f(y')) \right). \end{aligned}$$

Hence,

$$\begin{aligned} \int_W \sum_{i=1}^N \frac{\partial u_i}{\partial x_i}(y', y_N + f(y')) dy &= - \int_{R'} u_N(y', f(y')) dy' \\ &\quad - \sum_{i=1}^{N-1} \int_W \frac{\partial u_i}{\partial x_N}(y', y_N + f(y')) \frac{\partial f}{\partial y_i}(y') dy. \end{aligned}$$

Note that the change of variables Φ given in (9.51) is invertible, with inverse $\Psi := \Phi^{-1}$ given by

$$x = \Psi(y) = (y', y_N + f(y')).$$

Moreover, we have

$$J_\Psi(y) = \begin{pmatrix} I_{N-1} & 0 \\ -\nabla_{y'} f(y') & 1 \end{pmatrix},$$

which implies that $\det J_\Psi(y) = 1$. Hence, by changing variables (see Corollary 9.51),

$$(9.52) \quad \int_\Omega \operatorname{div} u(x) dx = - \int_{R'} u_N(x', f(x')) dx' - \sum_{i=1}^{N-1} \int_\Omega \frac{\partial u_i}{\partial x_N}(x) \frac{\partial f}{\partial x_i}(x') dx.$$

By Fubini's theorem, the fundamental theorem of calculus (see Theorem 3.20), and the fact that $u = 0$ near $x_N = b_N$,

$$\begin{aligned} \int_\Omega \frac{\partial u_i}{\partial x_N}(x) \frac{\partial f}{\partial x_i}(x') dx &= \int_{R'} \left(\int_{f(x')}^{b_N} \frac{\partial u_i}{\partial x_N}(x', x_N) dx_N \right) \frac{\partial f}{\partial x_i}(x') dx' \\ &= \int_{R'} (0 - u_i(x', f(x'))) \frac{\partial f}{\partial x_i}(x') dx', \end{aligned}$$

and so from (9.52),

$$\begin{aligned}
 \int_{\Omega} \operatorname{div} u(x) dx &= - \int_{R'} u_N(x', f(x')) dx_N + \sum_{i=1}^{N-1} \int_{R'} u_i(x', f(x')) \frac{\partial f}{\partial x_i}(x') dx' \\
 &= \int_{R'} u(x', f(x')) \cdot (-1, \nabla f(x')) \frac{\sqrt{1 + \|\nabla f(x')\|_{N-1}^2}}{\sqrt{1 + \|\nabla f(x')\|_{N-1}^2}} dx' \\
 &= \int_{\partial\Omega} u(x) \cdot \nu(x) d\mathcal{H}^{N-1}(x),
 \end{aligned}$$

where in the last equality we have used formula (9.47). This concludes the proof in this case.

Step 3: Assume that there exists a rigid motion T such that setting $y =: T(x) = Ax + c$, where A is an $N \times N$ rotation matrix and $c \in \mathbb{R}^N$, we can write

$$T(\Omega) = \{(y', y_N) \in R' \times \mathbb{R} : f(y') < y_N < b_N\}.$$

Note that $\operatorname{div} u = \operatorname{trace} J_u$. Hence, if we define $v(y) := Au(T^{-1}(y))$, we have that

$$J_v(y) = AJ_u(T^{-1}(y))A^t$$

and so

$$\operatorname{div}_y v(y) = \operatorname{trace} J_v(y) = \operatorname{trace} J_u(T^{-1}(y)) = \operatorname{div}_x u(T^{-1}(y)).$$

By applying Step 2 to v we get

$$\begin{aligned}
 \int_{T(\Omega)} \operatorname{div}_x u(T^{-1}(y)) dy &= \int_{T(\Omega)} \operatorname{div}_y v(y) dy = \int_{\partial T(\Omega)} v(y) \cdot \nu(y) d\mathcal{H}^{N-1}(y) \\
 &= \int_{\partial T(\Omega)} Au(T^{-1}(y)) \cdot \nu(y) d\mathcal{H}^{N-1}(y) \\
 &= \int_{\partial T(\Omega)} u(T^{-1}(y)) \cdot A^t \nu(y) d\mathcal{H}^{N-1}(y).
 \end{aligned}$$

Since $\det A^t = \det A = 1$, we can change variables (see Corollary 9.51) to get

$$\int_{T(\Omega)} \operatorname{div} u(T^{-1}(y)) dy = \int_{\Omega} \operatorname{div} u(x) dx.$$

On the other hand, since the Hausdorff measure is invariant by rotation (why?) we get

$$\int_{\partial T(\Omega)} u(T^{-1}(y)) \cdot A^t \nu(y) d\mathcal{H}^{N-1}(y) = \int_{\partial\Omega} u(x) \cdot \nu(x) d\mathcal{H}^{N-1}(x).$$

Step 4: We finally consider the general case. By definition of Lipschitz continuous boundary and Remark 9.58 for every point $x_0 \in \partial\Omega$ there exist a rigid motion $T_{x_0} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ with $T_{x_0}(x_0) = 0$, a Lipschitz continuous

function $f_{x_0} : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ and $r_{x_0} > 0$ such that, setting $y := T_{x_0}(x)$, we have

$$(9.53) \quad T_{x_0}(\Omega \cap Q_{T_{x_0}}(x_0, r_{x_0})) = \{y \in Q(0, r_{x_0}) : y_N > f_{x_0}(y')\},$$

where $Q_{T_{x_0}}(x_0, r_{x_0})$ is an open cube centered at x_0 and of side-length $r_{x_0} > 0$. On the other hand, if $x_0 \in \Omega$, which is open, then there exists $Q(x_0, r_{x_0}) \subseteq \Omega$. Since the union of all these open cubes as x_0 varies in $\overline{\Omega}$ covers the compact set $\overline{\Omega}$, we can find $Q_1, \dots, Q_n$ which still cover $\overline{\Omega}$. Let $\psi_1, \dots, \psi_\ell$ be a partition of unity subordinated to the family of open sets $Q_1, \dots, Q_n$ (see Theorem C.21). For every $k = 1, \dots, \ell$, we have that the function $\psi_k u$ is zero outside some Q_j and since Q_j is either contained in Ω or $\Omega \cap Q_k$ satisfies (9.53), we can apply either Step 1 or Step 3 to conclude that

$$(9.54) \quad \begin{aligned} \int_{\Omega} \operatorname{div}(\psi_k u) dx &= \int_{\Omega \cap Q_j} \operatorname{div}(\psi_k u) dx = \int_{\partial(\Omega \cap Q_j)} \psi_k u \cdot \nu d\mathcal{H}^{N-1} \\ &= \int_{\partial\Omega} \psi_k u \cdot \nu d\mathcal{H}^{N-1}, \end{aligned}$$

where we have used the fact that $\psi_k u$ is zero outside Q_j . Summing (9.54) over k and using the fact that $\sum_{k=1}^{\ell} \psi_k = 1$ in $\overline{\Omega}$, we have

$$\begin{aligned} \int_{\Omega} \operatorname{div} u dx &= \int_{\Omega} \operatorname{div} \left(\sum_{k=1}^{\ell} \psi_k u \right) dx = \sum_{k=1}^{\ell} \int_{\Omega} \operatorname{div}(\psi_k u) dx \\ &= \sum_{k=1}^{\ell} \int_{\partial\Omega} \psi_k u \cdot \nu d\mathcal{H}^{N-1} = \int_{\partial\Omega} \sum_{k=1}^{\ell} \psi_k u \cdot \nu d\mathcal{H}^{N-1} \\ &= \int_{\partial\Omega} u \cdot \nu d\mathcal{H}^{N-1}, \end{aligned}$$

which is what we wanted. $\square$

Remark 9.65. In physics $\int_{\partial\Omega} u(x) \cdot \nu(x) d\mathcal{H}^{N-1}(x)$ represents the *outward flux* of a vector field u across the boundary of a region Ω .

Corollary 9.66 (Integration by parts). *Let $\Omega \subset \mathbb{R}^N$ be an open, bounded set with Lipschitz continuous boundary, let $u : \overline{\Omega} \rightarrow \mathbb{R}$, $v : \overline{\Omega} \rightarrow \mathbb{R}$ be Lipschitz continuous. Then for every $i = 1, \dots, N$,*

$$\int_{\Omega} \partial_i u(x) v(x) dx = - \int_{\Omega} \partial_i v(x) u(x) dx + \int_{\partial\Omega} u(x) v(x) \nu_i(x) d\mathcal{H}^{N-1}(x),$$

where $\nu(x)$ is the outward unit normal to $\partial\Omega$ at x .

Proof. Apply the divergence theorem to the function $w : \overline{\Omega} \rightarrow \mathbb{R}^N$ given by $w_j := 0$ for $j \neq i$ and $w_i := uv$. $\square$

Exercise 9.67. Let $m \in \mathbb{N}$, let $\Omega \subset \mathbb{R}^N$ be an open, bounded set with Lipschitz continuous boundary, let $u : \overline{\Omega} \rightarrow \mathbb{R}$, $v : \overline{\Omega} \rightarrow \mathbb{R}$ be such that u and v are bounded and continuous in $\overline{\Omega}$ and there exist $\partial^\alpha u$ and $\partial^\alpha v$ in Ω for every multi-index α with $1 \leq |\alpha| \leq m$ and they are continuous and bounded in Ω .

(i) Prove that for every multi-index α with $1 \leq |\alpha| \leq m$,

$$\sum_{\beta+e_i=\alpha} \frac{\beta_i + 1}{|\beta| + 1} = 1.$$

(i) Prove that for every multi-index α with $1 \leq |\alpha| \leq m$,

$$(9.55) \quad \int_{\Omega} \partial^\alpha u(x) v(x) dx = (-1)^{|\alpha|} \int_{\Omega} \partial^\alpha v(x) u(x) dx \\ + \sum_{\beta+\gamma+e_i=\alpha} (-1)^{|\gamma|} \frac{|\beta|! |\gamma|!}{|\alpha|!} \frac{\alpha!}{\beta! \gamma!} \int_{\partial\Omega} \partial^\beta u(x) \partial^\gamma v(x) \nu_i(x) d\mathcal{H}^{N-1}(x).$$

Distributions

Prospective Grad Students, IV: “Can you really live comfortably in this major metropolitan area with that stipend, or will I find myself living out of a closet working part time as a shoe salesman?”

— Jorge Cham, www.phdcomics.com

Throughout this chapter the implicit space is the Euclidean space $\mathbb{R}^N$ and $\Omega \subseteq \mathbb{R}^N$ is an open set, not necessarily bounded.

10.1. The Spaces $\mathcal{D}_K(\Omega)$, $\mathcal{D}(\Omega)$, and $\mathcal{D}'(\Omega)$

In this section, given an open set $\Omega \subseteq \mathbb{R}^N$, we construct a topology on the space $C_c^\infty(\Omega)$. We begin by considering the space of all functions with support in a given compact set $K \subset \Omega$. In this subspace the topology should be consistent with the natural notion of convergence, which is uniform convergence of the functions and of all their partial derivatives of any order.

Consider an open set $\Omega \subseteq \mathbb{R}^N$ and fix a compact set $K \subset \Omega$. Let $\mathcal{D}_K(\Omega)$ be the set of all functions in $C_c^\infty(\Omega)$ whose support is contained in K , that is,

$$\mathcal{D}_K(\Omega) := \{\phi \in C_c^\infty(\Omega) : \text{supp } \phi \subseteq K\}.$$

For each $j \in \mathbb{N}_0$ define the norm $\|\cdot\|_{K,j}$ on $\mathcal{D}_K(\Omega)$ by

$$(10.1) \quad \|\phi\|_{K,j} := \sup\{|\partial^\alpha \phi(x)| : x \in K, \alpha \in \mathbb{N}_0^N \text{ with } |\alpha| \leq j\}.$$

By Theorem A.27, the family of norms $\{\|\cdot\|_{K,j}\}_j$ turns $\mathcal{D}_K(\Omega)$ into a locally convex space and a base for the topology τ_K is given by all sets of the form

$$\{\phi \in \mathcal{D}_K(\Omega) : \|\phi\|_{K,j_1} < 1/\ell_1, \dots, \|\phi\|_{K,j_k} < 1/\ell_k\},$$

where $j_1, \dots, j_k \in \mathbb{N}_0$ and $\ell_1, \dots, \ell_k \in \mathbb{N}$, $k \in \mathbb{N}$. Taking $j := \max\{j_1, \dots, j_k\}$ and $\ell := \max\{\ell_1, \dots, \ell_k\}$, it follows that

$$(10.2) \quad \begin{aligned} V_{K,j,\ell} &:= \{\phi \in \mathcal{D}_K(\Omega) : \|\phi\|_{K,j} < 1/\ell\} \\ &\subseteq \{\phi \in \mathcal{D}_K(\Omega) : \|\phi\|_{K,j_1} < 1/\ell_1, \dots, \|\phi\|_{K,j_k} < 1/\ell_k\}, \end{aligned}$$

and so it suffices to consider as a local base for the topology τ_K the family of sets $V_{K,j,\ell}$, where $j \in \mathbb{N}_0$ and $\ell \in \mathbb{N}$.

Exercise 10.1. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $K \subset \Omega$ be a compact set. Define

$$(10.3) \quad d_K(\phi, \psi) := \max_{j \in \mathbb{N}_0} \frac{1}{2^j} \frac{\|\phi - \psi\|_{K,j}}{1 + \|\phi - \psi\|_{K,j}}, \quad \phi, \psi \in \mathcal{D}_K(\Omega).$$

- (i) Prove that d_K is a metric.
- (ii) Prove that the topology τ_K is determined by the metric d_K .
- (iii) Prove that $\mathcal{D}_K(\Omega)$ is complete.

To construct a topology on $C_c^\infty(\Omega)$, let $\mathcal{B}_0$ be the collection of all balanced,¹ convex sets $U \subseteq C_c^\infty(\Omega)$ such that

$$(10.4) \quad U \cap \mathcal{D}_K(\Omega) \in \tau_K$$

for every compact set $K \subset \Omega$.

Theorem 10.2. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. The family

$$\mathcal{B} := \{\phi + V : \phi \in C_c^\infty(\Omega), V \in \mathcal{B}_0\}$$

is a base for a locally convex Hausdorff topology τ on $C_c^\infty(\Omega)$ that turns $C_c^\infty(\Omega)$ into a topological vector space.

Proof. Step 1: We first prove that $\mathcal{B}_0$ is nonempty. Indeed, for every $j \in \mathbb{N}_0$ and $\ell \in \mathbb{N}$ define the norms

$$(10.5) \quad \|\phi\|_j := \sup\{|\partial^\alpha \phi(x)| : x \in \Omega, \alpha \in \mathbb{N}_0^N \text{ with } |\alpha| \leq j\}$$

and

$$(10.6) \quad V_{j,\ell} := \{\phi \in C_c^\infty(\Omega) : \|\phi\|_j < 1/\ell\}.$$

Then $V_{j,\ell}$ is balanced, convex, and

$$(10.7) \quad V_{j,\ell} \cap \mathcal{D}_K(\Omega) = \{\phi \in \mathcal{D}_K(\Omega) : \|\phi\|_{K,j} < 1/\ell\} = V_{K,j,\ell} \in \tau_K$$

by (10.2), and so $V_{j,\ell}$ belongs to $\mathcal{B}_0$.

Step 2: To prove that $\mathcal{B}$ is a base for a topology, it suffices to verify the following two conditions:

- (i) for every $\phi \in C_c^\infty(\Omega)$ there exists $U \in \mathcal{B}$ such that $\phi \in U$;

¹We recall that a subset E of a vector space X is *balanced* if $tx \in E$ for all $x \in E$ and $t \in [-1, 1]$.

- (ii) for every $U_1, U_2 \in \mathcal{B}$, with $U_1 \cap U_2 \neq \emptyset$, and for every $\phi \in U_1 \cap U_2$ there exists $U_3 \in \mathcal{B}$ such that $\phi \in U_3$ and $U_3 \subseteq U_1 \cap U_2$.

To prove (i), let $\phi_0 \in C_c^\infty(\Omega)$. Fix $j \in \mathbb{N}_0$ and $\ell \in \mathbb{N}$ and consider the set $V_{j,\ell}$ defined in (10.6). Then $V_{j,\ell} \in \mathcal{B}_0$, and so $U := \phi_0 + V_{j,\ell} \in \mathcal{B}$.

To verify property (ii), let $\phi_1, \phi_2 \in C_c^\infty(\Omega)$ and $V_1, V_2 \in \mathcal{B}_0$ be such that

$$(\phi_1 + V_1) \cap (\phi_2 + V_2) \neq \emptyset$$

and fix $\phi \in (\phi_1 + V_1) \cap (\phi_2 + V_2)$. Since the supports of ϕ_1, ϕ_2 , and ϕ are compact sets contained in Ω , we may find a compact set $K \subset \Omega$ such that

$$K \supseteq \text{supp } \phi_1 \cup \text{supp } \phi_2 \cup \text{supp } \phi.$$

Note that if $(\phi - \phi_i)(x) \neq 0$ for some $x \in \Omega$ and for some $i \in \{1, 2\}$, then necessarily $x \in \text{supp } \phi_i \cup \text{supp } \phi$, and so $\phi - \phi_i \in \mathcal{D}_K(\Omega)$, $i = 1, 2$. Since $\phi - \phi_i \in V_i \cap \mathcal{D}_K(\Omega) \in \tau_K$, $i = 1, 2$, using the continuity of scalar multiplication in $\mathcal{D}_K(\Omega)$, we may find $\theta \in (0, 1)$ such that

$$\phi - \phi_i \in (1 - \theta)(V_i \cap \mathcal{D}_K(\Omega)) \subseteq (1 - \theta)V_i$$

for $i = 1, 2$. By the convexity of the sets V_i we have that

$$\phi - \phi_i + \theta V_i \subseteq (1 - \theta)V_i + \theta V_i = V_i$$

for $i = 1, 2$, so that

$$\phi + \theta(V_1 \cap V_2) \subseteq (\phi_1 + V_1) \cap (\phi_2 + V_2).$$

Thus, $\mathcal{B}$ is a base for a topology τ given by all unions of members of $\mathcal{B}$.

Step 3: Next we show that $(C_c^\infty(\Omega), \tau)$ is a topological vector space.

To prove the continuity of scalar multiplication at a point $(t_0, \phi_0) \in \mathbb{R} \times C_c^\infty(\Omega)$, consider an (open) neighborhood $U \in \tau$ of $t_0\phi_0$. Since $\mathcal{B}$ is a base for the topology, we may find $V \in \mathcal{B}_0$ such that $t_0\phi_0 + V \subseteq U$. Let $K := \text{supp } \phi_0$. Then $\phi_0 \in \mathcal{D}_K(\Omega)$ and since $V \cap \mathcal{D}_K(\Omega) \in \tau_K$, by the continuity of scalar multiplication in $\mathcal{D}_K(\Omega)$ we may find $\delta > 0$ so small that

$$\delta\phi_0 \in \frac{1}{2}(V \cap \mathcal{D}_K(\Omega)) \subseteq \frac{1}{2}V.$$

Let $s := 1/[2(|t_0| + \delta)]$. Then for every $|t - t_0| < \delta$ and $\phi \in \phi_0 + sV$ we have that

$$t\phi - t_0\phi_0 = t(\phi - \phi_0) + (t - t_0)\phi_0 \in tsV + \frac{1}{2}V \subseteq \frac{1}{2}V + \frac{1}{2}V = V,$$

where we have used the fact that V is convex and balanced. Hence, $t\phi \in t_0\phi_0 + V \subseteq U$ for every $|t - t_0| < \delta$ and every $\phi \in \phi_0 + sV$, which proves continuity of scalar multiplication at the point (t_0, ϕ_0) .

To prove the continuity of addition at a point $(\phi_1, \phi_2) \in C_c^\infty(\Omega) \times C_c^\infty(\Omega)$, consider a neighborhood $U \in \tau$ of $\phi_1 + \phi_2$. Since $\mathcal{B}$ is a base for the topology,

we may find $V \in \mathcal{B}_0$ such that $\phi_1 + \phi_2 + V \subseteq U$. The convexity of $V \in \mathcal{B}_0$ implies that

$$(\phi_1 + \tfrac{1}{2}V) + (\phi_2 + \tfrac{1}{2}V) = \phi_1 + \phi_2 + V.$$

By (10.4) for every compact set $K \subset \Omega$ we have that $V \cap \mathcal{D}_K(\Omega) \in \tau_K$, and since the topology τ_K turns $\mathcal{D}_K(\Omega)$ into a topological vector space, it also follows that $\tfrac{1}{2}V \cap \mathcal{D}_K(\Omega) \in \tau_K$. Thus, $\tfrac{1}{2}V \in \mathcal{B}_0$ and, in turn, $\phi_1 + \tfrac{1}{2}V$, $\phi_2 + \tfrac{1}{2}V \in \mathcal{B}$.

Step 4: Finally, to prove that $(C_c^\infty(\Omega), \tau)$ is a Hausdorff topological vector space, it suffices to show that singletons are closed. Let $\phi_1, \phi_2 \in C_c^\infty(\Omega)$ be two distinct elements and define

$$V := \{\phi \in C_c^\infty(\Omega) : \sup_{x \in \Omega} |\phi(x)| < \sup_{x \in \Omega} |\phi_1(x) - \phi_2(x)|\}.$$

In view of (10.6) the set V belongs to $\mathcal{B}_0$ and $\phi_1 \notin \phi_2 + V$. Hence, $\{\phi_1\}$ is closed and the proof is completed. $\square$

The space $C_c^\infty(\Omega)$ endowed with the topology τ is denoted $\mathcal{D}(\Omega)$ and its elements are called *testing functions*.

A natural question is why we define τ in such a convoluted way, instead of directly considering the locally convex topology generated by the family of norms $\{\|\cdot\|_j\}_{j \in \mathbb{N}_0}$ defined in (10.5). The problem is that this topology would not be complete.

Exercise 10.3. Take $N = 1$, $\Omega = \mathbb{R}$, and consider a function $\phi \in C_c^\infty(\mathbb{R})$ with support in $[0, 1]$ such that $\phi > 0$ in $(0, 1)$. Prove that the sequence

$$\phi_n(x) := \phi(x-1) + \frac{1}{2}\phi(x-2) + \cdots + \frac{1}{n}\phi(x-n), \quad x \in \mathbb{R},$$

is a Cauchy sequence in the topology generated by the family of norms $\{\|\cdot\|_j\}_{j \in \mathbb{N}_0}$ defined in (10.5), but its limit does not have compact support, and so it does not belong to $C_c^\infty(\mathbb{R})$.

Exercise 10.4. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Let $\{K_n\}_{n \in \mathbb{N}_0}$ be an increasing sequence of compact sets of Ω , with $K_0 := \emptyset$, such that $\text{dist}(K_n, \partial K_{n+1}) > 0$ and $\Omega = \bigcup_{n=1}^\infty K_n$. Given two sequences $\mathbf{m} := \{m_n\}_{n \in \mathbb{N}_0}$ in $\mathbb{N}_0$ and $\mathbf{a} := \{a_n\}_{n \in \mathbb{N}_0}$ in $(0, \infty)$, with $m_n \rightarrow \infty$ and $a_n \rightarrow 0$, for every $\phi \in \mathcal{D}(\Omega)$ define

$$p_{\mathbf{m}, \mathbf{a}}(\phi) := \sup_{n \in \mathbb{N}_0} \sup_{x \in \Omega \setminus K_n} \frac{1}{a_n} \sum_{|\alpha| \leq m_n} |\partial^\alpha \phi(x)|.$$

- (i) Prove that $p_{\mathbf{m}, \mathbf{a}}$ is a seminorm.
- (ii) Prove that the family of seminorms $\{p_{\mathbf{m}, \mathbf{a}}\}_{\mathbf{m}, \mathbf{a}}$, where $\mathbf{m}$ and $\mathbf{a}$ vary among all sequences as above, generates the topology τ defined in Theorem 10.2.

We now show that the topology τ , when restricted to $\mathcal{D}_K(\Omega)$, for some compact set $K \subset \Omega$, does not produce more open sets than the ones in τ_K .

Theorem 10.5. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Then for every compact set $K \subset \Omega$ the topology τ_K coincides with the relative topology of $\mathcal{D}_K(\Omega)$ as a subset of $\mathcal{D}(\Omega)$.*

Proof. Fix a compact set $K \subset \Omega$ and let $U \in \tau$. We claim that $U \cap \mathcal{D}_K(\Omega)$ belongs to τ_K . To see this, it suffices to consider the case in which $U \cap \mathcal{D}_K(\Omega)$ is nonempty. Let $\phi \in U \cap \mathcal{D}_K(\Omega)$. Since $\mathcal{B}$ is a base for τ , there exists $V \in \mathcal{B}_0$ such that $\phi + V \subseteq U$. Hence,

$$\phi + (V \cap \mathcal{D}_K(\Omega)) \subseteq U \cap \mathcal{D}_K(\Omega).$$

Since $\phi \in \mathcal{D}_K(\Omega)$ and $V \cap \mathcal{D}_K(\Omega) \in \tau_K$, we have that $\phi + (V \cap \mathcal{D}_K(\Omega)) \in \tau_K$. This shows that every point of $U \cap \mathcal{D}_K(\Omega)$ is an interior point with respect to τ_K , and so $U \cap \mathcal{D}_K(\Omega) \in \tau_K$.

Conversely, let $U \in \tau_K$. We claim that

$$(10.8) \quad U = V \cap \mathcal{D}_K(\Omega)$$

for some $V \in \tau$. Since the family of sets $V_{K,j,\ell}$, where $j \in \mathbb{N}_0$ and $\ell \in \mathbb{N}$, is a local base for the topology τ_K (see (10.2)), for every $\phi \in U$ we may find $j_\phi \in \mathbb{N}_0$ and $\ell_\phi \in \mathbb{N}$ such that $\phi + V_{K,j_\phi,\ell_\phi} \subseteq U$. Let V_{j_ϕ,ℓ_ϕ} be defined as in (10.6). By Step 1 of the proof of Theorem 10.2,

$$(\phi + V_{j_\phi,\ell_\phi}) \cap \mathcal{D}_K(\Omega) = \phi + V_{K,j_\phi,\ell_\phi} \subseteq U$$

and $\phi + V_{j_\phi,\ell_\phi} \in \mathcal{B}$. In turn, the set $V := \bigcup_{\phi \in U} (\phi + V_{j_\phi,\ell_\phi})$ belongs to τ and (10.8) holds. $\square$

Exercise 10.6. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Prove that for every $x_0 \in \Omega$, for every compact set $K \subset \Omega$, and for every $r > 0$, the set

$$U := \{\phi \in \mathcal{D}_K(\Omega) : |\phi(x_0)| < r\}$$

is open with respect to τ_K .

Next we study topologically bounded sets in $\mathcal{D}(\Omega)$ (see Definition A.20). The following result will be used to study convergence with respect to the topology τ .

Theorem 10.7. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $W \subseteq \mathcal{D}(\Omega)$ be a topologically bounded set. Then there exists a compact set $K \subset \Omega$ such that $W \subseteq \mathcal{D}_K(\Omega)$. Moreover, for every $j \in \mathbb{N}_0$ there exists a constant $M_j > 0$ such that $\|\phi\|_j \leq M_j$ for all $\phi \in W$, where $\|\cdot\|_j$ is defined in (10.5).*

Proof. Assume by contradiction that W is not contained in $\mathcal{D}_K(\Omega)$ for any compact set $K \subset \Omega$. Let $\{K_n\}_n$ be an increasing sequence of compact sets of Ω such that $\text{dist}(K_n, \partial K_{n+1}) > 0$ and $\Omega = \bigcup_{n=1}^{\infty} K_n$. Then for each $n \in \mathbb{N}$

we may find a function $\phi_n \in W$ and a point $x_n \in K_{n+1} \setminus K_n$ such that $\phi_n(x_n) \neq 0$. Define

$$U := \{\phi \in \mathcal{D}(\Omega) : |\phi(x_n)| < |\phi_n(x_n)|/n \text{ for all } n \in \mathbb{N}\}.$$

Since each compact set $K \subset \Omega$ contains only finitely many x_n , by the previous exercise we have that $U \cap \mathcal{D}_K(\Omega) \in \tau_K$, and so $U \in \tau$. Using the fact that the set W is topologically bounded, we may find $t > 0$ such that $W \subseteq tU$. Consider an integer $n \geq t$. Then $\phi_n(x_n) \neq 0$ and $|\phi_n(x_n)|/t \geq |\phi_n(x_n)|/n$, which implies that $t^{-1}\phi_n \notin U$, or, equivalently, that $\phi_n \notin tU$, which is a contradiction. This shows that $W \subseteq \mathcal{D}_K(\Omega)$ for some compact set $K \subset \Omega$.

To prove the final part of the statement, note that by Theorem 10.5 the set $W = W \cap \mathcal{D}_K(\Omega)$ is bounded with respect to the topology τ_K , and so, by Corollary A.28, for each $j \in \mathbb{N}_0$ the set $\{\|\phi\|_j : \phi \in W\}$ is bounded in $\mathbb{R}$. $\square$

We are now ready to characterize convergence with respect to the topology τ and to prove the completeness of $\mathcal{D}(\Omega)$ (recall Exercise 10.3).

Theorem 10.8. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Then the space $\mathcal{D}(\Omega)$ is complete. Moreover, a sequence $\{\phi_n\}_n$ in $\mathcal{D}(\Omega)$ converges to $\phi \in \mathcal{D}(\Omega)$ with respect to τ if and only if*

- (i) *there exists a compact set $K \subset \Omega$ such that the support of every ϕ_n is contained in K ,*
- (ii) *$\lim_{n \rightarrow \infty} \partial^\alpha \phi_n = \partial^\alpha \phi$ uniformly on K for every multi-index α .*

Proof. Let $\{\phi_n\}_n$ be a Cauchy sequence in $\mathcal{D}(\Omega)$. By Proposition A.22, the set $\{\phi_n : n \in \mathbb{N}\}$ is topologically bounded. Hence, we may apply Theorem 10.7 to find a compact set $K \subset \Omega$ such that $\{\phi_n : n \in \mathbb{N}\}$ is contained in $\mathcal{D}_K(\Omega)$. In turn, by Theorem 10.5, we have that $\{\phi_n\}_n$ is a Cauchy sequence in $\mathcal{D}_K(\Omega)$. Hence, for all integers $j \in \mathbb{N}_0$ and $\ell \in \mathbb{N}$ there exists an integer $\bar{n} \in \mathbb{N}$ such that

$$\phi_n - \phi_k \in V_{K,j,\ell} = \{\phi \in \mathcal{D}_K(\Omega) : \|\phi\|_{K,j} < 1/\ell\}$$

for all $k, n \geq \bar{n}$; that is, for every multi-index α , with $|\alpha| \leq j$, we have that

$$\sup_{x \in K} |\partial^\alpha \phi_n(x) - \partial^\alpha \phi_k(x)| \leq 1/\ell.$$

This implies that $\{\partial^\alpha \phi_n\}_n$ is a Cauchy sequence in the space of continuous bounded functions, and so it converges uniformly in Ω to a function ψ_α with support in K . In particular, taking $\alpha = 0$, we have that ϕ_n converges uniformly in Ω to a function ψ_0 with support in K and $\partial^\alpha \psi_0 = \psi_\alpha$ for every multi-index α , with $|\alpha| \leq j$ (why?). Given the arbitrariness of $j \in \mathbb{N}_0$, we conclude that $\psi_0 \in \mathcal{D}_K(\Omega)$ and that the sequence $\{\phi_n\}_n$ converges to $\psi_0 \in \mathcal{D}(\Omega)$ with respect to τ . Thus, $\mathcal{D}(\Omega)$ is complete.

The proof of the second part of the statement is left as an exercise. $\square$

Exercise 10.9. Let $\Omega \subseteq \mathbb{R}^N$ be an open set.

- (i) Prove that for every compact $K \subset \Omega$, the space $\mathcal{D}_K(\Omega)$ is closed and has empty interior in $\mathcal{D}(\Omega)$.
- (ii) Prove that $\mathcal{D}(\Omega)$ is not metrizable.

In the previous exercise we have proved that $\mathcal{D}(\Omega)$ is not a metrizable space. Despite this fact, we can still prove that linear functionals defined on $\mathcal{D}(\Omega)$ are continuous if and only if they are sequentially continuous. Precisely, the following result holds.

Theorem 10.10. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T : \mathcal{D}(\Omega) \rightarrow \mathbb{R}$ be linear. Then the following properties are equivalent:*

- (i) *T is continuous.*
- (ii) *T is bounded.*
- (iii) *If $\{\phi_n\}_n$ in $\mathcal{D}(\Omega)$ converges to $\phi \in \mathcal{D}(\Omega)$ with respect to τ , then*

$$\lim_{n \rightarrow \infty} T(\phi_n) = T(\phi).$$

- (iv) *The restriction of T to $\mathcal{D}_K(\Omega)$ is continuous for every compact set $K \subset \Omega$.*
- (v) *For every compact set $K \subset \Omega$ there exist an integer $j \in \mathbb{N}_0$ and a constant $c_K > 0$ such that*

$$(10.9) \quad |T(\phi)| \leq c_K \|\phi\|_{K,j} \quad \text{for all } \phi \in \mathcal{D}_K(\Omega).$$

Proof. (i) $\Rightarrow$ (ii) If T is continuous, then T is bounded by Theorem A.31.

(ii) $\Rightarrow$ (iii) Assume that T is bounded and let $\{\phi_n\}_n$ in $\mathcal{D}(\Omega)$ converge to $\phi \in \mathcal{D}(\Omega)$ with respect to τ . Since $\mathcal{D}(\Omega)$ is a topological vector space, by replacing $\{\phi_n\}$ with $\{\phi_n - \phi\}$, without loss of generality, we may assume that $\{\phi_n\}_n$ converges to 0 with respect to τ . By Proposition A.22, the set $\{\phi_n : n \in \mathbb{N}\}$ is topologically bounded. Since T is bounded, it follows that the set $\{T(\phi_n) : n \in \mathbb{N}\}$ is bounded.

By Exercise 10.1 and Theorem 10.8 there is a compact set $K \subset \Omega$ such that $d_K(\phi_n, 0) \rightarrow 0$ as $n \rightarrow \infty$. Since $\mathcal{D}_K(\Omega)$ is metrizable, it follows by Theorem A.31 that $T(\phi_n) \rightarrow 0$.

(iii) $\Rightarrow$ (iv) Fix a compact set $K \subset \Omega$ and assume that $\{\phi_n\}$ is in $\mathcal{D}_K(\Omega)$ and such that $d_K(\phi_n, 0) \rightarrow 0$ as $n \rightarrow \infty$. By Theorem 10.8 we have that $\{\phi_n\}$ converges to ϕ with respect to τ . Hence, by property (iii), $T(\phi_n) \rightarrow 0$ as $n \rightarrow \infty$. Using Theorem A.31 once more, we get that the restriction of T to $\mathcal{D}_K(\Omega)$ is continuous.

(iv) $\Rightarrow$ (i) For every $\varepsilon > 0$ and for every compact set $K \subset \Omega$ the restriction of T to $\mathcal{D}_K(\Omega)$ is continuous at zero, and so $T^{-1}((-\varepsilon, \varepsilon)) \cap \mathcal{D}_K(\Omega) \in \tau_K$. Hence, $T^{-1}((-\varepsilon, \varepsilon)) \in \tau$, which shows that T is continuous at zero and, by linearity, everywhere.

(iv) $\Leftrightarrow$ (v) Assume that (iv) holds and fix a compact set $K \subset \Omega$. Since T restricted to $\mathcal{D}_K(\Omega)$ is continuous at the origin, given $\varepsilon = 1$ there exist $j \in \mathbb{N}_0$ and $\ell \in \mathbb{N}$ such that $V_{K,j,\ell} \subseteq T^{-1}((-1, 1))$, that is,

$$|T(\phi)| \leq 1$$

for all $\phi \in \mathcal{D}_K(\Omega)$ with $\|\phi\|_{K,j} < 1/\ell$. If $\phi \in \mathcal{D}_K(\Omega)$ and $\phi \neq 0$, then $\|\phi\|_{K,j} \neq 0$ and

$$\left\| \frac{1}{2\ell} \frac{\phi}{\|\phi\|_{K,j}} \right\|_{K,j} < \frac{1}{\ell}.$$

By the linearity of T it follows that $|T(\phi)| \leq 2\ell\|\phi\|_{K,j}$, which gives (iv). Conversely, if (v) holds, then by taking ℓ sufficiently large, we have that $|T(\phi)| \leq \varepsilon$ for all $\phi \in V_{K,j,\ell}$, which shows the continuity of T restricted to $\mathcal{D}_K(\Omega)$. $\square$

The dual of $\mathcal{D}(\Omega)$ is denoted $\mathcal{D}'(\Omega)$ and its elements are called *distributions*. We often use the duality notation $\langle T, \phi \rangle$ to denote $T(\phi)$. The space $\mathcal{D}'(\Omega)$ is given the weak star topology (see Section A.5 in Appendix A), so that a sequence $\{T_n\}_n$ in $\mathcal{D}'(\Omega)$ converges to $T \in \mathcal{D}'(\Omega)$ if $T_n(\phi) \rightarrow T(\phi)$ for every $\phi \in \mathcal{D}(\Omega)$. In this case we say that $\{T_n\}_n$ *converges to T in the sense of distributions*.

Exercise 10.11. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$.

- (i) Prove that if $\phi, \psi \in C^\infty(\Omega)$, then for every multi-index β the *Leibnitz formula*

$$\partial^\alpha(\phi\psi) = \sum_{\alpha \leq \beta} c_{\alpha\beta} \partial^{\beta-\alpha} \phi \partial^\alpha \psi$$

holds for some $c_{\alpha\beta} \in \mathbb{R}$.

- (ii) Prove that if $\psi \in C^\infty(\Omega)$, then the linear functional $\psi T : \mathcal{D}(\Omega) \rightarrow \mathbb{R}$, defined by

$$(\psi T)(\phi) := T(\psi\phi), \quad \phi \in \mathcal{D}(\Omega),$$

is a distribution.

10.2. Order of a Distribution

In this section we define the order of a distribution. We begin by observing that the integer $j \in \mathbb{N}_0$ in (10.9) may change with the compact set $K \subset \Omega$. If the same integer will do for all compact sets $K \subset \Omega$, then the smallest

integer $j \in \mathbb{N}_0$ for which (10.9) holds for all compact sets $K \subset \Omega$ is called the *order* of the distribution T . If no such integer exists, then the distribution T is said to have *infinite order*.

Example 10.12. Let $\Omega \subseteq \mathbb{R}^N$ be an open set.

- (i) Let λ be a signed Radon measure on Ω . The functional

$$T_\lambda(\phi) := \int_\Omega \phi d\lambda, \quad \phi \in \mathcal{D}(\Omega),$$

is a distribution of order zero.

- (ii) Fix $x_0 \in \Omega$. The functional δ_{x_0} , defined by $\delta_{x_0}(\phi) := \phi(x_0)$, $\phi \in \mathcal{D}(\Omega)$, is a distribution and is called the *delta Dirac with mass at x_0* . The distribution δ_{x_0} has order zero.

- (iii) Let $u \in L^1_{\text{loc}}(\Omega)$. The functional

$$T_u(\phi) := \int_\Omega \phi(x)u(x) dx, \quad \phi \in \mathcal{D}(\Omega),$$

is a distribution of order zero.

Actually, it turns out that all distributions of order zero may be identified with measures.

Theorem 10.13. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$.

- (i) If T is positive, that is, if $T(\phi) \geq 0$ for all nonnegative functions $\phi \in \mathcal{D}(\Omega)$, then there exists a unique Radon measure $\mu : \mathcal{B}(\Omega) \rightarrow [0, \infty]$ such that

$$T(\phi) = \int_\Omega \phi d\mu \quad \text{for all } \phi \in \mathcal{D}(\Omega).$$

- (ii) If T has order zero, then there exist two Radon measure $\mu_1, \mu_2 : \mathcal{B}(\Omega) \rightarrow [0, \infty]$ such that

$$T(\phi) = \int_\Omega \phi d\mu_1 - \int_\Omega \phi d\mu_2 \quad \text{for all } \phi \in \mathcal{D}(\Omega).$$

Proof. (i) We claim that T has order zero. Fix a compact set $K \subset \Omega$ and find an open set U such that $K \subset U \Subset \Omega$. Construct a smooth cut-off function $\varphi \in C^\infty(\Omega)$ such that $\varphi \equiv 1$ on K , $\text{supp } \varphi \subset U$, and $0 \leq \varphi \leq 1$ (see Exercise C.23). In particular, $\varphi \in \mathcal{D}(\Omega)$. Since $\varphi \equiv 1$ on K and $\varphi \geq 0$, for every $\phi \in \mathcal{D}_K(\Omega)$ we have that $|\phi(x)| \leq \|\phi\|_{K,0} \varphi(x)$ for all $x \in \Omega$, and so

$$T(\|\phi\|_{K,0} \varphi - \phi) \geq 0, \quad T(\phi + \|\phi\|_{K,0} \varphi) \geq 0;$$

that is, by the linearity of T ,

$$|T(\phi)| \leq \|\phi\|_{K,0} T(\varphi) \quad \text{for all } \phi \in \mathcal{D}_K(\Omega),$$

which shows that T has order zero.

Let $\{\Omega_i\}_i$ be an increasing sequence of bounded open sets of Ω such that $\Omega_i \Subset \Omega_{i+1}$ and $\Omega = \bigcup_{i=1}^{\infty} \Omega_i$. We start by showing that for every fixed $i \in \mathbb{N}$ the distribution T can be extended in a unique way as a linear continuous map on $C_c(\Omega_i)$. Since $K_i := \overline{\Omega_i}$ is a compact set contained in Ω and T has order zero, by what we just proved there exists a constant $c_i > 0$ such that

$$(10.10) \quad |T(\phi)| \leq c_i \|\phi\|_{K_i,0}$$

for all $\phi \in \mathcal{D}_{K_i}(\Omega)$. If $\phi \in C_c(\Omega_i)$, then $\text{dist}(\text{supp } \phi, \partial\Omega_i) > 0$, and thus if we consider $\phi_n := \varphi_{1/n} * \phi$, where $\varphi_{1/n}$ are standard mollifiers (with $\varepsilon := 1/n$) and $1/n < \text{dist}(\text{supp } \phi, \partial\Omega_i)$, we have that $\{\phi_n\}_n$ is contained in $\mathcal{D}_{K_i}(\Omega)$ and $\phi_n \rightarrow \phi$ uniformly on K_i . It follows by (10.10) that

$$|T(\phi_n - \phi_l)| \leq c_i \|\phi_n - \phi_l\|_{K_i,0} \rightarrow 0$$

as $l, n \rightarrow \infty$. Hence, $\{T(\phi_n)\}_n$ is a Cauchy sequence, and therefore it converges to a limit that we denote by $T_i(\phi)$. Moreover, if $\phi \geq 0$, then $\phi_n \geq 0$ also, and so $T_i(\phi) \geq 0$. Note that, again by (10.10), $T_i(\phi)$ is independent of the choice of the approximating sequence $\{\phi_n\}_n$.

By the linearity of T it follows that $T_i : C_c(\Omega_i) \rightarrow \mathbb{R}$ is linear and positive, while by (10.10) we have that $|T_i(\phi)| \leq c_i \|\phi\|_{C_c(\Omega_i)}$ for all $\phi \in C_c(\Omega_i)$. Since $\Omega_i \subset \Omega_{i+1}$, it follows that $T_{i+1}(\phi) = T_i(\phi)$ for all $\phi \in C_c(\Omega_i)$. Thus $\{T_i\}_i$ defines a unique linear positive extension of T to the union of all $C_c(\Omega_i)$, which coincides with $C_c(\Omega)$. The result now follows from the Riesz representation theorem in $C_c(\Omega)$ for positive linear functionals (see Theorem B.113).

(ii) The second part of the proof of (i) continues to hold in this case. By (10.10) and the fact that $\{\Omega_i\}_i$ covers Ω , we have that the extended functional is locally bounded, and so the result now follows from the Riesz representation theorem in $C_c(\Omega)$ for locally bounded linear functionals (see Theorem B.113). $\square$

10.3. Derivatives of Distributions and Distributions as Derivatives

We now define the notion of a derivative of a distribution.

Definition 10.14. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$. Given a multi-index $\alpha \in \mathbb{N}_0^N \setminus \{0\}$, we define the α th *derivative* of T as

$$\frac{\partial^{|\alpha|} T}{\partial x^\alpha}(\phi) := (-1)^{|\alpha|} \left(\frac{\partial^{|\alpha|} \phi}{\partial x^\alpha} \right), \quad \phi \in \mathcal{D}(\Omega).$$

For $j \in \mathbb{N}$ the symbol $\nabla^j T$ stands for the collection of all α th distributional derivatives of T with $|\alpha| = j$.

For simplicity we often write ∂^α for $\frac{\partial^{|\alpha|}}{\partial x^\alpha}$ and ∂_i^n for $\frac{\partial^n}{\partial x_i^n}$.

Remark 10.15. It can be verified that $\partial^\alpha T$ is still a distribution. Indeed, let $K \subset \Omega$ be a compact set. By Theorem 10.10 there exist an integer $j \in \mathbb{N}_0$ and a constant $c_K > 0$ such that

$$|T(\phi)| \leq c_K \|\phi\|_{K,j}$$

for all $\phi \in \mathcal{D}_K(\Omega)$. It follows that

$$|\partial^\alpha T(\phi)| = |T(\partial^\alpha \phi)| \leq c_K \|\partial^\alpha \phi\|_{K,j} \leq c_K \|\phi\|_{K,j+|\alpha|}$$

for all $\phi \in \mathcal{D}_K(\Omega)$, which shows that $\partial^\alpha T \in \mathcal{D}'(\Omega)$, again by Theorem 10.10.

In particular, if $u \in L^1_{\text{loc}}(\Omega)$ and α is a multi-index, then the α th *weak*, or *distributional*, *derivative* of u is the distribution $\partial^\alpha T_u$.

In a similar way we may define directional derivatives of a distribution (see Section 9.1).

Definition 10.16. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$. Given a unit vector $\nu \in \mathbb{R}^N \setminus \{0\}$ and $n \in \mathbb{N}$, we define the n th *directional derivative* of T in the direction ν as

$$\frac{\partial^n T}{\partial \nu^n}(\phi) := (-1)^n T\left(\frac{\partial^n \phi}{\partial \nu^n}\right), \quad \phi \in \mathcal{D}(\Omega).$$

In particular, if $u \in L^1_{\text{loc}}(\Omega)$, $\nu \in \mathbb{R}^N \setminus \{0\}$ and $n \in \mathbb{N}$, then the n th *weak*, or *distributional*, *directional derivative* of u in the direction ν is the distribution $\frac{\partial^n T_u}{\partial \nu^n}$.

Example 10.17. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in L^1_{\text{loc}}(\Omega)$. The *Laplacian* of u in the sense of distribution is the distribution $T_{\Delta u} := \sum_{i=1}^N \partial_i^2 T_u$, that is,

$$\begin{aligned} T_{\Delta u}(\phi) &= \sum_{i=1}^N \partial_i^2 T_u(\phi) = \sum_{i=1}^N T_u(\partial_i^2 \phi) \\ &= \sum_{i=1}^N \int_{\Omega} u \partial_i^2 \phi \, dx = \int_{\Omega} u \Delta \phi \, dx \end{aligned}$$

for all $\phi \in \mathcal{D}(\Omega)$. So a function $u \in L^1_{\text{loc}}(\Omega)$ is *subharmonic* if “ $\Delta u \geq 0$ ”, that is,

$$T_{\Delta u}(\phi) = \int_{\Omega} u \Delta \phi \, dx \geq 0$$

for all $\phi \in \mathcal{D}(\Omega)$ with $\phi \geq 0$. By Theorem 10.13 there exists a unique (positive) Radon measure $\mu : \mathcal{B}(\Omega) \rightarrow [0, \infty]$ such that

$$T_{\Delta u}(\phi) = \int_{\Omega} u \Delta \phi \, dx = \int_{\Omega} \phi \, d\mu \quad \text{for all } \phi \in \mathcal{D}(\Omega).$$

Thus, with an abuse of notation, we may write $\Delta u = \mu$.

Similarly, for $u \in L^1_{\text{loc}}(\Omega; \mathbb{R}^N)$ one can define the divergence of u in the sense of distributions.

Definition 10.18. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $u \in L^1_{\text{loc}}(\Omega)$, and let $\alpha \in \mathbb{N}_0^N \setminus \{0\}$ be a multi-index. If there exists a function $v_\alpha \in L^1_{\text{loc}}(\Omega)$ such that

$$T_{v_\alpha}(\phi) = \partial^\alpha T_u(\phi) \quad \text{for all } \phi \in \mathcal{D}(\Omega),$$

then v_α is called the α th *weak*, or *distributional*, *derivative* of T_u . We write $\partial^\alpha u = \frac{\partial^{|\alpha|} u}{\partial x^\alpha} := v_\alpha$.

Thus, a function $v_\alpha \in L^1_{\text{loc}}(\Omega)$ is the α th weak derivative of $u \in L^1_{\text{loc}}(\Omega)$ if

$$(10.11) \quad \int_{\Omega} \phi v_\alpha dx = (-1)^{|\alpha|} \int_{\Omega} u \partial^\alpha \phi dx$$

for all $\phi \in C_c^\infty(\Omega)$. Note that this is consistent with the integration by parts (9.55).

Exercise 10.19. Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be defined as follows:

$$u(x) := \begin{cases} \cos x & \text{if } -\pi \leq x \leq 0, \\ 1 - \frac{x}{\pi} & \text{if } 0 < x \leq \pi, \\ 0 & \text{otherwise.} \end{cases}$$

- (i) Calculate the derivative of u in the sense of distributions and find its order.
- (ii) Let v be the restriction of u in the interval $(-\pi, \pi)$. Calculate the first and second derivative of v in the sense of distributions in $(-\pi, \pi)$ and find their orders.
- (iii) Let $w : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable in $\mathbb{R} \setminus \{a\}$ and assume that there exist

$$\lim_{x \rightarrow a^-} w(x) \in \mathbb{R}, \quad \lim_{x \rightarrow a^+} w(x) \in \mathbb{R}.$$

Calculate the derivative of w in the sense of distributions. What can you say about its order? Under what conditions can you conclude that it has order zero?

Exercise 10.20. Assume that $u \in L^1((-\infty, \delta) \cup (\delta, \infty))$ for every $\delta > 0$ and define the *principal value integral*

$$\text{PV} \int_{-\infty}^{\infty} u(x) dx := \lim_{\delta \rightarrow 0^+} \left(\int_{\delta}^{\infty} u(x) dx + \int_{-\infty}^{-\delta} u(x) dx \right)$$

whenever the limit exists. For $\phi \in \mathcal{D}(\mathbb{R})$ define

$$T(\phi) := \int_{-\infty}^{\infty} \phi(x) \log |x| dx.$$

Prove that

$$T'(\phi) = \text{PV} \int_{-\infty}^{\infty} \phi(x) \frac{1}{x} dx, \quad T''(\phi) = -\text{PV} \int_{-\infty}^{\infty} \frac{\phi(x) - \phi(0)}{x^2} dx.$$

Exercise 10.21. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T_n \in \mathcal{D}'(\Omega)$ be such that the limit

$$T(\phi) := \lim_{n \rightarrow \infty} T_n(\phi)$$

exists in $\mathbb{R}$ for every $\phi \in \mathcal{D}(\Omega)$.

(i) Prove that $T \in \mathcal{D}'(\Omega)$.

(ii) Prove that for every multi-index α and for every $\phi \in \mathcal{D}(\Omega)$,

$$\partial^\alpha T(\phi) = \lim_{n \rightarrow \infty} \partial^\alpha T_n(\phi).$$

In the next few theorems we characterize distributions as weak derivatives of continuous functions.

Theorem 10.22. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$. Then for every compact set $K \subset \Omega$ there exist a continuous function $u : \Omega \rightarrow \mathbb{R}$ and a multi-index α (both depending on K) such that

$$T(\phi) = (-1)^{|\alpha|} \int_{\Omega} u \partial^\alpha \phi dx \quad \text{for all } \phi \in \mathcal{D}_K(\Omega).$$

Proof. In what follows, we use the notation (E.2) in Appendix E. Without loss of generality we may assume that $K \subseteq Q := [0, 1]^N$. By the mean value theorem, for every $\psi \in \mathcal{D}_Q(\mathbb{R}^N)$, for all $i = 1, \dots, N$, and for all $x \in Q$ we have

$$\begin{aligned} (10.12) \quad |\psi(x)| &= |\psi(x'_i, x_i) - \psi(x'_i, 0)| \\ &= |\partial_i \psi(x'_i, t)(x_i - 0)| \leq \max_Q |\partial_i \psi|. \end{aligned}$$

For $x \in Q$ set

$$Q(x) := \{y \in Q : 0 \leq y_i \leq x_i, i = 1, \dots, N\}.$$

Then for $\psi \in \mathcal{D}_Q(\mathbb{R}^N)$ and $x \in Q$ we have

$$\begin{aligned} \psi(x) &= \psi(0, x_2, \dots, x_N) + \int_0^{x_1} \partial_1 \psi(y_1, x_2, \dots, x_N) dy_1 \\ &= \int_0^{x_1} \partial_1 \psi(y_1, x_2, \dots, x_N) dy_1 \\ &= \int_0^{x_1} \left[\partial_1 \psi(y_1, 0, x_3, \dots, x_N) + \int_0^{x_2} \partial_{1,2}^2 \psi(y_1, y_2, x_3, \dots, x_N) dy_2 \right] dy_1 \\ &= \int_0^{x_1} \int_0^{x_2} \partial_{1,2}^2 \psi(y_1, y_2, x_3, \dots, x_N) dy_2 dy_1 = \int_{Q(x)} \partial^\beta \psi(y) dy, \end{aligned}$$

where $\beta := (1, \dots, 1)$, and so

$$(10.13) \quad \psi(x) = \int_{Q(x)} \partial^\beta \psi(y) dy \quad \text{for all } x \in Q.$$

Fix an integer $j \in \mathbb{N}_0$ (to be determined later) and let α be a multi-index with $|\alpha| \leq j$. By repeated applications of (10.12) we obtain that

$$(10.14) \quad \max_Q |\partial^\alpha \psi| \leq \max_Q |\partial^{j\beta} \psi| \leq \int_Q |\partial^{(j+1)\beta} \psi(y)| dy,$$

where $j\beta = (j, \dots, j)$ and where in the last inequality we have used (10.13).

By (10.13) the linear operator

$$\begin{aligned} D^\beta : \mathcal{D}_K(\Omega) &\rightarrow \mathcal{D}_K(\Omega) \\ \phi &\mapsto \partial^\beta \phi \end{aligned}$$

is one-to-one, and hence so is the linear operator

$$\begin{aligned} L := D^{j\beta} : \mathcal{D}_K(\Omega) &\rightarrow \mathcal{D}_K(\Omega) \\ \phi &\mapsto \partial^{(j+1)\beta} \phi, \end{aligned}$$

since $D^{j\beta} = D^\beta \circ \dots \circ D^\beta$, where the composition is done j times. Let $Y := L(\mathcal{D}_K(\Omega))$ and define the linear functional $T_1 : Y \rightarrow \mathbb{R}$ as follows. Given $\psi \in Y$, there exists a unique $\phi \in \mathcal{D}_K(\Omega)$ such that

$$\psi = L(\phi) = \partial^{(j+1)\beta} \phi.$$

Define $T_1(\psi) := T(\phi)$. Since $T \in \mathcal{D}'(\Omega)$, by Theorem 10.10 there exist an integer $j \in \mathbb{N}_0$ and a constant $c_K > 0$ such that for all $\phi \in \mathcal{D}_K(\Omega)$,

$$|T(\phi)| \leq c_K \|\phi\|_{K,j} \leq c_K \int_K |\partial^{(j+1)\beta} \phi| dx,$$

where in the last inequality we have used (10.14). It follows from the definition of T_1 that

$$|T_1(\psi)| \leq c_K \int_K |\psi| dx$$

for all $\psi \in Y$, and thus we may apply the Hahn–Banach theorem (see Theorem A.32) to extend T_1 as a continuous linear functional defined in $L^1(K)$. By the Riesz representation theorem in $L^1(K)$ (see Theorem B.93) there exists a function $v \in L^\infty(K)$ such that

$$T_1(\psi) = \int_K v \psi dx$$

for all $\psi \in L^1(K)$. In particular, if $\phi \in \mathcal{D}_K(\Omega)$, then

$$T(\phi) = T_1(\partial^{(j+1)\beta} \phi) = \int_K v \partial^{(j+1)\beta} \phi dx.$$

Extend v by zero outside K and define

$$u(x) := \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_N} v(y) dy_N \cdots dy_1.$$

Then u is continuous and by integrating by parts N times we have that for all $\phi \in \mathcal{D}_K(\Omega)$,

$$T(\phi) = (-1)^N \int_K u \partial^{(j+2)\beta} \phi dx.$$

To complete the proof, we may define $\alpha := (j+2)\beta$ and, if needed, replace u with $-u$. $\square$

The previous result is local in the sense that the function u and the multi-index α change with K . Next we show that if T has compact support, then the previous local result becomes global.

Definition 10.23 (Support of a distribution). Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$. If $U \subseteq \Omega$ is open, then we write that $T = 0$ in U if $T(\phi) = 0$ for all $\phi \in \mathcal{D}(U)$. The *support* of T is the complement of V relative to Ω , where V is the union of all open subsets $U \subseteq \Omega$ in which $T = 0$.

The support of T will be written as $\text{supp } T$.

Exercise 10.24. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$. Show that $T = 0$ in the (possibly empty) open set $\Omega \setminus \text{supp } T$. Prove also that $\partial^\alpha T = 0$ in $\Omega \setminus \text{supp } T$ for every multi-index α .

We now prove a global version of Theorem 10.22 for distributions with compact support.

Theorem 10.25. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$ be such that $\text{supp } T$ is a compact set of Ω . Then

- (i) there exist an integer $j \in \mathbb{N}_0$ and a constant $c > 0$ such that $|T(\phi)| \leq c \|\phi\|_j$ for all $\phi \in \mathcal{D}(\Omega)$ (in particular, T has finite order $\ell \leq j$), where $\|\cdot\|_j$ is defined in (10.5),
- (ii) if U is an open set, with $\text{supp } T \subset U \subseteq \Omega$, then for each multi-index α , with $\alpha \leq \beta := (\ell+2, \dots, \ell+2)$, there exists a function $v_\alpha \in C(\Omega)$, with $\text{supp } v_\alpha \subset U$, such that

$$T(\phi) = \sum_{\alpha \leq \beta} \partial^\alpha T_{v_\alpha}(\phi) = \sum_{\alpha} (-1)^{|\alpha|} \int_{\Omega} v_\alpha \partial^\alpha \phi dx \quad \text{for all } \phi \in \mathcal{D}(\Omega).$$

Proof. (i) Consider an open set U , with $\text{supp } T \subset U \subseteq \Omega$, and (see Exercise C.23) construct a function $\psi \in \mathcal{D}(\Omega)$ such that $\psi = 1$ on U . We claim that $\psi T = T$. Indeed, if $\phi \in \mathcal{D}(\Omega)$, then since $\psi = 1$ on U , we have that $\phi - \psi\phi = 0$ on U , and so $\text{supp}(\phi - \psi\phi) \cap \text{supp } T = \emptyset$, which implies that

$T(\phi - \psi\phi) = 0$, or, equivalently, $T(\phi) = T(\psi\phi) = (\psi T)(\phi)$. Hence, the claim holds.

Since $T \in \mathcal{D}'(\Omega)$, by Theorem 10.10 there exist an integer $j \in \mathbb{N}_0$ and a constant $c_K > 0$ such that $|T(\phi)| \leq c_K \|\phi\|_{K,j}$ for all $\phi \in \mathcal{D}_K(\Omega)$, where $K := \text{supp } \psi$. On the other hand, if $\phi \in \mathcal{D}(\Omega)$, then $\psi\phi \in \mathcal{D}_K(\Omega)$, and so, since $\psi T = T$,

$$|T(\phi)| = |T(\psi\phi)| \leq c_K \|\psi\phi\|_{K,j}.$$

By the Leibnitz formula (see Exercise 10.11), $\|\psi\phi\|_{K,j} \leq c_\psi \|\phi\|_{K,j}$, and so $|T(\phi)| \leq c_K c_\psi \|\phi\|_{K,j}$, which shows part (i).

(ii) Consider an open set W , with $\text{supp } T \subset W \Subset U$. By Theorem 10.22 with $\overline{W}$ in place of K there exists a continuous function $u : \Omega \rightarrow \mathbb{R}$ such that

$$T(\phi) = (-1)^{|\beta|} \int_{\Omega} u \partial^\beta \phi \, dx$$

for all $\phi \in \mathcal{D}_{\overline{W}}(\Omega)$, where $\beta := (\ell + 2, \dots, \ell + 2)$. Consider an open set V , with $\text{supp } T \subset V \Subset W$, and as in (i) construct a function $\psi \in \mathcal{D}(\Omega)$ such that $\psi = 1$ on V and $K := \text{supp } \psi \subset W$. Then, by (i), for all $\phi \in \mathcal{D}(\Omega)$ we have that $\psi\phi \in \mathcal{D}_{\overline{W}}(\Omega)$ and, by the Leibnitz formula (see Exercise 10.11) again,

$$\begin{aligned} T(\phi) &= T(\psi\phi) = (-1)^{|\beta|} \int_{\Omega} u \partial^\beta (\psi\phi) \, dx \\ &= \sum_{\alpha \leq \beta} c_{\alpha\beta} (-1)^{|\beta|} \int_{\Omega} u \partial^{\beta-\alpha} \psi \partial^\alpha \phi \, dx, \end{aligned}$$

and so it suffices to take $v_\alpha := c_{\alpha\beta} (-1)^{|\beta-\alpha|} u \partial^{\beta-\alpha} \psi$. This concludes the proof. $\square$

Finally, using partitions of unity, we have a similar representation for every distribution.

Theorem 10.26. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$. Then for each multi-index α there exists a function $v_\alpha \in C(\Omega)$ such that*

- (i) *each compact set $K \subset \Omega$ intersects the support of finitely many v_α ,*
- (ii) *for all $\phi \in \mathcal{D}(\Omega)$,*

$$T(\phi) = \sum_{\alpha} \partial^\alpha T_{v_\alpha}(\phi) = \sum_{\alpha} (-1)^{|\alpha|} \int_{\Omega} v_\alpha \partial^\alpha \phi \, dx.$$

If T has finite order, then only finitely many v_α different from zero are needed.

Proof. (i) Let S be a countable dense set in Ω , e.g., $S := \{x \in \mathbb{Q}^N \cap \Omega\}$, and consider the countable family $\mathcal{F}$ of open cubes

$$\mathcal{F} := \{Q(x, r) : r \in \mathbb{Q}, 0 < r < 1, x \in S, Q(x, r) \subseteq \Omega\}.$$

Since $\mathcal{F}$ is countable, we may write $\mathcal{F} = \{Q(x_n, r_n)\}_{n \in \mathbb{N}}$. By the density of S and of the rational numbers we have that $\Omega = \bigcup_{n=1}^{\infty} \overline{Q(x_n, r_n/2)}$. Note that every compact set $K \subset \Omega$ intersects only finitely many cubes $Q(x_n, r_n)$. For each $n \in \mathbb{N}$ construct a function $\phi_n \in \mathcal{D}(\Omega)$ such that $\phi_n = 1$ on $\overline{Q(x_n, r_n/2)}$ and $\text{supp } \phi_n \subset Q(x_n, r_n)$. Use this family to construct a partition of unity $\{\psi_n\}_n$ in $\mathcal{D}(\Omega)$ with $\text{supp } \psi_n \subset Q(x_n, r_n)$ for each $n \in \mathbb{N}$ (see the proof of Theorem C.21). For each $n \in \mathbb{N}$ the distribution $\psi_n T$ has support contained in $Q(x_n, r_n)$, and so by Theorem 10.25 it has finite order ℓ_n and we may find finitely many functions $\{v_{\alpha,n}\}$ in $C(\Omega)$, with $\alpha \leq \beta_n := (\ell_n + 2, \dots, \ell_n + 2)$ and $\text{supp } v_{\alpha,n} \subset Q(x_n, r_n)$, such that

$$(\psi_n T)(\phi) = \sum_{\alpha \leq \beta_n} (-1)^{|\alpha|} \int_{\Omega} v_{\alpha,n} \partial^{\alpha} \phi \, dx$$

for all $\phi \in \mathcal{D}(\Omega)$. For every multi-index α for which $\alpha \leq \beta_n$ fails define $v_{\alpha,n} := 0$. Thus,

$$(\psi_n T)(\phi) = \sum_{\alpha} (-1)^{|\alpha|} \int_{\Omega} v_{\alpha,n} \partial^{\alpha} \phi \, dx$$

for all $\phi \in \mathcal{D}(\Omega)$.

Hence, for every multi-index α we may define $v_{\alpha} := \sum_{n=1}^{\infty} v_{\alpha,n}$. Since each compact set intersects the support of only finitely many $v_{\alpha,n}$, the function v_{α} is continuous and (i) holds.

To prove (ii), let $\phi \in \mathcal{D}(\Omega)$. Since $\{\psi_n\}_n$ is a partition of unity, we have that $\phi = \sum_{n=1}^{\infty} \psi_n \phi$, where the sum is actually finite since ϕ has compact support. Since the sum is finite and T is linear, we have that

$$\begin{aligned} T(\phi) &= \sum_{n=1}^{\infty} T(\psi_n \phi) = \sum_{n=1}^{\infty} (\psi_n T)(\phi) = \sum_{n=1}^{\infty} \sum_{\alpha} (-1)^{|\alpha|} \int_{\Omega} v_{\alpha,n} \partial^{\alpha} \phi \, dx \\ &= \sum_{\alpha} (-1)^{|\alpha|} \int_{\Omega} \sum_{n=1}^{\infty} v_{\alpha,n} \partial^{\alpha} \phi \, dx = \sum_{\alpha} (-1)^{|\alpha|} \int_{\Omega} v_{\alpha} \partial^{\alpha} \phi \, dx. \end{aligned}$$

This shows (ii).

Finally, if T has finite order ℓ , then $\psi_n T$ has finite order $\ell_n \leq \ell$, and so it suffices to consider only multi-indices $\alpha \leq \beta := (\ell + 2, \dots, \ell + 2)$. $\square$

Exercise 10.27. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Prove that every continuous linear functional on $C^{\infty}(\Omega)$ is of the form $\phi \mapsto T(\phi)$, where T is a distribution with compact support.

Exercise 10.28. Let $\Omega = (a_1, b_1) \times \cdots \times (a_N, b_N)$.

- (i) Prove that $\phi \in \mathcal{D}(\Omega)$ is such that $\int_{\Omega} \phi(x) dx = 0$ if and only if $\phi = \sum_{i=1}^N \partial_i \phi_i$ for some $\phi_1, \dots, \phi_N \in \mathcal{D}(\Omega)$. Hint: Use induction on N and look at Step 1 of the proof of Lemma 7.4.
- (ii) Prove that if $T \in \mathcal{D}'(\Omega)$ is such that $\frac{\partial T}{\partial x_i} = 0$ for all $i = 1, \dots, N$, then there exists a constant $c \in \mathbb{R}$ such that $T(\phi) = c \int_{\Omega} \phi(x) dx$ for all $\phi \in \mathcal{D}(\Omega)$, i.e., T is constant.
- (iii) Prove that if $\Omega \subseteq \mathbb{R}^N$ is an open connected set and $T \in \mathcal{D}'(\Omega)$ is such that $\partial_i T = 0$ for all $i = 1, \dots, N$, then T is constant.

10.4. Rapidly Decreasing Functions and Tempered Distributions

In view of the applications to Fourier transforms in this section we consider complex-valued functions. We recall that for a complex number $z = x + iy$, where $x, y \in \mathbb{R}$, the complex conjugate of z is the number $\bar{z} := x - iy$ and the norm of z is

$$\|z\| := \sqrt{x^2 + y^2}.$$

Definition 10.29. The space of rapidly decreasing functions $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is the space of all functions $\phi : \mathbb{R}^N \rightarrow \mathbb{C}$ of class C^∞ such that for all multi-indices $\alpha, \beta \in \mathbb{N}_0^N$,

$$(10.15) \quad \|\phi\|_{\alpha, \beta} := \sup_{x \in \mathbb{R}^N} \|x^\alpha \partial^\beta \phi(x)\| < \infty.$$

Thus, $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ consists of all functions that, together with all their derivatives, decay to zero faster than any polynomial.

Remark 10.30. The space $C_c^\infty(\mathbb{R}^N; \mathbb{C})$ of all C^∞ functions $\phi : \mathbb{R}^N \rightarrow \mathbb{C}$ with compact support is contained in $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$. The function $\phi(x) := e^{-\|x\|^2}$, $x \in \mathbb{R}^N$, is an example of a function in $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ without compact support.

Theorem 10.31. The space $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ with the topology induced by the family of seminorms $\|\cdot\|_{\alpha, \beta}$ is a Fréchet space.

Proof. Since there are countably many seminorms $\|\cdot\|_{\alpha, \beta}$, the space $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is metrizable, with metric

$$d(\phi, \psi) := \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{\|\phi - \psi\|_{\alpha_n, \beta_n}}{1 + \|\phi - \psi\|_{\alpha_n, \beta_n}}.$$

It remains to show that it is complete. Let $\{\phi_n\}_n$ be a Cauchy sequence in $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$. Then $\{x^\alpha \partial^\beta \phi_n\}_n$ is a Cauchy sequence in $C_b(\mathbb{R}^N; \mathbb{C})$ for every $\alpha, \beta \in \mathbb{N}_0^+$ and thus it converges uniformly to a function $\psi_{\alpha, \beta}$. Let $\phi := \psi_{0,0}$.

By the fundamental theorem of calculus for $x \in \mathbb{R}^N$, $t \in \mathbb{R}$ and $i = 1, \dots, N$,

$$\phi_n(x + te_i) = \phi_n(x) + \int_0^t \frac{\partial \phi_n}{\partial x_i}(x + se_i) ds.$$

Letting $n \rightarrow \infty$ it follows by uniform convergence that

$$\phi(x + te_i) = \phi(x) + \int_0^t \psi_{0,e_i}(x + se_i) ds.$$

Hence, there exists $\frac{\partial \phi}{\partial x_i} = \psi_{0,e_i}$. This proves that ϕ is of class C^1 . In a similar way we can show that ϕ is of class C^∞ with $\psi_{\alpha,\beta} = x^\alpha \partial^\beta \phi$. Thus, $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ and since $\|\phi_n - \phi\|_{\alpha,\beta} \rightarrow 0$ for all $\alpha, \beta \in \mathbb{N}_0^+$, it follows that $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is complete. $\square$

Definition 10.32. The dual of $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is called the *space of tempered distributions* and is denoted $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

The following theorem is important for applications. For $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ and $m, n \in \mathbb{N}_0$ we define

$$\|\phi\|_{m,n} := \sum_{|\alpha|=m} \sum_{|\beta|=n} \|\phi\|_{\alpha,\beta}.$$

Theorem 10.33. A linear functional $T : \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathbb{C}$ is continuous if and only if there exist a constant $c > 0$ and $m, n \in \mathbb{N}_0$ such that

$$(10.16) \quad \|T(\phi)\| \leq c \|\phi\|_{m,n} \quad \text{for every } \phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C}).$$

Proof. Exercise. $\square$

In view of the previous theorem, the restriction of a tempered distribution $T : \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathbb{C}$ to $\mathcal{D}(\mathbb{R}^N; \mathbb{C})$ defines a distribution.

Corollary 10.34. Given $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$, its restriction to $\mathcal{D}(\mathbb{R}^N; \mathbb{C})$ is a distribution.

Proof. By the previous theorem there exist a constant $c > 0$ and $m, n \in \mathbb{N}_0$ such that $\|T(\phi)\| \leq c \|\phi\|_{m,n}$ for every $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$. Given a compact set $K \subset \mathbb{R}^N$, let $R > 0$ be such that $K \subseteq \overline{B(0, R)}$. If $\phi \in \mathcal{D}_K(\mathbb{R}^N; \mathbb{C})$, then $\sup_{x \in \mathbb{R}^N} \|x^\alpha \partial^\beta \phi(x)\| \leq (1 + R^m) \sup_{x \in K} \|\partial^\beta \phi(x)\|$ and so

$$|T(\phi)| \leq (1 + R^m) \|\phi\|_{K,n}$$

for all $\phi \in \mathcal{D}_K(\mathbb{R}^N; \mathbb{C})$, where $\|\cdot\|_{K,n}$ is defined in (10.1). It follows from Theorem 10.10 that $T : \mathcal{D}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathbb{C}$ is continuous. $\square$

Next we show that $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is embedded in L^p for every p .

Theorem 10.35. For all $1 \leq p \leq \infty$, $\mathcal{S}(\mathbb{R}^N; \mathbb{C}) \hookrightarrow L^p(\mathbb{R}^N; \mathbb{C}) \hookrightarrow \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

Proof. We only need to consider the case $1 \leq p < \infty$. For $p = 1$, write

$$\begin{aligned} \int_{\mathbb{R}^N} \|\phi(x)\| dx &= \int_{\mathbb{R}^N} \frac{1 + \|x\|^{N+1}}{1 + \|x\|^{N+1}} \|\phi(x)\| dx \\ &\leq c \|\phi\|_{N+1,0} \int_{\mathbb{R}^N} \frac{1}{1 + \|x\|^{N+1}} dx. \end{aligned}$$

For $1 < p < \infty$ it is enough to observe that

$$\int_{\mathbb{R}^N} \|\phi(x)\|^p dx \leq \|\phi\|_{\infty}^{p-1} \int_{\mathbb{R}^N} \|\phi(x)\| dx \leq c \|\phi\|_{N+1,0}^p.$$

This shows that $\mathcal{S}(\mathbb{R}^N; \mathbb{C}) \hookrightarrow L^p(\mathbb{R}^N; \mathbb{C})$. Given $\psi \in L^p(\mathbb{R}^N; \mathbb{C})$, consider the linear functional $T_\psi : \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathbb{C}$ defined by

$$(10.17) \quad T_\psi(\phi) := \int_{\mathbb{R}^N} \phi \psi dx, \quad \phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C}).$$

Then by Hölder's inequality

$$\|T_\psi(\phi)\| \leq \|\phi\|_{L^{p'}} \|\psi\|_{L^p} \leq c \|\phi\|_{N+1,0} \|\psi\|_{L^p}.$$

Hence, by (10.16) the functional T_ψ belongs to $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ and the linear mapping $\psi \in L^p(\mathbb{R}^N; \mathbb{C}) \mapsto T_\psi$ is an embedding. $\square$

Remark 10.36. In what follows we identify ψ with T_ψ . Hence, $L^p(\mathbb{R}^N; \mathbb{C})$, and in particular $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$, can be thought as contained in $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

Remark 10.37. Note that if $f : \mathbb{R}^N \rightarrow \mathbb{C}$ is a polynomial, then

$$T_f(\phi) := \int_{\mathbb{R}^N} \phi f dx, \quad \phi \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C}),$$

is well-defined and $\|T_f(\phi)\| \leq c_f \|\phi\|_{d+N+1,0}$, where d is the degree of f and c_f is the supremum of the norms of the coefficients of f . By (10.16) the functional T_f belongs to $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

Example 10.38. Given a measure $\mu : \mathcal{B}(\mathbb{R}^N) \rightarrow [0, \infty]$ with the property that

$$\mu(\overline{B(0, r)}) \leq c_0(1 + r)^k$$

for some $c_0 > 0$, some $k \in \mathbb{N}$, and for all $r > 0$, the linear functional $T_\mu : \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathbb{C}$ defined by $T_\mu(\phi) := \int_{\mathbb{R}^N} \phi d\mu$ is well-defined and continuous.

Indeed, write

$$\begin{aligned} \int_{\mathbb{R}^N} \|\phi\| d\mu &= \int_{B(0,1)} \|\phi\| d\mu + \sum_{n=2}^{\infty} \int_{\overline{B(0,n)} \setminus B(0,n-1)} \|\phi\| d\mu \\ &\leq 2c_0 \|\phi\|_{\infty} + \sum_{n=2}^{\infty} \int_{B(0,n)} \frac{(1+\|x\|)^{2k}}{(1+\|x\|)^{2k}} \|\phi\| d\mu \\ &\leq 2c_0 \|\phi\|_{\infty} + cc_0 \|\phi\|_{0,2k} \sum_{n=1}^{\infty} \frac{(1+n)^k}{(1+n)^{2k}}. \end{aligned}$$

Hence by (10.16), $T_{\mu} \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

Exercise 10.39. Prove that the linear mapping

$$T(\phi) := \text{PV}\left(\frac{1}{x}\right)(\phi) := \lim_{\delta \rightarrow 0^+} \int_{\mathbb{R} \setminus [-\delta, \delta]} \frac{\phi(x)}{x} dx, \quad \phi \in \mathcal{S}(\mathbb{R}),$$

is well-defined and belongs to $\mathcal{S}'(\mathbb{R}; \mathbb{C})$.

Exercise 10.40. Let $\psi : \mathbb{R}^N \rightarrow \mathbb{C}$ be a function of class C^{∞} such that for every multi-index $\alpha \in \mathbb{N}_0^N$ there exist c_{α} and $n_{\alpha} \in \mathbb{N}$ such that

$$(10.18) \quad \|\partial^{\alpha} \psi(x)\| \leq c_{\alpha} (1 + \|x\|^2)^{n_{\alpha}}$$

for all $x \in \mathbb{R}^N$.

- (i) Prove that if $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ then $\phi\psi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$.
- (ii) Prove that if $h : \mathbb{R}^N \rightarrow \mathbb{C}$ is a Lebesgue measurable function such that $h\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ for all $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ and the mapping $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \mapsto h\phi$ is continuous, then h must be of class C^{∞} and satisfy (10.18).
- (iii) Given $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ prove that the linear functional $\psi T : \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathbb{C}$ defined by $(\psi T)(\phi) := T(\phi\psi)$, $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, belongs to $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

As in Definition 10.18 we can define the notion of a derivative of a tempered distribution.

Definition 10.41. Given $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ and a multi-index $\alpha \in \mathbb{N}_0^N \setminus \{0\}$, we define the α th derivative of T as the linear functional $\partial^{\alpha} T : \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathbb{C}$ defined by

$$\frac{\partial^{|\alpha|} T}{\partial x^{\alpha}}(\phi) := (-1)^{|\alpha|} T\left(\frac{\partial^{|\alpha|} \phi}{\partial x^{\alpha}}\right), \quad \phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C}).$$

For simplicity we often write ∂^{α} for $\frac{\partial^{|\alpha|}}{\partial x^{\alpha}}$ and ∂_i^n for $\frac{\partial^n}{\partial x_i^n}$.

Theorem 10.42. For every $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ and every multi-index α , the functional $\partial^{\alpha} T$ belongs to $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

Proof. Since $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$, by Theorem 10.33 there exist a constant $c > 0$ and $m, n \in \mathbb{N}_0$ such that $\|T(\phi)\| \leq c\|\phi\|_{m,n}$ for every $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$. In turn, since for $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, $\partial^\alpha \phi$ still belongs to $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$, with

$$\|(\partial^\alpha T)(\phi)\| = \|T(\partial^\alpha \phi)\| \leq c\|\partial^\alpha \phi\|_{m,n} \leq c\|\phi\|_{m,n+|\alpha|},$$

and so, again by Theorem 10.33 it follows that $\partial^\alpha T$ belongs to $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$. $\square$

Exercise 10.43. Prove that if $T \in \mathcal{D}'(\mathbb{R}^N; \mathbb{C})$ has compact support, then $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

Exercise 10.44. Prove that $\mathcal{D}(\mathbb{R}^N; \mathbb{C})$ is dense in $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ and that the inclusion

$$\begin{aligned} \mathcal{D}(\mathbb{R}^N; \mathbb{C}) &\rightarrow \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \\ T &\mapsto T \end{aligned}$$

is continuous.

Exercise 10.45. Prove that if f is a polynomial, $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, and $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$, then fT and ϕT belong to $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

10.5. Convolutions

Given two measurable functions $\phi : \mathbb{R}^N \rightarrow \mathbb{C}$ and $\psi : \mathbb{R}^N \rightarrow \mathbb{C}$, the *convolution* of ϕ and ψ is the function $\phi * \psi$ defined by

$$(10.19) \quad (\phi * \psi)(x) := \int_{\mathbb{R}^N} \phi(x - y)\psi(y) dy$$

for all $x \in \mathbb{R}^N$ for which the right-hand side is well-defined.

Theorem 10.46. *Given $\phi, \psi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, the function $\phi * \psi$ belongs to $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$.*

Proof. Fix $x \in \mathbb{R}^N$. For $m \in \mathbb{N}$ with $m > N$, we can write

$$\begin{aligned} \|(\phi * \psi)(x)\| &\leq \int_{\mathbb{R}^N} \|\phi(x - y)\| \|\psi(y)\| dy \\ &\leq c\|\psi\|_{0,m} \|\phi\|_{0,m} \int_{\mathbb{R}^N} \frac{1}{(1 + \|y\|)^m} \frac{1}{(1 + \|x - y\|)^m} dy. \end{aligned}$$

We now split $\mathbb{R}^N$ into the sets $E := \{y \in \mathbb{R}^N : \frac{1}{2}\|x\| \leq \|x - y\|\}$ and $\mathbb{R}^N \setminus E$. Then we have

$$\begin{aligned} &\int_E \frac{1}{(1 + \|y\|)^m} \frac{1}{(1 + \|x - y\|)^m} dy \\ &\leq \frac{2^m}{(2 + \|x\|)^m} \int_{\mathbb{R}^N} \frac{1}{(1 + \|y\|)^m} dy \leq \frac{c}{(2 + \|x\|)^m}, \end{aligned}$$

while in $\mathbb{R}^N \setminus E$, $\|y\| \geq \|x\| - \|x - y\| \geq \|x\| - \frac{1}{2}\|x\| = \frac{1}{2}\|x\|$, and so

$$\begin{aligned} & \int_{\mathbb{R}^N \setminus E} \frac{1}{(1 + \|y\|)^m} \frac{1}{(1 + \|x - y\|)^m} dy \\ & \leq \frac{2^m}{(2 + \|x\|)^m} \int_{\mathbb{R}^N} \frac{1}{(1 + \|x - y\|)^m} dy \leq \frac{c}{(2 + \|x\|)^m}, \end{aligned}$$

where $c = c(m, N) > 0$. Hence,

$$(2 + \|x\|)^m \|(\phi * \psi)(x)\| \leq c \|\psi\|_{0,m} \|\phi\|_{0,m}.$$

This shows that $\phi * \psi$ decays to zero faster than any power of $\|x\|$.

On the other hand, by differentiating under the integral sign, for every multi-index α ,

$$\frac{\partial^{|\alpha|}(\phi * \psi)}{\partial x^\alpha}(x) = \int_{\mathbb{R}^N} \frac{\partial^{|\alpha|}\phi}{\partial x^\alpha}(x - y) \psi(y) dy = \left(\frac{\partial^{|\alpha|}\phi}{\partial x^\alpha} * \psi \right)(x),$$

and so by repeating the same calculations above with ϕ replaced by $\frac{\partial^{|\alpha|}\phi}{\partial x^\alpha}$, we get that all derivatives of $\phi * \psi$ decay to zero faster than any power of $\|x\|$, which shows that $\phi * \psi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$. $\square$

Exercise 10.47. Prove that for every $\phi, \psi, \varphi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$,

$$(\phi * \psi) * \varphi = \phi * (\psi * \varphi).$$

Theorem 10.48 (Young's inequality). *Let $\phi \in L^p(\mathbb{R}^N; \mathbb{C})$, $1 \leq p \leq \infty$, and $\psi \in L^1(\mathbb{R}^N; \mathbb{C})$. Then $(\phi * \psi)(x)$ exists for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ and*

$$\|\phi * \psi\|_{L^p(\mathbb{R}^N)} \leq \|\phi\|_{L^p(\mathbb{R}^N)} \|\psi\|_{L^1(\mathbb{R}^N)}.$$

Proof. Consider two Borel functions ϕ_0 and ψ_0 such that $\phi_0(x) = \phi(x)$ and $\psi_0(x) = \psi(x)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. Since the integral in (10.19) is unchanged if we replace ϕ and ψ with ϕ_0 and ψ_0 , respectively, in what follows, without loss of generality, we may assume that ϕ and ψ are Borel functions.

Let $h : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{C}$ be the function defined by $h(x, y) := \phi(x - y)$, $(x, y) \in \mathbb{R}^N \times \mathbb{R}^N$. Then h is a Borel function, since it is the composition of the Borel function ϕ with the continuous function $\psi : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}^N$ given by $\psi(x, y) := x - y$. In turn, the function

$$(x, y) \in \mathbb{R}^N \times \mathbb{R}^N \mapsto \phi(x - y) \psi(y)$$

is Borel measurable. We are now in a position to apply Minkowski's inequality for integrals (Corollary B.83) and Tonelli's theorem to conclude

that

$$\begin{aligned}\|\phi * \psi\|_{L^p} &= \left\| \int_{\mathbb{R}^N} \|\phi(\cdot - y)\psi(y)\| dy \right\|_{L^p} \\ &\leq \int_{\mathbb{R}^N} \|\phi(\cdot - y)\psi(y)\|_{L^p} dy \\ &= \int_{\mathbb{R}^N} \|\psi(y)\| \|\phi(\cdot - y)\|_{L^p} dy = \|\phi\|_{L^p} \int_{\mathbb{R}^N} \|\psi(y)\| dy,\end{aligned}$$

where in the last equality we have used the fact that the Lebesgue measure is translation invariant. Hence, $\phi * \psi$ belongs to $L^p(\mathbb{R}^N; \mathbb{C})$, and so it is finite $\mathcal{L}^N$ -a.e. in $\mathbb{R}^N$. $\square$

The following is the generalized form of the previous inequality.

Theorem 10.49 (Young's inequality, general form). *Let $1 \leq p \leq q \leq \infty$ and let $\phi \in L^p(\mathbb{R}^N; \mathbb{C})$ and $\psi \in L^q(\mathbb{R}^N; \mathbb{C})$. Then $(\phi * \psi)(x)$ exists for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ and*

$$\|\phi * \psi\|_{L^r(\mathbb{R}^N)} \leq \|\phi\|_{L^p(\mathbb{R}^N)} \|\psi\|_{L^q(\mathbb{R}^N)},$$

where

$$(10.20) \quad 1/p + 1/q = 1 + 1/r.$$

Proof. If $r = \infty$, then q is the Hölder conjugate exponent of p and the result follows from Hölder's inequality and the translation invariance of the Lebesgue measure, while if $p = 1$, then $r = q$ and the result follows from the previous theorem (with ϕ and ψ interchanged).

In the remaining cases, write

$$\|\phi(x - y)\psi(y)\| = (\|\phi(x - y)\|^p \|\psi(y)\|^q)^{1/r} \|\phi(x - y)\|^{(r-p)/r} \|\psi(y)\|^{(r-q)/r}.$$

Define $p_1 := r$, $p_2 := pr/(r - p)$, $p_3 := qr/(r - q)$. Then $1/p_1 + 1/p_2 + 1/p_3 = 1$, and so by the extended Hölder inequality (see Exercise B.79) and the translation invariance of the Lebesgue measure,

$$\begin{aligned}\|(\phi * \psi)(x)\| &\leq \int_{\mathbb{R}^N} \|\phi(x - y)\psi(y)\| dy \\ &\leq \left(\int_{\mathbb{R}^N} \|\phi(x - y)\|^p \|\psi(y)\|^q dy \right)^{1/r} \|\phi\|_{L^p}^{1-p/r} \|\psi\|_{L^q}^{1-q/r}.\end{aligned}$$

Taking the norm in $L^r(\mathbb{R}^N)$ on both sides, we get

$$\|\phi * \psi\|_{L^r} \leq \left(\int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \|\phi(x - y)\|^p \|\psi(y)\|^q dy dx \right)^{1/r} \|\phi\|_{L^p}^{1-p/r} \|\psi\|_{L^q}^{1-q/r}.$$

Applying the previous theorem (with $p = 1$), we get that the right-hand side of the previous inequality is less than or equal to

$$(\|\phi\|_{L^1}^p \|\psi\|_{L^1}^q)^{1/r} \|\phi\|_{L^p}^{1-p/r} \|\psi\|_{L^q}^{1-q/r} = \|\phi\|_{L^p} \|\psi\|_{L^q}.$$

This concludes the proof. $\square$

10.6. Convolution of Distributions

In this section we define the convolution of a distribution (or a tempered distribution) T and a function φ . We begin with the case in which $T = T_u$ for some function $u \in L^1_{\text{loc}}(\mathbb{R}^N)$, where we recall that $T_u \in \mathcal{D}'(\mathbb{R}^N)$ is defined by

$$T_u(\phi) := \int_{\mathbb{R}^N} \psi(x)\phi(x) \, dx, \quad \phi \in \mathcal{D}(\mathbb{R}^N).$$

By Fubini's theorem

$$\begin{aligned} \int_{\mathbb{R}^N} (u * \varphi)(x)\phi(x) \, dx &= \int_{\mathbb{R}^N} \phi(x) \int_{\mathbb{R}^N} u(x-y)\varphi(y) \, dy \, dx \\ &= \int_{\mathbb{R}^N} u(\xi) \int_{\mathbb{R}^N} \varphi(x-\xi)\phi(x) \, dx \, d\xi \\ &= \int_{\mathbb{R}^N} u(\xi) \int_{\mathbb{R}^N} \tilde{\varphi}(\xi-x)\phi(x) \, dx \, d\xi \\ &= \int_{\mathbb{R}^N} u(\xi)(\tilde{\varphi} * \phi)(\xi) \, d\xi, \end{aligned}$$

where $\xi := x - y$ and $\tilde{\varphi}(x) := \varphi(-x)$. Hence, we have shown that $T_{u*\varphi}(\phi) = T_u(\tilde{\varphi} * \phi)$ for all $\phi \in \mathcal{D}(\mathbb{R}^N)$. Motivated by this formula we define:

Definition 10.50. If $T \in \mathcal{D}'(\mathbb{R}^N)$ and $\varphi \in \mathcal{D}(\mathbb{R}^N)$ the *convolution* of T and φ is the linear functional $T * \varphi : \mathcal{D}(\mathbb{R}^N) \rightarrow \mathbb{R}$ defined by $(T * \varphi)(\phi) := T(\tilde{\varphi} * \phi)$, where

$$(10.21) \quad \tilde{\varphi}(x) := \varphi(-x), \quad x \in \mathbb{R}^N.$$

It turns out that $T * \varphi$ can be identified with a function. Given $x \in \mathbb{R}^N$ and a function $\varphi : \mathbb{R}^N \rightarrow \mathbb{R}$ we define the function

$$(10.22) \quad \varphi^x(y) := \varphi(x - y), \quad y \in \mathbb{R}^N.$$

Exercise 10.51. Let $\phi, \varphi \in \mathcal{D}(\mathbb{R}^N)$. For $h > 0$ define

$$u_h(x) := h^N \sum_{y \in \mathbb{Z}^N} \varphi(x - hy)\phi(hy), \quad x \in \mathbb{R}^N.$$

Prove that $\{u_h\}_h$ converges uniformly to $\varphi * \phi$ as $h \rightarrow 0^+$. Hint: Use Riemann sums.

Theorem 10.52. Let $T \in \mathcal{D}'(\mathbb{R}^N)$ and $\phi, \psi \in \mathcal{D}(\mathbb{R}^N)$. Then $T * \varphi = T_{u_\varphi}$, where u_φ is the function given by $u_\varphi(x) := T(\varphi^x)$, $x \in \mathbb{R}^N$. Moreover,

- (i) $u_\varphi \in C^\infty(\mathbb{R}^N)$,
- (ii) $\text{supp } u_\varphi \subseteq \text{supp } T + \text{supp } \varphi$,

- (iii) for every multi-index $\alpha \in \mathbb{N}_0^N$, $\partial^\alpha u_\varphi = T * \partial^\alpha \varphi = \partial^\alpha T * \varphi$,
 (iv) $(T * \varphi) * \psi = T * (\varphi * \psi)$.

Proof. Step 1: Fix $\phi \in \mathcal{D}(\mathbb{R}^N)$. We approximate the function $\tilde{\varphi} * \phi$ (see (10.21)) with the Riemann sum

$$u_h(x) := h^N \sum_{y \in \mathbb{Z}^N} \tilde{\varphi}(x - hy) \phi(hy), \quad x \in \mathbb{R}^N,$$

where $h > 0$. Note that for every multi-index α ,

$$\frac{\partial^{|\alpha|} u_h}{\partial x^\alpha}(x) = h^N \sum_{y \in \mathbb{Z}^N} \frac{\partial^{|\alpha|} \tilde{\varphi}}{\partial x^\alpha}(x - hy) \phi(hy), \quad x \in \mathbb{R}^N.$$

Since $\left\{ \frac{\partial^{|\alpha|} u_h}{\partial x^\alpha} \right\}_h$ converges uniformly to $\frac{\partial^{|\alpha|} \tilde{\varphi}}{\partial x^\alpha} * \phi$ by the previous exercise and

$$\frac{\partial^{|\alpha|} \tilde{\varphi}}{\partial x^\alpha} * \phi = \frac{\partial^{|\alpha|}}{\partial x^\alpha} (\tilde{\varphi} * \phi)$$

(why?), by Theorem 10.8 we have that $\{u_h\}_h$ converges to $\tilde{\varphi} * \phi$ as $h \rightarrow 0$ with respect to the topology τ in $\mathcal{D}(\mathbb{R}^N)$.

Hence, by the continuity and linearity of T and by Theorem 10.10 we have that

$$\begin{aligned} (T * \varphi)(\phi) &= T(\tilde{\varphi} * \phi) = \lim_{h \rightarrow 0} T(u_h) = \lim_{h \rightarrow 0} h^N \sum_{y \in \mathbb{Z}^N} T(\tilde{\varphi}(\cdot - hy)) \phi(hy) \\ &= \lim_{h \rightarrow 0} h^N \sum_{y \in \mathbb{Z}^N} T(\varphi(hy - \cdot)) \phi(hy) = \int_{\mathbb{R}^N} T(\varphi^y) \phi(y) dy, \end{aligned}$$

where we have used the previous exercise. This shows that $T * \varphi = T_{u_\varphi}$. In what follows we identify $T * \varphi$ with u_φ .

Step 2: If $x_n \rightarrow x$ in $\mathbb{R}^N$, then for every $y \in \mathbb{R}^N$,

$$\varphi^{x_n}(y) = \varphi(x_n - y) \rightarrow \varphi(x - y) = \varphi^x(y)$$

and conditions (i) and (ii) of Theorem 10.8 are satisfied. Hence, $\{\varphi^{x_n}\}_n$ converges to φ^x with respect to τ , and so by Theorem 10.10,

$$(T * \varphi)(x_n) = T(\varphi^{x_n}) \rightarrow T(\varphi^x) = (T * \varphi)(x),$$

which proves that $T * \varphi$ is a continuous function.

To prove (ii), note that if $x \in \mathbb{R}^N$ is such that $\text{supp } \varphi^x \cap \text{supp } T = \emptyset$, then $(T * \varphi)(x) = 0$. Thus,

$$\text{supp}(T * \varphi) \subseteq \{x \in \mathbb{R}^N : \text{supp } \varphi^x \cap \text{supp } T \neq \emptyset\} = \text{supp } T + \text{supp } \varphi.$$

Next we prove (iii). Let e_i be an element of the canonical basis of $\mathbb{R}^N$ and for every $x \in \mathbb{R}^N$ and $h \neq 0$ consider the function

$$\varphi^{x,h,i}(y) := \frac{\varphi(x + he_i - y) - \varphi(x - y)}{h}, \quad y \in \mathbb{R}^N.$$

As $h \rightarrow 0$, we have that $\varphi^{x,h,i}(y) \rightarrow \frac{\partial \varphi}{\partial x_i}(x - y)$ for all $y \in \mathbb{R}^N$ and conditions (i) and (ii) of Theorem 10.8 are satisfied (why?). Hence, $\{\varphi^{x,h,i}\}_h$ converges to $(\partial_i \varphi)^x$ with respect to τ as $h \rightarrow 0$, and so, by the linearity of T and Theorem 10.10,

$$\frac{(T * \varphi)(x + he_i) - (T * \varphi)(x)}{h} = T(\varphi^{x,h,i}) \rightarrow T((\partial_i \varphi)^x)$$

as $h \rightarrow 0$, which proves that $\partial_i(T * \varphi) = T * \partial_i \varphi$.

Moreover, since for all $x, y \in \mathbb{R}^N$,

$$\left(\frac{\partial \varphi}{\partial x_i}\right)^x(y) = \frac{\partial \varphi}{\partial x_i}(x - y) = -\frac{\partial \varphi}{\partial y_i}(x - y) = -\frac{\partial \varphi^x}{\partial y_i}(y),$$

for all $x \in \mathbb{R}^N$ we have

$$\begin{aligned} \left(\frac{\partial T}{\partial y_i} * \varphi\right)(x) &= \frac{\partial T}{\partial y_i}(\varphi^x) = -T\left(\frac{\partial \varphi^x}{\partial y_i}\right) \\ &= T\left(-\left(\frac{\partial \varphi}{\partial y_i}\right)^x\right) = \left(T * \frac{\partial \varphi}{\partial x_i}\right)(x), \end{aligned}$$

which, together with an induction argument, gives (iii).

Finally, to prove (iv), we define

$$u_h(x) := h^N \sum_{y \in \mathbb{Z}^N} \varphi(x - hy) \psi(hy), \quad x \in \mathbb{R}^N,$$

where $h > 0$. As before we have that $\{u_h\}_h$ converges to $\varphi * \psi$ with respect to τ as $h \rightarrow 0$. It follows that for every $x \in \mathbb{R}^N$, $\{(u_h)^x\}_h$ converges to $(\varphi * \psi)^x$ with respect to τ as $h \rightarrow 0$. By the linearity of T and by Theorem 10.10 we have that

$$\begin{aligned} (T * (\varphi * \psi))(x) &= T((\varphi * \psi)^x) = \lim_{h \rightarrow 0} T((u_h)^x) = \lim_{h \rightarrow 0} (T * u_h)(x) \\ &= \lim_{h \rightarrow 0} h^N \sum_{y \in \mathbb{Z}^N} (T * \varphi)(x - hy) \psi(hy) \\ &= ((T * \varphi) * \psi)(x), \end{aligned}$$

where we have used the previous exercise. This completes the proof. $\square$

In what follows we identify $T * \varphi$ with u_φ . As a consequence of the previous theorem, we can approximate distributions with C^∞ functions.

Theorem 10.53. Let $T \in \mathcal{D}'(\mathbb{R}^N)$ and let $\{\varphi_\varepsilon\}_\varepsilon$, $\varepsilon > 0$, be a family of standard mollifiers². Then $\{T * \varphi_\varepsilon\}_\varepsilon$ converges to T in the sense of distributions as $\varepsilon \rightarrow 0^+$; that is,

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^N} (T * \varphi_\varepsilon)(x) \phi(x) dx = T(\phi)$$

for every $\phi \in \mathcal{D}(\mathbb{R}^N)$.

Proof. By Theorems C.16(i) and C.20 and Theorem 10.8 we have that for every $\phi \in \mathcal{D}(\mathbb{R}^N)$ the sequence $\{\varphi_\varepsilon * \phi\}$ converges to ϕ with respect to τ as $\varepsilon \rightarrow 0^+$. Then by Theorem 10.10,

$$\begin{aligned} T(\phi) &= (T * \tilde{\phi})(0) = T((\tilde{\phi})^0) = \lim_{\varepsilon \rightarrow 0^+} T((\varphi_\varepsilon * \tilde{\phi})^0) \\ &= \lim_{\varepsilon \rightarrow 0^+} (T * (\varphi_\varepsilon * \tilde{\phi}))(0) = \lim_{\varepsilon \rightarrow 0^+} ((T * \varphi_\varepsilon) * \tilde{\phi})(0) \\ &= \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^N} (T * \varphi_\varepsilon)(y) \tilde{\phi}(0 - y) dy = \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^N} (T * \varphi_\varepsilon)(y) \phi(y) dy, \end{aligned}$$

where we have used Theorem 10.52(iv). $\square$

Exercise 10.54. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $T \in \mathcal{D}'(\Omega)$.

- (i) Prove that there exists a sequence $\{T_n\}_n$ in $\mathcal{D}'(\Omega)$ such that each T_n has support compactly contained in Ω and $\{T_n\}$ converges to T in the sense of distributions.
- (ii) Prove that $C_c^\infty(\Omega)$ is dense in $\mathcal{D}'(\Omega)$ with respect to the weak star topology of $\mathcal{D}'(\Omega)$.

Similarly we can define the convolution of a tempered distribution and a function. We recall that $\tilde{\varphi}(x) := \varphi(-x)$, $x \in \mathbb{R}^N$ (see (10.21)).

Definition 10.55. If $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ and $\varphi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ the *convolution* of T and φ is the continuous linear functional $T * \varphi : \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathbb{C}$ defined by $(T * \varphi)(\phi) := T(\tilde{\varphi} * \phi)$, $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$.

The proof of the following theorem is similar to the one of Theorem 10.52 and is left as an exercise. We recall that $\varphi^x(y) = \varphi(x - y)$, $y \in \mathbb{R}^N$.

Theorem 10.56. If $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ and $\varphi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, then $T * \varphi = T_{u_\varphi}$, where u_φ is the function given by $u_\varphi(x) := T(\varphi^x)$, $x \in \mathbb{R}^N$. Moreover, $u_\varphi \in C^\infty(\mathbb{R}^N; \mathbb{C})$ and for every multi-index α there exist $c_\alpha > 0$ and $n_\alpha \in \mathbb{N}$ such that

$$\|\partial^\alpha u_\varphi(x)\| \leq c_\alpha (1 + \|x\|^2)^{n_\alpha}$$

for all $x \in \mathbb{R}^N$.

²See Appendix C.

In what follows we identify $T * \varphi$ with the function u_φ .

Exercise 10.57. Let $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ and $\varphi, \psi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$. Prove that $(T * \varphi) * \psi = (T * \psi) * \varphi$.

10.7. Fourier Transforms

Given $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, the *Fourier transform* of ϕ is the function

$$(10.23) \quad \widehat{\phi}(x) = \mathcal{F}(\phi)(x) := \int_{\mathbb{R}^N} e^{-2\pi i x \cdot y} \phi(y) dy, \quad x \in \mathbb{R}^N,$$

while the *inverse Fourier transform* of ϕ is the function

$$(10.24) \quad \phi^\vee(x) := \widehat{\phi}(-x) = \int_{\mathbb{R}^N} e^{2\pi i x \cdot y} \phi(y) dy, \quad x \in \mathbb{R}^N.$$

Since $\mathcal{S}(\mathbb{R}^N; \mathbb{C}) \subset L^1(\mathbb{R}^N; \mathbb{C})$ (see Theorem 10.35), the functions $\widehat{\phi}$ and $\phi^\vee$ are well-defined.

Theorem 10.58. *The Fourier transform $\mathcal{F}$ maps $\mathcal{S}(\mathbb{R}^N)$ into $\mathcal{S}(\mathbb{R}^N)$. Moreover, for every $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ and for every $\alpha, \beta \in \mathbb{N}_0^N$,*

$$(10.25) \quad x^\alpha \partial^\beta \widehat{\phi}(x) = \widehat{\psi_{\alpha, \beta}}(x) \quad \text{for every } x \in \mathbb{R}^N,$$

where $\psi_{\alpha, \beta}(x) := \partial^\alpha((-2\pi i x)^\beta \phi(x))$. In particular, $x^\alpha \widehat{\phi}(x) = \widehat{\partial^\alpha \phi}(x)$.

Proof. By differentiating under the integral sign we have that

$$\frac{\partial^{|\beta|} \widehat{\phi}}{\partial x^\beta}(x) = \int_{\mathbb{R}^N} \frac{\partial^{|\beta|}}{\partial x^\beta} (e^{-2\pi i x \cdot y}) \phi(y) dy = \int_{\mathbb{R}^N} (-2\pi i y)^\beta e^{-2\pi i x \cdot y} \phi(y) dy.$$

Hence,

$$\begin{aligned} x^\alpha \frac{\partial^{|\beta|} \widehat{\phi}}{\partial x^\beta}(x) &= \frac{1}{(-2\pi i)^{|\alpha|}} \int_{\mathbb{R}^N} (-2\pi i y)^\beta (-2\pi i x)^\alpha e^{-2\pi i x \cdot y} \phi(y) dy \\ &= \frac{1}{(-2\pi i)^{|\alpha|}} \int_{\mathbb{R}^N} (-2\pi i y)^\beta \phi(y) \frac{\partial^{|\alpha|}}{\partial y^\alpha} (e^{-2\pi i x \cdot y}) dy. \end{aligned}$$

By integrating by parts (see (9.55)) and using the fact that ϕ and its derivatives decay to zero at infinity we get

$$x^\alpha \frac{\partial^{|\beta|} \widehat{\phi}}{\partial x^\beta}(x) = \int_{\mathbb{R}^N} e^{-2\pi i x \cdot y} \frac{\partial^{|\alpha|}}{\partial y^\alpha} ((-2\pi i y)^\beta \phi(y)) dy,$$

which proves (10.25). It follows from Leibnitz rule (see Exercise 10.11) that

$$\begin{aligned}\|\widehat{\phi}\|_{\alpha,\beta} &\leq (2\pi)^{|\beta|} \int_{\mathbb{R}^N} \left\| \frac{\partial^{|\alpha|}}{\partial y^\alpha} ((-y)^\beta \phi(y)) \right\| dy \\ &= (2\pi)^{|\beta|} \int_{\mathbb{R}^N} \frac{1 + \|y\|^{N+1}}{1 + \|y\|^{N+1}} \left\| \frac{\partial^{|\alpha|}}{\partial y^\alpha} ((-y)^\beta \phi(y)) \right\| dy \\ &\leq c \|\phi\|_{N+1+|\alpha|,|\beta|},\end{aligned}$$

where $c = c(N, \alpha, \beta) > 0$, which shows that $\widehat{\phi} \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ and that the linear operator $\mathcal{F} : \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is continuous. $\square$

Exercise 10.59. Prove that the Fourier transform of the function $\phi(x) := e^{-\pi\|x\|^2}$, $x \in \mathbb{R}^N$, is ϕ .

Exercise 10.60. Prove that the Fourier transform of the function

$$\phi_\varepsilon(x) := e^{2\pi i x \cdot x_0} e^{-\pi \varepsilon^2 \|x\|^2}, \quad x \in \mathbb{R}^N,$$

where $\varepsilon > 0$ and $x_0 \in \mathbb{R}^N$, is

$$\widehat{\phi}_\varepsilon(x) = \frac{1}{\varepsilon^N} e^{-\pi \|(x-x_0)/\varepsilon\|^2}, \quad x \in \mathbb{R}^N.$$

Next we prove that $\mathcal{F}$ is invertible with inverse given by $\mathcal{F}^{-1}(\phi) = \phi^\vee$ (see (10.24)).

Proposition 10.61. *For every $\phi, \psi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, we have*

$$(10.26) \quad \int_{\mathbb{R}^N} \phi(x) \widehat{\psi}(x) dx = \int_{\mathbb{R}^N} \widehat{\phi}(x) \psi(x) dx.$$

Proof. By Fubini's theorem

$$\begin{aligned}\int_{\mathbb{R}^N} \phi(x) \widehat{\psi}(x) dx &= \int_{\mathbb{R}^N} \phi(x) \int_{\mathbb{R}^N} e^{-2\pi i x \cdot y} \psi(y) dy dx \\ &= \int_{\mathbb{R}^N} \psi(y) \int_{\mathbb{R}^N} e^{-2\pi i x \cdot y} \phi(x) dx dy = \int_{\mathbb{R}^N} \psi(y) \widehat{\phi}(y) dy,\end{aligned}$$

which shows (10.26). $\square$

Theorem 10.62 (Fourier inversion theorem). *For every $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$,*

$$(\widehat{\phi})^\vee = \widehat{(\phi^\vee)} = \phi.$$

In particular, the Fourier transform $\mathcal{F}$ is an isomorphism from $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ to $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ with inverse $\mathcal{F}^{-1}$ given by $\mathcal{F}^{-1}(\phi) = \phi^\vee$ for every $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$.

Proof. Fix $x_0 \in \mathbb{R}^N$ and $\varepsilon > 0$ and define $\psi_\varepsilon(x) := e^{2\pi i x \cdot x_0} e^{-\pi \varepsilon^2 \|x\|^2}$, $x \in \mathbb{R}^N$. By Exercise 10.60 we have that $\widehat{\psi}_\varepsilon(x) = \frac{1}{\varepsilon^N} e^{-\pi \|(x-x_0)/\varepsilon\|^2}$ and so, taking $\psi = \psi_\varepsilon$ in (10.26), we get

$$\int_{\mathbb{R}^N} \phi(x) \frac{1}{\varepsilon^N} e^{-\pi \|(x-x_0)/\varepsilon\|^2} dx = \int_{\mathbb{R}^N} e^{2\pi i y \cdot x_0} e^{-\pi \varepsilon^2 \|y\|^2} \widehat{\phi}(y) dy.$$

Note that $\widehat{\psi_\varepsilon}$ is a mollifier. Hence, the left-hand side converges to $\phi(x_0)$ by Exercise C.19 in Appendix C. On the other hand, by the Lebesgue dominated convergence theorem the right-hand side converges to $(\widehat{\phi})^\vee(x_0)$. Hence, $\phi(x_0) = (\widehat{\phi})^\vee(x_0)$, which shows that $(\widehat{\phi})^\vee = \phi$. Similarly we can show that, $(\phi^\vee)^\vee = \phi$.

Next observe that if $\widehat{\phi} = 0$, then $\phi = (\widehat{\phi})^\vee = 0^\vee = 0$, and so $\mathcal{F}$ is one-to-one. Since $(\widehat{\phi^\vee}) = \phi$, it follows that $\mathcal{F}$ is onto and that the inverse of $\mathcal{F}$ is $\mathcal{F}^{-1}(\phi) = \phi^\vee$. $\square$

We recall that for a complex number $z = x + iy$, where $x, y \in \mathbb{R}$, the complex conjugate of z is the number $\bar{z} := x - iy$.

Corollary 10.63 (Plancherel). *For every $\phi, h \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$,*

$$(10.27) \quad \int_{\mathbb{R}^N} \phi(x) \overline{h(x)} dx = \int_{\mathbb{R}^N} \widehat{\phi}(x) \overline{\widehat{h}(x)} dx$$

and

$$(10.28) \quad \int_{\mathbb{R}^N} \|\phi(x)\|^2 dx = \int_{\mathbb{R}^N} \|\widehat{\phi}(x)\|^2 dx = \int_{\mathbb{R}^N} \|\phi^\vee(x)\|^2 dx.$$

In particular, $\mathcal{F}$ extends uniquely to an isomorphism of $L^2(\mathbb{R}^N; \mathbb{C})$ onto itself.

The identity (10.27) is called *Parseval identity*, while the identity (10.28) is called *Plancherel identity*.

Proof. Let $\psi := \widehat{\widehat{h}}$. Then, using the facts that the cosine is even and the sine is odd, we have

$$\begin{aligned} \widehat{\psi}(x) &= \int_{\mathbb{R}^N} e^{-2\pi i y \cdot x} \overline{\widehat{h}(y)} dy = \int_{\mathbb{R}^N} \overline{e^{2\pi i y \cdot x} \widehat{h}(y)} dy \\ &= \int_{\mathbb{R}^N} \overline{e^{2\pi i y \cdot x} \widehat{h}(y)} dy = \overline{\int_{\mathbb{R}^N} e^{2\pi i y \cdot x} \widehat{h}(y) dy} = \overline{(\widehat{h})^\vee(x)} = \overline{h(x)}, \end{aligned}$$

where in the last equality we have used the inversion theorem (Theorem 10.62). Hence, Parseval's identity follows by (10.26). Taking $h = \phi$ and using the fact that $\phi(x)\overline{\phi(x)} = \|\phi(x)\|^2$ gives the first equality in Plancherel's identity. The second equality follows by replacing ϕ with $\phi^\vee$ and using the inversion theorem.

Since $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is dense in $L^2(\mathbb{R}^N; \mathbb{C})$, if $\{\phi_n\}_n$ is a sequence in $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ converging to ϕ in $L^2(\mathbb{R}^N; \mathbb{C})$, then by Plancherel's identity the sequence $\{\widehat{\phi_n}\}_n$ is a Cauchy sequence in $L^2(\mathbb{R}^N; \mathbb{C})$ and so it converges to a function $\psi \in L^2(\mathbb{R}^N; \mathbb{C})$. Again by Plancherel's identity, the function ψ does not depend on the particular sequence $\{\phi_n\}_n$. We define $\widehat{\phi} := \psi$. Similarly we can extend uniquely the inverse Fourier transform to $L^2(\mathbb{R}^N; \mathbb{C})$ and

reasoning as in the last part of the proof of the inversion theorem (Theorem 10.62) we have that the Fourier transform $\mathcal{F} : L^2(\mathbb{R}^N; \mathbb{C}) \rightarrow L^2(\mathbb{R}^N; \mathbb{C})$ is an isomorphism with inverse given by the extension of $\mathcal{F}^{-1}$ to $L^2(\mathbb{R}^N; \mathbb{C})$. $\square$

Remark 10.64 (Important). Note that the Fourier transform of a function ϕ in $L^2(\mathbb{R}^N; \mathbb{C})$ is obtained as a limit in $L^2(\mathbb{R}^N; \mathbb{C})$ of functions of the type (10.23), but in general we cannot say that $\widehat{\phi}$ has the form (10.23), since the integral in (10.23) is well-defined for functions in $L^1(\mathbb{R}^N; \mathbb{C})$ but not for functions in $L^2(\mathbb{R}^N; \mathbb{C})$. On the other hand, if $\phi \in L^1(\mathbb{R}^N; \mathbb{C})$, then (10.23) is well-defined. Hence, the Fourier transform of a function in $L^1(\mathbb{R}^N; \mathbb{C})$ is defined pointwise by (10.23), while the Fourier transform of a function in $L^2(\mathbb{R}^N; \mathbb{C})$ is defined as a limit in $L^2(\mathbb{R}^N; \mathbb{C})$.

Theorem 10.65 (Riemann–Lebesgue lemma). $\mathcal{F} : L^1(\mathbb{R}^N; \mathbb{C}) \rightarrow C_0(\mathbb{R}^N; \mathbb{C})$ with

$$(10.29) \quad \sup_{x \in \mathbb{R}^N} \|\mathcal{F}(\phi)(x)\| \leq \|\phi\|_{L^1(\mathbb{R}^N)}$$

In particular, $\lim_{\|x\| \rightarrow \infty} \|\widehat{\phi}(x)\| = 0$.

Proof. By (10.23), for every $\phi \in L^1(\mathbb{R}^N; \mathbb{C})$,

$$\|\widehat{\phi}(x)\| \leq \|\phi\|_{L^1}$$

for every $x \in \mathbb{R}^N$. Since $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is dense in $L^1(\mathbb{R}^N; \mathbb{C})$ let $\{\phi_n\}_n$ in $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ converge to ϕ in $L^1(\mathbb{R}^N; \mathbb{C})$. By the previous inequality

$$\sup_{x \in \mathbb{R}^N} \|\widehat{\phi_n}(x) - \widehat{\phi}(x)\| \leq \|\phi_n - \phi\|_{L^1}.$$

Hence, the sequence $\{\widehat{\phi_n}\}_n$ converges uniformly to $\widehat{\phi}$. On the other hand, by Theorem 10.58 we have that $\widehat{\phi_n} \in \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \subset C_0(\mathbb{R}^N; \mathbb{C})$ and hence, since $C_0(\mathbb{R}^N; \mathbb{C})$ is a closed under uniform convergence, it follows that $\widehat{\phi} \in C_0(\mathbb{R}^N; \mathbb{C})$. $\square$

Exercise 10.66. Consider the Fourier transform $\mathcal{F} : L^1(\mathbb{R}; \mathbb{C}) \rightarrow C_0(\mathbb{R}; \mathbb{C})$.

- (i) Prove that if $f \in L^1(\mathbb{R}; \mathbb{C})$ is such that $\widehat{f}$ is odd, then for every $b > 0$,

$$\left| \int_1^b \frac{\widehat{f}(x)}{x} dx \right| \leq c$$

for some constant $c > 0$ independent of b .

- (ii) Prove that there exists $g \in C_0(\mathbb{R}; \mathbb{C})$ such that g is not the Fourier transform of any function f in $L^1(\mathbb{R}; \mathbb{C})$.
 (iii) Prove that $\mathcal{F}(L^1(\mathbb{R}; \mathbb{C}))$ is dense in $C_0(\mathbb{R}; \mathbb{C})$.

Corollary 10.67 (Hausdorff–Young inequality). *Let $1 < p < 2$. Then $\mathcal{F}$ can be extended uniquely to $L^p(\mathbb{R}^N; \mathbb{C})$ and*

$$(10.30) \quad \|\mathcal{F}(\phi)\|_{L^{p'}(\mathbb{R}^N)} \leq \|\phi\|_{L^p(\mathbb{R}^N)} \quad \text{for every } \phi \in L^p(\mathbb{R}^N; \mathbb{C}).$$

Proof. By Plancherel's identity and (10.29),

$$\sup_{x \in \mathbb{R}^N} \|\mathcal{F}(\phi)(x)\| \leq \|\phi\|_{L^1}, \quad \|\mathcal{F}(\phi)\|_{L^2} = \|\phi\|_{L^2}$$

for every simple function ϕ vanishing at infinity. Hence, $\mathcal{F}$ is of strong type $(1, \infty)$ and $(2, 2)$ (see Definition B.95) and so we are in a position to apply the Riesz–Thorin theorem (see Theorem B.96) to obtain that $\mathcal{F}$ is of strong type (p, q) for every $\theta \in (0, 1)$, where

$$\frac{1}{p} = \frac{1-\theta}{1} + \frac{\theta}{2}, \quad \frac{1}{q} = \frac{1-\theta}{\infty} + \frac{\theta}{2}.$$

Hence, $p = \frac{2}{2-\theta} \in (1, 2)$ and $q = p' = 2/\theta$. Moreover, (10.30) holds. $\square$

Exercise 10.68. Let $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ and let $L : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be an orthogonal transformation.

(i) Prove that $(\widehat{\phi \circ L})(x) = \widehat{\phi}(L(x))$ for every $x \in \mathbb{R}^N$.

(ii) Prove that if ϕ is radial, then so is $\widehat{\phi}$.

Exercise 10.69. Prove that if $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is real-valued and $\phi(x) = \phi(-x)$ for every $x \in \mathbb{R}^N$, then $\widehat{\phi}$ is real-valued.

Exercise 10.70. Compute the Fourier transform of the function

$$\phi(x) := e^{(-a+ib)\|x\|^2}, \quad x \in \mathbb{R}^N,$$

where $a > 0$ and $b \in \mathbb{R}$.

Theorem 10.71. *For every $\phi, \psi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, $\widehat{\phi * \psi} = \widehat{\phi} \widehat{\psi}$.*

Proof. For $x \in \mathbb{R}^N$ by Fubini's theorem we have

$$\begin{aligned} (\widehat{\phi * \psi})(x) &= \int_{\mathbb{R}^N} e^{-2\pi i x \cdot y} (\phi * \psi)(y) dy = \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} e^{-2\pi i x \cdot y} \phi(y - \xi) \psi(\xi) d\xi dy \\ &= \int_{\mathbb{R}^N} \psi(\xi) \int_{\mathbb{R}^N} e^{-2\pi i x \cdot y} \phi(y - \xi) dy d\xi \\ &= \int_{\mathbb{R}^N} e^{-2\pi i x \cdot \xi} \psi(\xi) \int_{\mathbb{R}^N} e^{-2\pi i x \cdot (y - \xi)} \phi(y - \xi) dy d\xi \\ &= \int_{\mathbb{R}^N} e^{-2\pi i x \cdot \xi} \psi(\xi) \int_{\mathbb{R}^N} e^{-2\pi i x \cdot \eta} \phi(\eta) d\eta d\xi = \widehat{\psi}(x) \widehat{\phi}(x), \end{aligned}$$

where we have made the change of variables $\eta := y - \xi$. $\square$

Remark 10.72. The previous theorem continues to hold for $\phi \in L^1(\mathbb{R}^N; \mathbb{C})$ and $\psi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$.

As a corollary of Theorem 10.71 we can show how the support of the Fourier transform relates to the regularity of a function.

Theorem 10.73. *Let $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ be such that $\text{supp } \hat{\phi} \subseteq \overline{B(0, R)}$ for some $R > 0$. Then for every $1 \leq p \leq q \leq \infty$ and for every multi-index $\alpha \in \mathbb{N}_0^N$,*

$$\|\partial^\alpha \phi\|_{L^q(\mathbb{R}^N)} \leq cR^{|\alpha|+N/p-N/q} \|\phi\|_{L^p(\mathbb{R}^N)}$$

for some constant $c = c(|\alpha|, N, p, q) > 0$. In particular, for $\alpha = 0$,

$$\|\phi\|_{L^q(\mathbb{R}^N)} \leq cR^{N/p-N/q} \|\phi\|_{L^p(\mathbb{R}^N)}.$$

On the other hand, if $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ is such that $\text{supp } \hat{\phi} \subseteq \overline{B(0, Rr_2)} \setminus B(0, Rr_1)$ for some $R > 0$ and some $0 < r_1 < r_2$, then

$$c^{-1}R^m \|\phi\|_{L^p(\mathbb{R}^N)} \leq \sum_{|\alpha|=m} \|\partial^\alpha \phi\|_{L^p(\mathbb{R}^N)} \leq cR^m \|\phi\|_{L^p(\mathbb{R}^N)}$$

for some constant $c = c(N, m, p, r_1, r_2) > 0$.

Proof. Using the change of variables $x = Ry$, we can assume $R = 1$.

Step 1: Let $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ be such that $\text{supp } \hat{\phi} \subseteq \overline{B(0, 1)}$ and consider a function $\varphi \in \mathcal{D}(\mathbb{R}^N)$ such that $\varphi = 1$ in $\overline{B(0, 1)}$. Then by Theorem 10.71, $\hat{\phi} = \varphi \hat{\phi} = \widehat{\psi \phi} = \widehat{\psi} * \hat{\phi}$, where $\psi := \varphi^\vee$, which, by Theorem 10.62, implies that $\phi = \psi * \phi$. By differentiating inside the integral it follows from (10.19) that $\partial^\alpha \phi = (\partial^\alpha \psi) * \phi$, and so by Young's inequality (see Theorem 10.49)

$$\|\partial^\alpha \phi\|_{L^q} = \|(\partial^\alpha \psi) * \phi\|_{L^q} \leq \|\partial^\alpha \psi\|_{L^r} \|\phi\|_{L^p},$$

where $1/p + 1/r = 1 + 1/q$.

Step 2: Let $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$ be such that $\text{supp } \hat{\phi} \subseteq \overline{B(0, r_2)} \setminus B(0, r_1)$ and consider a function $\varphi \in \mathcal{D}(\mathbb{R}^N \setminus \{0\})$ such that $\varphi = 1$ in $\overline{B(0, r_2)} \setminus B(0, r_1)$. Writing

$$\|y\|^{2m} = \sum_{|\alpha|=m} c_\alpha (iy)^\alpha (-iy)^\alpha,$$

we have that

$$\hat{\phi}(y) = \sum_{|\alpha|=m} (iy)^\alpha \hat{\phi}(y) \psi_\alpha(y),$$

where $\psi_\alpha(y) := \varphi(y)(c_\alpha(-iy)^\alpha/\|y\|^{2m})$. Note that since $\varphi \in \mathcal{D}(\mathbb{R}^N \setminus \{0\})$, we have that $\psi_\alpha \in \mathcal{D}(\mathbb{R}^N \setminus \{0\})$. It follows from Theorems 10.58 and 10.71 that

$$\phi = \sum_{|\alpha|=m} \psi_\alpha^\vee * \partial^\alpha \phi,$$

and so by Young's inequality (see Theorem 10.49)

$$\|\phi\|_{L^p} \leq \sum_{|\alpha|=m} \|\psi_\alpha^\vee * \partial^\alpha \phi\|_{L^p} \leq \sum_{|\alpha|=m} \|\psi_\alpha^\vee\|_{L^1} \|\partial^\alpha \phi\|_{L^p} \leq c \sum_{|\alpha|=m} \|\partial^\alpha \phi\|_{L^p}.$$

The other inequality follows from Step 1. $\square$

Exercise 10.74. For $n, m \in \mathbb{N}$ let $f_n = \chi_{[-n, n]}$ and $g_{n, m} := \frac{1}{m} f_n * f_m$.

- (i) Compute the Fourier transform of f_n .
- (ii) Compute the Fourier transform of $g_{n, m}$.
- (iii) Calculate the Lebesgue integral $\int_{\mathbb{R}} \frac{\sin^2 x}{x^2} dx$.
- (iv) Compute the improper Riemann integral $\int_{\mathbb{R}} \frac{\sin x}{x} dx$.

We can also define the Fourier transform of tempered distributions. Given $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$, the *Fourier transform* $\widehat{T}$ of T is the tempered distribution given by

$$(10.31) \quad \widehat{T}(\phi) := T(\widehat{\phi}), \quad \phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C}).$$

Given $\psi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, consider the linear functional $T : \mathcal{S}(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathbb{C}$ defined by (10.17). By (10.26), for every $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, we have

$$(10.32) \quad T_{\widehat{\psi}}(\phi) = \int_{\mathbb{R}^N} \phi(x) \widehat{\psi}(x) dx = \int_{\mathbb{R}^N} \widehat{\phi}(x) \psi(x) dx = T_{\psi}(\widehat{\phi}) = \widehat{T}_{\psi}(\phi).$$

Since we are identifying ψ with T_{ψ} in $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$, this shows that the Fourier transform defined on $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ extends the Fourier transform defined in $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$.

Similarly, given $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$, the *inverse Fourier transform* of T is the tempered distribution given by

$$(10.33) \quad T^{\vee}(\phi) := T(\phi^{\vee}), \quad \phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C}).$$

Exercise 10.75. Let $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

- (i) Prove that $\widehat{T} \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.
- (ii) Prove that if $\{T_n\}_n$ is a sequence in $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ such that $T_n \xrightarrow{*} T$ in $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$, then $\widehat{T_n} \xrightarrow{*} \widehat{T}$.
- (iii) Prove that Theorem 10.62 continues to hold in $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.
- (iv) Prove that $\mathcal{F} : \mathcal{S}'(\mathbb{R}^N; \mathbb{C}) \rightarrow \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ is an isomorphism with inverse $\mathcal{F}^{-1}$ given by $\mathcal{F}^{-1}(T) = T^{\vee}$ for every $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$.

In view of Theorem 10.35 and the previous exercise, for every function $\psi \in L^p(\mathbb{R}^N)$ with $p > 2$, the Fourier transform $\widehat{\psi}$ of ψ is the Fourier transform $\widehat{T_{\psi}}$ of the tempered distribution T_{ψ} defined in (10.17). Hence, $\widehat{\psi}$ belongs to $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ but in general $\widehat{\psi}$ cannot be identified with a function. A simple example is given by $\psi = 1 \in L^{\infty}(\mathbb{R}^N)$. In this case

$$(10.34) \quad T_1(\phi) = \int_{\mathbb{R}^N} \phi dx, \quad \phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C}),$$

and so by Theorem 10.62,

$$\widehat{1}(\phi) = \widehat{T_1}(\phi) = \int_{\mathbb{R}^N} \widehat{\phi}(x) dx = \int_{\mathbb{R}^N} e^{2\pi i 0 \cdot y} \widehat{\phi}(x) dx = (\widehat{\phi})^\vee(0) = \phi(0),$$

which shows that $\widehat{1}$ is δ_0 .

Exercise 10.76. Given $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$, prove that the following properties are equivalent:

- (i) $\text{supp } \widehat{T} = \{0\}$,
- (ii) $T = T_f$ for some polynomial $f : \mathbb{R}^N \rightarrow \mathbb{C}$,
- (iii) $T = \sum_{|\alpha| \leq n} a_\alpha \partial^\alpha \delta_0$ for some $n \in \mathbb{N}$ and some $a_\alpha \in \mathbb{C}$, $\alpha \in \mathbb{N}_0^N$, $0 \leq |\alpha| \leq n$.

Exercise 10.77. Prove that Theorem 10.71 continues to hold, that is, that if $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ and $\varphi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$, then $\widehat{T * \varphi} = \widehat{\varphi} \widehat{T}$.

10.8. Littlewood–Paley Decomposition

Let $\omega_0 \in C_c^\infty(\mathbb{R}^N)$ be a nonnegative function such that

$$(10.35) \quad \text{supp } \omega_0 = \overline{B(0, 1)}, \quad \omega_0 = 1 \text{ in } B(0, 1/2),$$

and define

$$(10.36) \quad \varphi_0(x) := \omega_0(x/2) - \omega_0(x), \quad x \in \mathbb{R}^N.$$

Given $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ and $k \in \mathbb{Z}$ we define the tempered distributions

$$(10.37) \quad S_k(T) := (\omega_0(2^{-k} \cdot) \widehat{T})^\vee,$$

$$(10.38) \quad \dot{A}_k(T) := (\varphi_0(2^{-k} \cdot) \widehat{T})^\vee.$$

The tempered distribution $\dot{A}_k(T)$ is called the k th *dyadic block of the Littlewood–Paley decomposition of T* .

Observe that by Exercise 10.40, (10.31), (10.33) and (10.37), for every $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$,

$$(10.39) \quad \begin{aligned} S_k(T)(\phi) &= (\omega_0(2^{-k} \cdot) \widehat{T})^\vee(\phi) \\ &= T(\mathcal{F}(\omega_0(2^{-k} \cdot) \mathcal{F}^{-1}(\phi))) = T(S_k(\phi)), \end{aligned}$$

where the last equality follows from Theorems 10.62 and 10.71, and where $S_k(\phi) := (\omega_0(2^{-k} \cdot) \widehat{\phi})^\vee$. Similarly, for every $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$,

$$(10.40) \quad \dot{A}_k(T)(\phi) = T(\dot{A}_k(\phi)),$$

where $\dot{A}_k(\phi) := (\varphi_0(2^{-k} \cdot) \widehat{\phi})^\vee$.

Theorem 10.78 (Littlewood–Paley decomposition). *Let $T \in \mathcal{S}'(\mathbb{R}^N; \mathbb{C})$ and let $\omega_0 \in C_c^\infty(\mathbb{R}^N)$ be a nonnegative function satisfying (10.35). Then for every $l \in \mathbb{Z}$,*

$$(10.41) \quad T = S_l(T) + \sum_{k=l}^{\infty} \dot{\Delta}_k(T).$$

Moreover, if

$$(10.42) \quad S_k(T) \xrightarrow{*} 0 \quad \text{as } k \rightarrow -\infty \text{ in } \mathcal{S}'(\mathbb{R}^N; \mathbb{C}),$$

then

$$(10.43) \quad T = \sum_{k=-\infty}^{\infty} \dot{\Delta}_k(T).$$

The identity (10.43) is called the *homogeneous Littlewood–Paley decomposition* of T .

Proof. By (10.36), (10.37), and (10.38) for $n \in \mathbb{N}$ we have

$$S_l(T) + \sum_{k=l}^n \dot{\Delta}_k(T) = S_{n+1}(T)$$

and so, by (10.39), for every $\phi \in \mathcal{S}(\mathbb{R}^N; \mathbb{C})$,

$$S_{n+1}(T)(\phi) = T(S_{n+1}(\phi)).$$

We leave as an exercise to check that since $\omega_0(2^{-n-1}x) = 1$ for $x \in B(0, 2^n)$ by (10.35), we have that $\omega_0(2^{-n-1}\cdot)\hat{\phi} \rightarrow \hat{\phi}$ in $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ as $n \rightarrow \infty$. In turn, since $\mathcal{F}$ is an isomorphism from $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ to $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$ (see Theorem 10.62), it follows that $S_{n+1}(\phi) \rightarrow \phi$ in $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$, and so

$$S_l(T)(\phi) + \sum_{k=l}^n \dot{\Delta}_k(T)(\phi) = T(S_{n+1}(\phi)) \rightarrow T(\phi)$$

as $n \rightarrow \infty$, which shows (10.41).

If (10.42) holds, then (10.43) follows by letting $l \rightarrow -\infty$ in (10.41). $\square$

Remark 10.79. The previous theorem continues to hold if instead of (10.35) we take $\omega_0 \in C_c^\infty(\mathbb{R}^N)$ to be a nonnegative function such that

$$\text{supp } \omega_0 = \overline{B(0, r)}, \quad \omega_0 = 1 \text{ in } B(0, r/2)$$

for some $r > 0$.

The Littlewood–Paley decomposition (10.43) fails without (10.42). Indeed, by taking the functional T_1 corresponding to the constant function 1 (see (10.34)), we have that $\dot{\Delta}_k(T_1) = 0$ for all k , so (10.43) cannot hold.

Sobolev Spaces

Newton's first law of graduation: A grad student in procrastination tends to stay in procrastination unless an external force is applied to it.

— Jorge Cham, www.phdcomics.com

In this chapter we define Sobolev functions on domains of $\mathbb{R}^N$, $N \geq 1$, and begin to study their properties.

11.1. Definition and Main Properties

In view of the integration by parts formula (9.55) we can give the following definition.

Definition 11.1. Given an open set $\Omega \subseteq \mathbb{R}^N$, a multi-index $\alpha \in \mathbb{N}_0^N \setminus \{0\}$, and $1 \leq p \leq \infty$, we say that a function $u \in L_{\text{loc}}^1(\Omega; \mathbb{R}^M)$ admits a *weak* or *distributional* α th derivative in $L^p(\Omega; \mathbb{R}^M)$ if there exists a function $v_\alpha \in L^p(\Omega; \mathbb{R}^M)$ such that

$$(11.1) \quad \int_{\Omega} u \partial^\alpha \phi \, dx = (-1)^{|\alpha|} \int_{\Omega} v_\alpha \phi \, dx$$

for all $\phi \in C_c^\infty(\Omega)$. The function v_α is denoted $\partial^\alpha u$ or $\frac{\partial^{|\alpha|} u}{\partial x^\alpha}$.

A similar definition can be given when $L^p(\Omega; \mathbb{R}^M)$ is replaced by $L_{\text{loc}}^p(\Omega; \mathbb{R}^M)$. In terms of distributions the previous definition means that $\partial^\alpha T_u = T_{v_\alpha}$ (see Definition 10.18).

Remark 11.2. We leave as an exercise to check that weak derivatives are unique.

Similarly we can define weak directional derivatives in L^p (see Section 9.1).

Definition 11.3. Given an open set $\Omega \subseteq \mathbb{R}^N$, a vector $\nu \in \mathbb{R}^N \setminus \{0\}$, $n \in \mathbb{N}$, and $1 \leq p \leq \infty$, we say that a function $u \in L^1_{\text{loc}}(\Omega; \mathbb{R}^M)$ admits a *weak or distributional n th directional derivative in $L^p(\Omega; \mathbb{R}^M)$ in the direction ν* if there exists a function $v_{\nu,n} \in L^p(\Omega; \mathbb{R}^M)$ such that

$$\int_{\Omega} u \frac{\partial^n \phi}{\partial \nu^n} dx = (-1)^n \int_{\Omega} v_{\nu,n} \phi dx$$

for all $\phi \in C_c^\infty(\Omega)$. The function $v_{\nu,n}$ is denoted $\frac{\partial^n u}{\partial \nu^n}$.

Definition 11.4. Given an open set $\Omega \subseteq \mathbb{R}^N$, $m, M \in \mathbb{N}$, and $1 \leq p \leq \infty$, the *Sobolev space $W^{m,p}(\Omega; \mathbb{R}^M)$* is the space of all functions $u \in L^p(\Omega; \mathbb{R}^M)$ which admit weak derivatives $\partial^\alpha u$ in $L^p(\Omega; \mathbb{R}^M)$ for every $\alpha \in \mathbb{N}_0^N$ with $1 \leq |\alpha| \leq m$. The space $W^{m,p}(\Omega; \mathbb{R}^M)$ is endowed with the norm

$$\|u\|_{W^{m,p}(\Omega; \mathbb{R}^M)} := \|u\|_{L^p(\Omega; \mathbb{R}^M)} + \sum_{1 \leq |\alpha| \leq m} \|\partial^\alpha u\|_{L^p(\Omega; \mathbb{R}^M)}.$$

The space $W^{m,p}_{\text{loc}}(\Omega; \mathbb{R}^M)$ is defined as the space of all functions $u \in L^p_{\text{loc}}(\Omega; \mathbb{R}^M)$ which admit weak derivatives $\partial^\alpha u$ in $L^p_{\text{loc}}(\Omega; \mathbb{R}^M)$ for every $\alpha \in \mathbb{N}_0^N$ with $1 \leq |\alpha| \leq m$.

When $M = 1$ we write $W^{m,p}(\Omega)$ and $W^{m,p}_{\text{loc}}(\Omega)$ for $W^{m,p}(\Omega; \mathbb{R})$ and $W^{m,p}_{\text{loc}}(\Omega; \mathbb{R})$, respectively. For simplicity, unless otherwise specified, in what follows we will focus on the case $M = 1$. Also, when there is no possibility of confusion, we write $\|u\|_{L^p}$ and $\|u\|_{W^{m,p}}$ for $\|u\|_{L^p(\Omega)}$ and $\|u\|_{W^{m,p}(\Omega)}$, respectively.

Remark 11.5 (Important). Following the literature, we use the same notation to indicate the weak and the classical derivatives of a function. Unfortunately, this results in endless confusion for students. In the remainder of this book, when $u \in W^{m,p}(\Omega)$, unless otherwise specified, $\partial^\alpha u$ is the α th weak derivative of u .

Remark 11.6. Given $\Omega \subseteq \mathbb{R}^N$ open, $m \in \mathbb{N}$, with $m \geq 2$, and $1 \leq p \leq \infty$, $u \in W^{m,p}(\Omega)$, in view of (11.1) and Schwartz theorem on the order of differentiation for smooth functions (Theorem 9.6) applied to $\phi \in C_c^\infty(\Omega)$,

$$\int_{\Omega} \frac{\partial^2 u}{\partial x_i \partial x_j} \phi dx = \int_{\Omega} u \frac{\partial^2 \phi}{\partial x_i \partial x_j} dx = \int_{\Omega} u \frac{\partial^2 \phi}{\partial x_j \partial x_i} dx = \int_{\Omega} \frac{\partial^2 u}{\partial x_j \partial x_i} \phi dx,$$

which shows that $\frac{\partial^2 u}{\partial x_i \partial x_j} = \frac{\partial^2 u}{\partial x_j \partial x_i}$. Thus for distributional derivatives the order in which one differentiates is not important. This is in sharp contrast to classical derivatives.

For $u \in W^{m,p}(\Omega)$ and $1 \leq k \leq m$, we set $\nabla^k u$ to be the vector of components $\partial^\alpha u$, $|\alpha| = k$. Then an equivalent norm in $W^{m,p}(\Omega)$ is given by

$$(11.2) \quad \|u\|_{L^p(\Omega)} + \sum_{k=1}^m \|\nabla^k u\|_{L^p(\Omega; \mathbb{R}^{M_k})},$$

where M_k is the number of multi-indices $\alpha \in \mathbb{N}_0^N$ with $|\alpha| = k$. For simplicity in what follows we will write $\|\nabla^k u\|_{L^p(\Omega)}$ for $\|\nabla^k u\|_{L^p(\Omega; \mathbb{R}^{M_k})}$.

Another equivalent norm which we will sometimes use is given by

$$\|u\|_{W^{m,p}(\Omega)} := \left(\|u\|_{L^p(\Omega)}^p + \sum_{1 \leq |\alpha| \leq m} \|\partial^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}$$

for $1 \leq p < \infty$, and

$$\|u\|_{W^{m,\infty}(\Omega)} := \max\{\|\partial^\alpha u\|_{L^\infty(\Omega)} : 0 \leq |\alpha| \leq m\},$$

where $\partial^0 u := u$, for $p = \infty$.

Exercise 11.7. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, with $m \geq 2$, and let $1 \leq p \leq \infty$. Prove that $u \in L^p(\Omega)$ belongs to $W^{m,p}(\Omega)$ if and only if $\nabla u \in W^{m-1,p}(\Omega; \mathbb{R}^N)$.

Exercise 11.8. Show that the function $u : B(0, 1) \rightarrow \mathbb{R}$, defined by

$$u(x) = u(x_1, \dots, x_N) := \begin{cases} 1 & \text{if } x_N > 0, \\ 0 & \text{if } x_N < 0, \end{cases}$$

does not belong to $W^{1,p}(B(0, 1))$ for any $1 \leq p \leq \infty$.

Exercise 11.9. Let $\Omega \subset \mathbb{R}^N$ be an open bounded set, let $x_0 \in \Omega$, and let $1 \leq p < \infty$. Prove that if $u \in C(\overline{\Omega}) \cap C^1(\Omega \setminus \{x_0\})$ is such that the (classical) gradient ∇u belongs to $L^p(\Omega; \mathbb{R}^N)$, then $u \in W^{1,p}(\Omega)$.

Exercise 11.10. Let $\Omega \subset \mathbb{R}^N$ be an open set and let $C \subset \Omega$ be a closed set with $\mathcal{H}^{N-1}(C) = 0$, where we recall that $\mathcal{H}^{N-1}$ is the $(N-1)$ -dimensional Hausdorff measure (see Section C.7). Let $m \in \mathbb{N}$ and $1 \leq p \leq \infty$ and let $u \in W^{m,p}(\Omega \setminus C)$. Prove that $u \in W^{m,p}(\Omega)$. Hint: Consider first the case $m = 1$ and for $i = 1, \dots, N$ consider the projection Π_i into the hyperplane orthogonal to the x_i -axis. Prove that $\mathcal{H}^{N-1}(\Pi_i(C)) = 0$ and use Fubini's theorem.

Exercise 11.11. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, and let $1 \leq p \leq \infty$. Prove that if $u \in W^{m,p}(\Omega)$ and $T : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a rigid motion, then $u \circ T \in W^{m,p}(U; \mathbb{R}^N)$, where $U := T^{-1}(\Omega)$ and compute $\|u \circ T\|_{W^{m,p}(U)}$.

We now show that $W^{m,p}(\Omega)$ is a Banach space.

Theorem 11.12. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, and let $1 \leq p \leq \infty$.

Then

- (i) the space $W^{m,p}(\Omega)$ is a Banach space,
- (ii) the space $H^m(\Omega) := W^{m,2}(\Omega)$ is a Hilbert space with the inner product

$$\langle u, v \rangle_{H^1(\Omega)} := \int_{\Omega} uv \, dx + \sum_{1 \leq |\alpha| \leq m} \int_{\Omega} \partial^{\alpha} u \partial^{\alpha} v \, dx.$$

Proof. We only prove part (i). Let $\{u_n\}_n$ be a Cauchy sequence in $W^{m,p}(\Omega)$; that is,

$$\begin{aligned} 0 &= \lim_{l, n \rightarrow \infty} \|u_l - u_n\|_{W^{m,p}}. \\ &= \lim_{l, n \rightarrow \infty} \left(\|u_l - u_n\|_{L^p} + \sum_{1 \leq |\alpha| \leq m} \|\partial^{\alpha} u_l - \partial^{\alpha} u_n\|_{L^p} \right). \end{aligned}$$

Since $L^p(\Omega)$ is a Banach space, there exist $u, v_{\alpha} \in L^p(\Omega)$ such that

$$\lim_{n \rightarrow \infty} \|u_n - u\|_{L^p} = 0, \quad \lim_{n \rightarrow \infty} \|\partial^{\alpha} u_n - v_{\alpha}\|_{L^p} = 0$$

for every multi-index α with $1 \leq |\alpha| \leq m$. Fix a multi-index α with $1 \leq |\alpha| \leq m$. We claim that $\partial^{\alpha} u = v_{\alpha}$. To see this, let $\phi \in C_c^{\infty}(\Omega)$ and note that

$$\int_{\Omega} \phi \partial^{\alpha} u_n \, dx = (-1)^{|\alpha|} \int_{\Omega} u_n \partial^{\alpha} \phi \, dx.$$

Letting $n \rightarrow \infty$ in the previous equality yields

$$\int_{\Omega} \phi v_{\alpha} \, dx = (-1)^{|\alpha|} \int_{\Omega} u \partial^{\alpha} \phi \, dx$$

for all $\phi \in C_c^{\infty}(\Omega)$, which proves the claim. Thus, $u \in W^{m,p}(\Omega)$. This completes the proof. $\square$

Exercise 11.13. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $1 \leq p < \infty$.

- (i) Prove that a subset of a separable metric space is separable.
- (ii) Prove that $W^{m,p}(\Omega)$ is separable. Hint: Consider the mapping

$$\begin{aligned} W^{m,p}(\Omega) &\rightarrow L^p(\Omega) \times \cdots \times L^p(\Omega) \\ u &\mapsto (u, \partial^{\alpha_1} u, \dots, \partial^{\alpha_M} u), \end{aligned}$$

where $\alpha_1, \dots, \alpha_M$ are all the multi-indices α with $1 \leq |\alpha| \leq m$.

Exercise 11.14. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Prove that $W^{m,\infty}(\Omega)$ is not separable.

Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, and let $1 \leq p \leq \infty$. The space $W_0^{m,p}(\Omega)$ is defined as the closure of the space $C_c^{\infty}(\Omega)$ in $W^{m,p}(\Omega)$ (with respect to the topology of $W^{m,p}(\Omega)$).

Remark 11.15. It is important to observe that since uniform convergence preserves continuity, functions in $W_0^{m,\infty}(\Omega)$ are necessarily of class $C^m(\Omega)$. In particular, piecewise affine functions do not belong to $W_0^{1,\infty}(\Omega)$. This is the reason why some authors refer to $W_0^{1,\infty}(\Omega)$ as the closure of the space $C_c^\infty(\Omega)$ in $W^{1,\infty}(\Omega)$ with respect to the weak star topology of $W^{1,\infty}(\Omega)$, rather than the strong topology. These two definitions are not equivalent. Thus, caution is needed when applying results for $W_0^{1,\infty}(\Omega)$.

We conclude this section by observing that in several applications in partial differential equations and in the calculus of variations, when one works with domains Ω of infinite measure, the Sobolev space $W^{m,p}(\Omega)$, $m \in \mathbb{N}$, $1 \leq p \leq \infty$, may not be the right space to work with. Indeed, as the following exercise shows, there exist solutions of the Dirichlet problem for the Laplacian with the property that $u \in L_{\text{loc}}^p(\Omega)$, $\nabla u \in L^p(\Omega; \mathbb{R}^N)$, but u does not belong to $W^{1,p}(\Omega)$.

Exercise 11.16. Let $\Omega := \{x \in \mathbb{R}^N : \|x\| > 1\}$ and let $u(x) := 1 - \|x\|^{2-N}$, $x \in \Omega$, if $N \geq 3$, and $u(x) := \log \|x\|$, $x \in \Omega$, if $N = 2$.

(i) Prove that u is a classical solution of the Dirichlet problem

$$\begin{aligned} \Delta u &= 0 && \text{in } \Omega, \\ u &= 0 && \text{on } \partial\Omega. \end{aligned}$$

(ii) Prove that $u \in L_{\text{loc}}^q(\Omega)$ for all $1 \leq q < \infty$, but $u \notin L^q(\Omega)$ for any $1 \leq q < \infty$.

(iii) Prove that $\nabla u \in L^p(\Omega; \mathbb{R}^N)$ for all $\frac{N}{N-1} < p < \infty$.

(iv) Prove that $\frac{\partial^2 u}{\partial x_i \partial x_j} \in L^p(\Omega)$ for all $1 < p < \infty$.

To provide a functional setting for Dirichlet's problems as in the previous exercise, we introduce the space $\dot{W}^{m,p}(\Omega)$.

Definition 11.17. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, and let $1 \leq p < \infty$. The *homogeneous Sobolev space* $\dot{W}^{m,p}(\Omega)$ is the space of all functions $u \in L_{\text{loc}}^1(\Omega)$ whose α th weak derivative $\partial^\alpha u$ belongs to $L^p(\Omega)$ for every $\alpha \in \mathbb{N}_0^N$ with $|\alpha| = m$.

Note that the inclusion

$$W^{m,p}(\Omega) \subseteq \dot{W}^{m,p}(\Omega)$$

holds. We will see later on that as a consequence of Poincaré's inequality for sufficiently regular domains of finite measure the spaces $\dot{W}^{m,p}(\Omega)$ and $W^{m,p}(\Omega)$ actually coincide. However, for domains of infinite measure (see Exercise 11.16) or irregular domains with finite measure (see Exercise 12.15) this is not the case.

The space $\dot{W}^{m,p}(\Omega)$ is equipped with the seminorm

$$(11.3) \quad |u|_{\dot{W}^{m,p}(\Omega)} := \|\nabla^m u\|_{L^p(\Omega)}.$$

Sometimes we will also use the equivalent seminorm $u \mapsto \sum_{|\alpha|=m} \|\partial^\alpha u\|_{L^p(\Omega)}$. Observe that $|u|_{\dot{W}^{m,p}(\Omega)} = 0$ if and only if u is a polynomial of degree less than or equal to $m - 1$ in each connected component of Ω . When Ω has finitely many connected components and, in particular, when it is connected, then we can define a norm in $\dot{W}^{m,p}(\Omega)$.

Exercise 11.18. Let $\Omega \subseteq \mathbb{R}^N$ be an open set with finitely many connected components $\Omega_1, \dots, \Omega_\ell$, let $m \in \mathbb{N}$, and let $1 \leq p < \infty$. For each $i = 1, \dots, \ell$ consider a ball $B_i \Subset \Omega_i$ and let $U := \bigcup_{i=1}^\ell B_i$. Prove that $\dot{W}^{m,p}(\Omega)$ is a Banach space with the norm

$$(11.4) \quad \|u\|_{\dot{W}^{m,p}(\Omega)} := \sum_{k=0}^{m-1} \|\nabla^k u\|_{L^1(U)} + \|\nabla^m u\|_{L^p(\Omega)},$$

where $\nabla^0 u := u$.

The following exercise shows that when Ω has infinitely many connected components, then there is no norm in $\dot{W}^{m,p}(\Omega)$ which is compatible with the natural convergence in $\dot{W}^{m,p}(\Omega)$, which is strong convergence in $L^p(\Omega)$ for the partial derivatives of order m and convergence in $L^1_{\text{loc}}(\Omega)$ for the functions and their derivatives of order less than m .

Exercise 11.19. Let $\Omega \subseteq \mathbb{R}^N$ be an open set with infinitely many connected components Ω_n , $n \in \mathbb{N}$, and let $1 \leq p < \infty$. Prove that there cannot exist a norm $\|\cdot\|_{\dot{W}^{1,p}}$ in $\dot{W}^{1,p}(\Omega)$ compatible with the convergences $u_n \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$ and $\nabla u_n \rightarrow \nabla u$ in $L^p(\Omega; \mathbb{R}^N)$. Hint: Reason by contradiction and assume that $\|\cdot\|_{\dot{W}^{1,p}}$ exists. For each $n \in \mathbb{N}$ consider a ball $B_n \Subset \Omega_n$. Prove that there exist $\varepsilon > 0$ and $\ell \in \mathbb{N}$ such that

$$\int_{\bigcup_{n=1}^\ell B_n} |u| \, dx \geq \varepsilon$$

for all piecewise constant functions $u \in \dot{W}^{1,p}(\Omega)$ with norm $\|u\|_{\dot{W}^{1,p}} = 1$.

We recall that S^{N-1} is the unit sphere $\partial B(0, 1)$.

Exercise 11.20. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $m \in \mathbb{N}$.

- (i) Let $\nu_1, \nu_2 \in S^{N-1}$ and let $\nu := \frac{\nu_1 + t\nu_2}{\|\nu_1 + t\nu_2\|}$ for $t \in \mathbb{R}$, with $\nu_1 + t\nu_2 \neq 0$. Prove that

$$\|\nu_1 + t\nu_2\|^m \frac{\partial^m u}{\partial \nu^m} = \sum_{k=0}^m \binom{m}{k} \frac{\partial^{m-k}}{\partial \nu_1^{m-k}} \left(\frac{\partial^k u}{\partial \nu_2^k} \right) t^k$$

for every $u \in C^m(\Omega)$.

- (ii) By taking a finite number of different values of $t \in \mathbb{R}$ show that for each $k = 1, \dots, m$, $\frac{\partial^{m-k} u}{\partial \nu_1^{m-k}} \left(\frac{\partial^k u}{\partial \nu_2^k} \right)$ can be written as a linear combination of a finite number of directional derivatives $\frac{\partial^m u}{\partial \nu^m}$, with ν as in part (i).
- (iii) Prove that there exists a finite set $S_m \subset S^{N-1}$ such that for every $u \in C^m(\Omega)$,

$$\partial^\alpha u = \sum_{\nu \in S_m} c_{\alpha, \nu} \frac{\partial^m u}{\partial \nu^m}$$

for every multi-index $\alpha \in \mathbb{N}_0^N$, with $|\alpha| = m$, and for some constants $c_{\alpha, \nu} \in \mathbb{R}$ independent of u . Hint: Use induction on m and part (ii).

Exercise 11.21. Given $n \in \mathbb{N}$, let V be the vector space of all polynomials $P : [0, 1] \rightarrow \mathbb{R}$ of degree less than or equal to n .

- (i) Given $P \in V$, let $\|P\| := \max\{|a_0|, \dots, |a_n|\}$, where $P(t) = a_0 + \dots + a_n t^n$, $t \in [0, 1]$. Prove that $\|\cdot\|$ is a norm in V .
- (ii) Let $1 \leq p \leq \infty$ and $P \in V$ be such that $\|P\|_{L^p([0,1])} \leq 1$. Prove that there exists a constant $c = c(n) > 0$ such that $|a_k| \leq c$ for all $k = 1, \dots, n$. Hint: The vector space V has finite dimension n .

Exercise 11.22. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, and let $1 \leq p \leq \infty$. Prove that the following are equivalent seminorms in $\dot{W}^{m,p}(\mathbb{R}^N)$:

$$\left(\int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p}^p d\mathcal{H}^{N-1}(\nu) \right)^{1/p}, \quad \sup_{\nu \in S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p},$$

$$\sum_{\nu \in S_m} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p},$$

where S_m is the finite set of S^{N-1} given in the previous exercise. Hint: Use the previous exercises.

11.2. Density of Smooth Functions

The next exercise shows that Sobolev functions can behave quite badly.

Exercise 11.23. Let $\Omega := B(0, 1) \subset \mathbb{R}^N$.

- (i) Consider the function $u(x) := 1/\|x\|^a$, $x \neq 0$, where $a > 0$.
- (a) Prove that the (classical) gradient ∇u belongs to $L_{\text{loc}}^1(\Omega; \mathbb{R}^N)$ if and only if $a + 1 < N$.
- (b) Let $a + 1 < N$ and $1 \leq p < \infty$ and show that $u \in W^{1,p}(\Omega)$ if and only if $(a + 1)p < N$ (in particular, $u \notin W^{1,p}(\Omega)$ for all $p \geq N$).

(ii) Let $\{x_n : n \in \mathbb{N}\} \subset \Omega$ be a countable dense set and define

$$u(x) := \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{1}{\|x - x_n\|^a}, \quad x \in \Omega \setminus \bigcup_{n=1}^{\infty} \{x_n\},$$

where $0 < a < N - 1$. Prove that $u \in W^{1,p}(\Omega)$ if and only if $(a + 1)p < N$, but u is unbounded in each open subset of Ω .

In view of the previous exercise, it is important to understand density of well-behaved functions in $W^{m,p}(\Omega)$.

Theorem 11.24 (Meyers–Serrin). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, and let $1 \leq p < \infty$. Then the space $C^\infty(\Omega) \cap W^{m,p}(\Omega)$ is dense in $W^{m,p}(\Omega)$.*

We begin with an auxiliary result. We use standard mollifiers (see Appendix C).

Lemma 11.25. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, let $1 \leq p < \infty$, and let $u \in W^{m,p}(\Omega)$. For every $\varepsilon > 0$ define $u_\varepsilon := \varphi_\varepsilon * u$, where φ_ε is a standard mollifier, and let*

$$(11.5) \quad \Omega_\varepsilon := \{x \in \Omega : \text{dist}(x, \partial\Omega) > \varepsilon\}.$$

Then $\|u_\varepsilon - u\|_{W^{m,p}(\Omega_\varepsilon)} \rightarrow 0$ as $\varepsilon \rightarrow 0^+$. In particular, if $U \subset \Omega$, with $\text{dist}(U, \partial\Omega) > 0$, then $\|u_\varepsilon - u\|_{W^{m,p}(U)} \rightarrow 0$ as $\varepsilon \rightarrow 0^+$.

Proof. By Theorem C.20 we have that $u_\varepsilon \in C^\infty(\mathbb{R}^N)$ and for $x \in \Omega_\varepsilon$ and $\alpha \in \mathbb{N}_0^N$, with $1 \leq |\alpha| \leq m$,

$$\begin{aligned} \frac{\partial^{|\alpha|} u_\varepsilon}{\partial x^\alpha}(x) &= \int_{\Omega} \frac{\partial^{|\alpha|} \varphi_\varepsilon}{\partial x^\alpha}(x - y) u(y) dy = (-1)^{|\alpha|} \int_{\Omega} \frac{\partial^{|\alpha|} \varphi_\varepsilon}{\partial y^\alpha}(x - y) u(y) dy \\ &= \int_{\Omega} \varphi_\varepsilon(x - y) \frac{\partial^{|\alpha|} u}{\partial y^\alpha}(y) dy = \left(\varphi_\varepsilon * \frac{\partial^{|\alpha|} u}{\partial x^\alpha} \right)(x), \end{aligned}$$

where we have used (11.1) and the fact that for each $x \in \Omega_\varepsilon$ the function $\varphi_\varepsilon(x - \cdot)$ belongs to $C_c^\infty(\Omega)$, since $\text{supp } \varphi_\varepsilon(x - \cdot) \subseteq \overline{B(x, \varepsilon)} \subset \Omega$. The result now follows from Theorem C.16 applied to the functions u and $\frac{\partial^{|\alpha|} u}{\partial x^\alpha}$. $\square$

Remark 11.26. Note that if $\Omega = \mathbb{R}^N$, then $\Omega_\varepsilon = \mathbb{R}^N$. Hence, $u_\varepsilon \rightarrow u$ in $W^{m,p}(\mathbb{R}^N)$.

Exercise 11.27. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, and let $1 \leq p \leq \infty$. Prove that if $u \in W^{m,p}(\Omega)$ and $\varphi \in C_c^\infty(\mathbb{R}^N)$, then $\varphi u \in W^{m,p}(\Omega)$.

We now turn to the proof of the Meyers–Serrin theorem.

Proof of Theorem 11.24. Let $\Omega_i \subseteq \Omega_{i+1}$ be such that $\Omega = \bigcup_{i=1}^{\infty} \Omega_i$ and use Theorem C.21 to construct a smooth partition of unity $\mathcal{F}$ subordinated

to the open cover $\{\Omega_{i+1} \setminus \overline{\Omega_{i-1}}\}_i$, where $\Omega_{-1} = \Omega_0 := \emptyset$.¹ For each $i \in \mathbb{N}$ let ψ_i be the sum of all the finitely many $\psi \in \mathcal{F}$ such that $\text{supp } \psi \subset \Omega_{i+1} \setminus \overline{\Omega_{i-1}}$ and which have not already been selected at previous steps $j < i$. Then $\psi_i \in C_c^\infty(\Omega_{i+1} \setminus \overline{\Omega_{i-1}})$ and

$$(11.6) \quad \sum_{i=1}^{\infty} \psi_i = 1 \quad \text{in } \Omega.$$

Fix $\eta > 0$. For each $i \in \mathbb{N}$ we have that

$$(11.7) \quad \text{supp}(\psi_i u) \subset \Omega_{i+1} \setminus \overline{\Omega_{i-1}},$$

and so, by the previous lemma, we may find $\varepsilon_i > 0$ so small that

$$(11.8) \quad \text{supp}(\psi_i u)_{\varepsilon_i} \subset \Omega_{i+1} \setminus \overline{\Omega_{i-1}}$$

and

$$\|(\psi_i u)_{\varepsilon_i} - \psi_i u\|_{W^{m,p}(\Omega)} \leq \frac{\eta}{2^i},$$

where we have used the previous exercise.

Note that in view of (11.8), for every $U \Subset \Omega$ only finitely many $\Omega_{i+1} \setminus \overline{\Omega_{i-1}}$ cover U , and so the function $v := \sum_{i=1}^{\infty} (\psi_i u)_{\varepsilon_i}$ belongs to $C^\infty(\Omega)$. In particular, $v \in W_{\text{loc}}^{m,p}(\Omega)$.

For $x \in \Omega_\ell$, by (11.6), (11.7), and (11.8),

$$(11.9) \quad u(x) = \sum_{i=1}^{\ell} (\psi_i u)(x), \quad v(x) = \sum_{i=1}^{\ell} (\psi_i u)_{\varepsilon_i}(x).$$

Hence,

$$(11.10) \quad \|u - v\|_{W^{m,p}(\Omega_\ell)} \leq \sum_{i=1}^{\ell} \|(\psi_i u)_{\varepsilon_i} - \psi_i u\|_{W^{m,p}(\Omega)} \leq \sum_{i=1}^{\ell} \frac{\eta}{2^i} \leq \eta.$$

Letting $\ell \rightarrow \infty$, it follows from the Lebesgue monotone convergence theorem that $\|u - v\|_{W^{m,p}(\Omega)} \leq \eta$. This also implies that $u - v$ (and, in turn, v) belongs to the space $W^{m,p}(\Omega)$. $\square$

Remark 11.28. We will see later on that if $u \in \dot{W}^{m,p}(\Omega)$ (see Definition 11.17), then $u \in L_{\text{loc}}^p(\Omega)$. Hence, we can adapt the proof of the Meyers–Serrin theorem to show that if $u \in \dot{W}^{m,p}(\Omega)$, then for every $\varepsilon > 0$ there exists a function $v \in C^\infty(\Omega) \cap \dot{W}^{m,p}(\Omega)$ such that $\|u - v\|_{W^{m,p}(\Omega)} \leq \varepsilon$, despite the fact that neither u nor v need belong to $W^{m,p}(\Omega)$.

¹Note that we do not work simply with $\{\Omega_{i+1} \setminus \overline{\Omega_i}\}$, since, otherwise, $\partial\Omega_i$ would not be covered.

Exercise 11.29. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, let $1 \leq p, q < \infty$, and let $u \in L^q(\Omega) \cap \dot{W}^{m,q}(\Omega)$. Prove that for every $\varepsilon > 0$ there exists $v \in C^\infty(\Omega) \cap L^q(\Omega) \cap \dot{W}^{m,p}(\Omega)$ such that

$$\|u - v\|_{L^q(\Omega)} + \|\nabla^m u - \nabla^m v\|_{L^p(\Omega)} \leq \varepsilon.$$

Exercise 11.30. Prove that the function $u(x) := |x|$, $x \in (-1, 1)$, belongs to $W^{1,\infty}((-1, 1))$ but not to the closure of $C^\infty((-1, 1)) \cap W^{1,\infty}((-1, 1))$.

The previous exercise shows that the Meyers–Serrin theorem is false for $p = \infty$. This is intuitively clear, since if $\Omega \subseteq \mathbb{R}^N$ is an open set and $\{u_n\}_n$ is a sequence in $C^\infty(\Omega) \cap W^{1,\infty}(\Omega)$ such that $\|u_n - u\|_{W^{1,\infty}(\Omega)} \rightarrow 0$, then $u \in C^1(\Omega)$ (why?). However, the following weaker version of the Meyers–Serrin theorem holds.

Exercise 11.31. Let $\Omega \subset \mathbb{R}^N$ be an open set with finite measure and let $u \in W^{1,\infty}(\Omega)$. Modify the proof of the Meyers–Serrin theorem to show that there exists a sequence $\{u_n\}_n$ in $C^\infty(\Omega) \cap W^{1,\infty}(\Omega)$ such that $\|u_n - u\|_{L^\infty(\Omega)} \rightarrow 0$, $\|\nabla u_n\|_{L^\infty(\Omega)} \rightarrow \|\nabla u\|_{L^\infty(\Omega)}$, and $\nabla u_n(x) \rightarrow \nabla u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$ as $n \rightarrow \infty$.

We recall that given an open set $\Omega \subseteq \mathbb{R}^N$, the space $C^\infty(\overline{\Omega})$ is defined as the space of all functions in $C^\infty(\Omega)$ that can be extended to functions in $C^\infty(\mathbb{R}^N)$.

Exercise 11.32. Let $\Omega = B(0, 1) \setminus \{x \in \mathbb{R}^N : x_N = 0\}$. Show that the function $u : \Omega \rightarrow \mathbb{R}$, defined by

$$u(x) = u(x_1, \dots, x_N) := \begin{cases} 1 & \text{if } x_N > 0, \\ 0 & \text{if } x_N < 0, \end{cases}$$

belongs to $W^{1,p}(\Omega)$ for all $1 \leq p \leq \infty$, but it cannot be approximated by functions in $C^\infty(\overline{\Omega})$.

The previous exercise shows that in the Meyers–Serrin theorem for general open sets Ω we may not replace $C^\infty(\Omega)$ with $C^\infty(\overline{\Omega})$. The problem is that the domain lies on both sides of its boundary. This motivates the next definition.

Definition 11.33. An open set $\Omega \subseteq \mathbb{R}^N$ satisfies the *segment property* if for every $x_0 \in \partial\Omega$ there exist $r > 0$ and a vector $\nu \in \mathbb{R}^N \setminus \{0\}$ such that if $x \in \overline{\Omega} \cap B(x_0, r)$, then $x + \theta\nu \in \Omega$ for all $0 < \theta < 1$.

It turns out that the segment property is related to the regularity of the domain (see Definition 9.57). Indeed, we have the following result.

Theorem 11.34. *An open set $\Omega \subseteq \mathbb{R}^N$ satisfies the segment property if and only if $\partial\Omega$ is of class C .*

Proof. Assume that $\partial\Omega$ is of class C . Then for every $x_0 \in \partial\Omega$ there exist local coordinates $y = (y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R}$, with $y = 0$ at $x = x_0$, a continuous function $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$, and $r_{x_0} > 0$ such that $\Omega \cap B(x_0, r_{x_0})$ is given in local coordinates by the set

$$\{(y', y_N) \in B(0, r_{x_0}) : y_N > f(y')\}.$$

Note that $y = R(x - x_0)$, where $R : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is an orthogonal transformation. Define the sets

(11.11)

$$V(x_0, r, t) := \{(y', y_N) : y' \in Q_{N-1}(0, r), -t + f(y') < y_N < f(y') + t\},$$

(11.12)

$$U(x_0, r, t) := R^{-1}(V(x_0, r, t)).$$

Here, $Q_{N-1}(0, r) := (-r/2, r/2)^{N-1}$. By taking $r > 0$ smaller, if necessary, and $t > 0$ sufficiently small, we have that the open set $U(x_0, r, t)$ defined in (11.12) is contained in $B(x_0, r_{x_0})$. Then the neighborhood $U(x_0, r, t/2)$ of x_0 and the vector $\nu := R^{-1}((0, \dots, 0, t/2))$ satisfy the definition of segment property.

Conversely, assume that $\Omega \subseteq \mathbb{R}^N$ satisfies the segment property. Fix $x_0 \in \partial\Omega$ and find a neighborhood W of x_0 and a vector $\nu \in \mathbb{R}^N \setminus \{0\}$ such that if $x \in \overline{\Omega} \cap W$, then $x + \theta\nu \in \Omega$ for all $0 < \theta < 1$. Take local coordinates $y = (y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R}$, with $y = 0$ at $x = x_0$ and with $y = e_N$ at $x = x_0 + \frac{\nu}{\|\nu\|}$. For $x \in \overline{\Omega} \cap W$, let $y := L(x)$. The segment

$$L(\{x + \theta\nu : 0 < \theta < 1\}) = y + \{(0, \theta\|\nu\|) : 0 < \theta < 1\}$$

will be called *the arrow from y* .

From now on, for simplicity, we omit mention of L , and thus we write $0 \in \partial\Omega$, for example, in place of $0 \in L(\partial\Omega)$.

Consider the closed segment $S := \{0\} \times [-\eta, \eta]$, where $0 < \eta < \|\nu\|$ is chosen so small that $S \subseteq W$. Note that the points of S with $y_N > 0$ are in Ω because they are on the arrow from $0 \in \partial\Omega$, while the points of S with $y_N < 0$ are not in $\overline{\Omega}$ because the arrows from them contain 0 that is not in Ω . Let $S_l := \{0\} \times [-\eta, -\frac{1}{2}\eta]$, $S_m := \{0\} \times [-\frac{1}{2}\eta, \frac{1}{2}\eta]$, and $S_u := \{0\} \times [\frac{1}{2}\eta, \eta]$ (here the subscripts stand for lower, middle, upper). Since S_l , S_m , and S_u are compact sets, we may find finite open covers of cubes centered in S and contained in $(\mathbb{R}^N \setminus \overline{\Omega}) \cap W$, W , and $\Omega \cap W$, respectively. By taking $r > 0$ sufficiently small, we obtain that the union of all these cubes contains the open set

$$V := Q_{N-1}(0, r) \times (-\eta, \eta),$$

that $V_l := Q_{N-1}(0, r) \times (-\eta, -\frac{1}{2}\eta) \subset \mathbb{R}^N \setminus \overline{\Omega}$, and that $V_u := Q_{N-1}(0, r) \times (\frac{1}{2}\eta, \eta) \subseteq \Omega$. Define $f \in Q_{N-1}(0, r) \rightarrow \mathbb{R}$ by

$$f(y') := \inf\{y_N : (y', y_N) \in \overline{\Omega} \cap V\}.$$

Note that $-\frac{1}{2}\eta \leq f \leq \frac{1}{2}\eta$ because $V_l \subset \mathbb{R}^N \setminus \overline{\Omega}$ and $V_u \subset \Omega$, and the infimum is attained because $\overline{\Omega}$ is closed. Moreover, for every $y = (y', y_N) \in V$, if $y_N > f(y')$, then $y \in \Omega$, since y is on the arrow from $(y', f(y')) \in \overline{\Omega}$, while if $y_N < f(y')$, then $y \in \mathbb{R}^N \setminus \overline{\Omega}$ by the definition of $f(y')$. Thus, for every $y' \in Q_{N-1}(0, r)$ the point $(y', f(y'))$ belongs to $\partial\Omega$ and is the only such point in V .

It remains to prove that f is continuous. Let $y' \in Q_{N-1}(0, r)$ and let $0 < \varepsilon < \frac{1}{2}\eta$. Define

$$y_+ := (y', f(y') + \varepsilon), \quad y_- := (y', f(y') - \varepsilon).$$

By what we just proved and the bound $|f| \leq \frac{1}{2}\eta$, we have that $y_+ \in \Omega \cap V$ and $y_- \in (\mathbb{R}^N \setminus \overline{\Omega}) \cap V$. Since $\Omega \cap V$ and $(\mathbb{R}^N \setminus \overline{\Omega}) \cap V$ are open, there exist cubes $Q_N(y_+, \delta) \subseteq \Omega \cap V$ and $Q_N(y_-, \delta) \subseteq (\mathbb{R}^N \setminus \overline{\Omega}) \cap V$. We claim that

$$|f(y') - f(z')| \leq \varepsilon$$

for all $z' \in Q_N(y', \delta)$. Indeed, if $f(z') > f(y') + \varepsilon$ for some $z' \in Q_N(y', \delta)$, then $(z', t) \in Q_N(y_+, \delta) \subseteq \Omega \cap V$ for every $f(y') + \varepsilon < t < f(z')$ and this would contradict the minimum property of $f(z')$, while if $f(z') < f(y') - \varepsilon$, then the arrow from $(z', f(z')) \in \overline{\Omega}$ would contain points of $Q_N(y_-, \delta) \subset \mathbb{R}^N \setminus \overline{\Omega}$, which is again a contradiction. This completes the proof. $\square$

We now prove the density of functions $C_c^\infty(\mathbb{R}^N)$ in $W^{m,p}(\Omega)$ for domains with continuous boundary.

Theorem 11.35. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set whose boundary is of class C , let $m \in \mathbb{N}$, and let $1 \leq p < \infty$. Then the restriction to Ω of functions in $C_c^\infty(\mathbb{R}^N)$ is dense in $W^{m,p}(\Omega)$.*

To prove the theorem, we need an auxiliary result.

Lemma 11.36. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $1 \leq p < \infty$, and let $u \in L^p(\Omega)$. Extend u by zero outside Ω . Then for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$\int_{\Omega} |u(x+h) - u(x)|^p dx \leq \varepsilon$$

for all $h \in \mathbb{R}^N$, with $\|h\| \leq \delta$.

Proof. Exercise. $\square$

Exercise 11.37. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u, v \in C^m(\Omega)$, $m \in \mathbb{N}$. Prove that for every $\alpha \in \mathbb{N}_0^N$ with $1 \leq |\alpha| \leq m$,

$$\partial^\alpha(uv) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^{\alpha-\beta} u \partial^\beta v.$$

Hint: Use induction on N .

We now turn to the proof of Theorem 11.35.

Proof. Fix $u \in W^{m,p}(\Omega)$. By the Meyers–Serrin theorem (Theorem 11.24), without loss of generality, we may assume that $u \in C^\infty(\Omega) \cap W^{m,p}(\Omega)$.

Step 1: We first approximate u with a function with compact support. Consider a cut-off function $\varphi \in C_c^\infty(\mathbb{R}^N)$ such that $\text{supp } \varphi \subseteq \overline{B(0,2)}$, $\varphi = 1$ in $B(0,1)$ and $0 \leq \varphi \leq 1$. For $n \in \mathbb{N}$, define

$$u_n(x) := \varphi_n(x)u(x), \quad \varphi_n(x) := \varphi(x/n), \quad x \in \Omega.$$

By the Lebesgue dominated convergence theorem, we have that $u_n \rightarrow u$ in $L^p(\Omega)$ as $n \rightarrow \infty$, while by Exercise 11.37,

$$\partial^\alpha u_n(x) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta \varphi_n(x) \partial^{\alpha-\beta} u(x) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \frac{1}{n^{|\beta|}} \partial^\beta \varphi\left(\frac{x}{n}\right) \partial^{\alpha-\beta} u(x).$$

If $\beta = 0$, then again by the Lebesgue dominated convergence theorem, $\varphi_n \partial^\alpha u \rightarrow \partial^\alpha u$ in $L^p(\Omega)$ as $n \rightarrow \infty$, while for $\beta \neq 0$,

$$\frac{1}{n^{|\beta|p}} \int_{\Omega} |\partial^\beta \varphi(x/n) \partial^{\alpha-\beta} u(x)|^p dx \leq \frac{c}{n^{|\beta|p}} \int_{\Omega} |\partial^{\alpha-\beta} u(x)|^p dx \rightarrow 0$$

as $n \rightarrow \infty$. This concludes the proof.

Step 2: By Step 1, we can assume that $u = 0$ outside a compact set. For every $x_0 \in \partial\Omega$ there exist local coordinates $y = (y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R}$, with $y = 0$ at $x = x_0$, a continuous function $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$, and $r_{x_0} > 0$ such that $\Omega \cap B(x_0, r_{x_0})$ in local coordinates is given by

$$(11.13) \quad \{(y', y_N) \in B(0, r_{x_0}) : y_N > f(y')\}.$$

If the set $\Omega \setminus \bigcup_{x \in \partial\Omega} B(x, r_x/2)$ is nonempty, for every $x_0 \in \Omega \setminus \bigcup_{x \in \partial\Omega} B(x, r_x/2)$ let $B(x_0, r_{x_0})$ be any open ball contained in Ω . The family $\{B(x, r_x/2)\}_{x \in \overline{\Omega}}$ is an open cover of $\overline{\Omega}$. Since $u = 0$ outside a compact set K , we have that $\overline{\Omega} \cap K$ is compact. Hence, there is a finite number of balls $B_1, \dots, B_\ell$, where $B_n := B(x_n, r_{x_n}/2)$, that covers $\overline{\Omega} \cap K$. Let $\{\psi_n\}_{n=1}^\ell$ be a smooth partition of unity subordinated to $B_1, \dots, B_\ell$ (see Exercise C.22).

Fix $n \in \{1, \dots, \ell\}$ and define $u_n := u\psi_n \in W^{m,p}(\Omega)$ (see Exercise 11.27), where we extend u_n to be zero outside $\text{supp } \psi_n$. There are two cases.

If $\text{supp } \psi_n$ is contained in Ω , then we set $v_n := \phi_n u \in C_c^\infty(\mathbb{R}^N)$. If $\text{supp } \psi_n$ is not contained in Ω , let $x_n \in \partial\Omega$ be such that $\text{supp } \psi_n \subseteq B_n$. For

$t > 0$, using local coordinates in $B_n = B(x_n, r_{x_n}/2)$ (see Exercise 11.11) define the function $u_{n,t} : U_n^t \rightarrow \mathbb{R}$ by $u_{n,t}(y', y_N) := u_n(y', y_N + t)$, where

$$U_n^t := \{(y', y_N) \in B(0, r_{x_n}) : y_N > f(y') - t\}.$$

Note that for $t > 0$ sufficiently small U_n^t contains the set $\overline{\Omega} \cap B_n$ (written in local coordinates).

Fix $\eta > 0$. By the previous lemma there exists $t_n > 0$ so small that $\text{supp } \psi_n + B(0, t_n) \Subset U_n^{t_n}$ and

$$(11.14) \quad \|u_{n,t_n} - u_n\|_{W^{m,p}(\Omega)} \leq \eta/2^n.$$

Construct a function $\phi_n \in C^\infty(\mathbb{R})$ such that $\phi_n(y_N) = 1$ if $y_N > f(y') - t_n/2$ and $\phi_n(y_N) = 0$ if $y_N \leq f(y') - 3t_n/4$ and define $v_n := \phi_n u_{n,t_n}$. Then $v_n \in C_c^\infty(\mathbb{R}^N)$ (provided we define u_{n,t_n} to be zero, whenever ϕ_n is zero) and $\text{supp } v_n \subset U_n^{t_n}$. Note that $v_n = u_{n,t_n}$ in Ω , and so (11.14) may be rewritten as

$$(11.15) \quad \|v_n - u\psi_n\|_{W^{m,p}(\Omega)} \leq \eta/2^n.$$

Define the function $v := \sum_{n=1}^\ell v_n$. Then $v \in C_c^\infty(\mathbb{R}^N)$ and so, $v \in W^{m,p}(\Omega)$. Moreover,

$$\|u - v\|_{W^{m,p}(\Omega)} \leq \sum_{i=1}^\ell \|\psi_n u - v_n\|_{W^{m,p}(\Omega)} \leq \eta \sum_{i=1}^\ell 2^{-i} \leq \eta.$$

This concludes the proof. $\square$

Remark 11.38. Note that in view of Theorem 11.35 it follows that for all $m \in \mathbb{N}$ and $1 \leq p < \infty$,

$$W_0^{m,p}(\mathbb{R}^N) = W^{m,p}(\mathbb{R}^N).$$

In Exercise 11.48 below we will show that there exists an open bounded domain $\Omega \subset \mathbb{R}^2$, with $\partial\Omega = \partial(\overline{\Omega})$, such that $W^{1,p}(\Omega) \cap C(\overline{\Omega})$ is not dense in $W^{1,p}(\Omega)$, $1 \leq p < \infty$.

As a corollary of Theorem 11.35, we can prove that piecewise affine functions are dense in $W^{1,p}(\Omega)$, whenever $\Omega \subseteq \mathbb{R}^N$ has a boundary of class C .

Definition 11.39. An N -simplex Δ is the convex hull of $N + 1$ points $x_i \in \mathbb{R}^N$ (the *vertices* of Δ) that are not contained in a hyperplane, namely,

$$\Delta := \left\{ x \in \mathbb{R}^N : x = \sum_{j=1}^{N+1} \theta_j x_j, 0 \leq \theta_j \leq 1, \sum_{j=1}^{N+1} \theta_j = 1 \right\},$$

and for all $i \in \{1, \dots, N + 1\}$,

$$\{x_j - x_i : j \in \{1, \dots, N + 1\} \setminus \{i\}\} \text{ is a basis of } \mathbb{R}^N.$$

Let $\mathcal{PA}$ be the family of all continuous functions $u : \mathbb{R}^N \rightarrow \mathbb{R}$ for which there exists a finite number of N -simplexes $\Delta_1, \dots, \Delta_\ell$ with pairwise disjoint interiors such that the restriction of u to each Δ_i is affine and $u = 0$ outside $\bigcup_{i=1}^\ell \Delta_i$.

Theorem 11.40. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set whose boundary is of class C . Then the restriction to Ω of functions in $\mathcal{PA}$ is dense in $W^{1,p}(\Omega)$ for $1 \leq p < \infty$.*

Proof. Let $u \in W^{1,p}(\Omega)$. In view of Theorem 11.35 we may assume, without loss of generality, that $u \in C_c^\infty(\mathbb{R}^N)$. Let $K := \text{supp } u$ and consider $B(0, R) \supset K$. Fix $\alpha > 0$ and construct N -simplexes $\Delta_1, \dots, \Delta_\ell$ with pairwise disjoint interiors such that $K \subset \bigcup_{i=1}^\ell \Delta_i \subset B(0, R)$ and $\text{diam } \Delta_i \geq \alpha r_i$ for all $i = 1, \dots, \ell$, where r_i is the radius of the inscribed ball in Δ_i . Let v be the continuous piecewise affine function that coincides with u in all the vertices of the simplexes. Then by Taylor's formula (see Theorem 9.9),

$$\begin{aligned} \sup_{x \in \Delta_i} |u(x) - v(x)| &\leq ch^2 \sup_{x \in \Delta_i} \|\nabla^2 u(x)\| \leq ch^2 \sup_{x \in \mathbb{R}^N} \|\nabla^2 u(x)\|, \\ \sup_{x \in \Delta_i} \|\nabla u(x) - \nabla v(x)\| &\leq ch \sup_{x \in \Delta_i} \|\nabla^2 u(x)\| \leq ch \sup_{x \in \mathbb{R}^N} \|\nabla^2 u(x)\| \end{aligned}$$

for all $i = 1, \dots, \ell$, where $h := \max_{1 \leq i \leq \ell} \text{diam } \Delta_i$. Thus,

$$\begin{aligned} \int_{\Omega} \|\nabla u - \nabla v\|^p dx &\leq \int_{\mathbb{R}^N} \|\nabla u - \nabla v\|^p dx = \sum_{i=1}^{\ell} \int_{\Delta_i} \|\nabla u - \nabla v\|^p dx \\ &\leq c_u h^p \sum_{i=1}^{\ell} \mathcal{L}^N(\Delta_i) \leq c_u h^p \mathcal{L}^N(B(0, R)), \end{aligned}$$

where $c_u := c \sup_{x \in \mathbb{R}^N} \|\nabla^2 u(x)\|$. A similar estimate holds for $\int_{\Omega} |u - v|^p dx$. By choosing h sufficiently small (and so increasing the number ℓ of N -simplexes), we have the desired approximation. $\square$

Remark 11.41. If $\Omega \subseteq \mathbb{R}^N$ is an open set and $u \in W^{1,p}(\Omega)$ for $1 \leq p < \infty$, then for any $U \Subset \Omega$ we can construct a sequence $\{u_n\}_n$ in $\mathcal{PA}$ such that $u_n \rightarrow u$ in $W^{1,p}(U)$. Indeed, by the Meyers–Serrin theorem we may find $v_n \in C_c^\infty(\Omega) \cap W^{1,p}(\Omega)$ such that $v_n \rightarrow u$ in $W^{1,p}(\Omega)$. Construct a cut-off function $\psi \in C_c^\infty(\Omega)$ such that $\psi = 1$ in U . Since $\psi v_n \in C_c^\infty(\Omega)$, we may extend ψv_n to be zero outside Ω and then apply the previous theorem to find $u_n \in \mathcal{PA}$ such that $\|\psi v_n - u_n\|_{W^{1,p}(\Omega)} \leq 1/n$. Using the fact that $\psi = 1$ in U , it follows that

$$\begin{aligned} \|u - u_n\|_{W^{1,p}(U)} &\leq \|u - v_n\|_{W^{1,p}(\Omega)} + \|v_n - u_n\|_{W^{1,p}(U)} \\ &\leq \|u - v_n\|_{W^{1,p}(\Omega)} + \|\psi v_n - u_n\|_{W^{1,p}(\Omega)}. \end{aligned}$$

It suffices to let $n \rightarrow \infty$.

We conclude this section by discussing the density of $C_c^\infty(\mathbb{R}^N)$ functions in the homogeneous Sobolev space $\dot{W}^{m,p}(\mathbb{R}^N)$ (see Definition 11.17).

Exercise 11.42. Prove that if $f_0 \in L^p(\mathbb{R})$, $p > 1$, and $f_{n+1}(x) := \int_0^x f_n(t) dt$, for $n \in \mathbb{N}_0$, then the function $g_n(x) := f_n(x)|x|^{-n}$, $x \in \mathbb{R}$, belongs to $L^p(\mathbb{R})$ for every $n \in \mathbb{N}_0$. Hint: Use Hardy's inequality (see Theorem C.41).

Theorem 11.43. Let $u \in \dot{W}^{m,p}(\mathbb{R}^N)$, where $m \in \mathbb{N}$ and $1 \leq p < \infty$. Then there exists a sequence $\{u_n\}_n$ of functions in $C_c^\infty(\mathbb{R}^N)$ such that $\partial^\alpha u_n \rightarrow \partial^\alpha u$ in $L^p(\mathbb{R}^N)$ for every multi-index α with $|\alpha| = m$ if and only if $N \geq 2$ or $p > 1$.

Proof. Step 1: Assume first that $N = 1$ and $p = 1$. Let $\phi \in C_c^\infty(\mathbb{R})$ be such that $\int_{\mathbb{R}} \phi(t) dt \neq 0$ and let u be such that $u^{(m)} = \phi$. Then u belongs to $\dot{W}^{m,1}(\mathbb{R})$. Assume, by contradiction, that there exists a sequence $\{u_n\}_n$ in $C_c^\infty(\mathbb{R})$ such that $u_n^{(m)} \rightarrow u^{(m)}$ in $L^1(\mathbb{R})$. Then by the fundamental theorem of calculus,

$$0 = \int_{\mathbb{R}} u_n^{(m)}(t) dt \rightarrow \int_{\mathbb{R}} \phi(t) dt \neq 0,$$

which is a contradiction.

Step 2: We consider the case $N = 1$ and $1 < p < \infty$. Fix $u \in \dot{W}^{m,p}(\mathbb{R})$. Using standard mollifiers, we have that $C_c^\infty(\mathbb{R})$ is dense in $\dot{W}^{m,p}(\mathbb{R})$ with respect to the seminorm (11.3). Thus, without loss of generality, we may assume that $u \in C_c^\infty(\mathbb{R}) \cap \dot{W}^{m,p}(\mathbb{R})$. Consider a cut-off function $\varphi \in C_c^\infty(\mathbb{R})$ such that $\text{supp } \varphi \subseteq \overline{B(0,2)}$, $\varphi = 1$ in $B(0,1)$ and $0 \leq \varphi \leq 1$. For $n \in \mathbb{N}$, define $f_0 := u^{(m)}$, $f_{n+1}(x) := \int_0^x f_n(t) dt$, for $n \in \mathbb{N}_0$, and

$$u_n(x) := \varphi_n(x)f_m(x), \quad \varphi_n(x) := \varphi(x/n), \quad x \in \mathbb{R}.$$

Then $u_n \in C_c^\infty(\mathbb{R})$. Thus, it remains to show that $u_n^{(m)} \rightarrow u^{(m)}$ in $L^p(\mathbb{R})$. By Exercise 11.37 and the definition of f_m , we have

$$u_n^{(m)}(x) = \sum_{k=0}^m \binom{m}{k} \varphi_n^{(k)}(x) f_m^{(m-k)}(x) = \sum_{k=0}^m \binom{m}{k} \frac{1}{n^k} \varphi^{(k)}\left(\frac{x}{n}\right) f_k(x).$$

For $k = 0$ we have that $\varphi_n f_0 = \varphi_n u^{(m)} \rightarrow u^{(m)}$ in $L^p(\mathbb{R})$ by the Lebesgue dominated convergence theorem. On the other hand, for $1 \leq k \leq m$,

$$\begin{aligned} \frac{1}{n^{kp}} \int_{\mathbb{R}} |\varphi^{(k)}(x/n) f_k(x)|^p dx &\leq \frac{c}{n^{kp}} \int_{B(0,2n) \setminus B(0,n)} |f_k(x)|^p dx \\ &\leq c \int_{B(0,2n) \setminus B(0,n)} \frac{1}{|x|^{kp}} |f_k(x)|^p dx \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$, since the function $g_k(x) := f_k(x)|x|^{-k}$, $x \in \mathbb{R}$, belongs to $L^p(\mathbb{R})$ by Exercise 11.42.

Step 3: Finally, we study the case $N \geq 2$ and $1 \leq p < \infty$. Fix $u \in W^{m,p}(\mathbb{R}^N)$. As before we can assume that $u \in C^\infty(\mathbb{R}^N) \cap \overline{W^{m,p}(\mathbb{R}^N)}$. Consider a cut-off function $\varphi \in C_c^\infty(\mathbb{R}^N)$ such that $\text{supp } \varphi \subset \overline{B(0,2)}$, $\varphi = 1$ in $B(0,1)$ and $0 \leq \varphi \leq 1$. For $r > 0$, define the annulus $A_r := B(0,2r) \setminus \overline{B(0,r)}$. By Poincaré's inequality (see Theorem 13.27) applied to the annulus A_1 , there exists a constant $c = c(m, p, N) > 0$ such that for every $v \in W^{m,p}(A_1)$ and every $k = 0, \dots, m-1$,

$$\int_{A_1} \|\nabla^k v(x) - \nabla^k p_{A_1}(v)(x)\|^p dx \leq c \int_{A_1} \|\nabla^m v(x)\|^p dx,$$

where for every measurable set with positive finite measure $E \subset \mathbb{R}^N$, $p_E(v)$ is the polynomial of degree $m-1$ such that

$$(11.16) \quad \frac{1}{\mathcal{L}^N(E)} \int_E (\partial^\alpha v(x) - \partial^\alpha p_E(v)(x)) dx = 0$$

for every multi-index $\alpha \in \mathbb{N}_0^N$ with $0 \leq |\alpha| \leq m-1$. A rescaling argument (exercise) shows that for every $v \in W^{m,p}(A_r)$, $r > 0$,

$$(11.17) \quad \int_{A_r} \|\nabla^k v(x) - \nabla^k p_{A_r}(v)(x)\|^p dx \leq cr^{p(m-k)} \int_{A_r} \|\nabla^m v(x)\|^p dx.$$

For $n \in \mathbb{N}$, define

$$(11.18) \quad u_n(x) := \varphi_n(x)(u(x) - p_{A_n}(u)(x)), \quad \varphi_n(x) := \varphi(x/n), \quad x \in \mathbb{R}^N.$$

By Exercise 11.37, for every multi-index α with $|\alpha| = m$,

$$\begin{aligned} \partial^\alpha u_n(x) &= \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta \varphi_n(x) \partial^{\alpha-\beta}(u(x) - p_{A_n}(u)(x)) \\ &= \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \frac{1}{n^{|\beta|}} \partial^\beta \varphi\left(\frac{x}{n}\right) \partial^{\alpha-\beta}(u(x) - p_{A_n}(u)(x)). \end{aligned}$$

If $\beta = 0$, then by the Lebesgue dominated convergence theorem and the fact that $p_{A_n}(u)$ has degree $m-1$, $\varphi_n \partial^\alpha(u - p_{A_n}(u)) = \varphi_n \partial^\alpha u \rightarrow \partial^\alpha u$ in $L^p(\mathbb{R}^N)$ as $n \rightarrow \infty$, while for $\beta \neq 0$, by (11.17),

$$\begin{aligned} &\frac{1}{n^{|\beta|p}} \int_{A_n} |\partial^\beta \varphi(x/n) \partial^{\alpha-\beta}(u(x) - p_{A_n}(u)(x))|^p dx \\ &\leq \frac{c}{n^{|\beta|p}} \int_{A_n} |\partial^{\alpha-\beta}(u(x) - p_{A_n}(u)(x))|^p dx \\ &\leq \frac{cn^{p(m-|\alpha-\beta|)}}{n^{p|\beta|}} \int_{A_n} \|\nabla^m u(x)\|^p dx = c \int_{A_n} \|\nabla^m u(x)\|^p dx \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. This concludes the proof. $\square$

Exercise 11.44. Let $m \in \mathbb{N}$ and let $v \in L_{\text{loc}}^1(\mathbb{R}^N)$ be such that the function $w(x) := (1 + \|x\|^2)^{m/2} v(x)$, $x \in \mathbb{R}^N$, belongs to $L^2(\mathbb{R}^N)$.

- (i) Prove that there exists a function $u \in W^{m,2}(\mathbb{R}^N)$ whose Fourier transform is v .
- (ii) State and prove a similar result for $1 < p < \infty$.

11.3. Absolute Continuity on Lines

The next theorem relates weak partial derivatives with the (classical) partial derivatives. This result is the analog of Theorem 7.16.

In what follows, we use the notation (E.2) in Appendix E. Given $x'_i \in \mathbb{R}^{N-1}$ and a set $E \subseteq \mathbb{R}^N$, we write

$$(11.19) \quad E_{x'_i} := \{x_i \in \mathbb{R} : (x'_i, x_i) \in E\}.$$

Moreover, if $v : \Omega \rightarrow \mathbb{R}$ is Lebesgue integrable, with a slight abuse of notation, if $\Omega_{x'_i}$ is empty, we set

$$\int_{\Omega_{x'_i}} v(x'_i, x_i) dx_i := 0,$$

so that by Fubini's theorem

$$(11.20) \quad \int_{\Omega} v(x) dx = \int_{\mathbb{R}^{N-1}} \left(\int_{\Omega_{x'_i}} v(x'_i, x_i) dx_i \right) dx'_i.$$

Theorem 11.45 (Absolute continuity on lines). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $1 \leq p < \infty$. A function $u \in L^p(\Omega)$ belongs to the space $W^{1,p}(\Omega)$ if and only if it has a representative $\bar{u}$ that is absolutely continuous on $\mathcal{L}^{N-1}$ -a.e. line segments of Ω that are parallel to the coordinate axes and whose first-order (classical) partial derivatives belong to $L^p(\Omega)$. Moreover, the (classical) partial derivatives of $\bar{u}$ agree $\mathcal{L}^N$ -a.e. with the weak derivatives of u .*

Proof. Step 1: Assume that $u \in W^{1,p}(\Omega)$. Consider a sequence of standard mollifiers $\{\varphi_\varepsilon\}_{\varepsilon>0}$ and for every $\varepsilon > 0$ define $u_\varepsilon := u * \varphi_\varepsilon$ in $\Omega_\varepsilon := \{x \in \Omega : \text{dist}(x, \partial\Omega) > \varepsilon\}$. By Lemma 11.25,

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\Omega_\varepsilon} \|\nabla u_\varepsilon(x) - \nabla u(x)\|^p dx = 0.$$

It follows by Fubini's theorem and (11.20) that for all $i = 1, \dots, N$,

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^{N-1}} \left(\int_{(\Omega_\varepsilon)_{x'_i}} \|\nabla u_\varepsilon(x'_i, x_i) - \nabla u(x'_i, x_i)\|^p dx_i \right) dx'_i = 0,$$

where $(\Omega_\varepsilon)_{x'_i} := \{x_i \in \mathbb{R} : (x'_i, x_i) \in \Omega_\varepsilon\}$, and so we may find a subsequence $\{\varepsilon_n\}_n$ such that for all $i = 1, \dots, N$ and for $\mathcal{L}^{N-1}$ a.e. $x'_i \in \mathbb{R}^{N-1}$,

$$(11.21) \quad \lim_{n \rightarrow \infty} \int_{(\Omega_{\varepsilon_n})_{x'_i}} \|\nabla u_{\varepsilon_n}(x'_i, x_i) - \nabla u(x'_i, x_i)\|^p dx_i = 0.$$

Set $u_n := u_{\varepsilon_n}$ and $E := \{x \in \Omega : \lim_{n \rightarrow \infty} u_n(x) \text{ exists in } \mathbb{R}\}$. Define

$$\bar{u}(x) := \begin{cases} \lim_{n \rightarrow \infty} u_n(x) & \text{if } x \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Since by Theorem C.16, $\{u_n\}_n$ converges pointwise to u at every Lebesgue point of u , the set E contains every Lebesgue point of u . It follows from Corollary B.119 that $\mathcal{L}^N(\Omega \setminus E) = 0$, and so $\bar{u}$ is a representative of u . It remains to prove that $\bar{u}$ has the desired properties.

By Fubini's theorem and (11.20) for every $i = 1, \dots, N$ we have that

$$\int_{\mathbb{R}^{N-1}} \left(\int_{\Omega_{x'_i}} \|\nabla u(x'_i, x_i)\|^p dx_i \right) dx'_i < \infty$$

and

$$\int_{\mathbb{R}^{N-1}} \mathcal{L}^1(\{x_i \in \Omega_{x'_i} : (x'_i, x_i) \notin E\}) dx'_i = 0,$$

and so we may find a set $N_i \subset \mathbb{R}^{N-1}$, with $\mathcal{L}^{N-1}(N_i) = 0$, such that for all $x'_i \in \mathbb{R}^{N-1} \setminus N_i$ for which $\Omega_{x'_i}$ is nonempty we have that

$$(11.22) \quad \int_{\Omega_{x'_i}} \|\nabla u(x'_i, x_i)\|^p dx_i < \infty,$$

(11.21) holds and $(x'_i, x_i) \in E$ for $\mathcal{L}^1$ -a.e. $x_i \in \Omega_{x'_i}$.

Fix any such x'_i and let $I \subseteq \Omega_{x'_i}$ be a maximal interval. Fix $t_0 \in I$ such that $(x'_i, t_0) \in E$ and let $t \in I$. For all n large, the interval of endpoints t and t_0 is contained in $(\Omega_{\varepsilon_n})_{x'_i}$ and so, since $u_n \in C^\infty(\Omega_{\varepsilon_n})$, by the fundamental theorem of calculus,

$$u_n(x'_i, t) = u_n(x'_i, t_0) + \int_{t_0}^t \partial_i u_n(x'_i, s) ds.$$

Since $(x'_i, t_0) \in E$, we have $u_n(x'_i, t_0) \rightarrow \bar{u}(x'_i, t_0) \in \mathbb{R}$. On the other hand, by (11.21),

$$(11.23) \quad \lim_{n \rightarrow \infty} \int_{t_0}^t |\partial_i u_n(x'_i, s) - \partial_i u(x'_i, s)| ds = 0.$$

Hence as $n \rightarrow \infty$,

$$u_n(x'_i, t) \rightarrow \bar{u}(x'_i, t_0) + \int_{t_0}^t \partial_i u(x'_i, s) ds.$$

Note that by the definition of E and $\bar{u}$, this implies, in particular, that

$$(11.24) \quad (x'_i, t) \in E \quad \text{and} \quad \bar{u}(x'_i, t) = \bar{u}(x'_i, t_0) + \int_{t_0}^t \partial_i u(x'_i, s) ds$$

for all $t \in I$. Hence, by Theorem 3.16, the function $\bar{u}(x'_i, \cdot)$ is absolutely continuous in I and $\partial_i \bar{u}(x'_i, t) = \partial_i u(x'_i, t)$ for $\mathcal{L}^1$ -a.e. $t \in I$.

Step 2: Assume that u admits a representative $\bar{u}$ that is absolutely continuous on $\mathcal{L}^{N-1}$ -a.e. line segments of Ω that are parallel to the coordinate axes, and whose first-order (classical) partial derivatives belong to $L^p(\Omega)$. Fix $i = 1, \dots, N$ and let $x'_i \in \mathbb{R}^{N-1}$ be such that $\bar{u}(x'_i, \cdot)$ is absolutely continuous on every connected component of the open set $\Omega_{x'_i}$. Then for every function $\varphi \in C_c^\infty(\Omega)$, by the integration by parts formula for absolutely continuous functions, we have

$$\int_{\Omega_{x'_i}} \bar{u}(x'_i, t) \partial_i \varphi(x'_i, t) dt = - \int_{\Omega_{x'_i}} \partial_i \bar{u}(x'_i, t) \varphi(x'_i, t) dt.$$

Since this holds for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in \mathbb{R}^{N-1}$, integrating over $\mathbb{R}^{N-1}$ and using Fubini's theorem yields

$$\int_{\Omega} \bar{u}(x) \partial_i \varphi(x) dx = - \int_{\Omega} \partial_i \bar{u}(x) \varphi(x) dx,$$

which implies that $\frac{\partial \bar{u}}{\partial x_i} \in L^p(\Omega)$ is the weak partial derivative of $\bar{u}$ with respect to x_i . This shows that $u \in W^{1,p}(\Omega)$. $\square$

Remark 11.46. If instead of assuming that $u \in W^{1,p}(\Omega)$ we only assume that $u \in \dot{W}^{1,p}(\Omega)$, then Step 1 of the previous proof still proves that u has a representative $\bar{u}$ that is absolutely continuous on $\mathcal{L}^{N-1}$ -a.e. line segments of Ω that are parallel to the coordinate axes and whose first-order (classical) partial derivatives belong to $L^p(\Omega)$.

Exercise 11.47. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$ and let $1 \leq p < \infty$. Prove that if u belongs to $W^{m,p}(\Omega)$, then u has a representative $\bar{u}$ such that for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in \mathbb{R}^{N-1}$ the function $\bar{u}(x'_i, \cdot)$ is absolutely continuous in $\Omega_{x'_i}$ together with all its derivatives of order up to $m - 1$.

The next exercise shows that there exists an open bounded domain $\Omega \subset \mathbb{R}^2$, with $\partial\Omega = \partial(\bar{\Omega})$, such that $W^{1,p}(\Omega) \cap C(\bar{\Omega})$ is not dense in $W^{1,p}(\Omega)$, $1 \leq p < \infty$ (see Theorem 11.35).

Exercise 11.48. (i) Construct a sequence of open balls $B((t_n, 0), r_n) \subset \mathbb{R}^2$, with $0 < t_n < 1$ and $\sum_{n=1}^{\infty} r_n < \frac{1}{2}$, in such a way that $\bigcup_{n=1}^{\infty} (t_n - r_n, t_n + r_n)$ is dense in $(0, 1)$ and $\partial\Omega = \partial(\bar{\Omega})$.

(ii) Let

$$\Omega := B((1, 0), 1) \setminus \overline{\bigcup_{n=1}^{\infty} B((t_n, 0), r_n)}$$

and for $(x_1, x_2) \in \Omega$ define

$$u(x_1, x_2) := \begin{cases} 1 & \text{for } 0 < x_1 < 1/2, x_2 > 0, \\ 0 & \text{for } 0 < x_1 < 1/2, x_2 < 0, \\ \text{anything convenient} & \text{otherwise.} \end{cases}$$

Prove that $u \in W^{1,p}(\Omega)$ for all $1 \leq p < \infty$.

- (iii) Let Γ be the set of $x_1 \in (\frac{1}{4}, \frac{1}{2})$ with $(x_1, 0) \in \partial\Omega$. Assume by contradiction that there exists a sequence $\{u_n\}_n$ in $W^{1,p}(\Omega) \cap C(\overline{\Omega})$ that converges to u in $W^{1,p}(\Omega)$ and prove that for all $n \in \mathbb{N}$ and for all $x_2 \in (0, \frac{1}{2})$,

$$\int_{\Gamma} |u_n(x_1, x_2) - u_n(x_1, -x_2)| dx_1 \leq \int_{\Gamma \times (-x_2, x_2)} \left| \frac{\partial u_n}{\partial x_2}(x_1, s) \right| dx_1 ds.$$

- (iv) Using Fubini's theorem, prove that there is a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ such that for $\mathcal{L}^1$ -a.e. $x_2 \in (0, \frac{1}{2})$,

$$\lim_{k \rightarrow \infty} \int_{\Gamma} |u_{n_k}(x_1, x_2) - u_{n_k}(x_1, -x_2)| dx_1 = \int_{\Gamma} |u(x_1, x_2) - u(x_1, -x_2)| dx_1$$

and show that this is in contradiction to part (iii).

In the same spirit of Theorem 11.45, we discuss next Lipschitz functions and their relation with $W^{1,\infty}$. The next exercise shows that there exist functions that are in $W^{1,\infty}(\Omega)$ but are not Lipschitz continuous.

Exercise 11.49. Let $\Omega \subset \mathbb{R}^2$ be the open set defined in polar coordinates by $-\pi < \theta < \pi$ and $r > 1$. Prove that the function $u(r, \theta) = \theta$ belongs to $W^{1,\infty}(\Omega)$ but that it is not Lipschitz continuous.

The relation between Lipschitz functions and functions in $W^{1,\infty}$ is discussed in the next exercise.

Exercise 11.50 (Lipschitz functions and $W^{1,\infty}$). (i) Given a function $u \in L^1_{\text{loc}}(\mathbb{R}^N)$ prove that u has a representative $\bar{u}$ that is Lipschitz continuous if and only if its distributional gradient ∇u belongs to $L^\infty(\mathbb{R}^N; \mathbb{R}^N)$.

- (ii) Prove that a function $u \in L^1_{\text{loc}}(\mathbb{R}^N)$ has a representative $\bar{u}$ that is bounded and Lipschitz continuous if and only if $u \in W^{1,\infty}(\mathbb{R}^N)$.
- (iii) Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u : \Omega \rightarrow \mathbb{R}$ be bounded and Lipschitz continuous. Prove that $u \in W^{1,\infty}(\Omega)$.
- (iv) Let $\Omega \subseteq \mathbb{R}^N$ be an extension domain (see Definition 12.16) for $W^{1,\infty}(\Omega)$ and let $u \in W^{1,\infty}(\Omega)$. Prove that u has a representative $\bar{u}$ that is bounded and Lipschitz continuous.

As a consequence of Theorem 11.45, of the properties of absolutely continuous functions, and of the previous exercises, we have the following results.

Exercise 11.51. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $1 \leq p \leq \infty$.

- (i) (**Chain rule**) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be Lipschitz continuous and let $u \in W^{1,p}(\Omega)$. Assume that $f(0) = 0$ if Ω has infinite measure. Prove that $f \circ u \in W^{1,p}(\Omega)$ and that for all $i = 1, \dots, N$ and for $\mathcal{L}^N$ -a.e. $x \in \Omega$,

$$\partial_i(f \circ u)(x) = f'(u(x))\partial_i u(x),$$

where $f'(u(x))\partial_i u(x)$ is interpreted to be zero whenever $\partial_i u(x) = 0$.

- (ii) (**Product rule**) Let $u, v \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$. Prove that $uv \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$ for all $i = 1, \dots, N$ and that for $\mathcal{L}^N$ -a.e. $x \in \Omega$,

$$\partial_i(uv)(x) = v(x)\partial_i u(x) + u(x)\partial_i v(x).$$

- (iii) (**Reflection**) Let $\Omega = \mathbb{R}_+^N := \{(x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : x_N > 0\}$ and let $u \in W^{1,p}(\mathbb{R}_+^N)$. Prove that the function

$$v(x) := \begin{cases} u(x) & \text{if } x_N > 0, \\ u(x', -x_N) & \text{if } x_N < 0 \end{cases}$$

belongs to $W^{1,p}(\mathbb{R}^N)$ and that for all $i = 1, \dots, N$ and for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$,

$$\partial_i v(x) = \begin{cases} \partial_i u(x) & \text{if } x_N > 0, \\ (-1)^{\delta_{iN}} \partial_i u(x', -x_N) & \text{if } x_N < 0, \end{cases}$$

where δ_{iN} is the Kronecker delta, that is, $\delta_{iN} = 1$ if $i = N$ and $\delta_{iN} = 0$ otherwise.

- (iv) Let $E \subset \mathbb{R}$ be such that $\mathcal{L}^1(E) = 0$, let $u \in W_{\text{loc}}^{1,1}(\Omega)$, and let $\bar{u}$ be its precise representative given in Theorem 11.45. Prove that $\nabla u(x) = 0$ for $\mathcal{L}^N$ -a.e. $x \in (\bar{u})^{-1}(E)$.
- (v) Prove Exercise 11.23 using Theorem 11.45.
- (vi) Prove analogous versions of parts (i)–(v) in the case in which the functions u and v are in $\dot{W}^{1,p}(\Omega)$.

Exercise 11.52. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $1 \leq p \leq \infty$. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a Lipschitz continuous function such that the set

$$\Sigma^f := \{u \in \mathbb{R}^d : f \text{ is not differentiable at } u\}$$

is purely $\mathcal{H}^1$ -unrectifiable (see Definition 3.70). Assume that $f(0) = 0$ if Ω has infinite measure. Prove that $f \circ u \in W^{1,p}(\Omega)$ for all $u \in W^{1,p}(\Omega; \mathbb{R}^d)$ and that for all $i = 1, \dots, N$ and for $\mathcal{L}^N$ -a.e. $x \in \Omega$,

$$\frac{\partial(f \circ u)}{\partial x_i}(x) = \sum_{j=1}^d \frac{\partial f}{\partial u_j}(u(x)) \frac{\partial u_j}{\partial x_i}(x),$$

where $\frac{\partial f}{\partial u_j}(u(x)) \frac{\partial u_j}{\partial x_i}(x)$ is interpreted to be zero whenever $\frac{\partial u_j}{\partial x_i}(x) = 0$.

Theorem 11.53 (Change of variables). *Let $\Omega, U \subseteq \mathbb{R}^N$ be open sets, let $\Psi : U \rightarrow \Omega$ be invertible, with Ψ and Ψ^{-1} Lipschitz continuous functions, and let $u \in W^{1,p}(\Omega)$, $1 \leq p \leq \infty$. Then $u \circ \Psi \in W^{1,p}(U)$ and for all $i = 1, \dots, N$ and for $\mathcal{L}^N$ -a.e. $y \in U$,*

$$\frac{\partial(u \circ \Psi)}{\partial y_i}(y) = \sum_{j=1}^N \frac{\partial u}{\partial x_j}(\Psi(y)) \frac{\partial \Psi_j}{\partial y_i}(y).$$

Proof. We only prove the case $1 \leq p < \infty$. Let $\Omega_n \Subset \Omega_{n+1}$ be such that $\Omega = \bigcup_{n=1}^{\infty} \Omega_n$. Fix $n \in \mathbb{N}$ and for $0 < \varepsilon < \text{dist}(\Omega_n, \partial\Omega)$ define $u_\varepsilon := u * \varphi_\varepsilon$ in Ω_n . Let $U_n := \Psi^{-1}(\Omega_n)$ and for $y \in U_n$ set

$$(11.25) \quad v_\varepsilon(y) := u_\varepsilon(\Psi(y)).$$

By Rademacher's theorem the Lipschitz continuous function Ψ is differentiable $\mathcal{L}^N$ -a.e. in U_n . Since $u_\varepsilon \in C^\infty(\Omega_n)$, we conclude that for all $i = 1, \dots, N$ and for $\mathcal{L}^N$ -a.e. $y \in U_n$,

$$(11.26) \quad \frac{\partial v_\varepsilon}{\partial y_i}(y) = \sum_{j=1}^N \frac{\partial u_\varepsilon}{\partial x_j}(\Psi(y)) \frac{\partial \Psi_j}{\partial y_i}(y).$$

Hence, $\left| \frac{\partial v_\varepsilon}{\partial y_i}(y) \right| \leq \sum_{j=1}^N \text{Lip } \Psi_j \left| \frac{\partial u_\varepsilon}{\partial x_j}(\Psi(y)) \right|$, and so

$$\begin{aligned} \int_{U_n} \|\nabla v_\varepsilon(y)\|^p dy &\leq c \int_{\Psi^{-1}(\Omega_n)} \|\nabla u_\varepsilon(\Psi(y))\|^p dy \\ &\leq c \int_{\Psi^{-1}(\Omega_n)} \|\nabla u_\varepsilon(\Psi(y))\|^p |\det J_\Psi(y)| dy \\ &= c \int_{\Omega_n} \|\nabla u_\varepsilon(x)\|^p dx, \end{aligned}$$

where we have used the fact that $|\det J_\Psi|$ is bounded from below by a positive constant (since Ψ^{-1} is Lipschitz continuous) and Theorem 9.52. Taking a sequence $\varepsilon_j \rightarrow 0^+$, the previous inequality implies that

$$\int_{U_n} \|\nabla v_{\varepsilon_j}(y) - \nabla v_{\varepsilon_l}(y)\|^p dy \leq c \int_{\Omega_n} \|\nabla u_{\varepsilon_j}(y) - \nabla u_{\varepsilon_l}(y)\|^p dx$$

for all $j, l \in \mathbb{N}$. Similarly,

$$\int_{U_n} |v_{\varepsilon_j}(y) - v_{\varepsilon_l}(y)|^p dy \leq c \int_{\Omega_n} |u_{\varepsilon_j}(y) - u_{\varepsilon_l}(y)|^p dx$$

for all $j, l \in \mathbb{N}$. Since $u_{\varepsilon_j} \rightarrow u$ in $W^{1,p}(\Omega_n)$, it follows that $\{v_{\varepsilon_j}\}_j$ is a Cauchy sequence in $W^{1,p}(U_n)$, and so it converges to a function $v^{(n)} \in W^{1,p}(U_n)$. By extracting a subsequence, we can assume that v_{ε_j} and ∇v_{ε_j} converge, respectively, to $v^{(n)}$ and $\nabla v^{(n)}$ pointwise $\mathcal{L}^N$ -a.e. in U_n . Since Ψ^{-1} has the (N) property (see Exercise 9.54) and $u_{\varepsilon_j} \rightarrow u$ and $\nabla u_{\varepsilon_j} \rightarrow \nabla u$ pointwise

$\mathcal{L}^N$ -a.e. in Ω_n , it follows that $u_{\varepsilon_j} \circ \Psi \rightarrow u \circ \Psi$ and $(\nabla u_{\varepsilon_j}) \circ \Psi \rightarrow \nabla u \circ \Psi$ pointwise $\mathcal{L}^N$ -a.e. in U_n . In view of (11.25) and (11.26), we conclude that $v^{(n)}(y) = u(\Psi(y))$ for $\mathcal{L}^N$ -a.e. $y \in \Omega_n$ and that for all $i = 1, \dots, N$ and for $\mathcal{L}^N$ -a.e. $y \in U_n$,

$$(11.27) \quad \frac{\partial(u \circ \Psi)}{\partial y_i}(y) = \sum_{j=1}^N \frac{\partial u}{\partial x_j}(\Psi(y)) \frac{\partial \Psi_j}{\partial y_i}(y).$$

Since $u \circ \Psi = v^{(n)}$ in U_n , it follows that $u \circ \Psi \in W^{1,p}(U_n)$ for all $n \in \mathbb{N}$. Reasoning as in the first part of the proof, it follows from (11.27) and Theorem 9.52 that

$$\begin{aligned} \int_U \|\nabla(u \circ \Psi)(y)\|^p dy &\leq c \int_{\Psi^{-1}(\Omega)} \|\nabla u(\Psi(y))\|^p |\det J_\Psi(y)| dy \\ &= c \int_\Omega \|\nabla u(x)\|^p dx, \end{aligned}$$

and similarly

$$\int_U |(u \circ \Psi)(y)|^p dy \leq c \int_\Omega |u(x)|^p dx.$$

Since $u \circ \Psi \in W^{1,p}(U_n)$ for all $n \in \mathbb{N}$, applying either Theorem 11.45 or the definition of weak derivatives yields $u \circ \Psi \in W^{1,p}(U)$. $\square$

To extend the previous theorem to $W^{m,p}(\Omega)$ we need the extension of Faa di Bruno's formula (see [123], [219]) to calculate higher-order derivatives of a composition.

Theorem 11.54 (Faa di Bruno). *Let $\Omega \subseteq \mathbb{R}^N$ and $U \subseteq \mathbb{R}^M$ be open sets, let $u : \Omega \rightarrow \mathbb{R}$ be of class C^m and let $\Psi : U \rightarrow \Omega$ be of class C^m . Then $u \circ \Psi$ belongs to $C^m(U)$ and for every multi-index $\alpha \in \mathbb{N}_0^M$, with $1 \leq |\alpha| \leq m$,*

$$\frac{\partial^{|\alpha|}}{\partial y^\alpha}(u \circ \Psi)(y) = \sum c_{\alpha,\beta,\gamma,l} \frac{\partial^{|\beta|} u}{\partial x^\beta}(\Psi(y)) \prod_{i=1}^{|\beta|} \frac{\partial^{|\gamma_i|} \Psi_{l_i}}{\partial y^{\gamma_i}}(y),$$

where $c_{\alpha,\beta,\gamma,l} \in \mathbb{R}$, the sum is done over all $\beta \in \mathbb{N}_0^M$ with $1 \leq |\beta| \leq |\alpha|$, $\gamma = (\gamma_1, \dots, \gamma_{|\beta|})$, $\gamma_i \in \mathbb{N}_0^M$, with $|\gamma_i| > 0$ and $\sum_{i=1}^{|\beta|} \gamma_i = \alpha$, and $l = (l_1, \dots, l_{|\beta|})$, $l_i \in \{1, \dots, N\}$, $i = 1, \dots, |\beta|$.

Proof. The proof is by induction on m and is left as an exercise. $\square$

We remark that the multi-indices γ_i , $i = 1, \dots, |\beta|$, are not pairwise distinct so some of the derivatives $\frac{\partial^{|\gamma_i|} \Psi_{l_i}}{\partial y^{\gamma_i}}$ are repeated. The constants $c_{\alpha,\beta,\gamma,l}$ can be computed explicitly (see [70]).

Definition 11.55. Given an open set $\Omega \subseteq \mathbb{R}^N$, $m \in \mathbb{N}_0$, $M \in \mathbb{N}$, and $0 < \alpha \leq 1$, we define the space $C^{m,\alpha}(\overline{\Omega}; \mathbb{R}^M)$ as the space of all functions in $C^m(\overline{\Omega}; \mathbb{R}^M)$ such that

$$\|u\|_{C^{m,\alpha}(\overline{\Omega}; \mathbb{R}^M)} := \|u\|_{C^m(\overline{\Omega}; \mathbb{R}^M)} + \sum_{|\alpha|=m} |\partial^\alpha u|_{C^{0,\alpha}(\overline{\Omega}; \mathbb{R}^M)} < \infty.$$

We recall that for a function $v : \Omega \rightarrow \mathbb{R}^M$,

$$|v|_{C^{0,\alpha}(\overline{\Omega}; \mathbb{R}^M)} := \sup_{x, y \in \Omega, x \neq y} \frac{\|v(x) - v(y)\|_M}{\|x - y\|_N^\alpha}.$$

As usual, when $M = 1$ we write $C^{m,\alpha}(\overline{\Omega})$ for $C^{m,\alpha}(\overline{\Omega}; \mathbb{R})$.

Exercise 11.56. Given an open set $\Omega \subseteq \mathbb{R}^N$, $m \in \mathbb{N}_0$ and $0 < \alpha \leq 1$. Prove that the space $C^{m,\alpha}(\overline{\Omega})$, $0 < \alpha \leq 1$, is a Banach space.

Theorem 11.57. Let $\Omega, U \subseteq \mathbb{R}^N$ be open sets, let $m \in \mathbb{N}$, let $\Psi : U \rightarrow \Omega$ be invertible, with $\Psi \in C^{m-1,1}(\overline{U}; \mathbb{R}^N)$ and Ψ^{-1} Lipschitz continuous, and let $u \in W^{m,p}(\Omega)$, $1 \leq p \leq \infty$. Then $u \circ \Psi$ belongs to $W^{m,p}(U)$ and for every multi-index $\alpha \in \mathbb{N}_0^M$, with $0 < |\alpha| \leq m$, and for $\mathcal{L}^N$ -a.e. $y \in U$,

$$\frac{\partial^{|\alpha|}}{\partial y^\alpha} (u \circ \Psi)(y) = \sum c_{\alpha,\beta,\gamma,l} \frac{\partial^{|\beta|} u}{\partial x^\beta}(\Psi(y)) \prod_{i=1}^{|\beta|} \frac{\partial^{|\gamma_i|} \Psi_{l_i}}{\partial y^{\gamma_i}}(y),$$

where the summation is done as in Theorem 11.54.

Proof. The proof is similar to the one of Theorem 11.53 and we omit it. $\square$

Exercise 11.58. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Given $n \in \mathbb{N}$, let $u_1, \dots, u_n \in W^{1,p}(\Omega)$, $1 \leq p \leq \infty$. Prove that $v_n := \max_{1 \leq k \leq n} u_k \in W^{1,p}(\Omega)$ and that

$$|\partial_i v_n(x)| \leq \max_{1 \leq k \leq n} |\partial_i u_k(x)|$$

for $\mathcal{L}^N$ -a.e. $x \in \Omega$ and all $i = 1, \dots, N$.

Exercise 11.59. Let $u \in W^{1,p}(\mathbb{R}^N)$, $1 \leq p < \infty$, and for $r > 0$ let

$$\chi_r := \frac{1}{\alpha_N r^N} \chi_{B(0,r)},$$

where we recall that $\alpha_N = \mathcal{L}^N(B(0,1))$. Prove that $|u| * \chi_r$ belongs to $W^{1,p}(\mathbb{R}^N)$ and that $\partial_i(|u| * \chi_r) = \chi_r * \partial_i |u|$ for $i = 1, \dots, N$.

Exercise 11.60. Let $u \in W^{1,p}(\mathbb{R}^N)$, $1 < p < \infty$, and let $M(u)$ be the maximal function of u (see Definition C.25).

- (i) Let $\{r_n\}_n$ be an enumeration of the positive rationals and define $v_n := \max_{1 \leq k \leq n} |u| * \chi_{r_k}$. Prove that $v_n \in W^{1,p}(\mathbb{R}^N)$ and that $|\partial_i v_n(x)| \leq M(\partial_i u)(x)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. Deduce that the sequence $\{v_n\}_n$ is bounded in $W^{1,p}(\mathbb{R}^N)$.

- (ii) Prove that $M(u)$ belongs to $W^{1,p}(\mathbb{R}^N)$ and that $\partial_i(M(u)) = M(\partial_i u)$ for $i = 1, \dots, N$.

11.4. Duals and Weak Convergence

In this section we study the dual spaces of $W^{m,p}(\Omega)$ and $W_0^{m,p}(\Omega)$. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$. In this section it will be convenient to consider in $W^{m,p}(\Omega)$ the equivalent norm

$$(11.28) \quad \|u\|_{W^{m,p}(\Omega)} := \left(\sum_{0 \leq |\alpha| \leq m} \|\partial^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}$$

for $1 \leq p < \infty$ and

$$(11.29) \quad \|u\|_{W^{m,\infty}(\Omega)} := \max_{0 \leq |\alpha| \leq m} \|\partial^\alpha u\|_{L^\infty(\Omega)}$$

if $p = \infty$, where we recall that $\partial^0 u = u$.

For every $1 \leq q \leq \infty$ we consider the space $L^q(\Omega; \mathbb{R}^M)$, where M is the number of multi-indices α with $0 \leq |\alpha| \leq m$. Given $f \in L^q(\Omega; \mathbb{R}^M)$ we label the components of the vector-valued function f as f_α , with $0 \leq |\alpha| \leq m$, and we consider in $L^q(\Omega; \mathbb{R}^M)$ the equivalent norm

$$\|f\|_{L^q(\Omega; \mathbb{R}^M)} := \left(\sum_{0 \leq |\alpha| \leq m} \|f_\alpha\|_{L^q(\Omega)}^q \right)^{1/q}$$

if $1 \leq q < \infty$ and

$$\|f\|_{L^\infty(\Omega; \mathbb{R}^M)} := \max_{0 \leq |\alpha| \leq m} \|f_\alpha\|_{L^\infty(\Omega)}$$

if $q = \infty$.

The idea is to consider $W^{m,p}(\Omega)$ as a closed subspace of $L^p(\Omega; \mathbb{R}^M)$. In this way any element of $L \in (W^{m,p}(\Omega))'$ can be extended by the Hahn–Banach theorem (see Theorem A.32) to an element of $(L^p(\Omega; \mathbb{R}^M))'$. The Riesz representation theorem in $L^p(\Omega; \mathbb{R}^M)$ (see Theorem B.92) will give a representation for the extension of L and, in turn, for L .

We begin with the case $1 \leq p < \infty$. Observe that if p' is the Hölder conjugate exponent of p and $f \in L^{p'}(\Omega; \mathbb{R}^M)$, then the functional

$$(11.30) \quad L(u) := \sum_{0 \leq |\alpha| \leq m} \int_{\Omega} f_\alpha(x) \partial^\alpha u(x) dx, \quad u \in W^{m,p}(\Omega),$$

belongs to $(W^{m,p}(\Omega))'$. Indeed, by Hölder's inequality (for functions and for finite sums) we have that

$$|L(u)| \leq \sum_{0 \leq |\alpha| \leq m} \|f_\alpha\|_{L^{p'}(\Omega)} \|\partial^\alpha u\|_{L^p(\Omega)} \leq \|f\|_{L^{p'}(\Omega; \mathbb{R}^M)} \|u\|_{W^{m,p}(\Omega)}$$

for all $u \in W^{m,p}(\Omega)$, and thus by (11.28),

$$(11.31) \quad \|L\|_{(W^{m,p}(\Omega))'} = \sup_{u \in W^{m,p}(\Omega) \setminus \{0\}} \frac{|L(u)|}{\|u\|_{W^{m,p}(\Omega)}} \leq \|f\|_{L^{p'}(\Omega; \mathbb{R}^M)}.$$

Note that in general we do not have equality in the previous inequality (see Remark 11.63 below).

Exercise 11.61. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, and let $1 \leq p < \infty$. Using the Riesz representation theorem in $L^p(\Omega)$, prove that for every $L \in (L^p(\Omega; \mathbb{R}^M))'$ there exists a unique $f \in L^{p'}(\Omega; \mathbb{R}^M)$ such that

$$L(g) = \sum_{0 \leq |\alpha| \leq m} \int_{\Omega} f_{\alpha}(x) g_{\alpha}(x) dx \quad \text{for all } g \in L^p(\Omega; \mathbb{R}^M).$$

Prove also that $\|L\|_{(L^p(\Omega; \mathbb{R}^M))'} = \|f\|_{L^{p'}(\Omega; \mathbb{R}^M)}$.

Theorem 11.62 (Riesz's representation theorem in $W^{m,p}$). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, let $1 \leq p < \infty$, and let p' be its Hölder conjugate exponent. Then for every bounded linear functional $L \in (W^{m,p}(\Omega))'$ there exist $f \in L^{p'}(\Omega; \mathbb{R}^M)$ such that*

$$(11.32) \quad L(u) = \sum_{0 \leq |\alpha| \leq m} \int_{\Omega} f_{\alpha}(x) \partial^{\alpha} u(x) dx$$

for all $u \in W^{m,p}(\Omega)$ and

$$(11.33) \quad \|L\|_{(W^{m,p}(\Omega))'} = \|f\|_{L^{p'}(\Omega; \mathbb{R}^M)}.$$

Proof. Consider the application

$$(11.34) \quad \begin{aligned} T : W^{m,p}(\Omega) &\rightarrow L^p(\Omega; \mathbb{R}^M) \\ u &\mapsto (\partial^0 u, \partial^{\alpha_1} u, \dots, \partial^{\alpha_M} u), \end{aligned}$$

where $\alpha_1, \dots, \alpha_M$ are all the multi-indices α with $1 \leq |\alpha| \leq m$. The operator T is one-to-one and continuous and it preserves the norm; that is, $\|T(u)\|_{L^p(\Omega; \mathbb{R}^M)} = \|u\|_{W^{m,p}(\Omega)}$ for all $u \in W^{m,p}(\Omega)$. Hence, also by Theorem 11.12, the subspace $Y := T(W^{m,p}(\Omega))$ is closed in $L^p(\Omega; \mathbb{R}^M)$. Given $L \in (W^{m,p}(\Omega))'$, define the functional

$$L_1 : Y \rightarrow \mathbb{R}$$

$$g = (g_0, \dots, g_{\alpha_M}) \mapsto L(T^{-1}((g_0, \dots, g_{\alpha_M}))).$$

Since T preserves the norm, we have that L_1 is linear and continuous, with $\|L_1\|_{Y'} = \|L\|_{(W^{m,p}(\Omega))'}$. By the Hahn–Banach theorem (see Theorem A.32) we may extend L_1 as a continuous linear operator $L_2 : L^p(\Omega; \mathbb{R}^M) \rightarrow \mathbb{R}$ in such a way that $\|L_2\|_{(L^p(\Omega; \mathbb{R}^M))'} = \|L_1\|_{Y'} = \|L\|_{(W^{m,p}(\Omega))'}$. Note that since Y is closed, it is not dense in $L^p(\Omega; \mathbb{R}^M)$, and so the extension will not be unique.

By the previous exercise there exist a unique function $f \in L^{p'}(\Omega; \mathbb{R}^M)$ such that

$$L_1(g) = \sum_{0 \leq |\alpha| \leq m} \int_{\Omega} f_{\alpha}(x) g_{\alpha}(x) dx$$

for all $g \in L^p(\Omega; \mathbb{R}^M)$ and $\|L\|_{(W^{m,p}(\Omega))'} = \|L_2\|_{(L^p(\Omega; \mathbb{R}^M))'} = \|f\|_{L^{p'}(\Omega; \mathbb{R}^M)}$. It follows that (11.32) holds. $\square$

Remark 11.63. Note that the previous theorem does not imply that the dual of $W^{m,p}(\Omega)$ is $L^{p'}(\Omega; \mathbb{R}^M)$. Indeed, if f belongs to $L^{p'}(\Omega; \mathbb{R}^M)$, then we have shown that the functional L defined in (11.30) belongs to $(W^{m,p}(\Omega))'$ and that (11.31) holds. On the other hand, by Theorem 11.62 there exists h in $L^{p'}(\Omega; \mathbb{R}^M)$ (possibly different from f) such that $\|L\|_{(W^{m,p}(\Omega))'} = \|h\|_{L^{p'}(\Omega; \mathbb{R}^M)}$. Thus, we have that

$$\|L\|_{(W^{m,p}(\Omega))'} = \min\{\|h\|_{L^{p'}(\Omega)} : h \in L^{p'}(\Omega; \mathbb{R}^M) \text{ such that (11.32) holds}\}.$$

To explore the possible lack of uniqueness, assume that $h \in L^{p'}(\Omega; \mathbb{R}^M)$ is such that

$$\sum_{0 \leq |\alpha| \leq m} \int_{\Omega} f_{\alpha} \partial^{\alpha} u dx = \sum_{0 \leq |\alpha| \leq m} \int_{\Omega} h_{\alpha} \partial^{\alpha} u dx$$

for all $u \in W^{m,p}(\Omega)$. In particular, we obtain that

$$\sum_{0 \leq |\alpha| \leq m} \int_{\Omega} f_{\alpha} \partial^{\alpha} \phi dx = \sum_{0 \leq |\alpha| \leq m} \int_{\Omega} h_{\alpha} \partial^{\alpha} \phi dx$$

for all $\phi \in \mathcal{D}(\mathbb{R}^N)$. If we rewrite this identity in the sense of distributions (see Definition 10.14), we get

$$\sum_{0 \leq |\alpha| \leq m} (-1)^{|\alpha|} \partial_{\alpha} T_{f_{\alpha} \chi_{\Omega}}(\phi) = \sum_{0 \leq |\alpha| \leq m} (-1)^{|\alpha|} \partial_{\alpha} T_{h_{\alpha} \chi_{\Omega}}(\phi)$$

for all $\phi \in \mathcal{D}(\mathbb{R}^N)$, or, equivalently,

$$(11.35) \quad \sum_{0 \leq |\alpha| \leq m} (-1)^{|\alpha|} \partial_{\alpha} (f_{\alpha} \chi_{\Omega}) = \sum_{0 \leq |\alpha| \leq m} (-1)^{|\alpha|} \partial_{\alpha} (h_{\alpha} \chi_{\Omega})$$

for all $\phi \in \mathcal{D}(\mathbb{R}^N)$. Hence, we have shown that given $f \in L^{p'}(\Omega; \mathbb{R}^M)$, any solution $h \in L^{p'}(\Omega; \mathbb{R}^M)$ (in the sense of distributions) of the partial differential equation (11.35) in $\mathbb{R}^N$ will give rise to a different representation in (11.32). In particular, when $m = 1$ and Ω and $f_i \in L^{p'}(\Omega)$, $i = 1, \dots, N$ are sufficiently smooth, then any solution of the partial differential equation

$$\begin{cases} \operatorname{div} h = \operatorname{div} f & \text{in } \Omega, \\ h \cdot \nu = f \cdot \nu & \text{on } \partial\Omega, \end{cases}$$

where $f := (f_1, \dots, f_N)$ will do.

In view of Theorem 11.62, we can characterize weak convergence in $W^{m,p}(\Omega)$.

Exercise 11.64. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, let $1 \leq p < \infty$, and let $\{u_n\}_n$ be a sequence in $W^{m,p}(\Omega)$. Prove that $u_n \rightharpoonup u$ in $W^{m,p}(\Omega)$ if and only if $u_n \rightharpoonup u$ in $L^p(\Omega)$ and $\partial^\alpha u_n \rightharpoonup \partial^\alpha u$ in $L^p(\Omega)$ for every $1 \leq |\alpha| \leq m$.

If Ω has additional properties, in the previous exercise one can replace $u_n \rightharpoonup u$ in $L^p(\Omega)$ and $\partial^\alpha u_n \rightharpoonup \partial^\alpha u$ in $L^p(\Omega)$ with $u_n \rightarrow u$ in $L^p(\Omega)$ and $\partial^\alpha u_n \rightharpoonup \partial^\alpha u$ in $L^p(\Omega)$ for every $1 \leq |\alpha| \leq m-1$ (see Exercise 12.22).

Theorem 11.65 (Compactness). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, and let $1 < p < \infty$. Assume that $\{u_n\}$ is bounded in $W^{m,p}(\Omega)$. Then there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and $u \in W^{m,p}(\Omega)$ such that $u_{n_k} \rightharpoonup u$ in $W^{m,p}(\Omega)$.*

Proof. Since $\{u_n\}_n$ and $\{\partial^\alpha u_n\}_n$ are bounded in the reflexive Banach space $L^p(\Omega)$, we may select a subsequence $\{u_{n_k}\}$ such that $u_{n_k} \rightharpoonup u$ in $L^p(\Omega)$ and $\partial^\alpha u_{n_k} \rightharpoonup v_\alpha$ in $L^p(\Omega)$ for every $1 \leq |\alpha| \leq m$ and for some functions $u, v_\alpha \in L^p(\Omega)$, $1 \leq |\alpha| \leq m$. It remains to show that $u \in W^{m,p}(\Omega)$. For every $\phi \in C_c^\infty(\Omega)$, $1 \leq |\alpha| \leq m$, and $k \in \mathbb{N}$ we have

$$\int_{\Omega} u_{n_k} \partial^\alpha \phi \, dx = (-1)^{|\alpha|} \int_{\Omega} \partial^\alpha u_{n_k} \phi \, dx.$$

Letting $k \rightarrow \infty$ in the previous equality and using weak convergence in $L^p(\Omega)$ yields

$$\int_{\Omega} u \partial^\alpha \phi \, dx = (-1)^{|\alpha|} \int_{\Omega} v_\alpha \phi \, dx,$$

which shows that $\partial^\alpha u = v_\alpha$. Hence, $u \in W^{m,p}(\Omega)$. $\square$

Remark 11.66. The previous result fails for $p = 1$. Indeed, in this case, $L^1(\Omega)$ is not reflexive (so we do not expect weak sequential compactness of bounded sequences) and it is not the dual of a separable space (so we do not expect weak star sequential compactness of bounded sequences). To recover some compactness, we use the embedding

$$\begin{aligned} L^1(\Omega) &\rightarrow \mathcal{M}_b(\Omega) \\ u &\mapsto \lambda_u(E) \end{aligned}$$

where $\lambda_u(E) := \int_E u(x) \, dx$, $E \in \mathcal{B}(\Omega)$. Note that (why?) $\int_{\Omega} |u| \, dx = \|\lambda_u\|_{\mathcal{M}_b(\Omega)}$. Thus, given a bounded sequence $\{u_n\}_n$ in $L^1(\Omega)$, one can only conclude that there exist a subsequence $\{u_{n_k}\}_k$ and $\lambda \in \mathcal{M}_b(\Omega)$ such that $\lambda_{u_{n_k}} \xrightarrow{*} \lambda$ in $\mathcal{M}_b(\Omega)$. In particular, given a bounded sequence $\{u_n\}_n$ in $W^{1,1}(\Omega)$, one will only recover some compactness in the space $BV(\Omega)$. See Theorem 14.39.

Exercise 11.67. Let $\Omega = B(0, 1)$. Construct a bounded sequence $\{u_n\}_n$ in $W^{1,1}(\Omega)$ converging strongly in $L^1(\Omega)$ to the function given in Exercise 11.8.

Every functional $L \in (W^{m,p}(\Omega))'$ is the extension of a distribution $T \in \mathcal{D}'(\Omega)$. More precisely, if L has the form (11.32), then its restriction to $\mathcal{D}(\Omega)$ is given by

$$(11.36) \quad T(\phi) := \sum_{0 \leq |\alpha| \leq m} (-1)^{|\alpha|} \partial^\alpha T_{f_\alpha}(\phi) \quad \text{for all } \phi \in \mathcal{D}(\Omega).$$

Conversely, a distribution $T \in \mathcal{D}'(\Omega)$ of the form (11.36), where $f \in L^{p'}(\Omega; \mathbb{R}^M)$, can be extended to an element of $(W^{m,p}(\Omega))'$, but this extension may not be unique, unless $W^{m,p}(\Omega) = W_0^{m,p}(\Omega)$.

Definition 11.68. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, let $1 \leq p \leq \infty$, and let p' be its Hölder conjugate exponent. The dual space of $W_0^{m,p}(\Omega)$ is denoted by $W^{-m,p'}(\Omega)$.

Exercise 11.69. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, let $1 \leq p < \infty$, and let $T \in \mathcal{D}'(\Omega)$ be of the form (11.36), where $f \in L^{p'}(\Omega; \mathbb{R}^M)$. Prove that T may be uniquely extended to a functional $L \in W^{-m,p'}(\Omega)$.

In view of the previous exercise we have the following characterization of the dual of $W_0^{m,p}(\Omega)$.

Corollary 11.70 (Riesz's representation theorem in $W_0^{m,p}$). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $m \in \mathbb{N}$, and let $1 \leq p < \infty$. Then $W^{-m,p'}(\Omega)$ can be identified with the subspace of distributions T of the form*

$$T = \sum_{0 \leq |\alpha| \leq m} (-1)^{|\alpha|} \partial^\alpha T_{f_\alpha},$$

where $f_\alpha \in L^{p'}(\Omega; \mathbb{R}^M)$.

Next we study the case $p = \infty$. By Theorem B.94 the dual of $L^\infty(\Omega)$ can be identified with the space of all bounded finitely additive signed measures that are absolutely continuous with respect to the Lebesgue measure restricted to Ω . Thus, if λ_α , $0 \leq |\alpha| \leq m$, are any such measures, then the functional

$$L(u) := \sum_{0 \leq |\alpha| \leq m} \int_\Omega \partial^\alpha u \, d\lambda_\alpha, \quad u \in W^{m,\infty}(\Omega),$$

belongs to $(W^{m,\infty}(\Omega))'$. Indeed, since each λ_α belongs to $(L^\infty(\Omega))'$, $0 \leq |\alpha| \leq m$, we have that

$$\begin{aligned} |L(u)| &\leq \sum_{0 \leq |\alpha| \leq m} \|\lambda_\alpha\|_{(L^\infty(\Omega))'} \|\partial^\alpha u\|_{L^\infty(\Omega)} \\ &\leq \sum_{0 \leq |\alpha| \leq m} \|\lambda_\alpha\|_{(L^\infty(\Omega))'} \max_{0 \leq |\alpha| \leq m} \|\partial^\alpha u\|_{L^\infty(\Omega)} \end{aligned}$$

for all $u \in W^{m,\infty}(\Omega)$, and thus,

$$\|L\|_{(W^{m,\infty}(\Omega))'} = \sup_{u \in W^{m,\infty}(\Omega) \setminus \{0\}} \frac{|L(u)|}{\|u\|_{W^{m,\infty}(\Omega)}} \leq \sum_{0 \leq |\alpha| \leq m} \|\lambda_\alpha\|_{(L^\infty(\Omega))'}.$$

Reasoning exactly as in the proof of Theorem 11.62, we have the following result.

Theorem 11.71 (Riesz's representation theorem in $W^{m,\infty}$). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $m \in \mathbb{N}$. For every bounded linear functional $L \in (W^{m,\infty}(\Omega))'$ there exist bounded finitely additive signed measures λ_α , $0 \leq |\alpha| \leq m$, that are absolutely continuous with respect to the Lebesgue measure $\mathcal{L}^N$ restricted to Ω , such that*

$$(11.37) \quad L(u) = \sum_{0 \leq |\alpha| \leq m} \int_{\Omega} \partial^\alpha u \, d\lambda_\alpha$$

for all $u \in W^{m,\infty}(\Omega)$.

The proofs of the next results are left as an exercise.

Corollary 11.72 (Riesz's representation theorem in $W_0^{m,\infty}$). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $m \in \mathbb{N}$. The space $W^{-m,1}(\Omega)$ can be identified with the space of all distributions T of the form $T = \sum_{0 \leq |\alpha| \leq m} (-1)^{|\alpha|} \partial_\alpha T_{\lambda_\alpha}$, where λ_α , $0 \leq |\alpha| \leq m$, are bounded finitely additive measures that are absolutely continuous with respect to the Lebesgue measure $\mathcal{L}^N$ restricted to Ω .*

Theorem 11.73 (Compactness). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $m \in \mathbb{N}$. Assume that $\{u_n\}_n$ is bounded in $W^{m,\infty}(\Omega)$. Then there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and $u \in W^{m,\infty}(\Omega)$ such that $u_{n_k} \xrightarrow{*} u$ in $L^\infty(\Omega)$ and $\partial^\alpha u_{n_k} \xrightarrow{*} \partial^\alpha u$ in $L^\infty(\Omega)$ for every $1 \leq |\alpha| \leq m$.*

11.5. A Characterization of $W^{1,p}(\Omega)$

In this section we give a characterization of $W^{1,p}(\Omega)$ and $\dot{W}^{1,p}(\Omega)$ in terms of difference quotients. An analogous result has been given in Corollary 2.51 for functions of bounded pointwise variation in one variable. As we already mentioned, results of this type are often useful in the regularity theory for partial differential equations (see, e.g., [33] and [71]). Moreover,

they provide characterizations that do not involve derivatives and thus they can be used to extend the definition of Sobolev spaces to more abstract settings (see, e.g., [12] and [107]).

Let $\Omega \subseteq \mathbb{R}^N$ be an open set and for every $h \in \mathbb{R}^N \setminus \{0\}$, let

$$\Omega_h := \{x \in \Omega : x + th \in \Omega \text{ for all } t \in [0, 1]\}.$$

Exercise 11.74. (i) Prove that if $0 < p < 1$, then $(a + b)^p \leq a^p + b^p$ for all $a, b \geq 0$.

(ii) Prove that if $p \geq 1$ and $\varepsilon > 0$, then there exists a constant $c = c(\varepsilon, p) > 0$ such that

$$(a + b)^p \leq (1 + \varepsilon)a^p + cb^p$$

for all $a, b \geq 0$. Hint: Use the convexity of the function $g(t) = |t|^p$, $t \in \mathbb{R}$.

We recall that S^{N-1} is the unit sphere $\partial B(0, 1)$ and that

$$\beta_N = \mathcal{H}^{N-1}(S^{N-1}),$$

where we recall that $\mathcal{H}^{N-1}$ is the $(N - 1)$ -dimensional Hausdorff measure (see Section C.7 in Appendix C).

Theorem 11.75. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in \dot{W}^{1,p}(\Omega)$, $1 \leq p < \infty$. Then for every $h \in \mathbb{R}^N \setminus \{0\}$,

$$(11.38) \quad \int_{\Omega_h} \frac{|u(x + h) - u(x)|^p}{\|h\|^p} dx \leq \int_{\Omega} \|\nabla u(x)\|^p dx,$$

while

$$(11.39) \quad \kappa_{N,p} \int_{\Omega} \|\nabla u(x)\|^p dx \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x + h) - u(x)|^p}{\|h\|^p} dx,$$

where

$$(11.40) \quad \kappa_{N,p} := \frac{1}{\beta_N} \int_{S^{N-1}} |e_1 \cdot \xi|^p d\mathcal{H}^{N-1}(\xi).$$

Conversely, if $1 < p < \infty$ and $u \in L^1_{\text{loc}}(\Omega)$ is such that

$$(11.41) \quad \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x + h) - u(x)|^p}{\|h\|^p} dx < \infty,$$

then $u \in \dot{W}^{1,p}(\Omega)$.

Proof. Step 1: Let $u \in \dot{W}^{1,p}(\Omega)$ and $h \in \mathbb{R}^N \setminus \{0\}$. Let $U \Subset \Omega$ and for $0 < \varepsilon < \text{dist}(U, \partial\Omega)$ define $u_\varepsilon := \varphi_\varepsilon * u$, where φ_ε is a standard mollifier. By

the fundamental theorem of calculus for $x \in U_h$ we have

$$\begin{aligned} |u_\varepsilon(x+h) - u_\varepsilon(x)| &= \left| \int_0^1 \nabla u_\varepsilon(x+th) \cdot h \, dt \right| \\ &\leq \int_0^1 |\nabla u_\varepsilon(x+th) \cdot h| \, dt \leq \|h\| \int_0^1 \|\nabla u_\varepsilon(x+th)\| \, dt. \end{aligned}$$

Raising to the power p and integrating over U_h , by Hölder's inequality we get

$$\begin{aligned} \int_{U_h} |u_\varepsilon(x+h) - u_\varepsilon(x)|^p \, dx &\leq \|h\|^p \int_{U_h} \left(\int_0^1 \|\nabla u_\varepsilon(x+th)\| \, dt \right)^p \, dx \\ &\leq \|h\|^p \int_{U_h} \int_0^1 \|\nabla u_\varepsilon(x+th)\|^p \, dt \, dx \leq \|h\|^p \int_U \|\nabla u_\varepsilon(y)\|^p \, dy, \end{aligned}$$

where we have used Tonelli's theorem and the change of variables $y = x+th$. Since $\partial_i u_\varepsilon \rightarrow \partial_i u$ in $L^p(U)$ as $\varepsilon \rightarrow 0^+$ by Lemma 11.25 and $u_\varepsilon \rightarrow u$ pointwise for $\mathcal{L}^N$ -a.e. in Ω , letting $\varepsilon \rightarrow 0^+$ and using Fatou's lemma gives

$$\int_{U_h} \|u(x+h) - u(x)\|^p \, dx \leq \|h\|^p \int_U \|\nabla u(y)\|^p \, dy \leq \|h\|^p \int_\Omega \|\nabla u(y)\|^p \, dy.$$

To prove (11.38) it suffices to let $U \nearrow \Omega$ and use the Lebesgue monotone convergence theorem.

Step 2: We will prove (11.39) for $u \in C^\infty(\Omega) \cap \dot{W}^{1,p}(\Omega)$. Let

$$M := \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x+h) - u(x)|^p}{\|h\|^p} \, dx.$$

If $M = \infty$, then there is nothing to prove. Thus, in what follows we assume that $M < \infty$. Given $\delta > 0$ let $d_\delta > 0$ be so small that

$$\int_{\Omega_h} \frac{|u(x+h) - u(x)|^p}{\|h\|^p} \, dx \leq M + \delta$$

for every $0 < \|h\| < d_\delta$. Fix a compact set $K \subset \Omega$. Let

$$d := \frac{1}{2} \min\{\text{dist}(K, \partial\Omega), 1\}.$$

By Taylor's formula there exists a constant $c_K > 0$ (depending on u) such that

$$|u(x+h) - u(x) - \nabla u(x) \cdot h| \leq c_K \|h\|^2$$

for all $x \in K$ and $0 < \|h\| < d$. Hence,

$$|\nabla u(x) \cdot h| \leq |u(x+h) - u(x)| + c_K \|h\|^2$$

for all $x \in K$ and $0 < \|h\| < d$. By the previous exercise, for every $\delta > 0$ we have

$$|\nabla u(x) \cdot h|^p \leq (1 + \delta)|u(x + he) - u(x)|^p + c_{K,\delta}\|h\|^{2p}$$

for all $x \in K$ and $0 < \|h\| < d$. Fix $0 < \eta < \min\{d, d_\delta\}$. Dividing by $\|h\|^p$ and integrating over K yields

$$\begin{aligned} \int_K |\nabla u(x) \cdot \xi|^p dx &\leq (1 + \delta) \int_K \frac{|u(x + h) - u(x)|^p}{\|h\|^p} dx + c_{K,\delta} \mathcal{L}^N(K) \|h\|^p \\ &\leq (1 + \delta)(M + \delta) + c_{K,\delta} \mathcal{L}^N(K) \eta^p \end{aligned}$$

for all $x \in K$ and $0 < \|h\| < \eta$, and where $\xi := h/\|h\|$. Integrating in ξ over the unit sphere S^{N-1} and using the fact that for every $y \in \mathbb{R}^N \setminus \{0\}$, by a rotation,

$$\begin{aligned} \int_{S^{N-1}} |y \cdot \xi|^p d\mathcal{H}^{N-1}(\xi) &= \|y\|^p \int_{S^{N-1}} \left| \frac{y}{\|y\|} \cdot \xi \right|^p d\mathcal{H}^{N-1}(\xi) \\ &= \|y\|^p \int_{S^{N-1}} |e_1 \cdot \xi|^p d\mathcal{H}^{N-1}(\xi) =: \|y\|^p c_{N,p} \end{aligned}$$

gives

$$c_{N,p} \int_K \|\nabla u(x)\|^p dx \leq (1 + \delta)(M + \delta) \beta_N + c_{K,\delta} \beta_N \mathcal{L}^N(K) \eta^p.$$

Letting $\eta \rightarrow 0^+$, we obtain

$$c_{N,p} \int_K \|\nabla u(x)\|^p dx \leq (1 + \delta)(M + \delta) \beta_N.$$

If $\delta \rightarrow 0^+$ and $K \nearrow \Omega$, by the Lebesgue monotone convergence theorem we get

$$c_{N,p} \int_\Omega \|\nabla u(x)\|^p dx \leq \beta_N M = \beta_N \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x + h) - u(x)|^p}{\|h\|^p} dx.$$

Step 3: We will prove (11.39) for $u \in \dot{W}^{1,p}(\Omega)$. Let $U \Subset \Omega$ and for $0 < \varepsilon < \text{dist}(U, \partial\Omega)$ define $u_\varepsilon := \varphi_\varepsilon * u$, where φ_ε is a standard mollifier. By Theorem C.16(iii) applied to the function $x \in \Omega_h \mapsto u(x + h) - u(x)$, we have

$$\int_{U_h} |u_\varepsilon(x + h) - u_\varepsilon(x)|^p dx \leq \int_{\Omega_h} |u(x + he) - u(x)|^p dx.$$

Dividing by $\|h\|^p$, letting $h \rightarrow 0$ and using the previous step applied to u_ε gives

$$(11.42) \quad \kappa_{N,p} \int_U \|\nabla u_\varepsilon(x)\|^p dx \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x + h) - u(x)|^p}{\|h\|^p} dx.$$

Since $\partial_i u_\varepsilon \rightarrow \partial_i u$ in $L^p(U)$ as $\varepsilon \rightarrow 0^+$ by Lemma 11.25, we obtain

$$(11.43) \quad \kappa_{N,p} \int_U \|\nabla u(x)\|^p dx \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x + h) - u(x)|^p}{\|h\|^p} dx.$$

By letting $U \nearrow \Omega$ and using the Lebesgue monotone convergence theorem, we obtain (11.39).

Step 4: To prove the final statement of the theorem, let $1 < p < \infty$ and let $u \in L^1_{\text{loc}}(\Omega)$ be such that (11.41) holds. We claim that $u \in \dot{W}^{1,p}(\Omega)$. To see this, let $U \Subset \Omega$. Then reasoning as in Step 3, by (11.42) we get

$$\sup_{0 < \varepsilon < \text{dist}(U, \partial\Omega)} \int_U \|\nabla u_\varepsilon(x)\|^p dx \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x+h) - u(x)|^p}{\|h\|^p} dx < \infty.$$

Since $u_\varepsilon \rightarrow u$ in $L^1(U)$ by Theorem C.16, it follows from Theorem 11.65 that $u \in \dot{W}^{1,p}(U)$ and that (11.43) holds for every U . Given the arbitrariness of U , taking a sequence $U_j \Subset \Omega$ with $U_j \nearrow \Omega$, since (11.43) holds for all U_j , letting $j \rightarrow \infty$, we obtain (11.39), which shows that $u \in \dot{W}^{1,p}(\Omega)$. $\square$

Remark 11.76. Under the additional assumption that $u \in L^p(\Omega)$, the previous theorem gives a characterization of $W^{1,p}(\Omega)$.

The following exercise shows that the last part of the statement of the theorem is false for $p = 1$. See Theorem 14.53 for the correct result for $p = 1$.

Exercise 11.77. Prove that for $p = 1$ the last part of the statement of the theorem is false. Hint: It is enough to construct an example for $N = 1$.

Exercise 11.78. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and for every $i = 1, \dots, N$ and $h > 0$, let $\Omega_{h,i} := \{x \in \Omega : x + te_i \in \Omega \text{ for all } 0 < t \leq h\}$.

- (i) Let $u \in \dot{W}^{1,p}(\Omega)$, $1 \leq p < \infty$. Prove that for every $i = 1, \dots, N$ and $h > 0$,

$$\int_{\Omega_{h,i}} \frac{|u(x + he_i) - u(x)|^p}{h^p} dx \leq \int_{\Omega} |\partial_i u(x)|^p dx$$

and

$$\lim_{h \rightarrow 0^+} \int_{\Omega_{h,i}} \frac{|u(x + he_i) - u(x)|^p}{h^p} dx = \int_{\Omega} |\partial_i u(x)|^p dx.$$

- (ii) Prove that if $1 < p < \infty$ and $u \in L^1_{\text{loc}}(\Omega)$ is such that

$$\liminf_{h \rightarrow 0^+} \int_{\Omega_{h,i}} \frac{|u(x + he_i) - u(x)|^p}{h^p} dx < \infty$$

for every $i = 1, \dots, N$, then $u \in \dot{W}^{1,p}(\Omega)$.

The extension of Theorem 11.75 to $\dot{W}^{m,p}(\Omega)$ is given in Exercise 17.62.

Sobolev Spaces: Embeddings

Newton's second law of graduation: The age, $\mathbf{a}$, of a doctoral process is directly proportional to the flexibility, $\mathbf{f}$, given by the advisor and inversely proportional to the student's motivation, $\mathbf{m}$.

— Jorge Cham, www.phdcomics.com

In this chapter we are interested in finding an exponent q such that

$$(12.1) \quad \|u\|_{L^q(\mathbb{R}^N)} \leq c \|\nabla u\|_{L^p(\mathbb{R}^N)}$$

for all u in an appropriate subfamily of $W^{1,p}(\mathbb{R}^N)$ or $\dot{W}^{1,p}(\mathbb{R}^N)$.

Assume for simplicity that $u \in C_c^1(\mathbb{R}^N)$ and for $r > 0$ define the rescaled function $u_r(x) := u(rx)$, $x \in \mathbb{R}^N$. If (12.1) holds for u_r , we get

$$\begin{aligned} \left(\int_{\mathbb{R}^N} |u(rx)|^q dx \right)^{1/q} &= \left(\int_{\mathbb{R}^N} |u_r(x)|^q dx \right)^{1/q} \leq c \left(\int_{\mathbb{R}^N} \|\nabla u_r(x)\|^p dx \right)^{1/p} \\ &= c \left(r^p \int_{\mathbb{R}^N} \|\nabla u(rx)\|^p dx \right)^{1/p}, \end{aligned}$$

or, equivalently, after the change of variables $y := rx$,

$$\left(\frac{1}{r^N} \int_{\mathbb{R}^N} |u(y)|^q dy \right)^{1/q} \leq c \left(\frac{r^p}{r^N} \int_{\mathbb{R}^N} \|\nabla u(y)\|^p dy \right)^{1/p},$$

that is,

$$\left(\int_{\mathbb{R}^N} |u(y)|^q dy \right)^{1/q} \leq c r^{1-N/p+N/q} \left(\int_{\mathbb{R}^N} \|\nabla u(y)\|^p dy \right)^{1/p}.$$

If $1 - N/p + N/q > 0$, let $r \rightarrow 0^+$ to conclude that $u \equiv 0$, while if $1 - N/p + N/q < 0$, let $r \rightarrow \infty$ to conclude again that $u \equiv 0$. Hence, the only possible

case is when $1 - N/p + N/q = 0$. So in order for q to be positive, we need $p < N$, in which case,

$$(12.2) \quad q = p^* := \frac{Np}{N-p}.$$

The number p^* is called the *Sobolev critical exponent*.

In what follows, when there is no possibility of confusion, we write $\|u\|_{L^p}$ and $\|u\|_{W^{m,p}}$, for $\|u\|_{L^p(\Omega)}$ and $\|u\|_{W^{m,p}(\Omega)}$, respectively. We also write $\|\nabla^k u\|_{L^p(\Omega)}$ or $\|\nabla^k u\|_{L^p}$ for $\|\nabla^k u\|_{L^p(\Omega; \mathbb{R}^{M_k})}$. We recall, that β_N is the surface area of the unit sphere, that is,

$$(12.3) \quad \beta_N := \mathcal{H}^{N-1}(S^{N-1}) = \frac{2\pi^{N/2}}{\Gamma(N/2)}.$$

12.1. Embeddings: $mp < N$

In this section we prove inequality (12.1). We recall the following definition.

Definition 12.1. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u : E \rightarrow \mathbb{R}$ be a Lebesgue measurable function. The function u is said to *vanish at infinity* if for every $t > 0$,

$$(12.4) \quad \mathcal{L}^N(\{x \in E : |u(x)| > t\}) < \infty.$$

Note that the previous definition is automatically satisfied if E has finite measure, while if E has infinite measure and u does not satisfy (12.4), then it cannot belong to any $L^q(E)$ for $1 \leq q < \infty$.

Exercise 12.2. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u : E \rightarrow \mathbb{R}$ and $v : E \rightarrow \mathbb{R}$ be two functions vanishing at infinity. Prove that $u + v$ vanishes at infinity.

Exercise 12.3. Prove that a nonzero polynomial $P : \mathbb{R}^N \rightarrow \mathbb{R}$ cannot vanish at infinity.

We recall (see Definition A.36 and (A.2)) that given two normed spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$, the normed space X is *embedded* in the normed space Y , written $X \hookrightarrow Y$, if X is a subspace of Y and there exists a constant $c > 0$ such that

$$\|x\|_Y \leq c\|x\|_X \quad \text{for all } x \in X.$$

Theorem 12.4 (Sobolev–Gagliardo–Nirenberg’s embedding in $W^{1,p}$). *Let $1 \leq p < N$. Then there exists a constant $c = c(N, p) > 0$ such that for every function $u \in \dot{W}^{1,p}(\mathbb{R}^N)$ vanishing at infinity,*

$$(12.5) \quad \|u\|_{L^{p^*}(\mathbb{R}^N)} \leq c\|\nabla u\|_{L^p(\mathbb{R}^N)}.$$

In particular, $W^{1,p}(\mathbb{R}^N) \hookrightarrow L^q(\mathbb{R}^N)$ for all $p \leq q \leq p^$.*

The proof makes use of the following result.

Exercise 12.5. Prove that if $u \in L^p(\mathbb{R})$ for some $1 \leq p < \infty$, then for every representative v of u , $\liminf_{x \rightarrow -\infty} |v(x)| = 0$, $\liminf_{x \rightarrow \infty} |v(x)| = 0$. Prove that in general one cannot replace the limit inferiors with actual limits.

In what follows, we use the notation (E.2) in Appendix E. To be precise, given $x = (x_1, \dots, x_N) \in \mathbb{R}^N$, for every $i = 1, \dots, N$ we denote by x'_i the $(N-1)$ -dimensional vector obtained by removing the i th component from x and with an abuse of notation we write $x = (x'_i, x_i) \in \mathbb{R}^{N-1} \times \mathbb{R}$.

Lemma 12.6. Let $N \geq 2$ and let $u_i \in L^{N-1}(\mathbb{R}^{N-1})$, $i = 1, \dots, N$. Then the function

$$u(x) := u_1(x'_1)u_2(x'_2) \cdots u_N(x'_N), \quad x \in \mathbb{R}^N,$$

belongs to $L^1(\mathbb{R}^N)$ and

$$\|u\|_{L^1(\mathbb{R}^N)} \leq \prod_{i=1}^N \|u_i\|_{L^{N-1}(\mathbb{R}^{N-1})}.$$

Proof. The proof is by induction on N . If $N=2$, then $u(x) := u_1(x_2)u_2(x_1)$, $x = (x_1, x_2) \in \mathbb{R}^2$. Integrating both sides with respect to x and using Tonelli's theorem, we get

$$\int_{\mathbb{R}^2} |u(x)| dx = \int_{\mathbb{R}} |u_1(x_2)| dx_2 \int_{\mathbb{R}} |u_2(x_1)| dx_1.$$

Assume next that the result is true for N and let us prove it for $N+1$. Let

$$u(x) := u_1(x'_1)u_2(x'_2) \cdots u_{N+1}(x'_{N+1}), \quad x \in \mathbb{R}^{N+1},$$

where $u_i \in L^N(\mathbb{R}^N)$, $i = 1, \dots, N+1$. Fix $x_{N+1} \in \mathbb{R}$. Integrating both sides of the previous identity with respect to $x_1, \dots, x_N$ and using Hölder's inequality, we get

(12.6)

$$\begin{aligned} & \int_{\mathbb{R}^N} |u(x)| dx_1 \cdots dx_N \\ & \leq \|u_{N+1}\|_{L^N(\mathbb{R}^N)} \left(\int_{\mathbb{R}^N} \prod_{i=1}^N |u_i(x'_i)|^{N/(N-1)} dx_1 \cdots dx_N \right)^{(N-1)/N}. \end{aligned}$$

For every $i = 1, \dots, N$ we denote by x''_i the $(N-1)$ -dimensional vector obtained by removing the last component from x'_i and with an abuse of notation we write $x'_i = (x''_i, x_{N+1}) \in \mathbb{R}^{N-1} \times \mathbb{R}$. Since x_{N+1} is fixed, by the induction hypothesis applied to the functions $v_i(x''_i) := |u_i(x''_i, x_{N+1})|^{N/(N-1)}$,

$x_i'' \in \mathbb{R}^{N-1}$, $i = 1, \dots, N$, we obtain that

$$\begin{aligned} \int_{\mathbb{R}^N} \prod_{i=1}^N |u_i(x'_i)|^{N/(N-1)} dx_1 \cdots dx_N &\leq \prod_{i=1}^N \|v_i\|_{L^{N-1}(\mathbb{R}^{N-1})} \\ &= \prod_{i=1}^N \left(\int_{\mathbb{R}^{N-1}} |u_i(x''_i, x_{N+1})|^N dx''_i \right)^{1/(N-1)}, \end{aligned}$$

and so from (12.6),

$$\begin{aligned} \int_{\mathbb{R}^N} |u(x)| dx_1 \cdots dx_N \\ \leq \|u_{N+1}\|_{L^N(\mathbb{R}^N)} \prod_{i=1}^N \left(\int_{\mathbb{R}^{N-1}} |u_i(x''_i, x_{N+1})|^N dx''_i \right)^{1/N}. \end{aligned}$$

Integrating both sides with respect to x_{N+1} and using the extended Hölder inequality (see the Exercise B.79(i)), with $1 = 1/N + \cdots + 1/N$ where the sum has N addends, and Tonelli's theorem, we get

$$\int_{\mathbb{R}^{N+1}} |u(x)| dx_1 \cdots dx_{N+1} \leq \prod_{i=1}^{N+1} \|u_i\|_{L^N(\mathbb{R}^N)},$$

which concludes the proof. $\square$

We now turn to the proof of the Sobolev–Gagliardo–Nirenberg embedding theorem.

Proof of Theorem 12.4. Step 1: We prove (12.5) in the case $p = 1$ and under the additional hypothesis that $u \in L^{\frac{N}{N-1}}(\mathbb{R}^N) \cap C^1(\mathbb{R}^N)$ with $\nabla u \in L^1(\mathbb{R}^N; \mathbb{R}^N)$. Fix $i = 1, \dots, N$. By Fubini's theorem for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in \mathbb{R}^{N-1}$ the function $v(t) := u(x'_i, t)$, $t \in \mathbb{R}$, belongs to $L^{\frac{N}{N-1}}(\mathbb{R}) \cap C^1(\mathbb{R})$ and $v' \in L^1(\mathbb{R})$. Fix any such $x'_i \in \mathbb{R}^{N-1}$. By Exercise 12.5, $\liminf_{t \rightarrow -\infty} |v(t)| = 0$, and so we may find a sequence $t_n \rightarrow -\infty$ such that $v(t_n) \rightarrow 0$. Hence, for every $t \in \mathbb{R}$ we have that

$$v(t) = v(t_n) + \int_{t_n}^t v'(s) ds.$$

In turn,

$$|v(t)| \leq |v(t_n)| + \int_{\mathbb{R}} |v'(s)| ds$$

for all $t \in \mathbb{R}$ and $n \in \mathbb{N}$. Letting $n \rightarrow \infty$, we conclude that for each $x_i \in \mathbb{R}$ we have

$$|u(x'_i, x_i)| \leq \int_{\mathbb{R}} \left| \frac{\partial u}{\partial x_i}(x'_i, y_i) \right| dy_i.$$

Hence, we have shown that for all $i = 1, \dots, N$ and for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ we have

$$|u(x)| \leq \int_{\mathbb{R}} \left| \frac{\partial u}{\partial x_i}(x'_i, y_i) \right| dy_i.$$

Multiplying these N inequalities and raising to the power $\frac{1}{N-1}$, we get

$$|u(x)|^{N/(N-1)} \leq \prod_{i=1}^N \left(\int_{\mathbb{R}} \left| \frac{\partial u}{\partial x_i}(x'_i, y_i) \right| dy_i \right)^{1/(N-1)} =: \prod_{i=1}^N w_i(x'_i)$$

for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. We now apply the previous lemma to the function $w(x) := \prod_{i=1}^N w_i(x'_i)$, $x \in \mathbb{R}^N$, to obtain that

$$\begin{aligned} \int_{\mathbb{R}^N} |u(x)|^{N/(N-1)} dx &\leq \int_{\mathbb{R}^N} |w(x)| dx \leq \prod_{i=1}^N \|w_i\|_{L^{N-1}(\mathbb{R}^{N-1})} \\ &= \prod_{i=1}^N \left(\int_{\mathbb{R}^N} \left| \frac{\partial u}{\partial x_i}(x) \right| dx \right)^{1/(N-1)} \leq \left(\int_{\mathbb{R}^N} \|\nabla u(x)\| dx \right)^{N/(N-1)}, \end{aligned}$$

where we have used Tonelli's theorem. This gives the desired inequality for $p = 1$.

Step 2: Assume next that $1 < p < N$ and that $u \in L^{p^*}(\mathbb{R}^N) \cap C^1(\mathbb{R}^N)$ with $\nabla u \in L^p(\mathbb{R}^N; \mathbb{R}^N)$. Define $v := |u|^q$, $q := p(N-1)/(N-p)$. Note that since $q > 1$, we have that $v \in C^1(\mathbb{R}^N)$. Applying Step 1 to the function v , we get

$$\begin{aligned} \left(\int_{\mathbb{R}^N} |u|^{p^*} dx \right)^{(N-1)/N} &= \left(\int_{\mathbb{R}^N} |v|^{N/(N-1)} dx \right)^{(N-1)/N} \\ &\leq \int_{\mathbb{R}^N} \|\nabla v\| dx \leq q \int_{\mathbb{R}^N} |u|^{q-1} \|\nabla u\| dx \\ &\leq q \left(\int_{\mathbb{R}^N} |u|^{(q-1)p'} dx \right)^{1/p'} \left(\int_{\mathbb{R}^N} \|\nabla u\|^p dx \right)^{1/p}, \end{aligned}$$

where in the last inequality we have used Hölder's inequality.

Since $(q-1)p' = p^*$, if $u \neq 0$, we obtain

$$\begin{aligned} \left(\int_{\mathbb{R}^N} |u|^{p^*} dx \right)^{(N-1)/N - (p-1)/p} &= \left(\int_{\mathbb{R}^N} |u|^{p^*} dx \right)^{1/p^*} \\ &\leq q \left(\int_{\mathbb{R}^N} \|\nabla u\|^p dx \right)^{1/p}, \end{aligned}$$

which proves the result.

Step 3: Next consider the case $u \in L^{p^*}(\mathbb{R}^N)$ with distributional gradient $\nabla u \in L^p(\mathbb{R}^N; \mathbb{R}^N)$ for $1 \leq p < N$. Define $u_\varepsilon := u * \varphi_\varepsilon$, where φ_ε is a standard mollifier. Then by Theorems C.16 and C.20 we have that $u_\varepsilon \in$

$L^{p^*}(\mathbb{R}^N) \cap C^\infty(\mathbb{R}^N)$, while, reasoning as in Lemma 11.25 and using Theorem C.16 once more, we obtain that $\nabla u_\varepsilon \in L^p(\mathbb{R}^N; \mathbb{R}^N)$. Hence, by the previous two steps $\|u_\varepsilon\|_{L^{p^*}} \leq q \|\nabla u_\varepsilon\|_{L^p}$. Letting $\varepsilon \rightarrow 0^+$ and using Theorem C.16 and Lemma 11.25 one more time, we obtain the same inequality for u .

Step 4: Finally, we prove (12.5) in the general case. Assume that $u \in \dot{W}^{1,p}(\mathbb{R}^N)$ vanishes at infinity. For $n \in \mathbb{N}$ and $x \in \mathbb{R}^N$ define

$$v_n(x) := \begin{cases} |u(x)| - 1/n & \text{if } 1/n \leq |u(x)| \leq n, \\ 0 & \text{if } |u(x)| < 1/n, \\ n - 1/n & \text{if } |u(x)| > n. \end{cases}$$

By the chain rule (see Exercise 11.51(i) and (vi)) for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$,

$$\|\nabla v_n(x)\| = \begin{cases} \|\nabla u(x)\| & \text{if } 1/n < |u(x)| < n, \\ 0 & \text{otherwise,} \end{cases}$$

and so $\nabla v_n \in L^p(\mathbb{R}^N; \mathbb{R}^N)$, while

$$\begin{aligned} \int_{\mathbb{R}^N} |v_n|^{p^*} dx &= \int_{\{|u| > 1/n\}} |v_n|^{p^*} dx \\ &\leq (n - 1/n)^{p^*} \mathcal{L}^N(\{x \in \mathbb{R}^N : |u(x)| > 1/n\}) < \infty, \end{aligned}$$

since u is vanishing at infinity. Hence, by the previous step applied to v_n ,

$$\begin{aligned} \left(\int_{\{1/n \leq |u| \leq n\}} (|u(x)| - 1/n)^{p^*} dx \right)^{1/p^*} &\leq \left(\int_{\mathbb{R}^N} |v_n|^{p^*} dx \right)^{1/p^*} \\ &\leq q \left(\int_{\mathbb{R}^N} \|\nabla v_n\|^p dx \right)^{1/p} \leq q \left(\int_{\mathbb{R}^N} \|\nabla u\|^p dx \right)^{1/p}. \end{aligned}$$

Letting $n \rightarrow \infty$ and using Fatou's lemma, we obtain the desired result.

Step 5: To prove the last part of the theorem, assume that $u \in W^{1,p}(\mathbb{R}^N)$. Then by the previous steps we know that $u \in L^{p^*}(\mathbb{R}^N)$, and so we can now use Exercise B.79(ii) to conclude that $u \in L^q(\mathbb{R}^N)$ for all $p \leq q \leq p^*$. Indeed, assume that $p < q < p^*$ and write $1/q = \theta/p + (1-\theta)/p^*$ for some $0 < \theta < 1$. Then

$$\|u\|_{L^q} \leq \|u\|_{L^p}^\theta \|u\|_{L^{p^*}}^{1-\theta} \leq \|u\|_{L^p} + \|u\|_{L^{p^*}},$$

where we have used Young's inequality (see (B.17)) with exponents $1/\theta$ and $(1/\theta)'$. Hence,

$$\|u\|_{L^q} \leq \|u\|_{L^p} + \|u\|_{L^{p^*}} \leq \|u\|_{L^p} + c \|\nabla u\|_{L^p},$$

which shows that $W^{1,p}(\mathbb{R}^N) \hookrightarrow L^q(\mathbb{R}^N)$. □

Remark 12.7. (i) If $1 < p < N$, then the best constant in (12.5) is given by

$$c = \frac{1}{\pi^{1/2}} \frac{1}{N^{1/p}} \left(\frac{p-1}{N-p} \right)^{1-1/p} \left\{ \frac{\Gamma(1+N/2)\Gamma(N)}{\Gamma(N/p)\Gamma(1+N-N/p)} \right\}^{1/N}$$

and equality holds in (12.5) if $u(x) = (a + b\|x\|^{p/(p-1)})^{1-N/p}$, $x \in \mathbb{R}^N$, where a, b are positive constants (see [15], [228]).

If $p = 1 < N$, then the best constant in (12.5) is given by

$$c = \frac{1}{\pi^{1/2}} \frac{1}{N} (\Gamma(1+N/2))^{1/N}.$$

To see that this inequality is sharp, it suffices to take

$$u_n(x) := \begin{cases} 1 & \text{if } \|x\| \leq 1, \\ 1 + n - n\|x\| & \text{if } 1 < \|x\| \leq 1 + 1/n, \\ 0 & \text{if } \|x\| > 1 + 1/n, \end{cases}$$

which is a sequence of Lipschitz continuous functions that converges to the characteristic function of the unit ball. It is possible to show that equality does not hold in (12.5) for functions in $W^{1,1}(\mathbb{R}^N)$ but that (12.5) still holds in $BV(\mathbb{R}^N)$ (see Chapter 14), with the norm of ∇u in L^1 replaced by the total variation of the vectorial Radon measure Du , and that equality holds for the characteristic function of the unit ball.

(ii) In Steps 1 and 2 of the previous proof we could have used Theorem 11.45, and so avoid Step 3.

To extend the Sobolev–Gagliardo–Nirenberg embedding theorem to functions in $\dot{W}^{m,p}(\mathbb{R}^N)$, where $m \geq 2$ and $1 \leq p < \infty$ are such that $mp < N$, for every $k = 0, \dots, m$ we define the *Sobolev critical exponent*

$$(12.7) \quad p_{m,k}^* := \frac{Np}{N - (m-k)p}.$$

Note that $p_{m,m-1}^* = p^*$ and $p_{m,m}^* = p$.

Corollary 12.8 (Sobolev–Gagliardo–Nirenberg’s embedding in $W^{m,p}$). *Let $m \in \mathbb{N}$ and $1 \leq p < \infty$ be such $mp < N$. Then there exists a constant $c = c(N, m, p) > 0$ such that for every function $u \in W^{m,p}(\mathbb{R}^N)$ and for every $k = 0, \dots, m-1$ and*

$$(12.8) \quad \|\nabla^k u\|_{L^{p_{m,k}^*}(\mathbb{R}^N)} \leq c \|\nabla^m u\|_{L^p(\mathbb{R}^N)}.$$

In particular,

$$W^{m,p}(\mathbb{R}^N) \hookrightarrow L^{q_0}(\mathbb{R}^N) \cap W^{1,q_1}(\mathbb{R}^N) \cap \dots \cap W^{m-1,q_{m-1}}(\mathbb{R}^N)$$

for all $p \leq q_k \leq p_{m,k}^$, $k = 0, \dots, m-1$.*

Proof. The proof is by induction and is left as an exercise. $\square$

Next we consider the case of functions in $\dot{W}^{m,p}(\mathbb{R}^N)$ that do not vanish at infinity. As a corollary of Theorem 11.43, we have the following result.

Theorem 12.9 (Sobolev–Gagliardo–Nirenberg’s embedding in $\dot{W}^{m,p}$). *Let $m \in \mathbb{N}$ and $1 \leq p < \infty$ be such $mp < N$. Then for every function $u \in \dot{W}^{m,p}(\mathbb{R}^N)$ there exists a polynomial P_u of degree $m - 1$ such that*

$$(12.9) \quad \|\nabla^k(u - P_u)\|_{L^{p^*,k}(\mathbb{R}^N)} \leq c \|\nabla^m u\|_{L^p(\mathbb{R}^N)}$$

for every $k = 0, \dots, m - 1$, where $c = c(N, m, p) > 0$. Moreover,

$$(12.10) \quad P_u = u - \sum_{|\alpha|=m} (K_\alpha * \partial^\alpha u),$$

where $K_\alpha(x) := \frac{m}{\beta_N \alpha!} \frac{x^\alpha}{\|x\|^N}$, $x \in \mathbb{R}^N \setminus \{0\}$.

We recall that β_N is the surface area of the unit sphere (see (12.3)). We begin with a preliminary lemma.

Lemma 12.10. *Let $u \in C_c^m(\mathbb{R}^N)$, $m \in \mathbb{N}$. Then*

$$u = \sum_{|\alpha|=m} K_\alpha * \partial^\alpha u.$$

Proof. Fix $x \in \mathbb{R}^N$ and $\theta \in S^{N-1}$ and consider the function $g(r) := u(x + r\theta)$, $r \in \mathbb{R}$. Then by the fundamental theorem of calculus and by integrating by parts $m - 1$ times,

$$g(0) = - \int_0^\infty g'(r) dr = \frac{(-1)^m}{(m-1)!} \int_0^\infty r^{m-1} g^{(m)}(r) dr.$$

By the chain rule $g^{(m)}(r) = \sum_{|\alpha|=m} \frac{m!}{\alpha!} \theta^\alpha \partial^\alpha u(x + r\theta)$ and so

$$u(x) = m(-1)^m \sum_{|\alpha|=m} \frac{1}{\alpha!} \int_0^\infty r^{m-1} \theta^\alpha \partial^\alpha u(x + r\theta) dr.$$

Averaging both sides with respect to $\theta \in S^{N-1}$ and changing variables gives

$$\begin{aligned} u(x) &= \frac{m(-1)^m}{\beta_N} \sum_{|\alpha|=m} \frac{1}{\alpha!} \int_{S^{N-1}} \int_0^\infty r^{m-1} \theta^\alpha \partial^\alpha u(x + r\theta) dr d\mathcal{H}^{N-1}(\theta) \\ &= \frac{m(-1)^m}{\beta_N} \sum_{|\alpha|=m} \frac{1}{\alpha!} \int_{\mathbb{R}^N} \frac{1}{\|y - x\|^{N-m}} \frac{(y - x)^\alpha}{\|y - x\|^{|\alpha|}} \partial^\alpha u(y) dy, \end{aligned}$$

which gives the desired result. $\square$

We turn to the proof of Theorem 12.9.

Proof of Theorem 12.9. Given $u \in \dot{W}^{m,p}(\mathbb{R}^N)$, by Theorem 11.43, there exists a sequence $\{u_n\}_n$ in $C_c^\infty(\mathbb{R}^N)$ such that $\partial^\alpha u_n \rightarrow \partial^\alpha u$ in $L^p(\mathbb{R}^N)$ for every multi-index α with $|\alpha| = m$. Since for every $\ell, n \in \mathbb{N}$, $u_n - u_\ell \in W^{m,p}(\mathbb{R}^N)$, by the Sobolev–Gagliardo–Nirenberg theorem (see Theorem 12.8),

$$\|\nabla^k(u_n - u_\ell)\|_{L^{p_{m,k}}^*} \leq c \|\nabla^m(u_n - u_\ell)\|_{L^p}$$

for every $k = 0, \dots, m-1$. This shows that $\{\nabla^k u_n\}_n$ is a Cauchy sequence in $L^{p_{m,k}}^*(\mathbb{R}^N; \mathbb{R}^{M_k})$. Thus there exists $v \in L^{p_{m,0}}^*(\mathbb{R}^N) \cap \dots \cap L^{p_{m,m}}^*(\mathbb{R}^N)$ such that $\nabla^k u_n \rightarrow \nabla^k v$ in $L^{p_{m,k}}^*(\mathbb{R}^N; \mathbb{R}^{M_k})$. Since $\nabla^m u_n \rightarrow \nabla^m u$ in $L^p(\mathbb{R}^N; \mathbb{R}^{M_m})$, it follows that $\nabla^m u(x) = \nabla^m v(x)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. Hence, there exists a polynomial P_u of degree $m-1$ such that $u(x) = v(x) + P_u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. Again by the Sobolev–Gagliardo–Nirenberg theorem (see Theorem 12.8), this time applied to $u_n \in W^{m,p}(\mathbb{R}^N)$, $n \in \mathbb{N}$, we have $\|\nabla^k u_n\|_{L^{p_{m,k}}^*} \leq c \|\nabla^m u_n\|_{L^p}$ for every $k = 0, \dots, m-1$. Letting $n \rightarrow \infty$ gives (12.9).

It remains to show (12.10). By Lemma 12.10, $u_n = \sum_{|\alpha|=m} K_\alpha * \partial^\alpha u_n$. Since $|K_\alpha(x)| \leq \frac{c}{\|x\|^{\frac{N}{p}-m}}$ and $1 \leq p < N/m$, applying Proposition C.29 (where α and p there are here m and $pN/[pm + N(p-1)] \leq N/m$, respectively) we get that $\tilde{K}_\alpha * \psi \in L^{p'}(\mathbb{R}^N)$ for every $\psi \in C_c^\infty(\mathbb{R}^N)$, where $\tilde{K}_\alpha(x) := K_\alpha(-x)$. Note that when $p = 1$, one should use (C.21) and the fact that $\psi \in C_c^\infty(\mathbb{R}^N)$ to obtain that $\tilde{K}_\alpha * \psi \in L^\infty(\mathbb{R}^N)$. Hence, by Fubini's theorem

$$\begin{aligned} \int_{\mathbb{R}^N} u_n(x) \psi(x) dx &= \sum_{|\alpha|=m} \int_{\mathbb{R}^N} (K_\alpha * \partial^\alpha u_n)(x) \psi(x) dx \\ &= \sum_{|\alpha|=m} \int_{\mathbb{R}^N} \partial^\alpha u_n(y) (\tilde{K}_\alpha * \psi)(y) dy. \end{aligned}$$

Letting $n \rightarrow \infty$ and using Fubini's theorem we get

$$\begin{aligned} \int_{\mathbb{R}^N} (u(x) - P_u(x)) \psi(x) dx &= \sum_{|\alpha|=m} \int_{\mathbb{R}^N} \partial^\alpha u(y) (\tilde{K}_\alpha * \psi)(y) dy \\ &= \sum_{|\alpha|=m} \int_{\mathbb{R}^N} (K_\alpha * \partial^\alpha u)(x) \psi(x) dx. \end{aligned}$$

Since this holds for every $\psi \in C_c^\infty(\mathbb{R}^N)$, we obtain (12.10).

This completes the proof. $\square$

Next we show that if u vanishes at infinity, then $P_u = 0$.

Corollary 12.11. *Let $m \in \mathbb{N}$ and $1 \leq p < \infty$ be such $mp < N$. Then there exists a constant $c = c(N, m, p) > 0$ such that for every function*

$u \in \dot{W}^{m,p}(\mathbb{R}^N)$ vanishing at infinity, for every $k = 0, \dots, m-1$,

$$\|\nabla^k u\|_{L^{p_{m,k}}(\mathbb{R}^N)} \leq c \|\nabla^m u\|_{L^p(\mathbb{R}^N)}.$$

Proof. By the previous theorem we have that $u - P_u \in L^{p_{m,0}}(\mathbb{R}^N)$, which implies that $u - P_u$ vanishes at infinity. But since u vanishes at infinity, then by Exercise 12.2 so does P_u , which contradicts Exercise 12.3 unless $P_u = 0$. $\square$

The next exercises show that a similar result fails if $p \geq N$.

Exercise 12.12. Assume that $p > N$ and consider a function $u \in C^\infty(\mathbb{R}^N)$ such that $u(x) = \|x\|^\varepsilon$ for all $x \in \mathbb{R}^N \setminus B(0, 1)$, where $0 < \varepsilon < 1$. Prove that if $0 < \varepsilon < 1$ is chosen appropriately, then $u \in \dot{W}^{1,p}(\mathbb{R}^N)$, but there is no constant $a \in \mathbb{R}$ such that $u - a$ belongs to $L^q(\mathbb{R}^N)$ for some $1 \leq q \leq \infty$.

Exercise 12.13. Assume that $p = N > 1$ and consider a function $u \in C^\infty(\mathbb{R}^N)$ such that $u(x) = \log \log \|x\|$ for all $x \in \mathbb{R}^N \setminus B(0, e)$. Prove that $u \in \dot{W}^{1,N}(\mathbb{R}^N)$, but there is no constant $a \in \mathbb{R}$ such that $u - a$ belongs to $L^q(\mathbb{R}^N)$ for some $1 \leq q \leq \infty$.

Exercise 12.14. Prove that if $u \in \dot{W}^{1,1}(\mathbb{R})$, then there is a constant $a \in \mathbb{R}$ such that

$$\|u - a\|_{L^\infty(\mathbb{R})} \leq 2 \int_{\mathbb{R}} |u'(x)| dx.$$

Next we discuss the validity of the Sobolev–Gagliardo–Nirenberg embedding theorem for arbitrary domains.

Exercise 12.15 (Rooms and passages). Let $\{h_n\}_n$ and $\{\delta_{2n}\}_n$ be two sequences of positive numbers such that $\sum_{n=1}^\infty h_n = \ell < \infty$, $0 < \text{const.} \leq h_{n+1}/h_n \leq 1$, $0 < \delta_{2n} \leq h_{2n+1}$, and for $n \in \mathbb{N}$ let $c_n := \sum_{i=1}^n h_i$. Define $\Omega \subset \mathbb{R}^2$ to be the union of all sets of the form

$$\begin{aligned} R_j &:= (c_j - h_j, c_j) \times (-\tfrac{1}{2}h_j, \tfrac{1}{2}h_j), \\ P_{j+1} &:= [c_j, c_j + h_{j+1}] \times (-\tfrac{1}{2}\delta_{j+1}, \tfrac{1}{2}\delta_{j+1}), \end{aligned}$$

for $j = 1, 3, 5, \dots$ (see Figure 1).

- (i) Prove that $\partial\Omega$ is the range of a rectifiable curve but that it is not of class C .
- (ii) Let $h_n := 1/n^{3/2}$, $\delta_{2n} := 1/(2n)^{5/2}$ and for $j = 1, 3, 5, \dots$ define

$$u(x_1, x_2) := \begin{cases} \frac{j}{\log 2j} =: K_j & \text{in } R_j, \\ K_j + (K_{j+2} - K_j) \frac{x_1 - c_j}{h_{j+1}} & \text{in } P_{j+1}. \end{cases}$$

Prove that $u \in W^{1,2}(\Omega)$ but $u \notin L^q(\Omega)$ for any $q > 2$.

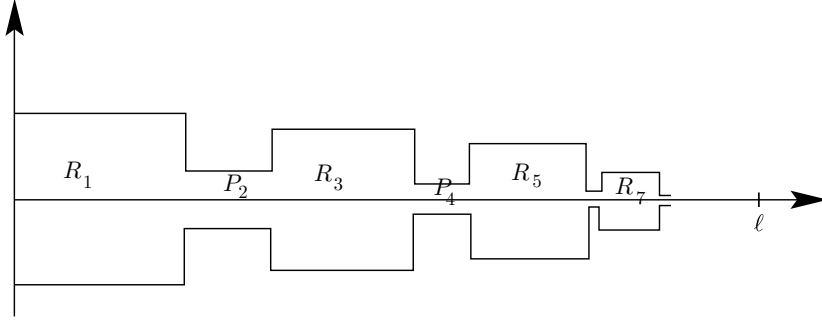


Figure 1. Rooms and passages.

- (iii) Let $p > 1$, $q \geq \frac{1}{2}(2p-1)$, $h_n := 1/n^p$, $\delta_{2n} := c/n^{2q+p}$, for some $c > 0$, and for $n \in \mathbb{N}$ define $u(x_1, x_2) := n^q$ in R_{2n-1} and $\nabla u(x_1, x_2) := (\frac{(n+1)^q - n^q}{1/n^p}, 0)$ in P_{2n} . Prove that $\nabla u \in L^2(\Omega; \mathbb{R}^{2 \times 2})$ but $u \notin L^2(\Omega)$.

The previous exercise shows that Theorem 12.4 fails for general domains. The problem is the regularity of the boundary.

Definition 12.16. Given $m \in \mathbb{N}$ and $1 \leq p \leq \infty$, an open set $\Omega \subseteq \mathbb{R}^N$ is called an *extension domain for the Sobolev space* $W^{m,p}(\Omega)$ if there exists a continuous linear operator

$$\mathcal{E} : W^{m,p}(\Omega) \rightarrow W^{m,p}(\mathbb{R}^N)$$

with the property that for all $u \in W^{m,p}(\Omega)$, $\mathcal{E}(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$.

Note that the *extension operator* $\mathcal{E}$ strongly depends on m and p . In the next chapter we will show that “nice” Lipschitz continuous domains are extension domains for *all* $m \in \mathbb{N}$ and *all* $1 \leq p \leq \infty$.

Proposition 12.17. Let $m \in \mathbb{N}$, $1 \leq p < \infty$ be such $mp < N$ and let $\Omega \subseteq \mathbb{R}^N$ be an extension domain for $W^{m,p}(\Omega)$. Then there exists a constant $c = c(N, m, p, \Omega) > 0$ such that for every $u \in W^{m,p}(\Omega)$, $\|\partial^\beta u\|_{L^q(\Omega)} \leq c\|u\|_{W^{m,p}(\Omega)}$ for every $k = 0, \dots, m-1$, for every multi-index β with $|\beta| = k$ and for all $p \leq q \leq p_{m,k}^*$.

Proof. Since $\Omega \subseteq \mathbb{R}^N$ is an extension domain for $W^{m,p}(\Omega)$, there exists a constant $c = c(N, m, p, \Omega) > 0$ such that $\|\mathcal{E}(u)\|_{W^{m,p}(\mathbb{R}^N)} \leq c\|u\|_{W^{m,p}(\Omega)}$ for all $u \in W^{m,p}(\Omega)$. On the other hand, by Theorem 12.4, for every $k = 0, \dots, m-1$ and for all $p \leq q \leq p_{m,k}^*$ we have that

$$\begin{aligned} \|\nabla^k u\|_{L^q(\Omega)} &= \|\nabla^k \mathcal{E}(u)\|_{L^q(\Omega)} \leq \|\nabla^k \mathcal{E}(u)\|_{L^q(\mathbb{R}^N)} \leq c\|\mathcal{E}(u)\|_{W^{m,p}(\mathbb{R}^N)} \\ &\leq c\|u\|_{W^{m,p}(\Omega)}, \end{aligned}$$

where as usual $c = c(N, m, p) > 0$ changed from line to line and we have used the fact that $\mathcal{E}(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$. $\square$

Next we show that if $p \leq q < p^*$ and Ω is an extension domain for $W^{1,p}(\Omega)$ with finite measure, then the embedding $W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$ is actually compact.

Theorem 12.18 (Rellich–Kondrachov’s compactness). *Let $1 \leq p < N$ and let $\Omega \subset \mathbb{R}^N$ be an extension domain for $W^{1,p}(\Omega)$ with finite measure. Let $\{u_n\}_n$ be a bounded sequence in $W^{1,p}(\Omega)$. Then there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and a function $u \in L^{p^*}(\Omega)$ such that $u_{n_k} \rightarrow u$ in $L^q(\Omega)$ for all $1 \leq q < p^*$.*

The proof makes use of the following auxiliary results.

Lemma 12.19. *Let $1 \leq p < \infty$ and let $u \in W^{1,p}(\mathbb{R}^N)$. Then for all $h \in \mathbb{R}^N \setminus \{0\}$,*

$$\int_{\mathbb{R}^N} |u(x+h) - u(x)|^p dx \leq \|h\|^p \int_{\mathbb{R}^N} \|\nabla u(x)\|^p dx.$$

Proof. Assume that $u \in W^{1,p}(\mathbb{R}^N) \cap C^\infty(\mathbb{R}^N)$. For $x \in \mathbb{R}^N$ and $h \in \mathbb{R}^N \setminus \{0\}$ by the fundamental theorem of calculus we have that

$$|u(x+h) - u(x)| = \left| \int_0^1 \frac{d}{dt}(u(x+th)) dt \right| \leq \|h\| \int_0^1 \|\nabla u(x+th)\| dt.$$

Raising both sides to the power p and integrating over $\mathbb{R}^N$, by Hölder’s inequality we get

$$\begin{aligned} \int_{\mathbb{R}^N} |u(x+h) - u(x)|^p dx &\leq \|h\|^p \int_{\mathbb{R}^N} \left(\int_0^1 \|\nabla u(x+th)\| dt \right)^p dx \\ &\leq \|h\|^p \int_{\mathbb{R}^N} \int_0^1 \|\nabla u(x+th)\|^p dt dx = \|h\|^p \int_0^1 \int_{\mathbb{R}^N} \|\nabla u(x+th)\|^p dx dt \\ &= \|h\|^p \int_{\mathbb{R}^N} \|\nabla u(y)\|^p dy, \end{aligned}$$

where we have used Tonelli’s theorem and the change of variables $y = x+th$.

To remove the additional hypothesis that $u \in C^\infty(\mathbb{R}^N)$, it suffices to apply the previous inequality to $u_\varepsilon := \varphi_\varepsilon * u$, where φ_ε is a standard mollifier, and to let $\varepsilon \rightarrow 0^+$ (see Theorem C.16 and Lemma 11.25). $\square$

Lemma 12.20. *Let $1 \leq p < \infty$ and let $u \in W^{1,p}(\mathbb{R}^N)$. Consider a standard mollifier φ as in (C.7). Then there exists a constant $c = c(N, p) > 0$ such that*

$$\int_{\mathbb{R}^N} |(u * \varphi_{1/k})(x) - u(x)|^p dx \leq \frac{c}{k^p} \int_{\mathbb{R}^N} \|\nabla u(x)\|^p dx.$$

Proof. By Hölder's inequality, (C.4), and the change of variables $h = y - x$, we have

$$(12.11) \quad \begin{aligned} |(u * \varphi_{1/k})(x) - u(x)|^p &\leq \int_{\mathbb{R}^N} \varphi_{1/k}(x - y) |u(y) - u(x)|^p dy \\ &\leq ck^N \int_{B(0, 1/k)} |u(x + h) - u(x)|^p dh. \end{aligned}$$

Hence,

$$(12.12) \quad \begin{aligned} \int_{\mathbb{R}^N} |(u * \varphi_{1/k})(x) - u(x)|^p dx \\ \leq ck^N \int_{B(0, 1/k)} \int_{\mathbb{R}^N} |u(x + h) - u(x)|^p dx dh. \end{aligned}$$

In turn, by the previous lemma and Tonelli's theorem we get

$$\begin{aligned} \int_{\mathbb{R}^N} |(u * \varphi_{1/k})(x) - u(x)|^p dx &\leq ck^N \int_{\mathbb{R}^N} \|\nabla u(x)\|^p dx \int_{B(0, 1/k)} \|h\|^p dh \\ &= \frac{c}{k^p} \int_{\mathbb{R}^N} \|\nabla u(x)\|^p dx, \end{aligned}$$

which concludes the proof. $\square$

We now turn to the proof of the Rellich–Kondrachov theorem.

Proof of Theorem 12.18. Since Ω is an extension domain for $W^{1,p}(\Omega)$, we can extend each u_n to a function $u_n \in W^{1,p}(\mathbb{R}^N)$ in such a way that the sequence $\{u_n\}_n$ remains bounded in $W^{1,p}(\mathbb{R}^N)$. It follows by Theorem 12.4 that the sequence $\{u_n\}_n$ is bounded in $L^{p^*}(\mathbb{R}^N)$. Since $p^* > 1$, by the reflexivity of $L^{p^*}(\mathbb{R}^N)$ (see Theorem B.92 and Corollary A.65) we may find a subsequence of $\{u_n\}_n$ (not relabeled) such that $u_n \rightharpoonup u$ in $L^{p^*}(\mathbb{R}^N)$. We claim that $u_n \rightarrow u$ in $L^p(\Omega)$.

For simplicity, for every $v \in L^p(\mathbb{R}^N)$ we set $v^{(k)} := v * \varphi_{1/k}$. By the previous lemma and the fact that $\{u_n\}_n$ is bounded in $W^{1,p}(\mathbb{R}^N)$, we get

$$\sup_{n \in \mathbb{N}} \int_{\mathbb{R}^N} |(u_n)^{(k)} - u_n|^p dx \leq \frac{c}{k^p} \sup_{n \in \mathbb{N}} \int_{\mathbb{R}^N} \|\nabla u_n\|^p dx \leq \frac{c}{k^p},$$

and so

$$(12.13) \quad \lim_{k \rightarrow \infty} \sup_{n \in \mathbb{N}} \int_{\mathbb{R}^N} |(u_n)^{(k)} - u_n|^p dx = 0.$$

By Minkowski's inequality,

$$\begin{aligned} \|u_n - u\|_{L^p(\Omega)} &\leq \|(u_n)^{(k)} - u_n\|_{L^p(\Omega)} \\ &\quad + \|(u_n)^{(k)} - u^{(k)}\|_{L^p(\Omega)} + \|u^{(k)} - u\|_{L^p(\Omega)}. \end{aligned}$$

Fix $\epsilon > 0$. By (12.13) and Theorem C.16 there exists $\bar{k}$ depending only on ϵ such that for all $k \geq \bar{k}$ and all $n \in \mathbb{N}$ the first and last terms in the previous inequality are both bounded by ϵ , and so

$$(12.14) \quad \|u_n - u\|_{L^p(\Omega)} \leq \|(u_n)^{(k)} - u^{(k)}\|_{L^p(\Omega)} + 2\epsilon$$

for all $k \geq \bar{k}$ and all $n \in \mathbb{N}$. Hence, to complete the proof, it suffices to show that

$$(12.15) \quad \lim_{n \rightarrow \infty} \|(u_n)^{(\bar{k})} - u^{(\bar{k})}\|_{L^p(\Omega)} = 0.$$

Since $u_n \rightharpoonup u$ in $L^{p^*}(\mathbb{R}^N)$, it follows that for all $x \in \mathbb{R}^N$,

$$(u_n)^{(\bar{k})}(x) = \int_{\mathbb{R}^N} \varphi_{1/\bar{k}}(x - y) u_n(y) dy \rightarrow \int_{\mathbb{R}^N} \varphi_{1/\bar{k}}(x - y) u(y) dy = u^{(\bar{k})}(x)$$

as $n \rightarrow \infty$. Moreover, reasoning somewhat as in (12.11) (with u replaced by u_n) and using the fact that $\{u_n\}_n$ is bounded in $L^p(\mathbb{R}^N)$, we get

$$|(u_n)^{(\bar{k})}(x) - u^{(\bar{k})}(x)|^p \leq c\bar{k}^N \int_{B(0,1/\bar{k})} |u_n(x+h) - u(x+h)|^p dh \leq c\bar{k}^N$$

for all $x \in \mathbb{R}^N$ and all $n \in \mathbb{N}$. Since Ω has finite measure, we are in a position to apply the Lebesgue dominated convergence theorem to conclude that (12.15) holds.

Hence, we have shown that $u_n \rightarrow u$ in $L^p(\Omega)$. Since $\{u_n\}_n$ is bounded in $L^{p^*}(\mathbb{R}^N)$, by Vitali's convergence theorem (see Theorem B.101) this implies that $u_n \rightarrow u$ in $L^q(\Omega)$ for all $1 \leq q < p^*$. $\square$

Remark 12.21. Note that when $p > 1$ the function u belongs to $W^{1,p}(\Omega)$. This follows from Theorem 11.65. On the other hand, when $p = 1$, we can only conclude that $u \in BV(\Omega)$ (see Theorem 14.39).

Exercise 12.22. Prove that if Ω is an extension domain for $W^{1,p}(\Omega)$, $1 \leq p < N$, with finite measure, then $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$ if and only if $u_n \rightarrow u$ in $L^p(\Omega)$ and $\nabla u_n \rightharpoonup \nabla u$ in $L^p(\Omega; \mathbb{R}^N)$.

Theorem 12.23 (Rellich–Kondrachov's compactness). *Let $m \in \mathbb{N}$, $1 \leq p < \infty$ be such $mp < N$, and let $\Omega \subseteq \mathbb{R}^N$ be an extension domain for $W^{m,p}(\Omega)$ with finite measure. Let $\{u_n\}_n$ be a bounded sequence in $W^{m,p}(\Omega)$. Then there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and a function*

$$u \in L^{p_{m,0}^*}(\Omega) \cap W^{1,p_{m,1}^*}(\Omega) \cap \dots \cap W^{m-1,p_{m,m-1}^*}(\Omega)$$

such that $\nabla^j u_{n_k} \rightarrow \nabla^j u$ in $L^q(\Omega; \mathbb{R}^{M_j})$ for every $j = 0, \dots, m-1$, and for all $p \leq q < p_{m,j}^$.*

We recall that $\nabla^0 u := u$. The proof is left as an exercise.

The following exercises show that compactness fails for $q = p^*$ even for nice domains and that for general domains even the embedding

$$W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$$

may fail to be compact.

Exercise 12.24. Let $1 \leq p < N$, and consider the sequence of functions $u_n : B(0, 1) \rightarrow \mathbb{R}$ defined by

$$u_n(x) := \begin{cases} n^{N/p-1}(1 - n\|x\|) & \text{if } \|x\| < 1/n, \\ 0 & \text{if } \|x\| \geq 1/n. \end{cases}$$

Prove that $\{u_n\}_n$ is bounded in $W^{1,p}(B(0, 1))$ but that it does not admit any subsequence strongly convergent in $L^{p^*}(B(0, 1))$.

Exercise 12.25. In Exercise 12.15 take $\delta_{2n} := (h_{2n})^a$, for some $a \geq 3$, and for $j = 1, 3, 5, \dots$, consider functions $u_j : \Omega \rightarrow \mathbb{R}$ such that

$$u_j(x_1, x_2) = \begin{cases} 1/h_j & \text{in } R_j, \\ 0 & \text{in } \Omega \setminus (P_{j-1} \cup R_j \cup P_{j+1}) \end{cases}$$

and

$$\nabla u_j(x_1, x_2) = \begin{cases} (1/(h_j h_{j-1}), 0) & \text{in } P_{j-1}, \\ -(1/(h_j h_{j-1}), 0) & \text{in } P_{j+1}. \end{cases}$$

Prove that the sequence $\{u_j\}_j$ is bounded in $W^{1,2}(\Omega)$ but that it does not admit any subsequence strongly convergent in $L^2(\Omega)$.

The previous exercise shows that the embedding $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ fails to be compact for arbitrary open bounded sets. The following exercise gives the only possible type of compactness that remains.

Exercise 12.26. Prove that if $\Omega \subset \mathbb{R}^N$ is an open set with finite measure and $1 < p < \infty$, then the embedding $W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$ is compact for all $1 \leq q < p$.

Exercise 12.27. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $1 \leq p < \infty$. Prove that the embedding $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ is compact if and only if

$$\lim_{n \rightarrow \infty} \sup \left\{ \int_{\Omega \setminus \Omega_n} |u|^p dx : \|u\|_{W^{1,p}(\Omega)} \leq 1 \right\} = 0,$$

where $\Omega_n := \{x \in \Omega : \text{dist}(x, \partial\Omega) > 1/n \text{ and } \|x\| < n\}$.

The Rellich–Kondrachov theorem and its variations hold for special domains with *finite* measure. The next two exercises show that if we restrict our attention to the class of radial functions, then we have compactness in the entire space $\mathbb{R}^N$ for all $N < q < p^*$, but not for $q = N$ or $q = p^*$.

Exercise 12.28 (Radial functions, I). Let $N \geq 2$, let $f \in C^1([0, \infty))$, and let

$$(12.16) \quad u(x) := f(\|x\|), \quad x \in \mathbb{R}^N.$$

(i) Let $1 \leq p < \infty$. Find necessary and sufficient conditions on f for u to be in $W^{1,p}(\mathbb{R}^N)$.

(ii) Let $a > 0$. Prove that for $r > 0$,

$$\begin{aligned} (r^{2a} f^2(r))' &\leq [(r^a f(r))']^2 + (r^a f(r))^2 \\ &= r^{2a} [(f'(r))^2 + f^2(r)] \\ &\quad + a(r^{2a-1} f^2(r))' - a(a-1)r^{2a-2} f^2(r). \end{aligned}$$

(iii) Prove that for every $N \in \mathbb{N}$ with $N \geq 3$ and for all $r > N-1$,

$$r^{N-1} f^2(r) \leq 2 \int_0^r t^{N-1} [(f'(t))^2 + f^2(t)] dt.$$

(iv) Prove that for all $r > 1$,

$$r f^2(r) \leq 2 \int_r^\infty t [(f'(t))^2 + f^2(t)] dt.$$

(v) Prove that if the function u defined in (12.16) belongs to $W^{1,2}(\mathbb{R}^N)$, $N \geq 2$, then

$$|u(x)| \leq c \|x\|^{-(N-1)/2} \|u\|_{W^{1,2}(\mathbb{R}^N)} \quad \text{for all } x \in \mathbb{R}^N, \text{ with } \|x\| > N-1.$$

Exercise 12.29 (Radial functions, II). Let $\{u_n\}_n$ be a sequence of radial functions in $W^{1,2}(\mathbb{R}^N) \cap C^1(\mathbb{R}^N)$ with $\sup_n \|u\|_{W^{1,2}(\mathbb{R}^N)} < \infty$.

(i) Prove that

$$\lim_{R \rightarrow \infty} \sup_n \int_{\mathbb{R}^N \setminus B(0,R)} |u_n(x)|^{2N/(N-2)} dx = 0.$$

(ii) Prove that there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and a function $u \in W^{1,2}(\mathbb{R}^N)$ such that $u_{n_k} \rightarrow u$ in $L^q(\mathbb{R}^N)$ for all $2 < q < \frac{2N}{N-2}$.

(iii) Let $\varphi \in C_c^1(\mathbb{R})$ be such that $\text{supp } \varphi \subseteq [0, 1]$, $\varphi \neq 0$, and define $u_n(x) := a_n \varphi(\|x\| - n)$, $x \in \mathbb{R}^N$. Find a_n in such a way that $\{u_n\}_n$ is bounded in $W^{1,2}(\mathbb{R}^N)$ but does not converge in $L^2(\mathbb{R}^N)$.

(iv) Let φ be as in part (iii) and define $u_n(x) = a_n \varphi(2^n \|x\| - 1)$, $x \in \mathbb{R}^N$. Find a_n in such a way that $\{u_n\}_n$ is bounded in $W^{1,2}(\mathbb{R}^N)$ but does not converge in $L^{2^*}(\mathbb{R}^N)$.

We conclude this section by showing that the Rellich–Kondrachov theorem continues to hold for bounded domains with boundary of class C (see Definition 9.57).

Theorem 12.30. *Let $1 \leq p < \infty$ and let $\Omega \subset \mathbb{R}^N$ be a bounded domain whose boundary is of class C . Then the embedding $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ is compact.*

Proof. For every $x_0 \in \partial\Omega$ there exist local coordinates $y = (y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R}$, with $y = 0$ at $x = x_0$, a continuous function $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$, and $r_{x_0} > 0$ such that $\Omega \cap B(x_0, r_{x_0})$ in local coordinates takes the form

$$\{(y', y_N) \in B(0, r_{x_0}) : y_N > f(y')\}.$$

By taking $r > 0$ and $t > 0$ sufficiently small, we have that the open set $U(x_0, r, t)$ defined in (11.12) is contained in $B(x_0, r_{x_0})$. Hence, $\Omega \cap U(x_0, r, t)$ in local coordinates becomes

$$(12.17) \quad \{(y', y_N) : y' \in Q_{N-1}(0, r), f(y') < y_N < f(y') + t\}.$$

Assume that $u \in C_c^\infty(\mathbb{R}^N)$. Fix $y' \in Q_{N-1}(0, r)$ and let

$$f(y') < y_N < f(y') + t, \quad f(y') + t/2 < \tau < f(y') + t.$$

By the fundamental theorem of calculus we have that

$$u(y', y_N) = u(y', \tau) - \int_{y_N}^{\tau} \partial_N u(y', s) ds,$$

and so by the convexity of $g(t) = |t|^p$, $t \in \mathbb{R}$, and Hölder's inequality,

$$\begin{aligned} |u(y', y_N)|^p &\leq 2^{p-1}|u(y', \tau)|^p + 2^{p-1} \left(\int_{y_N}^{f(y')+t} |\partial_N u(y', s)| ds \right)^p \\ &\leq 2^{p-1}|u(y', \tau)|^p + 2^{p-1} t^{p/p'} \int_{f(y')}^{f(y')+t} |\partial_N u(y', s)|^p ds. \end{aligned}$$

By averaging in τ over $(f(y') + t/2, f(y') + t)$, we obtain

$$|u(y', y_N)|^p \leq \frac{2^p}{t} \int_{f(y')+t/2}^{f(y')+t} |u(y', \tau)|^p d\tau + 2^{p-1} t^{p/p'} \int_{f(y')}^{f(y')+t} |\partial_N u(y', s)|^p ds.$$

Now we integrate in (y', y_N) over $Q_{N-1}(0, r) \times (f(y'), f(y') + \delta)$, where $0 < \delta \leq t$, to get

$$\begin{aligned} (12.18) \quad &\int_{Q_{N-1}(0, r)} \int_{f(y')}^{f(y')+\delta} |u(y', y_N)|^p dy_N dy' \\ &\leq \frac{2^p \delta}{t} \int_{Q_{N-1}(0, r)} \int_{f(y')+t/2}^{f(y')+t} |u(y', \tau)|^p d\tau dy' \\ &\quad + 2^{p-1} t^{p/p'} \delta \int_{Q_{N-1}(0, r)} \int_{f(y')}^{f(y')+t} |\partial_N u(y', s)|^p ds dy'. \end{aligned}$$

Since $\partial\Omega \subset \mathbb{R}^N$ is compact, $\partial\Omega \subset \bigcup_{i=1}^M U(x_i, r_i, t_i)$ for some $x_1, \dots, x_M \in \partial\Omega$. For every $\varepsilon > 0$ let $\Omega_\varepsilon := \{x \in \Omega : \text{dist}(x, \partial\Omega) > \varepsilon\}$. By taking

$0 < \delta < \min_{1 \leq i \leq M} t_i$ and $\varepsilon_\delta > 0$ sufficiently small, we have that $\Omega \setminus \Omega_{\varepsilon_\delta} \subseteq \bigcup_{i=1}^M U(x_i, r_i, \delta)$. Hence, by (12.18) we get that

$$\begin{aligned} \int_{\Omega \setminus \Omega_{\varepsilon_\delta}} |u(x)|^p dx &\leq \frac{2^p M \delta}{t_-} \int_{\Omega} |u(x)|^p dx \\ &\quad + 2^{p-1} M t_+^{p/p'} \delta \int_{\Omega} \|\nabla u(x)\|^p dx, \end{aligned}$$

where $t_- := \min_{1 \leq i \leq M} t_i$ and $t_+ := \max_{1 \leq i \leq M} t_i$.

Since the restriction to Ω of functions in $C_c^\infty(\mathbb{R}^N)$ is dense in $W^{1,p}(\Omega)$ by Theorem 11.35, the previous inequality continues to hold for all $u \in W^{1,p}(\Omega)$. In particular,

$$\sup_{\|u\|_{W^{1,p}(\Omega)} \leq 1} \int_{\Omega \setminus \Omega_{\varepsilon_\delta}} |u(x)|^p dx \leq \frac{2^p M \delta}{t_-} + 2^{p-1} M t_+^{p/p'} \delta \rightarrow 0$$

as $\delta \rightarrow 0$, and so the result follows from Exercise 12.27. $\square$

12.2. Embeddings: $mp = N$

In the previous section we have seen that $W^{1,p}(\mathbb{R}^N) \hookrightarrow L^{p^*}(\mathbb{R}^N)$, where $p^* = Np/(N-p)$. Since $p^* \rightarrow \infty$ as $p \rightarrow N^-$, one would be tempted to say that if $u \in W^{1,N}(\mathbb{R}^N)$, then $u \in L^\infty(\mathbb{R}^N)$. For $N = 1$ this is true since if $u \in W^{1,1}(\mathbb{R})$, then a representative $\bar{u}$ is absolutely continuous in $\mathbb{R}$ so that

$$\bar{u}(x) = \bar{u}(0) + \int_0^x \bar{u}'(s) ds,$$

and since $\bar{u}' = u' \in L^1(\mathbb{R})$, we have that $\bar{u}$ is bounded and continuous. For $N > 1$ this is not the case, as Exercise 12.13 shows.

For $N \geq 2$, $L^\infty(\mathbb{R}^N)$ should be replaced by $\text{BMO}(\mathbb{R}^N)$ (see Section C.5 in Appendix C). Indeed, we have the following result.

Theorem 12.31. *There exists a constant $c = c(N) > 0$ such that for every function $u \in \dot{W}^{1,N}(\mathbb{R}^N)$,*

$$(12.19) \quad \|u\|_{\text{BMO}(\mathbb{R}^N)} \leq c \|\nabla u\|_{L^N(\mathbb{R}^N)}.$$

In particular, $W^{1,N}(\mathbb{R}^N) \hookrightarrow \text{BMO}(\mathbb{R}^N)$.

Proof. Let $u \in \dot{W}^{1,N}(\mathbb{R}^N)$ and let Q_r be a cube of side-length $r > 0$ and sides parallel to the axes. By Poincaré's inequality for rectangles (see Theorem 13.34) we have

$$\|u - u_{Q_r}\|_{L^N(Q_r)} \leq N^{1-1/N} (\mathcal{L}^N(Q_r))^{1/N} \|\nabla u\|_{L^N(\mathbb{R}^N)},$$

where $u_{Q_r} := \frac{1}{\mathcal{L}^N(Q_r)} \int u(x) dx$, and so by Hölder's inequality,

$$\begin{aligned} \int_{Q_r} |u(x) - u_{Q_r}| dx &\leq \|u - u_{Q_r}\|_{L^N(Q_r)} (\mathcal{L}^N(Q_r))^{1/N'} \\ &\leq N^{1-1/N} \mathcal{L}^N(Q_r) \|\nabla u\|_{L^N(\mathbb{R}^N)}. \end{aligned}$$

By dividing by $\mathcal{L}^N(Q_r)$ and taking the supremum over all cubes, we get (12.19). $\square$

Next we extend the previous result to higher derivatives.

Theorem 12.32. *Let $m \in \mathbb{N}$, $m \geq 2$. Then for every $u \in \dot{W}^{m,N/m}(\mathbb{R}^N)$ there exists a polynomial P_u of degree $m-1$ such that*

$$(12.20) \quad \|u - P_u\|_{\text{BMO}(\mathbb{R}^N)} \leq c \|\nabla^m u\|_{L^{N/m}(\mathbb{R}^N)}$$

and

$$(12.21) \quad \|\nabla^k(u - P_u)\|_{L^{N/k}(\mathbb{R}^N)} \leq c \|\nabla^m u\|_{L^{N/m}(\mathbb{R}^N)}$$

for $k = 1, \dots, m-1$ and for some constant $c = c(N, m) > 0$. In particular,

$$\dot{W}^{m,N/m}(\mathbb{R}^N) \hookrightarrow \text{BMO}(\mathbb{R}^N) \cap W^{1,q_1}(\mathbb{R}^N) \cap \dots \cap W^{m-1,q_{m-1}}(\mathbb{R}^N)$$

for all $N/m \leq q_k \leq N/k$, $k = 1, \dots, m-1$.

Observe that N/k is the critical exponent $p_{m,k}^*$ given in (12.7) for $p = N/m$.

Proof. Given $u \in \dot{W}^{m,N/m}(\mathbb{R}^N)$, for every $i = 1, \dots, N$ we have that $\partial_i u \in \dot{W}^{m-1,N/m}(\mathbb{R}^N)$ and $(m-1)N/m < N$. Thus by the Sobolev–Gagliardo–Nirenberg's embedding in $\dot{W}^{m-1,N/m}(\mathbb{R}^N)$ (see Theorem 12.9) we can find a polynomial P_i of degree $m-2$ such that

$$(12.22) \quad \|\partial_i u - P_i\|_{L^N} \leq c \|\nabla^m u\|_{L^{N/m}}$$

and

$$(12.23) \quad \|\nabla^k(\partial_i u - P_i)\|_{L^{N/k}} \leq c \|\nabla^m u\|_{L^{N/m}}$$

for every $k = 1, \dots, m-2$, where $P_i = \partial_i u - \sum_{|\alpha|=m-1} K_\alpha * \partial^\alpha(\partial_i u)$. Using Remark 11.6 one can show (exercise) that $\partial_j P_i = \partial_i P_j$ and so we can find a polynomial P of degree $m-1$ such that $\partial_i P_i = P$. Since $\partial_i(u - P) \in L^N(\mathbb{R}^N)$, by the previous theorem and (12.22) we have that

$$\|u - P\|_{\text{BMO}} \leq c \sum_i \|\partial_i u - P_i\|_{L^N} \leq c \|\nabla^m u\|_{L^{N/m}},$$

while (12.21) follows from (12.23) and the fact that $\partial_i P_i = P$. $\square$

Next we study embeddings of the type $W^{1,N}(\mathbb{R}^N) \hookrightarrow L^q(\mathbb{R}^N)$.

Theorem 12.33. Let $m, N \in \mathbb{N}$ be such $N > m$. Then there exists a constant $c = c(N, m) > 0$ such that for every function $u \in W^{m, N/m}(\mathbb{R}^N)$,

$$(12.24) \quad \|u\|_{L^q(\mathbb{R}^N)} \leq cq^{1-m/N+1/q} \|u\|_{W^{m, N/m}(\mathbb{R}^N)}$$

for every $N/m < q < \infty$ and

$$(12.25) \quad \|\nabla^k u\|_{L^{N/k}(\mathbb{R}^N)} \leq c \|\nabla^m u\|_{L^{N/m}(\mathbb{R}^N)}$$

for every $k = 1, \dots, m-1$. In particular,

$$W^{m, N/m}(\mathbb{R}^N) \hookrightarrow L^{q_0}(\mathbb{R}^N) \cap W^{1, q_1}(\mathbb{R}^N) \cap \dots \cap W^{m-1, q_{m-1}}(\mathbb{R}^N)$$

for all $N/m < q_0 < \infty$ and all $N/m \leq q_k \leq N/k$, $k = 1, \dots, m-1$.

The proof makes use of the following exercise.

Exercise 12.34. Let $w_n : \mathbb{R}^N \rightarrow \mathbb{R}$, $n \in \mathbb{N}$, be Lebesgue measurable functions and let $w := \sum_{n=1}^{\infty} w_n$. Assume that there exists an integer $M \in \mathbb{N}$ such that for every $x \in \mathbb{R}^N$ at most M terms $w_n(x)$ are nonzero. Prove that for every $1 \leq p < \infty$,

$$\|w\|_{L^p(\mathbb{R}^N)} \leq M^{1/p'} \left(\sum_{n=1}^{\infty} \|w_n\|_{L^p(\mathbb{R}^N)}^p \right)^{1/p},$$

while $\|w\|_{L^\infty(\mathbb{R}^N)} \leq M \sup_n \|w_n\|_{L^\infty(\mathbb{R}^N)}$.

We turn to the proof of Theorem 12.33.

Proof of Theorem 12.33. It suffices to prove (12.24), since (12.25) follows from Theorem 12.8 and the fact that $\nabla^k u \in W^{m-k, N/m}(\mathbb{R}^N; \mathbb{R}^{M_k})$ for every $k = 1, \dots, m-1$ and $(m-k)N/m < N$.

Step 1: Assume first that $u \in C_c^\infty(\mathbb{R}^N)$. By Lemma 12.10,

$$|u(x)| \leq \frac{c_0}{\beta_N} \sum_{|\alpha|=m} \int_{B(x, d)} \frac{|\partial^\alpha u(y)|}{\|x-y\|^{N-m}} dy,$$

where d is the diameter of the support of u and $c_0 = c_0(N, m)$.

Let $N/m < q < \infty$. By Young's inequality (see Theorem 10.49), with $m/N + 1/s = 1 + 1/q$,

$$\begin{aligned} \|u\|_{L^q} &\leq \frac{c_0}{\beta_N} \left(\int_{B(0, d)} \frac{1}{\|w\|^{(N-m)s}} dw \right)^{1/s} \|\nabla^m u\|_{L^{N/m}} \\ &= c_0 \frac{1}{\beta_N^{1-1/s} (N - (N-m)s)^{1/s}} d^{[N - (N-m)s]/s} \|\nabla^m u\|_{L^{N/m}} \\ &\leq c_0 q^{1/s} d^{N/q} \|\nabla^m u\|_{L^{N/m}}, \end{aligned}$$

where in the last inequality we used the facts that $m/N + 1/s = 1 + 1/q$, $q > N/m > 1$, $\beta_N > 1$, $s > 1$.

Step 2: For every $z \in \mathbb{Z}^N$ consider the open cube $Q(z, 2)$ centered at z and of side-length 2. Then $\mathbb{R}^N = \bigcup_{z \in \mathbb{Z}^N} Q(z, 2)$, and so reasoning as in the proof of Theorem C.21 we can construct a smooth partition of unity $\{\psi_z\}_{z \in \mathbb{Z}^N}$ subordinated to $\{Q(z, 2)\}_{z \in \mathbb{Z}^N}$ with the property that $\|\partial^\alpha \psi_z\|_{L^\infty} \leq c(m, N)$ for every $|\alpha| \leq m$ and every z . Since for every $x \in \mathbb{R}^N$, there are at most 2^N cubes $Q(z, 2)$ that contain x , it follows from Exercise 12.34 that

$$\begin{aligned} \|u\|_{L^q} &= \left\| \sum_{z \in \mathbb{Z}^N} (u\psi_z) \right\|_{L^q} \leq 2^{N/q'} \left(\sum_{z \in \mathbb{Z}^N} \|u\psi_z\|_{L^q}^q \right)^{1/q} \\ &\leq 2^{N/q'} c_0 q^{1/s} (2\sqrt{N})^{N/q} \left(\sum_{z \in \mathbb{Z}^N} \|\nabla^m(u\psi_z)\|_{L^{N/m}}^q \right)^{1/q}, \end{aligned}$$

where in the last inequality we have used Step 1 with $d = 2\sqrt{N}$.

Using the fact that

$$\left(\sum_{z \in \mathbb{Z}^N} a_z^q \right)^{1/q} \leq \left(\sum_{z \in \mathbb{Z}^N} a_z^{N/m} \right)^{m/N}$$

for $a_z \geq 0$, we have that

$$\begin{aligned} \|u\|_{L^q} &\leq cq^{1/s} \left(\sum_{z \in \mathbb{Z}^N} \|\nabla^m(u\psi_z)\|_{L^{N/m}}^{N/m} \right)^{m/N} \\ &\leq cq^{1/s} \left(\sum_{z \in \mathbb{Z}^N} \int_{Q(z, 2)} \|\nabla^m(u\psi_z)\|_{L^{N/m}}^{N/m} dx \right)^{m/N} \\ &\leq cq^{1/s} \left(\sum_{k=0}^m \int_{\mathbb{R}^N} \|\nabla^k u\|_{L^{N/m}}^{N/m} dx \right)^{m/N} = cq^{1/s} \|u\|_{W^{m, N/m}}, \end{aligned}$$

where we used the facts that $\sum_{z \in \mathbb{Z}^N} \chi_{Q(z, 2)} \leq 2^N$ and that $\|\partial^\alpha \psi_z\|_{L^\infty} \leq c$ for every multi-index α with $|\alpha| \leq m$ and every z , and where the constant $c = c(m, N)$ has changed from line to line.

A standard mollification argument concludes the proof. $\square$

Remark 12.35. Unlike the case $mp < N$, Step 2 of the proof of (12.24) uses strongly the fact that both u and its derivatives belong to $L^p(\mathbb{R}^N)$.

Exercise 12.36. Using (12.25) in Step 2 of the previous proof shows that (12.24) can be replaced by the better estimate

$$\|u\|_{L^q(\mathbb{R}^N)} \leq cq^{1-m/N+1/q} \left(\|u\|_{L^{N/m}(\mathbb{R}^N)} + \|\nabla^m u\|_{L^{N/m}(\mathbb{R}^N)} \right).$$

The following exercise provides another proof of Theorem 12.33 when $m = 1$ but with worse constants.

Exercise 12.37. Let $u \in \dot{W}^{1,N}(\mathbb{R}^N)$, $N \geq 2$, with $u \in L^q(\mathbb{R}^N)$ for some $1 \leq q < \infty$. Let $v := |u|^t$, where $t > 1$ is such that $v \in W^{1,1}(\mathbb{R}^N)$.

(i) Prove that

$$\|u\|_{L^{tN/(N-1)}} \leq c \|u\|_{L^{(t-1)N/(N-1)}}^{(t-1)/t} \|\nabla u\|_{L^N}^{1/t}.$$

(ii) Taking inductively $t = n + q(N-1)/N$, $n \in \mathbb{N}$, prove that for all $q \leq r < \infty$,

$$\|u\|_{L^r} \leq c \|u\|_{L^q}^{q/r} \|\nabla u\|_{L^N}^{1-q/r}.$$

Note that, in general, if $u \in W^{m,N/m}(\mathbb{R}^N)$, we cannot conclude that $u \in L^q(\mathbb{R}^N)$ for $1 \leq q < N/m$. However, since $u \in L^q(\mathbb{R}^N)$ for all $q \geq N/m$, the function u must decay faster than algebraically at infinity. Indeed, we will show that it must have exponential decay. For every $\ell \in \mathbb{N}$ consider the function

$$(12.26) \quad \exp_\ell(s) := \sum_{n=\ell-1}^{\infty} \frac{1}{n!} s^n = \exp(s) - \sum_{n=0}^{\ell-2} \frac{1}{n!} s^n, \quad s \in \mathbb{R}.$$

The next theorem gives an embedding of the space $W^{m,N/m}(\mathbb{R}^N)$ into the Orlicz space generated by the function $\exp_{\lfloor N/m \rfloor + 1}(s^{(N/m)'})$. We introduce the notion of Orlicz space.

Definition 12.38. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $\Phi : [0, \infty) \rightarrow [0, \infty]$ be a convex, lower semicontinuous function such that $\Phi(0) = 0$ and Φ is not identically zero or infinity. The *Orlicz space* $L^\Phi(E)$ generated by the Orlicz function Φ is the space of all Lebesgue measurable functions $u : E \rightarrow \mathbb{R}$ such that $\int_E \Phi(|u(x)|/s) dx < \infty$ for some $s > 0$ (depending on u).

Exercise 12.39. Let E and Φ be as in the previous definition.

(i) Prove that $L^\Phi(E)$ is a normed space with the norm

$$(12.27) \quad \|u\|_\Phi := \inf \left\{ s > 0 : \int_E \Phi(|u(x)|/s) dx < \infty \right\}.$$

(ii) Prove that $L^\Phi(E)$ is a Banach space.

The following theorem is the main theorem of this section. Results of this kind in bounded domains are due to Pokhozhaev [189], Trudinger [234] and Yudovich [250] when $m = 1$ and to Strichartz [225] for $m \geq 2$ (see also [179] and see [141] for some recent results).

Theorem 12.40. *Let $m, N \in \mathbb{N}$ be such $N > m$. Then there exist two constants $c_1, c_2 > 0$ depending only on m and N such that*

$$(12.28) \quad \int_{\mathbb{R}^N} \exp_{[N/m]+1} \left(c_1 \frac{|u(x)|^{(N/m)'}}{\|u\|_{W^{m,N/m}(\mathbb{R}^N)}^{(N/m)'}} \right) dx \leq c_2$$

for all $u \in W^{m,N/m}(\mathbb{R}^N) \setminus \{0\}$. In particular, if

$$\Phi(s) := \exp_{[N/m]+1}(c_1 s^{(N/m)'}), s \geq 0,$$

then $W^{m,N/m}(\mathbb{R}^N) \hookrightarrow L^\Phi(\mathbb{R}^N)$.

Proof. Let $p := N/m$. By replacing u with $u/\|u\|_{W^{m,p}}$, we can assume that $\|u\|_{W^{m,p}} = 1$. Then by Theorem 12.33 with $q = np'$ for $a > 0$,

$$\begin{aligned} \int_{\mathbb{R}^N} \exp_{[p]+1}(a|u(x)|^{p'}) dx &= \sum_{n=[p]}^{\infty} \frac{a^n}{n!} \int_{\mathbb{R}^N} |u(x)|^{np'} dx \\ &\leq \sum_{n=[p]}^{\infty} \frac{a^n c^{np'}}{n!} (np')^{n+1}. \end{aligned}$$

By the ratio test,

$$\frac{\frac{a^{n+1} c^{(n+1)p'}}{(n+1)!} ((n+1)p')^{n+2}}{\frac{a^n c^{np'}}{n!} (np')^{n+1}} = ac^{p'} p' (1 + 1/n)^{n+1} \rightarrow ac^{p'} p' e < 1,$$

provided $a < 1/(c^{p'} p' e)$. Hence, the series converges. $\square$

Remark 12.41. When $m = 1$ we can take $\exp_N$ instead of $\exp_{N+1}$. Moreover, using symmetric rearrangement one can show (see Theorem 15.35) that (12.28) holds for any $c_1 \in (0, \gamma_N)$, where

$$(12.29) \quad \gamma_N := N\beta_N^{1/(N-1)},$$

and where β_N is the surface area of the unit sphere (see (12.3)). Moreover, the constant γ_N is sharp (see Exercise 15.37).

Remark 12.42. The analog of this result for bounded domain can be found in Exercise 13.23. See also Remark 13.24 for more references on this topic.

Exercise 12.43. Prove that for every function $u \in W^{N,1}(\mathbb{R}^N)$,

$$\|u\|_{L^\infty(\mathbb{R}^N)} \leq \left\| \frac{\partial^N u}{\partial x_1 \cdots \partial x_N} \right\|_{L^1(\mathbb{R}^N)}.$$

Prove also that for every function $u \in \dot{W}^{N,1}(\mathbb{R}^N)$ there exists a polynomial P_u of degree $N - 1$ such that

$$\|u - P_u\|_{L^\infty(\mathbb{R}^N)} \leq \left\| \frac{\partial^N u}{\partial x_1 \cdots \partial x_N} \right\|_{L^1(\mathbb{R}^N)}.$$

Theorem 12.44 (Rellich–Kondrachov). *Let $m, N \in \mathbb{N}$ be such $N > m$ and let $\Omega \subset \mathbb{R}^N$ be an extension domain for $W^{m, N/m}(\Omega)$ with finite measure. Let $\{u_n\}_n$ be a bounded sequence in $W^{m, N/m}(\Omega)$. Then there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and a function*

$$u \in L^{N/m}(\Omega) \cap W^{1, N}(\Omega) \cap \dots \cap W^{m, N/m}(\Omega)$$

such that $\partial^\beta u_{n_k} \rightarrow \partial^\beta u$ in $L^q(\Omega)$ for every multi-index β with $|\beta| = k$, $k = 0, \dots, m-1$, and for all $N/m \leq q < N/k$, where $N/0 := \infty$.

The proof is left as an exercise.

The following exercise shows that if $u \in W^{1, N}(\Omega)$, where $\Omega \subseteq \mathbb{R}^N$ is an open set, then one cannot have differentiability $\mathcal{L}^N$ -a.e. in Ω (if $u \in W^{1, p}(\Omega)$, with $p > N$, then this is true by Corollary 12.50).

Exercise 12.45 (Indian mystic's bed of nails). Let $Q := (0, 1)^2$ be the unit cube in $\mathbb{R}^2$, subdivide it into 4^k subcubes each having side length 2^{-k} , and let $C_k := \{x^{(1)}, \dots, x^{(4^k)}\}$ be the set of centers of the subcubes. Define

$$g_k(t) := \frac{\ln\left(\frac{t + e^{-2^k}}{1 + e^{-2^k}}\right)}{\ln\left(\frac{e^{-2^k}}{1 + e^{-2^k}}\right)}, \quad 0 \leq t \leq 1,$$

and

$$f_k(x) := \begin{cases} g_k(\varepsilon_k^{-1} \|x - x^{(n)}\|) & \text{if } \|x - x^{(n)}\| \leq \varepsilon_k \text{ for some } x^{(n)} \in C_k, \\ 0 & \text{otherwise,} \end{cases}$$

where $0 < \varepsilon_k < 4^{-k}$ are chosen sufficiently small. Define $u(x) := \sum_{k=1}^{\infty} \frac{k}{2^k} f_k(x)$, $x \in Q$.

- (i) Prove that $u \in W^{1, 2}(Q) \cap C(Q)$.
- (ii) Let $x \in Q$ be such that either $f_k(x) = 0$ for all k or $f_k(x) = 0$ for all k sufficiently large. Prove that u is not differentiable at x .
- (iii) Prove that if ε_k is chosen sufficiently small, then u is not differentiable on a set of positive measure.

12.3. Embeddings: $mp > N$

In this section we study the case $mp > N$. We begin by showing that $W^{m, p}(\mathbb{R}^N)$ is embedded in $L^\infty(\mathbb{R}^N)$.

Theorem 12.46. *Let $m \in \mathbb{N}$ and $1 \leq p < \infty$ be such $mp > N$. Then $W^{m, p}(\mathbb{R}^N) \hookrightarrow L^\infty(\mathbb{R}^N)$.*

We begin with an auxiliary lemma, which will also be used in the sequel.

Lemma 12.47. *Let $m \in \mathbb{N}$ and $1 \leq p < \infty$ be such $mp > N$, let Q_r be any cube with sides of length r parallel to the axes. Then for every $u \in W^{m,p}(Q_r) \cap C^\infty(Q_r)$,*

$$|u(x) - u(y)| \leq c \sum_{k=1}^{m-1} r^{k-N/p} \|\nabla^k u\|_{L^p(Q_r)} + cr^{m-N/p} \|\nabla^m u\|_{L^p(Q_r)}$$

for every $x, y \in Q_r$, and where $c > 0$ is a constant depending on m , N , and p . In particular, if $m = 1$, then

$$|u(x) - u(y)| \leq cr^{1-N/p} \|\nabla u\|_{L^p(Q_r)}.$$

Proof. Fix $x, y \in Q_r$. By Taylor's formula (see Theorem 9.9),

$$\begin{aligned} u(x) - u(y) &= \sum_{1 \leq |\alpha| \leq m-1} \frac{1}{\alpha!} \partial^\alpha u(y) (x - y)^\alpha \\ &\quad + \sum_{|\alpha|=m} \frac{m}{\alpha!} (x - y)^\alpha \int_0^1 (1-t)^{m-1} \partial^\alpha u(tx + (1-t)y) dt. \end{aligned}$$

Averaging in the y variable over Q_r yields

$$\begin{aligned} u(x) - u_{Q_r} &= \sum_{1 \leq |\alpha| \leq m-1} \frac{1}{\alpha!} \frac{1}{r^N} \int_{Q_r} \partial^\alpha u(y) (x - y)^\alpha dy \\ &\quad + \sum_{|\alpha|=m} \frac{m}{\alpha!} \frac{1}{r^N} \int_{Q_r} \int_0^1 (x - y)^\alpha (1-t)^{m-1} \partial^\alpha u(tx + (1-t)y) dt dy, \end{aligned}$$

where we recall that $u_{Q_r} := \frac{1}{r^N} \int_{Q_r} u(y) dy$. Hence,

$$\begin{aligned} |u(x) - u_{Q_r}| &\leq c \sum_{k=1}^{m-1} r^{k-N} \int_{Q_r} \|\nabla^k u(y)\| dy \\ &\quad + cr^{-N} \int_0^1 \int_{Q_r} \|x - y\|^m (1-t)^{m-1} \|\nabla^m u(tx + (1-t)y)\| dy dt \\ &\leq c \sum_{k=1}^{m-1} r^{k-N} \int_{Q_r} \|\nabla^k u(y)\| dy \\ &\quad + cr^{-N} \int_0^1 \int_{tx+Q_{(1-t)r}} \|z - x\|^m (1-t)^{-N-1} \|\nabla^m u(z)\| dz dt, \end{aligned}$$

where we have used the fact that $\|x - y\| \leq \sqrt{N}r$ in Q_r , Tonelli's theorem, the change of variables $z = tx + (1-t)y$ (so that $z - x = (1-t)(y - x)$ and $dz = (1-t)^N dy$). Using Tonelli's theorem once more, we have that the

integral in the second term on the right-hand side of the previous inequality can be bound from above by

$$\begin{aligned} & \int_{Q_r} \|z - x\|^m \|\nabla^m u(z)\| \int_0^{1-\|z-x\|/(r\sqrt{N})} (1-t)^{-N-1} dt dz \\ & \leq cr^N \int_{Q_r} \|z - x\|^{m-N} \|\nabla^m u(z)\| dz, \end{aligned}$$

and so

$$\begin{aligned} |u(x) - u_{Q_r}| & \leq c \sum_{k=1}^{m-1} r^{k-N} \int_{Q_r} \|\nabla^k u(y)\| dy \\ & \quad + c \int_{Q_r} \|y - x\|^{m-N} \|\nabla^m u(z)\| dy. \end{aligned}$$

By Hölder's inequality we now have

$$\begin{aligned} |u(x) - u_{Q_r}| & \leq c \sum_{k=1}^{m-1} r^{k-N+N/p'} \|\nabla^k u\|_{L^p(Q_r)} \\ & \quad + c \left(\int_{Q_r} \|y - x\|^{(m-N)p'} dy \right)^{1/p'} \|\nabla^m u\|_{L^p(Q_r)} \\ & \leq c \sum_{k=1}^{m-1} r^{k-N/p} \|\nabla^k u\|_{L^p(Q_r)} + cr^{m-N/p} \|\nabla^m u\|_{L^p(Q_r)}, \end{aligned}$$

where we used polar coordinates and the fact that $\|y - x\| \leq \sqrt{N}r$ to estimate

$$\begin{aligned} \int_{Q_r} \|y - x\|^{(m-N)p'} dy & \leq \int_{B(x, \sqrt{N}r)} \|y - x\|^{(m-N)p'} dy \\ & = \frac{\beta_N(p-1)}{mp-N} (\sqrt{N}r)^{(mp-N)/(p-1)}. \end{aligned}$$

Since this is true for all $y \in Q_r$, if $x, y \in Q_r$, then

$$\begin{aligned} |u(x) - u(y)| & \leq |u(x) - u_{Q_r}| + |u(y) - u_{Q_r}| \\ & \leq c \sum_{k=1}^{m-1} r^{k-N/p} \|\nabla^k u\|_{L^p(Q_r)} + cr^{m-N/p} \|\nabla^m u\|_{L^p(Q_r)}, \end{aligned}$$

which completes the proof. $\square$

We turn to the proof of Theorem 12.46.

Proof of Theorem 12.46. Assume that $u \in W^{m,p}(\mathbb{R}^N) \cap C^\infty(\mathbb{R}^N)$. Fix $x \in \mathbb{R}^N$ and let Q_1 be a cube with sides of length 1 parallel to the axes containing x . By Lemma 12.47 with $r = 1$,

$$|u(x)| \leq |u(y)| + c\|u\|_{W^{m,p}(Q_1)}$$

for every $y \in Q_1$. By averaging in y over Q_1 and using Hölder's inequality, we obtain

$$|u(x)| \leq c\|u\|_{W^{m,p}(Q_1)} \leq c\|u\|_{W^{m,p}(\mathbb{R}^N)}.$$

Now, if $u \in W^{m,p}(\mathbb{R}^N)$, let $u_\varepsilon := u * \varphi_\varepsilon$, where φ_ε is a standard mollifier. By the previous inequality applied to u_ε we get $|u_\varepsilon(x)| \leq c\|u_\varepsilon\|_{W^{m,p}(\mathbb{R}^N)}$. Since $\{u_\varepsilon\}$ converge pointwise at every Lebesgue point by Theorem C.16 and in $W^{m,p}(\mathbb{R}^N)$ by Lemma 11.25, it follows that $|u(x)| \leq c\|u\|_{W^{m,p}(\mathbb{R}^N)}$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. $\square$

The next theorem shows that if $p > N$, a function $u \in W^{1,p}(\mathbb{R}^N)$ has a representative in the space $C^{0,1-N/p}(\mathbb{R}^N)$ of bounded functions that are Hölder continuous with exponent $1 - N/p$.

Theorem 12.48 (Morrey's embedding in $W^{1,p}$). *Let $N < p < \infty$. Then $W^{1,p}(\mathbb{R}^N) \hookrightarrow C^{0,1-N/p}(\mathbb{R}^N)$. Moreover, if $u \in W^{1,p}(\mathbb{R}^N)$ and $\bar{u}$ is its representative in $C^{0,1-N/p}(\mathbb{R}^N)$, then*

$$\lim_{\|x\| \rightarrow \infty} \bar{u}(x) = 0.$$

Proof. Let $u \in W^{1,p}(\mathbb{R}^N) \cap C^\infty(\mathbb{R}^N)$. If $x, y \in \mathbb{R}^N$, consider a cube Q_r containing x and y and of side length $r := 2\|x - y\|$. By Lemma 12.47,

$$(12.30) \quad |u(x) - u(y)| \leq c\|x - y\|^{1-N/p} \|\nabla u\|_{L^p(Q_r)}.$$

Hence, u is Hölder continuous of exponent $1 - N/p$, while by Theorem 12.46, we get

$$(12.31) \quad |u(x)| \leq c\|u\|_{W^{1,p}(\mathbb{R}^N)}.$$

Next we remove the extra hypothesis that $u \in C^\infty(\mathbb{R}^N)$. Given any $u \in W^{1,p}(\mathbb{R}^N)$, let $x, y \in \mathbb{R}^N$ be two Lebesgue points of u and let $u_\varepsilon := u * \varphi_\varepsilon$, where φ_ε is a standard mollifier. By (12.30) we have that

$$|u_\varepsilon(x) - u_\varepsilon(y)| \leq c\|x - y\|^{1-N/p} \|\nabla u_\varepsilon\|_{L^p(Q_r)}.$$

Since $\{u_\varepsilon\}_\varepsilon$ converge at every Lebesgue point of u as $\varepsilon \rightarrow 0^+$ by Theorem C.16 and $\nabla u_\varepsilon = (\nabla u)_\varepsilon \rightarrow \nabla u$ in $L^p(\mathbb{R}^N; \mathbb{R}^N)$ by Theorems C.16 and C.20, letting $\varepsilon \rightarrow 0^+$, we get

$$(12.32) \quad |u(x) - u(y)| \leq c\|x - y\|^{1-N/p} \|\nabla u\|_{L^p(Q_r)}$$

for all Lebesgue points $x, y \in \mathbb{R}^N$ of u . This implies that $u : \{\text{Lebesgue points of } u\} \rightarrow \mathbb{R}$ can be uniquely extended to $\mathbb{R}^N$ as a Hölder continuous function $\bar{u}$ of exponent $1 - N/p$ in such a way that (12.32) holds for all $x, y \in \mathbb{R}^N$. With

a similar argument from (12.31) we conclude that $|\bar{u}(x)| \leq c\|u\|_{W^{1,p}(\mathbb{R}^N)}$ for all $x \in \mathbb{R}^N$. Hence,

$$\begin{aligned} \|\bar{u}\|_{C^{0,1-N/p}(\mathbb{R}^N)} &= \sup_{x \in \mathbb{R}^N} |\bar{u}(x)| + \sup_{x, y \in \mathbb{R}^N, x \neq y} \frac{|\bar{u}(x) - \bar{u}(y)|}{\|x - y\|^{1-N/p}} \\ &\leq c\|u\|_{W^{1,p}(\mathbb{R}^N)}. \end{aligned}$$

Finally, we prove that $\bar{u}(x) \rightarrow 0$ as $\|x\| \rightarrow \infty$. Let $\{u_n\}_n$ be any sequence in $C_c^\infty(\mathbb{R}^N)$ that converges to u in $W^{1,p}(\mathbb{R}^N)$. By Theorem 12.46,

$$\|u - u_n\|_{L^\infty(\mathbb{R}^N)} \leq c\|u - u_n\|_{W^{1,p}(\mathbb{R}^N)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Fix $\varepsilon > 0$ and find $\bar{n} \in \mathbb{N}$ such that $\|u - u_n\|_{L^\infty(\mathbb{R}^N)} \leq \varepsilon$ for all $n \geq \bar{n}$. Since $u_{\bar{n}} \in C_c^\infty(\mathbb{R}^N)$, there exists $R_{\bar{n}} > 0$ such that $u_{\bar{n}}(x) = 0$ for all $\|x\| \geq R_{\bar{n}}$. Hence, for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ with $\|x\| \geq R_{\bar{n}}$ we get

$$|\bar{u}(x)| = |\bar{u}(x) - u_{\bar{n}}(x)| \leq \|u - u_n\|_{L^\infty(\mathbb{R}^N)} \leq \varepsilon,$$

and, since $\bar{u}$ is continuous, we get that the previous inequality actually holds for all $x \in \mathbb{R}^N$ with $\|x\| \geq R_{\bar{n}}$. $\square$

Remark 12.49. The first part of the proof actually shows that if $u \in W^{1,p}(\mathbb{R}^N)$ for some $N < p < \infty$, then a representative $\bar{u}$ of u is Hölder continuous with exponent $1 - N/p$ and

$$|\bar{u}(x) - \bar{u}(y)| \leq c\|x - y\|^{1-N/p} \|\nabla u\|_{L^p(\mathbb{R}^N)}$$

for all $x, y \in \mathbb{R}^N$, where $c = c(N, p) > 0$.

As a consequence of Theorem 12.48 we obtain the following result.

Corollary 12.50. *If $u \in W^{1,p}(\mathbb{R}^N)$, $N < p < \infty$, and $\bar{u}$ is its representative in $C^{0,1-N/p}(\mathbb{R}^N)$, then $\bar{u}$ is differentiable at $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ and the weak partial derivatives of u coincide with the (classical) partial derivatives of $\bar{u}$ $\mathcal{L}^N$ -a.e. in $\mathbb{R}^N$.*

Proof. Let $x_0 \in \mathbb{R}^N$ be a p -Lebesgue point for ∇u (see Corollary B.120), that is,

$$\lim_{r \rightarrow 0^+} \frac{1}{r^N} \int_{Q(x_0, r)} \|\nabla u(x) - \nabla u(x_0)\|^p dx = 0,$$

where as usual $Q(x_0, r) := x_0 + (-\frac{r}{2}, \frac{r}{2})^N$, and define

$$v(x) := \bar{u}(x) - \bar{u}(x_0) - \nabla u(x_0) \cdot (x - x_0), \quad x \in \mathbb{R}^N.$$

Reasoning as in the first part of the proof of Theorem 12.48, if $x \in \mathbb{R}^N$ and $r := 2\|x - x_0\|$, we obtain that

$$|v(x) - v(x_0)| \leq cr^{1-N/p} \|\nabla v\|_{L^p(Q(x_0, r))},$$

or, equivalently,

$$\begin{aligned} & \frac{|\bar{u}(x) - \bar{u}(x_0) - \nabla u(x_0) \cdot (x - x_0)|}{\|x - x_0\|} \\ & \leq c \left(\frac{1}{r^N} \int_{Q(x_0, r)} \|\nabla u(y) - \nabla u(x_0)\|^p dy \right)^{1/p} \rightarrow 0 \end{aligned}$$

as $x \rightarrow x_0$. This completes the proof. $\square$

In view of Exercise 11.50 and Corollary 12.50 we can give another proof of Rademacher's theorem.

Exercise 12.51 (Rademacher). Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Prove that every Lipschitz continuous function $u : \Omega \rightarrow \mathbb{R}$ is differentiable at $\mathcal{L}^N$ -a.e. $x \in \Omega$.

The following result is another important consequence of Theorem 12.48.

Corollary 12.52. *Let $\Psi \in W^{1,p}(\mathbb{R}^N; \mathbb{R}^N)$, $N < p < \infty$, and let $\bar{\Psi}$ be its representative in $C^{0,1-N/p}(\mathbb{R}^N; \mathbb{R}^N)$. Then there exists a constant $c = c(N, p) > 0$ such that for every Lebesgue measurable set $E \subseteq \mathbb{R}^N$ the set $\bar{\Psi}(E)$ is Lebesgue measurable and*

$$\mathcal{L}^N(\bar{\Psi}(E)) \leq c(\mathcal{L}^N(E))^{1-N/p} \|J_\Psi\|_{L^p(E)}^N.$$

In particular, $\bar{\Psi}$ satisfies the (N) property; that is, it maps sets of $\mathcal{L}^N$ -measure zero into sets of $\mathcal{L}^N$ -measure zero.

Proof. In the proof of Theorem 12.48, we have seen that for every cube Q with sides parallel to the axes and for every $x, y \in Q$,

$$|\bar{\Psi}_i(x) - \bar{\Psi}_i(y)| \leq c(\text{diam } Q)^{1-N/p} \|\nabla \Psi_i\|_{L^p(Q)}$$

for all $i = 1, \dots, N$. Hence,

$$\begin{aligned} (12.33) \quad \mathcal{L}^N(\bar{\Psi}(Q)) & \leq 2^N (\text{diam } \bar{\Psi}(Q))^N \\ & \leq c(\text{diam } Q)^{(1-N/p)N} \|J_\Psi\|_{L^p(Q)}^N. \end{aligned}$$

Next, if E is an arbitrary Lebesgue measurable set, for every $\varepsilon > 0$ find a countable family $\{Q_n\}_n$ of pairwise disjoint cubes such that $E \subseteq \bigcup_n Q_n := E_0$ and

$$\sum_n \mathcal{L}^N(Q_n) \leq \mathcal{L}^N(E) + \varepsilon.$$

Then by (12.33) and Hölder's inequality (which one?) we have

$$\begin{aligned}
 \mathcal{L}_o^N(\bar{\Psi}(E)) &\leq \sum_n \mathcal{L}^N(\bar{\Psi}(Q_n)) \\
 &\leq c \sum_n (\text{diam } Q_n)^{(1-N/p)N} \|J_{\Psi}\|_{L^p(Q_n)}^N \\
 &\leq c \left(\sum_n (\text{diam } Q_n)^N \right)^{1-N/p} \left(\sum_n \|J_{\Psi}\|_{L^p(Q_n)}^p \right)^{N/p} \\
 &\leq c \left(\sum_n \mathcal{L}^N(Q_n) \right)^{1-N/p} \|J_{\Psi}\|_{L^p(E_0)}^N \\
 &\leq c(\mathcal{L}^N(E) + \varepsilon)^{1-N/p} \|J_{\Psi}\|_{L^p(E_0)}^N.
 \end{aligned}$$

Letting $\varepsilon \rightarrow 0^+$ yields

$$\mathcal{L}_o^N(\bar{\Psi}(E)) \leq c(\mathcal{L}^N(E))^{1-N/p} \|J_{\Psi}\|_{L^p(E)}^N.$$

This inequality implies, in particular, that $\bar{\Psi}$ has the (N) property.

It remains to show that $\bar{\Psi}(E)$ is Lebesgue measurable. Assume first that E has finite measure and write $E = E_{\infty} \cup E_1$, where E_{∞} is an F_{σ} set and E_1 is a set of Lebesgue measure zero. Then $\Psi(E) = \Psi(E_{\infty}) \cup \Psi(E_0)$. By the (N) property the set $\Psi(E_1)$ has Lebesgue measure zero, and so it is Lebesgue measurable by the completeness of the Lebesgue measure, while by the continuity of Ψ the set $\Psi(E_{\infty})$ is Lebesgue measurable. If E is not bounded, we may write it as the countable union of bounded Lebesgue measurable sets, and so by what we have just proved, the set $\Psi(E)$ is Lebesgue measurable, since it may be written as a countable union of Lebesgue measurable sets. $\square$

Remark 12.53. The previous corollary still holds if $\Psi \in \dot{W}^{1,p}(\mathbb{R}^N; \mathbb{R}^N)$ for some $N < p < \infty$.

Thus, from the previous corollary any such mapping can be used as a change of variables. More precisely, we have the following.

Exercise 12.54. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $\Psi : \Omega \rightarrow \mathbb{R}^N$ be one-to-one on a set whose complement in Ω has Lebesgue measure zero. Prove that if $\Psi \in W^{1,p}(\Omega; \mathbb{R}^N)$, $N < p \leq \infty$, then it has a representative $\bar{\Psi}$ for which all the hypotheses of Theorem 9.52 are satisfied.

Next we extend Morrey's theorem to $W^{m,p}(\mathbb{R}^N)$, $m \geq 2$.

Theorem 12.55 (Morrey's embedding in $W^{m,p}$). *Let $m \in \mathbb{N}$, $m \geq 2$, and $1 \leq p < \infty$ be such $mp > N$. Then $W^{m,p}(\mathbb{R}^N) \hookrightarrow C^{\ell,\theta}(\mathbb{R}^N)$, where if $m - N/p$ is not an integer,*

$$(12.34) \quad \ell := \lfloor m - N/p \rfloor, \quad \theta := m - \ell - N/p,$$

while if $m - N/p$ is an integer,

$$(12.35) \quad \ell := m - 1 - N/p, \quad \theta := \text{any number less than } 1.$$

Proof. Step 1: Assume that $m - N/p$ is not an integer. Then $\ell = \lfloor m - N/p \rfloor$ and $m - N/p = \ell + \theta$, where $\theta \in (0, 1)$. Hence, $(m - \ell - 1)p < N < (m - \ell)p$. We now distinguish two cases. If $\ell = m - 1$, then $p > N$ and $\theta = 1 - N/p$. Given $u \in W^{m,p}(\mathbb{R}^N)$, since $\partial^\alpha u \in W^{1,p}(\mathbb{R}^N)$ for every $|\alpha| \leq \ell$, we can apply Morrey's embedding theorem (Theorem 12.48) to $\partial^\alpha u$ to obtain that $\partial^\alpha u \in C^{0,1-N/p}(\mathbb{R}^N)$. It follows that $W^{m,p}(\mathbb{R}^N) \hookrightarrow C^{\ell,\theta}(\mathbb{R}^N)$.

If $\ell < m - 1$, then $p < N$. For every $n = 0, \dots, \ell$ and every β with $|\beta| = \ell - n$, we have that $\partial^\beta u \in W^{m-\ell+n,p}(\mathbb{R}^N)$, and since $(m - \ell + n)p > N$, by Theorem 12.46 we have that $\partial^\beta u$ belongs to $L^\infty(\mathbb{R}^N)$. Hence, $u \in C^{\ell-1,1}(\mathbb{R}^N)$ if $\ell \geq 1$ and $u \in L^\infty(\mathbb{R}^N)$ if $\ell = 0$. In particular, for $|\beta| = \ell$ we have that $\partial^\beta u \in L^\infty(\mathbb{R}^N) \cap L^p(\mathbb{R}^N)$ and, in turn, by Exercise B.79(ii), to all $L^q(\mathbb{R}^N)$ for $p \leq q \leq \infty$.

On the other hand, since $(m - \ell - 1)p < N$, using Sobolev–Gagliardo–Nirenberg's embedding theorem (Theorem 12.4) we have that $\partial^\alpha u \in L^{p_{m,\ell+1}^*}(\mathbb{R}^N)$ for every $|\alpha| = \ell + 1$, where $p_{m,\ell+1}^* = \frac{Np}{N-(m-\ell-1)p} > N$ (see (12.7)). It follows that $\partial^\beta u \in W^{1,p_{m,\ell+1}^*}(\mathbb{R}^N)$ for $|\beta| = \ell$, and since $p_{m,\ell+1}^* > N$, by Morrey's embedding theorem (Theorem 12.48), $\partial^\beta u \in C^{0,1-N/p_{m,\ell+1}^*}(\mathbb{R}^N)$. Note that $1 - N/p_{m,\ell+1}^* = m - \ell - N/p = \theta$. This shows that $W^{m,p}(\mathbb{R}^N) \hookrightarrow C^{\ell,\theta}(\mathbb{R}^N)$.

Step 2: Assume next that $m - N/p = n \in \mathbb{N}$. If $n = m - 1$, then $p = N$ and so by the critical embedding theorem (Theorem 12.33) applied to $\partial^\alpha u$ where $|\alpha| \leq m - 1$ we have that $\partial^\alpha u \in L^q(\mathbb{R}^N)$ for every $N \leq q < \infty$. In particular, $u \in W^{m-1,q}(\mathbb{R}^N)$ for every $N \leq q < \infty$. Hence, for every $q > N$ we have that $m - 2 < m - 1 - (N/q) < m - 1$ and so by Step 1, $W^{m,p}(\mathbb{R}^N) \hookrightarrow C^{\ell,\theta}(\mathbb{R}^N)$, where $\theta := m - N/p - (N/q)$.

If $n < m - 1$, then $p < N$. Since $p_{m,n+1}^* = N$ (see (12.7)), using Sobolev–Gagliardo–Nirenberg's embedding theorem (Theorem 12.4) we have that $\partial^\alpha u \in L^N(\mathbb{R}^N)$ for every $|\alpha| = n + 1$. In turn, by repeated applications of Exercise 12.37, $\partial^\beta u \in L^q(\mathbb{R}^N)$ for $0 \leq |\beta| \leq n$ for all $p \leq q < \infty$. Hence, $u \in W^{n,q}(\mathbb{R}^N)$ for every $p \leq q < \infty$. Taking $q > N$, by Step 1, $W^{m,p}(\mathbb{R}^N) \hookrightarrow C^{\ell,\theta}(\mathbb{R}^N)$, where $\theta := m - N/p - (N/q)$. $\square$

Remark 12.56. Let $m \in \mathbb{N}$, $m \geq 2$, and $1 \leq p < \infty$ be such $mp > N$ and $m - N/p$ is not an integer, and let $u \in \dot{W}^{m,p}(\mathbb{R}^N)$. As in Step 1 of the previous proof if $\ell = m - 1$, by Remark 12.49, we have that $\partial^\alpha u$ is Hölder continuous with exponent θ for every $|\alpha| = \ell$. If $\ell < m - 1$, consider $v := \partial^\beta u$ for $|\beta| = \ell$. Using Theorem 12.9 (with m replaced by $m - \ell - 1$ and u by $\partial_i v$) we can find a polynomial $P_{i,\beta}$ of degree $m - \ell - 2$ such that

$\partial_i v - P_{i,\beta} \in L^{p_{m,\ell+1}^*}(\mathbb{R}^N)$, where $P_{i,\beta} = \partial_i v - \sum_{|\alpha|=m-\ell-1} K_\alpha * \partial^\alpha(\partial_i v)$. Using Remark 11.6 one can show (exercise) that $\partial_j P_{i,\beta} = \partial_i P_{j,\beta}$ and so we can find a polynomial P_β of degree $m - \ell - 1$ such that $P_{i,\beta} = \partial_i P_\beta$. Since $\partial_i(v - P_\beta) \in L^{p_{m,\ell+1}^*}(\mathbb{R}^N)$ and $p_{m,\ell+1}^* = \frac{Np}{N-(m-\ell-1)p} > N$, by Remark 12.49, we have that $\partial^\beta u - P_\beta$ is Hölder continuous with exponent θ .

If $u \in L^q(\mathbb{R}^N)$ for some $1 \leq q < \infty$, then $P_\beta = 0$ (why?) and so $\partial^\beta u$ is Hölder continuous with exponent θ .

The next exercise provides an alternative proof of Step 1 in the case $\ell = 0$ and $1 < p < \infty$.

Exercise 12.57. Let $m \in \mathbb{N}$ and $1 < p < \infty$ be such $mp > N$ and $m - N/p \in (0, 1)$. Let $u \in C_c^\infty(\mathbb{R}^N)$.

(i) Use Lemma 12.10 to show that for $x, h \in \mathbb{R}^N$, with $h \neq 0$,

$$\begin{aligned} |u(x+h) - u(x)| &\leq \sum_{|\alpha|=m} \int_{B(x, 3\|h\|)} |K_\alpha(y-x-h) - K_\alpha(y-x)| |\partial^\alpha u(y)| dy \\ &\quad + \sum_{|\alpha|=m} \int_{\mathbb{R}^N \setminus B(x, 3\|h\|)} |K_\alpha(y-x-h) - K_\alpha(y-x)| |\partial^\alpha u(y)| dy \\ &=: \sum_{|\alpha|=m} I_\alpha + \sum_{|\alpha|=m} II_\alpha. \end{aligned}$$

(ii) Prove that $I_\alpha \leq c\|h\|^{m-N/p} \|\partial^\alpha u\|_{L^p}$ and $II_\alpha \leq c\|h\|^{m-N/p} \|\partial^\alpha u\|_{L^p}$.

(iii) Use parts (i) and (ii) to show that $|u|_{C^{0,m-N/p}} \leq c\|\nabla^m u\|_{L^p}$.

Exercise 12.58. Prove that $W^{N+1,1}(\mathbb{R}^N) \hookrightarrow C^{0,1}(\mathbb{R}^N)$.

Exercise 12.59. Let $1 \leq p < \infty$ be such that $p > N$ and let $\Omega \subseteq \mathbb{R}^N$ be an extension domain for $W^{1,p}(\Omega)$. Prove that $W^{1,p}(\Omega) \hookrightarrow C^{0,1-N/p}(\overline{\Omega})$ and that if $u \in W^{1,p}(\Omega)$ and $\bar{u}$ is its representative in $C^{0,1-N/p}(\overline{\Omega})$, then $\bar{u}$ is differentiable at $\mathcal{L}^N$ -a.e. $x \in \Omega$ and, if Ω is unbounded, $\bar{u}(x) \rightarrow 0$ as $\|x\| \rightarrow \infty$.

Exercise 12.60. Let $m \in \mathbb{N}$, $m \geq 2$, and $1 \leq p < \infty$ be such $mp > N$, and let $\Omega \subseteq \mathbb{R}^N$ be an extension domain for $W^{m,p}(\Omega)$. Prove that $W^{m,p}(\Omega) \hookrightarrow C^{\ell,\theta}(\overline{\Omega})$, where ℓ and θ are given in (12.34) and (12.35), respectively.

Theorem 12.61 (Rellich–Kondrachov). *Let $1 \leq p < \infty$ be such that $p > N$ and let $\Omega \subseteq \mathbb{R}^N$ be a bounded extension domain for $W^{1,p}(\Omega)$. Then for all $0 < \alpha < 1 - N/p$ the embedding $W^{1,p}(\Omega) \hookrightarrow C^{0,\alpha}(\overline{\Omega})$ is compact.*

Proof. Exercise. □

Exercise 12.62. Let $1 \leq p < \infty$ be such that $p > N$ and let $\Omega \subseteq \mathbb{R}^N$ be an extension domain for $W^{1,p}(\Omega)$ with finite measure. Prove that the embedding $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ is compact.

Exercise 12.63. Let $m \in \mathbb{N}$, $m \geq 2$, and $1 \leq p < \infty$ be such that $mp > N$, and let $\Omega \subseteq \mathbb{R}^N$ be a bounded extension domain for $W^{m,p}(\Omega)$. Prove that for all $0 < \alpha < \theta$ the embedding $W^{m,p}(\Omega) \hookrightarrow C^{\ell,\alpha}(\overline{\Omega})$ is compact. Here, ℓ and θ are given in (12.34) and (12.35), respectively.

Exercise 12.64. Let $m \in \mathbb{N}$ and $1 \leq p < \infty$ be such that $mp > N$ or $m \geq N$ and $p = 1$ and let $\Omega \subseteq \mathbb{R}^N$ be an extension domain for $W^{m,p}(\Omega)$. Prove that if $u, v \in W^{m,p}(\Omega)$, then uv belongs to $W^{m,p}(\Omega)$ and

$$\|uv\|_{W^{m,p}(\Omega)} \leq c\|u\|_{W^{m,p}(\Omega)}\|v\|_{W^{m,p}(\Omega)},$$

where $c = c(m, N, p, \Omega) > 0$.

Exercise 12.65. Let $1 \leq p < \infty$ and let $u \in C^\infty(\mathbb{R}^N)$ be such that $u \in W^{m,p}(\mathbb{R}^N)$ for every $m \in \mathbb{N}$. Prove that $u(x) \rightarrow 0$ as $\|x\| \rightarrow \infty$.

Exercise 12.66. Let $u \in C(\mathbb{R}^N) \cap W^{1,2}(\mathbb{R}^N)$. Is it true that $u(x) \rightarrow 0$ as $\|x\| \rightarrow \infty$?

12.4. Superposition

In Theorem 3.55 we have seen that given a function $f : \mathbb{R} \rightarrow \mathbb{R}$, a necessary and sufficient condition for $f \circ u$ to be locally absolutely continuous for *all* $u \in AC_{\text{loc}}(I)$ is that f is locally Lipschitz continuous. In this section we establish similar results for $u \in W^{m,p}(\mathbb{R}^N)$.

The following proposition will be used to construct counterexamples in the proofs of the necessary conditions.

Proposition 12.67. *Let $m \in \mathbb{N}$, $1 \leq p < \infty$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(0) = 0$ and $f \circ u$ belongs to $W^{m,p}(\mathbb{R}^N)$ for every $u \in W^{m,p}(\mathbb{R}^N)$. Then for every cube $Q \subset \mathbb{R}^N$ there exist $\delta > 0$ and $M > 0$ such that if $u \in W^{m,p}(\mathbb{R}^N)$ has support in Q and $\|u\|_{W^{m,p}(\mathbb{R}^N)} \leq \delta$, then $\|f \circ u\|_{W^{m,p}(\mathbb{R}^N)} \leq M$.*

Proof. Assume for simplicity that $Q = Q(0, 1)$. Arguing by contradiction, if the statement does not hold, for every $n \in \mathbb{N}$ we can find $u_n \in W^{m,p}(\mathbb{R}^N)$, with $\text{supp } u_n \subset Q$, such that $\|u_n\|_{W^{m,p}} \leq 1/2^n$ and $\|f \circ u_n\|_{W^{m,p}} \geq n$. Write $\mathbb{Z}^N = \{y_n : n \in \mathbb{N}\}$. Then the function $v_n(x) := u_n(x + y_n)$, $x \in \mathbb{R}^N$, is supported in $Q(y_n, 1)$, and by a change of variables $\|v_n\|_{W^{m,p}} \leq 1/2^n$ and $\|f \circ v_n\|_{W^{m,p}} \geq n$. Define $u := \sum_n v_n$. Then $u \in W^{m,p}(\mathbb{R}^N)$ and so $\|f \circ u\|_{W^{m,p}} < \infty$. On the other hand, since the cubes $Q(y_n, 1)$ are pairwise

disjoint, by a change of variables,

$$\begin{aligned} \infty &> \sum_{k=0}^m \int_{\mathbb{R}^N} \|\nabla^k(f \circ u)\|^p dx = \sum_n \sum_{k=0}^m \int_{Q(y_n, 1)} \|\nabla^k(f \circ v_n)\|^p dx \\ &= \sum_n \sum_{k=0}^m \int_Q \|\nabla^k(f \circ u_n)\|^p dx \geq \sum_n n = \infty, \end{aligned}$$

which is a contradiction and completes the proof. $\square$

Exercise 12.68. Let $m \in \mathbb{N}$ and $N \geq 2$, let $0 < \varepsilon < 1/2$, and let $v_\varepsilon(x) := \log \|x\| / \log \varepsilon$, $x \in B(0, 1) \setminus B(0, \varepsilon)$. Let $g \in C^\infty([0, 1])$ be zero near $t = 0$ and one near $t = 1$, and let $u_\varepsilon(x) := g(v_\varepsilon(x))$ for $x \in B(0, 1) \setminus B(0, \varepsilon)$, $u := 1$ in $B(0, \varepsilon)$, and $u := 0$ outside $B(0, 1)$. Prove that $u_\varepsilon \in W^{m, N/m}(\mathbb{R}^N)$ with $\|u_\varepsilon\|_{W^{m, N/m}(\mathbb{R}^N)} \leq c |\log \varepsilon|^{m/N-1}$.

We begin with the case $m = 1$. The following theorem is due to Marcus and Mizel [161] and [162]. The proof is taken from a paper of Bourdaud [31].

Theorem 12.69. Let $1 \leq p < \infty$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$.

- (i) If $p > N$ or $p = N = 1$, then $f \circ u \in W^{1,p}(\mathbb{R}^N)$ for all functions $u \in W^{1,p}(\mathbb{R}^N)$ if and only if $f(0) = 0$ and f is locally Lipschitz continuous.
- (ii) If $p < N$ or $p = N > 1$, then $f \circ u \in W^{1,p}(\mathbb{R}^N)$ for all functions $u \in W^{1,p}(\mathbb{R}^N)$ if and only if $f(0) = 0$ and f is Lipschitz continuous.

Proof. We first prove the sufficient conditions in (i) and (ii).

Step 1: Assume that $p > N$ or $p = N = 1$ and that $f(0) = 0$ and f is locally Lipschitz continuous. Given $u \in W^{1,p}(\mathbb{R}^N)$, by Morrey's theorem (Theorem 12.48) if $p > N$ or by Theorem 7.16 and the fundamental theorem of calculus for functions in $AC_{\text{loc}}(\mathbb{R})$, if $p = N = 1$, we have that u is continuous and bounded.

Using the notation (E.2), given $u \in W^{1,p}(\mathbb{R}^N)$ and $i = 1, \dots, N$, by Theorem 11.45, u has a representative $\bar{u}$ such that $\bar{u}(x'_i, \cdot)$ is absolutely continuous in $\mathbb{R}$ for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in \mathbb{R}^{N-1}$ and whose first-order (classical) partial derivatives belong to $L^p(\mathbb{R}^N)$. Moreover, the (classical) partial derivatives of $\bar{u}$ agree $\mathcal{L}^N$ -a.e. with the weak derivatives of u . Let $x'_i \in \mathbb{R}^{N-1}$ be such that $\bar{u}(x'_i, \cdot)$ is absolutely continuous in $\mathbb{R}$ and $\bar{u}(x'_i, \cdot), \partial_i \bar{u}(x'_i, \cdot) \in L^p(\mathbb{R})$. Since f is locally Lipschitz continuous, by Theorem 3.55, setting $v := \bar{u}(x'_i, \cdot)$, we have that $f \circ v$ is locally absolutely continuous and

$$(12.36) \quad (f \circ v)'(x_i) = f'(v(x_i))v'(x_i) = f'(\bar{u}(x'_i, x_i))\partial_i \bar{u}(x'_i, x_i)$$

for $\mathcal{L}^1$ -a.e. $x_i \in \mathbb{R}$, where $f'(v(x_i))v'(x_i)$ is interpreted to be zero whenever $v'(x_i) = 0$ (even if f' is not differentiable at $v(x_i)$).

Since u is bounded, so is $\bar{u}$, and so, since f is locally Lipschitz continuous, by (12.36), $(f \circ v)' \in L^p(\mathbb{R})$. Moreover, since $f(0) = 0$,

$$|(f \circ v)(x_i)| = |f(v(x_i)) - f(0)| \leq L_u |v(x_i) - 0|,$$

where L_u is the Lipschitz constant of f in $[-\|u\|_{L^\infty}, \|u\|_{L^\infty}]$. Since $v \in L^p(\mathbb{R})$, it follows that $f \circ v$ belongs to $L^p(\mathbb{R})$. Using Theorem 11.45 once more, we conclude that $f \circ u$ belongs to $W^{1,p}(\mathbb{R}^N)$.

Step 2: Assume that $p < N$ or $p = N > 1$ and that $f(0) = 0$ and f is Lipschitz continuous. We can now proceed as in the previous step to show that $f \circ u \in W^{1,p}(\mathbb{R}^N)$ for every function $u \in W^{1,p}(\mathbb{R}^N)$.

Step 3: Assume that $p > N$ or $p = N = 1$ and that $f \circ u \in W^{1,p}(\mathbb{R}^N)$ for all functions $u \in W^{1,p}(\mathbb{R}^N)$. Since $0 \in W^{1,p}(\mathbb{R}^N)$, we have that $f(0) \in W^{1,p}(\mathbb{R}^N)$, which implies that $f(0) = 0$.

Given $v \in W^{1,p}((-1/4, 1/4))$, by Theorem 7.16 we can assume that v is absolutely continuous in $[-1/4, 1/4]$. Extend v to $[-1/2, 1/2]$ by setting $v(t) := v(-1/4)$ for $t \leq -1/4$ and $v(t) := v(1/4)$ for $t \geq 1/4$. Let $\varphi \in C_c^\infty(\mathbb{R}^N)$ be such that $\varphi(x) = 1$ for all $x \in Q(0, 1/2)$ and $\varphi = 0$ outside $Q(0, 1)$. Then the function $u(x) := \varphi(x)v(x_1)$, $x \in \mathbb{R}^N$, belongs to $W^{1,p}(\mathbb{R}^N)$, and so $f \circ u \in W^{1,p}(\mathbb{R}^N)$. Since for $x \in Q(0, 1/2)$, $f(u(x)) = f(v(x_1))$, by Theorem 11.45, we have that $f \circ v \in W^{1,p}((-1/4, 1/4))$. Since this holds for every $v \in W^{1,p}((-1/4, 1/4))$, it follows by Theorem 7.23 that f is locally Lipschitz continuous.

Step 4: Assume that $p < N$ and that $f \circ u \in W^{1,p}(\mathbb{R}^N)$ for all functions $u \in W^{1,p}(\mathbb{R}^N)$. As in the previous step we have that $f(0) = 0$.

Let $\varphi, \psi \in C_c^\infty(\mathbb{R}^N)$ be such that $\varphi(x) = 1$ and $\psi(x) = x_1$ for all $x \in Q(0, 1/2)$ and $\text{supp } \varphi, \text{supp } \psi \subset Q(0, 1)$. Given $0 < r, R < 1$, $t_0 \in \mathbb{R}$, $\lambda > 0$, and $\ell \in \mathbb{N}$, define

$$(12.37) \quad u(x) := \lambda \sum_k \psi(x/r - k) + t_0 \varphi(x/R), \quad x \in \mathbb{R}^N,$$

where the sum is done over all $k \in \mathbb{Z}^N$ such that $|k_i| \leq \ell$ for all $i = 1, \dots, N$. By the change of variables $y = x/R$ and the facts that $R < 1$ and $p < N$,

$$(12.38) \quad \|\varphi(\cdot/R)\|_{W^{1,p}} \leq R^{(N-p)/p} \|\varphi\|_{W^{1,p}}.$$

Similarly,

$$(12.39) \quad \left\| \sum_k \psi(\cdot/r - k) \right\|_{W^{1,p}} \leq c_1 \ell^{N/p} r^{(N-p)/p},$$

where c_1 depends on $\|\psi\|_{W^{1,p}}$. Let $M, \delta > 0$ be the numbers given in Proposition 12.67 corresponding to $Q(0, 1)$. Take R so small that

$$|t_0| R^{(N-p)/p} \|\varphi\|_{W^{1,p}} \leq \delta/2$$

and let $r = r(\ell) > 0$ be such that

$$(12.40) \quad \lambda c_1 \ell^{N/p} r^{(N-p)/p} = \delta/2.$$

Then $\|u\|_{W^{1,p}} \leq \delta$. Writing

$$r\ell = (\delta/(2c_1\lambda))^{p/(N-p)} \ell^{-p/(N-p)}$$

and using the fact that $p < N$, we can take ℓ so large that $8r\ell \leq R$. In turn, all the cubes $\overline{Q(rk, r)}$ are contained in $Q(0, R/2)$, and thus $\text{supp } u \subset Q(0, 1)$. It follows from Proposition 12.67 that $\|f \circ u\|_{W^{1,p}} \leq M$.

Note that the cubes $Q(rk, r)$ are pairwise disjoint. Indeed, if $j, k \in \mathbb{Z}^N$ are two distinct numbers such that $|j_i|, |k_i| \leq \ell$ for all $i = 1, \dots, N$, then there exists $l \in \{1, \dots, N\}$ such that $j_l \neq k_l$, say, $j_l < k_l$. If $x \in Q(rj, r)$, then $x_l < rj_l + r/2 = r(j_l + 1) - r/2 \leq rk_l - r/2$, which shows that x cannot belong to $Q(rk, r)$.

Since $\varphi(x) = 1$ and $\psi(x) = x_1$ for all $x \in Q(0, 1/2)$, it follows that on the cube $Q(rk, r/2)$ we have that $(f \circ u)(x) = f(\lambda x_1/r - \lambda k_1 + t_0)$. Since $f \circ u \in W^{1,p}(\mathbb{R}^N)$, for $\mathcal{L}^N$ -a.e. $x \in Q(rk, r/2)$ there exists

$$(\lambda/r)f'(\lambda x_1/r - \lambda k_1 + t_0) = \partial_1(f \circ u)(x).$$

In turn, by Fubini's theorem, the change of variables $t = \lambda x_1/r - \lambda k_1$, and (12.40),

$$\begin{aligned} M^p &\geq \int_{\mathbb{R}^N} |\partial_1(f \circ u)(x)|^p dx \\ &\geq \sum_k (\lambda/r)^p \int_{Q(rk, r/2)} |f'(\lambda x_1/r - \lambda k_1 + t_0)|^p dx \\ &\geq c\ell^N \lambda^{p-1} r^{N-p} \int_{-\lambda/4}^{\lambda/4} |f'(t + t_0)|^p dt = \frac{c\delta^p (2c_1)^{-p}}{\lambda} \int_{-\lambda/4}^{\lambda/4} |f'(t + t_0)|^p dt. \end{aligned}$$

Letting $\lambda \rightarrow 0^+$ we obtain $c_2 \geq |f'(t_0)|^p$ for all $t_0 \in \mathbb{R}$ which are Lebesgue points of f' . It follows that f is Lipschitz continuous.

Step 5: Assume that $p = N > 1$ and that $f \circ u \in W^{1,N}(\mathbb{R}^N)$ for all functions $u \in W^{1,N}(\mathbb{R}^N)$. In this case we need to modify the function in (12.37) since in (12.38) we cannot make $\|\varphi(\cdot/R)\|_{W^{1,N}}$ small. We consider instead the function

$$(12.41) \quad u(x) := \lambda \sum_k \psi(x/r - k) + t_0 u_\varepsilon(x), \quad x \in \mathbb{R}^N,$$

where u_ε is given in Exercise 12.68. Take $0 < \varepsilon < 1$ so small that

$$|t_0| \|u_\varepsilon\|_{W^{1,N}} \leq c|t_0| |\log \varepsilon|^{1/N-1} \leq \delta/2$$

and for $0 < \lambda < \delta(4c_1)^{-1}$, where c_1 is the constant in (12.39), define $\ell := \lfloor \delta(2c_1\lambda)^{-1} \rfloor > 1$. Then $\delta(4c_1)^{-1} \leq \lambda\ell \leq \delta(2c_1)^{-1}$. Finally, by taking r

so small that $r\sqrt{N}(\ell + 1/2) \leq \varepsilon$, we have that all the cubes $\overline{Q(rk, r)}$ are contained in $B(0, \varepsilon)$, and thus $\text{supp } u \subset Q(0, 1)$. It follows from Proposition 12.67 that $\|f \circ u\|_{W^{1,N}} \leq M$. Since $\psi(x) = x_1$ for all $x \in Q(0, 1/2)$ and $u_\varepsilon = 1$ in $B(0, \varepsilon)$, we can proceed as in the previous step to obtain

$$\begin{aligned} M^N &\geq \int_{\mathbb{R}^N} |\partial_1(f \circ u)(x)|^N dx \\ &\geq \sum_k (\lambda/r)^N \int_{Q(rk, r/2)} |f'(\lambda x_1/r - \lambda k_1 + t_0)|^N dx \\ &= c\ell^N \lambda^{N-1} \int_{-\lambda/4}^{\lambda/4} |f'(t + t_0)|^N dt \geq \frac{c_2}{\lambda} \int_{-\lambda/4}^{\lambda/4} |f'(t + t_0)|^N dt, \end{aligned}$$

where in the last inequality we used the fact that $\delta(4c_1)^{-1} \leq \lambda\ell$. Letting $\lambda \rightarrow 0^+$ we obtain $M^N/c_2 \geq |f'(t_0)|^N$ for all $t_0 \in \mathbb{R}$ which are Lebesgue points of f' . It follows that f is Lipschitz continuous. $\square$

Next we consider the case $m \geq 2$. We begin with the subcritical case, $mp < N$. The following theorem is due to Dahlberg [55]. The proof is taken from a paper of Bourdaud [31].

Theorem 12.70. *Let $m \in \mathbb{N}$ and $1 \leq p < \infty$ be such that $1 + 1/p < m < N/p$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$. Then $f \circ u \in W^{m,p}(\mathbb{R}^N)$ for all functions $u \in W^{m,p}(\mathbb{R}^N)$ if and only if f is a linear function.*

Proof. If $f(t) = ct$ for some $c \in \mathbb{R}$ and $t \in \mathbb{R}$, then $f \circ u \in W^{m,p}(\mathbb{R}^N)$ for all functions $u \in W^{m,p}(\mathbb{R}^N)$.

Conversely, assume that $f \circ u \in W^{m,p}(\mathbb{R}^N)$ for all functions $u \in W^{m,p}(\mathbb{R}^N)$. We proceed as in Step 4 of the proof of Theorem 12.69. Let $\psi \in C_c^\infty(\mathbb{R}^N)$ be such that $\psi(x) = x_1$ for all $x \in Q(0, 1/2)$ and $\text{supp } \psi \subset Q(0, 1)$. Given $0 < r < 1$, $\lambda > 0$, and $\ell \in \mathbb{N}$, define

$$u(x) := \lambda \sum_k \psi(x/r - k), \quad x \in \mathbb{R}^N,$$

where the sum is done over all $k \in \mathbb{Z}^N$ such that $|k_i| \leq \ell$ for all $i = 1, \dots, N$. By the change of variables $y = x/r - k$ and the facts that $r < 1$ and $m < N/p$,

$$\left\| \sum_k \psi(\cdot/r - k) \right\|_{W^{m,p}} \leq c_1 \ell^{N/p} r^{(N-mp)/p},$$

where c_1 depends on $\|\psi\|_{W^{m,p}}$. Let $M, \delta > 0$ be the numbers given in Proposition 12.67 corresponding to $Q(0, 1)$. Take $r = r(\ell) > 0$ such that

$$(12.42) \quad \lambda c_1 \ell^{N/p} r^{(N-mp)/p} = \delta.$$

Then $\|u\|_{W^{m,p}} \leq \delta$. Writing

$$r\ell = (\delta/(c_1\lambda))^{p/(N-mp)} \ell^{-mp/(N-mp)}$$

and using the fact that $pm < N$, we can take ℓ so large that $8r\ell \leq 1$. In turn, all the cubes $\overline{Q(rk, r)}$ are contained in $Q(0, 1)$, and thus $\text{supp } u \subset Q(0, 1)$. It follows from Proposition 12.67 that $\|f \circ u\|_{W^{1,p}} \leq M$. As in Step 4 of the proof of Theorem 12.69, we have that the cubes $Q(rk, r)$ are pairwise disjoint. Since $\psi(x) = x_1$ for all $x \in Q(0, 1/2)$, it follows that on the cube $Q(rk, r/2)$ we have that $(f \circ u)(x) = f(\lambda x_1/r - \lambda k_1)$. Since $f \circ u \in W^{m,p}(\mathbb{R}^N)$, for $\mathcal{L}^N$ -a.e. $x \in Q(rk, r/2)$ there exists

$$(\lambda/r)^m f^{(m)}(\lambda x_1/r - \lambda k_1) = \partial_1^m (f \circ u)(x).$$

In turn, by Fubini's theorem, the change of variables $t = \lambda x_1/r - \lambda k_1$, and (12.42),

$$\begin{aligned} M^p &\geq \int_{\mathbb{R}^N} |\partial_1^m (f \circ u)(x)|^p dx \\ &\geq \sum_k (\lambda/r)^{mp} \int_{Q(rk, r/2)} |f^{(m)}(\lambda x_1/r - \lambda k_1)|^p dx \\ &\geq c\ell^N \lambda^{mp-1} r^{N-mp} \int_{-\lambda/4}^{\lambda/4} |f^{(m)}(t)|^p dt \\ &= c\delta^p c_1^{-p} \lambda^{mp-p-1} \int_{-\lambda/4}^{\lambda/4} |f^{(m)}(t)|^p dt. \end{aligned}$$

Hence,

$$(12.43) \quad \frac{c_2}{\lambda^{mp-p-1}} \geq \int_{-\lambda/4}^{\lambda/4} |f^{(m)}(t)|^p dt,$$

where $c_2 > 0$ is independent of λ . Letting $\lambda \rightarrow \infty$ and using the fact that $1 + 1/p < m$ we obtain that $\int_{\mathbb{R}} |f^{(m)}(t)|^p dt = 0$. It follows that f is a polynomial of degree less than m , say $f(t) = c_1 t + \cdots + c_k t^k$, where $c_k \neq 0$. Assume by contradiction that $k > 1$. Since $m < N/p$ we can find $\theta > 0$ such that $\theta + m < N/p < 2\theta + m$. Let $\psi \in C_c^\infty(\mathbb{R}^N)$ be such that $\psi(x) = 1$ for all $x \in B(0, 1)$ and define $v(x) := \|x\|^{-\theta} \psi(x)$. Since $\theta + m < N/p$, it follows that $v \in W^{m,p}(\mathbb{R}^N)$. On the other hand, for $x \in B(0, 1)$,

$$f(v(x)) = c_1 \|x\|^{-\theta} + \cdots + c_k c_1 \|x\|^{-k\theta},$$

and since $N/p < 2\theta + m$, passing to spherical coordinates, it can be checked that $f \circ v$ does not belong to $W^{m,p}(\mathbb{R}^N)$. The details are left as an exercise. Thus, $k = 1$, which completes the proof. $\square$

Exercise 12.71. Let $m \in \mathbb{N}$ and $1 \leq p < \infty$ be such that $1 + 1/p < m < N/p$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $f \circ u \in W^{m,p}(\mathbb{R}^N)$ for all functions $u \in W^{m,p}(\mathbb{R}^N)$.

- (i) Construct two sequences $\{a_n\}_n$ and $\{\varepsilon_n\}_n$ of positive numbers such that $\varepsilon_n < 1 < a_n$ for every n , $\varepsilon_n \rightarrow 0^+$ and $a_n \rightarrow \infty$ as $n \rightarrow \infty$, and

$$\sum_{n=1}^{\infty} a_n^p \varepsilon_n^{N-mp} < \infty, \quad \sum_{n=1}^{\infty} a_n^{mp-1} \varepsilon_n^{N-mp} = \infty.$$

- (ii) Let $\{y_n\}_n$ be a sequence of points in $\mathbb{R}^N$ such that $\|y_n - y_\ell\| \geq 10$ for $n \neq \ell$ and let $\psi \in C_c^\infty(\mathbb{R}^N)$ be such that $\psi(x) = x_1$ for all $x \in B(0, 1)$ and $\psi = 0$ outside $B(0, 2)$. Let

$$u(x) := \sum_{n=1}^{\infty} a_n \psi((x - y_n)/\varepsilon_n), \quad x \in \mathbb{R}^N.$$

Prove that $u \in W^{m,p}(\mathbb{R}^N)$.

- (iii) Assume by contradiction that $\text{ess inf}_{a < t < b} |f^{(m)}(t)| > 0$ for some $a < b$. Write $y_n = (y_{n,1}, \dots, y_{n,N})$ and consider the set

$$E_n := \{x \in \mathbb{R}^N : \|x - y_n\| < \varepsilon_n, a\varepsilon_n < a_n(x_1 - y_{n,1}) < b\varepsilon_n\}.$$

Prove that there exists a constant $c > 0$ such that $\mathcal{L}^N(E_n) \geq c\varepsilon_n^N/a_n$ for all n sufficiently large and deduce that $f \circ u$ cannot belong to $W^{m,p}(\mathbb{R}^N)$.

Next we consider the supercritical case $mp > N$. The remaining theorems of this section are due to Bourdaud [31].

Theorem 12.72. *Let $m \in \mathbb{N}$, $m \geq 2$, and $1 \leq p < \infty$ be such either $mp > N$ or $m = N$ and $p = 1$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$. Then $f \circ u \in W^{m,p}(\mathbb{R}^N)$ for all functions $u \in W^{m,p}(\mathbb{R}^N)$ if and only if $f(0) = 0$ and $f \in W_{\text{loc}}^{m,p}(\mathbb{R})$.*

We begin with some auxiliary results.

Lemma 12.73. *Let $1 \leq p < \infty$, let $f \in W_{\text{loc}}^{2,p}(\mathbb{R})$ be such that $f(0) = 0$. Then*

$$\int_{\mathbb{R}} |f'' \circ u|^p |u'|^{2p} dx \leq c \|f''\|_{L^p([- \ell_u, \ell_u])}^p \|u\|_{L^\infty(\mathbb{R})}^{p-1} \|u''\|_{L^p(\mathbb{R})}^p$$

for every $u \in C_c^\infty(\mathbb{R})$, where $\ell_u := \|u\|_{L^\infty(\mathbb{R})}$.

Proof. Let

$$M_u := \int_{-\ell_u}^{\ell_u} |f''(s)|^p ds, \quad g(t) := \int_t^{\ell_u} |f''(s)|^p ds.$$

Integrating by parts and using Hölder's inequality if $p > 1$, we have

$$\begin{aligned} \int_{\mathbb{R}} |f'' \circ u|^p |u'|^{2p} dx &= (2p-1) \int_{\mathbb{R}} (g \circ u) |u'|^{2p-2} u'' dx \\ &\leq (2p-1) M_u \|u'\|_{L^{2p}(\mathbb{R})}^{2(p-1)} \|u''\|_{L^p(\mathbb{R})}. \end{aligned}$$

By Theorem 7.41 with $k = 1$, $m = 2$, $r = 2p$, $q = \infty$, we have

$$\|u'\|_{L^{2p}(\mathbb{R})} \leq c\|u\|_{L^\infty(\mathbb{R})}^{1/2}\|u''\|_{L^p(\mathbb{R})}^{1/2}.$$

The result follows by combining these inequalities. $\square$

We now turn to the proof of Theorem 12.72.

Proof of Theorem 12.72. Step 1: We only prove the “only if” part of the statement for $m = 2$. Assume that $m = 2$ and that either $2p > N$ or $N = 2$ and $p = 1$. Let $f \in W_{\text{loc}}^{2,p}(\mathbb{R})$ be such that $f(0) = 0$. Given $u \in C_c^\infty(\mathbb{R}^N)$, by Exercise 12.43 or Theorem 12.46, $\ell_u := \|u\|_{L^\infty} \leq c\|u\|_{W^{2,p}}$. Define $L_u := \|f'\|_{L^\infty([-\ell_u, \ell_u])}$ and $M_u := \|f''\|_{L^p([-\ell_u, \ell_u])}^p$. By Theorem 3.59,

$$\partial_i(f \circ u) = (f' \circ u)\partial_i u,$$

$$\partial_j \partial_i(f \circ u) = (f'' \circ u)\partial_i u \partial_j u + (f' \circ u)\partial_j \partial_i u,$$

$\mathcal{L}^N$ -a.e. in $\mathbb{R}^N$, where $f''(u(x))\partial_j u(x)$ is interpreted to be zero whenever $\partial_j u(x) = 0$ (even if f' is not differentiable at $u(x)$). Hence, since $f(0) = 0$,

$$|f \circ u|^p = |f \circ u - f(0)|^p \leq L_u^p |u|^p, \quad |\partial_i(f \circ u)|^p \leq L_u^p |\partial_i u|^p,$$

$$|\partial_j \partial_i(f \circ u)|^p \leq c|(f'' \circ u)|^p |\partial_i u|^{2p} + c|(f'' \circ u)|^p |\partial_j u|^{2p} + cL_u^p |\partial_j \partial_i u|^p,$$

where $c = c(p) > 0$. Using the notation (E.2) in Appendix E and the previous lemma,

$$\int_{\mathbb{R}} |f''(u(x'_i, x_i))|^p |\partial_i u(x'_i, x_i)|^{2p} dx_i \leq cM_u^p \|u(x'_i, \cdot)\|_{L^\infty(\mathbb{R})}^{p-1} \int_{\mathbb{R}} |\partial_i^2 u(x'_i, x_i)|^p dx_i.$$

Integrating in x'_i over $\mathbb{R}^{N-1}$ and using Tonelli's theorem gives

$$\int_{\mathbb{R}^N} |f'' \circ u|^p |\partial_i u|^{2p} dx \leq cM_u^p \|u\|_{L^\infty}^{p-1} \int_{\mathbb{R}^N} |\partial_i^2 u|^p dx.$$

Note that when $p = 1$ the previous inequality reduces to

$$(12.44) \quad \int_{\mathbb{R}^N} |f'' \circ u| |\partial_i u|^2 dx \leq cM_u \int_{\mathbb{R}^N} |\partial_i^2 u| dx.$$

This implies that $f \circ u$ belongs to $W^{2,p}(\mathbb{R}^N)$.

Step 2: Assume that $f \circ u \in W^{m,p}(\mathbb{R}^N)$ for all functions $u \in W^{m,p}(\mathbb{R}^N)$. Since $0 \in W^{m,p}(\mathbb{R}^N)$, we have that $f(0) \in W^{m,p}(\mathbb{R}^N)$, which implies that $f(0) = 0$. Next fix $a < b$ and consider a function $u \in C_c^\infty(\mathbb{R}^N)$ such that $u(x) = x_1$ for all $x \in \mathbb{R}^N$ with $x_1 \in [a, b]$ and $|x_i| \leq 1$ for $i = 2, \dots, N$. Then $(f \circ u)(t, x_2, \dots, x_N) = f(t)$ for all $t \in [a, b]$ and $|x_i| \leq 1$ for $i = 2, \dots, N$. Since $f \circ u \in W^{m,p}(\mathbb{R}^N)$, by Theorem 11.45 and an induction argument, it follows that $f \in W^m((a, b))$ with

$$f^{(n)}(t) = \partial_1^n (f \circ u)(t, x_2, \dots, x_N)$$

for $\mathcal{L}^1$ -a.e. $t \in [a, b]$ and for $\mathcal{L}^{N-1}$ -a.e. $(x_2, \dots, x_N) \in \mathbb{R}^{N-1}$ with $|x_i| \leq 1$ for $i = 2, \dots, N$. $\square$

Exercise 12.74. Prove that in Step 1 of the previous proof the additional hypothesis that $u \in C_c^\infty(\mathbb{R}^N)$ can be removed.

Example 12.75. Prove that given m positive numbers $a_1, \dots, a_m$ and m exponents $p_1, \dots, p_m$, with $\sum_{i=1}^m 1/p_i = 1$, we have $\prod_{i=1}^m a_i \leq \sum_{i=1}^m a_i^{p_i}/p_i$.

Exercise 12.76. Use Faa di Bruno's theorem (see Theorem 11.54) and Exercise 12.75 to prove Step 1 of the previous proof for $m \geq 2$.

Theorem 12.77. Let $N > 2$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$. Then $f \circ u \in W^{2,1}(\mathbb{R}^N)$ for all functions $u \in W^{2,1}(\mathbb{R}^N)$ if and only if $f(0) = 0$ and $f \in \dot{W}^{2,1}(\mathbb{R})$.

Proof. Step 1: Assume that $f(0) = 0$ and $f \in \dot{W}^{2,1}(\mathbb{R})$. Since $f'' \in L^1(\mathbb{R})$, by the fundamental theorem of calculus, $f' \in L^\infty(\mathbb{R})$. Given $u \in C_c^\infty(\mathbb{R}^N)$, by (12.44),

$$\int_{\mathbb{R}^N} |f'' \circ u| |\partial_i u|^2 dx \leq c \int_{\mathbb{R}} |f''(s)| ds \int_{\mathbb{R}^N} |\partial_i^2 u|^2 dx.$$

Thus we may continue as in Step 1 of Theorem 12.72 to conclude that $f \circ u$ belongs to $W^{2,1}(\mathbb{R}^N)$.

Step 2: Assume that $f \circ u \in W^{2,1}(\mathbb{R}^N)$ for all functions $u \in W^{2,1}(\mathbb{R}^N)$. Since $0 \in W^{2,1}(\mathbb{R}^N)$, as usual we deduce that $f(0) = 0$. We proceed as in the proof of Theorem 12.70 until (12.43), which now becomes $c_2 \geq \int_{-\lambda/4}^{\lambda/4} |f''(t)| dt$. Letting $\lambda \rightarrow \infty$ shows that $f'' \in L^1(\mathbb{R})$. $\square$

Finally, we study the critical case $mp = N$.

Theorem 12.78. Let $m \in \mathbb{N}$ be such that $N > m \geq 2$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$. Then $f \circ u \in W^{m, N/m}(\mathbb{R}^N)$ for all functions $u \in W^{m, m/N}(\mathbb{R}^N)$ if and only if $f(0) = 0$, $f \in W_{\text{loc}}^{m, N/m}(\mathbb{R})$ with $f' \in L^\infty(\mathbb{R})$ and

$$(12.45) \quad \sup_{t \in \mathbb{R}} \int_{t-1}^{t+1} |f^{(m)}(s)|^{N/m} ds < \infty.$$

We begin with an auxiliary result.

Lemma 12.79. Let $m \in \mathbb{N}$, $m \geq 2$, and $1 \leq p < \infty$, let $g \in W_{\text{loc}}^{m,p}(\mathbb{R})$ be such that $g \in L^\infty(\mathbb{R})$ and

$$(12.46) \quad M := \sup_{t \in \mathbb{R}} \int_{t-1}^{t+1} |g^{(m)}(s)| ds < \infty.$$

Then $\|g^{(n)}\|_{L^\infty(\mathbb{R})} \leq c(2\|g\|_{L^\infty(\mathbb{R})} + M)$ for all $1 \leq n \leq m-1$ and for some constant $c > 0$ depending only on m .

Proof. Fix $t_0 \in \mathbb{R}$. By Taylor's formula with integral remainder for $t \in [0, 1]$,

$$g(t + t_0) = g(t_0) + \sum_{n=1}^{m-1} \frac{1}{n!} g^{(n)}(t_0) t^n + \frac{t^m}{(m-1)!} \int_0^1 (1-s)^{m-1} g^{(m)}(t_0 + st) ds.$$

Hence, by the change of variables $y = t_0 + st$ and (12.46),

$$\begin{aligned} \left| \sum_{n=1}^{m-1} \frac{1}{n!} g^{(n)}(t_0) t^n \right| &\leq |g(t + t_0)| + |g(t_0)| + \frac{t^m}{(m-1)!} \int_0^1 |g^{(m)}(t_0 + st)| ds \\ &\leq 2\|g\|_{L^\infty(\mathbb{R})} + \frac{t^{m-1}M}{(m-1)!}. \end{aligned}$$

It follows from Exercise 11.21 that there exists a constant $c_m > 0$ depending only on m such that

$$|g^{(n)}(t_0)| \leq c_m n! (2\|g\|_{L^\infty(\mathbb{R})} + M)$$

for all $n = 1, \dots, m-1$ and for all $t_0 \in \mathbb{R}$. □

We are now ready to prove Theorem 12.78.

Proof of Theorem 12.78. Step 1: We only prove the “only if” part of the statement for $m = 2$. Let $m = 2$, let $f \in W_{\text{loc}}^{2,N/2}(\mathbb{R})$ be such that $f(0) = 0$, $f' \in L^\infty(\mathbb{R})$, and (12.45) holds, and let $u \in C_c^\infty(\mathbb{R}^N)$. We follow the proof of Step 1 of Theorem 12.72. The only change is in the estimate for the term $|f'' \circ u|^{N/2} |\partial_i u|^N$. Let $\psi \in C^\infty([-1, 0])$ be a nonnegative function such that $\psi = 0$ in $[-1, -3/4]$, $\psi = 1$ in $[-1/4, 0]$, and let $\psi(t) := 1 - \psi(t-1)$ for $t \in [0, 1]$. Extend ψ to be 0 outside $[-1, 1]$. Then the function $\varphi := \sqrt{\psi}$ belongs to $C_c^\infty(\mathbb{R})$, $\text{supp } \varphi \subseteq [-1, 1]$, and $\varphi^2(t) + \varphi^2(t-1) = 1$ for all $t \in [0, 1]$. In turn,

$$(12.47) \quad \sum_{k \in \mathbb{Z}} \varphi^2(t - k) = 1$$

for every $t \in \mathbb{R}$. Indeed, given $t \in \mathbb{R}$, $\varphi^2(t - k) = 0$ for all $k \neq \lfloor t \rfloor - 1, \lfloor t \rfloor$, and so the series reduces to $\varphi^2(t - 1 - \lfloor t \rfloor) + \varphi^2(t - \lfloor t \rfloor) = 1$. Similarly, we have that

$$(12.48) \quad \sum_{k \in \mathbb{Z}} \varphi(t - k) \leq c, \quad \sum_{k \in \mathbb{Z}} |\varphi'(t - k)| \leq c$$

for every $t \in \mathbb{R}$ and for some constant $c > 0$.

Using (12.47) by Tonelli's theorem we can write

$$\int_{\mathbb{R}^N} |f''(u(x))|^{N/2} |\partial_i u(x)|^N dx = \sum_{k \in \mathbb{Z}} \int_{\mathbb{R}^N} h_k(u(x)) \varphi(u(x) - k) |\partial_i u(x)|^N dx,$$

where $h_k(t) := |f''(t)|^{N/2} \varphi(t-k)$, $t \in \mathbb{R}$. Define $g_k(t) := \int_t^\infty h_k(s) ds$, $t \in \mathbb{R}$. Integrating by parts we have

$$\begin{aligned} & \sum_{k \in \mathbb{Z}} \int_{\mathbb{R}^N} (h_k \circ u) \varphi \circ (u-k) |\partial_i u|^N dx \\ &= \sum_{k \in \mathbb{Z}} \int_{\mathbb{R}^N} (g_k \circ u) \partial_i (\varphi \circ (u-k)) |\partial_i u|^{N-2} \partial_i u dx \\ &= \sum_{k \in \mathbb{Z}} \int_{\mathbb{R}^N} (g_k \circ u) \varphi' \circ (u-k) |\partial_i u|^N dx \\ &+ (N-1) \sum_{k \in \mathbb{Z}} \int_{\mathbb{R}^N} (g_k \circ u) \varphi \circ (u-k) |\partial_i u|^{N-2} \partial_i^2 u dx =: I + II. \end{aligned}$$

By (12.45) and the fact that $\text{supp } \varphi \subseteq [-1, 1]$, we have

$$0 \leq g_k(t) \leq \|h_k\|_{L^1(\mathbb{R})} \leq c$$

for all $t \in \mathbb{R}$ and $k \in \mathbb{Z}$ and for some constant $c > 0$. In turn, using (12.48) we obtain

$$\begin{aligned} (12.49) \quad |I| &\leq c \int_{\mathbb{R}^N} |\partial_i u|^N dx, \\ |II| &\leq c \int_{\mathbb{R}^N} |\partial_i u|^{N-2} |\partial_i^2 u| dx \leq c \|\partial_i u\|_{L^N}^{2(N/2-1)} \|\partial_i^2 u\|_{L^{N/2}}, \end{aligned}$$

where we used Hölder's inequality if $N > 2$. Since $(N/2)^* = N$, it follows by the Sobolev–Gagliardo–Nirenberg embedding theorem (Theorem 12.4) applied to $\partial_i u$ that $\|\partial_i u\|_{L^N} < \infty$.

Step 2: Let $m \geq 2$ and let $f \in W_{\text{loc}}^{m, N/m}(\mathbb{R})$ be such that $f(0) = 0$, $f' \in L^\infty(\mathbb{R})$, and (12.45) holds and let $u \in C_c^\infty(\mathbb{R}^N)$. By Faa di Bruno's theorem (see Theorem 11.54) for every multi-index $\alpha \in \mathbb{N}_0^N$, with $1 \leq |\alpha| \leq m$,

$$\frac{\partial^{|\alpha|}}{\partial x^\alpha} (f \circ u) = \sum c_{\alpha, n, \gamma} (f^{(n)} \circ u) \prod_{i=1}^n \frac{\partial^{\gamma_i} u}{\partial x^{\gamma_i}},$$

where $c_{\alpha, n, \gamma, l} \in \mathbb{R}$, the sum is done over all $n \in \mathbb{N}$ with $1 \leq n \leq |\alpha|$, $\gamma = (\gamma_1, \dots, \gamma_n)$, $\gamma_i \in \mathbb{N}$, with $\sum_{i=1}^{|\beta|} \gamma_i = \alpha$, $i = 1, \dots, n$. Using exercise 12.75 with $p_i = |\alpha|/|\gamma_i|$ we get

$$|\partial^\alpha (f \circ u)(x)|^{N/m} \leq \sum d_{\alpha, n, \gamma, m} |f^{(n)} \circ u|^{N/m} \sum_{i=1}^n |\partial^{\gamma_i} u|^{N|\alpha|/(|\gamma_i|m)}.$$

It follows from Lemma 12.79 applied to $g = f' \in L^\infty(\mathbb{R})$ and with m replaced by $m-1$, that $f^{(n)} \in L^\infty(\mathbb{R})$ for every $n = 2, \dots, m-1$, and so if $1 \leq n \leq |\alpha| \leq m-1$, we need to estimate terms of the form

$$(12.50) \quad \|f^{(n)}\|_{L^\infty}^{N/m} |\partial^{\gamma_i} u|^{N|\alpha|/(|\gamma_i|m)},$$

while if $|\alpha| = m$, then $|\gamma_i| = 1$ for all $i = 1, \dots, m$, and so we have terms of the form $|f^{(m)} \circ u|^{N/m} |\partial_i u|^N$. For the latter, we proceed as in Step 1 with the only change that (12.49) now becomes

$$(12.51) \quad |I| \leq c \int_{\mathbb{R}^N} |\partial_i u|^N dx, \quad |II| \leq c \int_{\mathbb{R}^N} |\partial_i u|^{N-2} |\partial_i^2 u| dx.$$

By the Sobolev–Gagliardo–Nirenberg embedding theorem (Theorem 12.8), for every multi-index β with $|\beta| = 2$, $\partial^\beta u \in L^{N/2}(\mathbb{R}^N)$ and $\partial_i u \in L^N(\mathbb{R}^N)$. Since $(N-2)(N/2)' = N$, it follows that the right-hand sides of both inequalities in (12.51) are finite.

To estimate (12.50), again by Sobolev–Gagliardo–Nirenberg embedding theorem (Theorem 12.8), $\partial^{\gamma_i} u \in L^{N/|\gamma_i|}(\mathbb{R}^N)$, and since $N/m \leq N|\alpha|/(|\gamma_i|m) < N/|\gamma_i|$, by Exercise B.79(ii) we have that $\partial^{\gamma_i} u \in L^{N|\alpha|/(|\gamma_i|m)}(\mathbb{R}^N)$.

Step 3: The proof of the necessity part of the theorem is similar to the previous ones, using functions of the type (12.37) and is given in the following exercises. $\square$

Exercise 12.80. Let $m \in \mathbb{N}$ be such that $N > m \geq 2$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $f \circ u \in W^{m, N/m}(\mathbb{R}^N)$ for all functions $u \in W^{m, m/N}(\mathbb{R}^N)$. Let $\psi \in C_c^\infty(\mathbb{R}^N)$ be such that $\psi(x) = x_1$ for all $x \in Q(0, 1/2)$ and $\text{supp } \psi \subset Q(0, 1)$. Given $0 < r < 1$, $t_0 \in \mathbb{R}$, $\lambda > 0$, and $0 < \varepsilon < 1/2$, define

$$(12.52) \quad u(x) := \lambda \psi(x/r) + t_0 u_\varepsilon(x), \quad x \in \mathbb{R}^N,$$

where u_ε is the function given in Exercise 12.68. Reasoning as in the proofs of Theorems 12.69 and 12.70 prove that

$$M^{N/m} \geq c \lambda^{N-1} \int_{-\lambda/4}^{\lambda/4} |f^{(m)}(t + t_0)|^{N/m} dt$$

and deduce that (12.45) holds.

Exercise 12.81. Let $m \in \mathbb{N}$ be such that $N > m \geq 2$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $f \circ u \in W^{m, N/m}(\mathbb{R}^N)$ for all functions $u \in W^{m, m/N}(\mathbb{R}^N)$. Prove inductively that

$$(12.53) \quad \sup_{t \in \mathbb{R}} \int_{t-1}^{t+1} |f^{(n)}(s)|^{N/m} ds < \infty$$

for all $n = 2, \dots, m-1$ as follows:

- (i) Assume that (12.45) holds for all $k = n+1, \dots, m$ and take u as in (12.52), where now $\psi(x) = x_1 x_2^\ell$ for all $x \in Q(0, 1/2)$, where $\ell \in \mathbb{N}$ is to be determined. Let $v(x) := f(\lambda x_1 x_2^\ell + a)$ and prove that for

$$\ell \geq m/n - 1,$$

$$\begin{aligned} \partial^{ne_1 + (m-n)e_2} v(x) &= c_n \lambda^n f^{(n)}(\lambda x_1 x_2^\ell + a) x_2^{\ell n + n - m} \\ &\quad + \sum_{k=n+1}^m \lambda^k f^{(k)}(\lambda x_1 x_2^\ell + a) P_k(x_1, x_2), \end{aligned}$$

where $c_n \neq 0$ and P_k are polynomials in x_1 and x_2 .

(ii) Let $K = [-1/4, 1/4] \times [1/8, 1/4]$. Prove that for $k \in \{n+1, \dots, m\}$,

$$\int_K |\lambda^k f^{(k)}(\lambda x_1 x_2^\ell + a) P_k(x_1, x_2)|^{N/m} dx_1 dx_2 \leq c \sup_{t \in \mathbb{R}} \int_{t-1}^{t+1} |f^{(k)}(s)|^{N/m} ds.$$

Exercise 12.82. Let $m \in \mathbb{N}$ be such that $N > m \geq 2$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $f \circ u \in W^{m, N/m}(\mathbb{R}^N)$ for all functions $u \in W^{m, m/N}(\mathbb{R}^N)$. Let u be as in (12.41), where now $\psi(x) = x_1^m$ for all $x \in Q(0, 1/2)$ and prove that f is Lipschitz continuous. Hint: Compute $\partial^{me_1} v$, where $v(x) := f(\lambda x_1^m + a)$, and use the previous two exercises.

12.5. Interpolation Inequalities in $\mathbb{R}^N$

This section is devoted to interpolation inequalities. These are particularly useful in partial differential equations or in calculus of variations in unbounded domains (see, e.g., [33], [71], [94], [142]). In those contexts one gains control of the L^p norm of the highest derivatives $\nabla^m u$ of a function u (e.g., the solution of a PDE or a critical point of an energy functional) and has some additional information on u (e.g., boundary values, constraints) that allows them to conclude that u belongs to L^q . Then interpolation inequalities give bounds on the intermediate derivatives $\nabla^k u$, $1 \leq k \leq m-1$ and additional information on the function u itself.

Thus the problem we consider in this section is the following: Assume that $u \in L^q(\mathbb{R}^N)$ and all its distributional derivatives of order m belong to $L^p(\mathbb{R}^N)$, for some $1 \leq p, q \leq \infty$. What can we say about the intermediate derivatives $\nabla^k u$, $1 \leq k \leq m-1$?

To give a unified treatment, only in this section, we use a different notation for L^r norms. To be precise, given $r \in [-\infty, \infty]$, $r \neq 0$, and a function $u : \mathbb{R}^N \rightarrow \mathbb{R}$, we define

$$(12.54) \quad |u|_r := \begin{cases} \|u\|_{L^r(\mathbb{R}^N)} & \text{if } r > 0, \\ \|\nabla^n u\|_{L^\infty(\mathbb{R}^N)} & \text{if } r < 0 \text{ and } a = 0, \\ |\nabla^n u|_{C^{0,a}(\mathbb{R}^N)} & \text{if } r < 0 \text{ and } 0 < a < 1, \end{cases}$$

where if $r < 0$ we set $n := \lfloor -N/r \rfloor$ and $a := -n - N/r \in [0, 1)$, provided the right-hand sides are well-defined.

We begin with the case $m = 1$.

Theorem 12.83 (Gagliardo–Nirenberg interpolation, $m = 1$). *Let $1 \leq p, q \leq \infty$, let $0 \leq \theta \leq 1$, and let r be such that*

$$(12.55) \quad (1 - \theta) \left(\frac{1}{p} - \frac{1}{N} \right) + \frac{\theta}{q} = \frac{1}{r} \in (-\infty, 1].$$

Then there exists a constant $c = c(N, p, q, \theta) > 0$ such that

$$(12.56) \quad |u|_r \leq c \|u\|_{L^q(\mathbb{R}^N)}^\theta \|\nabla u\|_{L^p(\mathbb{R}^N)}^{1-\theta}$$

for every $u \in L^q(\mathbb{R}^N) \cap \dot{W}^{1,p}(\mathbb{R}^N)$, with the exceptions that if $p < N$ and $q = \infty$ we assume that u vanishes at infinity, while if $p = N > 1$ we take $0 < \theta \leq 1$.

The proof relies on the following interpolation result.

Proposition 12.84. *Let $u : \mathbb{R}^N \rightarrow \mathbb{R}$ be Hölder continuous with exponent $0 < \alpha < 1$ and such that $u \in L^q(\mathbb{R}^N)$ for some $1 \leq q \leq \infty$. Then u belongs to $L^r(\mathbb{R}^N)$ for all $q \leq r \leq \infty$ and to $C^{0,\beta}(\mathbb{R}^N)$ for all $0 < \beta \leq \alpha$, with*

$$(12.57) \quad \begin{aligned} \|u\|_{L^r(\mathbb{R}^N)} &\leq c \|u\|_{L^q(\mathbb{R}^N)}^{\theta_1} |u|_{C^{0,\alpha}(\mathbb{R}^N)}^{1-\theta_1}, \\ |u|_{C^{0,\beta}(\mathbb{R}^N)} &\leq c \|u\|_{L^q(\mathbb{R}^N)}^{\theta_2} |u|_{C^{0,\alpha}(\mathbb{R}^N)}^{1-\theta_2}, \end{aligned}$$

for some constant $c = c(\alpha, N, q) > 0$, where $\theta_1 := \frac{\alpha+N/r}{\alpha+N/q}$ and $\theta_2 := \frac{\alpha-\beta}{\alpha+N/q}$.

Proof. We begin by showing that u is bounded. If $q = \infty$, there is nothing to prove, thus assume that $q < \infty$. If $|u|_{C^{0,\alpha}} = 0$, then u is a constant and so it must be zero since $\|u\|_{L^q} < \infty$. Thus assume $|u|_{C^{0,\alpha}} > 0$. Given $R > 0$ and $x \in \mathbb{R}^N$, let $y \in B(x, R) \setminus \{x\}$. Then

$$|u(x)| \leq |u(y)| + |u(x) - u(y)| \leq |u(y)| + R^\alpha |u|_{C^{0,\alpha}}.$$

Averaging both sides in y over $B(x, R)$ (and using Hölder's inequality if $q > 1$) we get

$$\begin{aligned} |u(x)| &\leq \frac{1}{\alpha_N R^N} \int_{B(x,R)} |u(y)| dy + R^\alpha |u|_{C^{0,\alpha}} \\ &\leq (\alpha_N R^N)^{-1/q} \|u\|_{L^q} + R^\alpha |u|_{C^{0,\alpha}}. \end{aligned}$$

Minimizing the function $g(t) := \alpha_N^{-1/q} t^{-N/q} \|u\|_{L^q} + t^\alpha |u|_{C^{0,\alpha}}$ over all $t > 0$ gives

$$(12.58) \quad \|u\|_{L^\infty} \leq c \|u\|_{L^q}^{\theta_1} |u|_{C^{0,\alpha}}^{1-\theta_1}, \quad \theta_1 := \alpha/(\alpha + N/q).$$

In turn, by Exercise B.79(ii), for all $q < r < \infty$,

$$\begin{aligned} \|u\|_{L^r} &\leq c \|u\|_{L^\infty}^{\theta_3} \|u\|_{L^q}^{1-\theta_3} \leq c (\|u\|_{L^q}^{\theta_1} |u|_{C^{0,\alpha}}^{1-\theta_1})^{\theta_3} \|u\|_{L^q}^{1-\theta_3} \\ &= c \|u\|_{L^q}^{(1-\theta_3)+\theta_1\theta_3} |u|_{C^{0,\alpha}}^{(1-\theta_1)\theta_3}, \end{aligned}$$

where $\theta_3 := 1 - q/r$. This proves (12.57)₁.

To prove (12.57)₂ observe that if $0 < \beta < \alpha$, then for $x, y \in \mathbb{R}^N$, with $x \neq y$,

$$\begin{aligned} |u(y) - u(x)| &\leq 2^{1-\beta/\alpha} \|u\|_{L^\infty}^{1-\beta/\alpha} |u(y) - u(x)|^{\beta/\alpha} \\ &\leq 2^{1-\beta/\alpha} \|u\|_{L^\infty}^{1-\beta/\alpha} |u|_{C^{0,\alpha}}^{\beta/\alpha} \|x - y\|^\beta. \end{aligned}$$

Hence, by (12.58),

$$\begin{aligned} |u|_{C^{0,\beta}} &\leq c \|u\|_{L^\infty}^{1-\beta/\alpha} |u|_{C^{0,\alpha}}^{\beta/\alpha} \leq c (\|u\|_{L^q}^{\theta_1} |u|_{C^{0,\alpha}}^{1-\theta_1})^{1-\beta/\alpha} |u|_{C^{0,\alpha}}^{\beta/\alpha} \\ &= c \|u\|_{L^q}^{\theta_1(1-\beta/\alpha)} |u|_{C^{0,\alpha}}^{\beta/\alpha + (1-\theta_1)(1-\beta/\alpha)}, \end{aligned}$$

which completes the proof. $\square$

We now turn to the proof of Theorem 12.83.

Proof of Theorem 12.83. If $p < N$ and $q < \infty$, then since $u \in L^q(\mathbb{R}^N)$ we have that u vanishes at infinity, while if $q = \infty$, then u vanishes at infinity by hypothesis. Thus, in both cases we are in a position to apply the Sobolev–Gagliardo–Nirenberg embedding theorem (Theorem 12.4) to obtain

$$(12.59) \quad \|u\|_{L^{p^*}} \leq c \|\nabla u\|_{L^p},$$

where $p^* = pN/(N-p)$. Note that when $\theta = 0$ in (12.55) we get that r is exactly p^* . Since $u \in L^{p^*}(\mathbb{R}^N) \cap L^q(\mathbb{R}^N)$, we can now apply Exercise B.79(ii) to obtain

$$\|u\|_{L^r} \leq c \|u\|_{L^q}^\theta \|u\|_{L^{p^*}}^{1-\theta} \leq c \|u\|_{L^q}^\theta \|\nabla u\|_{L^p}^{1-\theta},$$

where we used (12.59). In view of (12.54), this proves (12.56) when $p < N$.

If $p = N \geq 2$, then by Exercise 12.37,

$$\|u\|_{L^r} \leq c \|u\|_{L^q}^{q/r} \|\nabla u\|_{L^N}^{1-q/r}$$

for all $q < r < \infty$, which, again by (12.54), gives (12.56) since (12.55) reduces to $\theta/q = 1/r$ for all $0 < \theta \leq 1$. The case $p = N = 1$ is left as an exercise.

If $p > N$, then by Remark 12.49, a representative $\bar{u}$ of u is Hölder continuous with exponent $1 - N/p$ and $|u|_{C^{0,1-N/p}} \leq c \|\nabla u\|_{L^p}$. We are now in a position to apply Proposition 12.84 to obtain (12.57), which by (12.54) gives

$$|u|_r \leq c \|u\|_{L^q}^\theta |u|_{C^{0,1-N/p}}^{1-\theta} \leq \|u\|_{L^q}^\theta \|\nabla u\|_{L^p}^{1-\theta},$$

and so (12.56) holds even in this case. $\square$

Next we prove an interpolation result to estimate the L^r norm of lower order derivatives with the L^q norm of u and the L^p norm of the derivatives of order m of u .

Theorem 12.85 (Gagliardo–Nirenberg interpolation, I). *Let $1 \leq p, q \leq \infty$, $m \in \mathbb{N}$, $k \in \mathbb{N}$, with $1 \leq k < m$, r be such that*

$$(12.60) \quad \frac{k}{m} \frac{1}{p} + \left(1 - \frac{k}{m}\right) \frac{1}{q} = \frac{1}{r}.$$

Then there exists a constant $c = c(k, m, N, p, q) > 0$ such that

$$(12.61) \quad \|\nabla^k u\|_{L^r(\mathbb{R}^N)} \leq c \|u\|_{L^q(\mathbb{R}^N)}^{1-k/m} \|\nabla^m u\|_{L^p(\mathbb{R}^N)}^{k/m}$$

for every $u \in L^q(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$. In particular, if $1 \leq p, q \leq \infty$, and r is given by

$$(12.62) \quad \frac{1}{2p} + \frac{1}{2q} = \frac{1}{r},$$

then there exists a constant $c = c(N, p, q) > 0$ such that

$$(12.63) \quad \|\nabla u\|_{L^r(\mathbb{R}^N)} \leq c \|u\|_{L^q(\mathbb{R}^N)}^{1/2} \|\nabla^2 u\|_{L^p(\mathbb{R}^N)}^{1/2}$$

for every $u \in L^q(\mathbb{R}^N) \cap \dot{W}^{2,p}(\mathbb{R}^N)$.

In the proof we use the notation (E.2) in Appendix E.

Proof. Step 1: For simplicity we present the proof first for $k = 1$ and $m = 2$. We treat the case $p, q, r < \infty$ and leave the other cases as an exercise. Assume first that $u \in C^\infty(\mathbb{R}^N)$. Let $i \in \{1, \dots, N\}$. For $x'_i \in \mathbb{R}^{N-1}$, we have that the one-dimensional function $u(x'_i, \cdot)$ satisfies the hypotheses of Theorem 7.41 and so

$$\int_{\mathbb{R}} |\partial_i u(x'_i, t)|^r dt \leq c \left(\int_{\mathbb{R}} |u(x'_i, t)|^q dt \right)^{r/(2q)} \left(\int_{\mathbb{R}} |\partial_i^2 u(x'_i, t)|^p dt \right)^{r/(2p)}.$$

Integrate the previous inequality in x'_i over $\mathbb{R}^{N-1}$ and use Tonelli's theorem to get

$$\begin{aligned} \int_{\mathbb{R}^N} |\partial_i u|^r dx &\leq c \int_{\mathbb{R}^{N-1}} \left(\int_{\mathbb{R}} |u(x'_i, t)|^q dt \right)^{r/(2q)} \left(\int_{\mathbb{R}} |\partial_i^2 u(x'_i, t)|^p dt \right)^{r/(2p)} dx'_i \\ &\leq c \|u\|_{L^q}^{r/2} \|\partial_i^2 u\|_{L^p}^{r/2}, \end{aligned}$$

where in the last inequality we have used Hölder's inequality and (12.62). The general case $u \in L^q(\mathbb{R}^N) \cap \dot{W}^{2,p}(\mathbb{R}^N)$ follows by a mollification argument. We omit the details.

Step 2: To prove the general case, fix a direction $\nu \in S^{N-1}$ and consider an orthonormal basis $\{b_1, \dots, b_{N-1}, \nu\}$. Given $u \in C^\infty(\mathbb{R}^N)$, define $v(y) = u(R(y))$, $y \in \mathbb{R}^N$, where R is an orthogonal transformation that maps $\{e_1, \dots, e_{N-1}, e_N\}$ into $\{b_1, \dots, b_{N-1}, \nu\}$. Then $\frac{\partial^k u}{\partial \nu^k}(R(y)) = \frac{\partial^k v}{\partial y_N^k}(y)$.

Fix $y' \in \mathbb{R}^{N-1}$ and apply Theorem 7.41 to the one-dimensional function $v(y', \cdot)$ to get

$$\int_{\mathbb{R}} \left| \frac{\partial^k v}{\partial y_N^k}(y', t) \right|^r dt \leq c \left(\int_{\mathbb{R}} |v(y', t)|^q dt \right)^{r(m-k)/(mq)} \left(\int_{\mathbb{R}} \left| \frac{\partial^m v}{\partial y_N^m}(y', t) \right|^p dt \right)^{rk/(mp)}.$$

Integrating the previous inequality in y' over $\mathbb{R}^{N-1}$ and reasoning exactly as in Step 1, using Hölder's inequality and (12.60), we obtain, after a change of variables,

$$\left\| \frac{\partial^k u}{\partial \nu^k} \right\|_{L^r} \leq c \|u\|_{L^q}^{1-k/m} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p}^{k/m}.$$

The result now follows from Exercise 11.20. $\square$

As a corollary of the previous theorem we obtain the following important result.

Corollary 12.86. *Let $m \in \mathbb{N}$ and $1 \leq p \leq \infty$. Then*

$$W^{m,p}(\mathbb{R}^N) = L^p(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$$

with equivalence of the norms

$$\|u\|_{L^p(\mathbb{R}^N)} + \sum_{k=1}^m \|\nabla^k u\|_{L^p(\mathbb{R}^N)} \quad \text{and} \quad \|u\|_{L^p(\mathbb{R}^N)} + \|\nabla^m u\|_{L^p(\mathbb{R}^N)}.$$

Proof. It is enough to take $p = q$ in (12.60). $\square$

Next we consider the general case. The following theorem is due to Nirenberg [184] in this generality and to Gagliardo [91] in the case $r \geq 1$.

Theorem 12.87 (Gagliardo–Nirenberg interpolation, general case). *Let $1 \leq p, q \leq \infty$, $m \in \mathbb{N}$, $k \in \mathbb{N}_0$, with $0 \leq k < m$, and let θ, r be such that*

$$(12.64) \quad 0 \leq \theta \leq 1 - k/m$$

and

$$(12.65) \quad (1 - \theta) \left(\frac{1}{p} - \frac{m - k}{N} \right) + \theta \left(\frac{1}{q} + \frac{k}{N} \right) = \frac{1}{r} \in (-\infty, 1].$$

Then there exists a constant $c = c(m, N, p, q, \theta, k) > 0$ such that

$$(12.66) \quad |\nabla^k u|_r \leq c \|u\|_{L^q(\mathbb{R}^N)}^\theta \|\nabla^m u\|_{L^p(\mathbb{R}^N)}^{1-\theta}$$

for every $u \in L^q(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$, with the following exceptional cases:

- (i) *If $k = 0$, $mp < N$, and $q = \infty$, we assume that u vanishes at infinity.*
- (ii) *If $1 < p < \infty$ and $m - k - N/p$ is a nonnegative integer, then (12.66) only holds for $0 < \theta \leq 1 - k/m$.*

The strategy of the proof is to treat first the cases $\theta = 0$ and $\theta = 1 - k/m$ and then to use the following interpolation proposition, whose proof is due to Hormander [121], to treat the values $0 < \theta < 1 - k/m$.

Proposition 12.88. *Let $-\infty < \lambda < \mu < \nu < \infty$. Then there exists a constant $c = c(\lambda, \mu, \nu, N) > 0$ such that*

$$(12.67) \quad |u|_{1/\mu} \leq c |u|_{1/\nu}^{1-\theta} |u|_{1/\lambda}^{\theta}$$

for all $u \in C^\infty(\mathbb{R}^N)$, with $|u|_{1/\lambda} < \infty$ and $|u|_{1/\nu} < \infty$, where $\theta := (\nu - \mu)/(\nu - \lambda)$.

Proof. We only give a sketch of the proof.

Step 1: We begin with the case $\lambda < \mu < \nu < 0$ and $-N\mu \in \mathbb{N}$. We claim that if $|u|_{1/\lambda} \leq 1$ and $|u|_{1/\nu} \leq 1$, then there exists a constant $c = c(\lambda, \mu, \nu, N) > 0$ such that $|u|_{1/\mu} \leq c$. Assume that $0 \leq -N\nu < 1$. Let $a := -N\nu$, let $m := \lfloor -N\lambda \rfloor$, and let $b := -N\lambda - m \in [0, 1)$. Assume first that $0 < a, b < 1$. Then by (12.54), $|u|_{1/\nu} = |u|_{C^{0,a}}$ and $|u|_{1/\lambda} = |\nabla^m u|_{C^{0,b}}$. Thus, we need to show that if $|u|_{C^{0,a}} \leq 1$ and $|\nabla^m u|_{C^{0,b}} \leq 1$, then there exists a constant $c = c(a, b, m, N) > 0$ such that $\|\nabla^n u\|_{L^\infty} \leq c$ for every $n = 1, \dots, m$. Fix $x_0 \in \mathbb{R}^N$ and let $B := B(x_0, 1)$. Then

$$(12.68) \quad |u(x) - u(y)| \leq |u|_{C^{0,a}} \|x - y\|^a \leq 1$$

for every $x, y \in B$, while by Taylor's formula (see Theorem 9.9),

$$u(x) = u(y) + P_y(x) + R_y(x),$$

where

$$(12.69) \quad P_y(x) := \sum_{1 \leq |\alpha| \leq m} \frac{1}{\alpha!} \partial^\alpha u(y) (x - y)^\alpha,$$

(12.70)

$$R_y(x) := \sum_{|\alpha|=m} \frac{m}{\alpha!} (x - y)^\alpha \int_0^1 (1-t)^{m-1} (\partial^\alpha u(tx + (1-t)y) - \partial^\alpha u(y)) dt.$$

Since $|\nabla^m u|_{C^{0,b}} \leq 1$,

$$(12.71) \quad \begin{aligned} |\partial^\alpha u(tx + (1-t)y) - \partial^\alpha u(y)| &\leq |\nabla^m u|_{C^{0,b}} \|tx + (1-t)y - y\|^b \\ &\leq t^b \|x - y\|^b \leq t^b (\|x - x_0\| + \|y - x_0\|)^b \leq 2^b, \end{aligned}$$

and so, also by (12.68),

$$\begin{aligned} |P_y(x)| &\leq |u(x) - u(y)| \\ &\quad + 2^b \sum_{|\alpha|=m} \frac{m}{\alpha!} \|x - y\|^m \int_0^1 (1-t)^{m-1} dt \leq 1 + 2^b \sum_{|\alpha|=m} \frac{1}{\alpha!} \end{aligned}$$

for all $x, y \in B$. Fix $y \in B$. Since the previous inequality holds for *all* $x \in B$, we leave as an exercise to show that this implies that there exists a constant $c = c(b, m, N) > 0$ (independent of y) such that $|\partial^\alpha u(y)| \leq c$ for every $1 \leq |\alpha| \leq m$ (see Exercise 11.21). Taking $y = x_0$, this concludes the proof of the claim in the case $0 \leq -N\nu < 1$ and $0 < a, b < 1$, since x_0 is arbitrary.

If $a = 0$, then $|u|_{1/\nu} = \|u\|_{L^\infty}$ by (12.54), and so in place of (12.68) we use the estimate $|u(x) - u(y)| \leq 2\|u\|_{L^\infty} \leq 2$. Similarly, if $b = 0$, then $|u|_{1/\lambda} = \|\nabla^m u\|_{L^\infty}$. In this case in the Taylor polynomial P_y (see (12.69)) one should replace $\sum_{1 \leq |\alpha| \leq m}$ with $\sum_{1 \leq |\alpha| \leq m-1}$ and replace the remainder term R_y with

$$\bar{R}_y(x) = \sum_{|\alpha|=m} \frac{m}{\alpha!} (x-y)^\alpha \int_0^1 (1-t)^{m-1} (\partial^\alpha u(tx + (1-t)y)) dt.$$

Then $|\bar{R}_y(x)| \leq \sum_{|\alpha|=m} \frac{1}{\alpha!} \|x-y\|^m \|\nabla^m u\|_{L^\infty} \leq \sum_{|\alpha|=m} \frac{1}{\alpha!}$ and we obtain the same bound on $|P_y(x)|$. We can then continue as before to prove that $\|\nabla^n u\|_{L^\infty} \leq c$ for every $n = 1, \dots, m-1$.

Finally, if $-N\nu \geq 1$, it is enough to apply what we just proved to the function $v = \partial^\alpha u$, where $|\alpha| = \lfloor -N\nu \rfloor$.

Step 2: We will prove (12.67) in the case $\lambda < \mu < \nu < 0$. We will give a full proof in the case $-N\mu \in \mathbb{N}$ and leave the other cases as an exercise. As in Step 1 assume first that $0 \leq -N\nu < 1$, and that $0 < a, b < 1$. If $|u|_{C^{0,a}} = 0$, then u is a constant, and so $\nabla^n u = 0$ for all $n = 1, \dots, m$ and there is nothing to prove. Thus, assume $|u|_{C^{0,a}} > 0$. If $|\nabla^m u|_{C^{0,b}} = 0$, then u is a polynomial of degree less than or equal to m , but since $|u|_{C^{0,a}} < \infty$, necessarily u must be a constant. Once more we have that $\nabla^n u = 0$ for all $n = 1, \dots, m$ and there is nothing to prove. Thus, in what follows, we assume that $|u|_{C^{0,a}} > 0$ and $|\nabla^m u|_{C^{0,b}} > 0$. Fix $x_0 \in \mathbb{R}^N$ and $R > 0$ and consider the function $w(x) := u(x_0 + R(x - x_0))$, $x \in \mathbb{R}^N$. By the chain rule we have that

$$|w|_{C^{0,a}} \leq R^a |u|_{C^{0,a}}, \quad |\nabla^m w|_{C^{0,b}} \leq R^{m+b} |\nabla^m u|_{C^{0,b}}.$$

Since $|u|_{C^{0,a}} > 0$ and $|\nabla^m u|_{C^{0,b}} > 0$, we can choose $R > 0$ in such a way that $R^{m+b} |\nabla^m u|_{C^{0,b}} = R^a |u|_{C^{0,a}} =: \kappa$. Then $|w|_{C^{0,a}} \leq \kappa$ and $|\nabla^m w|_{C^{0,b}} \leq \kappa$ and so by Step 1 applied to the function w/κ , for every multi-index $\alpha \in \mathbb{N}_0^N$ with $|\alpha| = -N\mu$,

$$R^{-N\mu} |\partial^\alpha u(x_0)| = |\partial^\alpha w(x_0)| \leq c\kappa = c(R^a |u|_{C^{0,a}})^{1-\theta} (R^{m+b} |\nabla^m u|_{C^{0,b}})^\theta.$$

Since $-N\mu = -(1-\theta)N\nu - \theta N\lambda = -(1-\theta)a - \theta(m+b)$ the powers of R cancel to give

$$|\partial^\alpha u(x_0)| \leq c |u|_{C^{0,a}}^{1-\theta} |\nabla^m u|_{C^{0,b}}^\theta,$$

which completes the proof of the claim in the case $0 < a, b < 1$. The cases $a = 0$ and $b = 0$ are similar and we omit them.

Step 3: Next we study the case $\lambda < \mu = 0 < \nu \leq 1$. Let $m := \lfloor -N\lambda \rfloor$, and let $b := -N\lambda - m \in [0, 1)$. Assume first that $m \geq 1$ and $0 < b < 1$. Thus, setting $q := 1/\nu$, by (12.54), we need to show that there exists a constant $c = c(b, N, q) > 0$ such that

$$\|u\|_{L^\infty} \leq c \|u\|_{L^q}^{1-\theta} |\nabla^m u|_{C^{0,b}}^\theta.$$

If $\|u\|_{L^q} = 0$, then $u = 0$. Thus, assume $\|u\|_{L^q} > 0$. If $|\nabla^m u|_{C^{0,b}} = 0$, then u is a polynomial of degree less than or equal to m , but since $\|u\|_{L^q} < \infty$, necessarily u must be a zero. Thus, in what follows, we assume that $\|u\|_{L^q} > 0$ and $|\nabla^m u|_{C^{0,b}} > 0$.

Fix $x_0 \in \mathbb{R}^N$ and let $B := B(x_0, 1)$ as before. Consider the vector space V of all polynomials $P : \mathbb{R}^N \rightarrow \mathbb{R}$ of degree less than or equal to m . Then V is a finite-dimensional space. Hence, all norms in V are equivalent (exercise). It follows that there exist two constants $c_1 > 0$ and $c_2 > 0$ such that

$$(12.72) \quad \|P\|_{L^\infty(B)} \leq c_1 \|P\|_{L^1(B)} \leq c_2 \|P\|_{L^q(B)}$$

for all $P \in V$.

By rescaling, we can assume that $|\nabla^m u|_{C^{0,b}} = 1$. Fix $R > 0$ and let w be the function defined in Step 2. Reasoning somewhat as in Step 1 with w in place of u we have that for all $x, y \in B$,

$$(12.73) \quad |w(x) - Q_y(x)| \leq c |\nabla^m w|_{C^{0,b}} \leq c R^{m+b} |\nabla^m u|_{C^{0,b}} = c R^{m+b},$$

where $Q_y(x) := \sum_{0 \leq |\alpha| \leq m} \frac{1}{\alpha!} \partial^\alpha w(y) (x - y)^\alpha$. Hence, by Minkowski's inequality and a change of variables

$$\|Q_y\|_{L^q(B)} \leq \|w\|_{L^q(B)} + c R^{m+b} (\mathcal{L}^N(B))^{1/q} \leq R^{-N/q} \|u\|_{L^q} + c R^{m+b} \alpha_N^{1/q}.$$

Combining (12.72) applied to Q_y , (12.73), and the previous inequality we obtain

$$\begin{aligned} |u(x_0)| &= |w(x_0)| \leq |Q_y(x_0)| + |w(x_0) - Q_y(x_0)| \\ &\leq c_2 \|Q_y\|_{L^q(B)} + c R^{m+b} \leq c_2 R^{-N/q} \|u\|_{L^q} + c R^{m+b}. \end{aligned}$$

It suffices to choose R so that $\|u\|_{L^q} = R^{m+b+N/q}$. The cases $m = 0$ and $b = 0$ are similar and are left as an exercise (see Proposition 12.84).

Step 4: In Step 3 we proved (12.67) for $\mu = 0$. To prove (12.67) for $0 < -N\mu < 1$ it suffices to use Step 2 to estimate $|u|_{C^{0,-N\mu}}$ in terms of $\|u\|_{L^\infty}$ and $|\nabla^m u|_{C^{0,b}}$ and Step 3 to estimate $\|u\|_{L^\infty}$ in terms of $\|u\|_{L^q}$ and $|\nabla^m u|_{C^{0,b}}$. We leave the remaining cases as an exercise. $\square$

We are now ready to prove Theorem 12.87.

Proof of Theorem 12.87. Step 1: We begin by considering the case $\theta = 0$. Then (12.65) reduces to $\frac{1}{p} - \frac{m-k}{N} = \frac{1}{r}$. We claim that

$$(12.74) \quad |\nabla^k u|_r \leq c \|\nabla^m u\|_{L^p}$$

for every $u \in L^q(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$ with the exception of the case $1 < p < \infty$ and $m - k - N/p$ is a nonnegative integer.

If $(m - k)p < N$, then $r = p_{m,k}^*$ (see (12.7)) and so (12.74) follows from the Sobolev–Gagliardo–Nirenberg embedding theorem (see Corollary 12.11) (using hypothesis (i)).

If $(m - k)p = N$ and $p = 1$, then $r = \infty$ and (12.74) is a consequence of Exercise 12.43. The case $(m - k)p = N$ and $p > 1$ is excluded by (ii).

If $(m - k)p > N$ and $m - k - N/p$ is not a nonnegative integer, then $1/r < 0$ and so by (12.54),

$$|\nabla^k u|_r = |\nabla^{k+n} u|_{C^{0,a}} = |\nabla^{\lfloor m-N/p \rfloor} u|_{C^{0,a}},$$

where $n := \lfloor -N/r \rfloor = \lfloor m - k - N/p \rfloor$ and $a := -n - N/r = m - N/p - \lfloor m - N/p \rfloor \in (0, 1)$. We can now use Remark 12.56 to obtain (12.74).

Step 2: Next we prove (12.66) in the case in which $0 < \theta < 1 - k/m$ but we are not in case (ii), that is, either $m - k - N/p$ is not a nonnegative integer or if it is, then $p = 1$. By Step 1 and Theorem 12.85 for $\theta = 0$ and $\theta = 1 - k/m$, respectively, we have

$$|\nabla^k u|_{r_0} \leq c \|\nabla^m u\|_{L^p}, \quad |\nabla^k u|_{r_1} \leq c \|u\|_{L^q}^{1-k/m} \|\nabla^m u\|_{L^p}^{k/m},$$

where

$$\frac{1}{p} - \frac{m-k}{N} = \frac{1}{r_0}, \quad \frac{k}{m} \frac{1}{p} + \left(1 - \frac{k}{m}\right) \frac{1}{q} = \frac{1}{r_1}.$$

Given $0 < \theta < 1 - k/m$ we have that $1/r$ is in between $1/r_0$ and $1/r_1$ and thus by Proposition 12.84,

$$\begin{aligned} |\nabla^k u|_r &\leq c |\nabla^k u|_{r_0}^{\theta_1} |\nabla^k u|_{r_1}^{1-\theta_1} \\ &\leq c \|u\|_{L^q}^{(1-k/m)\theta_1} \|\nabla^m u\|_{L^p}^{\theta_1 + (k/m)(1-\theta_1)} = c \|u\|_{L^q}^\theta \|\nabla^m u\|_{L^p}^{1-\theta}, \end{aligned}$$

where $\theta_1 := (1/r_1 - 1/r)/(1/r_1 - 1/r_0)$ and $\theta := (1 - k/m)\theta_1$.

Step 3: Finally, we prove (12.66) in the case in which $0 < \theta < 1 - k/m$, $1 < p < \infty$, and $m - k - N/p$ is a nonnegative integer. In view of Theorem 12.83 we can assume that $m \geq 2$. If $p = N$, then by Theorem 12.85,

$$\|\nabla^{m-1} u\|_{L^{r_m}} \leq c \|u\|_{L^q}^{1-(m-1)/m} \|\nabla^m u\|_{L^N}^{(m-1)/m},$$

where $\frac{m-1}{m} \frac{1}{N} + \frac{1}{m} \frac{1}{q} = \frac{1}{r_m}$. In turn, by Exercise 12.37,

$$\|\nabla^{m-1} u\|_{L^s} \leq c \|\nabla^{m-1} u\|_{L^{r_m}}^{r_m/s} \|\nabla^m u\|_{L^N}^{1-r_m/s}$$

for all $r_m < s < \infty$, which implies (12.66) for $k = m - 1$.

If $p < N$, then necessarily $k < m - 1$, since $m - 1$ is not an exceptional case. By applying Step 2 to $m - 1$ we obtain

$$(12.75) \quad \|\nabla^{m-1}u\|_{L^s} \leq c\|u\|_{L^q}^{\theta_1} \|\nabla^m u\|_{L^p}^{1-\theta_1}$$

where $(1 - \theta_1)\left(\frac{1}{p} - \frac{1}{N}\right) + \theta_1\left(\frac{1}{q} + \frac{m-1}{N}\right) = \frac{1}{s}$ for all $0 \leq \theta_1 \leq \frac{1}{m}$. Hence, $\nabla^{m-1}u \in L^s(\mathbb{R}^N; \mathbb{R}^{M_{m-1}})$ for all s between p^* and $(N + (m - 1)q)/(Nq)$. To conclude the proof it suffices to observe that for $k < m - 1$ we can write (12.65) as

$$\frac{1}{r} = (1 - \theta)\left(\frac{1}{p^*} - \frac{m-1-k}{N}\right) + \theta\left(\frac{1}{q} + \frac{k}{N}\right).$$

Since $\theta = 0$ is not allowed, we can write $\frac{1}{r}$ as a convex combination of $1/s - (m-1-k)/N$ and $1/q + k/N$ for some s between p^* and $(N + (m-1)q)/(Nq)$ with the property that $m-1-k-N/s$ is not a nonnegative integer. We can now apply Step 2 to estimate $\|\nabla^k u\|_{L^r}$ in terms of $\|u\|_{L^q}$ and $\|\nabla^{m-1}u\|_{L^s}$, which, together with (12.75), completes the proof. $\square$

Exercise 12.89. Let $u(x) = \sin(\lambda x_1)\varphi(x)$, $x \in \mathbb{R}^N$, where $\varphi \in C_c^\infty(\mathbb{R}^N)$ and $\lambda > 0$ is large. Prove that (12.64) is a necessary condition for (12.66) to hold, namely, that (12.66) fails if $1 - \frac{k}{m} < \theta \leq 1$.

Exercise 12.90. Let $u \in C^\infty(\mathbb{R}^N)$ and let φ_R is a standard mollifier, $R > 0$.

(i) Prove that for every $x, y \in \mathbb{R}^N$ and $m \in \mathbb{N}$,

$$\begin{aligned} u(x) &= \sum_{0 \leq |\alpha| \leq m} \frac{1}{\alpha!} (-1)^{|\alpha|} \int_{\mathbb{R}^N} u(y) \frac{\partial^\alpha}{\partial y^\alpha} (\varphi_R(y-x)(x-y)^\alpha) dy \\ &+ \sum_{|\alpha|=m} \frac{m}{\alpha!} \int_{\mathbb{R}^N} \varphi_R(y-x) \int_0^1 (x-y)^\alpha (1-t)^{m-1} (\partial^\alpha u(tx + (1-t)y) \\ &- \partial^\alpha u(y)) dt dy. \end{aligned}$$

(ii) Using part (i) prove that for every $0 < b < 1$ there exists a constant $c = c(b, m, N) > 0$ such that for every $x \in \mathbb{R}^N$,

$$|u(x)| \leq cR^{-N} \int_{B(x,R)} |u(y)| dy + cR^{m+b} |\nabla^m u|_{C^{0,b}(B(x,R))}.$$

Use this inequality to give an alternative proof to Step 3 of Proposition 12.88.

- (iii) Prove that for every $x, y \in \mathbb{R}^N$, $m \in \mathbb{N}$, and every multi-index $\beta \in \mathbb{N}_0^N$ with $|\beta| \geq 1$,

$$\begin{aligned} \partial^\beta u(x) &= \sum_{0 \leq |\alpha| \leq m} \frac{1}{\alpha!} (-1)^{|\alpha|+|\beta|} \int_{\mathbb{R}^N} u(y) \frac{\partial^{\alpha+\beta}}{\partial y^{\alpha+\beta}} (\varphi_R(y-x)(x-y)^\alpha) dy \\ &+ \sum_{|\alpha|=m} \frac{m}{\alpha!} \int_{\mathbb{R}^N} \varphi_R(y-x) \int_0^1 (x-y)^\alpha (1-t)^{m-1} (\partial^{\alpha+\beta} u(tx + (1-t)y) \\ &- \partial^{\alpha+\beta} u(y)) dt dy. \end{aligned}$$

- (iv) Using part (iii) to give an alternative proof to Step 1 of Proposition 12.88. Hint: Note that $\int_{B(x,R)} \frac{\partial^{\alpha+\beta}}{\partial y^{\alpha+\beta}} (\varphi_R(y-x)(x-y)^\alpha) dy = 0$.

Exercise 12.91. Let $u \in L^1(\mathbb{R}^N) \cap \dot{W}^{1,2}(\mathbb{R}^N)$.

- (i) Prove that for every $R > 0$,

$$\int_{\mathbb{R}^N \setminus B(0,R)} |\hat{u}|^2 dx \leq R^{-2} \|\nabla u\|_{L^2(\mathbb{R}^N)}^2,$$

where $\hat{u}$ is the Fourier transform of u .

- (ii) Prove that for every $R > 0$,

$$\int_{B(0,R)} |\hat{u}|^2 dx \leq \alpha_N R^N \|u\|_{L^1(\mathbb{R}^N)}^2.$$

- (iii) Prove that

$$\|u\|_{L^2(\mathbb{R}^N)} \leq c \|u\|_{L^1(\mathbb{R}^N)}^{2/(N+2)} \|\nabla u\|_{L^2(\mathbb{R}^N)}^{N/(N+2)}.$$

Sobolev Spaces: Further Properties

Newton's third law of graduation: For every action towards graduation there is an equal and opposite distraction.

— Jorge Cham, www.phdcomics.com

13.1. Extension Domains

As we have seen in the previous chapter, several embeddings that are valid for the entire space $\mathbb{R}^N$ continue to hold for extension domains. The next exercise shows that in general an arbitrary open set is not an extension domain for $W^{1,p}$. Thus, an important problem is to characterize the class of domains that are extension domains for $W^{m,p}$. To the author's knowledge this problem is still open, with the exception of the case $m = 1$, $N = 2$, and $p = 2$ (see the paper of Jones [122]).

Exercise 13.1. Let $1 \leq p < \infty$, let

$$\Omega = \{x = (x_1, x_2) \in \mathbb{R}^2 : 0 < x_1 < 1, 0 < x_2 < x_1^\alpha\},$$

where $\alpha > 1$, and let $u : \Omega \rightarrow \mathbb{R}$ be defined by $u(x) := x_1^{1-\beta}$, $x \in \Omega$, where $1 < \beta < (\alpha + 1)/p$. Prove that $u \in W^{1,p}(\Omega)$ and that if β is sufficiently close to $(\alpha + 1)/p$, then u cannot be extended to a function in $W^{1,p}(\mathbb{R}^2)$. Hint: Consider the three cases $1 \leq p < 2$, $p = 2$, and $p > 2$.

Next we show that extension domains strongly depend on p .

Exercise 13.2. Let Ω be as in the previous exercise, and using polar coordinates (r, θ) , let $u : \mathbb{R}^2 \setminus \overline{\Omega} \rightarrow \mathbb{R}$ be defined as

$$u(x_1, x_2) := r^{1-\beta} \psi(\theta) \varphi(r \cos \theta, r \sin \theta),$$

where $\beta > 0$, $\psi \in C^\infty([0, 2\pi])$, with $\psi = 1$ for θ small and $\psi = 0$ for $\theta \in [\pi, 2\pi]$, and $\varphi \in C_c^\infty(\mathbb{R}^2)$ with $\varphi = 1$ for $0 \leq r < 1$.

- (i) Let $1 < p < \infty$ and find for which values of β the function u belongs to $W^{1,p}(\mathbb{R}^2 \setminus \overline{\Omega})$.
- (ii) Prove that if $0 < \beta < 2/p$ is sufficiently close to $2/p$, then u cannot be extended to a function in $W^{1,p}(\mathbb{R}^2)$.
- (iii) For every function $v \in W^{1,1}(\mathbb{R}^2 \setminus \overline{\Omega})$, with $v = 0$ for $x_1 > \frac{1}{2}$, define

$$\mathcal{E}_0(v)(x) := \begin{cases} v(x) & \text{if } x \notin \Omega, \\ v^-(x) + \frac{x_2}{x_1^\alpha}(v^+(x) - v^-(x)) & \text{if } x \in \Omega, \end{cases}$$

where $v^-(x) := v(x_1, -x_2)$ and $v^+(x) := v(x_1, 2x_1^\alpha - x_2)$. Prove that $\mathcal{E}_0(v) \in W^{1,1}(\mathbb{R}^2)$ and that $\|\mathcal{E}_0(v)\|_{W^{1,1}(\mathbb{R}^2)} \leq c\|v\|_{W^{1,1}(\mathbb{R}^2 \setminus \overline{\Omega})}$.

- (iv) Prove $\mathbb{R}^2 \setminus \overline{\Omega}$ is an extension domain for $W^{1,1}$. Hint: Use cut-off functions to reduce to part (iii) and away from the origin use Theorem 13.4 below.

The previous exercise shows that $\mathbb{R}^2 \setminus \overline{\Omega}$ is an extension domain for $W^{1,1}$ but not for $W^{1,p}$ with $p > 1$. We now prove that if $\partial\Omega$ is sufficiently regular, then it is possible to construct an extension operator $\mathcal{E}$ that works for all Sobolev spaces $W^{m,p}(\Omega)$.

We begin with the case in which Ω is the half space $\mathbb{R}_+^N$ (see Exercise 7.24).

Exercise 13.3. Given $m \in \mathbb{N}$, and $1 \leq p \leq \infty$, let $u \in W^{m,p}(\mathbb{R}_+^N)$. Prove that there exist $c_1, \dots, c_{m+1} \in \mathbb{R}$ such that the function

$$v(x) := \begin{cases} \sum_{n=1}^{m+1} c_n u(x', -nx_N) & \text{if } x_N < 0, \\ u(x) & \text{if } x_N > 0, \end{cases}$$

is well-defined and belongs to $W^{m,p}(\mathbb{R}^N)$. Prove also that for every $0 \leq k \leq m$, $\|\nabla^k v\|_{L^p(\mathbb{R}^N)} \leq c\|\nabla^k u\|_{L^p(\mathbb{R}_+^N)}$ for some constant $c = c(m, N, p) > 0$.

Next we consider the important special case in which Ω lies above the graph of a Lipschitz continuous function.

Theorem 13.4. Let $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ be a Lipschitz continuous function and let

$$(13.1) \quad \Omega := \{(x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : x_N > f(x')\}.$$

Then for all $1 \leq p \leq \infty$ there exists a continuous linear operator $\mathcal{E} : W^{1,p}(\Omega) \rightarrow W^{1,p}(\mathbb{R}^N)$ such that for all $u \in W^{1,p}(\Omega)$, $\mathcal{E}(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$ and

$$(13.2) \quad \|\mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} = 2\|u\|_{L^p(\Omega)}, \quad \|\partial_N \mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} = \|\partial_N u\|_{L^p(\Omega)},$$

$$(13.3) \quad \|\partial_i \mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} \leq 2\|\partial_i u\|_{L^p(\Omega)} + 2\text{Lip } f \|\partial_N u\|_{L^p(\Omega)}$$

for all $i = 1, \dots, N$.

Proof. The idea of the proof is to first flatten the boundary to reduce to the case in which $\Omega = \mathbb{R}_+^N$ and then use a reflection argument (see Exercise 11.51). We only prove the case $1 \leq p < \infty$ and leave the easier case $p = \infty$ as an exercise. Consider the transformation $\Psi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ given by $\Psi(y) := (y', y_N + f(y'))$. Note that Ψ is invertible, with inverse given by $\Psi^{-1}(x) = (x', x_N - f(x'))$. Moreover, for all $y, z \in \mathbb{R}^N$,

$$\begin{aligned} \|\Psi(y) - \Psi(z)\| &= \|(y' - z', f(y') - f(z') + y_N - z_N)\| \\ &\leq \sqrt{\|y' - z'\|^2 + (\text{Lip } f) \|y' - z'\| + |y_N - z_N|)^2} \\ &\leq 2(1 + \text{Lip } f) \|y - z\|, \end{aligned}$$

which shows that Ψ (and similarly Ψ^{-1}) is Lipschitz continuous. Since f is Lipschitz continuous, by Rademacher's theorem (see Theorem 9.14) it is differentiable for $\mathcal{L}^{N-1}$ -a.e. $y' \in \mathbb{R}^{N-1}$, and so for any such $y' \in \mathbb{R}^{N-1}$ and for all $y_N \in \mathbb{R}$ we have

$$J_\Psi(y) = \begin{pmatrix} I_{N-1} & 0 \\ \nabla_{y'} f(y') & 1 \end{pmatrix},$$

which implies that $\det J_\Psi(y) = 1$. Note that $\Psi(\mathbb{R}_+^N) = \Omega$.

Given a function $u \in W^{1,p}(\Omega)$, $1 \leq p < \infty$, define the function

$$w(y) := u(\Psi(y)) = u(y', y_N + f(y')), \quad y \in \mathbb{R}_+^N.$$

By Theorem 11.53 the function w belongs to $W^{1,p}(\mathbb{R}_+^N)$ and the usual chain rule formula for the partial derivatives holds. By Exercise 11.51 the function $\hat{w} : \mathbb{R}^N \rightarrow \mathbb{R}$, defined by

$$\hat{w}(y) := \begin{cases} w(y) & \text{if } y_N > 0, \\ w(y', -y_N) & \text{if } y_N < 0, \end{cases}$$

belongs to $W^{1,p}(\mathbb{R}^N)$ and the usual chain rule formula for the partial derivatives holds.

Define the function $v : \mathbb{R}^N \rightarrow \mathbb{R}$ by

$$(13.4) \quad v(x) := (\hat{w} \circ \Psi^{-1})(x) = \begin{cases} u(x) & \text{if } x_N > f(x'), \\ u(x', 2f(x') - x_N) & \text{if } x_N < f(x'). \end{cases}$$

Again by Theorem 11.53, we have that $v \in W^{1,p}(\mathbb{R}^N)$ and the usual chain rule formula for the partial derivatives holds.

Using Theorem 9.52 and the fact that $\det \nabla \Psi = \det \nabla \Psi^{-1} = 1$, we have that

$$\int_{\mathbb{R}^N \setminus \overline{\Omega}} |v(x)|^p dx = \int_{\mathbb{R}^N \setminus \overline{\Omega}} |u(x', 2f(x') - x_N)|^p dx = \int_{\Omega} |u(y)|^p dy.$$

Since for all $i = 1, \dots, N-1$ and for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N \setminus \overline{\Omega}$,

$$(13.5) \quad \partial_i v(x) = \partial_i u(x', 2f(x') - x_N) + 2\partial_N u(x', 2f(x') - x_N) \partial_i f(x'),$$

again by Theorem 9.52 we have that

$$\begin{aligned} \left(\int_{\mathbb{R}^N \setminus \bar{\Omega}} |\partial_i v(x)|^p dx \right)^{1/p} &\leq \left(\int_{\mathbb{R}^N \setminus \bar{\Omega}} |\partial_i u(x', 2f(x') - x_N)|^p dx \right)^{1/p} \\ &\quad + 2 \operatorname{Lip} f \left(\int_{\mathbb{R}^N \setminus \bar{\Omega}} |\partial_N u(x', 2f(x') - x_N)|^p dx \right)^{1/p} \\ &\leq \left(\int_{\Omega} |\partial_i u(y)|^p dy \right)^{1/p} + 2 \operatorname{Lip} f \left(\int_{\Omega} |\partial_N u(y)|^p dy \right)^{1/p}. \end{aligned}$$

Similarly, using the fact that $\partial_N v(x) = -\partial_N u(x', 2f(x') - x_N)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N \setminus \bar{\Omega}$, we obtain

$$\begin{aligned} \int_{\mathbb{R}^N \setminus \bar{\Omega}} |\partial_N v(x)|^p dx &= \int_{\mathbb{R}^N \setminus \bar{\Omega}} |\partial_N u(x', 2f(x') - x_N)|^p dx \\ &= \int_{\Omega} |\partial_N u(y)|^p dy. \end{aligned}$$

Hence, the linear extension operator $u \in W^{1,p}(\Omega) \mapsto \mathcal{E}(u) := v \in W^{1,p}(\mathbb{R}^N)$ is continuous and satisfies (13.2) and (13.3). $\square$

Remark 13.5. Note that the operator $\mathcal{E}$ defined in the previous theorem does not depend on p . However, it has the disadvantage that it cannot be used for higher-order Sobolev spaces, unless one assumes that f is more regular (see Theorem 11.57).

In order to construct an extension operator that can be used for higher-order Sobolev spaces, we need to introduce the notion of *regularized distance*. The construction given in the following theorem is due to Lieberman [143]. An alternative construction was obtained earlier by Stein [220] and is sketched in Exercise 13.7 below.

Theorem 13.6 (Regularized distance). *Let $V \subset \mathbb{R}^N$ be an open set with nonempty boundary and let*

$$d(x) := \begin{cases} \operatorname{dist}(x, \partial V) & \text{if } x \in V, \\ -\operatorname{dist}(x, \partial V) & \text{if } x \notin V. \end{cases}$$

Then there exists a Lipschitz continuous function $d_{\operatorname{reg}} : \mathbb{R}^N \rightarrow \mathbb{R}$ such that $d_{\operatorname{reg}} \in C^\infty(\mathbb{R}^N \setminus \partial V)$,

$$(13.6) \quad \frac{1}{2} \leq \frac{d(x)}{d_{\operatorname{reg}}(x)} \leq \frac{3}{2}$$

for all $x \in \mathbb{R}^N \setminus \partial V$, and

$$(13.7) \quad |\partial^\alpha d_{\operatorname{reg}}(x)| \leq \frac{c_\alpha}{|d_{\operatorname{reg}}(x)|^{|\alpha|-1}}$$

for all $x \in \mathbb{R}^N \setminus \partial V$, for every multi-index α , $|\alpha| \geq 1$, and for some constant $c_\alpha > 0$.

Proof. Step 1: Consider a nonnegative function $\varphi \in C_c^\infty(\mathbb{R}^N)$ such that

$$(13.8) \quad \text{supp } \varphi \subseteq \overline{B(0, 1)}, \quad \int_{\mathbb{R}^N} \varphi(x) dx = 1$$

and define

$$(13.9) \quad G(x, t) := \int_{\mathbb{R}^N} d(x - \tfrac{t}{2}y) \varphi(y) dy, \quad x \in \mathbb{R}^N, \quad t \in \mathbb{R}.$$

Since d is 1-Lipschitz continuous, for all $x \in \mathbb{R}^N$ and for all $t_1, t_2 \in \mathbb{R}$, by (13.8) we have

$$(13.10) \quad |G(x, t_1) - G(x, t_2)| \leq \int_{B(0, 1)} |d(x - \tfrac{t_1}{2}y) - d(x - \tfrac{t_2}{2}y)| \varphi(y) dy \\ \leq \tfrac{1}{2}|t_1 - t_2| \int_{B(0, 1)} \varphi(y) dy = \tfrac{1}{2}|t_1 - t_2|,$$

while for all $t \in \mathbb{R}$ and $x_1, x_2 \in \mathbb{R}^N$,

$$(13.11) \quad |G(x_1, t) - G(x_2, t)| \leq \int_{B(0, 1)} |d(x_1 - \tfrac{t}{2}y) - d(x_2 - \tfrac{t}{2}y)| \varphi(y) dy \leq \|x_1 - x_2\|.$$

Step 2: By the change of variables $z := x - \tfrac{t}{2}y$, for $t \neq 0$ we can rewrite G as

$$(13.12) \quad G(x, t) = 2^N \int_{\mathbb{R}^N} t^{-N} \varphi(2t^{-1}(x - z)) d(z) dz = (d * \varphi_{t/2})(x).$$

Using Theorem C.20 and the Lebesgue dominated convergence theorem we have that G belongs to $C^\infty(\mathbb{R}^N \times (\mathbb{R} \setminus \{0\}))$ with

$$(13.13) \quad \frac{\partial^{|\alpha|} G}{\partial(x, t)^\alpha}(x, t) = 2^N \int_{\mathbb{R}^N} \frac{\partial^{|\alpha|}}{\partial(x, t)^\alpha} (t^{-N} \varphi(2t^{-1}(x - z))) d(z) dz$$

for every multi-index $\alpha \in \mathbb{N}_0^{N+1}$. Moreover, by Exercise 11.37, an induction argument, and the fact that $\varphi \in C_c^\infty(\mathbb{R}^N)$,

$$(13.14) \quad \left| \frac{\partial^{|\alpha|}}{\partial(x, t)^\alpha} (t^{-N} \varphi(2t^{-1}(x - z))) \right| \leq \frac{a_\alpha}{|t|^{N+|\alpha|}}$$

for all $x \in \mathbb{R}^N$ and $t \neq 0$ and for some constant a_α depending on φ .

Since $2^N \int_{\mathbb{R}^N} t^{-N} \varphi(2t^{-1}(x - z)) dz = 1$ by (13.8), by differentiating we obtain

$$\int_{\mathbb{R}^N} \frac{\partial^{|\alpha|}}{\partial(x, t)^\alpha} (t^{-N} \varphi(2t^{-1}(x - z))) dz = 0$$

and so

$$\frac{\partial^{|a|} G}{\partial(x, t)^\alpha}(x, t) = 2^N \int_{B(x, t/2)} \frac{\partial^{|a|}}{\partial(x, t)^\alpha} (t^{-N} \varphi(2t^{-1}(x - z))) (d(z) - d(x)) dz.$$

Together with (13.14) and the fact that d is 1-Lipschitz continuous, this gives

$$(13.15) \quad \left| \frac{\partial^{|a|} G}{\partial(x, t)^\alpha}(x, t) \right| \leq \frac{2^N a_\alpha}{|t|^{N+|\alpha|}} \mathcal{L}^N(B(x, t/2)) \frac{|t|}{2} \leq \frac{b_\alpha}{|t|^{|\alpha|-1}}.$$

Step 3: By (13.10) and Banach's fixed point theorem (Theorem A.14) for every $x \in \mathbb{R}^N$ there exists a unique $t_x \in \mathbb{R}$ such that $t_x = G(x, t_x)$. Defining $d_{\text{reg}}(x) := t_x$, from (13.10) and (13.11) we obtain that $|d_{\text{reg}}(x_1) - d_{\text{reg}}(x_2)| \leq 2\|x_1 - x_2\|$ for all $x_1, x_2 \in \mathbb{R}^N$. We claim that d_{reg} is of class $C^\infty(\mathbb{R}^N \setminus \partial V)$. Consider the function $H(x, t) = t - G(x, t)$ for $(x, t) \in \mathbb{R}^N \times (\mathbb{R} \setminus \{0\})$. Then

$$\partial_t H(x, t) = 1 - \partial_t G(x, t) \geq 1 - \frac{1}{2},$$

since $|\partial_t G(x, t)| \leq \frac{1}{2}$ by (13.10). By the implicit function theorem for every point $(x_0, t_{x_0}) \in \mathbb{R}^N \times (\mathbb{R} \setminus \{0\})$ there are a neighborhood of (x_0, t_{x_0}) and a unique function f of class C^∞ such that $H(x, f(x)) = 0$. But by the uniqueness of t_x , necessarily, $f = d_{\text{reg}}$, which proves the claim.

Step 4: We will prove (13.6). By (13.8), $G(x, 0) = \int_{\mathbb{R}^N} d(x) \varphi(y) dy = d(x)$. Using the fact that $d_{\text{reg}}(x) = G(x, d_{\text{reg}}(x))$, (13.9), and (13.10), we have

$$|d_{\text{reg}}(x) - d(x)| = |G(x, d_{\text{reg}}(x)) - G(x, 0)| \leq \frac{1}{2} |d_{\text{reg}}(x) - 0|,$$

which proves (13.6).

Step 5: In this step we will prove (13.7). By differentiating the equality $d_{\text{reg}}(x) = G(x, d_{\text{reg}}(x))$ with respect to x_i , we have

$$\partial_i d_{\text{reg}}(x) = \partial_i G(x, d_{\text{reg}}(x)) + \partial_t G(x, d_{\text{reg}}(x)) \partial_i d_{\text{reg}}(x).$$

Hence,

$$(13.16) \quad \partial_i d_{\text{reg}}(x) = \frac{\partial_i G(x, d_{\text{reg}}(x))}{1 - \partial_t G(x, d_{\text{reg}}(x))}.$$

From the inequalities $|\partial_i G(x, t)| \leq 1$ and $|\partial_t G(x, t)| \leq \frac{1}{2}$ (see (13.10) and (13.11)), we derive that $|\partial_i d_{\text{reg}}(x)| \leq 2$. By differentiating (13.16) and using (13.15), again the fact that $|\partial_t G| \leq \frac{1}{2}$, and an induction argument on $|\alpha|$ we obtain (13.7). We leave the details as an exercise. $\square$

Another way to construct a regularized distance is to use Whitney's decomposition theorem (see Theorem 9.44) and is due to Stein (see [220]).

Exercise 13.7 (Regularized distance). Let $V \subseteq \mathbb{R}^N$ be an open set, let $\{\overline{Q(x_n, r_n)}\}_n$ be a Whitney decomposition of V (see Theorem 9.44), and let $0 < \varepsilon < \frac{1}{4}$. Consider a nonnegative function $\varphi \in C_c^\infty(\mathbb{R}^N)$ such that

$\varphi(x) = 1$ for all $x \in Q(0, 1)$ and $\varphi(x) = 0$ for all $x \in \mathbb{R}^N \setminus Q(0, 1 + \varepsilon)$ and for each n set $\varphi_n(x) := \varphi((x - x_n)/r_n)$, $x \in \mathbb{R}^N$. Define the *regularized distance*

$$d_{\text{reg}}(x) := \sum_n \text{diam } Q(x_n, r_n) \varphi_n(x), \quad x \in \mathbb{R}^N.$$

- (i) Prove that $c_1 \text{dist}(x, \mathbb{R}^N \setminus V) \leq d_{\text{reg}}(x) \leq c_2 \text{dist}(x, \mathbb{R}^N \setminus V)$ for all $x \in V$ and for some constants $c_1, c_2 > 0$ depending only on N .
- (ii) Prove that $d_{\text{reg}} \in C^\infty(V)$.
- (iii) Prove that for every multi-index α ,

$$|\partial^\alpha d_{\text{reg}}(x)| \leq \frac{c_\alpha}{|d_{\text{reg}}(x)|^{|\alpha|-1}}$$

for all $x \in V$ and for some constant $c_\alpha > 0$ depending only on N and α .

Using the regularized distance we can show that special Lipschitz continuous domains of the form (13.1) are $W^{m,p}$ extension domains. The following results are due to Stein (see [220]).

Theorem 13.8 (Stein). *Let $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ be a Lipschitz continuous function and let*

$$\Omega := \{(x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : x_N > f(x')\}.$$

Then for all $m \in \mathbb{N}$ and $1 \leq p \leq \infty$ there exists a continuous linear operator $\mathcal{E} : W^{m,p}(\Omega) \rightarrow W^{m,p}(\mathbb{R}^N)$ such that for all $u \in W^{m,p}(\Omega)$, $\mathcal{E}(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$, and

$$(13.17) \quad \|\mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} \leq c \|u\|_{L^p(\Omega)},$$

$$(13.18) \quad \|\nabla^k \mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} \leq c(1 + (\text{Lip } f)^k) \|\nabla^k u\|_{L^p(\Omega)}$$

for every $1 \leq k \leq m$ and for some constant $c = c(m, N, p) > 0$.

For simplicity we give the proof for $m = 2$ and then indicate the changes in the general case $m \geq 2$.

Proof for $m = 2$. Step 1: Let $L := \text{Lip } f$. We claim that for every $x = (x', x_N) \in \mathbb{R}^N \setminus \overline{\Omega}$,

$$(13.19) \quad \frac{1}{\sqrt{1+L^2}}(f(x') - x_N) \leq \text{dist}(x, \overline{\Omega}) \leq f(x') - x_N.$$

Observe that $\text{dist}(x, \overline{\Omega}) = \text{dist}(x, \partial\Omega)$. Indeed, for every $y \in \Omega$, the segment of endpoints x and y must intersect $\partial\Omega$ at some point z and so $\|x - y\| \geq \|x - z\|$.

Next, given $y = (y', f(y')) \in \partial\Omega$, we have that

$$0 \leq f(x') - x_N = f(x') - f(y') + f(y') - x_N \leq L\|x' - y'\|_{N-1} + f(y') - x_N.$$

In turn,

$$\begin{aligned}
& (f(x') - x_N)^2 \\
& \leq L^2 \|x' - y'\|_{N-1}^2 + (f(y') - x_N)^2 + 2L \|x' - y'\|_{N-1} (f(y') - x_N) \\
& \leq L^2 \|x' - y'\|_{N-1}^2 + (f(y') - x_N)^2 + \|x' - y'\|_{N-1}^2 + L^2 (f(y') - x_N)^2 \\
& = (1 + L^2) (\|x' - y'\|_{N-1}^2 + (f(y') - x_N)^2) = (1 + L^2) \|x - y\|^2,
\end{aligned}$$

which shows that

$$\frac{1}{\sqrt{1+L^2}} (f(x') - x_N) \leq \|x - y\|$$

for all $y \in \partial\Omega$. Hence, the first inequality in (13.19) holds. The second inequality follows by observing that $(x', f(x')) \in \overline{\Omega}$. Thus, we have proved (13.19).

Using (13.6) and (13.19) for all $x \in \mathbb{R}^N \setminus \overline{\Omega}$ we obtain

$$(13.20) \quad 0 < \frac{2}{3\sqrt{1+L^2}} (f(x') - x_N) \leq d_{\text{reg}}(x) \leq 2(f(x') - x_N).$$

Define the function

$$(13.21) \quad d_1(x) := 3\sqrt{1+L^2} d_{\text{reg}}(x).$$

Then by (13.21) and (13.20),

$$(13.22) \quad 0 < 2(f(x') - x_N) \leq d_1(x) \leq 6\sqrt{1+L^2}(f(x') - x_N).$$

Moreover, by (13.7),

$$(13.23) \quad |d_1(x)|^{|\alpha|-1} |\partial^\alpha d_1(x)| \leq c_\alpha (1 + L^{|\alpha|})$$

for all $x \in \mathbb{R}^N \setminus \overline{\Omega}$ and for every multi-index α .

Step 2: Let $\phi : [1, \infty) \rightarrow \mathbb{R}$ be a continuous function such that

$$(13.24) \quad \int_1^\infty \phi(t) dt = 1, \quad \int_1^\infty t^n \phi(t) dt = 0, \quad \lim_{t \rightarrow \infty} t^n \phi(t) = 0$$

for all $n \in \mathbb{N}$. Given $u \in C_c^\infty(\mathbb{R}^N)$, let $v : \mathbb{R}^N \rightarrow \mathbb{R}$ be the function defined by

$$(13.25) \quad v(x) := \int_1^\infty \phi(t) u(g^t(x)) dt,$$

where

$$(13.26) \quad g^t(x) := (x', x_N + td_1(x)).$$

Note that in view of (13.22), if $x \in \mathbb{R}^N \setminus \overline{\Omega}$ and $t \geq 1$, then

$$x_N + td_1(x) \geq x_N + 2(f(x') - x_N) > x_N + f(x') - x_N = f(x'),$$

and so $x_t \in \Omega$. Moreover, if $x \in \partial\Omega$, then $d_1(x) = 0$ and so by (13.24)₁, $v(x) = u(x)$.

By Theorem B.52 and the facts that $u \in C_c^\infty(\mathbb{R}^N)$ and $d_{\text{reg}} \in C^\infty(\mathbb{R}^N \setminus \partial\Omega)$ (see Theorem 13.6), we have that $v \in C^\infty(\mathbb{R}^N \setminus \partial\Omega)$ with

$$(13.27) \quad \partial^\alpha v(x) = \int_1^\infty \phi(t) \partial^\alpha (u \circ g^t)(x) dt$$

for every multi-index $\alpha \in \mathbb{N}_0^N$ with $|\alpha| \leq 2$ and every $x \in \mathbb{R}^N \setminus \partial\Omega$. We claim that

$$(13.28) \quad \lim_{x \rightarrow x_0} \partial^\alpha v(x) = \partial^\alpha u(x_0) \quad \text{for every } x_0 \in \partial\Omega.$$

By (13.26), for all $i = 1, \dots, N$ and $x \in \mathbb{R}^N \setminus \partial\Omega$,

$$(13.29) \quad \partial_i(u \circ g^t)(x) = \partial_i u(g^t(x)) + \partial_N u(g^t(x)) t \partial_i d_1(x).$$

By (13.23), $|\partial_i d_1(x)| \leq c(1 + L)$ and so by the Lebesgue dominated convergence theorem, the fact that $d_{\text{reg}}(x) \rightarrow 0$ as $x \rightarrow x_0 \in \partial\Omega$, (13.24)₁ and (13.24)₂ with $n = 1, 2$, (13.27), and (13.29), $\partial_i v(x) \rightarrow \partial_i u(x_0)$ as $x \rightarrow x_0$.

The situation is more complicated for multi-indices of length greater than one. For all $i, j = 1, \dots, N$ and $x \in \mathbb{R}^N \setminus \partial\Omega$,

$$(13.30) \quad \begin{aligned} \partial_{j,i}^2(u \circ g^t) &= \partial_{j,i}^2 u \circ g^t + (\partial_{i,N}^2 u \circ g^t) t \partial_j d_1 + (\partial_{j,N}^2 u \circ g^t) t \partial_i d_1 \\ &\quad + (\partial_{N,N}^2 u \circ g^t) t^2 \partial_i d_1 \partial_j d_1 + (\partial_N u \circ g^t) t \partial_{j,i}^2 d_1. \end{aligned}$$

The first four terms can be treated as before using (13.24)₂ with $n = 1, 2$ and the fact that d_1 is a Lipschitz continuous function. However, the term $\partial_{j,i}^2 d_1$ is unbounded. By the fundamental theorem of calculus applied to the function $h(t) := \partial_N u(g^t(x))$, we have

$$(13.31) \quad \partial_N u(g^t(x)) = \partial_N u(x', x_N + d_1(x)) + d_1(x) \int_1^t \partial_{N,N}^2 u(x', x_N + s d_1(x)) ds.$$

Hence, by (13.24)₂ with $n = 1$,

$$(13.32) \quad \begin{aligned} \int_1^\infty \phi(t) \partial_N u(g^t(x)) t \partial_{j,i}^2 d_1(x) dt \\ = \int_1^\infty \phi(t) t d_1(x) \partial_{j,i}^2 d_1(x) \int_1^t \partial_{N,N}^2 u(x', x_N + s d_1(x)) ds dt. \end{aligned}$$

Since $u \in C_c^\infty(\mathbb{R}^N)$ and $d_1 \partial_{j,i}^2 d_1$ is bounded (see (13.23)), we can apply the Lebesgue dominated convergence theorem to show that the right-hand side of the previous integral goes to zero by (13.24) as $x \rightarrow x_0 \in \partial\Omega$. Thus, (13.28) holds. It follows that the function

$$(13.33) \quad \mathcal{E}(u)(x) := \begin{cases} v(x) & \text{for } x \in \mathbb{R}^N \setminus \Omega, \\ u(x) & \text{for } x \in \Omega, \end{cases}$$

is of class $C^2(\mathbb{R}^N)$ (why?).

Step 3: We will prove (13.17). By (13.24)₃ we have that $|\phi(t)| \leq ct^{-1}$ for all $t \geq 1$ and for some $c > 0$. Using the change of variables $r = x_N + d_1(x)t$,

from (13.25) we get

(13.34)

$$\begin{aligned} |v(x)| &\leq c \int_1^\infty \frac{1}{t} |u(x', x_N + d_1(x)t)| dt = c \int_{x_N + d_1(x)}^\infty \frac{1}{r - x_N} |u(x', r)| dr \\ &\leq c \int_{2f(x') - x_N}^\infty \frac{1}{r - f(x')} |u(x', r)| dr, \end{aligned}$$

where in the last inequality we used (13.22) and the fact that $r - x_N > r - f(x')$ for all $x \in \mathbb{R}^N \setminus \overline{\Omega}$. Raising both sides to the power p , changing variables, and using Hardy's inequality (see Theorem C.41) yield

$$\begin{aligned} \int_{-\infty}^{f(x')} |v(x', x_N)|^p dx_N &\leq c \int_{-\infty}^{f(x')} \left(\int_{2f(x') - x_N}^\infty \frac{1}{r - f(x')} |u(x', r)| dr \right)^p dx_N \\ &= c \int_0^\infty \left(\int_t^\infty \frac{1}{s} |u(x', s + f(x'))| ds \right)^p dt \\ &\leq c \int_0^\infty |u(x', t + f(x'))|^p dt = c \int_{f(x')}^\infty |u(x', x_N)|^p dx_N, \end{aligned}$$

where c changed from line to line, as usual. Integrating with respect to $x' \in \mathbb{R}^{N-1}$ gives (13.17).

Step 4: We show that

$$(13.35) \quad \|\nabla^k v\|_{L^p(\mathbb{R}^N \setminus \overline{\Omega})} \leq c(1 + L^k) \|\nabla^k u\|_{L^p(\Omega)}$$

for $k = 1, 2$ and for some $c > 0$ depending only on ϕ and p .

By (13.27), (13.29), and the fact that $\|\partial_i d_1\|_{L^\infty(\mathbb{R}^N)} \leq c(1 + L)$, for all $i = 1, \dots, N$, and $x \in \mathbb{R}^N \setminus \partial\Omega$, we have that

$$|\partial_i v(x)| \leq \int_1^\infty t |\phi(t)| |\partial_i u(g^t(x))| dt + c(1 + L) \int_1^\infty t |\phi(t)| |\partial_N u(g^t(x))| dt.$$

We may proceed as in the previous step with u replaced by $\partial_i u$ and $\partial_N u$, to obtain

$$\|\partial_i v\|_{L^p(\mathbb{R}^N \setminus \overline{\Omega})} \leq c \|\partial_i u\|_{L^p(\Omega)} + c(1 + L) \|\partial_N u\|_{L^p(\Omega)}.$$

Concerning the second-order derivatives of v , the first four terms on the right-hand side of (13.30) can be treated as in Step 3, again using the fact that $\|\partial_i d_1\|_{L^\infty(\mathbb{R}^N)} \leq c(1 + L)$ for all $i = 1, \dots, N$. Note that the fourth term will contribute to a factor of $(1 + L^2)$. It remains to study the last term on the right-hand side of (13.30). By (13.23), (13.32) and the fact that

$|\phi(t)| \leq ct^{-3}$ for all $t \geq 1$ (see (13.24)₃),

$$\begin{aligned} & \int_1^\infty |\phi(t) \partial_N u(g^t(x)) t \partial_{j,i}^2 d_1(x)| dt \\ & \leq c(1 + L^2) \int_1^\infty \frac{1}{t^2} \int_1^t |\partial_{N,N}^2 u(x', x_N + s d_1(x))| ds dt \\ & = c(1 + L^2) \int_1^\infty \frac{1}{s} |\partial_{N,N}^2 u(x', x_N + s d_1(x))| ds, \end{aligned}$$

where in the last equality we used Tonelli's theorem. Using the change of variables $r = x_N + s d_1(x)$ we can now continue as in (13.34). $\square$

We now indicate the main changes in the case $m \geq 2$.

Proof for $m \geq 2$. Step 1: We claim that (13.28) holds for every multi-index α , with $|\alpha| \leq m$. As in Step 2 of the previous proof,

$$(13.36) \quad \partial^\alpha v(x) = \int_1^\infty \phi(t) \partial^\alpha (u \circ g^t)(x) dt$$

for every $x \in \mathbb{R}^N \setminus \partial\Omega$. By Faa di Bruno's theorem (see Theorem 11.54),

$$(13.37) \quad \partial^\alpha (u \circ g^t)(x) = \sum c_{\alpha,\beta,\gamma,l} \partial^\beta u(g^t(x)) \prod_{i=1}^{|\beta|} \partial^{\gamma_i} g_{l_i}^t(x),$$

where $c_{\alpha,\beta,\gamma,l} \in \mathbb{R}$, the sum is done over all $\beta \in \mathbb{N}_0^N$ with $1 \leq |\beta| \leq |\alpha|$, $\gamma = (\gamma_1, \dots, \gamma_{|\beta|})$, $\gamma_i \in \mathbb{N}_0^N$, with $|\gamma_i| \geq 1$ and $\sum_{i=1}^{|\beta|} \gamma_i = \alpha$, and $l = (l_1, \dots, l_{|\beta|})$, $l_i \in \{1, \dots, N\}$, $i = 1, \dots, |\beta|$.

Since $g^t(x) = x + t d_1(x) e_N$, if $l_i < N$, then $\partial^{\gamma_i} g_{l_i}^t(x) = 0$ unless $\gamma_i = e_i$ in which case $\partial^{\gamma_i} g_{l_i}^t(x) = 1$, while $\partial^{\gamma_i} g_N^t(x) = \partial^{\gamma_i}(x_N) + t \partial^{\gamma_i} d_1(x)$. Moreover, if $|\beta| = |\alpha|$, then $|\gamma_i| = 1$ for all i and thus by (13.23) and the fact that $t \geq 1$,

$$(13.38) \quad \left| \partial^\beta u(g^t(x)) \prod_{i=1}^{|\alpha|} \partial^{\gamma_i} g_{l_i}^t(x) \right| \leq c(1 + L^{|\alpha|}) t^{|\alpha|} |\partial^\beta u(g^t(x))|.$$

In particular, if $\beta = \alpha$, $l_i = i$ and $\gamma_i = e_i$ for every $i = 1, \dots, N$, then the corresponding coefficient $c_{\alpha,\alpha,\gamma,l}$ is one. This can be shown by induction on $|\alpha|$ (see (13.29) and (13.30) for $|\alpha| = 1$ and $|\alpha| = 2$).

If $|\beta| < |\alpha|$ we apply Taylor's formula at $t = 1$ (see Theorem 9.9) to the function $h(t) := \partial^\beta u(g^t(x))$ to write

$$\begin{aligned} \partial^\beta u(g^t(x)) &= \sum_{n=0}^{\ell_\beta} \frac{(d_1(x))^n}{n!} \partial^{\beta+n e_N} u(x', x_N + d_1(x)) (t-1)^n \\ &\quad + \frac{(d_1(x))^{\ell_\beta}}{\ell_\beta!} \int_1^t (t-s)^{\ell_\beta} \partial^{\beta+\ell_\beta e_N} u(x', x_N + s d_1(x)) ds, \end{aligned}$$

where $\ell_\beta := |\alpha| - |\beta|$. By the binomial theorem and (13.24),

$$\int_1^\infty (t-1)^n t^k \phi(t) dt = \sum_{i=0}^n \binom{n}{i} (-1)^{n-i} \int_1^\infty t^{k+i} \phi(t) dt = 0,$$

and so

$$\begin{aligned} (13.39) \quad &\int_1^\infty t^k \phi(t) \partial^\beta u(g^t(x)) dt \\ &= \frac{(d_1(x))^{\ell_\beta}}{\ell_\beta!} \int_1^\infty t^k \phi(t) \int_1^t (t-s)^{\ell_\beta} \partial^{\beta+\ell_\beta e_N} u(x', x_N + s d_1(x)) ds dt. \end{aligned}$$

By (13.23) and the facts that $\ell_\beta = |\alpha| - |\beta| = \sum_{i=1}^{|\beta|} (|\gamma_i| - 1)$ and $t \geq 1$,

$$(13.40) \quad (d_1)^{\ell_\beta} \prod_{i=1}^{|\beta|} |\partial^{\gamma_i} g_{t_i}^t| \leq \prod_{i=1}^{|\beta|} c_i t (d_1)^{|\gamma_i|-1} (1 + |\partial^{\gamma_i} d_1|) \leq c(1 + L^{|\alpha|}) t^{|\beta|}.$$

Hence, by (13.24) and (13.36)–(13.40), as in the previous steps, we can apply the Lebesgue dominated convergence theorem to obtain

$$\lim_{x \rightarrow x_0} \partial^\alpha v(x) = \lim_{x \rightarrow x_0} \int_1^\infty \phi(t) \partial^\alpha u \circ g^t(x) dt = \partial^\alpha u(x_0),$$

which gives (13.28).

Step 2: Next we prove (13.35) for every multi-index α , with $|\alpha| \leq m$. If $|\beta| = |\alpha|$, using (13.38) and the fact that $|\phi(t)| \leq c t^{-|\alpha|-1}$ for all $t \geq 1$ (see (13.24)₃), we have

$$\begin{aligned} &\int_1^\infty \left| \phi(t) \partial^\beta u(g^t(x)) \prod_{i=1}^{|\beta|} \partial^{\gamma_i} g_{t_i}^t(x) \right| dt \\ &\leq c(1 + L^{|\alpha|}) \int_1^\infty \frac{1}{t} |\partial^\beta u(x', x_N + t d_1(x))| dt, \end{aligned}$$

while if $|\beta| < |\alpha|$, using (13.39), (13.40), and the fact that $|\phi(t)| \leq ct^{-|\alpha|-2}$ for all $t \geq 1$ we have

$$\begin{aligned}
 & \int_1^\infty \left| \phi(t) \partial^\beta u(g^t(x)) \prod_{i=1}^{|\beta|} \partial^{\gamma_i} g_{l_i}^t(x) \right| dt \\
 & \leq c(1 + L^{|\alpha|}) \int_1^\infty \frac{1}{t^{|\alpha|+2-|\beta|}} \int_1^t (t-s)^{|\alpha|-|\beta|} |\partial^{\beta+\ell_{\beta} e_N} u(x', x_N + sd_1(x))| ds dt \\
 & \leq c(1 + L^{|\alpha|}) \int_1^\infty \frac{1}{t^2} \int_1^t |\partial^{\beta+\ell_{\beta} e_N} u(x', x_N + sd_1(x))| ds dt \\
 & = c(1 + L^{|\alpha|}) \int_1^\infty \frac{1}{s} |\partial^{\beta+\ell_{\beta} e_N} u(x', x_N + sd_1(x))| ds,
 \end{aligned}$$

where the last equality follows from Tonelli's theorem. Using the change of variables $r = x_N + sd_1(x)$ we can now continue as in (13.34).

Step 3: We have proved that the linear operator $\mathcal{E} : C_c^\infty(\mathbb{R}^N) \rightarrow W^{m,p}(\mathbb{R}^N)$ defined by

$$\mathcal{E}(u)(x) := \begin{cases} v(x) & \text{for } x \in \mathbb{R}^N \setminus \Omega, \\ u(x) & \text{for } x \in \Omega, \end{cases}$$

satisfies (13.17) and (13.18). Since functions in $C_c^\infty(\mathbb{R}^N)$ are dense in $W^{m,p}(\Omega)$ (see Theorem 11.35) we can extend $\mathcal{E}$ to a linear operator defined on $W^{m,p}(\Omega)$ and with the same properties. $\square$

Remark 13.9. Note that the operator $\mathcal{E}$ does not depend on m and p .

Exercise 13.10. Prove that the function $\phi(t) := \frac{e}{\pi t} \operatorname{Im}(e^{-\lambda(t-1)^{1/4}})$, $t \geq 1$, where $\lambda := e^{i\pi/4}$ and $\operatorname{Im} z$ stands for the imaginary part of a complex number $z \in \mathbb{C}$, satisfies properties (13.24).

Next we extend the previous theorem to more general Lipschitz continuous domains.

Definition 13.11. The boundary $\partial\Omega$ of an open set $\Omega \subseteq \mathbb{R}^N$ is *uniformly Lipschitz continuous* if there exist $\varepsilon, L > 0$, $M \in \mathbb{N}$, and a locally finite countable open cover $\{\Omega_n\}_n$ of $\partial\Omega$ such that

- (i) if $x \in \partial\Omega$, then $B(x, \varepsilon) \subseteq \Omega_n$ for some $n \in \mathbb{N}$,
- (ii) no point of $\mathbb{R}^N$ is contained in more than M of the Ω_n 's,
- (iii) for each n there exist local coordinates $y = (y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R}$ and a Lipschitz continuous function $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ (both depending on n), with $\operatorname{Lip} f \leq L$, such that $\Omega_n \cap \Omega = \Omega_n \cap V_n$, where V_n is given in local coordinates by

$$\{(y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : y_N > f(y')\}.$$

Remark 13.12. Similarly, given $m \in \mathbb{N}_0$ and $0 < \alpha \leq 1$ we can define open sets $\Omega \subseteq \mathbb{R}^N$ whose boundary is *uniformly of class C^m* (respectively, *of class $C^{m,\alpha}$* , see Definition 11.55) with parameters $\varepsilon, L > 0, M$ provided (i), (ii), and (iii) hold but with f of class C^m (respectively of class $C^{m,\alpha}$) and with $\|f\|_{C^m(\mathbb{R}^{N-1})} \leq L$ (respectively, $\|f\|_{C^{m,\alpha}(\mathbb{R}^{N-1})} \leq L$).

Exercise 13.13. Let $\Omega \subseteq \mathbb{R}^N$ be an open set such that $\partial\Omega$ is bounded. Prove that $\partial\Omega$ is uniformly Lipschitz continuous if and only if it is Lipschitz continuous.

Exercise 13.14. Let $\Omega \subseteq \mathbb{R}$ be such that $\Omega = \bigcup_n I_n$, where the I_n are open intervals such that $\text{length } I_n \geq \delta$ for all $n \in \mathbb{N}$ and $\text{dist}(I_n, I_k) \geq \delta$ for all $n, k \in \mathbb{N}$ with $n \neq k$ and for some $\delta > 0$.

- (i) Prove that $\partial\Omega$ is uniformly Lipschitz continuous.
- (ii) Prove that the condition $\text{length } I_n \geq \delta$ for all $n \in \mathbb{N}$ is necessary to have an extension operator from $W^{1,1}(\Omega)$ to $W^{1,p}(\mathbb{R})$.
- (iii) Prove that the condition $\text{dist}(I_n, I_k) \geq \delta$ for all $n, k \in \mathbb{N}$ is necessary to have an extension operator from $W^{1,\infty}(\Omega)$ to $W^{1,\infty}(\mathbb{R})$.

In the next exercise, by a *finite cone* having vertex $v \in \mathbb{R}^N$, length ℓ , and vertex angle $2\alpha \in (0, \pi)$ we mean a set of the form $K_{v,R} := v + RK$, where R is an orthogonal $N \times N$ matrix and

$$K := \{(x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : x_N > x' \cot \alpha\} \cap B(0, \ell).$$

Exercise 13.15. Let $\Omega \subseteq \mathbb{R}^N$ be an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous.

- (i) Prove that for every $x \in \partial\Omega$ there exists a finite cone $K_{x,R}$, with vertex x , length ε , and vertex angle depending on L , such that $K_{x,R} \setminus \{x\} \subseteq \Omega$.
- (ii) Prove that if Ω is unbounded, then $\mathcal{L}^N(\Omega) = \infty$.

Exercise 13.16. Prove that for every $x = (x_1, \dots, x_N) \in \mathbb{R}^N$ and for every $n \in \mathbb{N}$,

$$(x_1 + \dots + x_N)^n = \sum_{|\alpha|=n} \frac{n!}{\alpha!} x^\alpha.$$

The next theorem shows that domains with uniformly Lipschitz continuous boundary are extension domains for $W^{m,p}(\Omega)$ for all $1 \leq p \leq \infty$.

Theorem 13.17 (Stein). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set with uniformly Lipschitz continuous boundary. Then for all $1 \leq p \leq \infty$ and $m \in \mathbb{N}$ there exists a continuous linear operator $\mathcal{E} : W^{m,p}(\Omega) \rightarrow W^{m,p}(\mathbb{R}^N)$ such that for all*

$u \in W^{m,p}(\Omega)$, $\mathcal{E}(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$, and

$$(13.41) \quad \|\mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} \leq c(1+M)\|u\|_{L^p(\Omega)},$$

$$(13.42) \quad \|\nabla^k \mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} \leq c(1+M(1+L^k)) \sum_{i=0}^k (M/\varepsilon)^{k-i} \|\nabla^i u\|_{L^p(\Omega)}$$

for every multi-index $\alpha \in \mathbb{N}_0^N$ with $1 \leq |\alpha| \leq m$.

For simplicity we first prove the case $m = 1$ and then treat the general case.

Proof for $m = 1$. Step 1: We only prove the case $1 \leq p < \infty$ and leave the easier case $p = \infty$ as an exercise. For every set $E \subseteq \mathbb{R}^N$ and every $r > 0$, we define $E^r := \{x \in \mathbb{R}^N : B(x, r) \subseteq E\}$. We observe that $E^r \subseteq E$ and that condition (i) in Definition 13.11 reads $\partial\Omega \subset \bigcup_n \Omega_n^\varepsilon$. Define the regularized functions

$$(13.43) \quad \phi_n := \varphi_{\varepsilon/4} * \chi_{\Omega_n^{3\varepsilon/4}},$$

where $\varphi_{\varepsilon/4}$ is a standard mollifier. Then

$$(13.44) \quad \text{supp } \phi_n \subset \Omega_n, \quad \phi_n = 1 \text{ in } \Omega_n^{\varepsilon/2}.$$

By Theorem C.20 for every multi-index $\alpha \in \mathbb{N}_0^N$ we have that

$$\partial^\alpha \phi_n = \partial^\alpha \varphi_{\varepsilon/4} * \chi_{\Omega_n^{3\varepsilon/4}},$$

and so for all $x \in \Omega_n$ we have that

$$(13.45) \quad |\partial^\alpha \phi_n(x)| \leq \int_{\mathbb{R}^N} |\partial^\alpha \varphi_{\varepsilon/4}(y-x)| dy \leq c_\alpha \varepsilon^{-|\alpha|}.$$

Next consider the three open sets

$$(13.46) \quad \begin{aligned} \Omega_0 &:= \{x \in \mathbb{R}^N : \text{dist}(x, \Omega) < \varepsilon/4\}, \\ \Omega_+ &:= \{x \in \mathbb{R}^N : \text{dist}(x, \partial\Omega) < 3\varepsilon/4\}, \\ \Omega_- &:= \{x \in \Omega : \text{dist}(x, \partial\Omega) > \varepsilon/4\}, \end{aligned}$$

and define the regularized functions

$$(13.47) \quad \phi_0 := \varphi_{\varepsilon/4} * \chi_{\Omega_0}, \quad \phi_\pm := \varphi_{\varepsilon/4} * \chi_{\Omega_\pm}.$$

Then $\phi_0 = 1$ in $\overline{\Omega}$, $\phi_+(x) = 1$ if $x \in \mathbb{R}^N$ and $\text{dist}(x, \partial\Omega) \leq \varepsilon/2$, and $\phi_-(x) = 1$ if $x \in \Omega$ and $\text{dist}(x, \partial\Omega) \geq \varepsilon/2$. Moreover, the supports of ϕ_0 , ϕ_+ , and ϕ_- are contained, respectively, in an $\varepsilon/2$ neighborhood of Ω , in an ε neighborhood of $\partial\Omega$, and in Ω . Finally, reasoning as in (13.45), we have that

$$(13.48) \quad \|\partial^\alpha \phi_0\|_\infty \leq c\varepsilon^{-|\alpha|}, \quad \|\partial^\alpha \phi_\pm\|_\infty \leq c\varepsilon^{-|\alpha|}.$$

Note that

$$(13.49) \quad \text{supp } \phi_0 \subseteq \{x \in \mathbb{R}^N : \phi_+(x) + \phi_-(x) \geq 1\}.$$

Thus, with a slight abuse of notation, we may define

$$(13.50) \quad \psi_+ := \phi_0 \frac{\phi_+}{\phi_+ + \phi_-}, \quad \psi_- := \phi_0 \frac{\phi_-}{\phi_+ + \phi_-},$$

where we interpret the right-hand sides to be zero whenever $\phi_0 = 0$. By (13.48) and (13.49) we have that all the derivatives of order α of $\psi_{\pm}$ are bounded by $c|\varepsilon|^{-|\alpha|}$. Also, $\psi_+ + \psi_- = 1$ in $\bar{\Omega}$ and $\psi_+ = \psi_- = 0$ outside an $\varepsilon/2$ neighborhood of Ω .

We are finally ready to construct the linear extension operator. Given a function $u \in C_c^\infty(\mathbb{R}^N)$, $1 \leq p < \infty$, since $\text{supp}(\phi_n u) \subset \Omega_n$ for each n by (13.44), by condition (i) in Definition 13.11 and Theorem 13.8 we can extend $\phi_n u$ to a function $v_n \in W^{m,p}(\mathbb{R}^N)$ in such a way that

$$(13.51) \quad \begin{aligned} \|v_n\|_{L^p(\mathbb{R}^N)} &= 2\|\phi_n u\|_{L^p(\Omega \cap \Omega_n)}, \\ \|\nabla^k v_n\|_{L^p(\mathbb{R}^N)} &\leq c(1 + L^k)\|\nabla^k(\phi_n u)\|_{L^p(\Omega \cap \Omega_n)} \end{aligned}$$

for every multi-index k with $1 \leq k \leq m$. Again with a slight abuse of notation we define

$$(13.52) \quad \mathcal{E}(u)(x) := \psi_+(x) \frac{\sum \phi_n(x) v_n(x)}{\sum_k \phi_k^2(x)} + \psi_-(x) u(x), \quad x \in \mathbb{R}^N.$$

Note that if $x \in \mathbb{R}^N$ is such that $\text{dist}(x, \partial\Omega) \leq \varepsilon/2$, then there exists an n such that $y \in \Omega_n^{\varepsilon/2}$, and so $\phi_n(x) = 1$ by (13.44). In particular, since all the functions ϕ_n are nonnegative, it follows that

$$(13.53) \quad \text{if } x \in \text{supp } \psi_+, \text{ then } \sum_k \phi_k^2(x) \geq 1,$$

and thus the first term on the right-hand side of (13.52) is well-defined, provided we interpret it to be zero whenever $\psi_+ = 0$. Similarly, since $\text{supp } \psi_- \subseteq \text{supp } \phi_- \subset \Omega$, the term $\psi_-(x)u(x)$ is well-defined, provided we set it to be zero outside Ω .

It remains to show that the linear operator $\mathcal{E}(u)$ is an extension operator and that it is bounded. For the former, it suffices to observe that if $x \in \Omega$, then $v_n(x) = \phi_n(x)u(x)$, and so

$$\mathcal{E}(u)(x) = \psi_+(x)u(x) + \psi_-(x)u(x) = u(x)$$

by (13.50). To prove that $\mathcal{E}(u)$ is bounded in $L^p(\mathbb{R}^N)$, we observe that by (13.52), condition (ii) in Definition 13.11, Exercise 12.34, (13.53), the fact

that $0 \leq \psi_{\pm}, \phi_n \leq 1$, and (13.51), in this order, we have

$$\begin{aligned}
 \|\mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} &\leq M^{1/p'} \left(\sum_n \int_{\Omega_n} |v_n|^p dx \right)^{1/p} + \|u\|_{L^p(\Omega)} \\
 (13.54) \quad &\leq cM^{1/p'} \left(\int_{\Omega} |u|^p \sum_n |\phi_n|^p dx \right)^{1/p} + \|u\|_{L^p(\Omega)} \\
 &\leq c(1+M)\|u\|_{L^p(\Omega)},
 \end{aligned}$$

where in the last inequality we have used the fact that $\sum_n |\phi_n|^p \leq M$.

Step 2: In this step we will estimate $\partial_i \mathcal{E}(u)$. Since $\{\Omega_n\}_n$ is locally finite, any small neighborhood of every point $x \in \mathbb{R}^N$ intersects only finitely many Ω_n 's. Hence, by (13.52) we can calculate

$$\begin{aligned}
 (13.55) \quad \partial_i \mathcal{E}(u) &= \frac{\sum_n \phi_n v_n}{\sum_k \phi_k^2} \partial_i \psi_+ + \psi_+ \frac{\sum_n (v_n \partial_i \phi_n + \phi_n \partial_i v_n)}{\sum_k \phi_k^2} \\
 &\quad - 2\psi_+ \frac{\left(\sum_n \phi_n v_n \right) \left(\sum_l \phi_l \partial_i \phi_l \right)}{\left(\sum_k \phi_k^2 \right)^2} + u \partial_i \psi_- + \psi_- \partial_i u.
 \end{aligned}$$

Using (13.51)–(13.53), condition (ii) in Definition 13.11, Exercise 12.34, and the facts that $0 \leq \psi_{\pm}, \phi_n \leq 1$, $\|\partial_i \psi_{\pm}\|_{\infty}, \|\partial_i \phi_n\|_{\infty} \leq c\varepsilon^{-1}$, and

$$(13.56) \quad \left| \sum_n \phi_l \partial_i \phi_l \right| \leq cM\varepsilon^{-1}, \quad \sum_n |\phi_n|^p \leq M, \quad \sum_n |\partial_i \phi_n|^p \leq cM\varepsilon^{-p},$$

we obtain

$$\begin{aligned}
 \|\nabla \mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} &\leq cM^{1/p'} \left(\sum_n \int_{\Omega_n} (M^p \varepsilon^{-p} |v_n|^p + \|\nabla v_n\|^p) dx \right)^{1/p} \\
 &\quad + c\varepsilon^{-1} \|u\|_{L^p(\Omega)} + c\|\nabla u\|_{L^p(\Omega)} \\
 &\leq cM^{1/p'} (1+L) \left(\sum_n \int_{\Omega} (M^p \varepsilon^{-p} |\phi_n u|^p + \|\nabla(\phi_n u)\|^p) dx \right)^{1/p} \\
 &\quad + c\varepsilon^{-1} \|u\|_{L^p(\Omega)} + c\|\nabla u\|_{L^p(\Omega)} \\
 &\leq c(1+M(1+L))(M\varepsilon^{-1} \|u\|_{L^p(\Omega)} + \|\nabla u\|_{L^p(\Omega)}).
 \end{aligned}$$

To conclude the proof, it suffices to observe that since $\{\Omega_n\}_n$ is locally finite, in a neighborhood of every point the infinite sum in (13.52) is finite. $\square$

We now turn to the proof for $m \geq 2$.

Proof for $m \geq 2$. It remains to estimate $\|\nabla^k \mathcal{E}(u)\|_{L^p(\mathbb{R}^N)}$ for $k \geq 2$. Let $\alpha \in \mathbb{N}_0^N$ be a multi-index with $|\alpha| \geq 2$. We begin with $\partial^\alpha \psi_\pm$. By Exercise 11.37,

$$\begin{aligned} \partial^\alpha \psi_\pm &= \partial^\alpha (\phi_0 \phi_\pm (\phi_+ + \phi_-)^{-1}) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta (\phi_0 \phi_\pm) \partial^{\alpha-\beta} (\phi_+ + \phi_-)^{-1} \\ &= \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \sum_{\delta \leq \beta} \binom{\beta}{\delta} \partial^\delta \phi_0 \partial^{\beta-\delta} \phi_\pm \partial^{\alpha-\beta} (\phi_+ + \phi_-)^{-1}, \end{aligned}$$

while by Theorem 11.54 with u replaced by $t \mapsto t^{-1}$ and Ψ by $\phi_+ + \phi_-$,

$$\partial^{\alpha-\beta} (\phi_+ + \phi_-)^{-1} = \sum a_{\alpha-\beta, l, \gamma} (\phi_+ + \phi_-)^{-l-1} \prod_{i=1}^l \partial^{|\gamma_i|} (\phi_+ + \phi_-),$$

where $a_{\alpha-\beta, l, \gamma} \in \mathbb{R}$, the sum is done over all $l \in \mathbb{N}_0$ with $1 \leq l \leq |\alpha| - |\beta|$, $\gamma = (\gamma_1, \dots, \gamma_l)$, $\gamma_i \in \mathbb{N}_0^N$, with $|\gamma_i| \geq 1$ and $\sum_{i=1}^l \gamma_i = \alpha - \beta$. Using the facts that $\beta \leq \alpha$, $\delta \leq \beta$, $\sum_{i=1}^l \gamma_i = \alpha - \beta$, and (13.48), we have that

$$(13.57) \quad \|\partial^\alpha \psi_\pm\|_\infty \leq c\varepsilon^{-|\alpha|}.$$

By (13.52) and Exercise 11.37,

$$\begin{aligned} (13.58) \quad \partial^\alpha \mathcal{E}(u) &= \sum_n \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta v_n \partial^{\alpha-\beta} \left(\psi_+ \phi_n \left(\sum_k \phi_k^2 \right)^{-1} \right) \\ &\quad + \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta u \partial^{\alpha-\beta} \psi_- \end{aligned}$$

and

$$\begin{aligned} \partial^{\alpha-\beta} \left(\psi_+ \phi_n \left(\sum_k \phi_k^2 \right)^{-1} \right) &= \sum_{\delta \leq \alpha-\beta} \binom{\alpha-\beta}{\delta} \partial^{\alpha-\beta-\delta} (\psi_+ \phi_n) \partial^\delta \left(\sum_k \phi_k^2 \right)^{-1} \\ &= \sum_{\delta \leq \alpha-\beta} \binom{\alpha-\beta}{\delta} \sum_{\eta \leq \alpha-\beta-\delta} \binom{\alpha-\beta-\delta}{\eta} \partial^\eta \psi_+ \partial^{\alpha-\beta-\delta-\eta} \phi_n \partial^\delta \left(\sum_k \phi_k^2 \right)^{-1}, \end{aligned}$$

while by Theorem 11.54 with u replaced by $t \mapsto t^{-1}$ and Ψ by $\sum_k \phi_k^2$,

$$\begin{aligned} \partial^\delta \left(\sum_k \phi_k^2 \right)^{-1} &= \sum b_{\delta, l, \gamma} \left(\sum_k \phi_k^2 \right)^{-l-1} \prod_{i=1}^l \partial^{\gamma_i} \left(\sum_k \phi_k^2 \right) \\ &= \sum b_{\delta, l, \gamma} \left(\sum_k \phi_k^2 \right)^{-l-1} \prod_{i=1}^l \sum_k \sum_{\alpha_i \leq \gamma_i} \binom{\gamma_i}{\alpha_i} \partial^{\alpha_i} \phi_k \partial^{\gamma_i - \alpha_i} \phi_k, \end{aligned}$$

where $b_{\delta, l, \gamma} \in \mathbb{R}$ are constant, the sum is done over all $l \in \mathbb{N}_0$ with $1 \leq l \leq |\delta|$, $\gamma = (\gamma_1, \dots, \gamma_l)$, $\gamma_i \in \mathbb{N}_0^N$, with $|\gamma_i| \geq 1$ and $\sum_{i=1}^l \gamma_i = \delta$.

Using (13.45) and (13.53), the fact that for each $x \in \mathbb{R}^N$ at most M terms in the sum are different from zero (see condition (ii) in Definition 13.11), Exercise 12.34, and the facts that $0 \leq \psi_{\pm}, \phi_n \leq 1$, $\left\| \partial^{\delta} \left(\sum_k \phi_k^2 \right)^{-1} \right\|_{\infty} \leq cM^{|\delta|} \varepsilon^{-|\delta|}$, and so by (13.45) and (13.57),

$$\left\| \partial^{\alpha-\beta} \left(\psi_+ \phi_n \left(\sum_k \phi_k^2 \right)^{-1} \right) \right\|_{\infty} \leq cM^{|\alpha|-|\beta|} \varepsilon^{-(|\alpha|-|\beta|)}.$$

Hence, by (13.58),

$$\|\nabla^k \mathcal{E}(u)\| \leq c \sum_n \sum_{i=0}^k M^{k-i} \varepsilon^{-(k-i)} \|\nabla^i v_n\| + c \sum_{i=0}^k \varepsilon^{-(k-i)} \|\nabla^i u\|.$$

By Exercise 12.34 and the fact that for each $x \in \mathbb{R}^N$ at most M terms in the sum are different from zero (see condition (ii) in Definition 13.11), we have

$$\begin{aligned} \|\nabla^k \mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} &\leq cM^{1/p'} \left(\sum_n \sum_{i=0}^k M^{(k-i)p} \varepsilon^{-(k-i)p} \int_{\Omega_n} \|\nabla^i v_n\|^p dx \right)^{1/p} \\ &\quad + c \sum_{i=0}^k \varepsilon^{-(k-i)} \|\nabla^i u\|_{L^p(\Omega)}. \end{aligned}$$

Finally, by (13.51), if $i = 0$,

$$\begin{aligned} \sum_n \int_{\Omega_n} |v_n|^p dx &\leq c \sum_n \int_{\Omega \cap \Omega_n} |\phi_n u|^p dx \leq c \int_{\Omega} \sum_n |\phi_n u|^p dx \\ &\leq cM \int_{\Omega} |u|^p dx, \end{aligned}$$

where we have used the facts that $\text{supp } \phi_n \subseteq \Omega_n$ and that $\sum_n |\phi_n|^p \leq M$. Similarly, if $k \geq 1$, by (13.51),

$$\begin{aligned} \sum_n \int_{\Omega_n} \|\nabla^k v_n\|^p dx &\leq c(1 + L^k)^p \int_{\Omega} \sum_n \|\nabla^k(\phi_n u)\|^p dx \\ &\leq cM(1 + L^k)^p \sum_{i=1}^k \varepsilon^{-(k-i)p} \int_{\Omega} \|\nabla^i u\|^p dx, \end{aligned}$$

where we have used Exercise 11.37 and the fact that $\sum_n \|\nabla^{k-i} \phi_n\|^p \leq cM \varepsilon^{-(k-i)p}$ to estimate

$$\sum_n \|\nabla^k(\phi_n u)\|^p \leq c \sum_n \sum_{i=1}^k \|\nabla^{k-i} \phi_n\|^p \|\nabla^i u\|^p \leq cM \sum_{i=1}^k \varepsilon^{-(k-i)p} \|\nabla^i u\|^p.$$

In turn,

$$\begin{aligned} \|\nabla^k \mathcal{E}(u)\|_{L^p(\mathbb{R}^N)} &\leq cM(1 + L^k) \sum_{i=0}^k M^{k-i} \varepsilon^{-(k-i)} \|\nabla^i u\|_{L^p(\Omega)} \\ &\quad + c \sum_{i=0}^k \varepsilon^{-(k-i)} \|\nabla^i u\|_{L^p(\Omega)}. \end{aligned}$$

Thus we have shown (13.41) and (13.42) for $u \in C_c^\infty(\mathbb{R}^N)$. Since functions in $C_c^\infty(\mathbb{R}^N)$ are dense in $W^{m,p}(\Omega)$, (see Theorem 11.35), we can now extend $\mathcal{E}$ to a linear operator defined on $W^{m,p}(\Omega)$ and with the properties (13.41) and (13.42). $\square$

Remark 13.18. Using Poincaré's inequality (see the next section), in the case of bounded Lipschitz continuous domains and $m = 1$ one can show that in the inequality (13.42) it is possible to drop the term $M\varepsilon^{-1}\|u\|_{L^p(\Omega)}$ (see Exercise 13.33 below) at the price of replacing $\mathbb{R}^N$ with any large ball containing Ω .

13.2. Poincaré Inequalities

The next result provides an equivalent norm in $W_0^{m,p}(\Omega)$ for a large class of domains Ω . We recall that $\nabla^0 u := u$.

Theorem 13.19 (Poincaré's inequality in $W_0^{m,p}(\Omega)$). *Let $\Omega \subset \mathbb{R}^N$ be an open set with finite width, that is, Ω lies between two parallel hyperplanes, and let $m \in \mathbb{N}$ and $1 \leq p < \infty$. Then there exists a constant $c = c(m, N, p) > 0$ such that for all $u \in W_0^{m,p}(\Omega)$ and $0 \leq k \leq m - 1$,*

$$\int_{\Omega} \|\nabla^k u(x)\|^p dx \leq c \frac{d^{p(m-k)}}{p^{m-k}} \int_{\Omega} \|\nabla^m u(x)\|^p dx,$$

where d is the distance between the two hyperplanes. In particular, if $m = 1$, for all $u \in W_0^{1,p}(\Omega)$,

$$\int_{\Omega} |u(x)|^p dx \leq c \frac{d^p}{p} \int_{\Omega} \|\nabla u(x)\|^p dx.$$

Proof. Without loss of generality, up to a rotation and translation, we may assume that Ω lies between the two parallel hyperplanes $x_N = 0$ and $x_N = d > 0$. For $u \in C_c^\infty(\Omega)$, by the fundamental theorem of calculus and Hölder's inequality, we have

$$\begin{aligned} |u(x', x_N)| &= |u(x', x_N) - u(x', 0)| = \left| \int_0^{x_N} \partial_N u(x', t) dt \right| \\ &\leq x_N^{1/p'} \left(\int_0^d |\partial_N u(x', t)|^p dt \right)^{1/p}. \end{aligned}$$

Extend u to be zero outside $\mathbb{R}^N \setminus \Omega$. Raising to the power p and integrating over $\mathbb{R}^{N-1} \times [0, d]$, by Tonelli's theorem we get

$$\begin{aligned} \int_{\Omega} |u(x)|^p dx &= \int_{\mathbb{R}^{N-1} \times [0, d]} |u(x', x_N)|^p dx \\ &\leq \int_{\mathbb{R}^{N-1}} \int_0^d x_N^{p/p'} \int_0^d |\partial_N u(x', t)|^p dt dx_N dx' \\ &= \int_{\Omega} |\partial_N u(y)|^p dy \int_0^d x_N^{p-1} dx_N = \frac{d^p}{p} \int_{\Omega} |\partial_N u(y)|^p dy. \end{aligned}$$

This proves the case $m = 1$. If $m \geq 2$ and α is a multi-index with $0 \leq |\alpha| < m - 1$, by repeated applications of the case $m = 1$ with u replaced by $\partial^{\alpha + ne_N} u(x)$, $0 \leq n \leq m - |\alpha| - 1$, we get

$$\begin{aligned} \int_{\Omega} |\partial^{\alpha} u(x)|^p dx &\leq \frac{d^p}{p} \int_{\Omega} |\partial^{\alpha + e_N} u(x)|^p dx \\ &\leq \dots \leq \frac{d^{p(m-|\alpha|)}}{p^{m-|\alpha|}} \int_{\Omega} |\partial^{\alpha + e_N} u(x)|^p dx. \end{aligned}$$

This concludes the proof. $\square$

Remark 13.20. It follows from the previous theorem that if an open set $\Omega \subset \mathbb{R}^N$ has finite width, then the seminorm $\|\nabla^m u\|_{L^p(\Omega)}$ is actually an equivalent norm in the space $W_0^{m,p}(\Omega)$.

Exercise 13.21. Show that if the open set $\Omega \subseteq \mathbb{R}^N$ contains a sequence of balls $B(x_n, r_n)$, where $x_n \in \mathbb{R}^N$ and $r_n \rightarrow \infty$, then the previous Poincaré inequality fails.

Exercise 13.22. Let $\Omega \subset \mathbb{R}^N$ be an open bounded set and let $u \in W_0^{1,p}(\Omega)$, $1 \leq p \leq \infty$. Prove that

$$\int_{\Omega} |u(x)|^p dx \leq \frac{p^p}{N^p} \inf_{x_0 \in \mathbb{R}^N} \sup_{y \in \Omega} \|y - x_0\| \int_{\Omega} \|\nabla u(x)\|^p dx.$$

Hint: Write $N = \operatorname{div}(x - x_0)$ and use the divergence theorem.

Exercise 13.23. Let $\Omega \subset \mathbb{R}^N$ be an open bounded set, let $m \in \mathbb{N}$ and $1 < p < \infty$ be such $mp = N$.

- (i) Prove that there exist two constants $c_1, c_2 > 0$ depending only on m and N such that for all $u \in W_0^{1,N}(\Omega) \setminus \{0\}$,

$$(13.59) \quad \int_{\Omega} \exp\left(c_1 \frac{|u(x)|^{p'}}{\|u\|_{W^{m,p}(\mathbb{R}^N)}^{p'}}\right) dx \leq c_2 \mathcal{L}^N(\Omega).$$

- (ii) Prove that when $m = 1$ the previous inequality fails if $c_1 > \gamma_N$, where γ_N is the constant given in (12.29).

Remark 13.24. The inequality (13.59) was established by Pokhozhaev [189], Trudinger [234] and Yudovich [250] when $m = 1$ and by Strichartz [225] for $m \geq 2$. In [179], Moser proved that for bounded domains the inequality (13.59) also holds for $\gamma = \gamma_N$. The proof is not immediate and we refer to [179] for more details. See also the recent paper of Li and Ruf [141] and the bibliography contained therein for some recent results on unbounded domains.

Exercise 13.25. Assume that the open set $\Omega \subset \mathbb{R}^N$ has finite width, let $m \in \mathbb{N}$, and let $1 \leq p < \infty$. Prove that for every functional $L \in W^{-m,p'}(\Omega)$ there exist $f_\alpha \in L^{p'}(\Omega)$, $\alpha \in \mathbb{N}_0^N$, with $|\alpha| = m$, such that $L(u) = \int_\Omega \sum_{|\alpha|=m} \partial^\alpha u(x) f_\alpha(x) dx$ for all $u \in W_0^{m,p}(\Omega)$.

Another form of Poincaré's inequality is

$$\int_\Omega |u(x) - u_E|^p dx \leq c \int_\Omega \|\nabla u(x)\|^p dx$$

for $u \in W^{1,p}(\Omega)$, where $E \subseteq \Omega$ is a Lebesgue measurable set with finite positive measure and

$$(13.60) \quad u_E := \frac{1}{\mathcal{L}^N(E)} \int_E u(x) dx.$$

Exercise 13.26. Let $\Omega \subset \mathbb{R}^N$ be an open bounded set, let $E \subseteq \Omega$ be a Lebesgue measurable set with finite positive measure, let $m \in \mathbb{N}$, and let $1 \leq p \leq \infty$. Prove that given $u \in W^{m,p}(\Omega)$ there exists a polynomial $p_E(u)$ of degree $m-1$ such that for every multi-index $\alpha \in \mathbb{N}_0^N$, with $0 \leq |\alpha| \leq m-1$,

$$(13.61) \quad \int_E (\partial^\alpha u(x) - \partial^\alpha p_E(u)(x)) dx = 0.$$

Theorem 13.27 (Poincaré's inequality in $W^{m,p}(\Omega)$). *Let $m \in \mathbb{N}$, let $1 \leq p < \infty$, and let $\Omega \subset \mathbb{R}^N$ be a connected extension domain for $W^{m,p}(\Omega)$ with finite measure. Let $E \subseteq \Omega$ be a Lebesgue measurable set with positive measure. Then there exists a constant $c = c(m, N, p, \Omega, E) > 0$ such that for all $u \in W^{m,p}(\Omega)$,*

$$\sum_{k=0}^{m-1} \|\nabla^k(u - p_E(u))\|_{L^p(\Omega)} \leq c \|\nabla^m u\|_{L^p(\Omega)}.$$

In particular, if $m = 1$, for all $u \in W^{1,p}(\Omega)$,

$$\|u - u_E\|_{L^p(\Omega)} \leq c \|\nabla u\|_{L^p(\Omega)}.$$

Proof. Assume by contradiction that the result is false. Then we may find a sequence $\{u_n\}_n$ in $W^{m,p}(\Omega)$ such that

$$\sum_{k=0}^{m-1} \|\nabla^k(u_n - p_E(u_n))\|_{L^p(\Omega)} > n \|\nabla^m u_n\|_{L^p(\Omega)}.$$

Define

$$v_n := \frac{u_n - p_E(u_n)}{\|u_n - p_E(u_n)\|_{W^{m-1,p}(\Omega)}},$$

where $\|\cdot\|_{W^{0,p}(\Omega)} := \|\cdot\|_{L^p(\Omega)}$. Then $v_n \in W^{m,p}(\Omega)$ and by (13.61),

$$(13.62) \quad \|v_n\|_{W^{m-1,p}(\Omega)} = 1, \quad p_E(v_n) = 0, \quad \|\nabla^m v_n\|_{L^p(\Omega)} \leq 1/n.$$

By the Rellich–Kondrachov theorem (Theorems 12.18, 12.44 and Exercise 12.63) there exists a subsequence $\{v_{n_k}\}_k$ such that $v_{n_k} \rightarrow v$ in $W^{m-1,p}(\Omega)$ for some function $v \in W^{m-1,p}(\Omega)$ with $\|v\|_{W^{m-1,p}(\Omega)} = 1$, $p_E(v) = 0$. Moreover, for every $\psi \in C_c^\infty(\Omega)$ and every multi-index β with $|\beta| = m$, by Hölder's inequality and (13.62),

$$\begin{aligned} \left| \int_{\Omega} v \partial^\beta \psi \, dx \right| &= \lim_{k \rightarrow \infty} \left| \int_{\Omega} v_{n_k} \partial^\beta \psi \, dx \right| = \lim_{k \rightarrow \infty} \left| \int_{\Omega} \psi \partial^\beta v_{n_k} \, dx \right| \\ &\leq \lim_{k \rightarrow \infty} \|\partial^\beta v_{n_k}\|_{L^p(\Omega)} \|\psi\|_{L^{p'}(\Omega)} = 0, \end{aligned}$$

and so $v \in W^{m,p}(\Omega)$ with $\partial^\beta v = 0$. Since Ω is connected, this implies that v is a polynomial of degree $m-1$, but since $p_E(v) = 0$, by (13.61) necessarily $v = 0$. This contradicts the fact that $\|v\|_{W^{m-1,p}(\Omega)} = 1$ and completes the proof. $\square$

Remark 13.28. In view of Theorem 13.17, it follows from the previous theorem that Poincaré's inequality holds for uniformly Lipschitz domains with finite measure.

Exercise 13.29. Let $1 \leq p < N$ and let $\Omega \subset \mathbb{R}^N$ be a connected extension domain for $W^{1,p}(\Omega)$ with finite measure. Let $E \subseteq \Omega$ be a Lebesgue measurable set with positive measure. Prove that there exists a constant $c = c(N, p, \Omega, E) > 0$ such that for all $u \in W^{1,p}(\Omega)$, $\|u - u_E\|_{L^{p^*}(\Omega)} \leq c \|\nabla u\|_{L^p(\Omega)}$.

Exercise 13.30. Let $\Omega \subset \mathbb{R}^N$ be an open connected set with finite measure and let $1 \leq p < \infty$. Assume that the embedding $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ is compact and prove that there exists a constant $c = c(N, p, \Omega) > 0$ such that for all $u \in W^{1,p}(\Omega)$, $\|u - u_\Omega\|_{L^p(\Omega)} \leq c \|\nabla u\|_{L^p(\Omega)}$.

Exercise 13.31. Let $\Omega \subset \mathbb{R}^3$ be a connected bounded extension domain for $W^{1,2}(\Omega)$ and let $\mathcal{S}$ be the subset of all $u \in W^{1,2}(\Omega)$ such that $\int_{\Omega} u^3(x) \, dx = 0$.

- (i) Prove there exists a constant $c = c(\Omega) > 0$ such that for all $u \in \mathcal{S}$, $\|u\|_{L^2(\Omega)} \leq c \|\nabla u\|_{L^2(\Omega)}$.

(ii) What properties of the subset $\mathcal{S}$ did you use in the proof?

Corollary 13.32. *Let $m \in \mathbb{N}$, let $1 \leq p < \infty$, and let $\Omega \subset \mathbb{R}^N$ be a connected bounded domain whose boundary is of class C . Let $E \subseteq \Omega$ be a Lebesgue measurable set with positive measure. Then there exists a constant $c = c(m, N, p, \Omega, E) > 0$ such that for all $u \in W^{m,p}(\Omega)$,*

$$(13.63) \quad \sum_{k=0}^{m-1} \|\nabla^k(u - p_E(u))\|_{L^p(\Omega)} \leq c \|\nabla^m u\|_{L^p(\Omega)}.$$

In particular, if $m = 1$, for all $u \in W^{1,p}(\Omega)$,

$$(13.64) \quad \|u - u_E\|_{L^p(\Omega)} \leq c \|\nabla u\|_{L^p(\Omega)}.$$

Proof. For $m = 1$, in view of Theorem 12.30 we are in a position to apply Exercise 13.30. This gives the proof for $m = 1$.

If $m \geq 2$ and α is a multi-index with $0 \leq |\alpha| < m - 1$, in view of (13.61) we can apply repeatedly inequality (13.64) with $u - u_E$ replaced by $\partial^{\alpha + ne_i}(u - p_E(u))$, $0 \leq n \leq m - |\alpha| - 1$, to get (13.63). $\square$

Exercise 13.33. Let $\Omega \subset \mathbb{R}^N$ be an open bounded domain with Lipschitz continuous boundary and let B be a ball with $\Omega \Subset B$. Prove that there exists a continuous linear operator $\mathcal{E}_B : W^{1,p}(\Omega) \rightarrow W^{1,p}(B)$ such that for all $u \in W^{1,p}(\Omega)$, $\mathcal{E}_B(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$ and

$$\|\mathcal{E}_B(u)\|_{L^p(B)} \leq c \|u\|_{L^p(\Omega)}, \quad \|\nabla \mathcal{E}_B(u)\|_{L^p(B)} \leq c \|\nabla u\|_{L^p(\Omega)}$$

for some constant $c = c(m, N, \Omega, B)$. Hint: Let $\mathcal{E}$ be the extension operator given in Theorem 13.17 and let $\mathcal{E}_1(u)$ be the restriction of $\mathcal{E}(u)$ to B . Define $\mathcal{E}_B(u) := \mathcal{E}_1(u - u_\Omega) + u_\Omega$.

In many applications it is of interest to have an estimate of the constant $c(m, N, p, \Omega, E)$. In the remainder of the chapter we study important special classes of domains for which there are explicit bounds on the Poincaré constant $c(m, N, p, \Omega, E)$. We begin by considering rectangles.

Proposition 13.34 (Poincaré's inequality for rectangles). *Let $m \in \mathbb{N}$, let $1 \leq p < \infty$, and let $R = (0, a_1) \times \cdots \times (0, a_N) \subset \mathbb{R}^N$. Then there exists a constant $c = c(m, N, p) > 0$ such that for all $u \in W^{m,p}(R)$, and every $0 \leq k \leq m - 1$,*

$$\|\nabla^k(u - p_R(u))\|_{L^p(R)} \leq c(\max\{a_1, \dots, a_N\})^{(m-k)} \|\nabla^m u\|_{L^p(R)}.$$

In particular, if $m = 1$, for all $u \in W^{1,p}(R)$,

$$(13.65) \quad \|u - u_R\|_{L^p(R)} \leq c(\max\{a_1, \dots, a_N\}) \|\nabla u\|_{L^p(R)}.$$

Proof. Step 1: Assume $m = 1$. Without loss of generality, by Theorem 11.35 we may assume that $u \in C^\infty(\overline{R})$, and, by eventually replacing u with $v(y) := u(\lambda y)$, $y \in \frac{1}{\lambda}R$ for an appropriate $\lambda > 0$, we may assume that

$$(13.66) \quad \max\{a_1, \dots, a_N\} = 1.$$

By the fundamental theorem of calculus, for all $x, y \in R$,

$$\begin{aligned} |u(x) - u(y)| &\leq |u(x) - u(x_1, \dots, x_{N-1}, y_N)| \\ &\quad + \dots + |u(x_1, y_2, \dots, y_N) - u(y)| \\ &\leq \sum_{i=1}^N \int_0^{a_i} |\partial_i u(x_1, \dots, x_{i-1}, t, y_{i+1}, \dots, y_N)| dt. \end{aligned}$$

Raising both sides to power p , by Hölder's inequality, (13.66), and the convexity of the function $g(t) = |t|^p$,

$$\begin{aligned} |u(x) - u(y)|^p &\leq \left(\sum_{i=1}^N \left(\int_0^{a_i} |\partial_i u(x_1, \dots, x_{i-1}, t, y_{i+1}, \dots, y_N)|^p dt \right)^{\frac{1}{p}} \right)^p \\ &\leq N^{p-1} \sum_{i=1}^N \int_0^{a_i} |\partial_i u(x_1, \dots, x_{i-1}, t, y_{i+1}, \dots, y_N)|^p dt. \end{aligned}$$

Hence, again by Hölder's inequality and the fact that $u(x) - u_R = (u(x) - u)_R$,

$$\begin{aligned} \int_R |u(x) - u_R|^p dx &\leq \frac{1}{(\mathcal{L}^N(R))^p} \int_R \left(\int_R |u(x) - u(y)| dy \right)^p dx \\ &\leq \frac{1}{(\mathcal{L}^N(R))^{p-p/p'}} \int_R \int_R |u(x) - u(y)|^p dy dx \\ &\leq \frac{N^{p-1}}{\mathcal{L}^N(R)} \sum_{i=1}^N \int_R \int_R \int_0^{a_i} |\partial_i u(x_1, \dots, x_{i-1}, t, y_{i+1}, \dots, y_N)|^p dt dy dx \\ &= N^{p-1} \sum_{i=1}^N a_i \int_R |\partial_i u(z)|^p dz \leq N^{p-1} \sum_{i=1}^N \int_R |\partial_i u(z)|^p dz, \end{aligned}$$

where we have used the fact that $p - p/p' = 1$, Fubini's theorem, and (13.66). This concludes the proof for $m = 1$.

Step 2: If $m \geq 2$ and α is a multi-index with $0 \leq |\alpha| < m - 1$, in view of (13.61) we can apply repeatedly inequality (13.65) with $u - u_R$ replaced by $\partial^{\alpha + ne_i}(u - p_R(u))$, $0 \leq n \leq m - |\alpha| - 1$ to conclude the proof. $\square$

Exercise 13.35. Let R be as in the previous theorem and let $u \in \dot{W}^{m,p}(R)$. Prove that $u \in W^{m,p}(R)$. Hint: Prove first that $p_R(u)$ can be replaced by $p_{R_1}(u)$, where R_1 is rectangle compactly contained in R .

Next we study convex domains. In the literature there are simpler proofs (see, e.g., [94]). The present one gives a sharper constant and the approach is quite interesting.

Theorem 13.36 (Poincaré's inequality for convex sets). *Let $m \in \mathbb{N}$, let $1 \leq p < \infty$ and let $\Omega \subset \mathbb{R}^N$ be an open bounded convex set. Then there exists a constant $c = c(m, N, p) > 0$ such that for all $u \in W^{m,p}(\Omega)$, and every $0 \leq k \leq m - 1$,*

$$\|\nabla^k(u - p_\Omega(u))\|_{L^p(\Omega)} \leq c(\text{diam } \Omega)^{m-k} \|\nabla^m u\|_{L^p(\Omega)}.$$

In particular, if $m = 1$, for all $u \in W^{1,p}(\Omega)$,

$$(13.67) \quad \|u - u_\Omega\|_{L^p(\Omega)} \leq c \text{diam } \Omega \|\nabla u\|_{L^p(\Omega)}.$$

We begin with some preliminary results.

Exercise 13.37. Let $D \subseteq \mathbb{R}^N$ be a convex set.

- (i) Prove that the interior and the closure of D are convex.
- (ii) Let $u : D \rightarrow \mathbb{R}$ be a continuous function such that $u(\frac{x_1+x_2}{2}) \geq \frac{u(x_1)+u(x_2)}{2}$ for all $x_1, x_2 \in D$. Prove that u is concave.

Lemma 13.38. *Let $\Omega \subset \mathbb{R}^N$ be an open bounded convex set. Then for every small $\delta > 0$, the set $\Omega_\delta := \{x \in \Omega : \text{dist}(x, \partial\Omega) > \delta\}$ is a convex set with $\overline{\Omega}_\delta \subset \Omega$.*

Proof. We begin by showing that the function $d(x) := \text{dist}(x, \partial\Omega)$, $x \in \Omega$, is concave in Ω . To see this, consider $x_1, x_2 \in \Omega$ and let $d_1 := d(x_1)$ and $d_2 := d(x_2)$. Fix any direction $\xi \in S^{N-1}$ and consider the trapezoid with vertices x_1, x_2, y_1, y_2 , where $y_1 := x_1 + d_1\xi$ and $y_2 := x_2 + d_2\xi$. Since the open balls $B(x_i, d_i)$, $i = 1, 2$, are contained in Ω by the definition of d_i , it follows that $y_i \in \overline{\Omega}$, $i = 1, 2$, and so the trapezoid also lies in $\overline{\Omega}$ by the convexity of $\overline{\Omega}$ (see Exercise 13.37). In particular, if $x := \frac{x_1+x_2}{2}$ and $d_0 := \frac{d_1+d_2}{2}$, then the point $y := \frac{y_1+y_2}{2} = x + d_0\xi$ lies on the edge of the trapezoid and therefore also in $\overline{\Omega}$. As the vector ξ varies in S^{N-1} , the corresponding point $y = x + d_0\xi$ fills the surface of the ball $B(x, d_0)$. Hence, $B(x, d_0) \subset \overline{\Omega}$, and so $d(x) \geq d_0$, that is, $d(\frac{x_1+x_2}{2}) \geq \frac{d(x_1)+d(x_2)}{2}$. Since d is Lipschitz continuous by the previous exercise, it follows from Exercise 13.37 that d is concave.

Next we show that Ω_δ is convex. Let $x_1, x_2 \in \Omega_\delta$ and let $\theta \in (0, 1)$. By the concavity of d ,

$$d(\theta x_1 + (1 - \theta)x_2) \geq \theta d(x_1) + (1 - \theta)d(x_2) > \delta,$$

and so $\theta x_1 + (1 - \theta)x_2 \in \Omega_\delta$. □

Lemma 13.39. *Let $\Omega \subset \mathbb{R}^N$ be an open bounded convex set and let $u \in L^1(\Omega)$ be such that $\int_{\Omega} u(x) dx = 0$. Then for every $\delta > 0$ there exists a decomposition of Ω into a finite number of pairwise disjoint convex domains Ω_n , $n = 1, \dots, \ell$, such that*

$$\overline{\Omega} = \bigcup_{n=1}^{\ell} \overline{\Omega}_n, \quad \int_{\Omega_n} u(x) dx = 0, \quad n = 1, \dots, \ell,$$

and each Ω_n is thin in all but one direction, that is, in an appropriate coordinate system $y = (y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R}$,

$$\Omega_n \subset [0, \delta] \times \dots \times [0, \delta] \times [0, \text{diam } \Omega].$$

Proof. For each $\alpha \in [0, 2\pi]$ there is a unique hyperplane $H_{\alpha} \subset \mathbb{R}^N$ with normal $(\cos \alpha, \sin \alpha, 0, \dots, 0)$ that divides Ω into two convex sets U_{α} and V_{α} of equal volume. Let $I(\alpha) := \int_{U_{\alpha}} u(x) dx$. Since $I(\alpha) = -I(\alpha + \pi)$, by continuity there exists α_0 such that $I(\alpha_0) = 0$. Since $\int_{U_{\alpha_0}} u(x) dx = \int_{V_{\alpha_0}} u(x) dx = 0$, we now apply the same reasoning to each of the sets U_{α_0} and V_{α_0} . Continuing in this way, we can subdivide Ω into convex sets W_n with the property that each of these sets is contained between two hyperplanes with normal of the form $(\cos \beta_n, \sin \beta_n, 0, \dots, 0)$ and at distance δ and the average of u vanishes on each of these sets.

Fix any W_n . By a rotation we can assume that the normal of the two hyperplanes is $(1, 0, \dots, 0)$. In these new coordinates we apply the same reasoning using hyperplanes with normals of the form $(0, \cos \alpha, \sin \alpha, 0, \dots, 0)$. We continue in this way to obtain the desired conclusion. $\square$

Proof of Theorem 13.36. As in Proposition 13.34 it is enough to prove the case $m = 1$.

Step 1: We begin by showing that we may assume $u \in C^1(\overline{\Omega})$. By Lemma 13.38 for every $\delta > 0$, the set Ω_{δ} is convex and $\text{dist}(\overline{\Omega}_{\delta}, \partial\Omega) = \delta > 0$. Consider a sequence of standard mollifiers $\{\varphi_{\varepsilon}\}_{\varepsilon>0}$. For $0 < \varepsilon < \delta$ define $u_{\varepsilon} := u * \varphi_{\varepsilon}$ in Ω_{δ} . Then $u_{\varepsilon} \in C^{\infty}(\overline{\Omega}_{\delta})$ by Theorem C.20, and so, if (13.67) holds for the pair $u_{\varepsilon}, \Omega_{\delta}$, we get

$$\int_{\Omega_{\delta}} |u_{\varepsilon}(x) - (u_{\varepsilon})_{\Omega_{\delta}}|^p dx \leq c \text{diam } \Omega_{\delta} \sum_{i=1}^N \int_{\Omega_{\delta}} |\partial_i u_{\varepsilon}|^p dx.$$

Letting $\varepsilon \rightarrow 0^+$ and using Theorems C.16 and C.20 we obtain

$$\begin{aligned} \int_{\Omega_\delta} |u(x) - u_{\Omega_\delta}|^p dx &\leq c \operatorname{diam} \Omega_\delta \sum_{i=1}^N \int_{\Omega_\delta} |\partial_i u|^p dx \\ &\leq c \operatorname{diam} \Omega \sum_{i=1}^N \int_{\Omega} |\partial_i u|^p dx. \end{aligned}$$

It now suffices to let $\delta \rightarrow 0^+$.

Step 2: By the previous step and by replacing u with $u - u_\Omega$, we may assume that $u \in C^1(\overline{\Omega})$ and that $\int_{\Omega} u(x) dx = 0$. By uniform continuity, given $0 < \varepsilon < 1$, there exists $\delta > 0$ such that

$$(13.68) \quad ||u(x_1)|^p - |u(x_2)|^p| + ||\partial_i u(x_1)|^p - |\partial_i u(x_2)|^p| \leq \varepsilon,$$

$$(13.69) \quad |u(x_1) - u(x_2)| + |\partial_i u(x_1) - \partial_i u(x_2)| \leq \varepsilon$$

for all $x_1, x_2 \in \Omega$ with $\|x_1 - x_2\| \leq \delta$ and all $i = 1, \dots, N$. By Lemma 13.39 we can decompose Ω into a finite number of pairwise disjoint convex domains Ω_n , $n = 1, \dots, \ell$, such that

$$(13.70) \quad \overline{\Omega} = \bigcup_{n=1}^{\ell} \overline{\Omega}_n, \quad \int_{\Omega_n} u(x) dx = 0, \quad n = 1, \dots, \ell,$$

and for every n there is an appropriate coordinate system $y = (y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R}$ such that the projection of Ω_n into the y_N -axis is $(0, d_n)$, where $d_n \leq \operatorname{diam} \Omega$, and

$$(13.71) \quad \Omega_n \subset [0, \delta/\sqrt{N-1}] \times \dots \times [0, \delta/\sqrt{N-1}] \times [0, d_n].$$

In particular,

$$(13.72) \quad \|(y', y_N) - (0, y_N)\| \leq \delta \quad \text{for all } (y', y_N) \in \Omega_n.$$

Fix one such Ω_n . By (13.68), (13.69), (13.70), and (13.72),

$$(13.73) \quad \left| \int_{\Omega_n} u(y) dy - \int_{\Omega_n} u(0, y_N) dy \right| \leq \varepsilon \mathcal{L}^N(\Omega_n),$$

$$(13.74) \quad \left| \int_{\Omega_n} |u(y)|^p dy - \int_{\Omega_n} |u(0, y_N)|^p dy \right| \leq \varepsilon \mathcal{L}^N(\Omega_n),$$

$$(13.75) \quad \left| \int_{\Omega_n} |\partial_N u(y)|^p dy - \int_{\Omega_n} |\partial_N u(0, y_N)|^p dy \right| \leq \varepsilon \mathcal{L}^N(\Omega_n).$$

Let $g(y_N) := \int_{\mathbb{R}^{N-1}} \chi_{\Omega_n}(y', y_N) dy'$. By Fubini's theorem,

$$(13.76) \quad \int_{\Omega_n} u(0, y_N) dy = \int_0^{d_n} u(0, y_N) g(y_N) dy_N,$$

$$(13.77) \quad \int_{\Omega_n} |u(0, y_N)|^p dy = \int_0^{d_n} |u(0, y_N)|^p g(y_N) dy_N,$$

$$(13.78) \quad \int_{\Omega_n} |\partial_N u(0, y_N)|^p dy = \int_0^{d_n} |\partial_N u(0, y_N)|^p g(y_N) dy_N.$$

By Minkowski's inequality with respect to the measure $g(y_N) dy_N$ we have that

$$(13.79) \quad \begin{aligned} & \left(\int_0^{d_n} |u(0, y_N)|^p g(y_N) dy_N \right)^{1/p} \\ & \leq \left(\int_0^{d_n} |u(0, y_N) - \bar{u}_{(0, d_n)}|^p g(y_N) dy_N \right)^{1/p} \\ & \quad + \left(\int_0^{d_n} |\bar{u}_{(0, d_n)}|^p g(y_N) dy_N \right)^{1/p} =: I + II, \end{aligned}$$

where

$$\bar{u}_{(0, d_n)} := \frac{\int_0^{d_n} u(0, y_N) g(y_N) dy_N}{\int_0^{d_n} g(y_N) dy_N}.$$

Since the function $g^{\frac{1}{N-1}}$ is concave by Remark C.9, we are in a position to apply Proposition 7.33 to conclude that

$$(13.80) \quad \begin{aligned} I & \leq c d_n \left(\int_0^{d_n} |\partial_N u(0, y_N)|^p g(y_N) dy_N \right)^{1/p} \\ & = c d_n \left(\int_{\Omega_n} |\partial_N u(0, y_N)|^p dy \right)^{1/p} \\ & \leq c \operatorname{diam} \Omega \left[\left(\int_{\Omega_n} |\partial_N u(y)|^p dy \right)^{1/p} + (\varepsilon \mathcal{L}^N(\Omega_n))^{1/p} \right], \end{aligned}$$

where we have used (13.78), (13.75), the inequality

$$(13.81) \quad (a + b)^{1/p} \leq a^{1/p} + b^{1/p} \quad \text{for } a, b \geq 0,$$

and the fact that $d_n \leq \operatorname{diam} \Omega$, in this order.

To estimate II , we distinguish two cases. If $\int_0^{d_n} g(y_N) dy_N > \varepsilon \mathcal{L}^N(\Omega_n)$, then by (13.70) and (13.73),

$$\begin{aligned} II &= \left| \int_0^{d_n} u(0, y_N) g(y_N) dy_N \right| \left(\int_0^{d_n} g(y_N) dy_N \right)^{-1+1/p} \\ &= \left| \int_0^{d_n} u(0, y_N) g(y_N) dy_N - \int_{\Omega_n} u(y) dy \right| \left(\int_0^{d_n} g(y_N) dy_N \right)^{-1+1/p} \\ &\leq \varepsilon \mathcal{L}^N(\Omega_n) \left(\int_0^{d_n} g(y_N) dy_N \right)^{-1+1/p} \leq (\varepsilon \mathcal{L}^N(\Omega_n))^{1/p}, \end{aligned}$$

while if $\int_0^{d_n} g(y_N) dy_N \leq \varepsilon \mathcal{L}^N(\Omega_n)$, then

$$\begin{aligned} II &= \left| \frac{\int_0^{d_n} u(0, y_N) g(y_N) dy_N}{\int_0^{d_n} g(y_N) dy_N} \right| \left(\int_0^{d_n} g(y_N) dy_N \right)^{1/p} \\ &\leq M \left(\int_0^{d_n} g(y_N) dy_N \right)^{1/p} \leq M (\varepsilon \mathcal{L}^N(\Omega_n))^{1/p}, \end{aligned}$$

where $M := \max_{\overline{\Omega}} |u| + \max_{\overline{\Omega}} |\partial u_N| + 1 < \infty$, since $u \in C^1(\overline{\Omega})$. Combining both cases, we have that

$$(13.82) \quad II \leq (1 + M)(\varepsilon \mathcal{L}^N(\Omega_n))^{1/p}.$$

It now follows from (13.79)–(13.82) that

$$\begin{aligned} \left(\int_0^{d_n} |u(0, y_N)|^p g(y_N) dy_N \right)^{1/p} &\leq c \operatorname{diam} \Omega \left(\int_{\Omega_n} |\partial_N u(y)|^p dy \right)^{1/p} \\ &\quad + c(\varepsilon \mathcal{L}^N(\Omega_n))^{1/p}, \end{aligned}$$

where the constant $c = c(M, N, p)$ changes from line to line. By (13.74), (13.81), and the previous inequality,

$$\begin{aligned} \left(\int_{\Omega_n} |u|^p dy \right)^{1/p} &\leq \left(\int_0^{d_n} |u(0, y_N)|^p g(y_N) dy_N \right)^{1/p} + (\varepsilon \mathcal{L}^N(\Omega_n))^{1/p} \\ &\leq c \operatorname{diam} \Omega \left(\int_{\Omega_n} |\partial_N u|^p dy \right)^{1/p} + c(\varepsilon \mathcal{L}^N(\Omega_n))^{1/p}. \end{aligned}$$

Using a discrete Minkowski inequality in the form

$$\left(\sum_{n=1}^{\ell} (a_n + b_n)^p \right)^{1/p} \leq \left(\sum_{n=1}^{\ell} a_n^p \right)^{1/p} + \left(\sum_{n=1}^{\ell} b_n^p \right)^{1/p},$$

where $a_n, b_n \geq 0$, it follows by summing over all n that

$$\begin{aligned}
 \left(\int_{\Omega} |u|^p dx \right)^{1/p} &= \left(\sum_{n=1}^{\ell} \int_{\Omega_n} |u|^p dy \right)^{1/p} \\
 &\leq c \left(\sum_{n=1}^{\ell} \left(\text{diam } \Omega \left(\int_{\Omega_n} |\partial_N u|^p dy \right)^{1/p} + (\varepsilon \mathcal{L}^N(\Omega_n))^{1/p} \right)^p \right)^{1/p} \\
 &\leq c \left(\text{diam } \Omega \left(\sum_{n=1}^{\ell} \int_{\Omega_n} |\partial_N u|^p dy \right)^{1/p} + \varepsilon^{1/p} \left(\sum_{n=1}^{\ell} \mathcal{L}^N(\Omega_n) \right)^{1/p} \right) \\
 &\leq c \left(\text{diam } \Omega \left(\int_{\Omega} |\partial_N u|^p dy \right)^{1/p} + \varepsilon^{1/p} (\mathcal{L}^N(\Omega))^{1/p} \right).
 \end{aligned}$$

Since c does not depend on ε , it suffices to let $\varepsilon \rightarrow 0^+$. $\square$

Remark 13.40. By choosing an appropriate norm in $W^{1,p}(\Omega)$, it can be shown that for $p = 1$ and $p = 2$ one can take the constant c in (13.67) to be $\frac{1}{2}$ and $\frac{1}{\pi^2}$, respectively (see [22] and [2], respectively), and that there exist convex domains for which these constants are optimal.

Next we consider star-shaped sets. We recall that a set $E \subseteq \mathbb{R}^N$ is *star-shaped with respect to a point* $x_0 \in E$ if $\theta x + (1 - \theta)x_0 \in E$ for all $\theta \in (0, 1)$ and for all $x \in E$.

Theorem 13.41 (Poincaré's inequality for star-shaped sets). *Let $m \in \mathbb{N}$, let $1 \leq p < \infty$, and let $\Omega \subset \mathbb{R}^N$ be an open star-shaped set with respect to $x_0 \in \Omega$ and such that*

$$(13.83) \quad Q(x_0, 4r) \subseteq \Omega \subseteq B(x_0, R)$$

for some $r, R > 0$. Then there exists a constant $c = c(m, N, p) > 0$ such that for all $u \in W^{m,p}(\Omega)$, and every $0 \leq k \leq m - 1$,

$$\|\nabla^k(u - p_{\Omega}(u))\|_{L^p(\Omega)} \leq c R^{m-k} (R/r)^{(N-1)(m-k)/p} \|\nabla^m u\|_{L^p(\Omega)}.$$

In particular, if $m = 1$, for all $u \in W^{1,p}(\Omega)$,

$$(13.84) \quad \|u - u_{\Omega}\|_{L^p(\Omega)} \leq c R (R/r)^{(N-1)/p} \|\nabla u\|_{L^p(\Omega)}.$$

Proof. As in Proposition 13.34 it is enough to prove the case $m = 1$. By the Meyers–Serrin theorem (Theorem 11.24), without loss of generality, we may assume that $u \in C^{\infty}(\Omega) \cap W^{1,p}(\Omega)$ and that $x_0 = 0$. Hence, we may write

$$\Omega = \{\rho y \in \mathbb{R}^N : y \in S^{N-1}, 0 \leq \rho < f(y)\},$$

where $f : S^{N-1} \rightarrow [0, \infty)$ is the *radial function* of Ω , that is, $f(y) := \sup\{\rho \geq 0 : \rho y \in \Omega\}$. Note that for all $y \in S^{N-1}$ we have that $f(y)y \in \partial\Omega$ and $2r \leq f(y) \leq R$ by (13.83).

Step 1: Assume that u vanishes in $B(0, r)$. Since $u(ry) = 0$ for all $y \in S^{N-1}$, using polar coordinates and the fundamental theorem of calculus, for all $r \leq \rho < f(y)$ we have that

$$u(\rho y) = u(\rho y) - u(ry) = \int_r^\rho \frac{\partial u}{\partial \rho}(ty) dt,$$

and so by Hölder's inequality,

$$\begin{aligned} |u(\rho y)|^p &= \left(\int_r^\rho \left| \frac{\partial u}{\partial \rho}(ty) \right| dt \right)^p \leq \rho^{p/p'} \int_r^\rho \left| \frac{\partial u}{\partial \rho}(ty) \right|^p dt \\ &\leq c \rho^{p/p'} \int_r^\rho \|\nabla u(ty)\|^p dt. \end{aligned}$$

Hence, by Tonelli's theorem

$$\begin{aligned} \int_0^{f(y)} |u(\rho y)|^p \rho^{N-1} d\rho &\leq c \int_0^{f(y)} \rho^{p/p' + N-1} \int_r^\rho \|\nabla u(ty)\|^p dt d\rho \\ &\leq c \int_r^{f(y)} \|\nabla u(ty)\|^p \left(\int_0^R \rho^{p + N-2} d\rho \right) dt \\ &= c \frac{R^{p+N-1}}{p+N-1} \int_r^{f(y)} \|\nabla u(ty)\|^p dt \\ &\leq c R^p (R/r)^{N-1} \int_r^{f(y)} \|\nabla u(ty)\|^p t^{N-1} dt. \end{aligned}$$

Integrating with respect to y over S^{N-1} and using spherical coordinates yields

$$\int_\Omega |u(x)|^p dx \leq c R^p (R/r)^{N-1} \int_\Omega \|\nabla u(x)\|^p dx.$$

Step 2: Assume next that $u \in C^\infty(\Omega) \cap W^{1,p}(\Omega)$ is such that $u_{Q(0,4r)} = 0$. Then by Proposition 13.34,

$$(13.85) \quad \int_{Q(0,4r)} |u(x)|^p dx \leq c r^p \int_{Q(0,4r)} \|\nabla u(x)\|^p dx.$$

Consider the function

$$\varphi(x) := \begin{cases} 0 & \text{if } x \in B(0, r), \\ (\|x\| - r)/r & \text{if } x \in B(0, 2r) \setminus B(0, r), \\ 1 & \text{if } x \notin B(0, 2r) \end{cases}$$

and write $u = (1 - \varphi)u + \varphi u =: u_1 + u_2$. Since u_2 vanishes in $B(0, r)$, by the previous step

$$\begin{aligned}
 \int_{\Omega} |u_2|^p dx &\leq cR^p(R/r)^{N-1} \int_{\Omega} \|\nabla u_2\|^p dx \\
 &= cR^p(R/r)^{N-1} \int_{\Omega} \|u \nabla \varphi + \varphi \nabla u\|^p dx \\
 (13.86) \quad &\leq cR^p(R/r)^{N-1} r^{-p} \int_{Q(0, 4r)} |u|^p dx + cR^p(R/r)^{N-1} \int_{\Omega} \|\nabla u\|^p dx \\
 &\leq cR^p(R/r)^{N-1} \int_{\Omega} \|\nabla u\|^p dx,
 \end{aligned}$$

where we have used the facts that $\partial_i \varphi = 0$ outside $B(0, 2r)$ and $\|\partial_i \varphi\|_{\infty} \leq cr^{-1}$, and (13.85). On the other hand, since u_1 is zero outside $Q(0, 4r)$,

$$\begin{aligned}
 \int_{\Omega} |u_1|^p dx &= \int_{Q(0, 4r)} |u_1|^p dx \\
 &\leq c \int_{Q(0, 4r)} |u|^p dx + c \int_{Q(0, 4r)} |u_2|^p dx \\
 &\leq cr^p \int_{Q(0, 4r)} \|\nabla u\|^p dx + cR^p(R/r)^{N-1} \int_{\Omega} \|\nabla u\|^p dx \\
 &\leq cR^p(R/r)^{N-1} \int_{\Omega} \|\nabla u\|^p dx,
 \end{aligned}$$

where we have used (13.85) and (13.86). Hence, by Minkowski's and Hölder's inequalities

$$\begin{aligned}
 \int_{\Omega} |u - u_{\Omega}|^p dx &\leq c \int_{\Omega} |u|^p dx \leq c \int_{\Omega} |u_1|^p dx + c \int_{\Omega} |u_2|^p dx \\
 &\leq cR^p(R/r)^{N-1} \int_{\Omega} \|\nabla u\|^p dx.
 \end{aligned}$$

Step 3: Finally, if $u \in C^{\infty}(\Omega) \cap W^{1,p}(\Omega)$ is such that $u_{Q(0, 4r)} \neq 0$, apply the previous step to the function $u - u_{Q(0, 4r)}$ to conclude that

$$\begin{aligned}
 \int_{\Omega} |u - u_{\Omega}|^p dx &= \int_{\Omega} |u - u_{Q(0, 4r)} - (u - u_{Q(0, 4r)})_{\Omega}|^p dx \\
 &\leq cR^p(R/r)^{N-1} \int_{\Omega} \|\nabla u(x)\|^p dx.
 \end{aligned}$$

This concludes the proof. $\square$

Exercise 13.42. Let Ω be as in the previous theorem and let $u \in \dot{W}^{m,p}(\Omega)$. Prove that $u \in W^{m,p}(\Omega)$.

For $p > 1$ the constant $R^p(R/r)^{N-1}$ in (13.84) can be significantly improved using Muckenhoupt's weighted norm inequality for the maximal function [180].

Theorem 13.43 (Muckenhoupt). *Let $J \subseteq \mathbb{R}$ be an interval, let $1 < p < \infty$, and let $w : J \rightarrow [0, \infty)$ be a Lebesgue measurable function. Then there is a constant $c > 0$ such that for all Lebesgue measurable functions $u : J \rightarrow \mathbb{R}$,*

$$(13.87) \quad \int_J |(M_1 u)(x)|^p w(x) dx \leq c \int_J |u(x)|^p w(x) dx$$

if and only if there is a constant $K > 0$ such that for all intervals $I \subseteq J$,

$$\int_I w(x) dx \left(\int_I (w(x))^{-1/(p-1)} dx \right)^{p-1} \leq K(\text{length } I)^p.$$

Furthermore, if c and K are the least constants for which the previous inequalities are true, then $K \leq c \leq 2^p K$.

In the previous theorem,

$$(13.88) \quad (M_1 u)(x) := \sup_{y \in J \setminus \{x\}} \frac{1}{y-x} \int_x^y |u(t)| dt, \quad x \in J.$$

The proof of this result is rather involved and goes beyond the purposes of this book. We refer to [180] for more details.

Exercise 13.44. Let $0 < r < R/2$ and let

$$(13.89) \quad w(t) := \min\{(r + |t|)^{N-1}, R^{N-1}\}, \quad t \in \mathbb{R}.$$

Prove that for all $1 < p < \infty$,

$$(13.90)$$

$$\begin{aligned} K(r, R, p, N) &:= \sup_{I \text{ interval}} \frac{1}{(\mathcal{L}^1(I))^p} \int_I w(x) dx \left(\int_I (w(x))^{-1/(p-1)} dx \right)^{p-1} \\ &\leq \begin{cases} (R/r)^{N-p} & \text{if } 1 < p < N, \\ \log^{N-1}(1 + R/r) & \text{if } p = N, \\ 1 & \text{if } p > N. \end{cases} \end{aligned}$$

Corollary 13.45. *Let $1 \leq p < \infty$ and let $\Omega \subset \mathbb{R}^N$ be an open star-shaped set with respect to $x_0 \in \Omega$ and such that $Q(x_0, 4r) \subseteq \Omega \subseteq B(x_0, R)$ for some $r, R > 0$. Then there exists a constant $c = c(m, N, p) > 0$ such that for all $u \in W^{m,p}(\Omega)$, and every $0 \leq k \leq m-1$,*

$$\|\nabla^k(u - p_\Omega(u))\|_{L^p(\Omega)} \leq cR^{(m-k)}(K(r, R, p, N))^{(m-k)/p} \|\nabla^m u\|_{L^p(\Omega)},$$

where K is defined in (13.90). In particular, if $m = 1$, for all $u \in W^{1,p}(\Omega)$,

$$\|u - u_\Omega\|_{L^p(\Omega)} \leq cR(K(r, R, p, N))^{1/p} \|\nabla u\|_{L^p(\Omega)}.$$

Proof. The only change is in Step 1 of the proof of Theorem 13.41, where instead of using Hölder's inequality, we write

$$\begin{aligned} |u(\rho y)|^p &= \left(\int_r^\rho \left| \frac{\partial u}{\partial \rho}(ty) \right| dt \right)^p \leq R^p \left(\frac{1}{\rho - r} \int_r^\rho \left| \frac{\partial u}{\partial \rho}(ty) \right| dt \right)^p \\ &\leq R^p |(M_1 g)(\rho)|^p, \end{aligned}$$

where

$$g(\rho) := \left| \frac{\partial u}{\partial \rho}(\rho y) \right| \chi_{(r, f(y))}(\rho), \quad \rho > 0,$$

and $M_1 g$ is defined in (13.88). In turn, since $u = 0$ in $B(0, r)$,

$$\begin{aligned} \int_0^{f(y)} |u(\rho y)|^p \rho^{N-1} d\rho &= \int_r^{f(y)} |u(\rho y)|^p \rho^{N-1} d\rho \\ &\leq R^p \int_r^{f(y)} |(M_1 g)(\rho)|^p \rho^{N-1} d\rho \\ &\leq R^p \int_r^{f(y)} |(M_1 g)(\rho)|^p w(\rho) d\rho \\ &\leq R^p K(r, R, p, N) \int_r^{f(y)} \left| \frac{\partial u}{\partial \rho}(\rho y) \right|^p w(\rho) d\rho \\ &= R^p K(r, R, p, N) \int_r^{f(y)} \left| \frac{\partial u}{\partial \rho}(\rho y) \right|^p \rho^{N-1} d\rho, \end{aligned}$$

where w and $K(r, R, p, N)$ are defined in (13.89) and (13.90), respectively, and we have used (13.87). Integrating with respect to y over S^{N-1} yields

$$\int_\Omega |u(x)|^p dx \leq c R^p K(r, R, p, N) \int_\Omega \|\nabla u(x)\|^p dx$$

and completes the proof. $\square$

The proof of Theorem 13.41 can be adapted to the case in which Ω is the subgraph of a lower semicontinuous function.

Theorem 13.46. *Let $m \in \mathbb{N}$, let $1 \leq p < \infty$, let $0 < \delta \leq r < R$, let $f : Q_{N-1}(0, \delta) \rightarrow [r, R]$ be a lower semicontinuous function, and let*

$$(13.91) \quad \Omega := \{(x', x_N) \in Q_{N-1}(0, \delta) \times \mathbb{R} : -2r < x_N < f(x')\}.$$

Then there exists a constant $c = c(m, N, p) > 0$ such that for all $u \in W^{m,p}(\Omega)$, and every $0 \leq k \leq m-1$,

$$\|\nabla^k(u - p_\Omega(u))\|_{L^p(\Omega)} \leq c R^{(m-k)} \|\nabla^m u\|_{L^p(\Omega)}.$$

In particular, if $m = 1$, for all $u \in W^{1,p}(\Omega)$,

$$\|u - u_\Omega\|_{L^p(\Omega)} \leq c R \|\nabla u\|_{L^p(\Omega)}.$$

Proof. The proof is similar to the one of Theorem 13.41 and thus we indicate only the main changes. As in that theorem it suffices to prove the case $m = 1$ and to assume that $u \in C^\infty(\Omega) \cap W^{1,p}(\Omega)$.

Step 1: Assume that u vanishes in $Q_{N-1}(0, \delta) \times (-2r, -r)$. Since $u(x', -r) = 0$ for all $x' \in Q_{N-1}(0, \delta)$, using polar coordinates and the fundamental theorem of calculus, for all $-r \leq x_N < f(x')$ we have that $u(x', x_N) = \int_{-r}^{x_N} \partial_N u(x', t) dt$, and so by Hölder's inequality,

$$|u(x', x_N)|^p \leq x_N^{p/p'} \int_{-r}^{x_N} |\partial_N u(x', t)|^p dt.$$

Hence, by Tonelli's theorem and the fact that $f \leq R$,

$$\begin{aligned} \int_{-2r}^{f(x')} |u(x', x_N)|^p dx_N &\leq \int_{-2r}^{f(x')} x_N^{p/p'} \int_{-r}^{x_N} |\partial_N u(x', t)|^p dt dx_N \\ &\leq \int_{-2r}^{f(x')} |\partial_N u(x', t)|^p \left(\int_t^{f(x')} x_N^{p-1} dx_N \right) dt \\ &\leq R^p \int_{-2r}^{f(x')} |\partial_N u(x', t)|^p dt. \end{aligned}$$

Integrating with respect to x' over $Q_{N-1}(0, \delta)$ yields

$$\int_{\Omega} |u(x)|^p dx \leq R^p \int_{\Omega} |\partial_N u(x)|^p dx.$$

Step 2: Assume next that $u \in C^\infty(\Omega) \cap W^{1,p}(\Omega)$ is such that $u_T = 0$, where $T := Q_{N-1}(0, \delta) \times (-r, 0)$. Then by Proposition 13.34,

$$(13.92) \quad \int_T |u(x)|^p dx \leq cR^p \int_T \|\nabla u(x)\|^p dx.$$

Consider the function

$$\varphi(x) := \begin{cases} 0 & \text{if } x_N \leq -r, \\ (r + x_N)/r & \text{if } -r \leq x_N \leq 0, \\ 1 & \text{if } x_N \geq 0 \end{cases}$$

and write

$$u = (1 - \varphi)u + \varphi u =: u_1 + u_2.$$

Since u_2 vanishes in $Q_{N-1}(0, \delta) \times (-2r, -r)$, as in (13.86),

$$\int_{\Omega} |u_2|^p dx \leq cR^p \int_{\Omega} \|\nabla u\|^p dx,$$

where we have used Minkowski's inequality, the fact that $\partial_N \varphi = 0$ outside T , and (13.92). We can now continue as in Step 2 of the proof of Theorem 13.41 to conclude that

$$\int_{\Omega} |u_1|^p dx \leq cR^p \int_{\Omega} \|\nabla u\|^p dx,$$

and in turn,

$$\int_{\Omega} |u - u_{\Omega}|^p dx \leq cR^p \int_{\Omega} \|\nabla u\|^p dx.$$

Step 3: Finally, if $u \in C^\infty(\Omega) \cap W^{1,p}(\Omega)$ is such that $u_T \neq 0$, then we can reason as in Step 3 of the proof of Theorem 13.41 to conclude the proof. $\square$

Exercise 13.47. Let Ω be as in the previous theorem and let $u \in \dot{W}^{m,p}(\Omega)$. Prove that $u \in W^{m,p}(\Omega)$.

In Proposition 13.34 and Theorems 13.36, 13.41, 13.46 we have proved Poincaré's inequality for several important classes of special domains. We now consider the case in which Ω can be written as union of overlapping sets in which Poincaré's inequality holds. A typical example is the case in which Ω is an open bounded, connected set of class C or more regular (see Definition 9.57). Then we can write Ω as a union of sets of the type (13.91) and of balls or cubes. In each of these subdomains Poincaré's inequality holds. To obtain Poincaré's inequality in Ω we then use a chaining argument. Chaining techniques are very useful in treating more general domains such as s -John domains (see, e.g., [106]).

Theorem 13.48. Let $m \in \mathbb{N}$, let $1 \leq p < \infty$, and let $\Omega \subset \mathbb{R}^N$ be a bounded, connected, open set. Assume that Ω can be written as $\Omega = \bigcup_{n=1}^{\ell} \Omega_n$, where each Ω_n is a bounded, connected, open set, such that for all $u \in W^{m,p}(\Omega_n)$, and every $0 \leq k \leq m-1$,

$$(13.93) \quad \|\nabla^k(u - p_{\Omega_n}(u))\|_{L^p(\Omega_n)} \leq c_n \|\nabla^m u\|_{L^p(\Omega_n)}$$

for some constant $c_n = c_n(m, N, p, \Omega_n) > 0$. Then there is a constant $c = c(m, N, p, \Omega) > 0$ such that for all $u \in W^{m,p}(\Omega)$, and every $0 \leq k \leq m-1$,

$$(13.94) \quad \|\nabla^k(u - p_{\Omega}(u))\|_{L^p(\Omega_n)} \leq c \|\nabla^m u\|_{L^p(\Omega_n)}.$$

Proof. As in Proposition 13.34 it is enough to prove the case $m = 1$. For every $n = 1, \dots, \ell$, consider a ball $B_n \subseteq \Omega_n$. By Hölder's inequality and (13.81),

$$\begin{aligned} \|u - u_{\Omega}\|_{L^p(\Omega)} &\leq \|u - u_{B_1}\|_{L^p(\Omega)} + |u_{\Omega} - u_{B_1}|(\mathcal{L}^N(\Omega))^{1/p} \\ &\leq 2\|u - u_{B_1}\|_{L^p(\Omega)} \leq 2 \sum_{n=1}^{\ell} \|u - u_{B_1}\|_{L^p(\Omega_n)}. \end{aligned}$$

Thus, it suffices to prove there is $k_n > 0$ such that

$$(13.95) \quad \|u - u_{B_1}\|_{L^p(\Omega_n)} \leq k_n \|\nabla u\|_{L^p(\Omega_n)}.$$

Since Ω is pathwise connected, there is a polygonal path in Ω joining the center of B_1 with the center of B_n . In turn, we can find a chain of balls in Ω joining B_1 to B_n . To be precise, we can find balls B_k^* , $k = 1, \dots, \ell_n$, such

that $B_1^* = B_1$, $B_{\ell_n}^* = B_n$, and $B_k^* \cap B_{k+1}^* \neq \emptyset$ for every $k \in \{1, \dots, \ell_n - 1\}$. Then

$$\|u - u_{B_1}\|_{L^p(\Omega_n)} \leq \|u - u_{B_n}\|_{L^p(\Omega_n)} + \sum_{k=1}^{\ell_n} |u_{B_k^*} - u_{B_{k+1}^*}| (\mathcal{L}^N(\Omega_n))^{1/p}.$$

For the first term on the right-hand side of the previous inequality we use Hölder's inequality and (13.93) to conclude

$$\begin{aligned} \|u - u_{B_n}\|_{L^p(\Omega_n)} &\leq \|u - u_{\Omega_n}\|_{L^p(\Omega_n)} + |u_{\Omega_n} - u_{B_n}| (\mathcal{L}^N(\Omega_n))^{1/p} \\ &\leq 2\|u - u_{\Omega_n}\|_{L^p(\Omega_n)} \leq 2c_n \|\nabla u\|_{L^p(\Omega_n)}. \end{aligned}$$

On the other hand, fix a ball $B(x_k, r_k) \subset B_k^* \cap B_{k+1}^*$. Then by Hölder's inequality

$$\begin{aligned} &|u_{B_k^*} - u_{B_{k+1}^*}| (\mathcal{L}^N(\Omega_n))^{1/p} \\ &\leq |u_{B_k^*} - u_{B(x_k, r_k)}| (\mathcal{L}^N(\Omega_n))^{1/p} + |u_{B(x_k, r_k)} - u_{B_{k+1}^*}| (\mathcal{L}^N(\Omega_n))^{1/p} \\ &\leq \|u - u_{B(x_k, r_k)}\|_{L^p(B_k^*)} \left(\mathcal{L}^N(\Omega_n) / \mathcal{L}^N(B_k^*) \right)^{1/p} \\ &\quad + \|u - u_{B(x_k, r_k)}\|_{L^p(B_{k+1}^*)} \left(\mathcal{L}^N(\Omega_n) / \mathcal{L}^N(B_{k+1}^*) \right)^{1/p} \\ &\leq c \left[R_k (R_k / r_k)^{(N-1)/p} \left(\mathcal{L}^N(\Omega_n) / \mathcal{L}^N(B^*) \right)^{1/p} \right. \\ &\quad \left. + R_{k+1} (R_{k+1} / r_k)^{(N-1)/p} \left(\mathcal{L}^N(\Omega_n) / \mathcal{L}^N(B_{k+1}^*) \right)^{1/p} \right] \|\nabla u\|_{L^p(\Omega)}, \end{aligned}$$

where in the last inequality we used the fact that B_k^* and B_{k+1}^* are star-shaped with respect to $B(x_k, r_k)$ and Theorem 13.41, and where R_k is the radius of B_k^* . By combining the last three inequalities we obtain (13.95). $\square$

Remark 13.49. It follows from the proof of the previous theorem that for $m = 1$ the constant c in (13.94) can be bounded from above by

$$\begin{aligned} c_* &:= 2 \sum_{n=1}^{\ell} c_n + c_1(m, n, p) \sum_{n=1}^{\ell} \sum_{k=1}^{\ell_n-1} R_{k,n} (R_{k,n} / r_{k,n})^{(N-1)/p} \times \\ &\quad \left(\mathcal{L}^N(\Omega_n) / \mathcal{L}^N(B_{k,n}^*) \right)^{1/p}, \end{aligned}$$

where for each n , $R_{k,n}$ are the radii of the chain of balls $B_{k,n}^*$ joining B_1 to B_n and $r_{k,n}$ are the radii of the balls $B(x_{k,n}, r_{k,n}) \subset B_{k,n}^* \cap B_{k+1,n}^*$.

Remark 13.50. By combining Theorems 13.46 and 13.48 it follows that Poincaré's inequality holds for open, connected, bounded sets with continuous boundary.

13.3. Interpolation Inequalities in Domains

In this section, we extend some of the results of Section 12.5. We begin by extending Theorem 12.85 for $m = 2$ to a rectangle R . Note that if R is bounded and $u \in L^q(R) \cap \dot{W}^{2,p}(R)$, with $p \geq q$, then we can use Poincaré's inequality (see Proposition 13.34) to conclude that u and $\partial_i u$ belong to $L^p(R)$. Thus the interesting case is $p < q$.

Theorem 13.51. *Let $R := I_1 \times \cdots \times I_N \subseteq \mathbb{R}^N$, where each $I_i \subseteq \mathbb{R}$ is an open interval, let $1 \leq p, q, r \leq \infty$ be such that*

$$(13.96) \quad \frac{1}{r} = \frac{1}{2p} + \frac{1}{2q},$$

and let $u \in L^q(R) \cap \dot{W}^{2,p}(R)$. Then there is a constant $c = c(N, p, q) > 0$ such that

$$(13.97) \quad \|\partial_i u\|_{L^r(R)} \leq c \|u\|_{L^q(R)}^{1/2} \|\partial_i^2 u\|_{L^p(R)}^{1/2}$$

for all $i = 1, \dots, N$, provided all the intervals I_i have infinite length, while

$$(13.98) \quad \|\partial_i u\|_{L^r(R)} \leq c \ell^{-1} (\mathcal{L}^N(R))^{1/r-1/q} \|u\|_{L^q(R)} + c \|u\|_{L^q(R)}^{1/2} \|\partial_i^2 u\|_{L^p(R)}^{1/2}$$

if all the intervals I_i have finite length and $p \leq q$, where $\ell := \min_i \mathcal{L}^1(I_i)$.

We begin with a preliminary lemma.

Lemma 13.52. *Let $\Omega \subset \mathbb{R}^N$ be an open bounded set, let $1 \leq p, q, r \leq \infty$ be such that $p < q$ and (13.96) hold, and let $\ell > 0$. Assume that $u \in L^q(\Omega) \cap \dot{W}^{2,p}(\Omega)$ satisfies the inequality*

$$(13.99) \quad \begin{aligned} \|\nabla u\|_{L^r(\Omega)} &\leq c \ell^{1/r-1/q-1} \|u\|_{L^q(\Omega)}^{q/r} + c \ell^{-1-1/q} (\mathcal{L}^N(\Omega))^{1/r} \\ &\quad + c \|u\|_{L^q(\Omega)}^{1/2} \|\nabla^2 u\|_{L^p(\Omega)}^{1/2} \end{aligned}$$

for some constant $c = c(N, p, q) > 0$. Then

$$\begin{aligned} \|\nabla u\|_{L^r(\Omega)} &\leq c_1 \ell^{-1} (\mathcal{L}^N(\Omega))^{1/r-1/q} \|u\|_{L^q(\Omega)} \\ &\quad + c_1 \|u\|_{L^q(\Omega)}^{1/2} \|\nabla^2 u\|_{L^p(\Omega)}^{1/2} \end{aligned}$$

for some constant $c_1 = c_1(N, p, q) > 0$.

Proof. By replacing u with su in (13.99) we obtain

$$\begin{aligned} \|\nabla u\|_{L^r} &\leq c s^{q/r-1} \ell^{1/r-1/q-1} \|u\|_{L^q}^{q/r} \\ &\quad + c s^{-1} \ell^{-1-1/q} (\mathcal{L}^N(\Omega))^{1/r} + c \|u\|_{L^q}^{1/2} \|\nabla^2 u\|_{L^p}^{1/2}. \end{aligned}$$

We now distinguish two cases. If $\|\nabla^2 u\|_{L^p} \leq a\|u\|_{L^q}$, where $a > 0$, we take $s := b\|u\|_{L^q}^{-1}$, with $b > 0$, in the previous inequality to obtain

$$\begin{aligned} \|\nabla u\|_{L^r} &\leq \left[cb^{q/r-1} \ell^{1/r-1/q-1} \right. \\ &\quad \left. + cb^{-1} \ell^{-1-1/q} (\mathcal{L}^N(\Omega))^{1/r} + ca^{1/2} \right] \|u\|_{L^q}. \end{aligned}$$

On the other hand, if $\|\nabla^2 u\|_{L^p} > a\|u\|_{L^q(R)}$ then we take

$$s := d\|u\|_{L^q}^{(r-2q)/[2(q-r)]} \|\nabla^2 u\|_{L^p}^{r/[2(q-r)]},$$

where $d > 0$, to obtain

$$\begin{aligned} \|\nabla u\|_{L^r} &\leq cd^{-1} \ell^{-1-1/q} (\mathcal{L}^N(\Omega))^{1/r} \|u\|_{L^q}^{-(r-2q)/[2(q-r)]} \|\nabla^2 u\|_{L^p}^{-r/[2(q-r)]} \\ &\quad + c(d^{q/r-1} \ell^{1/r-1/q-1} + 1) \|u\|_{L^q}^{1/2} \|\nabla^2 u\|_{L^p}^{1/2} \\ &\leq cd^{-1} a^{-r/[2(q-r)]} \ell^{-1-1/q} (\mathcal{L}^N(\Omega))^{1/r} \|u\|_{L^q} \\ &\quad + c(d^{q/r-1} \ell^{1/r-1/q-1} + 1) \|u\|_{L^q}^{1/2} \|\nabla^2 u\|_{L^p}^{1/2}. \end{aligned}$$

We now choose

$$a := \ell^{-2} (\mathcal{L}^N(\Omega))^{2/r-2/q}, \quad b := \ell^{-1/q} (\mathcal{L}^N(\Omega))^{1/q}, \quad d := \ell^{(r+r/q-1)/(q-r)}$$

to obtain the desired inequality. $\square$

We turn to the proof of Theorem 13.51.

Proof of Theorem 13.51. We only give the proof in the case $p, q, r < \infty$ and leave the other cases as an exercise.

Step 1: Assume that R is bounded, that $u \in C^\infty(R)$, and that $p \leq q$. In this case $p \leq r \leq q$. Let $i \in \{1, \dots, N\}$ and let $R'_i \subseteq \mathbb{R}^{N-1}$ be obtained from R by removing the i th set in the cartesian product. Then for $x'_i \in R'_i$, the one-dimensional function $u(x'_i, \cdot)$ satisfies the hypotheses of Theorem 7.41, and so,

$$\begin{aligned} \int_{I_i} |\partial_i u(x'_i, x_i)|^r dx_i &\leq c \ell^{1-r-r/q} \left(\int_{I_i} |u(x'_i, x_i)|^q dx_i \right)^{r/q} \\ &\quad + c \left(\int_{I_i} |u(x'_i, x_i)|^q dx_i \right)^{r/(2q)} \left(\int_{I_i} |\partial_i^2 u(x'_i, x_i)|^p dx_i \right)^{r/(2p)}, \end{aligned}$$

where we have used the notation (E.2). If $r < q$ we use the inequality $t^{r/q} \leq t + 1$, which holds for all $t \geq 0$, to obtain

$$\begin{aligned} \int_{I_i} |\partial_i u(x'_i, x_i)|^r dx_i &\leq c \ell^{1-r-r/q} \int_{I_i} |u(x'_i, x_i)|^q dx_i + c \ell^{1-r-r/q} \\ &\quad + c \left(\int_{I_i} |u(x'_i, x_i)|^q dx_i \right)^{r/(2q)} \left(\int_{I_i} |\partial_i^2 u(x'_i, x_i)|^p dx_i \right)^{r/(2p)}. \end{aligned}$$

Integrate the previous inequality in x'_i over R'_i and use Tonelli's theorem to get

$$\begin{aligned} \int_R |\partial_i u|^r dx &\leq c\ell^{1-r-r/q} \int_R |u|^q dx + c\ell^{-r-r/q} \mathcal{L}^N(R) \\ &\quad + c \int_{R'_i} \left(\int_{I_i} |u(x'_i, x_i)|^q dx_i \right)^{r/(2q)} \left(\int_{I_i} |\partial_i^2 u(x'_i, x_i)|^p dx_i \right)^{r/(2p)} dx'_i \\ &\leq c\ell^{1-r-r/q} \int_R |u|^q dx + c\ell^{-r-r/q} \mathcal{L}^N(R) + c\|u\|_{L^q(R)}^{r/2} \|\partial_i^2 u\|_{L^p(R)}^{r/2}, \end{aligned}$$

where in the last inequality we have used Hölder's inequality and (13.96). We are now in a position to apply Lemma 13.52 with $\|\nabla^2 u\|_{L^p}^{r/2}$ replaced by $\|\partial_i^2 u\|_{L^p}^{r/2}$ to get (13.96). The additional hypothesis that $u \in C^\infty(R)$ can be removed using Exercise 11.29.

The proof is significantly simpler in the case $p = q$ or in the case of R unbounded and is left as an exercise. $\square$

Next we consider uniformly Lipschitz continuous domains (see Definition 13.11).

Exercise 13.53. Given $L > 0$, consider the sector

$$\Xi_L := S^{N-1} \cap \{x = (x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : L\|x'\|_{N-1} < x_N\}.$$

Prove that there exists a constant $c = c(L, N, p) > 0$ such that

$$\int_{\Xi_L} |x \cdot y|^p d\mathcal{H}^{N-1}(y) \geq c\|x\|^p \quad \text{for every } x \in \mathbb{R}^N.$$

Theorem 13.54 (Gagliardo–Nirenberg interpolation, $m = 2$). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set with uniformly Lipschitz continuous boundary (with parameters ε, L, M), let $0 < \ell < \varepsilon/(4(1 + L))$, and let $1 \leq p, q, r \leq \infty$ be such that $p \leq q$ and (13.96) holds. If $p < q$ assume further that Ω is bounded. Then there is a constant $c = c(L, N, p, q) > 0$ such that for every $u \in L^q(\Omega) \cap \dot{W}^{2,p}(\Omega)$, if $p < q$,*

$$(13.100) \quad \|\nabla u\|_{L^r(\Omega)} \leq c\ell^{-1}(\mathcal{L}^N(\Omega))^{1/r-1/q} \|u\|_{L^q(\Omega)} + c\|u\|_{L^q(\Omega)}^{1/2} \|\nabla^2 u\|_{L^p(\Omega)}^{1/2}$$

if $p < q$, while if $p = q$,

$$(13.101) \quad \|\nabla u\|_{L^p(\Omega)} \leq c\ell^{-1} \|u\|_{L^p(\Omega)} + c\|u\|_{L^p(\Omega)}^{1/2} \|\nabla^2 u\|_{L^p(\Omega)}^{1/2}.$$

Proof. Step 1: Assume that Ω is bounded. We only give the proof in the case $p < q < \infty$ and leave the case $q = \infty$ as an exercise. Assume first that $u \in C^\infty(\Omega)$. Given $\ell > 0$ and $\xi \in S^{N-1}$, we define $\Omega(\xi, \ell)$ as the set of all points $x \in \Omega$ belonging to a segment parallel to ξ , contained in

Ω , and with length greater than ℓ . We claim that there exists a constant $c = c(L, N, r) > 0$ such that

$$\int_{\Omega} \|\nabla u\|^r dx \leq c \int_{S^{N-1}} \int_{\Omega(\xi, \ell)} |\nabla u(x) \cdot \xi|^r dx d\mathcal{H}^{N-1}(\xi).$$

To see this, we use Tonelli's theorem to write

$$(13.102) \quad \int_{S^{N-1}} \int_{\Omega(\xi, \ell)} |\nabla u(x) \cdot \xi|^r dx d\mathcal{H}^{N-1}(\xi) = \int_{\Omega} \int_{G(x, \ell)} |\nabla u(x) \cdot \xi|^r d\mathcal{H}^{N-1}(\xi) dx,$$

where $G(x, \ell) := \{\xi \in S^{N-1} : x \in \Omega(\xi, \ell)\}$.

Fix $x \in \Omega$. There are two cases: If $\text{dist}(x, \partial\Omega) \geq \ell$, then $\overline{B(x, \ell)} \subset \Omega$, and so $G(x, \ell) = S^{N-1}$. In this case,

$$(13.103) \quad \int_{G(x, \ell)} |\nabla u(x) \cdot \xi|^r d\mathcal{H}^{N-1}(\xi) = c_{N,r} \|\nabla u(x)\|^r,$$

where we used the fact that for every $z \in \mathbb{R}^N \setminus \{0\}$, by a rotation,

$$\begin{aligned} \int_{S^{N-1}} |z \cdot \xi|^r d\mathcal{H}^{N-1}(\xi) &= \|z\|^r \int_{S^{N-1}} \left| \frac{z}{\|z\|} \cdot \xi \right|^r d\mathcal{H}^{N-1}(\xi) \\ &= \|z\|^r \int_{S^{N-1}} |e_1 \cdot \xi|^r d\mathcal{H}^{N-1}(\xi) =: \|z\|^r c_{N,r}. \end{aligned}$$

On the other hand, if $\text{dist}(x, \partial\Omega) < \ell$, then there exists $x_0 \in \partial\Omega$ with $\|x - x_0\| < \ell$. By parts (i) and (iii) of Definition 13.11, there exist $n \in \mathbb{N}$, local coordinates $y = (y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R}$, and a Lipschitz continuous function $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ (both depending on n), with $\text{Lip } f \leq L$, such that $B(x_0, \varepsilon) \subseteq \Omega_n$ and $\Omega_n \cap \Omega = \Omega_n \cap V_n$, where V_n is given in local coordinates by $\{(y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : y_N > f(y')\}$. In particular, $B(x_0, \varepsilon) \cap \Omega = B(x_0, \varepsilon) \cap V_n$. Since $\ell < \varepsilon/(4(1+L))$, we have that $x \in B(x_0, \varepsilon/[4(1+L)])$. Let $(\bar{y}', \bar{y}_N)$ be the local coordinates of x and consider the point $(\bar{y}', f(\bar{y}'))$. Since f is Lipschitz continuous, with $\text{Lip } f \leq L$, for $y' \in \mathbb{R}^{N-1}$, we have that

$$f(y') \leq f(\bar{y}') + L\|y' - \bar{y}'\|_{N-1} < \bar{y}_N + L\|y' - \bar{y}'\|_{N-1}.$$

Hence, the cone

$$\begin{aligned} K_x &= \{(y', y_N) \in B_{N-1}(\bar{y}', \varepsilon/[4(1+L)]) \times \mathbb{R} : \\ &\quad \bar{y}_N + L\|y' - \bar{y}'\|_{N-1} < y_N < \bar{y}_N + \varepsilon/4\} \end{aligned}$$

is contained in $B(x_0, \varepsilon) \cap \Omega$. Using local coordinates, consider the sector

$$\Xi_x := \{\xi = (\xi', \xi_N) \in S^{N-1} : L\|\xi'\|_{N-1} < \xi_N\}.$$

Since $\ell < \varepsilon/(4(1+L))$, if $t \in (0, \ell)$ and $\xi \in \Xi_x$, then the point $x + t\xi$ belongs to $K_x \subseteq \Omega$. Hence, $G(x, \ell) \supseteq \Xi_x$, and so

$$\int_{G(x, \ell)} |\nabla u(x) \cdot \xi|^r d\mathcal{H}^{N-1}(\xi) \geq \int_{\Xi_x} |\nabla u(x) \cdot \xi|^r d\mathcal{H}^{N-1}(\xi) \geq c_{N,r,L} \|\nabla u(x)\|^r,$$

where we have used the previous exercise. Combining this inequality with (13.102) and (13.103) proves the claim.

Step 2: Let Ω be bounded, $p < q < \infty$, and $u \in C^\infty(\Omega)$. Given $\xi \in S^{N-1}$ and $0 < \ell < \varepsilon/(4(1+L))$, consider the hyperplane $\Pi_\xi := \{x \in \mathbb{R}^N : x \cdot \xi = 0\}$ and for $y \in \Pi_\xi$ define the slice $\Omega(\xi, \ell, y) := \{t \in \mathbb{R} : y + t\xi \in \Omega(\xi, \ell)\}$. Note that if $y \in \Pi_\xi$ is such that $\Omega(\xi, \ell, y)$ is nonempty, then $\Omega(\xi, \ell, y)$ is the union of a finite family of pairwise disjoint open intervals with length greater than ℓ and we may define the function $u_{y,\xi} : \Omega(\xi, \ell, y) \rightarrow \mathbb{R}$ by $u_{y,\xi}(t) := u(y + t\xi)$. Let I_i be one such interval. Applying Theorem 7.41 to the function $u_{y,\xi}$ in I_i we obtain

$$\begin{aligned} \int_{I_i} |\nabla u(y + t\xi) \cdot \xi|^r dt &\leq c\ell^{1-r-r/q} \left(\int_{I_i} |u(y + t\xi)|^q dt \right)^{r/q} \\ &\quad + c \left(\int_{I_i} |u(y + t\xi)|^q dt \right)^{1/(2q)} \left(\int_{I_i} \left| \frac{\partial^2 u}{\partial \xi^2}(y + t\xi) \right|^p dt \right)^{1/(2p)}. \end{aligned}$$

Using the inequality $t^{r/q} \leq t + 1$, which holds for all $t \geq 0$, and the fact that $\mathcal{L}^1(I_i) \geq \ell$, we get

$$\begin{aligned} \int_{I_i} |\nabla u(y + t\xi) \cdot \xi|^r dt &\leq c\ell^{1-r-r/q} \int_{I_i} |u(y + t\xi)|^q dt + c\ell^{-r-r/q} \mathcal{L}^1(I_i) \\ &\quad + c \left(\int_{I_i} |u(y + t\xi)|^q dt \right)^{1/(2q)} \left(\int_{I_i} \|\nabla^2 u(y + t\xi)\|^p dt \right)^{1/(2p)}. \end{aligned}$$

Summing over all I_i in $\Omega(\xi, \ell, y)$ and using the inequality (7.17) gives

$$\begin{aligned} \int_{\Omega(\xi, \ell, y)} |\nabla u(y + t\xi) \cdot \xi|^r dt &\leq c\ell^{1-r-r/q} \int_{\Omega(\xi, \ell, y)} |u(y + t\xi)|^q dt + c\ell^{-r-r/q} \mathcal{L}^1(\Omega(\xi, \ell, y)) \\ &\quad + c \left(\int_{\Omega(\xi, \ell, y)} |u(y + t\xi) \cdot \xi|^q dt \right)^{1/(2q)} \left(\int_{\Omega(\xi, \ell, y)} \|\nabla^2 u(y + t\xi)\|^p dt \right)^{1/(2p)}. \end{aligned}$$

Integrating both sides of the previous inequality over all $y \in \Pi_\xi$ for which $\Omega(\xi, \ell, y)$ is nonempty, by Tonelli's theorem, Hölder's inequality and (13.96),

we get

$$\begin{aligned} \int_{\Omega(\xi, \ell)} |\nabla u(x) \cdot \xi|^r dx &\leq c\ell^{1-r-r/q} \int_{\Omega} |u(x)|^q dx + c\ell^{-r-r/q} \mathcal{L}^N(\Omega) \\ &\quad + c\|u\|_{L^q(\Omega)}^{r/2} \|\nabla^2 u\|_{L^p(\Omega)}^{r/2}. \end{aligned}$$

Integrating both sides of the previous inequality in the variable ξ over S^{N-1} and using Step 1 gives

$$\begin{aligned} \int_{\Omega} \|\nabla u\|^r dx &\leq c\ell^{1-r-r/q} \int_{\Omega} |u|^q dx + c\ell^{-r-r/q} \mathcal{L}^N(\Omega) \\ &\quad + c\|u\|_{L^q(\Omega)}^{r/2} \|\nabla^2 u\|_{L^p(\Omega)}^{r/2}. \end{aligned}$$

We are now in a position to apply Lemma 13.52 to obtain (13.100).

The additional hypothesis that $u \in C^\infty(\Omega)$ can be removed using Exercise 11.29. We omit the details.

Step 3: The case $q = p$ and Ω not necessarily bounded is simpler. Indeed, we do not need to use the inequality $t^{r/q} \leq t + 1$ and so the term $c\ell^{-r-r/q} \mathcal{L}^N(\Omega)$ does not appear. The details are left as an exercise. $\square$

Remark 13.55. Note that in the previous proof we only used a uniform cone property.

Remark 13.56. We remark that when $\Omega \subset \mathbb{R}^N$ is an open, bounded, connected set with Lipschitz boundary and if $u \in L^q(\Omega) \cap \dot{W}^{2,p}(\Omega)$, where $1 \leq q \leq p$, then it follows from Poincaré's inequality that u and ∇u are in $L^p(\Omega)$ and so in all $L^r(\Omega)$ for $1 \leq r \leq p$. This is why we only considered the interesting case $p < q$ in the previous theorem.

Remark 13.57. We recall that if $\Omega \subset \mathbb{R}^N$ is an open bounded set with Lipschitz continuous boundary, then in view of Exercise 13.13 there exist $\varepsilon > 0$, $L > 0$, and $M \in \mathbb{N}$ such that Ω has uniformly Lipschitz continuous boundary with parameters ε , L , and M .

Exercise 13.58. Let $\Omega \subset \mathbb{R}^N$ be an open bounded set with Lipschitz continuous boundary, let $0 < \ell < \varepsilon/(4(1+L))$, let $m \in \mathbb{N}$, with $m \geq 2$, and let $1 \leq p, q, r \leq \infty$ be such that $p < q$ and

$$\frac{1}{m} \frac{1}{p} + \left(1 - \frac{1}{m}\right) \frac{1}{q} = \frac{1}{r}.$$

Prove there exists a constant $c = c(m, N, p, q) > 0$ such that for every $u \in L^q(\Omega) \cap \dot{W}^{m,p}(\Omega)$,

$$\|\nabla u\|_{L^r(\Omega)} \leq c\ell^{-1} (\mathcal{L}^N(\Omega))^{1/r-1/q} \|u\|_{L^q(\Omega)} + c\|u\|_{L^q(\Omega)}^{1-1/m} \|\nabla^m u\|_{L^p(\Omega)}^{1/m}.$$

Hint: Repeat the proof of the previous theorem.

An important consequence of the previous theorem is the following result.

Corollary 13.59. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set with uniformly Lipschitz continuous boundary and let $1 \leq p \leq \infty$. Then $W^{2,p}(\Omega) = L^p(\Omega) \cap \dot{W}^{2,p}(\Omega)$ with equivalence of the norms $\sum_{k=0}^2 \|\nabla^k u\|_{L^p(\Omega)}$ and $\|u\|_{L^p(\Omega)} + \|\nabla^2 u\|_{L^p(\Omega)}$.*

Remark 13.60. Note that for sets of infinite measure the previous corollary cannot be obtained using the Poincaré inequality.

Next we consider the case $m \geq 2$.

Theorem 13.61 (Gagliardo–Nirenberg interpolation, $m \geq 2$). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set with uniformly Lipschitz continuous boundary (with parameters ε, L, M), let $0 < \ell < \varepsilon/(4(1+L))$, let $m, k \in \mathbb{N}$, with $m \geq 2$ and $1 \leq k < m$, and let $1 \leq p, q, r \leq \infty$ be such that $p \leq q$ and*

$$(13.104) \quad \frac{k}{m} \frac{1}{p} + \left(1 - \frac{k}{m}\right) \frac{1}{q} = \frac{1}{r}.$$

If $p < q$, assume further that Ω is bounded. Then for every $u \in L^q(\Omega) \cap \dot{W}^{m,p}(\Omega)$,

$$(13.105) \quad \|\nabla^k u\|_{L^r(\Omega)} \leq c \ell^{-k} (\mathcal{L}^N(\Omega))^{1/r-1/q} \|u\|_{L^q(\Omega)} + c \|u\|_{L^q(\Omega)}^{1-k/m} \|\nabla^m u\|_{L^p(\Omega)}^{k/m}$$

if $p < q$, while

$$(13.106) \quad \|\nabla^k u\|_{L^p(\Omega)} \leq c \ell^{-k} \|u\|_{L^p(\Omega)} + c \|u\|_{L^p(\Omega)}^{1-k/m} \|\nabla^m u\|_{L^p(\Omega)}^{k/m}$$

if $p = q$. Here, $c > 0$ is a constant depending on m, N, p, q .

Proof. We prove the case $p < q$. By hypothesis Ω is bounded. The proof is by induction on m . When $m = 2$ and $k = 1$, (13.105) follows from Theorem 14.47. Let $n \in \mathbb{N}$, with $n \geq 2$, and assume that (13.105) holds for all $1 \leq p, q \leq \infty$ with $p < q$ all $m \leq n$, all $1 \leq k < m$, all r given by (13.104), all $u \in L^q(\Omega) \cap \dot{W}^{m,p}(\Omega)$ and for every multi-index $\beta \in \mathbb{N}_0^N$, with $|\beta| = k$.

Fix $1 \leq p, q \leq \infty$ with $p < q$. We claim that (13.105) holds with $m = n + 1$. To be precise, let $1 \leq k < n + 1$ and let r be given by

$$(13.107) \quad \frac{k}{n+1} \frac{1}{p} + \left(1 - \frac{k}{n+1}\right) \frac{1}{q} = \frac{1}{r}$$

and let $u \in L^q(\Omega) \cap \dot{W}^{n+1,p}(\Omega)$. We claim that there exists a constant $c = c(k, n, N, p, q) > 0$ such that

$$(13.108) \quad \|\nabla^k u\|_{L^r} \leq c \ell^{-k} (\mathcal{L}^N(\Omega))^{1/r-1/q} \|u\|_{L^q} + c \|u\|_{L^q}^{1-k/(n+1)} \|\nabla^{n+1} u\|_{L^p}^{k/(n+1)}.$$

If $k = 1$, this follows from Exercise 13.58. Thus, assume that $k \geq 2$. Since $\frac{1}{r}$ belongs to the segment of endpoints $\frac{1}{q}$ and $\frac{1}{p}$ by (13.107), we can find $r_1 \geq 1$ such that

$$(13.109) \quad \frac{1}{k} \frac{1}{r} + \left(1 - \frac{1}{k}\right) \frac{1}{q} = \frac{1}{r_1}.$$

Using (13.107) and (13.109), one can verify that

$$(13.110) \quad \frac{k-1}{n} \frac{1}{p} + \left(1 - \frac{k-1}{n}\right) \frac{1}{r_1} = \frac{1}{r}.$$

By Exercise 13.58,

$$\|\nabla u\|_{L^{r_1}} \leq c\ell^{-1}(\mathcal{L}^N(\Omega))^{1/r_1-1/q}\|u\|_{L^q} + c\|u\|_{L^q}^{1-1/k}\|\nabla^k u\|_{L^r}^{1/k}.$$

Given a multi-index $\beta \in \mathbb{N}_0^N$, with $|\beta| = k$, let $i \in \{1, \dots, N\}$ be such that $\beta = e_i + \delta$, with $|\delta| = k-1$. Hence, by the induction hypothesis applied to the function $\partial_i u$ in place of u ,

$$\|\partial^\beta u\|_{L^r} \leq c\ell^{1-k}(\mathcal{L}^N(\Omega))^{1/r-1/r_1}\|\partial_i u\|_{L^{r_1}} + c\|\partial_i u\|_{L^{r_1}}^{1-(k-1)/n}\|\nabla^n(\partial_i u)\|_{L^p}^{(k-1)/n}.$$

Combining the last two inequalities and summing over all β with $|\beta| = k$ gives

$$\begin{aligned} \|\nabla^k u\|_{L^r} &\leq c\ell^{-k}(\mathcal{L}^N(\Omega))^{1/r-1/q}\|u\|_{L^q} \\ &\quad + c\ell^{1-k}(\mathcal{L}^N(\Omega))^{1/r-1/r_1}\|u\|_{L^q}^{1-1/k}\|\nabla^k u\|_{L^r}^{1/k} \\ &\quad + c\ell^{-1+(k-1)/n}(\mathcal{L}^N(\Omega))^{(1/r_1-1/q)(1-(k-1)/n)}\|u\|_{L^q}^{1-(k-1)/n}\|\nabla^{n+1}u\|_{L^p}^{(k-1)/n} \\ &\quad + c\|u\|_{L^q}^{(1-(k-1)/n)(1-1/k)}\|\nabla^k u\|_{L^r}^{(1/k)(1-(k-1)/n)}\|\nabla^{n+1}u\|_{L^p}^{(k-1)/n} \\ &=: I + II + III + IV. \end{aligned}$$

We now use Young's inequality in the form

$$ab \leq \varepsilon a^t + c_\varepsilon b^{t'}$$

for $a, b \geq 0$, $\varepsilon > 0$, $1 < t < \infty$. Taking $\varepsilon = \frac{1}{3}$ and $t = k$ we have

$$II \leq \frac{1}{3}\|\nabla^k u\|_{L^r} + c\ell^{-k}(\mathcal{L}^N(\Omega))^{1/r-1/q}\|u\|_{L^q},$$

where we used (13.109). Similarly, with $\varepsilon = \frac{1}{3}$ and $t = nk/(n+1-k)$ so that $t' = nk/[(n+1)(k-1)]$,

$$IV \leq \frac{1}{3}\|\nabla^k u\|_{L^r} + c\|u\|_{L^q}^{1-k/(n+1)}\|\nabla^{n+1}u\|_{L^p}^{k/(n+1)}.$$

Finally, with $\varepsilon = 1$ and $t = kn/(n+1-k)$, so that $t' = kn/[(n+1)(k-1)]$, and writing

$$\|u\|_{L^q}^{1-(k-1)/n} = \|u\|_{L^q}^{(n+1-k)/(nk)}\|u\|_{L^q}^{(1-1/k)(n+1-k)/n},$$

we get

$$III \leq c\ell^{-k}(\mathcal{L}^N(\Omega))^{1/r-1/q}\|u\|_{L^q} + c\|u\|_{L^q}^{1-k/(n+1)}\|\nabla^{n+1}u\|_{L^p}^{k/(n+1)},$$

where we used (13.109). Combining the last four inequalities gives (13.105).

The case $p = q$ is simpler and is left as an exercise. $\square$

As in the case $m = 2$, the inequality (13.105) implies the following important consequence.

Corollary 13.62. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set with uniformly Lipschitz continuous boundary, let $m \in \mathbb{N}$ with $m \geq 2$ and let $1 \leq p \leq \infty$. Then $W^{m,p}(\Omega) = L^p(\Omega) \cap \dot{W}^{m,p}(\Omega)$ with equivalence of the norms $\sum_{k=0}^m \|\nabla^k u\|_{L^p(\Omega)}$ and $\|u\|_{L^p(\Omega)} + \|\nabla^m u\|_{L^p(\Omega)}$.*

Functions of Bounded Variation

Living with P.Q.S, I: What is PQS? “Post-Quals-Slump” or PQS, affects 99% of all grad students. It is the #1 cause of delayed graduation dates and contributes to the Ph.D. degree drop rate.

— Jorge Cham, www.phdcomics.com

In this chapter we introduce the space of functions of bounded variation in domains of $\mathbb{R}^N$ and study some of their basic properties. What is covered here is just the tip of the iceberg. We refer the interested reader to the monographs [10], [72], [76], [95], [149], and [251] for more information on the subject.

14.1. Definition and Main Properties

Definition 14.1. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. We define the *space of functions of bounded variation* $BV(\Omega)$ as the space of all functions $u \in L^1(\Omega)$ whose distributional first-order partial derivatives are finite signed Radon measures; that is, for all $i = 1, \dots, N$ there exists a finite signed measure $\lambda_i : \mathcal{B}(\Omega) \rightarrow \mathbb{R}$ such that

$$(14.1) \quad \int_{\Omega} u \partial_i \phi \, dx = - \int_{\Omega} \phi \, d\lambda_i$$

for all $\phi \in C_c^\infty(\Omega)$. The measure λ_i is called the *weak*, or *distributional*, *partial derivative* of u with respect to x_i and is denoted $D_i u$.

Here $\mathcal{B}(\Omega)$ is the Borel σ -algebra in Ω . We define

$$BV_{\text{loc}}(\Omega) := \{u \in L^1_{\text{loc}}(\Omega) : u \in BV(U) \text{ for all open sets } U \Subset \Omega\}.$$

For $u \in BV(\Omega)$ we set $Du := (D_1u, \dots, D_Nu)$. Thus, if $u \in BV(\Omega)$, then $Du \in \mathcal{M}_b(\Omega; \mathbb{R}^N)$ (see (B.26)) and so the *total variation measure* of Du , defined by

$$(14.2) \quad \|Du\|(E) := \sup \left\{ \sum_{n=1}^{\infty} \|Du(E_n)\| \right\}, \quad E \in \mathcal{B}(\Omega),$$

where the supremum is taken over all partitions $\{E_n\}_n$ in $\mathcal{B}(\Omega)$ of E , is a finite Radon measure (see Proposition B.74). Moreover, since $\mathcal{M}_b(\Omega; \mathbb{R}^N)$ may be identified with the dual of $C_0(\Omega; \mathbb{R}^N)$ (see Theorem B.111), we have that

$$\begin{aligned} \|Du\|(\Omega) &= \|Du\|_{\mathcal{M}_b(\Omega; \mathbb{R}^N)} \\ &= \sup \left\{ \sum_{i=1}^N \int_{\Omega} \Phi_i dD_iu : \Phi \in C_0(\Omega; \mathbb{R}^N), \|\Phi\|_{C_0(\Omega)} \leq 1 \right\} < \infty. \end{aligned}$$

Definition 14.2. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in L^1_{\text{loc}}(\Omega)$. The *variation* of u in Ω is defined by

$$V(u, \Omega) := \sup \left\{ \sum_{i=1}^N \int_{\Omega} \partial_i \Phi_i u \, dx : \Phi \in C_c^\infty(\Omega; \mathbb{R}^N), \|\Phi\|_{C_0(\Omega)} \leq 1 \right\}.$$

Exercise 14.3. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in L^1_{\text{loc}}(\Omega)$. Prove the following.

- (i) If the distributional gradient Du of u belongs to $\mathcal{M}_b(\Omega; \mathbb{R}^N)$, then $\|Du\|(\Omega) = V(u, \Omega)$.
- (ii) If $V(u, \Omega) < \infty$, then the distributional gradient Du of u belongs to $\mathcal{M}_b(\Omega; \mathbb{R}^N)$. In particular, if $u \in L^1(\Omega)$, then u belongs to $BV(\Omega)$ if and only if $V(u, \Omega) < \infty$. Hint: Use the Riesz representation theorem in $C_0(\Omega; \mathbb{R}^N)$.
- (iii) If $\{u_n\}_n$ is a sequence of functions in $L^1_{\text{loc}}(\Omega)$ converging to u in $L^1_{\text{loc}}(\Omega)$, then

$$V(u, \Omega) \leq \liminf_{n \rightarrow \infty} V(u_n, \Omega).$$

Exercise 14.4. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Prove that the space $BV(\Omega)$ is a Banach space with the norm

$$\|u\|_{BV(\Omega)} := \|u\|_{L^1(\Omega)} + \|Du\|(\Omega).$$

It follows from the definition of $BV(\Omega)$ that

$$W^{1,1}(\Omega) \subset BV(\Omega).$$

We have already seen that for $N = 1$ the inclusion is strict. Exercise 14.5 shows that this is also the case in higher dimensions.

Exercise 14.5. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $E \subset \mathbb{R}^N$, $N \geq 2$, be a bounded set with C^2 boundary.

- (i) Prove that $\chi_E \notin W^{1,1}(\Omega)$.
- (ii) Prove that $\|D(\chi_E)\|(\Omega) \leq \mathcal{H}^{N-1}(\partial E \cap \Omega)$. Hint: Use Exercise 14.3(i).
- (iii) Prove that the normal ν to ∂E may be extended to a function $C^1(\mathbb{R}^N; \mathbb{R}^N)$.
- (iv) Prove that

$$(14.3) \quad \|D(\chi_E)\|(\Omega) = \mathcal{H}^{N-1}(\partial E \cap \Omega).$$

The previous example shows that characteristic functions of smooth sets belong to $BV(\Omega)$. More generally, we have the following.

Definition 14.6. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $\Omega \subseteq \mathbb{R}^N$ be an open set. The *perimeter* of E in Ω , denoted $P(E, \Omega)$, is the variation of χ_E in Ω , that is,

$$P(E, \Omega) := V(\chi_E, \Omega) \\ = \sup \left\{ \sum_{i=1}^N \int_E \partial_i \Phi_i dx : \Phi \in C_c^\infty(\Omega; \mathbb{R}^N), \|\Phi\|_{C_0(\Omega)} \leq 1 \right\}.$$

The set E is said to have *finite perimeter in Ω* if $P(E, \Omega) < \infty$.

If $\Omega = \mathbb{R}^N$, we write

$$(14.4) \quad P(E) := P(E, \mathbb{R}^N).$$

Remark 14.7. In view of Exercise 14.3, if $\Omega \subseteq \mathbb{R}^N$ is an open set and $E \subseteq \mathbb{R}^N$ is a Lebesgue measurable set with $\mathcal{L}^N(E \cap \Omega) < \infty$, then χ_E belongs to $BV(\Omega)$ if and only if $P(E, \Omega) < \infty$.

Note that Exercise 14.5 shows that for smooth sets E the perimeter of E in Ω is simply $\mathcal{H}^{N-1}(\partial E \cap \Omega)$. The next exercise shows that this fact no longer holds if the boundary of the set E is not (sufficiently) smooth.

Exercise 14.8. Let $\mathbb{Q}^N = \{x_n : n \in \mathbb{N}\}$ and let $U \subset \mathbb{R}^N$ be the open set

$$U := \bigcup_{n=1}^{\infty} B(x_n, 1/2^n).$$

- (i) Prove that $\mathcal{L}^N(U) < \infty$ but $\mathcal{L}^N(\partial U) = \infty$.
- (ii) Prove that $\mathcal{H}^{N-1}(\partial U) = \infty$.
- (iii) Prove that U has finite perimeter. Hint: Define $u_m := \chi_{U_m}$, where $U_m := \bigcup_{n=1}^m B(x_n, 1/2^n)$, and use Exercise 14.3(iii).

14.2. Approximation by Smooth Functions

Since every function $u \in C^\infty(\Omega) \cap BV(\Omega)$ belongs to $W^{1,1}(\Omega)$ (why?) and

$$\|Du\|(\Omega) = \int_{\Omega} \|\nabla u\| \, dx,$$

the closure of $C^\infty(\Omega) \cap BV(\Omega)$ in $BV(\Omega)$ is $W^{1,1}(\Omega)$. Thus, we cannot expect the Meyers–Serrin density theorem (see Theorem 11.24) to hold in $BV(\Omega)$. However, the following weaker version holds.

Theorem 14.9. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in BV(\Omega)$. Then there exists a sequence $\{u_n\}_n$ in $C^\infty(\Omega) \cap W^{1,1}(\Omega)$ such that $u_n \rightarrow u$ in $L^1(\Omega)$ and*

$$\lim_{n \rightarrow \infty} \int_{\Omega} \|\nabla u_n\| \, dx = \|Du\|(\Omega).$$

We begin with an auxiliary result.

Lemma 14.10. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in BV(\Omega)$. For every $\varepsilon > 0$ define $u_\varepsilon := \varphi_\varepsilon * u$, where φ_ε is a standard mollifier and let Ω_ε be defined as in (C.6). Then*

$$(14.5) \quad \lim_{\varepsilon \rightarrow 0^+} \int_{\Omega_\varepsilon} \|\nabla u_\varepsilon\| \, dx = \|Du\|(\Omega).$$

Moreover, for every open set $U \subset \Omega$, with $\text{dist}(U, \partial\Omega) > 0$ and $\|Du\|(\partial U) = 0$,

$$(14.6) \quad \lim_{\varepsilon \rightarrow 0^+} \int_U \|\nabla u_\varepsilon\| \, dx = \|Du\|(U).$$

Proof. Step 1: For every $i = 1, \dots, N$, define $\lambda_i := D_i u \in \mathcal{M}_b(\Omega)$. By Theorem C.20, for every $x \in \Omega_\varepsilon$ we have

$$(14.7) \quad \begin{aligned} \frac{\partial u_\varepsilon}{\partial x_i}(x) &= \int_{\Omega} \frac{\partial_i \varphi_\varepsilon}{\partial x_i}(x-y) u(y) \, dy = - \int_{\Omega} \frac{\partial \varphi_\varepsilon}{\partial y_i}(x-y) u(y) \, dy \\ &= \int_{\Omega} \varphi_\varepsilon(x-y) \, d\lambda_i(y), \end{aligned}$$

where we have used (14.1) and the fact that the function $\varphi_\varepsilon(x - \cdot)$ belongs to $C_c^\infty(\Omega)$, since $\text{supp } \varphi_\varepsilon(x - \cdot) \subseteq \overline{B(x, \varepsilon)}$.

Let $\Phi \in C_c^\infty(\Omega_\varepsilon; \mathbb{R}^N)$ be such that $\|\Phi\|_{C_0(\Omega_\varepsilon)} \leq 1$ and extend Φ to be zero outside Ω_ε . By (14.7) and an integration by parts, we have

$$\begin{aligned} \int_{\Omega_\varepsilon} u_\varepsilon(x) \frac{\partial \Phi_i}{\partial x_i}(x) dx &= - \int_{\Omega_\varepsilon} \frac{\partial u_\varepsilon}{\partial x_i}(x) \Phi_i(x) dx \\ &= - \int_{\Omega_\varepsilon} \int_{\Omega} \Phi_i(x) \varphi_\varepsilon(x-y) d\lambda_i(y) dx \\ &= - \int_{\Omega} (\Phi_i)_\varepsilon(y) d\lambda_i(y) = \int_{\Omega} \frac{\partial (\Phi_i)_\varepsilon}{\partial y_i}(y) u(y) dy, \end{aligned}$$

where we have used Fubini's theorem and (14.1). Hence,

$$(14.8) \quad \int_{\Omega_\varepsilon} u_\varepsilon \operatorname{div} \Phi dx = \int_{\Omega} u \operatorname{div} \Phi_\varepsilon dx.$$

Moreover, for each $x \in \Omega_\varepsilon$, by Jensen's inequality (see Theorem B.50), which can be applied since $\int_{\Omega} \varphi_\varepsilon(x-y) dy = 1$, and by the fact that $\|\Phi\| \leq 1$, we have

$$\begin{aligned} \sum_{i=1}^N ((\Phi_i)_\varepsilon)^2(x) &= \sum_{i=1}^N \left(\int_{\Omega} \Phi_i(y) \varphi_\varepsilon(x-y) dy \right)^2 \\ (14.9) \quad &\leq \sum_{i=1}^N \int_{\Omega} \Phi_i^2(y) \varphi_\varepsilon(x-y) dy \\ &= \int_{\Omega} \|\Phi(y)\|^2 \varphi_\varepsilon(x-y) dy \leq \int_{\Omega} \varphi_\varepsilon(x-y) dy = 1, \end{aligned}$$

and (since $\operatorname{supp} \Phi \subset \Omega_\varepsilon$)

$$(14.10) \quad \operatorname{supp} \Phi_\varepsilon \subset \{x \in \Omega : \operatorname{dist}(x, \operatorname{supp} \Phi) < \varepsilon\} \subset \Omega.$$

Thus, by (14.8) and Exercise 14.3,

$$\int_{\Omega_\varepsilon} u_\varepsilon \operatorname{div} \Phi dx = \int_{\Omega} u \operatorname{div} \Phi_\varepsilon dx \leq \|Du\|(\Omega).$$

Taking the supremum over all admissible Φ and using Exercise 14.3 once more, this time applied to Ω_ε , yields

$$(14.11) \quad \|Du_\varepsilon\|(\Omega_\varepsilon) \leq \|Du\|(\Omega).$$

In particular, $u_\varepsilon \in BV(\Omega_\varepsilon) \cap C^\infty(\Omega_\varepsilon)$. Letting $\varepsilon \rightarrow 0^+$, we conclude that

$$(14.12) \quad \limsup_{\varepsilon \rightarrow 0^+} \|Du_\varepsilon\|(\Omega_\varepsilon) \leq \|Du\|(\Omega).$$

To prove the converse inequality, fix an open set $U \subset \Omega$ with the property that $\operatorname{dist}(U, \partial\Omega) > 0$. Since $u_\varepsilon \rightarrow u$ in $L^1(U)$ as $\varepsilon \rightarrow 0^+$ (see Theorem C.16), by Exercise 14.3,

$$(14.13) \quad \|Du\|(U) \leq \liminf_{\varepsilon \rightarrow 0^+} \|Du_\varepsilon\|(U).$$

Since $\Omega_\varepsilon \supset U$ for all $0 < \varepsilon < \text{dist}(U, \partial\Omega)$, it follows that

$$\|Du\|(U) \leq \liminf_{\varepsilon \rightarrow 0^+} \|Du_\varepsilon\|(\Omega_\varepsilon).$$

Letting $U \nearrow \Omega$ and using Proposition B.9 gives

$$(14.14) \quad \|Du\|(\Omega) \leq \liminf_{\varepsilon \rightarrow 0^+} \|Du_\varepsilon\|(\Omega_\varepsilon),$$

which, together with (14.12), gives (14.5).

Step 2: Fix an open set $U \subset \Omega$, with $\text{dist}(U, \partial\Omega) > 0$ and $\|Du\|(\partial\Omega) = 0$. In view of (14.13), it remains to show that

$$\limsup_{\varepsilon \rightarrow 0^+} \|Du_\varepsilon\|(U) \leq \|Du\|(U).$$

We proceed as in the previous step, with the only change that we now consider $\Phi \in C_c^\infty(U; \mathbb{R}^N)$ such that $\|\Phi\|_{C_0(U)} \leq 1$. If $0 < \varepsilon < \text{dist}(U, \partial\Omega)$, then by (14.10),

$$\text{supp } \Phi_\varepsilon \subset U^\varepsilon := \{x \in \Omega : \text{dist}(x, U) < \varepsilon\} \subset \Omega,$$

and so (14.11) should be replaced by

$$\|Du_\varepsilon\|(U) \leq \|Du\|(U^\varepsilon).$$

Letting $\varepsilon \rightarrow 0^+$ and using Proposition B.9, we conclude that

$$\limsup_{\varepsilon \rightarrow 0^+} \|Du_\varepsilon\|(U) \leq \|Du\|(\overline{U}) = \|Du\|(U),$$

since by hypothesis $\|Du\|(\partial U) = 0$. □

Remark 14.11. If $\Omega = \mathbb{R}^N$, then $\Omega_\varepsilon = \mathbb{R}^N$, and so $u_\varepsilon \rightarrow u$ in $L^1(\mathbb{R}^N)$ and by (14.6),

$$(14.15) \quad \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^N} \|\nabla u_\varepsilon\| dx = \|Du\|(\mathbb{R}^N).$$

Note that the previous proof continues to hold if $u \in L^1_{\text{loc}}(\mathbb{R}^N)$ and its distributional gradient Du belongs to $\mathcal{M}_b(\mathbb{R}^N; \mathbb{R}^N)$. The only difference is that now $u_\varepsilon \rightarrow u$ in $L^1_{\text{loc}}(\mathbb{R}^N)$.

We observe that if $u = \chi_E$, then (14.15) gives

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^N} \|\nabla(\chi_E)_\varepsilon\| dx = P(E).$$

This identity can be used to define the perimeter of a set. This approach was taken by De Giorgi [58], [59], who considered a somewhat different family of mollifiers.

Exercise 14.12. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $u \in BV(\Omega)$, and let $\{u_n\}_n$ be a sequence of functions in $BV(\Omega)$ converging in $L^1_{\text{loc}}(\Omega)$ to u and such that

$$\|Du\|(\Omega) = \lim_{n \rightarrow \infty} \|Du_n\|(\Omega).$$

Prove that $D_i u_n \xrightarrow{*} D_i u$ in $\mathcal{M}_b(\Omega)$ for all $i = 1, \dots, N$ and that

$$\|Du\|(U) = \lim_{n \rightarrow \infty} \|Du_n\|(U)$$

for every open set $U \subseteq \Omega$, with $\|Du\|(\partial U \cap \Omega) = 0$.

Exercise 14.13. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in BV(\Omega)$. Fix $1 \leq i \leq N$.

(i) Prove that

$$|D_i u|(\Omega) = \sup \left\{ \int_{\Omega} \phi dD_i u : \phi \in C_c^\infty(\Omega), \|\phi\|_{C_0(\Omega)} \leq 1 \right\}.$$

(ii) Prove that

$$\int_{\Omega_\varepsilon} |\partial_i u_\varepsilon| dx \leq |D_i u|(\Omega)$$

for all $\varepsilon > 0$, that

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\Omega_\varepsilon} |\partial_i u_\varepsilon| dx = |D_i u|(\Omega),$$

and that for every open set $U \subset \Omega$, with $\text{dist}(U, \partial\Omega) > 0$ and $|D_i u|(\partial U) = 0$,

$$\lim_{\varepsilon \rightarrow 0^+} \int_U |\partial_i u_\varepsilon| dx = |D_i u|(U).$$

(iii) Prove that if $u \in L^1_{\text{loc}}(\mathbb{R}^N)$ and its distributional gradient Du belongs to $\mathcal{M}_b(\mathbb{R}^N; \mathbb{R}^N)$, then

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^N} |\partial_i u_\varepsilon| dx = |D_i u|(\mathbb{R}^N).$$

Exercise 14.14. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Prove that if $u \in BV(\Omega)$ and $\phi \in C_c^\infty(\Omega)$, then $\phi u \in BV(\Omega)$ and

$$\|D(\phi u)\|(\Omega) \leq \|\phi\|_{C_0(\Omega)} \|Du\|(U) + \|\nabla \phi\|_{C_0(\Omega)} \|u\|_{L^1(U)},$$

where $U \subseteq \Omega$ is any open set such that $\text{supp } \phi \subset U$.

We now turn to the proof of Theorem 14.9.

Proof of Theorem 14.9. Since the measure $\|Du\|$ is finite,

$$\lim_{j \rightarrow \infty} \|Du\|(\Omega \setminus \{x \in \Omega : \text{dist}(x, \partial\Omega) > 1/j, \|x\| < j\}) = 0.$$

Fix $\eta > 0$ and let $j_0 \in \mathbb{N}$ be so large that

$$(14.16) \quad \|Du\|(\Omega \setminus \{x \in \Omega : \text{dist}(x, \partial\Omega) > 1/j, \|x\| < j\}) \leq \eta$$

for all $j \geq j_0$. For $i \in \mathbb{N}$ define

$$\Omega_i := \{x \in \Omega : \text{dist}(x, \partial\Omega) > 1/(j_0 + i), \|x\| < j_0 + i\}.$$

As in the proof of the Meyers–Serrin density theorem (see Theorem 11.24), consider a smooth partition of unity $\mathcal{F}$ subordinated to the open cover $\{\Omega_{i+1} \setminus \overline{\Omega_{i-1}}\}$, where $\Omega_{-1} = \Omega_0 := \emptyset$. For each $i \in \mathbb{N}$ let ψ_i be the sum of all the finitely many $\psi \in \mathcal{F}$ such that $\text{supp } \psi \subset \Omega_{i+1} \setminus \overline{\Omega_{i-1}}$ and such that they have not already been selected at previous steps $j < i$. For every $i \in \mathbb{N}$ find $\varepsilon_i > 0$ so small that $\text{supp}(\psi_i u)_{\varepsilon_i} \subset \Omega_{i+1} \setminus \overline{\Omega_{i-1}}$ and, using properties of mollifiers (see Theorems C.16 and C.20),

$$(14.17) \quad \int_{\Omega} |(\psi_i u)_{\varepsilon_i} - \psi_i u| dx \leq \frac{\eta}{2^i}, \quad \int_{\Omega} \|(u \nabla \psi_i)_{\varepsilon_i} - u \nabla \psi_i\| dx \leq \frac{\eta}{2^i}.$$

Define $v_\eta := \sum_{i=1}^{\infty} (\psi_i u)_{\varepsilon_i}$, where we have extended each $(\psi_i u)_{\varepsilon_i}$ to be zero outside its support. As in the Meyers–Serrin theorem we have that $v_\eta \in C^\infty(\Omega)$ and (see the proof of (11.10)), $\int_{\Omega} |u - v_\eta| dx \leq \eta$. Thus, $v_\eta \rightarrow u$ in $L^1(\Omega)$ as $\eta \rightarrow 0^+$.

Next we prove that

$$(14.18) \quad \limsup_{\eta \rightarrow 0^+} \|Dv_\eta\|(\Omega) \leq \|Du\|(\Omega).$$

Let $\Phi \in C_c^\infty(\Omega; \mathbb{R}^N)$ be such that $\|\Phi\|_{C_0(\Omega)} \leq 1$. Since by the previous exercise $\psi_i u \in BV(\Omega)$ for every $i \in \mathbb{N}$, by (14.8) with $\psi_i u$ in place of u we have

$$\int_{\Omega} (\psi_i u)_{\varepsilon_i} \text{div } \Phi dx = \int_{\Omega} \psi_i u \text{div } \Phi_{\varepsilon_i} dx,$$

and so, using the facts that the support of Φ is compact and that the partition of unity is locally finite, we have that

$$(14.19) \quad \begin{aligned} \int_{\Omega} v_\eta \text{div } \Phi dx &= \sum_{i=1}^{\infty} \int_{\Omega} (\psi_i u)_{\varepsilon_i} \text{div } \Phi dx = \sum_{i=1}^{\infty} \int_{\Omega} \psi_i u \text{div } \Phi_{\varepsilon_i} dx \\ &= \sum_{i=1}^{\infty} \int_{\Omega} u [\text{div}(\psi_i \Phi_{\varepsilon_i}) - \nabla \psi_i \cdot \Phi_{\varepsilon_i}] dx =: I + II. \end{aligned}$$

Since $\|\psi_i \Phi_{\varepsilon_i}\|_{C_0(\Omega)} \leq 1$ by (14.9), it follows from Exercise 14.3 applied to Ω and to $\Omega_{i+1} \setminus \overline{\Omega_{i-1}}$ (see also (11.8)) that

$$\begin{aligned} I &= \int_{\Omega} u \operatorname{div}(\psi_1 \Phi_{\varepsilon_1}) \, dx + \sum_{i=2}^{\infty} \int_{\Omega} u \operatorname{div}(\psi_i \Phi_{\varepsilon_i}) \, dx \\ &\leq \|Du\|(\Omega) + \sum_{i=2}^{\infty} \|Du\|(\Omega_{i+1} \setminus \overline{\Omega_{i-1}}). \end{aligned}$$

Using telescopic series, the right-hand side of the previous inequality may be bounded by

$$\|Du\|(\Omega) + 3\|Du\|(\Omega \setminus \Omega_1) \leq \|Du\|(\Omega) + 3\eta$$

by (14.16), and so

$$(14.20) \quad I \leq \|Du\|(\Omega) + 3\eta.$$

To estimate II , note that by Fubini's theorem

$$II = - \sum_{i=1}^{\infty} \int_{\Omega} (u \nabla \psi_i)_{\varepsilon_i} \cdot \Phi \, dx = - \sum_{i=1}^{\infty} \int_{\Omega} [(u \nabla \psi_i)_{\varepsilon_i} - u \nabla \psi_i] \cdot \Phi \, dx,$$

where in the second identity we have used the fact that $\sum_{i=1}^{\infty} \nabla \psi_i = 0$ by (11.6) and the local finiteness of the partition of unity. We now use (14.17) and the fact that $\|\Phi\|_{C_0(\Omega)} \leq 1$ to conclude that $II \leq \eta$, which, together with (14.19) and (14.20), yields

$$\int_{\Omega} v_{\eta} \operatorname{div} \Phi \, dx \leq \|Du\|(\Omega) + 3\eta.$$

Taking the supremum over all admissible Φ and using Exercise 14.3 gives

$$\|Dv_{\eta}\|(\Omega) \leq \|Du\|(\Omega) + 3\eta.$$

It now suffices to let $\eta \rightarrow 0^+$ to obtain (14.18).

To prove the converse inequality, note that, since $v_{\eta} \rightarrow u$ in $L^1(\Omega)$ as $\eta \rightarrow 0^+$, by Exercise 14.3,

$$\|Du\|(\Omega) \leq \liminf_{\eta \rightarrow 0^+} \|Dv_{\eta}\|(\Omega).$$

This concludes the proof. $\square$

Remark 14.15. Note that the previous proof continues to hold if $u \in L^1_{\text{loc}}(\Omega)$ and its distributional gradient Du belongs to $\mathcal{M}_b(\Omega; \mathbb{R}^N)$. The only difference is that now $\{u_n\}_n$ is in $C^\infty(\Omega) \cap W^{1,1}_{\text{loc}}(\Omega)$ and $u_{\varepsilon} \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$.

Exercise 14.16. Prove that the sequence constructed in Theorem 14.9 has the property that

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\Omega} |\partial_i u_n| \, dx = |D_i u|(\Omega)$$

for every $i = 1, \dots, N$.

Remark 14.17. If $N \geq 2$ and $u \in L^1_{\text{loc}}(\mathbb{R}^N)$ is such that its distributional gradient Du belongs to $\mathcal{M}_b(\Omega; \mathbb{R}^N)$, then reasoning as in Step 3 of the proof of Theorem 11.43, one can find a sequence $\{u_n\}_n$ in $C_c^\infty(\mathbb{R}^N)$ such that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \|\nabla u_n\| dx = \|Du\|(\mathbb{R}^N).$$

14.3. Bounded Pointwise Variation on Lines

In Theorem 11.45 we have seen that if a function u belongs to $W^{1,p}(\Omega)$, then it admits a representative that is absolutely continuous on $\mathcal{L}^{N-1}$ -a.a. line segments of Ω that are parallel to the coordinate axes and whose first-order (classical) partial derivatives agree $\mathcal{L}^N$ -a.e. with the weak derivatives of u .

In this section we prove a similar result for a function in $BV(\Omega)$. In what follows, we use the notation (E.2) in Appendix E. Given an open rectangle $R = (a_1, b_1) \times \cdots \times (a_N, b_N) \subset \mathbb{R}^N$, following (E.2), we consider the rectangle R to be the Cartesian product of a rectangle $R'_i \subset \mathbb{R}^{N-1}$ of the variable x'_i and of an interval $R_i \subset \mathbb{R}$ in the x_i variable and we write $R = R'_i \times R_i$.

Given $x'_i \in \mathbb{R}^{N-1}$ and a set $E \subseteq \mathbb{R}^N$, we write

$$(14.21) \quad E_{x'_i} := \{x_i \in \mathbb{R} : (x'_i, x_i) \in E\}.$$

Moreover, if v is a real-valued function defined $\mathcal{L}^N$ -a.e. on an open set $\Omega \subseteq \mathbb{R}^N$, for all $x'_i \in \mathbb{R}^{N-1}$ for which $\Omega_{x'_i}$ is nonempty we let $\text{essVar}_{\Omega_{x'_i}} v(x'_i, \cdot)$ be the essential pointwise variation of the function $x_i \in \Omega_{x'_i} \mapsto v(x'_i, \cdot)$ (see Definition 7.8 and Exercise 7.10). If v is Lebesgue integrable in Ω , with a slight abuse of notation, if $\Omega_{x'_i}$ is empty, we set $\int_{\Omega_{x'_i}} v(x'_i, x_i) dx_i := 0$, so that by Fubini's theorem

$$(14.22) \quad \int_{\Omega} v(x) dx = \int_{\mathbb{R}^{N-1}} \int_{\Omega_{x'_i}} v(x'_i, x_i) dx_i dx'_i.$$

Finally, for every set F in some Euclidean space $\mathbb{R}^m$ and for every $\varepsilon > 0$ we set

$$(14.23) \quad F_\varepsilon := \{y \in F : \text{dist}(y, \partial F) > \varepsilon\}.$$

Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in BV(\Omega)$. Setting $\lambda := Du \in \mathcal{M}_b(\Omega; \mathbb{R}^N)$, by Theorem B.76, the vectorial measure λ can be expressed in the form

$$(14.24) \quad \lambda(E) = \int_E \Psi(x) dx + \lambda^s(E)$$

for every Borel set $E \subseteq \Omega$, where $\Psi \in L^1(\Omega; \mathbb{R}^N)$ and λ^s is a vector-valued measure singular with respect to the Lebesgue measure. Here Ψ is the

Radon–Nikodym derivative $\frac{dDu}{d\mathcal{L}^N|_\Omega}$ of the vectorial measure Du with respect to the Lebesgue measure $\mathcal{L}^N$ restricted to Ω .

The following result shows the relation between the weak derivatives of a function in $BV(\Omega)$ and the (classical) partial derivatives of its representative (when they exist).

Theorem 14.18. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $u \in BV(\Omega)$, and let $1 \leq i \leq N$. If v is any function equivalent to u for which the (classical) partial derivative $\partial_i v(x)$ exists in $\mathbb{R}$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$, then $\partial_i v(x) = \Psi_i(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$, where Ψ is the vectorial function given in (14.24).*

Proof. Consider a convex function $f : \mathbb{R} \rightarrow [0, \infty)$ such that

$$(14.25) \quad f(s+t) \leq f(s) + |t|$$

for all $s, t \in \mathbb{R}$. We claim that for every rectangle R whose closure is contained in Ω ,

$$(14.26) \quad \int_R f(\partial_i v) dx \leq \liminf_{\varepsilon \rightarrow 0^+} \int_{R_\varepsilon} f(\partial_i u_\varepsilon) dx.$$

To begin with, for every function $\phi \in C^\infty(\overline{R})$, for $x = (x'_i, x_i) \in R$ and for all $h > 0$ sufficiently small by Jensen's inequality (see Theorem B.50) one has

$$\begin{aligned} f\left(\frac{\phi(x'_i, x_i + h) - \phi(x'_i, x_i)}{h}\right) &= f\left(\frac{1}{h} \int_{x_i}^{x_i+h} \partial_i \phi(x'_i, s) ds\right) \\ &\leq \frac{1}{h} \int_{x_i}^{x_i+h} f(\partial_i \phi(x'_i, s)) ds. \end{aligned}$$

Integrating this inequality along the interval $(R_i)_h$ (see (14.23)) and using Tonelli's theorem yields

$$\begin{aligned} &\int_{(R_i)_h} f\left(\frac{\phi(x'_i, x_i + h) - \phi(x'_i, x_i)}{h}\right) dx_i \\ &\leq \frac{1}{h} \int_{(R_i)_h} \int_{x_i}^{x_i+h} f(\partial_i \phi(x'_i, s)) ds dx_i \\ &\leq \frac{1}{h} \int_{R_i} \int_{s-h}^s f(\partial_i \phi(x'_i, s)) dx_i ds = \int_{R_i} f(\partial_i \phi(x'_i, s)) ds. \end{aligned}$$

Integrating in x'_i and using Tonelli's theorem, one then obtains

$$\int_{R_h} f\left(\frac{\phi(x'_i, x_i + h) - \phi(x'_i, x_i)}{h}\right) dx \leq \int_R f(\partial_i \phi(x)) dx.$$

Setting $\phi = u_\varepsilon$ in the previous inequality, replacing R with R_ε , and then letting $\varepsilon \rightarrow 0^+$, we deduce from Fatou's lemma that

$$\int_{R_h} f\left(\frac{u(x'_i, x_i + h) - u(x'_i, x_i)}{h}\right) dx \leq \liminf_{\varepsilon \rightarrow 0^+} \int_{R_\varepsilon} f(\partial_i u_\varepsilon(x)) dx.$$

Since v is a representative of u , the previous inequality may be written as

$$\int_{R_h} f\left(\frac{v(x'_i, x_i + h) - v(x'_i, x_i)}{h}\right) dx \leq \liminf_{\varepsilon \rightarrow 0^+} \int_{R_\varepsilon} f(\partial_i u_\varepsilon(x)) dx.$$

If we now let $h \rightarrow 0^+$ and use Fatou's lemma, we obtain (14.26).

Next we prove that

$$(14.27) \quad \int_{R_\varepsilon} f(\partial_i u_\varepsilon) dx \leq \int_R f(\Psi_i) dx + |\lambda_i^s|(R),$$

where $|\lambda_i^s|$ is the total variation measure of λ_i^s . By (14.7) and (14.24) for $x \in R_\varepsilon$ we have

$$\partial_i u_\varepsilon(x) = \int_{\Omega} \varphi_\varepsilon(x - y) \Psi_i(y) dy + \int_{\Omega} \varphi_\varepsilon(x - y) d\lambda_i^s(y).$$

Therefore by virtue of (14.25) and Jensen's inequality (see Theorem B.50)

$$\begin{aligned} f(\partial_i u_\varepsilon(x)) &\leq f\left(\int_{\Omega} \varphi_\varepsilon(x - y) \Psi_i(y) dy\right) + \left|\int_{\Omega} \varphi_\varepsilon(x - y) d\lambda_i^s(y)\right| \\ &\leq \int_{\Omega} \varphi_\varepsilon(x - y) f(\Psi_i(y)) dy + \int_{\Omega} \varphi_\varepsilon(x - y) d|\lambda_i^s|(y). \end{aligned}$$

Integrating over R_ε and applying Tonelli's theorem yields (14.27).

Combining (14.26) and (14.27), we obtain

$$(14.28) \quad \int_R f(\partial_i v) dx \leq \int_R f(\Psi_i) dx + |\lambda_i^s|(R).$$

By Besicovitch's derivation theorem (see Theorem B.116)

$$\lim_{r \rightarrow 0^+} \frac{|\lambda_i^s|(Q(x_0, r))}{r^N} = 0$$

for $\mathcal{L}^N$ -a.e. $x_0 \in \Omega$. Let x_0 be any such point and assume that x_0 is also a Lebesgue point for $f(\partial_i v)$ and $f(\Psi_i)$. By replacing R with $Q(x_0, r)$ in (14.28), dividing by r^N , and letting $r \rightarrow 0^+$, we obtain

$$(14.29) \quad f(\partial_i v(x_0)) \leq f(\Psi_i(x_0)).$$

Now choose f to be the function

$$f_1(t) := \begin{cases} e^t & \text{if } t < 0, \\ t + 1 & \text{if } t \geq 0. \end{cases}$$

Since f_1 is monotone increasing, it follows from (14.29) that $\partial_i v(x) \leq \Psi_i(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$. Similarly, setting $f(t) := f_1(-t)$, we find that $\partial_i v(x) \geq \Psi_i(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$. $\square$

Next we prove that Theorem 11.45 continues to hold for $BV(\Omega)$.

Definition 14.19. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. A function $u : \Omega \rightarrow \mathbb{R}$ is called of *bounded pointwise variation in the sense of Cesari* if there exist an equivalent function v defined $\mathcal{L}^N$ -a.e. in Ω and N nonnegative functions $V_1, \dots, V_N \in L^1(\mathbb{R}^{N-1})$ such that

$$(14.30) \quad \text{essVar}_{\Omega_{x'_i}} v(x'_i, \cdot) \leq V_i(x'_i)$$

for all $x'_i \in \mathbb{R}^{N-1}$ for which $\Omega_{x'_i}$ is nonempty and for all $1 \leq i \leq N$.

Theorem 14.20 (Serrin). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in L^1(\Omega)$. Then $u \in BV(\Omega)$ if and only if u is of bounded pointwise variation in the sense of Cesari. Moreover, if $u \in BV(\Omega)$, then u has a representative $\bar{u}$ that admits (classical) partial derivatives $\mathcal{L}^N$ -a.e. in Ω and $\nabla \bar{u}(x) = \Psi(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$, where Ψ is the vectorial function given in (14.24).*

We begin with some preliminary results, which are of interest in themselves.

Lemma 14.21. *Let $R \subset \mathbb{R}^N$ be an open rectangle, let $i \in \{1, \dots, N\}$, and let $u : R \rightarrow \mathbb{R}$ be a Lebesgue measurable function that is monotone on $\mathcal{L}^{N-1}$ -a.e. line parallel to the x_i axis. Then the (classical) partial derivative $\partial_i u$ exists and is finite $\mathcal{L}^N$ -a.e. in R . Moreover, $\partial_i u$ is Lebesgue measurable.*

Proof. For $x = (x'_i, x_i) \in R$ and $n \in \mathbb{N}$ we define the upper right (respectively, left) sequential derivative

$$D^\pm u(x) := \limsup_{n \rightarrow \infty} \frac{u(x'_i, x_i \pm 1/n) - u(x'_i, x_i)}{\pm 1/n}$$

and the lower right (respectively, left) sequential derivative

$$D_\pm u(x) := \liminf_{n \rightarrow \infty} \frac{u(x'_i, x_i \pm 1/n) - u(x'_i, x_i)}{\pm 1/n}.$$

Then the four functions $D^\pm u$ and $D_\pm u$ are Lebesgue measurable. Moreover, by Lebesgue's differentiation theorem (Theorem 1.18), for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in R'_i$ we have that $\frac{\partial u}{\partial x_i}(x'_i, x_i)$ exists and is finite for $\mathcal{L}^1$ -a.e. $x_i \in R_i$. By Tonelli's theorem $D^\pm u$ and $D_\pm u$ are finite $\mathcal{L}^N$ -a.e. in R and

$$\begin{aligned} & \int_R |D^\pm u(x) - D_\pm u(x)| dx \\ &= \int_{R'_i} \int_{R_i} |D^\pm u(x'_i, x_i) - D_\pm u(x'_i, x_i)| dx_i dx'_i = 0. \end{aligned}$$

It follows that both limits

$$\lim_{n \rightarrow \infty} \frac{u(x'_i, x_i + 1/n) - u(x'_i, x_i)}{1/n}, \quad \lim_{n \rightarrow \infty} \frac{u(x'_i, x_i - 1/n) - u(x'_i, x_i)}{-1/n}$$

exist in $\mathbb{R}$ and are equal $\mathcal{L}^N$ -a.e. in R . Let $v(x)$ be the common limit where it exists; otherwise put $v(x) := \infty$. Let E be the set of points $x = (x'_i, x_i) \in R$ such that $v(x)$ is finite and $u(x'_i, \cdot)$ is monotone. Note that $\mathcal{L}^N(R \setminus E) = 0$. If $x = (x'_i, x_i) \in E$, with, say, $u(x'_i, \cdot)$ increasing, and $1/(n+1) \leq h \leq 1/n$, then

$$\begin{aligned} \frac{u(x'_i, x_i + 1/(n+1)) - u(x'_i, x_i)}{1/n} &\leq \frac{u(x'_i, x_i + h) - u(x'_i, x_i)}{h} \\ &\leq \frac{u(x'_i, x_i + 1/n) - u(x'_i, x_i)}{1/(n+1)}. \end{aligned}$$

Since both the right- and left-hand sides of the above inequality tend to $v(x)$ as $h \rightarrow 0^+$, we conclude that

$$\lim_{h \rightarrow 0^+} \frac{u(x'_i, x_i + h) - u(x'_i, x_i)}{h} = v(x).$$

Similarly, we have that

$$\lim_{h \rightarrow 0^-} \frac{u(x'_i, x_i + h) - u(x'_i, x_i)}{h} = v(x)$$

and so $\partial_i u(x) = v(x)$, which shows that $\partial_i u$ exists and is finite $\mathcal{L}^N$ -a.e. in R . Moreover, $\partial_i u$ is Lebesgue measurable. $\square$

Lemma 14.22. *Let $R \subset \mathbb{R}^N$ be an open rectangle and let u be a Lebesgue integrable real-valued function defined $\mathcal{L}^N$ -a.e. in R and such that for some $1 \leq i \leq N$ and for all $x'_i \in R'_i$,*

$$(14.31) \quad \text{essVar}_{R_i} u(x'_i, \cdot) \leq V_i(x'_i)$$

for some nonnegative Lebesgue integrable function V_i defined in R'_i . Then there exist two Lebesgue integrable functions u_1 and u_2 on R , each increasing in the variable x_i , such that

$$u(x) = u_1(x) - u_2(x)$$

for $\mathcal{L}^N$ -a.e. $x \in R$. In particular, the (classical) partial derivative $\partial_i u$ exists and is finite $\mathcal{L}^N$ -a.e. in R .

Proof. We construct a function v equivalent to u but with fewer discontinuities. Let S'_i be the set of points $x'_i \in R'_i$ such that $u(x'_i, \cdot)$ is defined for $\mathcal{L}^1$ -a.e. $x_i \in R_i$ and $V_i(x'_i) < \infty$. By hypothesis $\mathcal{L}^{N-1}(R'_i \setminus S'_i) = 0$.

For every $x'_i \in S'_i$, $\text{essVar}_{R_i} u(x'_i, \cdot) < \infty$ by (14.31) and the definition of S'_i . Hence, by Theorems 7.3 and 7.9 we may find a right continuous function $w(x'_i, \cdot)$ such that

$$(14.32) \quad w(x'_i, x_i) = u(x'_i, x_i) \quad \text{for } \mathcal{L}^1\text{-a.e. } x_i \in R_i$$

and $\text{Var}_{R_i} w(x'_i, \cdot) = \text{essVar}_{R_i} u(x'_i, \cdot) < \infty$. For $x = (x'_i, x_i) \in R$ define

$$(14.33) \quad v(x) := \begin{cases} w(x'_i, x_i) & \text{if } x'_i \in S'_i, \\ 0 & \text{otherwise.} \end{cases}$$

Since $\mathcal{L}^{N-1}(R'_i \setminus S'_i) = 0$, it follows by (14.32) and by Tonelli's theorem that $v(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in R$. By the completeness of the Lebesgue measure, the function v is Lebesgue measurable.

Now let $a \in R_i$ be such that

$$\int_{R'_i} |v(x'_i, a)| dx'_i < \infty$$

(such a point exists by Fubini's theorem), and for any fixed $x'_i \in R'_i$ let $\mathbf{V}(x'_i, \cdot)$ be the indefinite pointwise variation of the function $x_i \in R_i \mapsto v(x'_i, x_i)$, normalized so that $\mathbf{V}(x'_i, a) \equiv 0$, that is (see (2.3)),

$$\mathbf{V}(x'_i, x_i) := \begin{cases} \text{Var}_{[a, x_i]} v(x'_i, x_i) & \text{if } x_i \geq a, \\ -\text{Var}_{[x_i, a]} v(x'_i, x_i) & \text{if } x_i < a. \end{cases}$$

We claim that $\mathbf{V}$ is Lebesgue measurable. To see this, denote by P_k the partition of the interval R_i by $k+1$ equally spaced points $t_0 < \cdots < t_k$ and for $x = (x'_i, x_i) \in R$ define

$$\mathbf{V}_k(x'_i, x_i) := \sum |v(x'_i, t_j) - v(x'_i, t_{j-1})| \text{sgn}(t_j - a),$$

where the sum is extended over all partition points t_j which lie in the interval of endpoints a and x_i . Since $v(x'_i, \cdot)$ is right continuous, it follows by Proposition 2.21 that

$$\mathbf{V}(x'_i, x_i) = \lim_{k \rightarrow \infty} \mathbf{V}_k(x'_i, x_i)$$

for all $x = (x'_i, x_i) \in R$. Thus, to prove the measurability of $\mathbf{V}$, it remains to show that each $\mathbf{V}_k$ is Lebesgue measurable. This follows from the fact that, for each x'_i fixed, the function $\mathbf{V}_k(x'_i, \cdot)$ is a step function. More precisely, for x_i between any two consecutive points of P_k we have that

$$\mathbf{V}_k(x'_i, x_i) = \text{finite sum of Lebesgue measurable functions of } x'_i.$$

Hence, we have shown that $\mathbf{V}$ is Lebesgue measurable. We can now define

$$(14.34) \quad u_1(x) := \frac{1}{2}[\mathbf{V}(x) + v(x)], \quad u_2(x) := \frac{1}{2}[\mathbf{V}(x) - v(x)]$$

for $x = (x'_i, x_i) \in R$. The functions u_1 and u_2 are Lebesgue measurable, increasing in the variable x_i , and $u = u_1 - u_2$ $\mathcal{L}^N$ -a.e. in R . Furthermore, by Proposition 2.11,

$$\begin{aligned} |u_1(x)| + |u_2(x)| &\leq (\text{Var}_{R_i} u_1(x'_i, \cdot) + \tfrac{1}{2}|v(x'_i, a)|) \\ &\quad + (\text{Var}_{R_i} u_2(x'_i, \cdot) + \tfrac{1}{2}|v(x'_i, a)|) \\ &= \text{Var}_{R_i} v(x'_i, \cdot) + |v(x'_i, a)| \leq V_i(x'_i) + |v(x'_i, a)|, \end{aligned}$$

and so u_1 and u_2 are Lebesgue integrable.

The last part of the statement follows by applying the previous lemma to the functions u_1 and u_2 . $\square$

Remark 14.23. Note that by Proposition 2.21 for every $x'_i \in R'_i$, for every $x_i \in R_i$ the functions $u_1(x'_i, \cdot)$ and $u_2(x'_i, \cdot)$ defined in (14.34) cannot both be discontinuous at x_i .

We now turn to the proof of Theorem 14.20.

Proof of Theorem 14.20. Step 1: We prove that if u is of locally bounded pointwise variation in the sense of Cesari, then $u \in BV_{\text{loc}}(\Omega)$. Let $v, V_1, \dots, V_N$ be as in Definition 14.19. For $\phi \in C_c^1(\Omega)$ and for every fixed i , $1 \leq i \leq N$, set

$$L(\phi) := \int_{\Omega} u \partial_i \phi \, dx.$$

The functional L is linear. We claim that it is continuous with respect to the topology of $C_0(\Omega)$. To see this, let us first observe that, since $\text{essVar}_{\Omega_{x'_i}} v(x'_i, \cdot) \leq V_i(x'_i)$ for all $x'_i \in \mathbb{R}^{N-1}$ for which $\Omega_{x'_i}$ is nonempty, for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in \mathbb{R}^{N-1}$ we have that $V_i(x'_i)$ (and, in turn, $\text{essVar}_{R_i} v(x'_i, \cdot)$) is finite. Thus, by Theorem 7.9 for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in \mathbb{R}^{N-1}$ for which $\Omega_{x'_i}$ is nonempty the function $v(x'_i, \cdot)$ belongs to $BV(\Omega_{x'_i})$, and so there exists a measure $\lambda_{x'_i} \in \mathcal{M}_b(\Omega_{x'_i})$ such that $\partial_i v(x'_i, \cdot) = \lambda_{x'_i}$ and

$$|\lambda_{x'_i}|(\Omega_{x'_i}) = \text{essVar}_{\Omega_{x'_i}} v(x'_i, \cdot) \leq V_i(x'_i),$$

again by Theorem 7.9. Using (14.22), it follows that

$$\begin{aligned} L(\phi) &= \int_{\Omega} u \partial_i \phi \, dx = \int_{\mathbb{R}^{N-1}} \int_{\Omega_{x'_i}} u \partial_i \phi \, dx_i \, dx'_i \\ &= - \int_{\mathbb{R}^{N-1}} \int_{\Omega_{x'_i}} \phi \, d\lambda_{x'_i} \, dx'_i, \end{aligned}$$

and so

$$|L(\phi)| \leq \max_{\Omega} |\phi| \int_{\mathbb{R}^{N-1}} V_i \, dx'_i.$$

Since L is continuous on $C_0(\Omega)$, by the Hahn–Banach theorem (see Theorem A.32) it may be extended to a continuous linear functional on $C_0(\Omega)$. But then by the Riesz representation theorem in $C_0(\Omega)$ (see Theorem B.111) there exists a finite Radon measure $\lambda_i \in \mathcal{M}_b(\Omega)$ such that $L(\phi) = -\int_{\Omega} \phi d\lambda_i$ for all $\phi \in C_0(\Omega)$ and, by the previous inequality,

$$|\lambda_i|(\Omega) = \|L\|_{(C_0(\Omega))'} \leq \int_{\mathbb{R}^{N-1}} V_i dx'_i.$$

This shows that $u \in BV(\Omega)$.

Step 2: Assume that $u \in BV_{\text{loc}}(\Omega)$. We prove that u is of bounded pointwise variation in the sense of Cesari. Let $\{\varphi_{\varepsilon}\}_{\varepsilon>0}$ be a sequence of standard mollifiers and define $u_{\varepsilon} := u * \varphi_{\varepsilon}$ in Ω_{ε} . Set

$$(14.35) \quad E := \{x \in \Omega : \lim_{\varepsilon \rightarrow 0^+} u_{\varepsilon}(x) \text{ exists in } \mathbb{R}\}$$

and

$$(14.36) \quad v(x) := \begin{cases} \lim_{\varepsilon \rightarrow 0^+} u_{\varepsilon}(x) & \text{if } x \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Since $\{u_{\varepsilon}\}_{\varepsilon}$ converges pointwise to u at every Lebesgue point of u by Theorem C.16, we have that E contains every Lebesgue point of u . Moreover, since by Corollary B.119 the complement in Ω of the set of Lebesgue points has Lebesgue measure zero, it follows that $\mathcal{L}^N(\Omega \setminus E) = 0$. This shows that the function v is a representative of u .

Let i be a fixed integer, $1 \leq i \leq N$. By Exercise 14.13 we get

$$(14.37) \quad \int_{\Omega_{\varepsilon}} |\partial_i u_{\varepsilon}| dx \leq |D_i u|(\Omega).$$

We now take $\varepsilon_n := 1/n$ and for simplicity we set $u_n := u_{\varepsilon_n}$, $(\Omega_{x'_i})_n := (\Omega_{x'_i})_{\varepsilon_n}$, and so on.

For $x'_i \in \mathbb{R}^{N-1}$ define

$$(14.38) \quad V_i(x'_i) := \liminf_{n \rightarrow \infty} \int_{(\Omega_{x'_i})_n} |\partial_i u_n(x'_i, x_i)| dx_i$$

if $\Omega_{x'_i}$ is nonempty and $V_i(x'_i) := 0$ otherwise. The function V_i is Lebesgue measurable on $\mathbb{R}^{N-1}$. By Fatou's lemma, Fubini's theorem, and (14.37),

$$\begin{aligned} \int_{\mathbb{R}^{N-1}} V_i(x'_i) dx'_i &\leq \liminf_{n \rightarrow \infty} \int_{\mathbb{R}^{N-1}} \int_{(\Omega_{x'_i})_n} |\partial_i u_n(x'_i, x_i)| dx_i dx'_i \\ &= \liminf_{n \rightarrow \infty} \int_{\Omega_n} |\partial_i u_n| dx \leq |D_i u|(\Omega). \end{aligned}$$

Hence,

$$(14.39) \quad \int_{\mathbb{R}^{N-1}} V_i(x'_i) dx'_i \leq |D_i u|(\Omega).$$

Since $\mathcal{L}^N(\Omega \setminus E) = 0$, by Fubini's theorem

$$(14.40) \quad \mathcal{L}^1(\Omega_{x'_i} \setminus E_{x'_i}) = 0$$

for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in \mathbb{R}^{N-1}$ for which $\Omega_{x'_i}$ is nonempty. Fix any such $x'_i \in \mathbb{R}^{N-1}$ and let $R_i \subset \mathbb{R}$ be any maximal open interval such that $\{x'_i\} \times R_i \subset \Omega_{x'_i}$. We claim that

$$(14.41) \quad \text{essVar}_{R_i} v(x'_i, \cdot) \leq V_i(x'_i).$$

By (14.40), $(x'_i, x_i) \in E$ for $\mathcal{L}^1$ -a.e. $x_i \in R_i$. Thus, to prove (14.41), it is enough to consider partitions contained in $E_{x'_i}$ (see Definition 7.8 and Exercise 7.10). Let

$$\inf R_i < t_0 < t_1 < \cdots < t_k < \sup R_i,$$

where $t_j \in E_{x'_i}$ for all $j = 0, \dots, k$. Then $[t_0, t_k] \subset (R_i)_n$ for all n sufficiently large, and so, by the fundamental theorem of calculus,

$$\begin{aligned} \sum_{j=1}^k |u_n(x'_i, t_j) - u_n(x'_i, t_{j-1})| &\leq \sum_{j=1}^k \int_{t_{j-1}}^{t_j} |\partial_i u_n(x'_i, x_i)| dx_i \\ &\leq \int_{(R_i)_n} |\partial_i u_n(x'_i, x_i)| dx_i. \end{aligned}$$

Letting $n \rightarrow \infty$, by (14.35), (14.36), and (14.38) we get

$$\sum_{j=1}^k |v(x'_i, t_j) - v(x'_i, t_{j-1})| \leq V_i(x'_i).$$

Taking the supremum over all admissible partitions yields (14.41). By eventually redefining V_i to be infinite on a set of $\mathcal{L}^{N-1}$ -measure zero, we have proved that (14.30) holds. This shows that u is of bounded pointwise variation in the sense of Cesari.

Step 3: We prove the last part of the theorem. Since the function v defined in (14.36) satisfies the hypotheses of Lemma 14.22, we may find two Lebesgue integrable functions v_1 and v_2 on R , each increasing in the variable x_i , such that $v(x) = v_1(x) - v_2(x)$ for $\mathcal{L}^N$ -a.e. $x \in R$. Let S'_i be the set of points $x'_i \in R'_i$ such that

$$v_1(x'_i, x_i) = \lim_{\varepsilon \rightarrow 0^+} (v_1)_\varepsilon(x'_i, x_i), \quad v_2(x'_i, x_i) = \lim_{\varepsilon \rightarrow 0^+} (v_2)_\varepsilon(x'_i, x_i)$$

for $\mathcal{L}^1$ -a.e. $x_i \in R_i$. Note that $\mathcal{L}^{N-1}(R'_i \setminus S'_i) = 0$.

Fix $(x'_i, x_i) \in S'_i \times R_i$. We assert that if the limit $\lim_{\varepsilon \rightarrow 0^+} (v_1)_\varepsilon(x'_i, x_i)$ does not exist, then the limit $\lim_{\varepsilon \rightarrow 0^+} (v_2)_\varepsilon(x'_i, x_i)$ does exist. Indeed, since

$v_1(x'_i, \cdot)$ and $(v_1)_\varepsilon(x'_i, \cdot)$ are increasing, if the limit $\lim_{\varepsilon \rightarrow 0^+} (v_1)_\varepsilon(x'_i, x_i)$ does not exist, then (why?)

$$(v_1)_-(x'_i, x_i) \leq \liminf_{\varepsilon \rightarrow 0^+} (v_1)_\varepsilon(x'_i, x_i) < \limsup_{\varepsilon \rightarrow 0^+} (v_1)_\varepsilon(x'_i, x_i) \leq (v_1)_+(x'_i, x_i).$$

Thus, $v_1(x'_i, \cdot)$ is discontinuous at x_i . But then by Remark 14.23, $v_2(x'_i, \cdot)$ must be continuous at x_i , and hence

$$v_2(x'_i, x_i) = \lim_{\varepsilon \rightarrow 0^+} (v_2)_\varepsilon(x'_i, x_i).$$

We now define the function $\bar{u}(x) := \limsup_{\varepsilon \rightarrow 0^+} u_\varepsilon(x)$ for $x = (x'_i, x_i) \in R$. The function $\bar{u}$ is a precise representative of u and is independent of i . We also introduce the functions $\bar{u}_1$ and $\bar{u}_2$ by means of

$$\bar{u}_1(x) := \begin{cases} \limsup_{\varepsilon \rightarrow 0^+} (v_1)_\varepsilon(x'_i, x_i) & \text{if } x'_i \in S'_i, \\ v(x) & \text{otherwise,} \end{cases}$$

$$\bar{u}_2(x) := \begin{cases} \liminf_{\varepsilon \rightarrow 0^+} (v_2)_\varepsilon(x'_i, x_i) & \text{if } x'_i \in S'_i, \\ 0 & \text{otherwise.} \end{cases}$$

Then $\bar{u}_1$ and $\bar{u}_2$ are equivalent to v_1 and v_2 , respectively, and so they are increasing on $\mathcal{L}^{N-1}$ -a.e. lines parallel to the x_i -axis. Furthermore, from the identity

$$u_\varepsilon = (v_1 - v_2)_\varepsilon = (v_1)_\varepsilon - (v_2)_\varepsilon$$

and the assertion of the preceding paragraph it follows that¹ $\bar{u}(x) = \bar{u}_1(x) - \bar{u}_2(x)$ for all $x \in R$. By Lemma 14.21 the partial derivatives $\frac{\partial \bar{u}_1}{\partial x_i}$ and $\frac{\partial \bar{u}_2}{\partial x_i}$ exist and are finite $\mathcal{L}^N$ -a.e. in R . The same is true for $\frac{\partial \bar{u}}{\partial x_i}$. The fact that $\nabla \bar{u}(x) = \Psi(x)$ follows from Theorem 14.18. $\square$

The following exercises are the analog of Exercise 11.51 and Theorem 11.53.

Exercise 14.24. Let $u \in W^{1,1}(\mathbb{R}_+^N)$ and $v \in W^{1,1}(\mathbb{R}_-^N)$. Prove that the function $w : \mathbb{R}^N \rightarrow \mathbb{R}$, defined by

$$w(x) := \begin{cases} u(x) & \text{if } x_N > 0, \\ v(x) & \text{if } x_N < 0, \end{cases}$$

belongs to $BV(\mathbb{R}^N)$.

¹The key point here is that given two sequences of real numbers $\{a_n\}_n$ and $\{b_n\}_n$, in general one can only conclude that

$$\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n,$$

with the strict inequality possible. However, if one of the two sequences converges, say $\{a_n\}_n$, then

$$\limsup_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n.$$

Exercise 14.25. Let $\Omega, U \subseteq \mathbb{R}^N$ be open sets, let $\Phi : U \rightarrow \Omega$ be invertible, with Φ and Φ^{-1} Lipschitz continuous functions, and let $u \in BV(\Omega)$. Prove that $u \circ \Phi \in BV(U)$.

Exercise 14.26. Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $u \in BV_{\text{loc}}(\Omega)$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a Lipschitz continuous function with $f(0) = 0$ if $\mathcal{L}^N(\Omega) = \infty$. Prove that $f \circ u \in BV_{\text{loc}}(\Omega)$ and that

$$\|D(f \circ u)\|(U) \leq \text{Lip } f \|Du\|(U)$$

for every open set $U \subseteq \Omega$.

Remark 14.27. One can actually show that a chain rule formula holds in $BV(\Omega)$. We refer to the paper of Ambrosio and Dal Maso [9] (see also [140]).

14.4. Coarea Formula for BV Functions

In this section we prove the coarea formula for a function $u \in BV(\Omega)$. This formula relates the total variation measure $\|Du\|$ with the perimeter of its superlevel sets $\{x \in \Omega : u(x) > t\}$.

Theorem 14.28 (Coarea formula). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in L^1_{\text{loc}}(\Omega)$. Then*

$$V(u, \Omega) = \int_{\mathbb{R}} P(\{x \in \Omega : u(x) > t\}, \Omega) dt.$$

In particular, if $u \in BV(\Omega)$, then the set $\{x \in \Omega : u(x) > t\}$ has finite perimeter in Ω for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$ and

$$(14.42) \quad \|Du\|(\Omega) = \int_{\mathbb{R}} P(\{x \in \Omega : u(x) > t\}, \Omega) dt.$$

We begin by proving the theorem for affine functions.

Lemma 14.29. *Let $u : \mathbb{R}^N \rightarrow \mathbb{R}$ be an affine function of the form $u(x) = a + b \cdot x$, $x \in \mathbb{R}^N$, where $a \in \mathbb{R}$ and $b \in \mathbb{R}^N$. Then for every Lebesgue measurable set $E \subseteq \mathbb{R}^N$ the function $t \in \mathbb{R} \mapsto \mathcal{H}^{N-1}(E \cap u^{-1}(\{t\}))$ is Lebesgue measurable and*

$$(14.43) \quad \|b\| \mathcal{L}^N(E) = \int_{\mathbb{R}} \mathcal{H}^{N-1}(E \cap u^{-1}(\{t\})) dt.$$

Proof. If $b = 0$, then there is nothing to prove, since both sides of (14.43) are zero. Thus, assume that $b \neq 0$. Hence, $u^{-1}(\{t\}) = \{x \in \mathbb{R}^N : a + b \cdot x = t\}$ is the translate of a hyperplane in $\mathbb{R}^N$, and so the measure $\mathcal{H}^{N-1}$ restricted to $u^{-1}(\{t\})$ coincides with the measure $\mathcal{L}^{N-1}$. It follows by Tonelli's theorem that the function

$$t \in \mathbb{R} \mapsto \mathcal{H}^{N-1}(E \cap u^{-1}(\{t\}))$$

is Lebesgue measurable. Since $\mathcal{L}^N$ and $\mathcal{H}^{N-1}$ are both rotation invariant, without loss of generality, we may assume that $\frac{b}{\|b\|} = e_1$. By the change of variables $t = \|b\|s + a$ we obtain

$$\begin{aligned} \int_{\mathbb{R}} \mathcal{H}^{N-1}(E \cap u^{-1}(\{t\})) dt &= \|b\| \int_{\mathbb{R}} \mathcal{H}^{N-1}(E \cap u^{-1}(\{\|b\|s + a\})) ds \\ &= \|b\| \int_{\mathbb{R}} \mathcal{H}^{N-1}(\{x \in E : x_1 = s\}) ds \\ &= \|b\| \int_{\mathbb{R}} \mathcal{L}^{N-1}(\{x \in E : x_1 = s\}) ds = \|b\| \mathcal{L}^N(E), \end{aligned}$$

where in the last inequality we have used Tonelli's theorem. $\square$

Remark 14.30. Note that $\nabla u \equiv b$.

Exercise 14.31. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in L^1_{\text{loc}}(\Omega)$. Prove that the function $t \in \mathbb{R} \mapsto P(\{x \in \Omega : u(x) > t\}, \Omega)$ is Lebesgue measurable.

We are now ready to prove Theorem 14.28. In what follows, for $t \in \mathbb{R}$ we set

$$\Omega_t := \{x \in \Omega : u(x) > t\}.$$

Proof of Theorem 14.28. Step 1: We claim that if $\{u_n\}_n$ is a sequence in $L^1_{\text{loc}}(\Omega)$ converging to u in $L^1_{\text{loc}}(\Omega)$, then there exists a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ such that

$$P(\Omega_t, \Omega) \leq \liminf_{k \rightarrow \infty} P(\Omega_t^{n_k}, \Omega)$$

for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$ and

$$\int_{\mathbb{R}} P(\Omega_t, \Omega) dt \leq \liminf_{k \rightarrow \infty} \int_{\mathbb{R}} P(\Omega_t^{n_k}, \Omega) dt,$$

where $\Omega_t^n := \{x \in \Omega : u_n(x) > t\}$.

Indeed, for every $x \in \Omega$ and $n \in \mathbb{N}$,

$$\int_{\mathbb{R}} |\chi_{\Omega_t^n}(x) - \chi_{\Omega_t}(x)| dt = \int_{\min\{u_n(x), u(x)\}}^{\max\{u_n(x), u(x)\}} 1 dt = |u_n(x) - u(x)|.$$

Hence, by Fubini's theorem, for every $U \Subset \Omega$,

$$\int_{\mathbb{R}} \int_U |\chi_{\Omega_t^n}(x) - \chi_{\Omega_t}(x)| dx dt = \int_U |u_n(x) - u(x)| dx \rightarrow 0.$$

By considering an increasing sequence $U_j \nearrow \Omega$, with $U_j \Subset \Omega$ and using a diagonal argument, we may find a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ such that $\chi_{\Omega_t^{n_k}} \rightarrow \chi_{\Omega_t}$ in $L^1_{\text{loc}}(\Omega)$ for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$. It follows by Exercise 14.3 that

$$P(\Omega_t, \Omega) \leq \liminf_{k \rightarrow \infty} P(\Omega_t^{n_k}, \Omega)$$

for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$. In turn, by Fatou's lemma,

$$\int_{\mathbb{R}} P(\Omega_t, \Omega) dt \leq \liminf_{k \rightarrow \infty} \int_{\mathbb{R}} P(\Omega_t^{n_k}, \Omega) dt.$$

Step 2: We prove that if u is piecewise affine in the sense of Definition 11.39, then

$$\int_{\mathbb{R}} \|D(\chi_{\Omega_t})\|(\Omega) dt = \int_{\Omega} \|\nabla u\| dx.$$

To see this, let $\Delta_1, \dots, \Delta_\ell$ be N -simplexes with pairwise disjoint interiors such that the restriction of u to each Δ_i is affine, say $u(x) = a_i + b_i \cdot x$ for $x \in \Delta_i$, where $a_i \in \mathbb{R}$ and $b_i \in \mathbb{R}^N$, and $u = 0$ outside $\bigcup_{i=1}^\ell \Delta_i$. Then by the previous lemma, applied to the affine function $u_i(x) = a_i + b_i \cdot x$, $x \in \mathbb{R}^N$,

$$\int_{\Omega} \|\nabla u\| dx = \sum_{i=1}^{\ell} \|b_i\| \mathcal{L}^N(\Omega \cap \Delta_i) = \sum_{i=1}^{\ell} \int_{\mathbb{R}} \mathcal{H}^{N-1}(\Omega \cap \Delta_i \cap u^{-1}(\{t\})) dt.$$

Since by Exercise 14.5 (note that in $\Omega \cap \Delta_i$ the function u coincides with the smooth function $v(x) := a_i + b_i \cdot x$, $x \in \mathbb{R}^N$)

$$\|D(\chi_{\Omega_t})\|(\Omega \cap \Delta_i) = \mathcal{H}^{N-1}(\Omega \cap \Delta_i \cap u^{-1}(\{t\})),$$

we have

$$\begin{aligned} \sum_{i=1}^{\ell} \int_{\mathbb{R}} \mathcal{H}^{N-1}(\Omega \cap \Delta_i \cap u^{-1}(\{t\})) dt &= \sum_{i=1}^{\ell} \int_{\mathbb{R}} \|D(\chi_{\Omega_t})\|(\Omega \cap \Delta_i) dt \\ &= \int_{\mathbb{R}} \|D(\chi_{\Omega_t})\|(\Omega) dt. \end{aligned}$$

Step 3: We prove that if $u \in C^\infty(\Omega) \cap W_{\text{loc}}^{1,1}(\Omega)$, then

$$\int_{\mathbb{R}} P(\Omega_t, \Omega) dt \leq \int_{\Omega} \|\nabla u\| dx.$$

It suffices to assume that $\int_{\Omega} \|\nabla u\| dx < \infty$. Fix an open set $U \Subset \Omega$. By Remark 11.41 we may find a sequence $\{u_n\}_n$ in $W^{1,1}(U)$ of piecewise affine functions such that $u_n \rightarrow u$ in $W^{1,1}(U)$. By the previous two steps (with Ω replaced by U) we may find a subsequence (not relabeled) of $\{u_n\}_n$ such that

$$\begin{aligned} \int_{\mathbb{R}} \|D(\chi_{U_t})\|(U) dt &\leq \liminf_{n \rightarrow \infty} \int_{\mathbb{R}} \|D(\chi_{U_t^n})\|(U) dt \\ &= \lim_{n \rightarrow \infty} \int_U \|\nabla u_n\| dx = \int_U \|\nabla u\| dx \leq \int_{\Omega} \|\nabla u\| dx. \end{aligned}$$

Since $\chi_{U_t} = \chi_{\Omega_t}$ in U , we have that $\|D(\chi_{U_t})\|(U) = \|D(\chi_{\Omega_t})\|(U)$. Hence,

$$\int_{\mathbb{R}} \|D(\chi_{\Omega_t})\|(U) dt \leq \int_{\Omega} \|\nabla u\| dx.$$

By considering an increasing sequence $U_k \nearrow \Omega$, with $U_k \Subset \Omega$, in the previous inequality and using the Lebesgue monotone convergence theorem, we obtain the result.

Step 4: We prove that for $u \in L^1_{\text{loc}}(\Omega)$,

$$\int_{\mathbb{R}} P(\Omega_t, \Omega) dt \leq V(u, \Omega).$$

It suffices to assume that $V(u, \Omega) < \infty$. Hence, by Exercise 14.3 we have that the distributional gradient Du of u belongs to $\mathcal{M}_b(\Omega; \mathbb{R}^N)$. By Theorem 14.9 and Remark 14.15 there exists a sequence $\{u_n\}_n$ in $C^\infty(\Omega) \cap W^{1,1}_{\text{loc}}(\Omega)$ such that $u_n \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$ and

$$\lim_{n \rightarrow \infty} \int_{\Omega} \|\nabla u_n\| dx = \|Du\|(\Omega).$$

By Step 1 and Step 3 applied to each function u_n ,

$$\begin{aligned} \int_{\mathbb{R}} \|D(\chi_{\Omega_t})\|(\Omega) dt &\leq \liminf_{n \rightarrow \infty} \int_{\mathbb{R}} \|D(\chi_{\Omega_t^n})\|(\Omega) dt \\ &\leq \lim_{n \rightarrow \infty} \int_{\Omega} \|\nabla u_n\| dx = \|Du\|(\Omega). \end{aligned}$$

Step 5: We claim that for every $\Phi \in C_c^\infty(\Omega; \mathbb{R}^N)$ with $\|\Phi\|_{C_0(\Omega)} \leq 1$,

$$\int_{\Omega} u \operatorname{div} \Phi dx = \int_{\mathbb{R}} \int_{\Omega_t} \operatorname{div} \Phi dx dt.$$

To see this, assume first that $u \geq 0$. Then for $\mathcal{L}^N$ -a.e. $x \in \Omega$, $u(x) = \int_0^\infty \chi_{\Omega_t}(x) dt$, and so, by Fubini's theorem

$$\begin{aligned} \int_{\Omega} u(x) \operatorname{div} \Phi(x) dx &= \int_{\Omega} \int_0^\infty \chi_{\Omega_t}(x) \operatorname{div} \Phi(x) dt dx \\ &= \int_0^\infty \int_{\Omega} \chi_{\Omega_t}(x) \operatorname{div} \Phi(x) dx dt \\ &= \int_0^\infty \int_{\Omega_t} \operatorname{div} \Phi(x) dx dt. \end{aligned}$$

Similarly, if $u \leq 0$, then for $\mathcal{L}^N$ -a.e. $x \in \Omega$, $u(x) = \int_{-\infty}^0 (\chi_{\Omega_t}(x) - 1) dt$, and so

$$\begin{aligned} \int_{\Omega} u(x) \operatorname{div} \Phi(x) dx &= \int_{\Omega} \int_{-\infty}^0 (\chi_{\Omega_t}(x) - 1) \operatorname{div} \Phi(x) dt dx \\ &= \int_{-\infty}^0 \int_{\Omega} (\chi_{\Omega_t}(x) - 1) \operatorname{div} \Phi(x) dx dt \\ &= \int_{-\infty}^0 \int_{\Omega_t} \operatorname{div} \Phi(x) dx dt, \end{aligned}$$

where we have used the fact that $\int_{\Omega} \operatorname{div} \Phi \, dx = 0$ by the divergence theorem and the fact that $\Phi \in C_c^{\infty}(\Omega; \mathbb{R}^N)$.

In the general case, write $u = u^+ - u^-$. Note that for $t > 0$,

$$\Omega_t = \{x \in \Omega : u(x) > t\} = \{x \in \Omega : u^+(x) > t\},$$

while for $t < 0$,

$$\Omega_t = \{x \in \Omega : u(x) > t\} = \{x \in \Omega : -u^-(x) > t\}.$$

Hence,

$$\begin{aligned} \int_{\Omega} u(x) \operatorname{div} \Phi(x) \, dx &= \int_{\Omega} (u^+(x) - u^-(x)) \operatorname{div} \Phi(x) \, dx \\ &= \int_{\mathbb{R}} \int_{\Omega_t} \operatorname{div} \Phi(x) \, dx \, dt, \end{aligned}$$

which proves the claim.

Step 6: We show that for $u \in L_{\text{loc}}^1(\Omega)$,

$$V(u, \Omega) \leq \int_{\mathbb{R}} P(\Omega_t, \Omega) \, dt.$$

Let $\Phi \in C_c^{\infty}(\Omega; \mathbb{R}^N)$ with $\|\Phi\|_{C_0(\Omega)} \leq 1$. By the previous step and the definition of the variation of the perimeter of Ω_t in Ω ,

$$\int_{\Omega} u \operatorname{div} \Phi \, dx = \int_{\mathbb{R}} \int_{\Omega_t} \operatorname{div} \Phi \, dx \, dt \leq \int_{\mathbb{R}} P(\Omega_t, \Omega) \, dt.$$

This concludes the proof. $\square$

Remark 14.32. We refer to [72] for a different proof of Step 3, which does not make use of piecewise affine functions.

14.5. Embeddings and Isoperimetric Inequalities

In this section we prove that the Sobolev–Gagliardo–Nirenberg embedding theorem continues to hold for functions of bounded variation. We recall that for $N \geq 2$, $1^* = \frac{N}{N-1}$.

Theorem 14.33 (Sobolev–Gagliardo–Nirenberg embedding in BV). *Let $u \in L_{\text{loc}}^1(\mathbb{R}^N)$, $N \geq 2$, be a function vanishing at infinity such that its distributional gradient Du belongs to $\mathcal{M}_b(\mathbb{R}^N; \mathbb{R}^N)$. Then there exists a constant $c = c(N) > 0$ such that*

$$\|u\|_{L^{1^*}(\mathbb{R}^N)} \leq c \|Du\|(\mathbb{R}^N).$$

In particular, $BV(\mathbb{R}^N) \hookrightarrow L^q(\mathbb{R}^N)$ for all $1 \leq q \leq 1^$.*

Proof. Step 1: Assume that $u \in L^{1*}(\mathbb{R}^N)$. Consider a sequence of standard mollifiers $\{\varphi_\varepsilon\}_{\varepsilon>0}$ and define $u_\varepsilon := u * \varphi_\varepsilon$. By Lemma 14.10, Remark 14.11, and Theorem C.16, $u_\varepsilon \rightarrow u$ in $L^{1*}(\mathbb{R}^N)$ and

$$(14.44) \quad \lim_{\varepsilon \rightarrow 0^+} \|\nabla u_\varepsilon\|_{L^1(\mathbb{R}^N)} = \|Du\|(\mathbb{R}^N).$$

Applying the Sobolev–Gagliardo–Nirenberg embedding theorem (Theorem 12.4) to each u_ε , we obtain

$$\|u_\varepsilon\|_{L^{1*}(\mathbb{R}^N)} \leq c \|\nabla u_\varepsilon\|_{L^1(\mathbb{R}^N)}.$$

It now suffices to let $\varepsilon \rightarrow 0^+$ and to use (14.44) and Theorem C.16.

Step 2: To remove the additional hypothesis that $u \in L^{1*}(\mathbb{R}^N)$, we truncate u as in Step 4 of the proof of the Sobolev–Gagliardo–Nirenberg embedding theorem (Theorem 12.4) and use Exercise 14.26. $\square$

Remark 14.34. If $N \geq 2$ and $u \in L^1_{\text{loc}}(\mathbb{R}^N)$ is such that its distributional gradient Du belongs to $\mathcal{M}_b(\Omega; \mathbb{R}^N)$, then reasoning as in the proof of Theorem 12.9, one can show that there exists a unique constant $c_u \in \mathbb{R}$ (depending on u) such that

$$\|u - c_u\|_{L^{1*}(\mathbb{R}^N)} \leq c \|Du\|(\mathbb{R}^N),$$

where $c = c(N) > 0$. See Theorem 3.47 in [10] for a different proof of this result.

Definition 14.35. An open set $\Omega \subset \mathbb{R}^N$ is called an *extension domain* for the space $BV(\Omega)$ if there exists a continuous linear operator $\mathcal{E} : BV(\Omega) \rightarrow BV(\mathbb{R}^N)$ such that

- (i) for all $u \in BV(\Omega)$, $\mathcal{E}(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$,
- (ii) $\mathcal{E}(u) \in W^{1,1}(\mathbb{R}^N)$ whenever $u \in W^{1,1}(\Omega)$.

Next we prove that the Rellich–Kondrachov theorem continues to hold in $BV(\Omega)$.

Theorem 14.36 (Rellich–Kondrachov). *Let $\Omega \subset \mathbb{R}^N$, $N \geq 2$, be an extension domain for $BV(\Omega)$ with finite measure. Let $\{u_n\}_n$ be a bounded sequence in $BV(\Omega)$. Then there exist a subsequence $\{u_{n_k}\}_n$ of $\{u_n\}_n$ and a function $u \in BV(\Omega) \cap L^{1*}(\Omega)$ such that $u_{n_k} \rightarrow u$ in $L^q(\Omega)$ for all $1 \leq q < 1^*$.*

The proof is very similar to that of Theorem 12.18. The following result is the analog of Lemma 12.19.

Lemma 14.37. *Let $u \in BV(\mathbb{R}^N)$. Then for all $h \in \mathbb{R}^N \setminus \{0\}$,*

$$\int_{\mathbb{R}^N} |u(x+h) - u(x)| dx \leq \|h\| \|Du\|(\mathbb{R}^N).$$

Proof. Assume first that $u \in W^{1,1}(\mathbb{R}^N) \cap C^\infty(\mathbb{R}^N)$. Then by Lemma 12.19 we have that for all $h \in \mathbb{R}^N \setminus \{0\}$,

$$\int_{\mathbb{R}^N} |u(x+h) - u(x)| dx \leq \|h\| \int_{\mathbb{R}^N} \|\nabla u(x)\| dx.$$

To remove the additional hypothesis that $u \in C^\infty(\mathbb{R}^N)$, it suffices to apply the previous inequality to $u_\varepsilon := \varphi_\varepsilon * u$, where φ_ε is a standard mollifier, and to let $\varepsilon \rightarrow 0^+$ (see Theorem C.16 and Lemma 14.10). $\square$

The following result is the analog of Lemma 12.20.

Lemma 14.38. *Let $u \in BV(\mathbb{R}^N)$. Consider a standard mollifier φ as in (C.7). Then*

$$\int_{\mathbb{R}^N} |(u * \varphi_{1/k})(x) - u(x)| dx \leq ck^{-1} \|Du\|(\mathbb{R}^N)$$

for some constant $c = c(N) > 0$.

Proof. By Lemma 12.20 (see (12.12)) we have

$$\begin{aligned} \int_{\mathbb{R}^N} |(u * \varphi_{1/k})(x) - u(x)| dx \\ \leq ck^N \int_{B(0,1/k)} \int_{\mathbb{R}^N} |u(x+h) - u(x)| dx dh. \end{aligned}$$

In turn, by the previous lemma we get

$$\begin{aligned} \int_{\mathbb{R}^N} |(u * \varphi_{1/k})(x) - u(x)| dx &\leq ck^N \|Du\|(\mathbb{R}^N) \int_{B(0,1/k)} \|h\| dh \\ &= ck^{-1} \|Du\|(\mathbb{R}^N). \end{aligned}$$

$\square$

We now turn to the proof of the Rellich–Kondrachov theorem.

Proof of Theorem 14.36. The proof of the first statement is the same as that for Theorem 12.18 with the only difference that we use the previous lemma in place of Lemma 12.20. It remains to prove that u belongs to $BV(\Omega)$. For all $i = 1, \dots, N$ and $k \in \mathbb{N}$ define $\lambda_{k,i} := D_i u_{n_k}$. Then $\sup_{k \in \mathbb{N}} |\lambda_{k,i}|(\Omega) < \infty$ for all $i = 1, \dots, N$. Since $\mathcal{M}_b(\Omega)$ is the dual of $C_0(\Omega)$ and $C_0(\Omega)$ is separable, by Corollary A.54 we may select a further subsequence, not relabeled, such that for each $i = 1, \dots, N$,

$$\lambda_{k,i} \xrightarrow{*} \lambda_i \quad \text{in } \mathcal{M}_b(\Omega)$$

for some finite signed Radon measures $\lambda_1, \dots, \lambda_N \in \mathcal{M}_b(\Omega)$. For every $\phi \in C_c^\infty(\Omega)$, $i = 1, \dots, N$, and $k \in \mathbb{N}$ we have

$$\int_{\Omega} u_{n_k} \partial_i \phi \, dx = - \int_{\Omega} \partial_i u_{n_k} \phi \, dx.$$

Letting $k \rightarrow \infty$ in the previous equality yields

$$\int_{\Omega} u \partial_i \phi \, dx = - \int_{\Omega} \phi \, d\lambda_i,$$

which shows that $D_i u = \lambda_i$. Hence, $u \in BV(\Omega)$ and the proof is complete. $\square$

As a consequence of the previous theorem, we can prove a compactness result in $BV(\Omega)$ (see Remark 11.66).

Theorem 14.39 (Compactness). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Assume that $\{u_n\}_n$ is bounded in $BV(\Omega)$. Then there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and $u \in BV(\Omega)$ such that $u_{n_k} \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$ and $D_i u_{n_k} \xrightarrow{*} D_i u$ in $\mathcal{M}_b(\Omega)$ for all $i = 1, \dots, N$.*

Proof. Fix an open set $\Omega_1 \Subset \Omega$. Construct a cut-off function $\phi \in C_c^\infty(\Omega)$ such that $\phi \equiv 1$ in Ω_1 and for every $n \in \mathbb{N}$ define

$$v_n(x) := \begin{cases} \phi(x)u_n(x) & \text{if } x \in \Omega, \\ 0 & \text{otherwise.} \end{cases}$$

Since $\text{supp } v_n \subseteq \text{supp } \phi$ and the latter set is compact, we are in a position to apply the Rellich–Kondrachov theorem to find a subsequence $\{v_{n,1}\}_{n,1}$ of $\{v_n\}$ and a function $u^{(1)} \in BV(\Omega_1)$ such that $v_{n,1} \rightarrow u^{(1)}$ in $L^1(\Omega)$. Using the fact that $\phi \equiv 1$ in Ω_1 , it follows that $u_{n,1} \rightarrow u^{(1)}$ in $L^1(\Omega_1)$. By considering an increasing sequence of open sets $\Omega_j \Subset \Omega_{j+1} \Subset \Omega$ such that $\Omega = \bigcup_{j=1}^{\infty} \Omega_j$ and using a diagonal argument (see, e.g., the proof of the Helly theorem, Theorem 2.44), we may find a subsequence $\{u_{n_k}\}_k$ and a function $u \in BV_{\text{loc}}(\Omega)$ such that $u_{n_k} \rightarrow u$ in $L^1(\Omega_j)$ as $k \rightarrow \infty$ for every $j \in \mathbb{N}$.

By selecting a further subsequence, not relabeled, we may assume that $u_{n_k}(x) \rightarrow u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$. Hence, by Fatou's lemma

$$\int_{\Omega} |u(x)| \, dx \leq \liminf_{k \rightarrow \infty} \int_{\Omega} |u_{n_k}(x)| \, dx < \infty,$$

which implies that $u \in L^1(\Omega)$. Since $u_{n_k} \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$, it follows by Exercise 14.3 that

$$V(u, \Omega) \leq \liminf_{k \rightarrow \infty} V(u_{n_k}, \Omega) = \liminf_{k \rightarrow \infty} \|Du_{n_k}\|(\Omega) < \infty,$$

which implies that $u \in BV(\Omega)$. $\square$

Remark 14.40. In particular, if $\Omega \subseteq \mathbb{R}^N$ is an open set and $\{u_n\}$ is bounded in $W^{1,1}(\Omega)$, then it follows from the previous theorem that there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and $u \in BV(\Omega)$ such that $u_{n_k} \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$ and $\lambda_{n_k,i} \xrightarrow{*} D_i u$ in $\mathcal{M}_b(\Omega)$ for all $i = 1, \dots, N$, where $\lambda_{n_k,i}(E) := \int_E \partial_i u_{n_k} dx$, $E \in \mathcal{B}(\Omega)$.

Exercise 14.41. Let $\Omega \subseteq \mathbb{R}^N$ be an open set with uniformly Lipschitz continuous boundary. Prove that $\Omega \subseteq \mathbb{R}^N$ is an extension domain for $BV(\Omega)$. Hint: Use Theorems 13.17 and 14.9.

Exercise 14.42 (Poincaré's inequality). Let $\Omega \subset \mathbb{R}^N$ be a connected extension domain for $BV(\Omega)$ with finite measure. Let $E \subseteq \Omega$ be a Lebesgue measurable set with positive measure. Prove that there exists a constant $c = c(\Omega, E) > 0$ such that for all $u \in BV(\Omega)$,

$$\|u - u_E\|_{L^{1^*}(\Omega)} \leq c \|Du\|(\Omega),$$

where, we recall, $u_E := \frac{1}{\mathcal{L}^N(E)} \int_E u(x) dx$.

Exercise 14.43 (Poincaré's inequality for balls). Let $x_0 \in \mathbb{R}^N$ and $R > 0$. Prove that there exists a constant $c = c(N) > 0$ (independent of R and x_0) such that for all $u \in BV(B(x_0, R))$,

$$\|u - u_{B(x_0, R)}\|_{L^{1^*}(B(x_0, R))} \leq c \|Du\|(B(x_0, R)).$$

As a corollary of the Sobolev–Gagliardo–Nirenberg embedding theorem (see Theorem 14.33) we can prove the following isoperimetric inequality (see also Theorem C.12).

Theorem 14.44 (Isoperimetric inequality). *Let $E \subseteq \mathbb{R}^N$, $N \geq 2$, be a set of finite perimeter. Then either E or $\mathbb{R}^N \setminus E$ has finite Lebesgue measure and*

$$(14.45) \quad \min\{\mathcal{L}^N(E), \mathcal{L}^N(\mathbb{R}^N \setminus E)\}^{1/1^*} \leq c P(E)$$

for some constant $c = c(N) > 0$.

Proof. If $\mathcal{L}^N(E) < \infty$, then $\chi_E \in BV(\mathbb{R}^N)$, and so (14.45) follows from the Sobolev–Gagliardo–Nirenberg embedding theorem (Theorem 14.33) applied to χ_E . It remains to show that if E has finite perimeter, then either E or $\mathbb{R}^N \setminus E$ has finite Lebesgue measure. Consider a ball $B_R := B(0, R)$ and set $u := \chi_E$. We claim that

$$\min\{\mathcal{L}^N(B_R \cap E), \mathcal{L}^N(B_R \setminus E)\}^{1/1^*} \leq c \|D(\chi_E)\|(B_R).$$

To see this, note that

$$u_{B_R} = \frac{1}{\mathcal{L}^N(B_R)} \int_{B_R} u(x) dx = \frac{\mathcal{L}^N(B_R \cap E)}{\mathcal{L}^N(B_R)},$$

and so

$$\begin{aligned} \int_{B_R} |u - u_{B_R}|^{1*} dx &= \left(1 - \frac{\mathcal{L}^N(B_R \cap E)}{\mathcal{L}^N(B_R)}\right)^{1*} \mathcal{L}^N(B_R \cap E) \\ &\quad + \left(\frac{\mathcal{L}^N(B_R \cap E)}{\mathcal{L}^N(B_R)}\right)^{1*} \mathcal{L}^N(B_R \setminus E). \end{aligned}$$

If $\mathcal{L}^N(B_R \cap E) \leq \mathcal{L}^N(B_R \setminus E)$, then

$$\begin{aligned} \left(\int_{B_R} |u - u_{B_R}|^{1*} dx\right)^{1/1*} &\geq \frac{\mathcal{L}^N(B_R \setminus E)}{\mathcal{L}^N(B_R)} (\mathcal{L}^N(B_R \cap E))^{1/1*} \\ &\geq \frac{1}{2} (\mathcal{L}^N(B_R \cap E))^{1/1*} \\ &= \frac{1}{2} \min\{\mathcal{L}^N(B_R \cap E), \mathcal{L}^N(B_R \setminus E)\}^{1/1*}. \end{aligned}$$

The other case is analogous.

By applying Poincaré's inequality for balls (see the previous exercise), we get that the left-hand side of the previous inequality is bounded from above by $c\|D(\chi_E)\|(B_R)$, and so

$$\begin{aligned} \frac{1}{2} \min\{\mathcal{L}^N(B_R \cap E), \mathcal{L}^N(B_R \setminus E)\}^{1/1*} &\leq c\|D(\chi_E)\|(B_R) \\ &\leq c\|D(\chi_E)\|(\mathbb{R}^N). \end{aligned}$$

Hence, the claim is proved.

By letting $R \rightarrow \infty$ in the previous inequality and using Proposition B.9, it follows that either E or $\mathbb{R}^N \setminus E$ has finite Lebesgue measure. $\square$

Thus, we have shown that the Sobolev–Gagliardo–Nirenberg embedding theorem in BV implies the isoperimetric inequality. Next we show that the opposite is also true.

Theorem 14.45. *Assume that the isoperimetric inequality (14.45) holds for all sets with finite perimeter. Then there exists a constant $c = c(N) > 0$ such that*

$$\|u\|_{L^{1*}(\mathbb{R}^N)} \leq c\|Du\|(\mathbb{R}^N)$$

for all $u \in BV(\mathbb{R}^N)$.

Proof. Assume first that $u \geq 0$ and that $u \in C^\infty(\mathbb{R}^N) \cap W^{1,1}(\mathbb{R}^N)$. For $t \in \mathbb{R}$, define $A_t := \{x \in \mathbb{R}^N : u(x) > t\}$. Then by the coarea formula (14.42) and the isoperimetric inequality (14.45),

$$(14.46) \quad \int_{\mathbb{R}^N} \|\nabla u\| dx = \int_0^\infty P(A_t) dt \geq \frac{1}{c} \int_0^\infty (\mathcal{L}^N(A_t))^{1/1*} dt.$$

For every $t \geq 0$, consider the function $u_t := \min\{u, t\}$ and let $f(t) := \|u_t\|_{L^{1^*}(\mathbb{R}^N)}$. Note that f is finite, since

$$\int_{\mathbb{R}^N} (u_t(x))^{1^*} dx \leq t^{1^*-1} \int_{\{u \leq t\}} u(x) dx + t^{1^*} \mathcal{L}^N(A_t) < \infty.$$

Moreover, f is increasing in $[0, \infty)$ and by the Lebesgue monotone convergence theorem,

$$(14.47) \quad \lim_{t \rightarrow \infty} f(t) = \|u\|_{L^{1^*}(\mathbb{R}^N)},$$

while by the fact that $0 \leq (u_t(x))^{1^*} \leq (u_1(x))^{1^*}$ for all $t \in [0, 1]$, it follows by the Lebesgue dominated convergence theorem that

$$(14.48) \quad \lim_{t \rightarrow 0^+} f(t) = 0 = f(0).$$

Furthermore, for $h, t > 0$, by Minkowski's inequality,

$$0 \leq f(t+h) - f(t) \leq \|u_{t+h} - u_t\|_{L^{1^*}(\mathbb{R}^N)} \leq h(\mathcal{L}^N(A_t))^{1/1^*},$$

where in the last inequality we have used the facts that $u_{t+h}(x) = u(x) = u_t(x)$ if $u(x) \leq t$, while

$$|u_{t+h}(x) - u_t(x)| = \min\{u(x), t+h\} - t \leq t+h-t = h$$

if $u(x) > t$. Thus, the function f is locally Lipschitz continuous in $(0, \infty)$, with $f'(t) \leq (\mathcal{L}^N(A_t))^{1/1^*}$ for $\mathcal{L}^1$ -a.e. $t > 0$. It follows by the fundamental theorem of calculus (see Theorem 3.20), (14.46), (14.47), and (14.48) that

$$\begin{aligned} \|u\|_{L^{1^*}(\mathbb{R}^N)} &= \lim_{t \rightarrow \infty} f(t) = \int_0^\infty f'(s) ds \\ &\leq \int_0^\infty (\mathcal{L}^N(A_s))^{1/1^*} ds \leq c \int_{\mathbb{R}^N} \|\nabla u\| dx. \end{aligned}$$

The general case follows as in the proof of Theorem 14.33. $\square$

The proofs of the following theorems are left as an exercise.

Theorem 14.46 (Gagliardo–Nirenberg interpolation, $m = 2$). *Let $1 \leq q \leq \infty$ and let r be given by*

$$(14.49) \quad \frac{1}{2} + \frac{1}{2q} = \frac{1}{r}.$$

There exists $c = c(N, q) > 0$ such that

$$\|\nabla u\|_{L^r(\mathbb{R}^N)} \leq c \|u\|_{L^q(\mathbb{R}^N)}^{1/2} (\|D(\nabla u)\|(\mathbb{R}^N))^{1/2}$$

for every $u \in L^q(\mathbb{R}^N) \cap W_{\text{loc}}^{1,1}(\mathbb{R}^N)$ with $D_j(\partial_k u) \in \mathcal{M}_b(\mathbb{R}^N)$ for all $j, k = 1, \dots, N$.

Next we consider uniformly Lipschitz continuous domains (see Definition 13.11).

Theorem 14.47 (Gagliardo–Nirenberg interpolation, $m = 2$). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set with uniformly Lipschitz continuous boundary (with parameters ε, L, M), let $0 < \ell < \varepsilon/(4(1 + L))$, and let $1 \leq q, r \leq \infty$ be such that (14.49) holds. If $q > 1$ assume further that Ω is bounded. Then there exists a constant $c = c(N, q) > 0$ such that for every $u \in L^q(\Omega) \cap W_{\text{loc}}^{1,1}(\Omega)$ with $D_j(\partial_k u) \in \mathcal{M}_b(\Omega)$ for all $j, k = 1, \dots, N$, if $q > 1$,*

$$\|\nabla u\|_{L^r(\Omega)} \leq c\ell^{-1}(\mathcal{L}^N(\Omega))^{1/r-1/q}\|u\|_{L^q(\Omega)} + c\|u\|_{L^q(\Omega)}^{1/2}(\|D(\nabla u)\|(\Omega))^{1/2}$$

while if $q = 1$,

$$\|\nabla u\|_{L^1(\Omega)} \leq c\ell^{-1}\|u\|_{L^1(\Omega)} + c\|u\|_{L^1(\Omega)}^{1/2}(\|D(\nabla u)\|(\Omega))^{1/2}.$$

14.6. Density of Smooth Sets

In this section we prove that sets of finite perimeter can be approximated by smooth sets. The proof makes use of a theorem of Sard (see [171], [204]).

Theorem 14.48 (Sard). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $u \in C^\infty(\Omega)$, and let $\Sigma := \{x \in \Omega : \nabla u(x) = 0\}$. Then $\mathcal{L}^1(u(\Sigma)) = 0$.*

The proof uses the following covering result.

Lemma 14.49 (Vitali). *Let $E \subset \mathbb{R}^N$ be the union of a finite number of balls $B(x_i, r_i)$, $i = 1, \dots, \ell$. Then there exists a subset $\mathcal{I} \subseteq \{1, \dots, \ell\}$ such that the balls $B(x_i, r_i)$ with $i \in \mathcal{I}$ are pairwise disjoint and $E \subseteq \bigcup_{i \in \mathcal{I}} B(x_i, 3r_i)$.*

Proof. Without loss of generality, we may assume that $r_1 \geq r_2 \geq \dots \geq r_\ell$. Put $i_1 := 1$ and discard all the balls that intersect $B(x_1, r_1)$. Let i_2 be the first integer, if it exists, such that $B(x_{i_2}, r_{i_2})$ does not intersect $B(x_1, r_1)$. If i_2 does not exist, then set $\mathcal{I} := \{i_1\}$, while if i_2 exists, discard all the balls $B(x_i, r_i)$, with $i > i_2$, that intersect $B(x_{i_2}, r_{i_2})$. Let $i_3 > i_2$ be the first integer, if it exists, such that $B(x_{i_3}, r_{i_3})$ does not intersect $B(x_{i_2}, r_{i_2})$. If i_3 does not exist, then set $\mathcal{I} := \{i_1, i_2\}$; if i_3 exists, continue the process. Since there are only a finite number of balls, the process stops after a finite number of steps and we obtain the desired set $\mathcal{I} \subseteq \{1, \dots, \ell\}$.

By construction, the balls $B(x_i, r_i)$ with $i \in \mathcal{I}$ are pairwise disjoint. To prove the last statement, let $x \in E$. Then there exists $i = 1, \dots, \ell$ such that $x \in B(x_i, r_i)$. If $i \in \mathcal{I}$, then there is nothing to prove. If $i \notin \mathcal{I}$, then $B(x_i, r_i)$ is one of the balls that has been discarded and thus there exists $j \in \mathcal{I}$ such that $r_j \geq r_i$ and $B(x_j, r_j) \cap B(x_i, r_i) \neq \emptyset$. Let $z \in B(x_j, r_j) \cap B(x_i, r_i)$. Then

$$\|x - x_j\| \leq \|x - x_i\| + \|x_i - z\| + \|z - x_j\| < r_i + r_i + r_j \leq 3r_j,$$

and so $x \in B(x_j, 3r_j)$ and the proof is complete. $\square$

We are now ready to prove Theorem 14.48. We use the notation (E.3) in Appendix E.

Proof of Theorem 14.48. Step 1: Assume that Ω is bounded and let

$$\Sigma_N := \{x \in \Omega : \nabla u(x) = 0, \nabla^2 u(x) = 0, \dots, \nabla^N u(x) = 0\}.$$

We claim that $\mathcal{L}^1(u(\Sigma_N)) = 0$. To see this, fix $\varepsilon > 0$ and let $x_0 \in \Sigma_N$. By Taylor's formula centered at x_0 (see Theorem 9.9) we have

$$\begin{aligned} u(x) &= \sum_{|\alpha|=0}^N \frac{1}{\alpha!} \partial^\alpha u(x_0) (x - x_0)^\alpha + o(\|x - x_0\|^N) \\ &= u(x_0) + o(\|x - x_0\|^N) \end{aligned}$$

as $x \rightarrow x_0$, and thus there exists $r_{x_0} > 0$ such that

$$|u(x) - u(x_0)| \leq \varepsilon \|x - x_0\|^N$$

for all $x \in \overline{B(x_0, r_{x_0})} \subset \Omega$. In particular, for all $0 < r \leq r_{x_0}$,

$$(14.50) \quad \frac{\text{osc}}{B(x_0, r)} u \leq 2\varepsilon r^N.$$

Fix a compact set $K \subseteq \Sigma_N$. Since $\{B(x, r_x/3)\}_{x \in \Sigma_N}$ is an open cover for K , we may find a finite subcover. By the previous lemma there exists a disjoint finite subfamily $\{B(x_i, r_i/3)\}_{i=1}^n$, such that $K \subset \bigcup_{i=1}^n B(x_i, r_i)$. Hence,

$$\begin{aligned} \mathcal{L}^1(u(K)) &\leq \sum_{i=1}^n \mathcal{L}^1(u(K \cap B(x_i, r_i))) \\ &\leq 2\varepsilon \sum_{i=1}^n r_i^N \leq 2 \cdot 3^N \alpha_N^{-1} \varepsilon \sum_{i=1}^n \alpha_N \left(\frac{r_i}{3}\right)^N \leq c\varepsilon \mathcal{L}^N(\Omega), \end{aligned}$$

where $c = c(N)$ and we have used (14.50) and the fact that $\{B(x_i, r_i/3)\}_{i=1}^n$ is a disjoint family contained in Ω . Given the arbitrariness of $\varepsilon > 0$, we conclude that $\mathcal{L}^1(u(K)) = 0$.

Now, let $\{K_n\}_n$ be an increasing sequence of compact sets such that $\Omega = \bigcup_{n=1}^\infty K_n$. Then $\Sigma_N \cap K_n$ is a compact set (since u is of class $C^\infty(\Omega)$), and so, by what we just proved, $\mathcal{L}^1(u(\Sigma_N \cap K_n)) = 0$ for every $n \in \mathbb{N}$. Thus, $\mathcal{L}^1(u(\Sigma_N)) = 0$.

The case in which Ω is unbounded follows by considering an increasing sequence of bounded open sets whose union is Ω and applying what we just proved in each open bounded set.

Step 2: We now prove the theorem by induction on N . The previous step shows that the result is true for $N = 1$. Thus we assume that the result is true for dimension $N - 1$ and we prove it for dimension N . Again by the previous step it remains to show that $\mathcal{L}^1(u(\Sigma \setminus \Sigma_N)) = 0$. If $x \in \Sigma \setminus \Sigma_N$,

then there exist a multi-index α , with $|\alpha| < N$, and $i \in \{1, \dots, N\}$ such that

$$(14.51) \quad \partial^\alpha u(x) = 0, \quad \partial_i(\partial^\alpha u)(x) \neq 0.$$

Let $E_{\alpha,i}$ be the set of points $x \in \Sigma \setminus \Sigma_N$ for which (14.51) holds. Since

$$\Sigma \setminus \Sigma_N = \bigcup_{|\alpha| < N, i \in \{1, \dots, N\}} E_{\alpha,i},$$

to prove that $\mathcal{L}^1(u(\Sigma \setminus \Sigma_N)) = 0$, it is enough to show that $\mathcal{L}^1(u(E_{\alpha,i})) = 0$. Without loss of generality, we may assume that $i = N$. Fix $x_0 = (x'_0, x_{0,N}) \in E_{\alpha,N}$ and set $v := \partial^\alpha u$. Then $v(x_0) = 0$ and $\partial_N v(x_0) \neq 0$ and thus, by the implicit function theorem, we may find a cube

$$Q(x_0, r) = Q_{N-1}(x'_0, r) \times (x_{0,N} - r/2, x_{0,N} + r/2) \subset \Omega$$

and a function $\psi \in C^\infty(Q_{N-1}(x'_0, r))$ such that

$$\begin{aligned} E_{\alpha,N} \cap Q(x_0, r) &= \{x \in Q(x_0, r) : v(x) = 0\} \\ &\subseteq \{x' \in Q_{N-1}(x'_0, r) : x_N = \psi(x')\}. \end{aligned}$$

The union of all these cubes as x_0 varies in $E_{\alpha,N}$ gives an open cover for $E_{\alpha,N}$. Since from this open cover we can always select a countable subcover (why?), it suffices to prove that

$$\mathcal{L}^1(u(E_{\alpha,N} \cap Q(x_0, r))) = 0.$$

Define the function $f(x') := u(x', \psi(x'))$, $x' \in Q_{N-1}(x'_0, r)$, and let $E' := \{x' \in Q_{N-1}(x'_0, r) : (x', \psi(x')) \in E_{\alpha,N} \cap Q(x_0, r)\}$. Since $\nabla u = 0$ in $E_{\alpha,N}$, it follows by the chain rule and the definition of f that $\nabla_{x'} f(x') = 0$ for all $x' \in E'$. By the induction hypothesis $\mathcal{L}^1(f(E')) = 0$. But

$$\begin{aligned} f(E') &= \{u(x', \psi(x')) : (x', \psi(x')) \in E_{\alpha,N} \cap Q(x_0, r)\} \\ &= u(E_{\alpha,N} \cap Q(x_0, r)) \end{aligned}$$

and so $\mathcal{L}^1(u(E_{\alpha,N} \cap Q(x_0, r))) = 0$ and the proof is complete. $\square$

The following result is an important consequence of the previous theorem:

Corollary 14.50 (Sard). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in C^\infty(\Omega)$. Then for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$ the sets $\{x \in \Omega : u(x) = t\}$ are C^∞ -manifolds.*

Proof. Let $\Sigma := \{x \in \Omega : \nabla u(x) = 0\}$. By the previous theorem $\mathcal{L}^1(u(\Sigma)) = 0$. Hence, if $t \in \mathbb{R} \setminus u(\Sigma)$, then by the implicit function theorem $\{x \in \Omega : u(x) = t\}$ is a C^∞ -manifold. $\square$

We now present the density result mentioned at the beginning of this section. We recall that Δ is the symmetric difference of sets.

Theorem 14.51 (Density of smooth sets). *Let $E \subseteq \mathbb{R}^N$, $N \geq 2$, be a set of finite perimeter. Then there exists a sequence of sets $\{E_n\}_n$ with smooth boundary such that*

$$\mathcal{L}^N(E_n \Delta E) \rightarrow 0 \quad \text{and} \quad P(E_n) \rightarrow P(E).$$

Proof. By the isoperimetric inequality (14.45), either E or its complement have finite Lebesgue measure. Since $P(E) = P(\mathbb{R}^N \setminus E)$ (why?), without loss of generality, we may assume that $\mathcal{L}^N(E) < \infty$. Hence, $\chi_E \in BV(\mathbb{R}^N)$. For every $\varepsilon > 0$ define $u_\varepsilon := \varphi_\varepsilon * \chi_E$, where φ_ε is a standard mollifier. By Lemma 14.10, $u_\varepsilon \rightarrow \chi_E$ in $L^1(\mathbb{R}^N)$ and

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^N} \|\nabla u_\varepsilon\| dx = \|D(\chi_E)(\mathbb{R}^N)\| = P(E).$$

Note that since $0 \leq \chi_E \leq 1$, then $0 \leq u_{\varepsilon_n} \leq 1$. By Step 1 of the proof of Theorem 14.28 there exists a sequence $\varepsilon_n \rightarrow 0^+$ such that

$$\liminf_{n \rightarrow \infty} P(A_t^n) \geq P(A_t)$$

for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$, where for $t \in \mathbb{R}$,

$$A_t := \{x \in \mathbb{R}^N : \chi_E(x) > t\}, \quad A_t^n := \{x \in \mathbb{R}^N : u_{\varepsilon_n}(x) > t\}.$$

Since

$$A_t = \begin{cases} E & \text{if } t < 1, \\ \emptyset & \text{if } t \geq 1, \end{cases}$$

it follows that

$$(14.52) \quad \liminf_{n \rightarrow \infty} P(A_t^n) \geq P(E)$$

for $\mathcal{L}^1$ -a.e. $t \in [0, 1]$.

On the other hand, by the coarea formula, Fatou's lemma, and the fact that $0 \leq u_{\varepsilon_n} \leq 1$,

$$P(E) = \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \|\nabla u_\varepsilon\| dx = \lim_{n \rightarrow \infty} \int_0^1 P(A_t^n) dt \geq \int_0^1 \liminf_{n \rightarrow \infty} P(A_t^n) dt.$$

Together with (14.52), this implies that

$$\liminf_{n \rightarrow \infty} P(A_t^n) = P(E)$$

for $\mathcal{L}^1$ -a.e. $t \in [0, 1]$.

Since $u_{\varepsilon_n} \in C^\infty(\mathbb{R}^N)$, it follows by Sard's theorem that the sets

$$\{x \in \mathbb{R}^N : u_{\varepsilon_n}(x) = t\}$$

are C^∞ -manifolds for $\mathcal{L}^1$ -a.e. $t \in [0, 1]$. Hence, we may find $t \in (0, 1)$ such that

$$\liminf_{n \rightarrow \infty} P(A_t^n) = P(E)$$

and the sets $\{x \in \mathbb{R}^N : u_{\varepsilon_n}(x) = t\}$ are C^∞ -manifolds for all $n \in \mathbb{N}$.

Define $E_n := A_t^n$. It remains to show that $\mathcal{L}^N(E_n \Delta E) \rightarrow 0$. To see this, note that $u_{\varepsilon_n}(x) - \chi_E(x) > t$ for $x \in E_n \setminus E$, while $\chi_E(x) - u_{\varepsilon_n}(x) \geq 1 - t$ for $x \in E \setminus E_n$. Hence,

$$\begin{aligned} \int_{\mathbb{R}^N} |u_{\varepsilon_n} - \chi_E| dx &\geq \int_{E_n \setminus E} |u_{\varepsilon_n} - \chi_E| dx + \int_{E \setminus E_n} |u_{\varepsilon_n} - \chi_E| dx \\ &\geq t\mathcal{L}^N(E_n \setminus E) + (1 - t)\mathcal{L}^N(E \setminus E_n), \end{aligned}$$

which implies that $\mathcal{L}^N(E_n \Delta E) \rightarrow 0$, since $0 < t < 1$. This concludes the proof. $\square$

Exercise 14.52. Prove that, in general, we cannot approximate E with smooth sets contained outside E (or inside). Hint: Consider the set A given in Exercise 14.8.

14.7. A Characterization of $BV(\Omega)$

In this section we give a characterization of $BV(\Omega)$ in terms of difference quotients. This is the extension of Corollary 2.51 in higher dimension. A similar characterization has been given in Section 11.5 for the Sobolev space $W^{1,p}(\Omega)$, $1 < p < \infty$.

Let $\Omega \subseteq \mathbb{R}^N$ be an open set and for every $h \in \mathbb{R}^N \setminus \{0\}$, let

$$\Omega_h := \{x \in \Omega : x + th \in \Omega \text{ for all } t \in [0, 1]\}.$$

We recall that S^{N-1} is the unit sphere $\partial B(0, 1)$ and that $\beta_N = \mathcal{H}^{N-1}(S^{N-1})$.

Theorem 14.53. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in L^1_{\text{loc}}(\Omega)$ be such that $Du \in \mathcal{M}_b(\Omega; \mathbb{R}^N)$. Then for every $h \in \mathbb{R}^N \setminus \{0\}$,*

$$(14.53) \quad \int_{\Omega_h} \frac{|u(x+h) - u(x)|}{\|h\|} dx \leq \|Du\|(\Omega),$$

while

$$(14.54) \quad \kappa_{N,1} \|Du\|(\Omega) \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x+h) - u(x)|}{\|h\|} dx,$$

where

$$(14.55) \quad \kappa_{N,1} := \frac{1}{\beta_N} \int_{S^{N-1}} |e_1 \cdot \xi| d\mathcal{H}^{N-1}(\xi).$$

Conversely, if $u \in L^1_{\text{loc}}(\Omega)$ is such that

$$(14.56) \quad \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x+h) - u(x)|}{\|h\|} dx < \infty,$$

then $Du \in \mathcal{M}_b(\Omega; \mathbb{R}^N)$.

Proof. The proof is very similar to the one of Theorem 11.75 and we only indicate the main changes. In Step 1 we assume that U satisfies the additional hypothesis $\|Du\|(\partial U) = 0$. We proceed as before to get

$$\int_{U_h} |u_\varepsilon(x+h) - u_\varepsilon(x)| dx \leq \|h\| \int_U \|\nabla u_\varepsilon(y)\| dy.$$

We are now in a position to apply Lemma 14.10 together with the fact that $u_\varepsilon \rightarrow u$ pointwise for $\mathcal{L}^N$ -a.e. in Ω , and Fatou's lemma to obtain

$$\int_{U_h} |u(x+h) - u(x)| dx \leq \|h\| \|Du\|(U) \leq \|h\| \|Du\|(\Omega).$$

To prove (14.53) it suffices to let $U \nearrow \Omega$ and use the Lebesgue monotone convergence theorem.

Step 2 of the proof of Theorem 11.75 remains unchanged.

In Step 3 we assume that U satisfies the additional hypothesis $\|Du\|(\partial U) = 0$. We proceed as before to obtain (11.42), which now reads

$$(14.57) \quad \kappa_{N,1} \int_U \|\nabla u_\varepsilon(x)\| dx \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x+h) - u(x)|}{\|h\|} dx.$$

We are now in a position to apply Lemma 14.10 to obtain

$$(14.58) \quad \kappa_{N,1} \|Du\|(U) \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x+h) - u(x)|}{\|h\|} dx.$$

By letting $U \nearrow \Omega$ along a sequence and using Proposition B.9, we have that

$$(14.59) \quad \kappa_{N,1} \|Du\|(\Omega) \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x+h) - u(x)|}{\|h\|} dx,$$

which completes Step 3.

Step 4: To prove the final statement of the theorem, let $u \in L^1_{\text{loc}}(\Omega)$ be such that (14.56) holds. We claim that $Du \in \mathcal{M}_b(\Omega; \mathbb{R}^N)$. To see this, let $U \Subset \Omega$. Then reasoning as in Step 4, by (14.57) we get

$$\sup_{0 < \varepsilon < \text{dist}(U, \partial\Omega)} \int_U \|\nabla u_\varepsilon(x)\| dx \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|u(x+h) - u(x)|}{\|h\|} dx < \infty.$$

Since $u_\varepsilon \rightarrow u$ in $L^1(U)$ by Theorem C.16, it follows from Theorem 14.39 that $u \in BV(U)$ and that (14.58) holds for every $i = 1, \dots, N$. Given the arbitrariness of U , taking a sequence $U_j \Subset \Omega$ with $U_j \nearrow \Omega$, we conclude that $u \in BV_{\text{loc}}(\Omega)$. Since (14.58) holds for all U_j , letting $j \rightarrow \infty$, we obtain (14.59), which shows that $Du \in \mathcal{M}_b(\Omega; \mathbb{R}^N)$. $\square$

Exercise 14.54. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and for every $i = 1, \dots, N$ and $h > 0$, let

$$\Omega_{h,i} := \{x \in \Omega : x + t e_i \in \Omega \text{ for all } t \in [0, h]\}.$$

- (i) Let $u \in L^1_{\text{loc}}(\Omega)$ be such that $Du \in \mathcal{M}_b(\Omega; \mathbb{R}^N)$. Prove that for every $i = 1, \dots, N$ and $h > 0$,

$$\int_{\Omega_{h,i}} |u(x + he_i) - u(x)| \, dx \leq h |D_i u|(\Omega)$$

and

$$\lim_{h \rightarrow 0^+} \int_{\Omega_{h,i}} \frac{|u(x + he_i) - u(x)|}{h} \, dx = |D_i u|(\Omega).$$

- (ii) Let $u \in L^1_{\text{loc}}(\Omega)$ be such that

$$\liminf_{h \rightarrow 0^+} \int_{\Omega_{h,i}} \frac{|u(x + he_i) - u(x)|}{h} \, dx < \infty$$

for every $i = 1, \dots, N$. Prove that $Du \in \mathcal{M}_b(\Omega; \mathbb{R}^N)$.

Sobolev Spaces: Symmetrization

Living with P.Q.S, II: What are the symptoms of PQS? Symptoms include: Inability to maintain interest in research area; Advisor Avoidance; Increased web-surfing; Healthy sleeping-pattern; Cynicism towards Academia.

— Jorge Cham, www.phdcomics.com

In this chapter we introduce and study the notion of spherically symmetric rearrangement of a function. For more information on this topic and on its applications to partial differential equations we refer the reader to the monographs of Kawohl [130] and Kesavan [131] and the review paper of Talenti [230]. Except for the last two sections, most of the results of this chapter continue to hold if we consider a measure space $(X, \mathfrak{M}, \mu)$ and replace the Lebesgue measurable set $E \subseteq \mathbb{R}^N$ and the Lebesgue measure $\mathcal{L}^N$ by a set $E \in \mathfrak{M}$ of X and by μ , respectively. We leave the details as an exercise.

In what follows, given a set $E \subseteq \mathbb{R}^N$, a function $u : E \rightarrow [-\infty, \infty]$, and $s \in \mathbb{R}$, when there is no possibility of confusion we write $\{u > s\}$, $\{u \geq s\}$ for the sets $\{x \in E : u(x) > s\}$, $\{x \in E : u(x) \geq s\}$, respectively.

15.1. Symmetrization in L^p Spaces

As in Chapter 4, given a Lebesgue measurable set $E \subseteq \mathbb{R}^N$ and a Lebesgue measurable function $u : E \rightarrow [0, \infty]$, the *distribution function* of u is the function $\varrho_u : [0, \infty) \rightarrow [0, \mathcal{L}^N(E)]$, defined by

$$(15.1) \quad \varrho_u(s) := \mathcal{L}^N(\{u > s\}), \quad s \geq 0.$$

Similarly, we define

$$(15.2) \quad \varsigma_u(s) := \mathcal{L}^N(\{u \geq s\}), \quad s \geq 0.$$

We recall that $u : E \rightarrow [0, \infty]$ *vanishes at infinity* if u is Lebesgue measurable and $\varrho_u(s) < \infty$ for $s > 0$ ($\varrho_u(0)$ can be infinite).

The proof of the following proposition is very similar to that of Proposition 4.1 and is left as an exercise.

Proposition 15.1. *Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u, v, u_n : E \rightarrow [0, \infty]$, $n \in \mathbb{N}$, be Lebesgue measurable functions. Then the following properties hold:*

- (i) *The function $\varrho_u : [0, \infty) \rightarrow [0, \mathcal{L}^N(E)]$ is decreasing and right continuous, while $\varsigma_u : [0, \infty) \rightarrow [0, \mathcal{L}^N(E)]$ is decreasing.*
- (ii) *If u vanishes at infinity, then ς_u is left continuous,*

$$\lim_{s \rightarrow \infty} \varrho_u(s) = \lim_{s \rightarrow \infty} \varsigma_u(s) = 0,$$

and ϱ_u and ς_u are continuous at $s > 0$ if and only if $\mathcal{L}^N(\{u = s\}) = 0$. In particular, $\varsigma_u(s) = \varrho_u(s)$ for all but countably many $s \geq 0$.

- (iii) *If $u(x) \leq v(x)$ for $\mathcal{L}^N$ -a.e. $x \in E$, then $\varrho_u \leq \varrho_v$. In particular, if $u(x) = v(x)$ for $\mathcal{L}^N$ -a.e. $x \in E$, then $\varrho_u = \varrho_v$.*
- (iv) *If $u_n(x) \nearrow u(x)$ for $\mathcal{L}^N$ -a.e. $x \in E$, then $\varrho_{u_n} \nearrow \varrho_u$.*

Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty]$ be a Lebesgue measurable function. As in Chapter 4, the function $u^* : [0, \infty) \rightarrow [0, \infty]$, defined by

$$(15.3) \quad u^*(t) := \inf\{s \in [0, \infty) : \varrho_u(s) \leq t\}, \quad t \geq 0,$$

is the *decreasing rearrangement* of u .

Proposition 15.2. *Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty]$ be a measurable function. Then the following properties hold:*

- (i) *The function u^* is decreasing, right continuous, and $u^*(0) = \operatorname{ess\,sup}_E u$.*
- (ii) *For all $s, t \geq 0$, $u^*(t) > s$ if and only if $\varrho_u(s) > t$.*
- (iii) *If u vanishes at infinity, then for all $0 < t < \varrho_u(0)$, $t \leq \varsigma_u(u^*(t))$.*
- (iv) *If u vanishes at infinity, then for all $t > 0$, $\varrho_u(u^*(t)) \leq t$, and if $\varrho_u(u^*(t_1)) < t_1$ for some $t_1 > 0$, then u^* is constant on $[t_0, t_1]$, where $t_0 := \varrho_u(u^*(t_1))$.*
- (v) *The functions u and u^* are equimeasurable; that is, for all $s \geq 0$,*

$$\mathcal{L}^1(\{t \in E^* : u^*(t) > s\}) = \mathcal{L}^N(\{x \in E : u(x) > s\}),$$

where $E^ := [0, \mathcal{L}^1(E))$.*

Proof. Properties (i), (ii), and (v) follow as in the proof of Proposition 4.3.

(iii) If $\varrho_u(0) > t > 0$, then $u^*(t) > 0$ by property (ii). Fix $0 \leq s < u^*(t)$. Then $\varrho_u(s) > t$ by part (ii), and since $\varrho_u \leq \varsigma_u$, we have $\varsigma_u(s) > t$. Using the fact that ς_u is left continuous, letting $s \nearrow u^*(t)$, we get $\varsigma_u(u^*(t)) \geq t$.

(iv) If $u^*(t) \leq s$, then by part (ii), $\varrho_u(s) \leq t$. Taking $s = u^*(t)$ gives $\varrho_u(u^*(t)) \leq t$. Assume next that $t_0 := \varrho_u(u^*(t_1)) < t_1$ for some $t_1 > 0$. Then for $t_0 \leq t \leq t_1$ we have $\varrho_u(u^*(t_1)) = t_0 \leq t$ and hence $u^*(t_1)$ is an admissible s in the definition of $u^*(t)$, and so $u^*(t) \leq u^*(t_1)$. But since u^* is decreasing, this implies that $u^*(t) = u^*(t_1)$ for all $t \in [t_0, t_1]$. $\square$

Remark 15.3. If in the previous proposition we assume that u is also bounded, then by part (i), $u^*(0) < \infty$. Hence, reasoning as in part (iv), we have that $\varrho_u(u^*(0)) = 0$.

Exercise 15.4. State and prove the analogous statement of part (iv) of Proposition 15.2 for the function ς_u .

Definition 15.5. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty]$ be a Lebesgue measurable function. The function $u^\sharp : \mathbb{R}^N \rightarrow [0, \infty]$, defined by

$$(15.4) \quad u^\sharp(y) := u^*(\alpha_N \|y\|^N) = \inf\{s \in [0, \infty) : \varrho_u(s) \leq \alpha_N \|y\|^N\}$$

for $y \in \mathbb{R}^N$, is the *spherically symmetric rearrangement* of u .

We recall that α_N is the Lebesgue measure of the unit ball. The function $u^\sharp$ is also called the *Schwarz symmetric rearrangement* of u . Note that if u vanishes at infinity, then by Proposition 15.1(ii), $u^\sharp(y) < \infty$ for all $y \neq 0$.

Given a Lebesgue measurable set $F \subset \mathbb{R}^N$, with $0 < \mathcal{L}^N(F) < \infty$, we define the *spherically symmetric rearrangement of the set F* to be the open ball centered at the origin and with the same measure of F , precisely $F^\sharp := B(0, r)$, where

$$(15.5) \quad r := (\mathcal{L}^N(F)/\alpha_N)^{1/N}.$$

If F has measure zero, then we take $F^\sharp$ to be the empty set, while if F has infinite measure, then we take $F^\sharp$ to be $\mathbb{R}^N$. Note that $F^\sharp$ is always open even if F is not. Hence, in all cases,

$$(15.6) \quad \mathcal{L}^N(F) = \mathcal{L}^N(F^\sharp).$$

The next proposition collects some elementary properties of symmetrization.

Proposition 15.6. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u : E \rightarrow [0, \infty]$ be a measurable function. Then the following properties hold:

- (i) For all $s \geq 0$, $\{y \in \mathbb{R}^N : u^\sharp(y) > s\} = \{x \in E : u(x) > s\}^\sharp$.

(ii) If u vanishes at infinity, then so does $u^\sharp$. Moreover,

$$\mathcal{L}^N(\{y \in E^\sharp : u^\sharp(y) = 0\}) \leq \mathcal{L}^N(\{x \in E : u(x) = 0\})$$

with equality holding if and only if either $\mathcal{L}^N(\{u > 0\}) < \infty$ or $\mathcal{L}^N(\{u > 0\}) = \infty$ and $\mathcal{L}^N(\{u = 0\}) = 0$.

(iii) The function $u^\sharp$ is lower semicontinuous and $u^\sharp(0) = \text{ess sup}_E u$.

(iv) If $v : E \rightarrow [0, \infty]$ is another Lebesgue measurable function with $u \leq v$ $\mathcal{L}^N$ -a.e. on E , then $u^\sharp \leq v^\sharp$ on $\mathbb{R}^N$.

(v) If $\{u_n\}_n$ is an increasing sequence of nonnegative Lebesgue measurable functions $u_n : E \rightarrow [0, \infty]$ such that $u_n(x) \rightarrow u(x)$ as $n \rightarrow \infty$ for all $x \in E$, then $u_n^\sharp(y) \rightarrow u^\sharp(y)$ as $n \rightarrow \infty$ for all $y \in \mathbb{R}^N$.

Proof. (i) For $s \geq 0$, $\{y \in \mathbb{R}^N : u^\sharp(y) > s\} = \{y \in \mathbb{R}^N : u^*(\alpha_N \|y\|^N) > s\}$. By Proposition 15.2 we have that $u^*(\alpha_N \|y\|^N) > s$ if and only if $\varrho_u(s) > \alpha_N \|y\|^N$, and so

$$\begin{aligned} \{y \in \mathbb{R}^N : u^\sharp(y) > s\} &= \{y \in \mathbb{R}^N : \|y\| < (\varrho_u(s)/\alpha_N)^{1/N}\} \\ &= \{y \in \mathbb{R}^N : \|y\| < (\mathcal{L}^N(\{x \in E : u(x) > s\})/\alpha_N)^{1/N}\} \\ &= \{x \in E : u(x) > s\}^\sharp, \end{aligned}$$

where in the last equality we have used (15.5).

The proof of part (ii) is the same as the one of Proposition 4.3(v).

(iii) To prove that $u^\sharp$ is lower semicontinuous, it suffices to show that the set $\{y \in \mathbb{R}^N : u^\sharp(y) > s\}$ is open for all $s \in \mathbb{R}$. If $s \geq 0$, then by part (i) the set $\{y \in \mathbb{R}^N : u^\sharp(y) > s\}$ is an open ball or empty or $\mathbb{R}^N$, while if $s < 0$, then $\{y \in \mathbb{R}^N : u^\sharp(y) > s\} = \mathbb{R}^N$ since $u^\sharp \geq 0$.

Properties (iv) and (v) follow as in the proof of Proposition 4.11(i) and (ii), respectively. $\square$

Corollary 15.7. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u \in L^\infty(E)$ be nonnegative. Then $u^\sharp$ belongs to $L^\infty(E^\sharp)$ and $\|u^\sharp\|_{L^\infty(E^\sharp)} = \|u\|_{L^\infty(E)}$.

Proof. By (15.4) and Propositions 15.2 and 15.6 we have that

$$\|u^\sharp\|_{L^\infty(E^\sharp)} = u^\sharp(0) = u^*(0) = \|u\|_{L^\infty(E)} < \infty. \quad \square$$

Exercise 15.8 (Uniqueness of Schwarz symmetrization). (i) Let $f_1, f_2 : (0, \infty) \rightarrow [0, \infty]$ be two decreasing right continuous functions such that

$$\mathcal{L}^N(\{x \in \mathbb{R}^N : f_1(\|x\|) > s\}) = \mathcal{L}^N(\{x \in \mathbb{R}^N : f_2(\|x\|) > s\})$$

for all $s > 0$. Prove that $f_1 = f_2$.

- (ii) Let $u : \mathbb{R}^N \rightarrow [0, \infty]$ be a function vanishing at infinity such that $u(x) = f(\|x\|)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ for some decreasing function $f : (0, \infty) \rightarrow [0, \infty)$. Prove that there exists a unique function $v : \mathbb{R}^N \rightarrow [0, \infty]$ vanishing at infinity such that
- (a) $v(x) = \tilde{f}(\|x\|)$ for all $x \in \mathbb{R}^N \setminus \{0\}$ for some decreasing right continuous function $\tilde{f} : (0, \infty) \rightarrow [0, \infty)$,
 - (b) $v(0) = \lim_{t \rightarrow 0^+} \tilde{f}(t)$,
 - (c) $u(x) = v(x)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$.

Prove also that the function v is lower semicontinuous.

- (iii) Let $u : \mathbb{R}^N \rightarrow [0, \infty]$ be as in part (ii). Prove that $u(x) = u^\sharp(x)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$.

The proofs of the following results are very similar to the analogous ones in Chapter 4 and are left as an exercise.

Theorem 15.9 (Hardy–Littlewood’s inequality). *Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u, v : E \rightarrow [0, \infty)$ be two Lebesgue measurable functions. Then*

$$\int_E u(x)v(x) dx \leq \int_{E^\sharp} u^\sharp(y)v^\sharp(y) dy = \int_0^{\mathcal{L}^N(E)} u^*(t)v^*(t) dt.$$

Theorem 15.10. *Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set, let $u : E \rightarrow [0, \infty)$ be a function vanishing at infinity, and let $f : [0, \infty) \rightarrow [0, \infty)$ be a Borel function. Then*

$$(15.7) \quad \int_{E^\sharp} f(u^\sharp(y)) dy \leq \int_E f(u(x)) dx,$$

with equality holding if $f(0) = 0$ or $\mathcal{L}^N(\{u > 0\}) < \infty$ or $\mathcal{L}^N(\{u > 0\}) = \infty$ and $\mathcal{L}^N(\{u = 0\}) = 0$. In particular, for any $p > 0$,

$$\int_{E^\sharp} (u^\sharp(y))^p dy = \int_E (u(x))^p dx.$$

Theorem 15.11. *Let $\Psi : \mathbb{R} \rightarrow [0, \infty)$ be a convex function such that $\Psi(0) = 0$, let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set, and let $u, v : E \rightarrow [0, \infty)$ be two functions vanishing at infinity. Then*

$$\int_{E^\sharp} \Psi(u^\sharp(y) - v^\sharp(y)) dy \leq \int_E \Psi(u(x) - v(x)) dx.$$

In particular,

$$\int_{E^\sharp} |u^\sharp(y) - v^\sharp(y)|^p dy \leq \int_E |u(x) - v(x)|^p dx$$

for all $1 \leq p < \infty$ and the operator $u \mapsto u^\sharp$ is a continuous operator from $L^p(E)$ into $L^p(E^\sharp)$.

Remark 15.12. It is important to observe that it is also possible to define the decreasing rearrangement and spherically symmetric rearrangement of a measurable function $u : X \rightarrow [0, \infty]$, where $(X, \mathfrak{M}, \mu)$ is a measure space.

Remark 15.13. Given a Lebesgue measurable set $E \subseteq \mathbb{R}^N$ and a Lebesgue measurable function $u : E \rightarrow [-\infty, \infty]$, we can define the *decreasing rearrangement* u^* of u to be the decreasing rearrangement of $|u|$, that is, $u^* := (|u|)^*$. In turn, we can define the *spherically symmetric rearrangement* $u^\sharp$ of u to be the spherically symmetric rearrangement of u .

15.2. Lorentz Spaces

In this section we introduce the Lorentz spaces $L^{p,q}$. We will see in Section 16.1 of Chapter 16 that $L^{p,q}$ is an interpolation space between L^1 and L^∞ .

Definition 15.14. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $0 < p < \infty$ and $0 < q \leq \infty$. The *Lorentz space* $L^{p,q}(E)$ is defined as the space of all measurable functions $u : E \rightarrow \mathbb{R}$ such that

$$(15.8) \quad \|u\|_{L^{p,q}(E)} := \left(\int_0^\infty (t^{1/p} u^*(t))^q \frac{dt}{t} \right)^{1/q} < \infty$$

if $q < \infty$ and

$$(15.9) \quad \|u\|_{L^{p,\infty}(E)} := \sup\{t^{1/p} u^*(t) : t > 0\} < \infty$$

if $q = \infty$, where u^* is defined in Remark 15.13.

In view of Proposition 15.1, we can identify functions in $L^{p,q}(E)$ that coincide up to a set of measure zero. Observe that

$$L^p(E) = L^{p,p}(E).$$

Indeed, if $p < \infty$, then by Theorem 15.10,

$$\|u\|_{L^{p,p}(E)} = \|u^*\|_{L^p([0,\infty))} = \|u^*\|_{L^p(E^\sharp)} = \|u\|_{L^p(E)}.$$

On the other hand, if $p = \infty$, then since u^* is decreasing by Proposition 15.2,

$$\|u\|_{L^{\infty,\infty}(E)} = \sup\{u^*(t) : t > 0\} = u^*(0) = \operatorname{ess\,sup}_E |u|.$$

Exercise 15.15. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set. Prove that if $q < \infty$, $L^{\infty,q}(E) = \{0\}$.

Proposition 15.16. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $0 < p < \infty$ and $0 < q \leq \infty$. Then $L^{p,q}(E)$ is a quasi-Banach space.

Proof. We only prove that $\|\cdot\|_{L^{p,q}}$ is a quasi-norm and leave completeness as an exercise. Let $u, v \in L^{p,q}(E)$ and let $t > 0$. We claim that

$$(15.10) \quad (u + v)^*(t) \leq u^*(t/2) + v^*(t/2).$$

To see this, consider the three sets

$$E_{u+v} := \{s \in [0, \infty) : \varrho_{|u+v|}(s) \leq t\},$$

$$E_u := \{s \in [0, \infty) : \varrho_{|u|}(s) \leq t/2\}, \quad E_v := \{s \in [0, \infty) : \varrho_{|v|}(s) \leq t/2\}.$$

Let $s_1 \in E_u$ and $s_2 \in E_v$. Then $\varrho_{|u|}(s_1) = \mu(\{|u| > s_1\}) \leq t/2$ and $\varrho_{|v|}(s_2) = \mu(\{|v| > s_2\}) \leq t/2$. But since

$$\{|u+v| > s_1 + s_2\} \subseteq \{|u| > s_1\} \cup \{|v| > s_2\},$$

it follows that

$$\varrho_{|u+v|}(s_1 + s_2) = \mu(\{|u+v| > s_1 + s_2\}) \leq \varrho_{|u|}(s_1) + \varrho_{|v|}(s_2) \leq t.$$

Hence, $s_1 + s_2 \in E_{u+v}$, and so

$$(u+v)^*(t) = \inf E_{u+v} \leq s_1 + s_2.$$

By taking first the infimum over all $s_1 \in E_u$ and then the infimum over all $s_2 \in E_v$, it follows that

$$(u+v)^*(t) \leq \inf E_u + \inf E_v = u^*(t/2) + v^*(t/2).$$

Multiplying both sides by $t^{1/p}$ we get

$$t^{1/p}(u+v)^*(t) \leq t^{1/p}u^*(t/2) + t^{1/p}v^*(t/2).$$

If $q \geq 1$, we can apply Minkowski's inequality with respect to the measure $\frac{1}{t} dt$ to obtain

$$\begin{aligned} \left(\int_0^\infty (t^{1/p}(u+v)^*(t))^q \frac{dt}{t} \right)^{1/q} &\leq \left(\int_0^\infty (t^{1/p}u^*(t/2) + t^{1/p}v^*(t/2))^q \frac{dt}{t} \right)^{1/q} \\ &\leq \left(\int_0^\infty (t^{1/p}u^*(t/2))^q \frac{dt}{t} \right)^{1/q} + \left(\int_0^\infty (t^{1/p}v^*(t/2))^q \frac{dt}{t} \right)^{1/q}. \end{aligned}$$

To obtain the triangle inequality we use the change of variables $s = t/2$.

On the other hand, if $q \leq 1$, then

$$\begin{aligned} \int_0^\infty (t^{1/p}(u+v)^*(t))^q \frac{dt}{t} &\leq \int_0^\infty (t^{1/p}u^*(t/2) + t^{1/p}v^*(t/2))^q \frac{dt}{t} \\ &\leq \int_0^\infty (t^{1/p}u^*(t/2))^q \frac{dt}{t} + \int_0^\infty (t^{1/p}v^*(t/2))^q \frac{dt}{t}. \end{aligned}$$

We now make the change of variables $s = t/2$, raise everything to power $1/q \geq 1$, and then use the inequality $(a+b)^{1/q} \leq 2^{1/q-1}a^{1/q} + 2^{1/q-1}b^{1/q}$. $\square$

Exercise 15.17. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u : E \rightarrow \mathbb{R}$ and $v : E \rightarrow \mathbb{R}$ be two Lebesgue measurable functions. Prove that for every $0 < \varepsilon < 1$ and every $t > 0$,

$$(u+v)^*(t) \leq u^*((1-\varepsilon)t) + v^*(\varepsilon t).$$

Remark 15.18. We will see in Section 16.1 of Chapter 16 that for $1 < p < \infty$ and $1 \leq q \leq \infty$, $L^{p,q}(E)$ is a Banach space.

Exercise 15.19. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $0 < p \leq \infty$ and $0 < q \leq r \leq \infty$. Prove that $L^{p,q}(E) \subseteq L^{p,r}(E)$, with $\|u\|_{L^{p,r}(E)} \leq c\|u\|_{L^{p,q}(E)}$ for all $u \in L^{p,q}(E)$ and for some constant $c = c(p, q, r) > 0$. In particular, $L^p(E) \subseteq L^{p,r}(E) \subseteq L^{p,\infty}(E)$ for all $p \leq r \leq \infty$.

Exercise 15.20. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $0 < p < \infty$. The *weak space* $L_w^p(E)$ is defined as the space of all measurable functions $u : E \rightarrow \mathbb{R}$ such that

$$\|u\|_{L_w^p(E)} := \sup\{s(\varrho_u(s))^{1/p} : s > 0\} < \infty.$$

Prove that $L_w^p(E) = L^{p,\infty}(E)$.

Exercise 15.21. Given $a > 0$, let $u(x) := \|x\|^{-a}$, where $x \in \mathbb{R}^N \setminus \{0\}$. Find for which p and q (if any) $u \in L^{p,q}(\mathbb{R}^N)$.

Exercise 15.22 (Hölder inequality). Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $1 \leq p, q \leq \infty$. Prove that if $u \in L^{p,q}(E)$ and $v \in L^{p',q'}(E)$, then $uv \in L^1(E)$ and

$$\int_E |u(x)v(x)| dx \leq \|u\|_{L^{p,q}(E)} \|v\|_{L^{p',q'}(E)}.$$

15.3. Symmetrization of $W^{1,p}$ and BV Functions

In this section we prove that if $u \in W^{1,p}(\mathbb{R}^N)$ is nonnegative, $1 \leq p \leq \infty$, then $u^\# \in W^{1,p}(\mathbb{R}^N)$ and

$$(15.11) \quad \|u^\#\|_{L^p(\mathbb{R}^N)} = \|u\|_{L^p(\mathbb{R}^N)}, \quad \|\nabla u^\#\|_{L^p(\mathbb{R}^N)} \leq \|\nabla u\|_{L^p(\mathbb{R}^N)}.$$

Theorem 15.23. Let $u \in W^{1,p}(\mathbb{R}^N)$, $1 \leq p \leq \infty$, be a nonnegative function. If $p = \infty$ assume also that u vanishes at infinity. Then for all $0 < a < b$ we have

$$\alpha_N^{1/N} (u^*(a) - u^*(b)) a^{(N-1)/N} \leq \int_{\{u^*(b) < u < u^*(a)\}} \|\nabla u\| dx.$$

In particular, u^* is locally absolutely continuous in $(0, \infty)$.

Exercise 15.24. Let $u \in W^{1,1}(\mathbb{R}^N)$.

- (i) Prove that if $N = 1$, then $u(x) = \int_0^\infty u'(x-t) dt$ for $\mathcal{L}^1$ -a.e. $x \in \mathbb{R}$.
- (ii) Prove that if $N \geq 2$, then for every $z \in S^{N-1}$ and for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$,

$$u(x) = \int_0^\infty \nabla u(x-tz) \cdot z dt.$$

- (iii) Prove that if $N \geq 2$, then for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$,

$$u(x) = \frac{1}{\beta_N} \sum_{i=1}^N \int_{\mathbb{R}^N} \frac{\partial u}{\partial x_i}(x-y) \frac{y_i}{\|y\|^N} dy.$$

We begin with a preliminary result that is of interest in and of itself.

Proposition 15.25. *Let $u \in W^{1,1}(\mathbb{R}^N)$, $N \geq 2$, be a nonnegative function. Then*

$$\alpha_N^{1/N} \sup_{s>0} s[\mathcal{L}^N(\{u \geq s\})]^{(N-1)/N} \leq \int_{\mathbb{R}^N} \|\nabla u\| dx.$$

Proof. Step 1: We claim that for every Lebesgue measurable set E with finite measure and every $y \in \mathbb{R}^N$,

$$\int_E \|x - y\|^{1-N} dx \leq N\alpha_N^{1-1/N} (\mathcal{L}^N(E))^{1/N}.$$

To see this, let $r > 0$ be such that $\mathcal{L}^N(E) = \mathcal{L}^N(B) = \alpha_N r^N$, where $B := B(y, r)$. Then $\mathcal{L}^N(B \setminus E) = \mathcal{L}^N(E \setminus B)$ and so

$$\int_{E \setminus B} \|x - y\|^{1-N} dx \leq \int_{E \setminus B} r^{1-N} dx = \int_{B \setminus E} r^{1-N} dx \leq \int_{B \setminus E} \|x - y\|^{1-N} dx.$$

Adding $\int_{E \cap B} \|x - y\|^{1-N} dx$ to both sides and using spherical coordinates and the fact that $\mathcal{L}^N(E) = \mathcal{L}^N(B)$ gives

$$\int_E \|x - y\|^{1-N} dx \leq \int_B \|x - y\|^{1-N} dx = N\alpha_N r = N\alpha_N^{1-1/N} (\mathcal{L}^N(E))^{1/N},$$

which proves the claim.

Step 2: By the previous exercise and a change of variables, for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$,

$$|u(x)| \leq \frac{1}{N\alpha_N} \int_{\mathbb{R}^N} \|\nabla u(y)\| \frac{1}{\|x - y\|^{N-1}} dy.$$

Let $K \subseteq \{u \geq s\}$ be a compact set. Integrating the previous inequality over K and using Tonelli's theorem yields

$$s\mathcal{L}^N(K) \leq \int_K |u(x)| dx \leq \frac{1}{N\alpha_N} \int_{\mathbb{R}^N} \|\nabla u(y)\| \int_K \frac{1}{\|x - y\|^{N-1}} dx dy,$$

which, together with the previous step, gives

$$s\mathcal{L}^N(K) \leq \alpha_N^{-1/N} (\mathcal{L}^N(K))^{1/N} \int_{\mathbb{R}^N} \|\nabla u(y)\| dy.$$

Since $\mathcal{L}^N(K) < \infty$,

$$s(\mathcal{L}^N(K))^{1-1/N} \leq \alpha_N^{-1/N} \int_{\mathbb{R}^N} \|\nabla u(y)\| dy.$$

Since this holds for every compact set $K \subseteq \{u \geq s\}$, by standard properties of the Lebesgue measure we get that the previous inequality holds with E_s in place of K . It now suffices to take the supremum over all $s > 0$. $\square$

Exercise 15.26. Prove Step 1 of the previous theorem using the Hardy–Littlewood inequality (see Theorem 15.9).

We are now ready to prove Theorem 15.23.

Proof of Theorem 15.23. Step 1: Let $0 \leq s_1 < s_2$ be such that $\mathcal{L}^N(\{s_1 < u < s_2\}) < \infty$. For $0 \leq s_1 < s_2$ define

$$f(s) := \begin{cases} 0 & \text{if } s \leq s_1, \\ s - s_1 & \text{if } s_1 < s < s_2, \\ s_2 - s_1 & \text{if } s \geq s_2. \end{cases}$$

Since $L^p(E) \subset L^1(E)$ for sets of finite measure, $\mathcal{L}^N(\{s_1 < u < s_2\}) < \infty$, $\mathcal{L}^N(\{u \geq s_2\}) < \infty$, and f is a Lipschitz continuous function, it follows by the chain rule (see Exercise 11.51) that the function $v := f \circ u$ belongs to $W^{1,1}(\mathbb{R}^N)$ with

$$\partial_i v(x) = f'(u(x)) \partial_i u(x) = \begin{cases} \partial_i u(x) & \text{if } s_1 < u(x) < s_2, \\ 0 & \text{otherwise,} \end{cases}$$

for all $i = 1, \dots, N$ and for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. Applying the previous proposition to v , we obtain

$$\alpha_N^{1/N} \sup_{s>0} s [\mathcal{L}^N(\{v \geq s\})]^{(N-1)/N} \leq \int_{\mathbb{R}^N} \|\nabla v\| dx = \int_{\{s_1 < u < s_2\}} \|\nabla u\| dx.$$

Since

$$\sup_{s>0} s [\mathcal{L}^N(\{v \geq s\})]^{(N-1)/N} \geq (s_2 - s_1) [\mathcal{L}^N(\{u \geq s_2\})]^{(N-1)/N},$$

we get that

$$(15.12) \quad \alpha_N^{1/N} (s_2 - s_1) [\zeta_u(s_2)]^{(N-1)/N} \leq \int_{\{s_1 < u < s_2\}} \|\nabla u\| dx.$$

Let $0 < a < b$. By Proposition 15.2,

$$(15.13) \quad \mathcal{L}^N(\{u^*(b) < u < u^*(a)\}) = [\varrho_u(u^*(b)) - \varsigma(u^*(a))] \leq b - a,$$

and so we are in a position to take $s_1 := u^*(b)$, $s_2 := u^*(a)$ in (15.12). Then by Proposition 15.2(iii) and (15.12),

$$(15.14) \quad \begin{aligned} & \alpha_N^{1/N} (u^*(a) - u^*(b)) a^{(N-1)/N} \\ & \leq \alpha_N^{1/N} (u^*(a) - u^*(b)) [\varsigma(u^*(a))]^{(N-1)/N} \\ & \leq \int_{\{u^*(b) < u < u^*(a)\}} \|\nabla u\| dx. \end{aligned}$$

Step 2: We show that the function u^* is absolutely continuous on closed subintervals of $(0, \infty)$. Let $[a, b] \subset (0, \infty)$ and fix $\varepsilon > 0$. Since $\nabla u \in L^p(\mathbb{R}^N; \mathbb{R}^N)$, we may find $\delta > 0$ such that

$$(15.15) \quad \int_E \|\nabla u\| dx \leq \varepsilon \alpha_N^{1/N} a^{(N-1)/N}$$

for every Lebesgue measurable set $E \subset \mathbb{R}^N$ with $\mathcal{L}^N(E) \leq \delta$. Consider any finite number of nonoverlapping intervals (a_k, b_k) , $k = 1, \dots, \ell$, with $[a_k, b_k] \subseteq [a, b]$ and $\sum_{k=1}^{\ell} (b_k - a_k) \leq \delta$. By (15.13),

$$\begin{aligned} & \mathcal{L}^N \left(\bigcup_{k=1}^{\ell} \{u^*(b_k) < u < u^*(a_k)\} \right) \\ & \leq \sum_{k=1}^{\ell} \mathcal{L}^N(\{u^*(b_k) < u < u^*(a_k)\}) \leq \sum_{k=1}^{\ell} (b_k - a_k) \leq \delta, \end{aligned}$$

and so, by (15.14) and (15.15),

$$\begin{aligned} \sum_{k=1}^{\ell} |u^*(b_k) - u^*(a_k)| & \leq \alpha_N^{-1/N} a^{-(N-1)/N} \\ & \times \int_{\bigcup_{k=1}^{\ell} \{u^*(b_k) < u < u^*(a_k)\}} \|\nabla u\| dx \leq \varepsilon. \end{aligned}$$

This proves the absolute continuity of u^* . $\square$

Remark 15.27. If $N = 1$, the local absolute continuity of u^* follows from Step 1 of Theorem 4.22.

Remark 15.28. The proof of Theorem 15.23 uses only elementary tools but the constant is not sharp. By the coarea formula (Theorem 14.28) and the (sharp) isoperimetric inequality (Theorem C.12),

$$\begin{aligned} \int_{\{u^*(b) < u < u^*(a)\}} \|\nabla u\| dx & = \int_{u^*(b)}^{u^*(a)} P(\{u > s\}) ds \geq N \alpha_N^{1/N} \\ & \times \int_{u^*(b)}^{u^*(a)} [\mathcal{L}^N(\{u \geq s\})]^{(N-1)/N} ds \\ & \geq N \alpha_N^{1/N} (u^*(a) - u^*(b)) a^{(N-1)/N}. \end{aligned}$$

We are now ready to prove (15.11) for $1 \leq p < \infty$.

Theorem 15.29. *Let $u \in W^{1,p}(\mathbb{R}^N)$, $1 \leq p \leq \infty$, be a nonnegative function. If $p = \infty$ assume also that u vanishes at infinity. Then $u^\sharp \in W^{1,p}(\mathbb{R}^N)$ and (15.11) holds.*

Proof. The equality (15.11)₁ follows from Theorem 15.10. It remains to prove the inequality (15.11)₂.

Step 1: Assume that $1 \leq p < \infty$ and let $u \in W^{1,p}(\mathbb{R}^N)$. Then by Theorem 15.23 the function u^* is locally absolutely continuous in $(0, \infty)$. In turn, the function $u^\sharp(y) = u^*(\alpha_N \|y\|^N)$, $y \in \mathbb{R}^N$, belongs to $W_{\text{loc}}^{1,1}(\mathbb{R}^N)$ with

$$(15.16) \quad \partial_i u^\sharp(y) = (u^*)'(\alpha_N \|y\|^N) N \alpha_N \|y\|^{N-2} y_i$$

for $\mathcal{L}^N$ -a.e. $y \in \mathbb{R}^N$ (why?). Using spherical coordinates, the fact that $\beta_N = N\alpha_N$, and a change of variables, we have that

$$(15.17) \quad \begin{aligned} \int_{\mathbb{R}^N} \|\nabla u^\sharp(y)\|^p dy &= (N\alpha_N)^{p+1} \int_0^\infty |(u^*)'(\alpha_N r^N)|^p r^{(N-1)(p+1)} dr \\ &= N^p \alpha_N^{p/N} \int_0^\infty |(u^*)'(t)|^p t^{p(N-1)/N} dt. \end{aligned}$$

Since the function u^* is decreasing (see Proposition 15.2), the function

$$g(t) := \int_{\{u > u^*(t)\}} \|\nabla u\|^p dx, \quad t \geq 0,$$

is increasing. Hence by Lebesgue's differentiation theorem (see Theorem 1.21) g is differentiable $\mathcal{L}^1$ -a.e. in $(0, \infty)$. We claim that

$$(15.18) \quad N^p \alpha_N^{p/N} |(u^*)'(t)|^p t^{p(N-1)/N} \leq g'(t)$$

for $\mathcal{L}^1$ -a.e. $t \in (0, \infty)$. The proof follows closely the one of Theorem 4.22 and we indicate only the main changes.

Since u^* is decreasing by Proposition 15.2 and g is increasing, it follows from Lebesgue's differentiation theorem that u^* and g are differentiable at $\mathcal{L}^1$ -a.e. $t \in (0, \infty)$. Fix a point $t_0 \in (0, \infty)$ at which both u^* and g are differentiable. If $(u^*)'(t_0) = 0$, then (15.18) holds for t_0 because $g'(t_0) \geq 0$. Thus assume $(u^*)'(t_0) < 0$. Assume also that t_0 is a Lebesgue point of $(u^*)'$. As in the proof of Theorem 4.22 we can show that there exists a sequence $h_n \rightarrow 0^+$ such that

$$(15.19) \quad \mathcal{L}^N(\{u^*(t_0 + h_n) < u \leq u^*(t_0)\}) = h_n$$

for all n . In turn, by Jensen's inequality (see Theorem B.50) and Remark 15.28, for all $h_n > 0$ sufficiently small,

$$\begin{aligned} N\alpha_N^{1/N} t_0^{(N-1)/N} \frac{u^*(t_0) - u^*(t_0 + h_n)}{h_n} &\leq \frac{1}{h_n} \int_{\{u^*(t_0 + h_n) < u \leq u^*(t_0)\}} \|\nabla u\| dx \\ &\leq \left(\frac{g(t_0 + h_n) - g(t_0)}{h_n} \right)^{1/p}. \end{aligned}$$

Letting $n \rightarrow \infty$ and using the fact that both u^* and g are differentiable at t_0 gives (15.18) at t_0 .

This shows that the claim (15.18) holds. By integrating (15.18) over $[a, b] \subset (0, \infty)$ and using Corollary 1.25 we get

$$\begin{aligned} N^p \alpha_N^{p/N} \int_a^b |(u^*)'(t)|^p t^{p(N-1)/N} dt &\leq \int_a^b g'(t) dt \\ &\leq g(b) - g(a) \leq \int_{\mathbb{R}^N} \|\nabla u(x)\|^p dx. \end{aligned}$$

To obtain (15.11)₂, it now suffices to let $a \rightarrow 0^+$ and $b \rightarrow \infty$ and use (15.17).

Step 2: Assume that $u \in W^{1,\infty}(\mathbb{R}^N)$ is a nonnegative function vanishing at infinity. Let t_0 be as in the previous step. By Remark 15.28 and (15.19), for all $h > 0$ sufficiently small

$$\begin{aligned} N\alpha_N^{1/N} t_0^{(N-1)/N} \frac{u^*(t_0) - u^*(t_0 + h_n)}{h_n} &\leq \frac{1}{h_n} \int_{\{u^*(t_0 + h_n) < u < u^*(t_0)\}} \|\nabla u\| \, dx \\ &\leq \|\nabla u\|_{L^\infty}. \end{aligned}$$

Letting $n \rightarrow \infty$ and using the fact that u^* is differentiable at t_0 gives

$$N\alpha_N^{1/N} |(u^*)'(t)| t^{(N-1)/N} \leq \|\nabla u\|_{L^\infty}$$

for $\mathcal{L}^1$ -a.e. $t \in (0, \infty)$. In turn, taking $t = \alpha_N \|y\|^N$, we get

$$N\alpha_N |(u^*)'(\alpha_N \|y\|^N)| \|y\|^{(N-1)} \leq \|\nabla u\|_{L^\infty}$$

for $\mathcal{L}^N$ -a.e. $y \in \mathbb{R}^N$, which, in view of (15.16), shows that (15.11)₂ holds for $p = \infty$. $\square$

Remark 15.30. Note that (15.19) is only needed for $p > 1$ in order to apply Jensen's inequality (see Theorem B.50).

Exercise 15.31. Let $u \in W^{1,\infty}(\mathbb{R}^N)$ be a nonnegative function vanishing at infinity and let $L := \|\nabla u\|_{L^\infty}$.

- (i) Let $y_1, y_2 \in \mathbb{R}^N$, with $\|y_1\| < \|y_2\|$, and define $\eta := u^\sharp(y_1) - u^\sharp(y_2) \geq 0$. Prove that

$$u^{-1}([u^\sharp(y_1), \infty)) + B(0, \eta/L) \subseteq u^{-1}((u^\sharp(y_2), \infty)).$$

- (ii) Use the Brunn–Minkowski inequality to prove that

$$\begin{aligned} (\mathcal{L}^N(u^{-1}([u^\sharp(y_1), \infty))))^{1/N} + (\mathcal{L}^N(B(0, \eta/L)))^{1/N} \\ \leq (\mathcal{L}^N(u^{-1}((u^\sharp(y_2), \infty))))^{1/N}. \end{aligned}$$

- (iii) Prove that $0 \leq u^\sharp(y_1) - u^\sharp(y_2) \leq L\|y_2 - y_1\|$.

Next we consider the case $u \in BV(\mathbb{R}^N)$.

Theorem 15.32. Let $u \in BV(\mathbb{R}^N)$ be a nonnegative function. Then $u^\sharp \in BV(\mathbb{R}^N)$ and $\|u^\sharp\|_{L^1(\mathbb{R}^N)} = \|u\|_{L^1(\mathbb{R}^N)}$, $\|Du^\sharp\|(\mathbb{R}^N) \leq \|Du\|(\mathbb{R}^N)$.

Proof. Let $u_\varepsilon := u * \varphi_\varepsilon$, where φ is a standard mollifier. By Remark 14.15, $u_\varepsilon \rightarrow u$ in $L^1(\mathbb{R}^N)$ and $\|\nabla u_\varepsilon\|_{L^1} \rightarrow \|Du\|(\mathbb{R}^N)$. Then by Theorem 15.29, $u_\varepsilon^\sharp \in W^{1,1}(\mathbb{R}^N)$ and (15.11) holds for u_ε . On the other hand, since $u_n^\sharp \rightarrow u^\sharp$ in $L^1(\mathbb{R}^N)$ by Theorem 15.11, it follows by Exercise 14.3 and by extracting a subsequence that $u^\sharp \in BV(\mathbb{R}^N)$ with

$$\|Du^\sharp\|(\mathbb{R}^N) \leq \liminf_{\varepsilon \rightarrow 0^+} \|\nabla u_\varepsilon^\sharp\|_{L^1} \leq \liminf_{\varepsilon \rightarrow 0^+} \|\nabla u_\varepsilon\|_{L^1} = \|Du\|(\mathbb{R}^N).$$

This concludes the proof. $\square$

15.4. Sobolev Embeddings Revisited

As an application of the decreasing rearrangement, we prove a sharper version of the Sobolev–Gagliardo–Nirenberg embedding theorem (Theorem 12.4). We recall that for $1 \leq p < N$, the Sobolev critical exponent is given by $p^* = pN/(N-p)$ and that by Exercise 15.19, $L^{p^*,p}(\mathbb{R}^N) \subseteq L^{p^*}(\mathbb{R}^N)$. One can show that the inclusion is strict. Thus the following theorem improves upon Theorem 12.4.

Theorem 15.33 (Sobolev–Gagliardo–Nirenberg embedding). *Let $1 \leq p < N$. Then there exists a constant $c = c(N, p) > 0$ such that for every function $u \in \dot{W}^{1,p}(\mathbb{R}^N)$ vanishing at infinity,*

$$\|u\|_{L^{p^*,p}(\mathbb{R}^N)} \leq c \|\nabla u\|_{L^p(\mathbb{R}^N)}.$$

Proof. Step 1: Assume that $\ell := \mathcal{L}^N(\{|u| > 0\}) < \infty$. Let $t_n := 2^{1-n}\ell$. We claim that

$$\sum_{n=1}^{\infty} t_n^{p/p^*} (u^*(t_{n+1}) - u^*(t_n))^p \leq c \|\nabla u\|_{L^p}^p,$$

where $c > 0$.

Let $v \in W^{1,1}(\mathbb{R}^N) \cap \dot{W}^{1,p}(\mathbb{R}^N)$ be a nonnegative function with $\mathcal{L}^N(\{v > 0\}) < \infty$. If $p > 1$, by Holder's inequality and Proposition 15.25, for $s > 0$,

$$(15.20) \quad \begin{aligned} \alpha_N^{1/N} s [\mathcal{L}^N(\{v \geq s\})]^{(N-1)/N} &\leq \|\nabla v\|_{L^1(\{v>0\})} \\ &\leq \|\nabla v\|_{L^p(\{v>0\})} [\mathcal{L}^N(\{v > 0\})]^{1/p'}. \end{aligned}$$

If $v \in W^{1,1}(\mathbb{R}^N)$ and $p = 1$, we only consider the first inequality in (15.20).

Let $0 < s_1 < s_2$ and for $0 < s_1 < s_2$ define

$$f(s) := \begin{cases} 0 & \text{if } |s| \leq s_1, \\ |s| - s_1 & \text{if } s_1 < |s| < s_2, \\ s_2 - s_1 & \text{if } |s| \geq s_2. \end{cases}$$

Since f is a Lipschitz continuous function, it follows by the chain rule (see Exercise 11.51) and the fact that u vanishes at infinity that the function $v := f \circ u$ belongs to $W^{1,1}(\mathbb{R}^N) \cap \dot{W}^{1,p}(\mathbb{R}^N)$ (see Step 4 of the proof of Theorem 12.4 for more details) with

$$\|\nabla v(x)\| = \begin{cases} \|\nabla u(x)\| & \text{if } s_1 < |u(x)| < s_2, \\ 0 & \text{otherwise.} \end{cases}$$

Hence, by applying (15.20) with $s = s_2 - s_1$ we get

$$\begin{aligned} \alpha_N^{1/N} (s_2 - s_1) [\mathcal{L}^N(\{u \geq s_2\})]^{(N-1)/N} \\ \leq \|\nabla u\|_{L^p(\{s_1 < |u| < s_2\})} [\mathcal{L}^N(\{|u| > s_1\})]^{1/p'}, \end{aligned}$$

where $[\mathcal{L}^N(\{|u| > s_1\})]^{1/p'} := 1$ if $p = 1$. Using the fact that

$$\mathcal{L}^N(\{|u| > u^*(t_n)\}) \leq t_n \leq \mathcal{L}^N(\{|u| \geq u^*(t_n)\})$$

(see Proposition 15.2(iii) and (iv)) and the previous inequality with $s_1 = u^*(t_n)$ and $s_2 = u^*(t_{n+1})$, we get

$$\alpha_N^{1/N} (u^*(t_{n+1}) - u^*(t_n)) t_{n+1}^{(N-1)/N} \leq \|\nabla u\|_{L^p(\{u^*(t_n) < |u| < u^*(t_{n+1})\})} t_n^{1/p'}.$$

Since $2t_{n+1} = t_n$ and $(N-1)/N - 1/p' = 1/p^*$ we have that

$$\alpha_N^{1/N} 2^{(N-1)/N} (u^*(t_{n+1}) - u^*(t_n)) t_n^{1/p^*} \leq \|\nabla u\|_{L^p(\{u^*(t_n) < |u| < u^*(t_{n+1})\})}.$$

Raising both sides to the power p and summing over n proves the claim.

Step 2: Assume that $\mathcal{L}^N(\{x \in \mathbb{R}^N : |u(x)| > 0\}) < \infty$ and that $u \in W^{1,p}(\mathbb{R}^N) \cap L^\infty(\mathbb{R}^N)$. Given $\varepsilon > 0$ by the convexity of $t \mapsto t^p$ we have that

$$b^p \leq (1 + \varepsilon^{-1})^{p-1} (b - a)^p + (1 + \varepsilon)^{p-1} a^p$$

for $0 \leq a \leq b$. Hence, using also the facts that $u^*(t_1) = 0$ and that $2t_{n+1} = t_n$,

$$\begin{aligned} 2^{p/p^*} \sum_{k=1}^{\infty} t_{k+1}^{p/p^*} (u^*(t_k))^p &= \sum_{n=1}^{\infty} t_{n+1}^{p/p^*} (u^*(t_{n+1}))^p \\ &\leq (1 + \varepsilon^{-1})^{p-1} \sum_{n=1}^{\infty} t_{n+1}^{p/p^*} (u^*(t_{n+1}) - u^*(t_n))^p \\ &\quad + (1 + \varepsilon)^{p-1} \sum_{n=1}^{\infty} t_{n+1}^{p/p^*} (u^*(t_n))^p. \end{aligned}$$

Since $u^*(t_n) \leq \|u\|_{L^\infty}$ and $2t_{n+1} = t_n$, the series $\sum_{n=1}^{\infty} t_{n+1}^{p/p^*} (u^*(t_n))^p$ converges and so, by taking ε so that $(1 + \varepsilon)^{p-1} < 2^{p/p^* - 1}$, from the previous inequality we obtain

$$\begin{aligned} c_1 \sum_{n=1}^{\infty} t_{n+1}^{p/p^*} (u^*(t_n))^p &\leq (1 + \varepsilon^{-1})^{p-1} \sum_{n=1}^{\infty} t_{n+1}^{p/p^*} (u^*(t_{n+1}) - u^*(t_n))^p \\ &= (1 + \varepsilon^{-1})^{p-1} 2^{p/p^*} \sum_{n=1}^{\infty} t_n^{p/p^*} (u^*(t_{n+1}) - u^*(t_n))^p \\ &\leq c \|\nabla u\|_{L^p}^p, \end{aligned}$$

where in the last inequality we used Step 1. Recalling that u^* is decreasing, it follows from (15.8) and the previous inequality that

$$\begin{aligned} \int_0^\infty t^{p/p^*} (u^*(t))^p \frac{dt}{t} &= \sum_{k=1}^{\infty} \int_{t_{k+1}}^{t_k} t^{p/p^*} (u^*(t))^p \frac{dt}{t} \leq 2 \sum_{k=1}^{\infty} t_k^{p/p^*} (u^*(t_{k+1}))^p \\ &\leq c \|\nabla u\|_{L^p}^p, \end{aligned}$$

where we used once more the fact that $2t_{n+1} = t_n$.

Step 3: To remove the extra hypotheses that $\mathcal{L}^N(\{|u| > 0\}) < \infty$ and that $u \in L^\infty(\mathbb{R}^N)$, we use a truncation argument. We leave this as an exercise. $\square$

The next exercise shows how in the critical case $p = N/m$, if $\nabla^m u$ belongs to the slightly smaller Lorentz space $L^{N/m,1}(\mathbb{R}^N; \mathbb{R}^{M_m})$ rather than the larger space $L^{N/m}(\mathbb{R}^N; \mathbb{R}^{M_m})$, then the embeddings in Section 12.2 of Chapter 12 can be improved.

Exercise 15.34. Let $m \in \mathbb{N}$ and $N \geq 2$. Prove that there exists a constant $c = c(m, N) > 0$ such that for all $u \in C_c^\infty(\mathbb{R}^N)$ with $\nabla^m u \in L^{N/m,1}(\mathbb{R}^N; \mathbb{R}^{M_m})$,

$$\|u\|_{C_b(\mathbb{R}^N)} \leq c \|\nabla^m u\|_{L^{N/m,1}(\mathbb{R}^N)}.$$

Hint: Use Lemma 12.10 and Exercise 15.22.

Next we can give the sharp version of the embedding Theorem 12.40 for $m = 1$. We recall that for every $\ell \in \mathbb{N}$, $\exp_\ell(s) := \sum_{n=\ell-1}^\infty \frac{1}{n!} s^n$, $s \in \mathbb{R}$, (see (12.26)) and that $\gamma_N := N\beta_N^{1/(N-1)}$ (see (12.29)), where β_N is the surface measure of the unit sphere.

Theorem 15.35. Suppose $N \geq 2$. Then for every $\gamma \in (0, \gamma_N)$ there exists a constant $c = c(N, \gamma) > 0$ such that

$$(15.21) \quad \int_{\mathbb{R}^N} \exp_N\left(\gamma \frac{|u(x)|^{N'}}{\|\nabla u\|_{L^N(\mathbb{R}^N; \mathbb{R}^N)}^{N'}}\right) dx \leq c \frac{\|u\|_{L^N(\mathbb{R}^N)}^N}{\|\nabla u\|_{L^N(\mathbb{R}^N; \mathbb{R}^N)}^N}$$

for all $u \in W^{1,N}(\mathbb{R}^N) \setminus \{0\}$. In particular, if $\Phi_\gamma(s) := \exp_N(\gamma s^{N'})$, $s \geq 0$, then

$$(15.22) \quad \|u\|_{\Phi_\gamma} \leq c \|\nabla u\|_{L^N(\mathbb{R}^N; \mathbb{R}^N)}$$

for all $u \in W^{1,N}(\mathbb{R}^N)$.

Proof. Fix $u \in W^{1,N}(\mathbb{R}^N) \setminus \{0\}$ and define $v(x) := |u(x)|/\|\nabla u\|_{L^N}$, $x \in \mathbb{R}^N$. Then $\|\nabla v\|_{L^N} = 1$, and (15.21) reduces to

$$(15.23) \quad \int_{\mathbb{R}^N} \exp_N(\gamma(v(x))^{N'}) dx \leq c \|v\|_{L^N}^N.$$

Let v^* be the spherically symmetric rearrangement of v . Then $v^*(x) = w(\|x\|)$, $x \in \mathbb{R}^N$, where w is nonnegative, decreasing, and locally absolutely continuous by Theorem 15.29. Hence, $\nabla v^*(x) = w'(\|x\|)x/\|x\|$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. Using spherical coordinates and Theorem 15.29, it follows that

$$(15.24) \quad \beta_N \int_0^\infty |w'(r)|^N r^{N-1} dr = \|\nabla v^*\|_{L^N}^N \leq \|\nabla v\|_{L^N}^N = 1.$$

Define

$$(15.25) \quad r_0 := \inf\{r \geq 0 : w(r) \leq 1\}.$$

Since $w(r) \rightarrow 0$ as $r \rightarrow \infty$, we have that r_0 must be finite.

Using spherical coordinates and Theorem 15.10, we have that

$$(15.26) \quad \begin{aligned} \int_{\mathbb{R}^N} \exp_N(\gamma(v(x))^{N'}) dx &= \int_{\mathbb{R}^N} \exp_N(\gamma(v^*(x))^{N'}) dx \\ &= \beta_N \int_0^\infty \exp_N(\gamma(w(r))^{N'}) r^{N-1} dr \\ &= \beta_N \int_0^{r_0} \exp_N(\gamma(w(r))^{N'}) r^{N-1} dr \\ &\quad + \beta_N \int_{r_0}^\infty \exp_N(\gamma(w(r))^{N'}) r^{N-1} dr =: I + II. \end{aligned}$$

To estimate I , it is enough to consider the case $r_0 > 0$, so that $w(r_0) = 1$ by (15.25). Since w is locally absolutely continuous, by the fundamental theorem of calculus (see Theorem 3.20), Hölder's inequality, (15.24), and (15.25), for $0 < r < r_0$ we have that

$$\begin{aligned} w(r) &= w(r_0) - \int_r^{r_0} w'(\tau) d\tau \leq 1 + \int_r^{r_0} |w'(\tau)| \frac{\tau^{N'}}{\tau^{N'}} d\tau \\ &\leq 1 + \left(\int_0^\infty |w'(\tau)|^N \tau^{N-1} d\tau \right)^{1/N} (\log(r_0/r))^{1/N'} \\ &\leq 1 + \beta_N^{-\frac{1}{N}} (\log(r_0/r))^{1/N'}. \end{aligned}$$

By the convexity of the function $s \mapsto s^{N'}$, for every $\varepsilon > 0$ we may find a constant $c_\varepsilon = c_\varepsilon(N) > 0$ such that $(1+s)^{N'} \leq (1+\varepsilon)s^{N'} + c_\varepsilon$ for all $s \geq 0$. Hence,

$$(15.27) \quad (w(r))^{N'} \leq (1+\varepsilon)\beta_N^{-1/(N-1)} \log(r_0/r) + c_\varepsilon$$

for all $0 < r < r_0$. Since $\gamma < \gamma_N$, we may take ε so small that $\gamma(1+\varepsilon) < \gamma_N = N\beta_N^{1/(N-1)}$, and so $\gamma(1+\varepsilon)\beta_N^{-1/(N-1)} < N$. Hence, also by (15.26) and (15.27),

$$(15.28) \quad \begin{aligned} I &\leq \beta_N e^{\gamma c_\varepsilon} \int_0^{r_0} \exp(\gamma(1+\varepsilon)\beta_N^{-1/(N-1)} \log(r_0/r)) r^{N-1} dr \\ &= \beta_N e^{\gamma c_\varepsilon} r_0^{\gamma(1+\varepsilon)\beta_N^{-1/(N-1)}} \int_0^{r_0} r^{N-1-\gamma(1+\varepsilon)\beta_N^{-1/(N-1)}} dr \\ &= \frac{\beta_N e^{\gamma c_\varepsilon}}{N - \gamma(1+\varepsilon)\beta_N^{-1/(N-1)}} r_0^N =: c_1(N, \gamma) r_0^N. \end{aligned}$$

On the other hand, by the Lebesgue monotone convergence theorem and the fact that $w(r) \leq 1$ for all $r \geq r_0$ by (15.25), we have that

$$\begin{aligned} II &= \beta_N \sum_{n=N-1}^{\infty} \frac{1}{n!} \gamma^n \int_{r_0}^{\infty} (w(r))^{nN/(N-1)} r^{N-1} dr \\ &\leq \beta_N \sum_{n=N-1}^{\infty} \frac{1}{n!} \gamma^n \int_{r_0}^{\infty} (w(r))^N r^{N-1} dr \\ &= \beta_N (\exp_N \gamma) \int_{r_0}^{\infty} (w(r))^N r^{N-1} dr. \end{aligned}$$

Combining this estimate with (15.26) and (15.28), we get

$$(15.29) \quad \int_{\mathbb{R}^N} \exp_N(\gamma(v(x))^{N'}) dx \leq c_1 r_0^N + (\exp_N \gamma) \beta_N \int_{r_0}^{\infty} (w(r))^N r^{N-1} dr.$$

Using spherical coordinates and Theorem 15.10, we have that

$$\beta_N \int_{r_0}^{\infty} (w(r))^N r^{N-1} dr \leq \int_{\mathbb{R}^N} (v^*(x))^N dx = \int_{\mathbb{R}^N} (v(x))^N dx.$$

Thus, to obtain (15.23), it remains to estimate r_0^N in the case that $r_0 > 0$. By (15.25) and the fact that w is decreasing, we have that $w(r) > w(r_0) = 1$ if and only if $0 \leq r < r_0$. Hence,

$$\{x \in \mathbb{R}^N : v^*(x) > 1\} = \{x \in \mathbb{R}^N : w(\|x\|) > 1\} = B(0, r_0).$$

By Proposition 15.6, we have that

$$\begin{aligned} \alpha_N r_0^N &= \mathcal{L}^N(\{x \in \mathbb{R}^N : v^*(x) > 1\}) \\ &= \mathcal{L}^N(\{x \in \mathbb{R}^N : v(x) > 1\}) \leq \int_{\{v>1\}} (v(x))^N dx, \end{aligned}$$

and so we have proved (15.23) and, in turn, (15.21).

By (15.21), the number $s = \|\nabla u\|_{L^N}^{N'}$ is admissible in the definition of $\|u\|_{\Phi_\gamma}$ (see (12.27)), and so (15.22) follows. $\square$

Remark 15.36. Note that the inequality (15.21) is scale invariant; that is, if for $r > 0$ we define the rescaled function $u_r(x) := u(rx)$, $x \in \mathbb{R}^N$, then $\|u_r\|_{L^N}^N = \frac{1}{r^N} \|u\|_{L^N}^N$, $\|\nabla u_r\|_{L^N} = \|\nabla u\|_{L^N}$, and

$$\begin{aligned} \int_{\mathbb{R}^N} \exp_N \left(c_1 \frac{|u_r(x)|^{N'}}{\|\nabla u_r\|_{L^N}^{N'}} \right) dx \\ = \frac{1}{r^N} \int_{\mathbb{R}^N} \exp_N \left(c_1 \frac{|u(x)|^{N'}}{\|\nabla u\|_{L^N}^{N'}} \right) dx. \end{aligned}$$

The previous theorem is complemented by the following exercise, which shows that (15.21) fails for all $\gamma \geq \gamma_N$.

Exercise 15.37. Suppose $N \geq 2$ and let $r > 0$.

- (i) Construct a sequence $\{u_n\}_n$ in $W^{1,N}(\mathbb{R}^N)$ of nonnegative radial functions such that $\text{supp } u_n \subseteq \overline{B(0, r)}$, $\|\nabla v_n\|_{L^N(\mathbb{R}^N; \mathbb{R}^N)} = 1$, and

$$\frac{\int_{\mathbb{R}^N} \exp_N(\gamma_N(v_n(x))^{N'}) dx}{\|v_n\|_{L^N(\mathbb{R}^N)}^N} \rightarrow \infty.$$

- (ii) Prove that for every open set $\Omega \subseteq \mathbb{R}^N$ and for every $c > 0$, the inequality

$$\int_{\Omega} \exp_N\left(\gamma_N \frac{|u(x)|^{N'}}{\|\nabla u\|_{L^N(\Omega; \mathbb{R}^N)}^{N'}}\right) dx \leq c \left(\frac{\|u\|_{L^N(\Omega)}}{\|\nabla u\|_{L^N(\Omega; \mathbb{R}^N)}}\right)^N$$

fails for some $u \in W_0^{1,N}(\Omega) \setminus \{0\}$.

Interpolation of Banach Spaces

Living with P.Q.S, III: How do I treat PQS? Experimental treatment such as Advisor-Pressure (AP), Spousal-Income-Frustration (SIF), Lack-of-Savings-Realization (LSR) and “cocktails” of these have known to cause a remission of the disease long enough to at least defend a thesis. More severe treatments are required to actually finish writing the thesis.

— Jorge Cham, www.phdcomics.com

In this chapter we give a brief overview of the real method of interpolation. We refer to [7], [24], [25], [40], [146], [232], and [233] for more details about interpolation theory. In Chapter 17 we will use this theory to interpolate between $L^p(\mathbb{R}^N)$ and $W^{m,p}(\mathbb{R}^N)$ to obtain the Besov spaces $B_q^{s,p}(\mathbb{R}^N)$.

16.1. Interpolation: K -Method

Definition 16.1. We say that two normed spaces $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ are an *admissible pair* if they are embedded into a common Hausdorff topological vector space X .

It can be checked (exercise) that the vector space $X_0 \cap X_1$ endowed with the norm

$$(16.1) \quad \|x\|_{X_0 \cap X_1} := \max\{\|x\|_{X_0}, \|x\|_{X_1}\}$$

is a normed space. The vector space

$$X_0 + X_1 := \{x \in X : x = x_0 + x_1, x_0 \in X_0, x_1 \in X_1\}$$

is also a normed space when endowed with the norm

$$(16.2) \quad \|x\|_{X_0 + X_1} := \inf\{\|x_0\|_{X_0} + \|x_1\|_{X_1}\},$$

where the infimum is taken over all possible decompositions $x = x_0 + x_1$, $x_0 \in X_0$, $x_1 \in X_1$. Moreover, if X_0 and X_1 are Banach spaces, then so are $X_0 \cap X_1$ and $X_0 + X_1$.

Exercise 16.2. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair of Banach spaces.

- (i) Prove that the set $D := \{(x, -x) : x \in X_0 \cap X_1\}$ is closed in the Banach space $X_0 \times X_1$, endowed with the norm $\|(x_0, x_1)\|_{X_0 \times X_1} := \|x_0\|_{X_0} + \|x_1\|_{X_1}$.
- (ii) Prove that $X_0 + X_1$ is isometric to the quotient space $(X_0 \times X_1)/D$.
- (iii) Prove that $X_0 + X_1$ is a Banach space.

Given $t > 0$, in $X_0 + X_1$ we can also consider the equivalent norm

$$(16.3) \quad x \in X_0 + X_1 \mapsto K(x, t) := \inf\{\|x_0\|_{X_0} + t\|x_1\|_{X_1}\},$$

where as before the infimum is taken over all possible decompositions $x = x_0 + x_1$, $x_0 \in X_0$, $x_1 \in X_1$. To highlight the dependence of K on X_0 and X_1 , when needed, we write

$$(16.4) \quad K(x, t; X_0, X_1) := K(x, t).$$

Remark 16.3. The function $K(\cdot, t)$ can be extended to $X \setminus (X_0 + X_1)$ by setting $K(x, t) := \infty$ if $x \in X \setminus (X_0 + X_1)$, where X is the Hausdorff topological vector space in Definition 16.1.

Exercise 16.4. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair. Prove that for every $x \in X_0 + X_1$, the function $t \mapsto K(x, t)$ is an increasing, concave function and such that $t^{-1}K(x, t) = K(x, t^{-1})$ and

$$K(x, t) \leq \max\{1, t/\tau\}K(x, \tau)$$

for every $t > 0$ and $\tau > 0$.

For $0 < \sigma < 1$ and $1 \leq q \leq \infty$ we can define the *real interpolation space*

$$(X_0, X_1)_{\sigma, q} := \{x \in X_0 + X_1 : \|x\|_{\sigma, q} < \infty\},$$

where if $1 \leq q < \infty$,

$$(16.5) \quad \|x\|_{\sigma, q} := \left(\int_0^\infty (K(x, t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q},$$

while if $q = \infty$,

$$(16.6) \quad \|x\|_{\sigma, \infty} := \sup_{t>0} t^{-\sigma} K(x, t).$$

Theorem 16.5. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair and let $1 \leq q \leq \infty$ and $0 < \sigma < 1$. Then the following embeddings hold:

$$X_0 \cap X_1 \hookrightarrow (X_0, X_1)_{\sigma, q} \hookrightarrow X_0 + X_1.$$

Moreover, if X_0 and X_1 are Banach spaces, then so is $(X_0, X_1)_{\sigma, q}$.

Proof. We leave as an exercise to check that $\|\cdot\|_{\sigma, q}$ is a norm. In view of (16.1) and (16.3) for every $x \in X_0 \cap X_1$,

$$K(x, t) \leq \|x\|_{X_0} \leq \|x\|_{X_0 \cap X_1}, \quad K(x, t) \leq t\|x\|_{X_1} \leq t\|x\|_{X_0 \cap X_1},$$

and so $K(x, t) \leq \min\{1, t\}\|x\|_{X_0 \cap X_1}$. In turn, if $1 \leq q < \infty$, by (16.5),

$$\|x\|_{\sigma, q} \leq \|x\|_{X_0 \cap X_1} \left(\int_0^\infty \min\{1, t^q\} \frac{dt}{t^{1+\sigma q}} \right)^{1/q} = c_q \|x\|_{X_0 \cap X_1},$$

where $c_q = 1/(q(1-\sigma))^{1/q} + 1/(\sigma q)^{1/q}$. If $q = \infty$, then

$$t^{-\sigma} K(x, t) \leq \min\{t^{-\sigma}, t^{1-\sigma}\} \|x\|_{X_0 \cap X_1} \leq \|x\|_{X_0 \cap X_1},$$

where we used the fact that $\sup_{t>0} \min\{t^{-\sigma}, t^{1-\sigma}\} = 1$, and so by (16.6), $\|x\|_{\sigma, \infty} \leq \|x\|_{X_0 \cap X_1}$. Thus, in both cases, we have shown that $X_0 \cap X_1 \hookrightarrow (X_0, X_1)_{\sigma, q}$.

On the other hand, by (16.2) and (16.3), for every $x \in (X_0, X_1)_{\sigma, q}$,

$$(16.7) \quad \min\{1, t\}\|x\|_{X_0+X_1} \leq K(x, t) \leq \max\{1, t\}\|x\|_{X_0+X_1}$$

and thus as before, for $1 \leq q \leq \infty$,

$$(16.8) \quad c_q \|x\|_{X_0+X_1} \leq \|x\|_{\sigma, q},$$

where $c_\infty := 1$, which proves that $(X_0, X_1)_{\sigma, q} \hookrightarrow X_0 + X_1$.

Next we claim that $(X_0, X_1)_{\sigma, q}$ is a Banach space. Let $\{x_n\}_n$ be a Cauchy sequence in $(X_0, X_1)_{\sigma, q}$. In view of (16.8), $\{x_n\}_n$ is a Cauchy sequence in $X_0 + X_1$, and so, since $X_0 + X_1$ is a Banach space (see Exercise 16.2), $x_n \rightarrow x$ in $X_0 + X_1$ for some $x \in X_0 + X_1$.

Given $\varepsilon > 0$ we can find $n_\varepsilon \in \mathbb{N}$ such that $\|x_m - x_n\|_{\sigma, q} \leq \varepsilon$ for all $m, n \geq n_\varepsilon$. Since $K(\cdot, t)$ is a norm in $X_0 + X_1$, by the triangle inequality and (16.7)₂,

$$(16.9) \quad t^{-\sigma} K(x - x_n, t) \leq t^{-\sigma} K(x_m - x_n, t) + \max\{t^{-\sigma}, t^{1-\sigma}\} \|x - x_m\|_{X_0+X_1}.$$

In turn, if $1 \leq q < \infty$, for every $\ell \in \mathbb{N}$,

$$\begin{aligned} \left(\int_{1/\ell}^\ell (K(x - x_n, t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} &\leq \|x_m - x_n\|_{\sigma, q} + c_{\ell, q, \sigma} \|x - x_m\|_{X_0+X_1} \\ &\leq \varepsilon + c_{\ell, q, \sigma} \|x - x_m\|_{X_0+X_1}, \end{aligned}$$

where

$$c_{\ell, q, \sigma} := \left(\int_{1/\ell}^\ell (\max\{t^{-\sigma}, t^{1-\sigma}\})^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} < \infty.$$

Letting first $m \rightarrow \infty$ and then $\ell \rightarrow \infty$ in the previous inequality gives $\|x - x_n\|_{\sigma, q} \leq \varepsilon$ for all $n \geq n_\varepsilon$, which shows that $(X_0, X_1)_{\sigma, q}$ is a Banach space.

If $q = \infty$, then by (16.6) and (16.9),

$$t^{-\sigma} K(x - x_n, t) \leq \varepsilon + \max\{t^{-\sigma}, t^{1-\sigma}\} \|x - x_n\|_{X_0 + X_1}.$$

Letting $m \rightarrow \infty$ shows that $t^{-\sigma} K(x - x_n, t) \leq \varepsilon$ for all $t > 0$ and for all $n \geq n_\varepsilon$. Hence, $\|x - x_n\|_{\sigma, \infty} \leq \varepsilon$ for all $n \geq n_\varepsilon$, and the proof is complete. $\square$

Remark 16.6. If $X_0 = X_1$, then it follows from the previous theorem that $X_0 = X_0 \cap X_1 = (X_0, X_1)_{\sigma, q} = X_0 + X_1 = X_0$.

On the other hand, if $X_0 \cap X_1 = \{0\}$, then for every $x \in X_0 + X_1$ there exist unique $x_0 \in X_0$ and $x_1 \in X_1$ such that $x = x_0 + x_1$. In turn, by (16.3), $K(x, t) = \|x_0\|_{X_0} + t\|x_1\|_{X_1}$ and so for $1 \leq q < \infty$,

$$\|x\|_{\sigma, q} = \left(\int_0^\infty (\|x_0\|_{X_0} + t\|x_1\|_{X_1})^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} = \infty$$

unless $x_0 = x_1 = 0$. Similarly, $\|x\|_{\sigma, \infty} = \sup_{t>0} t^{-\sigma} (\|x_0\|_{X_0} + t\|x_1\|_{X_1}) = \infty$ unless $x_0 = x_1 = 0$. Thus, $(X_0, X_1)_{\sigma, q} = \{0\}$ for every $1 \leq q \leq \infty$.

Exercise 16.7. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair, with $X_1 \hookrightarrow X_0$, and let $1 \leq q \leq \infty$ and $0 < \sigma < 1$.

(i) Prove that $\|\cdot\|_{X_0}$ is an equivalent norm in $X_0 + X_1$.

(ii) Prove that for every $T > 0$,

$$x \mapsto \|x\|_{X_0} + \left(\int_0^T (K(x, t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q}$$

is an equivalent norm in $(X_0, X_1)_{\sigma, q}$ for $1 \leq q < \infty$, while

$$x \mapsto \|x\|_{X_0} + \sup_{0 < t < T} t^{-\sigma} K(x, t)$$

is an equivalent norm in $(X_0, X_1)_{\sigma, \infty}$.

Exercise 16.8. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair, let $1 \leq q < \infty$ and $0 < \sigma < 1$. Prove that for every $x \in (X_0, X_1)_{\sigma, q}$ and $t > 0$,

$$K(x, t) \leq (\sigma q)^{1/q} t^\sigma \|x\|_{\sigma, q}.$$

Exercise 16.9. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair, let $1 \leq q \leq \infty$, and let $0 < \sigma < 1$. Prove that $x \in X_0 + X_1$ belongs to $(X_0, X_1)_{\sigma, q}$ if and only if the sequence $\{2^{-k\sigma} K(x, 2^k)\}_{k \in \mathbb{Z}}$ belongs to $L^q(\mathbb{Z}, \mathcal{H}^0)$, where $\mathcal{H}^0$ is the counting measure. Prove also that

$$x \in (X_0, X_1)_{\sigma, q} \mapsto \|\{2^{-k\sigma} K(x, 2^k)\}_{k \in \mathbb{Z}}\|_{L^q(\mathbb{Z}, \mathcal{H}^0)}$$

is an equivalent norm in $(X_0, X_1)_{\sigma, q}$.

Exercise 16.10. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair, let $1 \leq q \leq \infty$, and let $0 < \sigma < 1$. Prove that $(X_0, X_1)_{\sigma, q} = (X_1, X_0)_{1-\sigma, q}$.

Next we study the inclusions of different interpolation spaces.

Theorem 16.11. *Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair and let $1 \leq q_1 < q_2 \leq \infty$ and $0 < \sigma < 1$. Then the following embeddings hold:*

$$(X_0, X_1)_{\sigma, q_1} \hookrightarrow (X_0, X_1)_{\sigma, q_2} \hookrightarrow (X_0, X_1)_{\sigma, \infty}.$$

Proof. By Exercise 16.8 for $x \in (X_0, X_1)_{\sigma, q}$, $1 \leq q < \infty$, and $\tau > 0$,

$$\tau^{-\sigma} K(x, \tau) \leq (\sigma q)^{1/q} \|x\|_{\sigma, q}.$$

Taking the supremum over all $\tau > 0$ gives

$$(16.10) \quad \|x\|_{\sigma, \infty} \leq (\sigma q)^{1/q} \|x\|_{\sigma, q}.$$

This proves the embedding $(X_0, X_1)_{\sigma, q} \hookrightarrow (X_0, X_1)_{\sigma, \infty}$.

Now, if $1 \leq q_1 < q_2 < \infty$, then for $x \in (X_0, X_1)_{\sigma, q_1}$,

$$\begin{aligned} & \left(\int_0^\infty (K(x, t))^{q_2} \frac{dt}{t^{1+\sigma q_2}} \right)^{1/q_2} \\ & \leq \left(\sup_{\tau > 0} \tau^{-\sigma} K(x, \tau) \right)^{(q_2 - q_1)/q_2} \left(\int_0^\infty (K(x, t))^{q_1} \frac{dt}{t^{1+\sigma q_1}} \right)^{1/q_2} \\ & \leq (\sigma q_1)^{(q_2 - q_1)/(q_1 q_2)} \|x\|_{\sigma, q_1}^{(q_2 - q_1)/q_2 + q_1/q_2} = c \|x\|_{\sigma, q_1}, \end{aligned}$$

where in the last inequality we used (16.10) with q_1 in place of q . $\square$

An important property of interpolation spaces is given by the following theorem.

Theorem 16.12. *Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ and $(Y_0, \|\cdot\|_{Y_0})$, $(Y_1, \|\cdot\|_{Y_1})$ be two admissible pairs and let $T : X_0 + X_1 \rightarrow Y_0 + Y_1$ be a linear operator such that $T : X_0 \rightarrow Y_0$ and $T : X_1 \rightarrow Y_1$ are continuous. Then for every $1 \leq q \leq \infty$ and $0 < \sigma < 1$, $T : (X_0, X_1)_{\sigma, q} \rightarrow (Y_0, Y_1)_{\sigma, q}$ with*

$$\|T\|_{L((X_0, X_1)_{\sigma, q}; (Y_0, Y_1)_{\sigma, q})} \leq \|T\|_{L(X_0; Y_0)}^{1-\sigma} \|T\|_{L(X_1; Y_1)}^\sigma.$$

Proof. Let $c_0, c_1 > 0$ be such that

$$\|T(x_0)\|_{Y_0} \leq c_0 \|x_0\|_{X_0}, \quad \|T(x_1)\|_{Y_1} \leq c_1 \|x_1\|_{X_1}$$

for all $x_0 \in X_0$ and $x_1 \in X_1$. If $x \in (X_0, X_1)_{\sigma, q}$ and $x = x_0 + x_1$, with $x_0 \in X_0$ and $x_1 \in X_1$, it follows by the linearity of T that $T(x) = T(x_0) + T(x_1)$, with $T(x_0) \in Y_0$ and $T(x_1) \in Y_1$. Hence, by (16.3) and (16.4),

$$\begin{aligned} K(T(x), t; Y_0, Y_1) & \leq \|T(x_0)\|_{Y_0} + t \|T(x_1)\|_{Y_1} \\ & \leq c_0 \|x_0\|_{X_0} + t c_1 \|x_1\|_{X_1} = c_0 (\|x_0\|_{X_0} + t c_1 c_0^{-1} \|x_1\|_{X_1}). \end{aligned}$$

It follows that $K(T(x), t; Y_0, Y_1) \leq c_0 K(x, tc_1 c_0^{-1}; X_0, X_1)$, and so, if $1 \leq q < \infty$, by (16.5) and the change of variables $\tau = tc_1 c_0^{-1}$,

$$\begin{aligned} \|T(x)\|_{\sigma, q} &\leq c_0 \left(\int_0^\infty (K(x, tc_1 c_0^{-1}; X_0, X_1))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} \\ &= c_0 (c_1 c_0^{-1})^\sigma \left(\int_0^\infty (K(x, \tau; X_0, X_1))^q \frac{d\tau}{\tau^{1+\sigma q}} \right)^{1/q} \\ &= c_0^{1-\sigma} c_1^\sigma \|x\|_{\sigma, q}. \end{aligned}$$

Similarly, if $q = \infty$, $\|T(x)\|_{\sigma, \infty} \leq c_0^{1-\sigma} c_1^\sigma \|x\|_{\sigma, \infty}$. It remains to let $c_0 \searrow \|T\|_{L(X_0; Y_0)}$ and $c_1 \searrow \|T\|_{L(X_1; Y_1)}$. $\square$

Exercise 16.13. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair, let $1 \leq q \leq \infty$ and $0 < \sigma < 1$. Prove that $\|x\|_{\sigma, q} \leq c \|x\|_{X_0}^{1-\sigma} \|x\|_{X_1}^\sigma$ for every $x \in X_0 \cap X_1$ and for some constant $c = c(q, \sigma) > 0$.

Exercise 16.14. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair of Banach spaces, let $(Y, \|\cdot\|_Y)$ be a Banach space and let $T : X_0 + X_1 \rightarrow Y$ be a linear operator such that $T : X_0 \rightarrow Y$ is compact and $T : X_1 \rightarrow Y$ is continuous. Given $1 \leq q \leq \infty$ and $0 < \sigma < 1$, prove that $T : (X_0, X_1)_{\sigma, q} \rightarrow Y$ is compact. Hint: Given $\varepsilon > 0$ take $t > 0$ so large that $t^\sigma \leq \varepsilon t$ and use Exercise 16.8.

The following theorem tells us that the interpolation of two interpolation spaces may be realized as an interpolation between the original spaces. This is one of the key results in interpolation theory.

Theorem 16.15 (Reiteration). *Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair, let $1 \leq q \leq \infty$, and let $0 \leq \sigma_0 < \sigma_1 \leq 1$. Then for every $0 < \sigma < 1$,*

$$(X_{\sigma_0}, X_{\sigma_1})_{\sigma, q} = (X_0, X_1)_{\theta, q},$$

where $\theta := (1 - \sigma)\sigma_0 + \sigma\sigma_1$, $X_{\sigma_0} := (X_0, X_1)_{\sigma_0, q_0}$ if $\sigma_0 > 0$ for some $1 \leq q_0 \leq \infty$, $X_{\sigma_1} := (X_0, X_1)_{\sigma_1, q_1}$ if $\sigma_1 < 1$ for some $1 \leq q_1 \leq \infty$.

Proof. Step 1: We write $\bar{K} := K(\cdot, \cdot; X_{\sigma_0}, X_{\sigma_1})$ and $K := K(\cdot, \cdot; X_0, X_1)$. Given $x \in X_{\sigma_0} + X_{\sigma_1}$, write $x = x_0 + x_1$, where $x_0 \in X_{\sigma_0}$ and $x_1 \in X_{\sigma_1}$. Since $K(\cdot, t)$ is a norm in $X_0 + X_1$ and $X_{\sigma_i} \hookrightarrow X_0 + X_1$ for $i = 0, 1$ (see Theorem 16.5),

$$\begin{aligned} K(x, t) &\leq K(x_0, t) + K(x_1, t) \leq ct^{\sigma_0} \|x_0\|_{X_{\sigma_0}} + ct^{\sigma_1} \|x_1\|_{X_{\sigma_1}} \\ &= ct^{\sigma_0} (\|x_0\|_{X_{\sigma_0}} + t^{\sigma_1 - \sigma_0} \|x_1\|_{X_{\sigma_1}}), \end{aligned}$$

where we used (16.3) if $\sigma_0 = 0$ or $\sigma_1 = 1$, and Exercise 16.8 if $\sigma_0 > 0$ and $q_0 < \infty$ or $\sigma_1 < 1$ and $q_1 < \infty$. Taking the infimum over all such decompositions of x , gives

$$K(x, t) \leq ct^{\sigma_0} \bar{K}(x, t^{\sigma_1 - \sigma_0}).$$

Hence, if $q < \infty$, by the change of variables $\tau := t^{\sigma_1 - \sigma_0}$ and the fact that $\theta = (1 - \sigma)\sigma_0 + \sigma\sigma_1$,

$$\begin{aligned}\|x\|_{\theta,q} &\leq c \left(\int_0^\infty (t^{\sigma_0} \bar{K}(x, t^{\sigma_1 - \sigma_0}))^q \frac{dt}{t^{1+\theta q}} \right)^{1/q} \\ &= c \left(\int_0^\infty (\bar{K}(x, \tau))^q \frac{d\tau}{\tau^{1+\sigma q}} \right)^{1/q} = c \|x\|_{\sigma,q}^-, \end{aligned}$$

where $\|\cdot\|_{\sigma,q}^-$ is the norm of $(X_{\sigma_0}, X_{\sigma_1})_{\sigma,q}$. The proof is simpler when $q = \infty$ and is left as an exercise.

Step 2: Assume that $0 < \sigma_0 < \sigma_1 < 1$. Given $x \in X_0 + X_1$ and $t > 0$, let $x_0 \in X_0$ and $x_1 \in X_1$ be such that $x = x_0 + x_1$ and (see (16.3))

$$(16.11) \quad \|x_0\|_{X_0} + t\|x_1\|_{X_1} \leq 2K(x, t).$$

(if $K(x, t) = 0$ we take $x_0 = 0$ and $x_1 = 0$). Let

$$(16.12) \quad f_i(x, t) := \int_0^t K(x, \tau) \frac{d\tau}{\tau^{1+\sigma_i}}, \quad g_i(x, t) := \int_t^\infty K(x, \tau) \frac{d\tau}{\tau^{1+\sigma_i}},$$

$i = 0, 1$. We claim that if $f_0(x, t)$ and $g_1(x, t)$ are finite, then $x \in X_{\sigma_0} + X_{\sigma_1}$ and

$$(16.13) \quad \bar{K}(x, t^{\sigma_1 - \sigma_0}) \leq cf_0(x, t) + ct^{\sigma_1 - \sigma_0} g_1(x, t).$$

Since $x_0 \in X_0$ and $x_1 \in X_1$, by (16.3) and (16.11),

$$(16.14) \quad K(x_0, \tau) \leq \|x_0\|_{X_0} \leq 2K(x, t), \quad K(x_1, \tau) \leq \tau\|x_1\|_{X_1} \leq 2\tau t^{-1}K(x, t).$$

Since K is increasing as a function of t , using the previous inequalities we get

$$\begin{aligned}(16.15) \quad g_0(x_0, t) &= \int_t^\infty K(x_0, \tau) \frac{d\tau}{\tau^{1+\sigma_0}} \leq \frac{2}{\sigma_0 t^{\sigma_0}} K(x, t) \\ &= \frac{2(1 - \sigma_0)}{\sigma_0 t} K(x, t) \int_0^t \frac{d\tau}{\tau^{\sigma_0}} \leq \frac{2(1 - \sigma_0)}{\sigma_0} f_0(x, t), \end{aligned}$$

where in the last inequality we used (16.12) and the fact that $K(x, t)/t \leq K(x, \tau)/\tau$ (see Exercise 16.4). Moreover, again by (16.14) and the fact that $K(x, t)/t \leq K(x, \tau)/\tau$,

$$f_0(x_1, t) = \int_0^t K(x_1, \tau) \frac{d\tau}{\tau^{1+\sigma_0}} \leq 2 \int_0^t K(x, \tau) \frac{\tau}{t} \frac{d\tau}{\tau^{1+\sigma_0}} \leq 2f_0(x, t).$$

Since $x_0 = x - x_1$ and $K(\cdot, \tau)$ is a norm, it follows that

$$(16.16) \quad f_0(x_0, t) \leq f_0(x, t) + f_0(x_1, t) \leq 3f_0(x, t).$$

Thus, by (16.15) and (16.16),

$$\|x_0\|_{\sigma_0,1} = f_0(x_0, t) + g_0(x_0, t) \leq cf_0(x, t) < \infty.$$

Similarly, one can show that

$$\|x_1\|_{\sigma_1,1} = f_1(x_1, t) + g_1(x_1, t) \leq cg_1(x, t) < \infty.$$

By Theorem 16.11 and the last two inequalities it follows that

$$\|x_0\|_{\sigma_0,q_0} \leq c\|x_0\|_{\sigma_0,1} \leq cf_0(x, t), \quad \|x_1\|_{\sigma_1,q_1} \leq c\|x_1\|_{\sigma_1,1} \leq cg_1(x, t).$$

This proves that $x_0 \in X_{\sigma_0}$ and $x_1 \in X_{\sigma_1}$. In turn, $x \in X_{\sigma_0} + X_{\sigma_1}$, and so

$$\begin{aligned} \bar{K}(x, t^{\sigma_1-\sigma_0}) &\leq \|x_0\|_{\sigma_0,q_0} + t^{\sigma_1-\sigma_0}\|x_1\|_{\sigma_1,q_1} \\ &\leq cf_0(x, t) + ct^{\sigma_1-\sigma_0}g_1(x, t). \end{aligned}$$

Step 3: Assume that $0 < \sigma_0 < \sigma_1 < 1$. Given $x \in X_0 + X_1$, if $q < \infty$ we apply Hardy's inequalities (see Theorem C.41) to obtain

$$\begin{aligned} &\left(\int_0^\infty \left(\int_0^t K(x, \tau) \frac{d\tau}{\tau^{1+\sigma_0}} \right)^q \frac{dt}{t^{1+\sigma(\sigma_1-\sigma_0)q}} \right)^{1/q} \\ &\leq \frac{1}{\sigma(\sigma_1 - \sigma_0)} \left(\int_0^\infty (K(x, t))^q \frac{dt}{t^{1+\theta q}} \right)^{1/q} \end{aligned}$$

and

$$\begin{aligned} &\left(\int_0^\infty \left(\int_t^\infty K(x, \tau) \frac{d\tau}{\tau^{1+\sigma_1}} \right)^q \frac{dt}{t^{1-(1-\sigma)(\sigma_1-\sigma_0)q}} \right)^{1/q} \\ &\leq \frac{1}{(1-\sigma)(\sigma_1 - \sigma_0)} \left(\int_0^\infty (K(x, t))^q \frac{dt}{t^{1+\theta q}} \right)^{1/q}, \end{aligned}$$

where we used the fact that $\theta = (1-\sigma)\sigma_0 + \sigma\sigma_1$. It follows from (16.12) that if $\|x\|_{\theta,q} < \infty$, then for $\mathcal{L}^1$ -a.e. $t > 0$, $f_0(x, t)$ and $g_1(x, t)$ are finite and so we are in a position to apply Step 2 to get that $x \in X_{\sigma_0} + X_{\sigma_1}$ and that (16.13) holds. Hence, with the change of variables $r := t^{\sigma_1-\sigma_0}$, by (16.12), (16.13), and the previous inequalities,

$$\begin{aligned} \|x\|_{\sigma,q}^- &= (\sigma_1 - \sigma_0) \left(\int_0^\infty (\bar{K}(x, t^{\sigma_1-\sigma_0}))^q \frac{dt}{t^{1+\sigma(\sigma_1-\sigma_0)q}} \right)^{1/q} \\ &\leq c \left(\int_0^\infty (f_0(x, t) + ct^{\sigma_1-\sigma_0}g_1(x, t))^q \frac{dt}{t^{1+\sigma(\sigma_1-\sigma_0)q}} \right)^{1/q} \leq c\|x\|_{\theta,q}, \end{aligned}$$

which concludes the proof in the case $0 < \sigma_0 < \sigma_1 < 1$. The cases $\sigma_0 = 0$ and $\sigma_1 = 1$ are similar and are left as an exercise. $\square$

We conclude this section by studying the limits

$$\lim_{\sigma \rightarrow 0^+} (\sigma(1-\sigma)q)^{1/q} \|x\|_{\sigma,q} \quad \text{and} \quad \lim_{\sigma \rightarrow 1^-} (\sigma(1-\sigma)q)^{1/q} \|x\|_{\sigma,q}.$$

The following result will play an important role in Section 17.5 in Chapter 17 to show that the norm in $L^p(\mathbb{R}^N)$ and a suitable seminorm in $W^{m,p}(\mathbb{R}^N)$ can be regarded as limits of seminorms in the Besov space $B_q^{s,p}(\mathbb{R}^N)$ as $s \rightarrow 0^+$ and $s \rightarrow m^-$, respectively.

Theorem 16.16. *Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair and let $1 \leq q < \infty$. Assume that $x \in (X_0, X_1)_{\sigma_0, q}$ for some $0 < \sigma_0 < 1$. Then*

$$\begin{aligned} \liminf_{t \rightarrow \infty} K(x, t) &\leq \liminf_{\sigma \rightarrow 0^+} (\sigma(1 - \sigma)q)^{1/q} \|x\|_{\sigma, q} \\ &\leq \limsup_{\sigma \rightarrow 0^+} (\sigma(1 - \sigma)q)^{1/q} \|x\|_{\sigma, q} \leq \limsup_{t \rightarrow \infty} K(x, t) \end{aligned}$$

and

$$\begin{aligned} \liminf_{t \rightarrow 0^+} \frac{K(x, t)}{t} &\leq \liminf_{\sigma \rightarrow 1^-} (\sigma(1 - \sigma)q)^{1/q} \|x\|_{\sigma, q} \\ &\leq \limsup_{\sigma \rightarrow 1^-} (\sigma(1 - \sigma)q)^{1/q} \|x\|_{\sigma, q} \leq \limsup_{t \rightarrow 0^+} \frac{K(x, t)}{t}. \end{aligned}$$

Let $0 < q \leq \infty$ and $\sigma \in (0, 1)$ and for every nonnegative Lebesgue measurable function $g : [0, \infty) \rightarrow [0, \infty)$ define for $0 < q < \infty$,

$$(16.17) \quad \|g\|_{\sigma, q}^{\sim} := \left(\int_0^\infty (g(t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q},$$

while for $q = \infty$,

$$\|g\|_{\sigma, \infty}^{\sim} := \sup_{t > 0} \frac{g(t)}{t^\sigma}.$$

Theorem 16.16 follows by applying the following lemma to the function $g(t) = K(x, t)$, $t \geq 0$.

Lemma 16.17. *Let $0 < q < \infty$ and let $g : [0, \infty) \rightarrow [0, \infty)$ be a Lebesgue measurable function such that $\|g\|_{\sigma_0, q} < \infty$ for some $\sigma_0 \in (0, 1)$. Then*

$$\begin{aligned} (16.18) \quad \liminf_{t \rightarrow \infty} g(t) &\leq \liminf_{\sigma \rightarrow 0^+} (\sigma(1 - \sigma)q)^{1/q} \|g\|_{\sigma, q}^{\sim} \\ &\leq \limsup_{\sigma \rightarrow 0^+} (\sigma(1 - \sigma)q)^{1/q} \|g\|_{\sigma, q}^{\sim} \leq \limsup_{t \rightarrow \infty} g(t) \end{aligned}$$

and

$$\begin{aligned} (16.19) \quad \liminf_{t \rightarrow 0^+} \frac{g(t)}{t} &\leq \liminf_{\sigma \rightarrow 1^-} (\sigma(1 - \sigma)q)^{1/q} \|g\|_{\sigma, q}^{\sim} \\ &\leq \limsup_{\sigma \rightarrow 1^-} (\sigma(1 - \sigma)q)^{1/q} \|g\|_{\sigma, q}^{\sim} \leq \limsup_{t \rightarrow 0^+} \frac{g(t)}{t}. \end{aligned}$$

Proof. We prove only the first inequality in (16.18) and leave the second and (16.19) as exercises. Let $\ell := \liminf_{t \rightarrow \infty} g(t)$. Note that $\ell \in [0, \infty]$. If $\ell = 0$, then there is nothing to prove. Thus, assume that $\ell > 0$ and fix $0 < \ell_1 < \ell$. Then there exists $T > 0$ such that

$$(16.20) \quad g(t) \geq \ell_1$$

for all $t \geq T$. Write

$$\sigma(1-\sigma)q(\|g\|_{\sigma,q}^\sim)^q = \sigma(1-\sigma)q \int_0^T \frac{g^q(t)}{t^{1+\sigma q}} dt + \sigma(1-\sigma)q \int_T^\infty \frac{g^q(t)}{t^{1+\sigma q}} dt =: I + II.$$

Then for all $0 < \sigma \leq \sigma_0$,

$$I \leq \sigma(1-\sigma)qT^{(\sigma_0-\sigma)q} \int_0^T \frac{g^q(t)}{t^{1+\sigma_0 q}} dt \leq \sigma(1-\sigma)qT^{(\sigma_0-\sigma)q}(\|g\|_{\sigma_0,q}^\sim)^q,$$

and so letting $\sigma \rightarrow 0^+$, we obtain $\lim_{\sigma \rightarrow 0^+} I = 0$. On the other hand, by (16.20),

$$II \geq \sigma(1-\sigma)q\ell_1^q \int_T^\infty \frac{1}{t^{1+\sigma q}} dt = (1-\sigma)\frac{1}{T^{\sigma q}}\ell_1^q.$$

Hence, $\liminf_{\sigma \rightarrow 0^+} II \geq \ell_1^q$. Letting $\ell_1 \nearrow \ell$ gives the desired result. $\square$

Exercise 16.18. Prove (16.18)₂ and (16.19).

Exercise 16.19. Let $g : [0, \infty) \rightarrow [0, \infty)$ be a Lebesgue measurable function such that $\|g\|_{\sigma_0, \infty}^\sim < \infty$ for some $\sigma_0 \in (0, 1)$.

(i) Prove that if g is increasing, then there exists

$$\lim_{\sigma \rightarrow 0^+} \|g\|_{\sigma, \infty}^\sim = \lim_{t \rightarrow \infty} g(t).$$

What happens if we remove the hypothesis that g is increasing?

(ii) Prove that if $g(t)/t$ is decreasing, then there exists

$$\lim_{\sigma \rightarrow 1^-} \|g\|_{\sigma, \infty}^\sim = \lim_{t \rightarrow 0^+} g(t)/t.$$

What happens if we remove the hypothesis that $g(t)/t$ is decreasing?

16.2. Interpolation: J -Method

In this section we introduce an equivalent way to introduce the interpolation space $(X_0, X_1)_{\sigma, q}$.

Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair. We have seen that $X_0 \cap X_1$ is a normed space with the norm given in (16.1). Given $t > 0$, we can also consider the equivalent norm

$$(16.21) \quad x \in X_0 \cap X_1 \mapsto J(x, t) := \max\{\|x\|_{X_0}, t\|x\|_{X_1}\}.$$

To highlight the dependence of J on X_0 and X_1 , when needed, we write

$$(16.22) \quad J(x, t; X_0, X_1) := J(x, t).$$

Exercise 16.20. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair. Prove that for every $x \in X_0 \cap X_1$, the function $t \mapsto J(x, t)$ is increasing, convex, and such that for every $t > 0$ and $\tau > 0$,

$$J(x, t) \leq \max\{1, t/\tau\}J(x, \tau), \quad K(x, t) \leq \min\{1, t/\tau\}J(x, \tau).$$

For $0 < \sigma < 1$ and $1 \leq q \leq \infty$ we can define the *real interpolation space* $(X_0, X_1)_{\sigma, q, J}$ as the space of all $x \in X_0 + X_1$ such that $\|x\|_{\sigma, q, J} < \infty$, where if $1 \leq q < \infty$,

$$(16.23) \quad \|x\|_{\sigma, q, J} := \inf \left(\int_0^\infty (J(u(t), t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q},$$

while if $q = \infty$,

$$(16.24) \quad \|x\|_{\sigma, \infty, J} := \inf \sup_{t>0} t^{-\sigma} J(u(t), t),$$

where the infimum in (16.23) and (16.24) is done over all Bochner integrable functions (see Section 8.1 in Chapter 8) $u : (0, \infty) \rightarrow X_0 \cap X_1$ satisfying

$$(16.25) \quad x = \int_0^\infty u(t) \frac{dt}{t},$$

where the Bochner integral is considered with respect to the normed space $X_0 + X_1$.

The next theorem shows that the spaces $(X_0, X_1)_{\sigma, q}$ and $(X_0, X_1)_{\sigma, q, J}$ coincide.

Theorem 16.21. *Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair, let $1 \leq q \leq \infty$, and let $0 < \sigma < 1$. Then there exists a constant $c > 0$ such that*

$$c^{-1} \|x\|_{\sigma, q} \leq \|x\|_{\sigma, q, J} \leq c \|x\|_{\sigma, q}$$

for every $x \in X_0 + X_1$. In particular, $(X_0, X_1)_{\sigma, q} = (X_0, X_1)_{\sigma, q, J}$, with equivalent norms.

Proof. We only prove the case $1 \leq q < \infty$ and leave the case $q = \infty$ as an exercise.

Step 1: We claim that there exists a constant $c > 0$ such that $\|x\|_{\sigma, q} \leq c \|x\|_{\sigma, q, J}$ for every $x \in X_0 + X_1$. Without loss of generality, we may assume that $\|x\|_{\sigma, q, J} < \infty$, since otherwise there is nothing to prove. Hence, there exists a Bochner integrable function $u : (0, \infty) \rightarrow X_0 \cap X_1$ satisfying (16.25). Since $K(\cdot, t)$ is an equivalent norm in $X_0 + X_1$, by Theorem 8.9 and Exercise 16.20,

$$\begin{aligned} K(x, t) &= K\left(\int_0^\infty u(\tau) \frac{d\tau}{\tau}, t\right) \leq \int_0^\infty K(u(\tau), t) \frac{d\tau}{\tau} \\ &\leq \int_0^\infty \min\{1, t/\tau\} J(u(\tau), \tau) \frac{d\tau}{\tau} \\ &= \int_0^t J(u(\tau), \tau) \frac{d\tau}{\tau} + t \int_t^\infty J(u(\tau), \tau) \frac{d\tau}{\tau^2}. \end{aligned}$$

In turn, by Hardy's inequalities (see Theorem C.41),

$$\begin{aligned} \left(\int_0^\infty (K(x, t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} &\leq \left(\int_0^\infty \left(\int_0^t J(u(\tau), \tau) \frac{d\tau}{\tau} \right)^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} \\ &\quad + \left(\int_0^\infty \left(\int_t^\infty J(u(\tau), \tau) \frac{d\tau}{\tau^2} \right)^q \frac{dt}{t^{1-q+\sigma q}} \right)^{1/q} \\ &\leq \left(\frac{1}{\sigma} + \frac{1}{1-\sigma} \right) \left(\int_0^\infty (J(u(t), t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q}. \end{aligned}$$

By taking the infimum over all admissible functions u , the claim follows.

Step 2: We claim that if $x \in X_0 + X_1$ is such that

$$(16.26) \quad \lim_{t \rightarrow 0^+} K(x, t) = 0, \quad \lim_{t \rightarrow \infty} K(x, t)/t = 0,$$

then there exists a strongly measurable function $u : (0, \infty) \rightarrow X_0 \cap X_1$ (see Definition 8.1) satisfying

$$(16.27) \quad \int_\varepsilon^{1/\varepsilon} u(t) \frac{dt}{t} \rightarrow x$$

as $\varepsilon \rightarrow 0^+$ (where the limit is in the space $X_0 + X_1$) and such that

$$(16.28) \quad J(u(t), t) \leq 4eK(x, t) \quad \text{for all } t > 0.$$

To see this, by the definition of $K(x, t)$ (see (16.3)) for every $k \in \mathbb{Z}$ we can find $x_k \in X_0$ and $y_k \in X_1$ such that

$$(16.29) \quad x = x_k + y_k, \quad \|x_k\|_{X_0} + e^k \|y_k\|_{X_1} \leq 2K(x, e^k)$$

(if $K(x, e^k) = 0$ we take $x_k = 0$ and $y_k = 0$). In turn, by (16.26),

$$(16.30) \quad \lim_{k \rightarrow -\infty} \|x_k\|_{X_0} = \lim_{k \rightarrow \infty} \|y_k\|_{X_1} = 0.$$

Define $u(t) := x_{k+1} - x_k = y_k - y_{k+1} \in X_0 \cap X_1$ for $t \in [e^k, e^{k+1})$. Then u is strongly measurable and for every $\ell \in \mathbb{N}$, using telescopic sums,

$$\int_{e^{-\ell}}^{e^\ell} u(t) \frac{dt}{t} = \sum_{k=-\ell}^{\ell-1} (x_{k+1} - x_k) = x_\ell - x_{-\ell} = x - y_\ell - x_{-\ell} \rightarrow x$$

as $\ell \rightarrow \infty$ by (16.2), (16.29), and (16.30). This shows (16.27).

Moreover, for $e^k \leq t < e^{k+1}$, by the definition of J (see (16.21)) and of $u(t)$, Exercise 16.4, (16.29), and

$$\begin{aligned} J(u(t), t) &\leq \max\{\|x_k\|_{X_0} + \|x_{k+1}\|_{X_0}, t\|y_k\|_{X_1} + t\|y_{k+1}\|_{X_1}\} \\ &\leq e(\|x_k\|_{X_0} + e^k\|y_k\|_{X_1}) + \|x_{k+1}\|_{X_0} + e^{k+1}\|y_{k+1}\|_{X_1} \\ &\leq 2eK(x, e^k) + 2K(x, e^{k+1}) \leq 2eK(x, t) + 2\max\{1, e^{k+1}/t\}K(x, t) \\ &\leq 4eK(x, t), \end{aligned}$$

which proves (16.28).

Step 3: We claim that there exists a constant $c > 0$ such that $\|x\|_{\sigma,q,J} \leq c\|x\|_{\sigma,q}$ for every $x \in X_0 + X_1$. Without loss of generality, we may assume that $\|x\|_{\sigma,q} < \infty$, since otherwise there is nothing to prove. Then by Exercise 16.8, $K(x, t) \leq (\sigma q)^{1/q} t^\sigma \|x\|_{\sigma,q}$, which implies that (16.26) holds. Thus, we are in a position to apply Step 2 to find a strongly measurable function $u : (0, \infty) \rightarrow X_0 \cap X_1$ satisfying (16.27) and (16.28). By (16.28),

$$\left(\int_0^\infty (J(u(t), t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} \leq 4e \left(\int_0^\infty (K(x, t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} = 4e\|x\|_{\sigma,q} < \infty.$$

Moreover, since by the definition of J (see (16.21)) for $y \in X_0 \cap X_1$,

$$\|y\|_{X_0+X_1} \leq \min\{\|y\|_{X_0}, \|y\|_{X_1}\} \leq \min\{1, 1/t\} J(y, t),$$

when $1 < q < \infty$ it follows from Hölder's inequality and the fact that $0 < \sigma < 1$,

$$\begin{aligned} \int_0^\infty \|u(t)\|_{X_0+X_1} \frac{dt}{t} &\leq \int_0^\infty \min\{1, 1/t\} J(u(t), t) \frac{dt}{t} \\ &\leq \left(\int_0^\infty t^{\sigma q'} (\min\{1, 1/t\})^{q'} \frac{dt}{t} \right)^{1/q'} \left(\int_0^\infty (J(u(t), t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} < \infty. \end{aligned}$$

On the other hand, if $q = 1$, then instead of Hölder's inequality we use the fact that $\min\{1, 1/t\} \leq 1/t^\sigma$ for every $t > 0$. This shows that u is Bochner integrable, and so it is admissible in the definition of $\|\cdot\|_{\sigma,q,J}$ (see (16.23)). $\square$

Exercise 16.22. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair of Banach spaces, let $1 \leq q < \infty$, and let $0 < \sigma < 1$. Prove that $X_0 \cap X_1$ is dense in $(X_0, X_1)_{\sigma,q}$. Hint: Modify the proof of Steps 2 and 3 of the previous theorem.

Exercise 16.23. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair, with $X_1 \hookrightarrow X_0$, and let $1 \leq q \leq \infty$ and $0 < \sigma < 1$. Prove that for every $T > 0$,

$$x \mapsto \|x\|_{X_0} + \inf \left(\int_0^T (J(u(t), t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q}$$

is an equivalent norm in $(X_0, X_1)_{\sigma,q}$ for $1 \leq q < \infty$, while

$$x \mapsto \|x\|_{X_0} + \inf \sup_{0 < t < T} t^{-\sigma} J(u(t), t)$$

is an equivalent norm in $(X_0, X_1)_{\sigma,\infty}$, where the infimum is done over all Bochner integrable functions $u : (0, r) \rightarrow X_0 \cap X_1$ satisfying $x = \int_0^T u(t) \frac{dt}{t}$, where the Bochner integral is considered with respect to the normed space $X_0 + X_1$. Hint: See Exercise 16.7.

Exercise 16.24. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair, let $1 \leq q \leq \infty$, and let $0 < \sigma < 1$. Prove that $x \in X_0 + X_1$ belongs to $(X_0, X_1)_{\sigma,q}$ if and only if it can be written as $x = \sum_{k \in \mathbb{Z}} x_k$, where $x_k \in X_0 \cap X_1$, the

series converges in $X_0 + X_1$, and the sequence $\{2^{-k\sigma} J(x_k, 2^k)\}_{k \in \mathbb{Z}}$ belongs to $L^q(\mathbb{Z}, \mathcal{H}^0)$, where $\mathcal{H}^0$ is the counting measure. Prove also that

$$x \in (X_0, X_1)_{\sigma, q} \mapsto \inf \{ \|\{2^{-k\sigma} J(x_k, 2^k)\}_{k \in \mathbb{Z}}\|_{L^q(\mathbb{Z}, \mathcal{H}^0)} \},$$

where the infimum is done over all such representations, is an equivalent norm in $(X_0, X_1)_{\sigma, q}$. Hint: See Steps 2 and 3 in the proof of the previous theorem.

Exercise 16.25. Let $(Y_0, \|\cdot\|_{Y_0})$, $(Y_1, \|\cdot\|_{Y_1})$ be an admissible pair of Banach spaces, let $(X, \|\cdot\|_X)$ be a Banach space and let $T : X \rightarrow Y_0 + Y_1$ be a linear operator such that $T : X \rightarrow Y_0$ is compact and $T : X \rightarrow Y_1$ is continuous. Given $1 \leq q \leq \infty$ and $0 < \sigma < 1$, prove that $T : X \rightarrow (Y_0, Y_1)_{\sigma, q}$ is compact. Hint: Given $\varepsilon > 0$ take $t > 0$ so small that $\varepsilon t \leq t^\sigma$ and use (16.23) and (16.24).

Exercise 16.26. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair of Banach spaces with X_1 compactly embedded in X_0 . Given $1 \leq q_1, q_2 \leq \infty$ and $0 < \sigma_1 < \sigma_2 < 1$, prove that $(X_0, X_1)_{\sigma_2, q_2}$ is compactly embedded in $(X_0, X_1)_{\sigma_1, q_1}$. Hint: Use Exercise 16.14 with the identity operator to prove that $X_{\sigma_2} := (X_0, X_1)_{\sigma_2, q_2}$ is compactly embedded in X_0 , then use the previous exercise to show that X_{σ_2} is compactly embedded in $(X_0, X_{\sigma_2})_{\sigma, q}$, and finally use the reiteration theorem (Theorem 16.15).

16.3. Duality

In this section we study the dual space of $(X_0, X_1)_{\sigma, q}$. Given an admissible pair $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ of Banach spaces, in general their duals will not be an admissible pair. To overcome this problem we assume that $X_0 \cap X_1$ is dense in both X_0 and X_1 . Given $L \in X'_0$, we denote by L^* its restriction to $X_0 \cap X_1$. Then for every $x \in X_0 \cap X_1$,

$$|L^*(x)| \leq \|L\|_{X'_0} \|x\|_{X_0} \leq \|L\|_{X'_0} \|x\|_{X_0 \cap X_1},$$

where in the last inequality we used (16.1). It follows that $L^* \in (X_0 \cap X_1)'$. Moreover, using the first inequality,

$$(16.31) \quad \|L^*\|_{X_0^*} := \sup\{|L^*(x)| : x \in X_0 \cap X_1, \|x\|_{X_0} \leq 1\} \leq \|L\|_{X'_0} < \infty.$$

Conversely, given $L^* \in (X_0 \cap X_1)'$ such that $\|L^*\|_{X_0^*} < \infty$, since $X_0 \cap X_1$ is dense in X_0 , we can uniquely extend L^* to a linear continuous function $L : X_0 \rightarrow \mathbb{R}$ with $\|L\|_{X'_0} = \|L^*\|_{X_0^*}$. Thus, we can identify X'_0 with the space X_0^* of all $L^* \in (X_0 \cap X_1)'$ for which $\|L^*\|_{X_0^*} < \infty$. Similarly, we can identify X'_1 with the space X_1^* , which is defined analogously to X_0^* with the norm in X_1 in place of the one in X_0 in the definition of $\|L^*\|_{X_1^*}$. With this identification, X'_0 and X'_1 can be embedded into $(X_0 \cap X_1)'$, and thus they are an admissible pair of Banach spaces.

Theorem 16.27 (Duality). *Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair of Banach spaces and assume that $X_0 \cap X_1$ is dense in both X_0 and X_1 . Then $(X'_0, \|\cdot\|_{X'_0})$, $(X'_1, \|\cdot\|_{X'_1})$ is an admissible pair and for $1 \leq q < \infty$ and $0 < \sigma < 1$, the dual of $(X_0, X_1)_{\sigma, q}$ is given by $(X'_0, X'_1)_{\sigma, q'}$.*

We begin with some preliminary results.

Exercise 16.28. Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair of Banach spaces and let $t > 0$. Prove that the dual space of $X_0 \times X_1$ equipped with the norm $\|(x_0, x_1)\|_{X_0 \times X_1} := \max\{\|x_0\|_{X_0}, t\|x_1\|_{X_1}\}$, $x_0 \in X_0$, $x_1 \in X_1$, is given by $X'_0 \times X'_1$, equipped with the norm $\|L_0\|_{X'_0} + t^{-1}\|L_1\|_{X'_1}$, $L_0 \in X'_0$, $L_1 \in X'_1$.

Lemma 16.29. *Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair of Banach spaces and assume that $X_0 \cap X_1$ is dense in both X_0 and X_1 . If the space $X_0 + X_1$ is equipped with the norm $K(\cdot, t; X_0, X_1)$, where $t > 0$, then its dual space is $X'_0 \cap X'_1$ equipped with the norm $J(\cdot, t^{-1}; X'_0, X'_1)$, while if the space $X_0 \cap X_1$ is equipped with the norm $J(\cdot, t; X_0, X_1)$, where $t > 0$, then its dual space is $X'_0 + X'_1$ equipped with the norm $K(\cdot, t^{-1}; X'_0, X'_1)$.*

Proof. Step 1: We prove the first part of the statement. Assume that $X_0 + X_1$ is equipped with the norm $K(\cdot, t; X_0, X_1)$ and $X'_0 \cap X'_1$ with the norm $J(\cdot, t^{-1}; X'_0, X'_1)$. Given $x \in X_0 + X_1$, write $x = x_0 + x_1$, where $x_0 \in X_0$ and $x_1 \in X_1$. Then for $L \in X'_0 \cap X'_1$, we can extend L to $X_0 + X_1$ by defining $L(x) := L(x_0) + L(x_1)$ (why is this a good definition?). It follows that

$$\begin{aligned} |L(x)| &\leq |L(x_0)| + |L(x_1)| \leq \|L\|_{X'_0} \|x_0\|_{X_0} + t t^{-1} \|L\|_{X'_1} \|x_1\|_{X_1} \\ &\leq \max\{\|L\|_{X'_0}, t^{-1} \|L\|_{X'_1}\} (\|x_0\|_{X_0} + t \|x_1\|_{X_1}) \\ &= J(L, t^{-1}; X'_0, X'_1) (\|x_0\|_{X_0} + t \|x_1\|_{X_1}), \end{aligned}$$

where we used (16.22). Taking the infimum over all such decompositions of x and using (16.4) gives

$$|L(x)| \leq J(L, t^{-1}; X'_0, X'_1) K(x, t; X_0, X_1),$$

which shows that $\|L\|_{(X_0 + X_1)'} \leq J(L, t^{-1}; X'_0, X'_1)$.

Conversely, given $L \in (X_0 + X_1)'$, define

$$(16.32) \quad M := \sup\{|L(x)| : x \in X_0 + X_1, K(x, t; X_0, X_1) = 1\}.$$

If $x_0 \in X_0$, then

$$|L(x_0)| \leq M K(x_0, t; X_0, X_1) \leq M \|x_0\|_{X_0}$$

by (16.3) and, similarly, if $x_1 \in X_1$,

$$|L(x_1)| \leq M K(x_1, t; X_0, X_1) \leq M t \|x_1\|_{X_1}.$$

Thus, $L \in X'_0 \cap X'_1$. To estimate its norm, given $\varepsilon > 0$ find $x_0 \in X_0$ with $\|x_0\|_{X_0} = 1$ and $x_1 \in X_1$ with $\|x_1\|_{X_1} = 1$ such that

$$\|L\|_{X'_0} \leq (1 + \varepsilon)L(x_0), \quad \|L\|_{X'_1} \leq (1 + \varepsilon)L(x_1).$$

Given $a, b > 0$, by replacing x_0 with ax_0 and x_1 with bx_1 , we have that $\|x_0\|_{X_0} = a$, $\|x_1\|_{X_1} = b$ and

$$a\|L\|_{X'_0} \leq (1 + \varepsilon)L(x_0), \quad b\|L\|_{X'_1} \leq (1 + \varepsilon)L(x_1).$$

Hence, by (16.3) the vector $x = x_0 + x_1$ satisfies

$$K(x, t; X_0, X_1) \leq a + tb,$$

while

$$L(x) = L(x_0) + L(x_1) \geq (1 + \varepsilon)^{-1}(a\|L\|_{X'_0} + b\|L\|_{X'_1}).$$

In turn,

$$\|L\|_{(X_0+X_1)'} \geq (1 + \varepsilon)^{-1} \frac{a\|L\|_{X'_0} + b\|L\|_{X'_1}}{a + tb}.$$

Since ε , a and b are arbitrary, by letting $\varepsilon \rightarrow 0^+$ and then separately $a \rightarrow 0^+$ and $b \rightarrow 0^+$ we obtain

$$\|L\|_{(X_0+X_1)'} \geq \max\{\|L\|_{X'_0}, t^{-1}\|L\|_{X'_1}\} = J(L, t^{-1}; X'_0, X'_1).$$

Step 2: Assume that $X_0 \cap X_1$ is equipped with the norm $J(\cdot, t; X_0, X_1)$ and $X'_0 + X'_1$ with the norm $K(\cdot, t^{-1}; X'_0, X'_1)$. Given $L \in X'_0 + X'_1$, let $L = L_0 + L_1$, where $L_0 \in X'_0$ and $L_1 \in X'_1$. Then for $x \in X_0 \cap X_1$ we have

$$\begin{aligned} |L(x)| &\leq |L_0(x)| + |L_1(x)| \leq (\|L_0\|_{X'_0} + t^{-1}\|L_1\|_{X'_1}) \max\{\|x\|_{X_0}, t\|x\|_{X_1}\} \\ &= (\|L_0\|_{X'_0} + t^{-1}\|L_1\|_{X'_1})J(x, t; X_0, X_1). \end{aligned}$$

Taking the infimum over all such decompositions of L gives

$$|L(x)| \leq K(L, t^{-1}; X'_0, X'_1)J(x, t; X_0, X_1).$$

In turn, $L \in (X_0 \cap X_1)'$ with $\|L\|_{(X_0 \cap X_1)'} \leq K(L, t^{-1}; X'_0, X'_1)$.

Conversely, let $L \in (X_0 \cap X_1)'$ and consider the linear function $T : Y \rightarrow \mathbb{R}$ defined by

$$T(x_0, x_1) = L(x_0 + x_1)/2,$$

where Y is the diagonal $Y := \{(x_0, x_1) \in X_0 \times X_1 : x_0 = x_1\}$ and we endow $X_0 \times X_1$ with the norm $\|(x_0, x_1)\|_{X_0 \times X_1} := \max\{\|x_0\|_{X_0}, t\|x_1\|_{X_1}\}$ (see Exercise 16.28). Since $\|(x, x)\|_{X_0 \times X_1} = J(x, t; X_0, X_1)$, we have that

$$|T(x_0, x_1)| \leq \|L\|_{(X_0 \cap X_1)'} \|(x_0, x_1)\|_{X_0 \times X_1}$$

for all $(x_0, x_1) \in Y$. Thus, by the Hahn–Banach theorem (see Theorem A.32) we can extend T to a continuous linear function $T_1 : X_0 \times X_1 \rightarrow \mathbb{R}$ with $\|T_1\|_{(X_0 \times X_1)'} \leq \|L\|_{(X_0 \cap X_1)'}$. By Exercise 16.28 we can find $L_0 \in X'_0$ and

$L_1 \in X'_1$ such that $T_1(x_0, x_1) = L_0(x_0) + L_1(x_1)$ for all $(x_0, x_1) \in X_0 \times X_1$ and $\|T_1\|_{(X_0 \times X_1)'} = \|L_0\|_{X'_0} + t^{-1}\|L_1\|_{X'_1}$. Hence,

$$L(x) = L_0(x) + L_1(x)$$

for every $x \in X_0 \cap X_1$ and $\|L_0\|_{X'_0} + t^{-1}\|L_1\|_{X'_1} \leq \|L\|_{(X_0 \cap X_1)'}$, which shows that $K(L, t^{-1}; X'_0, X'_1) \leq \|L\|_{(X_0 \cap X_1)'}$. $\square$

We are now ready to prove Theorem 16.27.

Proof of Theorem 16.27. Step 1: We will show that $(X'_0, X'_1)_{\sigma, q'} \hookrightarrow (X_0, X_1)'_{\sigma, q}$, where $(X_0, X_1)'_{\sigma, q}$ is the dual of $(X_0, X_1)_{\sigma, q}$. Let $L \in (X'_0, X'_1)_{\sigma, q'}$ and let $u : (0, \infty) \rightarrow X'_0 \cap X'_1$ be such that

$$(16.33) \quad L = \int_0^\infty u(t) \frac{dt}{t},$$

where the Bochner integral is considered with respect to the normed space $X'_0 + X'_1$. By the previous lemma we have that $X'_0 + X'_1$ coincides with $(X_0 \cap X_1)'$ and for every $t > 0$,

$$J(u(t), t; X'_0, X'_1) = \sup\{|u(t)(x)| : x \in X_0 + X_1, K(x, t^{-1}; X_0, X_1) = 1\}.$$

Since $(X_0, X_1)_{\sigma, q} \hookrightarrow X_0 + X_1$ (see Theorem 16.5), it follows that for every $x \in (X_0, X_1)_{\sigma, q}$,

$$|u(t)(x)| \leq J(u(t), t; X'_0, X'_1) K(x, t^{-1}; X_0, X_1).$$

In turn, by (16.33) and Hölder's inequality,

$$\begin{aligned} |L(x)| &\leq \int_0^\infty |u(t)(x)| \frac{dt}{t} \leq \int_0^\infty J(u(t), t; X'_0, X'_1) K(x, t^{-1}; X_0, X_1) \frac{dt}{t} \\ &\leq \left(\int_0^\infty (J(u(t), t; X'_0, X'_1))^{q'} \frac{dt}{t^{1+\sigma q'}} \right)^{1/q'} \left(\int_0^\infty (K(x, t^{-1}; X_0, X_1))^q \frac{dt}{t^{1-\sigma q}} \right)^{1/q}. \end{aligned}$$

By taking the infimum over all representations (16.33) (see (16.23)) and by the change of variables $t = \tau^{-1}$ in the last integral, we obtain

$$|L(x)| \leq \|L\|_{\sigma, q', J} \|x\|_{\sigma, q}$$

for every $x \in (X_0, X_1)_{\sigma, q}$. This shows that L belongs to the dual $(X_0, X_1)'_{\sigma, q}$ of $(X_0, X_1)_{\sigma, q}$ with

$$\|L\|_{(X_0, X_1)'_{\sigma, q}} \leq \|L\|_{\sigma, q', J}.$$

It follows by Theorem 16.21 that $\|L\|_{(X_0, X_1)'_{\sigma, q}} \leq c\|L\|_{\sigma, q'}$.

Step 2: We will show that $(X_0, X_1)'_{\sigma, q} \hookrightarrow (X'_0, X'_1)_{\sigma, q'}$. Let $L \in (X_0, X_1)'_{\sigma, q}$. Since $X_0 \cap X_1 \hookrightarrow (X_0, X_1)_{\sigma, q}$ by Theorem 16.5, it follows that the restriction of L to $X_0 \cap X_1$ is linear and continuous and thus belongs to $(X_0 \cap X_1)'$. Hence, by the previous lemma, for every $t > 0$,

$$K(L, t; X'_0, X'_1) = \sup\{|L(x)| : x \in X_0 \cap X_1, J(x, t^{-1}; X_0, X_1) = 1\}.$$

In turn, for every $\varepsilon > 0$ and $k \in \mathbb{Z}$ we can find $x_k \in X_0 \cap X_1$ such that $J(x_k, 2^k; X_0, X_1) = 1$ and

$$(16.34) \quad L(x_k) \geq K(L, 2^{-k}; X'_0, X'_1) - \varepsilon \min\{1, 2^{-k}\}.$$

Let $a := \{a_k\}_{k \in \mathbb{Z}}$ be a sequence of nonnegative numbers such that the sequence $\{2^{-\sigma k} a_k\}_{k \in \mathbb{Z}}$ belongs to $L^q(\mathbb{Z}, \mathcal{H}^0)$, where $\mathcal{H}^0$ is the counting measure. Define $z_a := \sum_k a_k x_k$. By the properties of a norm, $J(a_k x_k, 2^k; X_0, X_1) = a_k J(x_k, 2^k; X_0, X_1) = a_k$ and so

$$\|\{2^{-k\sigma} J(a_k x_k, 2^k)\}_{k \in \mathbb{Z}}\|_{L^q(\mathbb{Z}, \mathcal{H}^0)} = \|\{2^{-k\sigma} a_k\}_{k \in \mathbb{Z}}\|_{L^q(\mathbb{Z}, \mathcal{H}^0)}.$$

It follows from Exercise 16.24 (see also Step 3 of the proof of Theorem 16.21) that z_a belongs to $(X_0, X_1)_{\sigma, q}$ with

$$\|z_a\|_{\sigma, q, J} \leq \|\{2^{-k\sigma} a_k\}_{k \in \mathbb{Z}}\|_{L^q(\mathbb{Z}, \mathcal{H}^0)}.$$

Using the fact that $L \in (X_0, X_1)'_{\sigma, q}$, the linearity of L , (16.34), and the previous inequality, we get

$$\begin{aligned} \|L\|_{(X_0, X_1)'_{\sigma, q}} \|\{2^{-k\sigma} a_k\}_{k \in \mathbb{Z}}\|_{L^q(\mathbb{Z}, \mathcal{H}^0)} &\geq \|L\|_{(X_0, X_1)'_{\sigma, q}} \|z_a\|_{\sigma, q, J} \\ &\geq L(z_a) = \sum_k a_k L(x_k) \\ &\geq \sum_k a_k K(L, 2^{-k}; X'_0, X'_1) - \varepsilon \sum_k a_k \min\{1, 2^{-k}\}. \end{aligned}$$

Since $\sum_k a_k \min\{1, 2^{-k}\} < \infty$ (why?), letting $\varepsilon \rightarrow 0^+$ gives

$$\|L\|_{(X_0, X_1)'_{\sigma, q}} \|\{2^{-k\sigma} a_k\}_{k \in \mathbb{Z}}\|_{L^q(\mathbb{Z}, \mathcal{H}^0)} \geq \sum_k a_k K(L, 2^{-k}; X'_0, X'_1)$$

for every sequence $\{a_k\}_{k \in \mathbb{Z}}$ of nonnegative numbers such that $\{2^{-\sigma k} a_k\}_{k \in \mathbb{Z}}$ belongs to $L^q(\mathbb{Z}, \mathcal{H}^0)$, or, equivalently,

$$\|L\|_{(X_0, X_1)'_{\sigma, q}} \|\{d_k\}_{k \in \mathbb{Z}}\|_{L^q(\mathbb{Z}, \mathcal{H}^0)} \geq \sum_k d_k 2^{\sigma k} K(L, 2^{-k}; X'_0, X'_1)$$

for every sequence $\{d_k\}_{k \in \mathbb{Z}}$ of nonnegative numbers which belongs to $L^q(\mathbb{Z}, \mathcal{H}^0)$. It follows by Corollary B.81 that

$$\|L\|_{(X_0, X_1)'_{\sigma, q}} \geq \|\{2^{k\sigma} K(L, 2^{-k}; X'_0, X'_1)\}_{k \in \mathbb{Z}}\|_{L^{q'}(\mathbb{Z}, \mathcal{H}^0)}.$$

Using Exercise 16.4 and (16.3) we obtain that

$$c \|L\|_{(X_0, X_1)'_{\sigma, q}} \geq \|L\|_{(X'_0, X'_1)_{\sigma, q'}},$$

which concludes the proof. $\square$

Next we show that if X_0 and X_1 are reflexive Banach spaces, then so is $(X_0, X_1)_{\sigma, q}$ for $q < \infty$. For simplicity, we consider here a very special case, which is all we need in the next chapter.

Theorem 16.30 (Reflexivity). *Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be an admissible pair of Banach spaces with X_0 and X_1 reflexive and assume that $X_1 \hookrightarrow X_0$ and that X_1 is dense in X_0 . Then for $1 \leq q < \infty$ and $0 < \sigma < 1$, the space $(X_0, X_1)_{\sigma, q}$ is reflexive.*

Proof. Step 1: Let X_0^* and X_1^* be the spaces introduced before Theorem 16.27. Since $X_1 \hookrightarrow X_0$, we have that $X_1 \cap X_0 = X_1$ and so $X_1^* = X_1'$. We claim that X_0^* is dense in X_1' . In view of the second geometric form of the Hahn–Banach theorem (Theorem A.35), it is enough to show that if $T \in X_1''$ is such that $T(L^*) = 0$ for all $L^* \in X_0^*$, then $T = 0$. Since X_1 is reflexive, there exists $x_1 \in X_1$ such that $T(L) = L(x_1)$ for all $L \in X_1'$. It follows by the hypothesis on T that $L^*(x_1) = 0$ for $L^* \in X_0^*$. Using the fact that $X_1 \hookrightarrow X_0$, we have that $x_1 \in X_0$. Moreover, every $L^* \in X_0^*$ can be uniquely extended to a linear continuous function $L \in X_0'$ with $\|L\|_{X_0'} = \|L^*\|_{X_0^*}$, where $\|L^*\|_{X_0^*}$ is defined in (16.31). Thus, $L(x_1) = 0$ for every $L \in X_0'$. Again by the second geometric form of the Hahn–Banach theorem (Theorem A.35) this implies that $x_1 = 0$, and in turn, that $T = 0$.

Step 2: By Theorem 16.27, the dual of $(X_0, X_1)_{\sigma, q}$ is given by $(X_0^*, X_1^*)_{\sigma, q'}$. In turn, by Step 1, we are in a position to apply Theorem 16.27 once more, to conclude that the dual of $(X_0^*, X_1^*)_{\sigma, q'}$ is given by $((X_0^*)^*, (X_1^*)^*)_{\sigma, q}$. By Step 1, $(X_0^*)^*$ and $(X_1^*)^*$ can be identified with $(X_0')'$ and $(X_1')'$. On the other hand, by the density of X_1 in X_0 , we have that X_0^* and X_1^* can be identified with X_0' and X_1' , respectively. Hence, $(X_0^*)^*$ and $(X_1^*)^*$ can be identified with X_0'' and X_1'' , respectively. Since X_0 and X_1 are reflexive, it follows that $((X_0^*)^*, (X_1^*)^*)_{\sigma, q}$ can be identified with $(X_0, X_1)_{\sigma, q}$. This completes the proof. $\square$

Remark 16.31. More generally, if $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ are an admissible pair of Banach spaces with $X_0 \cap X_1$ dense in X_0 and X_1 , one can show that for $1 \leq q < \infty$, $(X_0, X_1)_{\sigma, q}$ is reflexive if and only if the inclusion $i : X_0 \cap X_1 \rightarrow X_0 + X_1$ is weakly compact (see, e.g., [21] and [152]). We will not need this result.

16.4. Lorentz Spaces as Interpolation Spaces

As an application of the interpolation theory, we prove that the interpolation space between $L^1(E)$ and $L^\infty(E)$ is the Lorentz space $L^{\sigma m, p}(E)$ (see Section 15.2 in Chapter 15).

Theorem 16.32. *Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set, let $1 < p < \infty$, and let $1 \leq q \leq \infty$. Then*

$$(L^1(E), L^\infty(E))_{\sigma, q} = L^{p, q}(E),$$

where $\sigma := 1 - 1/p$. Moreover, a norm in $L^{p,q}(E)$ is given by

$$\|u\|_{\sigma,q} = \left(\int_0^\infty \left(\int_0^t u^*(\tau) d\tau \right)^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q},$$

where u^* is the decreasing rearrangement of u .

Proof. The fact that $L^1(E)$ and $L^\infty(E)$ are an admissible pair follows from Exercise B.105. Given a Lebesgue measurable function $u : E \rightarrow \mathbb{R}$ and $t > 0$, we claim that

$$(16.35) \quad K(u, t) = \int_0^t u^*(\tau) d\tau.$$

We begin by proving $\int_0^t u^* d\tau \leq K(u, t)$. Without loss of generality, we may assume that $K(u, t) < \infty$ (see Remark 16.3), so that $u \in L^1(E) + L^\infty(E)$. Write $u = v + w$, where $v \in L^1(E)$ and $w \in L^\infty(E)$. Then by Exercise 15.17, a change of variables and the fact that w^* is decreasing

$$\begin{aligned} \int_0^t u^*(\tau) d\tau &\leq \int_0^t v^*((1-\varepsilon)\tau) d\tau + \int_0^t w^*(\varepsilon\tau) d\tau \\ &\leq (1-\varepsilon)^{-1} \int_0^\infty v^*(r) dr + t w^*(0) \\ &= (1-\varepsilon)^{-1} \|v\|_{L^1(E)} + t \|w\|_{L^\infty(E)}, \end{aligned}$$

where the last equality follows from Corollary 4.7 and Theorem 4.16 (which continue to hold without changes when $N \geq 1$). Letting $\varepsilon \rightarrow 0^+$ shows that $\int_0^t u^* d\tau \leq \|v\|_{L^1(E)} + t \|w\|_{L^\infty(E)}$. Since this holds for every decomposition of u , it follows that $\int_0^t u^* d\tau \leq K(u, t)$.

To prove the converse inequality, assume that $\int_0^t u^* d\tau < \infty$ and define

$$v(x) := \max\{|u(x)| - u^*(t), 0\} \operatorname{sgn} u(x), \quad w(x) := u(x) - v(x).$$

Let $E_t := \{x \in E : |u(x)| > u^*(t)\}$ and $F_t := \{\tau > 0 : u^*(\tau) > u^*(t)\}$. By Proposition 15.2(iv), $\mathcal{L}^N(E_t) \leq t$ and u^* is constant on $[\mathcal{L}^N(E_t), t]$. Hence,

$$\begin{aligned} \|v\|_{L^1(E)} &= \int_{E_t} (|u(x)| - u^*(t)) dx = \int_0^{\mathcal{L}^N(E_t)} (u^*(\tau) - u^*(t)) d\tau \\ &\leq \int_0^t (u^*(\tau) - u^*(t)) d\tau, \end{aligned}$$

where in the second equality we used Proposition 15.2(v) and Theorem B.60. On the other hand, $|w(x)| = u^*(t)$ in E_t and $|w(x)| \leq u^*(t)$ outside E_t . It follows that

$$\|v\|_{L^1(E)} + t \|w\|_{L^\infty(E)} \leq \int_0^t (u^*(\tau) - u^*(t)) d\tau + t u^*(t) = \int_0^t u^*(\tau) d\tau,$$

which shows that $K(u, t) \leq \int_0^t u^* d\tau$. This proves (16.35).

In turn, since u^* is decreasing, $K(u, t) \geq tu^*(t)$ and so if $q < \infty$, using the fact that $\sigma = 1 - 1/p$,

$$\|u\|_{\sigma, q}^q = \int_0^\infty (K(x, t))^q \frac{dt}{t^{1+\sigma q}} \geq \int_0^\infty (tu^*(t))^q \frac{dt}{t^{1+\sigma q}} = \|u\|_{L^{p, q}(E)}^q.$$

On the other hand, by Hardy's inequality (see Theorem C.41), (16.35), and again the fact that $\sigma = 1 - 1/p$,

$$\begin{aligned} \|u\|_{\sigma, q} &= \left(\int_0^\infty (K(x, t))^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} = \left(\int_0^\infty \left(\int_0^t u^* d\tau \right)^q \frac{dt}{t^{1+\sigma q}} \right)^{1/q} \\ &\leq \sigma \left(\int_0^\infty (u^*(t))^q \frac{dt}{t^{1+\sigma q - q}} \right)^{1/q} = \sigma \|u\|_{L^{p, q}(E)}. \end{aligned}$$

The case $q = \infty$ is similar and is left as an exercise. $\square$

Exercise 16.33. Let $(X, \mathfrak{M}, \mu)$ be a measure space. Extend the previous theorem to the case in which the Lebesgue measurable set $E \subseteq \mathbb{R}^N$ and the Lebesgue measure $\mathcal{L}^N$ are replaced by a set $E \in \mathfrak{M}$ of X and by μ , respectively. Do you need any hypotheses on μ ?

Exercise 16.34. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set, let $1 \leq q \leq \infty$, and let $0 < \sigma < 1$. Prove that if $p_0 \neq p_1$,

$$(L^{p_0, q_0}(E), L^{p_1, q_1}(E))_{\sigma, q} = L^{p, q}(E),$$

with equivalent norms, where $\frac{1}{p} = \frac{1-\sigma}{p_0} + \frac{\sigma}{p_1}$ and deduce that

$$(L^{p_0}(E), L^{p_1}(E))_{\sigma, q} = L^{p, q}(E).$$

Exercise 16.35. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set, let $1 < p < r < s < \infty$ and let $1 \leq q \leq \infty$. Prove that if $u \in L^{p, \infty}(E) \cap L^{s, \infty}(E)$, then $u \in L^{r, q}(E)$ and estimate its $L^{r, q}(E)$ norm.

Exercise 16.36. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set, let $1 < p < \infty$ and let $1 \leq q < \infty$. Prove that $L^1(E) \cap L^\infty(E)$ is dense in $L^{p, q}(E)$. Deduce that simple functions are dense in $L^{p, q}(E)$.

Exercise 16.37. Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set, let $1 < p < \infty$ and let $1 \leq q_1 < q_2 < \infty$. Prove that $L^{p, q_1}(E) \hookrightarrow L^{p, q_2}(E)$.

Exercise 16.38. Let $1 < p \leq 2$ and let $1 \leq q \leq \infty$. Prove that for every $u \in L^{p, q}(\mathbb{R}^N)$, its Fourier transform $\widehat{u}$ (see (10.23)) belongs to $L^{p', q}(\mathbb{R}^N)$ with $\|\widehat{u}\|_{L^{p', q}(\mathbb{R}^N)} \leq c\|u\|_{L^{p, q}(\mathbb{R}^N)}$ (compare this with the Hausdorff–Young inequality (10.30)).

Exercise 16.39. Let $1 < r < \infty$, let $1 < p < r'$, let $1 \leq q \leq \infty$, and let $\frac{1}{s} = \frac{1}{p} + \frac{1}{r} - 1$. Prove that if $u \in L^{p, q}(\mathbb{R}^N)$ and $v \in L^r(\mathbb{R}^N)$, then $u * v$ belongs to $L^{s, q}(\mathbb{R}^N)$ with $\|u * v\|_{L^{s, q}(\mathbb{R}^N)} \leq c\|u\|_{L^{p, q}(\mathbb{R}^N)}\|v\|_{L^r(\mathbb{R}^N)}$ (compare this inequality with Young's inequality, Theorem 10.49).

Besov Spaces

Another principal concept in Quantum Gradnamics is the observation that graduate students do not move toward graduation in a steady and continuous manner. Rather they make progress through discrete bursts of random productivity called “wanta” (short for “want data”) whose energy is proportional to the frequency of meetings with their advisor.

— Jorge Cham, www.phdcomics.com

17.1. Besov Spaces $B_q^{s,p}$

In this section we define the Besov spaces $B_q^{s,p}(\mathbb{R}^N)$. Given a function $u : \mathbb{R}^N \rightarrow \mathbb{R}$, for every $h \in \mathbb{R}^N$, $m \in \mathbb{N}$, and $x \in \mathbb{R}^N$, we define inductively the forward difference operator

$$(17.1) \quad \Delta_h^1 u(x) := \Delta_h u(x) := u(x+h) - u(x)$$

for $m = 1$ and

$$(17.2) \quad \Delta_h^m u(x) := \Delta_h(\Delta_h^{m-1} u(x))$$

for $m \geq 2$. In particular,

$$(17.3) \quad \Delta_h^2 u(x) = u(x+2h) - 2u(x+h) + u(x).$$

We recall that the symbol $\lfloor s \rfloor$ stands for the integer part of s .

Definition 17.1. Let $1 \leq p, q \leq \infty$ and let $s > 0$. A Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ belongs to the *Besov space* $B_q^{s,p}(\mathbb{R}^N)$ if

$$(17.4) \quad \|u\|_{B_q^{s,p}(\mathbb{R}^N)} := \|u\|_{L^p(\mathbb{R}^N)} + |u|_{B_q^{s,p}(\mathbb{R}^N)} < \infty,$$

where for $q < \infty$,

$$(17.5) \quad |u|_{B_q^{s,p}(\mathbb{R}^N)} := \left(\int_{\mathbb{R}^N} \|\Delta_h^{\lfloor s \rfloor + 1} u\|_{L^p(\mathbb{R}^N)}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q},$$

while for $q = \infty$,

$$(17.6) \quad |u|_{B_{\infty}^{s,p}(\mathbb{R}^N)} := \sup_{h \in \mathbb{R}^N \setminus \{0\}} \frac{1}{\|h\|^s} \|\Delta_h^{\lfloor s \rfloor + 1} u\|_{L^p(\mathbb{R}^N)}.$$

Remark 17.2. (i) Given a Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$, to see that the seminorm $|u|_{B_q^{s,p}(\mathbb{R}^N)}$ is well-defined, consider a representative $\bar{u}$ of u that is Borel measurable and let $w : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}$ be the function defined by

$$w(x, h) := \bar{u}(x + h).$$

Then w is a Borel function, since it is the composition of $\bar{u}$ with the continuous function $g : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}^N$ given by $g(x, h) := x + h$. Thus, $(x, h) \mapsto \Delta_h^1 u(x)$ is Lebesgue measurable and, in turn, so is $(x, h) \mapsto \Delta_h^m u(x)$.

(ii) Observe that given a function $u \in L_{\text{loc}}^1(\mathbb{R}^N)$, we have that

$$\|\Delta_h u\|_{L^p(\mathbb{R}^N)} = 0$$

if and only if u is equivalent to a constant. More generally,

$$\|\Delta_h^m u\|_{L^p(\mathbb{R}^N)} = 0$$

if and only if u has a representative that is a polynomial of degree less than m (see (17.25) below).

(iii) In the sequel we will often use the fact that if $u \in L^p(\mathbb{R}^N)$, $1 \leq p \leq \infty$, then for every $h \in \mathbb{R}^N$, and $m \in \mathbb{N}$,

$$(17.7) \quad \|\Delta_h^m u\|_{L^p(\mathbb{R}^N)} \leq 2^m \|u\|_{L^p(\mathbb{R}^N)}.$$

This follows by induction on m from (17.1), (17.2), Minkowski's inequality, and the change of variables $y = x + h$.

(iv) If $u : \mathbb{R}^N \rightarrow \mathbb{R}$ is Lebesgue measurable, by a change of variables, we have that

$$\|\Delta_h^2 u\|_{L^p(\mathbb{R}^N)} = \|\tilde{\Delta}_h^2 u\|_{L^p(\mathbb{R}^N)},$$

where for $h \in \mathbb{R}^N$ and $x \in \mathbb{R}^N$,

$$(17.8) \quad \tilde{\Delta}_h^2 u(x) := u(x + h) - 2u(x) + u(x - h).$$

If $q = p$, we write

$$(17.9) \quad B^{s,p}(\mathbb{R}^N) := B_p^{s,p}(\mathbb{R}^N).$$

Remark 17.3. If $p = q < \infty$ and $0 < s < 1$, then (17.5) takes the simpler form

$$|u|_{B^{s,p}(\mathbb{R}^N)} = \left(\int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \frac{|u(x + h) - u(x)|^p}{\|h\|^{N+sp}} dx dh \right)^{1/p},$$

while if $p = q < \infty$ and $s = 1$,

$$|u|_{B^{1,p}(\mathbb{R}^N)} = \left(\int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \frac{|u(x+h) - 2u(x) + u(x-h)|^p}{\|h\|^{N+p}} dx dh \right)^{1/p}.$$

On the other hand, if $p = q = \infty$ and $0 < s < 1$, then

$$|u|_{B^{s,\infty}(\mathbb{R}^N)} = \sup_{h \in \mathbb{R}^N} \operatorname{ess\,sup}_{x \in \mathbb{R}^N} \frac{|u(x+h) - u(x)|}{\|h\|^s},$$

and thus a Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ is such that $|u|_{B^{s,\infty}(\mathbb{R}^N)} < \infty$ if and only if it has a Hölder continuous representative of exponent s . In particular, $u \in L^\infty(\mathbb{R}^N)$ belongs to $B^{s,\infty}(\mathbb{R}^N)$ if and only if it has a representative in $C^{0,s}(\mathbb{R}^N)$.

If $p = q = 1$, then

$$|u|_{B^{1,\infty}(\mathbb{R}^N)} = \sup_{h \in \mathbb{R}^N} \operatorname{ess\,sup}_{x \in \mathbb{R}^N} \frac{|u(x+h) - 2u(x) + u(x-h)|}{\|h\|}.$$

The space $B^{1,\infty}(\mathbb{R}^N)$ is known as the *Zygmund space* $\Lambda_1(\mathbb{R}^N)$. The next exercise shows that a function in $B^{1,\infty}(\mathbb{R}^N)$ admits a representative which is Hölder continuous of exponent s for every $0 < s < 1$.

Exercise 17.4. Let $u \in B^{1,\infty}(\mathbb{R}^N)$. Prove that for a precise representative of u ,

(i) for every $n \in \mathbb{N}$ and every $x, y \in \mathbb{R}^N$,

$$|2^n \Delta_{2^{-n}y} u(x) - 2^{n-1} \Delta_{2^{-n+1}y} u(x)| \leq |u|_{B^{1,\infty}(\mathbb{R}^N)} \|y\|,$$

(ii) for every $l \in \mathbb{N}$ and every $x, h \in \mathbb{R}^N$, with $2^{-l-1} \leq \|h\| < 2^{-l}$,

$$|\Delta_h u(x)| \leq 2^{-l+1} \|u\|_{L^\infty(\mathbb{R}^N)} + l |u|_{B^{1,\infty}(\mathbb{R}^N)} \|h\|,$$

(iii) there exists a constant $c > 0$ independent of u such that

$$|u(x+h) - u(x)| \leq c \|u\|_{B^{1,\infty}(\mathbb{R}^N)} \|h\| (1 + |\log \|h\||)$$

for every $x \in \mathbb{R}^N$ and $h \in \mathbb{R}^N \setminus \{0\}$. Deduce that $B^{1,\infty}(\mathbb{R}^N) \hookrightarrow B^{s,\infty}(\mathbb{R}^N)$ for $0 < s < 1$.

Note that $\Lambda_1(\mathbb{R}^N) \subset C^{0,1}(\mathbb{R}^N)$ with strict inclusion. Indeed, Lipschitz continuous functions are differentiable $\mathcal{L}^N$ -a.e. in their domain by Rademacher's theorem (see Theorem 9.14), while the Weierstrass nowhere differentiable function (1.2) belongs to $\Lambda_1(\mathbb{R})$ (exercise).

The next exercise shows that without measurability, there are everywhere discontinuous functions with $\Delta_h^2 u(x) \equiv 0$.

Exercise 17.5. Prove that there exists an everywhere discontinuous function $u : \mathbb{R} \rightarrow \mathbb{R}$ such that $u(x+h) - 2u(x) + u(x-h) = 0$ for every $x, h \in \mathbb{R}$. Hint: Consider $\mathbb{R}$ as a vector space over the rationals and take a Hamel basis.

Next we show that $B_q^{s,p}(\mathbb{R}^N)$ is a Banach space.

Proposition 17.6 (Completeness). *Let $1 \leq p, q \leq \infty$ and $s > 0$. Then the Besov space $B_q^{s,p}(\mathbb{R}^N)$ is a Banach space.*

Proof. We prove the result for $q < \infty$ and leave the case $q = \infty$ as an exercise. Let $\{u_n\}_n$ be a Cauchy sequence in $B_q^{s,p}(\mathbb{R}^N)$. Then by (17.4), $\{u_n\}_n$ is a Cauchy sequence of $L^p(\mathbb{R}^N)$ and so it converges to a function u in $L^p(\mathbb{R}^N)$. On the other hand, as in the proof of Theorem 8.28, the functions $v_n : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}$, defined by

$$v_n(x, h) := \frac{1}{\|h\|^{N/q+s}} \Delta_h^m u_n(x),$$

can be identified with functions in $L^q(\mathbb{R}^N; L^p(\mathbb{R}^N))$. It follows from (17.4) that $\{v_n\}_n$ is a Cauchy sequence in $L^q(\mathbb{R}^N; L^p(\mathbb{R}^N))$. By Theorem 8.15 we may find $v \in L^q(\mathbb{R}^N; L^p(\mathbb{R}^N))$ such that $v_n \rightarrow v$ in $L^q(\mathbb{R}^N; L^p(\mathbb{R}^N))$. Extract a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ such that $u_{n_k}(x) \rightarrow u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ and $v_{n_k}(x, h) \rightarrow v(x, h)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ and $\mathcal{L}^N$ -a.e. $h \in \mathbb{R}^N$. It follows that for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ and $\mathcal{L}^N$ -a.e. $h \in \mathbb{R}^N$ the function $v(x, h)$ coincides with the function

$$w(x, h) := \frac{1}{\|h\|^{N/q+s}} \Delta_h^m u(x).$$

Hence, $u \in B_q^{s,p}(\mathbb{R}^N)$ and $\|u_n - u\|_{B_q^{s,p}} \rightarrow 0$. □

Exercise 17.7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a Lipschitz continuous function and let $u : \mathbb{R}^N \rightarrow \mathbb{R}$ be a Lebesgue measurable function such that $|u|_{B_q^{s,p}(\mathbb{R}^N)} < \infty$ for some $1 \leq p, q \leq \infty$ and $0 < s < 1$. Prove that

$$|f \circ u|_{B_q^{s,p}(\mathbb{R}^N)} \leq \text{Lip } f |u|_{B_q^{s,p}(\mathbb{R}^N)}.$$

Remark 17.8. Given a function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ and $h \in \mathbb{R}^N$ consider the translation operator T_h defined by

$$(17.10) \quad T_h(u)(x) := u(x+h), \quad x \in \mathbb{R}^N.$$

Note that $T_h \circ T_\xi = T_\xi \circ T_h$ for every $h, \xi \in \mathbb{R}^N$. Moreover, $\Delta_h := T_h - I$ and for $m \in \mathbb{N}$, with $m \geq 2$,

$$\Delta_h^m = (T_h - I)^m := \underbrace{(T_h - I) \circ \cdots \circ (T_h - I)}_{m \text{ times}}.$$

Here, I is the identity operator. Hence, by the binomial theorem,

$$(17.11) \quad \Delta_h^m u(x) = \sum_{j=0}^m (-1)^{m-j} \binom{m}{j} u(x + jh), \quad x \in \mathbb{R}^N.$$

In particular, setting $y := x + mh$ and $k := m - j$, we have

$$(17.12) \quad \begin{aligned} \Delta_h^m u(x) &= \sum_{k=0}^m (-1)^k \binom{m}{m-k} u(y - kh) \\ &= (-1)^m \sum_{k=0}^m (-1)^{m-k} \binom{m}{k} u(y - kh) = (-1)^m \Delta_{-h}^m u(x + mh). \end{aligned}$$

Exercise 17.9. Let $1 \leq p < \infty$, $0 < s < N/p$, and $a > 0$ and consider the function

$$u(x) := \|x\|^{s-N/p} |\log \|x\||^{-a}$$

if $x \in \mathbb{R}^N$ with $\|x\| < 1/2$, with u extended to $\mathbb{R}^N$ in a smooth way and with compact support. Find the values of q (if any) for which $u \in B_q^{s,p}(\mathbb{R}^N)$.

In the following exercises, given an open set $\Omega \subseteq \mathbb{R}^N$, $1 \leq p, q < \infty$, and $s > 0$, for a Lebesgue measurable function $u : \Omega \rightarrow \mathbb{R}$ and $1 \leq q < \infty$ we define

$$(17.13) \quad |u|_{B_q^{s,p}(\Omega)} := \left(\int_{\mathbb{R}^N} \|\Delta_h^{\lfloor s \rfloor + 1} u\|_{L^p(\Omega_h)}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q},$$

while for $q = \infty$,

$$(17.14) \quad |u|_{B_\infty^{s,p}(\Omega)} := \sup_{h \in \mathbb{R}^N \setminus \{0\}} \frac{1}{\|h\|^s} \|\Delta_h^{\lfloor s \rfloor + 1} u\|_{L^p(\Omega_h)},$$

where $\Omega_h := \{x \in \Omega : x + h \in \Omega\}$.

Exercise 17.10. Let $0 < s < 1$, let $I = (0, 1)$, and let

$$u(x) = \sum_{n=1}^{\infty} 2^{-ns} n^{-1/2} \cos(2^n x), \quad x \in I.$$

- (i) Prove that $|u|_{B_q^{s,2}(I)} < \infty$ if $q > 2$. Hint: Given h find m such that $2^m |h| \leq 1$ and $2^{m+1} |h| > 1$ and break the series in the sum from 1 to m and from $m+1$ to infinity (see the proof of Theorem 1.15).
- (ii) Prove that $|u|_{B_q^{s,2}(I)} = \infty$ if $q \leq 2$.

Exercise 17.11. Let $1 \leq p, q \leq \infty$ and let $0 < s < 1$. Given a function $u \in L_{\text{loc}}^1(\mathbb{R}_+^N)$ such that $|u|_{B_q^{s,p}(\mathbb{R}_+^N)} < \infty$ (see (17.13) and (17.14)), define

$$v(x) := \begin{cases} u(x', x_N) & \text{if } x_N > 0, \\ u(x', -x_N) & \text{if } x_N < 0. \end{cases}$$

Prove that $|v|_{B_q^{s,p}(\mathbb{R}^N)} \leq c|u|_{B_q^{s,p}(\mathbb{R}_+^N)} < \infty$. Can you do something similar for $s \geq 1$? Hint: See Exercise 13.3.

We conclude this section by discussing the density of functions in $C^\infty(\mathbb{R}^N)$.

Proposition 17.12. *Let $1 \leq p, q \leq \infty$ and $s > 0$. For any $u \in L_{\text{loc}}^1(\mathbb{R}^N)$, let $u_\varepsilon := \varphi_\varepsilon * u$, where φ_ε is a standard mollifier. Then*

$$(17.15) \quad |u_\varepsilon|_{B_q^{s,p}(\mathbb{R}^N)} \leq |u|_{B_q^{s,p}(\mathbb{R}^N)}$$

for all $\varepsilon > 0$ and

$$(17.16) \quad \lim_{\varepsilon \rightarrow 0^+} |u_\varepsilon|_{B_q^{s,p}(\mathbb{R}^N)} = |u|_{B_q^{s,p}(\mathbb{R}^N)}.$$

Moreover, if $1 \leq p, q < \infty$ and $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ is such that $|u|_{B_q^{s,p}(\mathbb{R}^N)} < \infty$, then

$$(17.17) \quad \lim_{\varepsilon \rightarrow 0^+} |u_\varepsilon - u|_{B_q^{s,p}(\mathbb{R}^N)} = 0.$$

In particular, if $1 \leq p, q < \infty$, then $C^\infty(\mathbb{R}^N) \cap B_q^{s,p}(\mathbb{R}^N)$ is dense in $B_q^{s,p}(\mathbb{R}^N)$.

Proof. Given $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ observe that for $m \in \mathbb{N}$, $h \in \mathbb{R}^N$, and $\varepsilon > 0$, $\Delta_h^m u_\varepsilon = \varphi_\varepsilon * \Delta_h^m u$. Hence, by Theorem C.16 we have that

$$(17.18) \quad \|\Delta_h^m u_\varepsilon\|_{L^p} \leq \|\Delta_h^m u\|_{L^p}$$

and

$$(17.19) \quad \|\Delta_h^m u_\varepsilon\|_{L^p} \rightarrow \|\Delta_h^m u\|_{L^p}$$

as $\varepsilon \rightarrow 0^+$. It follows from (17.18) that (17.15) holds. In turn,

$$\limsup_{\varepsilon \rightarrow 0^+} |u_\varepsilon|_{B_q^{s,p}} \leq |u|_{B_q^{s,p}}.$$

To prove the opposite inequality, assume first that $q < \infty$. For $h \in \mathbb{R}^N$ and $\varepsilon > 0$ we define

$$f_\varepsilon(h) := \frac{1}{\|h\|^{N+sq}} \|\Delta_h^m u_\varepsilon\|_{L^p}^q, \quad f(h) := \frac{1}{\|h\|^{N+sq}} \|\Delta_h^m u\|_{L^p}^q.$$

Since $f_\varepsilon(h) \rightarrow f(h)$ for every h by (17.19), by Fatou's lemma we have that

$$\int_{\mathbb{R}^N} f(h) dh = \int_{\mathbb{R}^N} \lim_{\varepsilon \rightarrow 0^+} f_\varepsilon(h) dh \leq \liminf_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^N} f_\varepsilon(h) dh.$$

Thus, (17.16) holds.

If $q = \infty$, then since $\|\Delta_h^m u_\varepsilon\|_{L^p} \rightarrow \|\Delta_h^m u\|_{L^p}$ for all $h \in \mathbb{R}^N$ by (17.19), we have that

$$\frac{1}{\|h\|^s} \|\Delta_h^m u\|_{L^p} = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\|h\|^s} \|\Delta_h^m u_\varepsilon\|_{L^p} \leq \liminf_{\varepsilon \rightarrow 0^+} \sup_{h \in \mathbb{R}^N} \frac{1}{\|h\|^s} \|\Delta_h^m u_\varepsilon\|_{L^p}$$

for all $h \in \mathbb{R}^N$. It follows that

$$\sup_{h \in \mathbb{R}^N} \frac{1}{\|h\|^s} \|\Delta_h^m u\|_{L^p} \leq \liminf_{\varepsilon \rightarrow 0^+} \sup_{h \in \mathbb{R}^N} \frac{1}{\|h\|^s} \|\Delta_h^m u_\varepsilon\|_{L^p},$$

and so (17.16) is satisfied.

Assume next that $p < \infty$, $q < \infty$, and $u \in L^1_{\text{loc}}(\mathbb{R}^N)$ is such that $|u|_{B_q^{s,p}(\mathbb{R}^N)} < \infty$. By Theorem C.16(iv),

$$(17.20) \quad \lim_{\varepsilon \rightarrow 0^+} \|\Delta_h^m u_\varepsilon - \Delta_h^m u\|_{L^p} = 0.$$

For $h > 0$ and $\varepsilon > 0$ define the functions

$$g_\varepsilon(h) := \frac{1}{\|h\|^{N+sq}} \|\Delta_h^m u_\varepsilon - \Delta_h^m u\|_{L^p}^q, \quad g(h) := \frac{1}{\|h\|^{N+sq}} \|\Delta_h^m u\|_{L^p}^q.$$

By hypothesis $g \in L^1(\mathbb{R}^N)$, and by (17.18), Minkowski's inequality, and the convexity of the function $|y|^q$ we have that $g_\varepsilon(h) \leq 2^q g(h)$ for all h . Since $g_\varepsilon(h) \rightarrow 0$ for all h by (17.20), we are in a position to apply the Lebesgue dominated convergence theorem to conclude that (17.17) holds. $\square$

Remark 17.13. If $u \in B_q^{s,p}(\mathbb{R}^N)$ with $1 \leq p, q < \infty$, in the previous theorem, instead of taking $\varphi \in C_c^\infty(\mathbb{R}^N)$, it is enough to consider $\varphi \in \mathcal{S}(\mathbb{R}^N)$.

Exercise 17.14. Let $u \in L^1_{\text{loc}}(\mathbb{R})$ be such that

$$\int_{\mathbb{R}^2} \frac{|u(x) - u(y)|}{|x - y|^2} dx dy < \infty.$$

Prove that u is constant. Hint: Let $g(t)$ be the integral over the set $\{(x, y) \in \mathbb{R}^2 : |x - y| \leq t\}$ and show that $g(t) \leq g(t/2)$.

Exercise 17.15. Let $1 \leq p, q \leq \infty$ and let $0 < s < 1$. Prove that if $u \in B_q^{s,p}(\mathbb{R}^N)$ and $\psi \in C^{0,1}(\mathbb{R}^N)$, then $\psi u \in B_q^{s,p}(\mathbb{R}^N)$ and estimate $\|\psi u\|_{B_q^{s,p}(\mathbb{R}^N)}$ in terms of $\|u\|_{B_q^{s,p}(\mathbb{R}^N)}$ and $\|\psi\|_{C^{0,1}(\mathbb{R}^N)}$.

Exercise 17.16. Let $1 \leq p, q \leq \infty$ and let $s > 0$. Prove that if $u \in \mathcal{S}(\mathbb{R}^N)$, then $u \in B_q^{s,p}(\mathbb{R}^N)$.

17.2. Some Equivalent Seminorms

In this section we introduce some equivalent seminorms, which are useful in different applications. We begin by showing that in the definition of $B_q^{s,p}$, one can replace $\Delta_h^{[s]+1}$ with Δ_h^n for any $n \in \mathbb{N}$ with $n > [s] + 1$. To be precise, let $1 \leq p, q \leq \infty$, let $s > 0$, and let $n \in \mathbb{N}$ be such that $n \geq [s] + 1$. For every Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ if $q < \infty$ define

$$(17.21) \quad |u|_{B_q^{s,p}(\mathbb{R}^N)}^{(n)} := \left(\int_{\mathbb{R}^N} \|\Delta_h^n u\|_{L^p(\mathbb{R}^N)}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q},$$

while for $q = \infty$,

$$|u|_{B_{\infty}^{s,p}(\mathbb{R}^N)}^{(n)} := \sup_h \frac{1}{\|h\|^s} \|\Delta_h^n u\|_{L^p(\mathbb{R}^N)}.$$

Proposition 17.17. *Let $1 \leq p, q \leq \infty$, let $s > 0$, and let $n \in \mathbb{N}$ be such that $n \geq \lfloor s \rfloor + 1$. Then there exist two constants $c_1, c_2 > 0$, depending only on n, N, p, q , and s , such that for every Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ with $|u|_{B_q^{s,p}(\mathbb{R}^N)} < \infty$,*

$$c_1 |u|_{B_q^{s,p}(\mathbb{R}^N)}^{(n)} \leq |u|_{B_q^{s,p}(\mathbb{R}^N)} \leq c_2 |u|_{B_q^{s,p}(\mathbb{R}^N)}^{(n)}.$$

We begin with a preliminary result.

Lemma 17.18. *Given $u : \mathbb{R}^N \rightarrow \mathbb{R}$, $h \in \mathbb{R}^N \setminus \{0\}$, and $n \in \mathbb{N}$, the following identity holds:*

$$(17.22) \quad \Delta_h^n u = 2^{-n} \Delta_{2h}^n u + P_{n-1}(T_h) \circ \Delta_h^{n+1} u,$$

where T_h is the translation operator given in (17.10) and P_{n-1} is a polynomial of degree $n - 1$.

Proof. We use the following identity for polynomials

$$\begin{aligned} (t-1)^n &= 2^{-n}(t^2-1)^n + (t-1)^n - 2^{-n}(t^2-1)^n \\ &= 2^{-n}(t^2-1)^n + P_{n-1}(t)(t-1)^{n+1}, \quad t \in \mathbb{R}, \end{aligned}$$

where by the binomial theorem,

$$\begin{aligned} P_{n-1}(t) &= -2^{-n} \frac{(t+1)^n - 2^n}{t-1} = -2^{-n} \frac{((t-1)+2)^n - 2^n}{t-1} \\ &= - \sum_{j=1}^n 2^{-j} \binom{n}{j} (t-1)^{j-1}, \quad t \in \mathbb{R}. \end{aligned}$$

By replacing t with the translation operator T_h given in (17.10) we get (17.22). $\square$

Exercise 17.19. Let $u : \mathbb{R}^N \rightarrow \mathbb{R}$, $h \in \mathbb{R}^N \setminus \{0\}$, and $n \in \mathbb{N}$.

(i) Prove that there exists a polynomial P such that for every $t \in \mathbb{R}$,

$$(t-1)^n = -2^{-2n-1}(t^4-1)^n + 2^{-n-1}3(t^2-1)^n + P(t)(t-1)^{n+2}.$$

(ii) Prove that

$$\Delta_h^n u = -2^{-2n-1} \Delta_{4h}^n u + 2^{-n-1} 3 \Delta_{2h}^n u + P(T_h) \circ \Delta_h^{n+2} u.$$

We turn to the proof of Proposition 17.17.

Proof of Proposition 17.17. Using an induction argument, it is enough to prove that the seminorms $|u|_{B_q^{s,p}}^{(n)}$ and $|u|_{B_q^{s,p}}^{(n+1)}$ are equivalent for some $n \geq \lfloor s \rfloor + 1$ whenever $|u|_{B_q^{s,p}}^{(n)} < \infty$. By (17.7) for $m = 1$ we have that

$$\|\Delta_h^{n+1}u\|_{L^p} = \|\Delta_h^1(\Delta_h^n u)\|_{L^p} \leq 2\|\Delta_h^n u\|_{L^p}.$$

Thus, it remains to prove the opposite inequality.

Step 1: We consider first the case $0 < s < 1$ and $n = 1$. Fix a Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ with $|u|_{B_q^{s,p}} < \infty$. For $x \in \mathbb{R}^N$ and $h \in \mathbb{R}^N$, we have

$$\begin{aligned} 2(u(x+h) - u(x)) &= (u(x+2h) - u(x)) \\ &\quad - (u(x+2h) - 2u(x+h) + u(x)), \end{aligned}$$

and so

$$(17.23) \quad 2\Delta_h^1 u(x) = \Delta_{2h}^1 u(x) - \Delta_h^2 u(x).$$

If $q < \infty$, it follows by Minkowski's inequality that

$$\begin{aligned} 2 \left(\int_{\mathbb{R}^N} \|\Delta_h^1 u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q} &\leq \left(\int_{\mathbb{R}^N} \|\Delta_{2h}^1 u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q} \\ &\quad + \left(\int_{\mathbb{R}^N} \|\Delta_h^2 u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q} = 2^s \left(\int_{\mathbb{R}^N} \|\Delta_\eta^1 u\|_{L^p}^q \frac{d\eta}{\|\eta\|^{N+sq}} \right)^{1/q} \\ &\quad + \left(\int_{\mathbb{R}^N} \|\Delta_h^2 u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q}, \end{aligned}$$

where we have made the change of variables $h = \eta/2$. The case $q = \infty$ is similar. Since $|u|_{B_q^{s,p}} < \infty$, we have

$$(2 - 2^s)|u|_{B_q^{s,p}} \leq |u|_{B_q^{s,p}}^{(2)}.$$

Step 2: Assume next that $n \geq 2$. Fix a Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ with $|u|_{B_q^{s,p}}^{(n)} < \infty$. By (17.22) we get

$$\begin{aligned} \|\Delta_h^n u\|_{L^p} &\leq 2^{-n} \|\Delta_{2h}^n u\|_{L^p} + \|P_{n-1}(T_h) \circ \Delta_h^{n+1} u\|_{L^p} \\ &\leq 2^{-n} \|\Delta_{2h}^n u\|_{L^p} + c \|\Delta_h^{n+1} u\|_{L^p}, \end{aligned}$$

where we have used the fact that $\|T_h(v)\|_{L^p} = \|v\|_{L^p}$. If $q < \infty$, it follows by Minkowski's inequality that

$$\begin{aligned} \left(\int_{\mathbb{R}^N} \|\Delta_h^n u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q} &\leq 2^{-n} \left(\int_{\mathbb{R}^N} \|\Delta_{2h}^n u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q} \\ &\quad + c \left(\int_{\mathbb{R}^N} \|\Delta_h^{n+1} u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q} \\ &= 2^{s-n} \left(\int_{\mathbb{R}^N} \|\Delta_\eta^n u\|_{L^p}^q \frac{d\eta}{\|\eta\|^{N+sq}} \right)^{1/q} \\ &\quad + c \left(\int_{\mathbb{R}^N} \|\Delta_h^{n+1} u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q}, \end{aligned}$$

where we have made the change of variables $h = \eta/2$. Therefore, since $|u|_{B_q^{s,p}}^{(n)} < \infty$ we get

$$\begin{aligned} (1 - 2^{s-n}) \left(\int_{\mathbb{R}^N} \|\Delta_h^n u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q} \\ \leq c \left(\int_{\mathbb{R}^N} \|\Delta_h^{n+1} u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q}. \end{aligned}$$

Note that here it is important that $n \geq [s] + 1 > s$. The case $q = \infty$ is similar. $\square$

Remark 17.20. It is important to observe that the restriction $|u|_{B_q^{s,p}(\mathbb{R}^N)} < \infty$ is needed. Indeed, if u is a polynomial of degree $d > [s] + 1$, then $|u|_{B_q^{s,p}(\mathbb{R}^N)} = \infty$, while $\Delta_h^n u(x) = 0$ for all $x \in \mathbb{R}^N$ if $n > d$.

Another useful seminorm is the following: Let $1 \leq p, q \leq \infty$ and $s > 0$. For every Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ and for $1 \leq q < \infty$ define

$$(17.24) \quad |u|_{B_q^{s,p}(\mathbb{R}^N)}^\infty := \left(\int_0^\infty \sup_{\|h\| \leq t} \|\Delta_h^{[s]+1} u\|_{L^p}^q \frac{dt}{t^{1+qs}} \right)^{1/q},$$

while for $q = \infty$,

$$|u|_{B_\infty^{s,p}(\mathbb{R}^N)}^\infty := \sup_{t>0} t^{-s} \sup_{\|h\| \leq t} \|\Delta_h^{[s]+1} u\|_{L^p(\mathbb{R}^N)}.$$

Proposition 17.21. *Let $1 \leq p, q \leq \infty$ and let $s > 0$. Then there exist two constants $c_1, c_2 > 0$ depending only on N, p, q , and s such that for every Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$,*

$$c_1 |u|_{B_q^{s,p}(\mathbb{R}^N)}^\infty \leq |u|_{B_q^{s,p}(\mathbb{R}^N)} \leq c_2 |u|_{B_q^{s,p}(\mathbb{R}^N)}^\infty.$$

We begin with a preliminary result.

Lemma 17.22. *Let $1 \leq p \leq \infty$ and let $m \in \mathbb{N}$. Then there exists a constant $c = c(m, N, p) > 0$ such that for every Lebesgue measurable function $u : \mathbb{R}^N \rightarrow \mathbb{R}$ and every $t > 0$,*

$$\sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p} \leq c \int_{B(0,t)} \|\Delta_\zeta^m u\|_{L^p} \frac{d\zeta}{\|\zeta\|^N}.$$

Proof. By the binomial theorem for $y, z \in \mathbb{R}$, we have

$$\begin{aligned} \sum_{k=0}^m (-1)^{m-k} \binom{m}{k} (y^k - 1)^m z^{m-k} &= \sum_{k=0}^m \sum_{j=0}^m (-1)^{j+k} \binom{m}{k} \binom{m}{j} y^{jk} z^{m-k} \\ &= \sum_{j=0}^m (-1)^{m-j} \binom{m}{j} (y^j - z)^m, \end{aligned}$$

which, after relabeling the summation index in the last sum, gives

$$(z - 1)^m = \sum_{k=1}^m (-1)^{m-k} \binom{m}{k} [(y^k - 1)^m z^{m-k} - (-1)^m (z - y^k)^m].$$

By replacing y and z with the operators $T_{(1/m)\xi}$ and T_h (see (17.10)), respectively, where $h, \xi \in \mathbb{R}^N$, we get

$$\begin{aligned} \Delta_h^m u(x) &= \sum_{k=1}^m (-1)^{m-k} \binom{m}{k} [\Delta_{(k/m)\xi}^m u(x + (m-k)h) \\ &\quad - (-1)^m \Delta_{h-(k/m)\xi}^m u(x + k\xi)]. \end{aligned}$$

In turn,

$$\|\Delta_h^m u\|_{L^p} \leq c \sum_{k=1}^m [\|\Delta_{(k/m)\xi}^m u\|_{L^p} + \|\Delta_{h-(k/m)\xi}^m u\|_{L^p}].$$

Now, let $h \neq 0$ and $\xi \in B(h/2, \|h\|/2)$. Then $(k/m)\xi, h - (k/m)\xi \in B(0, \|h\|)$, and so by averaging in the variable ξ over $B(h/2, \|h\|/2)$ we obtain

$$\begin{aligned} \|\Delta_h^m u\|_{L^p} &\leq \sum_{k=1}^m \frac{c}{\|h\|^N} \int_{B(h/2, \|h\|/2)} [\|\Delta_{(k/m)\xi}^m u\|_{L^p} + \|\Delta_{h-(k/m)\xi}^m u\|_{L^p}] d\xi \\ &\leq \frac{c}{\|h\|^N} \int_{B(0, \|h\|)} \|\Delta_\zeta^m u\|_{L^p} d\zeta, \end{aligned}$$

where we have used the changes of variables $\zeta = (k/m)\xi$ and $\zeta = h - (k/m)\xi$. It follows that

$$\sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p} \leq c \int_{B(0,t)} \|\Delta_\zeta^m u\|_{L^p} \frac{d\zeta}{\|\zeta\|^N},$$

which concludes the proof. $\square$

We turn to the proof of Proposition 17.21.

Proof of Proposition 17.21. We only prove the case $1 \leq q < \infty$ and leave the case $q = \infty$ as an exercise. Let $m := \lfloor s \rfloor + 1$. By Lemma 17.22 and using spherical coordinates we get

$$\begin{aligned} \int_0^\infty \sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p}^q \frac{dt}{t^{1+qs}} &\leq c \int_0^\infty \left(\int_{B(0,t)} \|\Delta_\zeta^m u\|_{L^p} \frac{d\zeta}{\|\zeta\|^N} \right)^q \frac{dt}{t^{1+qs}} \\ &= c \int_0^\infty \left(\int_0^t \int_{S^{N-1}} \|\Delta_{\rho\sigma}^m u\|_{L^p} d\mathcal{H}^{N-1}(\sigma) \frac{d\rho}{\rho} \right)^q \frac{dt}{t^{1+qs}}. \end{aligned}$$

By Hardy's inequality (see Theorem C.41) and Hölder's inequality, up to a multiplicative constant, we can bound from above the right-hand side of the previous inequality by

$$\begin{aligned} c \int_0^\infty \left(\int_{S^{N-1}} \|\Delta_{t\sigma}^m u\|_{L^p} d\mathcal{H}^{N-1}(\sigma) \right)^q \frac{dt}{t^{1+qs}} \\ \leq c \int_0^\infty \int_{S^{N-1}} \|\Delta_{t\sigma}^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\sigma) \frac{dt}{t^{1+qs}} = c \int_{\mathbb{R}^N} \|\Delta_y^m u\|_{L^p}^q \frac{dy}{\|y\|^{N+qs}}. \end{aligned}$$

This shows that $|u|_{B^{s,p}}^\infty \leq c|u|_{B^{s,p}}$. On the other hand, again by spherical coordinates,

$$\begin{aligned} \int_{\mathbb{R}^N} \|\Delta_h^m u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} &= \int_0^\infty \int_{S^{N-1}} \|\Delta_{t\sigma}^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\sigma) \frac{dt}{t^{1+qs}} \\ &\leq \int_0^\infty \int_{S^{N-1}} \sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\sigma) \frac{dt}{t^{1+qs}} \\ &= \beta_N \int_0^\infty \sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p}^q \frac{dt}{t^{1+qs}}, \end{aligned}$$

where we recall that $\beta_N = \mathcal{H}^{N-1}(S^{N-1})$. This concludes the proof. $\square$

Exercise 17.23. Let $1 \leq p, q \leq \infty$ and $s > 0$, and let $0 < R < \infty$.

(i) Prove that

$$u \mapsto \|u\|_{L^p(\mathbb{R}^N)} + |u|_{B_q^{s,p}(\mathbb{R}^N)}^{[R]}$$

is an equivalent norm in $B_q^{s,p}(\mathbb{R}^N)$, where for $q < \infty$,

$$|u|_{B_q^{s,p}(\mathbb{R}^N)}^{[R]} := \left(\int_{B(0,R)} \|\Delta_h^{\lfloor s \rfloor + 1} u\|_{L^p(\mathbb{R}^N)}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q},$$

while for $q = \infty$,

$$|u|_{B_\infty^{s,p}(\mathbb{R}^N)}^{[R]} := \sup_{h \in B(0,R) \setminus \{0\}} \frac{1}{\|h\|^s} \|\Delta_h^{\lfloor s \rfloor + 1} u\|_{L^p(\mathbb{R}^N)}.$$

- (ii) Similarly, prove that given $n \geq \lfloor s \rfloor + 1$, an equivalent norm in $B_q^{s,p}(\mathbb{R}^N)$ for $q < \infty$ is given by

$$u \mapsto \|u\|_{L^p(\mathbb{R}^N)} + \left(\int_0^R \sup_{\|h\| \leq t} \|\Delta_h^n u\|_{L^p}^q \frac{dt}{t^{1+qs}} \right)^{1/q}$$

and for $q = \infty$ by

$$u \mapsto \|u\|_{L^p(\mathbb{R}^N)} + \sup_{t \in (0, R)} \frac{1}{t^s} \sup_{\|h\| \leq t} \|\Delta_h^{\lfloor s \rfloor + 1} u\|_{L^p(\mathbb{R}^N)}.$$

17.3. Besov Spaces as Interpolation Spaces

In this section we show that the interpolation space between $L^p(\mathbb{R}^N)$ and $W^{m,p}(\mathbb{R}^N)$ is the Besov space $B_q^{\sigma m, p}(\mathbb{R}^N)$. We recall that interpolation spaces have been defined in Chapter 16.

Theorem 17.24. *Let $m \in \mathbb{N}$, $1 \leq p, q \leq \infty$, and $0 < \sigma < 1$. Then*

$$(L^p(\mathbb{R}^N), W^{m,p}(\mathbb{R}^N))_{\sigma, q} = B_q^{\sigma m, p}(\mathbb{R}^N).$$

We begin with a preliminary result, which will be used also in the sequel.

Lemma 17.25. *Let $u \in C^\infty(\mathbb{R}^N)$ and let $m \in \mathbb{N}$. Then for every $h \in \mathbb{R}^N \setminus \{0\}$ and $x \in \mathbb{R}^N$,*

$$\begin{aligned} (17.25) \quad \Delta_h^m u(x) &= \|h\|^m \int_0^m \frac{\partial^m u}{\partial \nu^m}(x + th) \chi_m(t) dt \\ &= \|h\|^m \sum_{|\alpha|=m} \frac{1}{\alpha!} \nu^\alpha \int_0^m \partial^\alpha u(x + th) \chi_m(t) dt, \end{aligned}$$

where $\nu := h/\|h\|$, $\chi_1 := \chi_{(0,1)}$ and $\chi_m := \chi_{(0,1)} * \cdots * \chi_{(0,1)}$ (m times) for $m \geq 2$. In particular, there is a constant $c = c(m, N) > 0$ such that for every $x \in \mathbb{R}^N$,

$$(17.26) \quad |\Delta_h^m u(x)| \leq c \|h\|^m \int_0^m t^{m-1} \|\nabla^m u(x + th)\| dt.$$

Proof. Fix $x \in \mathbb{R}^N$ and $h \in \mathbb{R}^N \setminus \{0\}$ and define $f(t) := u(x + th)$, $t \in [0, m]$. Then $\Delta_h^m u(x) = \Delta_1^m f(0)$. By the fundamental theorem of calculus and induction on m we have that

$$\begin{aligned} \Delta_1^m f(0) &= \int_0^1 \cdots \int_0^1 f^{(m)}(t_1 + \cdots + t_m) dt_1 \cdots dt_m \\ &= \int_{\mathbb{R}} \cdots \int_{\mathbb{R}} \chi_I(t_1) \cdots \chi_I(t_m) f^{(m)}(t_1 + \cdots + t_m) dt_1 \cdots dt_m \\ &= \int_{\mathbb{R}} f^{(m)}(t) \chi_m(t) dt. \end{aligned}$$

Writing $f(t) = u(x + \|h\|t\nu)$, it follows by the chain rule and an induction argument that

$$f^{(m)}(t) = \|h\|^m \frac{\partial^m u}{\partial \nu^m}(x + th) = \|h\|^m \sum_{|\alpha|=m} \frac{1}{\alpha!} \nu^\alpha \partial^\alpha u(x + th),$$

which, together with the previous identity, shows (17.25). Inequality (17.26) now follows from the facts that $\chi_m(t) = 0$ for $t \leq 0$ and $t \geq m$ and $\chi_m(t) \leq t^{m-1}$ for all $0 \leq t \leq m$. $\square$

We turn to the proof of Theorem 17.24.

Proof of Theorem 17.24. Step 1: Since $W^{m,p}(\mathbb{R}^N) \subset L^p(\mathbb{R}^N)$, by Theorem 16.5 we have that

$$(L^p(\mathbb{R}^N), W^{m,p}(\mathbb{R}^N))_{\sigma,q} \hookrightarrow L^p(\mathbb{R}^N) + W^{m,p}(\mathbb{R}^N) = L^p(\mathbb{R}^N).$$

Let $u \in (L^p(\mathbb{R}^N), W^{m,p}(\mathbb{R}^N))_{\sigma,q}$ and let $v \in L^p(\mathbb{R}^N)$ and $w \in W^{m,p}(\mathbb{R}^N)$ be such that $u = v + w$. Then for every $h \in \mathbb{R}^N \setminus \{0\}$, by (17.7), (17.26), Tonelli's theorem and a change of variables,

$$\begin{aligned} \|\Delta_h^m u\|_{L^p} &\leq \|\Delta_h^m v\|_{L^p} + \|\Delta_h^m w\|_{L^p} \\ &\leq 2^m \|v\|_{L^p} + c \|h\|^m \|\nabla^m w\|_{L^p}. \end{aligned}$$

Taking the infimum over all possible decompositions of u and using (16.3) gives

$$(17.27) \quad \|\Delta_h^m u\|_{L^p} \leq cK(u, \|h\|^m).$$

In turn, if $1 \leq q < \infty$, by (16.5) and (17.5), using spherical coordinates and the change of variables $r^m = t$,

$$\begin{aligned} |u|_{B_q^{\sigma m,p}} &\leq c \left(\int_{\mathbb{R}^N} (K(u, \|h\|^m))^q \frac{dh}{\|h\|^{N+\sigma m q}} \right)^{1/q} \\ &= c \left(\int_0^\infty (K(u, r^m))^q \frac{dr}{r^{1+\sigma m q}} \right)^{1/q} = c \|u\|_{\sigma,q}, \end{aligned}$$

where as usual the constant c changed from expression to expression. On the other hand, if $q = \infty$, then by multiplying both sides of (17.27) by $\|h\|^{-m\sigma}$ it follows from (16.6) and (17.24) that $|u|_{B_\infty^{\sigma m,p}} \leq c \|u\|_{\sigma,q}$.

Step 2: For simplicity we present the proof of the opposite inequality first for $m = 1$. In view of Exercise 16.7, it suffices to consider $0 < t < 1$. Given $u \in B_q^{\sigma,p}(\mathbb{R}^N)$ and $t > 0$, for $x \in \mathbb{R}^N$ write

$$\begin{aligned} u(x) &= -\frac{1}{t^N} \int_{Q(0,t)} \Delta_y u(x) dy + \frac{1}{t^N} \int_{Q(0,t)} u(x+y) dy \\ &=: v_t(x) + w_t(x). \end{aligned}$$

By Minkowski's inequality for integrals (see Corollary B.83),

$$\|v_t\|_{L^p} \leq \frac{1}{t^N} \int_{Q(0,t)} \|\Delta_y u\|_{L^p} dy \leq \sup_{\|h\| \leq \sqrt{N}t} \|\Delta_h u\|_{L^p}.$$

Hence, by (17.24) and the change of variables $\tau = \sqrt{N}t$,

$$\int_0^1 \|v_t\|_{L^p}^q \frac{dt}{t^{1+\sigma q}} \leq \int_0^1 \sup_{\|h\| \leq \sqrt{N}t} \|\Delta_h u\|_{L^p}^q \frac{dt}{t^{1+\sigma q}} \leq c(|u|_{B_q^{\sigma,p}}^\infty)^q.$$

To estimate w_t , given $i = 1, \dots, N$, by a change of variables and using the notation (E.2) we have

$$\int_{-t/2}^{t/2} u(x'_i + y'_i, x_i + y_i) dy_i = \int_{x_i-t/2}^{x_i+t/2} u(x'_i + y'_i, \tau_i) d\tau_i.$$

Hence, for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in \mathbb{R}^{N-1}$,

$$\frac{\partial}{\partial x_i} \int_{-t/2}^{t/2} u(x'_i + y'_i, x_i + y_i) dy_i = \Delta_{te_i} u(x'_i + y'_i, x_i - t/2),$$

which by Fubini's theorem and differentiation under the integral sign implies that

$$\partial_i w_t(x) = \frac{1}{t^N} \int_{Q_{N-1}(0,t)} \Delta_{te_i} u(x'_i + y'_i, x_i - t/2) dy'_i.$$

By Minkowski's inequality for integrals (see Corollary B.83) and a change of variables,

$$\|\partial_i w_t\|_{L^p} \leq \frac{1}{t^N} \int_{Q_{N-1}(0,t)} \|\Delta_{te_i} u\|_{L^p} dy'_i \leq \frac{1}{t} \sup_{\|h\| \leq \sqrt{N}t} \|\Delta_h u\|_{L^p}.$$

In turn,

$$\int_0^\infty t^q \|\partial_i w_t\|_{L^p}^q \frac{dt}{t^{1+\sigma q}} \leq \int_0^\infty \sup_{\|h\| \leq \sqrt{N}t} \|\Delta_h u\|_{L^p}^q \frac{dt}{t^{1+\sigma q}} = c(|u|_{B_q^{\sigma,p}}^\infty)^q.$$

On the other hand, by (C.8), $\|w_t\|_{L^p} \leq \|u\|_{L^p}$ and so

$$\int_0^1 t^q \|w_t\|_{L^p}^q \frac{dt}{t^{1+\sigma q}} \leq \|u\|_{L^p}^q \int_0^1 t^{q(1-\sigma)-1} dt = \frac{1}{q(1-\sigma)} \|u\|_{L^p}^q.$$

Note that this is the only estimate which uses heavily the fact that $0 < t < 1$.

In conclusion we have shown that $v_t \in L^p(\mathbb{R}^N)$, $w_t \in W^{1,p}(\mathbb{R}^N)$, and

$$\int_0^1 (K(u, t))^q \frac{dt}{t^{1+\sigma q}} \leq \int_0^1 (\|v_t\|_{L^p}^q + t^q \|w_t\|_{W^{1,p}}^q) \frac{dt}{t^{1+\sigma q}} \leq c\|u\|_{L^p}^q + c(|u|_{B_q^{\sigma,p}}^\infty)^q.$$

The case $q = \infty$ is simpler and is left as an exercise.

Step 3: Next we study the case $m = 2$. Given $u \in B_q^{2\sigma,p}(\mathbb{R}^N)$ and $t > 0$, for $x \in \mathbb{R}^N$ write

$$\begin{aligned}
 u(x) &= \frac{1}{t^{2N}} \int_{Q(0,t)} \int_{Q(0,t)} \Delta_{y+z}^2 u(x) dy dz \\
 (17.28) \quad &- \frac{1}{t^{2N}} \int_{Q(0,t)} \int_{Q(0,t)} (u(x + 2(y+z)) - 2u(x+y+z)) dy dz \\
 &=: v_t(x) + w_t(x).
 \end{aligned}$$

By Minkowski's inequality for integrals (see Corollary B.83),

$$\|v_t\|_{L^p} \leq \frac{1}{t^{2N}} \int_{Q(0,t)} \int_{Q(0,t)} \|\Delta_{y+z}^2 u\|_{L^p} dy dz \leq \sup_{\|h\| \leq 2\sqrt{N}t} \|\Delta_h^2 u\|_{L^p}.$$

Hence, by (17.24) and the change of variables $\tau = 2\sqrt{N}t$,

$$\int_0^1 \|v_t\|_{L^p}^q \frac{dt}{t^{1+2\sigma q}} \leq \int_0^1 \sup_{\|h\| \leq 2\sqrt{N}t} \|\Delta_h^2 u\|_{L^p}^q \frac{dt}{t^{1+2\sigma q}} \leq c(|u|_{B_q^{2\sigma,p}}^\infty)^q.$$

To estimate w_t , define

$$\begin{aligned}
 f_t(x) &:= \frac{1}{t^{2N}} \int_{Q(0,t)} \int_{Q(0,t)} u(x + 2(y+z)) dy dz, \\
 g_t(x) &:= \frac{1}{t^{2N}} \int_{Q(0,t)} \int_{Q(0,t)} u(x+y+z) dy dz.
 \end{aligned}$$

Given $i = 1, \dots, N$, by a change of variables and using the notation (E.2) and Fubini's theorem, we have

$$\begin{aligned}
 &\int_{-t/2}^{t/2} \int_{-t/2}^{t/2} u(x'_i + y'_i + z'_i, x_i + y_i + z_i) dy_i dz_i \\
 &= \int_{x_i-t/2}^{x_i+t/2} \int_{\xi_i-t/2}^{\xi_i+t/2} u(x'_i + y'_i + z'_i, \tau_i) d\tau_i d\xi_i.
 \end{aligned}$$

Hence, for $\mathcal{L}^{N-1}$ -a.e. $x'_i \in \mathbb{R}^{N-1}$,

$$\frac{\partial^2}{\partial x_i^2} \int_{-t/2}^{t/2} \int_{-t/2}^{t/2} u(x'_i + y'_i + z'_i, x_i + y_i + z_i) dy_i dz_i \sim \Delta_{te_i}^2 u(x'_i + y'_i + z'_i, x_i),$$

which implies that

$$(17.29) \quad \partial_i^2 g_t(x) = \frac{1}{t^{2N}} \int_{Q_{N-1}(0,t)} \int_{Q_{N-1}(0,t)} \sim \Delta_{te_i}^2 u(x'_i + y'_i + z'_i, x_i) dy'_i dz'_i.$$

Similarly,

(17.30)

$$\partial_{i,j}^2 g_t(x) = \frac{1}{t^{2N}} \int_{Q_{N-1}(0,t)} \int_{Q_{N-1}(0,t)} \Delta_{te_j}(\Delta_{te_i} u(x''_{i,j} + y''_{i,j} + z''_{i,j}, x_i + z_i - t/2, x_j + z_j - t/2)) dy'_i dz'_j,$$

where for $N > 2$, $x''_{i,j} \in \mathbb{R}^{N-2}$ is the vector obtained from x by removing the i th and j th components and with a slight abuse of notation we write $x = (x''_{i,j}, x_i, x_j)$. Similar computations can be done for f_t . In view of Exercise 17.26 for every $i, j = 1, \dots, N$,

$$\|\Delta_{te_j}(\Delta_{te_i} u)\|_{L^p} \leq \sup_{\|h\| \leq ct} \|\Delta_h^2 u\|_{L^p}.$$

By Minkowski's inequality for integrals (see Corollary B.83), it follows from (17.29), (17.30), and the analogous identities for f_t that,

$$\|\nabla^2 w_t\|_{L^p} \leq ct^{-2} \sup_{\|h\| \leq ct} \|\Delta_h^2 u\|_{L^p}.$$

In turn,

$$\int_0^1 t^{2q} \|\nabla^2 w_t\|_{L^p}^q \frac{dt}{t^{1+2\sigma q}} \leq c \int_0^1 \sup_{\|h\| \leq ct} \|\Delta_h^2 u\|_{L^p}^q \frac{dt}{t^{1+2\sigma q}} \leq c(|u|_{B_q^{2\sigma,p}}^\infty)^q.$$

On the other hand, by (C.8), $\|w_t\|_{L^p} \leq \|u\|_{L^p}$, and so

$$\int_0^1 t^{2q} \|w_t\|_{L^p}^q \frac{dt}{t^{1+2\sigma q}} \leq \|u\|_{L^p}^q \int_0^1 t^{2q(1-\sigma)-1} dt = \frac{1}{2q(1-\sigma)} \|u\|_{L^p}^q.$$

Note that this is the only estimate which uses heavily the fact that $0 < t < 1$.

In conclusion we have shown that $v_t \in L^p(\mathbb{R}^N)$, $w_t \in W^{2,p}(\mathbb{R}^N)$, and

$$\begin{aligned} \int_0^1 (K(u, r))^q \frac{dr}{r^{1+\sigma q}} &= 2 \int_0^1 (K(u, t^2))^q \frac{dt}{t^{1+2\sigma q}} \\ &\leq c \int_0^1 (\|v_t\|_{L^p}^q + t^{2q} \|w_t\|_{W^{2,p}}^q) \frac{dt}{t^{1+2\sigma q}} \leq c \|u\|_{L^p}^q + c(|u|_{B_q^{2\sigma,p}}^\infty)^q. \end{aligned}$$

Step 4: The case $m > 2$ is similar to Step 3. The main difference is that in (17.28) to define the function v_t we use m repeated integrals over $Q(0, t)$, rescale by t^{mN} , replace $\Delta_{y+z}^2 u(x)$ with $\Delta_{h_1+\dots+h_m}^m u(x)$ where $h_i \in \mathbb{R}^N$ for $i = 1, \dots, m$, and use Exercise 17.26. We leave the details as an exercise. $\square$

Exercise 17.26. Let $h_1, \dots, h_m \in \mathbb{R}^N$ and let T_{h_i} be the translation operator defined in (17.10).

(i) Prove that

$$\Delta_{2h_2} \circ \Delta_{h_1} = -T_{h_1} \circ \Delta_{-h_1}^2 - T_{2h_2} \circ \Delta_{-h_2}^2 + T_{h_1+2h_2} \circ \Delta_{-h_1-h_2}^2.$$

(ii) Prove that for every $0 \leq \ell \leq m$,

$$(T_{(m-\ell)h_m} - I) \circ \cdots \circ (T_{(1-\ell)h_1} - I) = \sum_E (-1)^{m-\text{card } E} T_{h_E^*} \circ T_{-h_E}^m,$$

where the sum is taken over all subsets $E \subseteq \{1, \dots, m\}$, $h_E^* := \sum_{k \in E} kh_k$ and $h_E := \sum_{k \in E} h_k$.

(iii) Prove that

$$\Delta_{mh_m} \circ \cdots \circ \Delta_{h_1} = \sum_E (-1)^{\text{card } E} T_{h_E^*} \circ \Delta_{-h_E}^m.$$

Remark 17.27. Steps 2–4 of the previous proof show that for $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ with $|u|_{B_q^{\sigma m, p}(\mathbb{R}^N)} < \infty$ it is always possible to decompose $u = v + w$ with $v \in L^p(\mathbb{R}^N)$, $w \in \dot{W}^{m, p}(\mathbb{R}^N)$. The previous proof actually shows that for every $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ and every $t > 0$,

$$c_1 \dot{K}(u, t^m) \leq \sup_{\|h\| \leq ct} \|\Delta_h^m u\|_{L^p(\mathbb{R}^N)} \leq c_2 \dot{K}(u, t^m),$$

where

$$\dot{K}(u, t) := \inf\{\|v\|_{L^p(\mathbb{R}^N)} + t\|w\|_{\dot{W}^{m, p}(\mathbb{R}^N)}\},$$

where the infimum is taken over all possible decompositions $u = v + w$, $v \in L^p(\mathbb{R}^N)$, $w \in \dot{W}^{m, p}(\mathbb{R}^N)$. In turn, by the change of variables $r^m = t$,

$$c^{-1}\|u\|_{\sigma, q} \leq |u|_{B_q^{\sigma m, p}(\mathbb{R}^N)} \leq c\|u\|_{\sigma, q},$$

where $\|\cdot\|_{\sigma, q}$ is defined as in (16.5) with $\dot{K}$ in place of K and $|\cdot|_{B_q^{\sigma m, p}(\mathbb{R}^N)}$ is given in (17.24).

Motivated by the previous remark we give the following definition.

Definition 17.28. Let $1 \leq p, q \leq \infty$ and let $s > 0$. A function $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ belongs to the *homogeneous Besov space* $\dot{B}_q^{s, p}(\mathbb{R}^N)$ if $|u|_{B_q^{s, p}(\mathbb{R}^N)} < \infty$, where $|\cdot|_{B_q^{s, p}(\mathbb{R}^N)}$ is given in (17.5) and (17.6).

Remark 17.29. By Remark 17.27 every function $u \in \dot{B}_q^{s, p}(\mathbb{R}^N)$, where $1 \leq p, q \leq \infty$, and $s > 0$, can be written as the sum of a function v in $L^p(\mathbb{R}^N)$ and a function w in $\dot{W}^{m, p}(\mathbb{R}^N)$ where $m > s$ can be taken arbitrarily large. In particular, if $mp > N$, then by Remark 12.56, the function w has polynomial growth. It follows that u can be identified with a tempered distribution (see Definition 10.32).

In view of Remark 17.27 we have the following interpolation result.

Theorem 17.30. Let $m \in \mathbb{N}$, $1 \leq p, q \leq \infty$, and $0 < \sigma < 1$. Then

$$(L^p(\mathbb{R}^N), \dot{W}^{m, p}(\mathbb{R}^N))_{\sigma, q} = \dot{B}_q^{\sigma m, p}(\mathbb{R}^N).$$

Remark 17.31 (Important). We observe that strictly speaking we cannot apply the interpolation theory to the space $\dot{W}^{m,p}(\mathbb{R}^N)$ since $|\cdot|_{W^{m,p}(\mathbb{R}^N)}$ is only a seminorm and not a norm. One could still define interpolation spaces starting from an admissible pair of seminormed spaces instead of normed spaces in the definition of the function K in (16.3). With this modified definition, Theorem 17.30 would be legitimate. However, many theorems in Chapter 16 rely on the fact that the admissible pairs of spaces are endowed with a norm. This difficulty can be overcome by replacing $\dot{W}^{m,p}(\mathbb{R}^N)$ with the quotient space $\dot{W}^{m,p}(\mathbb{R}^N)/\mathcal{P}_{m-1}(\mathbb{R}^N)$, where $\mathcal{P}_{m-1}(\mathbb{R}^N)$ is the family of polynomials $P : \mathbb{R}^N \rightarrow \mathbb{R}$ of degree less than or equal to $m-1$. In this quotient space $|\cdot|_{W^{m,p}(\mathbb{R}^N)}$ becomes a norm. In the sequel we will sometimes identify $\dot{W}^{m,p}(\mathbb{R}^N)$ with $\dot{W}^{m,p}(\mathbb{R}^N)/\mathcal{P}_{m-1}(\mathbb{R}^N)$, without further notice.

Exercise 17.32. Prove that for $0 < s < 1$,

$$(C(\mathbb{R}^N), C^1(\mathbb{R}^N))_{s,\infty} = C^{0,s}(\mathbb{R}^N).$$

Hint: Follow the proof of Theorem 17.24.

In view of Theorems 16.5 and 17.24, the fact that $W^{m,p}(\mathbb{R}^N) \hookrightarrow L^p(\mathbb{R}^N)$, and Exercise 16.7 we have the following embedding.

Theorem 17.33. Let $1 \leq p, q \leq \infty$, let $s > 0$, and let $m \in \mathbb{N}$ be such that $m \geq \lfloor s \rfloor + 1$. Then there exists a constant $c = c(m, N, p, q, s) > 0$ such that

$$(17.31) \quad |u|_{B_q^{s,p}(\mathbb{R}^N)} \leq c \|u\|_{L^p(\mathbb{R}^N)} + c \|\nabla^m u\|_{L^p(\mathbb{R}^N; \mathbb{R}^{M_m})}$$

for all $u \in W^{m,p}(\mathbb{R}^N)$. In particular, $W^{m,p}(\mathbb{R}^N) \hookrightarrow B_q^{s,p}(\mathbb{R}^N)$.

Remark 17.34. Note that if we had used the seminorm in Exercise 17.23, then the term $c \|u\|_{L^p(\mathbb{R}^N)}$ would not be needed on the right-hand side of (17.31).

Using Lemma 17.25 it is possible to give a direct proof of this theorem.

Exercise 17.35. Let $1 \leq p, q \leq \infty$, let $s > 0$, and let $m \in \mathbb{N}$ be such that $m \geq \lfloor s \rfloor + 1$.

(i) If $1 \leq q < \infty$, prove that for all $u \in W^{m,p}(\mathbb{R}^N)$,

$$(17.32) \quad \left(\int_{\mathbb{R}^N} \|\Delta_h^m u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q} \\ \leq \frac{1}{[(m-s)q]^{1/q}} \left(\int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} + \frac{2^m \beta_N^{1/q}}{(sq)^{1/q}} \|u\|_{L^p},$$

where $\beta_N = \mathcal{H}^{N-1}(S^{N-1})$. Hint: Write $\int_{\mathbb{R}^N} = \int_{B(0,1)} + \int_{\mathbb{R}^N \setminus B(0,1)}$ and use Lemma 17.25.

(ii) If $q = \infty$, prove that for all $u \in W^{m,p}(\mathbb{R}^N)$,

$$\sup_{h \in \mathbb{R}^N \setminus \{0\}} \frac{1}{\|h\|^s} \|\Delta_h^m u\|_{L^p} \leq \sup_{\|\nu\|=1} \left\| \frac{\partial^\nu u}{\partial \nu^m} \right\|_{L^p} + 2^m \|u\|_{L^p}.$$

Exercise 17.36. Let $0 < s < 1$ and $1 \leq q \leq \infty$. Prove that there exists a constant $c = c(N, q, s) > 0$ such that $|u|_{B_q^{s,1}(\mathbb{R}^N)} \leq c \|u\|_{BV(\mathbb{R}^N)}$ for all $u \in BV(\mathbb{R}^N)$.

As a corollary of Theorem 17.33 we can prove the density of $C_c^\infty(\mathbb{R}^N)$ in $B_q^{s,p}(\mathbb{R}^N)$ for $1 \leq p, q < \infty$.

Theorem 17.37. Let $1 \leq p, q < \infty$ and $s > 0$. Then $C_c^\infty(\mathbb{R}^N)$ is dense in $B_q^{s,p}(\mathbb{R}^N)$ and $\dot{B}_q^{s,p}(\mathbb{R}^N)$.

Proof. Step 1: By Theorem 17.24, $(L^p(\mathbb{R}^N), W^{m,p}(\mathbb{R}^N))_{\sigma,q} = B_q^{s,p}(\mathbb{R}^N)$, where $m > s$, $\sigma := s/m$. In turn, by Exercise 16.22 we have that $W^{m,p}(\mathbb{R}^N)$ is dense in $B_q^{s,p}(\mathbb{R}^N)$. By the density of $C_c^\infty(\mathbb{R}^N)$ in $W^{m,p}(\mathbb{R}^N)$ (see Theorem 11.35) and (17.31), it follows that $C_c^\infty(\mathbb{R}^N)$ is dense in $B_q^{s,p}(\mathbb{R}^N)$.

Step 2: Since $W^{m,p}(\mathbb{R}^N) = L^p(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$ by Corollary 12.86, again by Exercise 16.22 and Theorem 17.30 we have that $W^{m,p}(\mathbb{R}^N)$ is dense in $\dot{B}_q^{s,p}(\mathbb{R}^N)$. We can continue as in the previous step to conclude that $C_c^\infty(\mathbb{R}^N)$ is dense in $\dot{B}_q^{s,p}(\mathbb{R}^N)$. $\square$

Theorems 16.11 and 17.24 give the following embedding.

Theorem 17.38. Let $1 \leq p \leq \infty$, let $1 \leq q_1 < q_2 \leq \infty$, and let $s > 0$. Then there exists a constant $c = c(N, p, q_1, q_2, s) > 0$ such that

$$|u|_{B_{q_2}^{s,p}(\mathbb{R}^N)} \leq c |u|_{B_{q_1}^{s,p}(\mathbb{R}^N)}$$

for all $u \in L_{\text{loc}}^1(\mathbb{R}^N)$. In particular, $\dot{B}_{q_1}^{s,p}(\mathbb{R}^N) \hookrightarrow \dot{B}_{q_2}^{s,p}(\mathbb{R}^N)$ and $B_{q_1}^{s,p}(\mathbb{R}^N) \hookrightarrow B_{q_2}^{s,p}(\mathbb{R}^N)$.

Using the seminorm $|\cdot|_{B_{s,p}^\infty}$ defined in (17.24), it is possible to give an alternative proof of Theorem 17.38, as described in the next two exercises.

Exercise 17.39. Let $1 \leq q_1 < q_2 < \infty$.

(i) Given a sequence $\{a_k\}_{k \in \mathbb{Z}}$ of nonnegative numbers, prove that

$$\left(\sum_{k \in \mathbb{Z}} a_k^{q_2} \right)^{1/q_2} \leq \left(\sum_{k \in \mathbb{Z}} a_k^{q_1} \right)^{1/q_1}.$$

(ii) Given an increasing function $g : [0, \infty) \rightarrow [0, \infty)$ and $r > 0$, prove that there exists a constant $c > 0$ independent of g and q_2 such that

$$\left(\int_0^\infty \left(\frac{g(t)}{t^r} \right)^{q_2} \frac{dt}{t} \right)^{1/q_2} \leq c \left(\int_0^\infty \left(\frac{g(t)}{t^r} \right)^{q_1} \frac{dt}{t} \right)^{1/q_1}.$$

(iii) Using part (ii), prove that

$$\sup_{t>0} \frac{g(t)}{t^r} \leq c \left(\int_0^\infty \left(\frac{g(t)}{t^r} \right)^{q_1} \frac{dt}{t} \right)^{1/q_1}.$$

Exercise 17.40. Use the previous exercise to give an alternative proof of Theorem 17.38 using the equivalent seminorm $|\cdot|_{B_{s,p}^\infty}$ defined in (17.24).

In view of the duality theorem for interpolation spaces (see Theorem 16.27) and of Theorem 16.30, we can identify the dual of $B_q^{s,p}(\mathbb{R}^N)$.

Theorem 17.41 (Duality). *Let $m \in \mathbb{N}$, $1 \leq p \leq \infty$, $1 \leq q < \infty$, and $0 < \sigma < 1$. Then the dual of $B_q^{\sigma m, p}(\mathbb{R}^N)$ can be identified with*

$$(L^{p'}(\mathbb{R}^N), (W^{m,p}(\mathbb{R}^N))')_{\sigma, q'}.$$

Moreover, if $1 < p < \infty$, then $B_q^{\sigma m, p}(\mathbb{R}^N)$ is reflexive.

As a consequence of the reiteration theorem (Theorem 16.15) and of Theorems 17.24 and 17.30 we have the following result.

Corollary 17.42. *Let $m \in \mathbb{N}$, $1 \leq p, q, q_0, q_1 \leq \infty$, $0 < \sigma < 1$, and $0 < s_0 < s_1$. Then*

$$\begin{aligned} (\dot{B}_{q_0}^{s_0, p}(\mathbb{R}^N), \dot{B}_{q_1}^{s_1, p}(\mathbb{R}^N))_{\sigma, q} &= \dot{B}_q^{s, p}(\mathbb{R}^N), \\ (B_{q_0}^{s_0, p}(\mathbb{R}^N), B_{q_1}^{s_1, p}(\mathbb{R}^N))_{\sigma, q} &= B_q^{s, p}(\mathbb{R}^N), \end{aligned}$$

where $s := (1 - \sigma)s_0 + \sigma s_1$.

In view of the previous corollary and Exercise 16.13 we have the following inequality, which is similar in spirit to the Gagliardo–Nirenberg interpolation theorem (see Theorem 12.87).

Corollary 17.43. *Let $m \in \mathbb{N}$, $1 \leq p, q, q_0, q_1 \leq \infty$, $0 < \sigma < 1$, and $0 < s_0 < s_1$. Then there is a constant $c = c(m, p, q, q_0, q_1, \sigma, s_0, s_1) > 0$ such that*

$$|u|_{B_q^{s, p}(\mathbb{R}^N)} \leq c |u|_{B_{q_0}^{s_0, p}(\mathbb{R}^N)}^{1-\sigma} |u|_{B_{q_1}^{s_1, p}(\mathbb{R}^N)}^\sigma$$

for all $u \in \dot{B}_{q_0}^{s_0, p}(\mathbb{R}^N) \cap \dot{B}_{q_1}^{s_1, p}(\mathbb{R}^N)$, where $s := (1 - \sigma)s_0 + \sigma s_1$.

A similar result holds for $B_q^{s, p}(\mathbb{R}^N)$.

We conclude this section with a few remarks on Besov spaces defined on open sets $\Omega \subseteq \mathbb{R}^N$. In view of Theorem 17.24, given $1 \leq p, q \leq \infty$ and $s > 0$, one can define the Besov spaces $B_q^{s, p}(\Omega)$ and $\dot{B}_q^{s, p}(\Omega)$ as

$$\begin{aligned} B_q^{s, p}(\Omega) &:= (L^p(\Omega), W^{m, p}(\Omega))_{s/m, q}, \\ \dot{B}_q^{s, p}(\Omega) &:= (L^p(\Omega), \dot{W}^{m, p}(\Omega))_{s/m, q}, \end{aligned}$$

where $m \in \mathbb{N}$ is such that $m > s$. If $\partial\Omega$ is uniformly Lipschitz continuous, then it follows from the proof of Theorems 13.8 and 13.17 (see (13.25)) that there exists a continuous linear operator $\mathcal{E} : L^p(\Omega) \rightarrow L^p(\mathbb{R}^N)$ with the property that for all $u \in L^p(\Omega)$, $\mathcal{E}(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$. Moreover, $\mathcal{E} : W^{m,p}(\Omega) \rightarrow W^{m,p}(\mathbb{R}^N)$ is linear and continuous. It follows by Theorem 16.12 that

$$(17.33) \quad \mathcal{E} : B_q^{s,p}(\Omega) \rightarrow B_q^{s,p}(\mathbb{R}^N)$$

with

$$(17.34) \quad \|\mathcal{E}\|_{L(B_q^{s,p}(\Omega); B_q^{s,p}(\mathbb{R}^N))} \leq \|\mathcal{E}\|_{L(L^p(\Omega); L^p(\mathbb{R}^N))}^{1-\sigma} \|\mathcal{E}\|_{L(W^{m,p}(\Omega); W^{m,p}(\mathbb{R}^N))}^{\sigma}.$$

Since for all $u \in L^p(\Omega)$, $\mathcal{E}(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$, it follows in particular that for all $u \in B_q^{s,p}(\Omega)$, $\mathcal{E}(u)(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in \Omega$. We remark that (17.33) can be used to obtain embedding theorems in $B_q^{s,p}(\Omega)$. Indeed, one can first prove embeddings for $B_q^{s,p}(\mathbb{R}^N)$ (see the following section) and then use (17.33) to obtain the same embeddings for $B_q^{s,p}(\Omega)$.

Exercise 17.44 (Compactness). Let $\Omega \subset \mathbb{R}^N$ be an open bounded set with Lipschitz continuous boundary, let $1 \leq p, q \leq \infty$, and let $0 < s < 1$. Prove that the embedding $B_q^{s,p}(\Omega) \hookrightarrow L^p(\Omega)$ is compact. Hint: Use the fact that the embedding $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ is compact (see Theorem 12.30) and Exercises 16.10 and 16.14.

Exercise 17.45 (Compactness). Let $\Omega \subset \mathbb{R}^N$ be an open bounded set with Lipschitz continuous boundary, let $1 \leq p, q \leq \infty$, and let $0 < s_1 < s_2 < 1$. Prove that the embedding $B_q^{s_2,p}(\Omega) \hookrightarrow B_q^{s_1,p}(\Omega)$ is compact. Hint: Use the fact that the embedding $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ is compact (see Theorem 12.30) and Exercise 16.26.

Exercise 17.46. Let $\Omega, U \subseteq \mathbb{R}^N$ be open sets, let $\Psi : U \rightarrow \Omega$ be invertible, with Ψ and Ψ^{-1} Lipschitz continuous functions, let $1 \leq p, q \leq \infty$, and let $0 < s < 1$. Prove that $u \in B_q^{s,p}(\Omega)$ if and only if $u \circ \Psi \in B_q^{s,p}(U)$ and estimate $\|u \circ \Psi\|_{B_q^{s,p}(U)}$. Hint: Use Theorems 11.53 and 16.12.

Exercise 17.47. Let $\Omega \subseteq \mathbb{R}^N$ be an open set with bounded Lipschitz continuous boundary, let $1 \leq p, q \leq \infty$, and let $0 < s < 1$. Prove that $u \mapsto \|u\|_{L^p(\Omega)} + |u|_{B_q^{s,p}(\Omega)}$, where $|\cdot|_{B_q^{s,p}(\Omega)}$ is defined in (17.13), is an equivalent norm in $B_q^{s,p}(\Omega)$. Hint: Consider first the case in which $\Omega = \mathbb{R}_+^N$ and use Exercise 17.11, then use a partition of unity together with Exercises 17.15 and 17.46.

Exercise 17.48 (Poincaré's inequality). Let B be an open ball of radius $r > 0$, let $1 \leq p < \infty$, and $0 < s < 1$. Prove that there exists a constant $c = c(N, p, s) > 0$ such that for every $u \in \dot{B}^{s,p}(B)$,

$$\|u - u_B\|_{L^p(B)} \leq cr^s |u|_{B^{s,p}(B)},$$

where we recall that (see (13.60)) for a measurable set $E \subset \mathbb{R}^N$ of positive finite measure, $u_E := \frac{1}{\mathcal{L}^N(E)} \int_E u(x) dx$. Hint: Use Poincaré's inequality for balls (see, e.g. Theorem 13.41) and Theorem 16.12.

17.4. Sobolev Embeddings

In this section we use interpolation theory to extend the embeddings for Sobolev spaces to Besov spaces. We begin with the Sobolev–Gagliardo–Nirenberg embedding theorem (Theorem 12.4). Given $s > 0$ and $1 \leq p < N/s$ we define the *Sobolev critical exponent*.

$$(17.35) \quad p_s^* := \frac{Np}{N - sp}.$$

Note that $p_1^* = p^*$, which is the Sobolev critical exponent for $W^{1,p}(\mathbb{R}^N)$ (see (12.2)). In what follows Remark 17.31 will be of particular relevance.

Theorem 17.49 (Sobolev–Gagliardo–Nirenberg embedding in $B_q^{s,p}$). *Let $s > 0$, $1 \leq p < N/s$, and $1 \leq q \leq \infty$. Then for every function $u \in B_q^{s,p}(\mathbb{R}^N)$ there exists a polynomial P_u such that*

$$(17.36) \quad \|u - P_u\|_{L^{p_s^*,q}(\mathbb{R}^N)} \leq c|u|_{B_q^{s,p}(\mathbb{R}^N)}$$

for some constant $c = c(N, p, q, s) > 0$. In particular, $\dot{B}^{s,p}(\mathbb{R}^N)/\mathcal{P}(\mathbb{R}^N) \hookrightarrow L^{p_s^*}(\mathbb{R}^N)$ and $B_q^{s,p}(\mathbb{R}^N) \hookrightarrow L^r(\mathbb{R}^N)$ for all $p \leq r \leq p_s^*$ and $1 \leq q \leq p_s^*$.

Proof. Step 1: Let $m \in \mathbb{N}$ be such that $mp > N$. By the Gagliardo–Nirenberg interpolation theorem (see Theorem 12.87) with $k = 0$, $\beta = 0$, $q = p < r < \infty$, and $\theta = 1 - \sigma$, where $\sigma := (1/p - 1/r)N/m \in (0, 1)$,

$$\|v\|_{L^r} \leq c\|v\|_{L^p}^{1-\sigma} |v|_{W^{m,p}}^\sigma$$

for every $v \in L^p(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$. Since by (16.21) for every $t > 0$, $\dot{J}(v, t) := \max\{\|v\|_{L^p}, t|v|_{W^{m,p}}\}$, from the previous inequality we deduce that

$$\|v\|_{L^r} \leq ct^{-\sigma} \dot{J}(v, t).$$

Step 2: Let $m \in \mathbb{N}$ be such that $mp > N$ and let $0 < \sigma < 1$. Given $u \in \dot{B}_1^{\sigma m, p}(\mathbb{R}^N)$ let $v : (0, \infty) \rightarrow L^p(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$ be a Bochner integrable function such that

$$u(x) = \int_0^\infty (v(t))(x) \frac{dt}{t}, \quad x \in \mathbb{R}^N.$$

By Minkowski's inequality for integrals (see Corollary B.83),

$$\|u\|_{L^r} \leq \left\| \int_0^\infty |v(t)(\cdot)| \frac{dt}{t} \right\|_{L^r} \leq \int_0^\infty \|v(t)\|_{L^r} \frac{dt}{t} \leq c \int_0^\infty \dot{J}(v(t), t) \frac{dt}{t^{1+\sigma}},$$

where in the last inequality we applied Step 1 to the function $v(t) \in L^p(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$ for every $t > 0$. By taking the infimum over all such functions v and using (16.23) we get

$$\|u\|_{L^r} \leq c\|u\|_{\sigma,1,J} \leq c\|u\|_{\sigma,1} \leq c|u|_{\dot{B}_1^{\sigma m,p}}^\infty,$$

where in the last two inequalities we applied Theorems 16.21 and 17.24 and Remark 17.27. This shows that

$$\dot{B}_1^{\sigma m,p}(\mathbb{R}^N) \hookrightarrow L^r(\mathbb{R}^N).$$

Step 3: By taking $m \in \mathbb{N}$ such that $m > N/p > s$ we have that $\sigma := s/m \in (0, 1)$ and thus, by the previous step

$$(17.37) \quad \dot{B}_1^{s,p}(\mathbb{R}^N) \hookrightarrow L^r(\mathbb{R}^N),$$

where $s = (1/p - 1/r)N$, that is, $r = p_s$, where p_s is the critical exponent given in (17.35). To obtain (17.36), consider $0 < s_0 < s < s_1 < N/p$ and let $\sigma \in (0, 1)$ be such that $s = (1 - \sigma)s_0 + \sigma s_1$. By (17.37) for s_0 and s_1 ,

$$\dot{B}_1^{s_0,p}(\mathbb{R}^N) \hookrightarrow L^{p_{s_0}}(\mathbb{R}^N), \quad \dot{B}_1^{s_1,p}(\mathbb{R}^N) \hookrightarrow L^{p_{s_1}}(\mathbb{R}^N).$$

In turn, by Exercise 16.34, Theorem 16.12 and Corollary 17.42,

$$\begin{aligned} \dot{B}_q^{s,p}(\mathbb{R}^N) &= (\dot{B}_1^{s_0,p}(\mathbb{R}^N), \dot{B}_1^{s_1,p}(\mathbb{R}^N))_{\sigma,q} \\ &\hookrightarrow (L^{p_{s_0}}(\mathbb{R}^N), L^{p_{s_1}}(\mathbb{R}^N))_{\sigma,q} = L^{r,q}(\mathbb{R}^N), \end{aligned}$$

where $1/r = (1 - \sigma)/p_{s_0} + \sigma/p_{s_1} = 1/p - s/N$. $\square$

Exercise 17.50. Let $1 \leq p < \infty$. Prove that for every $u \in \dot{B}_1^{N/p,p}(\mathbb{R}^N)$ there exists a polynomial P_u such that

$$\|u - P_u\|_{L^\infty(\mathbb{R}^N)} \leq c|u|_{\dot{B}_1^{N/p,p}(\mathbb{R}^N)}$$

for some constant $c = c(N, p) > 0$. Hint: Use (12.54) with $r < 0$ and $a = 0$, in Step 1.

Exercise 17.51. Let $\Omega \subset \mathbb{R}^N$ be an open bounded set with Lipschitz continuous boundary, let $0 < s < 1$, let $1 \leq p < N/s$, and let $1 \leq q \leq \infty$. Prove that the embedding $B_q^{s,p}(\Omega) \hookrightarrow L^r(\Omega)$ is compact for every $1 \leq r < p_s^*$ and $1 \leq q \leq p_s^*$. Hint: Use the previous theorem, (17.33), and Exercise 17.44.

Next we extend Morrey's embedding theorem (see Theorems 12.48 and 12.55) to Besov spaces.

Theorem 17.52 (Morrey embedding in $B_q^{s,p}$). *Let $1 \leq p < \infty$ and $s > 0$ be such that $0 < s - N/p < 1$, and let $1 \leq q \leq \infty$. Then for every function $u \in \dot{B}_q^{s,p}(\mathbb{R}^N)$ there exists a polynomial P_u such that*

$$(17.38) \quad \|u - P_u\|_{C^{0,s-N/p}(\mathbb{R}^N)} \leq c|u|_{\dot{B}_q^{s,p}(\mathbb{R}^N)}$$

for some constant $c = c(N, p, q, s) > 0$. In particular,

$$B_q^{s,p}(\mathbb{R}^N) \hookrightarrow C^{0,s-N/p}(\mathbb{R}^N).$$

Proof. Step 1: Let $m \in \mathbb{N}$ be such that $m > s$. By the Gagliardo–Nirenberg interpolation theorem (see Theorem 12.87) with $k = 0$, $\beta = 0$, $q = p$, $\theta = 1 - s/m$, and $1/r = 1/p - s/N < 0$,

$$|v|_{C^{0,s-N/p}} \leq c \|v\|_{L^p}^{1-s/m} |v|_{W^{m,p}}^{s/m}$$

for every $v \in L^p(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$ (see (12.54)). Since by (16.21) for every $t > 0$, $\dot{J}(v, t) := \max\{\|v\|_{L^p}, t|v|_{W^{m,p}}\}$, from the previous inequality we deduce that

$$|v|_{C^{0,s-N/p}} \leq ct^{-s/m} \dot{J}(v, t).$$

Step 2: Given $u \in \dot{B}_1^{s,p}(\mathbb{R}^N)$ let $v : (0, \infty) \rightarrow L^p(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$ be a Bochner integrable function such that

$$u(x) = \int_0^\infty (v(t))(x) \frac{dt}{t}, \quad x \in \mathbb{R}^N.$$

Then for $x, y \in \mathbb{R}^N$,

$$\begin{aligned} |u(x) - u(y)| &\leq \int_0^\infty |(v(t))(x) - (v(t))(y)| \frac{dt}{t} \\ &\leq \|x - y\|^{s-N/p} \int_0^\infty |v(t)|_{C^{0,s-N/p}} \frac{dt}{t} \\ &\leq c \|x - y\|^{s-N/p} \int_0^\infty \dot{J}(v(t), t) \frac{dt}{t^{1+s/m}}, \end{aligned}$$

where in the last inequality we applied Step 1 to the function $v(t) \in L^p(\mathbb{R}^N) \cap \dot{W}^{m,p}(\mathbb{R}^N)$ for every $t > 0$. By dividing by $\|x - y\|^{s-N/p}$ for $x \neq y$, taking the supremum over all such x, y , then taking the infimum over all such functions v , and using (16.23) we get

$$|u|_{C^{0,s-N/p}} \leq c \|u\|_{s/m,1,j} \leq c \|u\|_{s/m,1} \leq c |u|_{\dot{B}_1^{s,p}}^\infty,$$

where in the last two inequalities we applied Theorems 16.21 and 17.24 and Remark 17.27. This shows that

$$(17.39) \quad \dot{B}_1^{sm,p}(\mathbb{R}^N) \hookrightarrow \dot{B}^{s-N/p,\infty}(\mathbb{R}^N).$$

To obtain (17.38), consider $N/p < s_0 < s < s_1 < 1 + N/p$ and let $\sigma \in (0, 1)$ be such that $s = (1 - \sigma)s_0 + \sigma s_1$. By (17.39) for s_0 and s_1 ,

$$\dot{B}_1^{s_0,p}(\mathbb{R}^N) \hookrightarrow \dot{B}^{s_0-N/p,\infty}(\mathbb{R}^N), \quad \dot{B}_1^{s_1,p}(\mathbb{R}^N) \hookrightarrow \dot{B}^{s_1-N/p,\infty}(\mathbb{R}^N).$$

In turn, by Theorem 16.12 and Corollary 17.42,

$$\begin{aligned} \dot{B}_q^{s,p}(\mathbb{R}^N) &= (\dot{B}_1^{s_0,p}(\mathbb{R}^N), \dot{B}_1^{s_1,p}(\mathbb{R}^N))_{\sigma,q} \\ &\hookrightarrow (\dot{B}^{s_0-N/p,\infty}(\mathbb{R}^N), \dot{B}^{s_1-N/p,\infty}(\mathbb{R}^N))_{\sigma,q} = \dot{B}_q^{s_1-N/p,\infty}(\mathbb{R}^N), \end{aligned}$$

where $1/r = (1 - \sigma)/p_{s_0} + \sigma/p_{s_1} = 1/p - s/N$. $\square$

Exercise 17.53. Let $1 \leq p < \infty$, let $s = 1 + N/p$, let $\theta \in (0, 1)$, and let $1 \leq q \leq \infty$. Prove that for every $u \in \dot{B}_q^{1+N/p, p}(\mathbb{R}^N)$ there exists a polynomial P_u such that

$$|u - P_u|_{C^{0, \theta}(\mathbb{R}^N)} \leq c|u|_{B_q^{1+N/p, p}(\mathbb{R}^N)}$$

for some constant $c = c(N, p, q, s, \theta) > 0$. Hint: Use (12.54) to estimate the Lipschitz norm of v in Step 1.

Exercise 17.54. Let $\Omega \subset \mathbb{R}^N$ be an open bounded set with Lipschitz continuous boundary, let $1 \leq p < \infty$ and $0 < s < 1$ be such that $0 < s - N/p < 1$, and let $1 \leq q \leq \infty$. Prove that the embedding $B_q^{s, p}(\Omega) \hookrightarrow C^{0, \theta}(\Omega)$ is compact for every $0 < \theta < s - N/p$. Hint: Use the previous theorem, (17.33), and Proposition 12.84.

See Exercises 17.70 and 17.71 for the case $s - N/p > 1$.

Finally, we consider the critical case $s = N/p$. As in the Sobolev case (see, e.g., Theorems 12.33 or 12.40), we will need the full Besov norm and not just the seminorm. Indeed, we will make use of the equivalent norm given in Exercise 16.23.

Theorem 17.55. Let $1 \leq p < \infty$ and $1 \leq q \leq \infty$. Then for every $p \leq r < \infty$ there exists a constant $c = c(N, p, q, r) > 0$ such that for every $u \in B_q^{N/p, p}(\mathbb{R}^N)$,

$$\|u\|_{L^r(\mathbb{R}^N)} \leq c\|u\|_{B_q^{N/p, p}(\mathbb{R}^N)}.$$

In particular, $B_q^{N/p, p}(\mathbb{R}^N) \hookrightarrow L^r(\mathbb{R}^N)$ for all $p \leq r < \infty$.

Proof. We only prove the case $q < \infty$. Let $m \in \mathbb{N}$ be such that $mp > N$. As in Step 1 of the proof of Theorem 17.49, for every $p < r < \infty$ we have that

$$(17.40) \quad \|v\|_{L^r} \leq ct^{-\sigma} J(v, t)$$

for every $v \in W^{m, p}(\mathbb{R}^N)$, where $\sigma := (1/p - 1/r)N/m \in (0, 1)$.

Let $\sigma_0 := N/(mp) \in (0, 1)$. Then $(L^p(\mathbb{R}^N), W^{m, p}(\mathbb{R}^N))_{\sigma_0, q} = B_q^{N/p, p}(\mathbb{R}^N)$ by Theorem 17.24. Moreover, since $W^{m, p}(\mathbb{R}^N) \hookrightarrow L^m(\mathbb{R}^N)$, we can apply Exercise 16.23. Reasoning as in Steps 2 and 3 of the proof of Theorem 16.21, given $u \in B_q^{N/p, p}(\mathbb{R}^N)$ we can find a Bochner integrable function $v : (0, 1) \rightarrow W^{m, p}(\mathbb{R}^N)$ such that

$$u(x) = \int_0^1 (v(t))(x) \frac{dt}{t}, \quad x \in \mathbb{R}^N,$$

and $J(v(t), t) \leq cK(u, t)$ for all $t \in (0, 1)$.

Reasoning as in Step 2 of the proof of Theorem 17.49, we get

$$\begin{aligned} \|u\|_{L^r} &\leq c \int_0^1 J(v(t), t) \frac{dt}{t^{1+\sigma}} \leq c \int_0^1 K(u, t) \frac{dt}{t^{1+\sigma}} \\ &\leq c \left(\int_0^1 t^{(\sigma_0 - \sigma)q' - 1} dt \right)^{1/q'} \left(\int_0^1 (K(u, t))^q \frac{dt}{t^{1+\sigma_0 q}} \right)^{1/q} \\ &\leq \frac{c}{(\sigma_0 - \sigma)^{1/q'}} \|u\|_{B_q^{N/p, p}}, \end{aligned}$$

where we have used Hölder's inequality for $1 < q < \infty$ and as usual c changed from line to line. Note that $\sigma_0 - \sigma = N/(rm) > 0$. $\square$

Exercise 17.56. Let $\Omega \subset \mathbb{R}^N$ be an open bounded set with Lipschitz continuous boundary, let $1 \leq q \leq \infty$, and let $0 < s < 1$. Prove that the embedding $B_q^{s, N/s}(\Omega) \hookrightarrow L^r(\Omega)$ is compact for every $1 \leq r < \infty$. Hint: Use the previous theorem and (17.33).

17.5. The Limit of $B_q^{s,p}$ as $s \rightarrow 0^+$ and $s \rightarrow m^-$

In this section we prove that the norm in $L^p(\mathbb{R}^N)$ and a suitable seminorm in $W^{m,p}(\mathbb{R}^N)$ can be regarded as limits of seminorms in $B_q^{s,p}(\mathbb{R}^N)$ as $s \rightarrow 0^+$ and $s \rightarrow m^-$, respectively. We begin with the case $s \rightarrow 0^+$. For simplicity we take $m = 1$ and leave the other cases as an exercise. We recall that $\beta_N = \mathcal{H}^{N-1}(S^{N-1})$.

Theorem 17.57. Let $1 \leq p, q < \infty$, let $0 < s_0 < 1$ and let $u \in \dot{B}_q^{s_0, p}(\mathbb{R}^N)$ vanish at infinity. Then $u \in L^p(\mathbb{R}^N)$ if and only if

$$\liminf_{s \rightarrow 0^+} [s(1-s)q]^{1/q} \|u\|_{B_q^{s,p}(\mathbb{R}^N)} < \infty.$$

Moreover, if $u \in L^p(\mathbb{R}^N)$, then

$$(17.41) \quad \lim_{s \rightarrow 0^+} [s(1-s)q]^{1/q} \|u\|_{B_q^{s,p}(\mathbb{R}^N)} = 2\beta_N^{1/q} \|u\|_{L^p(\mathbb{R}^N)}.$$

The idea of the proof is to use Lemma 16.17.

Lemma 17.58. Let $1 \leq p, q < \infty$ and let $u \in L^p(\mathbb{R}^N)$. Then

$$(17.42) \quad \lim_{t \rightarrow \infty} \int_{S^{N-1}} \|\Delta_{t\nu} u\|_{L^p(\mathbb{R}^N)}^q d\mathcal{H}^{N-1}(\nu) = 2^q \beta_N \|u\|_{L^p(\mathbb{R}^N)}^q.$$

Proof. Since $u \in L^p(\mathbb{R}^N)$, by Theorem C.24 for every $\varepsilon > 0$ we may find a function $v \in C_c(\mathbb{R}^N)$ such that

$$(17.43) \quad \|u - v\|_{L^p} \leq \varepsilon.$$

Writing

$$\Delta_{t\nu} v(x) = v(x + t\nu) - u(x + t\nu) + \Delta_{t\nu} u(x) + u(x) - v(x),$$

from (17.43), a change of variables, and Minkowski's inequality, it follows that

$$(17.44) \quad \|\Delta_{t\nu}v\|_{L^p} \leq \|\Delta_{t\nu}u\|_{L^p} + 2\|u - v\|_{L^p} \leq \|\Delta_{t\nu}u\|_{L^p} + 2\varepsilon,$$

and, similarly,

$$(17.45) \quad \|\Delta_{t\nu}u\|_{L^p} \leq \|\Delta_{t\nu}v\|_{L^p} + 2\varepsilon.$$

Using (17.44), (17.45), and Minkowski's inequality we get

$$\begin{aligned} & \left(\int_{S^{N-1}} \|\Delta_{t\nu}v\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} - 2\beta_N^{1/q}\varepsilon \\ & \leq \left(\int_{S^{N-1}} \|\Delta_{t\nu}u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} \\ & \leq \left(\int_{S^{N-1}} \|\Delta_{t\nu}v\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} + 2\beta_N^{1/q}\varepsilon. \end{aligned}$$

Thus, also by (17.43), it suffices to prove (17.42) for the function v .

Next, we write

$$\begin{aligned} & \int_{\mathbb{R}^N} |\Delta_{t\nu}v(x)|^p dx \\ & = \int_{E_{t,\nu}} |\Delta_{t\nu}v(x)|^p dx + \int_{\mathbb{R}^N \setminus E_{t,\nu}} |\Delta_{t\nu}v(x)|^p dx \\ & = \int_{E_{t,\nu}} |\Delta_{t\nu}v(x)|^p dx + \int_{F_{t,\nu}} |\Delta_{-t\nu}v(y)|^p dy =: I_{t,\nu} + II_{t,\nu}, \end{aligned}$$

where $E_{t,\nu} := \{x \in \mathbb{R}^N : \|x + t\nu\| \geq \|x\|\}$ and $F_{t,\nu} := \{y \in \mathbb{R}^N : \|y - t\nu\| > \|y\|\}$. Since $v \in C_c(\mathbb{R}^N)$, there exists $r > 0$ such that $v(x) = 0$ for all $x \in \mathbb{R}^N$ with $\|x\| \geq r$. Note that if $\|x + t\nu\| \geq \|x\| \geq r$, then $v(x + t\nu) = 0$. Hence we may write

$$I_{t,\nu} = \int_{B(0,r)} \chi_{E_{t,\nu}(x)} |\Delta_{t\nu}v(x)|^p dx.$$

Since $\chi_{E_{t,\nu}(x)} \Delta_{t\nu}v(x) \rightarrow -v(x)$ as $t \rightarrow \infty$ and $|\Delta_{t\nu}v(x)| \leq 2\|v\|_\infty$ for all t , ν and x , we may apply Lebesgue's dominated convergence theorem (twice) to conclude that

$$\begin{aligned} & \lim_{t \rightarrow \infty} \int_{S^{N-1}} I_{t,\nu}^{q/p} d\mathcal{H}^{N-1}(\nu) \\ & = \int_{S^{N-1}} \left(\int_{B(0,r)} \lim_{t \rightarrow \infty} \chi_{E_{t,\nu}(x)} |\Delta_{t\nu}v(x)|^p dx \right)^{q/p} d\mathcal{H}^{N-1}(\nu) \\ & = \int_{S^{N-1}} \left(\int_{B(0,r)} |v(x)|^p dx \right)^{q/p} d\mathcal{H}^{N-1}(\nu) = \beta_N \|v\|_{L^p}^q. \end{aligned}$$

The term $II_{t,\nu}$ can be treated in the same way. We omit the details. $\square$

Exercise 17.59. Let $u : \mathbb{R}^N \rightarrow \mathbb{R}$ be a Lebesgue measurable function vanishing at infinity and let $1 \leq p, q < \infty$. Prove that

$$\liminf_{t \rightarrow \infty} \int_{S^{N-1}} \|\Delta_{t\nu} u\|_{L^p(\mathbb{R}^N)}^q d\mathcal{H}^{N-1}(\nu) \geq 2^q \beta_N \|u\|_{L^p(\mathbb{R}^N)}^q.$$

Hint: Assume first that $u \in C_c(\mathbb{R}^N)$, then that $u \in L^p(\mathbb{R}^N)$, and then do the general case.

We turn to the proof of Theorem 17.57.

Proof of Theorem 17.57. Let $u \in \dot{B}_q^{s_0,p}(\mathbb{R}^N)$ vanish at infinity. Using spherical coordinates we have

$$\int_{\mathbb{R}^N} \|\Delta_h u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} = \int_0^\infty \int_{S^{N-1}} \|\Delta_{t\nu} u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \frac{dt}{t^{1+sq}},$$

and so by (16.17) and (17.5), $|u|_{B_q^{s,p}} = \|g\|_{s,q}^\sim$, where for $t > 0$,

$$g(t) := \left(\int_{S^{N-1}} \|\Delta_{t\nu} u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q}.$$

Hence, if $u \in L^p(\mathbb{R}^N)$, in view of Lemma 17.58, we are in a position to apply Lemma 16.17 to obtain (17.41). On the other hand, if

$$\liminf_{s \rightarrow 0^+} (s(1-s)pq)^{1/q} |u|_{B_q^{s,p}} < \infty,$$

then by Lemma 16.17 and Exercise 17.59, we obtain

$$\liminf_{s \rightarrow 0^+} [s(1-s)q]^{1/q} |u|_{B_q^{s,p}} \geq 2\beta_N^{1/q} \|u\|_{L^p}.$$

Thus, $u \in L^p(\mathbb{R}^N)$. This completes the proof. $\square$

Next we study the case $s \rightarrow m^-$.

Theorem 17.60. Let $m \in \mathbb{N}$, let $1 < p < \infty$ and $1 \leq q < \infty$, let $m - 1 < s_0 < m$, and let $u \in \dot{B}_q^{s_0,p}(\mathbb{R}^N)$ vanish at infinity. Then u belongs to $\dot{W}^{m,p}(\mathbb{R}^N)$ if and only if

$$\liminf_{s \rightarrow m^-} [s(m-s)q]^{1/q} |u|_{B_q^{s,p}(\mathbb{R}^N)} < \infty.$$

Moreover, if $u \in \dot{W}^{m,p}(\mathbb{R}^N)$, then
(17.46)

$$\lim_{s \rightarrow m^-} [s(m-s)q]^{1/q} |u|_{B_q^{s,p}(\mathbb{R}^N)} = \left(m \int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p(\mathbb{R}^N)}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q}.$$

The overall strategy of the proof is similar to the one of Theorem 11.75. We begin with a preliminary lemma.

Lemma 17.61. *Let $m \in \mathbb{N}$ and $1 \leq p, q < \infty$. Then for every $u \in W_{\text{loc}}^{m,p}(\mathbb{R}^N)$,*

$$(17.47) \quad \liminf_{t \rightarrow 0^+} \frac{1}{t^{mq}} \int_{S^{N-1}} \|\Delta_{t\nu}^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \\ \geq \int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p}^q d\mathcal{H}^{N-1}(\nu).$$

Moreover, if $u \in W^{m,p}(\mathbb{R}^N)$, then

$$(17.48) \quad \lim_{t \rightarrow 0^+} \frac{1}{t^{mq}} \int_{S^{N-1}} \|\Delta_{t\nu}^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) = \int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p}^q d\mathcal{H}^{N-1}(\nu).$$

Proof. Step 1: Assume that $u \in C^\infty(\mathbb{R}^N) \cap W_{\text{loc}}^{m,p}(\mathbb{R}^N)$. Since $\int_0^m \chi_m(t) dt = 1$, by (17.25),

$$(17.49) \quad \frac{\partial^m u}{\partial \nu^m}(x) = t^{-m} \Delta_{t\nu}^m u(x) + \int_0^m \left(\frac{\partial^m u}{\partial \nu^m}(x) - \frac{\partial^m u}{\partial \nu^m}(x + t\tau\nu) \right) \chi_m(\tau) d\tau.$$

Let $B_r := B(0, r)$, $r > 0$. Taking the L^p norm in the x variable over B_r on both sides and using Minkowski's inequality for integrals (see Corollary B.83) gives

$$(17.50) \quad \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p(B_r)} \leq t^{-m} \|\Delta_{t\nu}^m u\|_{L^p} \\ + c \int_0^m \left\| \frac{\partial^m u}{\partial \nu^m} - \frac{\partial^m u}{\partial \nu^m}(\cdot + t\tau\nu) \right\|_{L^p(B_r)} d\tau.$$

By mollification the previous inequality holds for $u \in W_{\text{loc}}^{m,p}(\mathbb{R}^N)$. Taking the $L^q(S^{N-1}; \mathcal{H}^{N-1})$ norm in the ν variable on both sides of (17.50) and using Minkowski's inequality and Minkowski's inequality for integrals (see Corollary B.83) yields

$$\left(\int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p(B_r)}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} \leq t^{-m} \left(\int_{S^{N-1}} \|\Delta_{t\nu}^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} \\ + c \int_0^m \left(\int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} - \frac{\partial^m u}{\partial \nu^m}(\cdot + t\tau\nu) \right\|_{L^p(B_r)}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} d\tau.$$

Letting $t \rightarrow 0^+$ and using the facts that the L^p norm is continuous with respect to translations and that $u \in W_{\text{loc}}^{m,p}(\mathbb{R}^N)$, it follows from the Lebesgue dominated convergence theorem that

$$\left(\int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p(B_r)}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} \\ \leq \liminf_{t \rightarrow 0^+} t^{-m} \left(\int_{S^{N-1}} \|\Delta_{t\nu}^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q}.$$

Letting $r \rightarrow \infty$ and using the Lebesgue monotone convergence theorem gives (17.47).

Step 2: To complete the proof it is enough to assume that $u \in W^{m,p}(\mathbb{R}^N)$ (why?). Then, reasoning in a similar way, from (17.49) we obtain

$$\frac{1}{t^m} \|\Delta_{t\nu}^m u\|_{L^p} \leq \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p} + c \int_0^m \left\| \frac{\partial^m u}{\partial \nu^m}(\cdot + t\tau\nu) - \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p} d\tau,$$

and in turn

$$\begin{aligned} \frac{1}{t^m} \left(\int_{S^{N-1}} \|\Delta_{t\nu} v\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} &\leq \left(\int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} \\ &+ c \int_0^m \left(\int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m}(\cdot + t\tau\nu) - \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} d\tau. \end{aligned}$$

Letting $t \rightarrow 0^+$ and using the fact that the L^p norm is continuous with respect to translations, it follows by the Lebesgue dominated convergence theorem that

$$\begin{aligned} \limsup_{t \rightarrow 0^+} \frac{1}{t^m} \left(\int_{S^{N-1}} \|\Delta_{t\nu} v\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} \\ \leq \left(\int_{S^{N-1}} \left\| \frac{\partial^m u}{\partial \nu^m} \right\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q}. \end{aligned}$$

Combining this inequality with (17.47) completes the proof. $\square$

We turn to the proof of Theorem 17.60.

Proof of Theorem 17.60. Let $u \in \dot{B}_q^{s_0,p}(\mathbb{R}^N)$ vanish at infinity. Using spherical coordinates and the change of variables $\rho = t^{1/m}$ we have

$$\begin{aligned} \int_{\mathbb{R}^N} \|\Delta_h^m u\|_{L^p}^q \frac{dh}{\|h\|^{N+sq}} &= \int_0^\infty \int_{S^{N-1}} \|\Delta_{\rho\nu}^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \frac{d\rho}{\rho^{1+sq}} \\ &= \frac{1}{m} \int_0^\infty \int_{S^{N-1}} \|\Delta_{t^{1/m}\nu}^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \frac{d\rho}{t^{1+sq/m}} \end{aligned}$$

and so by (16.17) and (17.5), $|u|_{B_q^{s,p}} = \|g\|_{s/m,q}^\sim$, where for $t > 0$,

$$g(t) := \left(\int_{S^{N-1}} \|\Delta_{t^{1/m}\nu}^m u\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q}.$$

Hence, if $u \in \dot{W}^{m,p}(\mathbb{R}^N)$, in view of Lemma 17.61, we are in a position to apply Lemma 16.17 to obtain (17.46). On the other hand, if

$$M := \liminf_{s \rightarrow m^-} [s(m-s)q]^{1/q} |u|_{B_q^{s,p}} < \infty,$$

let $u_\varepsilon := \varphi_\varepsilon * u$, where φ_ε is a standard mollifier. By (17.18), $\|\Delta_h^m u_\varepsilon\|_{L^p} \leq \|\Delta_h^m u\|_{L^p}^q$, and so by Lemma 16.17 and (17.47), we obtain

$$\left(m \int_{S^{N-1}} \left\| \frac{\partial^m u_\varepsilon}{\partial \nu^m} \right\|_{L^p}^q d\mathcal{H}^{N-1}(\nu) \right)^{1/q} \leq M.$$

Since $p > 1$ and $u_\varepsilon \rightarrow u$ in $L_{\text{loc}}^1(\mathbb{R}^N)$, it follows from Exercise 11.22 that $u \in \dot{W}^{m,p}(\mathbb{R}^N)$. This completes the proof. $\square$

By adapting the proofs of Theorem 11.75 and of the previous theorem we can show the following result, which extends Theorem 11.75 to $\dot{W}^{m,p}(\Omega)$, $m \geq 2$.

Exercise 17.62. Let $\Omega \subseteq \mathbb{R}^N$ be an open set and for every $h \in \mathbb{R}^N \setminus \{0\}$, let

$$\Omega_h := \{x \in \Omega : x + th \in \Omega \text{ for all } t \in [0, 1]\}.$$

(i) Let $m \in \mathbb{N}$, $1 \leq p < \infty$, and let $u \in \dot{W}^{m,p}(\Omega)$. Prove that

$$\int_{\Omega_h} |\Delta_h u(x)|^p dx \leq \|h\|^{mp} \int_{\Omega} \|\nabla^m u(x)\|^p dx,$$

and that there exists a constant $\kappa = \kappa(m, N, p) > 0$ such that

$$\kappa \int_{\Omega} \|\nabla^m u(x)\|^p dx \leq \limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|\Delta_h u(x)|^p}{\|h\|^{mp}} dx.$$

(ii) Let $m \in \mathbb{N}$, $1 < p < \infty$, and let $u \in L_{\text{loc}}^1(\Omega)$ be such that

$$\limsup_{h \rightarrow 0} \int_{\Omega_h} \frac{|\Delta_h u(x)|^p}{\|h\|^{mp}} dx < \infty.$$

Prove that $u \in \dot{W}^{m,p}(\Omega)$.

Next we consider the case $p = 1$.

Theorem 17.63. Let $m \in \mathbb{N}$, let $1 \leq q < \infty$, and let $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ be a function vanishing at infinity such that $|u|_{B_q^{s_0,1}(\mathbb{R}^N)} < \infty$ for some $m - 1 < s_0 < m$. Then u belongs to $W_{\text{loc}}^{m-1,1}(\mathbb{R}^N)$ and the distributional gradient $D(\partial^\alpha u)$ of $\partial^\alpha u$ belongs to $\mathcal{M}_b(\mathbb{R}^N; \mathbb{R}^N)$ for every multi-index α with $|\alpha| = m - 1$ if and only if

$$\liminf_{s \rightarrow m^-} [s(m-s)q]^{1/q} |u|_{B_q^{s,1}(\mathbb{R}^N)} < \infty.$$

The proof is left as an exercise.

Exercise 17.64. Given a function $u \in L_{\text{loc}}^1(\mathbb{R}^N)$, a multi-index $\alpha \in \mathbb{N}_0^N$, and $h \in \mathbb{R}^N$, define

$$\Delta_h^\alpha u(x) := \Delta_{h_N e_N}^{\alpha_N} (\cdots \Delta_{h_2 e_2}^{\alpha_2} (\Delta_{h_1 e_1}^{\alpha_1} u(x)))$$

and for $1 \leq p \leq \infty$, $1 \leq q < \infty$, and $s > 0$,

$$|u|_{B_q^{s,p}(\mathbb{R}^N)}^e := \sum_{|\alpha|=\lfloor s \rfloor + 1} \left(\int_{\mathbb{R}^N} \|\Delta_h^\alpha u\|_{L^p(\mathbb{R}^N)}^q \frac{dh}{\|h\|^{N+sq}} \right)^{1/q}.$$

(i) Prove that $|\cdot|_{B_q^{s,p}(\mathbb{R}^N)}^e$ is an equivalent seminorm.

(ii) Given $m \in \mathbb{N}$, $1 \leq p < \infty$, and $m - 1 < s_0 < m$, let $u \in \dot{B}_q^{s_0,p}(\mathbb{R}^N)$ vanish at infinity. Prove that if $u \in \dot{W}^{m,p}(\mathbb{R}^N)$, then the limit

$$\lim_{s \rightarrow m^-} [s(m-s)q]^{1/q} |u|_{B_q^{s,p}(\mathbb{R}^N)}^e$$

exists, and compute it.

Exercise 17.65. Given $m \in \mathbb{N}$, $1 \leq p < \infty$, and $m - 1 < s_0 < m$, let $u \in \dot{B}_q^{s_0,p}(\mathbb{R}^N)$ vanish at infinity. Prove that if $u \in \dot{W}^{m,p}(\mathbb{R}^N)$, then the limit

$$\lim_{s \rightarrow m^-} [s(m-s)q]^{1/q} |u|_{B_q^{s,p}(\mathbb{R}^N)}^\infty$$

exists, and compute it. Here, $|\cdot|_{B_q^{s,p}(\mathbb{R}^N)}^\infty$ is the seminorm defined in (17.24).

17.6. Besov Spaces and Derivatives

In this section we study the regularity of functions in Besov spaces. We begin by considering the space $B_q^{s,p}(\mathbb{R}^N)$ when $s \in \mathbb{N}$. In this case we can show that

$$B_1^{s,p}(\mathbb{R}^N) \hookrightarrow W^{s,p}(\mathbb{R}^N).$$

Theorem 17.66. *Let $m \in \mathbb{N}$ and let $1 \leq p \leq \infty$. If $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ is such that*

$$(17.51) \quad \int_0^1 \sup_{\|h\| \leq t} \|\Delta_h^{m+1} u\|_{L^p} \frac{dt}{t^{1+m}} < \infty,$$

then $u \in \dot{W}^{m,p}(\mathbb{R}^N)$. If, in addition, $u \in B_1^{m,p}(\mathbb{R}^N)$, then

$$(17.52) \quad \|\nabla^m u\|_{L^p(\mathbb{R}^N)} \leq c \|u\|_{B_1^{m,p}(\mathbb{R}^N)}$$

for some constant $c = c(N, m) > 0$. In particular, $B_1^{m,p}(\mathbb{R}^N) \hookrightarrow W^{m,p}(\mathbb{R}^N)$.

Proof. Step 1: We consider first the case $m = 1$. Assume that u satisfies (17.51) with $m = 1$. Then for every $\nu \in S^{N-1}$ and $x \in \mathbb{R}^N$, using telescopic

sums for $k, l \in \mathbb{N}$ we can write

$$\begin{aligned}
 & 2^{l+k} \Delta_{2^{-l-k}\nu} u(x) - 2^k \Delta_{2^{-k}\nu} u(x) \\
 &= - \sum_{n=k}^{l+k-1} [2^n (u(x + 2^{-n}\nu) - u(x)) - 2^{n+1} (u(x + 2^{-n-1}\nu) - u(x))] \\
 &= - \sum_{n=k}^{l+k-1} 2^n \Delta_{2^{-n-1}\nu}^2 u(x).
 \end{aligned}$$

Taking the L^p norm in x and making a change of variables we obtain

$$\begin{aligned}
 \|2^{l+k} \Delta_{2^{-l-k}\nu} u - 2^k \Delta_{2^{-k}\nu} u\|_{L^p} &\leq \sum_{n=k}^{\infty} 2^n \|\Delta_{2^{-n-1}\nu}^2 u\|_{L^p} \\
 &\leq \sum_{n=k}^{\infty} \int_{2^{-n-1}}^{2^{-n}} \sup_{\|h\| \leq t} \|\Delta_h^2 u\|_{L^p} \frac{dt}{t^2} \\
 &= \int_0^{2^{-k}} \sup_{\|h\| \leq t} \|\Delta_h^2 u\|_{L^p} \frac{dt}{t^2}.
 \end{aligned}$$

By (17.51) the right-hand side of the previous inequality goes to zero as $k \rightarrow \infty$. Hence, the sequence $\{2^k \Delta_{2^{-k}\nu} u\}_k$ is a Cauchy sequence in $L^p(\mathbb{R}^N)$, and thus it converges to some function $v_\nu \in L^p(\mathbb{R}^N)$. For every $\varphi \in C_c^\infty(\mathbb{R}^N)$, by a change of variables we have that

$$\int_{\mathbb{R}^N} 2^k \Delta_{2^{-k}\nu} u(x) \varphi(x) dx = \int_{\mathbb{R}^N} u(y) \frac{\varphi(y - 2^{-k}\nu) - \varphi(y)}{2^{-k}} dy.$$

Letting $k \rightarrow \infty$ and using the facts that $2^k \Delta_{2^{-k}\nu} u \rightarrow v_\nu$ in $L^p(\mathbb{R}^N)$ on the left-hand side and the Lebesgue dominated convergence theorem on the right-hand side, we obtain

$$\int_{\mathbb{R}^N} v_\nu(x) \varphi(x) dx = - \int_{\mathbb{R}^N} u(y) \frac{\partial \varphi}{\partial \nu}(y) dy,$$

which shows that v_ν is the distributional directional derivative $\frac{\partial u}{\partial \nu}$. Thus, $u \in \dot{W}^{1,p}(\mathbb{R}^N)$.

Step 2: If $u \in B_1^{1,p}(\mathbb{R}^N) \cap C^\infty(\mathbb{R}^N)$, then

$$\frac{\partial u}{\partial \nu}(x) = \lim_{n \rightarrow \infty} \frac{u(x + 2^{-n}\nu) - u(x)}{2^{-n}},$$

and so using telescopic sums as in the previous step we can write

$$\Delta_\nu u(x) - \frac{\partial u}{\partial \nu}(x) = \sum_{n=0}^{\infty} 2^n \Delta_{2^{-n-1}\nu}^2 u(x).$$

Reasoning as in the previous step we obtain

$$\begin{aligned} \left\| \frac{\partial u}{\partial \nu} \right\|_{L^p} &\leq \|\Delta_\nu u\|_{L^p} + \sum_{n=0}^{\infty} 2^n \|\Delta_{2^{-n-1}\nu}^2 u\|_{L^p} \\ &\leq 2\|u\|_{L^p} + \sum_{n=0}^{\infty} \int_{2^{-n-1}}^{2^{-n}} \sup_{\|h\| \leq t} \|\Delta_h^2 u\|_{L^p} \frac{dt}{t^2} \\ &\leq 2\|u\|_{L^p} + |u|_{B_1^{1,p}}, \end{aligned}$$

where we used (17.7) and (17.24). Using a mollification argument, it follows from Proposition 17.12 that the same inequality holds for $u \in B_1^{1,p}(\mathbb{R}^N)$. Hence, $u \in W^{1,p}(\mathbb{R}^N)$.

Step 3: Assume next that $m \geq 2$ and that u satisfies (17.51). By Lemma 17.18 we can write

$$\Delta_h^m u = 2^{-m} \Delta_{2h}^m u + P_{m-1}(T_h) \Delta_h^{m+1} u,$$

where T_h is the operator given in (17.10) and P_{m-1} is a polynomial of degree $m-1$. Hence,

$$2^{(n+1)m} \Delta_{2^{-n-1}\nu}^m u - 2^{nm} \Delta_{2^{-n}\nu}^m u = 2^{(n+1)m} P_{m-1}(T_{2^{-n-1}\nu}) \Delta_{2^{-n-1}\nu}^{m+1} u,$$

and so using telescopic sums for $k, l \in \mathbb{N}$ we can write

$$\begin{aligned} 2^{(l+k)m} \Delta_{2^{-l-k}\nu}^m u - 2^{km} \Delta_{2^{-k}\nu}^m u \\ = \sum_{n=k}^{l+k-1} 2^{(n+1)m} P_{m-1}(T_{2^{-n-1}\nu}) \Delta_{2^{-n-1}\nu}^{m+1} u. \end{aligned}$$

Taking the L^p norm in x and making a change of variables we obtain

$$\begin{aligned} \|2^{(l+k)m} \Delta_{2^{-l-k}\nu}^m u - 2^{km} \Delta_{2^{-k}\nu}^m u\|_{L^p} &\leq c \sum_{n=k}^{\infty} 2^{nm} \|\Delta_{2^{-n-1}\nu}^{m+1} u\|_{L^p} \\ &\leq c \sum_{n=k}^{\infty} \int_{2^{-n-1}}^{2^{-n}} \sup_{\|h\| \leq t} \|\Delta_h^{m+1} u\|_{L^p} \frac{dt}{t^{1+m}} \\ &= c \int_0^{2^{-k}} \sup_{\|h\| \leq t} \|\Delta_h^{m+1} u\|_{L^p} \frac{dt}{t^{1+m}}. \end{aligned}$$

We can now continue as in Step 1 to conclude that if $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ satisfies (17.51), then for every $\nu \in S^{N-1}$ there exists the distributional directional derivative $\frac{\partial^m u}{\partial \nu^m} \in L^p(\mathbb{R}^N)$. In view of Exercise 11.22, we have that $u \in \dot{W}^{m,p}(\mathbb{R}^N)$.

Step 4: In the case $m \geq 2$ and $u \in B_1^{m,p}(\mathbb{R}^N) \cap C^\infty(\mathbb{R}^N)$ we use (17.25) to show that for every $\nu \in S^{N-1}$ and $x \in \mathbb{R}^N$ we have

$$\frac{\partial^m u}{\partial \nu^m}(x) = \lim_{n \rightarrow \infty} \frac{\Delta_{2^{-n}\nu}^m u(x)}{2^{-nm}}.$$

We can now proceed as in Step 2 using Step 3 in place of Step 1. We omit the details.

Step 5: The embedding $B_1^{m,p}(\mathbb{R}^N) \hookrightarrow W^{m,p}(\mathbb{R}^N)$ follows from the Gagliardo–Nirenberg interpolation theorem (see Theorem 12.87) and (17.52). $\square$

Exercise 17.67. Let $m \in \mathbb{N}$, let $1 \leq p \leq \infty$, and let $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ be such that

$$\int_0^1 \sup_{\|h\| \leq t} \|\Delta_h^{m+2} u\|_{L^p} \frac{dt}{t^{1+m}} < \infty.$$

Imitate the proof of Steps 1 and 3 to show that the sequence

$$\{2^{km} \Delta_{2^{-k}\nu}^m u + 2^{(k-2)m-1} \Delta_{2^{-k+2}\nu}^m u\}_{k \in \mathbb{N}}$$

is a Cauchy sequence in $L^p(\mathbb{R}^N)$, and that for every $\nu \in S^{N-1}$ there exists the distributional directional derivative $\frac{\partial^m u}{\partial \nu^m} \in L^p(\mathbb{R}^N)$. Hint: Use Exercise 17.19.

As a consequence of the previous theorem we can prove the inclusion $B_q^{s,p}(\mathbb{R}^N) \subseteq W^{\lfloor s \rfloor, p}(\mathbb{R}^N)$, where $s > 1$ is not an integer.

Corollary 17.68. Let $1 \leq p, q \leq \infty$ and let $s \in (1, \infty) \setminus \mathbb{N}$. If $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ is such that

$$\int_0^1 \sup_{\|h\| \leq t} \|\Delta_h^{\lfloor s \rfloor + 1} u\|_{L^p}^q \frac{dt}{t^{1+sq}} < \infty,$$

then $u \in \dot{W}^{\lfloor s \rfloor, p}(\mathbb{R}^N)$. Moreover, if $u \in B_q^{s,p}(\mathbb{R}^N)$, then there exists a constant $c = c(N, p, q, s) > 0$ such that

$$\|\nabla^{\lfloor s \rfloor} u\|_{L^p(\mathbb{R}^N)} \leq c \|u\|_{B_q^{m,p}(\mathbb{R}^N)}.$$

In particular, $B_q^{s,p}(\mathbb{R}^N) \hookrightarrow W^{\lfloor s \rfloor, p}(\mathbb{R}^N)$.

Proof. Reasoning as in Step 3 of the previous proof with $m = \lfloor s \rfloor$ we get

$$\begin{aligned} \|2^{(l+k)m} \Delta_{2^{-l-k}\nu}^m u - 2^{km} \Delta_{2^{-k}\nu}^m u\|_{L^p} &\leq c \int_0^{2^{-k}} \sup_{\|h\| \leq t} \|\Delta_h^{m+1} u\|_{L^p} \frac{dt}{t^{1+m}} \\ &\leq c \left(\int_0^{2^{-k}} \sup_{\|h\| \leq t} \|\Delta_h^{m+1} u\|_{L^p}^q \frac{dt}{t^{1+sq}} \right)^{1/q} \left(\int_0^{2^{-k}} \frac{dt}{t^{1-(s-m)q'}} \right)^{1/q'}, \end{aligned}$$

where we used Hölder's inequality. We can now continue as in the previous proof. We omit the details. $\square$

In conclusion, also by Theorem 17.33, we have shown that if $s > 0$ is not an integer, then

$$W^{\lfloor s \rfloor + 1, p}(\mathbb{R}^N) \hookrightarrow B_q^{s, p}(\mathbb{R}^N) \hookrightarrow W^{\lfloor s \rfloor, p}(\mathbb{R}^N),$$

where $W^{0, p}(\mathbb{R}^N) := L^p(\mathbb{R}^N)$, while if $s = m \in \mathbb{N}$, with $m \geq 2$, then

$$B_1^{m, p}(\mathbb{R}^N) \hookrightarrow W^{m, p}(\mathbb{R}^N) \hookrightarrow B_q^{m-1, p}(\mathbb{R}^N).$$

Finally, when $s = 1$,

$$B_1^{1, p}(\mathbb{R}^N) \hookrightarrow W^{1, p}(\mathbb{R}^N).$$

In what follows we say that a function $u \in L_{\text{loc}}^1(\mathbb{R}^N; \mathbb{R}^M)$ belongs to $\dot{B}_q^{s, p}(\mathbb{R}^N; \mathbb{R}^M)$ if all its components u_i belong to $\dot{B}_q^{s-\ell, p}(\mathbb{R}^N)$ for every $i = 1, \dots, M$ and we set $|u|_{B_q^{s, p}(\mathbb{R}^N)} := \sum_{i=1}^M |u_i|_{B_q^{s, p}(\mathbb{R}^N)}$. We define $B_q^{s, p}(\mathbb{R}^N; \mathbb{R}^M)$ analogously.

The next theorem is the main result of this section.

Theorem 17.69. *Given $1 \leq p, q \leq \infty$ and $s > 1$, let $\ell \in \mathbb{N}$ be such that $\ell \leq \max\{n \in \mathbb{N} : n < s\}$. Then $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ belongs to $\dot{B}_q^{s, p}(\mathbb{R}^N)$ if and only if the weak derivative $\nabla^\ell u$ belongs to $\dot{B}_q^{s-\ell, p}(\mathbb{R}^N; \mathbb{R}^{M_\ell})$. Moreover, there exists a constant $c = c(N, p, q, s) > 0$ such that*

$$c^{-1} |u|_{B_q^{s, p}(\mathbb{R}^N)} \leq |\nabla^\ell u|_{B_q^{s-\ell, p}(\mathbb{R}^N)} \leq c |u|_{B_q^{s, p}(\mathbb{R}^N)}.$$

Proof. Step 1: Let $\ell, m \in \mathbb{N}$ be such that $1 \leq \ell < m$ and let $u \in W_{\text{loc}}^{\ell, p}(\mathbb{R}^N)$. We claim that

$$(17.53) \quad |u|_{W^{\ell, p}} \leq c \int_0^\infty \sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p} \frac{dt}{t^{1+\ell}}.$$

Without loss of generality, we may assume that the right-hand side is finite. By Remark 17.27 for every $k \in \mathbb{Z}$ we can find $v_k \in L^p(\mathbb{R}^N)$ and $w_k \in W^{m, p}(\mathbb{R}^N)$ such that $u = v_k + w_k$ and

$$\|v_k\|_{L^p} + 2^{km} |w_k|_{W^{m, p}} \leq c_1 \sup_{\|h\| \leq c2^k} \|\Delta_h^m u\|_{L^p}.$$

Multiplying both sides by $2^{-k\ell}$ and summing over k gives

$$\begin{aligned} \sum_{k \in \mathbb{Z}} (2^{-k\ell} \|v_k\|_{L^p} + 2^{k(m-\ell)} |w_k|_{W^{m, p}}) &\leq c_1 \sum_{k \in \mathbb{Z}} 2^{-k\ell} \sup_{\|h\| \leq c2^k} \|\Delta_h^m u\|_{L^p} \\ &\leq c_1 \int_0^\infty \sup_{\|h\| \leq ct} \|\Delta_h^m u\|_{L^p} \frac{dt}{t^{1+\ell}}, \end{aligned}$$

where the last inequality follows since the function $t \mapsto \sup_{\|h\| \leq ct} \|\Delta_h^m u\|_{L^p}$ is increasing. Since the right-hand side of the previous inequality is finite and $m > \ell$, we have that $\|v_k\|_{L^p} \rightarrow 0$ as $k \rightarrow -\infty$ and $|w_k|_{W^{m, p}} \rightarrow 0$ as $k \rightarrow \infty$. As $u = v_k + w_k$, we obtain that $\|u - w_k\|_{L^p} \rightarrow 0$ as $k \rightarrow -\infty$.

Moreover, writing $u = v_k + w_k = v_{k+1} + w_{k+1}$, we have that $f_k := v_{k+1} - v_k = w_k - w_{k+1} \in W^{m,p}(\mathbb{R}^N)$ with

$$\|f_k\|_{L^p} + 2^{km}|f_k|_{W^{m,p}} \leq c_1 \sup_{\|h\| \leq c2^{k+1}} \|\Delta_h^m u\|_{L^p}.$$

In turn, by the Gagliardo–Nirenberg interpolation theorem (see Theorem 12.87), Young's inequality with exponent m/ℓ , and the previous inequality,

$$\begin{aligned} |f_k|_{W^{\ell,p}} &\leq \|f_k\|_{L^p}^{1-\ell/m} |f_k|_{W^{m,p}}^{\ell/m} \leq 2^{-k\ell} \|f_k\|_{L^p} + 2^{k(m-\ell)} |f_k|_{W^{m,p}} \\ &\leq c_1 2^{-k\ell} \sup_{\|h\| \leq c2^{k+1}} \|\Delta_h^m u\|_{L^p}. \end{aligned}$$

Hence,

$$\begin{aligned} \sum_{k \in \mathbb{Z}} |f_k|_{W^{\ell,p}} &\leq c_1 \sum_{k \in \mathbb{Z}} 2^{-k\ell} \sup_{\|h\| \leq c2^{k+1}} \|\Delta_h^m u\|_{L^p} \\ &\leq c_1 \int_0^\infty \sup_{\|h\| \leq ct} \|\Delta_h^m u\|_{L^p} \frac{dt}{t^{1+\ell}} < \infty. \end{aligned}$$

Fix $n \in \mathbb{N}$. Since

$$s_j := \sum_{k=-j}^{n-1} f_k = w_{-j} - w_n \rightarrow u - w_n \quad \text{in } L^p(\mathbb{R}^N)$$

as $j \rightarrow \infty$ and for $j > i$, $|s_j - s_i|_{W^{\ell,p}} \leq \sum_{k=-j}^{-i} |f_k|_{W^{\ell,p}} \rightarrow 0$ as $i \rightarrow \infty$, it follows from Corollary 12.86 that the sequence $\{s_j\}_{j \in \mathbb{N}}$ is a Cauchy sequence in $W^{\ell,p}(\mathbb{R}^N)$, but since it converges to $u - w_n$ in $L^p(\mathbb{R}^N)$, necessarily, $s_j \rightarrow u - w_n$ in $W^{\ell,p}(\mathbb{R}^N)$ as $j \rightarrow \infty$. In turn, $\sum_{k=-\infty}^{n-1} f_k = u - w_n$ in $W^{\ell,p}(\mathbb{R}^N)$ and

$$(17.54) \quad |u - w_n|_{W^{\ell,p}} \leq \sum_{k=-\infty}^{n-1} |f_k|_{W^{\ell,p}} \leq c_1 \int_0^\infty \sup_{\|h\| \leq ct} \|\Delta_h^m u\|_{L^p} \frac{dt}{t^{1+\ell}}.$$

For $i < n$ we have $|w_i - w_n|_{W^{\ell,p}} \leq \sum_{k=i}^{n-1} |f_k|_{W^{\ell,p}} \rightarrow 0$ as $i \rightarrow \infty$. Hence, $\{w_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $W^{\ell,p}(\mathbb{R}^N)$, but since $|w_n|_{W^{m,p}} \rightarrow 0$ as $n \rightarrow \infty$, we have that $\{w_n\}_{n \in \mathbb{N}}$ converges in $W^{\ell,p}(\mathbb{R}^N)$ to a polynomial P_u . Since $|P_u|_{W^{\ell,p}} < \infty$, P_u must have degree less than ℓ , so that $|P_u|_{W^{\ell,p}} = 0$. In turn, letting $n \rightarrow \infty$ in (17.54) we conclude that (17.53) holds.

Step 2: Let $\ell, m \in \mathbb{N}$ be such that $1 \leq \ell < m$ and let $u \in W_{\text{loc}}^{\ell,p}(\mathbb{R}^N)$. We claim that for every $\alpha \in \mathbb{N}_0^N$ with $|\alpha| = \ell$ and every $t > 0$,

$$(17.55) \quad \sup_{\|\xi\| \leq t} \|\Delta_\xi^m(\partial^\alpha u)\|_{L^p} \leq c \int_0^t \sup_{\|h\| \leq 2\tau} \|\Delta_h^m u\|_{L^p} \frac{d\tau}{\tau^{1+\ell}}.$$

By the previous step

$$\|\partial^\alpha u\|_{L^p} \leq c \int_0^\infty \sup_{\|h\| \leq \tau} \|\Delta_h^m u\|_{L^p} \frac{d\tau}{\tau^{1+\ell}}.$$

Replacing u with $\Delta_\xi^m u$, where $\|\xi\| \leq t$, gives

$$\|\Delta_\xi^m(\partial^\alpha u)\|_{L^p} = \|\partial^\alpha(\Delta_\xi^m u)\|_{L^p} \leq c \int_0^\infty \sup_{\|h\| \leq \tau} \|\Delta_h^m(\Delta_\xi^m u)\|_{L^p} \frac{d\tau}{\tau^{1+\ell}}.$$

By (17.7), if $\|h\| \leq \tau$ and $\|\xi\| \leq t$,

$$\sup_{\|h\| \leq \tau} \|\Delta_h^m(\Delta_\xi^m u)\|_{L^p} \leq 2^m \sup_{\|\zeta\| \leq \min\{\tau, t\}} \|\Delta_\zeta^m u\|_{L^p},$$

and so

$$\begin{aligned} \|\Delta_\xi^m(\partial^\alpha u)\|_{L^p} &\leq c \int_0^t \sup_{\|h\| \leq \tau} \|\Delta_h^m u\|_{L^p} \frac{d\tau}{\tau^{1+\ell}} + c \sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p} \int_t^\infty \frac{d\tau}{\tau^{1+\ell}} \\ &= c \int_0^t \sup_{\|h\| \leq \tau} \|\Delta_h^m u\|_{L^p} \frac{d\tau}{\tau^{1+\ell}} + \frac{c}{t^\ell} \sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p} \\ &\leq c \int_0^t \sup_{\|h\| \leq 2\tau} \|\Delta_h^m u\|_{L^p} \frac{d\tau}{\tau^{1+\ell}}, \end{aligned}$$

where in the last inequality we used the fact that for $t > 0$,

$$\frac{1}{t^\ell} \sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p} \leq c \int_{t/2}^t \sup_{\|h\| \leq 2\tau} \|\Delta_h^m u\|_{L^p} \frac{d\tau}{\tau^{1+\ell}},$$

since the function $\tau \mapsto \sup_{\|h\| \leq \tau} \|\Delta_h^m u\|_{L^p}$ is increasing.

Step 3: Let $u \in C^\infty(\mathbb{R}^N)$ be such that $|u|_{B_q^{s,p}(\mathbb{R}^N)} < \infty$ and let $\ell := \max\{n \in \mathbb{N} : n < s\}$ and $m := \lfloor s \rfloor + 1$. Assume that $1 \leq q < \infty$. For every $\alpha \in \mathbb{N}_0^N$ with $|\alpha| = \ell$, by the previous step,

$$\begin{aligned} |\partial^\alpha u|_{B_q^{s-\ell,p}(\mathbb{R}^N)}^\infty &= \left(\int_0^\infty \sup_{\|\xi\| \leq t} \|\Delta_\xi^m(\partial^\alpha u)\|_{L^p}^q \frac{dt}{t^{1+q(s-\ell)}} \right)^{1/q} \\ &\leq c \left(\int_0^\infty \left(\int_0^t \sup_{\|h\| \leq 2\tau} \|\Delta_h^m u\|_{L^p} \frac{d\tau}{\tau^{1+\ell}} \right)^q \frac{dt}{t^{1+q(s-\ell)}} \right)^{1/q}. \end{aligned}$$

By Hardy's inequality (see Theorem C.41) the right-hand of the previous inequality side is bounded from above by $c|u|_{B_q^{s,p}(\mathbb{R}^N)}^\infty$. The hypothesis that $u \in C^\infty(\mathbb{R}^N)$ can be removed by mollification (see the proof of Proposition 17.12).

Next, if $u \in \dot{B}_q^{s,p}(\mathbb{R}^N)$ and s is not an integer, then $u \in \dot{W}^{\lfloor s \rfloor, p}(\mathbb{R}^N)$ by Corollary 17.68. On the other hand, if $s \in \mathbb{N}$, then $\ell = s - 1$ and $m = s + 1$ and again by Hardy's inequality the right-hand side of the previous inequality is finite, and so

$$\int_0^t \sup_{\|h\| \leq 2\tau} \|\Delta_h^{\ell+2} u\|_{L^p} \frac{d\tau}{\tau^{1+\ell}} < \infty$$

for $\mathcal{L}^1$ -a.e. $t > 0$. It follows by Exercise 17.67 that $u \in \dot{W}^{\ell,p}(\mathbb{R}^N)$. Hence, in both cases by the first part of this step we get $|\nabla^\ell u|_{B_q^{s-\ell,p}(\mathbb{R}^N)}^\infty \leq c|u|_{B_q^{s,p}(\mathbb{R}^N)}^\infty$.

Conversely, if $u \in W_{\text{loc}}^{\ell,p}(\mathbb{R}^N)$ is such that $|\nabla^\ell u|_{B_q^{s-\ell,p}(\mathbb{R}^N)}^\infty < \infty$, then by Lemma 17.25 applied to $\Delta_h^{m-\ell}u$,

$$\begin{aligned} \|\Delta_h^m u\|_{L^p} &= \|\Delta_h^\ell(\Delta_h^{m-\ell}u)\|_{L^p} \leq c\|h\|^\ell \|\nabla^\ell(\Delta_h^{m-\ell}u)\|_{L^p} \\ &\leq ct^\ell \sup_{\|h\| \leq t} \|\Delta_h^{m-\ell}(\nabla^\ell u)\|_{L^p}. \end{aligned}$$

Integrating both sides gives

$$\begin{aligned} \left(\int_0^\infty \sup_{\|h\| \leq t} \|\Delta_h^m u\|_{L^p}^q \frac{dt}{t^{1+qs}} \right)^{1/q} &\leq c \left(\int_0^\infty \sup_{\|h\| \leq t} \|\Delta_h^{m-\ell}(\nabla^\ell u)\|_{L^p}^q \frac{dt}{t^{1+(q-\ell)s}} \right)^{1/q} \\ &\leq c|\nabla^\ell u|_{B_q^{s-\ell,p}(\mathbb{R}^N)}^\infty, \end{aligned}$$

which concludes the proof. $\square$

Exercise 17.70. Let $s > 0$ and $1 \leq p \leq \infty$ be such that $m < s - N/p < m + 1$ for some $m \in \mathbb{N}$. Prove that for every $u \in \dot{B}_q^{s,p}(\mathbb{R}^N)$ there exists a polynomial P_u such that

$$|\nabla^m(u - P_u)|_{C^{0,s-m-N/p}(\mathbb{R}^N)} \leq c|u|_{B_q^{s,p}(\mathbb{R}^N)}$$

for some constant $c = c(N, p, q, s) > 0$.

Exercise 17.71. Let $s > 0$ and $1 \leq p \leq \infty$ be such that $s - N/p = m + 1$ for some $m \in \mathbb{N}$, let $\theta \in (0, 1)$, and let $1 \leq q \leq \infty$. Prove that for every $u \in \dot{B}_q^{s,p}(\mathbb{R}^N)$ there exists a polynomial P_u such that

$$|\nabla^m(u - P_u)|_{C^{0,\theta}(\mathbb{R}^N)} \leq c|u|_{B_q^{s,p}(\mathbb{R}^N)}$$

for some constant $c = c(N, p, q, s, \theta) > 0$.

Exercise 17.72. Given $1 \leq p, q \leq \infty$ and $s > 1$, let $\ell \in \mathbb{N}$ be such that $\ell \leq \max\{n \in \mathbb{N} : n < s\}$. Prove that $u \in L^p(\mathbb{R}^N)$ belongs to $B_q^{s,p}(\mathbb{R}^N)$ if and only if $\nabla^\ell u$ belongs to $B_q^{s-\ell,p}(\mathbb{R}^N; \mathbb{R}^{M_\ell})$.

17.7. Yet Another Equivalent Norm

In this section we introduce an equivalent norm in $B_q^{s,p}(\mathbb{R}^N)$, which relies on the Littlewood–Paley decomposition (see Theorem 10.78). Given $a > 0$, let $\varphi_0 \in C_c^\infty(\mathbb{R}^N)$ be a nonnegative radial function such that

$$(17.56) \quad \text{supp } \varphi_0 = \overline{B(0, 2a)} \setminus B(0, a/2), \quad \varphi_0 > 0 \text{ in } B(0, 2a) \setminus \overline{B(0, a/2)},$$

$$(17.57) \quad \sum_{k \in \mathbb{Z}} \varphi_0(2^{-k}x) = 1 \text{ for every } x \in \mathbb{R}^N \setminus \{0\}.$$

Note that for every $x \in \mathbb{R}^N \setminus \{0\}$ we can find a unique $k_x \in \mathbb{Z}$ such that $2^{k_x-1}a < \|x\| \leq 2^{k_x}a$, and since

$$(17.58) \quad \text{supp } \varphi_0(2^{-k} \cdot) = \overline{B(0, 2^{k+1}a)} \setminus B(0, 2^{k-1}a)$$

for every $k \in \mathbb{Z}$, we have that

$$(17.59) \quad 1 = \sum_{k \in \mathbb{Z}} \varphi_0(2^{-k}x) = \varphi_0(2^{-k_x-1}x) + \varphi_0(2^{-k_x}x) + \varphi_0(2^{-k_x+1}x).$$

By Theorem 10.62 the inverse Fourier transform of the function $\varphi_0(2^{-k} \cdot)$ is given by

$$(17.60) \quad \begin{aligned} (\varphi_0(2^{-k} \cdot))^\vee(x) &= \int_{\mathbb{R}^N} e^{2\pi i x \cdot y} \varphi_0(2^{-k}y) dy \\ &= 2^{kN} \int_{\mathbb{R}^N} e^{2\pi i x 2^k \cdot z} \varphi_0(z) dz = 2^{kN} \varphi_0^\vee(2^k x) =: \psi_k(x), \end{aligned}$$

where we used the change of variables $z = 2^{-k}y$. Since φ_0 is radial, it follows that ψ_k is real-valued (see Exercise 10.69). Note that ψ_k belongs to the space of real-valued rapidly decreasing functions $\mathcal{S}(\mathbb{R}^N)$ (see Definition 10.29).

By Young's inequality (Theorem 10.48) for every $u \in L^p(\mathbb{R}^N)$, $1 \leq p \leq \infty$, we have that $\psi_k * u \in L^p(\mathbb{R}^N)$, and so if $1 \leq q < \infty$ and $s \in \mathbb{R}$ we may define

$$(17.61) \quad |u|_{B_q^{s,p}(\mathbb{R}^N)}^\vee := \left(\sum_{k \in \mathbb{Z}} 2^{kqs} \|u * \psi_k\|_{L^p(\mathbb{R}^N)}^q \right)^{1/q},$$

while if $q = \infty$,

$$(17.62) \quad |u|_{B_\infty^{s,p}(\mathbb{R}^N)}^\vee := \sup_{k \in \mathbb{Z}} 2^{ks} \|u * \psi_k\|_{L^p(\mathbb{R}^N)}.$$

Exercise 17.73. Prove that there exists a radial function $\varphi_0 \in C_c^\infty(\mathbb{R}^N)$ satisfying (17.56)–(17.57). Hint: Let $\varphi \in C_c^\infty(\mathbb{R}^N)$ satisfy (17.56) and take

$$\varphi_0(x) := \frac{\varphi(x)}{\sum_{k \in \mathbb{Z}} \varphi(2^{-k}x)}, \quad x \in \mathbb{R}^N.$$

Remark 17.74. To relate (17.56)–(17.57) to the Littlewood–Paley decomposition of u (see Theorem 10.78), define

$$(17.63) \quad \omega_0(x) := 1 - \sum_{k=0}^{\infty} \varphi_0(2^{-k}x), \quad x \in \mathbb{R}^N.$$

Then by (17.58), $\omega_0 = 1$ in $B(0, 2^{-1}a)$, while by (17.56) and (17.59), $\text{supp } \omega_0 = \overline{B(0, a)}$. Moreover,

$$\omega_0(x/2) - \omega_0(x) = \sum_{k=0}^{\infty} \varphi_0(2^{-k}x) - \sum_{k=0}^{\infty} \varphi_0(2^{-k-1}x) = \varphi_0(x),$$

and so (10.36) holds. Hence, (10.35) holds. In turn, given $u \in L^p(\mathbb{R}^N)$, by (10.37) and (10.38) and Theorems 10.71 and 10.62 and Exercises 10.75 and 10.77, we have that

$$(17.64) \quad S_k(u) = (\omega_0(2^{-k}\cdot)\hat{u})^\vee = u * (\omega_0(2^{-k}\cdot))^\vee,$$

$$(17.65) \quad \dot{\Lambda}_k(u) = (\varphi_0(2^{-k}\cdot)\hat{u})^\vee = u * \psi_k.$$

Observe that the seminorm 17.61 has been defined also for $s \leq 0$. More generally, given a real-valued tempered distribution $T \in \mathcal{S}'(\mathbb{R}^N)$ (see Definition 10.32), by Theorem 10.56 we have that the tempered distribution $T * \psi_k$ can be identified with the C^∞ function $x \mapsto T(\psi_k^x)$, where $\psi_k^x(y) := \psi_k(x-y)$, $y \in \mathbb{R}^N$. Thus, with a slight abuse of notation, given $1 \leq p, q \leq \infty$ and $s \in \mathbb{R}$, for $T \in \mathcal{S}'(\mathbb{R}^N)$ and $1 \leq q < \infty$ we can define

$$(17.66) \quad |T|_{B_q^{s,p}(\mathbb{R}^N)}^\vee := \left(\sum_{k \in \mathbb{Z}} 2^{kqs} \|T * \psi_k\|_{L^p(\mathbb{R}^N)}^q \right)^{1/q},$$

while for $q = \infty$,

$$(17.67) \quad |T|_{B_\infty^{s,p}(\mathbb{R}^N)}^\vee := \sup_{k \in \mathbb{Z}} 2^{ks} \|T * \psi_k\|_{L^p(\mathbb{R}^N)}.$$

Exercise 17.75. Let $1 \leq p, q \leq \infty$ and $s \in \mathbb{R}$, and let $\varphi_0 \in C_c^\infty(\mathbb{R}^N)$ satisfy (17.56)–(17.57). Given $T \in \mathcal{S}'(\mathbb{R}^N)$, prove that $|T|_{B_q^{s,p}(\mathbb{R}^N)}^\vee = 0$ if and only if T can be identified with a polynomial.

We have seen in Remark 17.29 that a function $u \in \dot{B}_q^{s,p}(\mathbb{R}^N)$, $s > 0$, can always be identified with a tempered distribution T_u . In the next theorem we prove the opposite implication, that is, that if $T \in \mathcal{S}'(\mathbb{R}^N)$ is such that $|T|_{B_q^{s,p}(\mathbb{R}^N)}^\vee < \infty$ for $s > 0$, then T can be identified with a function in $L_{\text{loc}}^1(\mathbb{R}^N)$.

Theorem 17.76. *Let $1 \leq p, q < \infty$ and $s > 0$. If $T \in \mathcal{S}'(\mathbb{R}^N)$ is such that $|T|_{B_q^{s,p}(\mathbb{R}^N)}^\vee < \infty$, then there exists a function $u \in L_{\text{loc}}^1(\mathbb{R}^N)$ such that $T(\phi) = \int_{\mathbb{R}^N} \phi u \, dx$ for every $\phi \in C_c^\infty(\mathbb{R}^N)$.*

Proof. In view of Theorem 10.78 and using the function ω_0 in Remark 17.74 we can write

$$T = T * \omega_0^\vee + \sum_{k=0}^{\infty} T * \psi_k.$$

If $q > 1$, by Hölder's and Minkowski's inequalities, then

$$(17.68) \quad \begin{aligned} \sum_{k=0}^{\infty} \|T * \psi_k\|_{L^p} &\leq \left(\sum_{k=0}^{\infty} 2^{-kq's} \right)^{1/q'} \left(\sum_{k=0}^{\infty} 2^{kqs} \|T * \psi_k\|_{L^p}^q \right)^{1/q} \\ &\leq c |T|_{B_q^{s,p}}^\vee. \end{aligned}$$

The same inequality holds if $q = 1$. On the other hand, since $\omega_0(2^{-k}\cdot) \in \mathcal{S}(\mathbb{R}^N)$, $T*(\omega_0(2^{-k}\cdot))^\vee$ can be identified with a C^∞ function with polynomial growth by Theorem 10.56. Hence, T is given by the sum of a C^∞ function with polynomial growth and an L^p function. Thus, it can be identified with a function $u \in L^1_{\text{loc}}(\mathbb{R}^N)$. This concludes the proof. $\square$

In the following theorem we will prove that $\|\cdot\|_{B_q^{s,p}(\mathbb{R}^N)}^\vee$ is an equivalent norm in $B_q^{s,p}(\mathbb{R}^N)$. In the statement the norms are obtained by adding the L^p norm to the corresponding seminorm.

Theorem 17.77. *Let $1 \leq p, q \leq \infty$ and $s > 0$, and let $\varphi_0 \in C_c^\infty(\mathbb{R}^N)$ satisfy (17.56)–(17.57). Then there exists a constant $c = c(N, p, q, s) > 0$ such that for every $u \in B_q^{s,p}(\mathbb{R}^N)$,*

$$c^{-1}\|u\|_{B_q^{s,p}(\mathbb{R}^N)} \leq \|u\|_{B_q^{s,p}(\mathbb{R}^N)}^\vee \leq c\|u\|_{B_q^{s,p}(\mathbb{R}^N)}^\infty.$$

Proof. Step 1: Let $u \in B_q^{s,p}(\mathbb{R}^N)$ and let $m \in \mathbb{N}$ be so large that $mp > N$ and $m > s$. Write $u = v + w$, where $v \in L^p(\mathbb{R}^N)$ and $w \in W^{m,p}(\mathbb{R}^N)$. By Young's inequality (Theorem 10.48), (17.60), and the change of variables $x = 2^k y$,

$$\|v * \psi_k\|_{L^p} \leq \|\psi_k\|_{L^1} \|v\|_{L^p} = \|\varphi_0^\vee\|_{L^1} \|v\|_{L^p}.$$

On the other hand, by Theorem 10.71 and (17.60), $\widehat{w * \psi_k} = \widehat{w} \widehat{\psi_k} = \widehat{w} \varphi_0(2^{-k}\cdot)$. It follows from (17.58) that $\text{supp } \widehat{w * \psi_k} \subseteq \overline{B(0, 2^{k+1}a)} \setminus B(0, 2^{k-1}a)$ and thus we are in a position to apply Theorem 10.73 to get

$$\begin{aligned} \|w * \psi_k\|_{L^p} &\leq c2^{-km} \|\nabla^m(w * \psi_k)\|_{L^p} = c2^{-km} \|\nabla^m w * \psi_k\|_{L^p} \\ &\leq c2^{-km} \|\psi_k\|_{L^1} \|\nabla^m w\|_{L^p} = c2^{-km} \|\varphi_0^\vee\|_{L^1} |w|_{W^{m,p}}, \end{aligned}$$

where we used again Young's inequality (Theorem 10.48), (17.60), and the change of variables $x = 2^k y$.

Combining the estimates for v and w and using Minkowski's inequality we get

$$\begin{aligned} \|u * \psi_k\|_{L^p} &\leq \|v * \psi_k\|_{L^p} + \|w * \psi_k\|_{L^p} \\ &\leq \|\varphi_0^\vee\|_{L^1} \|v\|_{L^p} + c2^{-km} \|\varphi_0^\vee\|_{L^1} |w|_{W^{m,p}}. \end{aligned}$$

Taking the infimum over all such decomposition and using Remark 17.27 gives

$$(17.69) \quad \|u * \psi_k\|_{L^p} \leq c\dot{K}(u, 2^{-km}).$$

In turn, assuming $q < \infty$,

$$\begin{aligned} \sum_{k \in \mathbb{Z}} 2^{kqs} \|u * \psi_k\|_{L^p}^q &\leq c \sum_{k \in \mathbb{Z}} 2^{kqs} (\dot{K}(u, 2^{-km}))^q \\ &\leq c \int_0^\infty (\dot{K}(u, \tau^m))^q \frac{d\tau}{\tau^{1+qs}}, \end{aligned}$$

where we used Exercise 16.9. Since $\dot{K}(u, \tau^m) \leq c \sup_{\|h\| \leq c\tau} \|\Delta_h^m u\|_{L^p(\mathbb{R}^N)}$ (see Remark 17.27) it follows that

$$\sum_{k \in \mathbb{Z}} 2^{kqs} \|u * \psi_k\|_{L^p}^q \leq c \int_0^\infty \sup_{\|h\| \leq c\tau} \|\Delta_h^m u\|_{L^p(\mathbb{R}^N)}^q \frac{d\tau}{\tau^{1+qs}},$$

and so $|u|_{B_q^{s,p}}^\vee \leq c|u|_{B_q^{s,p}}^\infty$. The case $q = \infty$ is left as an exercise.

Step 2: Let $u \in L^p(\mathbb{R}^N)$ be such that $|u|_{B_q^{s,p}}^\vee < \infty$. In this step we will use the J -method of interpolation (see Section 16.2 in Chapter 16), especially Exercise 16.23. Let $m \in \mathbb{N}$ be so large that $mp > N$ and $m > s$. By Exercise 10.77 and (17.60), $\widehat{u * \psi_k} = \varphi_0(2^{-k} \cdot) \hat{u}$. Hence, by (17.58),

$$\text{supp } \widehat{u * \psi_k} \subseteq \overline{B(0, 2^{k+1}a)}$$

and thus we are in a position to apply Theorem 10.73 to conclude that $u * \psi_k \in W^{m,p}(\mathbb{R}^N)$, with

$$(17.70) \quad \|\nabla^m(u * \psi_k)\|_{L^p} \leq c 2^{km} \|u * \psi_k\|_{L^p}.$$

Moreover, if $k \in \mathbb{N}_0$, then $\|u * \psi_k\|_{L^p} \leq 2^{km} \|u * \psi_k\|_{L^p}$. Hence, for $k \in \mathbb{N}_0$,

$$J(u * \psi_k, 2^{-km}) \leq c \|u * \psi_k\|_{L^p},$$

where for $v \in W^{m,p}(\mathbb{R}^N)$ and $t > 0$,

$$J(v, t) := \max\{\|v\|_{L^p}, t\|v\|_{W^{m,p}}\}.$$

Similar estimates can be obtained for $u * \omega_0$ (see Remark 17.74). Set $g_0 := \psi_0 + \omega_0^\vee$ and $g_k := \psi_k$ for $k \in \mathbb{N}$. Define $u(t) := (u * g_k)/\log 2$ for $t \in (2^{-(k+1)m}, 2^{-km}]$, $k \in \mathbb{N}_0$. Then

$$\begin{aligned} \int_0^1 (J(u(t), t))^q \frac{dt}{t^{1+qs/m}} &\leq \sum_{k \in \mathbb{N}_0} \int_{2^{-(k+1)m}}^{2^{-km}} (J(u(t), t))^q \frac{dt}{t^{1+qs/m}} \\ (17.71) \quad &\leq \frac{1}{(\log 2)^q} \sum_{k \in \mathbb{N}_0} \int_{2^{-(k+1)m}}^{2^{-km}} (J(u * g_k, 2^{-km}))^q \frac{dt}{t^{1+qs/m}} \\ &\leq c \sum_{k \in \mathbb{N}_0} 2^{kqs} \|u * g_k\|_{L^p}^q < \infty, \end{aligned}$$

where we used the fact that $J(\cdot, t)$ is a norm and that $J(v, \cdot)$ is increasing (see Exercise 16.20). Moreover,

$$(17.72) \quad \int_{2^{-\ell m}}^1 u(t) \frac{dt}{t} = \sum_{k=0}^{\ell-1} u * g_k.$$

In view of (17.68),

$$\sum_{k=0}^{\infty} \|u * g_k\|_{L^p} \leq c \|u\|_{B_q^{s,p}}^{\vee} < \infty.$$

In turn, by (10.41), (17.72) (see also the Step 3 of the proof of Theorem 16.21),

$$u = \sum_{k \in \mathbb{N}_0} u * g_k = \int_0^1 u(t) \frac{dt}{t},$$

where the series converges in $L^p(\mathbb{R}^N)$ (and so also in $L^p(\mathbb{R}^N) + W^{m,p}(\mathbb{R}^N)$). It follows by Theorem 16.21, Exercise 16.23, Remark 17.27, and (17.71), that $\|u\|_{B_q^{s,p}(\mathbb{R}^N)}^{\infty} \leq c \|u\|_{B_q^{s,p}}^{\vee}$, which concludes the proof. $\square$

Remark 17.78. We remark that Step 1 of the previous proof actually shows that

$$(17.73) \quad |u|_{B_q^{s,p}(\mathbb{R}^N)}^{\vee} \leq c |u|_{B_q^{s,p}(\mathbb{R}^N)}$$

for every $u \in \dot{B}_q^{s,p}(\mathbb{R}^N)$. Indeed, if we write $u = v + w$, where $v \in L^p(\mathbb{R}^N)$ and $w \in \dot{W}^{m,p}(\mathbb{R}^N)$, since w has polynomial growth (see Remark 12.56), the convolution of w (and, in turn, of u) with a function in $\mathcal{S}(\mathbb{R}^N)$ is well-defined. Thus we can continue as in the proof of Step 1 to obtain (17.73).

Conversely, for every $T \in \mathcal{S}'(\mathbb{R}^N)$ such that $|T|_{B_q^{s,p}(\mathbb{R}^N)}^{\vee} < \infty$, consider the Banach space $\dot{W}^{m,p}(\mathbb{R}^N)/\mathcal{P}_{m-1}(\mathbb{R}^N)$ endowed with the norm $|\cdot|_{W^{m,p}(\mathbb{R}^N)}$, where $\mathcal{P}_{m-1}(\mathbb{R}^N)$ is the family of polynomials of degree less than or equal to $m-1$. Then in view of (17.70), for every $k \in \mathbb{Z}$,

$$\dot{J}(T * \psi_k, 2^{-km}) \leq c \|T * \psi_k\|_{L^p},$$

where for $v \in W^{m,p}(\mathbb{R}^N)$ and $t > 0$, $\dot{J}(v, t) := \max\{\|v\|_{L^p}, t|v|_{W^{m,p}}\}$. Define $u(t)$ as in Step 2 but now with $k \in \mathbb{Z}$. Then (17.71) becomes

$$\int_0^{\infty} (\dot{J}(u(t), t))^q \frac{dt}{t^{1+qs/m}} \leq c \sum_{k \in \mathbb{Z}} 2^{kqs} \|T * \psi_k\|_{L^p}^q,$$

while (17.72) should be replaced by

$$\int_{2^{-\ell m}}^{2^{\ell m}} u(t) \frac{dt}{t} = \sum_{k=-\ell}^{\ell-1} T * \psi_k.$$

In view of (17.68),

$$\sum_{k=0}^{\infty} \|T * \psi_k\|_{L^p + \dot{W}^{m,p}/\mathcal{P}_{m-1}} \leq \sum_{k=0}^{\infty} \|T * \psi_k\|_{L^p} \leq c|T|_{B_q^{s,p}}^{\vee} < \infty,$$

where $\|\cdot\|_{L^p + \dot{W}^{m,p}/\mathcal{P}_{m-1}}$ is defined as in (16.2). On the other hand, if $q > 1$, by (17.70), the fact that $m > s$ and Hölder's and Minkowski's inequalities, we get

$$\begin{aligned} \sum_{n=1}^{\infty} \|T * \psi_{-n}\|_{L^p + \dot{W}^{m,p}/\mathcal{P}_{m-1}} &\leq \sum_{n=1}^{\infty} |T * \psi_{-n}|_{W^{m,p}} \leq c \sum_{n=1}^{\infty} 2^{-nm} \|T * \psi_{-n}\|_{L^p} \\ &\leq \left(\sum_{n=1}^{\infty} 2^{-n(m-s)q'} \right)^{1/q'} \left(\sum_{n=1}^{\infty} 2^{-nqs} \|T * \psi_{-n}\|_{L^p}^q \right)^{1/q} \leq c|T|_{B_q^{s,p}}^{\vee}. \end{aligned}$$

If $q = 1$, the same inequality holds since $2^{-nm} \leq 2^{-ns}$. Hence, we have shown that

$$\sum_{k \in \mathbb{Z}} \|T * \psi_k\|_{L^p + \dot{W}^{m,p}/\mathcal{P}_{m-1}} < \infty.$$

In turn, by (17.72) (see also the Step 3 of the proof of Theorem 16.21),

$$v := \sum_{k \in \mathbb{Z}} T * \psi_k = \int_0^{\infty} u(t) \frac{dt}{t},$$

where the series converges in $L^p(\mathbb{R}^N) + \dot{W}^{m,p}(\mathbb{R}^N)/\mathcal{P}_{m-1}(\mathbb{R}^N)$. It follows by Theorem 16.21, Remark 17.27, and (17.71),

$$|v|_{B_q^{s,p}(\mathbb{R}^N)}^{\infty} \leq c|T|_{B_q^{s,p}}^{\vee}.$$

The following exercises give an alternative proof of Step 1 in the cases $0 < s < 1$ and $s = 1$.

Exercise 17.79. Let $1 \leq p \leq \infty$, $1 \leq q < \infty$, and $0 < s < 1$.

- (i) Prove that $(u * \psi_k)(x) = \int_{\mathbb{R}^N} \psi_k(y) \Delta_{-y} u(x) dy$ for every $u \in \dot{B}_q^{s,p}(\mathbb{R}^N)$ and $x \in \mathbb{R}^N$.
- (ii) Prove that $|u|_{B_q^{s,p}}^{\vee} \leq c|u|_{B_q^{s,p}}$. Hint: Write

$$\int_{\mathbb{R}^N} = \int_{B(0,2^{-k})} + \int_{\mathbb{R}^N \setminus B(0,2^{-k})}$$

and use Minkowski's inequality for integrals.

Exercise 17.80. Let $1 \leq p \leq \infty$, $1 \leq q < \infty$, and $s = 1$.

- (i) Prove that $(u * \psi_k)(x) = \frac{1}{2} \int_{\mathbb{R}^N} \psi_k(y) \sim \Delta_y^2 u(x) dy$ for every $u \in \dot{B}_q^{1,p}(\mathbb{R}^N)$ and $x \in \mathbb{R}^N$, where $\sim \Delta_y^2 u$ is defined in (17.8). Hint: ψ_k is even.

(ii) Prove that $|u|_{B_q^{1,p}}^\vee \leq c|u|_{B_q^{1,p}}$.

Exercise 17.81. Let $u \in \mathcal{S}(\mathbb{R}^N)$ and let $h \in \mathbb{R}^N$ and $m \in \mathbb{N}$. Prove that

$$\widehat{\Delta_h u}(x) = (e^{2\pi i h \cdot x} - 1)\widehat{u}(x), \quad x \in \mathbb{R}^N,$$

and that

$$\widehat{\Delta_h^m u}(x) = (e^{2\pi i h \cdot x} - 1)^m \widehat{u}(x), \quad x \in \mathbb{R}^N.$$

17.8. And More Embeddings

In this section we show how we can use the seminorm (17.61) to give simple proofs of more embeddings for Besov spaces.

Theorem 17.82. Let $1 \leq p, q \leq \infty$ and $s_2 > s_1 > 0$. Then there exists a constant $c = c(N, p, q, s_1, s_2) > 0$ such that

$$|u|_{B_q^{s_1,p}(\mathbb{R}^N)}^\vee \leq |u|_{B_q^{s_2,p}(\mathbb{R}^N)}^\vee + c\|u\|_{L^p(\mathbb{R}^N)}$$

for all $u \in B_q^{s_2,p}(\mathbb{R}^N)$. In particular, $B_q^{s_2,p}(\mathbb{R}^N) \hookrightarrow B_q^{s_1,p}(\mathbb{R}^N)$.

Proof. Assume that $1 \leq q < \infty$. Given $u \in B_q^{s_2,p}(\mathbb{R}^N)$, by Young's inequality (Theorem 10.48), (17.60), and the change of variables $x = 2^k y$, we have

$$\|u * \psi_k\|_{L^p} \leq \|\psi_k\|_{L^1} \|u\|_{L^p} = \|\varphi_0^\vee\|_{L^1} \|u\|_{L^p}.$$

Hence,

$$(17.74) \quad \sum_{k=-\infty}^0 2^{kqs_1} \|u * \psi_k\|_{L^p}^q \leq \|\varphi_0^\vee\|_{L^1}^q \|u\|_{L^p}^q \sum_{k=-\infty}^0 2^{kqs_1} = c\|u\|_{L^p}^q.$$

On the other hand, since $s_1 < s_2$, we get

$$\sum_{k=0}^{\infty} 2^{kqs_1} \|u * \psi_k\|_{L^p}^q \leq \sum_{k=0}^{\infty} 2^{kqs_2} \|u * \psi_k\|_{L^p}^q.$$

Combining the last two inequalities and using (17.61) completes the proof in the case $1 \leq q < \infty$. The case $q = \infty$ is similar and is left as an exercise. $\square$

Exercise 17.83. Prove the previous theorem in the special case $\lfloor s_1 \rfloor = \lfloor s_2 \rfloor$ using the seminorms (17.21).

Next we show that by slightly lowering the regularity parameter s we can drastically change q . To be precise, we have the following result.

Theorem 17.84. Let $1 \leq p \leq \infty$, let $1 \leq q_1, q_2 \leq \infty$, and let $s > 0$. Then for every $0 < \varepsilon < s$ there exists a constant $c = c(\varepsilon, N, p, q_1, q_2, s) > 0$ such that

$$|u|_{B_{q_2}^{s-\varepsilon,p}(\mathbb{R}^N)}^\vee \leq c|u|_{B_{q_1}^{s,p}(\mathbb{R}^N)}^\vee + c\|u\|_{L^p(\mathbb{R}^N)}$$

for all $u \in B_{q_1}^{s,p}(\mathbb{R}^N)$. In particular, $B_{q_1}^{s,p}(\mathbb{R}^N) \hookrightarrow B_{q_2}^{s-\varepsilon,p}(\mathbb{R}^N)$.

Observe that there is no relationship between q_1 and q_2 .

Proof. If $q_1 \leq q_2$, then by Theorems 17.38 and 17.82,

$$B_{q_1}^{s,p}(\mathbb{R}^N) \hookrightarrow B_{q_1}^{s-\varepsilon,p}(\mathbb{R}^N) \hookrightarrow B_{q_2}^{s-\varepsilon,p}(\mathbb{R}^N).$$

Thus it remains to study the case $q_1 > q_2$. By (17.74), with q and s replaced here by q_2 and $s - \varepsilon$, respectively,

$$\sum_{k=-\infty}^0 2^{q_2 k(s-\varepsilon)} \|u * \psi_k\|_{L^p}^{q_2} \leq c \|u\|_{L^p}^{q_2}.$$

On the other hand, by Hölder's inequality with exponents q_1/q_2 ,

$$\begin{aligned} \sum_{k=0}^{\infty} 2^{q_2 k(s-\varepsilon)} \|u * \psi_k\|_{L^p}^{q_2} &\leq \left(\sum_{k=0}^{\infty} 2^{q_1 k s} \|u * \psi_k\|_{L^p}^{q_1} \right)^{q_2/q_1} \\ &\quad \times \left(\sum_{k=0}^{\infty} 2^{-\varepsilon k(q_1/q_2)'} \right)^{1/(q_1/q_2)'}. \end{aligned}$$

This concludes the proof. $\square$

The next theorem shows that by lowering the regularity parameter s of functions in Besov spaces, one can increase their integrability parameter p .

Theorem 17.85. *Let $1 \leq p_2 < p_1 \leq \infty$ and $0 < s_1 < s_2$ be such that*

$$(17.75) \quad s_1 + \frac{N}{p_2} = s_2 + \frac{N}{p_1}$$

and let $1 \leq q_2 \leq q_1 \leq \infty$. Then there exists a constant $c = c(N, p_1, p_2, q_1, q_2, s_1, s_2) > 0$ such that

$$|u|_{B_{q_1}^{s_1, p_1}(\mathbb{R}^N)}^{\vee} \leq c |u|_{B_{q_2}^{s_2, p_2}(\mathbb{R}^N)}^{\vee}$$

for every $u \in \mathcal{S}(\mathbb{R}^N)$.

Proof. Assume that $q_1 < \infty$. By Theorems 10.62 and 10.71,

$$\widehat{u * \psi_k}(x) = \widehat{u}(x) \widehat{\psi_k}(x) = \widehat{u}(x) \varphi_0(2^{-k}x).$$

Hence, by (17.56), $\text{supp } \widehat{u * \psi_k} \subseteq \overline{B(0, 2^{k+1})}$. It follows from Theorem 10.73 and (17.75) that

$$\|u * \psi_k\|_{L^{p_1}} \leq c 2^{kN(1/p_2 - 1/p_1)} \|u * \psi_k\|_{L^{p_2}} = c 2^{k(s_2 - s_1)} \|u * \psi_k\|_{L^{p_2}}.$$

Hence,

$$2^{k s_1} \|u * \psi_k\|_{L^{p_1}} \leq c 2^{k s_2} \|u * \psi_k\|_{L^{p_2}}.$$

Using Exercise 17.39 when $q_2 < q_1$, we get

$$\left(\sum_{k \in \mathbb{Z}} 2^{q_1 k s_1} \|u * \psi_k\|_{L^{p_1}}^{q_1} \right)^{1/q_1} \leq c \left(\sum_{k \in \mathbb{Z}} 2^{q_2 k s_2} \|u * \psi_k\|_{L^{p_2}}^{q_2} \right)^{1/q_2},$$

which completes the proof in this case. The case $q_1 = \infty$ is left as an exercise. $\square$

Exercise 17.86. Use Theorem 17.85 to give an alternative proof of Theorem 17.52.

We conclude this section with an alternative proof of (a slightly weaker version of) Theorem 17.49.

Theorem 17.87 (Sobolev–Gagliardo–Nirenberg embedding in $B_q^{s,p}$). *Let $s > 0$, $1 \leq p < N/s$, and $1 \leq q \leq p$. Then there exists a constant $c = c(N, p, q, s) > 0$ such that*

$$(17.76) \quad \|u\|_{L^{p_s^*}} \leq c |u|_{B_q^{s,p}(\mathbb{R}^N)}^\vee$$

for every $u \in \mathcal{S}(\mathbb{R}^N)$. In particular, $B_q^{s,p}(\mathbb{R}^N) \hookrightarrow L^r(\mathbb{R}^N)$ for all $p \leq r \leq p_s^*$.

We begin with a preliminary result, which is of interest in itself. We recall that the seminorm (17.61) has been defined also for negative values of s .

Lemma 17.88. *Let $1 \leq p < r < \infty$ and $s > 0$. Then there exists a constant $c = c(N, p, r, s) > 0$ such that*

$$\|u\|_{L^r(\mathbb{R}^N)} \leq c \left(|u|_{B^{-\theta,\infty}(\mathbb{R}^N)}^\vee \right)^{1-p/r} \left(|u|_{B^{s,p}(\mathbb{R}^N)}^\vee \right)^{p/r}$$

for every $u \in \mathcal{S}(\mathbb{R}^N)$, where $\theta := sp/(r-p)$.

Proof. Given a function $v : \mathbb{R}^N \rightarrow \mathbb{R}$ and $t \in \mathbb{R}$ for brevity we write $\{v > t\}$ for the set $\{x \in \mathbb{R}^N : v(x) > t\}$. Since $|u|_{B^{-\theta,\infty}}^\vee < \infty$, by (17.62) we have

$$\|u * \psi_k\|_{L^\infty} \leq 2^{k\theta} |u|_{B^{-\theta,\infty}}^\vee,$$

and so for $l \in \mathbb{Z}$,

$$(17.77) \quad \sum_{k=-\infty}^l \|u * \psi_k\|_{L^\infty} \leq |u|_{B^{-\theta,\infty}}^\vee \sum_{k=-\infty}^l 2^{k\theta} = |u|_{B^{-\theta,\infty}}^\vee 2^{l\theta}.$$

By rescaling we may assume that $|u|_{B^{-\theta,\infty}}^\vee = 1$. It follows from Theorem 10.78, Remark 17.74 and (17.77) that

$$(17.78) \quad u = S_l(u) + (u - S_l(u)) = \sum_{k=-\infty}^l u * \psi_k + \sum_{k=l+1}^{\infty} u * \psi_k.$$

Hence, for every $t > 0$,

$$(17.79) \quad \{|u| > t\} \subseteq \{|S_l(u)| > t/2\} \cup \{|u - S_l(u)| > t/2\}.$$

Let $l_t \in \mathbb{Z}$ be such that

$$(17.80) \quad 2^{-1}(t/2)^{1/\theta} \leq 2^{l_t} \leq (t/2)^{1/\theta}.$$

Then by (17.77), $\|S_{l_t}(u)\|_{L^\infty} \leq 2^{l_t\theta} \leq t/2$, which implies that the set $\{|S_{l_t}(u)| > t/2\}$ has Lebesgue measure zero. Thus, by (17.79),

$$\mathcal{L}^N(\{|u| > t\}) \leq \mathcal{L}^N(\{|u - S_{l_t}(u)| > t/2\}),$$

and so by Tonelli's theorem,

$$\begin{aligned} \|u\|_{L^r}^r &= r \int_0^\infty t^{r-1} \mathcal{L}^N(\{|u| > t\}) dt \\ (17.81) \quad &\leq r \int_0^\infty t^{r-1} \mathcal{L}^N(\{|u - S_{l_t}(u)| > t/2\}) dt \\ &\leq c \int_0^\infty t^{r-p-1} \|u - S_{l_t}(u)\|_{L^p}^p dt, \end{aligned}$$

where we used the inequality

$$\tau^p \mathcal{L}^N(\{|v| > \tau\}) \leq \int_{\{|v| > \tau\}} |v(x)|^p dx \leq \int_{\mathbb{R}^N} |v(x)|^p dx.$$

To estimate $\|u - S_{l_t}(u)\|_{L^p}$, observe by Minkowski's and Hölder's inequalities that

$$\begin{aligned} \|u - S_{l_t}(u)\|_{L^p} &\leq \sum_{k=l_t+1}^\infty \|u * \psi_k\|_{L^p} = \sum_{k=l_t+1}^\infty 2^{-ka} 2^{ka} \|u * \psi_k\|_{L^p} \\ &\leq \left(\sum_{k=l_t+1}^\infty 2^{-kap'} \right)^{1/p'} \left(\sum_{k=l_t+1}^\infty 2^{kap} \|u * \psi_k\|_{L^p}^p \right)^{1/p} \\ &= 2^{-(l_t+1)a} \left(\sum_{k=l_t+1}^\infty 2^{kap} \|u * \psi_k\|_{L^p}^p \right)^{1/p}, \end{aligned}$$

where $0 < a < \theta(r-p)/p = s$. Combining the previous inequality with (17.80) and (17.81) gives

$$\begin{aligned} \|u\|_{L^r}^r &\leq c \int_0^\infty t^{r-p-1-ap/\theta} \sum_{k=l_t+1}^\infty 2^{kap} \|u * \psi_k\|_{L^p}^p dt \\ &\leq c \sum_{k=-\infty}^\infty 2^{kap} \|u * \psi_k\|_{L^p}^p \int_0^{2^{k\theta+1}} t^{r-p-1-ap/\theta} dt \\ &= \frac{c 2^{r-p-ap/\theta}}{r-p-ap/\theta} \sum_{k=-\infty}^\infty 2^{k\theta(r-p)} \|u * \psi_k\|_{L^p}^p, \end{aligned}$$

where we used Fubini's theorem and the fact that if $k \geq l_t+1$, then $t \leq 2^{k\theta+1}$. Since $\theta(r-p) = sp$, the proof is concluded in view of (17.61). $\square$

We turn to the proof of Theorem 17.87.

Proof of Theorem 17.87. Since $q \leq p$ it follows by Theorem 17.38 that

$$|u|_{B^{s,p}}^\vee \leq c|u|_{B_q^{s,p}}^\vee.$$

Thus it suffices to show (17.76) for $q = p$. By the general form of Young's inequality (see Theorem 10.49) with $1/p_s^* + 1/(p_s^*)' = 1 + 1/\infty$ (see (10.20)),

$$\|u * \psi_k\|_{L^\infty} \leq \|\psi_k\|_{L^{(p_s^*)}' } \|u\|_{L^{p_s^*}} \leq c2^{kN/p_s^*} \|u\|_{L^{p_s^*}},$$

where we used (17.60), and a change of variables to estimate

$$\begin{aligned} \|\psi_k\|_{L^{(p_s^*)}' } &= 2^{kN} \left(\int_{\mathbb{R}^N} (\varphi_0^\vee(2^k y))^{(p_s^*)'} dy \right)^{1/(p_s^*)'} \\ &\leq 2^{kN(1-1/(p_s^*)')} \left(\int_{\mathbb{R}^N} (\varphi_0^\vee(x))^{(p_s^*)'} dx \right)^{1/(p_s^*)'}. \end{aligned}$$

Hence, $u \in B^{-N/p_s^*, \infty}(\mathbb{R}^N)$, with

$$|u|_{B^{-N/p_s^*, \infty}}^\vee \leq c\|u\|_{L^{p_s^*}}.$$

We are now in a position to apply the previous lemma to conclude that

$$\begin{aligned} \|u\|_{L^{p_s^*}} &\leq c \left(|u|_{B^{-N/p_s^*, \infty}}^\vee \right)^{1-p/p_s^*} \left(|u|_{B^{s,p}}^\vee \right)^{p/p_s^*} \\ &\leq c \|u\|_{L^{p_s^*}}^{1-p/p_s^*} \left(|u|_{B^{s,p}}^\vee \right)^{p/p_s^*}. \end{aligned}$$

Since $\|u\|_{L^{p_s^*}} < \infty$, if $u \neq 0$, we can divide by $\|u\|_{L^{p_s^*}}^{1-p/p_s^*}$ to obtain (17.76). $\square$

Sobolev Spaces: Traces

Grad students, or “p-ons” as Einstein called them, can only occupy a discrete number of energy states: working, thinking about working, sleeping.

— Jorge Cham, www.phdcomics.com

Sobolev functions $u \in W^{m,p}(\Omega)$ have been defined using integration by parts (see (11.1)) with test functions $\phi \in C_c^\infty(\Omega)$ vanishing on the boundary. To drop this requirement we need to make sense of the “restriction” of u (and of its normal derivatives when $m \geq 2$) to the boundary $\partial\Omega$ (see (9.55)). Since Sobolev functions are L^p functions, they are equivalence classes of functions, and thus talking about their pointwise value does not make sense in general. One possibility would be to find a good representative. Indeed, in the supercritical case $mp > N$ and if Ω is a $W^{m,p}$ extension domain, we can extend u to $W^{m,p}(\mathbb{R}^N)$ and then apply Morrey’s theorem (see Theorem 12.55) to conclude that u has a Hölder continuous representative $\bar{u}$. Thus, the value of u on the boundary of Ω is well-defined.

The situation is quite different in the subcritical and critical cases $mp \leq N$ (unless $N = 1$). In this case we will prove that if $\partial\Omega$ is sufficiently regular, say, Lipschitz continuous, we can introduce a linear operator

$$\text{Tr} : W^{m,p}(\Omega) \rightarrow L^p_{\text{loc}}(\partial\Omega, \mathcal{H}^{N-1})$$

such that $\text{Tr}(u) = u$ on $\partial\Omega$ for all $u \in W^{m,p}(\Omega) \cap C(\bar{\Omega})$ and for which integration by parts holds, that is,

$$(18.1) \quad \int_{\Omega} u \partial_i \psi \, dx = - \int_{\Omega} \psi \partial_i u \, dx + \int_{\partial\Omega} \psi \, \text{Tr}(u) \nu_i \, d\mathcal{H}^{N-1}$$

for all $u \in W^{m,p}(\Omega)$, $\psi \in C_c^1(\mathbb{R}^N)$, and $i = 1, \dots, N$, where ν is the outward unit normal to $\partial\Omega$. We will study the continuity properties of this operator.

In what follows, we will use the abbreviations

$$(18.2) \quad L^p(\partial\Omega) := L^p(\partial\Omega, \mathcal{H}^{N-1}), \quad L_{\text{loc}}^p(\partial\Omega) := L_{\text{loc}}^p(\partial\Omega, \mathcal{H}^{N-1}),$$

$$\|\cdot\|_{L^p(\partial\Omega)} := \|\cdot\|_{L^p(\partial\Omega, \mathcal{H}^{N-1})}.$$

18.1. The Trace Operator

In this section we establish the existence of a trace operator. For simplicity we will consider separately the cases $m = 1$ and $m \geq 2$. The parameter $\varepsilon > 0$ in the estimate (18.3) will be used to prove compactness of the trace operator.

Theorem 18.1. *Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is Lipschitz continuous, let $1 \leq p < \infty$. There exists a unique linear operator*

$$\text{Tr} : W^{1,p}(\Omega) \rightarrow L_{\text{loc}}^p(\partial\Omega)$$

such that

- (i) $\text{Tr}(u) = u$ on $\partial\Omega$ for all $u \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$,
- (ii) the integration by parts formula (18.1) holds for all $u \in W^{1,p}(\Omega)$, all $\psi \in C_c^1(\mathbb{R}^N)$, and all $i = 1, \dots, N$,
- (iii) for every $R > 0$ there exist two constants $c_R, \varepsilon_R > 0$ depending on R, Ω and p such that

$$(18.3) \quad \int_{B(0,R) \cap \partial\Omega} |\text{Tr}(u)|^p d\mathcal{H}^{N-1} \leq c_R \varepsilon^{-1} \int_{B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)} |u|^p dx$$

$$+ c_R \varepsilon^{p-1} \int_{B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)} \|\nabla u\|^p dx$$

for every $0 < \varepsilon \leq \varepsilon_R$, where $\Omega_\varepsilon := \{x \in \Omega : \text{dist}(x, \partial\Omega) > \varepsilon\}$.

The function $\text{Tr}(u)$ is called the *trace* of u on $\partial\Omega$.

Proof. Since $\partial\Omega$ is Lipschitz continuous, by Theorem 11.35 the restriction to Ω of functions in $C_c^\infty(\mathbb{R}^N)$ is dense in $W^{1,p}(\Omega)$. Let $u \in C_c^\infty(\mathbb{R}^N)$. We begin by proving (iii). Let $x_0 \in \partial\Omega$. Since $\partial\Omega$ is Lipschitz continuous (see Definition 9.57) there exist a rigid motion $T : \mathbb{R}^N \rightarrow \mathbb{R}^N$ with $T(x_0) = 0$, a Lipschitz continuous function $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$, and $r_{x_0} > 0$ such that, setting $y := T(x)$, we have

$$T(\Omega \cap Q_T(x_0, r_{x_0})) = \{y \in Q(0, r_{x_0}) : y_N > f(y')\},$$

where $Q_T(x_0, r_{x_0})$ is an open cube centered at x_0 and of side-length $r_{x_0} > 0$. Fix $0 < \varepsilon < r \leq r_{x_0}/[2(\text{Lip } f + 1)\sqrt{N-1}]$. Setting $v(y) := u(T^{-1}(y))$, by the fundamental theorem of calculus we can write

$$|v(y', f(y'))| \leq |v(y', y_N)| + \int_{f(y')}^{y_N} |\partial_N v(y', s)| ds$$

for $y' \in Q_{N-1}(0, r)$ and $f(y') < y_N < f(y') + \varepsilon$. Raising both sides to the power p and using Young's and Hölder's inequalities if $p > 1$ gives

$$\begin{aligned} |v(y', f(y'))|^p &\leq 2^{p-1} |v(y', y_N)|^p + 2^{p-1} (y_N - f(y'))^{p-1} \int_{f(y')}^{y_N} |\partial_N v(y', s)|^p ds \\ &\leq 2^{p-1} |v(y', y_N)|^p + 2^{p-1} \varepsilon^{p-1} \int_{f(y')}^{f(y')+\varepsilon} |\partial_N v(y', s)|^p ds. \end{aligned}$$

Multiplying both sides by $\sqrt{1 + \|\nabla_{y'} f(y')\|_{N-1}^2}$, averaging in y_N over the interval $(f(y'), f(y') + \varepsilon)$, and integrating in y' over $Q_{N-1}(0, r)$ gives

$$\begin{aligned} (18.4) \quad &\int_{Q_{N-1}(0, r)} |v(y', f(y'))|^p \sqrt{1 + \|\nabla_{y'} f(y')\|_{N-1}^2} dy' \\ &\leq 2^{p-1} (1 + \text{Lip } f) \varepsilon^{-1} \int_{Q_{N-1}(0, r)} \int_{f(y')}^{f(y')+\varepsilon} |v(y', y_N)|^p dy_N dy' \\ &\quad + 2^{p-1} (1 + \text{Lip } f) \varepsilon^{p-1} \int_{Q_{N-1}(0, r)} \int_{f(y')}^{f(y')+\varepsilon} |\partial_N v(y', s)|^p ds dy'. \end{aligned}$$

By a change of variables, we obtain that for every $0 < \varepsilon < r$,

$$\begin{aligned} \int_{Q_T(x_0, r) \cap \partial\Omega} |u|^p d\mathcal{H}^{N-1} &\leq 2^{p-1} (1 + \text{Lip } f) \varepsilon^{-1} \int_{Q_T(x_0, r) \cap (\Omega \setminus \Omega_\varepsilon)} |u|^p dx \\ &\quad + 2^{p-1} (1 + \text{Lip } f) \varepsilon^{p-1} \int_{Q_T(x_0, r) \cap (\Omega \setminus \Omega_\varepsilon)} \|\nabla u\|^p dx. \end{aligned}$$

By a compactness argument, given $R > 0$ we can cover $\overline{B(0, R)} \cap \partial\Omega$ with a finite number of cubes as above, say $Q_1, \dots, Q_\ell$. Then

$$\begin{aligned} \int_{B(0, R) \cap \partial\Omega} |u|^p d\mathcal{H}^{N-1} &\leq 2^{p-1} (1 + L) \varepsilon^{-1} \ell \int_{B(0, R) \cap (\Omega \setminus \Omega_\varepsilon)} |u|^p dx \\ &\quad + 2^{p-1} (1 + L) \varepsilon^{p-1} \ell \int_{B(0, R) \cap (\Omega \setminus \Omega_\varepsilon)} \|\nabla u\|^p dx, \end{aligned}$$

where L is the maximum of the Lipschitz constants in each cube and ε is taken to be smaller than every side-length of the cubes. This shows that (iii) holds for every $u \in C_c^\infty(\mathbb{R}^N)$.

Now, if $u \in W^{1,p}(\Omega)$, consider a sequence $\{u_n\}_n$ in $C_c^\infty(\mathbb{R}^N)$ such that $u_n \rightarrow u$ in $W^{1,p}(\Omega)$. Applying (18.3) to $u_n - u_m$, we obtain that the sequence of functions u_n restricted to $B(0, R) \cap \partial\Omega$ is a Cauchy sequence in $L^p(B(0, R) \cap \partial\Omega)$ and thus it converges to a function $w_R \in L^p(B(0, R) \cap \partial\Omega)$. By uniqueness, we have that $w_R(x) = w_S(x)$ for $\mathcal{H}^{N-1}$ -a.e. $x \in B(0, R) \cap \partial\Omega$ for $R < S$ and thus, by taking a sequence $R_k \rightarrow \infty$ we obtain that the functions u_n restricted to $\partial\Omega$ converge on bounded sets of $\partial\Omega$ to a function in $L_{\text{loc}}^p(\partial\Omega)$ that we denote $\text{Tr}(u)$. Since (18.1) holds for each u_n and

$\psi \in C_c^\infty(\mathbb{R}^N)$ in (see (11.1)), letting $n \rightarrow \infty$ and using the fact that $u_n \rightarrow u$ in $W^{1,p}(\Omega)$ and $\{u_n\}_n$ restricted to $\partial\Omega$ converges to $\text{Tr}(u)$ in $L^p(\text{supp } \psi \cap \partial\Omega)$, we obtain that (18.1) holds for u .

Finally, the uniqueness of the trace operator follows from (18.3). We omit the details. $\square$

Exercise 18.2. Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is Lipschitz continuous, and let $1 \leq p < \infty$. Prove that the previous theorem continues to hold in $W^{1,p}(\Omega)$. Hint: Use Poincaré's inequality.

Exercise 18.3. Let $\Omega \subset \mathbb{R}^2$ and $u \in W^{1,2}(\Omega)$ be given as in Exercise 12.15. Prove that the trace of u on the line segment $\Gamma := (0, \ell) \times \{0\}$ is not in $L^q(\Gamma, \mathcal{H}^1)$ for any $q \geq 1$.

An important corollary of the previous theorem is compactness of the trace operator for $p > 1$.

Corollary 18.4 (Compactness of traces). *Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open bounded set whose boundary $\partial\Omega$ is Lipschitz continuous, let $1 < p < \infty$, and let $\{u_n\}_n$ be a bounded sequence in $W^{1,p}(\Omega)$. Then there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and a function $u \in W^{1,p}(\Omega)$ such that $u_{n_k} \rightarrow u$ in $L^p(\Omega)$ and $\text{Tr}(u_{n_k}) \rightarrow \text{Tr}(u)$ in $L^p(\partial\Omega)$.*

Proof. Since Ω is bounded, by (18.3) we have

$$\int_{\partial\Omega} |\text{Tr}(v)|^p d\mathcal{H}^{N-1} \leq c\varepsilon^{-1} \int_{\Omega} |v|^p dx + c\varepsilon^{p-1} \int_{\Omega} \|\nabla v\|^p dx$$

for every $v \in W^{1,p}(\Omega)$ and for every $\varepsilon > 0$ sufficiently small (depending only on Ω and p). By the Rellich–Kondrachov theorem (Theorems 12.18, 12.44 and 12.63) there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and a function $u \in W^{1,p}(\Omega)$ such that $u_{n_k} \rightarrow u$ in $L^p(\Omega)$. Hence,

$$\int_{\partial\Omega} |\text{Tr}(u_{n_k}) - \text{Tr}(u)|^p d\mathcal{H}^{N-1} \leq c\varepsilon^{-1} \int_{\Omega} |u_{n_k} - u|^p dx + c_1 \varepsilon^{p-1}$$

for some constant $c_1 > 0$, where we used the fact that $\{u_n\}_n$ is bounded in $W^{1,p}(\Omega)$. First, letting $k \rightarrow \infty$ and then $\varepsilon \rightarrow 0^+$ completes the proof. $\square$

Exercise 18.5. Prove that the previous corollary fails for $p = 1$.

Exercise 18.6. Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open bounded set whose boundary $\partial\Omega$ is Lipschitz continuous. Prove that if $u_n \rightharpoonup u$ in $W^{1,1}(\Omega)$, then $\text{Tr}(u_n)$ converges to $\text{Tr}(u)$ in $L^1(\partial\Omega)$. Hint: Use equi-integrability.

Next we show that if the domain Ω is sufficiently regular, we may characterize $W_0^{1,p}(\Omega)$ as the subspace of functions in $W^{1,p}(\Omega)$ with trace zero.

Theorem 18.7 (Traces and $W_0^{1,p}$). *Let $\Omega \subset \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is Lipschitz continuous, let $1 \leq p < \infty$, and let $u \in W^{1,p}(\Omega)$. Then $\text{Tr}(u) = 0$ if and only if $u \in W_0^{1,p}(\Omega)$.*

Proof. If $u \in C_c^\infty(\Omega)$, then $\text{Tr}(u) = 0$, and so, since $W_0^{1,p}(\Omega)$ is the closure of $C_c^\infty(\Omega)$ with respect to the norm $\|\cdot\|_{W^{1,p}(\Omega)}$ it follows from (18.3) that $\text{Tr}(u) = 0$ for all $u \in W_0^{1,p}(\Omega)$.

To prove the converse implication, let $u \in W^{1,p}(\Omega)$ be such that $\text{Tr}(u) = 0$. Using a partition of unity and flattening out $\partial\Omega$, we may assume that $\Omega = \mathbb{R}_+^N$ and that $u = 0$ for all $x \in \mathbb{R}_+^N$ with $\|x\| \geq R$ (exercise). For $x \in \mathbb{R}^N$ define

$$(18.5) \quad v(x) := \begin{cases} u(x) & \text{if } x_N > 0, \\ 0 & \text{if } x_N \leq 0. \end{cases}$$

By Theorem 18.1 and the fact that $\text{Tr}(u) = 0$, for all $\psi \in C_c^1(\mathbb{R}^N)$ and $i = 1, \dots, N$ we have that

$$\int_{\mathbb{R}^N} v \partial_i \psi \, dx = \int_{\mathbb{R}_+^N} u \partial_i \psi \, dx = - \int_{\mathbb{R}_+^N} \psi \partial_i u \, dx,$$

which shows that $v \in W^{1,p}(\mathbb{R}^N)$, with

$$\partial_i v(x) = \begin{cases} \partial_i u(x) & \text{if } x_N > 0, \\ 0 & \text{if } x_N < 0, \end{cases}$$

$i = 1, \dots, N$. We now translate v upwards. To be precise, for $t > 0$ and $x \in \mathbb{R}^N$ define $v_t(x) := v(x', x_N - t)$. Note that the support of v_t is a compact set contained in $\mathbb{R}^{N-1} \times (t/2, \infty)$. For each $\eta > 0$, by Lemma 11.36 we may find t so small that $\|u - v_t\|_{W^{1,p}(\mathbb{R}_+^N)} \leq \eta$. It now suffices to consider a sequence of mollifications $\{(v_t)_\varepsilon\}_{\varepsilon>0}$ of v_t . Find $0 < \varepsilon < t/4$ so small that $\|(v_t)_\varepsilon - v_\varepsilon\|_{W^{1,p}(\mathbb{R}_+^N)} \leq \eta$. Since, for $x \in \mathbb{R}^N$,

$$(v_t)_\varepsilon(x) = \int_{\mathbb{R}^N} \varphi_\varepsilon(x - y) v_t(y) \, dy = \int_{\mathbb{R}^{N-1} \times (t/2, \infty)} \varphi_\varepsilon(x - y) v_t(y) \, dy$$

and $\text{supp } \varphi_\varepsilon \subseteq \overline{B(0, \varepsilon)}$, it follows that if $0 < x_N < \varepsilon$, then $(v_t)_\varepsilon(x) = 0$. Thus $v_\varepsilon \in C_c^\infty(\mathbb{R}_+^N)$. This shows that $u \in W_0^{1,p}(\mathbb{R}_+^N)$. $\square$

We now consider the case $m \geq 2$. Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is Lipschitz continuous. Given a function $u \in W^{m,p}(\Omega)$, $1 \leq p < \infty$, and a multi-index $\alpha \in \mathbb{N}_0^N$, with $1 \leq |\alpha| \leq m - 1$, since $\partial^\alpha u \in W^{1,p}(\Omega)$, by Theorem 18.1 there exists $\text{Tr}(\partial^\alpha u) \in L_{\text{loc}}^p(\partial\Omega)$. Similarly, for every direction ξ and $1 \leq n \leq m - 1$ we have that there exists $\text{Tr}(\frac{\partial^n u}{\partial \xi^n}) \in$

$L^p_{\text{loc}}(\partial\Omega)$. Moreover, since by (9.4) for $u \in C^\infty_c(\mathbb{R}^N)$ and for every $x \in \mathbb{R}^N$ we have that

$$\frac{\partial^n u}{\partial \xi^n}(x) = \sum_{|\alpha|=n} \frac{1}{\alpha!} \partial^\alpha u(x) \xi^\alpha,$$

by Theorem 18.1(i) it follows that

$$(18.6) \quad \text{Tr}\left(\frac{\partial^n u}{\partial \xi^n}\right) = \sum_{|\alpha|=n} \frac{1}{\alpha!} \text{Tr}(\partial^\alpha u) \xi^\alpha.$$

We have seen in Chapter 9 that, in view of Rademacher's theorem, at $\mathcal{H}^{N-1}$ -a.e. $x_0 \in \partial\Omega$ there exist the tangent space $T_{\partial\Omega}(x_0)$ to $\partial\Omega$ at x_0 and the outward unit normal ν to $\partial\Omega$ at x_0 (see Definitions 9.62 and 9.63). Thus for $1 \leq n \leq m-1$, in view of (18.6), we can define

$$(18.7) \quad \text{Tr}\left(\frac{\partial^n u}{\partial \nu^n}\right) := \sum_{|\alpha|=n} \frac{1}{\alpha!} \text{Tr}(\partial^\alpha u) \nu^\alpha.$$

We will study the linear mapping

$$(18.8) \quad u \in W^{m,p}(\Omega) \mapsto \text{Tr}_m(u) := \left(\text{Tr}(u), \text{Tr}\left(\frac{\partial u}{\partial \nu}\right), \dots, \text{Tr}\left(\frac{\partial^{m-1} u}{\partial \nu^{m-1}}\right) \right).$$

The reason we are only interested in the traces of the normal derivatives is that all tangential derivatives depend only on $\text{Tr}(u)$. To be precise, given a function $v : \partial\Omega \rightarrow \mathbb{R}$ differentiable at x_0 and a tangent vector $t \in T_{\partial\Omega}(x_0)$, there exists a function $\psi : (-\delta, \delta) \rightarrow \mathbb{R}^N$ differentiable at 0 such that $\psi((-\delta, \delta)) \subseteq \partial\Omega$, $\psi(0) = x_0$ and $\psi'(0) = t$. Hence, by the chain rule, the function $v \circ \psi : (-\delta, \delta) \rightarrow \mathbb{R}$ is differentiable at 0, with $(v \circ \psi)'(0) = dv_{x_0}(t)$. In particular, if $v : \partial\Omega \rightarrow \mathbb{R}$ is of class C^1 , then it can be extended to a function $u \in C^1(\mathbb{R}^N)$ and so $dv_{x_0}(t) = \frac{\partial u}{\partial t}(x_0) = \sum_{i=1}^N \partial_i u(x_0) t_i$. It follows that, given a function $u \in C^1(\mathbb{R}^N)$, for every tangent vector $t \in T_{\partial\Omega}(x_0)$, the directional derivative $\frac{\partial u}{\partial t}(x_0)$ is completely determined by the restriction of u to $\partial\Omega$, that is, by $\text{Tr}(u)$. Since $T_{\partial\Omega}(x_0)$ is a vector space of dimension $N-1$, by taking $N-1$ linearly independent tangent unit vectors $b_1, \dots, b_{N-1}$ and adding to them the outward unit normal ν to $\partial\Omega$ at x_0 , we obtain a basis in $\mathbb{R}^N$. In particular every partial and directional derivative of u at x_0 can be written as a linear combination of $\frac{\partial u}{\partial b_i}(x_0)$, $i = 1, \dots, N-1$, and of the normal derivative $\frac{\partial u}{\partial \nu}(x_0)$.

To extend Theorem 18.7 we assume that $\partial\Omega$ is of class $C^{m-1,1}$ (see Definition 11.55).

Theorem 18.8 (Traces and $W_0^{m,p}$). *Let $m \in \mathbb{N}$, $m \geq 2$, let $\Omega \subset \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is of class $C^{m-1,1}$, let $1 \leq p < \infty$, and let $u \in W^{m,p}(\Omega)$. Then $\text{Tr}(u) = 0$, $\text{Tr}(\frac{\partial u}{\partial \nu}) = 0$, $\dots$, $\text{Tr}(\frac{\partial^{m-1} u}{\partial \nu^{m-1}}) = 0$ if and only if $u \in W_0^{m,p}(\Omega)$.*

Proof. We only give the proof for $m = 2$ and leave the case $m > 2$ as an exercise.

Step 1: Let $u \in W^{2,p}(\Omega)$. Let $x_0 \in \partial\Omega$. Since $\partial\Omega$ is of class $C^{1,1}$ there exist a rigid motion $T : \mathbb{R}^N \rightarrow \mathbb{R}^N$ with $T(x_0) = 0$, a function $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ of class $C^{1,1}$, and $r_{x_0} > 0$ such that, setting $y := T(x)$, we have

$$T(\Omega \cap Q_T(x_0, r_{x_0})) = \{y \in Q(0, r_{x_0}) : y_N > f(y')\},$$

where $Q_T(x_0, r_{x_0})$ is an open cube centered at x_0 and of side-length $r_{x_0} > 0$. Fix $0 < \varepsilon < r \leq r_{x_0}/[2(\text{Lip } f + 1)\sqrt{N-1}]$. Setting $w(y) := u(T^{-1}(y))$ for $y \in Q(0, r)$, by Theorem 11.53, $w \in W^{2,p}(Q(0, r))$. We claim that $\text{Tr}(w) \in W^{1,p}(Q_{N-1}(0, r))$. If $u \in C_c^\infty(\mathbb{R}^N)$, then $g(y') := \text{Tr}(w)(y', f(y')) = w(y', f(y'))$ and thus we can differentiate with respect to y_i for $i = 1, \dots, N-1$ to obtain that for $\mathcal{L}^{N-1}$ -a.e. $y' \in Q_{N-1}(0, r)$,

$$\frac{\partial g}{\partial y_i}(y') = \frac{\partial w}{\partial y_i}(y', f(y')) + \frac{\partial w}{\partial y_N}(y', f(y')) \frac{\partial f}{\partial y_i}(y').$$

Hence,

$$\left| \frac{\partial g}{\partial y_i}(y') \right| \leq \left| \frac{\partial w}{\partial y_i}(y', f(y')) \right| + \text{Lip } f \left| \frac{\partial w}{\partial y_N}(y', f(y')) \right|.$$

Using (18.4) with w replaced by $\frac{\partial w}{\partial y_i}$ and by $\frac{\partial w}{\partial y_N}$ respectively, we obtain

$$\begin{aligned} & \int_{Q_{N-1}(0, r)} \left| \frac{\partial g}{\partial y_i}(y') \right|^p \sqrt{1 + \|\nabla_{y'} f(y')\|_{N-1}^2} dy' \\ & \leq c \int_{Q_{N-1}(0, r)} \int_{f(y')}^{f(y')+\varepsilon} \left(\left| \frac{\partial w}{\partial y_i}(y) \right|^p + \left| \frac{\partial w}{\partial y_N}(y) \right|^p \right) dy_N dy' \\ & \quad + c \int_{Q_{N-1}(0, r)} \int_{f(y')}^{f(y')+\varepsilon} \left(\left| \frac{\partial^2 w}{\partial y_N \partial y_i}(y) \right|^p + \left| \frac{\partial^2 w}{\partial y_N^2}(y) \right|^p \right) dy_N dy'. \end{aligned}$$

It follows by (18.4) and the previous inequality that $g \in W^{1,p}(Q_{N-1}(0, r))$. Now, if $u \in W^{2,p}(\Omega)$, then by Theorem 11.35 there exists a sequence $\{u_n\}$ in $C_c^\infty(\mathbb{R}^N)$ such that u_n restricted to Ω converges to u in $W^{2,p}(\Omega)$. Setting $w_n(y) := u_n(T^{-1}(y))$ and $g_n(y') := w_n(y', f(y'))$, by (18.4) and the previous inequality applied to $g_n - g_\ell$, it follows that $\{g_n\}$ is a Cauchy sequence in $W^{1,p}(Q_{N-1}(0, r))$ and thus it converges to a function g in $W^{1,p}(Q_{N-1}(0, r))$. This proves the claim.

Step 2: If $u \in C_c^\infty(\Omega)$, then $\text{Tr}(u) = 0$ and $\text{Tr}(\partial_i u) = 0$ for all $i = 1, \dots, N$ and so $\text{Tr}(\frac{\partial u}{\partial \nu}) = 0$ by (18.7). Since $W_0^{2,p}(\Omega)$ is the closure of $C_c^\infty(\Omega)$ with respect to the norm $\|\cdot\|_{W^{2,p}(\Omega)}$ it follows from (18.3) applied to u and to $\partial_i u$ that $\text{Tr}(u) = \text{Tr}(\partial_i u) = 0$ for all $i = 1, \dots, N$ and for all $u \in W_0^{2,p}(\Omega)$. In turn, $\text{Tr}(\frac{\partial u}{\partial \nu}) = 0$ by (18.7).

Conversely, assume that $u \in W^{2,p}(\Omega)$ is such that $\text{Tr}(u) = 0$, $\text{Tr}(\frac{\partial u}{\partial \nu}) = 0$. Let f , r , and w be as in Step 1. Since $\text{Tr}(u) = 0$, we have that

$\text{Tr}(w)(y', f(y')) = 0$ for all $y' \in Q_{N-1}(0, r)$. Since $\text{Tr}(w) \in W^{1,p}(Q_{N-1}(0, r))$, by differentiating with respect to y_i for $i = 1, \dots, N-1$, for $\mathcal{L}^{N-1}$ -a.e. $y' \in Q_{N-1}(0, r)$ we have that

$$(18.9) \quad \text{Tr}\left(\frac{\partial w}{\partial y_i}\right)(y', f(y')) + \text{Tr}\left(\frac{\partial w}{\partial y_N}\right)(y', f(y')) \frac{\partial f}{\partial y_i}(y') = 0.$$

On the other hand, since $\text{Tr}\left(\frac{\partial u}{\partial \nu}\right)(y', f(y')) = 0$ and the outward unit normal ν to $\partial\Omega$ at $(y', f(y'))$ is given by

$$(18.10) \quad \nu(y', f(y')) = \left(\frac{\nabla_{y'} f(y')}{\sqrt{1 + \|\nabla_{y'} f(y')\|_{N-1}^2}}, \frac{-1}{\sqrt{1 + \|\nabla_{y'} f(y')\|_{N-1}^2}} \right),$$

by (18.7) we get

$$(18.11) \quad \sum_{i=1}^{N-1} \text{Tr}\left(\frac{\partial w}{\partial y_i}\right)(y', f(y')) \frac{\partial f}{\partial y_i}(y') - \text{Tr}\left(\frac{\partial w}{\partial y_N}\right)(y', f(y')) = 0.$$

The $N-1$ equalities in (18.9) and (18.11) give a system of N equations in the unknowns $\text{Tr}\left(\frac{\partial w}{\partial y_i}\right)(y', f(y'))$, $i = 1, \dots, N$. Since the determinant is different from zero (exercise), it follows that $\text{Tr}\left(\frac{\partial w}{\partial y_i}\right)(y', f(y')) = 0$, $i = 1, \dots, N$.

It follows that $\text{Tr}\left(\frac{\partial u}{\partial x_i}\right) = 0$ on $\partial\Omega$. We can now continue as in the proof of Theorem 18.7 to prove that the function v defined in (18.5) belongs to $W^{2,p}(\mathbb{R}^N)$ and then proceed as in that proof to show that $u \in W_0^{2,p}(\Omega)$. $\square$

In what follows we characterize the image of the trace operator. We will see that the cases $p = 1$ and $p > 1$ are quite different. We start with $p = 1$.

18.2. Traces of Functions in $W^{1,1}(\Omega)$

In this section we show that if Ω is sufficiently regular, then

$$\text{Tr}(W^{1,1}(\Omega)) = L^1(\partial\Omega).$$

We begin with the simplest case in which Ω is the half-space

$$\mathbb{R}_+^N = \{x = (x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : x_N > 0\}.$$

Theorem 18.9. *Let $N \geq 2$. Then for all functions $u \in \dot{W}^{1,1}(\mathbb{R}_+^N)$ vanishing at infinity,*

$$(18.12) \quad \|\text{Tr}(u)(\cdot, 0)\|_{L^1(\mathbb{R}^{N-1})} \leq \|\partial_N u\|_{L^1(\mathbb{R}_+^N)}.$$

Proof. Step 1: Assume first that $u \in L^{1*}(\mathbb{R}^N) \cap C^1(\mathbb{R}^N)$ with $\nabla u \in L^1(\mathbb{R}^N; \mathbb{R}^N)$. Reasoning as in the first step of the proof of Theorem 12.4,

for every $x' \in \mathbb{R}^{N-1}$ we have that

$$|u(x', 0)| \leq \int_0^\infty |\partial_N u(x', x_N)| dx_N.$$

Integrate both sides with respect to x' and use Tonelli's theorem to conclude that

$$\int_{\mathbb{R}^{N-1}} |u(x', 0)| dx' \leq \int_{\mathbb{R}_+^N} |\partial_N u(x)| dx.$$

Step 2: To remove the additional assumption that $u \in L^{1*}(\mathbb{R}^N) \cap C^1(\mathbb{R}^N)$ with $\nabla u \in L^1(\mathbb{R}^N; \mathbb{R}^N)$, note that given a function $u \in \dot{W}^{1,1}(\mathbb{R}_+^N)$ vanishing at infinity, using reflection (see Exercise 11.51), we can extend u to a function $u \in \dot{W}^{1,1}(\mathbb{R}^N)$ vanishing at infinity. In turn, by Theorem 12.4, we have that $u \in L^{1*}(\mathbb{R}^N)$. Let $u_\varepsilon := \varphi_\varepsilon * u$, where φ_ε is a standard mollifier. As in the proof of Lemma 11.25, we obtain that

$$\lim_{\varepsilon \rightarrow 0^+} \|u_\varepsilon - u\|_{L^{1*}(\mathbb{R}^N)} = 0, \quad \lim_{\varepsilon \rightarrow 0^+} \|\nabla u_\varepsilon - \nabla u\|_{L^1(\mathbb{R}^N)} = 0.$$

Since, by Step 1,

$$\int_{\mathbb{R}^{N-1}} |u_\varepsilon(x', 0)| dx' \leq \int_{\mathbb{R}^N} |\partial_N u_\varepsilon(x)| dx$$

for all $\varepsilon > 0$, by letting $\varepsilon \rightarrow 0^+$ we conclude that (18.12) holds. $\square$

Remark 18.10. In particular, it follows from the previous theorem that the linear operator

$$\begin{aligned} \text{Tr} : W^{1,1}(\mathbb{R}_+^N) &\rightarrow L^1(\mathbb{R}^{N-1}) \\ u &\mapsto \text{Tr}(u)(\cdot, 0) \end{aligned}$$

is continuous.

Exercise 18.11. Let $u, v \in W^{1,1}(\mathbb{R}_+^N)$. Prove that for all $i = 1, \dots, N$,

$$\int_{\mathbb{R}_+^N} u \partial_i v dx = - \int_{\mathbb{R}_+^N} v \partial_i u dx + \int_{\mathbb{R}^{N-1}} \text{Tr}(u) \text{Tr}(v) \nu_i dx',$$

where $\nu = -e_N$.

Exercise 18.12. Let $u \in W^{1,1}(\mathbb{R}_+^N)$ and $v \in W^{1,1}(\mathbb{R}_-^N)$, where

$$\mathbb{R}_-^N := \{(x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : x_N < 0\}.$$

(i) Prove that the function $w : \mathbb{R}^N \rightarrow \mathbb{R}$, defined by

$$w(x) := \begin{cases} u(x) & \text{if } x \in \mathbb{R}_+^N, \\ v(x) & \text{if } x \in \mathbb{R}_-^N, \end{cases}$$

belongs to $BV_{\text{loc}}(\mathbb{R}^N)$.

(ii) Prove that the function w belongs to $W^{1,1}(\mathbb{R}^N)$ if and only if $\text{Tr}(u) = \text{Tr}(v)$.

Next we prove that the operator Tr is onto. We use the abbreviation 18.2. The following theorem is due to Gagliardo [89]. The proof we present here is due to Mironescu [172].

Theorem 18.13 (Gagliardo). *Let $g \in L^1(\mathbb{R}^{N-1})$, $N \geq 2$. Then for every $0 < \delta < 1$ there exists a function $u \in W^{1,1}(\mathbb{R}_+^N)$ such that $\text{Tr}(u) = g$ and*

$$\|u\|_{L^1(\mathbb{R}_+^N)} \leq \delta \|g\|_{L^1(\mathbb{R}^{N-1})}, \quad \|\nabla u\|_{L^1(\mathbb{R}_+^N)} \leq (1 + \delta) \|g\|_{L^1(\mathbb{R}^{N-1})}.$$

Proof. Step 1: Assume that $g \in C_c^\infty(\mathbb{R}^{N-1}) \setminus \{0\}$. Fix $\delta > 0$ and let $\varphi \in C_c^\infty([0, \infty))$ be such that $\varphi(0) = 1$ and $\int_{\mathbb{R}_+} |\varphi'(t)| dt < 1 + \delta$. Note that the Lipschitz continuous function $\varphi_0(t) = (1 - t)^+$, $t \geq 0$, satisfies $\int_{\mathbb{R}_+} |\varphi_0'(t)| dt = 1$. Hence, to obtain φ it suffices to regularize φ_0 . For $n \in \mathbb{N}$ and $x = (x', x_N) \in \mathbb{R}^{N-1} \times [0, \infty)$ define $u_n(x) := g(x')\varphi(nx_N)$. Then $u_n \in C^\infty(\mathbb{R}^{N-1} \times [0, \infty))$, $u_n(x', 0) = g(x')$ for every $x' \in \mathbb{R}^{N-1}$, while

$$\partial_i u_n(x) = \partial_i g(x')\varphi(nx_N), \quad \partial_N u_n(x) = ng(x')\varphi'(nx_N)$$

for $x \in \mathbb{R}_+^N$, $i = 1, \dots, N - 1$. Moreover, by Fubini's theorem and the change of variables $t = nx_N$,

$$\int_{\mathbb{R}_+^N} |u_n(x)| dx = \frac{1}{n} \int_{\mathbb{R}^{N-1}} |g(x')| dx' \int_{\mathbb{R}_+} |\varphi(t)| dt \rightarrow 0$$

as $n \rightarrow \infty$. Similarly, for $i = 1, \dots, N - 1$,

$$\int_{\mathbb{R}_+^N} |\partial_i u_n(x)| dx = \frac{1}{n} \int_{\mathbb{R}^{N-1}} |\partial_i g(x')| dx' \int_{\mathbb{R}_+} |\varphi(t)| dt \rightarrow 0$$

as $n \rightarrow \infty$, while

$$\int_{\mathbb{R}_+^N} |\partial_N u_n(x)| dx = \int_{\mathbb{R}^{N-1}} |g(x')| dx' \int_{\mathbb{R}_+} |\varphi'(t)| dt \leq (1 + \delta) \int_{\mathbb{R}^{N-1}} |g(x')| dx'.$$

By taking n large enough we obtain the desired result in the case $g \in C_c^\infty(\mathbb{R}^{N-1})$.

Step 2: In the general case, it suffices to use the continuity of the trace operator and the density of smooth functions. We omit the details. $\square$

Remark 18.14. (i) Note that from the proof we actually obtain a stronger estimate than the one in the statement of the theorem, to be precise,

$$\|\partial_i u\|_{L^1(\mathbb{R}_+^N)} \leq \delta \|g\|_{L^1(\mathbb{R}^{N-1})}, \quad \|\partial_N u\|_{L^1(\mathbb{R}_+^N)} \leq (1 + \delta) \|g\|_{L^1(\mathbb{R}^{N-1})}$$

for $i = 1, \dots, N - 1$. Moreover, by taking n sufficiently large, given $\varepsilon > 0$ we can assume that

$$\text{supp } u \subseteq \text{supp } g \times [-\varepsilon, \varepsilon].$$

(ii) Peetre [193] has proved that there does not exist a bounded linear operator

$$\begin{aligned} L : L^1(\mathbb{R}^{N-1}) &\rightarrow W^{1,1}(\mathbb{R}_+^N) \\ g &\mapsto L(g) \end{aligned}$$

with the property that $\text{Tr}(L(g)) = g$.

Next we extend the previous theorem to special Lipschitz continuous domains.

Theorem 18.15. *Let $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ be a Lipschitz continuous function, $N \geq 2$, and let*

$$(18.13) \quad \Omega := \{(x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : x_N > f(x')\}.$$

Then for all $u \in W^{1,1}(\Omega)$,

$$(18.14) \quad \|\text{Tr}(u)\|_{L^1(\partial\Omega)} \leq (1 + \text{Lip } f) \|\partial_N u\|_{L^1(\Omega)}.$$

Moreover, for every $g \in L^1(\partial\Omega)$ and $0 < \delta < 1$ there exists a function $u \in W^{1,1}(\Omega)$ such that $\text{Tr}(u) = g$ and

$$\|u\|_{L^1(\Omega)} \leq \delta \|g\|_{L^1(\partial\Omega)}, \quad \|\nabla u\|_{L^1(\Omega; \mathbb{R}^N)} \leq 2(1 + \delta)(1 + \text{Lip } f) \|g\|_{L^1(\partial\Omega)}.$$

Proof. We only prove the second part of the statement and leave the first as an exercise. Given $g \in L^1(\partial\Omega)$, let $h(x') := g(x', f(x'))$, $x' \in \mathbb{R}^{N-1}$. Then

$$(18.15) \quad \int_{\mathbb{R}^{N-1}} |h(y')| dy' \leq \int_{\mathbb{R}^{N-1}} |g(x', f(x'))| \sqrt{1 + \|\nabla_{x'} f(x')\|_{N-1}^2} dx' < \infty,$$

and so by Theorem 18.13 for $0 < \delta < 1$ there exists $w \in W^{1,1}(\mathbb{R}_+^N)$ such that $\text{Tr}(w) = h$ and

$$(18.16) \quad \begin{aligned} \int_{\mathbb{R}_+^N} |w| dy &\leq \delta \int_{\mathbb{R}^{N-1}} |h| dy', \quad \int_{\mathbb{R}_+^N} |\partial_i w| dy \leq \delta \int_{\mathbb{R}^{N-1}} |h| dy' \\ \int_{\mathbb{R}_+^N} |\partial_N w| dy &\leq (1 + \delta) \int_{\mathbb{R}^{N-1}} |h| dy' \end{aligned}$$

for $i = 1, \dots, N-1$. Set

$$u(x) := w(\Phi(x)) = w(x', x_N - f(x')), \quad x \in \Omega.$$

As in the proof of Theorem 13.4 we have that Φ is invertible, with inverse given by $\Phi^{-1}(y', y_N) = (y', y_N + f(y'))$, and that Φ and Φ^{-1} are Lipschitz continuous, $\Phi(\Omega) = \mathbb{R}_+^N$, and $\det \nabla \Phi(x', x_N) = 1$ for $\mathcal{L}^{N-1}$ -a.e. $x' \in \mathbb{R}^{N-1}$

and for all $x_N \in \mathbb{R}$. Hence, by Exercise 11.51(iv), we have that $u \in W^{1,1}(\Omega)$, with

$$(18.17) \quad \begin{aligned} \frac{\partial u}{\partial x_i}(x) &= \frac{\partial w}{\partial y_i}(x', x_N - f(x')) - \frac{\partial w}{\partial y_N}(x', x_N - f(x')) \frac{\partial f}{\partial y_i}(y'), \\ \frac{\partial u}{\partial x_N}(x) &= \frac{\partial w}{\partial y_N}(x', x_N - f(x')). \end{aligned}$$

In turn, by Theorems 9.52 and 11.53, (18.15), (18.16),

$$\begin{aligned} \int_{\Omega} |\partial_i u(x)| dx &\leq \int_{\Omega} |\partial_i w(\Phi(x))| dx + \text{Lip } f \int_{\Omega} |\partial_N w(\Phi(x))| dx \\ &= \int_{\mathbb{R}_+^N} |\partial_i w(y)| dy + \text{Lip } f \int_{\mathbb{R}_+^N} |\partial_N w(y)| dy \\ &\leq (\delta + (1 + \delta) \text{Lip } f) \int_{\partial\Omega} |g| d\mathcal{H}^{N-1}, \end{aligned}$$

where we have used the fact that $\det \nabla \Phi = \det \nabla \Phi^{-1} = 1$. Similarly,

$$\int_{\Omega} |\partial_N u(x)| dx = \int_{\mathbb{R}_+^N} |\partial_N w(y)| dy \leq (1 + \delta) \int_{\partial\Omega} |g| d\mathcal{H}^{N-1}$$

and

$$\int_{\Omega} |u(x)| dx = \int_{\mathbb{R}_+^N} |w(y)| dy \leq \delta \int_{\partial\Omega} |g| d\mathcal{H}^{N-1},$$

which concludes the proof. $\square$

Remark 18.16. In view of Remark 18.14, given $\varepsilon > 0$ we can assume that

$$(18.18) \quad \text{supp } u \subseteq \{x \in \mathbb{R}^N : x' \in \text{supp } g, -\varepsilon + f(x') \leq x_N \leq f(x') + \varepsilon\}.$$

The first part of the previous theorem continues to hold with $W^{1,1}(\Omega)$ replaced with $\dot{W}^{1,1}(\Omega)$.

Exercise 18.17. Let $\Omega \subset \mathbb{R}^N$ be as in (18.13) and let $u \in W^{1,1}(\Omega)$ and $\phi \in C_c^1(\mathbb{R}^N)$. Prove that for $\mathcal{L}^{N-1}$ -a.e. $x' \in \mathbb{R}^{N-1}$,

$$\text{Tr}(u\phi)(x', f(x')) = \phi(x', f(x')) \text{Tr}(u)(x', f(x')).$$

Finally, we extend the previous result to open sets $\Omega \subset \mathbb{R}^N$ with uniformly Lipschitz continuous boundary (see Definition 13.11).

Theorem 18.18. *Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous. Then*

$$\|\text{Tr}(u)\|_{L^1(\partial\Omega)} \leq cM\varepsilon^{-1}(1 + L)\|u\|_{L^1(\Omega)} + c(1 + L)\|\nabla u\|_{L^1(\Omega)}$$

for all $u \in W^{1,1}(\Omega)$, where $c = c(N) > 0$ and $\varepsilon, L > 0$, and $M \in \mathbb{N}$ are given in Definition 13.11.

Moreover, for every $g \in L^1(\partial\Omega)$ there exists a function $u \in W^{1,1}(\Omega)$ such that $\text{Tr}(u) = g$ and

$$\|u\|_{L^1(\Omega)} \leq \|g\|_{L^1(\partial\Omega)}, \quad \|\nabla u\|_{L^1(\Omega)} \leq 4(1+L)\|g\|_{L^1(\partial\Omega)}.$$

Proof. Step 1: Let $u \in W^{1,1}(\Omega)$. Define $\phi_n, \Omega_0, \Omega_\pm, \phi_0, \phi_\pm, \psi_\pm$ as in (13.43), (13.46), (13.47), (13.50), respectively. Let

$$(18.19) \quad \psi_n(x) := \frac{\psi_+(x)\phi_n(x)}{\sum_k \phi_k(x)}.$$

Using the facts that $0 \leq \psi_\pm, \phi_n \leq 1$, $\|\nabla \psi_\pm\|_\infty \leq c\varepsilon^{-1}$, and (13.56), we get

$$(18.20) \quad \sum_n |\psi_n| \leq M, \quad \sum_n \|\nabla \psi_n\| \leq Mc\varepsilon^{-1}.$$

Then in Ω we may write (see (13.52))

$$(18.21) \quad u = \sum_n \psi_n u + \psi_- u =: \sum_n u_n + \psi_- u.$$

Observe that since $\{\Omega_n\}_n$ is locally finite, any bounded neighborhood of every point $x \in \Omega$ intersects only finitely many Ω_n 's.

By (13.50) the support of ψ_- is contained in Ω . Hence, $\text{Tr}(\psi_- u) = 0$ by Theorem 18.7.

Next we study the functions u_n . Fix n . By property (iii) of Definition 13.11 there exist local coordinates $y = (y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R}$ and a Lipschitz continuous function $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ (both depending on n), with $\text{Lip } f \leq L$, such that

$$(18.22) \quad \Omega_n \cap \Omega = \Omega_n \cap V_n,$$

where V_n in local coordinates is given by

$$\{(y', y_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : y_N > f(y')\}.$$

Since by (13.44) the support of u_n is contained in Ω_n , we may extend u_n to be zero in $V_n \setminus \Omega_n$. Thus, we are in a position to apply Theorem 18.15 to find that $\text{Tr}(u_n)$ satisfies (18.14) with V_n in place of Ω . Since the support of ψ_n is contained in Ω_n , it follows from (18.22) that $\psi_n = 0$ on $\partial V_n \setminus \partial\Omega$. Hence, the same holds for $\text{Tr}(u_n)$. This shows that

$$(18.23) \quad \begin{aligned} \int_{\partial\Omega \cap \partial V_n} |\text{Tr}(u_n)| d\mathcal{H}^{N-1} &\leq (1+L) \int_{V_n} \|\nabla u_n\| dx \\ &= (1+L) \int_{\Omega \cap \Omega_n} \|\nabla u_n\| dx. \end{aligned}$$

Since by (18.21), in $\Omega \cap \Omega_n$,

$$(18.24) \quad \nabla u_n = \nabla \psi_n u + \psi_n \nabla u,$$

using (18.20), we get that in $\Omega \cap \Omega_n$,

$$\|\nabla u_n\| \leq cM\varepsilon^{-1}|u| + \|\nabla u\|.$$

Hence, by Exercise 18.17 and (18.23),

$$\begin{aligned} \int_{\partial\Omega} |\operatorname{Tr}(u)| d\mathcal{H}^{N-1} &\leq \sum_n \int_{\partial\Omega \cap \partial V_n} |\operatorname{Tr}(u_n)| d\mathcal{H}^{N-1} \\ &\leq cM\varepsilon^{-1}(1+L) \int_{\Omega} |u| dx + (1+L) \int_{\Omega} \|\nabla u\| dx. \end{aligned}$$

Step 2: Given $g \in L^1(\partial\Omega)$, since $\psi_+ = 1$ on $\partial\Omega$, we can write $g = \sum_n \psi_n g$.

By extending $\psi_n g$ to be zero on $\partial V_n \setminus \partial\Omega_n$, we are in a position to apply Theorem 18.15 to find a function $u_n \in W^{1,1}(V_n)$ such that $\operatorname{Tr}(u_n) = \psi_n g$ on ∂V_n and

$$\begin{aligned} \int_{V_n} |u_n| dx &\leq \int_{\partial V_n} \psi_n |g| d\mathcal{H}^{N-1} = \int_{\partial\Omega} \psi_n |g| d\mathcal{H}^{N-1}, \\ \int_{V_n} \|\nabla u_n\| dx &\leq 4(1+L) \int_{\partial\Omega} \psi_n |g| d\mathcal{H}^{N-1}, \end{aligned}$$

where we used the fact that $\psi_n = 0$ in $\partial V_n \setminus \partial\Omega$. Moreover, by (18.18) we can assume that $\operatorname{supp} u_n \subseteq \Omega_n$. Let $u := \sum_n u_n$. Since $\{\Omega_n\}_n$ is locally finite we have that $u \in W_{\operatorname{loc}}^{1,1}(\Omega)$ and $\operatorname{Tr}(u) = g$. Moreover,

$$\int_{\Omega} |u| dx \leq \sum_n \int_{V_n} |u_n| dx \leq \sum_n \int_{\partial\Omega} \psi_n |g| d\mathcal{H}^{N-1} = \int_{\partial\Omega} |g| d\mathcal{H}^{N-1},$$

and similarly

$$\int_{\Omega} \|\nabla u\| dx \leq \sum_n \int_{V_n} \|\nabla u_n\| dx \leq 4(1+L) \int_{\partial\Omega} |g| d\mathcal{H}^{N-1}.$$

This completes the proof. $\square$

Remark 18.19. Note that by Exercise 13.15, if $\Omega \subseteq \mathbb{R}^N$ is an unbounded open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous, then Ω has infinite measure. On the other hand, if $\Omega \subset \mathbb{R}^N$ is an unbounded open set with finite measure, then $\partial\Omega$ is unbounded. In this case the function 1 belongs to $W^{1,1}(\Omega)$, but its trace does not belong to $L^1(\partial\Omega)$.

Exercise 18.20. Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous and let $u \in W^{1,1}(\Omega)$ and $v \in W^{1,1}(\mathbb{R}^N \setminus \overline{\Omega})$.

(i) Prove that the function $w : \mathbb{R}^N \rightarrow \mathbb{R}$, defined by

$$w(x) := \begin{cases} u(x) & \text{if } x \in \Omega, \\ v(x) & \text{if } x \in \mathbb{R}^N \setminus \overline{\Omega}, \end{cases}$$

belongs to $BV(\mathbb{R}^N)$.

- (ii) Prove that the function w belongs to $W^{1,1}(\mathbb{R}^N)$ if and only if $\text{Tr}(u) = \text{Tr}(v)$.

18.3. Traces of Functions in $BV(\Omega)$

In this section we prove that every function in $BV(\Omega)$ has a trace in $L^1(\partial\Omega)$, provided Ω is sufficiently regular.

Theorem 18.21. *Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is Lipschitz continuous. There exists a unique linear operator*

$$\text{Tr} : BV(\Omega) \rightarrow L^1_{\text{loc}}(\partial\Omega)$$

such that

- (i) $\text{Tr}(u) = u$ on $\partial\Omega$ for all $u \in BV(\Omega) \cap C(\overline{\Omega})$,
(ii) for all $\psi \in C_c^1(\mathbb{R}^N)$, $u \in BV(\Omega)$, and $i = 1, \dots, N$,
- $$(18.25) \quad \int_{\Omega} u \partial_i \psi \, dx = - \int_{\Omega} \psi D_i u \, dx + \int_{\partial\Omega} \psi \text{Tr}(u) \nu_i \, d\mathcal{H}^{N-1},$$
- where ν is the outward unit normal to $\partial\Omega$,
(iii) for all but countably many $R > 0$ there exist two constants c_R , $\varepsilon_R > 0$ depending on R and Ω such that

$$\begin{aligned} \int_{B(0,R) \cap \partial\Omega} |\text{Tr}(u)| \, d\mathcal{H}^{N-1} &\leq c_R \varepsilon^{-1} \int_{B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)} |u| \, dx \\ &\quad + c_R \|Du\|(B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)) \end{aligned}$$

for all but countably many $0 < \varepsilon \leq \varepsilon_R$, where $\Omega_\varepsilon := \{x \in \Omega : \text{dist}(x, \partial\Omega) > \varepsilon\}$.

Proof. By Theorem 14.9 for every $u \in BV(\Omega)$ there exists a sequence $\{u_n\}_n$ in $C^\infty(\Omega) \cap W^{1,1}(\Omega)$ such that $u_n \rightarrow u$ in $L^1(\Omega)$, $\nabla u_n \mathcal{L}^N \lfloor \Omega \xrightarrow{*} Du$ in the sense of measures and

$$\lim_{n \rightarrow \infty} \int_{\Omega} \|\nabla u_n\| \, dx = \|Du\|(\Omega).$$

It follows from Theorem 18.1,

$$\begin{aligned} \int_{B(0,R) \cap \partial\Omega} |\text{Tr}(u_n) - \text{Tr}(u_\ell)| \, d\mathcal{H}^{N-1} &\leq c_R \varepsilon^{-1} \int_{B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)} |u_n - u_\ell| \, dx \\ &\quad + c_R \int_{B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)} (\|\nabla u_n\| + \|\nabla u_\ell\|) \, dx \end{aligned}$$

for every $\ell, n \in \mathbb{N}$. By Exercise 14.12 and Proposition B.12, letting $n, \ell \rightarrow \infty$ first and then $\varepsilon \rightarrow 0^+$, we obtain that $\{\text{Tr}(u_n)\}_n$ is a Cauchy sequence in $L^1(B(0,R) \cap \partial\Omega)$ for all but countably many $R > 0$ and thus it converges in $L^1_{\text{loc}}(\partial\Omega)$ to a function $\text{Tr}(u)$.

Since (18.1) holds for all u_n , all $\psi \in C_c^1(\mathbb{R}^N)$, and all $i = 1, \dots, N$, letting $n \rightarrow \infty$ in (18.1) gives (ii).

Moreover, since

$$\begin{aligned} \int_{B(0,R) \cap \partial\Omega} |\operatorname{Tr}(u_n)| d\mathcal{H}^{N-1} &\leq c_R \varepsilon^{-1} \int_{B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)} |u_n| dx \\ &\quad + c_R \int_{B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)} \|\nabla u_n\| dx, \end{aligned}$$

if R and ε are such that $\|Du\|(\partial(B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)) \cap \Omega) = 0$, which happens for all but countably many ε and R , then by Exercise 14.12,

$$\lim_{n \rightarrow \infty} \int_{B(0,R) \cap (\Omega \setminus \Omega_\varepsilon)} \|\nabla u_n\| dx = \|Du\|(B(0,R) \cap (\Omega \setminus \Omega_\varepsilon))$$

and so, letting $n \rightarrow \infty$ in the previous inequality we obtain (iii). $\square$

The analog of Theorem 18.18 is given by the following theorem. The proof is very similar to the one of Theorem 18.18 and is left as an exercise.

Theorem 18.22. *Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous. Then*

$$\|\operatorname{Tr}(u)\|_{L^1(\partial\Omega)} \leq cM\varepsilon^{-1}(1+L)\|u\|_{L^1(\Omega)} + c(1+L)\|Du\|(\Omega)$$

for all $u \in BV(\Omega)$, where $\varepsilon, L > 0$, $M \in \mathbb{N}$ are given as in Definition 13.11.

Since $W^{1,1}(\Omega) \subset BV(\Omega)$, it follows from Theorem 18.18 that if $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, is an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous, then

$$\operatorname{Tr} : BV(\Omega) \rightarrow L^1(\Omega)$$

is onto.

Exercise 18.23 (Compactness of traces). Let $\Omega \subset \mathbb{R}^N$ be an open bounded set with Lipschitz continuous boundary. Prove that if $\{u_n\}_n$ is a sequence in $BV(\Omega)$ converging in $L^1(\Omega)$ to a function $u \in BV(\Omega)$ and such that $\nabla u_n \mathcal{L}^N \lfloor \Omega \xrightarrow{*} Du$ in the sense of measures and $\|Du_n\|(\Omega) \rightarrow \|Du\|(\Omega)$, then $\operatorname{Tr}(u_n) \rightarrow \operatorname{Tr}(u)$ in $L^1(\partial\Omega)$. Hint: See Exercise 18.6.

18.4. Traces of Functions in $W^{1,p}(\Omega)$, $p > 1$

In this section we study trace of functions in $W^{1,p}(\Omega)$ for $1 < p < N$. We will see that the situation is quite different from the case $p = 1$.

Theorem 18.24. *Let $1 < p < N$. Then there exists a constant $c = c(N, p) > 0$ such that for all functions $u \in \dot{W}^{1,p}(\mathbb{R}_+^N)$ vanishing at infinity,*

$$(18.26) \quad \|\operatorname{Tr}(u)(\cdot, 0)\|_{L^{p(N-1)/(N-p)}(\mathbb{R}^{N-1})} \leq c \|\nabla u\|_{L^p(\mathbb{R}_+^N)}.$$

Proof. Step 1: Assume first that $u \in L^{p^*}(\mathbb{R}^N) \cap C^1(\mathbb{R}^N)$ with $\nabla u \in L^p(\mathbb{R}^N; \mathbb{R}^N)$. Reasoning as in the first step of the proof of Theorem 12.4, for every $x' \in \mathbb{R}^{N-1}$ and for $r > 1$ we have that

$$|u(x', 0)|^r \leq r \int_0^\infty |u(x', x_N)|^{r-1} |\partial_N u(x', x_N)| dx_N.$$

Integrate both sides with respect to x' and use Tonelli's theorem to conclude that

$$(18.27) \quad \begin{aligned} \int_{\mathbb{R}^{N-1}} |u(x', 0)|^r dx' &\leq r \int_{\mathbb{R}_+^N} |u(x)|^{r-1} |\partial_N u(x)| dx \\ &\leq r \|u\|_{L^{(r-1)p'}(\mathbb{R}_+^N)}^{(r-1)} \|\partial_N u\|_{L^p(\mathbb{R}_+^N)}, \end{aligned}$$

where we have used Hölder's inequality. Taking $r := p(N-1)/(N-p)$, we have that $(r-1)p' = p^*$, and so, by Theorem 12.4,

$$\begin{aligned} \int_{\mathbb{R}^{N-1}} |u(x', 0)|^{p(N-1)/(N-p)} dx' &\leq c \|u\|_{L^{p^*}(\mathbb{R}_+^N)}^{p^*/p'} \|\partial_N u\|_{L^p(\mathbb{R}_+^N)} \\ &\leq c \|\nabla u\|_{L^p(\mathbb{R}_+^N)}^{1+p^*/p'}. \end{aligned}$$

Note that $1 + p^*/p = p(N-1)/(N-p)$, so that (18.26) holds.

Step 2: To remove the additional assumption that $u \in L^{p^*}(\mathbb{R}^N) \cap C^1(\mathbb{R}^N)$ with $\nabla u \in L^p(\mathbb{R}^N; \mathbb{R}^N)$, we proceed as in the proof of Theorem 18.9 by first extending a function $u \in \dot{W}^{1,p}(\mathbb{R}_+^N)$ vanishing at infinity to a function $u \in \dot{W}^{1,p}(\mathbb{R}^N)$ vanishing at infinity and such that $\|\nabla u\|_{L^p(\mathbb{R}^N)} \leq c \|\nabla u\|_{L^p(\mathbb{R}_+^N)}$ and then using mollifiers. We omit the details. $\square$

Remark 18.25. In particular, it follows from the previous theorem that the linear operator

$$\text{Tr} : W^{1,p}(\mathbb{R}_+^N) \rightarrow L^{p(N-1)/(N-p)}(\mathbb{R}^{N-1})$$

is continuous. Moreover, taking $r = p$ in (18.27) yields

$$\begin{aligned} \int_{\mathbb{R}^{N-1}} |u(x', 0)|^p dx' &\leq p \left(\int_{\mathbb{R}_+^N} |u(x)|^p dx \right)^{1/p'} \left(\int_{\mathbb{R}_+^N} |\partial_N u(x)|^p dx \right)^{1/p} \\ &\leq p \int_{\mathbb{R}_+^N} |u(x)|^p dx + p \int_{\mathbb{R}_+^N} |\partial_N u(x)|^p dx, \end{aligned}$$

where we have used Young's inequality (see (B.17) in Appendix B). Note that this inequality actually holds for every $1 \leq p < \infty$. Hence,

$$\int_{\mathbb{R}^{N-1}} |\text{Tr}(u)(x', 0)|^p dx' \leq p \int_{\mathbb{R}_+^N} |u(x)|^p dx + p \int_{\mathbb{R}_+^N} |\partial_N u(x)|^p dx,$$

for all $u \in W^{1,p}(\mathbb{R}_+^N)$.

Exercise 18.26. Prove that if $p = N$, then for every $N \leq q < \infty$ we have that

$$\text{Tr} : W^{1,N}(\mathbb{R}_+^N) \rightarrow L^q(\mathbb{R}^{N-1})$$

is a continuous operator.

Unlike the case $p = 1$, when $1 < p < \infty$, the trace operator

$$\text{Tr} : W^{1,p}(\mathbb{R}_+^N) \rightarrow L^p(\mathbb{R}^{N-1})$$

is not onto. We now show that if $u \in W^{1,p}(\mathbb{R}_+^N)$, $1 < p < \infty$, then its trace $\text{Tr}(u)$ belongs to the Besov space $B^{1-1/p,p}(\mathbb{R}^{N-1})$ (see Definition 17.1). The following theorem is due to Gagliardo [89].

Theorem 18.27 (Gagliardo). *Let $1 < p < \infty$ and let $N \geq 2$. Then there exists a constant $c = c(p, N) > 0$ such that for all $u \in \dot{W}^{1,p}(\mathbb{R}_+^N)$,*

$$(18.28) \quad |\text{Tr}(u)(\cdot, 0)|_{B^{1-1/p,p}(\mathbb{R}^{N-1})} \leq c \|\nabla u\|_{L^p(\mathbb{R}_+^N)}.$$

Proof. Assume that $u \in \dot{W}^{1,p}(\mathbb{R}_+^N) \cap C^\infty(\mathbb{R}^N)$. By Theorem 18.57 and Remark 18.60 below we have that

$$|u(\cdot, 0)|_{B^{1-1/p,p}(\mathbb{R}^{N-1})} \leq c \|\nabla u\|_{L^p(\mathbb{R}_+^N)}.$$

To remove the extra assumption that $u \in C^\infty(\mathbb{R}^N)$, we first use a reflection argument to extend every function $u \in \dot{W}^{1,p}(\mathbb{R}_+^N)$ to a function $u \in \dot{W}^{1,p}(\mathbb{R}^N)$ and then consider a sequence of $\{u_\varepsilon\}_\varepsilon$, where $u_\varepsilon := u * \varphi_\varepsilon$ and the φ_ε are standard mollifiers. Since $u_\varepsilon \rightarrow u$ in $L_{\text{loc}}^1(\mathbb{R}^N)$, $\nabla u_\varepsilon \rightarrow \nabla u$ in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, and $u_\varepsilon(\cdot, 0) \rightarrow \text{Tr}(u)$ in $L_{\text{loc}}^1(\mathbb{R}^{N-1})$, by selecting a subsequence, we may assume that $u_\varepsilon(x', 0) \rightarrow \text{Tr}(u)(x')$ for $\mathcal{L}^{N-1}$ -a.e. $x' \in \mathbb{R}^{N-1}$. Hence, by Fatou's lemma,

$$\begin{aligned} & \int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}^{N-1}} \frac{|\text{Tr}(u)(x', 0) - \text{Tr}(u)(y', 0)|^p}{\|x' - y'\|_{N-1}^{p+N-2}} dx' dy' \\ & \leq \liminf_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}^{N-1}} \frac{|u_\varepsilon(x', 0) - u_\varepsilon(y', 0)|^p}{\|x' - y'\|_{N-1}^{p+N-2}} dx' dy' \\ & \leq c \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}_+^N} \|\nabla u_\varepsilon(x)\|^p dx = c \int_{\mathbb{R}_+^N} \|\nabla u(x)\|^p dx. \end{aligned}$$

This concludes the proof. $\square$

The previous result shows in particular that

$$\text{Tr}(W^{1,p}(\mathbb{R}_+^N)) \subseteq B^{1-1/p,p}(\mathbb{R}^{N-1}).$$

To prove the opposite inclusion one could use Theorem 18.53 below, however, we present a different proof, which relies on the seminorm $|\cdot|_{B^{1-1/p,p}(\mathbb{R}^{N-1})}$.

Theorem 18.28 (Gagliardo). *Let $1 < p < \infty$, let $N \geq 2$, and let $g \in L^1_{\text{loc}}(\mathbb{R}^{N-1})$ be such that $|g|_{B^{1-1/p,p}(\mathbb{R}^{N-1})} < \infty$. Then there exists a function $v \in \dot{W}^{1,p}(\mathbb{R}^N_+)$ such that $\text{Tr}(v)(\cdot, 0) = g$ and*

$$(18.29) \quad \|\nabla v\|_{L^p(\mathbb{R}^N_+)} \leq c|g|_{B^{1-1/p,p}(\mathbb{R}^{N-1})},$$

where $c = c(N, p) > 0$.

Proof. Let $\varphi \in C_c^\infty(\mathbb{R}^{N-1})$ be such that

$$\text{supp } \varphi \subseteq \overline{B_{N-1}(0, 1)}$$

and

$$\int_{\mathbb{R}^{N-1}} \varphi(x') dx' = 1.$$

For $x' \in \mathbb{R}^{N-1}$ and $x_N > 0$ define

$$(18.30) \quad v(x) := \frac{1}{x_N^{N-1}} \int_{\mathbb{R}^{N-1}} \varphi((x' - y')/x_N) g(y') dy'.$$

By Theorem C.20 (where x_N plays the role of ε), for any $i = 1, \dots, N-1$ we have that

$$\begin{aligned} \frac{\partial v}{\partial x_i}(x) &= \frac{1}{x_N^N} \int_{\mathbb{R}^{N-1}} \frac{\partial \varphi}{\partial x_i}((x' - y')/x_N) g(y') dy' \\ &= \frac{1}{x_N^N} \int_{\mathbb{R}^{N-1}} \frac{\partial \varphi}{\partial x_i}((x' - y')/x_N) [g(y') - g(x')] dy', \end{aligned}$$

where in the second equality we used the fact that

$$\begin{aligned} 0 &= \frac{\partial}{\partial x_i}(1) = \frac{\partial}{\partial x_i} \left(\frac{1}{x_N^{N-1}} \int_{\mathbb{R}^{N-1}} \varphi((x' - y')/x_N) dy' \right) \\ &= \frac{1}{x_N^N} \int_{\mathbb{R}^{N-1}} \frac{\partial \varphi}{\partial x_i}((x' - y')/x_N) dy'. \end{aligned}$$

Since $\text{supp } \varphi \subseteq \overline{B_{N-1}(0, 1)}$, we have that

$$(18.31) \quad |\partial_i v(x)| \leq \frac{c}{x_N^N} \int_{B_{N-1}(x', x_N)} |g(y') - g(x')| dy'.$$

Raising both sides to the power p , integrating in x over $\mathbb{R}^N_+$, and using Hölder's inequality, we get

$$\begin{aligned} &\int_{\mathbb{R}^N_+} |\partial_i v(x)|^p dx \\ &\leq c \int_{\mathbb{R}^N_+} \frac{1}{x_N^{Np}} \left(\int_{B_{N-1}(x', x_N)} |g(y') - g(x')| dy' \right)^p dx \\ &\leq c \int_{\mathbb{R}^N_+} \frac{x_N^{(N-1)(p-1)}}{x_N^{Np}} \int_{B_{N-1}(x', x_N)} |g(y') - g(x')|^p dy' dx =: cI. \end{aligned}$$

By Tonelli's theorem, we get that

$$\begin{aligned}
 I &= \int_0^\infty \int_{\mathbb{R}^{N-1}} \int_{B_{N-1}(x', x_N)} \frac{1}{x_N^{N+p-1}} |g(y') - g(x')|^p dy' dx' dx_N \\
 &= \int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}^{N-1}} |g(y') - g(x')|^p \int_{\|x' - y'\|_{N-1}}^\infty \frac{1}{x_N^{N+p-1}} dx_N dy' dx' \\
 &= \frac{1}{N+p-2} \int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}^{N-1}} \frac{|g(y') - g(x')|^p}{\|x' - y'\|_{N-1}^{p+N-2}} dy' dx'.
 \end{aligned}$$

Hence, we have shown that

$$(18.32) \quad \int_{\mathbb{R}_+^N} |\partial_i v(x)|^p dx \leq c \int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}^{N-1}} \frac{|g(y') - g(x')|^p}{\|x' - y'\|_{N-1}^{p+N-2}} dy' dx'$$

for all $i = 1, \dots, N-1$.

Similarly, by Theorem B.52, we obtain that

$$\begin{aligned}
 \partial_N v(x) &= \int_{\mathbb{R}^{N-1}} \frac{\partial}{\partial x_N} \left(\frac{1}{x_N^{N-1}} \varphi((x' - y')/x_N) \right) g(y') dy' \\
 &= \int_{\mathbb{R}^{N-1}} \frac{\partial}{\partial x_N} \left(\frac{1}{x_N^{N-1}} \varphi((x' - y')/x_N) \right) [g(y') - g(x')] dy',
 \end{aligned}$$

where in the second equality we used the fact that

$$\begin{aligned}
 0 &= \frac{\partial}{\partial x_N} (1) = \frac{\partial}{\partial x_N} \left(\int_{\mathbb{R}^{N-1}} \frac{1}{x_N^{N-1}} \varphi((x' - y')/x_N) dy' \right) \\
 &= \int_{\mathbb{R}^{N-1}} \frac{\partial}{\partial x_N} \left(\frac{1}{x_N^{N-1}} \varphi((x' - y')/x_N) \right) dy'.
 \end{aligned}$$

Since $\text{supp } \varphi \subseteq \overline{B_{N-1}(0, 1)}$, we have that

$$\partial_N v(x) = \int_{B_{N-1}(x', x_N)} \frac{\partial}{\partial x_N} \left(\frac{1}{x_N^{N-1}} \varphi((x' - y')/x_N) \right) [g(y') - g(x')] dy'.$$

Now for $y' \in B_{N-1}(x', x_N)$,

$$\begin{aligned}
 |\partial_N (x_N^{1-N} \varphi((x' - y')/x_N))| &= \left| -(N-1) x_N^{-N} \varphi((x' - y')/x_N) \right. \\
 &\quad \left. - x_N^{1-N} \sum_{i=1}^{N-1} \partial_i \varphi((x' - y')/x_N) (x_i - y_i) x_N^{-2} \right| \\
 &\leq c x_N^{-N} + c \|x' - y'\|_{N-1} x_N^{-1-N} \leq c x_N^{-N}.
 \end{aligned}$$

In turn,

$$|\partial_N v(x)| \leq \frac{c}{x_N^N} \int_{B_{N-1}(x', x_N)} |g(y') - g(x')| dy'.$$

We can now continue as before (see (18.31)) to conclude that

$$(18.33) \quad \int_{\mathbb{R}_+^N} |\partial_N v(x)|^p dx \leq c \int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}^{N-1}} \frac{|g(y') - g(x')|^p}{\|x' - y'\|_{N-1}^{p+N-2}} dy' dx'.$$

Since $v \in C^1(\mathbb{R}_+^N)$, we have shown that $v \in \dot{W}^{1,p}(\mathbb{R}_+^N)$ and that (18.29) holds. To conclude the proof, it remains to prove that $\text{Tr}(v) = g$. Using standard mollifiers (exercise), we may find a sequence $\{g_n\}_n$ in $C^\infty(\mathbb{R}^{N-1})$ with $|g_n|_{B^{1-1/p,p}(\mathbb{R}^{N-1})} < \infty$ such that $|g - g_n|_{B^{1-1/p,p}(\mathbb{R}^{N-1})} \rightarrow 0$. Let v_n be defined as in (18.30) with g replaced by g_n . Reasoning as in the proof of Theorem C.16(i), we have that $v_n \in C(\mathbb{R}^{N-1} \times [0, \infty))$, with $v_n(x', 0) = g_n(x')$.

By (18.29) applied to $v_n - v$ and $g_n - g$ we have that $\partial_i v_n \rightarrow \partial_i v$ in $L^p(\mathbb{R}_+^N)$ for every $i = 1, \dots, N$. In turn, by (18.3) and Exercise 18.2, we obtain that $\text{Tr}(u)(\cdot, 0) = g$. $\square$

Corollary 18.29. *Let $1 < p < \infty$, let $N \geq 2$, and let $g \in B^{1-1/p,p}(\mathbb{R}^{N-1})$. Then for every $0 < \varepsilon \leq 1$ there exists a function $u \in W^{1,p}(\mathbb{R}_+^N)$ such that $\text{Tr}(u)(\cdot, 0) = g$, $\text{supp } u \subseteq \mathbb{R}^{N-1} \times [-\varepsilon, \varepsilon]$ and*

$$\begin{aligned} \|u\|_{L^p(\mathbb{R}_+^N)} &\leq \varepsilon^{1/p} \|g\|_{L^p(\mathbb{R}^{N-1})}, \\ |u|_{W^{1,p}(\mathbb{R}_+^N)} &\leq c\varepsilon^{-1/p'} \|g\|_{L^p(\mathbb{R}^{N-1})} + c|g|_{B^{1-1/p,p}(\mathbb{R}^{N-1})}, \end{aligned}$$

where $c = c(N, p) > 0$.

Proof. Let v be as in the previous theorem. By Theorem C.16 (with x_N in place of ε) we have that for all $x_N > 0$,

$$(18.34) \quad \int_{\mathbb{R}^{N-1}} |v(x', x_N)|^p dx' \leq \int_{\mathbb{R}^{N-1}} |g(x')|^p dx'.$$

Integrating in x_N in $(0, \varepsilon)$ gives

$$(18.35) \quad \int_0^\varepsilon \int_{\mathbb{R}^{N-1}} |v(x', x_N)|^p dx' dx_N \leq \varepsilon \int_{\mathbb{R}^{N-1}} |g(x')|^p dx'.$$

Let $\psi \in C^\infty([0, \infty))$ be a decreasing function such that $\psi = 1$ in $[0, \varepsilon/2]$, $\psi(x_N) = 0$ for $x_N \geq \varepsilon$ and $\|\psi'\|_\infty \leq c\varepsilon^{-1}$. For $x = (x', x_N) \in \mathbb{R}_+^N$ we define $u(x) := \psi(x_N)v(x)$.

By (18.35), Tonelli's theorem, and the fact that $\psi(x_N) = 0$ for $x_N \geq \varepsilon$ we have

$$\begin{aligned} (18.36) \quad \int_{\mathbb{R}_+^N} |u(x)|^p dx &= \int_0^\varepsilon (\psi(x_N))^p \int_{\mathbb{R}^{N-1}} |v(x)|^p dx' dx_N \\ &\leq \int_0^\varepsilon \int_{\mathbb{R}^{N-1}} |v(x)|^p dx' dx_N \leq \varepsilon \int_{\mathbb{R}^{N-1}} |g(x')|^p dx', \end{aligned}$$

while for $i = 1, \dots, N-1$,

$$|\partial_i u(x)| = |\psi(x_N) \partial_i v(x)| \leq |\partial_i v(x)|,$$

where we used the fact that $\|\psi\|_\infty \leq 1$. In turn, by (18.32), we obtain that

$$\int_{\mathbb{R}_+^N} |\partial_i u(x)|^p dx \leq c \int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}^{N-1}} \frac{|g(y') - g(x')|^p}{\|x' - y'\|_{N-1}^{p+N-2}} dy' dx'$$

for all $i = 1, \dots, N-1$. On the other hand,

$$\partial_N u(x) = \psi(x_N) \partial_N v(x) + \psi'(x_N) v(x),$$

and so, by (18.33), (18.35) and the fact that $\psi'(x_N) = 0$ for $x_N \geq \varepsilon$,

$$\begin{aligned} \left(\int_{\mathbb{R}_+^N} |\partial_N u|^p dx \right)^{1/p} &\leq \left(\int_{\mathbb{R}_+^N} |\partial_N v|^p dx \right)^{1/p} + c \left(\int_0^\varepsilon \varepsilon^{-p} \int_{\mathbb{R}^{N-1}} |v|^p dx' dx_N \right)^{1/p} \\ &\leq c \left(\int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}^{N-1}} \frac{|g(y') - g(x')|^p}{\|x' - y'\|_{N-1}^{p+N-2}} dy' dx' \right)^{1/p} + c \varepsilon^{-(p-1)/p} \left(\int_{\mathbb{R}^{N-1}} |g|^p dx' \right)^{1/p}. \end{aligned}$$

Reasoning as in the last part of the previous theorem we can show that $\text{Tr}(u) = g$. $\square$

Exercise 18.30. Prove that the previous corollary (without the condition $\text{supp } u \subseteq \mathbb{R}^{N-1} \times [-\varepsilon, \varepsilon]$) continues to hold if we consider the function $u(x) := e^{-x_N/p} v(x)$, $x \in \mathbb{R}_+^N$.

Exercise 18.31. Let $N = 2$ and $x = (x_1, x_2)$.

- (i) Prove that if $u \in \dot{W}^{1,2}(\mathbb{R}^2)$, then it has a trace on the line $\mathbb{R} \times \{0\}$ and

$$\int_{\mathbb{R}} \int_{\mathbb{R}} \frac{|v(x_1) - v(y_1)|^2}{|x_1 - y_1|^2} dx_1 dy_1 \leq \pi \int_{\mathbb{R}^2} \|\nabla u(x)\|^2 dx,$$

where $v(x_1) := \text{Tr}(u)(x_1, 0)$. Hint: Use Fourier transforms for v and u .

- (ii) Prove that if $u \in \dot{W}^{1,2}(B(0, r) \cap \mathbb{R}_+^2)$, then

$$\int_{-r}^r \int_{-r}^r \frac{|v(x_1) - v(y_1)|^2}{|x_1 - y_1|^2} dx_1 dy_1 \leq 2\pi \int_{\mathbb{R}^2} \|\nabla u(x)\|^2 dx,$$

where $v(x_1) := \text{Tr}(u)(x_1, 0)$.

- (iii) Prove that the constant 2π in (ii) is sharp. Hint: Use polar coordinates.

To characterize the traces of Sobolev functions in $W^{1,p}(\Omega)$ when $p > 1$, we need to extend the definition of Besov spaces (see Chapter 17) to $\partial\Omega$.

Definition 18.32. Let $\Omega \subset \mathbb{R}^N$ be as in (18.13). Given $1 \leq p < \infty$ and $0 < s < 1$, for every function $u \in L^1_{\text{loc}}(\partial\Omega, \mathcal{H}^{N-1})$ we set

$$|u|_{B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})} := \left(\int_{\partial\Omega} \int_{\partial\Omega} \frac{|u(x) - u(y)|^p}{\|x - y\|^{N-1+sp}} d\mathcal{H}^{N-1}(y) d\mathcal{H}^{N-1}(x) \right)^{1/p}$$

and we say that u belongs to the *Besov space* $B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})$ if

$$(18.37) \quad \|u\|_{B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})} := \|u\|_{L^p(\partial\Omega)} + |u|_{B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})} < \infty.$$

In what follows we use the abbreviations

$$(18.38) \quad B^{s,p}(\partial\Omega) := B^{s,p}(\partial\Omega, \mathcal{H}^{N-1}),$$

$$|\cdot|_{B^{s,p}(\partial\Omega)} := |\cdot|_{B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})}, \quad \|\cdot\|_{B^{s,p}(\partial\Omega)} := \|\cdot\|_{B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})}.$$

Exercise 18.33. Given $1 \leq p < \infty$, $0 < s < 1$, and Ω as in (18.13), prove that

$$(1 + \text{Lip } f)^{-((N+1)/p+s)} |w|_{B^{s,p}(\mathbb{R}^{N-1})} \leq |u|_{B^{s,p}(\partial\Omega)} \leq c(1 + \text{Lip } f)^{2/p} |w|_{B^{s,p}(\mathbb{R}^{N-1})},$$

where $w(x') := u(x', f(x'))$, $x' \in \mathbb{R}^{N-1}$.

Theorem 18.34. Let $f : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$ be a Lipschitz continuous function, $N \geq 2$, let

$$\Omega := \{(x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : x_N > f(x')\},$$

and let $1 < p < \infty$. Then there exists a constant $c = c(p, N) > 0$ such that

$$\|\text{Tr}(u)\|_{L^p(\partial\Omega)} \leq c(1 + \text{Lip } f)^{1/p} \|u\|_{W^{1,p}(\Omega)},$$

$$|\text{Tr}(u)|_{B^{1-1/p,p}(\partial\Omega)} \leq c(1 + \text{Lip } f)^{1+2/p} \|u\|_{W^{1,p}(\Omega)}$$

for all $u \in W^{1,p}(\Omega)$. Conversely, for every $g \in B^{1-1/p,p}(\partial\Omega)$ there exists a function $u \in W^{1,p}(\mathbb{R}_+^N)$ such that $\text{Tr}(u) = g$ and

$$\|u\|_{L^p(\Omega)} \leq c\|g\|_{L^p(\partial\Omega)},$$

$$\|\nabla u\|_{L^p(\Omega)} \leq (1 + \text{Lip } f)\|g\|_{L^p(\partial\Omega)} + c(1 + \text{Lip } f)^{2+N/p} |g|_{B^{1-1/p,p}(\partial\Omega)},$$

where $c = c(N, p) > 0$.

Proof. Step 1: If $u \in \dot{W}^{1,p}(\Omega)$, by Theorems 9.52 and 11.53 the function

$$w(y) := u(\Psi(y)) = u(y', y_N + f(y')), \quad y \in \mathbb{R}_+^N$$

belongs to $\dot{W}^{1,p}(\mathbb{R}_+^N)$ with

$$(18.39) \quad \frac{\partial w}{\partial y_i}(y) = \frac{\partial u}{\partial x_i}(y', y_N + f(y')) + \frac{\partial u}{\partial x_N}(y', y_N + f(y')) \frac{\partial f}{\partial x_i}(y'),$$

$$\frac{\partial w}{\partial y_N}(y) = \frac{\partial u}{\partial x_N}(y', y_N + f(y')).$$

Hence, by Theorem 18.27, Exercise 18.33, (18.39), and again Theorem 11.53,

$$\begin{aligned} |\operatorname{Tr}(u)|_{B^{1-1/p,p}(\partial\Omega)} &\leq (1 + \operatorname{Lip} f)^{2/p} |\operatorname{Tr}(w)|_{B^{1-1/p,p}(\mathbb{R}^{N-1})} \\ &\leq c(1 + \operatorname{Lip} f)^{2/p} \|\nabla_y w\|_{L^p(\mathbb{R}_+^N)} \\ &\leq c(1 + \operatorname{Lip} f)^{1+2/p} \|\nabla u\|_{L^p(\Omega)}, \end{aligned}$$

where $c = c(N, p)$. Moreover, if $u \in W^{1,p}(\Omega)$, then by (9.47), Remark 18.25, (18.39), and again Theorem 11.53,

$$\begin{aligned} \int_{\partial\Omega} |\operatorname{Tr}(u)(x')|^p d\mathcal{H}^{N-1} &\leq (1 + \operatorname{Lip} f) \int_{\mathbb{R}^{N-1}} |\operatorname{Tr}(w)(y')|^p dy' \\ &\leq p(1 + \operatorname{Lip} f) \int_{\mathbb{R}_+^N} |w(y)|^p dy + p(1 + \operatorname{Lip} f) \int_{\mathbb{R}_+^N} \left| \frac{\partial w}{\partial y_N}(y) \right|^p dy \\ &= p(1 + \operatorname{Lip} f) \int_{\Omega} |u(x)|^p dx + p(1 + \operatorname{Lip} f) \int_{\Omega} \left| \frac{\partial u}{\partial x_N}(x) \right|^p dx. \end{aligned}$$

Step 2: Conversely, given $g \in L_{\operatorname{loc}}^1(\partial\Omega)$ with $|g|_{B^{1-1/p,p}(\partial\Omega)} < \infty$, setting $h(x') := g(x', f(x'))$, by Theorem 18.28 there exists a function $w \in \dot{W}^{1,p}(\mathbb{R}_+^N)$ such that $\operatorname{Tr}(v) = h$ and

$$\begin{aligned} \|\nabla_y w\|_{L^p(\mathbb{R}_+^N)} &\leq c|h|_{B^{1-1/p,p}(\mathbb{R}^{N-1})} \\ &\leq c(1 + \operatorname{Lip} f)^{1+N/p} |g|_{B^{1-1/p,p}(\partial\Omega)}, \end{aligned}$$

where $c = c(N, p) > 0$ and in the last inequality we used Exercise 18.33. Setting $u(x) := w(x', x_N - f(x'))$, $x \in \Omega$, reasoning somewhat as in the first part of the proof we have that $u \in \dot{W}^{1,p}(\Omega)$, with

$$\begin{aligned} (18.40) \quad \|\nabla u\|_{L^p(\Omega)} &\leq c(1 + \operatorname{Lip} f) \|\nabla_y w\|_{L^p(\mathbb{R}_+^N)} \\ &\leq c(1 + \operatorname{Lip} f)^{2+N/p} |g|_{B^{1-1/p,p}(\partial\Omega)}. \end{aligned}$$

Moreover, by Corollary 18.29, if $g \in L^p(\partial\Omega)$, then we can assume that $w \in W^{1,p}(\mathbb{R}_+^N)$ with $\operatorname{supp} w \subseteq \mathbb{R}^{N-1} \times [-\varepsilon, \varepsilon]$ and

$$\begin{aligned} \|w\|_{L^p(\mathbb{R}_+^N)} &\leq c\varepsilon^{1/p} \|h\|_{L^p(\mathbb{R}^{N-1})} \leq c\varepsilon^{1/p} \|g\|_{L^p(\partial\Omega)}, \\ |w|_{W^{1,p}(\mathbb{R}_+^N)} &\leq c\varepsilon^{-1/p'} \|h\|_{L^p(\mathbb{R}^{N-1})} + c|h|_{B^{1-1/p,p}(\mathbb{R}^{N-1})} \\ &\leq c\varepsilon^{-1/p'} \|g\|_{L^p(\partial\Omega)} + c(1 + \operatorname{Lip} f)^{1+N/p} |g|_{B^{1-1/p,p}(\partial\Omega)}, \end{aligned}$$

where we used the fact that $1 \leq \sqrt{1 + \|\nabla_{x'} f(x')\|_{N-1}^2}$ (see (9.47)) and Exercise 18.33. In turn, by Theorem 11.53,

$$(18.41) \quad \|u\|_{L^p(\Omega)} = \|w\|_{L^p(\mathbb{R}_+^N)} \leq c\varepsilon^{1/p} \|g\|_{L^p(\partial\Omega)},$$

while as in (18.40),

$$(18.42) \quad |u|_{W^{1,p}(\Omega)} \leq c(1 + \text{Lip } f)|w|_{W^{1,p}(\mathbb{R}_+^N)} \leq c(1 + \text{Lip } f)\varepsilon^{-1/p'}\|g\|_{L^p(\partial\Omega)} \\ + c(1 + \text{Lip } f)^{2+N/p}|g|_{B^{1-1/p,p}(\partial\Omega)}.$$

This completes the proof. $\square$

Remark 18.35. Note that in the second part of the proof we have that

$$\text{supp } u \subseteq \{x \in \mathbb{R}^N : x' \in \text{supp } g, -\varepsilon + f(x') \leq x_N \leq f(x') + \varepsilon\}.$$

To extend the previous theorem to open sets with uniformly Lipschitz continuous boundary, we consider a different seminorm.

Definition 18.36. Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous. Given $1 \leq p < \infty$ and $0 < s < 1$, for every function $u \in L^1_{\text{loc}}(\partial\Omega)$ we set

$$|u|_{B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})}^\diamond := \left(\int_{\partial\Omega} \left(\int_{\partial\Omega \cap B(x, \varepsilon)} \frac{|u(x) - u(y)|^p}{\|x - y\|^{N-1+sp}} d\mathcal{H}^{N-1}(y) \right) d\mathcal{H}^{N-1}(x) \right)^{1/p}$$

and we say that u belongs to the *Besov space* $B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})$ if

$$(18.43) \quad \|u\|_{B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})} := \|u\|_{L^p(\partial\Omega)} + |u|_{B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})}^\diamond < \infty.$$

Here $\varepsilon > 0$ is the parameter given in Definition 13.11. In what follows we will use the abbreviations (18.38).

Exercise 18.37. Prove that if $\partial\Omega$ is bounded, then the norm (18.43) is equivalent to the norm (18.10).

Exercise 18.38. Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set of the form (18.13), let $1 \leq p < \infty$ and $0 < s < 1$. Prove that for every $\varepsilon > 0$ and every $u \in B^{s,p}(\partial\Omega)$,

$$|u|_{B^{s,p}(\partial\Omega)} \leq c(1 + \text{Lip } f)^{s+1/p}\varepsilon^{-s}\|u\|_{L^p(\partial\Omega)} + |u|_{B^{s,p}(\partial\Omega)}^\diamond,$$

where $c = c(N, p, s) > 0$.

Exercise 18.39. Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous, let $w_n : \partial\Omega \rightarrow \mathbb{R}$, $n \in \mathbb{N}$, be Lebesgue measurable functions, and let $w(x) := \sum_{n=1}^\infty w_n(x)$, $x \in \partial\Omega$, where we assume that there exists an integer $M \in \mathbb{N}$ such that for every $x \in \partial\Omega$ at most M terms $w_n(x)$ are nonzero. Prove that for every $1 \leq p < \infty$ and $0 < s < 1$,

$$\|w\|_{L^p(\partial\Omega)} \leq M^{1/p'} \left(\sum_{n=1}^\infty \|w_n\|_{L^p(\partial\Omega)}^p \right)^{1/p},$$

while

$$|w|_{B^{s,p}(\partial\Omega)}^\diamond \leq (2M)^{1/p'} \left(\sum_{n=1}^{\infty} (|w_n|_{B^{s,p}(\partial\Omega)}^\diamond)^p \right)^{1/p}.$$

Theorem 18.40. *Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous and let $1 < p < \infty$. Then*

$$\|\operatorname{Tr}(u)\|_{L^p(\partial\Omega)} \leq cM(1+L)^{1/p}(\varepsilon^{-1}\|u\|_{L^p(\Omega)} + \|\nabla u\|_{L^p(\Omega;\mathbb{R}^N)}),$$

$$|\operatorname{Tr}(u)|_{B^{1-1/p,p}(\partial\Omega)}^\diamond \leq cM(1+L)^{1+2/p}\varepsilon^{-1}(\varepsilon^{-1}\|u\|_{L^p(\Omega)} + \|\nabla u\|_{L^p(\Omega;\mathbb{R}^N)})$$

for all $u \in W^{1,p}(\Omega)$, where $c = c(N, p) > 0$ and $\varepsilon, L > 0$, and $M \in \mathbb{N}$ are given in Definition 13.11.

Moreover, for every $g \in B^{1-1/p,p}(\partial\Omega)$ there exist a constant $c = c(N, p)$ and a function $u \in W^{1,p}(\Omega)$ such that $\operatorname{Tr}(u) = g$,

$$\|u\|_{L^p(\Omega)} \leq M^{1/p'} \varepsilon^{1/p} \|g\|_{L^p(\partial\Omega)}$$

and

$$\begin{aligned} \|\nabla u\|_{L^p(\Omega;\mathbb{R}^N)} &\leq cM(1+L)^{3+N/p}\varepsilon^{-1/p'} \|g\|_{L^p(\partial\Omega)} \\ &\quad + cM(1+L)^{2+(N+1)/p} |g|_{B^{1-1/p,p}(\partial\Omega)}^\diamond. \end{aligned}$$

We begin with a preliminary result. Define $\phi_n, \Omega_0, \Omega_\pm, \phi_0, \phi_\pm, \psi_\pm$ as in (13.43), (13.46), (13.47), (13.50), respectively, and let ψ_n be as in (18.19).

Lemma 18.41. *Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous, let $1 \leq p < \infty$, and let $0 < s < 1$. Then there exists a constant $c = c(N, p, s) > 0$ such that for every function $u \in B^{s,p}(\partial\Omega)$,*

$$\begin{aligned} (18.44) \quad |u|_{B^{s,p}(\partial\Omega)}^\diamond &\leq cM^{1/p'}(1+L)^{1/p}\varepsilon^{-1/p-s} \left(\sum_n \|\psi_n u\|_{L^p(\partial V_n)}^p \right)^{1/p} \\ &\quad + cM^{1/p'} \left(\sum_n (|\psi_n u|_{B^{s,p}(\partial V_n)}^\diamond)^p \right)^{1/p}, \end{aligned}$$

while

$$\begin{aligned} (18.45) \quad &\left(\sum_n (|\psi_n u|_{B^{s,p}(\partial V_n)}^\diamond)^p \right)^{1/p} \\ &\leq cM^{1/p}(1+L)^{(1-s-(N-1)/p)^++1/p}\varepsilon^{-s} \|u\|_{L^p(\partial\Omega)} \\ &\quad + cM^{1/p} |u|_{B^{s,p}(\partial\Omega)}^\diamond. \end{aligned}$$

Proof. Step 1: We begin by proving that

$$|\psi_n u|_{B^{s,p}(\partial\Omega)}^\diamond \leq c(1+L)^{1/p}\varepsilon^{-1/p-s} \|\psi_n u\|_{L^p(\partial\Omega \cap \partial V_n)} + c|\psi_n u|_{B^{s,p}(\partial V_n)}^\diamond,$$

$$|\psi_n u|_{B^{s,p}(\partial V_n)}^\diamond \leq c(1+L)^{1/p}\varepsilon^{-1/p-s} \|\psi_n u\|_{L^p(\partial\Omega \cap \partial V_n)} + c|\psi_n u|_{B^{s,p}(\partial\Omega)}^\diamond,$$

where we recall that $\Omega_n \cap \Omega = \Omega_n \cap V_n$, where V_n is (up to a rigid motion) a special Lipschitz domain. It is enough to show the first inequality, since the second is similar. If $x, y \notin \Omega_n$, then $(\psi_n u)(x) = (\psi_n u)(y) = 0$ since the support of ψ_n is contained in Ω_n , while if $x, y \in \Omega_n$, then $\Omega_n \cap \partial\Omega = \Omega_n \cap \partial V_n$. Hence, by interchanging x and y if needed, and using the fact that $\chi_{B(x,\varepsilon)}(y) = \chi_{B(y,\varepsilon)}(x)$, we can write

$$\begin{aligned} & \int_{\partial\Omega} \int_{\partial\Omega \cap B(x,\varepsilon)} \frac{|(\psi_n u)(x) - (\psi_n u)(y)|^p}{\|x - y\|^{N-1+sp}} d\mathcal{H}^{N-1}(y) d\mathcal{H}^{N-1}(x) \\ &= \int_{\partial V_n \cap \Omega_n} \int_{\partial V_n \cap \Omega_n \cap B(x,\varepsilon)} \frac{|(\psi_n u)(x) - (\psi_n u)(y)|^p}{\|x - y\|^{N-1+sp}} d\mathcal{H}^{N-1}(y) d\mathcal{H}^{N-1}(x) \\ & \quad + 2 \int_{\partial\Omega \cap \Omega_n} |(\psi_n u)(x)|^p \int_{(\partial\Omega \setminus \Omega_n) \cap B(x,\varepsilon)} \frac{d\mathcal{H}^{N-1}(y)}{\|x - y\|^{N-1+sp}} d\mathcal{H}^{N-1}(x) \\ &=: I + II. \end{aligned}$$

It remains to estimate II . Note that we can replace $\int_{\partial\Omega \cap \Omega_n}$ with $\int_{\partial\Omega \cap \{\psi_n > 0\}}$. We recall that $\phi_n := \varphi_{\varepsilon/4} * \chi_{\Omega_n^{3\varepsilon/4}}$ (see (13.43)), where $\varphi_{\varepsilon/4}$ is a standard mollifier and $\Omega_n^{3\varepsilon/4} = \{x \in \mathbb{R}^N : B(x, 3\varepsilon/4) \subseteq \Omega_n\}$. Hence, if we assume that $\text{supp } \varphi \subseteq \overline{B(0, 1/2)}$, for every $x \in \partial\Omega \cap \{\psi_n > 0\}$, we have that $\text{dist}(x, \partial\Omega_n) \geq 3\varepsilon/8$. On the other hand, if $y \in (\partial\Omega \setminus \Omega_n) \cap B(x, \varepsilon)$, we have that

$$\|x - y\| \geq 3\varepsilon/8.$$

Moreover, by Definition 13.11(i), there exists k such that $B(x, \varepsilon) \subseteq \Omega_k$. Hence, up to a rotation, we can write

$$\begin{aligned} (18.46) \quad & \int_{(\partial\Omega \setminus \Omega_n) \cap B(x,\varepsilon)} \frac{d\mathcal{H}^{N-1}(y)}{\|x - y\|^{N-1+sp}} \\ & \leq c\varepsilon^{-N+1-sp} \int_{B_{N-1}(x', \varepsilon)} \sqrt{1 + \|\nabla_{y'} f_k(y')\|_{N-1}^2} dy' \\ & \leq c(1 + L)\varepsilon^{-1-sp}. \end{aligned}$$

Thus,

$$II \leq c(1 + L)\varepsilon^{-1-sp} \int_{\partial\Omega \cap \Omega_n} |(\psi_n u)(x)|^p d\mathcal{H}^{N-1}(x).$$

Step 2: Since $u = \sum_n \psi_n u$, by Exercise 18.39 and Step 1,

$$\begin{aligned} |u|_{B^{s,p}(\partial\Omega)}^\diamond &\leq (2M)^{1/p'} \left(\sum_{n=1}^{\infty} (|\psi_n u|_{B^{s,p}(\partial\Omega)}^\diamond)^p \right)^{1/p} \\ &\leq cM^{1/p'} (1+L)^{1/p} \varepsilon^{-1/p-s} \left(\sum_n \|\psi_n u\|_{L^p(\partial V_n)}^p \right)^{1/p} \\ &\quad + cM^{1/p'} \left(\sum_n (|\psi_n u|_{B^{s,p}(\partial V_n)}^\diamond)^p \right)^{1/p}, \end{aligned}$$

which proves (18.44).

Step 3: To prove (18.45), we first apply Step 1 to obtain

$$\begin{aligned} \left(\sum_n (|\psi_n u|_{B^{s,p}(\partial V_n)}^\diamond)^p \right)^{1/p} &\leq c(1+L)^{1/p} \varepsilon^{-1/p-s} \left(\sum_n \|\psi_n u\|_{L^p(\partial\Omega \cap \partial V_n)}^p \right)^{1/p} \\ &\quad + c \left(\sum_n (|\psi_n u|_{B^{s,p}(\partial\Omega)}^\diamond)^p \right)^{1/p}. \end{aligned}$$

We now estimate the second term on the right-hand side of the previous inequality. For $x \in \partial\Omega$ and $y \in \partial\Omega \cap B(x, \varepsilon)$ we have

$$\begin{aligned} |(\psi_n u)(x) - (\psi_n u)(y)|^p &\leq 2^{p-1} (\psi_n(y))^p |u(x) - u(y)|^p \\ &\quad + 2^{p-1} |u(x)|^p |\psi_n(x) - \psi_n(y)|^p \leq 2^{p-1} (\psi_n(y))^p |u(x) - u(y)|^p \\ &\quad + 2^{p-1} |u(x)|^p \|x - y\|^p \int_0^1 \|\nabla \psi_n(tx + (1-t)y)\|^p dt, \end{aligned}$$

where in the last term we used the fundamental theorem of calculus and Hölder's inequality. By (13.56),

$$(18.47) \quad \sum_n |\psi_n|^p \leq M, \quad \sum_n \|\nabla \psi_n\|^p \leq cM\varepsilon^{-p}.$$

Hence,

$$\begin{aligned} \sum_n |(\psi_n u)(x) - (\psi_n u)(y)|^p &\leq 2^{p-1} M |u(x) - u(y)|^p \\ &\quad + cM\varepsilon^{-p} |u(x)|^p \|x - y\|^p. \end{aligned}$$

In turn,

$$\begin{aligned} \sum_n (|\psi_n u|_{B^{s,p}(\partial\Omega)}^\diamond)^p &\leq cM (|u|_{B^{s,p}(\partial\Omega)}^\diamond)^p \\ &\quad + cM\varepsilon^{-p} \int_{\partial\Omega} |u(x)|^p \int_{\partial\Omega \cap B(x, \varepsilon)} \|x - y\|^{p(1-s)-N+1} d\mathcal{H}^{N-1}(y) d\mathcal{H}^{N-1}(x). \end{aligned}$$

By Definition 13.11(i), if $x \in \partial\Omega$ there exists k such that $B(x, \varepsilon) \subseteq \Omega_k$. Hence, up to a rotation, we can write

$$\begin{aligned} & \int_{\partial\Omega \cap B(x, \varepsilon)} \|x - y\|^{p(1-s)-N+1} d\mathcal{H}^{N-1}(y) \\ & \leq (1+L)^{(p(1-s)-(N-1))^+} \int_{B_{N-1}(x', \varepsilon)} \|x' - y'\|_{N-1}^{p(1-s)-(N-1)} J_k(y') dy' \\ & \leq c(1+L)^{(p(1-s)-(N-1))^++1} \varepsilon^{p(1-s)}, \end{aligned}$$

where $J_k(y') := \sqrt{1 + \|\nabla_{y'} f_k(y')\|_{N-1}^2}$ and we used the fact that $\|x' - y'\|_{N-1} \leq \|x - y\| \leq (1+L)\|x' - y'\|_{N-1}$. Combining the last two inequalities gives (18.45). $\square$

We turn to the proof of Theorem 18.40.

Proof of Theorem 18.40. The proof is similar to the one of Theorem 18.18 and we indicate only the main changes.

Step 1: Given $u \in W^{1,p}(\Omega)$, as in Theorem 18.18 we write u as in (18.21) and study the functions u_n . Since the support of ψ_n is contained in Ω_n and $\Omega_n \cap \Omega = \Omega_n \cap V_n$, where V_n is (up to a rigid motion) a special Lipschitz domain, by extending u_n to be zero in $V_n \setminus \Omega_n$, we are in a position to apply Theorem 18.34 to find that

$$\begin{aligned} (18.48) \quad & \|\text{Tr}(u_n)\|_{L^p(\partial V_n)} \leq c(1+L)^{1/p} \|u_n\|_{W^{1,p}(V_n)}, \\ & |\text{Tr}(u_n)|_{B^{1-1/p,p}(\partial V_n)}^\diamond \leq c(1+L)^{1+2/p} \|u_n\|_{W^{1,p}(V_n)}. \end{aligned}$$

Since $\psi_+ = 1$ on $\partial V_n \cap \partial\Omega$ and $\psi_n = 0$ on $\partial V_n \setminus \partial\Omega$, we have that $\text{Tr}(u_n) = \text{Tr}(\psi_n u)$ on ∂V_n , and so by (18.44) and (18.48),

$$\begin{aligned} |\text{Tr}(u)|_{B^{1-1/p,p}(\partial\Omega)}^\diamond & \leq cM^{1/p'} \varepsilon^{-1} (1+L)^{1/p} \left(\sum_n \|\text{Tr}(u_n)\|_{L^p(\partial V_n)}^p \right)^{1/p} \\ & \quad + cM^{1/p'} \left(\sum_n (|\text{Tr}(u_n)|_{B^{1-1/p,p}(\partial V_n)}^\diamond)^p \right)^{1/p} \\ & \leq cM^{1/p'} (1+L)^{1+2/p} \varepsilon^{-1} \left(\sum_n \|u_n\|_{W^{1,p}(\Omega_n \cap \Omega)}^p \right)^{1/p}. \end{aligned}$$

By (18.24), (18.47), the following pointwise inequalities hold in Ω ,

$$\begin{aligned} \sum_n \psi_n^p |u|^p & \leq M |u|^p, \\ \sum_n \|\nabla u_n\|^p & \leq 2^{p-1} \sum_n \|\nabla \psi_n\|^p |u|^p + 2^{p-1} \sum_n \psi_n^p \|\nabla u\|^p \\ & \leq cM \varepsilon^{-p} |u|^p + cM \|\nabla u\|^p, \end{aligned}$$

which, combined with the previous inequality, give

$$\begin{aligned} |\operatorname{Tr}(u)|_{B^{1-1/p,p}(\partial\Omega)}^\diamond &\leq cM(1+L)^{1+2/p}\varepsilon^{-2}\|u\|_{L^p(\Omega)} \\ &\quad + cM(1+L)^{1+2/p}\varepsilon^{-1}\|\nabla u\|_{L^p(\Omega;\mathbb{R}^N)}. \end{aligned}$$

Similarly, by Exercise 18.39 and (18.48),

$$\begin{aligned} \|\operatorname{Tr}(u)\|_{L^p(\partial\Omega)} &\leq M^{1/p'} \left(\sum_{n=1}^{\infty} \|\operatorname{Tr}(u_n)\|_{L^p(\partial\Omega)}^p \right)^{1/p} \\ &\leq M^{1/p'} (1+L)^{1/p} \left(\sum_n \|u_n\|_{W^{1,p}(\Omega_n \cap \Omega)}^p \right)^{1/p} \\ &\leq cM(1+L)^{1/p} (\varepsilon^{-1}\|u\|_{L^p(\Omega)} + \|\nabla u\|_{L^p(\Omega;\mathbb{R}^N)}). \end{aligned}$$

Step 2: Given $g \in B^{1-1/p,p}(\partial\Omega)$, since $\psi_+ = 1$ on $\partial\Omega$, we can write $g = \sum_n \psi_n g$. By extending $\psi_n g$ to be zero on $\partial V_n \setminus \partial\Omega_n$, we are in a position to apply Theorem 18.34 (see (18.41) and (18.42)) and Remark 18.35 to find a function $u_n \in W^{1,p}(V_n)$ such that $\operatorname{Tr}(u_n) = \psi_n g$ on ∂V_n , $\operatorname{supp} u_n \subset \Omega_n$, $\|u_n\|_{L^p(V_n)} \leq \varepsilon^{1/p} \|\psi_n g\|_{L^p(\partial V_n)}$, and

$$\begin{aligned} \|u_n\|_{W^{1,p}(V_n)} &\leq c(1+L)\varepsilon^{-1/p'} \|\psi_n g\|_{L^p(\partial V_n)} \\ &\quad + c(1+L)^{2+N/p} \|\psi_n g\|_{B^{1-1/p,p}(\partial V_n)} \\ &\leq c(1+L)^{3+N/p} \varepsilon^{-1/p'} \|\psi_n g\|_{L^p(\partial V_n)} \\ &\quad + c(1+L)^{2+N/p} \|\psi_n g\|_{B^{1-1/p,p}(\partial V_n)}^\diamond, \end{aligned}$$

where $c = c(N, p) > 0$ and in the second inequality we used Exercise 18.38.

Let $u := \sum_n u_n$. Since $\{\Omega_n\}_n$ is locally finite we have that $u \in W_{\text{loc}}^{1,p}(\Omega)$ and $\operatorname{Tr}(u) = g$. By Exercise 12.34 and the facts that $\operatorname{supp} u_n \subset \Omega_n$, $\Omega_n \cap \Omega = \Omega_n \cap V_n$, $\psi_n = 0$ on $\partial V_n \setminus \partial\Omega$, and $\sum_{n=1}^{\infty} \psi_n = 1$ on $\partial\Omega$,

$$\begin{aligned} \|u\|_{L^p(\Omega)} &\leq M^{1/p'} \left(\sum_{n=1}^{\infty} \|u_n\|_{L^p(\Omega)}^p \right)^{1/p} \\ &\leq M^{1/p'} \varepsilon^{1/p} \left(\sum_{n=1}^{\infty} \|\psi_n g\|_{L^p(\partial\Omega)}^p \right)^{1/p} = M^{1/p'} \varepsilon^{1/p} \|g\|_{L^p(\partial\Omega)}. \end{aligned}$$

Similarly,

$$\begin{aligned}
|u|_{W^{1,p}(\Omega)} &\leq M^{1/p'} \left(\sum_{n=1}^{\infty} |u_n|_{W^{1,p}(\Omega)}^p \right)^{1/p} \leq M^{1/p'} \left(\sum_{n=1}^{\infty} |u_n|_{W^{1,p}(V_n)}^p \right)^{1/p} \\
&\leq cM^{1/p'} (1+L)^{3+N/p} \varepsilon^{-1/p'} \left(\sum_{n=1}^{\infty} \|\psi_n g\|_{L^p(\partial V_n)}^p \right)^{1/p} \\
&\quad + M^{1/p'} (1+L)^{2+N/p} \left(\sum_{n=1}^{\infty} (|\psi_n g|_{B^{1-1/p,p}(\partial V_n)})^p \right)^{1/p} \\
&\leq cM(1+L)^{3+N/p} \varepsilon^{-1/p'} \|g\|_{L^p(\partial\Omega)} \\
&\quad + cM(1+L)^{2+(N+1)/p} |g|_{B^{1-1/p,p}(\partial\Omega)},
\end{aligned}$$

where the last inequality follows from (18.45) and the fact that $-(N-2)/p)^+ = 0$. $\square$

As in Remark 18.19, if $\Omega \subseteq \mathbb{R}^N$ is Lipschitz continuous but $\partial\Omega$ is unbounded, then the results in the previous theorem continue to hold locally.

Exercise 18.42. Let $\Omega \subset \mathbb{R}^N$ be an open set whose boundary $\partial\Omega$ is uniformly Lipschitz continuous and let $u \in W^{1,p}(\Omega)$ and $v \in W^{1,p}(\mathbb{R}^N \setminus \overline{\Omega})$, $1 < p < \infty$. Prove that the function

$$w(x) := \begin{cases} u(x) & \text{if } x \in \Omega, \\ v(x) & \text{if } x \in \mathbb{R}^N \setminus \overline{\Omega} \end{cases}$$

belongs to $W^{1,p}(\mathbb{R}^N)$ if and only if $\text{Tr}(u) = \text{Tr}(v)$.

18.5. Traces of Functions in $W^{m,1}(\Omega)$

In this section given an integer $m \in \mathbb{N}$, $m \geq 2$, we characterize the image of the trace operator Tr_m in (18.8) when $p = 1$.

Theorem 18.43. *Let $N \geq 2$ and let $m \in \mathbb{N}$ with $m \geq 2$. Then the image of the linear mapping (18.8) is given by*

$$\text{Tr}_m(W^{m,1}(\mathbb{R}_+^N)) = \prod_{k=0}^{m-2} B^{m-k-1,1}(\mathbb{R}^{N-1}) \times L^1(\mathbb{R}^{N-1}).$$

Proof. We only prove the case $m = 2$.

Step 1: Given $u \in W^{2,1}(\mathbb{R}_+^N) \cap C^\infty(\mathbb{R}^{N-1} \times [0, \infty))$, by Theorem 18.57 below we have that

$$|\text{Tr}(u)(\cdot, 0)|_{B^{1,1}(\mathbb{R}^{N-1})} \leq c \|\nabla^2 u\|_{L^1(\mathbb{R}_+^N)}.$$

The general case follows by density (see Theorem 11.35) and is left as an exercise. Moreover, by Theorem 18.9,

$$\|\operatorname{Tr}(u)(\cdot, 0)\|_{L^1(\mathbb{R}^{N-1})} \leq \|\partial_N u\|_{L^1(\mathbb{R}_+^N)}.$$

Thus, $\operatorname{Tr}(W^{2,1}(\mathbb{R}_+^N)) \subseteq B^{1,1}(\mathbb{R}^{N-1})$.

To prove the opposite inclusion, let $g \in \mathcal{S}(\mathbb{R}^{N-1})$ and let $\varphi \in S(\mathbb{R}^{N-1})$ be a radial function such that $\operatorname{supp} \widehat{\varphi} \subseteq B_{N-1}(0, 1)$ and $\widehat{\varphi} = 1$ in $B_{N-1}(0, 1/2)$, where $\widehat{\varphi}$ is the Fourier transform of φ . Then by Theorems 17.77 and 18.53 below the function

$$v(x) := \frac{1}{x_N^N} \int_{\mathbb{R}^N} \varphi\left(\frac{x' - y'}{x_N}\right) g(y') dy', \quad x \in \mathbb{R}_+^N,$$

satisfies $v(x', 0) = g(x')$ for all $x' \in \mathbb{R}^{N-1}$, and

$$(18.49) \quad \|\nabla^2 v\|_{L^1(\mathbb{R}_+^N)} \leq c|g|_{B^{1,1}(\mathbb{R}^{N-1})}.$$

Moreover, (18.35) continues to hold with $p = 1$. Hence, by Theorem 13.51 with $p = q = 1$ (see (13.98)),

$$\begin{aligned} \|\nabla v\|_{L^1(R)} &\leq c\varepsilon^{-1} \|v\|_{L^1(R)} + c\|v\|_{L^1(R)}^{1/2} \|\nabla^2 v\|_{L^1(R)}^{1/2} \\ &\leq c\varepsilon^{-1} \|v\|_{L^1(R)} + c\varepsilon \|\nabla^2 v\|_{L^1(R)}, \end{aligned}$$

where $R = I_1 \times \cdots \times I_N$ is a rectangle with $\varepsilon := \min_i \mathcal{L}^1(I_i)$ and we used Young's inequality. By the Lebesgue monotone convergence theorem, letting $\mathcal{L}^1(I_i) \rightarrow \infty$ we get

$$\begin{aligned} (18.50) \quad \|\nabla v\|_{L^1(\mathbb{R}^{N-1} \times (0, \varepsilon))} &\leq c\varepsilon^{-1} \|v\|_{L^1(\mathbb{R}^{N-1} \times (0, \varepsilon))} + c\varepsilon \|\nabla^2 v\|_{L^1(\mathbb{R}_+^N)} \\ &\leq c\|g\|_{L^1(\mathbb{R}^{N-1})} + c\varepsilon |g|_{B^{1,1}(\mathbb{R}^{N-1})}, \end{aligned}$$

where the last inequality follows from (18.35) and (18.49).

Let $\psi \in C^\infty([0, \infty))$ be a decreasing function such that $\psi = 1$ in $[0, \varepsilon/2]$, $\psi(x_N) = 0$ for $x_N \geq \varepsilon$, $\|\psi'\|_\infty \leq c\varepsilon^{-1}$ and $\|\psi''\|_\infty \leq c\varepsilon^{-2}$ for some constant $c > 0$. For $x = (x', x_N) \in \mathbb{R}_+^N$ we define $u(x) := \psi(x_N)v(x)$. Then for $i, j = 1, \dots, N-1$,

$$\begin{aligned} \partial_i u(x) &= \psi(x_N) \partial_i v(x), \quad \partial_N u(x) = \psi(x_N) \partial_N v(x) + \psi'(x_N) v(x), \\ \partial_{i,j}^2 u(x) &= \psi(x_N) \partial_{i,j}^2 v(x), \quad \partial_{i,N}^2 u(x) = \psi(x_N) \partial_{i,N}^2 v(x) + \psi'(x_N) \partial_i v(x), \\ \partial_N^2 u(x) &= \psi(x_N) \partial_N^2 v(x) + 2\psi'(x_N) \partial_N v(x) + \psi''(x_N) v(x). \end{aligned}$$

Hence, also by (18.35), (18.36), (18.50), and the properties of ψ , we have that

$$\|\nabla u\|_{L^1(\mathbb{R}_+^N)} \leq c\|g\|_{L^1(\mathbb{R}^{N-1})} + c\varepsilon |g|_{B^{1,1}(\mathbb{R}^{N-1})},$$

while

$$\|\nabla^2 u\|_{L^1(\mathbb{R}_+^N)} \leq c\varepsilon^{-1} \|g\|_{L^1(\mathbb{R}^{N-1})} + c|g|_{B^{1,1}(\mathbb{R}^{N-1})}.$$

The general case follows by density.

Step 2: We prove that for every $h \in L^1(\mathbb{R}^{N-1})$ there exists a function $w \in W^{2,1}(\mathbb{R}_+^N)$ such that $\text{Tr}(w) = 0$ and $\text{Tr}(\partial_N w) = h$. Consider a sequence $\{h_n\}_n$ of functions in $C_c^2(\mathbb{R}^{N-1})$ such that $h_n \rightarrow h$ in $L^1(\mathbb{R}^{N-1})$. Since $\{h_n\}_n$ is a Cauchy sequence in $L^1(\mathbb{R}^{N-1})$, for each $k \in \mathbb{N}$ there exists N_k such that for all $n \geq N_k$ and $\ell \geq 1$,

$$\|h_{n+\ell} - h_n\|_{L^1(\mathbb{R}^{N-1})} \leq 2^{-k} \|h\|_{L^1(\mathbb{R}^{N-1})}.$$

Let $n_k := \max\{N_k, N_{k-1} + 1\}$. The subsequence $\{h_{n_k}\}_k$ has the property

$$\sum_{k=1}^{\infty} \|h_{n_{k+1}} - h_{n_k}\|_{L^1(\mathbb{R}^{N-1})} \leq 2 \|h\|_{L^1(\mathbb{R}^{N-1})}.$$

Without loss of generality, we assume that the entire sequence $\{h_n\}_n$ satisfies

$$(18.51) \quad \sum_{n=1}^{\infty} \|h_{n+1} - h_n\|_{L^1(\mathbb{R}^{N-1})} \leq 2 \|h\|_{L^1(\mathbb{R}^{N-1})}.$$

Define $h_0 := 0$ and construct a strictly decreasing sequence $\{t_n\}_n$ in $(0, 1)$ such that $t_n \rightarrow 0^+$ and

$$(18.52) \quad |t_n - t_{n+1}| \leq \frac{1}{2^n} \frac{\|h\|_{L^1(\mathbb{R}^{N-1})}}{\|h_{n+1}\|_{W^{2,1}(\mathbb{R}^{N-1})} + \|h_n\|_{W^{2,1}(\mathbb{R}^{N-1})} + 1}.$$

Note that $t_0 = \sum_{n=0}^{\infty} (t_n - t_{n+1})$. For $x' \in \mathbb{R}^{N-1}$ and $x_N \geq 0$ define

$$\omega(x', x_N) := \begin{cases} 0 & \text{if } x_N \geq t_0, \\ \frac{t_n - x_N}{t_n - t_{n+1}} h_{n+1}(x') + \frac{x_N - t_{n+1}}{t_n - t_{n+1}} h_n(x') & \text{if } t_{n+1} \leq x_N \leq t_n, \end{cases}$$

and

$$w(x', x_N) := \int_0^{x_N} \omega(x', s) ds.$$

Then

$$\lim_{x_N \rightarrow 0^+} \|\omega(\cdot, x_N) - h\|_{L^1(\mathbb{R}^{N-1})} = 0,$$

and so

$$w(x', 0) = 0, \quad \partial_N w(x', 0) = h(x').$$

Moreover, by Tonelli's theorem

$$\begin{aligned} \int_{\mathbb{R}_+^N} |\partial_N^2 w| dx &= \int_{\mathbb{R}_+^N} |\partial_N \omega| dx \\ &= \int_{\mathbb{R}^{N-1}} \sum_{n=0}^{\infty} \int_{t_{n+1}}^{t_n} \left| \frac{h_{n+1}(x') - h_n(x')}{t_n - t_{n+1}} \right| dx_N dx' \\ &\leq \sum_{n=0}^{\infty} \|h_{n+1} - h_n\|_{L^1(\mathbb{R}^{N-1})} \leq 2 \|h\|_{L^1(\mathbb{R}^{N-1})} \end{aligned}$$

by (18.51), while for $i = 1, \dots, N-1$,

$$\begin{aligned}
 \int_{\mathbb{R}_+^N} |\partial_{N,i}^2 w| dx &= \int_{\mathbb{R}_+^N} |\partial_i \omega| dx \\
 &= \int_{\mathbb{R}^{N-1}} \sum_{n=0}^{\infty} \int_{t_{n+1}}^{t_n} \left| \frac{t_n - x_N}{t_n - t_{n+1}} \partial_i h_{n+1}(x') + \frac{x_N - t_{n+1}}{t_n - t_{n+1}} \partial_i h_n(x') \right| dx_N dx' \\
 &\leq \sum_{n=0}^{\infty} (\|\nabla_{x'} h_{n+1}\|_{L^1(\mathbb{R}^{N-1})} + \|\nabla_{x'} h_n\|_{L^1(\mathbb{R}^{N-1})}) |t_{n+1} - t_n| \\
 &\leq 2\|h\|_{L^1(\mathbb{R}^{N-1})},
 \end{aligned}$$

by (18.52). Similarly, for $i, j = 1, \dots, N-1$,

$$\begin{aligned}
 \int_{\mathbb{R}_+^N} |\partial_{i,j}^2 w| dx' dx_N &= \int_{\mathbb{R}_+^N} \int_0^{x_N} |\partial_{i,j}^2 \omega| ds dx \\
 &\leq \int_{\mathbb{R}^{N-1}} \sum_{n=0}^{\infty} \int_{t_{n+1}}^{t_n} |\partial_{i,j}^2 h_{n+1}(x')| + |\partial_{i,j}^2 h_n(x')| dx_N dx' \\
 &\leq \sum_{n=0}^{\infty} (\|\nabla_{x'}^2 h_{n+1}\|_{L^1(\mathbb{R}^{N-1})} + \|\nabla_{x'}^2 h_n\|_{L^1(\mathbb{R}^{N-1})}) |t_{n+1} - t_n| \\
 &\leq 2\|h\|_{L^1(\mathbb{R}^{N-1})}
 \end{aligned}$$

by (18.52). Similar estimates hold for the first order derivatives of w and for w .

Step 3: To prove that

$$\mathrm{Tr}_2(W^{2,1}(\mathbb{R}_+^N)) = B^{1,1}(\mathbb{R}^{N-1}) \times L^1(\mathbb{R}^{N-1})$$

consider $g \in B^{1,1}(\mathbb{R}^{N-1})$ and $h \in L^1(\mathbb{R}^{N-1})$. By Step 1 there exists $v \in W^{2,1}(\mathbb{R}_+^N)$ such that $\mathrm{Tr}(v) = g$. Since $\partial_N v \in W^{1,1}(\mathbb{R}_+^N)$, we have that $\mathrm{Tr}(\partial_N v) \in L^1(\mathbb{R}^{N-1})$ by Theorem 18.9. In turn, by Step 2 we can find a function $w \in W^{2,1}(\mathbb{R}_+^N)$ such that $\mathrm{Tr}(w) = 0$ and $\mathrm{Tr}(\partial_N w) = h - \mathrm{Tr}(\partial_N v)$. Taking $u := v + w$, we have that $\mathrm{Tr}(u) = g$ and $\mathrm{Tr}(\partial_N u) = h$, as desired. $\square$

Exercise 18.44. Prove the case $m \geq 2$.

Remark 18.45. Step 1 of the previous proof actually shows that

$$\mathrm{Tr}(\dot{W}^{2,1}(\mathbb{R}_+^N)) = \dot{B}^{1,1}(\mathbb{R}^{N-1}).$$

Remark 18.46. By taking $t_0 \leq \varepsilon$ in Step 2 of the previous proof, we can assume that $\mathrm{supp} u \subseteq \mathbb{R}^{N-1} \times [-\varepsilon, \varepsilon]$ in Step 3.

Unlike the first order difference operator $\Delta_h u(x) = u(x+h) - u(x)$, there is no simple way (see the papers of Jonsson and Wallin [124], [125]) to define in an intrinsic way the second-order difference operator $\Delta_h^2 u(x) = u(x+2h) - 2u(x+h) + u(x)$ or the higher-order ones $\Delta_h^m u(x)$ for functions

defined on the boundary of special Lipschitz domains or uniformly Lipschitz domains. Thus, we first flatten the boundary and then use the definition on $\mathbb{R}^{N-1}$. However, this requires more regularity on the boundary. We recall Remark 13.12.

Definition 18.47. Given $1 \leq p < \infty$ and $s \geq 1$, let $m := \lfloor s \rfloor + 1$, let $f \in C^{m-1,1}(\mathbb{R}^{N-1})$, $N \geq 2$, and let

$$(18.53) \quad \Omega := \{(x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : x_N > f(x')\}.$$

For every function $u \in L^1_{\text{loc}}(\partial\Omega)$ we set

$$(18.54) \quad |u|_{B^{s,p}(\partial\Omega)} := |v|_{B^{s,p}(\mathbb{R}^{N-1})},$$

where $v(x') := u(x', f(x'))$, $x' \in \mathbb{R}^{N-1}$, and we say that u belongs to the *Besov space* $B^{s,p}(\partial\Omega)$ if the function v belongs to $B^{s,p}(\mathbb{R}^{N-1})$. We define

$$\|u\|_{B^{s,p}(\partial\Omega)} := \|u\|_{L^p(\partial\Omega)} + |u|_{B^{s,p}(\partial\Omega)}.$$

Given $1 \leq p < \infty$ and $s > 0$, if $\Omega \subset \mathbb{R}^N$, $N \geq 2$, is an open set for which there exists a rigid motion $T : \mathbb{R}^N \rightarrow \mathbb{R}^N$ such that $T(\Omega)$ can be written in the form (18.53), for every function $u \in L^1_{\text{loc}}(\partial\Omega)$ we set

$$|u|_{B^{s,p}(\partial\Omega)} := |u \circ T^{-1}|_{B^{s,p}(\partial T(\Omega))}$$

and we say that u belongs to the *Besov space* $B^{s,p}(\partial\Omega)$ if the function $u \circ T^{-1}$ belongs to $B^{s,p}(\partial T(\Omega))$. We define

$$\|u\|_{B^{s,p}(\partial\Omega)} := \|u\|_{L^p(\partial\Omega)} + |u|_{B^{s,p}(\partial\Omega)}.$$

Using partitions of unity we are now ready to extend Definition 18.47 to domains with Lipschitz continuous boundary.

Definition 18.48. Given $1 \leq p < \infty$ and $s \geq 1$, let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is of class $C^{m-1,1}$, where $m := \lfloor s \rfloor + 1$, and let $\{\psi_n\}_n$ be a partition of unity such that $\text{supp } \psi_n \subset \Omega_n$, with $\Omega_n \subset \mathbb{R}^N$ an open set such that $\Omega \cap \Omega_n = \Omega \cap V_n$, where V_n is, up to a rigid motion, a domain of the type (18.53). We say that a function $u \in L^p(\partial\Omega)$ belongs to the *Besov space* $B^{s,p}(\partial\Omega)$ if

$$\|u\|_{B^{s,p}_q(\partial\Omega)} := \|u\|_{L^p(\partial\Omega)} + |u|_{B^{s,p}_q(\partial\Omega)} < \infty,$$

where

$$(18.55) \quad |u|_{B^{s,p}_q(\partial\Omega)} := \left(\sum_n |\psi_n u|_{B^{s,p}(\partial V_n)}^p \right)^{1/p}.$$

We leave as an exercise to check that if $\partial\Omega$ is bounded, then the space $B^{s,p}(\partial\Omega)$ does not depend on the particular partition of unity and that a different partition of unity gives rise to an equivalent seminorm $|\cdot|_{B^{s,p}_q(\partial\Omega)}$. If $\partial\Omega$ is uniformly Lipschitz continuous, we take the sets Ω_n and V_n in the previous definition to be the ones in Definition 13.11 and ψ_n as in (18.19).

Theorem 18.49. *Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is uniformly of class $C^{m-1,1}$, where $m \in \mathbb{N}$ with $m \geq 2$. Then the image of the linear mapping (18.8) is given by*

$$\mathrm{Tr}_m(W^{m,1}(\Omega)) = \prod_{k=0}^{m-2} B^{m-k-1,1}(\partial\Omega) \times L^1(\partial\Omega).$$

The proof follows the lines of Theorems 18.34 and 18.40 using Theorem 18.43 in place of Theorems 18.27 and 18.28. I am out of time and out of space, so I am going to omit it.

18.6. Traces of Functions in $W^{m,p}(\Omega)$, $p > 1$

In this section given an integer $m \in \mathbb{N}$, we characterize the image of the trace operator Tr_m in (18.8) when $1 < p < \infty$.

Theorem 18.50. *Let $N \geq 2$, let $m \in \mathbb{N}$, with $m \geq 2$, and let $1 < p < \infty$. Then the image of the linear mapping (18.8) is given by*

$$\mathrm{Tr}_m(W^{m,p}(\mathbb{R}_+^N)) = \prod_{k=0}^{m-1} B^{m-k-1/p,p}(\mathbb{R}^{N-1}).$$

Proof. The proof follows by an induction argument of m using Theorems 17.69 and Theorems 18.27 and 18.28. We leave the details as an exercise. $\square$

In turn, reasoning as in Theorems 18.34 and 18.40 using Theorem 18.50 in place of Theorems 18.27 and 18.28 we have the following result, whose proof we omit.

Theorem 18.51. *Let $\Omega \subseteq \mathbb{R}^N$, $N \geq 2$, be an open set whose boundary $\partial\Omega$ is uniformly of class $C^{m-1,1}$, where $m \in \mathbb{N}$ with $m \geq 2$, and let $1 < p < \infty$. Then the image of the linear mapping (18.8) is given by*

$$\mathrm{Tr}_m(W^{m,p}(\Omega)) = \prod_{k=0}^{m-1} B^{m-k-1/p,1}(\partial\Omega).$$

Remark 18.52. The previous theorem was recently extended by Maz'ja, Mitrea, and Shaposhnikova [166] to bounded open sets with Lipschitz continuous boundary.

18.7. Besov Spaces and Weighted Sobolev Spaces

In this section we will show that a smooth function u belongs to $B_q^{s,p}(\mathbb{R}^{N-1})$ if and only if it is the trace of a function $v : \mathbb{R}_+^N \rightarrow \mathbb{R}$ in some weighted Sobolev space. Let $\varphi \in S(\mathbb{R}^{N-1})$ be a radial function such that $\mathrm{supp} \hat{\varphi} \subset$

$B_{N-1}(0, 1)$, and $\widehat{\varphi} = 1$ in $B_{N-1}(0, 1/2)$, where $\widehat{\varphi}$ is the Fourier transform of φ . In particular,

$$\int_{\mathbb{R}^{N-1}} \varphi(x') dx' = \widehat{\varphi}(0) = 1.$$

Given $u \in \mathcal{S}(\mathbb{R}^{N-1})$, for $x = (x', x_N) \in \mathbb{R}_+^N$ define

$$(18.56) \quad v(x) := \frac{1}{x_N^{N-1}} \int_{\mathbb{R}^{N-1}} \varphi\left(\frac{x' - y'}{x_N}\right) u(y') dy'.$$

Note that $v(x) = u_{x_N}(x') = (u * \varphi_{x_N})(x')$, that is, v is just the mollification of u with x_N in place of ε . The proof of the following theorem is due to Mironescu and Russ [173].

Theorem 18.53. *Let $1 \leq p \leq \infty$, $1 \leq q < \infty$, and $s > 0$. Given $u \in \mathcal{S}(\mathbb{R}^{N-1})$, let v be the function given in (18.56). Then for every $\alpha \in \mathbb{N}_0^N$, with $|\alpha| > s$,*

$$\left(\int_0^\infty \|\partial^\alpha v(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})}^q \frac{dx_N}{x_N^{1-q(|\alpha|-s)}} \right)^{1/q} \leq c |u|_{B_q^{s,p}(\mathbb{R}^{N-1})},$$

where $c = c(\alpha, p, q, s) > 0$.

We begin with a preliminary result.

Lemma 18.54. *Let $w \in C^\infty(\mathbb{R}_+^N)$ be such that*

$$(18.57) \quad w(x) = - \int_{x_N}^\infty \frac{\partial w}{\partial x_N}(x', r) dr$$

for every $x \in \mathbb{R}_+^N$. Let $1 \leq p \leq \infty$, $1 \leq q < \infty$, and $a > -1$. Then

$$\int_0^\infty x_N^a \|w(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})}^q dx_N \leq c \int_0^\infty x_N^{a+q} \left\| \frac{\partial w}{\partial x_N}(\cdot, x_N) \right\|_{L^p(\mathbb{R}^{N-1})}^q dx_N.$$

Proof. We have

$$\|w(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})} \leq \int_{x_N}^\infty \left\| \frac{\partial w}{\partial x_N}(\cdot, s) \right\|_{L^p(\mathbb{R}^{N-1})} ds$$

and so

$$\int_0^\infty x_N^a \|w(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})}^q dx_N \leq \int_0^\infty x_N^a \left(\int_{x_N}^\infty \left\| \frac{\partial w}{\partial x_N}(\cdot, r) \right\|_{L^p(\mathbb{R}^{N-1})} dr \right)^q dx_N.$$

We now apply Hardy's inequality (see Theorem C.41) to the function

$$f(x_N) := \left\| \frac{\partial w}{\partial x_N}(\cdot, x_N) \right\|_{L^p(\mathbb{R}^{N-1})}. \quad \square$$

We turn to the proof of Theorem 18.53.

Proof of Theorem 18.53. Step 1: Given an integer $\ell \in \mathbb{N}$ we consider $\frac{\partial^\ell v}{\partial x_N^\ell}$ and take the Fourier transform $\mathcal{F}_{x'}$ in the variable x' . By Theorem 10.71 and the fact that $\widehat{\varphi_{x_N}}(y') = \widehat{\varphi}(x_N y')$, for $y' \in \mathbb{R}^{N-1}$ we have

$$\begin{aligned} \mathcal{F}_{x'}\left(\frac{\partial^\ell v}{\partial x_N^\ell}\right)(y') &= \frac{\partial^\ell}{\partial x_N^\ell}(\widehat{\varphi_{x_N} * u})(y') = \frac{\partial^\ell}{\partial x_N^\ell}(\widehat{\varphi_{x_N}}(y')\widehat{u}(y')) \\ &= \frac{\partial^\ell}{\partial x_N^\ell}(\widehat{\varphi}(x_N y')\widehat{u}(y')). \end{aligned}$$

Consider two functions $\eta \in \mathcal{S}(\mathbb{R})$ and $\psi \in \mathcal{S}(\mathbb{R}^{N-1})$ such that

$$(18.58) \quad \widehat{\varphi}(y') = \psi(\|y'\|_{N-1}) \quad \text{and} \quad \widehat{\eta}(y') = \|y'\|_{N-1}^\ell \psi^{(\ell)}(\|y'\|_{N-1})$$

for all $y' \in \mathbb{R}^{N-1}$. Then we can rewrite the previous equality as

$$\begin{aligned} \mathcal{F}_{x'}\left(\frac{\partial^\ell v}{\partial x_N^\ell}\right)(y') &= \frac{\partial^\ell}{\partial x_N^\ell}(\psi(x_N \|y'\|_{N-1})\widehat{u}(y')) \\ &= \|y'\|_{N-1}^\ell \psi^{(\ell)}(x_N \|y'\|_{N-1})\widehat{u}(y') \\ &= \frac{1}{x_N^\ell} (x_N \|y'\|_{N-1})^\ell \psi^{(\ell)}(x_N \|y'\|_{N-1})\widehat{u}(y') = \frac{1}{x_N^\ell} \widehat{\eta}(x_N y')\widehat{u}(y'). \end{aligned}$$

Since $\text{supp } \widehat{\varphi} \subset B_{N-1}(0, 1)$, $\widehat{\varphi} = 1$ in $B_{N-1}(0, 1/2)$, by (18.58) we have that $\widehat{\eta}(y') = 0$ for $\|y'\|_{N-1} < 1/2$ or $\|y'\|_{N-1} > 1$. Hence, by (17.57) and (17.58), we can write

$$\begin{aligned} \mathcal{F}_{x'}\left(\frac{\partial^\ell v}{\partial x_N^\ell}\right)(y') &= \frac{1}{x_N^\ell} \widehat{\eta}(x_N y')\widehat{u}(y') = \frac{1}{x_N^\ell} \sum_{k \in \mathbb{Z}} \widehat{\eta}(x_N y') \varphi_0(2^{-k} y') \widehat{u}(y') \\ &= \frac{1}{x_N^\ell} \sum_{k \in I_{x_N}} \widehat{\eta}(x_N y') \varphi_0(2^{-k} y') \widehat{u}(y'), \end{aligned}$$

where

$$(18.59) \quad I_{x_N} := \{k \in \mathbb{Z} : \text{supp } \varphi_0(2^{-k} \cdot) \cap \text{supp } \eta(x_N \cdot) \neq \emptyset\},$$

and so, by Theorem 10.71 again,

$$\frac{\partial^\ell v}{\partial x_N^\ell}(x', x_N) = \frac{1}{x_N^\ell} \sum_{k \in I_{x_N}} (\eta_{x_N} * (u * \psi_k))(x'),$$

where ψ_k is the inverse Fourier transform of the function $\varphi_0(2^{-k} \cdot)$ (see 17.60) and we used Theorem 10.62. Now, if $\alpha \in \mathbb{N}_0^N$ is a multi-index of the form $\alpha = (\beta, \ell)$, we have

$$\frac{\partial^{|\alpha|} v}{\partial (x')^\beta \partial x_N^\ell}(x', x_N) = \frac{1}{x_N^\ell} \sum_{k \in I_{x_N}} (\eta_{x_N} * \partial^\beta (u * \psi_k))(x').$$

Hence,

$$\begin{aligned} \|\partial^\alpha v(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})} &\leq \frac{1}{x_N^\ell} \sum_{k \in I_{x_N}} \|\eta_{x_N} * \partial^\beta(u * \psi_k)\|_{L^p(\mathbb{R}^{N-1})} \\ &\leq \frac{c_\eta}{x_N^\ell} \sum_{k \in I_{x_N}} \|\partial^\beta(u * \psi_k)\|_{L^p(\mathbb{R}^{N-1})}, \end{aligned}$$

where we used the fact that $\|\eta_{x_N} * \partial^\beta(u * \psi_k)\|_{L^p(\mathbb{R}^{N-1})} \leq c_\eta \|\partial^\beta(u * \psi_k)\|_{L^p(\mathbb{R}^{N-1})}$ (see the proof of Theorem C.16(iii)). Since $\text{supp } u * \psi_k \subseteq \overline{B_{N-1}(0, 2^{k+1}a)}$ we can now apply Theorem 10.73 to get $\|\partial^\beta(u * \psi_k)\|_{L^p(\mathbb{R}^{N-1})} \leq c 2^{(k+1)|\beta|} \|u * \psi_k\|_{L^p(\mathbb{R}^{N-1})}$ and so

$$\begin{aligned} \|\partial^\alpha v(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})} &\leq \frac{c}{x_N^\ell} \sum_{k \in I_{x_N}} 2^{(k+1)|\beta|} \|u * \psi_k\|_{L^p(\mathbb{R}^{N-1})} \\ &\leq \frac{c}{x_N^{|\alpha|}} \sum_{k \in I_{x_N}} \|u * \psi_k\|_{L^p(\mathbb{R}^{N-1})}, \end{aligned}$$

where in the last inequality we used the facts that $|\alpha| = |\beta| + \ell$ and that by taking $a = 1/4$ in (17.56) we have that $\text{supp } \varphi_0(2^{-k}\cdot) = \overline{B(0, 2^{k-1})} \setminus B(0, 2^{k-3})$ (see (17.58)), and so the set I_{x_N} in (18.59) is contained in the set of $k \in \mathbb{Z}$ such that $1/x_N < 2^k < 8/x_N$. Since for every $x_N > 0$ there are at most three k such that $1/x_N < 2^k < 8/x_N$, we obtain

$$\begin{aligned} &\int_0^\infty \|\partial^\alpha v(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})}^q \frac{dx_N}{x_N^{1-q(|\alpha|-s)}} \\ &\leq c \int_0^\infty \frac{1}{x_N^{1+qs}} \left(\sum_{k \in I_{x_N}} \|u * \psi_k\|_{L^p(\mathbb{R}^{N-1})} \right)^q dx_N \\ &\leq c \int_0^\infty \frac{1}{x_N^{1+qs}} \sum_{k \in I_{x_N}} \|u * \psi_k\|_{L^p(\mathbb{R}^{N-1})}^q dx_N \\ &\leq c \sum_{k \in \mathbb{Z}} \|u * \psi_k\|_{L^p(\mathbb{R}^{N-1})}^q \int_{2^{-k}}^{2^{3-k}} \frac{1}{x_N^{1+qs}} dx_N = c \sum_{k \in \mathbb{Z}} 2^{qsk} \|u * \psi_k\|_{L^p(\mathbb{R}^{N-1})}^q, \end{aligned}$$

where in the third inequality we used Tonelli's theorem and as usual the constant c changed from line to line. In view of Theorem 17.77 we have proved the conclusion of the theorem for every multi-index $\alpha \in \mathbb{N}_0^N$ with $\alpha_N > 0$.

Step 2: It remains to study the case of a multi-index $\alpha \in \mathbb{N}_0^N$ with $\alpha_N = 0$ and $|\alpha| > s$. Write $\alpha = (\beta, 0)$, where $\beta \in \mathbb{N}_0^{N-1}$ is such that $|\beta| = |\alpha| > s$.

Then $a := q(|\beta| - s) - 1 > -1$, and so by Lemma 18.54 applied to $w := \partial^\alpha v$,

$$\begin{aligned} & \int_0^\infty \|\partial^\alpha v(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})}^q \frac{dx_N}{x_N^{1-q(|\alpha|-s)}} \\ & \leq \int_0^\infty \|\partial^{(\beta,1)} v(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})}^q \frac{dx_N}{x_N^{1-q(|\beta|+1-s)}}. \end{aligned}$$

It now suffices to apply Step 1 to $\partial^{(\beta,1)} v$. This concludes the proof. $\square$

Exercise 18.55. Prove that the previous theorem continues to hold for $q = \infty$.

Remark 18.56. Since the function v defined in (18.56) is given by $v(x', x_N) = (u * \varphi_{x_N})(x')$, we can define v also for a tempered distribution $T \in \mathcal{S}'(\mathbb{R}^{N-1})$, by setting (see Theorem 10.56), $v(x', x_N) = T(\varphi_{x_N}')$, where we recall that $\varphi_{x_N}'(y') := \varphi_{x_N}(x' - y')$, $y' \in \mathbb{R}^{N-1}$. Reasoning as in the proof of Theorem 10.52, one can show that $v \in C^\infty(\mathbb{R}_+^N)$. Reasoning as in Step 1 of the previous proof, we still have that

$$\left(\int_0^\infty \|\partial^\alpha v(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})}^q \frac{dx_N}{x_N^{1-q(|\alpha|-s)}} \right)^{1/q} \leq c |T|_{B_q^{s,p}(\mathbb{R}^{N-1})}^\vee$$

for every multi-index $\alpha \in \mathbb{N}_0^N$, with $\alpha_N \geq 1$ and $|\alpha| > s$. However, Lemma 18.54 in general fails in this case. As an example, taking the distribution T_p associated to a polynomial p (see Remark 10.37), one can show that $v(x', x_N) = p(x')$ for every $x_N > 0$ and $x' \in \mathbb{R}^{N-1}$ (exercise), and so $w := \partial^\alpha v = \partial^\alpha p$ does not satisfy (18.57).

We now prove the opposite of Theorem 18.53.

Theorem 18.57. *Let $1 \leq p \leq \infty$, $1 \leq q < \infty$, and $s > 0$. Given $v \in C^\infty(\mathbb{R}^{N-1} \times [0, \infty))$, let $u(x') := v(x', 0)$, $x' \in \mathbb{R}^{N-1}$. Then for every $m \in \mathbb{N}$, with $m > s$,*

$$|u|_{B_q^{s,p}(\mathbb{R}^{N-1})}^q \leq c \sum_{\alpha \in \Xi_m} \int_0^\infty \|\partial^\alpha v(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})}^q \frac{dx_N}{x_N^{1-q(m-s)}},$$

where $c = c(m, p, q, s) > 0$ and $\Xi_m := \{(\beta, 0) : \beta \in \mathbb{N}_0^{N-1}, |\beta| = m\} \cup \{(0, m)\}$.

We begin with a preliminary result.

Lemma 18.58. *Given $v \in C^\infty(\mathbb{R}^{N-1} \times [0, \infty))$, let $u(x') := v(x', 0)$, $x' \in \mathbb{R}^{N-1}$. Then for $x', h \in \mathbb{R}^{N-1}$ and $m \in \mathbb{N}$,*

$$\begin{aligned} \Delta_h^m u(x') &= \sum_{k=0}^m (-1)^k \binom{m}{k} \Delta_{\|h\|_{N-1} e_N}^m v(x' + kh, 0) \\ &\quad + \sum_{k=1}^m (-1)^{k+1} \binom{m}{k} \Delta_{(h,0)}^m v(x', k\|h\|_{N-1}). \end{aligned}$$

Proof. Set $r := \|h\|_{N-1}$. For every $x' \in \mathbb{R}^{N-1}$ we can write

$$u(x') = \sum_{k=0}^m (-1)^k \binom{m}{k} v(x', kr) + \sum_{k=1}^m (-1)^{k+1} \binom{m}{k} v(x', kr).$$

By applying the previous identity to $u(x' + jh)$ it follows from (17.11) that

$$\begin{aligned} \Delta_h^m u(x') &= \sum_{j=0}^m (-1)^{m-j} \binom{m}{j} u(x' + jh) \\ &= \sum_{j=0}^m (-1)^{m-j} \binom{m}{j} \sum_{k=0}^m (-1)^k \binom{m}{k} v(x' + jh, kr) \\ &\quad + \sum_{k=1}^m (-1)^{k+1} \binom{m}{k} \sum_{j=0}^m (-1)^{m-j} \binom{m}{j} v(x' + jh, kr) \\ &= \sum_{j=0}^m (-1)^j \binom{m}{j} \Delta_{\|h\|_{N-1} e_N}^m v(x' + jh, 0) \\ &\quad + \sum_{k=1}^m (-1)^{k+1} \binom{m}{k} \Delta_{(h,0)}^m v(x', kr), \end{aligned}$$

which concludes the proof. $\square$

We turn to the proof of Theorem 18.57.

Proof of Theorem 18.57. Reasoning as in the proof of Lemma 17.25 we can show that if $\xi = (h, \tau) \in \mathbb{R}^{N-1} \times [0, \infty)$ with either $h = 0$ or $\tau = 0$, then

$$(18.60) \quad |\Delta_\xi^m v(x)| \leq c \|\xi\|_N^m \sum_{\alpha \in \Xi_m} \int_0^m x_N^{m-1} |\partial^\alpha v(x + x_N \xi)| dx_N.$$

For $h \in \mathbb{R}^{N-1}$ set $r := \|h\|_{N-1}$. In view of Lemma 18.58 and (18.60) we have

$$\begin{aligned} |\Delta_h^m u(x')| &\leq c \sum_{k=0}^m |\Delta_{re_N}^m v(x' + kh, 0)| + \sum_{k=1}^m |\Delta_{(h,0)}^m v(x', kr)| \\ &\leq cr^m \sum_{k=0}^m \sum_{\alpha \in \Xi_m} \int_0^m x_N^{m-1} |\partial^\alpha v(x' + kh, x_N r)| dx_N \\ &\quad + cr^m \sum_{k=1}^m \sum_{\alpha \in \Xi_m} \int_0^m x_N^{m-1} |\partial^\alpha v(x' + x_N h, kr)| dx_N. \end{aligned}$$

Taking the L^p norm in x' on both sides and using Tonelli's theorem and a change of variables gives

$$\begin{aligned} \|\Delta_h^m u\|_{L^p(\mathbb{R}^{N-1})} &\leq cr^m \int_0^m x_N^{m-1} f(x_N r) dx_N + cr^m \sum_{k=0}^m f(kr), \\ &\leq c \int_0^{mr} s^{m-1} f(s) ds + cr^m \sum_{k=0}^m f(kr), \end{aligned}$$

where $f(x_N) := \sum_{\alpha \in \Xi_m} \|\partial^\alpha v(\cdot, x_N)\|_{L^p(\mathbb{R}^{N-1})}$. In turn,

$$\begin{aligned} &\int_{\mathbb{R}^{N-1}} \|\Delta_h^m u\|_{L^p(\mathbb{R}^{N-1})}^q \frac{dh}{\|h\|_{N-1}^{N-1+sq}} \\ &\leq c \int_{\mathbb{R}^{N-1}} \left(\int_0^{m\|h\|_{N-1}} \tau^{m-1} f(\tau) d\tau \right)^q \frac{dh}{\|h\|_{N-1}^{N-1+sq}} \\ &\quad + c \sum_{k=0}^m \int_{\mathbb{R}^{N-1}} (f(k\|h\|_{N-1}))^q \frac{dh}{\|h\|_{N-1}^{N-1-(m-s)q}} =: I + II. \end{aligned}$$

Using spherical coordinates and Hardy's inequality (see Theorem C.41) we have that

$$\begin{aligned} I &\leq c \int_0^\infty \left(\int_0^{mr} \tau^{m-1} f(\tau) d\tau \right)^q \frac{dr}{r^{1+sq}} \\ &\leq c \int_0^\infty (f(x_N))^q \frac{dx_N}{x_N^{1-q(m-s)}}. \end{aligned}$$

Similarly, again by spherical coordinates and a change of variables,

$$II \leq c \sum_{k=0}^m \int_0^\infty (f(kr))^q \frac{dr}{r^{1-(m-s)q}} \leq c \int_0^\infty (f(x_N))^q \frac{dx_N}{x_N^{1-q(m-s)}},$$

which concludes the proof. $\square$

Exercise 18.59. Prove that the previous theorem continues to hold for $q = \infty$.

Remark 18.60. If $m \in \mathbb{N}$, $1 \leq p = q < \infty$, and $s = m - 1/p > 0$, we obtain

$$|u|_{B^{m-1/p,p}(\mathbb{R}^{N-1})}^p \leq c \sum_{\alpha \in \Xi_m} \int_0^\infty \int_{\mathbb{R}^{N-1}} |\partial^\alpha v(x', x_N)|^p dx' dx_N.$$

Functional Analysis

When probed under pressure, grad students will either blurt out what they are doing (but won't know if it means anything), or they will blurt out what they plan to do (but won't know how to do it).

— Jorge Cham, www.phdcomics.com

In this chapter we present without proofs most of the results from functional analysis used in the text.

A.1. Topological Spaces

Definition A.1. Let X be a nonempty set. A collection $\tau \subseteq \mathcal{P}(X)$ is a *topology* if the following hold.

- (i) $\emptyset, X \in \tau$.
- (ii) If $U_i \in \tau$ for $i = 1, \dots, n$, $n \in \mathbb{N}$, then $U_1 \cap \dots \cap U_n \in \tau$.
- (iii) If $\{U_\alpha\}_{\alpha \in \Lambda}$ is an arbitrary collection of elements of τ , then

$$\bigcup_{\alpha \in \Lambda} U_\alpha \in \tau.$$

Example A.2. Given a nonempty set X , the smallest topology consists of $\{\emptyset, X\}$, while the largest topology contains all subsets as open sets.

The pair (X, τ) is called a *topological space* and the elements of τ are *open sets*. For simplicity, we often apply the term topological space only to X . A set $C \subseteq X$ is *closed* if its complement $X \setminus C$ is open. The *closure* $\overline{E}$ of a set $E \subseteq X$ is the smallest closed set that contains E . The *interior* E° of a set $E \subseteq X$ is the union of all its open subsets. A point $x_0 \in X$ is an *accumulation point* for a set $E \subseteq X$ if for every open set U that contains x_0 there exists $x \in E \cap U$, with $x \neq x_0$.

Given a set $E \subseteq X$ we say that a set $F \subseteq E$ is *relatively open* (respectively, *relatively closed*) if there exists an open set $U \subseteq X$ (respectively, a closed set $C \subseteq X$) such that $F = E \cap U$ (respectively, $F = E \cap C$).

A subset E of a topological space X is said to be *dense* if its closure is the entire space, i.e., $\overline{E} = X$. We say that a topological space is *separable* if it contains a countable dense subset.

Given a point $x \in X$, a *neighborhood*¹ of x is an open set $U \in \tau$ that contains x . Given a set $E \subseteq X$, a *neighborhood* of E is an open set $U \in \tau$ that contains E . A topological space is a *Hausdorff space* if for all $x, y \in X$ with $x \neq y$ there exist two disjoint neighborhoods of x and y .

Given a topological space X and a sequence $\{x_n\}_n$ in X , we say that $\{x_n\}_n$ *converges* to a point $x \in X$ if for every neighborhood U of x we have that $x_n \in U$ for all n sufficiently large. Note that unless the space is Hausdorff, the limit may not be unique.

We now introduce the notion of a base for a topology. Let (X, τ) be a topological space. A family β of open sets of X is a *base* for the topology τ if every open set $U \in \tau$ may be written as a union of elements of β . Given a point $x \in X$, a family β_x of neighborhoods of x is a *local base at x* if every neighborhood of x contains an element of β_x .

Proposition A.3. *Let X be a nonempty set and let $\beta \subseteq \mathcal{P}(X)$ be a family of sets such that*

- (i) *for every $x \in X$ there exists $B \in \beta$ such that $x \in B$,*
- (ii) *for every $B_1, B_2 \in \beta$, with $B_1 \cap B_2 \neq \emptyset$, and for every $x \in B_1 \cap B_2$ there exists $B_3 \in \beta$ such that $x \in B_3$ and $B_3 \subseteq B_1 \cap B_2$.*

Then the collection $\tau \subseteq \mathcal{P}(X)$ of arbitrary unions of members of β is a topology for which β is a base.

As we will see later on, the metrizability and the normability of a given topology depend on the properties of a base (see Theorems A.15 and A.42 below).

Definition A.4. Let (X, τ) be a topological space.

- (i) The space X satisfies the *first axiom of countability* if every $x \in X$ admits a countable local base at x .
- (ii) The space X satisfies the *second axiom of countability* if it has a countable base.

In the text we use several notions of compactness.

¹The reader should be warned that in some texts (e.g., [200]) the definition of neighborhood is different.

Definition A.5. Let X be a topological space.

- (i) A set $K \subseteq X$ is *compact* if for every open cover of K , i.e., for every collection $\{U_\alpha\}_{\alpha \in \Lambda}$ of elements of τ such that $K \subseteq \bigcup_{\alpha \in \Lambda} U_\alpha$, there exists a finite subcover (i.e., a finite subcollection of $\{U_\alpha\}_{\alpha \in \Lambda}$ whose union still contains K).
- (ii) A set $E \subseteq X$ is *relatively compact* (or *precompact*) if its closure $\overline{E}$ is compact.
- (iii) A set $E \subseteq X$ is σ -*compact* if it can be written as a countable union of compact sets.
- (iv) The topological space X is *locally compact* if every point $x \in X$ has a neighborhood whose closure is compact.

Remark A.6. A closed subset of a compact topological space is compact. On the other hand, a compact set of a Hausdorff space is closed.

Definition A.7. Let X be a topological space and let $\mathcal{F}$ be a collection of subsets of X . Then

- (i) $\mathcal{F}$ is *locally finite* if every $x \in X$ has a neighborhood meeting only finitely many $U \in \mathcal{F}$,
- (ii) $\mathcal{F}$ is σ -*locally finite* if $\mathcal{F} = \bigcup_{n=1}^{\infty} \mathcal{F}_n$, where each $\mathcal{F}_n$ is a locally finite collection in X .

Definition A.8. Let X, Y be two topological spaces, let $E \subseteq X$ and let $u : E \rightarrow Y$. We say that u is *continuous* if $u^{-1}(U)$ is relatively open in E for every open set $U \subseteq Y$. The space of all continuous functions $u : X \rightarrow Y$ is denoted by $C(X; Y)$.

If $Y = \mathbb{R}$, we write $C(X)$ in place of $C(X; \mathbb{R})$. If X is a topological space, we denote by $C_c(X)$ the space of all continuous functions $u : X \rightarrow \mathbb{R}$ whose support is compact.

A topological space is a *normal space* if for every pair of disjoint closed sets $C_1, C_2 \subset X$ there exist two disjoint neighborhoods of C_1 and C_2 . The next theorems give an important characterization of normal spaces.

Theorem A.9 (Urysohn). *A topological space X is normal if and only if for all disjoint nonempty closed sets $C_1, C_2 \subset X$ there exists a continuous function $u : X \rightarrow [0, 1]$ such that $u \equiv 1$ in C_1 and $u \equiv 0$ in C_2 .*

Theorem A.10 (Tietze's extension theorem). *A topological space X is normal if and only if for every closed set $C \subset X$ and every continuous function $u : C \rightarrow \mathbb{R}$ there exists a continuous function $v : X \rightarrow \mathbb{R}$ such that $v(x) = u(x)$ for all $x \in C$. Moreover, if $u(C) \subseteq [a, b]$, then v may be constructed so that $v(X) \subseteq [a, b]$.*

A.2. Metric Spaces

Definition A.11. A *metric* on a set X is a map $d : X \times X \rightarrow [0, \infty)$ such that

- (i) $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$,
- (ii) $d(x, y) = d(y, x)$ for all $x, y \in X$,
- (iii) $d(x, y) = 0$ if and only if $x = y$.

A *metric space* (X, d) is a set X endowed with a metric d . When there is no possibility of confusion, we abbreviate by saying that X is a metric space.

If $r > 0$, the (*open*) *ball* of center $x_0 \in X$ and radius r is the set $B(x_0, r) := \{x \in X : d(x_0, x) < r\}$. If $x \in X$ and $E \subseteq X$, the *distance* of x from the set E is defined by $\text{dist}(x, E) := \inf\{d(x, y) : y \in E\}$, while the distance between two sets $E_1, E_2 \subseteq X$ is defined by

$$\text{dist}(E_1, E_2) := \inf\{d(x, y) : x \in E_1, y \in E_2\}.$$

A sequence $\{x_n\}_n$ in X *converges (strongly)* to $x \in X$ if

$$\lim_{n \rightarrow \infty} d(x_n, x_0) = 0.$$

A *Cauchy sequence* in a metric space is a sequence $\{x_n\}_n$ in X such that

$$\lim_{n, m \rightarrow \infty} d(x_n, x_m) = 0.$$

A metric space X is said to be *complete* if every Cauchy sequence is convergent.

Theorem A.12 (Baire category theorem). *Let (X, d) be a complete metric space. Then the intersection of a countable family of open dense sets in X is still dense in X .*

Corollary A.13. *Let (X, d) be a nonempty complete metric space. If $X = \bigcup_{n=1} C_n$, where C_n is closed for every $n \in \mathbb{N}$, then at least one C_n has nonempty interior.*

Theorem A.14 (Banach's fixed point). *Let (X, d) be a nonempty complete metric space and let $u : X \rightarrow X$ be a contraction, that is there exists $0 < L < 1$ such that $d(u(x), u(y)) \leq Ld(x, y)$ for all $x, y \in X$. Then u has a unique fixed point; that is, there is a unique $x \in X$ such that $u(x) = x$.*

A metric space (X, d) can always be turned into a topological space (X, τ) by taking as a base for the topology τ the family of all open balls. We then say that τ is *determined* by d . Note that (X, τ) is a Hausdorff normal space.

A topological space X is *metrizable* if its topology can be determined by a metric.

Theorem A.15. *A topological space is metrizable if and only if the following properties hold:*

- (i) *Singletons are closed.*
- (ii) *For every closed set $C \subset X$ and for every $x \notin C$ there exist disjoint open neighborhoods of C and x .*
- (iii) *X has a σ -locally finite base.*

A.3. Topological Vector Spaces

Definition A.16. A *vector space*, or *linear space*, over $\mathbb{R}$ is a nonempty set X , whose elements are called *vectors*, together with two operations, *addition* and *multiplication by scalars*,

$$\begin{aligned} X \times X &\rightarrow X, & \text{and} & & \mathbb{R} \times X &\rightarrow X, \\ (x, y) &\mapsto x + y, & & & (t, x) &\mapsto tx, \end{aligned}$$

with the properties that

- (i) $(X, +)$ is a commutative group, that is,
 - (a) $x + y = y + x$ for all $x, y \in X$ (commutative property),
 - (b) $x + (y + z) = (x + y) + z$ for all $x, y, z \in X$ (associative property),
 - (c) there is a vector $0 \in X$, called *zero*, such that $x + 0 = 0 + x = x$ for all $x \in X$,
 - (d) for every $x \in X$ there exists a vector in X , called the *opposite* of x and denoted $-x$, such that $x + (-x) = 0$,
- (ii) for all $x, y \in X$ and $s, t \in \mathbb{R}$,
 - (a) $s(tx) = (st)x$,
 - (b) $1x = x$,
 - (c) $s(x + y) = (sx) + (sy)$,
 - (d) $(s + t)x = (sx) + (tx)$.

Let X be a vector space over $\mathbb{R}$ and let $E \subseteq X$. The set E is said to be *balanced* if $tx \in E$ for all $x \in E$ and $t \in [-1, 1]$. We say that $E \subseteq X$ is *absorbing* if for every $x \in X$ there exists $t > 0$ such that $sx \in E$ for all $0 \leq s \leq t$.

Definition A.17. Given a vector space X over $\mathbb{R}$ endowed with a topology τ , the pair (X, τ) is called a *topological vector space* if the functions

$$\begin{aligned} X \times X &\rightarrow X, & \text{and} & & \mathbb{R} \times X &\rightarrow X, \\ (x, y) &\mapsto x + y, & & & (t, x) &\mapsto tx, \end{aligned}$$

are continuous.

Remark A.18. (i) In a topological vector space a set U is open if and only if $x + U$ is open for all $x \in X$. Hence, to give a base, it is enough to give a local base at the origin.

(ii) Using the continuity of addition and scalar multiplication, it is possible to show that each neighborhood U of the origin is absorbing and it contains a neighborhood W of zero such that $W + W \subseteq U$ and $\overline{W} \subseteq U$, as well as a balanced neighborhood of zero.

As a corollary of Theorem A.15 we have the following:

Corollary A.19. *A topological vector space X is metrizable if and only if*

- (i) *singletons are closed,*
- (ii) *X has a countable local base at 0.*

Definition A.20. Let X be a topological vector space. A set $E \subseteq X$ is said to be *topologically bounded* if for each neighborhood U of 0 there exists $t > 0$ such that $E \subseteq tU$.

Note that when the topology τ is generated by a metric d , sets bounded in the topological sense and in the metric sense may be different. To see this, it suffices to observe that the metric $d_1 := \frac{d}{d+1}$ generates the same topology as d , but since $d_1 \leq 1$, every set in X is bounded with respect to d_1 .

We now define Cauchy sequences in a topological vector space.

Definition A.21. Let X be a topological vector space. A sequence $\{x_n\}_n$ in X is called a *Cauchy sequence* if for every neighborhood U of the origin there exists an integer $\bar{n} \in \mathbb{N}$ such that

$$x_n - x_m \in U$$

for all $n, m \geq \bar{n}$. The space X is *complete* if every Cauchy sequence is convergent.

Note that Cauchy (and hence convergent) sequences are bounded in the topological sense.

Proposition A.22. *Let X be a topological vector space and let $\{x_n\}_n$ be a Cauchy sequence in X . Then the set $\{x_n : n \in \mathbb{N}\}$ is topologically bounded.*

Topologically bounded sets play an important role in the normability of locally convex topological vector spaces (see Theorem A.42 below).

Definition A.23. A topological vector space X is *locally convex* if every point $x \in X$ has a neighborhood that is convex.

Proposition A.24. *A locally convex topological vector space admits a local base at the origin consisting of balanced convex neighborhoods of zero.*

Let X be a vector space over $\mathbb{R}$ and let $E \subseteq X$. The function $p_E : X \rightarrow \mathbb{R}$ defined by

$$p_E(x) := \inf\{t > 0 : x \in tE\}, \quad x \in X,$$

is called the *gauge* or *Minkowski functional* of E .

Definition A.25. Let X be a vector space over $\mathbb{R}$. A function $p : X \rightarrow \mathbb{R}$ is called a *seminorm* if

$$p(x + y) \leq p(x) + p(y)$$

for all $x, y \in X$ and $p(tx) = |t|p(x)$ for all $x \in X$ and $t \in \mathbb{R}$.

Remark A.26. Let X be a vector space over $\mathbb{R}$ and let $E \subseteq X$. The gauge p_E of E is a seminorm if and only if E is balanced, absorbing, and convex.

Theorem A.27. *If $\mathcal{F}$ is a balanced, convex local base of 0 for a locally convex topological vector space X , then the family $\{p_U : U \in \mathcal{F}\}$ is a family of continuous seminorms. Conversely, given a family $\mathcal{P}$ of seminorms on a vector space X , the collection of all finite intersections of sets of the form*

$$V(p, n) := \{x \in X : p(x) < 1/n\}, \quad p \in \mathcal{P}, n \in \mathbb{N},$$

is a balanced, convex local base of 0 for a topology τ that turns X into a locally convex topological vector space such that each p is continuous with respect to τ .

We now give some necessary and sufficient conditions for the topology τ given in the previous theorem to be Hausdorff and for a set to be topologically bounded.

Corollary A.28. *Let $\mathcal{P}$ be a family of seminorms on a vector space X and let τ be the locally convex topology generated by $\mathcal{P}$. Then*

- (i) τ is Hausdorff if and only if $p(x) = 0$ for all $p \in \mathcal{P}$ implies that $x = 0$,
- (ii) a set $E \subseteq X$ is topologically bounded if and only if the set $p(E)$ is bounded in $\mathbb{R}$ for all $p \in \mathcal{P}$.

A topological vector space X is called *seminormable* if the topology of the space can be induced by a single seminorm p (see Theorem A.27). The pair (X, p) is called a *seminormed space*.

We now introduce the notion of dual space.

Definition A.29. Let X and Y be two vector spaces. A map $L : X \rightarrow Y$ is called a *linear operator* if

- (i) $L(x_1 + x_2) = L(x_1) + L(x_2)$ for all $x_1, x_2 \in X$,
- (ii) $L(tx) = tL(x)$ for all $x \in X$ and $t \in \mathbb{R}$.

If X and Y are topological vector spaces, then the vector space of all continuous linear operators from X to Y is denoted by $\mathcal{L}(X; Y)$. In the special case $Y = \mathbb{R}$, the space $\mathcal{L}(X; \mathbb{R})$ is called the (*topological*) *dual space* of X and it is denoted by X' . The elements of X' are also called continuous linear *functionals*.

The bilinear (i.e., linear in each variable) mapping

$$(A.1) \quad \begin{aligned} \langle \cdot, \cdot \rangle_{X', X} : X' \times X &\rightarrow \mathbb{R} \\ (L, x) &\mapsto L(x) \end{aligned}$$

is called the *duality pairing*.

The dual space $\mathcal{L}(X'; \mathbb{R})$ of X' is called the *bidual space* of X and it is denoted by X'' .

Definition A.30. Let X, Y be topological vector spaces. An operator $L : X \rightarrow Y$ is *bounded* if it sends topologically bounded sets of X into topologically bounded sets of Y .

Theorem A.31. Let X, Y be topological vector spaces and let $L : X \rightarrow Y$ be a linear operator. Consider the following properties:

- (i) L is continuous.
- (ii) L is bounded.
- (iii) If $\{x_n\}_n$ is a sequence in X and $x_n \rightarrow 0$, then $\{L(x_n)\}_n$ is a topologically bounded set.
- (iv) If $\{x_n\}_n$ is a sequence in X and $x_n \rightarrow 0$, then $L(x_n) \rightarrow 0$.

Then (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). Moreover, if X is metrizable, then (iii) $\Rightarrow$ (iv) $\Rightarrow$ (i) so that all four properties are equivalent.

Theorem A.32 (Hahn–Banach theorem, analytic form). Let X be a vector space, let Y be a subspace of X , and let $p : X \rightarrow \mathbb{R}$ be a convex function. Then for every linear functional $L : Y \rightarrow \mathbb{R}$ such that $L(x) \leq p(x)$ for all $x \in Y$, there exists a linear functional $L_1 : X \rightarrow \mathbb{R}$ such that $L_1(x) = L(x)$ for all $x \in Y$ and $L_1(x) \leq p(x)$ for all $x \in X$.

Theorem A.33 (Hahn–Banach theorem, first geometric form). Let X be a topological vector space and let $E, F \subset X$ be nonempty disjoint convex sets. Assume that E has an interior point. Then there exist a continuous linear functional $L : X \rightarrow \mathbb{R}$, $L \neq 0$, and a number $a \in \mathbb{R}$ such that $L(x) \geq a$ for all $x \in E$ and $L(x) \leq a$ for all $x \in F$.

Corollary A.34. Let X be a topological vector space. Then the dual X' of X is not the null space if and only if X has a convex neighborhood of the origin strictly contained in X .

In view of the previous corollary it is natural to restrict our attention to locally convex topological vector spaces.

Theorem A.35 (Hahn–Banach theorem, second geometric form). *Let X be a locally convex topological vector space and let $C, K \subset X$ be nonempty disjoint convex sets, with C closed and K compact. Then there exist a continuous linear functional $L : X \rightarrow \mathbb{R}$ and two numbers $a \in \mathbb{R}$ and $\varepsilon > 0$ such that $L(x) \leq a - \varepsilon$ for all $x \in C$ and $L(x) \geq a + \varepsilon$ for all $x \in K$.*

Definition A.36. We say that a topological vector space (X, τ_X) is *embedded* in a topological vector space (Y, τ_Y) and we write $X \hookrightarrow Y$ if X is a subspace of Y and the *immersion* $i : X \rightarrow Y$, given by $i(x) := x$, is continuous.

Remark A.37 (Important). The definition of embedding changes with books. In some books an embedding only means that X is a subspace of Y and our definition is called a continuous embedding.

A.4. Normed Spaces

Definition A.38. A *norm* on a vector space X is a map $\|\cdot\| : X \rightarrow [0, \infty)$ such that

- (i) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$,
- (ii) $\|tx\| = |t| \|x\|$ for all $x \in X$ and $t \in \mathbb{R}$,
- (iii) $\|x\| = 0$ implies $x = 0$.

A *normed space* $(X, \|\cdot\|)$ is a vector space X endowed with a norm $\|\cdot\|$. For simplicity, we often say that X is a normed space.

If for every $x, y \in X$ we define $d(x, y) := \|x - y\|$, then (X, d) is a metric space. We say that a normed space X is a *Banach space* if it is complete as a metric space.

Example A.39 (Euclidean space). The most important example of Banach space in this book is the Euclidean space $\mathbb{R}^N$. Given $x = (x_1, \dots, x_N) \in \mathbb{R}^N$ the *Euclidean norm* of x is the number

$$\|x\| := \sqrt{x_1^2 + \dots + x_N^2}.$$

When working with Euclidean spaces of different dimension, we will sometimes use the notation $\|x\|_N$ for $\|x\|$.

Remark A.40. A seminormable space (X, p) induces a normed space on the quotient space X/Y , where $Y := \{x \in X : p(x) = 0\}$. The induced norm on X/Y is given by $\|[x]\| := p(x)$, where $[x] := x + Y$.

Example A.41. Let X be a topological space and consider the space $C_c(X)$ of all continuous functions $u : X \rightarrow \mathbb{R}$ whose support is compact. For $u \in C_c(X)$ define $\|u\|_\infty := \max_{x \in X} |u(x)|$. Then $\|\cdot\|_\infty$ is a norm. In general $C_c(X)$ is not complete. We denote by $C_0(X)$ the completion of $C_c(X)$ relative to the supremum norm $\|\cdot\|_\infty$. If X is compact, then $C_0(X) = C_c(X) = C(X)$. If X is a locally compact Hausdorff space, then it can be shown that $u \in C_0(X)$ if and only if $u \in C(X)$ and for every $\varepsilon > 0$ there exists a compact set $K \subseteq X$ such that $|u(x)| < \varepsilon$ for all $x \in X \setminus K$. We also define $C_c(X; \mathbb{R}^m)$ and $C_0(X; \mathbb{R}^m)$ as the spaces of all functions $u : X \rightarrow \mathbb{R}^m$ whose components belong to $C_c(X)$ and $C_0(X)$, respectively.

A topological vector space is *normable* if its topology can be determined by a norm.

Theorem A.42. *A topological vector space X is normable if and only if it is Hausdorff, locally convex and it has a topologically bounded neighborhood of 0.*

Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on the same vector space X are *equivalent* if there exists a positive constant $c > 0$ such that

$$c^{-1}\|x\|_1 \leq \|x\|_2 \leq c\|x\|_1 \quad \text{for all } x \in X.$$

Equivalent norms induce the same topology on X .

Proposition A.43. *Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces.*

(i) *A linear operator $L : X \rightarrow Y$ is continuous if and only if*

$$\|L\|_{\mathcal{L}(X;Y)} := \sup_{x \in X \setminus \{0\}} \frac{\|L(x)\|_Y}{\|x\|_X} < \infty.$$

(ii) *The mapping $L \in \mathcal{L}(X;Y) \mapsto \|L\|_{\mathcal{L}(X;Y)}$ is a norm.*

(iii) *If Y is a Banach space, then so is $\mathcal{L}(X;Y)$. Conversely, if $X \neq \{0\}$ and $\mathcal{L}(X;Y)$ is a Banach space, then so is Y .*

As a corollary of the Hahn–Banach theorem (Theorem A.32) one has the following result.

Corollary A.44. *Let X be a normed space. Then for all $x \in X$,*

$$\|x\|_X = \max_{L \in X', \|L\|_{X'} \leq 1} \frac{|L(x)|}{\|x\|_X}.$$

Definition A.45. Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces. A continuous linear operator $L \in \mathcal{L}(X;Y)$ is said to be *compact* if it maps every bounded subset of X onto a relatively compact subset of Y .

In particular, if L is compact, then from every bounded sequence $\{x_n\}_n$ in X we may extract a subsequence $\{x_{n_k}\}_k$ such that $\{L(x_{n_k})\}_k$ converges in Y .

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces with X embedded in Y (see Definition A.36). Since the immersion is linear, in view of Proposition A.43 the continuity of i is equivalent to requiring the existence of a constant $M > 0$ such that

$$(A.2) \quad \|x\|_Y \leq M\|x\|_X \quad \text{for all } x \in X.$$

We say that X is *compactly embedded* in Y if the immersion i is a compact operator.

Given a vector space X , a *quasi-norm* is a map $\|\cdot\| : X \rightarrow [0, \infty)$ such that it satisfies conditions (ii) and (iii) of Definition A.38 and such that $\|x + y\| \leq c(\|x\| + \|y\|)$ for all $x, y \in X$ and for some constant $c > 0$. A *quasi-normed space* $(X, \|\cdot\|)$ is a vector space X endowed with a quasi-norm $\|\cdot\|$. For simplicity, we often say that X is a quasi-normed space.

Setting $d(x, y) := \|x - y\|$, $x, y \in X$, we cannot conclude that d is a metric, since the triangle inequality is now replaced by $d(x, y) \leq c(d(x, z) + d(z, y))$. However, we can carry on most of the analysis done for metric space. In particular we can define limits and Cauchy sequences and define completeness. A quasi-normed space X is a *quasi-Banach space* if every Cauchy sequence has a limit in X .

A.5. Weak Topologies

Given a locally convex topological vector space X , for each $L \in X'$ the function $p_L : X \rightarrow [0, \infty)$ defined by

$$(A.3) \quad p_L(x) := |L(x)|, \quad x \in X,$$

is a seminorm. In view of Theorem A.27, the family of seminorms $\{p_L\}_{L \in X'}$ generates a locally convex topology $\sigma(X, X')$ on the space X , called the *weak topology*, such that each p_L is continuous with respect to $\sigma(X, X')$. In turn, this implies that every $L \in X'$ is $\sigma(X, X')$ continuous.

Theorem A.46. *Let X be a locally convex topological vector space and let $E \subseteq X$. Then*

- (i) *E is bounded with respect to the (strong) topology if and only if it is weakly bounded,*
- (ii) *if E is convex, then E is closed if and only if it is weakly closed.*

Definition A.47. Given a locally convex topological vector space X , a sequence $\{x_n\}_n$ in X *converges weakly* to $x \in X$ if it converges to x with respect to the weak topology $\sigma(X, X')$.

We write $x_n \rightharpoonup x$. In view of Theorem A.27 and (A.3), we have the following result.

Proposition A.48. *Let X be a locally convex topological vector space. A sequence $\{x_n\}_n$ in X converges weakly to $x \in X$ if and only if $\lim_{n \rightarrow \infty} L(x_n) = L(x)$ for every $L \in X'$.*

Similarly, given a locally convex topological vector space X , for each $x \in X$ the function $p_x : X' \rightarrow [0, \infty)$ defined by

$$(A.4) \quad p_x(L) := |L(x)|, \quad L \in X',$$

is a seminorm. In view of Theorem A.27, the family of seminorms $\{p_x\}_{x \in X}$ generates a locally convex topology $\sigma(X', X)$ on the space X' , called the *weak star topology*, such that each p_x is continuous with respect to $\sigma(X', X)$.

Definition A.49. Let X be a locally convex topological vector space. A sequence $\{L_n\}_n$ in X' is *weakly star convergent* to L in X' if it converges to L with respect to the weak star topology $\sigma(X', X)$. We write $L_n \xrightarrow{*} L$.

In view of Theorem A.27 and (A.4), we have the following result.

Proposition A.50. *Let X be a locally convex topological vector space. A sequence $\{L_n\}_n$ in X' converges weakly star to $L \in X'$ if and only if $\lim_{n \rightarrow \infty} L_n(x) = L(x)$ for every $x \in X$.*

Theorem A.51 (Banach–Alaoglu). *If U is a neighborhood of 0 in a locally convex topological vector space X , then*

$$K := \{L \in X' : |L(x)| \leq 1 \text{ for every } x \in U\}$$

is weak star compact.

Corollary A.52. *If X is a normed space, then the closed unit ball $\{L \in X' : \|L\|_{X'} \leq 1\}$ of X' is weak star compact.*

If X is separable, it actually turns out that weak star compact sets are metrizable, and thus one can work with the friendlier notion of sequential compactness.

Theorem A.53. *Let X be a separable locally convex topological vector space and let $K \subseteq X'$ be weak star compact. Then $(K, \sigma(X', X))$ is metrizable.*

Hence, also in view of the Banach–Alaoglu theorem, we have the following:

Corollary A.54. *Let U be a neighborhood of 0 in a separable locally convex topological vector space X and let $\{L_n\}_n$ be a sequence in X' be such that*

$$\sup_n \sup_{x \in U} |L_n(x)| < \infty.$$

Then there exists a subsequence $\{L_{n_k}\}_k$ that is weak star convergent. In particular, if X is a separable Banach space and $\{L_n\}_n$ is any bounded sequence in X' , then there exists a subsequence that is weak star convergent.

For Banach spaces the converse of Theorem A.53 holds:

Theorem A.55. *Let X be a Banach space. Then the unit ball $B(0; 1)$ in X' endowed with the weak star topology is metrizable if and only if X is separable.*

Proposition A.56. *Let X be a Banach space. If a sequence $\{L_n\}_n$ in X' converges weakly star to $L \in X'$, then it is bounded and*

$$\|L\|_{X'} \leq \liminf_{n \rightarrow \infty} \|L_n\|_{X'}.$$

Proposition A.57. *Let X be a Banach space. If X' is separable, then so is X .*

The converse is false in general (take, for example, the separable space $L^1(\mathbb{R}^N)$ and its dual $L^\infty(\mathbb{R}^N)$).

We now study analogous results for the weak topology. An infinite-dimensional Banach space when endowed with the weak topology is never metrizable. However, we have the following:

Theorem A.58. *Let X be a Banach space whose dual X' is separable. Then the unit ball $B(0; 1)$ endowed with the weak topology is metrizable.*

Definition A.59. Let X be a locally convex topological vector space. A set $K \subseteq X$ is called *sequentially weakly compact* if every sequence $\{x_n\}_n$ in K has a subsequence converging weakly to a point in K .

Theorem A.60. *Let X be a Banach space. If $K \subseteq X$ is weakly compact, then it is weakly sequentially compact.*

Using Banach–Alaoglu’s theorem, one can prove the following theorem:

Theorem A.61 (Eberlein–Šmulian). *Let E be a subset of a Banach space X . Then the weak closure of E is weakly compact if and only if for every sequence $\{x_n\}_n$ in E there exists a subsequence weakly convergent to some element of X .*

As an immediate application of the Hahn–Banach theorem we have the following:

Proposition A.62. *Let X be a normed space and consider the linear operator mapping $J : X \rightarrow X''$ defined by $J(x)(L) := L(x)$, $L \in X'$. Then $\|J(x)\|_{X''} = \|x\|_X$ for all $x \in X$. In particular, J is injective and continuous.*

Definition A.63. A normed space X is *reflexive* if $J(X) = X''$.

In this case it is possible to identify X with its bidual X'' .

Theorem A.64 (Kakutani). *A Banach space is reflexive if and only if the closed unit ball $\{x \in X : \|x\| \leq 1\}$ is weakly compact.*

In view of the previous theorem and the Eberlein–Šmulian theorem we have the following corollary:

Corollary A.65. *Let X be a reflexive Banach space and let $\{x_n\}_n$ be a bounded sequence in X . Then there exists a subsequence that is weakly convergent.*

Proposition A.66. *A normed space X is reflexive if and only if X' is reflexive.*

The following proposition is used throughout the text, sometimes without mention.

Proposition A.67. *Let X be a Banach space. If a sequence $\{x_n\}_n$ in X converges weakly to $x \in X$, then it is bounded and*

$$\|x\| \leq \liminf_{n \rightarrow \infty} \|x_n\|.$$

A.6. Hilbert Spaces

Definition A.68. An *inner product* on a vector space X is a map $(\cdot, \cdot) : X \times X \rightarrow \mathbb{R}$ such that

- (i) $(x, y) = (y, x)$ for all $x, y \in X$,
- (ii) $(sx + ty, z) = s(x, z) + t(y, z)$ for all $x, y, z \in X$ and $s, t \in \mathbb{R}$,
- (iii) $(x, x) \geq 0$ for every $x \in X$ and $(x, x) = 0$ if and only if $x = 0$.

If for every $x \in X$ we define

$$(A.5) \quad \|x\| := \sqrt{(x, x)},$$

then X becomes a normed space and the following inequality, called *Cauchy's inequality*, holds

$$|(x, y)| \leq \|x\| \|y\|$$

for all $x, y \in X$. We say that X is a *Hilbert space* if it is a Banach space when endowed with the norm (A.5).

Theorem A.69. *Let $(X, \|\cdot\|)$ be a normed space. Then there exists an inner product $(\cdot, \cdot) : X \times X \rightarrow \mathbb{R}$ such that $\|x\| = \sqrt{(x, x)}$ for all $x \in X$ if and only if $\|\cdot\|$ satisfies the parallelogram law*

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

for all $x, y \in X$.

Remark A.70. If $\|\cdot\|$ satisfies the parallelogram law, then the inner product in the previous theorem is defined as

$$(x, y) := \frac{1}{4}[\|x + y\|^2 - \|x - y\|^2]$$

for all $x, y \in X$.

Example A.71. The most important example of Hilbert space in this book is the Euclidean space $\mathbb{R}^N$. Given $x = (x_1, \dots, x_N) \in \mathbb{R}^N$ and $y = (y_1, \dots, y_N) \in \mathbb{R}^N$, the *Euclidean inner product of x and y* is the real number defined by

$$x \cdot y := \sum_{i=1}^N x_i y_i.$$

Measures

ANOVA: Analysis of Value. Is your research worth anything? Significance is determined by comparing one's research with the Dull Hypothesis: $H_0: \mu_1 = \mu_2$? where H_0 : the Dull Hypothesis, μ_1 : significance of your research, μ_2 : significance of a monkey typing randomly on a typewriter in a forest where no one can hear it.

— Jorge Cham, www.phdcomics.com

In this chapter we present, without proofs, all the results in abstract measure theory and Lebesgue integration that we used in the text. This chapter draws upon [83], to which we refer for the proofs of all the results that cannot be found in classical texts (such as [61], [72], [82], [199]). Here, we prove only a few results that are used in this text and that cannot be found in [83] or in classical textbooks.

B.1. Outer Measures and Measures

Definition B.1. Let X be a nonempty set. A map $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ is called an *outer measure* if

- (i) $\mu^*(\emptyset) = 0$,
- (ii) $\mu^*(E) \leq \mu^*(F)$ for all $E \subseteq F \subseteq X$,
- (iii) $\mu^*(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} \mu^*(E_n)$ for every countable collection $\{E_n\}_n$ in $\mathcal{P}(X)$.

To construct outer measures, we usually start with a family $\mathcal{G}$ of “elementary sets” (e.g., cubes in $\mathbb{R}^N$) for which we have an “elementary notion of measure” $\rho : \mathcal{G} \rightarrow [0, \infty]$ (i.e., in $\mathbb{R}^N$ the volume of a cube $Q(x, r)$ is r^N).

Proposition B.2. Let X be a nonempty set and let $\mathcal{G} \subseteq \mathcal{P}(X)$ be such that $\emptyset \in \mathcal{G}$ and there exists $\{X_n\}_n$ in $\mathcal{G}$ with $X = \bigcup_{n=1}^{\infty} X_n$. Let $\rho : \mathcal{G} \rightarrow [0, \infty]$

be such that $\rho(\emptyset) = 0$. Then the map $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$, defined by

$$\mu^*(E) := \inf \left\{ \sum_{n=1}^{\infty} \rho(E_n) : \{E_n\}_n \text{ in } \mathcal{G}, E \subseteq \bigcup_{n=1}^{\infty} E_n \right\}, \quad E \subseteq X,$$

is an outer measure.

Remark B.3. Note that if $E \in \mathcal{G}$, then taking $E_1 := E$, $E_n := \emptyset$ for all $n \geq 2$, it follows from the definition of μ^* that $\mu^*(E) \leq \rho(E)$, with the strict inequality possible. However, if ρ is countably subadditive, that is, if $\rho(E) \leq \sum_{n=1}^{\infty} \rho(E_n)$ for all $E \subseteq \bigcup_{n=1}^{\infty} E_n$ with $E \in \mathcal{G}$ and $\{E_n\}_n$ in $\mathcal{G}$, then $\mu^* = \rho$ on $\mathcal{G}$.

The main problem with outer measures is that they are not additive on disjoint sets. To circumvent this problem, we restrict an outer measure $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ to a smaller class of subsets for which additivity of disjoint sets holds.

Definition B.4. Let X be a nonempty set and let $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ be an outer measure. A set $E \subseteq X$ is said to be μ^* -measurable if

$$\mu^*(F) = \mu^*(F \cap E) + \mu^*(F \setminus E)$$

for all sets $F \subseteq X$.

We will see below in Theorem B.13 that the restriction of μ^* to the class

$$\mathfrak{M}^* := \{E \subseteq X : E \text{ is } \mu^*\text{-measurable}\}$$

is additive, actually countably additive, and that the class $\mathfrak{M}^*$ has some important properties; to be precise, it is a σ -algebra.

Definition B.5. Let X be a nonempty set. A collection $\mathfrak{M} \subseteq \mathcal{P}(X)$ is an *algebra* if the following properties hold:

- (i) $\emptyset \in \mathfrak{M}$.
- (ii) If $E \in \mathfrak{M}$, then $X \setminus E \in \mathfrak{M}$.
- (iii) If $E_1, E_2 \in \mathfrak{M}$, then $E_1 \cup E_2 \in \mathfrak{M}$.

$\mathfrak{M}$ is said to be a σ -algebra if it satisfies (i)–(ii) and

- (iii)' if $E_n \in \mathfrak{M}$ for every $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathfrak{M}$.

If $\mathfrak{M}$ is a σ -algebra, then the pair $(X, \mathfrak{M})$ is called a *measurable space*.

Let X be a nonempty set. Given a subset $\mathcal{F} \subseteq \mathcal{P}(X)$, the smallest (in the sense of inclusion) σ -algebra that contains $\mathcal{F}$ is given by the intersection of all σ -algebras on X that contain $\mathcal{F}$.

If X is a topological space, then the *Borel σ -algebra* $\mathcal{B}(X)$ is the smallest σ -algebra containing all open subsets of X .

Definition B.6. Let X be a nonempty set. A collection $\mathfrak{M} \subseteq \mathcal{P}(X)$ is a *semiring* if the following properties hold:

- (i) $\emptyset \in \mathfrak{M}$.
- (ii) If $E, F \in \mathfrak{M}$, with $E \subseteq F$, then there exist $E_0, \dots, E_n$ in $\mathfrak{M}$, with $E_0 = E$ and $E_n = F$, such that $E_{i-i} \subseteq E_i$ and $E_i \setminus E_{i-1} \in \mathfrak{M}$ for every $i = 1, \dots, n$.
- (iii) If $E, F \in \mathfrak{M}$, then $E \cap F \in \mathfrak{M}$.

$\mathfrak{M}$ is said to be a *ring* if for every $E, F \in \mathfrak{M}$, we have that $E \cup F \in \mathfrak{M}$ and $E \setminus F \in \mathfrak{M}$.

The family of intervals $(a, b]$ is a semiring.

Definition B.7. Let X be a nonempty set and let $\mathfrak{M} \subseteq \mathcal{P}(X)$ be an *algebra*. A map $\mu : \mathfrak{M} \rightarrow [0, \infty]$ is called a *(positive) finitely additive measure* if

$$\mu(\emptyset) = 0, \quad \mu(E_1 \cup E_2) = \mu(E_1) + \mu(E_2)$$

for all $E_1, E_2 \in \mathfrak{M}$ with $E_1 \cap E_2 = \emptyset$.

Definition B.8. Let X be a nonempty set, let $\mathfrak{M} \subseteq \mathcal{P}(X)$ be a σ -algebra. A map $\mu : \mathfrak{M} \rightarrow [0, \infty]$ is called a *(positive) measure* if

$$\mu(\emptyset) = 0, \quad \mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n)$$

for every countable collection $\{E_n\}_n$ of pairwise disjoint sets in $\mathfrak{M}$. The triple $(X, \mathfrak{M}, \mu)$ is said to be a *measure space*.

Given a measure space $(X, \mathfrak{M}, \mu)$, where X is a topological space, the measure μ is called a *Borel measure* if $\mathfrak{M}$ contains $\mathcal{B}(X)$.

One of the most important properties of measures is given in the next proposition.

Proposition B.9. Let $(X, \mathfrak{M}, \mu)$ be a measure space.

- (i) If $\{E_n\}_n$ is an increasing sequence of subsets of $\mathfrak{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

- (ii) If $\{E_n\}_n$ is a decreasing sequence of subsets of $\mathfrak{M}$ and $\mu(E_1) < \infty$, then

$$\mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

Definition B.10. Let $(X, \mathfrak{M}, \mu)$ be a measure space.

- (i) The measure μ is said to be *complete* if for every $E \in \mathfrak{M}$ with $\mu(E) = 0$ it follows that every $F \subseteq E$ belongs to $\mathfrak{M}$.
- (ii) A set $E \in \mathfrak{M}$ has σ -finite μ -measure if it can be written as a countable union of measurable sets of finite measure; μ is said to be σ -finite if X has σ -finite μ -measure; μ is said to be *finite* if $\mu(X) < \infty$.
- (iii) A set $E \in \mathfrak{M}$ of positive measure is said to be an *atom* if for every $F \in \mathfrak{M}$, with $F \subseteq E$, either $\mu(F) = 0$ or $\mu(F) = \mu(E)$. The measure μ is said to be *nonatomic* if there are no atoms, that is, if for every set $E \in \mathfrak{M}$ of positive measure there exists $F \in \mathfrak{M}$, with $F \subseteq E$, such that $0 < \mu(F) < \mu(E)$. The measure μ is *purely atomic*, if every set of positive measure contains an atom.

Analogous definitions can be given for finitely additive measures.

Remark B.11. The Hausdorff measure $\mathcal{H}^s$ (see Appendix C) is not σ -finite.

The following is a useful property of finite measures.

Proposition B.12. *Let $(X, \mathfrak{M}, \mu)$ be a measure space with μ finite and let $\{E_\lambda\}_{\lambda \in \Lambda}$ be an arbitrary family of pairwise disjoint subsets in $\mathfrak{M}$. Then $\mu(E_\lambda) = 0$ for all but at most countably many $\lambda \in \Lambda$.*

Theorem B.13 (Carathéodory). *Let X be a nonempty set and let $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ be an outer measure. Then*

$$(B.1) \quad \mathfrak{M}^* := \{E \subseteq X : E \text{ is } \mu^*\text{-measurable}\}$$

is a σ -algebra and $\mu^ : \mathfrak{M}^* \rightarrow [0, \infty]$ is a complete measure.*

From Proposition B.2 and Remark B.3 we have the following.

Corollary B.14. *Let X be a nonempty set, let $\mathfrak{M} \subseteq \mathcal{P}(X)$ be an algebra, and let $\mu : \mathfrak{M} \rightarrow [0, \infty]$ be a finitely additive measure. Let μ^* be the outer measure defined in Proposition B.2 (with $\mathcal{G} := \mathfrak{M}$ and $\rho := \mu$). Then every element of $\mathfrak{M}$ is μ^* -measurable. Moreover, if μ is countably additive, then $\mu^* = \mu$ on $\mathfrak{M}$.*

Remark B.15. Note that the previous result implies that every countably additive measure $\mu : \mathfrak{M} \rightarrow [0, \infty]$ defined on an algebra $\mathfrak{M}$ may be extended as a measure to a σ -algebra that contains $\mathfrak{M}$. It actually turns out that when μ is σ -finite, this extension is unique.

Corollary B.16. *Let $I \subseteq \mathbb{R}$ be an open interval and let $\mu, \nu : \mathcal{B}(I) \rightarrow [0, \infty]$ be two measures such that $\mu((a, b]) = \nu((a, b]) < \infty$ for all intervals $(a, b]$ with $a, b \in I$. Then $\mu = \nu$.*

Remark B.17. In the previous corollary, we could have used any other family (e.g., open, closed) of intervals that generates $\mathcal{B}(I)$.

Using Carathéodory's theorem, we have created a large class of complete measures. The next problem is to understand the class $\mathfrak{M}^*$ of the μ^* -measurable sets. To do this, we consider a special class of outer measures.

Definition B.18. Let X be a metric space and let $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ be an outer measure. Then μ^* is said to be a *metric outer measure* if

$$\mu^*(E \cup F) = \mu^*(E) + \mu^*(F)$$

for all sets $E, F \subseteq X$, with $\text{dist}(E, F) > 0$.

Proposition B.19. Let X be a metric space and let $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ be a metric outer measure. Then every Borel set is μ^* -measurable.

B.2. Measurable and Integrable Functions

In this section we introduce the notions of measurable and integrable functions.

Definition B.20. Let X and Y be nonempty sets and let $\mathfrak{M}$ and $\mathfrak{N}$ be algebras on X and Y , respectively. A function $u : X \rightarrow Y$ is said to be *measurable* if $u^{-1}(F) \in \mathfrak{M}$ for every set $F \in \mathfrak{N}$.

If X and Y are topological spaces, $\mathfrak{M} := \mathcal{B}(X)$ and $\mathfrak{N} := \mathcal{B}(Y)$, then a measurable function $u : X \rightarrow Y$ will be called a *Borel function*, or *Borel measurable*. If in a topological space (usually, $\mathbb{R}^N$, $\mathbb{R}$, or $[-\infty, \infty]$) the σ -algebra is not specified, it is understood that we take the Borel σ -algebra $\mathcal{B}(X)$.

Proposition B.21. If $\mathfrak{M}$ is a σ -algebra on a set X and $\mathfrak{N}$ is the smallest σ -algebra that contains a given family $\mathcal{G}$ of subsets of a set Y , then $u : X \rightarrow Y$ is measurable if and only if $u^{-1}(F) \in \mathfrak{M}$ for every set $F \in \mathcal{G}$.

Remark B.22. In particular, if in the previous proposition Y is a topological space and $\mathfrak{N} = \mathcal{B}(Y)$, then it suffices to verify that $u^{-1}(V) \in \mathfrak{M}$ for every open set $V \subseteq Y$. Moreover, if $Y = \mathbb{R}$ (respectively, $Y = [-\infty, \infty]$), then it suffices to check that $u^{-1}((a, \infty)) \in \mathfrak{M}$ (respectively, $u^{-1}((a, \infty]) \in \mathfrak{M}$) for every $a \in \mathbb{R}$.

Remark B.23. If X and Y are topological spaces, then, in view of the previous remark, every continuous function $u : X \rightarrow Y$ is a Borel function.

Proposition B.24. Let $(X, \mathfrak{M})$, $(Y, \mathfrak{N})$, $(Z, \mathfrak{O})$ be measurable spaces and let $u : X \rightarrow Y$ and $v : Y \rightarrow Z$ be two measurable functions. Then $v \circ u : X \rightarrow Z$ is measurable.

Corollary B.25. *Let $(X, \mathfrak{M})$ be a measurable space and let $u : X \rightarrow \mathbb{R}$ (respectively, $u : X \rightarrow [-\infty, \infty]$) be a measurable function. Then u^2 , $|u|$, u^+ , u^- , cu , where $c \in \mathbb{R}$, are measurable.*

Remark B.26. If $c = 0$ and $u : X \rightarrow [-\infty, \infty]$, the function cu is defined to be identically equal to zero.

Given two measurable spaces $(X, \mathfrak{M})$ and $(Y, \mathfrak{N})$, we denote by $\mathfrak{M} \otimes \mathfrak{N} \subseteq \mathcal{P}(X \times Y)$ the smallest σ -algebra that contains all sets of the form $E \times F$, where $E \in \mathfrak{M}$, $F \in \mathfrak{N}$. Then $\mathfrak{M} \otimes \mathfrak{N}$ is called the *product σ -algebra* of $\mathfrak{M}$ and $\mathfrak{N}$.

Proposition B.27. *Let $(X, \mathfrak{M})$, $(Y_1, \mathfrak{N}_1), \dots, (Y_n, \mathfrak{N}_n)$ be measurable spaces and consider*

$$(Y_1 \times \cdots \times Y_n, \mathfrak{N}_1 \otimes \cdots \otimes \mathfrak{N}_n).$$

Then the vector-valued function $u : X \rightarrow Y_1 \times \cdots \times Y_n$ is measurable if and only if its components $u_i : X \rightarrow Y_i$ are measurable functions for all $i = 1, \dots, n$.

Corollary B.28. *Let $(X, \mathfrak{M})$ be a measurable space and let $u : X \rightarrow \mathbb{R}$ and $v : X \rightarrow \mathbb{R}$ be two measurable functions. Then $u + v$, uv , $\min\{u, v\}$, $\max\{u, v\}$ are measurable.*

Remark B.29. The previous corollary continues to hold if $\mathbb{R}$ is replaced by $[-\infty, \infty]$, provided $u + v$ are well-defined, i.e., $(u(x), v(x)) \neq (\pm\infty, \mp\infty)$ for all $x \in X$. Concerning uv , we define $(uv)(x) := 0$ whenever $u(x)$ or $v(x)$ is zero.

Proposition B.30. *Let $(X, \mathfrak{M})$ be a measurable space and let $u_n : X \rightarrow [-\infty, \infty]$, $n \in \mathbb{N}$, be measurable functions. Then the functions*

$$\sup_n u_n, \inf_n u_n, \liminf_{n \rightarrow \infty} u_n, \limsup_{n \rightarrow \infty} u_n$$

are measurable.

Remark B.31. The previous proposition uses in a crucial way the fact that $\mathfrak{M}$ is a σ -algebra.

Let $(X, \mathfrak{M}, \mu)$ be a measure space. We will see later on that the Lebesgue integration does not “see” sets of μ -measure zero. Also, several properties (existence of pointwise limit of a sequence of functions, convergence of a series of functions) do not hold at every point $x \in X$. Thus, it is important to work with functions u that are defined only on $X \setminus E$ with $\mu(E) = 0$. For this reason, we extend Definition B.20 to read as follows.

Definition B.32. Let $(X, \mathfrak{M})$ and $(Y, \mathfrak{N})$ be two measurable spaces and let $\mu : \mathfrak{M} \rightarrow [0, \infty]$ be a measure. Given a function $u : X \setminus E \rightarrow Y$ where

$\mu(E) = 0$, u is said to be *measurable over X* if $u^{-1}(F) \in \mathfrak{M}$ for every set $F \in \mathfrak{N}$.

In general, measurability of u on $X \setminus E$ does not entail the measurability of an arbitrary extension of u to X unless μ is complete. However, if we define

$$w(x) := \begin{cases} u(x) & \text{if } x \in X \setminus E, \\ y_1 & \text{if } x \in E, \end{cases}$$

where $y_1 \in Y$, then w is measurable. Hence, in general, there are extensions of u that are measurable and others that are not. The next result shows that if the measure μ is complete, then this cannot happen.

In what follows, if μ is a measure, we write that a property holds μ -a.e. on a measurable set E if there exists a measurable set $F \subseteq E$ such that $\mu(F) = 0$ and the property holds everywhere on the set $E \setminus F$.

Proposition B.33. *Let $(X, \mathfrak{M})$ and $(Y, \mathfrak{N})$ be two measurable spaces and let $u : X \rightarrow Y$ be a measurable function. Let $\mu : \mathfrak{M} \rightarrow [0, \infty]$ be a complete measure. If $v : X \rightarrow Y$ is a function such that $u(x) = v(x)$ for μ -a.e. $x \in X$, then v is measurable.*

Going back to the setting in which $u : X \setminus E \rightarrow Y$ with $\mu(E) = 0$, since Lebesgue integration does not take into account sets of measure zero, we will see that integration of u depends mostly on its measurability on $X \setminus E$.

Corollary B.34. *Let $(X, \mathfrak{M})$ be a measurable space and let $u_n : X \rightarrow [-\infty, \infty]$, $n \in \mathbb{N}$, be measurable functions. Let $\mu : \mathfrak{M} \rightarrow [0, \infty]$ be a complete measure. If there exists $\lim_{n \rightarrow \infty} u_n(x)$ for μ -a.e. $x \in X$, then $\lim_{n \rightarrow \infty} u_n$ is measurable.*

We are now in a position to introduce the notion of integral.

Definition B.35. Let X be a nonempty set and let $\mathfrak{M}$ be an algebra on X . A *simple function* is a measurable function $s : X \rightarrow \mathbb{R}$ whose range consists of finitely many points.

If $c_1, \dots, c_\ell$ are the distinct values of s , then we write

$$s = \sum_{n=1}^{\ell} c_n \chi_{E_n},$$

where $E_n := \{x \in X : s(x) = c_n\}$ and for every set $E \subseteq X$, χ_E is the *characteristic function* of the set E , i.e.,

$$\chi_E(x) := \begin{cases} 1 & \text{if } x \in E, \\ 0 & \text{otherwise.} \end{cases}$$

If μ is a finitely additive (positive) measure on X and $s \geq 0$, then for every measurable set $E \in \mathfrak{M}$ we define the *Lebesgue integral* of s over E as

$$(B.2) \quad \int_E s \, d\mu := \sum_{n=1}^{\ell} c_n \mu(E_n \cap E),$$

where if $c_n = 0$ and $\mu(E_n \cap E) = \infty$, then we use the convention $c_n \mu(E_n \cap E) := 0$.

Theorem B.36. *Let X be a nonempty set, let $\mathfrak{M}$ be an algebra on X , and let $u : X \rightarrow [0, \infty]$ be a measurable function. Then there exists a sequence $\{s_n\}_n$ of simple functions such that*

$$0 \leq s_1(x) \leq s_2(x) \leq \cdots \leq s_n(x) \rightarrow u(x)$$

for every $x \in X$. The convergence is uniform on every set on which u is bounded from above.

Corollary B.37. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u : X \rightarrow [0, \infty]$ be a measurable function. If the set $\{x \in X : u(x) > 0\}$ has σ -finite measure and u is finite μ -a.e., then there exists a sequence $\{s_n\}_n$ of simple functions, each of them bounded and vanishing outside a set of finite measure (depending on n), such that*

$$0 \leq s_1(x) \leq s_2(x) \leq \cdots \leq s_n(x) \rightarrow u(x)$$

for every $x \in X$.

In the remainder of this section, $\mathfrak{M}$ is a σ -algebra and μ a (countably additive) measure. In view of the previous theorem, if $u : X \rightarrow [0, \infty]$ is a measurable function, then we define its (*Lebesgue*) *integral* over a measurable set E as

$$\int_E u \, d\mu := \sup \left\{ \int_E s \, d\mu : s \text{ simple, } 0 \leq s \leq u \right\}.$$

We list below some basic properties of Lebesgue integration for nonnegative functions.

Proposition B.38. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u, v : X \rightarrow [0, \infty]$ be two measurable functions.*

- (i) *If $0 \leq u \leq v$, then $\int_E u \, d\mu \leq \int_E v \, d\mu$ for every measurable set E .*
- (ii) *If $c \in [0, \infty]$, then $\int_E cu \, d\mu = c \int_E u \, d\mu$ (here we set $0\infty := 0$).*
- (iii) *If $E \in \mathfrak{M}$, then $\int_E u \, d\mu = 0$ if and only if $u(x) = 0$ for μ -a.e. $x \in E$ (even if $\mu(E) = \infty$).*
- (iv) *If $E \in \mathfrak{M}$ and $\mu(E) = 0$, then $\int_E u \, d\mu = 0$ (even if $u \equiv \infty$ in E).*
- (v) *$\int_E u \, d\mu = \int_X \chi_E u \, d\mu$ for every measurable set E .*

The next results are central to the theory of integration of nonnegative functions.

Theorem B.39 (Lebesgue's monotone convergence). *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u_n : X \rightarrow [0, \infty]$ be a sequence of measurable functions such that*

$$0 \leq u_1(x) \leq u_2(x) \leq \cdots \leq u_n(x) \rightarrow u(x)$$

for every $x \in X$. Then u is measurable and

$$\lim_{n \rightarrow \infty} \int_X u_n d\mu = \int_X u d\mu.$$

Remark B.40. The previous theorem continues to hold if we assume that $u_n(x) \rightarrow u(x)$ for μ -a.e. $x \in X$. Indeed, in view of Proposition B.38(iv), it suffices to redefine u_n and u to be zero in the set of measure zero in which there is no pointwise convergence.

Corollary B.41. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u, v : X \rightarrow [0, \infty]$ be two measurable functions. Then*

$$\int_X (u + v) d\mu = \int_X u d\mu + \int_X v d\mu.$$

Corollary B.42. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u_n : X \rightarrow [0, \infty]$ be a sequence of measurable functions. Then*

$$\sum_{n=1}^{\infty} \int_X u_n d\mu = \int_X \sum_{n=1}^{\infty} u_n d\mu.$$

Example B.43. Given a doubly indexed sequence $\{a_{k,n}\}_{k,n \in \mathbb{N}}$, with $a_{k,n} \geq 0$ for all $n, k \in \mathbb{N}$, we have

$$\sum_{n=1}^{\infty} \sum_{k=1}^{\infty} a_{k,n} = \sum_{k=1}^{\infty} \sum_{n=1}^{\infty} a_{k,n}.$$

To see this, it suffices to consider $X = \mathbb{N}$ with the *counting measure*¹ and to define $u_n : \mathbb{N} \rightarrow [0, \infty]$ by $u_n(k) := a_{k,n}$. Then

$$\int_X u_n d\mu = \sum_{k=1}^{\infty} a_{k,n},$$

and the result now follows from the previous corollary.

¹Given a set X , the *counting measure* $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ is defined by

$$\mu(E) := \begin{cases} \text{the number of elements of } E & \text{if } E \text{ is a finite set,} \\ \infty & \text{otherwise} \end{cases}$$

for every $E \subseteq X$.

Lemma B.44 (Fatou). *Let $(X, \mathfrak{M}, \mu)$ be a measure space. If $u_n : X \rightarrow [0, \infty]$ is a sequence of measurable functions, then*

$$\int_X \liminf_{n \rightarrow \infty} u_n d\mu \leq \liminf_{n \rightarrow \infty} \int_X u_n d\mu.$$

In order to extend the notion of integral to functions of arbitrary sign, consider $u : X \rightarrow [-\infty, \infty]$. Note that $u = u^+ - u^-$ and $|u| = u^+ + u^-$ and that u is measurable if and only if u^+ and u^- are measurable. Also, if u is bounded, then so are u^+ and u^- , and in view of Theorem B.36, u is then the uniform limit of a sequence of simple functions.

Definition B.45. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u : X \rightarrow [-\infty, \infty]$ be a measurable function. Given a measurable set $E \in \mathfrak{M}$, if at least one of the two integrals $\int_E u^+ d\mu$ and $\int_E u^- d\mu$ is finite, then we define the (*Lebesgue*) *integral of u over the measurable set E* by

$$\int_E u d\mu := \int_E u^+ d\mu - \int_E u^- d\mu.$$

If both $\int_E u^+ d\mu$ and $\int_E u^- d\mu$ are finite, then u is said to be (*Lebesgue*) *integrable over the measurable set E* .

A measurable function $u : X \rightarrow [-\infty, \infty]$ is Lebesgue integrable over the measurable set E if and only if $\int_E |u| d\mu < \infty$.

Remark B.46. If $(X, \mathfrak{M}, \mu)$ is a measure space and $u : X \rightarrow [-\infty, \infty]$ is Lebesgue integrable, then the set $\{x \in X : |u(x)| = \infty\}$ has measure zero, while the set $\{x \in X : |u(x)| > 0\}$ is σ -finite.

If $(X, \mathfrak{M}, \mu)$ is a measure space, with X a topological space, and if $\mathfrak{M}$ contains $\mathcal{B}(X)$, then $u : X \rightarrow [-\infty, \infty]$ is said to be *locally integrable* if it is Lebesgue integrable over every compact set.

Proposition B.47. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u, v : X \rightarrow [-\infty, \infty]$ be two integrable functions.*

(i) *If $\alpha, \beta \in \mathbb{R}$, then $\alpha u + \beta v$ is integrable and*

$$\int_X (\alpha u + \beta v) d\mu = \alpha \int_X u d\mu + \beta \int_X v d\mu.$$

(ii) *If $u(x) = v(x)$ for μ -a.e. $x \in X$, then*

$$(B.3) \quad \int_X u d\mu = \int_X v d\mu.$$

(iii)

$$\left| \int_X u d\mu \right| \leq \int_X |u| d\mu.$$

Part (ii) of the previous proposition motivates the next definition.

Let $(X, \mathfrak{M}, \mu)$ be a measure space. If $F \in \mathfrak{M}$ is such that $\mu(F) = 0$ and $u : X \setminus F \rightarrow [-\infty, \infty]$ is a measurable function in the sense of Definition B.32, then we define the (Lebesgue) integral of u over the measurable set E as the Lebesgue integral of the function

$$v(x) := \begin{cases} u(x) & \text{if } x \in X \setminus F, \\ 0 & \text{otherwise,} \end{cases}$$

provided $\int_E v d\mu$ is well-defined. Note that in this case $\int_E v d\mu = \int_E \tilde{v} d\mu$, where

$$\tilde{v}(x) := \begin{cases} u(x) & \text{if } x \in X \setminus F, \\ w(x) & \text{otherwise,} \end{cases}$$

and w is an arbitrary measurable function defined on F . If the measure μ is complete, then $\int_E v d\mu$ is well-defined if and only if $\int_{E \setminus F} u d\mu$ is well-defined.

For functions of arbitrary sign we have the following convergence result.

Theorem B.48 (Lebesgue's dominated convergence). *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u_n : X \rightarrow [-\infty, \infty]$ be a sequence of measurable functions such that $\lim_{n \rightarrow \infty} u_n(x) = u(x)$ for μ -a.e. $x \in X$. If there exists a Lebesgue integrable function v such that $|u_n(x)| \leq v(x)$ for μ -a.e. $x \in X$ and all $n \in \mathbb{N}$, then u is Lebesgue integrable and*

$$\lim_{n \rightarrow \infty} \int_X |u_n - u| d\mu = 0.$$

In particular,

$$\lim_{n \rightarrow \infty} \int_X u_n d\mu = \int_X u d\mu.$$

Corollary B.49. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u_n : X \rightarrow [-\infty, \infty]$ be a sequence of measurable functions. If $\sum_{n=1}^{\infty} \int_X |u_n| d\mu < \infty$, then the series $\sum_{n=1}^{\infty} u_n(x)$ converges for μ -a.e. $x \in X$, the function $u(x) := \sum_{n=1}^{\infty} u_n(x)$, defined for μ -a.e. $x \in X$, is integrable, and*

$$\sum_{n=1}^{\infty} \int_X u_n d\mu = \int_X \sum_{n=1}^{\infty} u_n d\mu.$$

Theorem B.50 (Jensen's inequality). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex and let $(X, \mathfrak{M}, \mu)$ be a measure space, where $\mu(X) = 1$. Then for every integrable function $u : X \rightarrow \mathbb{R}$,*

$$(B.4) \quad f\left(\int_X u d\mu\right) \leq \int_X f \circ u d\mu.$$

B.3. Integrals Depending on a Parameter

In this section we study the continuity and the differentiability of functions of the type

$$F(y) = \int_X f(x, y) d\mu(x), \quad y \in Y.$$

We begin by studying continuity.

Theorem B.51. *Let $(X, \mathfrak{M}, \mu)$ be a measure space, let Y be a metric space, and let $f : X \times Y \rightarrow \mathbb{R}$ be a function. Assume that for each fixed $y \in Y$ the function $x \in X \mapsto f(x, y)$ is measurable and that there exists $y_0 \in Y$ such that*

$$\lim_{y \rightarrow y_0} f(x, y) = f(x, y_0)$$

for every $x \in X$. Assume also that there exists an integrable function $g : X \rightarrow [0, \infty)$ such that $|f(x, y)| \leq g(x)$ for μ -a.e. $x \in X$ and for all $y \in Y$. Then the function $F : Y \rightarrow \mathbb{R}$, defined by

$$F(y) = \int_X f(x, y) d\mu(x), \quad y \in Y,$$

is well-defined and is continuous at y_0 .

Next we study the differentiability of F .

Theorem B.52. *Let Y be an interval of $\mathbb{R}$ and assume that for each fixed $x \in X$ the function $y \in Y \mapsto f(x, y)$ is differentiable and that for each fixed $y \in Y$ the functions $x \in X \mapsto f(x, y)$ and $x \in X \mapsto \frac{\partial f}{\partial y}(x, y)$ are measurable. Assume also that for some $y_0 \in Y$ the function $x \in X \mapsto f(x, y_0)$ is integrable and that there exists an integrable function $h : X \rightarrow [0, \infty)$ such that $|\frac{\partial f}{\partial y}(x, y)| \leq h(x)$ for μ -a.e. $x \in X$ and for all $y \in Y$. Then the function $F : Y \rightarrow \mathbb{R}$, defined by*

$$F(y) = \int_X f(x, y) d\mu(x), \quad y \in Y,$$

is well-defined and differentiable, with

$$F'(y) = \int_X \frac{\partial f}{\partial y}(x, y) d\mu(x).$$

The next exercise shows that, without the integrability of g , Theorem B.51 fails.

Exercise B.53. Consider the function

$$f(x, y) = \begin{cases} \frac{|y| - |x|}{y^2} & \text{if } |y| < |x|, \\ 0 & \text{if } |y| \geq |x|. \end{cases}$$

Prove that the function $F(y) = \int_{\mathbb{R}} f(x, y) dx$, $y \in \mathbb{R}$, is well-defined and is not continuous at $y = 0$.

B.4. Product Spaces

We recall that, given two measurable spaces $(X, \mathfrak{M})$ and $(Y, \mathfrak{N})$, we denote by $\mathfrak{M} \otimes \mathfrak{N} \subseteq \mathcal{P}(X \times Y)$ the smallest σ -algebra that contains all sets of the form $E \times F$, where $E \in \mathfrak{M}$, $F \in \mathfrak{N}$. Then $\mathfrak{M} \otimes \mathfrak{N}$ is called the *product σ -algebra* of $\mathfrak{M}$ and $\mathfrak{N}$.

Exercise B.54. Let X and Y be topological spaces and let $\mathcal{B}(X)$ and $\mathcal{B}(Y)$ be their respective Borel σ -algebras. Prove that $\mathcal{B}(X) \otimes \mathcal{B}(Y) \subseteq \mathcal{B}(X \times Y)$. Show also that if X and Y are separable metric spaces, then

$$\mathcal{B}(X) \otimes \mathcal{B}(Y) = \mathcal{B}(X \times Y).$$

In particular, $\mathcal{B}(\mathbb{R}^N) = \mathcal{B}(\mathbb{R}) \otimes \cdots \otimes \mathcal{B}(\mathbb{R})$.

Let $(X, \mathfrak{M}, \mu)$ and $(Y, \mathfrak{N}, \nu)$ be two measure spaces. For every $E \in X \times Y$ define

$$(B.5) \quad (\mu \times \nu)^*(E) := \inf \left\{ \sum_{n=1}^{\infty} \mu(F_n) \nu(G_n) : F_n \in \mathfrak{M}, G_n \in \mathfrak{N}, \right. \\ \left. E \subseteq \bigcup_{n=1}^{\infty} (F_n \times G_n) \right\}.$$

By Proposition B.2, $(\mu \times \nu)^* : \mathcal{P}(X) \rightarrow [0, \infty]$ is an outer measure, called the *product outer measure* of μ and ν . By Carathéodory's theorem, the restriction of $(\mu \times \nu)^*$ to the σ -algebra $\mathfrak{M} \times \mathfrak{N}$ of $(\mu \times \nu)^*$ -measurable sets is a complete measure, denoted by $\mu \times \nu$ and called the *product measure* of μ and ν .

Note that $\mathfrak{M} \times \mathfrak{N}$ is, in general, larger than the product σ -algebra $\mathfrak{M} \otimes \mathfrak{N}$.

Theorem B.55. Let $(X, \mathfrak{M}, \mu)$ and $(Y, \mathfrak{N}, \nu)$ be two measure spaces.

(i) If $F \in \mathfrak{M}$ and $G \in \mathfrak{N}$, then $F \times G$ is $(\mu \times \nu)^*$ -measurable and

$$(B.6) \quad (\mu \times \nu)(F \times G) = \mu(F) \nu(G).$$

(ii) If μ and ν are complete and E has σ -finite $(\mu \times \nu)$ -measure, then for μ -a.e. $x \in X$ the section $E_x := \{y \in Y : (x, y) \in E\}$ belongs to the σ -algebra $\mathfrak{N}$ and for ν -a.e. $y \in Y$ the section $E_y := \{x \in X : (x, y) \in E\}$ belongs to the σ -algebra $\mathfrak{M}$. Moreover, the functions $y \mapsto \mu(E_y)$ and $x \mapsto \nu(E_x)$ are measurable and

$$(\mu \times \nu)(E) = \int_Y \mu(E_y) d\nu(y) = \int_X \nu(E_x) d\mu(x).$$

The previous result is a particular case of Tonelli's theorem in the case that $u = \chi_E$.

Theorem B.56 (Tonelli). *Let $(X, \mathfrak{M}, \mu)$ and $(Y, \mathfrak{N}, \nu)$ be two measure spaces. Assume that μ and ν are complete and σ -finite and let $u : X \times Y \rightarrow [0, \infty]$ be an $(\mathfrak{M} \times \mathfrak{N})$ -measurable function. Then for μ -a.e. $x \in X$ the function $u(x, \cdot)$ is measurable and the function $\int_Y u(\cdot, y) d\nu(y)$ is measurable. Similarly, for ν -a.e. $y \in Y$ the function $u(\cdot, y)$ is measurable and the function $\int_X u(x, \cdot) d\mu(x)$ is measurable. Moreover,*

$$\begin{aligned} \int_{X \times Y} u(x, y) d(\mu \times \nu)(x, y) &= \int_X \left(\int_Y u(x, y) d\nu(y) \right) d\mu(x) \\ &= \int_Y \left(\int_X u(x, y) d\mu(x) \right) d\nu(y). \end{aligned}$$

Throughout this book, for simplicity, we write

$$\int_X \int_Y u(x, y) d\nu(y) d\mu(x) := \int_X \left(\int_Y u(x, y) d\nu(y) \right) d\mu(x).$$

Exercise B.57. Prove that in the case that $u : X \times Y \rightarrow [0, \infty]$ is $(\mathfrak{M} \otimes \mathfrak{N})$ -measurable, then Tonelli's theorem still holds even if the measures μ and ν are not complete, and the statements are satisfied for every $x \in X$ and $y \in Y$ (as opposed to for μ -a.e. $x \in X$ and for ν -a.e. $y \in Y$).

Theorem B.58 (Fubini). *Let $(X, \mathfrak{M}, \mu)$ and $(Y, \mathfrak{N}, \nu)$ be two measure spaces. Assume that μ and ν are complete and let $u : X \times Y \rightarrow [-\infty, \infty]$ be $(\mu \times \nu)$ -integrable. Then for μ -a.e. $x \in X$ the function $u(x, \cdot)$ is ν -integrable and the function $\int_Y u(\cdot, y) d\nu(y)$ is μ -integrable.*

Similarly, for ν -a.e. $y \in Y$ the function $u(\cdot, y)$ is μ -integrable and the function $\int_X u(x, \cdot) d\mu(x)$ is ν -integrable. Moreover,

$$\begin{aligned} \int_{X \times Y} u(x, y) d(\mu \times \nu)(x, y) &= \int_X \left(\int_Y u(x, y) d\nu(y) \right) d\mu(x) \\ &= \int_Y \left(\int_X u(x, y) d\mu(x) \right) d\nu(y). \end{aligned}$$

Exercise B.59. Prove that in the case that $u : X \times Y \rightarrow [-\infty, \infty]$ is $(\mathfrak{M} \otimes \mathfrak{N})$ -measurable, then Fubini's theorem still holds even if the measures μ and ν are not complete.

The following result is a consequence of Tonelli's theorem.

Theorem B.60. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u : X \rightarrow [0, \infty]$ be a measurable function. Then*

$$(B.7) \quad \int_X u d\mu = \int_0^\infty \mu(\{x \in X : u(x) > t\}) dt.$$

B.5. Radon–Nikodym’s and Lebesgue’s Decomposition Theorems

Definition B.61. Let $(X, \mathfrak{M})$ be a measurable space and let $\mu, \nu : \mathfrak{M} \rightarrow [0, \infty]$ be two measures. The measure ν is said to be *absolutely continuous with respect to μ* , and we write $\nu \ll \mu$, if for every $E \in \mathfrak{M}$ with $\mu(E) = 0$ we have $\nu(E) = 0$.

Proposition B.62. Let $(X, \mathfrak{M})$ be a measurable space and let $\mu, \nu : \mathfrak{M} \rightarrow [0, \infty]$ be two measures with ν finite. Then ν is absolutely continuous with respect to μ if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$(B.8) \quad \nu(E) \leq \varepsilon$$

for every measurable set $E \subseteq X$ with $\mu(E) \leq \delta$.

Proposition B.63. Let $(X, \mathfrak{M})$ be a measurable space and let $\mu, \nu : \mathfrak{M} \rightarrow [0, \infty]$ be two measures. For every $E \in \mathfrak{M}$ define

$$(B.9) \quad \nu_{ac}(E) := \sup \left\{ \int_E u \, d\mu : u : X \rightarrow [0, \infty] \text{ is measurable,} \right. \\ \left. \int_F u \, d\mu \leq \nu(F) \text{ for all } F \subseteq E, F \in \mathfrak{M} \right\}.$$

Then ν_{ac} is a measure, with $\nu_{ac} \ll \mu$, and for each $E \in \mathfrak{M}$ the supremum in the definition of ν_{ac} is actually attained by a function u admissible for $\nu_{ac}(E)$. Moreover, if ν_{ac} is σ -finite, then u may be chosen independently of the set E .

Theorem B.64 (Radon–Nikodym). Let $(X, \mathfrak{M})$ be a measurable space and let $\mu, \nu : \mathfrak{M} \rightarrow [0, \infty]$ be two measures, with μ σ -finite and ν absolutely continuous with respect to μ . Then there exists a measurable function $u : X \rightarrow [0, \infty]$ such that $\nu(E) = \int_E u \, d\mu$ for every $E \in \mathfrak{M}$. The function u is unique up to a set of μ -measure zero.

The function u is called the *Radon–Nikodym derivative* of ν with respect to μ , and we write $u = \frac{d\nu}{d\mu}$.

Definition B.65. Let $(X, \mathfrak{M})$ be a measurable space. Two measures $\mu, \nu : \mathfrak{M} \rightarrow [0, \infty]$ are said to be *mutually singular*, and we write $\nu \perp \mu$, if there exist two disjoint sets $X_\mu, X_\nu \in \mathfrak{M}$ such that $X = X_\mu \cup X_\nu$ and for every $E \in \mathfrak{M}$ we have

$$\mu(E) = \mu(E \cap X_\mu), \quad \nu(E) = \nu(E \cap X_\nu).$$

Proposition B.66. Let $(X, \mathfrak{M})$ be a measurable space and let $\mu, \nu : \mathfrak{M} \rightarrow [0, \infty]$ be two measures. For every $E \in \mathfrak{M}$ define

$$(B.10) \quad \nu_s(E) := \sup \{ \nu(F) : F \subseteq E, F \in \mathfrak{M}, \mu(F) = 0 \}.$$

Then ν_s is a measure and for each $E \in \mathfrak{M}$ the supremum in the definition of ν_s is actually attained by a measurable set.

Moreover, if ν_s is σ -finite, then $\nu_s \perp \mu$.

Theorem B.67 (Lebesgue's decomposition). *Let $(X, \mathfrak{M})$ be a measurable space and let $\mu, \nu : \mathfrak{M} \rightarrow [0, \infty]$ be two measures, with μ σ -finite. Then*

$$(B.11) \quad \nu = \nu_{ac} + \nu_s,$$

where ν_{ac} and ν_s are defined in (B.9) and (B.10), respectively, and $\nu_{ac} \ll \mu$. Moreover, if ν is σ -finite, then $\nu_s \perp \mu$ and the decomposition (B.11) is unique, that is, if $\nu = \overline{\nu}_{ac} + \overline{\nu}_s$, for some measures $\overline{\nu}_{ac}, \overline{\nu}_s$, with $\overline{\nu}_{ac} \ll \mu$ and $\overline{\nu}_s \perp \mu$, then $\nu_{ac} = \overline{\nu}_{ac}$ and $\nu_s = \overline{\nu}_s$.

B.6. Signed Measures

Definition B.68. Let X be a nonempty set and let $\mathfrak{M}$ be an algebra on X . A *finitely additive signed measure* is a function $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty]$ such that

- (i) $\lambda(\emptyset) = 0$,
- (ii) λ takes at most one of the two values ∞ and $-\infty$, that is, either $\lambda : \mathfrak{M} \rightarrow (-\infty, \infty]$ or $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty)$,
- (iii) if $E_1, E_2 \in \mathfrak{M}$ are disjoint sets, then $\lambda(E_1 \cup E_2) = \lambda(E_1) + \lambda(E_2)$.

Definition B.69. Let $(X, \mathfrak{M})$ be a measurable space. A *signed measure* is a function $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty]$ such that

- (i) $\lambda(\emptyset) = 0$,
- (ii) λ takes at most one of the two values ∞ and $-\infty$, that is, either $\lambda : \mathfrak{M} \rightarrow (-\infty, \infty]$ or $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty)$,
- (iii) for every countable collection $\{E_n\}_n$ of pairwise disjoint sets in $\mathfrak{M}$,

$$\lambda\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \lambda(E_n).$$

Proposition B.70. *Let $(X, \mathfrak{M})$ be a measurable space and let $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty]$ be a signed measure. For every $E \in \mathfrak{M}$ define*

$$(B.12) \quad \lambda^+(E) := \sup\{\lambda(F) : F \subseteq E, F \in \mathfrak{M}\},$$

$$(B.13) \quad \lambda^-(E) := -\inf\{\lambda(F) : F \subseteq E, F \in \mathfrak{M}\}.$$

Then λ^+ and λ^- are measures. Moreover, if $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty)$, then for every $E \in \mathfrak{M}$ we have

$$(B.14) \quad \lambda^+(E) = \sup\{\lambda(F) : F \subseteq E, F \in \mathfrak{M}, \lambda^-(F) = 0\},$$

λ^+ is finite, and $\lambda = \lambda^+ - \lambda^-$.

Theorem B.71 (Jordan's decomposition). *Let $(X, \mathfrak{M})$ be a measurable space and let $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty]$ be a signed measure. Then there exists a unique pair λ^+, λ^- of mutually singular (nonnegative) measures, one of which is finite, such that $\lambda = \lambda^+ - \lambda^-$.*

The measures λ^+, λ^- are called, respectively, the *upper* and *lower variation* of λ , while the measure $|\lambda| := \lambda^+ + \lambda^-$ is called the *total variation* of λ . We say that λ is *bounded*, or *finite*, if the measure $|\lambda|$ is finite.

Proposition B.72. *Let $(X, \mathfrak{M})$ be a measurable space and let $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty]$ be a signed measure. Then for every $E \in \mathfrak{M}$,*

$$|\lambda|(E) = \sup \left\{ \sum_{n=1}^{\infty} |\lambda(E_n)| \right\},$$

where the supremum is taken over all partitions $\{E_n\}_n$ in $\mathfrak{M}$ of E .

Definition B.73. Let $(X, \mathfrak{M})$ be a measurable space and let Y be a normed space. A set function $\lambda : \mathfrak{M} \rightarrow Y$ is a *vector-valued measure* if $\lambda(\emptyset) = 0$ and for every countable collection $\{E_n\}_n$ of pairwise disjoint sets in $\mathfrak{M}$,

$$\lambda \left(\bigcup_{n=1}^{\infty} E_n \right) = \sum_{n=1}^{\infty} \lambda(E_n).$$

Proposition B.74. *Let $(X, \mathfrak{M})$ be a measurable space, let $(Y, \|\cdot\|_Y)$ be a normed space and let $\lambda : \mathfrak{M} \rightarrow Y$ be a vector-valued measure. Then the function $\|\lambda\| : \mathfrak{M} \rightarrow [0, \infty]$, defined by*

$$\|\lambda\|(E) := \sup \left\{ \sum_{n=1}^{\infty} \|\lambda(E_n)\|_Y \right\}, \quad E \in \mathfrak{M},$$

where the supremum is taken over all partitions $\{E_n\}_n$ in $\mathfrak{M}$ of E , is a measure.

Definition B.75. Let $(X, \mathfrak{M})$ be a measurable space, let $\mu : \mathfrak{M} \rightarrow [0, \infty]$ be a measure, and let $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty]$ and $\rho : \mathfrak{M} \rightarrow [-\infty, \infty]$ be signed measures.

- (i) λ is said to be *absolutely continuous with respect to μ* , and we write $\lambda \ll \mu$, if $\lambda(E) = 0$ whenever $E \in \mathfrak{M}$ and $\mu(E) = 0$.
- (ii) λ and ρ are said to be *mutually singular*, and we write $\lambda \perp \rho$, if there exist two disjoint sets $X_\lambda, X_\rho \in \mathfrak{M}$ such that $X = X_\lambda \cup X_\rho$ and for every $E \in \mathfrak{M}$ we have

$$\lambda(E) = \lambda(E \cap X_\lambda), \quad \rho(E) = \rho(E \cap X_\rho).$$

Note that if $\lambda \ll \mu$, then $\lambda^+ \ll \mu$ and $\lambda^- \ll \mu$.

Theorem B.76 (Lebesgue's decomposition). *Let $(X, \mathfrak{M})$ be a measurable space, let $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty]$ be a signed measure, and let $\mu : \mathfrak{M} \rightarrow [0, \infty]$ be a σ -finite measure. Then $\lambda = \lambda_{ac} + \lambda_s$, with $\lambda_{ac} \ll \mu$, and there exists a measurable function $u : X \rightarrow [-\infty, \infty]$ such that $\lambda_{ac}(E) = \int_E u d\mu$ for all $E \in \mathfrak{M}$. Moreover, if λ is σ -finite, then $\lambda_s \perp \mu$ and the decomposition is unique; that is, if $\lambda = \bar{\lambda}_{ac} + \bar{\lambda}_s$, for some signed measures $\bar{\lambda}_{ac}, \bar{\lambda}_s$, with $\bar{\lambda}_{ac} \ll \mu$ and $\bar{\lambda}_s \perp \mu$, then $\lambda_{ac} = \bar{\lambda}_{ac}$ and $\lambda_s = \bar{\lambda}_s$.*

It is implicit in the statement of the theorem that $\int_E u d\mu$ is well-defined for every $E \in \mathfrak{M}$, which implies that u^+ or u^- is Lebesgue integrable.

We call λ_{ac} and λ_s , respectively, the *absolutely continuous part* and the *singular part* of λ with respect to μ , and often we write $u = \frac{d\lambda_{ac}}{d\mu}$.

B.7. L^p Spaces

Let $(X, \mathfrak{M}, \mu)$ be a measure space. Given two measurable functions $u, v : X \rightarrow [-\infty, \infty]$, we say that u is *equivalent* to v and we write

$$(B.15) \quad u \sim v \text{ if } u(x) = v(x) \text{ for } \mu\text{-a.e. } x \in X.$$

Note that $\sim$ is an equivalence relation in the class of measurable functions. With an abuse of notation, from now on we identify a measurable function $u : X \rightarrow [-\infty, \infty]$ with its equivalence class $[u]$.

Definition B.77. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $0 < p < \infty$. Then

$$L^p(X, \mathfrak{M}, \mu) := \{u : X \rightarrow [-\infty, \infty] : u \text{ is measurable, } \|u\|_{L^p(X, \mathfrak{M}, \mu)} < \infty\},$$

where

$$\|u\|_{L^p(X, \mathfrak{M}, \mu)} := \left(\int_X |u|^p d\mu \right)^{1/p}.$$

If $p = \infty$, then

$$L^\infty(X, \mathfrak{M}, \mu) := \{u : X \rightarrow [-\infty, \infty] : u \text{ is measurable, } \|u\|_{L^\infty(X, \mathfrak{M}, \mu)} < \infty\},$$

where $\|u\|_{L^\infty(X, \mathfrak{M}, \mu)}$ is the *essential supremum* $\text{ess sup } |u|$ of the function $|u|$, that is,

$$\text{ess sup } |u| := \inf \{t \geq 0 : |u(x)| \leq t \text{ for } \mu\text{-a.e. } x \in X\}.$$

For simplicity, and when there is no possibility of confusion, we denote the spaces $L^p(X, \mathfrak{M}, \mu)$ simply by $L^p(X, \mu)$ or $L^p(X)$ and the norms $\|u\|_{L^p(X, \mathfrak{M}, \mu)}$ by $\|u\|_{L^p(X)}$, or $\|u\|_{L^p}$.

We denote by $L^p(X, \mathfrak{M}, \mu; \mathbb{R}^m)$ (or more simply by $L^p(X; \mathbb{R}^m)$) the space of all functions $u : X \rightarrow \mathbb{R}^m$ whose components are in $L^p(X, \mathfrak{M}, \mu)$. We will

endow $L^p(X, \mathfrak{M}, \mu; \mathbb{R}^m)$ with the norm

$$\|u\|_{L^p(X, \mathfrak{M}, \mu; \mathbb{R}^m)} := \left(\sum_{i=1}^m \|u_i\|_{L^p(X, \mathfrak{M}, \mu)}^p \right)^{1/p}$$

for $1 \leq p < \infty$ and

$$\|u\|_{L^\infty(X, \mathfrak{M}, \mu; \mathbb{R}^m)} := \max_i \|u_i\|_{L^\infty(X, \mathfrak{M}, \mu)}$$

if $p = \infty$. For $1 \leq p < \infty$ sometimes it will be more convenient to use the equivalent norm

$$\|u\|_{L^p(X, \mathfrak{M}, \mu; \mathbb{R}^m)} := \left(\int_X \|u\|^p d\mu \right)^{1/p}.$$

Given $1 \leq p \leq \infty$, the *Hölder conjugate exponent* of p is the extended real number $p' \in [1, \infty]$, defined by

$$p' := \begin{cases} \frac{p}{p-1} & \text{if } 1 < p < \infty, \\ \infty & \text{if } p = 1, \\ 1 & \text{if } p = \infty. \end{cases}$$

Note that, with an abuse of notation, we have

$$\frac{1}{p} + \frac{1}{p'} = 1.$$

Theorem B.78 (Hölder's inequality). *Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $1 \leq p \leq \infty$, and let p' be its Hölder conjugate exponent. If $u, v : X \rightarrow [-\infty, \infty]$ are measurable functions, then*

$$(B.16) \quad \|uv\|_{L^1} \leq \|u\|_{L^p} \|v\|_{L^{p'}}.$$

In particular, if $u \in L^p(X)$ and $v \in L^{p'}(X)$, then $uv \in L^1(X)$.

The proof of Hölder's inequality for $1 \leq p < \infty$ is based on *Young's inequality*

$$(B.17) \quad ab \leq \frac{1}{p} a^p + \frac{1}{p'} b^{p'},$$

which holds for all $a, b \geq 0$ and whose proof is left as an exercise.

Exercise B.79. Let $(X, \mathfrak{M}, \mu)$ be a measure space.

- (i) Let $1 \leq p_1, \dots, p_n, p \leq \infty$, with $\frac{1}{p_1} + \dots + \frac{1}{p_n} = \frac{1}{p}$, and $u_i \in L^{p_i}(X)$, $i = 1, \dots, n$. Prove that

$$\left\| \prod_{i=1}^n u_i \right\|_{L^p} \leq \prod_{i=1}^n \|u_i\|_{L^{p_i}}.$$

(ii) Let $u : X \rightarrow \mathbb{R}$ be a measurable function. Prove that

$$\|u\|_{L^q} \leq \|u\|_{L^p}^\theta \|u\|_{L^r}^{1-\theta},$$

where $0 < p \leq q \leq r \leq \infty$ and $\frac{1}{q} = \frac{\theta}{p} + \frac{1-\theta}{r}$.

(iii) Let $u : X \rightarrow \mathbb{R}$ be a measurable function. Prove that for every $R > 0$,

$$\|u\|_{L^q} \leq R^{-1/q+1/p} \|u\|_{L^p} + R^{-1/q+1/r} \|u\|_{L^r},$$

where $0 < p \leq q \leq r \leq \infty$.

Proposition B.80. Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $1 \leq p \leq \infty$, and let p' be the Hölder conjugate exponent of p . If $p = \infty$, assume that μ is σ -finite. Then for every function $u \in L^p(X)$,

(B.18)

$$\|u\|_{L^p} = \begin{cases} \max \left\{ \int_X |uv| d\mu : v \in L^{p'}(X), \|v\|_{L^{p'}} \leq 1 \right\} & \text{if } p < \infty, \\ \sup \left\{ \int_X |uv| d\mu : v \in L^1(X), \|v\|_{L^1} \leq 1 \right\} & \text{if } p = \infty. \end{cases}$$

Corollary B.81. Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $1 \leq p \leq \infty$, and let p' be the Hölder conjugate exponent of p . Assume that μ is σ -finite. Then for every measurable function $u : X \rightarrow \mathbb{R}$,

$$(B.19) \quad \|u\|_{L^p} = \sup \left\{ \int_X |uv| d\mu : v \in L^{p'}(X), \|v\|_{L^{p'}} \leq 1 \right\}.$$

Exercise B.82. Prove the previous corollary.

As a corollary of Proposition B.80 we have the following result.

Corollary B.83 (Minkowski's inequality for integrals). Let $(X, \mathfrak{M}, \mu)$ and $(Y, \mathfrak{N}, \nu)$ be two measure spaces. Assume that μ and ν are complete and σ -finite. Let $u : X \times Y \rightarrow [0, \infty]$ be an $(\mathfrak{M} \times \mathfrak{N})$ -measurable function and let $1 \leq p \leq \infty$. Then

$$\left\| \int_X |u(x, \cdot)| d\mu(x) \right\|_{L^p(Y, \mathfrak{N}, \nu)} \leq \int_X \|u(x, \cdot)\|_{L^p(Y, \mathfrak{N}, \nu)} d\mu(x).$$

Exercise B.84. Prove the previous corollary. Hint: Use Proposition B.80 and Tonelli's theorem.

We now turn to the relation between different L^p spaces.

Theorem B.85. Let $(X, \mathfrak{M}, \mu)$ be a measure space. Suppose that $1 \leq p < q < \infty$. Then

- (i) $L^p(X)$ is not contained in $L^q(X)$ if and only if X contains measurable sets of arbitrarily small positive measure²,

²By this we mean that for every $\varepsilon > 0$ there is a measurable set of positive measure less than ε .

- (ii) $L^q(X)$ is not contained in $L^p(X)$ if and only if X contains measurable sets of arbitrarily large finite measure³.

Corollary B.86. *Let $(X, \mathfrak{M}, \mu)$ be a measure space. Suppose that $1 \leq p < q \leq \infty$. If $\mu(X) < \infty$, then $L^q(X) \subseteq L^p(X)$.*

Theorem B.87 (Minkowski's inequality). *Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $1 \leq p \leq \infty$, and let $u, v : X \rightarrow [-\infty, \infty]$ be measurable functions. Then,*

$$\|u + v\|_{L^p} \leq \|u\|_{L^p} + \|v\|_{L^p}.$$

In particular, if $u, v \in L^p(X)$, then $u + v \in L^p(X)$.

By identifying functions with their equivalence classes $[u]$, it follows from Minkowski's inequality that $\|\cdot\|_{L^p}$ is a norm on $L^p(X)$.

Theorem B.88. *Let $(X, \mathfrak{M}, \mu)$ be a measure space. Then $L^p(X)$ is a Banach space for $1 \leq p \leq \infty$.*

Next we study some density results for $L^p(X)$ spaces.

Theorem B.89. *Let $(X, \mathfrak{M}, \mu)$ be a measure space. Then the family of all simple functions in $L^p(X)$ is dense in $L^p(X)$ for $1 \leq p \leq \infty$.*

The next result gives conditions on X and μ that ensure the density of continuous functions in $L^p(X)$.

Theorem B.90. *Let $(X, \mathfrak{M}, \mu)$ be a measure space, with X a normal space and μ a Borel measure such that*

$\mu(E) = \sup\{\mu(C) : C \text{ is closed, } C \subseteq E\} = \inf\{\mu(U) : U \text{ is open, } U \supseteq E\}$
for every set $E \in \mathfrak{M}$ with finite measure. Then $L^p(X) \cap C_b(X)$ is dense in $L^p(X)$ for $1 \leq p < \infty$.

Definition B.91. Let $(X, \mathfrak{M}, \mu)$ be a measure space, with X a topological space, $\mu : \mathfrak{M} \rightarrow [0, \infty]$ a Borel measure, and $1 \leq p \leq \infty$. A measurable function $u : X \rightarrow [-\infty, \infty]$ is said to belong to $L^p_{\text{loc}}(X)$ if $u \in L^p(K)$ for every compact set $K \subseteq X$. A sequence $\{u_n\}_n$ in $L^p_{\text{loc}}(X)$ is said to converge to u in $L^p_{\text{loc}}(X)$ if $u_n \rightarrow u$ in $L^p(K)$ for every compact set $K \subseteq X$.

Theorem B.92 (Riesz's representation theorem in L^p). *Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $1 < p < \infty$, and let p' be its Hölder conjugate exponent. Then every bounded linear functional $L : L^p(X) \rightarrow \mathbb{R}$ is represented by a unique $v \in L^{p'}(X)$ in the sense that*

$$(B.20) \quad L(u) = \int_X uv \, d\mu \quad \text{for every } u \in L^p(X).$$

³By this we mean that for every $M > 0$ there is a measurable set of finite positive measure greater than M .

Moreover, the norm of L coincides with $\|v\|_{L^{p'}}$. Conversely, every functional of the form (B.20), where $v \in L^{p'}(X)$, is a bounded linear functional on $L^p(X)$. Thus, the dual of $L^p(X)$ may be identified with $L^{p'}(X)$. In particular, $L^p(X)$ is reflexive.

Theorem B.93 (Riesz's representation theorem in L^1). *Let $(X, \mathfrak{M}, \mu)$ be a measure space with μ σ -finite. Then the dual of $L^1(X)$ may be identified with $L^\infty(X)$.*

The next result shows that $L^1(X)$ is not reflexive, since in general its bidual is larger than $L^1(X)$.

Let $(X, \mathfrak{M}, \mu)$ be a measure space. The dual of $L^\infty(X)$ may be identified with the space $\text{ba}(X, \mathfrak{M}, \mu)$ of all *bounded finitely additive signed measures absolutely continuous with respect to μ* , that is, all maps $\lambda : \mathfrak{M} \rightarrow \mathbb{R}$ such that

- (i) λ is a finitely additive signed measure,
- (ii) λ is *bounded*, that is, its *total variation norm*

$$\|\lambda\| := \sup \left\{ \sum_{n=1}^l |\lambda(E_n)| \right\},$$

where the supremum is taken over all finite partitions $\{E_n\}_{n=1}^l \subseteq \mathfrak{M}$ of X , $l \in \mathbb{N}$, is finite,

- (iii) $\lambda(E) = 0$ whenever $E \in \mathfrak{M}$ and $\mu(E) = 0$.

Given a measure space $(X, \mathfrak{M}, \mu)$, $\lambda \in \text{ba}(X, \mathfrak{M}, \mu)$, and $u \in L^\infty(X)$, by Theorem B.36 we may find a sequence $\{s_n\}_n$ of simple functions in $L^\infty(X)$ that converges uniformly to u . For every $E \in \mathfrak{M}$, we define $\int_E s_n d\lambda$ as in (B.2) with λ and s_n in place of μ and s , respectively. Then

$$\left| \int_E (s_m - s_n) d\lambda \right| \leq \|s_m - s_n\|_{L^\infty(X)} |\lambda|(E)$$

for all $m, n \in \mathbb{N}$, and so there exists the limit

$$(B.21) \quad \int_E u d\lambda := \lim_{n \rightarrow \infty} \int_E s_n d\lambda.$$

It may be verified that the integral $\int_E u d\lambda$ does not depend on the particular approximating sequence $\{s_n\}_n$.

Theorem B.94 (Riesz's representation theorem in L^∞). *Let $(X, \mathfrak{M}, \mu)$ be a measure space. Then every bounded linear functional $L : L^\infty(X) \rightarrow \mathbb{R}$ is represented by a unique $\lambda \in \text{ba}(X, \mathfrak{M}, \mu)$ in the sense that*

$$(B.22) \quad L(u) = \int_X u d\lambda \quad \text{for every } u \in L^\infty(X).$$

Moreover, the norm of L coincides with $\|\lambda\|$. Conversely, every functional of the form (B.22), where $\lambda \in \text{ba}(X, \mathfrak{M}, \mu)$, is a bounded linear functional on $L^\infty(X)$.

Definition B.95. Let $(X, \mathfrak{M}, \mu)$ and $(Y, \mathfrak{N}, \nu)$ be two measure spaces, let $L^0(X) := \{u : X \rightarrow \mathbb{C} \text{ measurable}\}$, $L^0(Y) := \{v : Y \rightarrow \mathbb{C} \text{ measurable}\}$, let $V \subseteq L^0(X)$ be a subspace. We say that $T : V \rightarrow L^0(Y)$ is of *strong type* (p, q) if it is bounded from $L^p(X, \mu) \cap V$ into $L^q(Y, \nu)$, that is,

$$\|T(u)\|_{L^q(Y)} \leq c_{p,q} \|u\|_{L^p(X)}$$

for all $u \in L^p(X, \mu) \cap V$ and for some constant $c_{p,q} > 0$.

Let $(X, \mathfrak{M}, \mu)$ and $(Y, \mathfrak{N}, \nu)$ be two measure spaces, and let $\mathcal{S}$ be the family of simple functions in $L^0(X)$ vanishing at infinity.

Theorem B.96 (Riesz–Thorin). *Let $(X, \mathfrak{M}, \mu)$ and $(Y, \mathfrak{N}, \nu)$ be two measure spaces, let $1 \leq p_1, q_1, p_2, q_2 \leq \infty$, let $T : \mathcal{S} \rightarrow L^0(Y)$ be a linear operator of strong type (p_1, q_1) and (p_2, q_2) . Then for every $\theta \in [0, 1]$, T is of strong type (p, q) , with*

$$\|T(u)\|_{L^q(Y)} \leq c_{p_1, q_1}^{1-\theta} c_{p_2, q_2}^\theta \|u\|_{L^{p_1}(X)}$$

for all $u \in \mathcal{S}$, with

$$\frac{1}{p} = \frac{1-\theta}{p_1} + \frac{\theta}{p_2}, \quad \frac{1}{q} = \frac{1-\theta}{q_1} + \frac{\theta}{q_2}.$$

In particular, T can be uniquely extended to $L^p(X)$ as a linear operator of strong type (p, q) .

B.8. Modes of Convergence

In this section we study different modes of convergence and their relation to one another.

Definition B.97. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u_n, u : X \rightarrow \mathbb{R}$ be measurable functions.

- (i) $\{u_n\}_n$ is said to *converge to u pointwise μ -a.e.* if there exists a set $E \in \mathfrak{M}$ such that $\mu(E) = 0$ and $\lim_{n \rightarrow \infty} u_n(x) = u(x)$ for all $x \in X \setminus E$.
- (ii) $\{u_n\}_n$ is said to *converge to u almost uniformly* if for every $\varepsilon > 0$ there exists a set $E \in \mathfrak{M}$ such that $\mu(E) \leq \varepsilon$ and $\{u_n\}_n$ converges to u uniformly in $X \setminus E$, that is,

$$\lim_{n \rightarrow \infty} \sup_{x \in X \setminus E} |u_n(x) - u(x)| = 0.$$

- (iii) $\{u_n\}_n$ is said to *converge to u in measure* if for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mu(\{x \in X : |u_n(x) - u(x)| > \varepsilon\}) = 0.$$

The next theorem relates the types of convergence introduced in Definition B.97 to convergence in $L^p(X)$.

Theorem B.98. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u_n, u : X \rightarrow \mathbb{R}$ be measurable functions.*

- (i) *If $\{u_n\}_n$ converges to u almost uniformly, then it converges to u in measure and pointwise μ -a.e.*
- (ii) *If $\{u_n\}_n$ converges to u in measure, then there exists a subsequence $\{u_{n_k}\}_k$ such that $\{u_{n_k}\}_k$ converges to u almost uniformly (and hence pointwise μ -a.e.).*
- (iii) *If $\{u_n\}_n$ converges to u in $L^p(X)$, $1 \leq p < \infty$, then it converges to u in measure and there exist a subsequence $\{u_{n_k}\}_k$ and an integrable function v such that $\{u_{n_k}\}_k$ converges to u almost uniformly (and hence pointwise μ -a.e.) and $|u_{n_k}(x)|^p \leq v(x)$ for μ -a.e. $x \in X$ and for all $k \in \mathbb{N}$.*

Theorem B.99 (Egoroff). *Let $(X, \mathfrak{M}, \mu)$ be a measure space with μ finite and let $u_n : X \rightarrow \mathbb{R}$ be measurable functions converging pointwise μ -a.e. Then $\{u_n\}_n$ converges almost uniformly (and hence in measure).*

In order to characterize convergence in L^p , we need to introduce the notion of p -equi-integrability.

Definition B.100. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $1 \leq p < \infty$. A family $\mathcal{F}$ of measurable functions $u : X \rightarrow [-\infty, \infty]$ is said to be p -equi-integrable if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\int_E |u|^p d\mu \leq \varepsilon$$

for all $u \in \mathcal{F}$ and for every measurable set $E \subseteq X$ with $\mu(E) \leq \delta$.

When $p = 1$, we refer to 1-*equi-integrability* simply as *equi-integrability*.

Theorem B.101 (Vitali's convergence). *Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $1 \leq p < \infty$, and let $u_n, u \in L^p(X)$, $n \in \mathbb{N}$. Then $\{u_n\}_n$ converges to u in $L^p(X)$ if and only if the following conditions hold:*

- (i) $\{u_n\}_n$ converges to u in measure.
- (ii) $\{u_n\}_n$ is p -equi-integrable.
- (iii) For every $\varepsilon > 0$ there exists $E \subseteq X$ with $E \in \mathfrak{M}$ such that $\mu(E) < \infty$ and

$$\int_{X \setminus E} |u_n|^p d\mu \leq \varepsilon$$

for all n .

Remark B.102. Note that condition (iii) is automatically satisfied when X has finite measure.

Theorem B.103 (Dunford–Pettis). *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $u_n, u \in L^1(X)$, $n \in \mathbb{N}$, with $\sup_n \int_X |u_n| d\mu < \infty$. Then there exist a subsequence $\{u_{n_k}\}_k$ of $\{u_n\}_n$ and an integrable function u such that $u_{n_k} \rightharpoonup u$ in $L^1(X)$ if and only if $\{u_n\}_n$ is equi-integrable and satisfies property (iii) of Vitali’s convergence theorem (with $p = 1$).*

In view of Vitali’s convergence theorem it becomes important to understand equi-integrability.

Theorem B.104. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $\mathcal{F}$ be a family of integrable functions $u : X \rightarrow [-\infty, \infty]$. Consider the following conditions:*

(i) $\mathcal{F}$ is equi-integrable.

(ii)

$$(B.23) \quad \lim_{t \rightarrow \infty} \sup_{u \in \mathcal{F}} \int_{\{|u| > t\}} |u| d\mu = 0.$$

(iii) (**De la Vallée Poussin**) *There exists an increasing function $g : [0, \infty) \rightarrow [0, \infty]$ with*

$$(B.24) \quad \lim_{t \rightarrow \infty} \frac{g(t)}{t} = \infty$$

such that

$$(B.25) \quad \sup_{u \in \mathcal{F}} \int_X g(|u|) d\mu < \infty.$$

Then (ii) and (iii) are equivalent and either one implies (i). If in addition we assume that

$$\sup_{u \in \mathcal{F}} \int_X |u| d\mu < \infty,$$

then (i) implies (ii) (and so all three conditions are equivalent in this case).

Exercise B.105. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $L^0(X)$ be the space of all (equivalence classes) of measurable functions $u : X \rightarrow \mathbb{R}$. Given $u, v \in L^0(X)$ define $d_0(u, v) := \min\{d(u, v), 1\}$, where

$$d(u, v) := \inf_{\varepsilon > 0} [\varepsilon + \mu(\{x \in X : |u(x) - v(x)| > \varepsilon\})].$$

(i) Prove that $L^0(X)$ is a complete metric space and that a sequence $\{u_n\}_n$ of functions in $L^0(X)$ converges if and only if it converges in measure.

(ii) Prove that $L^1(X)$ and $L^\infty(X)$ are embedded in $L^0(X)$.

B.9. Radon Measures

In this section we study regularity properties for measures.

Definition B.106. Let X be a topological space and let $\mu : \mathcal{B}(X) \rightarrow [0, \infty]$ be a Borel measure.

- (i) A Borel set $E \subseteq X$ is said to be *inner regular* if

$$\mu(E) = \sup\{\mu(K) : K \subseteq E, K \text{ is compact}\},$$

and it is *outer regular* if

$$\mu(E) = \inf\{\mu(U) : U \supseteq E, U \text{ is open}\}.$$

- (ii) A Borel set $E \subseteq X$ is said to be *regular* if it is both inner and outer regular.

Definition B.107. Let $(X, \mathfrak{M}, \mu)$ be a measure space. If X is a topological space, then the following hold.

- (i) μ is a *Borel regular measure* if μ is a Borel measure and if for every set $E \in \mathfrak{M}$ there exists a Borel set F such that $E \subseteq F$ and $\mu(E) = \mu(F)$.
- (ii) A Borel measure $\mu : \mathfrak{M} \rightarrow [0, \infty]$ is a *Radon measure* if
- (a) $\mu(K) < \infty$ for every compact set $K \subseteq X$,
 - (b) every open set $U \subseteq X$ is inner regular,
 - (c) every set $E \in \mathfrak{M}$ is outer regular.

Hausdorff measures $\mathcal{H}^s$, $s > 0$, represent an important class of regular Borel measures that are not Radon measures.

Proposition B.108. Let X be a locally compact Hausdorff space such that every open set is σ -compact. Let $\mu : \mathcal{B}(X) \rightarrow [0, \infty]$ be a measure finite on compact sets. Then μ is a Radon measure and every Borel set E is inner regular.

Theorem B.109 (Lusin). Let $(X, \mathfrak{M}, \mu)$ be a measure space, with X a topological space and μ a Radon measure, and let $u : X \rightarrow \mathbb{R}$ be a measurable function. Then for every set $E \in \mathfrak{M}$ with finite measure and every $\varepsilon > 0$ there exists a closed set $C \subseteq E$ with $\mu(E \setminus C) \leq \varepsilon$ such that $u : C \rightarrow \mathbb{R}$ is continuous.

Definition B.110. Let $(X, \mathfrak{M})$ be a measurable space and let $\lambda : \mathfrak{M} \rightarrow [-\infty, \infty]$ be a signed measure. If X is a topological space, then λ is a signed Radon measure if $|\lambda| : \mathfrak{M} \rightarrow [0, \infty]$ is a Radon measure.

If X is a topological space, then $\mathcal{M}_b(X; \mathbb{R})$ is the space of all signed finite Radon measures $\lambda : \mathcal{B}(X) \rightarrow \mathbb{R}$ endowed with the total variation norm. It

can be verified that $\mathcal{M}_b(X; \mathbb{R})$ is a Banach space with the norm

$$\|\lambda\|_{\mathcal{M}_b(X; \mathbb{R})} := |\lambda|(X).$$

Similarly, if $(X, \mathfrak{M})$ is a measurable space and X is a topological space, then $\lambda : \mathfrak{M} \rightarrow \mathbb{R}^m$ is a vectorial *Radon measure* if each component $\lambda_i : \mathfrak{M} \rightarrow \mathbb{R}$ is a signed Radon measure.

The space $\mathcal{M}_b(X; \mathbb{R}^m)$ of all vectorial Radon measures $\lambda : \mathcal{B}(X) \rightarrow \mathbb{R}^m$ is a Banach space with the norm

$$(B.26) \quad \|\lambda\|_{\mathcal{M}_b(X; \mathbb{R}^m)} := \|\lambda\|(X).$$

Theorem B.111 (Riesz's representation theorem in C_0). *Let X be a locally compact Hausdorff space. Then for every bounded linear functional $L : C_0(X; \mathbb{R}^m) \rightarrow \mathbb{R}$ there exists a unique $\lambda \in \mathcal{M}_b(X; \mathbb{R}^m)$ such that*

$$(B.27) \quad L(u) = \int_X u d\lambda \quad \text{for every } u \in C_0(X; \mathbb{R}^m).$$

Moreover, the norm of L coincides with $\|\lambda\|_{\mathcal{M}_b(X; \mathbb{R}^m)}$. Conversely, every functional of the form (B.27), where $\lambda \in \mathcal{M}_b(X; \mathbb{R}^m)$, is a bounded linear functional on $C_0(X; \mathbb{R}^m)$.

Remark B.112. A similar result holds for bounded linear functionals $L : C_0(X; \mathbb{C}) \rightarrow \mathbb{R}$.

Theorem B.113 (Riesz's representation theorem in C_c). *Let X be a locally compact Hausdorff space and let $L : C_c(X) \rightarrow \mathbb{R}$ be a linear functional. Then*

- (i) *if L is positive, that is, $L(v) \geq 0$ for all $v \in C_c(X)$, then there exists a unique (positive) Radon measure $\mu : \mathcal{B}(X) \rightarrow [0, \infty]$ such that*

$$L(u) = \int_X u d\mu \quad \text{for every } u \in C_c(X),$$

- (ii) *if L is locally bounded, that is, for every compact set $K \subseteq X$ there exists a constant $c_K > 0$ such that*

$$|L(u)| \leq c_K \|u\|_{C_c(X)}$$

for all $u \in C_c(X)$ with $\text{supp } u \subseteq K$, then there exist two (positive) Radon measures $\mu_1, \mu_2 : \mathcal{B}(X) \rightarrow [0, \infty]$ such that

$$L(u) = \int_X u d\mu_1 - \int_X u d\mu_2 \quad \text{for every } u \in C_c(X).$$

Note that since both μ_1 and μ_2 could have infinite measure, their difference is not defined in general, although on every compact set it is a well-defined finite signed measure.

B.10. Covering Theorems in $\mathbb{R}^N$

Theorem B.114 (Besicovitch's covering). *There exists a constant ℓ , depending only on the dimension N of $\mathbb{R}^N$, such that for every collection $\mathcal{F}$ of (nondegenerate) closed balls with*

$$(B.28) \quad \sup\{\text{diam } \overline{B} : \overline{B} \in \mathcal{F}\} < \infty$$

there exist $\mathcal{F}_1, \dots, \mathcal{F}_\ell \subseteq \mathcal{F}$ such that each $\mathcal{F}_n$, $n = 1, \dots, \ell$, is a countable family of disjoint balls in $\mathcal{F}$ and

$$E \subseteq \bigcup_{n=1}^{\ell} \bigcup_{\overline{B} \in \mathcal{F}_n} \overline{B},$$

where E is the set of centers of balls in $\mathcal{F}$.

The following result is an important consequence of the Besicovitch covering theorem.

Theorem B.115 (Vitali–Besicovitch's covering). *Let $E \subseteq \mathbb{R}^N$ be a Borel set and let $\mathcal{F}$ be a family of closed balls such that each point of E is the center of arbitrarily small balls, that is, $\inf\{r : \overline{B}(x, r) \in \mathcal{F}\} = 0$ for every $x \in E$. Let $\mu : \mathcal{B}(\mathbb{R}^N) \rightarrow [0, \infty]$ be a Radon measure. Then there exists a countable family $\{\overline{B}_n\}_n$ in $\mathcal{F}$ of disjoint closed balls such that $\mu(E \setminus \bigcup_n \overline{B}_n) = 0$.*

Theorem B.116 (Besicovitch's derivation). *Let*

$$\mu, \nu : \mathcal{B}(\mathbb{R}^N) \rightarrow [0, \infty]$$

be two Radon measures. Then there exists a Borel set $E_0 \subseteq \mathbb{R}^N$, with $\mu(E_0) = 0$, such that for every $x \in \mathbb{R}^N \setminus E_0$,

$$(B.29) \quad \frac{d\nu_{ac}}{d\mu}(x) = \lim_{r \rightarrow 0^+} \frac{\nu(\overline{B}(x, r))}{\mu(\overline{B}(x, r))} \in \mathbb{R} \quad \text{and} \quad \lim_{r \rightarrow 0^+} \frac{\nu_s(\overline{B}(x, r))}{\mu(\overline{B}(x, r))} = 0,$$

where $\nu = \nu_{ac} + \nu_s$, $\nu_{ac} \ll \mu$, $\nu_s \perp \mu$.

Remark B.117. In the previous theorems one can use closed cubes instead of balls.

Theorem B.118. *Let $\mu : \mathcal{B}(\mathbb{R}^N) \rightarrow [0, \infty]$ be a Radon measure, let $u : \mathbb{R}^N \rightarrow [-\infty, \infty]$ be a locally integrable function. Then there exists a Borel set $E_0 \subseteq \mathbb{R}^N$, with $\mu(E_0) = 0$, such that for every $x \in \mathbb{R}^N \setminus E_0$,*

$$\lim_{r \rightarrow 0^+} \frac{1}{\mu(\overline{B}(x, r))} \int_{\overline{B}(x, r)} u(y) d\mu(y) = u(x).$$

By enlarging the “bad” set E_0 , we can strengthen the conclusion of the previous theorem.

Corollary B.119. *Let $\mu : \mathcal{B}(\mathbb{R}^N) \rightarrow [0, \infty]$ be a Radon measure and let $u : \mathbb{R}^N \rightarrow [-\infty, \infty]$ be a locally integrable function. Then there exists a Borel set $E_0 \subseteq \mathbb{R}^N$, with $\mu(E_0) = 0$, such that for every $x \in \mathbb{R}^N \setminus E_0$,*

$$(B.30) \quad \lim_{r \rightarrow 0^+} \frac{1}{\mu(\overline{B(x, r)})} \int_{\overline{B(x, r)}} |u(y) - u(x)| d\mu(y) = 0.$$

A point $x \in \mathbb{R}^N$ for which (B.30) holds is called a *Lebesgue point* of u .

Corollary B.120. *Let $\mu : \mathcal{B}(\mathbb{R}^N) \rightarrow [0, \infty]$ be a Radon measure, let $1 \leq p < \infty$, and let $u \in L^p(\mathbb{R}^N)$. Then there exists a Borel set $E_0 \subseteq \mathbb{R}^N$, with $\mu(E_0) = 0$, such that for every $x \in \mathbb{R}^N \setminus E_0$,*

$$(B.31) \quad \lim_{r \rightarrow 0^+} \frac{1}{\mu(\overline{B(x, r)})} \int_{\overline{B(x, r)}} |u(y) - u(x)|^p d\mu(y) = 0.$$

A point $x \in \mathbb{R}^N$ for which (B.31) holds is called a *p-Lebesgue point* of u .

By applying Theorem B.118 to χ_E , we obtain the following result.

Corollary B.121. *Let $\mu : \mathcal{B}(\mathbb{R}^N) \rightarrow [0, \infty]$ be a Radon measure and let $E \subseteq \mathbb{R}^N$ be a Borel set. Then there exists a Borel set $E_0 \subseteq \mathbb{R}^N$, with $\mu(E_0) = 0$, such that for every $x \in \mathbb{R}^N \setminus E_0$,*

$$\lim_{r \rightarrow 0^+} \frac{\mu(\overline{B(x, r)} \cap E)}{\mu(\overline{B(x, r)})} = \chi_E(x).$$

A point $x \in E$ for which the previous limit is one is called a *point of density one* for E . More generally, for any $t \in [0, 1]$ a point $x \in \mathbb{R}^N$ such that

$$(B.32) \quad \lim_{r \rightarrow 0^+} \frac{\mu(B(x, r) \cap E)}{\mu(\overline{B(x, r)})} = t$$

is called a *point of density t* for E .

The Lebesgue and Hausdorff Measures

The Analysis of Value must be used carefully to avoid the following two types of errors. Type I: You incorrectly believe your research is not dull. Type II: No conclusions can be made. Good luck graduating.

— Jorge Cham, www.phdcomics.com

C.1. The Lebesgue Measure

For each set $E \subseteq \mathbb{R}^N$ define

$$\mathcal{L}_o^N(E) := \inf \left\{ \sum_{n=1}^{\infty} r_n^N : E \subseteq \bigcup_{n=1}^{\infty} Q(x_n, r_n) \right\},$$

where for $x \in \mathbb{R}^N$ and $r > 0$, $Q(x, r) := x + (-\frac{r}{2}, \frac{r}{2})^N$. By Proposition B.2, $\mathcal{L}_o^N$ is an outer measure, called the N -dimensional *Lebesgue outer measure*. Using Remark B.3, it can be shown that

$$(C.1) \quad \mathcal{L}_o^N(Q(x, r)) = r^N$$

and that $\mathcal{L}_o^N$ is translation-invariant, i.e., $\mathcal{L}_o^N(x + E) = \mathcal{L}_o^N(E)$ for all $x \in \mathbb{R}^N$ and all $E \subseteq \mathbb{R}^N$. The class of all $\mathcal{L}_o^N$ -measurable subsets of $\mathbb{R}^N$ is called the σ -algebra of *Lebesgue measurable sets*, and by Carathéodory's theorem (Theorem B.13), $\mathcal{L}_o^N$ restricted to this σ -algebra is a complete measure, called the N -dimensional *Lebesgue measure* and denoted by $\mathcal{L}^N$. Since $\mathcal{L}^N(\mathbb{R}^N) \geq \mathcal{L}_o^N(Q(x, r)) = r^N$, by sending $r \rightarrow \infty$, we conclude that $\mathcal{L}^N$ is not a finite measure. However, it is σ -finite, since $\mathbb{R}^N = \bigcup_{n=1}^{\infty} Q(0, n)$ and $\mathcal{L}^N(Q(0, n)) = n^N < \infty$.

Exercise C.1. Prove that for every $\delta > 0$ and for each set $E \subseteq \mathbb{R}^N$,

$$\mathcal{L}_o^N(E) = \inf \left\{ \sum_{n=1}^{\infty} r_n^N : r_n < \delta, E \subseteq \bigcup_{n=1}^{\infty} Q(x_n, r_n) \right\}.$$

Conclude that $\mathcal{L}_o^N$ is a metric outer measure.

It follows from the previous exercise and Proposition B.19 that every Borel subset of $\mathbb{R}^N$ is Lebesgue measurable. It may be proved that there are Lebesgue measurable sets that are not Borel sets (see Exercise 1.33). Hence, $\mathcal{L}^N : \mathcal{B}(\mathbb{R}^N) \rightarrow [0, \infty]$ is not a complete measure.

Using the axiom of choice, it is also possible to construct sets that are not Lebesgue measurable.

Exercise C.2 (A non-Lebesgue measurable set). On the real line consider the equivalence relation $x \sim y$ if $x - y \in \mathbb{Q}$. By the axiom of choice we may construct a set $E \subseteq (0, 1)$ that contains exactly one element from each equivalence class. Let

$$F := \bigcup_{r \in (-1, 1) \cap \mathbb{Q}} (r + E) \subseteq (-1, 2).$$

- (i) Prove that $F \supseteq (0, 1)$.
- (ii) Prove that if $r, q \in \mathbb{Q}$, with $r \neq q$, then $(r + E) \cap (q + E) = \emptyset$.
- (iii) Prove that E is not Lebesgue measurable.

We observe that the Lebesgue measure is a Radon measure. Indeed, outer regularity of arbitrary sets follows from (C.1) and the definition of $\mathcal{L}_o^N$, while inner regularity of open sets is an immediate consequence of the fact that each open set can be written as a countable union of closed cubes with pairwise disjoint interiors.

Proposition C.3. $\mathcal{L}^N$ is a Radon measure. Moreover, every Lebesgue measurable set E is the union of a Borel set and a set of Lebesgue measure zero. In addition, for every set $F \subseteq \mathbb{R}^N$ there exists a Lebesgue measurable set E containing F with $\mathcal{L}^N(E) = \mathcal{L}_o^N(F)$.

Using the previous proposition and Proposition B.9 we have:

Proposition C.4. Let $\{E_n\}_n$ be an increasing sequence of subsets of $\mathbb{R}^N$, then

$$\mathcal{L}_o^N \left(\bigcup_{n=1}^{\infty} E_n \right) = \lim_{n \rightarrow \infty} \mathcal{L}_o^N(E_n).$$

Using the notation introduced in Section B.4, we have

Proposition C.5. *Let $N = n + m$, where $N, n, m \in \mathbb{N}$. Then $(\mathcal{L}^n \times \mathcal{L}^m)^* = \mathcal{L}_o^N$.*

If $E \subseteq \mathbb{R}^N$ is a Lebesgue measurable set, $\mathfrak{M}$ is the σ -algebra of all Lebesgue measurable subsets of E , Y is a nonempty set, and $\mathfrak{N}$ is an algebra on Y , then a measurable function $u : E \rightarrow Y$ is called *Lebesgue measurable*.

Proposition C.6. *Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set and let $u : E \rightarrow \mathbb{R}$ be a Lebesgue measurable function. Then there exists a Borel function $v : E \rightarrow \mathbb{R}$ such that $v(x) = u(x)$ for $\mathcal{L}^N$ -a.e. $x \in E$.*

If $E \subseteq \mathbb{R}^N$ is a Lebesgue measurable set and $u : E \rightarrow [-\infty, \infty]$ is a Lebesgue measurable function, then, whenever it is well-defined, we denote $\int_E u d\mathcal{L}^N$ simply by $\int_E u dx$. If $N = 1$ and I is an interval of endpoints a and b , we also write $\int_a^b u dx$ for $\int_I u dx$.

C.2. The Brunn–Minkowski Inequality

We leave the following preliminary result as an exercise.

Exercise C.7. Let $E, F \subset \mathbb{R}^N$ be two compact sets.

- (i) Construct decreasing sequences of sets $\{E_n\}_n$ and $\{F_n\}_n$ such that $E = \bigcap_{n=1}^{\infty} E_n$, $F = \bigcap_{n=1}^{\infty} F_n$, and all E_n and F_n consist of finite unions of rectangles with sides parallel to the axes.
- (ii) Prove that $\mathcal{L}^N(E_n + F_n) \rightarrow \mathcal{L}^N(E + F)$ as $n \rightarrow \infty$.

Theorem C.8 (Brunn–Minkowski's inequality). *Let $E, F \subset \mathbb{R}^N$ be two Lebesgue measurable sets such that*

$$E + F := \{x + y : x \in E, y \in F\}$$

is also Lebesgue measurable. Then

$$(\mathcal{L}^N(E))^{1/N} + (\mathcal{L}^N(F))^{1/N} \leq (\mathcal{L}^N(E + F))^{1/N}.$$

Proof. Step 1: Assume that E and F are rectangles whose sides are parallel to the axes and let x_i and y_i , $i = 1, \dots, N$, be their respective side lengths. Then

$$\mathcal{L}^N(E) = \prod_{i=1}^N x_i, \quad \mathcal{L}^N(F) = \prod_{i=1}^N y_i, \quad \mathcal{L}^N(E + F) = \prod_{i=1}^N (x_i + y_i).$$

By the arithmetic-geometric mean inequality

$$\left(\prod_{i=1}^N \frac{x_i}{x_i + y_i} \right)^{1/N} + \left(\prod_{i=1}^N \frac{y_i}{x_i + y_i} \right)^{1/N} \leq \frac{1}{N} \sum_{i=1}^N \frac{x_i}{x_i + y_i} + \frac{1}{N} \sum_{i=1}^N \frac{y_i}{x_i + y_i} = 1,$$

which gives the Brunn–Minkowski inequality for rectangles.

Step 2: We now suppose that E and F are finite unions of rectangles as in Step 1. The proof is by induction on the sum of the numbers of rectangles in E and in F . By interchanging E with F , if necessary, we may assume that E has at least two rectangles. By translating E if necessary, we may also assume that a coordinate hyperplane, say $\{x_N = 0\}$, separates two rectangles in E . Let E^+ and E^- denote the union of the rectangles formed by intersecting the rectangles in E with the half spaces $\{x_N \geq 0\}$ and $\{x_N \leq 0\}$, respectively. Translate F so that $\frac{\mathcal{L}^N(E^\pm)}{\mathcal{L}^N(E)} = \frac{\mathcal{L}^N(F^\pm)}{\mathcal{L}^N(F)}$, where F^+ and F^- are defined analogously to E^+ and E^- . Then $E^+ + F^+ \subseteq \{x_N \geq 0\}$ and $E^- + F^- \subseteq \{x_N \leq 0\}$, and the numbers of rectangles in $E^+ \cup F^+$ and in $E^- \cup F^-$ are both smaller than the number of rectangles in $E \cup F$. By induction and Step 1 we obtain

$$\begin{aligned} \mathcal{L}^N(E + F) &\geq \mathcal{L}^N(E^+ + F^+) + \mathcal{L}^N(E^- + F^-) \\ &\geq ((\mathcal{L}^N(E^+))^{1/N} + (\mathcal{L}^N(F^+))^{1/N})^N \\ &\quad + ((\mathcal{L}^N(E^-))^{1/N} + (\mathcal{L}^N(F^-))^{1/N})^N \\ &= \mathcal{L}^N(E^+) \left(1 + \frac{(\mathcal{L}^N(F))^{1/N}}{(\mathcal{L}^N(E))^{1/N}}\right)^N + \mathcal{L}^N(E^-) \left(1 + \frac{(\mathcal{L}^N(F))^{1/N}}{(\mathcal{L}^N(E))^{1/N}}\right)^N \\ &= \mathcal{L}^N(E) \left(1 + \frac{(\mathcal{L}^N(F))^{1/N}}{(\mathcal{L}^N(E))^{1/N}}\right)^N = ((\mathcal{L}^N(E))^{1/N} + (\mathcal{L}^N(F))^{1/N})^N. \end{aligned}$$

Step 3: Assume next that E and F are compact sets. Let $\{E_n\}_n$ and $\{F_n\}_n$ be as in the previous exercise. Then for all $n \in \mathbb{N}$,

$$(\mathcal{L}^N(E_n))^{1/N} + (\mathcal{L}^N(F_n))^{1/N} \leq (\mathcal{L}^N(E_n + F_n))^{1/N}.$$

It now suffices to let $n \rightarrow \infty$. Finally, if E and F are Lebesgue measurable sets, with $E + F$ measurable, fix two compact sets $K_1 \subseteq E$, $K_2 \subseteq F$. Then $K_1 + K_2 \subseteq E + F$, and so

$$(\mathcal{L}^N(K_1))^{1/N} + (\mathcal{L}^N(K_2))^{1/N} \leq (\mathcal{L}^N(K_1 + K_2))^{1/N} \leq (\mathcal{L}^N(E + F))^{1/N}.$$

Using the inner regularity of the Lebesgue measure, we obtain the desired result. $\square$

Remark C.9. (i) Note that the hypothesis that $E + F$ is measurable cannot be omitted. Indeed, Sierpiński [211] constructed an example of two Lebesgue measurable sets whose sum is not measurable. However, if E and F are Borel sets, then $E + F$ is an analytic set, and so it is measurable. We refer to [83] for more information on this topic.

- (ii) Fix $\theta \in (0, 1)$. By replacing E with θE and F with $(1 - \theta)F$ and using the N -homogeneity of the Lebesgue measure, we obtain that

$$\begin{aligned} \theta(\mathcal{L}^N(E))^{1/N} + (1 - \theta)(\mathcal{L}^N(F))^{1/N} \\ &= (\mathcal{L}^N(\theta E))^{1/N} + (\mathcal{L}^N((1 - \theta)F))^{1/N} \\ &\leq (\mathcal{L}^N(\theta E + (1 - \theta)F))^{1/N}. \end{aligned}$$

Thus, the function $f(t) := (\mathcal{L}^N(tE + (1 - t)F))^{1/N}$ is concave in $[0, 1]$. In particular, if $C \subseteq \mathbb{R}^{N+1}$ is a convex set and E and F are the intersections of C with the hyperplanes $\{x_1 = 1\}$ and $\{x_1 = 0\}$ (we write $x = (x_1, x_2, \dots, x_{N+1}) \in \mathbb{R}^{N+1}$), then $tE + (1 - t)F$ is the intersection of the convex hull of E and F with the hyperplane $x_1 = t$ and is therefore contained in the intersection of C with the hyperplane $\{x_1 = t\}$. Hence, the function giving the n th root of the volume of parallel hyperplane sections of an $(n + 1)$ -dimensional convex set is concave (see [92]). We used this fact in the proof of Poincaré's inequality in convex domains (see Theorem 13.36).

Exercise C.10. Let $C = \{x \in \mathbb{R}^3 : \sqrt{x_2^2 + x_3^2} \leq \frac{1}{2}(x_1 + 1), 0 \leq x_1 \leq 1\}$. Prove that C is convex but the function

$$g(x_1) := \int_{\mathbb{R}^2} \chi_E(x_1, x_2, x_3) dx_2 dx_3$$

is not concave (although its square root is by the previous remark).

Definition C.11. Given a set $E \subseteq \mathbb{R}^N$ and an integer $0 \leq n \leq N$, we define the n -dimensional upper Minkowski content of E ,

$$\mathcal{M}^{*n}(E) := \limsup_{\varepsilon \rightarrow 0^+} \frac{\mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, E) < \varepsilon\})}{\alpha_{N-n} \varepsilon^{N-n}},$$

and the n -dimensional lower Minkowski content of E ,

$$\mathcal{M}_*^n(E) := \liminf_{\varepsilon \rightarrow 0^+} \frac{\mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, E) < \varepsilon\})}{\alpha_{N-n} \varepsilon^{N-n}}.$$

When the n -dimensional upper and lower Minkowski contents of E coincide, their common value is called the n -dimensional Minkowski content of E and is denoted $\mathcal{M}^n(E)$.

Next we prove the isoperimetric inequality (see also Theorem 14.44).

Theorem C.12 (Isoperimetric inequality). *Let $E \subseteq \mathbb{R}^N$ be a Lebesgue measurable set with $\mathcal{L}^N(\overline{E}) < \infty$. Then*

$$\alpha_N^{-(N-1)/N} (\mathcal{L}^N(\overline{E}))^{(N-1)/N} \leq \beta_N^{-1} \mathcal{M}_*^{N-1}(\partial E).$$

Proof. Since

$$\{x \in \mathbb{R}^N : \text{dist}(x, E) < \varepsilon\} = \overline{E} + B(0, \varepsilon),$$

by the Brunn–Minkowski inequality

$$\begin{aligned} \mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, E) < \varepsilon\}) &\geq \left((\mathcal{L}^N(\overline{E}))^{1/N} + \alpha_N^{1/N} \varepsilon \right)^N \\ &\geq \mathcal{L}^N(\overline{E}) + N(\mathcal{L}^N(\overline{E}))^{(N-1)/N} \alpha_N^{1/N} \varepsilon, \end{aligned}$$

where we have used the inequality $(a+b)^N \geq a^N + Na^{N-1}b$ for $a, b \geq 0$. Hence,

$$\begin{aligned} \text{(C.2)} \quad \mathcal{L}^N(\{x \in \mathbb{R}^N : 0 < \text{dist}(x, E) < \varepsilon\}) \\ &= \mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, E) < \varepsilon\}) - \mathcal{L}^N(\overline{E}) \\ &\geq N(\mathcal{L}^N(\overline{E}))^{(N-1)/N} \alpha_N^{1/N} \varepsilon. \end{aligned}$$

On the other hand,

$$E^\circ \supseteq \{x \in \mathbb{R}^N : \text{dist}(x, \mathbb{R}^N \setminus E) \geq \varepsilon\} + B(0, \varepsilon),$$

and so, again by the Brunn–Minkowski inequality and reasoning as before,

$$\begin{aligned} \mathcal{L}^N(E^\circ) &\geq \mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, \mathbb{R}^N \setminus E) \geq \varepsilon\}) \\ &\quad + N(\mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, \mathbb{R}^N \setminus E) \geq \varepsilon\}))^{(N-1)/N} \alpha_N^{1/N} \varepsilon. \end{aligned}$$

It follows that

$$\begin{aligned} \text{(C.3)} \quad \mathcal{L}^N(\{x \in \mathbb{R}^N : 0 < \text{dist}(x, \mathbb{R}^N \setminus E) < \varepsilon\}) \\ &= \mathcal{L}^N(E^\circ) - \mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, \mathbb{R}^N \setminus E) \geq \varepsilon\}) \\ &\geq N(\mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, \mathbb{R}^N \setminus E) \geq \varepsilon\}))^{(N-1)/N} \alpha_N^{1/N} \varepsilon. \end{aligned}$$

Since the sets

$$\{x \in \mathbb{R}^N : 0 < \text{dist}(x, E) < \varepsilon\}$$

and

$$\{x \in \mathbb{R}^N : 0 < \text{dist}(x, \mathbb{R}^N \setminus E) < \varepsilon\}$$

are disjoint subsets of $\{x \in \mathbb{R}^N : \text{dist}(x, \partial E) < \varepsilon\}$, combining the inequalities (C.2) and (C.3), we obtain

$$\begin{aligned} &\underline{\mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, \partial E) < \varepsilon\})} \\ &\quad \geq N(\mathcal{L}^N(\overline{E}))^{(N-1)/N} \alpha_N^{1/N} \\ &\quad \quad + N(\mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, \mathbb{R}^N \setminus E) \geq \varepsilon\}))^{(N-1)/N} \alpha_N^{1/N}. \end{aligned}$$

Letting $\varepsilon \rightarrow 0^+$ and using the facts that $\alpha_1 = 2$ and

$$\mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, \mathbb{R}^N \setminus E) \geq \varepsilon\}) \nearrow \mathcal{L}^N(E^\circ),$$

we deduce

$$\begin{aligned} 2\mathcal{M}_*^{N-1}(\partial E) &= \liminf_{\varepsilon \rightarrow 0^+} \frac{\mathcal{L}^N(\{x \in \mathbb{R}^N : \text{dist}(x, \partial E) < \varepsilon\})}{\varepsilon} \\ &\geq N\alpha_N^{1/N} [(\mathcal{L}^N(\overline{E}))^{(N-1)/N} + (\mathcal{L}^N(E^\circ))^{(N-1)/N}]. \end{aligned}$$

To conclude, we now observe that if $\mathcal{M}_*^{N-1}(\partial E) < \infty$, then, necessarily, $\mathcal{L}^N(\partial E) = 0$. $\square$

Exercise C.13. Let $Q' := (0, 1)^{N-1}$, let $f \in C^1(\overline{Q'})$, and let

$$E := \{(x', x_N) \in Q' \times \mathbb{R} : f(x') < x_N < f(x') + 1\}.$$

Prove that $\mathcal{M}_*^{N-1}(\partial E) < \infty$ and give an explicit formula.

C.3. Mollifiers

Given a nonnegative bounded function $\varphi \in L^1(\mathbb{R}^N)$ with

$$(C.4) \quad \text{supp } \varphi \subseteq \overline{B(0, 1)}, \quad \int_{\mathbb{R}^N} \varphi(x) dx = 1,$$

for every $\varepsilon > 0$ we define

$$\varphi_\varepsilon(x) := \frac{1}{\varepsilon^N} \varphi\left(\frac{x}{\varepsilon}\right), \quad x \in \mathbb{R}^N.$$

The functions φ_ε are called *mollifiers*. Note that $\text{supp } \varphi_\varepsilon \subseteq \overline{B(0, \varepsilon)}$. Hence, given an open set $\Omega \subseteq \mathbb{R}^N$ and a function $u \in L^1_{\text{loc}}(\Omega)$, we may define

$$(C.5) \quad u_\varepsilon(x) := (u * \varphi_\varepsilon)(x) = \int_{\Omega} \varphi_\varepsilon(x - y) u(y) dy$$

for $x \in \Omega_\varepsilon$, where the open set Ω_ε is given by

$$(C.6) \quad \Omega_\varepsilon := \{x \in \Omega : \text{dist}(x, \partial\Omega) > \varepsilon\}.$$

The function $u_\varepsilon : \Omega_\varepsilon \rightarrow \mathbb{R}$ is called a *mollification* of u .

Note that if $x \in \Omega$, then $u_\varepsilon(x)$ is well-defined for all $0 < \varepsilon < \text{dist}(x, \partial\Omega)$. Thus, it makes sense to talk about $\lim_{\varepsilon \rightarrow 0^+} u_\varepsilon(x)$. We will use this fact without further mention.

Moreover, if $u \in L^1(\Omega)$, then $u_\varepsilon(x)$ is well-defined for every $x \in \mathbb{R}^N$.

Remark C.14. Note that if $\Omega = \mathbb{R}^N$, then $\Omega_\varepsilon = \mathbb{R}^N$; thus u_ε is defined in the entire space $\mathbb{R}^N$.

Remark C.15. In the applications we will consider two important types of mollifiers:

- (i) φ is the (renormalized) characteristic function of the unit ball, that is,

$$\varphi(x) := \frac{1}{\alpha_N} \chi_{B(0,1)}(x), \quad x \in \mathbb{R}^N,$$

(ii) φ is the function of class C_c^∞ defined by

$$(C.7) \quad \varphi(x) := \begin{cases} c \exp\left(\frac{1}{\|x\|^2-1}\right) & \text{if } \|x\| < 1, \\ 0 & \text{if } \|x\| \geq 1, \end{cases}$$

where we choose $c > 0$ so that (C.4) is satisfied. In this case, the functions φ_ε are called *standard mollifiers*.

The following theorem is the first main result of this subsection.

Theorem C.16. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $\varphi \in L^1(\mathbb{R}^N)$ be a nonnegative bounded function satisfying (C.4), and let $u \in L^1_{\text{loc}}(\Omega)$.*

- (i) *If $u \in C(\Omega)$, then $u_\varepsilon \rightarrow u$ as $\varepsilon \rightarrow 0^+$ uniformly on compact subsets of Ω .*
- (ii) *For every Lebesgue point $x \in \Omega$ (and so for $\mathcal{L}^N$ -a.e. $x \in \Omega$), $u_\varepsilon(x) \rightarrow u(x)$ as $\varepsilon \rightarrow 0^+$. Moreover, if $u \in L^p(\Omega)$, $1 \leq p \leq \infty$, then $u_\varepsilon(x) \rightarrow 0$ for every $x \in \mathbb{R}^N \setminus \overline{\Omega}$.*
- (iii) *If $u \in L^p(\Omega)$, $1 \leq p \leq \infty$, then*

$$(C.8) \quad \|u_\varepsilon\|_{L^p(\mathbb{R}^N)} \leq \|u\|_{L^p(\Omega)}$$

for every $\varepsilon > 0$ and

$$(C.9) \quad \|u_\varepsilon\|_{L^p(\mathbb{R}^N)} \rightarrow \|u\|_{L^p(\Omega)} \text{ as } \varepsilon \rightarrow 0^+.$$

- (iv) *If $u \in L^p(\Omega)$, $1 \leq p < \infty$, then*

$$\lim_{\varepsilon \rightarrow 0^+} \|u_\varepsilon - u\|_{L^p(\Omega)} = 0.$$

Proof. (i) Let $K \subset \Omega$ be a compact set. For any fixed $0 < \eta < \text{dist}(K, \partial\Omega)$, let $K_\eta := \{x \in \mathbb{R}^N : \text{dist}(x, K) \leq \eta\}$, so that $K_\eta \subset \Omega$. Note that for $\varepsilon > 0$ sufficiently small we have that $K_\eta \subset \Omega_\varepsilon$. Since K_η is compact and u is uniformly continuous on K_η , for every $\rho > 0$ there exists $\delta = \delta(\eta, K, \rho) > 0$ such that

$$(C.10) \quad |u(x) - u(y)| \leq \rho$$

for all $x, y \in K_\eta$, with $\|x - y\| \leq \delta$. Let $0 < \varepsilon < \min\{\delta, \eta\}$. Then for all $x \in K$,

$$(C.11) \quad \begin{aligned} |u_\varepsilon(x) - u(x)| &= \left| \int_{\Omega} \varphi_\varepsilon(x-y)u(y) dy - u(x) \right| \\ &= \frac{1}{\varepsilon^N} \left| \int_{B(x, \varepsilon)} \varphi\left(\frac{x-y}{\varepsilon}\right)[u(y) - u(x)] dy \right| \\ &\leq \|\varphi\|_\infty \frac{1}{\varepsilon^N} \int_{B(x, \varepsilon)} |u(y) - u(x)| dy, \end{aligned}$$

where we have used (C.4) and the fact that $\text{supp } \varphi_\varepsilon(\cdot - y) \subseteq \overline{B(x, \varepsilon)}$. It follows by (C.10) that $|u_\varepsilon(x) - u(x)| \leq \rho \alpha_N \|\varphi\|_\infty$ for all $x \in K$, and so $\|u_\varepsilon - u\|_{C(K)} \leq \rho \alpha_N \|\varphi\|_\infty$.

(ii) Let $x \in \Omega$ be a Lebesgue point of u , that is,

$$\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon^N} \int_{B(x, \varepsilon)} |u(y) - u(x)| dy = 0,$$

then from (C.11) it follows that $u_\varepsilon(x) \rightarrow u(x)$ as $\varepsilon \rightarrow 0^+$.

(iii) To prove (C.8) it is enough to assume that $u \in L^p(\Omega)$. If $1 \leq p < \infty$, then by Hölder's inequality and (C.4) for all $x \in \mathbb{R}^N$,

$$\begin{aligned} |u_\varepsilon(x)| &= \left| \int_{\Omega} (\varphi_\varepsilon(x - y))^{1/p'} (\varphi_\varepsilon(x - y))^{1/p} u(y) dy \right| \\ (C.12) \quad &\leq \left(\int_{\Omega} \varphi_\varepsilon(x - y) dy \right)^{1/p'} \left(\int_{\Omega} \varphi_\varepsilon(x - y) |u(y)|^p dy \right)^{1/p} \\ &\leq \left(\int_{\Omega} \varphi_\varepsilon(x - y) |u(y)|^p dy \right)^{1/p}, \end{aligned}$$

and so by Fubini's theorem and (C.4) once more

$$\begin{aligned} \int_{\mathbb{R}^N} |u_\varepsilon(x)|^p dx &\leq \int_{\mathbb{R}^N} \int_{\Omega} \varphi_\varepsilon(x - y) |u(y)|^p dy dx \\ &= \int_{\Omega} |u(y)|^p \left(\int_{\mathbb{R}^N} \varphi_\varepsilon(x - y) dx \right) dy = \int_{\Omega} |u(y)|^p dy. \end{aligned}$$

On the other hand, if $p = \infty$, then for every $x \in \mathbb{R}^N$,

$$\begin{aligned} |u_\varepsilon(x)| &\leq \int_{\Omega} \varphi_\varepsilon(x - y) |u(y)| dy \\ &\leq \|u\|_{L^\infty(\Omega)} \int_{\Omega} \varphi_\varepsilon(x - y) dy \leq \|u\|_{L^\infty(\Omega)} \end{aligned}$$

again by (C.4), and so (C.8) holds for all $1 \leq p \leq \infty$.

In particular,

$$\limsup_{\varepsilon \rightarrow 0^+} \|u_\varepsilon\|_{L^p(\mathbb{R}^N)} \leq \|u\|_{L^p(\Omega)}.$$

To prove the opposite inequality, assume first that $1 \leq p < \infty$. By part (ii), $u_\varepsilon(x) \rightarrow u(x)$ as $\varepsilon \rightarrow 0^+$ for $\mathcal{L}^N$ a.e. $x \in \Omega$, and so by Fatou's lemma

$$\int_{\Omega} |u(x)|^p dx = \int_{\Omega} \lim_{\varepsilon \rightarrow 0^+} |u_\varepsilon(x)|^p dx \leq \liminf_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^N} |u_\varepsilon(x)|^p dx.$$

If $p = \infty$, then again by part (ii) $u_\varepsilon(x) \rightarrow u(x)$ as $\varepsilon \rightarrow 0^+$ for $\mathcal{L}^N$ a.e. $x \in \Omega$. Hence,

$$|u(x)| = \lim_{\varepsilon \rightarrow 0^+} |u_\varepsilon(x)| \leq \liminf_{\varepsilon \rightarrow 0^+} \|u_\varepsilon\|_{L^\infty(\mathbb{R}^N)}$$

for $\mathcal{L}^N$ a.e. $x \in \Omega$. It follows that

$$\|u\|_{L^\infty(\Omega)} \leq \liminf_{\varepsilon \rightarrow 0^+} \|u_\varepsilon\|_{L^\infty(\mathbb{R}^N)}.$$

Hence, (C.9) holds also in this case.

(iv) Fix $\rho > 0$ and find a function $v \in C_c(\Omega)$ such that $\|u - v\|_{L^p(\Omega)} \leq \rho$. Since $K := \text{supp } v$ is compact, it follows from part (i) that for every $0 < \eta < \text{dist}(K, \partial\Omega)$, the mollification v_ε of v converges to v uniformly in the compact set

$$K_\eta := \{x \in \mathbb{R}^N : \text{dist}(x, K) \leq \eta\}.$$

Since $v_\varepsilon = v = 0$ in $\Omega \setminus K_\eta$ for $0 < \varepsilon < \eta$, we have that

$$\int_\Omega |v_\varepsilon - v|^p dx = \int_{K_\eta} |v_\varepsilon - v|^p dx \leq \|v_\varepsilon - v\|_{C(K_\eta)}^p \mathcal{L}^N(K_\eta) \leq \rho,$$

provided $\varepsilon > 0$ is sufficiently small. By Minkowski's inequality,

$$\begin{aligned} \|u_\varepsilon - u\|_{L^p(\Omega)} &\leq \|u_\varepsilon - v_\varepsilon\|_{L^p(\Omega)} + \|v_\varepsilon - v\|_{L^p(\Omega)} + \|v - u\|_{L^p(\Omega)} \\ &\leq 2\|u - v\|_{L^p(\Omega)} + \|v_\varepsilon - v\|_{L^p(\Omega)} \leq 3\rho, \end{aligned}$$

where we have used (C.8) for the function $u - v$. $\square$

Remark C.17. If we do not assume that $u \in L^p(\Omega)$ in part (iii), then we can still prove that $\|u_\varepsilon\|_{L^p(\Omega_\varepsilon)} \leq \|u\|_{L^p(\Omega)}$ and that $\|u_\varepsilon\|_{L^p(\Omega_\varepsilon)} \rightarrow \|u\|_{L^p(\Omega)}$ as $\varepsilon \rightarrow 0^+$.

Exercise C.18. Let

$$(C.13) \quad \widehat{\mathcal{O}}(\mathbb{R}^N) := \{\widehat{\phi} : \phi \in C_c^\infty(\mathbb{R}^N), 0 \notin \text{supp } \phi\} \subset \mathcal{S}(\mathbb{R}^N),$$

where we recall that $\widehat{\phi}$ is the Fourier transform of ϕ (see (10.23)). Prove that $\widehat{\mathcal{O}}(\mathbb{R}^N)$ is dense in $L^p(\mathbb{R}^N)$ for $1 \leq p < \infty$.

Exercise C.19. Let $\varphi \in L^1(\mathbb{R}^N)$ be a nonnegative function satisfying $\int_{\mathbb{R}^N} \varphi(x) dx = 1$.

- (i) Prove that for every $\rho > 0$ there exist $\psi \in \mathcal{S}(\mathbb{R}^N)$ and $\delta = \delta(\rho) > 0$ such that $\|\varphi - \psi\|_{L^1(\mathbb{R}^N)} \leq \rho$, $\widehat{\psi}(x) = 1$ for $\|x\| \leq \delta$ and $\widehat{\psi}(x) = 0$ for $\|x\| > \delta$.
- (ii) Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $u \in L^p(\Omega)$, $1 \leq p < \infty$. Prove that $u_\varepsilon \rightarrow u$ in $L^p(\Omega)$ as $\varepsilon \rightarrow 0^+$. Hint: Use part (i) and Exercise C.18.
- (iii) Let Ω , u and p be as in part (ii). Prove that $u_\varepsilon(x) \rightarrow u(x)$ as $\varepsilon \rightarrow 0^+$ for every Lebesgue point $x \in \Omega$ of u .

More can be said about the regularity of u_ε if we restrict our attention to standard mollifiers.

Theorem C.20. *Let $\Omega \subseteq \mathbb{R}^N$ be an open set, let $\varphi \in L^1(\mathbb{R}^N)$ be defined as in (C.7), and let $u \in L^1_{\text{loc}}(\Omega)$. Then $u_\varepsilon \in C^\infty(\Omega_\varepsilon)$ for all $0 < \varepsilon < 1$, and for every multi-index α ,*

$$(C.14) \quad \partial^\alpha u_\varepsilon(x) = (u * \partial^\alpha \varphi_\varepsilon)(x) = \int_{\mathbb{R}^N} \frac{\partial^{|\alpha|} \varphi_\varepsilon}{\partial x^\alpha}(x-y) u(y) dy$$

for all $x \in \Omega_\varepsilon$. Moreover, if $u \in L^p(\mathbb{R}^N)$ for $1 \leq p \leq \infty$, then $u_\varepsilon \in C^\infty(\mathbb{R}^N)$.

Proof. Fix $x \in \Omega_\varepsilon$ and $0 < \eta < \text{dist}(x, \partial\Omega) - \varepsilon$. Let $e_i, i = 1, \dots, N$, be an element of the canonical basis of $\mathbb{R}^N$ and for every $h \in \mathbb{R}$, with $0 < |h| \leq \eta$, consider

$$\begin{aligned} & \frac{u_\varepsilon(x + he_i) - u_\varepsilon(x)}{h} - \int_{\Omega} \frac{\partial \varphi_\varepsilon}{\partial x_i}(x-y) u(y) dy \\ &= \int_{\Omega} \left(\frac{\varphi_\varepsilon(x + he_i - y) - \varphi_\varepsilon(x - y)}{h} - \frac{\partial \varphi_\varepsilon}{\partial x_i}(x - y) \right) u(y) dy \\ &= \int_{\Omega} \left(\frac{1}{h} \int_0^h \frac{\partial \varphi_\varepsilon}{\partial x_i}(x - y + te_i) dt - \frac{\partial \varphi_\varepsilon}{\partial x_i}(x - y) \right) u(y) dy \\ &= \frac{1}{h} \int_0^h \int_{B(x, \varepsilon + \eta)} \left(\frac{\partial \varphi_\varepsilon}{\partial x_i}(x - y + te_i) - \frac{\partial \varphi_\varepsilon}{\partial x_i}(x - y) \right) u(y) dy dt, \end{aligned}$$

where we have used Fubini's theorem and the fact that $\text{supp } \varphi_\varepsilon \subseteq \overline{B(0, \varepsilon)}$.

Since $\varphi_\varepsilon \in C_c^\infty(\mathbb{R}^N)$, its partial derivatives are uniformly continuous. Hence, for every $\rho > 0$ there exists $\delta = \delta(\eta, x, \rho, \varepsilon) > 0$ such that

$$\left| \frac{\partial \varphi_\varepsilon}{\partial x_i}(z) - \frac{\partial \varphi_\varepsilon}{\partial x_i}(w) \right| \leq \frac{\rho}{1 + \|u\|_{L^1(B(x, \varepsilon + \eta))}}$$

for all $z, w \in B(x, \varepsilon + \eta)$, with $\|z - w\| \leq \delta$. Then for $0 < |h| \leq \min\{\eta, \delta\}$ we have

$$\left| \frac{u_\varepsilon(x + he_i) - u_\varepsilon(x)}{h} - \int_{\Omega} \frac{\partial \varphi_\varepsilon}{\partial x_i}(x - y) u(y) dy \right| \leq \rho,$$

which shows that

$$\frac{\partial u_\varepsilon}{\partial x_i}(x) = \int_{\Omega} \frac{\partial \varphi_\varepsilon}{\partial x_i}(x - y) u(y) dy.$$

Note that the only properties that we have used on the function φ_ε are that $\varphi_\varepsilon \in C_c^\infty(\mathbb{R}^N)$ with $\text{supp } \varphi_\varepsilon \subseteq \overline{B(0, \varepsilon)}$. Hence, the same proof carries over if we replace φ_ε with $\psi_\varepsilon := \frac{\partial \varphi_\varepsilon}{\partial x_i}$. Thus, by induction we may prove that for every multi-index α there holds

$$\partial^\alpha u_\varepsilon(x) = (u * \partial^\alpha \varphi_\varepsilon)(x) = \int_{\Omega} \frac{\partial^{|\alpha|} \varphi_\varepsilon}{\partial x^\alpha}(x - y) u(y) dy.$$

This completes the first part of the proof. We leave as an exercise to modify the previous proof to show that if $u \in L^p(\mathbb{R}^N)$ for $1 \leq p \leq \infty$, then $u_\varepsilon \in C^\infty(\mathbb{R}^N)$. $\square$

An important application of the theory of mollifiers is the existence of smooth partitions of unity.

Theorem C.21 (Smooth partition of unity). *Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $\{U_\alpha\}_{\alpha \in \Lambda}$ be an open cover of Ω . Then there exists a sequence $\{\psi_n\}_n$ of nonnegative functions in $C_c^\infty(\Omega)$ such that*

- (i) *each ψ_n has support in some U_α ,*
- (ii) *$\sum_{n=1}^\infty \psi_n(x) = 1$ for all $x \in \Omega$,*
- (iii) *for every compact set $K \subset \Omega$ there exists an integer $\ell \in \mathbb{N}$ and an open set U , with $K \subset U \subseteq \Omega$, such that $\sum_{n=1}^\ell \psi_n(x) = 1$ for all $x \in U$.*

Proof. Let S be a countable dense set in Ω , e.g., $S := \{x \in \mathbb{Q}^N \cap \Omega\}$, and consider the countable family $\mathcal{F}$ of closed balls

$$(C.15) \quad \mathcal{F} := \{\overline{B(x, r)} : r \in \mathbb{Q}, 0 < r < 1, x \in S, \\ \overline{B(x, r)} \subseteq U_\alpha \cap \Omega \text{ for some } \alpha \in \Lambda\}.$$

Since $\mathcal{F}$ is countable, we may write $\mathcal{F} = \{\overline{B(x_n, r_n)} : n \in \mathbb{N}\}$. Since $\{U_\alpha\}_{\alpha \in \Lambda}$ is an open cover of Ω , by the density of S and of the rational numbers we have that

$$(C.16) \quad \Omega = \bigcup_{n=1}^\infty B(x_n, r_n/2).$$

For each $n \in \mathbb{N}$ consider $\phi_n := \varphi_{r_n/4} * \chi_{B(x_n, 3r_n/4)}$, where the $\varphi_{r_n/4}$ are standard mollifiers (with $\varepsilon := r_n/4$). By Theorem C.20, $\phi_n \in C^\infty(\mathbb{R}^N)$. Moreover, if $x \in B(x_n, r_n/2)$, then

$$\begin{aligned} \phi_n(x) &= \int_{\mathbb{R}^N} \varphi_{r_n/4}(x-y) \chi_{B(x_n, 3r_n/4)}(y) dy \\ &= \int_{B(x, r_n/4)} \varphi_{r_n/4}(x-y) \chi_{B(x_n, 3r_n/4)}(y) dy \\ &= \int_{B(x, r_n/4)} \varphi_{r_n/4}(x-y) dy = 1, \end{aligned}$$

where we have used (C.4) and the fact that if $x \in B(x_n, r_n/2)$, then $B(x, r_n/4) \subseteq B(x_n, 3r_n/4)$. Since $0 \leq \chi_{B(x_n, 3r_n/4)} \leq 1$, a similar calculation shows that

$0 \leq \phi_n \leq 1$. On the other hand, if $x \notin \overline{B(x_n, r_n)}$, then

$$\begin{aligned}\phi_n(x) &= \int_{\mathbb{R}^N} \varphi_{r_n/4}(x-y) \chi_{B(x_n, 3r_n/4)}(y) dy \\ &= \int_{B(x, r_n/4)} \varphi_{r_n/4}(x-y) \chi_{B(x_n, 3r_n/4)}(y) dy = 0,\end{aligned}$$

where we have used the fact that if $x \notin \overline{B(x_n, r_n)}$, then $B(x, r_n/4) \cap B(x_n, 3r_n/4) = \emptyset$. In particular, $\phi_n \in C_c^\infty(\mathbb{R}^N)$ and $\text{supp } \phi_n \subseteq \overline{B(x_n, r_n)}$. Note that in view of the definition of $\mathcal{F}$, $\text{supp } \phi_n \subset U_\alpha \cap \Omega$ for some $\alpha \in \Lambda$.

Define $\psi_1 := \phi_1$ and

$$(C.17) \quad \psi_n := (1 - \phi_1) \cdots (1 - \phi_{n-1}) \phi_n$$

for $n \geq 2$, $n \in \mathbb{N}$. Since $0 \leq \phi_k \leq 1$ and $\text{supp } \phi_k \subseteq \overline{B(x_k, r_k)}$ for all $k \in \mathbb{N}$, we have that $0 \leq \psi_n \leq 1$ and $\text{supp } \psi_n \subseteq \overline{B(x_n, r_n)}$. This gives (i). To prove (ii), we prove by induction that

$$(C.18) \quad \psi_1 + \cdots + \psi_n = 1 - (1 - \phi_1) \cdots (1 - \phi_n)$$

for all $n \in \mathbb{N}$. The relation (C.18) is true for $n = 1$, since $\psi_1 := \phi_1$. Assume that (C.18) holds for n . Then by (C.17),

$$\begin{aligned}\psi_1 + \cdots + \psi_n + \psi_{n+1} &= 1 - (1 - \phi_1) \cdots (1 - \phi_n) + \psi_{n+1} \\ &= 1 - (1 - \phi_1) \cdots (1 - \phi_n) + (1 - \phi_1) \cdots (1 - \phi_n) \phi_{n+1} \\ &= 1 - (1 - \phi_1) \cdots (1 - \phi_{n+1}).\end{aligned}$$

Hence, (C.18) holds for all $n \in \mathbb{N}$.

Since $\phi_k = 1$ in $B(x_k, r_k/2)$ for all $k \in \mathbb{N}$, it follows from (C.18) that

$$(C.19) \quad \psi_1(x) + \cdots + \psi_n(x) = 1 \quad \text{for all } x \in \bigcup_{k=1}^n B(x_k, r_k/2).$$

Thus, in view of (C.16), property (ii) holds.

Finally, if $K \subset \Omega$ is compact, again by (C.16), we may find $\ell \in \mathbb{N}$ so large that $\bigcup_{k=1}^{\ell} B(x_k, r_k/2) \supset K$, and so (iii) follows by (C.19). $\square$

Exercise C.22. Prove that if $\Lambda = \{1, \dots, \ell\}$, then in the previous theorem we can construct a partition of unity consisting of ℓ functions.

Exercise C.23 (Cut-off function). ; Let $\Omega \subseteq \mathbb{R}^N$ be an open set and let $K \subset \Omega$ be a compact set. Prove that there exists a function $\varphi \in C_c^\infty(\Omega)$ such that $0 \leq \varphi \leq 1$ and $\varphi = 1$ on K .

The function φ constructed in the previous exercise is usually called a *cut-off function*.

Finally, using mollifiers, it is possible to give the following density result in L^p spaces.

Theorem C.24. Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Then the space $C_c^\infty(\Omega)$ is dense in $L^p(\Omega)$ for $1 \leq p < \infty$.

C.4. Maximal Functions

In this section we introduce the notion of maximal function and study its properties. Throughout this section we consider L^p spaces in the case that the underlying measure is the Lebesgue measure.

Definition C.25. Let $u \in L^1_{\text{loc}}(\mathbb{R}^N)$. The (Hardy–Littlewood) maximal function of u is defined by

$$M(u)(x) := \sup_{r>0} \frac{1}{\mathcal{L}^N(B(x,r))} \int_{B(x,r)} |u(y)| dy$$

for all $x \in \mathbb{R}^N$.

Exercise C.26. Prove that the set $\{x \in \mathbb{R}^N : M(u)(x) > t\}$ is open.

Theorem C.27. Let $u \in L^p(\mathbb{R}^N)$, $1 \leq p \leq \infty$. Then

- (i) $M(u)(x) < \infty$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$,
- (ii) if $p = 1$, then for any $t > 0$,

$$(C.20) \quad \mathcal{L}^N(\{x \in \mathbb{R}^N : M(u)(x) > t\}) \leq \frac{3^N}{t} \int_{\mathbb{R}^N} |u(x)| dx,$$

- (iii) if $1 < p \leq \infty$, then $M(u) \in L^p(\mathbb{R}^N)$ and $\|M(u)\|_{L^p(\mathbb{R}^N)} \leq c\|u\|_{L^p(\mathbb{R}^N)}$ for some constant $c = c(N, p) > 0$.

Exercise C.28. Let $u, v \in L^1(\mathbb{R}^N)$. Assume that $|u(x)| \leq v(x)$ for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$, where $v \in L^1(\mathbb{R}^N)$ has the form $v(x) = v_1(\|x\|)$ for some nonnegative, decreasing function v_1 . Prove that

$$\left| \int_{\mathbb{R}^N} u(x)v(x-y) dy \right| \leq \|v\|_{L^1(\mathbb{R}^N)} M(u)(x)$$

for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. Hint: Consider first the case in which $v_1 = \chi_{[0,r]}$ for some $r > 0$.

Using the previous exercise and Theorem C.27, we can prove the following result.

Proposition C.29. Let $0 < \alpha < N$, $1 < p < N/\alpha$. Then for every $u \in L^p(\mathbb{R}^N)$,

$$\left(\int_{\mathbb{R}^N} \left| \int_{\mathbb{R}^N} \frac{u(y)}{\|x-y\|^{N-\alpha}} dy \right|^q dx \right)^{1/q} \leq c\|u\|_{L^p(\mathbb{R}^N)}$$

for some constant $c = c(\alpha, p, N) > 0$, where $q := pN/(N - \alpha p)$.

Proof. Define

$$v_1(t) := \begin{cases} \frac{1}{t^{N-\alpha}} & \text{if } 0 \leq t \leq R, \\ 0 & \text{otherwise.} \end{cases}$$

Then by the previous exercise, for every $R > 0$ and for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$ we have that

$$\int_{B(x,R)} \frac{|u(y)|}{\|x - y\|^{N-\alpha}} dy \leq cR^\alpha M(u)(x),$$

and so

$$\begin{aligned} \int_{\mathbb{R}^N} \frac{|u(y)|}{\|x - y\|^{N-\alpha}} dy &\leq \int_{B(x,R)} \frac{|u(y)|}{\|x - y\|^{N-\alpha}} dy \\ &\quad + \int_{\mathbb{R}^N \setminus B(x,R)} \frac{|u(y)|}{\|x - y\|^{N-\alpha}} dy. \\ (C.21) \quad &\leq cR^\alpha M(u)(x) \\ &\quad + \|u\|_{L^p} \left(\int_{\mathbb{R}^N \setminus B(x,R)} \frac{1}{\|x - y\|^{(N-\alpha)p'}} dy \right)^{1/p'} \\ &= cR^\alpha M(u)(x) + c\|u\|_{L^p} R^{\alpha-N/p}, \end{aligned}$$

where we have used Hölder's inequality. Taking

$$R := \left(\frac{\|u\|_{L^p}}{M(u)(x) + \varepsilon} \right)^{p/N},$$

where $\varepsilon > 0$, we get

$$\int_{\mathbb{R}^N} \frac{|u(y)|}{\|x - y\|^{N-\alpha}} dy \leq c\|u\|_{L^p}^{\alpha p/N} (M(u)(x) + \varepsilon)^{1-\alpha p/N}$$

for $\mathcal{L}^N$ -a.e. $x \in \mathbb{R}^N$. Letting $\varepsilon \rightarrow 0^+$ and taking the norm in $L^q(\mathbb{R}^N)$ on both sides yields

$$\begin{aligned} \left(\int_{\mathbb{R}^N} \left| \int_{\mathbb{R}^N} \frac{u(y)}{\|x - y\|^{N-\alpha}} dy \right|^q dx \right)^{1/q} &\leq c\|u\|_{L^p}^{1-p/q} \left(\int_{\mathbb{R}^N} (M(u))^p dx \right)^{1/q} \\ &\leq c\|u\|_{L^p}, \end{aligned}$$

where we have used Theorem C.27. □

Exercise C.30. What happens if $p = 1$ or $p = N/\alpha$?

C.5. BMO Spaces

Definition C.31. We say that a function $u \in L^1_{\text{loc}}(\mathbb{R}^N)$ has *bounded mean oscillation*, and we write $u \in \text{BMO}(\mathbb{R}^N)$, if

$$(C.22) \quad |u|_{\text{BMO}} := \sup \frac{1}{\mathcal{L}^N(Q)} \int_Q |u(x) - u_Q| dx < \infty,$$

where the supremum is taken over all cubes Q .

Here we are using the notation

$$(C.23) \quad u_E := \frac{1}{\mathcal{L}^N(E)} \int_E u(x) dx.$$

The quantity $\frac{1}{\mathcal{L}^N(Q)} \int_Q |u(x) - u_Q| dx$ is called the *mean oscillation* of u over Q . It measures the average distance of u from u_Q over the same cube.

Remark C.32. The space $\text{BMO}(\mathbb{R}^N)$ is a vector space and $|\cdot|_{\text{BMO}}$ is a seminorm. Since $|u|_{\text{BMO}} = 0$ if and only if $u \equiv \text{const}$, $|\cdot|_{\text{BMO}}$ is a norm in the quotient space $\text{BMO}(\mathbb{R}^N)/\mathbb{R}$.

Remark C.33. In the sequel we will use the fact that $u \in L^1_{\text{loc}}(\mathbb{R}^N)$ belongs to $\text{BMO}(\mathbb{R}^N)$ if and only if for every cube Q there exists $c_{Q,u} \in \mathbb{R}$ such that

$$\sup \frac{1}{\mathcal{L}^N(Q)} \int_Q |u(x) - c_{Q,u}| dx < \infty.$$

To see this assume that the latter holds. Then

$$u(x) - u_Q = u(x) - c_{Q,u} + \frac{1}{\mathcal{L}^N(Q)} \int_Q (u(y) - c_{Q,u}) dy$$

and so

$$|u(x) - u_Q| \leq |u(x) - c_{Q,u}| + \frac{1}{\mathcal{L}^N(Q)} \int_Q |u(y) - c_{Q,u}| dy.$$

By averaging over Q in x we get

$$(C.24) \quad \frac{1}{\mathcal{L}^N(Q)} \int_Q |u(x) - u_Q| dx \leq 2 \frac{1}{\mathcal{L}^N(Q)} \int_Q |u(x) - c_{Q,u}| dx.$$

Note that if we define

$$|u|_{\text{BMO}}^* := \sup \frac{1}{\mathcal{L}^N(Q)} \inf_{c_Q \in \mathbb{R}} \int_Q |u(x) - c_Q| dx,$$

then by (C.24), $\frac{1}{2}|u|_{\text{BMO}} \leq |u|_{\text{BMO}}^* \leq |u|_{\text{BMO}}$.

Exercise C.34. Prove that $|\cdot|_{\text{BMO}}^*$ is a seminorm.

Using the previous property we can show that $L^\infty(\mathbb{R}^N) \subsetneq \text{BMO}(\mathbb{R}^N)$.

Theorem C.35. The space $L^\infty(\mathbb{R}^N)$ is strictly contained in $\text{BMO}(\mathbb{R}^N)$.

Proof. Given $u \in L^\infty(\mathbb{R}^N)$ take $c_{Q,u} := 0$. Then

$$\frac{1}{\mathcal{L}^N(Q)} \int_Q |u(x) - 0| dx \leq \|u\|_{L^\infty},$$

which, by Remark C.33, implies that $u \in \text{BMO}(\mathbb{R}^N)$.

Next we show that the unbounded function $u(x) := \log \|x\|$ belongs to $\text{BMO}(\mathbb{R}^N)$. Let $Q = Q(x_0, r)$ for some x_0 and $r > 0$. If $Q \cap B(0, 2r) = \emptyset$, take $c_{Q,u} := \log \|x_0\|$. By the mean value theorem for $x \in Q$,

$$\log \|x\| - c_{Q,u} = \log \|x\| - \log \|x_0\| = \frac{y \cdot (x - x_0)}{\|y\|^2}$$

for some y between x and x_0 , and so

$$|\log \|x\| - c_{Q,u}| \leq \frac{\|x - x_0\|}{\|y\|} \leq \frac{r\sqrt{N}}{2r} = \frac{\sqrt{N}}{2}.$$

It follows that

$$\frac{1}{\mathcal{L}^N(Q)} \int_Q |u(x) - c_{Q,u}| dx \leq \frac{\sqrt{N}}{2}.$$

On the other hand, if $Q \cap B(0, 2r) \neq \emptyset$, then $Q \subset B(0, 2r + \sqrt{N}r)$. In this case take $c_{Q,u} := \log r$. Then

$$\begin{aligned} \frac{1}{\mathcal{L}^N(Q)} \int_Q |u(x) - c_{Q,u}| dx &= \frac{1}{r^N} \int_Q |\log \|x\| - \log r| dx \\ &\leq \frac{1}{r^N} \int_{B(0, 2r + \sqrt{N}r)} |\log \|x\|/r| dx = \frac{\alpha_N}{r^N} \int_0^{2r + \sqrt{N}r} s^{N-1} |\log s/r| ds \\ &= \alpha_N \int_0^{2 + \sqrt{N}} t^{N-1} |\log t| dt < \infty, \end{aligned}$$

where $rt = s$. Hence, u belongs to $\text{BMO}(\mathbb{R}^N)$. □

Exercise C.36. Prove that for $N = 1$, the function

$$v(x) = \begin{cases} \log x & \text{if } x > 0, \\ 0 & \text{if } x \leq 0 \end{cases}$$

does not belong to $\text{BMO}(\mathbb{R})$. So multiplying a BMO function by a characteristic function does not preserve BMO.

Exercise C.37. Let $u, v \in \text{BMO}(\mathbb{R}^N)$.

- (i) Prove that the function $h := |u|$ belongs to $\text{BMO}(\mathbb{R}^N)$ and $|h|_{\text{BMO}}^* \leq |u|_{\text{BMO}}^*$.
- (ii) Prove that the functions $\min\{u, v\}$ and $\max\{u, v\}$ belong to $\text{BMO}(\mathbb{R}^N)$, with $|\min\{u, v\}|_{\text{BMO}}^* \leq |u|_{\text{BMO}}^* + |v|_{\text{BMO}}^*$ and $|\max\{u, v\}|_{\text{BMO}}^* \leq |u|_{\text{BMO}}^* + |v|_{\text{BMO}}^*$.
- (iii) Prove that for every $t > 0$ the truncated function $u_t(x) := \min\{t, \max\{u, -t\}\}$ belongs to $\text{BMO}(\mathbb{R}^N)$ with $|u_t|_{\text{BMO}}^* \leq |u|_{\text{BMO}}^*$.

Exercise C.38. Prove that $\text{BMO}(\mathbb{R}^N)/\mathbb{R}$ is a Banach space.

C.6. Hardy's Inequality

In this section we prove Hardy's inequality.

Theorem C.39. *Let $f : [0, \infty) \rightarrow [0, \infty)$ be a convex, strictly increasing function and let $v : [0, \infty) \rightarrow [0, \infty)$ and $g : [0, \infty) \rightarrow [0, \infty]$ be measurable functions. Then*

$$\int_0^\infty g(x) f\left(\frac{1}{x} \int_0^x v(y) dy\right) dx \leq \int_0^\infty f(v(y)) \left(\int_y^\infty g(x) \frac{dx}{x}\right) dy.$$

Proof. By Jensen's inequality (see Theorem B.50) for every measurable function $w : [0, \infty) \rightarrow [0, \infty)$ and every $x > 0$,

$$\begin{aligned} f\left(\frac{1}{x} \int_0^x (f^{-1} \circ w)(y) dy\right) &\leq \frac{1}{x} \int_0^x (f \circ (f^{-1} \circ w))(y) dy \\ &= \frac{1}{x} \int_0^x w(y) dy. \end{aligned}$$

Multiplying both sides by $g(x)$, integrating with respect to x , and using Tonelli's theorem gives

$$\begin{aligned} \int_0^\infty g(x) f\left(\frac{1}{x} \int_0^x (f^{-1} \circ w)(y) dy\right) dx &\leq \int_0^\infty g(x) \left(\frac{1}{x} \int_0^x w(y) dy\right) dx \\ &= \int_0^\infty w(y) \left(\int_y^\infty g(x) \frac{dx}{x}\right) dy. \end{aligned}$$

It suffices to replace w with $f \circ v$. □

Remark C.40. Taking $g(x) = \frac{1}{x}$ and $f(t) = t^p$, where $1 \leq p < \infty$ in the previous theorem gives

$$(C.25) \quad \int_0^\infty x^{-p-1} \left(\int_0^x v(y) dy\right)^p dx \leq \int_0^\infty x^{-1} v^p(x) dx.$$

An important corollary of the previous theorem is Hardy's inequality.

Theorem C.41 (Hardy's inequality). *Let $u : [0, \infty) \rightarrow [0, \infty]$ be a measurable function, where $0 \leq a < b \leq \infty$, and let $1 \leq p < \infty$ and $0 < s < \infty$. Then*

$$(C.26) \quad \left(\int_0^\infty x^{-s-1} \left(\int_0^x u(y) dy\right)^p dx\right)^{1/p} \leq \frac{p}{s} \left(\int_0^\infty x^{-s+p-1} u^p(x) dx\right)^{1/p}$$

and

$$(C.27) \quad \left(\int_0^\infty x^{s-1} \left(\int_x^\infty u(y) dy\right)^p dx\right)^{1/p} \leq \frac{p}{s} \left(\int_0^\infty x^{s+p-1} u^p(x) dx\right)^{1/p}.$$

Proof. Take $v(x) := u(x^{1/a})x^{1/a-1}$ in (C.25), where $a > 0$. By the changes of variables $t = y^{1/a}$ and $z = x^{1/a}$ we have

$$\begin{aligned} \int_0^\infty x^{-p-1} \left(\int_0^x v(y) dy \right)^p dx &= \int_0^\infty x^{-p-1} \left(\int_0^x u(y^{1/a}) y^{1/a-1} dy \right)^p dx \\ &= \int_0^\infty x^{-p-1} \left(a \int_0^{x^{1/a}} u(t) dt \right)^p dx \\ &= a \int_0^\infty z^{-pa-1} \left(a \int_0^z u(t) dt \right)^p dz, \end{aligned}$$

while by the changes of variables $z = x^{1/a}$,

$$\int_0^\infty x^{-1} v^p(x) dx = \int_0^\infty x^{-1} u^p(x^{1/a}) x^{p/a-p} dx = a \int_0^\infty z^{p-pa-1} u^p(z) dz.$$

Combining these two inequalities from (C.25) we get

$$\int_0^\infty z^{-pa-1} \left(a \int_0^z u(t) dt \right)^p dz \leq \int_0^\infty z^{p-pa-1} u^p(z) dz.$$

It suffices to take $a := s/p$. □

Exercise C.42. Prove (C.27). Hint: Take $u(x) = v(x^{-1})x^{-2}$ in (C.26).

C.7. Hausdorff Measures

In this section we introduce the Hausdorff measure in a metric space. In $\mathbb{R}^N$, loosely speaking, the Hausdorff measure is a measure that is adapted to measure sets of lower dimensions in $\mathbb{R}^N$, say, a curve in the plane or a surface in $\mathbb{R}^3$. It is also used to measure fractals.

Given a metric space (X, d) , for $0 < \delta \leq \infty$ and for each set $E \subseteq X$ we define

$$(C.28) \quad \mathcal{H}_\delta^s(E) := \inf \left\{ \sum_{n=1}^\infty c_s (\text{diam } E_n)^s : E \subseteq \bigcup_{n=1}^\infty E_n, \text{diam } E_n < \delta \right\},$$

where, when $s = 0$, we sum only over those $E_n \neq \emptyset$. Here, $c_s > 0$ is a renormalization constant. In many books c_s is taken to be 1. However, in $\mathbb{R}^N$ it is more convenient to take c_s as in (C.30) below. The advantage is that with this choice for $s \in \{1, \dots, N\}$ the Hausdorff measure coincides with the surface measure on a k -dimensional manifold if $s < N$ and with the Lebesgue measure $\mathcal{L}^N$ when $s = N$.

By Proposition B.2, $\mathcal{H}_\delta^s$ is an outer measure. We next define the Hausdorff outer measure. Since for each set $E \subseteq X$ the function $\delta \mapsto \mathcal{H}_\delta^s(E)$ is decreasing, there exists

$$(C.29) \quad \mathcal{H}_o^s(E) := \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^s(E) = \sup_{\delta > 0} \mathcal{H}_\delta^s(E).$$

$\mathcal{H}_o^s$ is called the *s-dimensional Hausdorff outer measure* of E .

Exercise C.43. Prove that if $X = \mathbb{R}^N$ in the definition (C.28) it is possible to restrict the class of admissible sets in the covers $\{E_n\}_n$ to closed and convex sets (open and convex, respectively) and that the condition $\text{diam } E_n < \delta$ can be replaced by $\text{diam } E_n \leq \delta$, without changing the value of $\mathcal{H}_o^s(E)$.

If X is a normed space, it follows from (C.28) and (C.29) that for all $E \subseteq X$, $x \in \mathbb{R}^N$, and $t > 0$, we have

$$\mathcal{H}_o^s(x + E) = \mathcal{H}_o^s(E), \quad \mathcal{H}_o^s(tE) = t^s \mathcal{H}_o^s(E).$$

Another useful property of $\mathcal{H}_o^s$ is the following result whose proof is left as an exercise.

Proposition C.44. Let (X, d_X) and (Y, d_Y) be metric spaces, let $E \subseteq X$ and let $u : E \rightarrow Y$ be a Lipschitz continuous function. Prove that for all $s > 0$,¹

$$\mathcal{H}_o^s(u(E)) \leq (\text{Lip } u)^s \mathcal{H}_o^s(E).$$

Proposition C.45. Let (X, d) be a metric space and let $0 \leq s < \infty$. Then $\mathcal{H}_o^s$ is an outer measure.

By Carathéodory's theorem, $\mathcal{H}_o^s$ restricted to the σ -algebra of all $\mathcal{H}_o^s$ -measurable subsets of X is a complete measure denoted $\mathcal{H}^s$ and called the *s-dimensional Hausdorff measure*.

Using Proposition B.19 we have

Proposition C.46. Let (X, d) be a metric space. For $0 \leq s < \infty$ the outer measure $\mathcal{H}_o^s$ is a metric outer measure, so that every Borel subset of X is $\mathcal{H}_o^s$ -measurable.

Remark C.47. We remark that unlike $\mathcal{H}_o^s$, $\mathcal{H}_\delta^s$ is not a metric outer measure in general. As an example, consider in $\mathbb{R}^N$ the outer measure $\mathcal{H}_\delta^0$, given by

$$\mathcal{H}_\delta^0(E) := \inf \left\{ \sum_{E_n \neq \emptyset} 1 : E \subseteq \bigcup_{n=1}^{\infty} E_n, \text{diam } E_n < \delta \right\}, \quad E \subseteq \mathbb{R}^N.$$

If we take $E_1 = \{x\}$ and $E_2 = \{y\}$, where $0 < \|x - y\| < \delta$, then to cover $E_1 \cup E_2$ it is enough to consider one set, so $\mathcal{H}_\delta^0(E_1 \cup E_2) = 1$, while $\mathcal{H}_\delta^0(E_1) = 1$, $\mathcal{H}_\delta^0(E_2) = 1$. Thus,

$$\mathcal{H}_\delta^0(E_1 \cup E_2) = 1 < 2 = \mathcal{H}_\delta^0(E_1) + \mathcal{H}_\delta^0(E_2).$$

Proposition C.48. Let (X, d) be a metric space, let $E \subseteq X$, and let $0 \leq s < t < \infty$.

¹Hence, on the right-hand side we have the Hausdorff outer measure in Y and on the left-hand side we have the Hausdorff outer measure in X . We use the same notation.

- (i) If $\mathcal{H}_\delta^s(E) = 0$ for some $0 < \delta \leq \infty$, then $\mathcal{H}_o^s(E) = 0$.
- (ii) If $\mathcal{H}_o^s(E) < \infty$, then $\mathcal{H}_o^t(E) = 0$.
- (iii) If $\mathcal{H}_o^t(E) > 0$, then $\mathcal{H}_o^s(E) = \infty$.

In view of Proposition C.48 the following definition makes sense.

Definition C.49. Let (X, d) be a metric space. The *Hausdorff dimension* of a set $E \subseteq X$ is defined by

$$\dim_{\mathcal{H}}(E) := \inf\{0 \leq s < \infty : \mathcal{H}_o^s(E) = 0\}.$$

Note that $\dim_{\mathcal{H}}$ may be any number in $[0, \infty]$, not necessarily an integer number. Moreover, if $t := \dim_{\mathcal{H}}(E)$, then by the previous proposition we have that $\mathcal{H}_o^s(E) = 0$ for all $s > t$, while if $t > 0$, then $\mathcal{H}_o^s(E) = \infty$ for all $0 < s < t$.

We now restrict our attention to the case $X = \mathbb{R}^N$ (with the Euclidean norm). For $0 \leq s < \infty$ we define

$$(C.30) \quad c_s := \frac{\pi^{s/2}}{2^s \Gamma(1 + s/2)},$$

where $\Gamma(t)$ is the Euler Gamma function

$$\Gamma(t) := \int_0^\infty e^{-x} x^{t-1} dx, \quad 0 < t < \infty.$$

Note that $\Gamma(n) = (n-1)!$ for all $n \in \mathbb{N}$. We remark that $2^N c_N = \alpha_N := \frac{\pi^{N/2}}{\Gamma(1+N/2)}$, where α_N is the Lebesgue measure of the unit ball in $\mathbb{R}^N$, so that $\mathcal{L}^N(B(x, r)) = \alpha_N r^N$ for every ball $B(x, r)$ in $\mathbb{R}^N$.

Theorem C.50. Let $X = \mathbb{R}^N$ and let $0 \leq s < \infty$. Then

- (i) $\mathcal{H}_o^0$ is the counting measure,
- (ii) $\mathcal{H}_o^N = \mathcal{L}_o^N$,
- (iii) $\mathcal{H}_o^s \equiv 0$ if $s > N$.

Remark C.51. In view of Theorem C.50, it follows that the Hausdorff outer measures $\mathcal{H}_o^s$ are of interest only when $0 < s < N$.

Proposition C.52. We remark that Proposition C.48(ii) and Theorem C.50(iii) imply that $\dim_{\mathcal{H}}(E) \leq N$ for every set $E \subseteq \mathbb{R}^N$.

Exercise C.53. Give a direct, simple proof of the fact that $\mathcal{H}_o^1 = \mathcal{L}_o^1$. Hint: Show that $\mathcal{H}_\delta^1 = \mathcal{L}_o^1$ for all $\delta > 0$.

As a consequence of Proposition C.48 we have the following result.

Proposition C.54. Let $X = \mathbb{R}^N$ and let $0 < s < N$. Then the Hausdorff measure $\mathcal{H}^s$ is not σ -finite.

Notes

Deciphering Academese, I: “To the best of the author’s knowledge” = “We were too lazy to do a real literature search.”

—Jorge Cham, www.phdcomics.com

Chapters 1, 2, and 3 draw upon the book of van Rooij and Schikhof [237]¹.

Chapter 1: For a proof of Lebesgue’s differentiation theorem which relies only on the definition of Lebesgue outer measure we refer to the paper of Faure [74] (see also [29] for another proof).

For more information and extensions on the Weierstrass function (see (1.2)) we refer to the paper of Hardy [108], Krantz [135], and the recent paper of Pinkus [194].

We refer to the paper of Dovgoshey, Martio, Ryazanov, and Vuorinen [65] and the references contained therein for an extensive treatment of the Cantor function. For Exercise 1.32 see [12]. For Exercise 1.38 see [127]. Theorem 1.37 is due to Freilich [85], while Exercise 1.39 is due to Takács [227] (see also [202] and [118]). Exercise 1.40 is due to Pompeiu ([190]).

Chapter 2: Exercises 2.25 and Exercise 2.26 are based on papers of Gehring [93] and Wiener [247], while Exercise 2.30 is due to Heuer [115]. For Exercise 2.33 see [23]. Theorem 2.40 is due to Weil [243] (see also Katznelson and Stromberg [129] for an explicit construction).

The proof of the Helly selection theorem (Theorem 2.44) follows [181]. For Exercises 2.43 and 2.50 see [5]. Corollary 2.51 is proved in [252]. Theorem 2.55 and Exercise 2.57 are due to Josephy [126].

Theorem 2.60 is due to Banach [18] in the real-valued case, while Theorem 2.59, Lemma 2.61 and Theorem 2.60 in the general case are due to Federer [75].

¹This is a really nice book that is not as well known as it should be. I highly recommend it.

Chapter 3: For Exercise 3.3 see [181]. Exercise 3.14 is due to Botsko [30].

The author would like to thank Jan Malý for useful conversations on the proof of Theorem 3.29.

Section 3.3 draws from papers of Goffman [96], Van Vleck [238], and Varberg [239]. Exercise 3.32 is based on a paper of Gehring [93]. For Exercises 3.45 and 3.46 see [181], while Exercise 3.50 is based on a paper of Lindner [144].

Theorem 3.59 was first proved by Krzyżewski [136] and then later independently by Serrin and Varberg [210] (see also [97] for some extensions).

Theorem 3.55 is adapted from a paper of Josephy [126]. An alternative proof of Lemma 3.60 in the Sobolev setting due to A. Ancona may be found in [28]. Example 3.69 and Theorem 3.73 are due to Marcus and Mizel [159] (see also [9], [140], and [175]). Theorem 3.75 and all its corollaries were proved by Serrin and Varberg [210].

The proof of Corollary 3.90 uses some ideas of [203]. We refer also to the papers of Morse [176] and of Kober [133] for more information on singular functions.

Chapter 4: Theorem 4.16 and Exercise 4.18 follow [102]. For Theorem 4.19 see [142]. The case in which Ψ is even was first proved independently by Chiti [43] and by Crandall and Tartar [53]. We refer to the paper of Hajaiej [103] for extensions to the case in which Ψ is real-valued. Exercise 4.20 is based on [43].

The proof of Theorem 4.22 is adapted from the review paper of Talenti [229] (see also the paper of Duff [66]).

Exercise 4.24 is based on a paper of Cianchi [47]. We refer to the papers of Cianchi [47] and Dahlberg [56] for more information on the regularity of $(u^*)'$. Corollary 4.25 is due to Novak [186].

Exercises 4.30–4.33 are based on a paper of Ryff [198], (see also [48] and [78]).

For more information on the topics of this chapter and for an extensive bibliography we refer to the monographs of Kawohl [130] and Kesavan [131]; see also the book of Lieb and Loss [142] and the recent review paper of Talenti [230].

Chapter 5: The first two sections of Chapter 5 draw upon the review paper of Cesari [42]. Theorem 5.8 is due to Hilbert [119]. Our proof is due to Moore [174]. The first example of this type was given by Peano in 1890 [188]. Theorem 5.19 is due to Almgren and Lieb [8]. For Theorem 5.38 see [12].

Section 5.4 draws upon the books of Ambrosio and Tilli [12] and of Falconer [73].

Our proof of the Jordan curve theorem (Theorem 5.45) is due to Maehara [148] (see also the paper of Tverberg [235] for a different proof which does not rely on Brouwer fixed point theorem).

Chapter 6: This chapter draws upon the books of Dinculeanu [63], Dunford and Schwartz [67], and Saks [201]. For Theorem 6.14 see the paper of Winter [248]. Exercises 6.16 and 6.17 were proved by Hewitt [117] to which we refer for extensions to the case of unbounded intervals.

Chapter 7 For more information on functions in $BV(\Omega)$ we refer to the monographs of Ambrosio, Fusco, and Pallara [10], Evans and Gariepy [72], and Ziemer [251].

The proof of Lemma 7.4 is adapted from [33]. For Exercise 7.6 see the lecture notes of Dal Maso [57]. For more information on Sobolev functions of one variable we refer to the monographs of Brezis [33] and Burenkov [39]. The inequality (7.17) can be found in the book of Hardy, Littlewood, and Pólya [109, page 29]. Exercise 7.22 is based on a paper of Helmberg [113]. Exercise 7.32 is due to Lu and Wheeden [147]. Proposition 7.33 was proved by Chua and Wheeden [45].

Section 7.3 draws upon the paper of Gabushin [88].

Chapter 8: Sections 8.1 and 8.2 draw upon [62], [63], [67], [69], [83], and [206]. Theorem 8.24 is due to Simon [213].

For more information on the material in Sections 8.3, 8.4, and 8.5 and its applications to abstract evolution equations we refer to the monographs of Barbu [19], Brezis [34], and Lions [145]. For the notion of metric derivative (see Definition 8.36) and its applications we refer to the books of Ambrosio, Gigli and Savare [11] and Ambrosio and Tilli [12].

Theorem 8.62 was proved by Aubin [14] and Lions [145, Chapter 5], under the additional hypothesis that Y_0 and Y_2 are reflexive, and by Simon [213] in the general case.

Chapter 9: Theorem 9.14 is due to Rademacher [196]. The proof presented here follows the lecture notes of Lang [137]. Theorem 9.16 is due to Stepanoff [223]. The proof presented here is due to Malý [154].

Theorem 9.17 is essentially due to Haslam-Jones [110]. The proof presented here is based on an argument of Federer (see [220, page 268]) and its details are carried out in [164].

Exercise 9.19 is taken from the paper of Marcus and Mizel [160] (see also the paper [208] of Serrin).

Sections 9.3 and 9.4 draw upon [155] (see also [10], [72], [149], [251]).

The proof of the Cauchy–Binet formula in Theorem 9.25 is due to Tao [231].

For a proof of Lemma 9.31 in the case $k = N$, which does not make use of Hausdorff measures, we refer to Flett [81].

Exercises 9.38–9.42 are taken from the paper of Lax [138] to which we refer for some historical background and references on the Brouwer fixed point theorem [36].

Theorems 9.43, 9.44, 9.46 are due to Whitney [244].

Theorem 9.52 is due to Varberg [241] (see also Rudin [199] and the paper of Schwartz [207]). We refer to the paper of Hajłasz [104] and the references therein for significant improvements.

In recent years there has been a renewed interest in absolutely continuous functions of several variables and the Lusin (N) property. In [153], Malý proposed an alternative definition. We refer to the papers of Csörnyei [54] and of Hencl and Malý [114] for more information and references on this topic.

Exercise 9.60 is taken from a paper of Fraenkel [84] to which we refer for more information about different definitions of regular domains (see also the monograph of Delfour and Zolésio [60]).

Chapter 10: The material on distributions is based on [69], [200], and [249]. For Exercise 10.4 see the lecture notes of Acquistapace [3]. The second part of the chapter, starting from Section 10.4 draws upon [16], [98], [99], [142], and [221].

Chapter 11: For more information on Sobolev functions we refer to the monographs [7], [39], [72], [165], and [251].

Exercise 11.16 is based on a paper of Simader [212]. Exercise 11.19 is due to Nicolás García Trillos and Matteo Rinaldi, Exercises 11.20 and 11.21 to Ian Tice.

Theorem 11.24 is due to Meyers and Serrin [169]. Theorem 11.34 follows a paper of Fraenkel [84] (see also the monograph of Delfour and Zolésio [60]). For the proof of Theorem 11.40 see Hildén [120]. Theorem 11.43 is due to Hajłasz and Kałamajska [105]. Exercise 11.48 is due to Kolsrud [134].

For Exercise 11.49 see [232]. Exercises 11.58, 11.59, and 11.60 are based on a paper of Kinnunen [132].

The first step of the proof of Theorem 11.75 is due to Stein (see the paper of Brezis [35]). Theorem 11.75 is quite important for two different reasons. This characterization is often used to prove higher regularity of solutions of partial differential equations (see, e.g., [33], [71], and [94]). It is also important because it allows us to give a definition of Sobolev spaces

that does not involve derivatives. Such characterizations have been studied extensively in recent years, since they can be used to define Sobolev spaces on metric spaces. We refer to the books of Ambrosio and Tilli [12] and Hajlasz and Koskela [107] and to the recent survey paper of Heinonen [112] for more details on this subject.

Chapter 12: Step 4 of the proof of the Sobolev–Gagliardo–Nirenberg embedding theorem follows the paper of Maggi and Villani [150]. For a proof of the results mentioned in Remark 12.7 we refer to the papers of Aubin [15] and Talenti [228]. For sharp forms of the Sobolev–Gagliardo–Nirenberg inequality we refer to the recent work of Cianchi, Fusco, Maggi, and Pratelli [50] and to the references contained therein.

Exercises 12.15 (i) and (ii) and 12.25 are based on [84]. For Exercise 12.15(iii) see [13].

The proof of the Rellich–Kondrachov theorem is adapted from a paper of Serrin [209]. For Exercise 12.24 see [209].

Exercise 12.28 and the first part of Exercise 12.29 are based on a paper of Strauss [224], while the second part of Exercise 12.29 is based on a paper of Ebihara and Schonbek [68]. The author would like to thank Bill Hrusa for suggesting the reference [224].

The proof of Theorem 12.33 is adapted from [165]. Results similar to Theorem 12.40 in bounded domains are due to Pokhozhaev [189], Trudinger [234] and Yudovich [250] when $m = 1$ and to Strichartz [225] for $m \geq 2$ (see also the work of Moser [179] and the recent paper of Li and Ruf [141] for some recent results).

Corollary 12.50 and Exercise 12.45 follow a paper of Serrin [208]. Corollary 12.52 follows a paper of Marcus and Mizel [158].

In [156] Malý and Martio have constructed a continuous transformation $\Psi \in W^{1,N}(\mathbb{R}^N; \mathbb{R}^N)$ that does not satisfy the (N) property and whose Jacobian is zero $\mathcal{L}^N$ -a.e. in $\mathbb{R}^N$.

Section 12.4 is based on paper of Bourdaud [31]. For Exercise 12.68 see [165]. Theorem 12.69 is due to Marcus and Mizel [161] and [162]. Theorem 12.70 and Exercise 12.71 are due to Dahlberg [55]. All the other theorems are due to Bourdaud [31].

Section 12.5 is based on the papers of Gagliardo [91] and Nirenberg [184]. The author would like to thank Ian Tice for his crucial help and endless discussions on this section. The proof of Proposition 12.88 is due to Hormander [121].

Chapter 13: A necessary and sufficient condition on the regularity of $\partial\Omega$ for the existence of an extension operator is still missing except in the two-dimensional case (see the paper of Jones [122]). For Exercises 13.1 and 13.2

see the book of Maz'ja [165]. The proof of Theorem 13.6 is adapted from a paper of Lieberman [143].

Exercises 13.7 and 13.14 and the proof of Theorem 13.17 follow the book of Stein [220].

Definition 13.11 is also from the book of Stein [220], but we added the condition that $\{\Omega_n\}_n$ is locally finite, since we were not able to show that conditions (i)–(iii) imply that $\{\Omega_n\}_n$ is locally finite and we used this property in the proof of Theorem 13.17.

We refer to the book of Ziemer [251] for an extensive treatment of Poincaré's inequality. Exercise 13.30, Proposition 13.34, and Theorems 13.41 and 13.46 are due to Smith and Stegenga [214]. Exercises 13.22 and 13.31 were suggested by Ian Tice and Bill Hrusa, respectively.

Theorem 13.36 follows a paper of Chua and Wheeden [46]. The idea of the proof comes from the paper of Bebendorf [22], who corrected a mistake in the original paper of Payne and Weinberger [187] in the case $p = 2$. See also the paper of Acosta and Durán [2] for the case $p = 1$.

Section 13.3 is based on the papers of Gagliardo [91] and Nirenberg [184]. In particular, Lemma 13.52 is adapted from [91]. The author would like to thank Ian Tice for several useful discussions on this section.

Chapter 14: As we already mentioned at the beginning of Chapter 14, what is covered here is just the tip of the iceberg. We refer the interested reader to the monographs [10], [72], [76], [95], [149], and [251] for more information on functions of bounded variation.

For Exercises 14.5 and 14.8 see the book of Giusti [95]. Section 14.3 is based on a paper of Serrin [208].

Theorem 14.28 is due to Fleming and Rishel [79].

For sharp forms of the Sobolev–Gagliardo–Nirenberg inequality in BV (Theorem 14.33) we refer to the recent work of Fusco, Maggi, and Pratelli [87] and to the references contained therein.

Theorem 14.48 is due to Sard [204], and the proof follows the book of Milnor [171].

Chapter 15: Exercise 15.8 is based on [102]. Theorems 15.23, 15.29 and 15.32 were proved independently by Hildén [120] and Talenti [228]. We follow here the approach of Talenti [229]. Step 1 of the proof of Proposition 15.25 is due to Malý and Pick [157], while Step 2 is due to Martin, Milman, and Pustylnik [163]. Theorem 15.32 has been significantly extended by Cianchi and Fusco [49].

The proof of Theorem 15.33 is due to Malý and Pick [157]. Exercise 15.34 was suggested by Ian Tice.

Theorem 15.35 and Exercise 15.37 are due to Adachi and Tanaka [4]. The simplified proof of Theorem 15.35 is due to Francesco Maggi.

For more information on the topics of this chapter and for an extensive bibliography we refer to the monographs of Kawohl [130] and Kesavan [131] and the review paper of Talenti [230]; see also the book of Lieb and Loss [142].

Chapter 16: This chapter draws upon the books [7], [24], [25], [40], [146], [232], and [233] to which we refer for more details about interpolation theory.

Theorem 16.16 is due to Karadzhov, Milman, and Xiao [128] and its generalization to abstract settings of papers of Bourgain, Brezis, and Mironescu [32] and Maz'ja and Shaposhnikova [167].

Some of the exercises at the end of Section 16.4 were suggested by Ian Tice. Exercises 16.38 and 16.39 are adapted from a paper of Maligranda and Persson [151].

Chapter 17: This chapter was written in collaboration with Ian Tice, whom I would like to thank for a wonderful summer spent discussing Besov spaces. Any mistake is purely due to me. What is written here is just the tip of the iceberg. We refer to the textbooks [7], [16], [24], [25], [26], [27], [39], [40], [146], [192], [220], [232], and [233] for more information on this subject.

Although I spent many months rewriting this chapter, I am still not satisfied with the results. My initial hope was to pick the seminorm 17.5 and give simple proofs of all the results using only that seminorm. Needless to say, I failed miserably and ended up using more seminorms that I would have liked.

Exercise 17.10 and similar counterexamples can be found in the paper of Taibleson [226] (see also [220]).

Exercise 17.14 is a particular case of more general results contained in a paper of Brezis [35]. The suggestion for the proof came from one of the questions in <http://math.stackexchange.com>

Theorem 17.24 is adapted from [24]. The embedding Theorems 17.49, 17.52, and 17.55 are based on a paper of Peetre [191].

Theorem 17.57 is due to Maz'ja and Shaposhnikova [167]. The present proof is based on that of Karadzhov, Milman, and Xiao [128]. Theorem 17.60 was first proved by Bourgain, Brezis, and Mironescu [32] and by Ponce [195] (see also the paper of Brezis [35]). The present proof is based on that of Karadzhov, Milman, and Xiao [128].

The proofs of Theorem 17.66 and of Corollary 17.68 were suggested by Ian Tice. Theorem 17.69 is adapted from [24].

For refined inequalities of the type in Lemma 17.88 in Lorentz spaces we refer to the recent paper [17] and the references therein.

Chapter 18: Theorems 18.13, 18.27, and 18.28 were first proved for bounded Lipschitz continuous domains by Gagliardo in [89]. The proof of Theorem 18.27 is due to Solonnikov [216]. The proof of Theorem 18.28 is adapted from a paper of Uspenskiĭ [236]. Exercise 18.3 is based on [84].

Appendix A: This chapter draws upon [83].

Appendix B: This chapter draws upon [83], to which we refer for the proofs of all the results that cannot be found in classical texts (such as [61], [72], [82], [199]).

Appendix C: The proof of the Brunn–Minkowski inequality (see Theorem C.8) follows a survey paper of Gardner [92], to which we refer for more information and an extensive bibliography. Exercise C.10 is based on a paper of Bebendorf [22]. Exercise C.28 is due to L. Tartar. The proof of Proposition C.29 follows a paper of Hedberg [111] but is presented here in a simplified form due L. Tartar.

Notation and List of Symbols

*Deciphering Academese, II: "...remains an open question" =
"we have no clue either."*

— Jorge Cham, www.phdcomics.com

Since the number of letters and symbols (and the author's imagination) are limited, sometimes, and when there is no possibility of confusion, we use the same letter or symbol for different objects (not in the same theorem, we hope). For example, the letter C is used for constants, but also for closed sets, while the letter α is used for multi-indices, but also as a real number.

A subscript on an equation number refers to that expression in the display. For example, given

$$(E.1) \qquad \text{Expression 1,} \qquad \text{Expression 2,}$$

$(E.1)_2$ refers to Expression 2 in (E.1).

Constants

- C, c : arbitrary constants that *can change from line to line* and from expression to expression that can be computed in terms of known quantities.

Sets

- X, Y : sets or spaces; $\text{card } X$: the cardinality of a set X ; $\mathcal{P}(X)$: the family of all subsets of a set X ; $\mathcal{F}, \mathcal{G}$: family of sets or of functions.
- $\subseteq$: inclusion between two sets, *with equality possible*, $\subset$: strict inclusion between two sets,
- $X \Delta Y = (X \setminus Y) \cup (Y \setminus X)$: symmetric difference between the sets X and Y .

Functional Analysis

- Ω, U, V : open sets; K : a compact set; B : a Borel set; C : a closed or convex set. For two open sets U, V , $U \subseteq V$ means that the closure of U is compact and contained in V .
- E, F, G : usually denote sets; ∂E : boundary of E ; E° : the interior of E ; $\overline{E}$: the closure of E .
- χ_E : characteristic function of the set E .
- τ : topology; $\|\cdot\|$: norm.
- dist: distance; diam: diameter.
- $\langle \cdot, \cdot \rangle_{X, X'}$: duality pairing.
- $\rightharpoonup$: weak convergence; $\overset{*}{\rightharpoonup}$: weak star convergence.
- $C(X)$: the space of all continuous functions; $C_c(X)$: the space of all continuous functions whose support is compact; $C_0(X)$: the closure of $C_c(X)$ in the sup norm.

Measure Theory

- $\mathfrak{M}, \mathfrak{N}$: algebras or σ -algebras; $\mathfrak{M} \otimes \mathfrak{N}$: product σ -algebra of $\mathfrak{M}$ and $\mathfrak{N}$ (not to be confused with $\mathfrak{M} \times \mathfrak{N}$); $\mathcal{B}(X)$: Borel σ -algebra, $\mathcal{B}_b(X)$ the family of bounded Borel sets, $\mathcal{B}_c(X)$ the family of Borel sets compactly contained in X .
- μ, ν : (positive) finitely additive measures or (positive) measures; $\nu \perp \mu$ means that μ, ν are mutually singular measures; $\nu \ll \mu$ means that the measure ν is absolutely continuous with respect to the measure μ ; $\frac{d\nu}{d\mu}$: the Radon–Nikodym derivative of ν with respect to μ ; μ^* : outer measure.
- λ : a finitely additive signed measure or a signed measure; $\lambda^+, \lambda^-, |\lambda|$: the upper, lower, and total variation measures of λ . For a vector-valued measure λ , $\|\lambda\|$ the total variation measure of λ .
- v , and w usually denote functions or variables.
- $f, g, \varphi, \psi, \phi, u$ usually denote functions; $\text{supp } u$: support of the function u ; $\text{Lip } u$: Lipschitz constant of the function u ; $\text{osc}(u; E)$: oscillation of u on E ; $u|_E$ the restriction of u to the set E ; $u * v$: convolution of the functions u and v , $\widehat{u}$ or $\mathcal{F}(u)$: the Fourier transform of u , $u^\vee$ the inverse Fourier transform; $M(u)$: maximal function of u ; ϱ_u : the distribution function of a function u ; u^* : the decreasing rearrangement of u ; $u^\sharp$: the spherically symmetric rearrangement (or Schwarz symmetric rearrangement) of a function u .
- $\mathcal{M}_b(X; \mathbb{R})$: the space of all signed finite Radon measures.

- $\text{ba}(X, \mathfrak{M}, \mu)$: space of all bounded finitely additive signed measures absolutely continuous with respect to μ .
- $L^p(X, \mathfrak{M}, \mu)$, $L^p(X, \mu)$, $L^p(X)$ are various notations for L^p spaces; p' : Hölder conjugate exponent of p ; $\|\cdot\|_{L^p(X, \mathfrak{M}, \mu)}$, $\|\cdot\|_{L^p(X)}$, $\|\cdot\|_{L^p}$, or $\|\cdot\|_p$ are various notations for the norm in L^p . Similarly, given $L^p(X; Y)$ we sometimes write $\|\cdot\|_{L^p(X)}$ or $\|\cdot\|_{L^p}$ for $\|\cdot\|_{L^p(X; Y)}$. $L^{p,q}(X)$ Lorentz spaces. $L^\Phi(X)$: an Orlicz space.
- Given a function u and a set E , $u_E = \frac{1}{\mathcal{L}^N(E)} \int_E u \, d\mu$ is the integral average of u over a set E .

Functions of One Variable

- $\mathbb{N}$: the set of positive integers; $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$; $\mathbb{Z}$: the integers; $\mathbb{Q}$: the rational numbers; $\mathbb{R}$: the real line; $\overline{\mathbb{R}} := [-\infty, \infty]$: the extended real line; $\mathbb{C}$: the complex numbers; if $x \in \mathbb{R}$, then $x^+ := \max\{x, 0\}$, $x^- := \max\{-x, 0\}$, $|x| := x^+ + x^-$; $\lfloor x \rfloor$: the integer part of x , $\text{sgn } x$ the sign of x , $\text{sgn } x := x/|x|$ for $x \neq 0$ and $\text{sgn } 0 := 0$.
- For a complex number $z = x + iy$, where $x, y \in \mathbb{R}$, the complex conjugate of z is the number $\bar{z} := x - iy$ and the norm of z is $\|z\| := \sqrt{x^2 + y^2}$.
- I : an interval; $\mathbb{D}$: the Cantor set.
- $\mathcal{L}_o^1$: the 1-dimensional Lebesgue outer measure; $\mathcal{L}^1$: the 1-dimensional Lebesgue measure.
- $BPV(I)$: the space of all functions of bounded pointwise variation on I ; $AC(I)$: the space of all absolutely continuous functions defined on I .
- Given a function u , $u_+(x)$ and $u_-(x)$ are the right and left limits of u at a point x , u_J is the jump function of u , u_C is the Cantor part of u , u_{AC} is the absolutely continuous part of u , $\text{Var } u$ is the pointwise variation of u , $\mathbf{V}$ (or $\mathbf{V}_u$) is the indefinite pointwise variation of u , $\text{PVar } u$ and $\text{NVar } u$ are the positive and negative pointwise variations of u , $\text{Var}_p u$ is the p -variation of u , $\text{essVar } u$ is the essential variation of u , $N_u(\cdot; E)$ is the Banach indicatrix (or counting function) of u over a set E , and $\underline{D}_-u(x)$, $\overline{D}_-u(x)$, $\overline{D}_+u(x)$, and $\underline{D}_+u$ are Dini's derivatives of u at x .
- μ_u : the Lebesgue–Stieltjes measure generated by an increasing function u ; λ_u : the Lebesgue–Stieltjes signed measure generated by a function u with bounded pointwise variation.
- $L(\gamma)$: the length of a curve γ .

Functions of Several Variables

- $\mathbb{R}^N$: the N -dimensional Euclidean space, $N \geq 1$, for $x = (x_1, \dots, x_N)$ and $y = (y_1, \dots, y_N) \in \mathbb{R}^N$ in $\mathbb{R}^N$,¹ $x \cdot y = \sum_{i=1}^N x_i y_i$ the Euclidean inner product of x and y , $\|x\| = \sqrt{x \cdot x}$ the Euclidean norm of x . When working with Euclidean spaces of different dimension, we will sometimes use the notation $\|x\|_N$ for $\|x\|$. $\mathbb{R}_+^N = \{x \in \mathbb{R}^N : x_N > 0\}$, $\mathbb{R}_-^N = \{x \in \mathbb{R}^N : x_N < 0\}$. Given $x = (x_1, \dots, x_N) \in \mathbb{R}^N$, for every $i = 1, \dots, N$ we denote by x'_i the $(N-1)$ -dimensional vector obtained by removing the i th component from x and with an abuse of notation we write

$$(E.2) \quad x = (x'_i, x_i) \in \mathbb{R}^{N-1} \times \mathbb{R}.$$

When $i = N$, we will also use the simpler notation

$$(E.3) \quad x = (x', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R}.$$

- δ_{ij} : the Kronecker delta; that is, $\delta_{ij} := 1$ if $i = j$ and $\delta_{iN} := 0$ otherwise.
- e_i , $i = 1, \dots, N$: the unit vectors of the standard (or canonical) orthonormal basis of $\mathbb{R}^N$.
- $\det$: determinant of a matrix or a linear mapping.
- Ω : an open set of $\mathbb{R}^N$ (not necessarily bounded); ν : the outward unit normal to $\partial\Omega$; $B_N(x, r)$ (or simply $B(x_0, r)$): open ball in $\mathbb{R}^N$ of center x_0 and radius r ; $Q_N(x_0, r) := x_0 + (-\frac{r}{2}, \frac{r}{2})^N$, (or simply $Q(x_0, r)$); S^{N-1} : unit sphere $\partial B(0, 1)$ in $\mathbb{R}^N$.
- $\mathcal{L}_o^N$: the N -dimensional Lebesgue outer measure; $\mathcal{L}^N$: the N -dimensional Lebesgue measure; $\alpha_N := \mathcal{L}^N(B(0, 1))$,
- $\mathcal{H}_o^s$: the s -dimensional Hausdorff outer measure; $\mathcal{H}^s$: the s -dimensional Hausdorff measure. β_N is the surface area of the unit sphere, that is, $\beta_N := \mathcal{H}^{N-1}(S^{N-1}) = \frac{2\pi^{N/2}}{\Gamma(N/2)}$. Hence, $\beta_N = N\alpha_N$.
- Given $E \subseteq \mathbb{R}^N$ and $\Psi : E \rightarrow \mathbb{R}^M$, $d\Psi(x)$ is the differential of Ψ at x , $\frac{\partial \Psi}{\partial x_i}(x)$ is the partial derivative of Ψ at x_0 with respect to x_i , $\frac{\partial \Psi}{\partial v}(x)$ is the directional derivative of Ψ at x in the direction $v \in S^{N-1}$, $\nabla \Psi(x)$ is the gradient of Ψ at x_0 , and for $M \geq 2$, $J_\Psi(x) = \nabla \Psi(x)$ is the Jacobian matrix at x and $||J_\Psi(x)|| := \sqrt{\det(J_\Psi^t(x) J_\Psi(x))}$ is the Jacobian of Ψ at x .
- $C^m(\Omega)$: the space of all functions that are continuous together with their partial derivatives up to order $m \in \mathbb{N}_0$, $C^\infty(\Omega) :=$

¹When there is no possibility of confusion, we will also use x_1, x_2 , etc., to denote different points of $\mathbb{R}^N$. Thus, depending on the context, x_i is either a point of $\mathbb{R}^N$ or the i th coordinate of the point $x \in \mathbb{R}^N$.

$\bigcap_{m=0}^{\infty} C^m(\Omega)$; $C_c^m(\Omega)$ and $C_c^\infty(\Omega)$: the subspaces of $C^m(\Omega)$ and $C^\infty(\Omega)$, respectively, consisting of all functions with compact support; $C^{0,a}(\overline{\Omega})$: the space of all bounded Hölder continuous functions with exponent $0 < a \leq 1$.

- For a multi-index $\alpha = (\alpha_1, \dots, \alpha_N) \in \mathbb{N}_0^N$, $\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial x^\alpha} := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_N^{\alpha_N}}$, $|\alpha| := \alpha_1 + \dots + \alpha_N$: the length of α ; $\nabla^m u$ the vector of components $\partial^\alpha u$, M_m is the number of all multi-indices α with $|\alpha| = m$; $\partial_i^n := \partial^{ne_i}$. Given two multi-indices $\alpha = (\alpha_1, \dots, \alpha_N)$ and $\beta = (\beta_1, \dots, \beta_N)$, we write $\alpha \leq \beta$ if $\alpha_i \leq \beta_i$ for all $i = 1, \dots, N$. Also, $\alpha! := \alpha_1! \dots \alpha_N!$ and

$$(E.4) \quad \binom{\alpha}{\beta} := \frac{\alpha!}{\beta!(\alpha - \beta)!}.$$

Given $x \in \mathbb{R}^N$ and $\alpha \in \mathbb{N}_0^N$ we write $x^\alpha := x_1^{\alpha_1} \dots x_N^{\alpha_N}$.

- $\nabla^k u$: to be the vector of components $\partial^\alpha u$, $|\alpha| = k$; M_k , $\nabla^0 u := u$.
- $\mathcal{D}_K(\Omega)$: the set of all functions in $C_c^\infty(\Omega)$ with support in the compact set K ; $\mathcal{D}(\Omega)$: the space $C_c^\infty(\Omega)$ endowed with a particular topology τ ; $\mathcal{D}'(\Omega)$: the topological dual of $\mathcal{D}(\Omega)$; $\mathcal{S}(\mathbb{R}^N)$: the space of rapidly decreasing functions; $\mathcal{S}'(\mathbb{R}^N)$: the topological dual of $\mathcal{S}(\mathbb{R}^N)$, called the space of tempered distributions, rapidly decreasing functions.
- In Chapter 10, T : a distribution or a tempered distribution; δ_{x_0} : the delta Dirac with mass at x_0 ; $\frac{\partial^\alpha T}{\partial x^\alpha}$ or $\partial^\alpha T$: the derivative of T with respect to the multi-index α ; $\text{supp } T$: the support of T ; $T * \phi$: the convolution of T with the function ϕ .
- $W^{m,p}(\Omega)$: a Sobolev space; $\|\cdot\|_{W^{m,p}(\Omega)}$: the norm in $W^{m,p}(\Omega)$; for $u \in W^{m,p}(\Omega)$, $\partial^\alpha u$ or $\frac{\partial^{|\alpha|} u}{\partial x^\alpha}$: the α th weak (or distributional) partial derivative of u , $\partial_i u$ or $\frac{\partial u}{\partial x_i}$: the weak (or distributional) partial derivative of u with respect to x_i ; $\|u\|_{W^{m,p}(\Omega)} := \|\nabla^m u\|_{L^p(\Omega; \mathbb{R}^{M_m})}$; $H^m(\Omega)$: the Hilbert space $W^{m,2}(\Omega)$; $\dot{W}^{m,p}(\Omega)$: the homogeneous Sobolev space; $(W^{m,p}(\Omega))'$: the topological dual of $W^{m,p}(\Omega)$; $W_0^{m,p}(\Omega)$: the closure of $C_c^\infty(\Omega)$ in $W^{m,p}(\Omega)$; $W^{-m,p'}(\Omega)$: the topological dual of $W_0^{m,p}(\Omega)$; $\mathcal{E} : W^{m,p}(\Omega) \rightarrow W^{m,p}(\mathbb{R}^N)$: an extension operator; $\text{Tr}(u)$: the trace of a Sobolev function; $p^* := \frac{Np}{N-p}$, $p_{m,k}^* := \frac{Np}{N-(m-k)p}$: Sobolev critical exponents.
- φ_ε : a standard mollifier; $u_\varepsilon := \varphi_\varepsilon * u$: the mollification of u .
- $\mathcal{PA}$: a particular family of continuous piecewise affine functions.
- d_{reg} : the regularized distance.

- $BV(\Omega)$: the space of functions of bounded variation. Given $u \in BV(\Omega)$, $D_i u$ is the weak (or distributional) partial derivative of u with respect to x_i , Du is the weak (or distributional) gradient of u , $\|Du\|$ is the total variation measure of the measure Du ; $V(u, \Omega)$: the variation of a function u in Ω ; $P(E, \Omega)$: the perimeter of a set E in Ω .
- For $x, h \in \mathbb{R}^N$, $\Delta^h u(x) := u(x+h) - u(x)$, the forward difference operator; $\Delta_h^m u(x) := \Delta_h(\Delta_h^{m-1} u(x))$; $\sim \Delta_h^2 u(x) := u(x+h) - 2u(x) + u(x-h)$; $B_q^{s,p}(\mathbb{R}^N)$: a Besov space; $\dot{B}_q^{s,p}(\mathbb{R}^N)$: a homogeneous Besov space; $|\cdot|_{B_q^{s,p}(\mathbb{R}^N)}$, $|\cdot|_{B_q^{s,p}(\mathbb{R}^N)}^{(n)}$, $|\cdot|_{B_q^{s,p}(\mathbb{R}^N)}^\infty$, $|\cdot|_{B_q^{s,p}(\mathbb{R}^N)}^{[R]}$, $|\cdot|_{B_q^{s,p}(\mathbb{R}^N)}^\vee$: various seminorms in $B_q^{s,p}(\mathbb{R}^N)$; $\Lambda_1(\mathbb{R}^N)$: the Zygmund space.

Bibliography

*In the end, we will remember not the words of our enemies, but
the silence of our friends.*

—Martin Luther King, Jr.

- [1] E. Acerbi, V. Chiadò Piat, G. Dal Maso, and D. Percivale, *An extension theorem from connected sets, and homogenization in general periodic domains*, Nonlinear Anal. **18** (1992), no. 5, 481–496.
- [2] G. Acosta and R.G. Durán, *An optimal Poincaré inequality in L^1 for convex domains*, Proc. Amer. Math. Soc. **132** (2004), 195–202.
- [3] P. Acquistapace, *Appunti di Analisi convessa*, 2005.
- [4] S. Adachi and K. Tanaka, *Trudinger type inequalities in R^N and their best exponents*, Proc. Amer. Math. Soc. **128** (2000), no. 7, 2051–2057.
- [5] C.R. Adams, *The space of functions of bounded variation and certain general spaces*, Trans. Amer. Math. Soc. **40** (1936), no. 3, 421–438.
- [6] R.A. Adams, *Sobolev spaces*, Pure and Applied Mathematics, 65. A Series of Monographs and Textbooks, New York–San Francisco–London: Academic Press, Inc., a subsidiary of Harcourt Brace Jovanovich, Publishers. XVIII, 1975.
- [7] R.A. Adams and J.J.F. Fournier, *Sobolev spaces*, Second edition, Academic Press (Elsevier), 2003.
- [8] F.J. Almgren and E.H. Lieb, *Symmetric decreasing rearrangement is sometimes continuous*, J. Amer. Math. Soc. **2** (1989), no. 4, 683–773.
- [9] L. Ambrosio and G. Dal Maso, *A general chain rule for distributional derivatives*, Proc. Amer. Math. Soc. **108** (1990), no. 3, 691–702.
- [10] L. Ambrosio, N. Fusco, and D. Pallara, *Functions of bounded variation and free discontinuity problems*, Oxford Mathematical Monographs, The Clarendon Press, Oxford University Press, New York, 2000.
- [11] L. Ambrosio, N. Gigli, and G. Savare, *Gradient flows*, Birkhäuser Basel, 2008.
- [12] L. Ambrosio and P. Tilli, *Topics on analysis in metric spaces*, Oxford Lecture Series in Mathematics and its Applications, 25, Oxford University Press, Oxford, 2004.
- [13] C.J. Amick, *Decomposition theorems for solenoidal vector fields*, J. London Math. Soc. (2) **15** (1977), no. 2, 288–296.
- [14] J. P. Aubin, *Un théorème de compacité*, C. R. Acad. Sci., **256** (1963), 5042–5044.

- [15] T. Aubin, *Problèmes isopérimétriques et espaces de Sobolev*, J. Differential Geometry **11** (1976), no. 4, 573–598.
- [16] H. Bahouri, J.Y. Chemin, and R. Danchin, *Fourier analysis and nonlinear partial differential equations*, Springer-Verlag Berlin Heidelberg, 2011.
- [17] H. Bahouri and A. Cohen, *Refined Sobolev inequalities in Lorentz spaces*, J. Fourier Anal. Appl. **17** (2011) 662–673.
- [18] S. Banach, *Sur les lignes rectifiables et les surfaces dont l’aire est finie*, Fundamenta Mathematicae **7** (1925), 225–236.
- [19] V. Barbu, *Nonlinear Semigroups and Differential Equations in Banach Spaces*, Noordhoff, 1976.
- [20] R.G. Bartle, *The elements of real analysis*, Second edition, John Wiley & Sons, New York–London–Sydney, 1976.
- [21] B. Beauzamy, *Espaces d’interpolation réels: Topologie et géométrie*, Lecture Notes in Mathematics, 666, Springer-Verlag Berlin Heidelberg, 1978.
- [22] M. Bebendorf, *A note on the Poincaré inequality for convex domains*, Z. Anal. Anwend. **22** (2003), no. 4, 751–756.
- [23] J.J. Benedetto, *Real variable and integration*. With historical notes, Mathematische Leitfäden. B. G. Teubner, Stuttgart, 1976.
- [24] C. Bennett and R. C. Sharpley, *Interpolation of operators*. Academic Press, Inc., Boston, MA, 1988.
- [25] J. Bergh and J. Lofstrom, *Interpolation spaces: An introduction*, Springer-Verlag, Berlin–New York, 1976.
- [26] O.V. Besov, V.P. Il’in, and S.M. Nikol’skiĭ, *Integral representations of functions and imbedding theorems*, Vol. I. Translated from the Russian, Scripta Series in Mathematics. Edited by Mitchell H. Taibleson, V. H. Winston & Sons, Washington, D.C.; Halsted Press [John Wiley & Sons], New York–Toronto, Ontario–London, 1978.
- [27] O.V. Besov, V.P. Il’in, and S.M. Nikol’skiĭ, *Integral representations of functions and imbedding theorems*, Vol. II. Translated from the Russian, Scripta Series in Mathematics. Edited by Mitchell H. Taibleson. V. H. Winston & Sons, Washington, D.C.; Halsted Press [John Wiley & Sons], New York–Toronto, Ontario–London, 1979.
- [28] L. Boccardo and F. Murat, *Remarques sur l’homogénéisation de certains problèmes quasi-linéaires*, Portugal. Math. **41** (1982), no. 1–4, 535–562 (1984).
- [29] M.W. Botsko, *An elementary proof of Lebesgue’s differentiation theorem*, Amer. Math. Monthly **110** (2003), no. 9, 834–838.
- [30] M.W. Botsko, *An elementary proof that a bounded a.e. continuous function is Riemann integrable*, Amer. Math. Monthly **95** (1988), no. 3, 249–252.
- [31] G. Bourdaud, *Le calcul fonctionnel dans les espaces de Sobolev*. (French) Invent. Math. **104** (1991) 435–446.
- [32] J. Bourgain, H. Brezis, and P. Mironescu, *Another look at Sobolev spaces*, J.L. Menaldi, E. Rofman and A. Sulem, eds. Optimal control and partial differential equations. In honour of Professor Alain Bensoussan’s 60th birthday. Proceedings of the conference, Paris, France, December 4, 2000. Amsterdam: IOS Press; Tokyo: Ohmsha. 439–455 (2001).
- [33] H. Brezis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations*, Springer-Verlag New York, 2011.

- [34] H. Brezis, *Opérateurs maximaux monotones et semi-groupes de contractions dans les espaces de Hilbert*, North-Holland, 1973.
- [35] H. Brezis, *How to recognize constant functions. A connection with Sobolev spaces*, Uspekhi Mat. Nauk **57** (2002), no. 4 (346), 59–74; translation in Russian Math. Surveys **57** (2002), no. 4, 693–708.
- [36] L.E.J. Brouwer, *Über Abbildung von Mannigfaltigkeiten*, Math. Ann. **71** (1911), no. 1, 97–115.
- [37] A. Bruckner, *Differentiation of real functions*, second edition, CRM Monograph Series, 5, American Mathematical Society, Providence, RI, 1994.
- [38] D. Burago, Y. Burago, and S. Ivanov, *A course in metric geometry*, Graduate Studies in Mathematics, 33, American Mathematical Society, Providence, RI, 2001.
- [39] V.I. Burenkov, *Sobolev spaces on domains*, Teubner-Texte zur Mathematik, 137. B. G. Teubner Verlagsgesellschaft mbH, Stuttgart, 1998.
- [40] P.L. Butzer and H. Berens, *Semi-groups of operators and approximation*, Springer-Verlag New York Inc., New York 1967.
- [41] F.S. Cater, *When total variation is additive*, Proc. Amer. Math. Soc. **84** (1982), no. 4, 504–508.
- [42] L. Cesari, *Rectifiable curves and the Weierstrass integral*, Amer. Math. Monthly **65** (1958), no. 7, 485–500.
- [43] G. Chiti, *Rearrangements of functions and convergence in Orlicz spaces*, Applicable Anal. **9** (1979), no. 1, 23–27.
- [44] M. Chlebík, A. Cianchi, and N. Fusco, *The perimeter inequality under Steiner symmetrization: Cases of equality*, Ann. of Math. (2) **162** (2005), no. 1, 525–555.
- [45] S.K. Chua and R.I. Wheeden, *Sharp conditions for weighted 1-dimensional Poincaré inequalities*, Indiana Univ. Math. J. **49** (2000), no. 1, 143–175.
- [46] S.K. Chua and R.I. Wheeden, *Estimates of best constants for weighted Poincaré inequalities on convex domains*, Proc. London Math. Soc. (3) **93** (2006), no. 1, 197–226.
- [47] A. Cianchi, *Second-order derivatives and rearrangements*, Duke Math. J. **105** (2000), no. 3, 355–385.
- [48] A. Cianchi and A. Ferone, *A strengthened version of the Hardy–Littlewood inequality*, J. Lond. Math. Soc. (2) **77** (2008), no. 3, 581–592.
- [49] A. Cianchi and N. Fusco, *Functions of bounded variation and rearrangements*, Arch. Ration. Mech. Anal. **165** (2002), no. 1, 1–40.
- [50] A. Cianchi, N. Fusco, F. Maggi, and A. Pratelli, *The sharp Sobolev inequality in quantitative form*, J. Eur. Math. Soc. (JEMS) **11** (2009), no. 5, 1105–1139.
- [51] J. Ciernoczołowski and W. Orlicz, *Composing functions of bounded φ -variation*, Proc. Amer. Math. Soc. **96** (1986), no. 3, 431–436.
- [52] D. Cioranescu and J. Saint Jean Paulin, *Homogenization in open sets with holes*, J. Math. Anal. Appl. **71** (1979), no. 2, 590–607.
- [53] M.G. Crandall and L. Tartar, *Some relations between nonexpansive and order preserving mappings*, Proc. Amer. Math. Soc. **78** (1980), no. 3, 385–390.
- [54] M. Csörnyei, *Absolutely continuous functions of Rado, Reichelderfer, and Malý*, J. Math. Anal. Appl. **252** (2000), no. 1, 147–166.
- [55] B.E.J. Dahlberg, *A note on Sobolev spaces*. Harmonic analysis in Euclidean spaces, Part 1, pp. 183–185, Proc. Sympos. Pure Math., XXXV, Part, Amer. Math. Soc., Providence, RI, 1979.

- [56] B.E.J. Dahlberg, *Total curvature and rearrangements*. Posthumous paper prepared for publication by Vilhelm Adolfsson and Peter Kumlin. Ark. Mat. **43** (2005), no. 2, 323–345.
- [57] G. Dal Maso, *BV functions*, SISSA.
- [58] E. De Giorgi, *Definizione ed espressione analitica del perimetro di un insieme*, Atti Accad. Naz. Lincei. Rend. Cl. Sci. Fis. Mat. Nat. (8) **14** (1953), 390–393.
- [59] E. De Giorgi, *Selected papers*, edited by Luigi Ambrosio, Gianni Dal Maso, Marco Forti, Mario Miranda, and Sergio Spagnolo. Springer-Verlag, Berlin, 2006.
- [60] M.C. Delfour and J.P. Zolésio, *Shapes and geometries. Analysis, differential calculus, and optimization*, Advances in Design and Control, 4. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2001.
- [61] E. DiBenedetto, *Real analysis*, Birkhäuser Advanced Texts: Basler Lehrbücher, Birkhäuser Boston, Inc., Boston, MA, 2002.
- [62] J. Diestel, J.J. Uhl, Jr., *Vector Measures*. With a foreword by B. J. Pettis. Mathematical Surveys, No. 15. American Mathematical Society, Providence, R.I. (1977).
- [63] N. Dinculeanu, *Vector Integration and Stochastic Integration in Banach Spaces*, John Wiley & Sons, Inc., New York (2000).
- [64] D.L. Donoho and P.B. Stark, *Rearrangements and smoothing*, Tech. Rep. 148, Dept. of Statist., Univ. of California, Berkeley, 1988.
- [65] O. Dovgoshey, O. Martio, V. Ryazanov, and M. Vuorinen, *The Cantor function*, Expo. Math. **24** (2006), no. 1, 1–37.
- [66] G.F.D. Duff, *Differences, derivatives, and decreasing rearrangements*, Canad. J. Math. **19** (1967), 1153–1178.
- [67] N. Dunford and J.T. Schwartz, *Linear Operators*. Part I. General Theory. With the assistance of William G. Bade and Robert G. Bartle. Reprint of the 1958 original. Wiley Classics Library. A Wiley-Interscience Publication. John Wiley & Sons, Inc., New York (1988).
- [68] Y. Ebihara and T.P. Schonbek, *On the (non)compactness of the radial Sobolev spaces*, Hiroshima Math. J. **16** (1986), 665–669.
- [69] R.E. Edwards, *Functional analysis. Theory and applications*, corrected reprint of the 1965 original, Dover Publications, Inc., New York, 1995.
- [70] L. Hernández Encinas and J Muñoz Masqué, *A short proof of the generalized Faà di Bruno's formula*, Appl. Math. Lett. **16** (2003), no. 6, 975–979.
- [71] L.C. Evans, *Partial differential equations*, Second Edition. Graduate Studies in Mathematics, 19 R, American Mathematical Society, Providence, RI, 2010.
- [72] L.C. Evans and R.F. Gariepy, *Measure theory and fine properties of functions*, Studies in Advanced Mathematics, CRC Press, Boca Raton, FL, 1992.
- [73] K.J. Falconer, *The geometry of fractal sets*, Cambridge Tracts in Mathematics, 85, Cambridge University Press, Cambridge, 1986.
- [74] C.A. Faure, *The Lebesgue differentiation theorem via the rising sun lemma*, Real Anal. Exchange **29** (2003/04), no. 2, 947–951.
- [75] H. Federer, *Surface area. I*, Trans. Amer. Math. Soc. **55** (1944), 420–437.
- [76] H. Federer, *Geometric measure theory*, Die Grundlehren der mathematischen Wissenschaften, Band 153, Springer-Verlag, Inc., New York, 1969.
- [77] C. Fefferman, *Whitney's extension problem for C^m* , Ann. of Math. (2) **164** (2006), no. 1, 313–359.

- [78] A. Ferone and R. Volpicelli, *Polar factorization and pseudo-rearrangements: Applications to Pólya–Szegő type inequalities*, Nonlinear Anal. **53** (2003), no. 7–8, 929–949.
- [79] W.E. Fleming and R. Rishel, *An integral formula for total gradient variation*, Arch. Math. **11** (1960), 218–222.
- [80] W.E. Fleming, *Functions of several variables*, second edition, Undergraduate Texts in Mathematics, Springer–Verlag, New York–Heidelberg, 1977.
- [81] T.M. Flett, *On transformations in R^n and a theorem of Sard*, Amer. Math. Monthly **71** (1964), 623–629.
- [82] G.B. Folland, *Real analysis. Modern techniques and their applications*, second edition, Pure and Applied Mathematics, A Wiley–Interscience Series of Texts, Monographs, and Tracts, New York, 1999.
- [83] I. Fonseca and G. Leoni, *Modern methods in the calculus of variations: L^p spaces*, Springer Monographs in Mathematics, Springer, New York, 2007.
- [84] L.E. Fraenkel, *On regularity of the boundary in the theory of Sobolev spaces*, Proc. London Math. Soc. (3) **39** (1979), no. 3, 385–427.
- [85] G. Freilich, *Increasing continuous singular functions*, Amer. Math. Monthly **80** (1973), 918–919.
- [86] K. O. Friedrichs, *On the boundary-value problems of the theory of elasticity and Korn’s inequality*, Ann. Math. **48** (1947) 441–471, .
- [87] N. Fusco, F. Maggi, and A. Pratelli, *The sharp quantitative Sobolev inequality for functions of bounded variation*, J. Funct. Anal. **244** (2007), no. 1, 315–341.
- [88] V.N. Gabushin, *Inequalities for the norms of a function and its derivatives in metric L_p* , Mat. Zametki **1** (1967), no. 3, 291–298, translation in Math. Notes **1** (1967), no. 1, 194–198
- [89] E. Gagliardo, *Caratterizzazioni delle tracce sulla frontiera relative ad alcune classi di funzioni in n variabili*, Rend. Sem. Mat. Univ. Padova **27** (1957), 284–305.
- [90] E. Gagliardo, *Proprietà di alcune classi di funzioni in più variabili*, Ricerche Mat. **7** (1958), 102–137.
- [91] E. Gagliardo, *Ulteriori proprietà di alcune classi di funzioni in più variabili*, Ricerche Mat. **8** (1959), 24–51.
- [92] R.J. Gardner, *The Brunn–Minkowski inequality*, Bull. Amer. Math. Soc. (N.S.) **39** (2002), no. 3, 355–405.
- [93] F.W. Gehring, *A study of α -variation*, Trans. Amer. Math. Soc. **76** (1954), 420–443.
- [94] D. Gilbarg and N.S. Trudinger, *Elliptic partial differential equations of second order*, reprint of the 1998 edition, Classics in Mathematics, Springer–Verlag, Berlin, 2001.
- [95] E. Giusti, *Minimal surfaces and functions of bounded variation*, Monographs in Mathematics, **80**, Birkhäuser Verlag, Basel, 1984.
- [96] C. Goffman, *On functions with summable derivative*, Amer. Math. Monthly **78** (1971), 874–875.
- [97] G.S. Goodman, *Integration by substitution*, Proc. Amer. Math. Soc. **70** (1978), no. 1, 89–91.
- [98] L. Grafakos, *Classical Fourier analysis*, third edition, Graduate Texts in Mathematics **249**, Springer–Verlag, New York, 2014.
- [99] L. Grafakos, *Modern Fourier Analysis*, third edition, Graduate Texts in Mathematics **249**, Springer–Verlag, New York, 2014.

- [100] L.M. Graves, *The theory of functions of real variables*, first edition, McGraw–Hill Book Company, Inc., New York and London, 1946.
- [101] M.E. Gurtin, *An introduction to continuum mechanics*, Vol. 158, Academic Press, Inc., New York, 1982.
- [102] H. Hajaiej and C.A. Stuart, *Symmetrization inequalities for composition operators of Carathéodory type*, Proc. London Math. Soc. (3) **87** (2003), no. 2, 396–418.
- [103] H. Hajaiej, *Cases of equality and strict inequality in the extended Hardy–Littlewood inequalities*, Proc. Roy. Soc. Edinburgh Sect. A **135** (2005), no. 3, 643–661.
- [104] P. Hajłasz, *Change of variables formula under minimal assumptions*, Colloq. Math. **64** (1993), no. 1, 93–101.
- [105] P. Hajłasz and A. Kalamajska, *Polynomial asymptotics and approximation of Sobolev functions*, Studia Math. **113** (1995), no. 1, 55–64.
- [106] P. Hajłasz and P. Koskela, *Isoperimetric inequalities and imbedding theorems in irregular domains*, J. London Math. Soc. (2) **58** (1998) 425–450.
- [107] P. Hajłasz and P. Koskela, *Sobolev met Poincaré*, Mem. Amer. Math. Soc. **145** (2000), no. 688,
- [108] G.H. Hardy, *Weierstrass’s non-differentiable function*, Trans. Amer. Math. Soc. **17** (1916), no. 3, 301–325.
- [109] G.H. Hardy, J.E. Littlewood, G. Pólya, *Inequalities*, Cambridge University Press, 1952.
- [110] U.S. Haslam-Jones, *Derivate planes and tangent planes of a measurable function*, Q. J. Math. (1932) 121–132.
- [111] L.I. Hedberg, *On certain convolution inequalities*, Proc. Amer. Math. Soc. **36** (1972), 505–510.
- [112] J. Heinonen, *Nonsmooth calculus*, Bull. Amer. Math. Soc. (N.S.) **44** (2007), no. 2, 163–232.
- [113] G. Helmsberg, *An absolutely continuous function in $L^1(\mathbb{R}) \setminus W^{1,1}(\mathbb{R})$* , Amer. Math. Monthly **114** (2007), no. 4, 356–357.
- [114] S. Hencl and J. Malý, *Absolutely continuous functions of several variables and diffeomorphisms*, Cent. Eur. J. Math. **1** (2003), no. 4, 690–705.
- [115] G.A. Heuer, *The derivative of the total variation function*, Amer. Math. Monthly **78** (1971), 1110–1112.
- [116] Y. Heurteaux, *Weierstrass functions in Zygmund’s class*, Proc. Amer. Math. Soc. **133** (1981), no. 9, 2711–2720.
- [117] E. Hewitt, *Integration by parts for Stieltjes integrals*, Amer. Math. Monthly **67** (1960), 419–423.
- [118] E. Hewitt and K. Stromberg, *Real and abstract analysis. A modern treatment of the theory of functions of a real variable*, Springer–Verlag, New York, 1965.
- [119] D. Hilbert, *Ueber die stetige Abbildung einer Linie auf ein Flächenstück*, Math. Ann. **38** (1891), no. 3, 459–460.
- [120] K. Hildén, *Symmetrization of functions in Sobolev spaces and the isoperimetric inequality*, Manuscripta Math. **18** (1976), no. 3, 215–235.
- [121] L. Hormander, *The boundary problems of physical geodesy*, Arch. Rational Mech. Anal. **62** (1976), no. 1, 1–52.
- [122] P.W. Jones, *Quasiconformal mappings and extendability of functions in Sobolev spaces*, Acta Math. **147** (1981), no. 1–2, 71–88.

- [123] W. Johnson, *The curious history of Faà di Bruno's formula*, Amer. Math. Monthly **109** (2002), no. 3, 217–234.
- [124] A. Jonsson and H. Wallin, *The trace to closed sets of functions in R^n with second difference of order $O(h)$* , J. Approx. Theory **26** (1979), no. 2, 159–184.
- [125] A. Jonsson and H. Wallin, *The trace to subsets of R^n of Besov spaces in the general case*, Anal. Math. **6** (1980), no. 3, 223–254.
- [126] M. Josephy, *Composing functions of bounded variation*, Proc. Amer. Math. Soc. **83** (1981), no. 2, 354–356.
- [127] W.J. Kaczor and M.T. Nowak, *Problems in mathematical analysis. II. Continuity and differentiation*, translated from the 1998 Polish original, revised and augmented by the authors. Student Mathematical Library, 12, American Mathematical Society, Providence, RI, 2001.
- [128] G.E. Karadzhov, M. Milman, and J. Xiao, *Limits of higher-order Besov spaces and sharp reiteration theorems*, J. Funct. Anal. **221** (2005), no. 2, 323–339.
- [129] Y. Katznelson and K. Stromberg, *Everywhere differentiable, nowhere monotone, functions*, Amer. Math. Monthly **81** (1974), 349–354.
- [130] B. Kawohl, *Rearrangements and convexity of level sets in PDE*, Lecture Notes in Mathematics, 1150, Springer-Verlag, Berlin, 1985.
- [131] S. Kesavan, *Symmetrization & applications*, Series in Analysis, 3, World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ, 2006.
- [132] J. Kinnunen, *The Hardy–Littlewood maximal function of a Sobolev function*, Israel J. Math. **100** (1997), 117–124.
- [133] H. Kober, *On singular functions of bounded variation*, J. London Math. Soc. **23** (1948), 222–229.
- [134] T. Kolsrud, *Approximation by smooth functions in Sobolev spaces, a counterexample*, Bull. London Math. Soc. **13** (1981), no. 2, 167–169.
- [135] S.G. Krantz, *Lipschitz spaces, smoothness of functions, and approximation theory*, Exposition. Math. **1** (1983), no. 3, 193–260.
- [136] K. Krzyżewski, *On change of variable in the Denjoy–Perron integral*, I, Colloq. Math. **9** (1962), 99–104.
- [137] U. Lang, *Introduction to geometric measure theory*, 2005.
- [138] P.D. Lax, *Change of variables in multiple integrals*, Amer. Math. Monthly **106** (1999), no. 6, 497–501.
- [139] P.D. Lax, *Change of variables in multiple integrals. II*, Amer. Math. Monthly **108** (2001), no. 2, 115–119.
- [140] G. Leoni and M. Morini, *Necessary and sufficient conditions for the chain rule in $W_{\text{loc}}^{1,1}(\mathbb{R}^N; \mathbb{R}^d)$ and $BV_{\text{loc}}(\mathbb{R}^N; \mathbb{R}^d)$* , J. Eur. Math. Soc. (JEMS) **9** (2007), no. 2, 219–252.
- [141] Y. Li and B. Ruf, *A sharp Trudinger–Moser type inequality for unbounded domains in R^n* , Indiana Univ. Math. J. **57** (2008), no. 1, 451–480.
- [142] E.H. Lieb and M. Loss, *Analysis*, second edition, Graduate Studies in Mathematics, 14, American Mathematical Society, Providence, RI, 2001.
- [143] G.M. Lieberman, *Regularized distance and its applications*, Pacific J. Math. **117** (1985), no. 2, 329–352.
- [144] S. Lindner, *Additional properties of the measure v_f* , Tatra Mt. Math. Publ. **28** (2004), part II, 199–205.

- [145] J.L. Lions, *Quelques méthodes de résolution des problèmes aux limites non linéaires*, Dunod-Gauthier Villars, Paris, 1969.
- [146] A. Lunardi, *An Introduction to interpolation theory*, Lecture notes. Unpublished, 2007.
- [147] G. Lu and R.I. Wheeden, *Poincaré inequalities, isoperimetric estimates, and representation formulas on product spaces*, Indiana Univ. Math. J. **47** (1998), no. 1, 123–151.
- [148] R. Maehara, *The Jordan curve theorem via the Brouwer fixed point theorem*, Amer. Math. Monthly **91** (1984), no. 10, 641–643.
- [149] F. Maggi, *Sets of finite perimeter and geometric variational problems*, Cambridge University Press, 2012.
- [150] F. Maggi and C. Villani, *Balls have the worst best Sobolev inequalities*, J. Geom. Anal. **15** (2005), no. 1, 83–121.
- [151] L. Maligranda and L.E. Persson, *Inequalities and interpolation*, Collect. Math. **44** (1993), 181–199.
- [152] L. Maligranda and A. Quevedo, *Interpolation of weakly compact operators*, Arch. Math. **55** (1990), 280–284.
- [153] J. Malý, *Absolutely continuous functions of several variables*, J. Math. Anal. Appl. **231** (1999), no. 2, 492–508.
- [154] J. Malý, *A simple proof of the Stepanov theorem on differentiability almost everywhere*, Exposition. Math. **17** (1999), no. 1, 59–61.
- [155] J. Malý, *Lectures on change of variables in integral*, Graduate School in Helsinki, 2001.
- [156] J. Malý and O. Martio, *Lusin's condition (N) and mappings of the class $W^{1,n}$* , J. Reine Angew. Math. **458** (1995), 19–36.
- [157] J. Malý and L. Pick, *An elementary proof of sharp Sobolev embeddings*, Proc. Amer. Math. Soc. **130** (2002), no. 2, 555–563.
- [158] M. Marcus and V.J. Mizel, *Transformations by functions in Sobolev spaces and lower semicontinuity for parametric variational problems*, Bull. Amer. Math. Soc. **79** (1973), 790–795.
- [159] M. Marcus and V.J. Mizel, *Absolute continuity on tracks and mappings of Sobolev spaces*, Arch. Rational Mech. Anal. **45** (1972), 294–320.
- [160] M. Marcus and V.J. Mizel, *Measurability of partial derivatives*, Proc. Amer. Math. Soc. **63** (1977), 236–238.
- [161] M. Marcus and V.J. Mizel, *Complete characterization of functions which act, via superposition, on Sobolev spaces*, Trans. Amer. Math. Soc. **251** (1979), 187–218.
- [162] M. Marcus and V.J. Mizel, *A characterization of first order nonlinear partial differential operators on Sobolev spaces*, J. Funct. Anal. **38** (1980), no. 1, 118–138.
- [163] J. Martin, M. Milman, and E. Pustylnik, *Sobolev inequalities: Symmetrization and self-improvement via truncation*, J. Funct. Anal. **252** (2007), no. 2, 677–695.
- [164] A. Matszangosz, *The Denjoy-Young-Saks theorem in higher dimensions*, Master's thesis, Eötvös Loránd University.
- [165] V.G. Maz'ja, *Sobolev spaces*, Springer-Verlag, Berlin Heidelberg, 2011.
- [166] V.G. Maz'ja, M. Mitrea, T. O. Shaposhnikova, *The Dirichlet problem in Lipschitz domains for higher order elliptic systems with rough coefficients*, J. Anal. Math. **110** (2010), no. 1, 167–239.

- [167] V.G. Maz'ja and T.O. Shaposhnikova, *On the Brezis and Mironescu conjecture concerning a Gagliardo–Nirenberg inequality for fractional Sobolev norms*, J. Math. Pures Appl. (9) **81** (2002), no. 9, 877–884. Erratum J. Funct. Anal. **201** (2003), no. 1, 298–300.
- [168] J. McCarthy, *An everywhere continuous nowhere differentiable function*, Amer. Math. Monthly **60** (1953), 709.
- [169] N.G. Meyers and J. Serrin, $H = W$, Proc. Nat. Acad. Sci. U.S.A. **51** (1964), 1055–1056.
- [170] N. Merentes, *On the composition operator in $AC[a, b]$* , Collect. Math. **42** (1991), no. 3, 237–243 (1992).
- [171] J.W. Milnor, *Topology from the differentiable viewpoint*, based on notes by David W. Weaver. Revised reprint of the 1965 original. Princeton Landmarks in Mathematics. Princeton University Press, Princeton, NJ, 1997.
- [172] P. Mironescu, *Note on Gagliardo's theorem $trW^{1,1} = L^1$* , Ann. Univ. Buchar. Math. Ser. **6** (LXIV) (2015), no 1, 99–103.
- [173] P. Mironescu and E. Russ, *Traces of weighted Sobolev spaces. Old and new*, Nonlinear Anal. **119** (2015) 354–381.
- [174] E.H. Moore, *On certain crinkly curves*, Trans. Amer. Math. Soc. **1** (1900), no. 1, 72–90. Errata, Trans. Amer. Math. Soc. **1** (1900), no. 4, 507.
- [175] M. Morini, *A note on the chain rule in Sobolev spaces and the differentiability of Lipschitz functions*, preprint.
- [176] A.P. Morse, *Convergence in variation and related topics*, Trans. Amer. Math. Soc. **41** (1937), no. 1, 48–83. Errata, Trans. Amer. Math. Soc. **41** (1937), no. 3, 482.
- [177] A.P. Morse, *A continuous function with no unilateral derivatives*, Trans. Amer. Math. Soc. **44** (1938), no. 3, 496–507.
- [178] A.P. Morse, *The behavior of a function on its critical set*, Ann. of Math. (2) **40** (1939), no. 1, 62–70.
- [179] J. Moser, *A sharp form of an inequality by N. Trudinger*, Indiana Univ. Math. J. **20** (1970/71), 1077–1092.
- [180] B. Muckenhoupt, *Weighted norm inequalities for the Hardy maximal function*, Trans. Amer. Math. Soc. **165** (1972), 207–226.
- [181] I.P. Natanson, *Theory of functions of a real variable*, translated by Leo F. Boron with the collaboration of Edwin Hewitt, Frederick Ungar Publishing Co., New York, 1955.
- [182] J. Něcas, *Les méthodes directes en théorie des équations elliptiques*, Masson et Cie, Éditeurs, Paris, Academia, Éditeurs, Prague, 1967.
- [183] J. A. Nitsche. *On Korn's second inequality*, RAIRO-Analyse numérique, **15** (1981), 237–248.
- [184] L. Nirenberg, *On elliptic partial differential equations*, Ann. Scuola Norm. Sup. Pisa (3) **13** (1959) 115–162.
- [185] L. Nirenberg, *An extended interpolation inequality*, Ann. Scuola Norm. Sup. Pisa (3) **20** (1966), 733–737.
- [186] E. Novak, *Two remarks on the decreasing rearrangement of a function*, J. Math. Anal. Appl. **122** (1987), no. 2, 485–486.
- [187] L.E. Payne and H.F. Weinberger, *An optimal Poincaré inequality for convex domains*, Arch. Rational Mech. Anal. **5** (1960), 286–292.

- [188] G. Peano, *Sur une courbe, qui remplit toute une aire plane*, Math. Ann. **36** (1890), no. 1, 157–160.
- [189] S. I. Pohožaev, *On the eigenfunctions of the equation $\Delta u + \lambda f(u) = 0$* . Translation from Dokl. Akad. Nauk SSSR **165** (1965) 36–39.
- [190] D. Pompeiu, *Sur les fonctions dérivées*, Math. Ann. **63** (1907), no. 3, 326–332.
- [191] J. Peetre, *Espaces d'interpolation et théorème de Soboleff*, Ann. Inst. Fourier (Grenoble) **16** (1966), 1, 279–317.
- [192] J. Peetre, *New thoughts on Besov spaces*, Mathematics Department, Duke University, 1976.
- [193] J. Peetre, *A counterexample connected with Gagliardo's trace theorem*, special issue dedicated to Władysław Orlicz on the occasion of his seventy-fifth birthday, Comment. Math. Special Issue **2** (1979), 277–282.
- [194] A. Pinkus, *Weierstrass and approximation theory*, J. Approx. Theory **107** (2000), no. 1, 1–66.
- [195] A.C. Ponce, *A new approach to Sobolev spaces and connections to Γ -convergence*, Calc. Var. Partial Differential Equations **19** (2004), no. 3, 229–255.
- [196] H. Rademacher, *Über partielle und totale differenzierbarkeit von Funktionen mehrerer Variablen und über die Transformation der Doppelintegrale*, Math. Ann. **79** (1919), no. 4, 340–359.
- [197] F. Riesz, *Sur l'existence de la dérivée des fonctions monotones et sur quelques problèmes qui s'y rattachent*, Acta Sci. Math. **5** (1930–1932), 208–221.
- [198] J.V. Ryff, *Measure preserving transformations and rearrangements*, J. Math. Anal. Appl. **31** (1970), 449–458.
- [199] W. Rudin, *Real and complex analysis*, third edition, McGraw–Hill Book Co., New York, 1987.
- [200] W. Rudin, *Functional analysis*, second edition, International Series in Pure and Applied Mathematics, McGraw–Hill, Inc., New York, 1991.
- [201] S. Saks, *Theory of the integral*, second revised edition, English translation by L.C. Young, with two additional notes by Stefan Banach, Dover Publications, Inc., New York 1964.
- [202] R. Salem, *On some singular monotonic functions which are strictly increasing*, Trans. Amer. Math. Soc. **53** (1943), 427–439.
- [203] R. Sandberg and R.A. Christianson, *Problems and solutions: Solutions of advanced problems: 6007*, Amer. Math. Monthly **83** (1976), no. 8, 663–664.
- [204] A. Sard, *The measure of the critical values of differentiable maps*, Bull. Amer. Math. Soc. **48** (1942), 883–890.
- [205] A. Sard, *Images of critical sets*, Ann. of Math. (2) **68** (1958), 247–259.
- [206] S. Schwabik and G. Ye, *Topics in Banach Space Integration*. Series in Real Analysis, 10. World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ (2005).
- [207] J. Schwartz, *The formula for change in variables in a multiple integral*, Amer. Math. Monthly **61** (1954), 81–85.
- [208] J. Serrin, *On the differentiability of functions of several variables*, Arch. Rational Mech. Anal. **7** (1961), 359–372.
- [209] J. Serrin, *Strong convergence in a product space*, Proc. Amer. Math. Soc. **13** (1962), 651–655.

- [210] J. Serrin and D.E. Varberg, *A general chain rule for derivatives and the change of variables formula for the Lebesgue integral*, Amer. Math. Monthly **76** (1969), 514–520.
- [211] W. Sierpiński, *Sur la question de la mesurabilité de la base de M. Hamel*, Fund. Math. **1** (1920), 105–111.
- [212] C.G. Simader, *Sobolev’s original definition of his spaces revisited and a comparison with nowadays definition*, Boundary value problems for elliptic and parabolic operators (Catania, 1998), Matematiche (Catania) **54** (1999), suppl., 149–178.
- [213] J. Simon, *Compact sets in the space $L^p(0, T; B)$* , Ann. Mat. Pura Appl. (4) **146** (1987), 65–96.
- [214] W. Smith and D.A. Stegenga, *Hölder domains and Poincaré domains*, Trans. Amer. Math. Soc. **319** (1990), no. 1, 67–100.
- [215] S.L. Sobolev, *On a theorem of functional analysis*, Transl., Ser. 2, Amer. Math. Soc. **34** (1963), 39–68.
- [216] V.A. Solonnikov, *A priori estimates for second-order parabolic equations*, Amer. Math. Soc., Transl., II. Ser. **65**, 1967, 51–137.
- [217] V.A. Solonnikov, *Inequalities for functions of the classes $W_p^{\vec{m}}(R^n)$* , J. Sov. Math. **3** (1975), 549–564.
- [218] E.J. Sperner, *Zur Symmetrisierung von Funktionen auf Sphären*, Math. Z. **134** (1973), 317–327.
- [219] K. Spindler, *A short proof of the formula of Faà di Bruno*, Elem. Math. **60** (2005), no. 1, 33–35.
- [220] E.M. Stein, *Singular integrals and differentiability properties of functions*, Princeton Mathematical Series, no. 30, Princeton University Press, Princeton, NJ, 1970.
- [221] E.M. Stein and G. Weiss, *Introduction to Fourier analysis on Euclidean spaces*, Princeton University Press, Princeton, NJ, 1971.
- [222] E.M. Stein and A. Zygmund, *On the differentiability of functions*, Studia Math. **23** (1963/1964), 247–283.
- [223] W. Stepanoff, *Über totale Differenzierbarkeit*, Math. Ann. **90** (1923), no. 3–4, 318–320.
- [224] W.A. Strauss, *Existence of solitary waves in higher dimensions*, Commun. Math. Phys. **55** (1977), 149–162.
- [225] R.S. Strichartz, *A note on Trudinger’s extension of Sobolev’s inequalities*, Indiana Univ. Math. J. **21** (1971/72), 841–842.
- [226] M. Taibleson, *On the theory of Lipschitz spaces of distributions on Euclidean n -space I. Principal properties*, Indiana Univ. Math. J. **13** No. 3 (1964), 407–479.
- [227] L. Takács, *An increasing continuous singular function*, Amer. Math. Monthly **85** (1978), no. 1, 35–37.
- [228] G. Talenti, *Best constant in Sobolev inequality*, Ann. Mat. Pura Appl. (4) **110** (1976), 353–372.
- [229] G. Talenti, *Inequalities in rearrangement invariant function spaces*. Pages 177–231 from Nonlinear Analysis, Function Spaces and Applications, Vol. 5. Prometheus Publishing House, Praha, 1994.
- [230] G. Talenti, *The art of rearranging*, Milan J. Math. **84** (2016), 105–157.
- [231] T. Tao, *Topics in random matrix theory*, Graduate Studies in Mathematics, 132, American Mathematical Society, Providence, RI, 2012.

- [232] L. Tartar, *An introduction to Sobolev spaces and interpolation spaces*, Lecture Notes of the Unione Matematica Italiana, 3. Springer, Berlin; UMI, Bologna, 2007.
- [233] H. Triebel, *Interpolation theory, function spaces, differential operators*, second edition. Johann Ambrosius Barth, Heidelberg, 1995.
- [234] N.S. Trudinger, *On imbeddings into Orlicz spaces and some applications*, J. Math. Mech. **17** (1967), 473–483.
- [235] H. Tverberg, *A proof of the Jordan curve theorem*, Bull. London Math. Soc. **12** (1980), no. 1, 34–38.
- [236] S.V. Uspenskii, *Imbedding theorems for weighted classes*, Amer. Math. Soc., Transl., II. Ser. 87, 1970, 121–145; translation from Trudy Mat. Inst. Steklov **60** (1961), 282–303.
- [237] A.C.M. van Rooij and W.H. Schikhof, *A second course on real functions*, Cambridge University Press, Cambridge–New York, 1982.
- [238] F.S. Van Vleck, *A remark concerning absolutely continuous functions*, Amer. Math. Monthly **80** (1973), 286–287.
- [239] D.E. Varberg, *On absolutely continuous functions*, Amer. Math. Monthly **72** (1965), 831–841.
- [240] D.E. Varberg, *On differentiable transformations in R^n* , Amer. Math. Monthly **73** (1966), no. 4, part II, 111–114.
- [241] D.E. Varberg, *Change of variables in multiple integrals*, Amer. Math. Monthly **78** (1971), 42–45.
- [242] K. Weierstrass, *Mathematische werke. II. Abhandlungen 2*, Georg Olms Verlagsbuchhandlung, Hildesheim; Johnson Reprint Corp., New York, 1967.
- [243] C. E. Weil, *On nowhere monotone functions*, Proc. Amer. Math. Soc. **56** (1976), 388–389.
- [244] H. Whitney, *Analytic extensions of differentiable functions defined in closed sets*, Trans. Amer. Math. Soc. **36** (1934), no. 1, 63–89.
- [245] H. Whitney, *A function not constant on a connected set of critical points*, Duke Math. J. **1** (1935), no. 4, 514–517.
- [246] H. Whitney, *Differentiable functions defined in arbitrary subsets of Euclidean space*, Trans. Amer. Math. Soc. **40** (1936), no. 2, 309–317.
- [247] N. Wiener, *The quadratic variation of a function and its Fourier coefficients*, Mass. J. of Math. **3** (1924), 72–94.
- [248] B.B. Winter, *Transformations of Lebesgue–Stieltjes integrals*, J. Math. Anal. Appl. **205** (1997), no. 2, 471–484.
- [249] K. Yosida, *Functional analysis*, reprint of the sixth (1980) edition, Classics in Mathematics, Springer–Verlag, Berlin, 1995.
- [250] V. I. Judovič, *Some estimates connected with integral operators and with solutions of elliptic equations*, Dokl. Akad. Nauk SSSR **138** (1961) 805–808. (Russian)
- [251] W.P. Ziemer, *Weakly differentiable functions. Sobolev spaces and functions of bounded variation*, Graduate Texts in Mathematics, 120, Springer–Verlag, New York, 1989.
- [252] P.R. Zingano and S.L. Steinberg, *On the Hardy–Littlewood theorem for functions of bounded variation*, SIAM J. Math. Anal. **33** (2002), no. 5, 1199–1210.

Index

- algebra, 652
- atom, 654

- ball
 - open, 638
- base
 - for a topology, 636
 - local, 636
- basis
 - orthonormal, 240
- boundary
 - Lipschitz, 273
 - uniformly of class C^m , 424
 - uniformly of class $C^{m,\alpha}$, 424
 - uniformly Lipschitz, 423

- Cantor diagonal argument, 49
- Cantor part of a function, 104
- Cauchy–Binet formula, 251
- change of variables, 166
 - for multiple integrals, 270
- coarea formula, 478
- cofactor, 260
- compact embedding, 366
- connected component, 147
 - exterior, 152
 - interior, 152
- convergence
 - almost everywhere, 673
 - almost uniform, 673
 - in measure, 673
 - in the sense of distributions, 288
 - weak, 645
 - weak star, 646
- convolution, 302
 - of a distribution, 305, 308
- coordinates
 - background, 273
 - local, 273
- countability
 - first axiom, 636
 - second axiom, 636
- curve, 133
 - arclength, 143
 - closed, 134
 - closed simple, 134
 - continuous, 133
 - equivalent, 133
 - length, 136
 - locally rectifiable, 137
 - parameter change, 133
 - parametric representation, 133
 - range, 134
 - rectifiable, 136
 - simple arc, 134
 - simple point, 134
- cut-off function, 693

- delta Dirac, 289
- derivative, 9
 - ∇^k , 321
 - Dini, 14
 - directional $\frac{\partial}{\partial \nu}$, 239
 - distributional, 183, 188, 229, 291, 319
 - directional, 320
 - distributional partial , 459

- $\frac{dm}{dx}$, 224
- of a distribution, 290
 - directional, 291
- partial ∂_i , 239
- Radon–Nikodym, 665
- weak, 183, 188, 229, 291, 319
 - directional, 320
 - weak partial, 459
- differentiability, 9
- distance, 638
 - regularized, 414
- distribution, 288
 - infinite order, 289
 - order of, 289
- dual spaces
 - $\mathcal{D}'(\Omega)$, 288
 - $\mathcal{M}_b(X; \mathbb{R})$, Radon measures, 676
 - of $W^{1,p}(\Omega)$, 345
 - $W^{-1,p'}(\Omega)$, 348
 - $\mathcal{S}'(\mathbb{R}^N; \mathbb{C})$, 299
- duality pairing, 642
- embedding, 356, 643
 - compact, 645
- equi-integrability, 674
- essential infimum, 112
- essential supremum, 112, 668
- evolution triple, 233
- extension domain
 - for $BV(\Omega)$, 483
 - for $W^{m,p}(\Omega)$, 365
- extension operator, 365
- finite cone, 424
- Fourier transform, 309
 - inverse, 309
 - of a tempered distribution, 315
 - inverse, 315
- function
 - absolutely continuous, 67
 - Banach indicatrix, 59
 - Bochner integrable, 208
 - Borel, 655
 - Cantor, 22
 - characteristic, χ , 657
 - continuous, 637
 - counting, 59
 - cut-off, 693
 - decreasing, 3
 - decreasing rearrangement, 502
 - distribution, 111, 497
 - equi-integrable, 82
 - equivalent, 668
 - Gamma, 701
 - Hölder continuous, 10
 - increasing, 3
 - inverse of a monotone, 6
 - jump, 5, 6
 - Lebesgue integrable, 660
 - Lebesgue measurable, 683
 - linear
 - adjoint, 249
 - diagonal, 250
 - orthogonal, 249
 - positive definite, 250
 - rotation, 250
 - symmetric, 250
 - Lipschitz continuous, 9, 242
 - locally absolutely continuous, 68
 - locally integrable, 660
 - maximal, 694
 - measurable, 655, 657
 - measure-preserving, 130
 - monotone, 3
 - of bounded pointwise variation, 30
 - in the sense of Cesari, 471
 - of bounded variation, 459
 - radial
 - of a star-shaped domain, 441
 - rearrangement
 - decreasing, 114, 498
 - Schwarz symmetric, 499, 502
 - spherically symmetric, 499
 - saltus, 5
 - simple, 205, 657
 - singular, 103
 - Sobolev, 188, 230, 320
 - strictly decreasing, 3
 - strictly increasing, 3
 - strongly measurable, 205
 - subharmonic, 291
 - testing, 284
 - vanishing at infinity, 111, 356, 498
 - weakly measurable, 205
 - weakly star measurable, 206
 - Weierstrass, 9
- function space
 - $AC_p([a, b])$, 83
 - $AC(I)$, 68
 - $AC_{loc}(I)$, 68
 - $AC_{loc}(I; Y)$, 68
 - $AC(\Omega; Y)$, 68
 - $AC(I; Y)$, 68

- $B_q^{s,p}(\mathbb{R}^N)$, 539
- Besov $B^{s,p}(\partial\Omega, \mathcal{H}^{N-1})$, 613, 615, 625
- $\dot{B}_q^{s,p}(\mathbb{R}^N)$, 556
- BMO, 695
- $BVP(I)$, 30
- $BVP(I; X)$, 30
- $BVP(\Omega; X)$, 31
- $BVP_{\text{loc}}(I)$, 30
- $BVP_{\text{loc}}(I; X)$, 30
- $BV(\Omega)$, 183, 459
- $BV_{\text{loc}}(\Omega)$, 188
- $C^{m,\alpha}(\overline{\Omega})$, 343
- $C(X; Y)$, 637
- $C_0(X)$, 644
- c_0 , 221
- $C_c(X)$, 644
- $C^\infty(\Omega)$, 242
- $C_c^\infty(\Omega)$, 242
- $C^m(\Omega)$, 242
- $C_c^m(\Omega)$, 242
- $C_c(X)$, 637
- $\mathcal{D}(\Omega)$, 284
- $\mathcal{D}_K(\Omega)$, 281
- $H^m(\Omega)$, 190
- $H^m(\Omega; \mathbb{R}^M)$, 190, 230
- $H^m(\Omega)$, 322
- $L^\infty(X)$, 668
- $L^{p,q}$, Lorentz, 502
- $L^p(X)$, 668
- L_{loc}^p , 671
- L_w^p , weak L^p , 504
- $L^\Phi(E)$, 376
- $\mathcal{PA}$, 333
- $\mathcal{S}(\mathbb{R}^N; \mathbb{C})$, 298
- $W^{m,p}(\Omega)$, 189, 320
- $W^{m,p}(\Omega; \mathbb{R}^M)$, 188, 320
- $W^{m,p}(\Omega; Y)$, 230
- $\dot{W}^{m,p}(\Omega)$, 323
- $W_{\text{loc}}^{m,p}(\Omega)$, 189, 320
- $W_{\text{loc}}^{m,p}(\Omega; \mathbb{R}^M)$, 189, 320
- $W_{\text{loc}}^{m,p}(\Omega; Y)$, 230
- $W_0^{m,p}(\Omega)$, 322
- Zygmund $\Lambda_1(\mathbb{R}^N)$, 541
- functional
 - locally bounded, 677
 - positive, 677
- gauge, 641
- Hausdorff dimension, 701
- Hölder's conjugate exponent, 669
- identity
 - Parseval, 311
 - Plancherel, 311
- immersion, 643
- inequality
 - Brunn–Minkowski, 683
 - Cauchy, 648
 - Hardy, 698
 - Hardy–Littlewood, 120, 501
 - Hölder, 669
 - isoperimetric, 486, 685
 - Jensen, 661
 - Minkowski, 671
 - for integrals, 670
 - Poincaré, 193
 - for continuous domains, 434
 - for convex sets, 436
 - for rectangles, 434
 - for star-shaped sets, 441
 - in BV , 486
 - in $W_0^{m,p}$, 430
 - in $W^{m,p}$, 432
 - weighted, 194
 - Young, 303, 669
 - Young, general form, 304
- infinite sum, 4
- inner product, 648
 - Euclidean, 649
- integral
 - Bochner, 208
 - Lebesgue
 - of a nonnegative function, 658
 - of a real-valued function, 660
 - of a simple function, 658
- integrals depending on a parameter, 662
- integration
 - by parts, 78, 278
 - Riemann, 73
- interval, 3
 - partition, 29
- Jacobian, 251
- Jacobian matrix, 251
- Laplacian, 291
- Leibnitz formula, 288
- lemma
 - Fatou, 210, 660
 - Riemann–Lebesgue, 312
- length distance, 146
- length of a curve
 - σ -finite, 137

-
- line integral, 151
 - Lipschitz constant, 10
 - Littlewood–Paley decomposition
 - dyadic block, $\tilde{\Lambda}_k$, 316
 - homogeneous, 317
 - locally finite, 637
 - Lusin (N) property, 84, 383
 - measure, 653
 - σ -finite, 654
 - absolute continuous, 665
 - absolutely continuous part, 668
 - Borel, 653
 - Borel regular, 676
 - complete, 654
 - counting, 659
 - finite, 654
 - finitely additive, 653
 - Hausdorff, 700
 - Lebesgue, 681
 - Lebesgue–Stieltjes, 163
 - lower variation, 667
 - metric outer, 655
 - nonatomic, 654
 - outer, 651
 - Hausdorff, 700
 - Lebesgue, 12, 681
 - Lebesgue–Stieltjes, 164
 - product, 663
 - purely atomic, 654
 - Radon, 157, 676
 - signed, 666
 - absolutely continuous, 667
 - bounded, 667
 - signed
 - finitely additive, 666
 - signed Radon, 676
 - singular part, 668
 - total variation, 460, 667
 - upper variation, 667
 - vector-valued, 667
 - Radon, 677
 - vector-valued Lebesgue–Stieltjes, 173
 - measure space, 653
 - measures
 - mutually singular, 665, 667
 - metric, 638
 - metric space
 - length space, 146
 - Minkowski content
 - lower, 685
 - upper, 685
 - Minkowski functional, 641
 - mollification, 687
 - mollifier, 687
 - standard, 688
 - multi-index, 241
 - multiplicity of a point, 134
 - N -simplex, 332
 - neighborhood, 636
 - norm, 643
 - equivalent, 644
 - Euclidean, 643
 - normal vector, 274
 - operator
 - bounded, 642
 - compact, 644
 - linear, 641
 - outer measure
 - product, 663
 - p -equi-integrability, 674
 - parallelogram law, 648
 - parameter of a curve, 133
 - partition of unity
 - smooth, 692
 - perimeter of a set, 461
 - point
 - accumulation, 635
 - Lebesgue, 679
 - of density one, 679
 - of density t , 679
 - p -Lebesgue, 679
 - pointwise variation, 29
 - essential, 187
 - indefinite, 36
 - locally bounded, 30
 - negative, 31
 - p -variation, 39
 - positive, 31
 - principal value integral, 292
 - quasi-norm, 645
 - Radon–Nikodym property, 224
 - regularized distance, 417
 - ring, 653
 - σ -algebra, 652
 - Borel, 652
 - product, 656, 663
 - σ -locally finite, 637
 - section, 663

- segment property, 328
- seminorm, 641
- semiring, 653
- sequence
 - Cauchy, 638, 640
 - convergent, 636, 638
- sequentially weakly compact set, 647
- set
 - absorbing, 639
 - balanced, 282, 639
 - Cantor , 21
 - closed, 635
 - closure, 635
 - compact, 637
 - connected, 147
 - dense, 636
 - disconnected, 147
 - F_σ , 20
 - finite width, 430
 - G_δ , 20
 - $\mathcal{H}^k$ -rectifiable, 98
 - inner regular, 676
 - interior, 635
 - Lebesgue measurable, 681
 - μ^* -measurable, 652
 - of finite perimeter, 461
 - open, 635
 - outer regular, 676
 - pathwise connected, 147
 - precompact, 637
 - purely $\mathcal{H}^k$ -unrectifiable , 98
 - regular, 676
 - relatively closed, 636
 - relatively compact , 637
 - relatively open, 636
 - σ -compact, 637
 - spherically symmetric rearrangement, 499
 - star-shaped, 441
 - symmetric difference, 169
 - topologically bounded, 640
- shortest distance, 146
- Sobolev critical exponent, 356, 361, 561
- space
 - Banach, 643
 - bidual, 642
 - complete, 638, 640
 - dual, 642
 - Hausdorff, 636
 - Hilbert, 648
 - locally compact, 637
 - locally convex, 640
 - measurable , 652
 - metric, 638
 - metrizable, 639
 - normable, 644
 - normal, 637
 - normed, 643
 - quasi-Banach, 645
 - quasi-normed, 645
 - $(Y_0, Y_1)_{s,q,J}$ real interpolation, 527
 - $(Y_0, Y_1)_{s,q}$ real interpolation, 518
 - reflexive, 647
 - seminormable, 641
 - seminormed, 641
 - separable, 636
 - topological, 635
 - topological vector, 639
 - vector, 639
- spherical coordinates, 271
- support of a distribution, 295
- tangent line, 138
- tangent space, 274
- tangent vector, 138, 274
- Taylor's formula, 242
- theorem
 - area formula, C^1 case, 253
 - area formula, the differentiable case, 269
 - Ascoli–Arzelà, 147
 - Aubin–Lions–Simon, 235
 - Baire category, 638
 - Banach, 61
 - Banach fixed point, 638
 - Banach–Alaoglu, 646
 - Besicovitch's covering, 678
 - Besicovitch's derivation, 678
 - Brouwer fixed point, 258
 - Carathéodory, 654
 - chain rule, 95, 99
 - change of variables, 100, 341
 - change of variables, the C^1 case, 257
 - change of variables, the differentiable case, 270
 - De la Vallée Poussin, 180, 675
 - decomposition, 250
 - divergence, 274
 - Dunford–Pettis, 675
 - Eberlein–Šmulian, 647
 - Egoroff, 211, 674
 - Faa di Bruno, 342
 - Fubini, 23, 664

- fundamental theorem of calculus, 77, 82
- Gagliardo, 600, 608, 609
- Gagliardo–Nirenberg interpolation, 400, 403, 451, 455, 488, 489
- Hahn–Banach
 - analytic form, 642
 - first geometric form, 642
 - second geometric form, 643
- Helly’s selection, 49
- Hilbert, 135
- integration by parts, 278
- Jordan’s curve, 152
- Jordan’s decomposition, 667
- Joseph, 55
- Kakutani, 648
- Lebesgue differentiation, 11
- Lebesgue dominated convergence, 210
- Lebesgue’s decomposition, 666, 668
- Lebesgue’s dominated convergence, 661
- Lebesgue’s monotone convergence, 659
- Littlewood–Paley decomposition, 317
- Lusin, 676
- Meyers–Serrin, 326
- Morrey’s embedding
 - in $B_q^{\sigma,p}$, 562
 - in $W^{1,p}$, 381
 - in $W^{m,p}$, 384
- Muckenhoupt, 444
- Peano, 135
- Pettis, 206
- Plancherel, 311
- Rademacher, 243
- Radon–Nikodym, 665
- reiteration, 522
- Rellich–Kondrachov, 483
- Rellich–Kondrachov’s compactness, 366, 368, 378, 386
 - for continuous domains, 371
- Riesz representation
 - in C_0 , 677
 - in C_c , 677
 - in L^1 , 672
 - in L^∞ , 672
 - in L^p , 671
 - in $L^p(X; Y)$, 213
 - in $W^{m,\infty}(\Omega)$, 349
 - in $W_0^{m,\infty}(\Omega)$, 349
 - in $W^{m,p}(\Omega)$, 345
 - in $W_0^{m,p}(\Omega)$, 348
- Riesz–Thorin, 673
- Sard, 489, 491
- Schwartz, 241
- Serrin, 471
- Simon, 215
- Sobolev–Gagliardo–Nirenberg
 - in $B_q^{s,p}$, 561, 587
 - in BV , 482
 - in $W^{1,p}$, 356, 510
 - in $\dot{W}^{m,p}$, 362
 - in $W^{m,p}$, 361
- Stepanoff, 245
- superposition, 388, 391, 393, 395
 - in AC_{loc} , 92, 191
- Tietze extension, 637
- Tonelli, 80, 137, 664
- Urysohn, 637
- Vitali’s convergence, 674
- Vitali’s covering, 489
- Vitali–Besicovitch’s covering, 678
- Weil, 45
- Whitney extension, 262
- Whitney extension, II, 268
- Whitney’s decomposition, 262
- topology, 635
 - weak, 645
 - weak star, 646
- total variation norm, 672
- trace of a function, 592
- unit sphere
 - S^{N-1} , 243
- vanishing at infinity, 356
- variation, 460
- vertex of a simplex, 332
- Whitney’s decomposition, 264

Selected Published Titles in This Series

- 181 **Giovanni Leoni**, *A First Course in Sobolev Spaces*, Second Edition, 2017
- 180 **Joseph J. Rotman**, *Advanced Modern Algebra: Third Edition, Part 2*, 2017
- 179 **Henri Cohen and Fredrik Strömberg**, *Modular Forms*, 2017
- 178 **Jeanne N. Clelland**, *From Frenet to Cartan: The Method of Moving Frames*, 2017
- 177 **Jacques Sauloy**, *Differential Galois Theory through Riemann-Hilbert Correspondence*, 2016
- 176 **Adam Clay and Dale Rolfen**, *Ordered Groups and Topology*, 2016
- 175 **Thomas A. Ivey and Joseph M. Landsberg**, *Cartan for Beginners: Differential Geometry via Moving Frames and Exterior Differential Systems*, Second Edition, 2016
- 174 **Alexander Kirillov Jr.**, *Quiver Representations and Quiver Varieties*, 2016
- 173 **Lan Wen**, *Differentiable Dynamical Systems*, 2016
- 172 **Jinho Baik, Percy Deift, and Toufic Suidan**, *Combinatorics and Random Matrix Theory*, 2016
- 171 **Qing Han**, *Nonlinear Elliptic Equations of the Second Order*, 2016
- 170 **Donald Yau**, *Colored Operads*, 2016
- 169 **András Vasy**, *Partial Differential Equations*, 2015
- 168 **Michael Aizenman and Simone Warzel**, *Random Operators*, 2015
- 167 **John C. Neu**, *Singular Perturbation in the Physical Sciences*, 2015
- 166 **Alberto Torchinsky**, *Problems in Real and Functional Analysis*, 2015
- 165 **Joseph J. Rotman**, *Advanced Modern Algebra: Third Edition, Part 1*, 2015
- 164 **Terence Tao**, *Expansion in Finite Simple Groups of Lie Type*, 2015
- 163 **Gérald Tenenbaum**, *Introduction to Analytic and Probabilistic Number Theory*, Third Edition, 2015
- 162 **Firas Rassoul-Agha and Timo Seppäläinen**, *A Course on Large Deviations with an Introduction to Gibbs Measures*, 2015
- 161 **Diane Maclagan and Bernd Sturmfels**, *Introduction to Tropical Geometry*, 2015
- 160 **Marius Overholt**, *A Course in Analytic Number Theory*, 2014
- 159 **John R. Faulkner**, *The Role of Nonassociative Algebra in Projective Geometry*, 2014
- 158 **Fritz Colonius and Wolfgang Kliemann**, *Dynamical Systems and Linear Algebra*, 2014
- 157 **Gerald Teschl**, *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, Second Edition, 2014
- 156 **Markus Haase**, *Functional Analysis*, 2014
- 155 **Emmanuel Kowalski**, *An Introduction to the Representation Theory of Groups*, 2014
- 154 **Wilhelm Schlag**, *A Course in Complex Analysis and Riemann Surfaces*, 2014
- 153 **Terence Tao**, *Hilbert's Fifth Problem and Related Topics*, 2014
- 152 **Gábor Székelyhidi**, *An Introduction to Extremal Kähler Metrics*, 2014
- 151 **Jennifer Schultens**, *Introduction to 3-Manifolds*, 2014
- 150 **Joe Diestel and Angela Spalsbury**, *The Joys of Haar Measure*, 2013
- 149 **Daniel W. Stroock**, *Mathematics of Probability*, 2013
- 148 **Luis Barreira and Yakov Pesin**, *Introduction to Smooth Ergodic Theory*, 2013
- 147 **Xingzhi Zhan**, *Matrix Theory*, 2013
- 146 **Aaron N. Siegel**, *Combinatorial Game Theory*, 2013
- 145 **Charles A. Weibel**, *The K-book*, 2013
- 144 **Shun-Jen Cheng and Weiqiang Wang**, *Dualities and Representations of Lie Superalgebras*, 2012

For a complete list of titles in this series, visit the
AMS Bookstore at www.ams.org/bookstore/gsmseries/.

This book is about differentiation of functions. It is divided into two parts, which can be used as different textbooks, one for an advanced undergraduate course in functions of one variable and one for a graduate course on Sobolev functions. The first part develops the theory of monotone, absolutely continuous, and bounded variation functions of one variable and their relationship with Lebesgue–Stieltjes measures and Sobolev functions. It also studies decreasing rearrangement and curves. The second edition includes a chapter on functions mapping time into Banach spaces.



Photo by Adella Guo

The second part of the book studies functions of several variables. It begins with an overview of classical results such as Rademacher’s and Stepanoff’s differentiability theorems, Whitney’s extension theorem, Brouwer’s fixed point theorem, and the divergence theorem for Lipschitz domains. It then moves to distributions, Fourier transforms and tempered distributions.

The remaining chapters are a treatise on Sobolev functions. The second edition focuses more on higher order derivatives and it includes the interpolation theorems of Gagliardo and Nirenberg. It studies embedding theorems, extension domains, chain rule, superposition, Poincaré’s inequalities and traces.

A major change compared to the first edition is the chapter on Besov spaces, which are now treated using interpolation theory.

ISBN 978-1-4704-2921-8



9 781470 429218

GSM/I8I



For additional information
and updates on this book, visit

www.ams.org/bookpages/gsm-181



www.ams.org